

Contents

	Preface	17
.1	Choices this book makes	18
.2	Code	18
.3	Sources	18
.4	How to Use This Book	19
	Parts and chapters	19
	Reading paths	19
	Prerequisites	19
	Exercises	19
.5	Notation	19
1	Chapter 1	21
	Matrix Algebra for Linear Models	21
1.0.1	Roadmap	21
1.1	Vectors, matrices and the operations we use	21
1.1.1	Conventions	22
1.1.2	Products as linear combinations	22
1.1.3	Inner products, lengths and the Cauchy–Schwarz inequality	23
1.1.4	Quadratic forms	23
1.2	Rank, column space and null space	24
1.2.1	Subspaces and dimension	24
1.2.2	The four subspaces of a matrix	24
1.2.3	Rank of products and of $\mathbf{A}^\top \mathbf{A}$	25
1.2.4	Nonsingular submatrices	26
1.2.5	Exercises	26
1.2.5.1	A. Check your understanding	26
1.2.5.2	B. Practice	27
1.3	Inverses and generalized inverses	27
1.3.1	The inverse of a square matrix	27
1.3.2	Generalized inverses	28
1.3.3	The Moore–Penrose inverse	29
1.4	Systems of equations and consistency	30
1.4.1	Exercises	32
1.4.1.1	A. Check your understanding	32
1.4.1.2	B. Practice	33
1.4.1.3	C. Going deeper	34

1.5	Orthogonality and orthogonal matrices	34
1.5.1	Orthogonal vectors and complements	34
1.5.2	Gram–Schmidt and the QR factorization	34
1.5.3	Orthogonal matrices	35
1.5.4	Exercises	36
1.5.4.1	A. Check your understanding	36
1.5.4.2	B. Practice	36
1.6	Trace, determinants and partitioned matrices	37
1.6.1	Trace	37
1.6.2	Determinants	38
1.6.3	Schur complements and block determinants	38
1.6.4	Inverses of partitioned matrices	40
1.6.5	Exercises	42
1.6.5.1	A. Check your understanding	42
1.6.5.2	B. Practice	42
1.7	Eigenvalues, the spectral theorem and positive definiteness	43
1.7.1	Eigenvalues	43
1.7.2	The spectral theorem	44
1.7.3	Positive definite and nonnegative definite matrices	45
1.7.4	Extremal properties of eigenvalues	48
1.7.5	Exercises	49
1.7.5.1	A. Check your understanding	49
1.7.5.2	B. Practice	50
1.7.5.3	C. Going deeper	50
1.8	The singular value decomposition and conditioning	51
1.8.1	The condition number	52
1.8.2	Exercises	54
1.8.2.1	B. Practice	54
1.9	Idempotent matrices	55
1.9.1	Exercises	57
1.9.1.1	A. Check your understanding	57
1.9.1.2	B. Practice	57
1.10	Kronecker products and the vec operator	57
1.11	Vector and matrix calculus	59
1.11.1	Gradients of linear and quadratic functions	59
1.11.2	Constrained minimization	60
1.11.3	Derivatives with respect to matrices and scalars	61
1.11.4	Exercises	62
1.11.4.1A	A. Check your understanding	62
1.11.4.2B	B. Practice	62
1.12	Summary and Notes	63
1.12.1	Summary	63
1.12.2	Notes and sources	64
1.12.3	References	65

2	Chapter 2	66
	Random Vectors: Expectation and Covariance Operators	66
2.0.1	Roadmap	66
2.1	Mean vectors and covariance matrices	66
2.1.1	Random matrices and their expectation	67
2.1.2	Mean vector and covariance matrix	68
2.1.3	Uncorrelated is not independent	69
2.1.4	Exercises	69
2.1.4.1	A. Check your understanding	69
2.1.4.2	B. Practice	69
2.2	Linear and affine transformations	70
2.2.1	The transformation rules	70
2.2.2	Covariance matrices are nonnegative definite	72
2.2.3	Standardization and Mahalanobis distance	73
2.2.4	Second moments in the linear model	75
2.2.5	Exercises	75
2.2.5.1	A. Check your understanding	75
2.2.5.2	B. Practice	76
2.2.5.3	C. Going deeper	77
2.3	Expectation of quadratic forms	77
2.3.1	The basic formula	77
2.3.2	The variance needs fourth moments	79
2.3.3	Exercises	81
2.3.3.1	A. Check your understanding	81
2.3.3.2	B. Practice	81
2.4	Correlation matrices	82
2.4.1	Exercises	84
2.4.1.1	A. Check your understanding	84
2.4.1.2	B. Practice	84
2.4.1.3	C. Going deeper	85
2.5	Moments of partitioned random vectors	85
2.5.1	Blocks of the mean and covariance	85
2.5.2	Linear prediction and residual covariance	86
2.5.3	Conditioning: total expectation and total covariance	88
2.5.4	Exercises	88
2.5.4.1	A. Check your understanding	88
2.5.4.2	B. Practice	89
2.5.4.3	C. Going deeper	89
2.6	Moment generating functions and independence	90
2.6.1	Exercises	92
2.6.1.1	A. Check your understanding	92
2.6.1.2	B. Practice	92
2.7	Sample moments and their computation	92
2.7.1	Exercises	94
2.7.1.1	B. Practice	94

2.8	Summary and Notes	94
2.8.1	Summary	94
2.8.2	Notes and sources	95
2.8.3	References	96
3	Chapter 3	97
	The Multivariate Normal Distribution	97
3.0.1	Roadmap	97
3.1	Definition through linear combinations	97
3.1.1	The univariate normal, including the degenerate case	98
3.1.2	The definition	98
3.1.3	Existence	99
3.1.4	Singular normal distributions	100
3.1.5	Why linear combinations suffice	101
3.1.6	Exercises	101
3.1.6.1	A. Check your understanding	101
3.1.6.2	B. Practice	101
3.2	Moment generating function and density	102
3.2.1	Moment generating functions of random vectors	102
3.2.2	The density	103
3.2.3	Contours and the Mahalanobis distance	104
3.2.4	Exercises	105
3.2.4.1	A. Check your understanding	105
3.2.4.2	B. Practice	106
3.2.4.3	C. Going deeper	107
3.3	Linear transformations and marginals	107
3.3.1	Affine maps	107
3.3.2	The rank of a normal vector	108
3.3.3	Independent normal vectors	109
3.3.4	Normal marginals are not enough	109
3.3.5	Exercises	111
3.3.5.1	A. Check your understanding	111
3.3.5.2	B. Practice	111
3.4	Independence and zero covariance	112
3.4.1	Exercises	114
3.4.1.1	A. Check your understanding	114
3.4.1.2	B. Practice	114
3.4.1.3	C. Going deeper	115
3.5	Conditional distributions	115
3.5.1	Partial covariance with a generalized inverse	115
3.5.2	The conditional distribution	116
3.5.3	What the theorem says	117
3.5.4	The conditional mean is the best predictor	120
3.5.5	Exercises	121
3.5.5.1	A. Check your understanding	121
3.5.5.2	B. Practice	121

3.5.5.3 C. Going deeper	122
3.6 Partial correlation	123
3.6.1 Definition and interpretations	123
3.6.2 Computing partial correlations	124
3.6.3 Sample partial correlations	125
3.6.4 Exercises	127
3.6.4.1 A. Check your understanding	127
3.6.4.2 B. Practice	127
3.7 Summary and Notes	128
3.7.1 Summary	128
3.7.2 Notes and sources	129
3.7.3 References	130
4 Chapter 4	131
Quadratic Forms and Their Distributions	131
4.0.1 Roadmap	131
4.1 Noncentral chi-squared, t and F distributions	131
4.1.1 Noncentral chi-squared	132
4.1.2 Noncentral F and t	135
4.1.3 Exercises	136
4.1.3.1 A. Check your understanding	136
4.1.3.2 B. Practice	137
4.2 Mean and variance of quadratic forms	138
4.2.1 The mean needs no normality	138
4.2.2 A canonical representation	139
4.2.3 Variance and covariances	140
4.2.4 Exercises	142
4.2.4.1 A. Check your understanding	142
4.2.4.2 B. Practice	142
4.2.4.3 C. Going deeper	143
4.3 When a quadratic form is chi-squared	144
4.3.1 Spherical normal vectors	144
4.3.2 General covariance matrices	145
4.3.3 Forms that are not chi-squared	148
4.3.4 Exercises	149
4.3.4.1 A. Check your understanding	149
4.3.4.2 B. Practice	149
4.3.4.3 C. Going deeper	150
4.4 Independence of quadratic forms and linear forms	150
4.4.1 The independence theorems	150
4.4.2 Orthogonal projections of a spherical normal vector	152
4.4.3 Preview: the F statistic for nested models	153
4.4.4 Exercises	155
4.4.4.1 A. Check your understanding	155
4.4.4.2 B. Practice	155

4.5	The Cochran theorem	156
4.5.1	The algebra	156
4.5.2	The distributional statement	157
4.5.3	Exercises	159
4.5.3.1	A. Check your understanding	159
4.5.3.2	B. Practice	159
4.5.3.3	C. Going deeper	160
4.6	Summary and Notes	160
4.6.1	Summary	160
4.6.2	Notes and sources	161
4.6.3	References	162
5	Chapter 5	163
	The Model, Least Squares and the Normal Equations	163
5.0.1	Roadmap	163
5.1	What a regression model claims	164
5.1.1	The regression function	164
5.1.2	What the linear model adds	165
5.1.3	Fixed and random regressors	167
5.1.4	What regression is used for	168
5.1.5	Exercises	168
5.1.5.1	A. Check your understanding	168
5.1.5.2	B. Practice	169
5.1.5.3	C. Going deeper	169
5.2	Simple linear regression as a warm-up	170
5.2.1	The least squares line	170
5.2.2	Regression toward the mean	172
5.2.3	A worked example	172
5.2.4	Regression through the origin	174
5.2.5	Exercises	174
5.2.5.1	A. Check your understanding	174
5.2.5.2	B. Practice	175
5.2.5.3	C. Going deeper	175
5.3	The multiple linear regression model	175
5.3.1	Matrix form	176
5.3.2	The intercept	177
5.3.3	Linear in the parameters, not in the regressors	177
5.3.4	Categorical regressors and indicator variables	179
5.3.5	Interactions	180
5.3.6	When is the model matrix of full rank?	180
5.3.7	Exercises	181
5.3.7.1	A. Check your understanding	181
5.3.7.2	B. Practice	181
5.3.7.3	C. Going deeper	182
5.4	Least squares and the normal equations	182
5.4.1	The least squares estimate	182

5.4.2	Fitted values, residuals and the hat matrix	184
5.4.3	Centring the regressors	185
5.4.4	Indicator variables	186
5.4.5	A worked example	187
5.4.6	Exercises	188
5.4.6.1	A. Check your understanding	188
5.4.6.2	B. Practice	188
5.4.6.3	C. Going deeper	189
5.5	Properties of the estimator under second-moment assumptions	189
5.5.1	Mean and covariance of the estimator	189
5.5.2	Fitted values and residuals	191
5.5.3	The straight line	191
5.5.4	Only two moments are needed	193
5.5.5	Random regressors	193
5.5.6	Consistency	194
5.5.7	What if the assumptions fail?	195
5.5.8	Exercises	195
5.5.8.1	A. Check your understanding	195
5.5.8.2	B. Practice	195
5.5.8.3	C. Going deeper	196
5.6	Estimating the error variance	197
5.6.1	The residual sum of squares	197
5.6.2	Dividing by n	199
5.6.3	The variance of s^2	200
5.6.4	When the mean is wrong	201
5.6.5	Exercises	202
5.6.5.1	A. Check your understanding	202
5.6.5.2	B. Practice	202
5.6.5.3	C. Going deeper	203
5.7	Interpreting coefficients	203
5.7.1	What a coefficient measures	203
5.7.2	Units, rescaling and reparameterization	204
5.7.3	Logarithms	205
5.7.4	Terms that move together	206
5.7.5	Indicator variables	206
5.7.6	Omitted regressors	207
5.7.7	Exercises	208
5.7.7.1	A. Check your understanding	208
5.7.7.2	B. Practice	208
5.7.7.3	C. Going deeper	209
5.8	A panorama of regression models to come	209
5.8.1	Three components of a regression model	210
5.8.2	The ladder	210
5.8.3	A first look beyond the linear model	211
5.8.4	Exercises	213
5.8.4.1	A. Check your understanding	213
5.8.4.2	B. Practice	213
5.8.4.3	C. Going deeper	213

5.9	Summary and Notes	214
5.9.1	Summary	214
5.9.2	Notes and sources	215
5.9.3	References	215
6	Chapter 6	217
	Projections: The Geometry of Least Squares	217
6.0.1	Roadmap	217
6.1	Least squares as a nearest-point problem	217
6.1.1	Two pictures of the same data	218
6.1.2	The least squares principle, restated	218
6.1.3	A picture with three observations	219
6.1.4	Plan of the chapter	220
6.2	The subspaces we need	220
6.2.1	Subspaces, spans and dimension	220
6.2.2	Inner products, orthogonality and complements	221
6.2.3	Orthonormal bases and Gram–Schmidt	222
6.2.4	Direct sums and the orthogonal decomposition	223
6.2.5	Exercises	224
6.2.5.1	A. Check your understanding	224
6.2.5.2	B. Practice	224
6.2.5.3	C. Going deeper	225
6.3	Orthogonal projection	225
6.3.1	Nearest points	225
6.3.2	The projection as a matrix	225
6.3.3	Characterizing projection matrices	226
6.3.4	Generalized inverses	227
6.3.5	Sums and products of projections	230
6.3.6	Exercises	230
6.3.6.1	A. Check your understanding	230
6.3.6.2	B. Practice	231
6.3.6.3	C. Going deeper	232
6.4	Least squares via projection	232
6.4.1	The characterization of least squares estimates	232
6.4.2	Fitted values and residuals	233
6.4.3	Coefficients are not unique; fitted values are	234
6.4.4	Exercises	236
6.4.4.1	A. Check your understanding	236
6.4.4.2	B. Practice	236
6.4.4.3	C. Going deeper	236
6.5	Nested models and sequential projection	237
6.5.1	Two nested subspaces	237
6.5.2	Chains of subspaces	239
6.5.3	Two factors and when the pieces are orthogonal	240
6.5.4	Reduced models defined by constraints	241
6.5.5	Exercises	241

6.5.5.1	A. Check your understanding	241
6.5.5.2	B. Practice	241
6.5.5.3	C. Going deeper	242
6.6	The Frisch–Waugh–Lovell theorem	242
6.6.1	Statement and proof	242
6.6.2	Consequences	244
6.6.3	Fixed effects are a projection	246
6.6.4	Exercises	247
6.6.4.1	A. Check your understanding	247
6.6.4.2	B. Practice	247
6.6.4.3	C. Going deeper	248
6.7	Reparameterization, centring and R^2	248
6.7.1	Same subspace, same fit	248
6.7.2	Centring as a projection	249
6.7.3	R^2 is a squared cosine	249
6.7.4	Exercises	251
6.7.4.1	A. Check your understanding	251
6.7.4.2	B. Practice	251
6.7.4.3	C. Going deeper	252
6.8	Leverage as geometry	252
6.8.1	Leverage as distance from the centre	253
6.8.2	Exercises	254
6.8.2.1	A. Check your understanding	254
6.8.2.2	B. Practice	255
6.9	Other inner products	255
6.9.1	Projection in a general inner product	255
6.9.2	Generalized least squares	256
6.9.3	Weighted least squares	258
6.9.4	Oblique projections	259
6.9.5	Exercises	260
6.9.5.1	A. Check your understanding	260
6.9.5.2	B. Practice	260
6.10	Computation	261
6.10.1	QR: Gram–Schmidt as a factorization	261
6.10.2	Conditioning	262
6.10.3	Rank deficiency and the singular value decomposition	263
6.10.4	Exercises	264
6.10.4.1A	A. Check your understanding	264
6.10.4.2B	B. Practice	264
6.10.4.3C	C. Going deeper	265
6.11	Projection in the population: best linear prediction	265
6.11.1	Random variables as vectors	265
6.11.2	Conditional expectation is also a projection	266
6.11.3	Least squares estimates the population projection	267
6.12	Summary and Notes	269
6.12.1	Summary	269

6.12.2	Notes and sources	270
6.12.3	References	271
7	Chapter 7	272
	Optimality: Gauss–Markov, Maximum Likelihood and Minimum Variance	272
7.0.1	Roadmap	272
7.1	Linear unbiased estimation	273
7.1.1	The setting	273
7.1.2	Linear and unbiased	273
7.1.3	The geometry of the unbiased class	274
7.1.4	Three estimators of a slope	275
7.1.5	Why these restrictions?	277
7.1.6	Exercises	277
7.1.6.1	A. Check your understanding	277
7.1.6.2	B. Practice	277
7.1.6.3	C. Going deeper	278
7.2	The Gauss–Markov theorem	278
7.2.1	The theorem	278
7.2.2	What the matrix form buys	280
7.2.3	Correlated errors: Aitken’s theorem	282
7.2.4	Design: when is a coefficient estimated best?	283
7.2.5	What the theorem does not say	284
7.2.6	Exercises	285
7.2.6.1	A. Check your understanding	285
7.2.6.2	B. Practice	285
7.2.6.3	C. Going deeper	286
7.3	Maximum likelihood under normality	287
7.3.1	The likelihood	288
7.3.2	The maximum likelihood estimates	288
7.3.3	The bias of the variance estimate	290
7.3.4	Profile likelihood	291
7.3.5	Exercises	294
7.3.5.1	A. Check your understanding	294
7.3.5.2	B. Practice	294
7.3.5.3	C. Going deeper	295
7.4	Sufficiency, completeness and minimum variance	296
7.4.1	Sufficiency	296
7.4.2	Rao–Blackwell and Lehmann–Scheffé	297
7.4.3	Best unbiased estimators in the normal linear model	298
7.4.4	Rao–Blackwellization by reflection	300
7.4.5	The Cramér–Rao bound	301
7.4.6	What changes without normality	302
7.4.7	Exercises	302
7.4.7.1	A. Check your understanding	302
7.4.7.2	B. Practice	303
7.4.7.3	C. Going deeper	303

7.5	Sampling distributions of the estimates	304
7.5.1	The main theorem	304
7.5.2	Standardizing: the t statistic	305
7.5.3	Several functions at once	306
7.5.4	Without normality	308
7.5.5	Exercises	309
7.5.5.1	A. Check your understanding	309
7.5.5.2	B. Practice	310
7.5.5.3	C. Going deeper	310
7.6	Bayesian thread I: the conjugate linear model	311
7.6.1	The normal-inverse-gamma family	311
7.6.2	The conjugate update	312
7.6.3	The posterior mean as shrinkage	313
7.6.4	The flat-prior limit	314
7.6.5	A worked example	315
7.6.6	Exercises	317
7.6.6.1	A. Check your understanding	317
7.6.6.2	B. Practice	318
7.6.6.3	C. Going deeper	318
7.7	Summary and Notes	319
7.7.1	Summary	319
7.7.2	Notes and sources	320
7.7.3	References	321
8	Chapter 8	323
	Estimability, Identifiability and Indicator Variables	323
8.0.1	Roadmap	323
8.1	Identifiability	324
8.1.1	What the distribution of the data can reveal	324
8.1.2	When the parameter is identifiable	325
8.1.3	Where rank deficiency comes from	326
8.1.4	Linear and nonlinear identifiable functions	328
8.1.5	Exercises	329
8.1.5.1	A. Check your understanding	329
8.1.5.2	B. Practice	329
8.1.5.3	C. Going deeper	330
8.2	Estimable functions	330
8.2.1	Linear unbiased estimators	330
8.2.2	Estimable is the same as identifiable	331
8.2.3	The mean at a new point	332
8.2.4	Many unbiased estimators, one least squares estimator	333
8.2.5	What estimability does not depend on	334
8.2.6	Exercises	334
8.2.6.1	A. Check your understanding	334
8.2.6.2	B. Practice	335
8.2.6.3	C. Going deeper	335

8.3	Characterizations of estimability	336
8.3.1	Eight equivalent conditions	336
8.3.2	The estimator and its variance	337
8.3.3	What a generalized inverse reports for a nonestimable function	337
8.3.4	Deciding estimability in floating point	340
8.3.5	Exercises	341
8.3.5.1	A. Check your understanding	341
8.3.5.2	B. Practice	341
8.3.5.3	C. Going deeper	342
8.4	Reparameterization and side conditions	342
8.4.1	Full-rank reparameterizations	343
8.4.2	Side conditions	344
8.4.3	What the constrained coefficients mean	345
8.4.4	What software does	348
8.4.5	Exercises	348
8.4.5.1	A. Check your understanding	348
8.4.5.2	B. Practice	349
8.4.5.3	C. Going deeper	349
8.5	Indicator variables and the coding of a factor	350
8.5.1	One binary regressor	350
8.5.2	A factor with g levels	350
8.5.3	The standard codings	352
8.5.4	Choosing a coding	354
8.5.5	Exercises	355
8.5.5.1	A. Check your understanding	355
8.5.5.2	B. Practice	355
8.5.5.3	C. Going deeper	356
8.6	Several factors, interactions and covariates	357
8.6.1	Two factors without interaction	357
8.6.2	Two factors with interaction	359
8.6.3	A factor and a numeric covariate	361
8.6.4	Broken lines	363
8.6.5	Exercises	363
8.6.5.1	A. Check your understanding	363
8.6.5.2	B. Practice	364
8.6.5.3	C. Going deeper	364
8.7	Contrasts	365
8.7.1	Definition and estimability	365
8.7.2	Sums of squares and orthogonal contrasts	366
8.7.3	Helmert contrasts	367
8.7.4	Polynomial contrasts	367
8.7.5	Coding matrices are not contrast matrices	369
8.7.6	Contrasts in the two-way additive model	370
8.7.7	Exercises	371
8.7.7.1	A. Check your understanding	371
8.7.7.2	B. Practice	371
8.7.7.3	C. Going deeper	371

8.8	Summary and Notes	372
8.8.1	Summary	372
8.8.2	Notes and sources	373
8.8.3	References	374
9	Chapter 9	375
	Sums of Squares and Orthogonal Decomposition	375
9.0.1	Roadmap	375
9.1	Decomposing the total sum of squares	376
9.1.1	Orthogonal decompositions of the model space	376
9.1.2	Degrees of freedom are dimensions	377
9.1.3	The regression analysis of variance table	378
9.1.4	Models without an intercept	380
9.1.5	Coarsening and refining	381
9.1.6	Exercises	381
9.1.6.1	A. Check your understanding	381
9.1.6.2	B. Practice	381
9.1.6.3	C. Going deeper	382
9.2	Extra sums of squares	382
9.2.1	Notation	382
9.2.2	Adding regressors to a fitted model	383
9.2.3	One column at a time	385
9.2.4	Sums of squares for linear hypotheses	386
9.2.5	Constraints that are not estimable	388
9.2.6	Exercises	389
9.2.6.1	A. Check your understanding	389
9.2.6.2	B. Practice	389
9.2.6.3	C. Going deeper	390
9.3	Sequential and partial sums of squares	390
9.3.1	Sequential sums of squares	391
9.3.2	Partial sums of squares: Types II and III	392
9.3.3	What each sum of squares tests	393
9.3.4	An unbalanced example	395
9.3.5	Which one to use	397
9.3.6	Exercises	398
9.3.6.1	A. Check your understanding	398
9.3.6.2	B. Practice	398
9.3.6.3	C. Going deeper	399
9.4	Orthogonal designs	399
9.4.1	When the order does not matter	400
9.4.2	Balanced and proportional layouts	401
9.4.3	Orthogonal polynomials	403
9.4.4	Single-degree-of-freedom decompositions	405
9.4.5	Exercises	406
9.4.5.1	A. Check your understanding	406
9.4.5.2	B. Practice	407
9.4.5.3	C. Going deeper	408

9.5	The coefficient of determination and its limits	408
9.5.1	A maximal correlation	408
9.5.2	R^2 when nothing is going on	409
9.5.3	Adjusted R^2	411
9.5.4	Partial R^2	412
9.5.5	What R^2 does not measure	414
9.5.6	Exercises	415
9.5.6.1	A. Check your understanding	415
9.5.6.2	B. Practice	415
9.5.6.3	C. Going deeper	416
9.6	Expected mean squares	416
9.6.1	Expected sums of squares for a decomposition	417
9.6.2	The logic of the F ratio	418
9.6.3	Expected mean squares in an unbalanced layout	419
9.6.4	Exercises	422
9.6.4.1	A. Check your understanding	422
9.6.4.2	B. Practice	422
9.6.4.3	C. Going deeper	423
9.7	Summary and notes	423
9.7.1	Summary	423
9.7.2	Notes and sources	424
9.7.3	References	425
10	Chapter 10	427
	Computation: QR, SVD and Conditioning	427
10.0.1	Roadmap	427
10.1	Why not invert the cross-product matrix	428
10.1.1	Floating-point numbers	428
10.1.2	Accumulating rounding errors	429
10.1.3	Conditioning and stability	430
10.1.4	What forming the cross-product matrix loses	431
10.1.5	Why not the inverse either	432
10.1.6	Counting the work	432
10.1.7	Exercises	433
10.1.7.1A	A. Check your understanding	433
10.1.7.2B	B. Practice	433
10.1.7.3C	C. Going deeper	434
10.2	The Cholesky factorization	434
10.2.1	The algorithm	434
10.2.2	Everything from one factorization	436
10.2.3	When the normal equations are good enough	437
10.2.4	The sweep operator	437
10.2.5	Forming cross-products: the centring problem	439
10.2.6	Exercises	441
10.2.6.1A	A. Check your understanding	441
10.2.6.2B	B. Practice	441
10.2.6.3C	C. Going deeper	442

10.3	Householder and Givens QR	442
10.3.1	Householder reflections	442
10.3.2	Regression quantities from the factorization	445
10.3.3	Givens rotations	446
10.3.4	Gram–Schmidt revisited	448
10.3.5	Exercises	449
10.3.5.1A.	Check your understanding	449
10.3.5.2B.	Practice	449
10.3.5.3C.	Going deeper	450
10.4	The singular value decomposition	450
10.4.1	Least squares through the SVD	450
10.4.2	Computing the SVD	451
10.4.3	Numerical rank	452
10.4.4	Truncated SVD as an estimator	453
10.4.5	Exercises	455
10.4.5.1A.	Check your understanding	455
10.4.5.2B.	Practice	455
10.4.5.3C.	Going deeper	455
10.5	Conditioning and perturbation bounds	456
10.5.1	The perturbation theorem	456
10.5.2	What the theorem says about algorithms	458
10.5.3	Column scaling	459
10.5.4	Numerical error and statistical error	460
10.5.5	Exercises	461
10.5.5.1A.	Check your understanding	461
10.5.5.2B.	Practice	461
10.5.5.3C.	Going deeper	461
10.6	Updating and downdating	462
10.6.1	Adding and deleting observations	462
10.6.2	Leave-one-out quantities from one fit	464
10.6.3	Adding and deleting variables	465
10.6.4	A moving window	466
10.6.5	Exercises	467
10.6.5.1A.	Check your understanding	467
10.6.5.2B.	Practice	468
10.6.5.3C.	Going deeper	468
10.7	Rank-revealing factorizations	469
10.7.1	QR with column pivoting	469
10.7.2	Basic solutions and aliased columns	472
10.7.3	Choosing a tolerance	473
10.7.4	Exercises	474
10.7.4.1A.	Check your understanding	474
10.7.4.2B.	Practice	475
10.7.4.3C.	Going deeper	475
10.8	Summary and Notes	475
10.8.1	Summary	475
10.8.2	Notes and sources	476

10.8.3	References	477
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Preface

Regression is taught under at least five different descriptions, and each one is correct. In one, regression is least squares together with the distribution theory of quadratic forms. In another, it is orthogonal projection onto a subspace, and everything else follows as a corollary. A third starts from likelihood and treats the normal linear model as the Gaussian member of the generalized linear family. A fourth defines regression as estimating a conditional distribution, with linearity one structural assumption among many. A fifth treats it as a data-analytic procedure whose assumptions must be checked, one at a time, against the data.

Most textbooks pick one of these descriptions and stay with it. This book uses all of them, in order, as rungs of a single ladder. Parts I–III set up the classical linear model and its exact theory. After that, *each part relaxes exactly one assumption*, and asks what survives, what must be rebuilt, and what new tools the relaxation requires.

Part	What changes	What we study
I	—	The apparatus: matrices, random vectors, the normal distribution, quadratic forms
II	—	$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $E(\boldsymbol{\varepsilon}) = \mathbf{0}$, $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$, as a projection problem
III	adds $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$	Exact finite-sample inference
IV	drops $\text{rank}(\mathbf{X}) = p$	Estimability, analysis of variance, designed structure
V	audits every assumption	What breaks, how to see it, what inference survives
VI	drops unbiasedness	Shrinkage, selection, regularization, $p > n$
VII	drops $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$	Generalized least squares, general Gauss–Markov, mixed models
VIII	drops normality of $\mathbf{Y}$	Exponential families and generalized linear models
IX	drops the linear predictor	Polynomials, splines, smoothing, additive models
X	drops modeling only the mean	Quantile regression and distributional models

Choices this book makes

Geometry before distribution theory. Least squares is introduced as a nearest-point problem ([Chapter 6](#)) before any sampling theory. Fitted values, residuals and sums of squares are projections. Coefficients are coordinates, and coordinates depend on the basis you choose. This is the only framing in which rank-deficient models need no special treatment.

Estimability is an estimation idea. It appears in Part II, not as a preliminary to the analysis of variance. Putting it there is what allows Part IV to treat ANOVA as ordinary subspace comparison.

Computation is part of the theory. Conditioning, QR and the singular value decomposition come immediately after least squares ([Chapter 10](#)), because numerical stability and statistical stability are closely linked.

The Bayesian treatment is a thread, not a quarantined chapter. Boxed “Bayesian thread” sections close Parts II, III, VII and VIII. Each develops the Bayesian counterpart of the result just proved.

Departures have a theory. Part V opens by deriving what omitted variables, wrong covariance, non-normality, outliers and collinearity actually do to the estimator. Diagnostics and remedies come after that.

Material most regression texts omit. This book adds chapters on bootstrap and permutation inference ([Chapter 23](#)), errors in variables ([Chapter 24](#)), the causal interpretation of coefficients ([Chapter 25](#)), high-dimensional regression ([Chapter 28](#)) and missing data ([Chapter 41](#)).

Code

Every figure and every number obtained by computation is produced by a Python script in the book’s `code/` directory, and the scripts check their own claims with assertions. The code in the text is taken from those scripts. On this website each code block is a live cell: press **Run** (or Shift+Enter) to execute it in your browser, edit it, and run it again. Nothing needs to be installed; the first run downloads a Python environment with numpy, scipy, pandas and statsmodels. Cells on a page share their variables, and running a cell first runs the cells above it that have not run yet.

Sources

The coverage of this book was planned against six standard texts: Rencher and Schaalje (2008), Seber and Lee (2003), Christensen (2020), Sen and Srivastava (1990), Agresti (2015) and Fahrmeir et al. (2021). Their chapters were mapped onto a single prerequisite graph, and the order of this book respects that graph. Each chapter’s *Notes and sources* section names the treatments it drew on and the primary literature. The exposition, proofs, examples and exercises are original to this book.

How to Use This Book

Parts and chapters

Parts I–VI (Chapters 1–30) are the classical linear model, complete in itself: estimation, exact inference, designed experiments, diagnostics, and the move from unbiased estimation to regularization. Parts VII–X (Chapters 31–45) climb the rest of the ladder: correlated errors, non-normal responses, nonlinear predictors, and models for entire conditional distributions. Chapters are published here as they are written; the ones not yet on the site are named in plain text.

Reading paths

- **A first course on linear models.** Chapters 5–9, 11–12, 15, 19–20 and 29. Consult Part I as needed.
- **The theory spine.** Part I, then Chapters 6, 7, 8, 9, 11 and 31. This is the shortest route to a full structural understanding of the linear model.
- **For implementers.** Chapters 5, 6, 10 and 20, Part VI and Chapter 34.
- **A course on generalized models.** Chapters 5, 6 and 11, then Parts VII–X.

Prerequisites

Each chapter opens with a list of the earlier chapters it uses, taken from the prerequisite graph the book was planned against. Part I is written as reference material. Read it quickly the first time and come back to it when a later chapter sends you there.

Exercises

Exercises come at the end of each section in three groups: **A. Check your understanding** (routine: a definition or computation has been understood), **B. Practice** (core: the result is used later in the book) and **C. Going deeper** (hard: may need an idea not in the text). Many have a worked solution, folded under the exercise; try the exercise before opening it.

Notation

Symbol	Meaning
$\mathbf{Y}, \mathbf{y}$	response vector (random, observed), $n \times 1$
$\mathbf{X}$	model matrix, $n \times p$; columns $\mathbf{x}_1, \dots, \mathbf{x}_p$
$\boldsymbol{\beta}, \hat{\boldsymbol{\beta}}$	coefficient vector and a least squares estimate
$\boldsymbol{\varepsilon}$	error vector, $\boldsymbol{\varepsilon} = \mathbf{Y} - \mathbf{X}\boldsymbol{\beta}$
$\hat{\mathbf{Y}}, \hat{\boldsymbol{\varepsilon}}$	fitted values $\mathbf{X}\hat{\boldsymbol{\beta}}$ and residuals $\mathbf{Y} - \hat{\mathbf{Y}}$
σ^2	error variance
$\mathbb{R}^n$	n -dimensional real space with $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v}$
$\mathcal{C}(\mathbf{A}), \mathcal{N}(\mathbf{A})$	column space and null space of $\mathbf{A}$
$\mathcal{S}^\perp$	orthogonal complement of a subspace $\mathcal{S}$
$\mathbf{A}^\top, \mathbf{A}^-$	transpose; a generalized inverse ($\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$)
$\mathbf{M}$	orthogonal projection onto $\mathcal{C}(\mathbf{X})$
$\mathbf{M}_0$	orthogonal projection onto the column space of a reduced model

Symbol	Meaning
rank, tr, diag	rank, trace, diagonal matrix
1 , 0 , I	vector of ones, zero vector, identity
$\ \cdot\ $	Euclidean norm unless stated otherwise
E, Var, Cov	expectation, variance, covariance (matrix)
$N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	n -variate normal distribution
$\chi^2(r, \gamma)$, $F(r, s, \gamma)$	noncentral distributions, noncentrality γ

Vectors are columns. Matrices and vectors are set in bold upright type and scalars in italic. A bare “projection” always means an orthogonal projection with respect to the Euclidean inner product. [Section 6.9](#) covers the other kinds.

Chapter 1

Matrix Algebra for Linear Models

A linear model is a statement about vectors and matrices, and nearly every argument in this book is an argument about them. This chapter collects the matrix algebra the book relies on: rank and the fundamental subspaces, inverses and generalized inverses, consistent systems of equations, orthogonal matrices, determinants and partitioned matrices, the spectral theorem and positive definiteness, the singular value decomposition, idempotent matrices, Kronecker products and the calculus of quadratic forms. It is a reference chapter, meant to be skimmed first and consulted later, but every central result is proved.

What you need. None beyond the Notation chapter. A first course in linear algebra helps, but is not assumed.

1.0.1 Roadmap

- [Vectors, matrices and the operations we use](#)
- [Rank, column space and null space](#)
- [Inverses and generalized inverses](#)
- [Systems of equations and consistency](#)
- [Orthogonality and orthogonal matrices](#)
- [Trace, determinants and partitioned matrices](#)
- [Eigenvalues, the spectral theorem and positive definiteness](#)
- [The singular value decomposition and conditioning](#)
- [Idempotent matrices](#)
- [Kronecker products and the vec operator](#)
- [Vector and matrix calculus](#)
- [Summary and notes](#)

1.1 Vectors, matrices and the operations we use

This chapter works in $\mathbb{R}^n$ and with real matrices only. It fixes the conventions, states the results later chapters cite, and proves them. A reader who already knows linear algebra can go straight

to the results that are specific to statistics: generalized inverses (Section 1.3 and Section 1.4), Schur complements and the Woodbury identity (Section 1.6), nonnegative definite matrices (Section 1.7), conditioning (Section 1.8) and idempotent matrices (Section 1.9). Orthogonal projections, the geometric heart of least squares, get only the minimum here. Chapter 6 develops them.

1.1.1 Conventions

An $m \times n$ matrix $\mathbf{A} = (a_{ij})$ has m rows and n columns. We write $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_n]$ for its columns and $\mathbf{a}_{(i)}^\top$ for its i th row. A **vector** is always a column, so $\mathbf{x} \in \mathbb{R}^n$ is an $n \times 1$ matrix and $\mathbf{x}^\top$ is a row. Bold upright letters denote vectors and matrices, and italic letters denote scalars. The special matrices we use constantly are the identity $\mathbf{I}_n$, the zero matrix $\mathbf{0}$, the vector of ones $\mathbf{1}_n$, the matrix of ones $\mathbf{J}_n = \mathbf{1}_n \mathbf{1}_n^\top$, and the coordinate vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$, the columns of $\mathbf{I}_n$. Subscripts giving sizes are dropped when the size is clear. $\text{diag}(d_1, \dots, d_n)$ is the diagonal matrix with the given diagonal entries. A square matrix is **symmetric** if $\mathbf{A}^\top = \mathbf{A}$, *upper triangular* if $a_{ij} = 0$ for $i > j$, and *lower triangular* if $\mathbf{A}^\top$ is upper triangular.

The **transpose** satisfies $(\mathbf{A}^\top)^\top = \mathbf{A}$, $(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top$ and, most usefully,

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top. \quad (1.1.1)$$

1.1.2 Products as linear combinations

If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times q$, the (i, j) entry of $\mathbf{AB}$ is $\sum_k a_{ik} b_{kj}$. Three equivalent readings of the product carry most of the arguments in this book:

$$\mathbf{Ab} = \sum_{j=1}^n b_j \mathbf{a}_j, \quad \mathbf{AB} = [\mathbf{Ab}_1, \dots, \mathbf{Ab}_q], \quad \mathbf{AB} = \sum_{k=1}^n \mathbf{a}_k \mathbf{b}_{(k)}^\top. \quad (1.1.2)$$

The first says that $\mathbf{Ab}$ combines the columns of $\mathbf{A}$, with weights taken from $\mathbf{b}$. The second says that each column of $\mathbf{AB}$ is such a combination. The third writes a product as a sum of n rank-one matrices (outer products), and is the form in which spectral and singular value decompositions are usually used. Transposing the first reading shows that each row of $\mathbf{AB}$ is a combination of the rows of $\mathbf{B}$.

Matrices partitioned into conformable blocks multiply as if the blocks were scalars, as long as the order of factors is kept:

$$\begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix} \begin{pmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11}\mathbf{B}_{11} + \mathbf{A}_{12}\mathbf{B}_{21} & \mathbf{A}_{11}\mathbf{B}_{12} + \mathbf{A}_{12}\mathbf{B}_{22} \\ \mathbf{A}_{21}\mathbf{B}_{11} + \mathbf{A}_{22}\mathbf{B}_{21} & \mathbf{A}_{21}\mathbf{B}_{12} + \mathbf{A}_{22}\mathbf{B}_{22} \end{pmatrix}.$$

Multiplication is associative and distributive but not commutative, and $\mathbf{AB} = \mathbf{0}$ does not force $\mathbf{A} = \mathbf{0}$ or $\mathbf{B} = \mathbf{0}$. The **Hadamard product** $\mathbf{A} \circ \mathbf{B} = (a_{ij}b_{ij})$ multiplies entrywise. It appears only occasionally.

1.1.3 Inner products, lengths and the Cauchy–Schwarz inequality

The **inner product** of $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v} = \sum_i u_i v_i$, and the **Euclidean norm** is $\|\mathbf{u}\| = (\mathbf{u}^\top \mathbf{u})^{1/2}$. The entries of $\mathbf{A}^\top \mathbf{A}$ are the inner products of the columns of $\mathbf{A}$, and those of $\mathbf{A}\mathbf{A}^\top$ are the inner products of its rows. In particular

$$\sum_{i,j} a_{ij}^2 = \sum_j \|\mathbf{a}_j\|^2, \quad (1.1.3)$$

so $\mathbf{A}^\top \mathbf{A} = \mathbf{0}$ forces every column to be zero, that is, $\mathbf{A} = \mathbf{0}$. The square root of Eq. 1.1.3 is the **Frobenius norm** $\|\mathbf{A}\|_F$.

Proposition 1.1.1 — Cauchy–Schwarz

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, $(\mathbf{u}^\top \mathbf{v})^2 \leq (\mathbf{u}^\top \mathbf{u})(\mathbf{v}^\top \mathbf{v})$, with equality iff one vector is a multiple of the other. Consequently $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

Proof. If $\mathbf{v} = \mathbf{0}$ there is nothing to prove. Otherwise, the quadratic $q(t) = \|\mathbf{u} - t\mathbf{v}\|^2 = \mathbf{u}^\top \mathbf{u} - 2t \mathbf{u}^\top \mathbf{v} + t^2 \mathbf{v}^\top \mathbf{v}$ is nonnegative for all t , so its discriminant $4(\mathbf{u}^\top \mathbf{v})^2 - 4(\mathbf{u}^\top \mathbf{u})(\mathbf{v}^\top \mathbf{v})$ is at most zero. Equality holds iff q has a real root, that is, iff $\mathbf{u} = t\mathbf{v}$ for some t . The triangle inequality follows by expanding $\|\mathbf{u} + \mathbf{v}\|^2$ and bounding the cross term. ■

1.1.4 Quadratic forms

A **quadratic form** in $\mathbf{x} \in \mathbb{R}^n$ is a function $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \sum_{i,j} a_{ij} x_i x_j$. Since $\mathbf{x}^\top \mathbf{A} \mathbf{x}$ is a scalar it equals its own transpose $\mathbf{x}^\top \mathbf{A}^\top \mathbf{x}$, so

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} = \mathbf{x}^\top \left(\frac{1}{2} (\mathbf{A} + \mathbf{A}^\top) \right) \mathbf{x}. \quad (1.1.4)$$

Every quadratic form therefore has a symmetric matrix, and that matrix is unique: if $\mathbf{x}^\top \mathbf{S} \mathbf{x} = 0$ for all $\mathbf{x}$ with $\mathbf{S}$ symmetric, then taking $\mathbf{x} = \mathbf{e}_i$ gives $s_{ii} = 0$, and $\mathbf{x} = \mathbf{e}_i + \mathbf{e}_j$ gives $2s_{ij} = 0$. From now on we always take the matrix in a quadratic form to be symmetric.

Example. (Sums of squares as quadratic forms)

The sum of squared deviations of $\mathbf{y} \in \mathbb{R}^n$ from its mean is

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \mathbf{y}^\top \mathbf{y} - n\bar{y}^2 = \mathbf{y}^\top \left(\mathbf{I}_n - \frac{1}{n} \mathbf{J}_n \right) \mathbf{y},$$

because $n\bar{y}^2 = (\mathbf{1}^\top \mathbf{y})^2 / n = \mathbf{y}^\top \mathbf{J}_n \mathbf{y} / n$. The **centring matrix** $\mathbf{I}_n - n^{-1} \mathbf{J}_n$ is symmetric and satisfies $(\mathbf{I}_n - n^{-1} \mathbf{J}_n)^2 = \mathbf{I}_n - n^{-1} \mathbf{J}_n$, because $\mathbf{J}_n^2 = n\mathbf{J}_n$. Every sum of squares

in an analysis of variance is a quadratic form whose matrix has these two properties (Section 1.9).

1.2 Rank, column space and null space

1.2.1 Subspaces and dimension

A nonempty set $\mathcal{S} \subseteq \mathbb{R}^n$ is a **subspace** if $\mathbf{u}, \mathbf{v} \in \mathcal{S}$ and $a, b \in \mathbb{R}$ imply $a\mathbf{u} + b\mathbf{v} \in \mathcal{S}$. Vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are **linearly independent** if $\sum_j c_j \mathbf{v}_j = \mathbf{0}$ only for $c_1 = \dots = c_k = 0$, that is, if the matrix $[\mathbf{v}_1, \dots, \mathbf{v}_k]$ sends only the zero vector to zero. Their **span** is the set of all their linear combinations. A **basis** of $\mathcal{S}$ is an independent set that spans it.

We use the following facts of elementary linear algebra without proof (Halmos 1958, chapter 1). Every subspace of $\mathbb{R}^n$ has a basis. All bases of a subspace have the same number of elements, its **dimension** $\dim \mathcal{S}$. In an r -dimensional subspace, any r independent vectors form a basis, and any independent set can be extended to a basis. If $\mathcal{S} \subseteq \mathcal{T}$ and $\dim \mathcal{S} = \dim \mathcal{T}$, then $\mathcal{S} = \mathcal{T}$. Finally, if $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$, the union of a basis of $\mathcal{S}$ and a basis of $\mathcal{T}$ is independent, so the sum $\mathcal{S} + \mathcal{T} = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in \mathcal{S}, \mathbf{v} \in \mathcal{T}\}$ has dimension $\dim \mathcal{S} + \dim \mathcal{T}$. Such a sum is called **direct** and written $\mathcal{S} \oplus \mathcal{T}$.

1.2.2 The four subspaces of a matrix

An $m \times n$ matrix $\mathbf{A}$ carries two subspaces of $\mathbb{R}^m$ and two of $\mathbb{R}^n$:

$$\begin{aligned} \mathcal{C}(\mathbf{A}) &= \{\mathbf{A}\mathbf{b} : \mathbf{b} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m, & \mathcal{N}(\mathbf{A}) &= \{\mathbf{b} \in \mathbb{R}^n : \mathbf{A}\mathbf{b} = \mathbf{0}\} \subseteq \mathbb{R}^n, \\ \mathcal{C}(\mathbf{A}^\top) &\subseteq \mathbb{R}^n, & \mathcal{N}(\mathbf{A}^\top) &\subseteq \mathbb{R}^m. \end{aligned}$$

By Eq. 1.1.2, $\mathcal{C}(\mathbf{A})$ is the span of the columns, the **column space**, and $\mathcal{C}(\mathbf{A}^\top)$ is the span of the rows, the **row space**. $\mathcal{N}(\mathbf{A})$ is the **null space**. The **rank** of $\mathbf{A}$ is $\text{rank}(\mathbf{A}) = \dim \mathcal{C}(\mathbf{A})$, the largest number of independent columns. The number of independent rows, $\dim \mathcal{C}(\mathbf{A}^\top)$, looks like a different quantity. It is not.

Theorem 1.2.1 — Row rank equals column rank

For every matrix $\mathbf{A}$, $\dim \mathcal{C}(\mathbf{A}^\top) = \dim \mathcal{C}(\mathbf{A})$. That is, $\text{rank}(\mathbf{A}^\top) = \text{rank}(\mathbf{A})$. Moreover, if $\text{rank}(\mathbf{A}) = r \geq 1$ there are an $m \times r$ matrix $\mathbf{B}$ and an $r \times n$ matrix $\mathbf{F}$, both of rank r , with $\mathbf{A} = \mathbf{B}\mathbf{F}$.

Proof. Let $r = \text{rank}(\mathbf{A})$ and let the columns of the $m \times r$ matrix $\mathbf{B}$ be a basis of $\mathcal{C}(\mathbf{A})$. Each column $\mathbf{a}_j$ is a combination $\mathbf{B}\mathbf{f}_j$ of these, so $\mathbf{A} = \mathbf{B}\mathbf{F}$ with $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_n]$. By the transposed reading of Eq. 1.1.2, every row of $\mathbf{A} = \mathbf{B}\mathbf{F}$ is a combination of the r rows of $\mathbf{F}$, so $\dim \mathcal{C}(\mathbf{A}^\top) \leq r = \dim \mathcal{C}(\mathbf{A})$. Applying this to $\mathbf{A}^\top$ gives the reverse inequality. Finally $\text{rank}(\mathbf{F}) \leq r$ because $\mathbf{F}$ has r rows, and $r = \dim \mathcal{C}(\mathbf{A}^\top) \leq \dim \mathcal{C}(\mathbf{F}^\top) = \text{rank}(\mathbf{F})$, so $\text{rank}(\mathbf{F}) = r$. ■

The factorization $\mathbf{A} = \mathbf{B}\mathbf{F}$ is a **rank factorization**: $\mathbf{B}$ has **full column rank** (its columns are independent) and $\mathbf{F}$ has **full row rank**. It is a convenient tool for proofs, as [Proposition 1.9.1](#) shows.

Theorem 1.2.2 — Rank-nullity

For an $m \times n$ matrix $\mathbf{A}$, $\text{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A}) = n$.

Proof. Let $\mathbf{v}_1, \dots, \mathbf{v}_k$ be a basis of $\mathcal{N}(\mathbf{A})$ and extend it to a basis $\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}_1, \dots, \mathbf{w}_{n-k}$ of $\mathbb{R}^n$. Every $\mathbf{A}\mathbf{b}$ is a combination of $\mathbf{A}\mathbf{w}_1, \dots, \mathbf{A}\mathbf{w}_{n-k}$, since $\mathbf{A}\mathbf{v}_i = \mathbf{0}$. These vectors are independent: if $\sum_j c_j \mathbf{A}\mathbf{w}_j = \mathbf{0}$, then $\sum_j c_j \mathbf{w}_j \in \mathcal{N}(\mathbf{A})$ is a combination of the $\mathbf{v}_i$, and independence of the full basis forces every $c_j = 0$. So $\text{rank}(\mathbf{A}) = n - k$. ■

In a linear model with $n \times p$ model matrix $\mathbf{X}$, $\dim \mathcal{N}(\mathbf{X}) = p - \text{rank}(\mathbf{X})$ counts the directions in which the coefficient vector can move without changing the mean $\mathbf{X}\boldsymbol{\beta}$. $\mathbf{X}$ has full column rank iff $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$.

1.2.3 Rank of products and of $\mathbf{A}^\top \mathbf{A}$

Proposition 1.2.3

Let $\mathbf{A}$ be $m \times n$ and $\mathbf{B}$ be $n \times q$.

- (a) $\mathcal{C}(\mathbf{AB}) \subseteq \mathcal{C}(\mathbf{A})$ and $\mathcal{N}(\mathbf{B}) \subseteq \mathcal{N}(\mathbf{AB})$, so $\text{rank}(\mathbf{AB}) \leq \min\{\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B})\}$.
- (b) If $\mathbf{B}$ has full row rank, then $\mathcal{C}(\mathbf{AB}) = \mathcal{C}(\mathbf{A})$. If $\mathbf{A}$ has full column rank, then $\mathcal{N}(\mathbf{AB}) = \mathcal{N}(\mathbf{B})$. So $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{A})$ in the first case and $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{B})$ in the second. In particular, multiplying by a nonsingular matrix does not change rank.
- (c) $\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A})$ and $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)$, so $\text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{AA}^\top) = \text{rank}(\mathbf{A})$.
- (d) $\text{rank}(\mathbf{A} + \mathbf{C}) \leq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{C})$ for $\mathbf{C}$ of the same size as $\mathbf{A}$.

Proof.

- (a) is [Eq. 1.1.2](#), together with $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{B}^\top \mathbf{A}^\top) \leq \text{rank}(\mathbf{B}^\top)$ by the first inclusion and [Theorem 1.2.1](#). (b) If $\mathbf{B}$ has full row rank, $\mathcal{C}(\mathbf{B}) = \mathbb{R}^n$ because it is an n -dimensional subspace of $\mathbb{R}^n$, so every $\mathbf{A}\mathbf{u}$ equals $\mathbf{A}\mathbf{B}\mathbf{v}$ for some $\mathbf{v}$. If $\mathbf{A}$ has full column rank, $\mathbf{A}\mathbf{B}\mathbf{v} = \mathbf{0}$ forces $\mathbf{B}\mathbf{v} = \mathbf{0}$. The rank statements follow from rank-nullity. (c) If $\mathbf{A}^\top \mathbf{A}\mathbf{v} = \mathbf{0}$ then $\|\mathbf{A}\mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{A}^\top \mathbf{A}\mathbf{v} = 0$, so $\mathbf{A}\mathbf{v} = \mathbf{0}$. Thus the null spaces agree, and by [Theorem 1.2.2](#) $\text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{A})$. Since $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) \subseteq \mathcal{C}(\mathbf{A}^\top)$ and both have dimension $\text{rank}(\mathbf{A})$, they are equal. Applying this to $\mathbf{A}^\top$ gives $\text{rank}(\mathbf{AA}^\top) = \text{rank}(\mathbf{A})$. (d) $\mathcal{C}(\mathbf{A} + \mathbf{C}) \subseteq \mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{C})$, and the dimension of a sum is at most the sum of the dimensions. ■

Part (c) is the reason the normal equations $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$ always have a solution: the right-hand side lies in $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$ (Section 1.4).

1.2.4 Nonsingular submatrices

Proposition 1.2.4

Let $\mathbf{A}$ have rank $r \geq 1$. Choose any r linearly independent rows, indexed by I , and any r linearly independent columns, indexed by J . Then the $r \times r$ submatrix $\mathbf{A}_{IJ}$ at their intersection is nonsingular, and

$$\mathbf{A} = \mathbf{A}_{\cdot J} \mathbf{A}_{IJ}^{-1} \mathbf{A}_{I\cdot}, \quad (1.2.1)$$

where $\mathbf{A}_{\cdot J}$ holds the columns in J and $\mathbf{A}_{I\cdot}$ the rows in I .

Proof. The columns in J are a basis of $\mathcal{C}(\mathbf{A})$, so $\mathbf{A} = \mathbf{A}_{\cdot J} \mathbf{F}$ for some $r \times n$ matrix $\mathbf{F}$. Keeping only the rows in I gives $\mathbf{A}_{I\cdot} = \mathbf{A}_{IJ} \mathbf{F}$. The left side has rank r , so by Proposition 1.2.3(a) the $r \times r$ matrix $\mathbf{A}_{IJ}$ has rank r and is nonsingular. Then $\mathbf{F} = \mathbf{A}_{IJ}^{-1} \mathbf{A}_{I\cdot}$, which gives Eq. 1.2.1. ■

Conversely, a nonsingular $k \times k$ submatrix makes the k columns of $\mathbf{A}$ that contain it linearly independent, so $\text{rank}(\mathbf{A}) \geq k$. In particular, a matrix has rank r iff it has a nonsingular $r \times r$ submatrix and no nonsingular larger one. This proposition gives an explicit generalized inverse in Theorem 1.3.2.

1.2.5 Exercises

1.2.5.1 A. Check your understanding

Exercise A1.

Show that $\sum_{i < j} (y_i - y_j)^2 = n \sum_i (y_i - \bar{y})^2$, and write both sides as $\mathbf{y}^\top \mathbf{A} \mathbf{y}$ with $\mathbf{A}$ symmetric. Find the symmetric matrix of the quadratic form $2x_1^2 - x_2^2 + 6x_1x_2 - 4x_2x_3$.

Exercise A2.

Show that $\mathbf{X}^\top \mathbf{X} \mathbf{B} = \mathbf{X}^\top \mathbf{X} \mathbf{C}$ implies $\mathbf{X} \mathbf{B} = \mathbf{X} \mathbf{C}$. Deduce that $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{0}$ iff $\mathbf{X} \mathbf{b} = \mathbf{0}$.

Solution. Put $\mathbf{D} = \mathbf{B} - \mathbf{C}$, so that $\mathbf{X}^\top \mathbf{X} \mathbf{D} = \mathbf{0}$. Then $(\mathbf{X} \mathbf{D})^\top (\mathbf{X} \mathbf{D}) = \mathbf{D}^\top \mathbf{X}^\top \mathbf{X} \mathbf{D} = \mathbf{0}$, and by Eq. 1.1.3 a matrix $\mathbf{Z}$ with $\mathbf{Z}^\top \mathbf{Z} = \mathbf{0}$ is zero. So $\mathbf{X} \mathbf{D} = \mathbf{0}$. The second statement is the case $\mathbf{C} = \mathbf{0}$, with the converse obvious.

Exercise A3.

For the additive two-way layout with two rows, three columns and one observation per cell, write the 6×6 model matrix $\mathbf{X} = [\mathbf{1}, \mathbf{R}, \mathbf{C}]$, where $\mathbf{R}$ and $\mathbf{C}$ hold the row and column indicators. Find $\text{rank}(\mathbf{X})$ and a basis of $\mathcal{N}(\mathbf{X})$. Compare $[\mathbf{R}, \mathbf{C}]^\top$ with the matrix $\mathbf{A}$ of [Example \(Filling a table from its margins\)](#).

1.2.5.2 B. Practice**Exercise B1.**

Let $\mathbf{A}$ be $m \times n$ and $\mathbf{B}$ be $n \times q$. Prove $\text{rank}(\mathbf{AB}) \geq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}) - n$. *Hint:* apply rank-nullity to $\mathbf{AB}_0$, where the columns of $\mathbf{B}_0$ are a basis of $\mathcal{C}(\mathbf{B})$.

Solution. Let the $n \times s$ matrix $\mathbf{B}_0$, $s = \text{rank}(\mathbf{B})$, have columns forming a basis of $\mathcal{C}(\mathbf{B})$. Then $\mathcal{C}(\mathbf{AB}) = \mathcal{C}(\mathbf{AB}_0)$, so by rank-nullity $\text{rank}(\mathbf{AB}) = s - \dim \mathcal{N}(\mathbf{AB}_0)$. The map $\mathbf{c} \mapsto \mathbf{B}_0 \mathbf{c}$ is one-to-one and sends $\mathcal{N}(\mathbf{AB}_0)$ into $\mathcal{N}(\mathbf{A})$, so $\dim \mathcal{N}(\mathbf{AB}_0) \leq \dim \mathcal{N}(\mathbf{A}) = n - \text{rank}(\mathbf{A})$. Hence $\text{rank}(\mathbf{AB}) \geq \text{rank}(\mathbf{B}) - n + \text{rank}(\mathbf{A})$.

Exercise B2.

Show that if $\mathbf{A} = \mathbf{B}_1 \mathbf{F}_1 = \mathbf{B}_2 \mathbf{F}_2$ are two rank factorizations, then $\mathbf{B}_2 = \mathbf{B}_1 \mathbf{T}$ and $\mathbf{F}_2 = \mathbf{T}^{-1} \mathbf{F}_1$ for a nonsingular $\mathbf{T}$.

1.3 Inverses and generalized inverses

1.3.1 The inverse of a square matrix

An $n \times n$ matrix $\mathbf{A}$ is **nonsingular** if there is a matrix $\mathbf{A}^{-1}$ with $\mathbf{AA}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$. The inverse is unique: if $\mathbf{B}$ and $\mathbf{C}$ both qualify, then $\mathbf{B} = \mathbf{B}(\mathbf{AC}) = (\mathbf{BA})\mathbf{C} = \mathbf{C}$.

Proposition 1.3.1

For an $n \times n$ matrix $\mathbf{A}$ the following are equivalent: (a) $\mathbf{A}$ is nonsingular; (b) $\text{rank}(\mathbf{A}) = n$; (c) $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$; (d) $\mathbf{BA} = \mathbf{I}$ for some $\mathbf{B}$; (e) $\mathbf{AC} = \mathbf{I}$ for some $\mathbf{C}$. When they hold, $\mathbf{B} = \mathbf{C} = \mathbf{A}^{-1}$, and

$$(\mathbf{A}^\top)^{-1} = (\mathbf{A}^{-1})^\top, \quad (\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

for nonsingular $\mathbf{A}, \mathbf{B}$ of the same size. A nonsingular symmetric matrix has a symmetric inverse.

Proof. (b) $\Leftrightarrow$ (c) is rank-nullity. (a) $\Rightarrow$ (d) and (a) $\Rightarrow$ (e) are trivial. (d) $\Rightarrow$ (c): $\mathbf{Ax} = \mathbf{0}$ gives $\mathbf{x} = \mathbf{BAx} = \mathbf{0}$. (e) $\Rightarrow$ (b): $\mathbf{I} = \mathbf{AC}$ has rank n , and $\text{rank}(\mathbf{AC}) \leq \text{rank}(\mathbf{A})$. (b) $\Rightarrow$ (a): since $\mathcal{C}(\mathbf{A}) = \mathbb{R}^n$, we can solve $\mathbf{Ac}_j = \mathbf{e}_j$ for each j , giving $\mathbf{AC} = \mathbf{I}$. Then

$\mathbf{A}(\mathbf{CA} - \mathbf{I}) = \mathbf{0}$, and because $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$ each column of $\mathbf{CA} - \mathbf{I}$ is zero. So $\mathbf{CA} = \mathbf{I}$ as well. If $\mathbf{BA} = \mathbf{I}$ then $\mathbf{B} = \mathbf{BAA}^{-1} = \mathbf{A}^{-1}$, and similarly for $\mathbf{C}$. The formulas are verified by multiplication: $\mathbf{A}^\top(\mathbf{A}^{-1})^\top = (\mathbf{A}^{-1}\mathbf{A})^\top = \mathbf{I}$ and $(\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{I}$. If $\mathbf{A} = \mathbf{A}^\top$, then $\mathbf{A}^{-1} = (\mathbf{A}^\top)^{-1} = (\mathbf{A}^{-1})^\top$.

■

A rectangular matrix has no two-sided inverse. If $\mathbf{X}$ is $n \times p$ with full column rank, then $\mathbf{X}^\top \mathbf{X}$ is nonsingular by Proposition 1.2.3(c), and $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ is a *left inverse*: it multiplies $\mathbf{X}$ on the left to give $\mathbf{I}_p$. This is the matrix that computes least squares coefficients.

1.3.2 Generalized inverses

When $\mathbf{A}$ is singular or rectangular we still want a matrix that inverts $\mathbf{A}$ on the vectors where inversion makes sense, namely those in $\mathcal{C}(\mathbf{A})$. The following weak requirement turns out to be exactly what solving equations needs.

Definition 1.3.1 — Generalized inverse

A **generalized inverse** of an $m \times n$ matrix $\mathbf{A}$ is any $n \times m$ matrix $\mathbf{A}^-$ with

$$\mathbf{AA}^-\mathbf{A} = \mathbf{A}.$$

Read columnwise, the definition says that $\mathbf{AA}^-$ acts as the identity on every column of $\mathbf{A}$, hence on all of $\mathcal{C}(\mathbf{A})$: if $\mathbf{u} = \mathbf{Ab}$ then $\mathbf{A}(\mathbf{A}^-\mathbf{u}) = \mathbf{AA}^-\mathbf{Ab} = \mathbf{u}$. So $\mathbf{A}^-\mathbf{u}$ is a solution of $\mathbf{Ax} = \mathbf{u}$ whenever one exists. If $\mathbf{A}$ is nonsingular, multiplying the definition by $\mathbf{A}^{-1}$ on both sides gives $\mathbf{A}^- = \mathbf{A}^{-1}$. In general there are many generalized inverses. Chapter 6 uses them to write projections, and Chapter 8 uses them to describe estimable functions in models of less than full rank.

Theorem 1.3.2 — Existence

Every matrix has a generalized inverse. Explicitly, if $\text{rank}(\mathbf{A}) = r \geq 1$ and $\mathbf{A}_{IJ}$ is the nonsingular $r \times r$ submatrix of Proposition 1.2.4, the $n \times m$ matrix $\mathbf{G}$ that holds $\mathbf{A}_{IJ}^{-1}$ in the rows indexed by J and the columns indexed by I , and zeros elsewhere, is a generalized inverse of $\mathbf{A}$.

Proof. For $\mathbf{A} = \mathbf{0}$ any matrix of the right size works. Otherwise let $\mathbf{E}_I$ and $\mathbf{E}_J$ be the columns of $\mathbf{I}_m$ and $\mathbf{I}_n$ indexed by I and J . Then $\mathbf{G} = \mathbf{E}_J \mathbf{A}_{IJ}^{-1} \mathbf{E}_I^\top$, $\mathbf{AE}_J = \mathbf{A}_{.J}$ and $\mathbf{E}_I^\top \mathbf{A} = \mathbf{A}_{I.}$. So $\mathbf{AGA} = \mathbf{A}_{.J} \mathbf{A}_{IJ}^{-1} \mathbf{A}_{I.} = \mathbf{A}$ by Eq. 1.2.1.

■

Proposition 1.3.3 — Properties of generalized inverses

Let $\mathbf{A}$ be $m \times n$ of rank r and let $\mathbf{G}$ be a generalized inverse of $\mathbf{A}$.

- (a) $\mathbf{AG}$ and $\mathbf{GA}$ are idempotent. Moreover $\mathcal{C}(\mathbf{AG}) = \mathcal{C}(\mathbf{A})$ and $\mathcal{N}(\mathbf{GA}) = \mathcal{N}(\mathbf{A})$, and $\text{rank}(\mathbf{AG}) = \text{rank}(\mathbf{GA}) = r \leq \text{rank}(\mathbf{G})$.
- (b) $\mathbf{G}^\top$ is a generalized inverse of $\mathbf{A}^\top$.
- (c) If $\mathbf{G}_1, \mathbf{G}_2$ are generalized inverses of $\mathbf{A}$, so is $\mathbf{G}_1\mathbf{AG}_2$. In particular $\mathbf{GAG}$ is a generalized inverse which also satisfies $(\mathbf{GAG})\mathbf{A}(\mathbf{GAG}) = \mathbf{GAG}$.
- (d) If $\mathbf{A}$ is symmetric, then $\mathbf{G}^\top$ and $\frac{1}{2}(\mathbf{G} + \mathbf{G}^\top)$ are generalized inverses of $\mathbf{A}$, so a symmetric generalized inverse always exists.
- (e) $\mathbf{G} + (\mathbf{I} - \mathbf{GA})\mathbf{U} + \mathbf{V}(\mathbf{I} - \mathbf{AG})$ is a generalized inverse of $\mathbf{A}$ for every $\mathbf{U}$ and $\mathbf{V}$ of size $n \times m$.

Proof.

- (a) $(\mathbf{AG})^2 = (\mathbf{AGA})\mathbf{G} = \mathbf{AG}$, and likewise for $\mathbf{GA}$. Next, $\mathcal{C}(\mathbf{A}) = \mathcal{C}(\mathbf{AGA}) \subseteq \mathcal{C}(\mathbf{AG}) \subseteq \mathcal{C}(\mathbf{A})$, and $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{GA}) \subseteq \mathcal{N}(\mathbf{AGA}) = \mathcal{N}(\mathbf{A})$. The ranks follow from these and rank-nullity, and $r = \text{rank}(\mathbf{AG}) \leq \text{rank}(\mathbf{G})$.
- (b) Transpose $\mathbf{AGA} = \mathbf{A}$. (c) $\mathbf{AG}_1\mathbf{AG}_2\mathbf{A} = \mathbf{AG}_2\mathbf{A} = \mathbf{A}$. The second statement is $\mathbf{GAGAGAG} = \mathbf{G}(\mathbf{AGA})\mathbf{GAG} = \mathbf{GAGAG} = \mathbf{GAG}$.
- (c) By (b), $\mathbf{G}^\top$ is a generalized inverse of $\mathbf{A}^\top = \mathbf{A}$. The condition $\mathbf{AGA} = \mathbf{A}$ is linear in $\mathbf{G}$, so it is preserved by averaging.
- (d) Multiply out, using $\mathbf{A}(\mathbf{I} - \mathbf{GA}) = \mathbf{0}$ and $(\mathbf{I} - \mathbf{AG})\mathbf{A} = \mathbf{0}$.

■

A generalized inverse with the extra property in (c), $\mathbf{GAG} = \mathbf{G}$, is called **reflexive**. [Exercise \(B1\)](#) shows that the reflexive ones are exactly those of rank r . Part (e) in fact produces *every* generalized inverse from a single one as $\mathbf{U}$ and $\mathbf{V}$ vary. Idempotence in (a) means that $\mathbf{AG}$ and $\mathbf{GA}$ are projections, generally not orthogonal ones. [Chapter 6](#) shows that the particular combination $\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$ is symmetric, does not depend on the choice of generalized inverse, and is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$.

1.3.3 The Moore–Penrose inverse

One generalized inverse is singled out by symmetry requirements.

Definition 1.3.2 — Moore–Penrose inverse

The **Moore–Penrose inverse** of $\mathbf{A}$ is the matrix $\mathbf{A}^+$ satisfying

$$\mathbf{AA}^+\mathbf{A} = \mathbf{A}, \quad \mathbf{A}^+\mathbf{AA}^+ = \mathbf{A}^+, \quad (\mathbf{AA}^+)^\top = \mathbf{AA}^+, \quad (\mathbf{A}^+\mathbf{A})^\top = \mathbf{A}^+\mathbf{A}.$$

There is at most one such matrix. If $\mathbf{G}$ and $\mathbf{H}$ both satisfy the four conditions, then

$$\mathbf{G} = \mathbf{GAG} = \mathbf{G}(\mathbf{AG})^\top = \mathbf{GG}^\top(\mathbf{AHA})^\top = \mathbf{G}(\mathbf{AG})^\top(\mathbf{AH})^\top = \mathbf{GAGAH} = \mathbf{GAH},$$

where the steps use the second and third conditions for $\mathbf{G}$, the first for $\mathbf{H}$, and the third for both. The same argument with the fourth condition in place of the third gives

$$\mathbf{H} = (\mathbf{HA})^\top\mathbf{H} = \mathbf{A}^\top\mathbf{H}^\top\mathbf{H} = (\mathbf{AGA})^\top\mathbf{H}^\top\mathbf{H} = (\mathbf{GA})^\top(\mathbf{HA})^\top\mathbf{H} = \mathbf{GAHAH} = \mathbf{GAH},$$

so $\mathbf{G} = \mathbf{H}$. Existence follows from the singular value decomposition, which gives $\mathbf{A}^+$ explicitly (Theorem 1.8.1).

1.4 Systems of equations and consistency

A system $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{A}$ of size $m \times n$ is **consistent** if it has at least one solution. Least squares leads to such systems in which $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ may be singular, so we need to know when solutions exist, what all of them look like, and which features are common to all of them.

Theorem 1.4.1 — Consistency and the general solution

Let $\mathbf{G}$ be any generalized inverse of $\mathbf{A}$. The following are equivalent:

- (a) $\mathbf{Ax} = \mathbf{b}$ is consistent;
- (b) $\mathbf{b} \in \mathcal{C}(\mathbf{A})$;
- (c) $\text{rank}[\mathbf{A}, \mathbf{b}] = \text{rank}(\mathbf{A})$;
- (d) $\mathbf{AGb} = \mathbf{b}$.

If the system is consistent, $\mathbf{x}_0 = \mathbf{Gb}$ is a solution, and the set of all solutions is

$$\{\mathbf{Gb} + (\mathbf{I} - \mathbf{GA})\mathbf{z} : \mathbf{z} \in \mathbb{R}^n\} = \mathbf{x}_0 + \mathcal{N}(\mathbf{A}). \quad (1.4.1)$$

Proof. (a) $\Leftrightarrow$ (b) is the first reading in Eq. 1.1.2. (b) $\Leftrightarrow$ (c): appending $\mathbf{b}$ as a column leaves the column space unchanged iff $\mathbf{b}$ is already in it, and a subspace of $\mathcal{C}([\mathbf{A}, \mathbf{b}])$ with the same dimension equals it. (d) $\Rightarrow$ (a): $\mathbf{Gb}$ is a solution. (b) $\Rightarrow$ (d): if $\mathbf{b} = \mathbf{Au}$, then $\mathbf{AGb} = \mathbf{AGAu} = \mathbf{Au} = \mathbf{b}$.

The solutions of a consistent system are $\mathbf{x}_0 + \mathcal{N}(\mathbf{A})$, since $\mathbf{Ax} = \mathbf{b}$ iff $\mathbf{A}(\mathbf{x} - \mathbf{x}_0) = \mathbf{0}$. It remains to show $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{I} - \mathbf{GA})$. The inclusion $\supseteq$ holds because $\mathbf{A}(\mathbf{I} - \mathbf{GA}) = \mathbf{0}$. Conversely, if $\mathbf{Av} = \mathbf{0}$, then $\mathbf{v} = (\mathbf{I} - \mathbf{GA})\mathbf{v}$.

■

Criterion (d) is useful in practice: it tests consistency with a single matrix product, for any generalized inverse. By Proposition 1.3.3(a) and rank-nullity, $\text{rank}(\mathbf{I} - \mathbf{GA}) = n - r$, so the solution set is an affine subspace of dimension $n - r$. The solution is unique iff $\text{rank}(\mathbf{A}) = n$, and then $\mathbf{Gb}$ is the same for every $\mathbf{G}$.

Idea. Any generalized inverse does three jobs for $\mathbf{Ax} = \mathbf{b}$. It tests consistency ($\mathbf{AGb} = \mathbf{b}$), produces one solution ($\mathbf{Gb}$), and parametrizes all of them ($\mathbf{Gb} + (\mathbf{I} - \mathbf{GA})\mathbf{z}$). Which $\mathbf{G}$ is used changes the particular solution but never the solution set, and a linear function $\mathbf{q}^\top \mathbf{x}$ is the same across solutions exactly when $\mathbf{q} \in \mathcal{C}(\mathbf{A}^\top)$.

Corollary 1.4.2 — Invariant linear functions

Let $\mathbf{Ax} = \mathbf{b}$ be consistent and let $\mathbf{q} \in \mathbb{R}^n$. The value $\mathbf{q}^\top \mathbf{x}$ is the same for every solution

$\mathbf{x}$ iff $\mathbf{q} \in \mathcal{C}(\mathbf{A}^\top)$, that is, iff $\mathbf{q}^\top = \mathbf{q}^\top \mathbf{G} \mathbf{A}$ for one (and then every) generalized inverse $\mathbf{G}$.

Proof. By Eq. 1.4.1, $\mathbf{q}^\top \mathbf{x}$ is constant over solutions iff $\mathbf{q}^\top (\mathbf{I} - \mathbf{G} \mathbf{A}) \mathbf{z} = 0$ for all $\mathbf{z}$, that is, iff $\mathbf{q}^\top = \mathbf{q}^\top \mathbf{G} \mathbf{A}$. That equation puts $\mathbf{q}$ in $\mathcal{C}(\mathbf{A}^\top)$. Conversely, if $\mathbf{q} = \mathbf{A}^\top \mathbf{w}$, then $\mathbf{q}^\top \mathbf{G} \mathbf{A} = \mathbf{w}^\top \mathbf{A} \mathbf{G} \mathbf{A} = \mathbf{w}^\top \mathbf{A} = \mathbf{q}^\top$. ■

Applied to the normal equations $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$, which are consistent by Proposition 1.2.3(c), the corollary says that $\mathbf{q}^\top \hat{\boldsymbol{\beta}}$ is the same for all least squares solutions iff $\mathbf{q} \in \mathcal{C}(\mathbf{X}^\top \mathbf{X}) = \mathcal{C}(\mathbf{X}^\top)$. This is the algebra behind estimable functions (Theorem 8.3.1).

Proposition 1.4.3 — Minimum-norm solution

If $\mathbf{A} \mathbf{x} = \mathbf{b}$ is consistent, then $\mathbf{A}^+ \mathbf{b}$ is its unique solution lying in $\mathcal{C}(\mathbf{A}^\top)$, and $\|\mathbf{A}^+ \mathbf{b}\| < \|\mathbf{x}\|$ for every other solution $\mathbf{x}$.

Proof. $\mathbf{A}^+ \mathbf{b}$ is a solution because $\mathbf{A}^+$ is a generalized inverse. From the second and fourth Penrose conditions, $\mathbf{A}^+ = (\mathbf{A}^+ \mathbf{A}) \mathbf{A}^+ = (\mathbf{A}^+ \mathbf{A})^\top \mathbf{A}^+ = \mathbf{A}^\top (\mathbf{A}^+)^{\top} \mathbf{A}^+$, so $\mathbf{A}^+ \mathbf{b} \in \mathcal{C}(\mathbf{A}^\top)$. Every vector of $\mathcal{C}(\mathbf{A}^\top)$ is orthogonal to every $\mathbf{v} \in \mathcal{N}(\mathbf{A})$, since $(\mathbf{A}^\top \mathbf{w})^\top \mathbf{v} = \mathbf{w}^\top \mathbf{A} \mathbf{v} = 0$. Any other solution is $\mathbf{x} = \mathbf{A}^+ \mathbf{b} + \mathbf{v}$ with $\mathbf{0} \neq \mathbf{v} \in \mathcal{N}(\mathbf{A})$, and then $\|\mathbf{x}\|^2 = \|\mathbf{A}^+ \mathbf{b}\|^2 + \|\mathbf{v}\|^2$. The same computation shows that $\mathbf{x} \notin \mathcal{C}(\mathbf{A}^\top)$, because $\mathbf{x}^\top \mathbf{v} = \|\mathbf{v}\|^2 \neq 0$. ■

Example. (Filling a table from its margins)

Suppose the six entries of a 2×3 table are unknown, but its row totals (9, 6) and column totals (4, 5, 6) are known. With the cells ordered $x_{11}, x_{12}, x_{13}, x_{21}, x_{22}, x_{23}$, the equations are $\mathbf{A} \mathbf{x} = \mathbf{b}$ with

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 9 \\ 6 \\ 4 \\ 5 \\ 6 \end{pmatrix}.$$

The rows of $\mathbf{A}$ satisfy one relation: the two row-total rows and the three column-total rows both add to $\mathbf{1}_6^\top$. So $\text{rank}(\mathbf{A}) = 4$ and there is a $6 - 4 = 2$ -dimensional family of solutions. The same relation must hold for $\mathbf{b}$. By Theorem 1.4.1(c), the system is consistent iff the row totals and the column totals have the same sum, and here both sums are 15.

The listing builds three generalized inverses: $\mathbf{G}_1$ from the nonsingular submatrix in rows 1–4 and columns 1–4 (Theorem 1.3.2), the Moore–Penrose inverse $\mathbf{G}_2 = \mathbf{A}^+$, and $\mathbf{G}_3 = \mathbf{G}_1 + (\mathbf{I} - \mathbf{G}_1 \mathbf{A}) \mathbf{Z}$ for an integer matrix $\mathbf{Z}$ (Proposition 1.3.3(e)). They give the solutions

$$\begin{aligned} \mathbf{G}_1 \mathbf{b} &= (-2, 5, 6, 6, 0, 0)^\top, & \|\mathbf{G}_1 \mathbf{b}\| &= 10.050, \\ \mathbf{G}_2 \mathbf{b} &= (5/2, 3, 7/2, 3/2, 2, 5/2)^\top, & \|\mathbf{G}_2 \mathbf{b}\| &= 6.325, \\ \mathbf{G}_3 \mathbf{b} &= (21, -39, 27, -17, 44, -21)^\top, & \|\mathbf{G}_3 \mathbf{b}\| &= 73.192. \end{aligned}$$

The individual cell x_{11} takes the values -2 , $5/2$ and 21 : margins do not determine cells. But $x_{12} + x_{13} - x_{21}$ equals 5 for all three, as [Corollary 1.4.2](#) predicts, because its coefficient vector is the first row of $\mathbf{A}$ minus the third. The minimum-norm solution has the additive form $x_{ij} = r_i/3 + c_j/2 - 15/6$ in terms of the row totals r_i and column totals c_j ([Exercise \(A2\)](#)). If the last column total is changed to 7 , then $\mathbf{A}\mathbf{G}\mathbf{b} \neq \mathbf{b}$ for each of the three matrices, and the script confirms that the test detects the inconsistency. The script also checks all 60 submatrices formed by 4 independent rows and 4 independent columns, and finds each one nonsingular, as [Proposition 1.2.4](#) requires.

```
from fractions import Fraction
from itertools import combinations
import numpy as np
m, n = A.shape
# cells ordered (1,1), (1,2), (1,3), (2,1), (2,2), (2,3)
A = np.array([[1, 1, 1, 0, 0, 0],      # row 1 total
              [0, 0, 0, 1, 1, 1],      # row 2 total
              [1, 0, 0, 1, 0, 0],      # column 1 total
              [0, 1, 0, 0, 1, 0],      # column 2 total
              [0, 0, 1, 0, 0, 1]],      # column 3 total
             dtype=float)
b = np.array([9, 6, 4, 5, 6], dtype=float) # both sets of totals add to 15
r = np.linalg.matrix_rank(A)
print("rank(A) =", r, " rank([A, b]) =", np.linalg.matrix_rank(np.column_stack([A, b
])))

def ginverse_from_submatrix(A, rows, cols):
    """Place the inverse of the nonsingular block A[rows, cols] at [cols, rows]."""
    G = np.zeros(A.T.shape)
    G[np.ix_(cols, rows)] = np.linalg.inv(A[np.ix_(rows, cols)])
    return G

rows, cols = [0, 1, 2, 3], [0, 1, 2, 3] # r independent rows and
# columns
G1 = ginverse_from_submatrix(A, rows, cols)
G2 = np.linalg.pinv(A) # Moore-Penrose inverse
rng = np.random.default_rng(1)
Z = rng.integers(-2, 3, size=A.T.shape).astype(float)
G3 = G1 + (np.eye(n) - G1 @ A) @ Z # another, non-reflexive
# one

for G in (G1, G2, G3):
    assert np.allclose(A @ G @ A, A) # each is a g-inverse
    assert np.allclose(A @ G @ b, b) # consistency test passes
    x = G @ b
    print(np.round(x, 4), " A x - b =", np.abs(A @ x - b).max())
```

1.4.1 Exercises

1.4.1.1 A. Check your understanding

Exercise A1.

Find all generalized inverses of a nonzero vector $\mathbf{a} \in \mathbb{R}^n$, regarded as an $n \times 1$ matrix. Which one is $\mathbf{a}^+$?

Exercise A2.

Generalize [Example \(Filling a table from its margins\)](#) to an $a \times b$ table with row totals r_i , column totals c_j and grand total $T = \sum_i r_i = \sum_j c_j$. Show that $x_{ij} = r_i/b + c_j/a - T/(ab)$ is a solution lying in $\mathcal{C}(\mathbf{A}^\top)$, and hence is the minimum-norm solution.

Solution. Row i of the proposed table sums to $r_i + \sum_j c_j/a - bT/(ab) = r_i$, and column j sums to $\sum_i r_i/b + c_j - T/b = c_j$. The rows of $\mathbf{A}$ are the row indicators and the column indicators, so $\mathcal{C}(\mathbf{A}^\top)$ consists of the tables of the form $x_{ij} = \alpha_i + \beta_j$. The proposed solution has this form. By [Proposition 1.4.3](#), it is $\mathbf{A}^+\mathbf{b}$. For $a = 2$, $b = 3$ and $T = 15$ this is the additive formula quoted in [Example \(Filling a table from its margins\)](#).

Exercise A3.

For which c and d is the system $x_1 + x_2 + x_3 = 2$, $x_1 - x_2 = c$, $3x_1 + x_2 + 2x_3 = d$ consistent? When it is, give its general solution in the form [Eq. 1.4.1](#).

1.4.1.2 B. Practice

Exercise B1.

Let $\mathbf{G}$ be a generalized inverse of $\mathbf{A}$. Show that $\mathbf{GAG} = \mathbf{G}$ iff $\text{rank}(\mathbf{G}) = \text{rank}(\mathbf{A})$.

Solution. If $\mathbf{GAG} = \mathbf{G}$, then $\text{rank}(\mathbf{G}) \leq \text{rank}(\mathbf{A})$ by [Proposition 1.2.3\(a\)](#), and $\text{rank}(\mathbf{A}) \leq \text{rank}(\mathbf{G})$ by [Proposition 1.3.3\(a\)](#). Conversely, suppose $\text{rank}(\mathbf{G}) = \text{rank}(\mathbf{A}) = r$. Since $\mathcal{C}(\mathbf{GA}) \subseteq \mathcal{C}(\mathbf{G})$ and $\text{rank}(\mathbf{GA}) = r$, the two column spaces coincide, so $\mathbf{G} = \mathbf{GAK}$ for some $\mathbf{K}$. Then $\mathbf{GAG} = \mathbf{GAGAK} = \mathbf{GAK} = \mathbf{G}$, using $\mathbf{AGA} = \mathbf{A}$.

Exercise B2.

Let $\mathbf{A} = \mathbf{BF}$ be a rank factorization. Show that $\mathbf{A}^+ = \mathbf{F}^\top(\mathbf{FF}^\top)^{-1}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top$. Deduce that $\mathbf{A}^+ = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top$ when $\mathbf{A}$ has full column rank.

Solution. Write $\mathbf{G}$ for the proposed matrix. Then $\mathbf{AG} = \mathbf{B}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top$ and $\mathbf{GA} = \mathbf{F}^\top(\mathbf{FF}^\top)^{-1}\mathbf{F}$, both symmetric. Also $\mathbf{AGA} = \mathbf{B}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top\mathbf{BF} = \mathbf{A}$ and $\mathbf{GAG} = \mathbf{F}^\top(\mathbf{FF}^\top)^{-1}\mathbf{FF}^\top(\mathbf{FF}^\top)^{-1}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top = \mathbf{G}$. By uniqueness $\mathbf{G} = \mathbf{A}^+$. If $\mathbf{A}$ has full column rank, take $\mathbf{B} = \mathbf{A}$ and $\mathbf{F} = \mathbf{I}$.

Exercise B3.

Let $\mathbf{A}$ be $m \times n$ of rank r and let $\mathbf{Ax} = \mathbf{b}$ be consistent with $\mathbf{b} \neq \mathbf{0}$. Show that the system has $n - r + 1$ linearly independent solutions and no more, and that every solution is an affine combination of them.

1.4.1.3 C. Going deeper

Exercise C1.

Show that $(\mathbf{A}^+)^+ = \mathbf{A}$, $(\mathbf{A}^\top)^+ = (\mathbf{A}^+)^{\top}$, $(\mathbf{A}^\top \mathbf{A})^+ = \mathbf{A}^+ (\mathbf{A}^+)^{\top}$ and $\mathbf{A}^+ = (\mathbf{A}^\top \mathbf{A})^+ \mathbf{A}^\top$.
Give an example with $(\mathbf{AB})^+ \neq \mathbf{B}^+ \mathbf{A}^+$.

1.5 Orthogonality and orthogonal matrices

1.5.1 Orthogonal vectors and complements

Vectors $\mathbf{u}, \mathbf{v}$ are **orthogonal**, written $\mathbf{u} \perp \mathbf{v}$, if $\mathbf{u}^\top \mathbf{v} = 0$. For orthogonal vectors the Pythagorean identity $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ holds. A set of nonzero, mutually orthogonal vectors is independent: if $\sum_j c_j \mathbf{v}_j = \mathbf{0}$, taking the inner product with $\mathbf{v}_k$ leaves $c_k \|\mathbf{v}_k\|^2 = 0$. A set is **orthonormal** if, in addition, each vector has unit length. The columns of an $n \times r$ matrix $\mathbf{Q}$ are orthonormal iff $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r$.

The **orthogonal complement** of a subspace $\mathcal{S} \subseteq \mathbb{R}^n$ is $\mathcal{S}^\perp = \{\mathbf{v} : \mathbf{v}^\top \mathbf{u} = 0 \text{ for all } \mathbf{u} \in \mathcal{S}\}$. The next proposition is all that this chapter needs about complements. Chapter 6 builds the theory of orthogonal projections on it.

Proposition 1.5.1

Let $\mathbf{A}$ be $m \times n$ and let $\mathcal{S} \subseteq \mathbb{R}^n$ be a subspace.

- (a) $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)^\perp$ and $\mathcal{N}(\mathbf{A}^\top) = \mathcal{C}(\mathbf{A})^\perp$.
- (b) $\dim \mathcal{S}^\perp = n - \dim \mathcal{S}$, and $\mathbb{R}^n = \mathcal{S} \oplus \mathcal{S}^\perp$. In particular $\mathbb{R}^n = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$ and $\mathbb{R}^m = \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A}^\top)$.
- (c) $(\mathcal{S}^\perp)^\perp = \mathcal{S}$.

Proof.

- (a) $\mathbf{A}\mathbf{v} = \mathbf{0}$ says that $\mathbf{v}$ is orthogonal to every row of $\mathbf{A}$, hence to their span. Apply this to $\mathbf{A}^\top$ for the second identity.
- (b) Put a basis of $\mathcal{S}$ (k vectors) into the columns of $\mathbf{B}$. Then $\mathcal{S}^\perp = \mathcal{N}(\mathbf{B}^\top)$, which has dimension $n - k$ by rank-nullity. If $\mathbf{u} \in \mathcal{S} \cap \mathcal{S}^\perp$ then $\mathbf{u}^\top \mathbf{u} = 0$, so the sum is direct and has dimension n , so it is $\mathbb{R}^n$.
- (c) $\mathcal{S} \subseteq (\mathcal{S}^\perp)^\perp$, and both have dimension $\dim \mathcal{S}$ by (b).

■

1.5.2 Gram–Schmidt and the QR factorization

Theorem 1.5.2 — Gram–Schmidt; QR factorization

Let $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_p]$ be $n \times p$ with full column rank. Define recursively

$$\mathbf{w}_j = \mathbf{x}_j - \sum_{k < j} (\mathbf{u}_k^\top \mathbf{x}_j) \mathbf{u}_k, \quad \mathbf{u}_j = \mathbf{w}_j / \|\mathbf{w}_j\|.$$

Then $\mathbf{u}_1, \dots, \mathbf{u}_p$ are orthonormal and, for each j , span the same subspace as $\mathbf{x}_1, \dots, \mathbf{x}_j$. Equivalently, $\mathbf{X} = \mathbf{Q}\mathbf{R}$ with $\mathbf{Q} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$ satisfying $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$ and $\mathbf{R}$ upper triangular with positive diagonal entries.

Proof. By induction on j . Suppose $\mathbf{u}_1, \dots, \mathbf{u}_{j-1}$ are orthonormal and span $\text{span}(\mathbf{x}_1, \dots, \mathbf{x}_{j-1})$. Then $\mathbf{w}_j \neq \mathbf{0}$, since otherwise $\mathbf{x}_j$ would lie in that span, contradicting independence. For $i < j$, $\mathbf{u}_i^\top \mathbf{w}_j = \mathbf{u}_i^\top \mathbf{x}_j - \sum_{k < j} (\mathbf{u}_k^\top \mathbf{x}_j) \mathbf{u}_k^\top \mathbf{u}_i = 0$. So $\mathbf{u}_1, \dots, \mathbf{u}_j$ are orthonormal, and they span $\text{span}(\mathbf{x}_1, \dots, \mathbf{x}_j)$ because $\mathbf{x}_j = \|\mathbf{w}_j\| \mathbf{u}_j + \sum_{k < j} (\mathbf{u}_k^\top \mathbf{x}_j) \mathbf{u}_k$. That last identity, read column by column, is $\mathbf{X} = \mathbf{Q}\mathbf{R}$ with $r_{jj} = \|\mathbf{w}_j\| > 0$ and $r_{kj} = \mathbf{u}_k^\top \mathbf{x}_j$ for $k < j$. ■

The factorization is unique ([Exercise \(B2\)](#)). Numerical software computes it by a more stable route, but the object is the same, and [Chapter 6](#) and [Chapter 10](#) rely on it. Two consequences are used repeatedly. First, any orthonormal set $\mathbf{u}_1, \dots, \mathbf{u}_r$ in $\mathbb{R}^n$ extends to an orthonormal basis: apply Gram–Schmidt to $\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{e}_1, \dots, \mathbf{e}_n$, discarding each vector that depends on its predecessors. Second, every subspace has an orthonormal basis.

1.5.3 Orthogonal matrices

A square matrix $\mathbf{Q}$ is **orthogonal** if $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$. By [Proposition 1.3.1](#), $\mathbf{Q}^\top$ is then the inverse of $\mathbf{Q}$, so also $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}$: the rows of an orthogonal matrix are orthonormal as well as its columns.

Proposition 1.5.3

Let $\mathbf{Q}$ and $\mathbf{P}$ be $n \times n$ orthogonal matrices.

- (a) $(\mathbf{Q}\mathbf{x})^\top (\mathbf{Q}\mathbf{v}) = \mathbf{x}^\top \mathbf{v}$ and $\|\mathbf{Q}\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x}, \mathbf{v}$.
- (b) $\mathbf{Q}^\top$ and $\mathbf{Q}\mathbf{P}$ are orthogonal.
- (c) Every entry satisfies $|q_{ij}| \leq 1$, and $\det \mathbf{Q} = \pm 1$.
- (d) If $[\mathbf{Q}_1, \mathbf{Q}_2]$ is orthogonal with $\mathbf{Q}_1$ of size $n \times r$, then $\mathbf{Q}_1 \mathbf{Q}_1^\top + \mathbf{Q}_2 \mathbf{Q}_2^\top = \mathbf{I}_n$, and $\mathcal{C}(\mathbf{Q}_2) = \mathcal{C}(\mathbf{Q}_1)^\perp$.

Proof.

- (a) $(\mathbf{Q}\mathbf{x})^\top (\mathbf{Q}\mathbf{v}) = \mathbf{x}^\top \mathbf{Q}^\top \mathbf{Q} \mathbf{v} = \mathbf{x}^\top \mathbf{v}$. (b) $(\mathbf{Q}\mathbf{P})^\top \mathbf{Q}\mathbf{P} = \mathbf{P}^\top \mathbf{P} = \mathbf{I}$.
- (b) Each column has unit length. For the determinant, see [Proposition 1.6.2](#): $1 = \det(\mathbf{Q}^\top \mathbf{Q}) = (\det \mathbf{Q})^2$. (d) The first statement is $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}$ written in blocks. For the second, $\mathbf{Q}_1^\top \mathbf{Q}_2 = \mathbf{0}$ gives $\mathcal{C}(\mathbf{Q}_2) \subseteq \mathcal{C}(\mathbf{Q}_1)^\perp$, and both have dimension $n - r$.

Part (a) says an orthogonal matrix is a rigid motion that fixes the origin: a rotation, a reflection, or a composition of the two. Changing coordinates by an orthogonal matrix changes neither lengths nor angles. This is why the distribution theory of [Chapter 4](#) can rotate a normal vector with independent components into a convenient coordinate system without changing its distribution.

Example. (Helmert matrices)

The $n \times n$ **Helmert matrix** H has first row $n^{-1/2}\mathbf{1}^\top$ and, for $k = 2, \dots, n$, k th row

$$\mathbf{h}_k^\top = \frac{1}{\sqrt{k(k-1)}} \underbrace{(1, \dots, 1)}_{k-1}, -(k-1), 0, \dots, 0).$$

Each row has unit length, since $(k-1) + (k-1)^2 = k(k-1)$. Row k is orthogonal to row 1 because its entries sum to zero. For $2 \leq j < k$, the nonzero entries of $\mathbf{h}_j$ sit in positions $1, \dots, j$, where $\mathbf{h}_k$ is constant, so $\mathbf{h}_j^\top \mathbf{h}_k$ is proportional to the sum of the entries of $\mathbf{h}_j$, which is zero. So H is orthogonal. For $\mathbf{y} \in \mathbb{R}^n$, $(H\mathbf{y})_1 = \sqrt{n}\bar{y}$ and, by [Proposition 1.5.3\(a\)](#),

$$\sum_{k=2}^n (\mathbf{h}_k^\top \mathbf{y})^2 = \|H\mathbf{y}\|^2 - n\bar{y}^2 = \sum_{i=1}^n (y_i - \bar{y})^2.$$

The sum of squared deviations is a sum of $n-1$ squares of orthonormal contrasts. With independent normal observations this gives a short proof that the sample mean and variance are independent ([Exercise \(B1\)](#)).

Two other families appear later. **Permutation matrices**, the identity with its rows reordered, are orthogonal. For a unit vector $\mathbf{u}$, the **Householder reflection** $\mathbf{I} - 2\mathbf{u}\mathbf{u}^\top$ is symmetric and orthogonal and reflects $\mathbb{R}^n$ in the hyperplane $\mathbf{u}^\perp$. Householder reflections are how QR factorizations are computed in practice ([Exercise \(B1\)](#)).

1.5.4 Exercises

1.5.4.1 A. Check your understanding

Exercise A1.

Show that every 2×2 orthogonal matrix is either a rotation $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ (determinant 1) or a reflection $\begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$ (determinant -1). Find the line fixed by the reflection.

1.5.4.2 B. Practice

Exercise B1.

Let $\mathbf{u}$ be a unit vector and $H = \mathbf{I} - 2\mathbf{u}\mathbf{u}^\top$. Show that H is symmetric and orthogonal, that $H\mathbf{u} = -\mathbf{u}$, and that $H\mathbf{v} = \mathbf{v}$ for $\mathbf{v} \perp \mathbf{u}$. Given x not a nonnegative multiple of

$\mathbf{e}_1$, find $\mathbf{u}$ with $\mathbf{H}\mathbf{x} = \|\mathbf{x}\|\mathbf{e}_1$. Explain how repeating this column by column produces a QR factorization.

Solution. $\mathbf{H}^\top = \mathbf{H}$, and $\mathbf{H}^\top \mathbf{H} = \mathbf{I} - 4\mathbf{u}\mathbf{u}^\top + 4\mathbf{u}(\mathbf{u}^\top \mathbf{u})\mathbf{u}^\top = \mathbf{I}$. Also $\mathbf{H}\mathbf{u} = \mathbf{u} - 2\mathbf{u} = -\mathbf{u}$, and $\mathbf{H}\mathbf{v} = \mathbf{v}$ when $\mathbf{u}^\top \mathbf{v} = 0$. Let $\mathbf{w} = \mathbf{x} - \|\mathbf{x}\|\mathbf{e}_1 \neq \mathbf{0}$ and $\mathbf{u} = \mathbf{w}/\|\mathbf{w}\|$. Then $\mathbf{w}^\top \mathbf{w} = 2\|\mathbf{x}\|^2 - 2\|\mathbf{x}\|x_1 = 2\mathbf{w}^\top \mathbf{x}$, so $\mathbf{H}\mathbf{x} = \mathbf{x} - 2\mathbf{w}(\mathbf{w}^\top \mathbf{x})/(\mathbf{w}^\top \mathbf{w}) = \mathbf{x} - \mathbf{w} = \|\mathbf{x}\|\mathbf{e}_1$. Applying such a reflection $\mathbf{H}_1$ to the first column of $\mathbf{X}$, then a reflection acting on coordinates $2, \dots, n$ to the second column of $\mathbf{H}_1\mathbf{X}$, and so on, gives $\mathbf{H}_p \cdots \mathbf{H}_1\mathbf{X} = \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$ with $\mathbf{R}$ upper triangular. Hence $\mathbf{X} = \mathbf{Q}\mathbf{R}$, where $\mathbf{Q}$ holds the first p columns of the orthogonal matrix $\mathbf{H}_1 \cdots \mathbf{H}_p$. In floating point one uses $\mathbf{w} = \mathbf{x} + \text{sgn}(x_1)\|\mathbf{x}\|\mathbf{e}_1$ instead, which avoids cancellation. It gives $\mathbf{H}\mathbf{x} = -\text{sgn}(x_1)\|\mathbf{x}\|\mathbf{e}_1$, so diagonal entries of $\mathbf{R}$ may come out negative; changing the sign of such a row of $\mathbf{R}$ and of the matching column of $\mathbf{Q}$ at the end restores the positive diagonal of Theorem 1.5.2.

Exercise B2.

Show that the QR factorization of a matrix with full column rank, with $\mathbf{R}$ having positive diagonal, is unique, and that $\mathbf{R}^\top$ is the Cholesky factor of $\mathbf{X}^\top \mathbf{X}$.

Solution. If $\mathbf{X} = \mathbf{Q}_1\mathbf{R}_1 = \mathbf{Q}_2\mathbf{R}_2$, then $\mathbf{X}^\top \mathbf{X} = \mathbf{R}_1^\top \mathbf{R}_1 = \mathbf{R}_2^\top \mathbf{R}_2$. Each $\mathbf{R}_k^\top$ is lower triangular with positive diagonal, so both are the Cholesky factor of $\mathbf{X}^\top \mathbf{X}$, which is unique by Theorem 1.7.4(d). So $\mathbf{R}_1 = \mathbf{R}_2$ and $\mathbf{Q}_1 = \mathbf{X}\mathbf{R}_1^{-1} = \mathbf{Q}_2$.

1.6 Trace, determinants and partitioned matrices

1.6.1 Trace

The **trace** of a square matrix is the sum of its diagonal entries, $\text{tr}(\mathbf{A}) = \sum_i a_{ii}$. It is linear, $\text{tr}(a\mathbf{A} + b\mathbf{B}) = a\text{tr}(\mathbf{A}) + b\text{tr}(\mathbf{B})$, and $\text{tr}(\mathbf{A}^\top) = \text{tr}(\mathbf{A})$.

Theorem 1.6.1 — Cyclic property

If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times m$, then $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$.

Proof. $\text{tr}(\mathbf{AB}) = \sum_{i=1}^m \sum_{k=1}^n a_{ik}b_{ki} = \sum_{k=1}^n \sum_{i=1}^m b_{ki}a_{ik} = \text{tr}(\mathbf{BA})$. ■

The two products need not have the same size. Applied repeatedly, the theorem allows cyclic permutations such as $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{CAB}) = \text{tr}(\mathbf{BCA})$, but not arbitrary reorderings. Three consequences occur again and again:

$$\text{tr}(\mathbf{A}^\top \mathbf{A}) = \sum_{i,j} a_{ij}^2 = \|\mathbf{A}\|_F^2, \quad \mathbf{x}^\top \mathbf{A} \mathbf{x} = \text{tr}(\mathbf{A} \mathbf{x} \mathbf{x}^\top), \quad \text{tr}(\mathbf{P}^{-1} \mathbf{A} \mathbf{P}) = \text{tr}(\mathbf{A}). \quad (1.6.1)$$

The middle one turns a scalar quadratic form into the trace of a matrix that is linear in $\mathbf{x} \mathbf{x}^\top$. Taking expectations then gives the mean of a random quadratic form ([Chapter 2](#)).

1.6.2 Determinants

The **determinant** of an $n \times n$ matrix is

$$\det \mathbf{A} = \sum_{\pi} \text{sgn}(\pi) a_{1\pi(1)} a_{2\pi(2)} \cdots a_{n\pi(n)},$$

the sum over all permutations π of $\{1, \dots, n\}$, where $\text{sgn}(\pi) = \pm 1$ according to whether π is a product of an even or odd number of transpositions. We collect the standard properties. Parts (a)–(d) are proved in any linear algebra text (Harville 1997, chapter 13), and the others follow from them.

Proposition 1.6.2 — Determinants

Let $\mathbf{A}, \mathbf{B}$ be $n \times n$.

- (a) $\det \mathbf{A}^\top = \det \mathbf{A}$.
- (b) $\det \mathbf{A}$ is a linear function of each column separately. It changes sign when two columns are interchanged, and is unchanged when a multiple of one column is added to another. The same holds for rows.
- (c) A triangular matrix has determinant equal to the product of its diagonal entries.
- (d) $\det(\mathbf{A}\mathbf{B}) = \det \mathbf{A} \det \mathbf{B}$.
- (e) $\mathbf{A}$ is nonsingular iff $\det \mathbf{A} \neq 0$, and then $\det \mathbf{A}^{-1} = 1/\det \mathbf{A}$.
- (f) $\det(c\mathbf{A}) = c^n \det \mathbf{A}$, and $\det \mathbf{Q} = \pm 1$ for orthogonal $\mathbf{Q}$.

Proof (Proof of (e) and (f)). If $\mathbf{A}$ is nonsingular, $1 = \det \mathbf{I} = \det \mathbf{A} \det \mathbf{A}^{-1}$ by (d). If $\mathbf{A}$ is singular, some column is a combination of the others. Subtracting that combination leaves a zero column without changing the determinant, by (b), and the determinant of a matrix with a zero column is zero, since every term of the sum contains a factor from that column. For (f), apply (b) to each of the n columns, and use (a) and (d) on $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$. ■

1.6.3 Schur complements and block determinants

Partitioned matrices arise whenever the regressors split into two groups, the model matrix $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ giving

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} \mathbf{X}_1^\top \mathbf{X}_1 & \mathbf{X}_1^\top \mathbf{X}_2 \\ \mathbf{X}_2^\top \mathbf{X}_1 & \mathbf{X}_2^\top \mathbf{X}_2 \end{pmatrix},$$

and whenever a random vector splits into two blocks (Chapter 3). In what follows

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}, \quad (1.6.2)$$

with $\mathbf{A}_{11}$ of size $k \times k$ and $\mathbf{A}_{22}$ of size $(n - k) \times (n - k)$.

Definition 1.6.1 — Schur complement

If $\mathbf{A}_{11}$ is nonsingular, the **Schur complement** of $\mathbf{A}_{11}$ in $\mathbf{A}$ is $\mathbf{A}_{22 \cdot 1} = \mathbf{A}_{22} - \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$.
If $\mathbf{A}_{22}$ is nonsingular, the Schur complement of $\mathbf{A}_{22}$ is $\mathbf{A}_{11 \cdot 2} = \mathbf{A}_{11} - \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21}$.

Everything in this subsection comes from one identity, block Gaussian elimination. If $\mathbf{A}_{11}$ is nonsingular, multiplying out shows

$$\mathbf{A} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{A}_{11} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22 \cdot 1} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}. \quad (1.6.3)$$

The outer factors are block triangular with identity blocks on the diagonal, and are nonsingular with inverses obtained by changing the sign of the off-diagonal block.

Theorem 1.6.3 — Block determinants

Let $\mathbf{A}$ be partitioned as in Eq. 1.6.2.

- (a) If $\mathbf{A}_{12} = \mathbf{0}$ or $\mathbf{A}_{21} = \mathbf{0}$, then $\det \mathbf{A} = \det \mathbf{A}_{11} \det \mathbf{A}_{22}$.
- (b) If $\mathbf{A}_{11}$ is nonsingular, $\det \mathbf{A} = \det \mathbf{A}_{11} \det \mathbf{A}_{22 \cdot 1}$. If $\mathbf{A}_{22}$ is nonsingular, $\det \mathbf{A} = \det \mathbf{A}_{22} \det \mathbf{A}_{11 \cdot 2}$.
- (c) For $m \times n$ matrices $\mathbf{U}$ and $n \times m$ matrices $\mathbf{V}$, $\det(\mathbf{I}_m + \mathbf{U}\mathbf{V}) = \det(\mathbf{I}_n + \mathbf{V}\mathbf{U})$. If $\mathbf{B}$ is $m \times m$ and nonsingular, $\det(\mathbf{B} + \mathbf{U}\mathbf{V}) = \det \mathbf{B} \det(\mathbf{I}_n + \mathbf{V}\mathbf{B}^{-1}\mathbf{U})$.

Proof.

- (a) Suppose $\mathbf{A}_{12} = \mathbf{0}$ (otherwise transpose). In the permutation expansion, a term that is nonzero cannot use an entry of the zero block $\mathbf{A}_{12}$. So each of the first k rows takes its entry from the first k columns, and π maps $\{1, \dots, k\}$ onto itself. Such a permutation splits into a permutation of the first k indices and one of the rest. Its sign is the product of their signs, so the sum factors as $\det \mathbf{A}_{11} \det \mathbf{A}_{22}$.
- (b) Take determinants in Eq. 1.6.3, using (a) and Proposition 1.6.2(d). The outer factors have determinant 1. The second statement follows in the same way from the elimination that starts with $\mathbf{A}_{22}$.
- (c) Apply (b) in both ways to $\begin{pmatrix} \mathbf{I}_m & -\mathbf{U} \\ \mathbf{V} & \mathbf{I}_n \end{pmatrix}$, whose Schur complements are $\mathbf{I}_n + \mathbf{V}\mathbf{U}$ and $\mathbf{I}_m + \mathbf{U}\mathbf{V}$. For the second statement, write $\mathbf{B} + \mathbf{U}\mathbf{V} = \mathbf{B}(\mathbf{I}_m + \mathbf{B}^{-1}\mathbf{U}\mathbf{V})$ and apply the first.

■

Part (c) with $n = 1$ is the **matrix determinant lemma** $\det(\mathbf{B} + \mathbf{u}\mathbf{v}^\top) = \det \mathbf{B} (1 + \mathbf{v}^\top \mathbf{B}^{-1} \mathbf{u})$.

1.6.4 Inverses of partitioned matrices

Theorem 1.6.4 — Partitioned inverse

Let $\mathbf{A}$ be partitioned as in Eq. 1.6.2 with $\mathbf{A}_{11}$ nonsingular. Then $\mathbf{A}$ is nonsingular iff $\mathbf{A}_{22 \cdot 1}$ is, and in that case

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{A}_{11}^{-1} + \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{A}_{22 \cdot 1}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & -\mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{A}_{22 \cdot 1}^{-1} \\ -\mathbf{A}_{22 \cdot 1}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & \mathbf{A}_{22 \cdot 1}^{-1} \end{pmatrix}. \quad (1.6.4)$$

If $\mathbf{A}_{22}$ is also nonsingular, the leading block of $\mathbf{A}^{-1}$ equals $\mathbf{A}_{11 \cdot 2}^{-1}$:

$$(\mathbf{A}_{11} - \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21})^{-1} = \mathbf{A}_{11}^{-1} + \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{A}_{22 \cdot 1}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1}. \quad (1.6.5)$$

Proof. The nonsingularity statement is Theorem 1.6.3(b). Inverting the three factors of Eq. 1.6.3 and multiplying them in reverse order gives

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{I} & -\mathbf{A}_{11}^{-1} \mathbf{A}_{12} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{A}_{11}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22 \cdot 1}^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{A}_{21} \mathbf{A}_{11}^{-1} & \mathbf{I} \end{pmatrix},$$

which multiplies out to Eq. 1.6.4. Eliminating with $\mathbf{A}_{22}$ first gives a second expression for $\mathbf{A}^{-1}$, whose leading block is $\mathbf{A}_{11 \cdot 2}^{-1}$. The inverse is unique, so the two leading blocks agree. ■

For a symmetric positive definite $\mathbf{X}^\top \mathbf{X}$ (Section 1.7), the theorem says that the block of $(\mathbf{X}^\top \mathbf{X})^{-1}$ belonging to $\mathbf{X}_1$ is $(\mathbf{X}_1^\top \mathbf{X}_1 - \mathbf{X}_1^\top \mathbf{X}_2 (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} \mathbf{X}_2^\top \mathbf{X}_1)^{-1}$. Chapter 6 derives the same formula geometrically, from the Frisch–Waugh–Lovell theorem.

Identity Eq. 1.6.5 is a disguised form of a result that deserves its own name. Rename $\mathbf{A}_{11} = \mathbf{B}$, $\mathbf{A}_{12} = \mathbf{U}$, $\mathbf{A}_{21} = \mathbf{V}^\top$ and $\mathbf{A}_{22} = -\mathbf{C}^{-1}$.

Theorem 1.6.5 — Sherman–Morrison–Woodbury

Let $\mathbf{B}$ ($n \times n$) and $\mathbf{C}$ ($k \times k$) be nonsingular, and let $\mathbf{U}, \mathbf{V}$ be $n \times k$. If $\mathbf{C}^{-1} + \mathbf{V}^\top \mathbf{B}^{-1} \mathbf{U}$ is nonsingular, then so is $\mathbf{B} + \mathbf{U} \mathbf{C} \mathbf{V}^\top$, and

$$(\mathbf{B} + \mathbf{U} \mathbf{C} \mathbf{V}^\top)^{-1} = \mathbf{B}^{-1} - \mathbf{B}^{-1} \mathbf{U} (\mathbf{C}^{-1} + \mathbf{V}^\top \mathbf{B}^{-1} \mathbf{U})^{-1} \mathbf{V}^\top \mathbf{B}^{-1}. \quad (1.6.6)$$

In particular, for vectors $\mathbf{u}, \mathbf{v}$ with $1 + \mathbf{v}^\top \mathbf{B}^{-1} \mathbf{u} \neq 0$,

$$(\mathbf{B} + \mathbf{u} \mathbf{v}^\top)^{-1} = \mathbf{B}^{-1} - \frac{\mathbf{B}^{-1} \mathbf{u} \mathbf{v}^\top \mathbf{B}^{-1}}{1 + \mathbf{v}^\top \mathbf{B}^{-1} \mathbf{u}}. \quad (1.6.7)$$

Proof. Apply Theorem 1.6.4 to $\mathbf{N} = \begin{pmatrix} \mathbf{B} & \mathbf{U} \\ \mathbf{V}^\top & -\mathbf{C}^{-1} \end{pmatrix}$. Both diagonal blocks are nonsingular. The Schur complement of $\mathbf{B}$ is $-(\mathbf{C}^{-1} + \mathbf{V}^\top \mathbf{B}^{-1} \mathbf{U})$, which is nonsingular by hypothesis,

so $\mathbf{N}$ is nonsingular. Then the Schur complement of the lower block, $\mathbf{B} + \mathbf{UCV}^\top$, is nonsingular by the same theorem with the roles of the blocks exchanged, and Eq. 1.6.5 becomes Eq. 1.6.6. For Eq. 1.6.7 take $k = 1$ and $\mathbf{C} = 1$.

■

Idea. Every result in this section comes from one block elimination, Eq. 1.6.3: a partitioned matrix is a unit triangular matrix times $\text{diag}(\mathbf{A}_{11}, \mathbf{A}_{22.1})$ times another. Taking determinants gives Theorem 1.6.3, inverting the factors gives Theorem 1.6.4, and eliminating in the other order and comparing gives the Woodbury identity. When a formula about a block is forgotten, redo the elimination.

The identity replaces the inversion of an $n \times n$ matrix by that of a $k \times k$ matrix. That matters when $\mathbf{B}^{-1}$ is already known or cheap, as in the following two examples.

Example. (Deleting an observation)

Removing the i th row $\mathbf{x}_i^\top$ from a model matrix changes $\mathbf{X}^\top \mathbf{X}$ into $\mathbf{X}^\top \mathbf{X} - \mathbf{x}_i \mathbf{x}_i^\top$. With $\mathbf{u} = -\mathbf{x}_i$, $\mathbf{v} = \mathbf{x}_i$ and $h_i = \mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i$, Eq. 1.6.7 gives

$$(\mathbf{X}^\top \mathbf{X} - \mathbf{x}_i \mathbf{x}_i^\top)^{-1} = (\mathbf{X}^\top \mathbf{X})^{-1} + \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i \mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1}}{1 - h_i},$$

valid whenever $h_i < 1$, and the determinant lemma gives $\det(\mathbf{X}^\top \mathbf{X} - \mathbf{x}_i \mathbf{x}_i^\top) = (1 - h_i) \det(\mathbf{X}^\top \mathbf{X})$. The quantity h_i is the leverage of observation i (Chapter 6), and these formulas are what make the deletion diagnostics of Chapter 20 cheap. For Brownlee's stack-loss data ($n = 21$, $p = 4$), the listing updates the inverse for each of the 21 deletions. The largest relative discrepancy from direct inversion is 1×10^{-13} . Observation 17 has the largest leverage, 0.412, and removing it multiplies $\det(\mathbf{X}^\top \mathbf{X})$ by 0.588.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.stackloss.load_pandas().data
X = np.column_stack([np.ones(len(data)), data[["AIRFLOW", "WATERTEMP", "ACIDCONC"]]])
n, p = X.shape

A_inv = np.linalg.inv(X.T @ X)
h = np.einsum("ij,jk,ik->i", X, A_inv, X)           # leverages  $x_{-i}' (X'X)^{-1} x_{-i}$ 
x_i

worst = 0.0
for i in range(n):
    x = X[i]
    # Sherman-Morrison with  $u = -x$ ,  $v = x$ :  $(X'X - x x')^{-1}$ 
    B_update = A_inv + np.outer(A_inv @ x, x @ A_inv) / (1 - h[i])
    X_del = np.delete(X, i, axis=0)
    B_direct = np.linalg.inv(X_del.T @ X_del)
    worst = max(worst, np.abs(B_update - B_direct).max() / np.abs(B_direct).max())
print(f"largest relative discrepancy over all {n} deletions: {worst:.1e}")
```

Example. (Diagonal plus low rank)

In factor models and in the linear mixed models of Chapter 32, covariance matrices have the form $\Sigma = D + LL^\top$ with D diagonal and positive and L of size $N \times k$, $k \ll N$. By Eq. 1.6.6 and Theorem 1.6.3(c),

$$\Sigma^{-1} = D^{-1} - D^{-1}L(I_k + L^\top D^{-1}L)^{-1}L^\top D^{-1}, \quad \det \Sigma = \det D \det(I_k + L^\top D^{-1}L).$$

Each needs only a $k \times k$ factorization. The script checks this with $N = 1500$ and $k = 4$. The solution of $\Sigma w = v$ agrees with a direct solve to a relative error of 8×10^{-14} .

Example. (Equicorrelation)

The matrix $E = (1 - \rho)I_n + \rho J_n$ has unit diagonal and every off-diagonal entry equal to ρ . It is the correlation matrix of exchangeable observations. Taking $B = (1 - \rho)I$ and $u = \rho \mathbf{1}$, $v = \mathbf{1}$, for $\rho \neq 1$ and $1 + (n - 1)\rho \neq 0$,

$$E^{-1} = \frac{1}{1 - \rho} \left(I_n - \frac{\rho}{1 + (n - 1)\rho} J_n \right), \quad \det E = (1 - \rho)^{n-1} (1 + (n - 1)\rho).$$

1.6.5 Exercises

1.6.5.1 A. Check your understanding

Exercise A1.

Show that $\text{tr}(ABC) = \text{tr}(ACB)$ when A, B, C are symmetric, and find 2×2 matrices for which the two traces differ.

Solution. For symmetric matrices, $\text{tr}(ABC) = \text{tr}((ABC)^\top) = \text{tr}(CBA) = \text{tr}(ACB)$, using transposition and then the cyclic property. For a counterexample take $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. Then $ABC = C$ has trace 1, while $AC = \mathbf{0}$ gives $\text{tr}(ACB) = 0$.

Exercise A2.

Let A be nonsingular, a a vector and c a scalar with $s = c - a^\top A^{-1}a \neq 0$. Write down the inverse and the determinant of $\begin{pmatrix} A & a \\ a^\top & c \end{pmatrix}$. Use it to update $(X^\top X)^{-1}$ when a column is added to X .

1.6.5.2 B. Practice

Exercise B1.

Let $\hat{\beta} = (X^\top X)^{-1}X^\top y$, $e_i = y_i - x_i^\top \hat{\beta}$ and $h_i = x_i^\top (X^\top X)^{-1}x_i < 1$. Show that the

least squares coefficients computed without observation i are

$$\hat{\beta}_{(i)} = \hat{\beta} - \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i e_i}{1 - h_i}.$$

Solution. Write $\mathbf{K} = (\mathbf{X}^\top \mathbf{X})^{-1}$ and $\mathbf{x} = \mathbf{x}_i$. Deleting row i turns $\mathbf{X}^\top \mathbf{y}$ into $\mathbf{X}^\top \mathbf{y} - \mathbf{x} y_i$, so by [Example \(Deleting an observation\)](#)

$$\hat{\beta}_{(i)} = \left(\mathbf{K} + \frac{\mathbf{K} \mathbf{x} \mathbf{x}^\top \mathbf{K}}{1 - h_i} \right) (\mathbf{X}^\top \mathbf{y} - \mathbf{x} y_i) = \hat{\beta} - \mathbf{K} \mathbf{x} y_i + \frac{\mathbf{K} \mathbf{x} (\mathbf{x}^\top \hat{\beta} - h_i y_i)}{1 - h_i} = \hat{\beta} - \mathbf{K} \mathbf{x} \frac{y_i - \mathbf{x}^\top \hat{\beta}}{1 - h_i},$$

since $(1 - h_i)y_i - \mathbf{x}^\top \hat{\beta} + h_i y_i = y_i - \mathbf{x}^\top \hat{\beta}$.

Exercise B2.

For an $n \times p$ matrix $\mathbf{X}$ and $\lambda > 0$, show that $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I}_n)^{-1}$. Explain why the right-hand side is preferable when p is much larger than n , and derive the identity a second way from [Theorem 1.6.5](#).

Solution. Both $\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}$ and $\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I}$ are positive definite. The identity $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p) \mathbf{X}^\top = \mathbf{X}^\top \mathbf{X} \mathbf{X}^\top + \lambda \mathbf{X}^\top = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I}_n)$ gives the result after multiplying by the two inverses. When $p \gg n$, the right-hand side inverts an $n \times n$ matrix instead of a $p \times p$ one. For the second derivation, apply [Theorem 1.6.5](#) with $\mathbf{B} = \lambda \mathbf{I}_p$, $\mathbf{U} = \mathbf{X}^\top$, $\mathbf{C} = \mathbf{I}_n$ and $\mathbf{V} = \mathbf{X}^\top$: $(\lambda \mathbf{I}_p + \mathbf{X}^\top \mathbf{X})^{-1} = \lambda^{-1} \mathbf{I}_p - \lambda^{-2} \mathbf{X}^\top \mathbf{W}^{-1} \mathbf{X}$ with $\mathbf{W} = \mathbf{I}_n + \mathbf{X} \mathbf{X}^\top / \lambda$. Multiplying on the right by $\mathbf{X}^\top$ gives $\lambda^{-1} \mathbf{X}^\top [\mathbf{I}_n - \mathbf{W}^{-1} (\mathbf{X} \mathbf{X}^\top / \lambda)] = \lambda^{-1} \mathbf{X}^\top \mathbf{W}^{-1} [\mathbf{W} - \mathbf{X} \mathbf{X}^\top / \lambda] = \lambda^{-1} \mathbf{X}^\top \mathbf{W}^{-1} = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I}_n)^{-1}$.

1.7 Eigenvalues, the spectral theorem and positive definiteness

1.7.1 Eigenvalues

A scalar λ is an **eigenvalue** of the $n \times n$ matrix $\mathbf{A}$, with **eigenvector** $\mathbf{x} \neq \mathbf{0}$, if $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$. Equivalently, $\mathbf{A} - \lambda\mathbf{I}$ is singular, that is, λ is a root of the **characteristic polynomial** $p(\lambda) = \det(\lambda\mathbf{I} - \mathbf{A})$. This polynomial has degree n and leading coefficient one, so over the complex numbers it factors as $\prod_{i=1}^n (\lambda - \lambda_i)$. The eigenvalues $\lambda_1, \dots, \lambda_n$ are listed with multiplicity and may be complex when $\mathbf{A}$ is not symmetric.

Proposition 1.7.1

Let $\mathbf{A}$ be $n \times n$ with eigenvalues $\lambda_1, \dots, \lambda_n$.

- (a) $\det \mathbf{A} = \prod_i \lambda_i$ and $\text{tr } \mathbf{A} = \sum_i \lambda_i$.
- (b) If $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$, then $f(\mathbf{A})\mathbf{x} = f(\lambda)\mathbf{x}$ for every polynomial f . If $\mathbf{A}$ is nonsingular, $\mathbf{A}^{-1}\mathbf{x} = \lambda^{-1}\mathbf{x}$.

- (c) $\mathbf{A}$ and $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ have the same characteristic polynomial for nonsingular $\mathbf{P}$.
 (d) If $\mathbf{C}$ is $m \times n$ and $\mathbf{D}$ is $n \times m$, then $\lambda^n \det(\lambda \mathbf{I}_m - \mathbf{C}\mathbf{D}) = \lambda^m \det(\lambda \mathbf{I}_n - \mathbf{D}\mathbf{C})$. So $\mathbf{C}\mathbf{D}$ and $\mathbf{D}\mathbf{C}$ have the same nonzero eigenvalues, with the same multiplicities.

Proof.

- (a) Setting $\lambda = 0$ in $\det(\lambda \mathbf{I} - \mathbf{A}) = \prod (\lambda - \lambda_i)$ gives $(-1)^n \det \mathbf{A} = (-1)^n \prod \lambda_i$. For the trace, compare coefficients of λ^{n-1} . On the right it is $-\sum \lambda_i$. On the left, in the permutation expansion of $\det(\lambda \mathbf{I} - \mathbf{A})$, only the product of diagonal entries $\prod_i (\lambda - a_{ii})$ contains λ^{n-1} , because every other term omits at least two diagonal entries. Its coefficient is $-\sum a_{ii}$.
 (b) $\mathbf{A}^k \mathbf{x} = \lambda^k \mathbf{x}$ by induction, and $\mathbf{x} = \mathbf{A}^{-1} \mathbf{A} \mathbf{x} = \lambda \mathbf{A}^{-1} \mathbf{x}$ with $\lambda \neq 0$. (c) $\det(\lambda \mathbf{I} - \mathbf{P}^{-1} \mathbf{A} \mathbf{P}) = \det(\mathbf{P}^{-1} (\lambda \mathbf{I} - \mathbf{A}) \mathbf{P}) = \det(\lambda \mathbf{I} - \mathbf{A})$.
 (c) For $\lambda \neq 0$, Theorem 1.6.3(c) gives $\det(\mathbf{I}_m - \lambda^{-1} \mathbf{C}\mathbf{D}) = \det(\mathbf{I}_n - \lambda^{-1} \mathbf{D}\mathbf{C})$. Multiplying by λ^{m+n} gives the identity for $\lambda \neq 0$. Both sides are polynomials, so it holds for all λ .

■

1.7.2 The spectral theorem

Symmetric matrices have the best possible eigenstructure, and almost every matrix whose eigenvalues matter in statistics is symmetric.

Theorem 1.7.2 — Spectral theorem

Let $\mathbf{A}$ be an $n \times n$ symmetric matrix. Then all eigenvalues of $\mathbf{A}$ are real, and there is an orthogonal matrix $\mathbf{Q} = [\mathbf{q}_1, \dots, \mathbf{q}_n]$ with

$$\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top = \sum_{i=1}^n \lambda_i \mathbf{q}_i \mathbf{q}_i^\top, \quad \mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n). \quad (1.7.1)$$

The columns of $\mathbf{Q}$ are an orthonormal basis of eigenvectors, $\mathbf{A} \mathbf{q}_i = \lambda_i \mathbf{q}_i$. Eigenvectors belonging to distinct eigenvalues are orthogonal.

Proof. Real eigenvalues. Let $\mathbf{A} \mathbf{z} = \lambda \mathbf{z}$ with $\mathbf{z} \in \mathbb{C}^n$ nonzero, and let $\bar{\mathbf{z}}$ be the complex conjugate. The scalar $s = \bar{\mathbf{z}}^\top \mathbf{A} \mathbf{z}$ is real, since its conjugate is $\mathbf{z}^\top \mathbf{A} \bar{\mathbf{z}} = (\mathbf{z}^\top \mathbf{A} \bar{\mathbf{z}})^\top = \bar{\mathbf{z}}^\top \mathbf{A} \mathbf{z} = s$ by symmetry. Also $s = \lambda \bar{\mathbf{z}}^\top \mathbf{z} = \lambda \sum_i |z_i|^2$, with the sum real and positive. So λ is real. The real singular matrix $\mathbf{A} - \lambda \mathbf{I}$ then has a real nonzero null vector, so there is a real eigenvector.

Diagonalization. By induction on n , the case $n = 1$ being trivial. Let $\mathbf{q}_1$ be a real unit eigenvector for λ_1 and extend it to an orthogonal matrix $[\mathbf{q}_1, \mathbf{Q}_2]$ (Section 1.5). Since $\mathbf{Q}_2^\top \mathbf{A} \mathbf{q}_1 = \lambda_1 \mathbf{Q}_2^\top \mathbf{q}_1 = \mathbf{0}$ and $\mathbf{A}$ is symmetric,

$$[\mathbf{q}_1, \mathbf{Q}_2]^\top \mathbf{A} [\mathbf{q}_1, \mathbf{Q}_2] = \begin{pmatrix} \lambda_1 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{Q}_2^\top \mathbf{A} \mathbf{Q}_2 \end{pmatrix}.$$

The $(n-1) \times (n-1)$ block is symmetric, so by induction $\mathbf{Q}_2^\top \mathbf{A} \mathbf{Q}_2 = \mathbf{P} \mathbf{\Lambda}_2 \mathbf{P}^\top$ with $\mathbf{P}$ orthogonal. Then $\mathbf{Q} = [\mathbf{q}_1, \mathbf{Q}_2 \mathbf{P}]$ is orthogonal (Proposition 1.5.3(b)) and $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \text{diag}(\lambda_1, \mathbf{\Lambda}_2)$.

Orthogonality. If $\mathbf{A} \mathbf{u} = \lambda \mathbf{u}$ and $\mathbf{A} \mathbf{v} = \mu \mathbf{v}$, then $\lambda \mathbf{u}^\top \mathbf{v} = \mathbf{u}^\top \mathbf{A} \mathbf{v} = \mu \mathbf{u}^\top \mathbf{v}$, so $\mathbf{u}^\top \mathbf{v} = 0$ when $\lambda \neq \mu$.

■

Unless stated otherwise, the eigenvalues of symmetric $\mathbf{A}$ are ordered as $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$, and we write $\lambda_{\max} = \lambda_1$ and $\lambda_{\min} = \lambda_n$. The following consequences are used without comment later on.

Corollary 1.7.3

Let $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top$ be symmetric.

- (a) $\text{rank}(\mathbf{A})$ counts the nonzero eigenvalues. $\mathcal{C}(\mathbf{A})$ is spanned by the $\mathbf{q}_i$ with $\lambda_i \neq 0$, and $\mathcal{N}(\mathbf{A})$ by those with $\lambda_i = 0$.
- (b) $\mathbf{A}^k = \mathbf{Q} \mathbf{\Lambda}^k \mathbf{Q}^\top$ for $k = 1, 2, \dots$. If $\mathbf{A}$ is nonsingular, $\mathbf{A}^{-1} = \mathbf{Q} \mathbf{\Lambda}^{-1} \mathbf{Q}^\top$, and $\mathbf{A}^+ = \mathbf{Q} \mathbf{\Lambda}^+ \mathbf{Q}^\top$ in general, where $\mathbf{\Lambda}^+$ inverts the nonzero diagonal entries.
- (c) The eigenspace $\{\mathbf{x} : \mathbf{A} \mathbf{x} = \lambda \mathbf{x}\}$ has dimension equal to the multiplicity of λ as a root of the characteristic polynomial.

Proof. Multiplying by the nonsingular $\mathbf{Q}$ and $\mathbf{Q}^\top$ changes neither rank (Proposition 1.2.3(b)) nor, after the change of variable $\mathbf{x} = \mathbf{Q} \mathbf{c}$, the structure of null spaces and eigenspaces. So each statement reduces to the diagonal matrix $\mathbf{\Lambda}$, where it is immediate. For (b), $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$ collapses the middle factors, and the four Penrose conditions for $\mathbf{Q} \mathbf{\Lambda}^+ \mathbf{Q}^\top$ reduce to those for $\mathbf{\Lambda}^+$.

■

Example. (Eigenvalues of the equicorrelation matrix)

For $\mathbf{E} = (1 - \rho) \mathbf{I} + \rho \mathbf{J}_n$ of Example (Equicorrelation), $\mathbf{E} \mathbf{1} = (1 + (n-1)\rho) \mathbf{1}$, and $\mathbf{E} \mathbf{v} = (1 - \rho) \mathbf{v}$ for every $\mathbf{v} \perp \mathbf{1}$. So the eigenvalues are $1 + (n-1)\rho$ once and $1 - \rho$ with multiplicity $n-1$, with eigenvectors $\mathbf{1}/\sqrt{n}$ and the last $n-1$ rows of the Helmert matrix (Example (Helmert matrices)). The determinant and inverse found earlier follow at once from Proposition 1.7.1 and Corollary 1.7.3(b).

1.7.3 Positive definite and nonnegative definite matrices

Definition 1.7.1

A symmetric matrix $\mathbf{A}$ is **positive definite** if $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ for every $\mathbf{x} \neq \mathbf{0}$, and **nonnegative definite** if $\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0$ for every $\mathbf{x}$. For symmetric $\mathbf{A}, \mathbf{B}$ of the same size we

write $\mathbf{A} \succeq \mathbf{B}$ if $\mathbf{A} - \mathbf{B}$ is nonnegative definite, and $\mathbf{A} \succ \mathbf{B}$ if it is positive definite (the **Loewner order**).

In this book, “nonnegative definite” includes positive definite. Some texts, Rencher and Schaalje among them, reserve “positive semidefinite” for matrices that are nonnegative definite but singular. We use the terms only for symmetric matrices. Covariance matrices are nonnegative definite (Chapter 2), and *Gram matrices* $\mathbf{X}^\top \mathbf{X}$ are too, since $\mathbf{x}^\top \mathbf{X}^\top \mathbf{X} \mathbf{x} = \|\mathbf{X} \mathbf{x}\|^2$.

Theorem 1.7.4 — Characterizations

For a symmetric $n \times n$ matrix $\mathbf{A}$ the following are equivalent:

- (a) $\mathbf{A}$ is positive definite;
- (b) every eigenvalue of $\mathbf{A}$ is positive;
- (c) $\mathbf{A} = \mathbf{B}^\top \mathbf{B}$ for some matrix $\mathbf{B}$ with n columns and full column rank;
- (d) $\mathbf{A} = \mathbf{L} \mathbf{L}^\top$ with $\mathbf{L}$ lower triangular with positive diagonal entries (the **Cholesky factorization**; $\mathbf{L}$ is unique);
- (e) every leading principal minor $\det \mathbf{A}_{[k]}$, $k = 1, \dots, n$, is positive, where $\mathbf{A}_{[k]}$ is the upper left $k \times k$ block.

Likewise, the following are equivalent: (a') $\mathbf{A}$ is nonnegative definite; (b') every eigenvalue is nonnegative; (c') $\mathbf{A} = \mathbf{B}^\top \mathbf{B}$ for some $\mathbf{B}$, which can be taken of size $\text{rank}(\mathbf{A}) \times n$; (e') every principal minor (determinant of a submatrix with the same row and column indices) is nonnegative.

Proof. Write $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top$. (a) $\Rightarrow$ (b): $\lambda_i = \mathbf{q}_i^\top \mathbf{A} \mathbf{q}_i > 0$. (b) $\Rightarrow$ (c): $\mathbf{B} = \mathbf{\Lambda}^{1/2} \mathbf{Q}^\top$ is nonsingular. (c) $\Rightarrow$ (a): $\mathbf{x}^\top \mathbf{B}^\top \mathbf{B} \mathbf{x} = \|\mathbf{B} \mathbf{x}\|^2$, which is positive for $\mathbf{x} \neq \mathbf{0}$ by full column rank. (d) $\Rightarrow$ (c) is immediate, with $\mathbf{B} = \mathbf{L}^\top$.

(a) $\Rightarrow$ (d), by induction on n . Write $\mathbf{A} = \begin{pmatrix} a & \mathbf{b}^\top \\ \mathbf{b} & \mathbf{A}_2 \end{pmatrix}$. Then $a = \mathbf{e}_1^\top \mathbf{A} \mathbf{e}_1 > 0$. The Schur complement $\mathbf{A}_2 - \mathbf{b} \mathbf{b}^\top / a$ is positive definite: for $\mathbf{v} \neq \mathbf{0}$, put $\mathbf{x} = (-\mathbf{b}^\top \mathbf{v} / a, \mathbf{v}^\top)^\top$ and expand to get $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \mathbf{v}^\top (\mathbf{A}_2 - \mathbf{b} \mathbf{b}^\top / a) \mathbf{v} > 0$. By induction $\mathbf{A}_2 - \mathbf{b} \mathbf{b}^\top / a = \mathbf{L}_2 \mathbf{L}_2^\top$, and $\mathbf{L} = \begin{pmatrix} \sqrt{a} & \mathbf{0}^\top \\ \mathbf{b} / \sqrt{a} & \mathbf{L}_2 \end{pmatrix}$ satisfies $\mathbf{L} \mathbf{L}^\top = \mathbf{A}$. For uniqueness, if $\mathbf{L} \mathbf{L}^\top = \mathbf{K} \mathbf{K}^\top$, then $\mathbf{K}^{-1} \mathbf{L} = \mathbf{K}^\top (\mathbf{L}^\top)^{-1}$ is both lower and upper triangular, hence diagonal, say $\mathbf{D}$. From $\mathbf{K}^{-1} \mathbf{L} = \mathbf{D}$ and $\mathbf{L} \mathbf{L}^\top = \mathbf{K} \mathbf{K}^\top$ we get $\mathbf{D} \mathbf{D}^\top = \mathbf{I}$, so $\mathbf{D} = \mathbf{I}$ because its diagonal is positive.

(a) $\Rightarrow$ (e): $\mathbf{A}_{[k]} = \mathbf{E}^\top \mathbf{A} \mathbf{E}$ with $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_k]$ is positive definite, since $\mathbf{x}^\top \mathbf{A}_{[k]} \mathbf{x} = (\mathbf{E} \mathbf{x})^\top \mathbf{A} (\mathbf{E} \mathbf{x})$. By (b) and Proposition 1.7.1(a) its determinant is positive. (e) $\Rightarrow$ (a), by induction on n . The block $\mathbf{A}_{[n-1]}$ satisfies (e) and so is positive definite, so by (a) $\Rightarrow$ (d), which is already proved, $\mathbf{A}_{[n-1]} = \mathbf{L} \mathbf{L}^\top$. Write $\mathbf{A} = \begin{pmatrix} \mathbf{A}_{[n-1]} & \mathbf{c} \\ \mathbf{c}^\top & d \end{pmatrix}$ and $s = d - \mathbf{c}^\top \mathbf{A}_{[n-1]}^{-1} \mathbf{c}$. By Theorem 1.6.3(b), $s = \det \mathbf{A} / \det \mathbf{A}_{[n-1]} > 0$. With $\mathbf{T} = \begin{pmatrix} \mathbf{I} & \mathbf{A}_{[n-1]}^{-1} \mathbf{c} \\ \mathbf{0}^\top & 1 \end{pmatrix}$, a direct check gives $\mathbf{A} = \mathbf{T}^\top \text{diag}(\mathbf{A}_{[n-1]}, s) \mathbf{T} = \mathbf{B}^\top \mathbf{B}$ with $\mathbf{B} = \text{diag}(\mathbf{L}^\top, \sqrt{s}) \mathbf{T}$ nonsingular, so $\mathbf{A}$ is positive definite by (c) $\Rightarrow$ (a).

The primed statements: (a') $\Leftrightarrow$ (b') $\Leftrightarrow$ (c') exactly as above, taking for $\mathbf{B}$ the rows of $\mathbf{\Lambda}^{1/2} \mathbf{Q}^\top$ that belong to nonzero eigenvalues. (a') $\Rightarrow$ (e') as above, with an arbitrary index set in place of $\{1, \dots, k\}$. For (e') $\Rightarrow$ (a'), fix $t > 0$. Each column of $\mathbf{A}_{[k]} + t \mathbf{I}$ is a column of $\mathbf{A}_{[k]}$ plus t times a coordinate vector. Expanding the determinant by linearity in each

column gives $\det(\mathbf{A}_{[k]} + t\mathbf{I}) = \sum_S t^{k-|S|} \det \mathbf{A}_{SS}$, the sum over all subsets $S \subseteq \{1, \dots, k\}$, with $\det \mathbf{A}_{\emptyset\emptyset} = 1$. Every term is nonnegative and the term for $S = \emptyset$ is $t^k > 0$. So $\mathbf{A} + t\mathbf{I}$ is positive definite by (e) $\Rightarrow$ (a), and $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \lim_{t \rightarrow 0} \mathbf{x}^\top (\mathbf{A} + t\mathbf{I}) \mathbf{x} \geq 0$.

■

Two warnings. Leading principal minors do not suffice for nonnegative definiteness: $\text{diag}(0, -1)$ has leading minors 0 and 0. And (c') holds with many matrices $\mathbf{B}$. The Cholesky factor and the symmetric square root below are two convenient choices.

Proposition 1.7.5

Let $\mathbf{A}$ be $n \times n$ nonnegative definite.

- (a) $a_{ii} \geq 0$, and if $a_{ii} = 0$ then row i and column i of $\mathbf{A}$ are zero.
- (b) $\mathbf{x}^\top \mathbf{A} \mathbf{x} = 0$ iff $\mathbf{A} \mathbf{x} = \mathbf{0}$.
- (c) $\mathbf{B}^\top \mathbf{A} \mathbf{B}$ is nonnegative definite for any $n \times q$ matrix $\mathbf{B}$. If $\mathbf{A}$ is positive definite, $\mathbf{B}^\top \mathbf{A} \mathbf{B}$ is positive definite iff $\mathbf{B}$ has full column rank.
- (d) Every principal submatrix of $\mathbf{A}$ is nonnegative definite, and positive definite if $\mathbf{A}$ is.
- (e) If $\mathbf{A}$ is positive definite, it is nonsingular and $\mathbf{A}^{-1}$ is positive definite. If also $\mathbf{A}$ is partitioned as in Eq. 1.6.2, both Schur complements $\mathbf{A}_{11.2}$ and $\mathbf{A}_{22.1}$ are positive definite.
- (f) If $\mathbf{A} \succeq \mathbf{B} \succ \mathbf{0}$, then $\mathbf{B}^{-1} \succeq \mathbf{A}^{-1}$.

Proof.

- (b) Write $\mathbf{A} = \mathbf{B}^\top \mathbf{B}$ by Theorem 1.7.4(c'). Then $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \|\mathbf{B} \mathbf{x}\|^2$, which vanishes iff $\mathbf{B} \mathbf{x} = \mathbf{0}$, which implies $\mathbf{A} \mathbf{x} = \mathbf{0}$.
- (c) $a_{ii} = \mathbf{e}_i^\top \mathbf{A} \mathbf{e}_i \geq 0$. If $a_{ii} = 0$ then $\mathbf{A} \mathbf{e}_i = \mathbf{0}$ by (b), so column i is zero, and row i by symmetry.
- (d) $\mathbf{x}^\top \mathbf{B}^\top \mathbf{A} \mathbf{B} \mathbf{x} = (\mathbf{B} \mathbf{x})^\top \mathbf{A} (\mathbf{B} \mathbf{x})$ is nonnegative, and when $\mathbf{A}$ is positive definite it is zero only if $\mathbf{B} \mathbf{x} = \mathbf{0}$.
- (e) is (c) with $\mathbf{B}$ a set of columns of $\mathbf{I}$.
- (f) The eigenvalues of $\mathbf{A}^{-1}$ are $1/\lambda_i > 0$. The inverse in Eq. 1.6.4 has $\mathbf{A}_{22.1}^{-1}$ as a principal submatrix, which is positive definite by (d). The other Schur complement follows by symmetry of the argument.
- (g) By Theorem 1.7.6 below, $\mathbf{B} = \mathbf{B}^{1/2} \mathbf{B}^{1/2}$. Then $\mathbf{C} = \mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2} \succeq \mathbf{I}$ by (c), so every eigenvalue of $\mathbf{C}$ is at least 1 and every eigenvalue of $\mathbf{C}^{-1} = \mathbf{B}^{1/2} \mathbf{A}^{-1} \mathbf{B}^{1/2}$ is at most 1. So $\mathbf{I} \succeq \mathbf{B}^{1/2} \mathbf{A}^{-1} \mathbf{B}^{1/2}$, and multiplying on both sides by $\mathbf{B}^{-1/2}$ gives $\mathbf{B}^{-1} \succeq \mathbf{A}^{-1}$ by (c).

■

Part (f) is used in Chapter 7: an estimator whose covariance matrix is smaller in the Loewner order has smaller variance for every linear combination of its components.

Theorem 1.7.6 — Square root

A nonnegative definite matrix $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$ has exactly one nonnegative definite matrix $\mathbf{B}$ with $\mathbf{B}^2 = \mathbf{A}$, namely $\mathbf{A}^{1/2} = \mathbf{Q}\mathbf{\Lambda}^{1/2}\mathbf{Q}^\top$. It satisfies $\mathcal{C}(\mathbf{A}^{1/2}) = \mathcal{C}(\mathbf{A})$. If $\mathbf{A}$ is positive definite, $\mathbf{A}^{1/2}$ is positive definite and its inverse is $\mathbf{A}^{-1/2} = \mathbf{Q}\mathbf{\Lambda}^{-1/2}\mathbf{Q}^\top$.

Proof. $\mathbf{A}^{1/2}$ is symmetric with nonnegative eigenvalues and $(\mathbf{Q}\mathbf{\Lambda}^{1/2}\mathbf{Q}^\top)^2 = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$. Its column space is spanned by the $\mathbf{q}_i$ with $\lambda_i > 0$, as is $\mathcal{C}(\mathbf{A})$ (Corollary 1.7.3(a)). For uniqueness, let $\mathbf{B}$ be nonnegative definite with $\mathbf{B}^2 = \mathbf{A}$, and write $\mathbf{B} = \mathbf{P}\mathbf{\Delta}\mathbf{P}^\top$ with $\mathbf{\Delta} = \text{diag}(\delta_i)$, $\delta_i \geq 0$. Then $\mathbf{A} = \mathbf{P}\mathbf{\Delta}^2\mathbf{P}^\top$. Let $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ and expand $\mathbf{v} = \sum_i c_i \mathbf{p}_i$. Comparing $\mathbf{A}\mathbf{v} = \sum_i c_i \delta_i^2 \mathbf{p}_i$ with $\lambda\mathbf{v}$ gives $c_i(\delta_i^2 - \lambda) = 0$, so $c_i \neq 0$ only if $\delta_i = \sqrt{\lambda}$. Hence $\mathbf{B}\mathbf{v} = \sum_i c_i \delta_i \mathbf{p}_i = \sqrt{\lambda}\mathbf{v}$. Applying this to each $\mathbf{q}_j$ gives $\mathbf{B}\mathbf{Q} = \mathbf{Q}\mathbf{\Lambda}^{1/2}$, that is, $\mathbf{B} = \mathbf{A}^{1/2}$. The last statement is Corollary 1.7.3(b). ■

1.7.4 Extremal properties of eigenvalues

Theorem 1.7.7 — Rayleigh quotient

Let $\mathbf{A}$ be symmetric with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n$ and orthonormal eigenvectors $\mathbf{q}_1, \dots, \mathbf{q}_n$. For every $\mathbf{x} \neq \mathbf{0}$,

$$\lambda_n \leq \frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} \leq \lambda_1, \quad (1.7.2)$$

with equality on the right at $\mathbf{x} = \mathbf{q}_1$ and on the left at $\mathbf{x} = \mathbf{q}_n$. More generally, the maximum of $\mathbf{x}^\top \mathbf{A} \mathbf{x} / \mathbf{x}^\top \mathbf{x}$ over nonzero $\mathbf{x}$ orthogonal to $\mathbf{q}_1, \dots, \mathbf{q}_{k-1}$ is λ_k , attained at $\mathbf{q}_k$.

Proof. Put $\mathbf{c} = \mathbf{Q}^\top \mathbf{x}$, so that $\mathbf{x}^\top \mathbf{x} = \mathbf{c}^\top \mathbf{c}$ and

$$\frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} = \frac{\sum_i \lambda_i c_i^2}{\sum_i c_i^2},$$

a weighted average of the eigenvalues with weights $c_i^2 / \sum_j c_j^2$. It lies between the smallest and largest eigenvalue, and equals λ_1 when all weight is on c_1 , that is, at $\mathbf{x} = \mathbf{q}_1$. The condition $\mathbf{x} \perp \mathbf{q}_1, \dots, \mathbf{q}_{k-1}$ means $c_1 = \dots = c_{k-1} = 0$, so the average is then over $\lambda_k, \dots, \lambda_n$, whose largest member is λ_k . ■

Figure 1.7.1 shows the theorem for $\mathbf{A} = \begin{pmatrix} 2.5 & 1 \\ 1 & 1.5 \end{pmatrix}$. The eigenvalues are 3.118 and 0.882, with eigenvectors at angles 31.7° and 121.7° . As $\mathbf{x}$ turns around the unit circle, the quotient swings between the two eigenvalues. The ellipse $\mathbf{x}^\top \mathbf{A} \mathbf{x} = 1$ has its principal axes along the eigenvectors, with semi-axis lengths $\lambda_i^{-1/2}$. Consequently the Rayleigh quotient is largest in the direction in which the ellipse is narrowest.

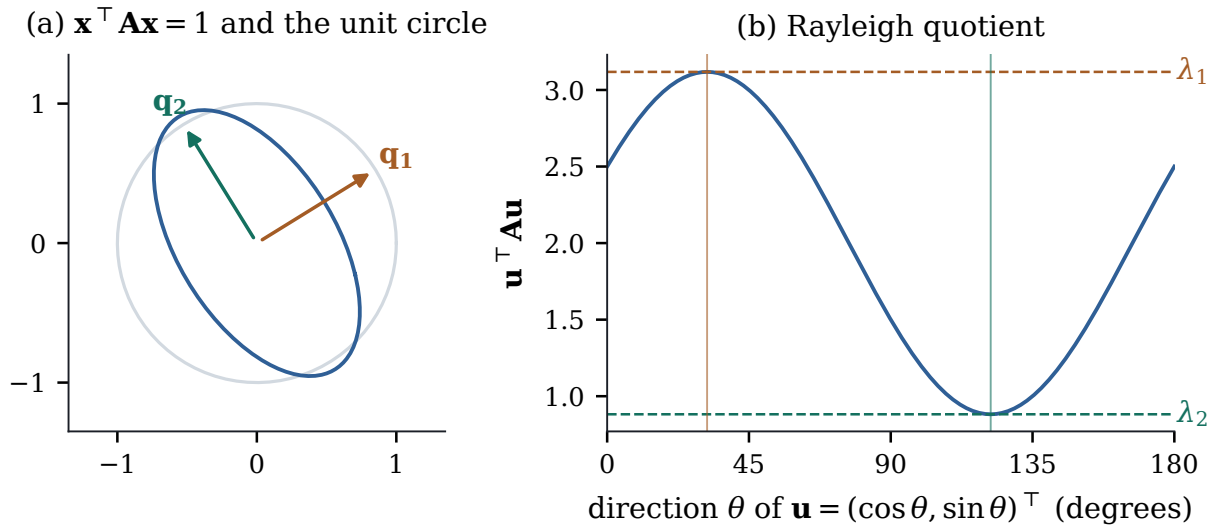


Figure 1.7.1: The Rayleigh quotient of a 2×2 symmetric matrix. (a) The level set $\mathbf{x}^T \mathbf{A} \mathbf{x} = 1$, an ellipse with axes along the orthonormal eigenvectors $\mathbf{q}_1, \mathbf{q}_2$, and the unit circle. (b) The quotient $\mathbf{u}^T \mathbf{A} \mathbf{u}$ along the unit circle. Its maximum λ_1 and minimum λ_2 are attained at the eigenvector directions (Theorem 1.7.7).

Corollary 1.7.8 — Relative to a positive definite matrix

Let $\mathbf{A}$ be symmetric and $\mathbf{B}$ positive definite, both $n \times n$.

- (a) $\max_{\mathbf{x} \neq \mathbf{0}} \mathbf{x}^T \mathbf{A} \mathbf{x} / \mathbf{x}^T \mathbf{B} \mathbf{x}$ is the largest eigenvalue of $\mathbf{B}^{-1} \mathbf{A}$, which has the same eigenvalues as the symmetric $\mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2}$.
- (b) There is a nonsingular $\mathbf{P}$ with $\mathbf{P}^T \mathbf{B} \mathbf{P} = \mathbf{I}$ and $\mathbf{P}^T \mathbf{A} \mathbf{P}$ diagonal.
- (c) For $\mathbf{a} \in \mathbb{R}^n$, $\max_{\mathbf{x} \neq \mathbf{0}} (\mathbf{a}^T \mathbf{x})^2 / \mathbf{x}^T \mathbf{B} \mathbf{x} = \mathbf{a}^T \mathbf{B}^{-1} \mathbf{a}$, attained at $\mathbf{x} = \mathbf{B}^{-1} \mathbf{a}$.

Proof. Substitute $\mathbf{x} = \mathbf{B}^{-1/2} \mathbf{z}$. The quotient in (a) becomes $\mathbf{z}^T \mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2} \mathbf{z} / \mathbf{z}^T \mathbf{z}$, to which Theorem 1.7.7 applies, and $\mathbf{B}^{-1} \mathbf{A} = \mathbf{B}^{-1/2} (\mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2}) \mathbf{B}^{1/2}$ is similar to it (Proposition 1.7.1(c)). For (b), if $\mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T$, take $\mathbf{P} = \mathbf{B}^{-1/2} \mathbf{Q}$. For (c), the matrix $\mathbf{A} = \mathbf{a} \mathbf{a}^T$ has $\mathbf{B}^{-1/2} \mathbf{a} \mathbf{a}^T \mathbf{B}^{-1/2}$ of rank at most one, whose only nonzero eigenvalue is its trace $\mathbf{a}^T \mathbf{B}^{-1} \mathbf{a}$. Substituting $\mathbf{x} = \mathbf{B}^{-1} \mathbf{a}$ attains it. ■

Part (c) is the Cauchy–Schwarz inequality in the inner product $\mathbf{x}^T \mathbf{B} \mathbf{v}$. It is the algebra behind simultaneous confidence intervals (Chapter 12), and part (b) is used for linear discriminants and for comparing two covariance matrices.

1.7.5 Exercises

1.7.5.1 A. Check your understanding

Exercise A1.

Show that $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ is positive definite iff $a > 0$ and $ac > b^2$, and nonnegative definite iff $a \geq 0$, $c \geq 0$ and $ac \geq b^2$. Why is $a \geq 0$ and $ac \geq b^2$ not enough?

1.7.5.2 B. Practice

Exercise B1.

Let $\mathbf{A} \succeq \mathbf{B} \succ \mathbf{0}$. Show that $\det \mathbf{A} \geq \det \mathbf{B}$ and $\operatorname{tr} \mathbf{A} \geq \operatorname{tr} \mathbf{B}$.

Exercise B2.

Let $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ have full column rank. Show that the leading block of $(\mathbf{X}^\top \mathbf{X})^{-1}$ satisfies $[(\mathbf{X}^\top \mathbf{X})^{-1}]_{11} \succeq (\mathbf{X}_1^\top \mathbf{X}_1)^{-1}$, with equality iff $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$. Interpret this for the variances of least squares coefficients when regressors are added to a model.

Solution. Write $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ in blocks. By Theorem 1.6.4, $[\mathbf{A}^{-1}]_{11} = \mathbf{A}_{11.2}^{-1}$. Now $\mathbf{A}_{11} - \mathbf{A}_{11.2} = \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21}$ is nonnegative definite by Proposition 1.7.5(c,e), and $\mathbf{A}_{11.2}$ is positive definite. So $\mathbf{A}_{11} \succeq \mathbf{A}_{11.2} \succ \mathbf{0}$, and Proposition 1.7.5(f) gives $\mathbf{A}_{11.2}^{-1} \succeq \mathbf{A}_{11}^{-1}$. Equality forces $\mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21} = \mathbf{0}$, hence $\mathbf{A}_{21} \mathbf{x} = \mathbf{0}$ for all $\mathbf{x}$ by Proposition 1.7.5(b), that is, $\mathbf{X}_2^\top \mathbf{X}_1 = \mathbf{0}$. When $\operatorname{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, the covariance matrix of the coefficients of $\mathbf{X}_1$ is $\sigma^2 [\mathbf{A}^{-1}]_{11}$ in the larger model and $\sigma^2 \mathbf{A}_{11}^{-1}$ in the smaller one. Adding regressors never decreases the variance of any linear combination of the original coefficients, and leaves it unchanged only when the new regressors are orthogonal to the old ones.

1.7.5.3 C. Going deeper

Exercise C1.

Let $\mathbf{A}$ be symmetric with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$. Prove the Courant–Fischer theorem

$$\lambda_k = \max_{\dim S=k} \min_{\mathbf{0} \neq \mathbf{x} \in S} \frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}.$$

Deduce that if $\mathbf{B}$ is obtained from $\mathbf{A}$ by deleting one row and the corresponding column, its eigenvalues $\mu_1 \geq \cdots \geq \mu_{n-1}$ interlace: $\lambda_k \geq \mu_k \geq \lambda_{k+1}$.

Exercise C2.

Let $\mathbf{S}$ be positive definite of size $p \times p$. Show that $g(\boldsymbol{\Sigma}) = -\log \det \boldsymbol{\Sigma} - \operatorname{tr}(\boldsymbol{\Sigma}^{-1} \mathbf{S})$, over positive definite $\boldsymbol{\Sigma}$, is maximized uniquely at $\boldsymbol{\Sigma} = \mathbf{S}$. *Hint:* write g in terms of the eigenvalues of $\mathbf{S}^{-1/2} \boldsymbol{\Sigma} \mathbf{S}^{-1/2}$.

Solution. Let $\mathbf{M} = \mathbf{S}^{-1/2} \boldsymbol{\Sigma} \mathbf{S}^{-1/2}$, which is positive definite with eigenvalues $\mu_1, \dots, \mu_p > 0$. Then $\det \boldsymbol{\Sigma} = \det \mathbf{S} \det \mathbf{M}$ and $\operatorname{tr}(\boldsymbol{\Sigma}^{-1} \mathbf{S}) = \operatorname{tr}(\mathbf{S}^{1/2} \boldsymbol{\Sigma}^{-1} \mathbf{S}^{1/2}) = \operatorname{tr}(\mathbf{M}^{-1})$. So

$$g(\boldsymbol{\Sigma}) = -\log \det \mathbf{S} - \sum_{i=1}^p \left(\log \mu_i + \frac{1}{\mu_i} \right).$$

The function $\mu \mapsto \log \mu + 1/\mu$ has derivative $(\mu - 1)/\mu^2$ and a unique minimum at $\mu = 1$.

So g is maximized iff every $\mu_i = 1$, that is, $\mathbf{M} = \mathbf{I}$ and $\mathbf{\Sigma} = \mathbf{S}$. The maximum is $-\log \det \mathbf{S} - p$.

1.8 The singular value decomposition and conditioning

The spectral theorem needs a symmetric matrix. A model matrix is rectangular, and the decomposition that fits it is the singular value decomposition.

Theorem 1.8.1 — Singular value decomposition

Let $\mathbf{A}$ be $m \times n$ of rank $r \geq 1$. There are an $m \times r$ matrix $\mathbf{U}$ and an $n \times r$ matrix $\mathbf{V}$, both with orthonormal columns, and numbers $\sigma_1 \geq \dots \geq \sigma_r > 0$, such that

$$\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{V}^\top = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top, \quad \mathbf{D} = \text{diag}(\sigma_1, \dots, \sigma_r). \quad (1.8.1)$$

The **singular values** σ_i are the positive square roots of the nonzero eigenvalues of $\mathbf{A}^\top \mathbf{A}$ (equivalently of $\mathbf{A} \mathbf{A}^\top$). The columns of $\mathbf{V}$ and $\mathbf{U}$ are eigenvectors of $\mathbf{A}^\top \mathbf{A}$ and $\mathbf{A} \mathbf{A}^\top$, with $\mathcal{C}(\mathbf{V}) = \mathcal{C}(\mathbf{A}^\top)$ and $\mathcal{C}(\mathbf{U}) = \mathcal{C}(\mathbf{A})$. Moreover $\mathbf{A}^+ = \mathbf{V} \mathbf{D}^{-1} \mathbf{U}^\top$ is the Moore–Penrose inverse of $\mathbf{A}$.

Proof. $\mathbf{A}^\top \mathbf{A}$ is nonnegative definite of rank r (Proposition 1.2.3(c)). By the spectral theorem its eigenvalues are $\sigma_1^2 \geq \dots \geq \sigma_r^2 > 0$ and $n-r$ zeros, with orthonormal eigenvectors $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_r]$ for the nonzero ones and $\mathbf{V}_0$ for the zeros. Define $\mathbf{U} = \mathbf{A} \mathbf{V} \mathbf{D}^{-1}$. Then $\mathbf{U}^\top \mathbf{U} = \mathbf{D}^{-1} \mathbf{V}^\top \mathbf{A}^\top \mathbf{A} \mathbf{V} \mathbf{D}^{-1} = \mathbf{D}^{-1} \mathbf{D}^2 \mathbf{D}^{-1} = \mathbf{I}_r$. For each column $\mathbf{v}$ of $\mathbf{V}_0$, $\|\mathbf{A} \mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{A}^\top \mathbf{A} \mathbf{v} = 0$, so $\mathbf{A} \mathbf{V}_0 = \mathbf{0}$. Since $\mathbf{V} \mathbf{V}^\top + \mathbf{V}_0 \mathbf{V}_0^\top = \mathbf{I}$ (Proposition 1.5.3(d)),

$$\mathbf{A} = \mathbf{A} \mathbf{V} \mathbf{V}^\top + \mathbf{A} \mathbf{V}_0 \mathbf{V}_0^\top = \mathbf{U} \mathbf{D} \mathbf{V}^\top.$$

Then $\mathbf{A} \mathbf{A}^\top \mathbf{U} = \mathbf{U} \mathbf{D} \mathbf{V}^\top \mathbf{V} \mathbf{D} \mathbf{U}^\top \mathbf{U} = \mathbf{U} \mathbf{D}^2$, so the columns of $\mathbf{U}$ are eigenvectors of $\mathbf{A} \mathbf{A}^\top$. $\mathcal{C}(\mathbf{U}) \subseteq \mathcal{C}(\mathbf{A})$ by definition of $\mathbf{U}$, and both have dimension r . The statement for $\mathbf{V}$ is the same argument applied to $\mathbf{A}^\top = \mathbf{V} \mathbf{D} \mathbf{U}^\top$. The four Penrose conditions for $\mathbf{V} \mathbf{D}^{-1} \mathbf{U}^\top$ follow from $\mathbf{U}^\top \mathbf{U} = \mathbf{V}^\top \mathbf{V} = \mathbf{I}_r$. For instance, $\mathbf{A} \mathbf{A}^+ = \mathbf{U} \mathbf{U}^\top$ and $\mathbf{A}^+ \mathbf{A} = \mathbf{V} \mathbf{V}^\top$ are symmetric. ■

Completing $\mathbf{U}$ and $\mathbf{V}$ to orthogonal matrices gives the *full* SVD $\mathbf{A} = \tilde{\mathbf{U}} \tilde{\mathbf{D}} \tilde{\mathbf{V}}^\top$, with $\tilde{\mathbf{D}}$ of size $m \times n$ carrying $\sigma_1, \dots, \sigma_r$ on its diagonal and zeros elsewhere. By convention, $\sigma_{r+1} = \dots = \sigma_{\min(m,n)} = 0$. Geometrically, $\mathbf{A}$ maps the unit ball of $\mathbb{R}^n$ onto a solid ellipsoid in $\mathcal{C}(\mathbf{A})$ with semi-axes $\sigma_i \mathbf{u}_i$. The input direction $\mathbf{v}_i$ is stretched by the factor σ_i and turned into $\mathbf{u}_i$. The Moore–Penrose inverse undoes the stretching on $\mathcal{C}(\mathbf{A})$ and sends $\mathcal{C}(\mathbf{A})^\perp$ to zero. The matrix $\mathbf{U} \mathbf{U}^\top$ is the orthogonal projection onto $\mathcal{C}(\mathbf{A})$, whose theory is in Chapter 6.

Proposition 1.8.2 — Norms and extremal properties

Let $\mathbf{A}$ be $m \times n$ with singular values $\sigma_1 \geq \sigma_2 \geq \dots$, and let $\|\mathbf{A}\|_2 = \max_{\mathbf{x} \neq \mathbf{0}} \|\mathbf{A}\mathbf{x}\|/\|\mathbf{x}\|$ be the **spectral norm**.

- (a) $\|\mathbf{A}\|_2 = \sigma_1$ and $\|\mathbf{A}\|_F^2 = \sum_i \sigma_i^2$.
- (b) If $\mathbf{A}$ has full column rank n , then $\min_{\mathbf{x} \neq \mathbf{0}} \|\mathbf{A}\mathbf{x}\|/\|\mathbf{x}\| = \sigma_n$, attained at $\mathbf{v}_n$.
- (c) (Eckart–Young.) For $k < r$, let $\mathbf{A}_k = \sum_{i \leq k} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$. Then $\|\mathbf{A} - \mathbf{B}\|_2 \geq \sigma_{k+1} = \|\mathbf{A} - \mathbf{A}_k\|_2$ for every $\mathbf{B}$ with $\text{rank}(\mathbf{B}) \leq k$.

Proof.

- (a) and (b): $\|\mathbf{A}\mathbf{x}\|^2/\|\mathbf{x}\|^2 = \mathbf{x}^\top \mathbf{A}^\top \mathbf{A} \mathbf{x} / \mathbf{x}^\top \mathbf{x}$, and the eigenvalues of $\mathbf{A}^\top \mathbf{A}$ are the σ_i^2 (padded with zeros), so [Theorem 1.7.7](#) applies. For the Frobenius norm, $\|\mathbf{A}\|_F^2 = \text{tr}(\mathbf{A}^\top \mathbf{A}) = \sum \sigma_i^2$ by [Proposition 1.7.1\(a\)](#).
- (b) $\mathbf{A} - \mathbf{A}_k = \sum_{i > k} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ has largest singular value σ_{k+1} . If $\text{rank}(\mathbf{B}) \leq k$, then $\dim \mathcal{N}(\mathbf{B}) \geq n - k$, and since $(n - k) + (k + 1) > n$, a basis of $\mathcal{N}(\mathbf{B})$ together with $\mathbf{v}_1, \dots, \mathbf{v}_{k+1}$ is dependent. So the two subspaces share a unit vector $\mathbf{x} = \sum_{i \leq k+1} c_i \mathbf{v}_i$. Then $\|(\mathbf{A} - \mathbf{B})\mathbf{x}\|^2 = \|\mathbf{A}\mathbf{x}\|^2 = \sum_{i \leq k+1} \sigma_i^2 c_i^2 \geq \sigma_{k+1}^2$.

■

The Frobenius version of (c), $\|\mathbf{A} - \mathbf{B}\|_F^2 \geq \sum_{i > k} \sigma_i^2$, also holds (Eckart and Young 1936). It is the basis of principal components.

1.8.1 The condition number

Definition 1.8.1 — Condition number

The **condition number** of an $m \times n$ matrix $\mathbf{A}$ of full column rank is

$$\kappa(\mathbf{A}) = \frac{\sigma_{\max}(\mathbf{A})}{\sigma_{\min}(\mathbf{A})} = \frac{\sigma_1}{\sigma_n}.$$

For a nonsingular square matrix this equals $\|\mathbf{A}\|_2 \|\mathbf{A}^{-1}\|_2$. A matrix without full column rank has $\kappa(\mathbf{A}) = \infty$ by convention.

By [Proposition 1.8.2](#), $\kappa(\mathbf{A})$ is the ratio of the largest to the smallest stretching that $\mathbf{A}$ applies to a unit vector. It is at least 1, is unchanged by scaling $\mathbf{A}$ or multiplying it on the left by a matrix with orthonormal columns, and equals 1 iff $\mathbf{A}^\top \mathbf{A} = c\mathbf{I}$, that is, iff the columns are orthogonal with equal lengths. Two further facts explain its role in regression. First, the singular values of $\mathbf{A}^\top \mathbf{A}$ are the σ_i^2 , so

$$\kappa(\mathbf{A}^\top \mathbf{A}) = \kappa(\mathbf{A})^2. \quad (1.8.2)$$

Second, κ bounds how much relative errors in the data can grow when a linear system is solved. If $\mathbf{A}$ is square and nonsingular, $\mathbf{A}\mathbf{x} = \mathbf{b}$ and $\mathbf{A}(\mathbf{x} + \delta\mathbf{x}) = \mathbf{b} + \delta\mathbf{b}$, then $\delta\mathbf{x} = \mathbf{A}^{-1}\delta\mathbf{b}$, and

$\|\mathbf{b}\| \leq \sigma_1 \|\mathbf{x}\|$ gives

$$\frac{\|\delta \mathbf{x}\|}{\|\mathbf{x}\|} \leq \frac{\|\mathbf{A}^{-1}\|_2 \|\delta \mathbf{b}\|}{\|\mathbf{b}\|/\sigma_1} = \kappa(\mathbf{A}) \frac{\|\delta \mathbf{b}\|}{\|\mathbf{b}\|}. \quad (1.8.3)$$

In floating point, relative errors of about 10^{-16} in the data can therefore become errors of about $\kappa \times 10^{-16}$ in the solution. Solving the normal equations works with $\mathbf{X}^\top \mathbf{X}$ and so, by Eq. 1.8.2, squares the damage. Chapter 6 and Chapter 10 take this up.

A small $\sigma_{\min}$ has a direct statistical reading. By Proposition 1.8.2(b), $\|\mathbf{X} \mathbf{v}_n\| = \sigma_n$ with $\|\mathbf{v}_n\| = 1$. The coefficients $\mathbf{v}_n$ combine the columns of $\mathbf{X}$ into a vector that is nearly zero, which is a near-dependence among the regressors. This is the basis of the collinearity diagnostics of Chapter 26. The size of κ depends on the units of the columns, so columns are usually rescaled before it is interpreted.

Example. (A macroeconomic design)

Take the 203 quarterly US observations of real GDP, real investment, real government spending, real disposable income, population and unemployment, with an intercept, as a model matrix with $p = 7$ columns. As recorded, $\kappa(\mathbf{X}) = 5.10 \times 10^5$. Most of that is units: GDP is measured in billions and unemployment in percent. Rescaling each column to unit length reduces it to 383, and centring the regressors before rescaling reduces it further to 124. By Eq. 1.8.2 the corresponding cross-product matrix has condition number 15386.

What remains is not an artefact of units. Figure 1.8.1(b) shows the right singular vector for the smallest singular value, $\sigma_{\min} = 0.0175$, of the centred and scaled design. Its weight is concentrated on GDP (0.78) and disposable income (-0.61). The combination of the scaled columns with these coefficients is a vector of length 0.0175, although every column has length one. The two series move almost in lockstep: their correlation is 0.9990. No rescaling can separate their coefficients well. Centring is not a neutral choice, however: it removes near-dependences that involve the intercept, and Belsley et al. (1980) recommend scaling without centring for that reason, so their diagnostics would use the unit-length figure 383. Chapter 26 returns to the choice. The listing computes the singular values, and the script also checks Eq. 1.8.2, the Penrose conditions for $\mathbf{V} \mathbf{D}^{-1} \mathbf{U}^\top$, and the Eckart–Young bound.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.macrodta.load_pandas().data
names = ["realgdp", "realinv", "realgovt", "realdpi", "pop", "unemp"]
labels = ["GDP", "investment", "government", "disp. income", "population", "unemployment"]
Z = data[names].to_numpy()
n = len(Z)

X = np.column_stack([np.ones(n), Z])                # as recorded
X_unit = X / np.linalg.norm(X, axis=0)              # unit-length columns
Xc = np.column_stack([np.ones(n), Z - Z.mean(axis=0)]) # regressors centred
X_std = Xc / np.linalg.norm(Xc, axis=0)              # ... then length 1

for name, M in [("raw", X), ("unit length", X_unit), ("centred", X_std)]:
    U, s, Vt = np.linalg.svd(M, full_matrices=False) # M = U diag(s) V'
```

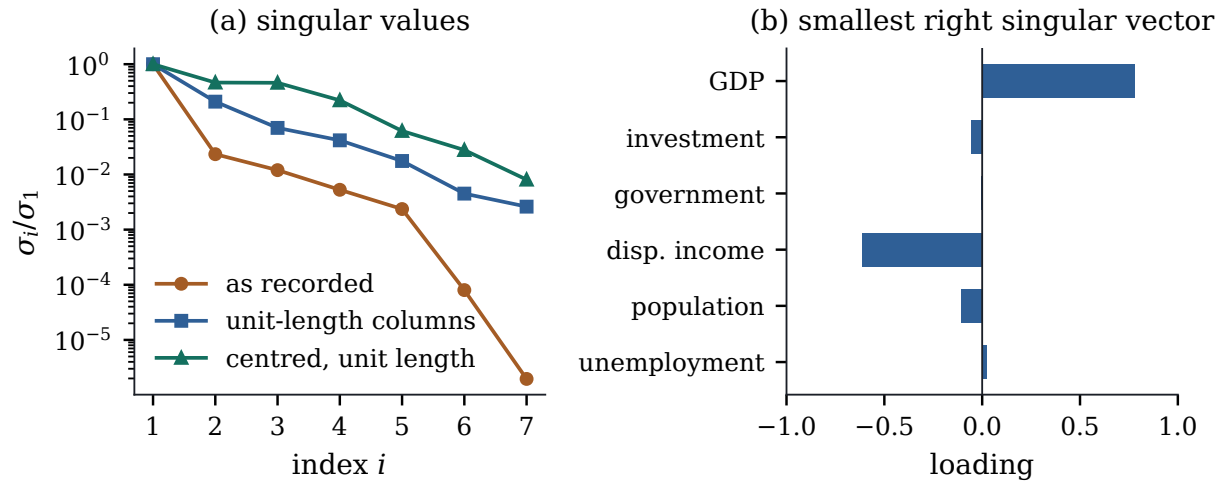


Figure 1.8.1: Singular value decomposition of a macroeconomic model matrix (Example (A macroeconomic design)). (a) Singular values relative to the largest, for the columns as recorded, rescaled to unit length, and centred then rescaled. The ratio of the first to the last is the condition number. (b) The right singular vector for the smallest singular value of the centred design. Its two large loadings reveal a near-dependence between GDP and disposable income.

```
print(f"{name:12s} kappa = {s[0] / s[-1]:10.1f}    singular values / s_1:",
      np.array2string(s / s[0], precision=4))
```

1.8.2 Exercises

1.8.2.1 B. Practice

Exercise B1.

Let $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2]$ with $\|\mathbf{x}_1\| = \|\mathbf{x}_2\| = 1$ and $\mathbf{x}_1^\top \mathbf{x}_2 = r$, $|r| < 1$. Show that $\kappa(\mathbf{X}) = \sqrt{(1 + |r|)/(1 - |r|)}$. How large must $|r|$ be for $\kappa(\mathbf{X}) > 30$?

Solution. $\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}$ has eigenvalues $1 \pm r$, with eigenvectors $(1, \pm 1)^\top$. The singular values of $\mathbf{X}$ are their square roots, so $\kappa(\mathbf{X})^2 = (1 + |r|)/(1 - |r|)$. For $\kappa > 30$ we need $(1 + |r|)/(1 - |r|) > 900$, that is, $|r| > 899/901 \approx 0.998$. For two columns, even $\kappa(\mathbf{X}) = 100$ requires $|r| > 9999/10001 \approx 0.9998$.

Exercise B2.

Show that $\kappa(\mathbf{X}) \geq \max_j \|\mathbf{x}_j\| / \min_k \|\mathbf{x}_k\|$, so that columns on very different scales force a large condition number. Show by example that rescaling the columns can also increase κ .

1.9 Idempotent matrices

A square matrix $\mathbf{A}$ is **idempotent** if $\mathbf{A}^2 = \mathbf{A}$. Applying it twice has the same effect as applying it once, which is how a projection behaves. Examples are $\mathbf{I}$, $\mathbf{0}$, the centring matrix of [Example \(Sums of squares as quadratic forms\)](#), the products $\mathbf{AG}$ and $\mathbf{GA}$ for a generalized inverse $\mathbf{G}$ ([Proposition 1.3.3\(a\)](#)), and the non-symmetric $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$. In linear models, sums of squares are quadratic forms in *symmetric* idempotent matrices, and their degrees of freedom are ranks.

Proposition 1.9.1 — General idempotent matrices

Let $\mathbf{A}$ be $n \times n$ and idempotent of rank r .

- (a) Every eigenvalue of $\mathbf{A}$ is 0 or 1.
- (b) $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A})$.
- (c) $\mathbf{I} - \mathbf{A}$ is idempotent, $\mathbf{A}(\mathbf{I} - \mathbf{A}) = (\mathbf{I} - \mathbf{A})\mathbf{A} = \mathbf{0}$, $\mathcal{C}(\mathbf{I} - \mathbf{A}) = \mathcal{N}(\mathbf{A})$, and $\text{rank}(\mathbf{A}) + \text{rank}(\mathbf{I} - \mathbf{A}) = n$.
- (d) An idempotent matrix that is nonsingular equals $\mathbf{I}$.
- (e) $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ is idempotent for nonsingular $\mathbf{P}$, and $\mathbf{A}^\top$ is idempotent.

Proof.

- (a) If $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$, then $\lambda\mathbf{x} = \mathbf{A}\mathbf{x} = \mathbf{A}^2\mathbf{x} = \lambda^2\mathbf{x}$, so $\lambda^2 = \lambda$.
- (b) If $r = 0$ both sides vanish. Otherwise take a rank factorization $\mathbf{A} = \mathbf{B}\mathbf{F}$ ([Theorem 1.2.1](#)). Then $\mathbf{B}\mathbf{F}\mathbf{B}\mathbf{F} = \mathbf{B}\mathbf{F}$. Since $\mathbf{B}$ has full column rank and $\mathbf{F}$ has full row rank, $\mathbf{B}$ has a left inverse and $\mathbf{F}$ a right inverse ([Section 1.3](#)), and cancelling them gives $\mathbf{F}\mathbf{B} = \mathbf{I}_r$. So $\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{B}\mathbf{F}) = \text{tr}(\mathbf{F}\mathbf{B}) = r$ by [Theorem 1.6.1](#).
- (c) The first two claims are one-line expansions. $\mathcal{C}(\mathbf{I} - \mathbf{A}) \subseteq \mathcal{N}(\mathbf{A})$ because $\mathbf{A}(\mathbf{I} - \mathbf{A}) = \mathbf{0}$, and $\mathbf{v} \in \mathcal{N}(\mathbf{A})$ gives $\mathbf{v} = (\mathbf{I} - \mathbf{A})\mathbf{v}$. The rank statement is then rank-nullity.
- (d) Multiply $\mathbf{A}^2 = \mathbf{A}$ by $\mathbf{A}^{-1}$.
- (e) $(\mathbf{P}^{-1}\mathbf{A}\mathbf{P})^2 = \mathbf{P}^{-1}\mathbf{A}^2\mathbf{P}$ and $(\mathbf{A}^\top)^2 = (\mathbf{A}^2)^\top$.

■

By (c), every $\mathbf{x}$ splits as $\mathbf{A}\mathbf{x} + (\mathbf{I} - \mathbf{A})\mathbf{x}$ with the first part in $\mathcal{C}(\mathbf{A})$ and the second in $\mathcal{N}(\mathbf{A})$. So $\mathbb{R}^n = \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A})$, and $\mathbf{A}$ projects onto $\mathcal{C}(\mathbf{A})$ along $\mathcal{N}(\mathbf{A})$. The projection is orthogonal exactly when the two subspaces are orthogonal, which happens iff $\mathbf{A}$ is symmetric ([Exercise \(B1\)](#)).

Theorem 1.9.2 — Symmetric idempotent matrices

Let $\mathbf{A}$ be an $n \times n$ symmetric matrix of rank r . Then $\mathbf{A}$ is idempotent iff all its eigenvalues are 0 or 1. In that case

- (a) $\mathbf{A} = \mathbf{Q}_1\mathbf{Q}_1^\top$, where the $n \times r$ matrix $\mathbf{Q}_1$ has orthonormal columns spanning $\mathcal{C}(\mathbf{A})$;
- (b) $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}) = r$, the multiplicity of the eigenvalue 1;
- (c) $\mathbf{A}$ and $\mathbf{I} - \mathbf{A}$ are nonnegative definite, and $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \|\mathbf{A}\mathbf{x}\|^2 \leq \|\mathbf{x}\|^2$.

Proof. If $\mathbf{A}$ is idempotent its eigenvalues are 0 or 1 by [Proposition 1.9.1\(a\)](#). Conversely, if $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$ with each $\lambda_i \in \{0, 1\}$, then $\mathbf{A}^2 = \mathbf{Q}\mathbf{\Lambda}^2\mathbf{Q}^\top = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top = \mathbf{A}$. In that case, order the eigenvalues so that the r ones come first and let $\mathbf{Q}_1$ hold the corresponding eigenvectors. Then $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top = \mathbf{Q}_1\mathbf{Q}_1^\top$, and $\mathcal{C}(\mathbf{A}) = \mathcal{C}(\mathbf{Q}_1)$ by [Corollary 1.7.3\(a\)](#). This is (a). The trace is the sum of the eigenvalues, which is (b). For (c), the eigenvalues of $\mathbf{A}$ and of $\mathbf{I} - \mathbf{A}$ are nonnegative, and $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \mathbf{x}^\top \mathbf{A}^\top \mathbf{A} \mathbf{x}$. Also $\|\mathbf{x}\|^2 - \|\mathbf{A} \mathbf{x}\|^2 = \mathbf{x}^\top (\mathbf{I} - \mathbf{A}) \mathbf{x} \geq 0$.

■

[Chapter 6](#) shows that symmetric idempotent matrices are exactly the orthogonal projections, and $\mathbf{Q}_1\mathbf{Q}_1^\top$ in (a) is the projection onto $\mathcal{C}(\mathbf{A})$. In [Chapter 4](#), part (b) becomes the degrees of freedom of a chi-squared distribution.

Theorem 1.9.3 — Sums of symmetric idempotent matrices

Let $\mathbf{A}_1, \dots, \mathbf{A}_k$ be symmetric $n \times n$ matrices and $\mathbf{A} = \sum_i \mathbf{A}_i$. Consider

- (i) $\mathbf{A}$ is idempotent;
- (ii) every $\mathbf{A}_i$ is idempotent;
- (iii) $\mathbf{A}_i \mathbf{A}_j = \mathbf{0}$ for all $i \neq j$.

Any two of these imply the third. Moreover, if $\mathbf{A} = \mathbf{I}_n$ and $\sum_i \text{rank}(\mathbf{A}_i) = n$, then (ii) and (iii) hold.

Proof.

(ii) and (iii) imply (i): $\mathbf{A}^2 = \sum_i \mathbf{A}_i^2 + \sum_{i \neq j} \mathbf{A}_i \mathbf{A}_j = \sum_i \mathbf{A}_i = \mathbf{A}$.

(iii) and (ii) imply (iii): By [Theorem 1.9.2\(c\)](#), $\|\mathbf{x}\|^2 \geq \mathbf{x}^\top \mathbf{A} \mathbf{x}$ for all $\mathbf{x}$. Fix j and take $\mathbf{x} = \mathbf{A}_j \mathbf{y}$, so that $\mathbf{A}_j \mathbf{x} = \mathbf{x}$ and $\mathbf{x}^\top \mathbf{A}_j \mathbf{x} = \|\mathbf{x}\|^2$. Then

$$\|\mathbf{x}\|^2 \geq \mathbf{x}^\top \mathbf{A} \mathbf{x} = \|\mathbf{x}\|^2 + \sum_{i \neq j} \mathbf{x}^\top \mathbf{A}_i \mathbf{x} = \|\mathbf{x}\|^2 + \sum_{i \neq j} \|\mathbf{A}_i \mathbf{x}\|^2.$$

So $\mathbf{A}_i \mathbf{A}_j \mathbf{y} = \mathbf{A}_i \mathbf{x} = \mathbf{0}$ for every $i \neq j$ and every $\mathbf{y}$.

(iv) and (iii) imply (ii): By (iii), $\mathbf{A} \mathbf{A}_i = \mathbf{A}_i^2$. Using $\mathbf{A}^2 = \mathbf{A}$, $\mathbf{A}_i^2 = \mathbf{A} \mathbf{A}_i = \mathbf{A}^2 \mathbf{A}_i = \mathbf{A} \mathbf{A}_i^2 = \mathbf{A}_i^3$. The eigenvalues of the symmetric $\mathbf{A}_i$ therefore satisfy $\lambda^2 = \lambda^3$, so they lie in $\{0, 1\}$, and $\mathbf{A}_i$ is idempotent by [Theorem 1.9.2](#).

For the last statement, let $r_i = \text{rank}(\mathbf{A}_i)$. For each i , $\mathbf{I} - \mathbf{A}_i = \sum_{j \neq i} \mathbf{A}_j$ has rank at most $\sum_{j \neq i} r_j = n - r_i$ ([Proposition 1.2.3\(d\)](#)). So the eigenvalue 1 of $\mathbf{A}_i$, whose eigenspace is $\mathcal{N}(\mathbf{I} - \mathbf{A}_i)$, has multiplicity at least r_i . But $\mathbf{A}_i$ has exactly r_i nonzero eigenvalues, so all of them equal 1 and $\mathbf{A}_i$ is idempotent. Now (i) holds trivially, and (iii) follows from the second part of the proof.

■

The last statement is the algebraic core of Cochran's theorem ([Chapter 4](#)): if $\mathbf{y}^\top \mathbf{y} = \sum_i \mathbf{y}^\top \mathbf{A}_i \mathbf{y}$ and the ranks add up to n , the quadratic forms are built from mutually orthogonal projections.

1.9.1 Exercises

1.9.1.1 A. Check your understanding

Exercise A1.

Let $\mathbf{A}, \mathbf{B}$ be idempotent of the same size. Show that $\mathbf{AB}$ is idempotent if $\mathbf{AB} = \mathbf{BA}$, and that $\mathbf{A} + \mathbf{B}$ is idempotent iff $\mathbf{AB} + \mathbf{BA} = \mathbf{0}$. Show that the latter forces $\mathbf{AB} = \mathbf{0}$ (symmetry is not needed).

1.9.1.2 B. Practice

Exercise B1.

Let $\mathbf{A}$ be idempotent. Show that $\mathbf{A}$ is symmetric iff $\mathcal{C}(\mathbf{A}) \perp \mathcal{N}(\mathbf{A})$.

Solution. Suppose $\mathbf{A}$ is symmetric. For $\mathbf{u} = \mathbf{Ax} \in \mathcal{C}(\mathbf{A})$ and $\mathbf{v} \in \mathcal{N}(\mathbf{A})$, $\mathbf{u}^\top \mathbf{v} = \mathbf{x}^\top \mathbf{A} \mathbf{v} = 0$. Conversely, suppose $\mathcal{C}(\mathbf{A}) \perp \mathcal{N}(\mathbf{A})$. For all $\mathbf{x}, \mathbf{y}$, $(\mathbf{I} - \mathbf{A})\mathbf{y} \in \mathcal{N}(\mathbf{A})$, so $\mathbf{x}^\top \mathbf{A}^\top (\mathbf{I} - \mathbf{A})\mathbf{y} = 0$. Hence $\mathbf{A}^\top (\mathbf{I} - \mathbf{A}) = \mathbf{0}$, that is, $\mathbf{A}^\top = \mathbf{A}^\top \mathbf{A}$. The right side is symmetric, so $\mathbf{A}^\top$ is symmetric, and so is $\mathbf{A}$.

1.10 Kronecker products and the vec operator

Balanced designs have model matrices with a product structure, and multivariate observations have covariance matrices with one. The Kronecker product expresses both.

Definition 1.10.1

The **Kronecker product** of an $m \times n$ matrix $\mathbf{A}$ and a $p \times q$ matrix $\mathbf{B}$ is the $mp \times nq$ block matrix $\mathbf{A} \otimes \mathbf{B} = (a_{ij}\mathbf{B})$. The **vec operator** places the n columns of an $m \times n$ matrix one below another, giving a vector of length mn : $\text{vec}(\mathbf{A}) = (\mathbf{a}_1^\top, \dots, \mathbf{a}_n^\top)^\top$.

Proposition 1.10.1

For matrices of compatible sizes:

- (a) $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = \mathbf{AC} \otimes \mathbf{BD}$ (mixed-product rule);
- (b) $(\mathbf{A} \otimes \mathbf{B})^\top = \mathbf{A}^\top \otimes \mathbf{B}^\top$, and $(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}$ for nonsingular $\mathbf{A}, \mathbf{B}$. More generally $\mathbf{A}^- \otimes \mathbf{B}^-$ is a generalized inverse of $\mathbf{A} \otimes \mathbf{B}$;
- (c) $\text{tr}(\mathbf{A} \otimes \mathbf{B}) = \text{tr}(\mathbf{A}) \text{tr}(\mathbf{B})$ and $\text{rank}(\mathbf{A} \otimes \mathbf{B}) = \text{rank}(\mathbf{A}) \text{rank}(\mathbf{B})$;
- (d) if $\mathbf{A}$ ($m \times m$) and $\mathbf{B}$ ($p \times p$) are symmetric with eigenvalues λ_i and μ_j , the eigenvalues of $\mathbf{A} \otimes \mathbf{B}$ are the products $\lambda_i \mu_j$, and $\det(\mathbf{A} \otimes \mathbf{B}) = (\det \mathbf{A})^p (\det \mathbf{B})^m$;

- (e) Kronecker products of orthogonal, symmetric, idempotent, or nonnegative definite matrices are of the same kind;
- (f) $\text{vec}(\mathbf{ABC}) = (\mathbf{C}^\top \otimes \mathbf{A}) \text{vec}(\mathbf{B})$ and $\text{tr}(\mathbf{A}^\top \mathbf{B}) = \text{vec}(\mathbf{A})^\top \text{vec}(\mathbf{B})$.

Proof.

- (a) The (i, j) block of the left side is $\sum_k a_{ik} c_{kj} \mathbf{B} \mathbf{D}$, which is the (i, j) block of the right side. (b) Transposition is direct, and the rest follows from (a), for example $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^- \otimes \mathbf{B}^-)(\mathbf{A} \otimes \mathbf{B}) = \mathbf{A} \mathbf{A}^- \mathbf{A} \otimes \mathbf{B} \mathbf{B}^- \mathbf{B}$.
- (b) With $\mathbf{A} = \mathbf{P} \mathbf{\Lambda} \mathbf{P}^\top$ and $\mathbf{B} = \mathbf{Q} \mathbf{M} \mathbf{Q}^\top$, (a) and (b) give $\mathbf{A} \otimes \mathbf{B} = (\mathbf{P} \otimes \mathbf{Q})(\mathbf{\Lambda} \otimes \mathbf{M})(\mathbf{P} \otimes \mathbf{Q})^\top$, a spectral decomposition because $\mathbf{P} \otimes \mathbf{Q}$ is orthogonal and $\mathbf{\Lambda} \otimes \mathbf{M}$ is diagonal with entries $\lambda_i \mu_j$. The determinant is the product of these. (c) The trace statement is direct. For the rank, by (a) $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^- \otimes \mathbf{B}^-) = \mathbf{A} \mathbf{A}^- \otimes \mathbf{B} \mathbf{B}^-$ is idempotent with the same rank as $\mathbf{A} \otimes \mathbf{B}$ (Proposition 1.3.3(a)). Its trace is $\text{tr}(\mathbf{A} \mathbf{A}^-) \text{tr}(\mathbf{B} \mathbf{B}^-) = \text{rank}(\mathbf{A}) \text{rank}(\mathbf{B})$ by Proposition 1.9.1(b).
- (c) follows from (a), (b) and (d). (f) The j th column of $\mathbf{ABC}$ is $\mathbf{A} \mathbf{B} \mathbf{c}_j = \sum_k c_{kj} \mathbf{A} \mathbf{b}_k$, which is the j th block of $(\mathbf{C}^\top \otimes \mathbf{A}) \text{vec}(\mathbf{B})$. The trace identity is $\sum_{ij} a_{ij} b_{ij}$ written two ways.

■

The determinant formula in (d) holds for all square $\mathbf{A}$ and $\mathbf{B}$, not only symmetric ones. We need only the symmetric case.

Example. (The balanced two-way layout)

Let observations y_{ij} , $i = 1, \dots, a$, $j = 1, \dots, b$, be stacked with j running inside i . The additive model $E(y_{ij}) = \mu + \alpha_i + \beta_j$ has model matrix

$$\mathbf{X} = [\mathbf{1}_a \otimes \mathbf{1}_b, \mathbf{I}_a \otimes \mathbf{1}_b, \mathbf{1}_a \otimes \mathbf{I}_b],$$

because $\mathbf{I}_a \otimes \mathbf{1}_b$ has a 1 in column i exactly for the b observations in row i . With $\bar{\mathbf{J}}_k = k^{-1} \mathbf{J}_k$, a symmetric idempotent averaging matrix, the mixed-product rule shows that $\mathbf{I}_a \otimes \bar{\mathbf{J}}_b$, $\bar{\mathbf{J}}_a \otimes \mathbf{I}_b$ and $\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$ replace each observation by its row mean, its column mean and the grand mean. The matrix $\mathbf{M} = \mathbf{I}_a \otimes \bar{\mathbf{J}}_b + \bar{\mathbf{J}}_a \otimes \mathbf{I}_b - \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$ is symmetric and idempotent, with trace $a + b - 1 = \text{rank}(\mathbf{X})$. It is the projection onto $\mathcal{C}(\mathbf{X})$ (Chapter 15). The script checks this for $a = 3$ and $b = 4$, together with each part of Proposition 1.10.1.

```
import numpy as np
rng = np.random.default_rng(11)
def vec(M):
    return M.reshape(-1, order="F") # stack the columns

A, B, C = rng.normal(size=(3, 4)), rng.normal(size=(4, 2)), rng.normal(size=(2, 5))
assert np.allclose(vec(A @ B @ C), np.kron(C.T, A) @ vec(B)) # vec(ABC)

# balanced two-way layout: a = 3 rows, b = 4 columns, one observation per cell
a, b = 3, 4
one = lambda k: np.ones((k, 1))
X = np.hstack([np.kron(one(a), one(b)), # intercept
               np.kron(np.eye(a), one(b)), # row indicators
```

```
np.kron(one(a), np.eye(b))]) # column indicators
print("rank(X) =", np.linalg.matrix_rank(X), "= a + b - 1 =", a + b - 1)
```

1.11 Vector and matrix calculus

1.11.1 Gradients of linear and quadratic functions

For a differentiable scalar function f of $\mathbf{x} \in \mathbb{R}^n$, the **gradient** $\partial f / \partial \mathbf{x}$ is the column vector with entries $\partial f / \partial x_i$, and the **Hessian** $\partial^2 f / \partial \mathbf{x} \partial \mathbf{x}^\top$ is the $n \times n$ matrix of second derivatives. For a vector-valued $\mathbf{g}(\mathbf{x}) \in \mathbb{R}^m$, the $m \times n$ *Jacobian* $\partial \mathbf{g} / \partial \mathbf{x}^\top$ has (i, j) entry $\partial g_i / \partial x_j$.

Theorem 1.11.1 — Derivatives of linear and quadratic forms

Let $\mathbf{a} \in \mathbb{R}^n$, let $\mathbf{A}$ be $n \times n$ and $\mathbf{B}$ be $m \times n$. Then

$$\frac{\partial(\mathbf{a}^\top \mathbf{x})}{\partial \mathbf{x}} = \mathbf{a}, \quad \frac{\partial(\mathbf{B}\mathbf{x})}{\partial \mathbf{x}^\top} = \mathbf{B}, \quad \frac{\partial(\mathbf{x}^\top \mathbf{A}\mathbf{x})}{\partial \mathbf{x}} = (\mathbf{A} + \mathbf{A}^\top)\mathbf{x}, \quad \frac{\partial^2(\mathbf{x}^\top \mathbf{A}\mathbf{x})}{\partial \mathbf{x} \partial \mathbf{x}^\top} = \mathbf{A} + \mathbf{A}^\top.$$

In particular $\partial(\mathbf{x}^\top \mathbf{A}\mathbf{x}) / \partial \mathbf{x} = 2\mathbf{A}\mathbf{x}$ when $\mathbf{A}$ is symmetric.

Proof. The first two are immediate from $\mathbf{a}^\top \mathbf{x} = \sum_i a_i x_i$ and $(\mathbf{B}\mathbf{x})_i = \sum_j b_{ij} x_j$. For the quadratic form, expand

$$(\mathbf{x} + \mathbf{h})^\top \mathbf{A}(\mathbf{x} + \mathbf{h}) = \mathbf{x}^\top \mathbf{A}\mathbf{x} + \mathbf{h}^\top (\mathbf{A} + \mathbf{A}^\top)\mathbf{x} + \mathbf{h}^\top \mathbf{A}\mathbf{h}.$$

The middle term is linear in $\mathbf{h}$ and the last is $O(\|\mathbf{h}\|^2)$, so $(\mathbf{A} + \mathbf{A}^\top)\mathbf{x}$ is the gradient. The gradient is linear in $\mathbf{x}$ with matrix $\mathbf{A} + \mathbf{A}^\top$, which is the Hessian by the second formula. ■

Example. (The least squares criterion)

Let $S(\mathbf{b}) = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \mathbf{y}^\top \mathbf{y} - 2\mathbf{y}^\top \mathbf{X}\mathbf{b} + \mathbf{b}^\top \mathbf{X}^\top \mathbf{X}\mathbf{b}$. By Theorem 1.11.1,

$$\frac{\partial S}{\partial \mathbf{b}} = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X}\mathbf{b} = -2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{b}), \quad \frac{\partial^2 S}{\partial \mathbf{b} \partial \mathbf{b}^\top} = 2\mathbf{X}^\top \mathbf{X}.$$

Setting the gradient to zero gives the normal equations $\mathbf{X}^\top \mathbf{X}\mathbf{b} = \mathbf{X}^\top \mathbf{y}$. The Hessian is nonnegative definite, so S is convex and every solution is a global minimizer. The next result proves this without calculus, and covers the rank-deficient case.

Proposition 1.11.2 — Minimizing a quadratic function

Let $\mathbf{A}$ be $n \times n$ nonnegative definite, $\mathbf{b} \in \mathbb{R}^n$ and $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A}\mathbf{x} - 2\mathbf{b}^\top \mathbf{x}$. If $\mathbf{b} \in \mathcal{C}(\mathbf{A})$, the minimizers of f are exactly the solutions of $\mathbf{A}\mathbf{x} = \mathbf{b}$, and the minimum value is

$-\mathbf{b}^\top \mathbf{A}^- \mathbf{b}$ for any generalized inverse $\mathbf{A}^-$. If $\mathbf{b} \notin \mathcal{C}(\mathbf{A})$, then f is unbounded below.

Proof. Let $\mathbf{b} \in \mathcal{C}(\mathbf{A})$ and let $\mathbf{x}_0$ solve $\mathbf{A}\mathbf{x}_0 = \mathbf{b}$. Completing the square,

$$f(\mathbf{x}) = (\mathbf{x} - \mathbf{x}_0)^\top \mathbf{A}(\mathbf{x} - \mathbf{x}_0) - \mathbf{x}_0^\top \mathbf{A}\mathbf{x}_0.$$

The first term is nonnegative, and by Proposition 1.7.5(b) it is zero iff $\mathbf{A}(\mathbf{x} - \mathbf{x}_0) = \mathbf{0}$, that is, iff $\mathbf{A}\mathbf{x} = \mathbf{b}$. The minimum is $-\mathbf{x}_0^\top \mathbf{A}\mathbf{x}_0 = -\mathbf{x}_0^\top \mathbf{A}\mathbf{A}^- \mathbf{A}\mathbf{x}_0 = -\mathbf{b}^\top \mathbf{A}^- \mathbf{b}$. If $\mathbf{b} \notin \mathcal{C}(\mathbf{A})$, write $\mathbf{b} = \mathbf{b}_1 + \mathbf{b}_2$ with $\mathbf{b}_1 \in \mathcal{C}(\mathbf{A})$ and $\mathbf{0} \neq \mathbf{b}_2 \in \mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A})$ (Proposition 1.5.1). Then $f(t\mathbf{b}_2) = -2t\|\mathbf{b}_2\|^2 \rightarrow -\infty$ as $t \rightarrow \infty$.

■

With $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ and $\mathbf{b} = \mathbf{X}^\top \mathbf{y}$, the proposition gives least squares for every model matrix: the minimizers are the solutions of the normal equations, and the minimum of S is $\mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top \mathbf{y}$.

1.11.2 Constrained minimization

To minimize a differentiable $f(\mathbf{x})$ subject to linear constraints $\mathbf{H}\mathbf{x} = \mathbf{h}$, with $\mathbf{H}$ of size $q \times n$, the method of **Lagrange multipliers** looks for stationary points of $f(\mathbf{x}) + 2\boldsymbol{\theta}^\top(\mathbf{H}\mathbf{x} - \mathbf{h})$ in $(\mathbf{x}, \boldsymbol{\theta})$. The factor 2 is only for convenience. For quadratic f the stationary point can be written down and checked directly.

Proposition 1.11.3

Let $\mathbf{A}$ be positive definite, $\mathbf{H}$ of size $q \times n$ with full row rank, and $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A}\mathbf{x} - 2\mathbf{b}^\top \mathbf{x}$. The unique minimizer of f subject to $\mathbf{H}\mathbf{x} = \mathbf{h}$ is

$$\mathbf{x}_* = \mathbf{x}_0 - \mathbf{A}^{-1} \mathbf{H}^\top (\mathbf{H} \mathbf{A}^{-1} \mathbf{H}^\top)^{-1} (\mathbf{H} \mathbf{x}_0 - \mathbf{h}), \quad \mathbf{x}_0 = \mathbf{A}^{-1} \mathbf{b}.$$

Proof. $\mathbf{H} \mathbf{A}^{-1} \mathbf{H}^\top$ is positive definite by Proposition 1.7.5(c,e), so $\mathbf{x}_*$ is well defined, and $\mathbf{H}\mathbf{x}_* = \mathbf{h}$ by direct substitution. Stationarity of the Lagrangian gives $2\mathbf{A}\mathbf{x} - 2\mathbf{b} + 2\mathbf{H}^\top \boldsymbol{\theta} = \mathbf{0}$. Indeed $\mathbf{A}\mathbf{x}_* - \mathbf{b} = -\mathbf{H}^\top \boldsymbol{\theta}_*$ with $\boldsymbol{\theta}_* = (\mathbf{H} \mathbf{A}^{-1} \mathbf{H}^\top)^{-1} (\mathbf{H} \mathbf{x}_0 - \mathbf{h})$. For any feasible $\mathbf{x}$, put $\mathbf{d} = \mathbf{x} - \mathbf{x}_*$, so that $\mathbf{H}\mathbf{d} = \mathbf{0}$. Then

$$f(\mathbf{x}) = f(\mathbf{x}_*) + 2\mathbf{d}^\top (\mathbf{A}\mathbf{x}_* - \mathbf{b}) + \mathbf{d}^\top \mathbf{A}\mathbf{d} = f(\mathbf{x}_*) - 2(\mathbf{H}\mathbf{d})^\top \boldsymbol{\theta}_* + \mathbf{d}^\top \mathbf{A}\mathbf{d} = f(\mathbf{x}_*) + \mathbf{d}^\top \mathbf{A}\mathbf{d},$$

which exceeds $f(\mathbf{x}_*)$ unless $\mathbf{d} = \mathbf{0}$.

■

With $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ and $\mathbf{b} = \mathbf{X}^\top \mathbf{y}$, this is least squares subject to a linear hypothesis $\mathbf{H}\boldsymbol{\beta} = \mathbf{h}$, the starting point of Chapter 11.

1.11.3 Derivatives with respect to matrices and scalars

Likelihoods for normal models involve $\log \det \Sigma$ and $\text{tr}(\Sigma^{-1}S)$. For a scalar function f of an $m \times n$ matrix $\mathbf{W}$ with functionally independent entries, $\partial f / \partial \mathbf{W}$ is the $m \times n$ matrix with entries $\partial f / \partial w_{ij}$.

Proposition 1.11.4 (a) $\partial \text{tr}(\mathbf{A}\mathbf{W}) / \partial \mathbf{W} = \mathbf{A}^\top$.

(b) If $\det \mathbf{W} > 0$, then $\partial \log \det \mathbf{W} / \partial \mathbf{W} = (\mathbf{W}^{-1})^\top$.

(c) If $\mathbf{A}(t)$ is a nonsingular matrix whose entries are differentiable functions of a scalar t , with derivative $\dot{\mathbf{A}} = \partial \mathbf{A} / \partial t$, then

$$\frac{\partial \mathbf{A}^{-1}}{\partial t} = -\mathbf{A}^{-1} \dot{\mathbf{A}} \mathbf{A}^{-1}, \quad \frac{\partial \log |\det \mathbf{A}|}{\partial t} = \text{tr}(\mathbf{A}^{-1} \dot{\mathbf{A}}).$$

Proof.

(a) $\text{tr}(\mathbf{A}\mathbf{W}) = \sum_{i,j} a_{ji} w_{ij}$. (c) Differentiate $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$ by the product rule to get $\dot{\mathbf{A}}\mathbf{A}^{-1} + \mathbf{A}\partial \mathbf{A}^{-1} / \partial t = \mathbf{0}$. For the determinant, $\mathbf{A}(t+h) = \mathbf{A}(t)(\mathbf{I} + h\mathbf{A}^{-1}\dot{\mathbf{A}} + o(h))$. In the permutation expansion of $\det(\mathbf{I} + h\mathbf{C})$ only the diagonal product contributes terms of first order in h , so $\det(\mathbf{I} + h\mathbf{C}) = 1 + h \text{tr} \mathbf{C} + O(h^2)$. Hence $\det \mathbf{A}(t+h) = \det \mathbf{A}(t)(1 + h \text{tr}(\mathbf{A}^{-1}\dot{\mathbf{A}}) + o(h))$, and taking logarithms gives the formula. (b) Apply (c) with $\mathbf{A}(t) = \mathbf{W} + t\mathbf{e}_i\mathbf{e}_j^\top$. Then $\text{tr}(\mathbf{W}^{-1}\mathbf{e}_i\mathbf{e}_j^\top) = \mathbf{e}_j^\top \mathbf{W}^{-1} \mathbf{e}_i$, the (j, i) entry of $\mathbf{W}^{-1}$.

■

When $\mathbf{W}$ is constrained to be symmetric, w_{ij} and w_{ji} are the same variable, so the derivatives with respect to off-diagonal entries double. Some books (Renchner and Schaalje 2008) report formulas of the type $2\mathbf{W}^{-1} - \text{diag}(\mathbf{W}^{-1})$ for this reason. The simplest way to avoid the issue is to differentiate along a symmetric direction $\mathbf{W} + t\mathbf{E}$, as in (c). This is the convenient route for the normal likelihoods of Chapter 32. The listing checks the gradient formulas against central finite differences, and the script also checks part (c) and Proposition 1.11.2. The largest discrepancy in the listing is 3×10^{-9} , which is the size one expects from the finite-difference approximation.

```
import numpy as np
rng = np.random.default_rng(5)
p = 4
def num_grad(f, x, h=1e-6):
    """Central-difference gradient of a scalar function of a vector or matrix."""
    g = np.zeros_like(x)
    for idx in np.ndindex(x.shape):
        e = np.zeros_like(x)
        e[idx] = h
        g[idx] = (f(x + e) - f(x - e)) / (2 * h)
    return g

a = rng.normal(size=p)
A = rng.normal(size=(p, p)) # not symmetric
x = rng.normal(size=p)
checks = {
    "a'x": (num_grad(lambda v: a @ v, x), a),
    "x'Ax": (num_grad(lambda v: v @ A @ v, x), (A + A.T) @ x),
```

```

}
X = rng.normal(size=(20, p))
y = rng.normal(size=20)
b = rng.normal(size=p)
grad_ls = -2 * X.T @ (y - X @ b)
checks["||y - Xb||^2"] = (num_grad(lambda v: np.sum((y - X @ v) ** 2), b), grad_ls)

S = rng.normal(size=(p, p)); S = S @ S.T + p * np.eye(p) # positive definite
checks["log det S"] = (num_grad(lambda W: np.linalg.slogdet(W)[1], S), np.linalg.inv(S).T)
checks["tr(AS)"] = (num_grad(lambda W: np.trace(A @ W), S), A.T)
for name, (numeric, formula) in checks.items():
    print(f"{name:14s} max error {np.abs(numeric - formula).max():.1e}")

```

1.11.4 Exercises

1.11.4.1 A. Check your understanding

Exercise A1.

Observations fall in a clusters of size b , and $\Sigma = \sigma^2 \mathbf{I}_a \otimes \mathbf{E}_b$, where $\mathbf{E}_b = (1 - \rho)\mathbf{I}_b + \rho\mathbf{J}_b$. Find the eigenvalues, determinant and inverse of Σ using [Proposition 1.10.1](#), [Example \(Equicorrelation\)](#) and [Example \(Eigenvalues of the equicorrelation matrix\)](#).

Exercise A2.

Use [Proposition 1.11.2](#) to show that $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda\|\mathbf{b}\|^2$ with $\lambda > 0$ has the unique minimizer $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$, whatever the rank of $\mathbf{X}$.

1.11.4.2 B. Practice

Exercise B1.

Let $\mathbf{X}$ have full column rank, $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$, and let $\mathbf{H}$ be $q \times p$ with full row rank. Show that the minimizer of $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ subject to $\mathbf{H}\mathbf{b} = \mathbf{h}$ is

$$\hat{\boldsymbol{\beta}}_H = \hat{\boldsymbol{\beta}} - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h}), \quad \mathbf{W} = \mathbf{H} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top,$$

and that the increase in the residual sum of squares is $(\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})$.

Solution. Apply [Proposition 1.11.3](#) with $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ and $\mathbf{b} = \mathbf{X}^\top \mathbf{y}$. Since $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \mathbf{y}^\top \mathbf{y} + f(\mathbf{b})$ and $\mathbf{x}_0 = \hat{\boldsymbol{\beta}}$, this gives $\hat{\boldsymbol{\beta}}_H$. By completing the square as in [Proposition 1.11.2](#), $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 + (\mathbf{b} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\mathbf{b} - \hat{\boldsymbol{\beta}})$ for every $\mathbf{b}$. With $\mathbf{b} - \hat{\boldsymbol{\beta}} = -(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})$, the increase is

$$(\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})^\top \mathbf{W}^{-1} \mathbf{H} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h}) = (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h}).$$

This quadratic form is the numerator of the F statistic in Chapter 11.

1.12 Summary and Notes

1.12.1 Summary

Idea.

1. Row rank equals column rank, and $\text{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A})$ is the number of columns. $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)$, so the normal equations are always consistent (Theorem 1.2.1, Theorem 1.2.2 and Proposition 1.2.3).
2. A generalized inverse satisfies $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$. Every matrix has one, built from any nonsingular $r \times r$ submatrix. The Moore–Penrose inverse is the unique one that is also reflexive and makes $\mathbf{A}\mathbf{A}^+$ and $\mathbf{A}^+\mathbf{A}$ symmetric (Definition 1.3.1, Theorem 1.3.2, Proposition 1.2.4 and Definition 1.3.2).
3. $\mathbf{A}\mathbf{x} = \mathbf{b}$ is consistent iff $\mathbf{A}\mathbf{A}^-\mathbf{b} = \mathbf{b}$, and then its solutions are $\mathbf{A}^-\mathbf{b} + (\mathbf{I} - \mathbf{A}^-\mathbf{A})\mathbf{z}$. A linear function $\mathbf{q}^\top \mathbf{x}$ is the same for all solutions iff $\mathbf{q} \in \mathcal{C}(\mathbf{A}^\top)$ (Theorem 1.4.1 and Corollary 1.4.2).
4. Orthogonal matrices preserve lengths and angles. Gram–Schmidt gives the QR factorization, and $\mathbb{R}^n = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$ (Proposition 1.5.3, Theorem 1.5.2 and Proposition 1.5.1).
5. $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$. Block elimination gives the determinant and inverse of a partitioned matrix through Schur complements, and the Sherman–Morrison–Woodbury identity as a special case (Theorem 1.6.1, Theorem 1.6.3, Theorem 1.6.4 and Theorem 1.6.5).
6. A symmetric matrix is $\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$ with $\mathbf{Q}$ orthogonal. Positive definiteness is equivalent to positive eigenvalues, to a Cholesky factorization, and to positive leading minors. Nonnegative definite matrices have a unique nonnegative definite square root (Theorem 1.7.2, Theorem 1.7.4 and Theorem 1.7.6).
7. The Rayleigh quotient $\mathbf{x}^\top \mathbf{A}\mathbf{x} / \mathbf{x}^\top \mathbf{x}$ ranges over $[\lambda_{\min}, \lambda_{\max}]$ (Theorem 1.7.7 and Corollary 1.7.8).
8. Every matrix is $\mathbf{U}\mathbf{D}\mathbf{V}^\top$. The condition number $\kappa = \sigma_{\max}/\sigma_{\min}$ bounds error growth, is squared by forming $\mathbf{X}^\top \mathbf{X}$, and its smallest singular vector exhibits near-dependences among the columns (Theorem 1.8.1, Definition 1.8.1 and Proposition 1.8.2).
9. Idempotent matrices have eigenvalues 0 and 1 and rank equal to trace. Symmetric ones are $\mathbf{Q}_1\mathbf{Q}_1^\top$. If symmetric matrices summing to $\mathbf{I}$ have ranks summing to n , they are idempotent and mutually orthogonal (Proposition 1.9.1, Theorem 1.9.2 and Theorem 1.9.3).
10. Kronecker products factor balanced designs. $\text{vec}(\mathbf{ABC}) = (\mathbf{C}^\top \otimes \mathbf{A})\text{vec}(\mathbf{B})$ (Proposition 1.10.1).
11. $\partial(\mathbf{x}^\top \mathbf{A}\mathbf{x})/\partial \mathbf{x} = (\mathbf{A} + \mathbf{A}^\top)\mathbf{x}$. Quadratic functions with nonnegative definite matrices are minimized by solving a linear system, with or without linear constraints (Theorem 1.11.1, Proposition 1.11.2 and Proposition 1.11.3).

1.12.2 Notes and sources

Coverage and framing. The selection of results follows what the rest of the book uses. Rencher and Schaalje (2008, chapter 2) was the checklist for topics: its sections on positive definite matrices, generalized inverses, orthogonal matrices, idempotent matrices and matrix calculus correspond to [Section 1.7](#), [Section 1.3](#), [Section 1.5](#), [Section 1.9](#) and [Section 1.11](#). Several results that Rencher and Schaalje state without proof or with a reference, such as the general solution of a consistent system, the characterizations of positive definiteness and the sums of idempotent matrices, are proved here. The vector-space point of view, in which column spaces, null spaces and orthogonal complements come first and matrices second, is that of Christensen (2020, appendices A and B). Seber and Lee (2003, appendix A) is a compact list of further matrix facts. The Kronecker product results in [Section 1.10](#) are also in Christensen’s appendix B.

Linear algebra for statisticians. Harville (1997) is the most complete treatment written for statisticians, with full proofs of the determinant facts assumed in [Proposition 1.6.2](#) and a long development of generalized inverses. Searle (1982) is more elementary and has many worked examples from linear models. Horn and Johnson (2013) is the standard reference for eigenvalue inequalities, positive definite matrices and norms. Halmos (1958) gives the coordinate-free view of the vector-space facts used in [Section 1.2](#).

Generalized inverses. The four conditions of [Definition 1.3.2](#) are due to Penrose (1955), who proved existence and uniqueness. E.H. Moore had described an equivalent inverse decades earlier. Rao (1973) made the weaker inverse $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$ the basic tool of linear-model theory, and it is his approach that [Section 1.4](#) and [Chapter 8](#) follow. The construction from a nonsingular submatrix in [Theorem 1.3.2](#) is a matrix-free way of stating an algorithm that is often given as a sequence of steps (Searle 1982).

Schur complements and updating formulas. The name “Schur complement” and the notation for it vary. Zhang (2005) collects the history and many applications. The rank-one formula [Eq. 1.6.7](#) is from Sherman and Morrison (1950), and the general identity [Eq. 1.6.6](#) from Woodbury (1950). Henderson and Searle (1981) trace earlier appearances and compare several derivations. Obtaining the identity by comparing two expressions for one block of a partitioned inverse, as in [Theorem 1.6.5](#), is a classical route.

Singular values and conditioning. Stewart (1993) tells the early history of the singular value decomposition, beginning with Beltrami and Jordan in the 1870s. The low-rank approximation theorem, [Proposition 1.8.2\(c\)](#) and its Frobenius form, is usually credited to Eckart and Young (1936), although Stewart points out that Schmidt had proved it earlier in the setting of integral equations. Golub and Van Loan (2013) is the reference for computing the decomposition and for perturbation bounds such as [Eq. 1.8.3](#), and Higham (2002) for their refinements. The use of scaled condition numbers and singular vectors to diagnose collinearity is due to Belsley et al. (1980). Chapter 26 returns to it.

Kronecker products and calculus. Henderson and Searle (1979) survey the vec operator and its uses in multivariate statistics. Magnus and Neudecker (2019) develop matrix calculus systematically through differentials, which avoid the bookkeeping of the entrywise definitions used in [Section 1.11](#) and handle symmetric matrix arguments cleanly.

1.12.3 References

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Chapter 2

Random Vectors: Expectation and Covariance Operators

A linear model is a statement about the first two moments of a random vector: its mean lies in a subspace and its covariance matrix has a given form. This chapter builds the calculus for such statements. Expectation and covariance become matrix operators that commute with linear maps, covariance matrices turn out to be exactly the nonnegative definite matrices, and a singular covariance means the vector is confined to a flat. The mean of a quadratic form needs only two moments; its variance needs four. Partitioning a vector gives cross-covariances, Schur complements and the best linear predictor, which foreshadow conditional distributions under normality.

What you need. [Chapter 1](#) (trace, spectral theorem, nonnegative definite matrices and their square roots, generalized inverses, partitioned matrices), and probability at the level of expectations, variances and independence of random variables.

2.0.1 Roadmap

- [Mean vectors and covariance matrices](#)
- [Linear and affine transformations](#)
- [Expectation of quadratic forms](#)
- [Correlation matrices](#)
- [Moments of partitioned random vectors](#)
- [Moment generating functions and independence](#)
- [Sample moments and their computation](#)
- [Summary and notes](#)

2.1 Mean vectors and covariance matrices

Random vectors enter regression in two different ways. The first is the response vector. In the model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

$\mathbf{Y} = (Y_1, \dots, Y_n)^\top$ has one entry per observational unit, and so does the unobservable error ε . The assumptions that do the real work in [Chapter 5](#), [Chapter 6](#) and [Chapter 7](#) are not about the shape of any distribution. They are statements about moments: $E(\varepsilon) = \mathbf{0}$, and the errors have equal variances and are uncorrelated. The second way is the measurement vector. When several quantities are recorded on the same unit, such as income, age and years of schooling, the unit contributes a vector whose components are usually correlated and measured in different units. Random regressors, errors-in-variables models and multivariate responses are all built from such vectors.

Both uses need the same toolkit: expectation and covariance as operations on vectors and matrices, their behaviour under linear maps, and the expected values of quadratic forms. Throughout, a random vector *has finite second moments* if every component has finite variance.

2.1.1 Random matrices and their expectation

Definition 2.1.1 — Expectation of a random matrix

A **random matrix** $\mathbf{Z} = (Z_{ij})$ is an $m \times k$ array of random variables. If every Z_{ij} has a finite expectation, the **expectation** of $\mathbf{Z}$ is the $m \times k$ matrix of expectations,

$$E(\mathbf{Z}) = (E(Z_{ij})).$$

A random vector is the case $k = 1$.

The definition works entry by entry, so every rule that is linear in the entries carries over at once.

Proposition 2.1.1 — Linearity of expectation

Let $\mathbf{Z}$ and $\mathbf{Z}_2$ be $m \times k$ random matrices with integrable entries, and let $\mathbf{A}$ ($l \times m$), $\mathbf{B}$ ($k \times s$) and $\mathbf{C}$ ($l \times s$) be constant matrices. Then

- (a) $E(\mathbf{AZB} + \mathbf{C}) = \mathbf{A} E(\mathbf{Z}) \mathbf{B} + \mathbf{C}$;
- (b) $E(\mathbf{Z} + \mathbf{Z}_2) = E(\mathbf{Z}) + E(\mathbf{Z}_2)$ and $E(\mathbf{Z}^\top) = E(\mathbf{Z})^\top$;
- (c) if $m = k$, then $E \operatorname{tr}(\mathbf{Z}) = \operatorname{tr} E(\mathbf{Z})$.

Proof. The (i, j) entry of $\mathbf{AZB} + \mathbf{C}$ is $\sum_r \sum_t a_{ir} Z_{rt} b_{tj} + c_{ij}$, a finite linear combination of integrable random variables with constant coefficients. Its expectation is $\sum_r \sum_t a_{ir} E(Z_{rt}) b_{tj} + c_{ij}$, which is the (i, j) entry of $\mathbf{A} E(\mathbf{Z}) \mathbf{B} + \mathbf{C}$. Parts (b) and (c) are the same argument applied to $Z_{ij} + Z_{2,ij}$, to Z_{ji} , and to $\sum_i Z_{ii}$. ■

Two warnings go with this proposition. The multipliers must be constant: if $\mathbf{A}$ is random, $E(\mathbf{AZ})$ is generally not $E(\mathbf{A})E(\mathbf{Z})$. And the expectation of a product of random matrices is generally not the product of expectations. It is when every entry of the first is independent of every entry of the second, because then $E(Z_{ir}W_{rj}) = E(Z_{ir})E(W_{rj})$ for each term.

2.1.2 Mean vector and covariance matrix

Definition 2.1.2 — Mean vector and covariance matrices

Let $\mathbf{Y}$ be a $p \times 1$ random vector with finite second moments. Its **mean vector** is $\boldsymbol{\mu} = \mathbf{E}(\mathbf{Y})$, and its **covariance matrix** is the $p \times p$ matrix

$$\text{Cov}(\mathbf{Y}) = \mathbf{E}[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^\top],$$

whose (i, j) entry is $\sigma_{ij} = \text{Cov}(Y_i, Y_j)$ and whose diagonal holds the variances $\sigma_{ii} = \text{Var}(Y_i)$. If $\mathbf{U}$ ($p \times 1$) and $\mathbf{V}$ ($q \times 1$) both have finite second moments, their **cross-covariance matrix** is the $p \times q$ matrix

$$\text{Cov}(\mathbf{U}, \mathbf{V}) = \mathbf{E}[(\mathbf{U} - \mathbf{E} \mathbf{U})(\mathbf{V} - \mathbf{E} \mathbf{V})^\top],$$

with (i, j) entry $\text{Cov}(U_i, V_j)$.

The expectations exist: $|U_i V_j| \leq \frac{1}{2}(U_i^2 + V_j^2)$, so finite second moments make every product integrable. We often write $\boldsymbol{\Sigma}$ for $\text{Cov}(\mathbf{Y})$, and $\text{Cov}(\mathbf{Y}) = \text{Cov}(\mathbf{Y}, \mathbf{Y})$. Some books write $\text{Var}(\mathbf{Y})$ or $D(\mathbf{Y})$ for the covariance matrix. We keep Var for scalars. The symbol Cov then has three uses: $\text{Cov}(Y_i, Y_j)$ is a number, $\text{Cov}(\mathbf{Y})$ is a symmetric square matrix, and $\text{Cov}(\mathbf{U}, \mathbf{V})$ is a rectangular matrix that is usually not symmetric even when $p = q$.

Proposition 2.1.2 — Elementary properties

Let $\mathbf{Y}, \mathbf{U}, \mathbf{V}$ have finite second moments, with $\boldsymbol{\mu} = \mathbf{E}(\mathbf{Y})$ and $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y})$, and let $\mathbf{a}, \mathbf{b}$ be constant vectors of matching sizes. Then:

- (a) $\boldsymbol{\Sigma}$ is symmetric;
- (b) $\text{Cov}(\mathbf{U}, \mathbf{V}) = \mathbf{E}(\mathbf{U}\mathbf{V}^\top) - \mathbf{E}(\mathbf{U})\mathbf{E}(\mathbf{V})^\top$, and in particular $\boldsymbol{\Sigma} = \mathbf{E}(\mathbf{Y}\mathbf{Y}^\top) - \boldsymbol{\mu}\boldsymbol{\mu}^\top$;
- (c) $\text{Cov}(\mathbf{V}, \mathbf{U}) = \text{Cov}(\mathbf{U}, \mathbf{V})^\top$;
- (d) $\text{Cov}(\mathbf{U} + \mathbf{a}, \mathbf{V} + \mathbf{b}) = \text{Cov}(\mathbf{U}, \mathbf{V})$;
- (e) $\mathbf{E}[(\mathbf{Y} - \mathbf{a})(\mathbf{Y} - \mathbf{a})^\top] = \boldsymbol{\Sigma} + (\boldsymbol{\mu} - \mathbf{a})(\boldsymbol{\mu} - \mathbf{a})^\top$, and hence $\mathbf{E}\|\mathbf{Y} - \mathbf{a}\|^2 = \text{tr}(\boldsymbol{\Sigma}) + \|\boldsymbol{\mu} - \mathbf{a}\|^2$.

Proof.

- (a) and (c) hold because $(\mathbf{u}\mathbf{v}^\top)^\top = \mathbf{v}\mathbf{u}^\top$ and expectation commutes with transposition. For (b), expand $(\mathbf{U} - \boldsymbol{\mu}_U)(\mathbf{V} - \boldsymbol{\mu}_V)^\top = \mathbf{U}\mathbf{V}^\top - \mathbf{U}\boldsymbol{\mu}_V^\top - \boldsymbol{\mu}_U\mathbf{V}^\top + \boldsymbol{\mu}_U\boldsymbol{\mu}_V^\top$ and take expectations with Proposition 2.1.1. The two middle terms each give $-\boldsymbol{\mu}_U\boldsymbol{\mu}_V^\top$. (d) holds because adding a constant does not change the centred vectors. For (e), write $\mathbf{Y} - \mathbf{a} = (\mathbf{Y} - \boldsymbol{\mu}) + (\boldsymbol{\mu} - \mathbf{a})$ and expand. The cross terms contain the factor $\mathbf{E}(\mathbf{Y} - \boldsymbol{\mu}) = \mathbf{0}$ and vanish. The trace identity follows from $\|\mathbf{v}\|^2 = \text{tr}(\mathbf{v}\mathbf{v}^\top)$ and Proposition 2.1.1(c). ■

Part (e) is the vector form of “mean squared error equals variance plus squared bias”. It shows that the mean is the constant vector closest to $\mathbf{Y}$ in expected squared distance, and that the

minimal value is $\text{tr}(\mathbf{\Sigma}) = \sum_i \text{Var}(Y_i)$. The matrix statement is stronger than the trace statement. It says that the second-moment matrix about any point exceeds $\mathbf{\Sigma}$ by a nonnegative definite matrix of rank at most one.

2.1.3 Uncorrelated is not independent

Random vectors $\mathbf{U}$ and $\mathbf{V}$ are **uncorrelated** if $\text{Cov}(\mathbf{U}, \mathbf{V}) = \mathbf{0}$. If $\mathbf{U}$ and $\mathbf{V}$ are independent, then $E(U_i V_j) = E(U_i)E(V_j)$ for all i, j , so they are uncorrelated. The converse is false, and it fails in the most innocent-looking cases.

Example. (A point on a circle)

Let Θ be uniform on $[0, 2\pi)$ and $\mathbf{Y} = (\cos \Theta, \sin \Theta)^\top$. Then $E(\cos \Theta) = E(\sin \Theta) = 0$, $E(\cos^2 \Theta) = E(\sin^2 \Theta) = \frac{1}{2}$, and $E(\cos \Theta \sin \Theta) = \frac{1}{2} E(\sin 2\Theta) = 0$. So $\boldsymbol{\mu} = \mathbf{0}$ and $\text{Cov}(\mathbf{Y}) = \frac{1}{2} \mathbf{I}_2$, exactly the moments of two independent coordinates. Yet $Y_1^2 + Y_2^2 = 1$: once Y_1 is known, Y_2 is known up to sign. Covariance measures *linear* association only, and the dependence here is entirely nonlinear.

For jointly normal vectors, zero covariance and independence coincide ([Theorem 3.4.1](#)). That equivalence rests on joint normality, not on normal marginals, and it is one of the main reasons normal theory is so tractable.

Scalar summaries. The **total variance** $\text{tr}(\mathbf{\Sigma})$ ignores correlation; the **generalized variance** $\det(\mathbf{\Sigma})$ does not. If $\lambda_1, \dots, \lambda_p$ are the eigenvalues of $\mathbf{\Sigma}$ ([Theorem 1.7.2](#)), these are $\sum_i \lambda_i$ and $\prod_i \lambda_i$, and both are unchanged by a rotation $\mathbf{Y} \mapsto \mathbf{QY}$. The determinant can be tiny even when every variance is large, if one eigenvalue is near zero. [Theorem 2.2.3](#) shows that a zero eigenvalue means some linear combination of $\mathbf{Y}$ does not vary at all.

2.1.4 Exercises

2.1.4.1 A. Check your understanding

Exercise A1.

Let $\mathbf{Z}$ be an $n \times p$ random matrix whose entries are independent, with mean 0 and variance σ^2 . Show that $E(\mathbf{Z}^\top \mathbf{Z}) = n\sigma^2 \mathbf{I}_p$ and $E(\mathbf{Z} \mathbf{Z}^\top) = p\sigma^2 \mathbf{I}_n$. For a constant $p \times p$ matrix $\mathbf{B}$, find $E(\mathbf{Z} \mathbf{B} \mathbf{Z}^\top)$.

2.1.4.2 B. Practice

Exercise B1.

Let X have a distribution symmetric about 0 with $\text{Var}(X) = 1$ and $E X^4 < \infty$, let S be independent of X with $\Pr(S = 1) = \Pr(S = -1) = \frac{1}{2}$, and put $\mathbf{Y} = (X, SX)^\top$. Show that the two components have the same distribution and are uncorrelated, and that $\text{Cov}(Y_1^2, Y_2^2) = \text{Var}(X^2)$. Deduce that Y_1 and Y_2 are independent iff X is itself a random sign. ([Chapter 3](#) takes X standard normal.)

Solution. Since S is independent of X and X is symmetric, $\Pr(SX \leq y) = \frac{1}{2} \Pr(X \leq y) + \frac{1}{2} \Pr(-X \leq y) = \Pr(X \leq y)$. By independence, $\text{Cov}(X, SX) = \text{E}(S) \text{E}(X^2) = 0$. Both squares equal X^2 , so $\text{Cov}(Y_1^2, Y_2^2) = \text{Var}(X^2)$. Functions of independent variables are independent, hence uncorrelated, so $\text{Var}(X^2) > 0$ makes Y_1 and Y_2 dependent. If $\text{Var}(X^2) = 0$, then $X^2 = \text{E} X^2 = 1$ with probability one, and by symmetry $X = \pm 1$ with probability $\frac{1}{2}$ each. Then $\Pr(Y_1 = a, Y_2 = b) = \Pr(X = a) \Pr(S = ab) = \frac{1}{4}$ for all signs a, b , which is the product of the marginal probabilities, as in [Example \(Pairwise but not mutually independent\)](#).

Exercise B2.

Show that $\text{E}[(\mathbf{Y} - \mathbf{a})(\mathbf{Y} - \mathbf{a})^\top] - \text{Cov}(\mathbf{Y})$ is nonnegative definite for every constant $\mathbf{a}$, with equality iff $\mathbf{a} = \boldsymbol{\mu}$. Deduce that $\mathbf{a} = \boldsymbol{\mu}$ minimizes $\text{E}[(\mathbf{Y} - \mathbf{a})^\top \mathbf{N}(\mathbf{Y} - \mathbf{a})]$ for every positive definite $\mathbf{N}$.

2.2 Linear and affine transformations

Almost every statistic in a linear model is a linear function of the response: least squares coefficients, fitted values, residuals, contrasts. Their means and covariance matrices follow from one theorem.

2.2.1 The transformation rules

Theorem 2.2.1 — Moments of affine transformations

Let $\mathbf{Y}$ be a $p \times 1$ random vector with finite second moments, $\text{E}(\mathbf{Y}) = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$. Let $\mathbf{A}$ be $k \times p$, $\mathbf{B}$ be $m \times p$, and $\mathbf{b}$ be $k \times 1$, all constant. Then

- (a) $\text{E}(\mathbf{A}\mathbf{Y} + \mathbf{b}) = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$;
- (b) $\text{Cov}(\mathbf{A}\mathbf{Y} + \mathbf{b}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top$;
- (c) $\text{Cov}(\mathbf{A}\mathbf{Y}, \mathbf{B}\mathbf{Y}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^\top$.

More generally, if $\mathbf{U}$ and $\mathbf{V}$ have finite second moments and $\mathbf{A}, \mathbf{B}, \mathbf{b}, \mathbf{c}$ are constants of conformable sizes, then $\text{Cov}(\mathbf{A}\mathbf{U} + \mathbf{b}, \mathbf{B}\mathbf{V} + \mathbf{c}) = \mathbf{A} \text{Cov}(\mathbf{U}, \mathbf{V}) \mathbf{B}^\top$.

Proof.

- (a) is [Proposition 2.1.1\(a\)](#). For the general statement, the centred vectors are $\mathbf{A}\mathbf{U} + \mathbf{b} - \text{E}(\mathbf{A}\mathbf{U} + \mathbf{b}) = \mathbf{A}(\mathbf{U} - \text{E}\mathbf{U})$ and similarly $\mathbf{B}(\mathbf{V} - \text{E}\mathbf{V})$. Hence

$$\text{Cov}(\mathbf{A}\mathbf{U} + \mathbf{b}, \mathbf{B}\mathbf{V} + \mathbf{c}) = \text{E}[\mathbf{A}(\mathbf{U} - \text{E}\mathbf{U})(\mathbf{V} - \text{E}\mathbf{V})^\top \mathbf{B}^\top] = \mathbf{A} \text{E}[(\mathbf{U} - \text{E}\mathbf{U})(\mathbf{V} - \text{E}\mathbf{V})^\top] \mathbf{B}^\top,$$

again by [Proposition 2.1.1\(a\)](#). Parts (c) and (b) are the cases $\mathbf{U} = \mathbf{V} = \mathbf{Y}$, and $\mathbf{B} = \mathbf{A}$ in (b).

For a single linear combination, $\mathbf{A} = \mathbf{a}^\top$ is a row vector, and (b) and (c) give

$$\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a} = \sum_{i=1}^p \sum_{j=1}^p a_i a_j \sigma_{ij}, \quad \text{Cov}(\mathbf{a}^\top \mathbf{Y}, \mathbf{b}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{b}. \quad (2.2.1)$$

Every variance and covariance enters $\text{Var}(\mathbf{a}^\top \mathbf{Y})$, weighted by the products of the coefficients. Conversely, $\boldsymbol{\Sigma}$ is determined by the variances of linear combinations, because $\sigma_{ij} = \frac{1}{2}[\text{Var}(Y_i + Y_j) - \text{Var}(Y_i) - \text{Var}(Y_j)]$.

Proposition 2.2.2 — Covariance of sums

Let $\mathbf{U}, \mathbf{U}_2, \mathbf{V}$ have finite second moments, with $\mathbf{U}$ and $\mathbf{U}_2$ of the same size in (a), and $\mathbf{U}$ and $\mathbf{V}$ of the same size in (b). Then

- (a) $\text{Cov}(\mathbf{U} + \mathbf{U}_2, \mathbf{V}) = \text{Cov}(\mathbf{U}, \mathbf{V}) + \text{Cov}(\mathbf{U}_2, \mathbf{V})$;
- (b) $\text{Cov}(\mathbf{U} + \mathbf{V}) = \text{Cov}(\mathbf{U}) + \text{Cov}(\mathbf{U}, \mathbf{V}) + \text{Cov}(\mathbf{V}, \mathbf{U}) + \text{Cov}(\mathbf{V})$;
- (c) if $\mathbf{Y}_1, \dots, \mathbf{Y}_n$ are pairwise uncorrelated $p \times 1$ vectors and $c_1, \dots, c_n$ are constants, then $\text{Cov}(\sum_i c_i \mathbf{Y}_i) = \sum_i c_i^2 \text{Cov}(\mathbf{Y}_i)$. In particular, if every $\text{Cov}(\mathbf{Y}_i) = \boldsymbol{\Sigma}$, then $\text{Cov}(\bar{\mathbf{Y}}) = \boldsymbol{\Sigma}/n$.

Proof.

- (a) Centre each vector and use linearity of the outer product and of E. (b) Apply (a) in each argument; note that the two cross terms are transposes of each other, not equal.
- (b) By (a) and (b), applied repeatedly, $\text{Cov}(\sum_i c_i \mathbf{Y}_i) = \sum_i \sum_j c_i c_j \text{Cov}(\mathbf{Y}_i, \mathbf{Y}_j)$, using $\text{Cov}(c\mathbf{U}, \mathbf{V}) = c \text{Cov}(\mathbf{U}, \mathbf{V})$ from [Theorem 2.2.1](#). The terms with $i \neq j$ vanish.

Example. (Centring a sample)

Let $\mathbf{Y}$ be $n \times 1$ with $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$, and write $\mathbf{J} = \mathbf{1}\mathbf{1}^\top$. The mean is $\bar{Y} = n^{-1} \mathbf{1}^\top \mathbf{Y}$, and the vector of deviations is $\mathbf{d} = (\mathbf{I} - n^{-1} \mathbf{J})\mathbf{Y}$. The matrix $\mathbf{I} - n^{-1} \mathbf{J}$ is symmetric and idempotent, so [Theorem 2.2.1](#) gives

$$\text{Cov}(\mathbf{d}) = \sigma^2 (\mathbf{I} - n^{-1} \mathbf{J}), \quad \text{Cov}(\bar{Y}, \mathbf{d}) = \sigma^2 n^{-1} \mathbf{1}^\top (\mathbf{I} - n^{-1} \mathbf{J}) = \mathbf{0}^\top.$$

The deviations have variances $\sigma^2(1 - 1/n)$ and are negatively correlated, with correlation $-1/(n-1)$ between each pair, because they are forced to sum to zero. And each deviation is uncorrelated with the mean. Under normality, uncorrelated becomes independent ([Example \(Sample mean and deviations\)](#)), which is the first half of [Corollary 4.4.5](#).

2.2.2 Covariance matrices are nonnegative definite

A variance cannot be negative, and by Eq. 2.2.1 this simple fact constrains the whole matrix. Recall from Definition 1.7.1 that a symmetric Σ is nonnegative definite if $\mathbf{a}^\top \Sigma \mathbf{a} \geq 0$ for all $\mathbf{a}$, and positive definite if in addition $\mathbf{a}^\top \Sigma \mathbf{a} > 0$ for $\mathbf{a} \neq \mathbf{0}$.

Theorem 2.2.3 — Covariance matrices

Let $\mathbf{Y}$ have finite second moments, mean $\boldsymbol{\mu}$ and covariance matrix Σ .

- (a) Σ is nonnegative definite.
- (b) Conversely, every $p \times p$ nonnegative definite matrix is the covariance matrix of some random vector.
- (c) For a constant $\mathbf{a}$, $\Sigma \mathbf{a} = \mathbf{0}$ iff $\mathbf{a}^\top \mathbf{Y} = \mathbf{a}^\top \boldsymbol{\mu}$ with probability one.
- (d) $\Pr(\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\Sigma)) = 1$. If $\mathcal{S}$ is any subspace with $\Pr(\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{S}) = 1$, then $\mathcal{C}(\Sigma) \subseteq \mathcal{S}$.

In particular, Σ is singular iff some nontrivial linear combination of $\mathbf{Y}$ is constant with probability one. Then $\mathbf{Y}$ lies, with probability one, in the affine subspace $\boldsymbol{\mu} + \mathcal{C}(\Sigma)$ of dimension $\text{rank}(\Sigma) < p$, and this is the smallest such flat.

Proof.

- (a) Σ is symmetric, and $\mathbf{a}^\top \Sigma \mathbf{a} = \text{Var}(\mathbf{a}^\top \mathbf{Y}) \geq 0$ by Eq. 2.2.1.
- (b) Let Σ be nonnegative definite and let $\Sigma^{1/2}$ be its symmetric nonnegative definite square root (Theorem 1.7.6). Let $\mathbf{Z}$ have independent components with mean 0 and variance 1, for instance independent standard normal variables, so that $\text{Cov}(\mathbf{Z}) = \mathbf{I}$. Then $\text{Cov}(\Sigma^{1/2} \mathbf{Z}) = \Sigma^{1/2} \mathbf{I} \Sigma^{1/2} = \Sigma$.
- (c) If $\Sigma \mathbf{a} = \mathbf{0}$, the random variable $\mathbf{a}^\top (\mathbf{Y} - \boldsymbol{\mu})$ has mean zero and variance $\mathbf{a}^\top \Sigma \mathbf{a} = 0$, so it is zero with probability one. Conversely, if $\mathbf{a}^\top (\mathbf{Y} - \boldsymbol{\mu}) = 0$ with probability one, then $\Sigma \mathbf{a} = \mathbb{E}[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^\top \mathbf{a}] = \mathbf{0}$.
- (d) Let $\mathbf{a}_1, \dots, \mathbf{a}_s$ be a basis of $\mathcal{N}(\Sigma)$. By (c), each event $\{\mathbf{a}_j^\top (\mathbf{Y} - \boldsymbol{\mu}) = 0\}$ has probability one, and so does their intersection. On that intersection $\mathbf{Y} - \boldsymbol{\mu}$ is orthogonal to $\mathcal{N}(\Sigma)$. Because Σ is symmetric, $\mathcal{N}(\Sigma)$ is the orthogonal complement of $\mathcal{C}(\Sigma)$, and so $\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\Sigma)$. Now let $\mathcal{S}$ be a subspace with $\Pr(\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{S}) = 1$. For every $\mathbf{a} \in \mathcal{S}^\perp$ we have $\mathbf{a}^\top (\mathbf{Y} - \boldsymbol{\mu}) = 0$ with probability one, hence $\mathbf{a} \in \mathcal{N}(\Sigma) = \mathcal{C}(\Sigma)^\perp$ by (c). So $\mathcal{S}^\perp \subseteq \mathcal{C}(\Sigma)^\perp$, and taking orthogonal complements gives $\mathcal{C}(\Sigma) \subseteq \mathcal{S}$. The final statements follow, with $\dim \mathcal{C}(\Sigma) = \text{rank}(\Sigma)$. For minimality among all flats, suppose $\Pr(\mathbf{Y} \in \mathbf{c} + \mathcal{S}) = 1$ for a vector $\mathbf{c}$ and a subspace $\mathcal{S}$. Every $\mathbf{a} \in \mathcal{S}^\perp$ has $\mathbf{a}^\top (\mathbf{Y} - \mathbf{c}) = 0$ with probability one, and taking expectations gives $\mathbf{a}^\top (\boldsymbol{\mu} - \mathbf{c}) = 0$. Hence $\boldsymbol{\mu} - \mathbf{c} \in \mathcal{S}$, so $\mathbf{c} + \mathcal{S} = \boldsymbol{\mu} + \mathcal{S}$, and the subspace case applies.

■

A covariance matrix is therefore positive definite exactly when no linear combination of the components is degenerate. Singular covariance matrices are not pathological. They arise whenever the components satisfy an exact linear constraint, as with the deviations of Example (Centring a sample), whose covariance matrix has rank $n - 1$ because $\mathbf{1}^\top \mathbf{d} = 0$. The residual vector of a

regression is another case: it is orthogonal to every column of $\mathbf{X}$, and when $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ its covariance matrix has rank $n - \text{rank}(\mathbf{X})$ (Section 2.2, Chapter 6).

Example. (Multinomial counts)

In m independent trials, each outcome falls into one of k categories with probabilities $\boldsymbol{\pi} = (\pi_1, \dots, \pi_k)^\top$, where $\sum_j \pi_j = 1$. Let $\mathbf{Z}_t$ be the indicator vector of trial t , with a single 1 in the position of its category. Then $\mathbb{E}(\mathbf{Z}_t) = \boldsymbol{\pi}$, and $\mathbf{Z}_t \mathbf{Z}_t^\top = \text{diag}(\mathbf{Z}_t)$ because the indicator has only one nonzero entry. So $\text{Cov}(\mathbf{Z}_t) = \text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi} \boldsymbol{\pi}^\top$. The count vector $\mathbf{Y} = \sum_t \mathbf{Z}_t$ is a sum of independent vectors, and Proposition 2.2.2(c) gives

$$\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y}) = m(\text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi} \boldsymbol{\pi}^\top).$$

Because $\boldsymbol{\Sigma} \mathbf{1} = m(\boldsymbol{\pi} - \boldsymbol{\pi} \boldsymbol{\pi}^\top \mathbf{1}) = \mathbf{0}$, the matrix is singular, and Theorem 2.2.3(c) recovers the constraint $\mathbf{1}^\top \mathbf{Y} = m$. For $m = 20$ and $\boldsymbol{\pi} = (0.5, 0.3, 0.2)^\top$ the eigenvalues are 7.762, 4.638 and 0. All 5000 simulated count vectors lie in the plane $\boldsymbol{\mu} + \mathcal{C}(\boldsymbol{\Sigma}) = \{\mathbf{y} : \mathbf{1}^\top \mathbf{y} = m\}$.

Linear maps cannot increase the rank of a covariance matrix, since $\text{rank}(\mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top) \leq \text{rank}(\boldsymbol{\Sigma})$ (Exercise (B2)). They can, however, discard degenerate directions. If $\mathbf{Y} = (Z, Z)^\top$ with $\text{Var}(Z) = 1$, then $\boldsymbol{\Sigma}$ has all entries equal to 1 and is singular, while $\mathbf{A} = (1, 0)$ gives $\text{Var}(\mathbf{A} \mathbf{Y}) = 1 > 0$. By Theorem 2.2.1, the covariance matrix of $\mathbf{A} \mathbf{Y}$ is $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top$. If $\boldsymbol{\Sigma}$ is positive definite and $\mathbf{A}$ has full row rank $k \leq p$, then for $\mathbf{b} \neq \mathbf{0}$ the vector $\mathbf{A}^\top \mathbf{b}$ is nonzero and $\mathbf{b}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top \mathbf{b} > 0$. So $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top$ is positive definite. If $k > p$, then $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top$ has rank at most p and is singular. A vector of more than p linear combinations of p variables always satisfies a linear constraint.

2.2.3 Standardization and Mahalanobis distance

A scalar with mean μ and variance $\sigma^2 > 0$ is standardized by $(Y - \mu)/\sigma$. The vector version must remove correlation as well as scale.

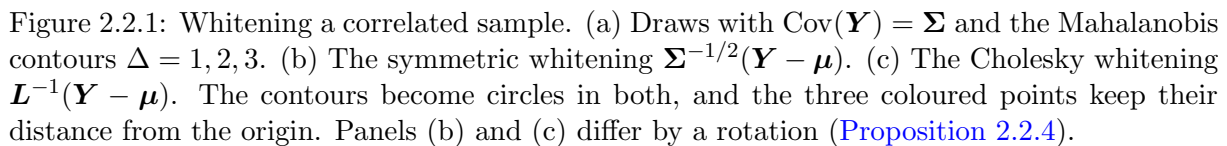
Proposition 2.2.4 — Whitening

Let $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y})$ be positive definite, with $\mathbb{E}(\mathbf{Y}) = \boldsymbol{\mu}$.

- (a) $\mathbf{Z} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{Y} - \boldsymbol{\mu})$ has $\mathbb{E}(\mathbf{Z}) = \mathbf{0}$ and $\text{Cov}(\mathbf{Z}) = \mathbf{I}_p$.
- (b) A $p \times p$ matrix $\mathbf{W}$ satisfies $\mathbf{W} \boldsymbol{\Sigma} \mathbf{W}^\top = \mathbf{I}_p$ iff $\mathbf{W} = \mathbf{Q} \boldsymbol{\Sigma}^{-1/2}$ for some orthogonal $\mathbf{Q}$. In particular, if $\boldsymbol{\Sigma} = \mathbf{L} \mathbf{L}^\top$ is the Cholesky factorization, $\mathbf{W} = \mathbf{L}^{-1}$ is such a matrix.
- (c) For every $\mathbf{W}$ as in (b), $\|\mathbf{W}(\mathbf{Y} - \boldsymbol{\mu})\|^2 = (\mathbf{Y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu})$.

Proof.

- (a) By Theorem 2.2.1, $\text{Cov}(\mathbf{Z}) = \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\Sigma} \boldsymbol{\Sigma}^{-1/2} = \mathbf{I}$, since $\boldsymbol{\Sigma}^{-1/2}$ is symmetric and commutes with $\boldsymbol{\Sigma}$. (b) If $\mathbf{W} = \mathbf{Q} \boldsymbol{\Sigma}^{-1/2}$, then $\mathbf{W} \boldsymbol{\Sigma} \mathbf{W}^\top = \mathbf{Q} \mathbf{Q}^\top = \mathbf{I}$. Conversely, if $\mathbf{W} \boldsymbol{\Sigma} \mathbf{W}^\top = \mathbf{I}$, put $\mathbf{Q} = \mathbf{W} \boldsymbol{\Sigma}^{1/2}$. Then $\mathbf{Q} \mathbf{Q}^\top = \mathbf{W} \boldsymbol{\Sigma} \mathbf{W}^\top = \mathbf{I}$, so the square matrix $\mathbf{Q}$ is orthogonal, and $\mathbf{W} = \mathbf{Q} \boldsymbol{\Sigma}^{-1/2}$. For the Cholesky factor, $\mathbf{L}^{-1} \boldsymbol{\Sigma} (\mathbf{L}^\top)^{-1} = \mathbf{L}^{-1} \mathbf{L} \mathbf{L}^\top (\mathbf{L}^\top)^{-1} = \mathbf{I}$. (c) $\mathbf{W}^\top \mathbf{W} = \boldsymbol{\Sigma}^{-1/2} \mathbf{Q}^\top \mathbf{Q} \boldsymbol{\Sigma}^{-1/2} = \boldsymbol{\Sigma}^{-1}$.


$$\Delta^2(\mathbf{Y}) = (\mathbf{Y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu}), \quad (2.2.2)$$

Example. (Two whitenings)

Take $\boldsymbol{\mu} = (3, 1)^\top$ and $\boldsymbol{\Sigma} = \begin{pmatrix} 4 & 2.4 \\ 2.4 & 2.25 \end{pmatrix}$, so the standard deviations are 2 and 1.5 and the correlation is 0.8. The eigenvalues are 5.680 and 0.570. The symmetric and Cholesky whitening matrices are

$$\Sigma^{-1/2} = \begin{pmatrix} 0.717 & -0.425 \\ -0.425 & 1.027 \end{pmatrix}, \quad L^{-1} = \begin{pmatrix} 0.500 & 0 \\ -0.667 & 1.111 \end{pmatrix}.$$

The Cholesky version standardizes Y_1 alone and then removes from Y_2 its linear dependence on Y_1 , in the spirit of Gram–Schmidt. The symmetric version treats both coordinates alike and moves points as little as possible (Exercise (B3)). Figure 2.2.1 applies both to 400 simulated points. The two whitened clouds differ by a rotation through 22.5° , and every point keeps its Mahalanobis distance.

```
import numpy as np
rng = np.random.default_rng(2)

mu = np.array([3.0, 1.0])
Sigma = np.array([[4.0, 2.4],
                  [2.4, 2.25]])
```

*# correlation 2.4 / (2 * 1.5) = 0.8*

```

lam, U = np.linalg.eigh(Sigma)                # spectral decomposition
W_sym = U @ np.diag(lam ** -0.5) @ U.T        # Sigma^{-1/2}
L = np.linalg.cholesky(Sigma)                # Sigma = L L', L lower triangular
W_chol = np.linalg.inv(L)                    # L^{-1}

n = 400
Y = mu + rng.standard_normal((n, 2)) @ L.T    # any draws with Cov = Sigma will
    do
Z_sym = (Y - mu) @ W_sym.T
Z_chol = (Y - mu) @ W_chol.T

Q = W_chol @ np.linalg.inv(W_sym)            # L^{-1} Sigma^{1/2}
print("Q'Q =", np.round(Q.T @ Q, 12))

```

Whitening runs in both directions. If $\text{Cov}(\mathbf{Z}) = \mathbf{I}$, then $\boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$, or $\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}$, has mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. This is how correlated vectors are simulated, and with $\mathbf{Z}$ standard normal it is how [Chapter 3](#) constructs $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. The Mahalanobis distance then appears in the exponent of the density ([Theorem 3.2.2](#)). When $\boldsymbol{\Sigma}$ is singular, no matrix whitens all of $\mathbf{Y}$, since a degenerate combination cannot be scaled to variance one. The best one can do is whiten within the support $\boldsymbol{\mu} + \mathcal{C}(\boldsymbol{\Sigma})$ using the Moore–Penrose inverse square root ([Exercise \(B4\)](#)).

2.2.4 Second moments in the linear model

Here is how the rules of this section enter regression. Suppose $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ an $n \times p$ constant matrix of rank p , $E(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$. Adding the constant $\mathbf{X}\boldsymbol{\beta}$ changes the mean but not the covariance, so $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$. The least squares estimator $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ is $\mathbf{A}\mathbf{Y}$ with $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$, so

$$E(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} = \boldsymbol{\beta}, \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

No distributional assumption was used, and none is needed for the Gauss–Markov theorem ([Theorem 7.2.1](#)). Fitted values and residuals are also linear in $\mathbf{Y}$, and their covariance matrices are computed the same way in [Chapter 5](#) and [Chapter 6](#). If instead $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$ with $\mathbf{V}$ positive definite, then $\mathbf{V}^{-1/2} \mathbf{Y} = \mathbf{V}^{-1/2} \mathbf{X} \boldsymbol{\beta} + \mathbf{V}^{-1/2} \boldsymbol{\varepsilon}$ is a model with $\text{Cov}(\mathbf{V}^{-1/2} \boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$. Whitening the data turns correlated errors back into the standard case. This is generalized least squares, which [Chapter 6](#) and [Chapter 31](#) develop.

2.2.5 Exercises

2.2.5.1 A. Check your understanding

Exercise A1.

Let $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_3$. Find the covariance and correlation matrices of the successive differences $(Y_2 - Y_1, Y_3 - Y_2)^\top$, and of $(Y_2 - Y_1, Y_3 - Y_2, Y_3 - Y_1)^\top$. Which is singular, and which linear combination is degenerate?

Exercise A2.

For $p = 2$ and $\Sigma = \mathbf{L}\mathbf{L}^\top$, show that the second component of $\mathbf{L}^{-1}(\mathbf{Y} - \boldsymbol{\mu})$ is a multiple of $Y_2 - \mu_2 - (\sigma_{12}/\sigma_{11})(Y_1 - \mu_1)$. Explain the connection with [Theorem 2.5.2](#).

2.2.5.2 B. Practice**Exercise B1.**

Let $\varepsilon_0, \dots, \varepsilon_n$ be uncorrelated with variance σ^2 , and let $Y_t = \varepsilon_t + \theta\varepsilon_{t-1}$ for $t = 1, \dots, n$. Find $\text{Cov}(\mathbf{Y})$, and show that it is positive definite for every real θ , including $\theta = \pm 1$.

Solution. $\mathbf{Y} = \mathbf{A}\boldsymbol{\varepsilon}$, where $\mathbf{A}$ is $n \times (n+1)$ with θ in column t and 1 in column $t+1$ of row t (columns indexed by $\varepsilon_0, \dots, \varepsilon_n$). So $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{A}\mathbf{A}^\top$ is tridiagonal, with $\sigma^2(1 + \theta^2)$ on the diagonal and $\sigma^2\theta$ next to it. The last n columns of $\mathbf{A}$ form a lower bidiagonal matrix with unit diagonal, which is nonsingular. So $\mathbf{A}$ has full row rank n , and $\mathbf{A}\mathbf{A}^\top$ is positive definite ([Section 2.2](#)) whatever the value of θ .

Exercise B2.

Show that $\mathcal{C}(\mathbf{A}\Sigma\mathbf{A}^\top) = \mathcal{C}(\mathbf{A}\Sigma)$ for every nonnegative definite Σ and conformable $\mathbf{A}$. Deduce that $\text{rank Cov}(\mathbf{A}\mathbf{Y}) \leq \text{rank Cov}(\mathbf{Y})$.

Exercise B3.

Let Σ be positive definite. Among all $\mathbf{W}$ with $\mathbf{W}\Sigma\mathbf{W}^\top = \mathbf{I}$, show that $\mathbb{E}\|\mathbf{W}(\mathbf{Y} - \boldsymbol{\mu}) - (\mathbf{Y} - \boldsymbol{\mu})\|^2$ is minimized uniquely by $\mathbf{W} = \Sigma^{-1/2}$. *Hint:* use [Proposition 2.2.4\(b\)](#) and show that $\text{tr}(\mathbf{Q}\Sigma^{1/2}) \leq \text{tr}(\Sigma^{1/2})$ for orthogonal $\mathbf{Q}$.

Solution. Write $\mathbf{W} = \mathbf{Q}\Sigma^{-1/2}$. By [Corollary 2.3.2\(a\)](#) with $\mathbf{A} = (\mathbf{W} - \mathbf{I})^\top(\mathbf{W} - \mathbf{I})$,

$$\mathbb{E}\|(\mathbf{W} - \mathbf{I})(\mathbf{Y} - \boldsymbol{\mu})\|^2 = \text{tr}((\mathbf{W} - \mathbf{I})\Sigma(\mathbf{W} - \mathbf{I})^\top) = \text{tr}(\mathbf{W}\Sigma\mathbf{W}^\top) - 2\text{tr}(\mathbf{W}\Sigma) + \text{tr}(\Sigma) = p + \text{tr}(\Sigma) - 2\text{tr}(\mathbf{Q}\Sigma^{1/2}).$$

Let $\Sigma^{1/2} = \mathbf{\Gamma}\mathbf{\Lambda}^{1/2}\mathbf{\Gamma}^\top$ and $\mathbf{G} = \mathbf{\Gamma}^\top\mathbf{Q}\mathbf{\Gamma}$, which is orthogonal, so each $|g_{ii}| \leq 1$. Then $\text{tr}(\mathbf{Q}\Sigma^{1/2}) = \text{tr}(\mathbf{G}\mathbf{\Lambda}^{1/2}) = \sum_i g_{ii}\lambda_i^{1/2} \leq \sum_i \lambda_i^{1/2}$. Since every $\lambda_i > 0$, equality forces every $g_{ii} = 1$. An orthogonal matrix with unit diagonal is $\mathbf{I}$, because each column has norm one. So $\mathbf{G} = \mathbf{I}$, which means $\mathbf{Q} = \mathbf{I}$.

Exercise B4.

Let $\Sigma = \text{Cov}(\mathbf{Y})$ have rank r , with spectral decomposition $\Sigma = \mathbf{\Gamma}_r\mathbf{\Lambda}_r\mathbf{\Gamma}_r^\top$, where $\mathbf{\Gamma}_r^\top\mathbf{\Gamma}_r = \mathbf{I}_r$ and $\mathbf{\Lambda}_r$ holds the positive eigenvalues. Show that $\mathbf{Z} = \mathbf{\Lambda}_r^{-1/2}\mathbf{\Gamma}_r^\top(\mathbf{Y} - \boldsymbol{\mu})$ has $\text{Cov}(\mathbf{Z}) = \mathbf{I}_r$ and that $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{\Gamma}_r\mathbf{\Lambda}_r^{1/2}\mathbf{Z}$ with probability one. Show that $(\mathbf{Y} - \boldsymbol{\mu})^\top\Sigma^+(\mathbf{Y} - \boldsymbol{\mu}) = \|\mathbf{Z}\|^2$, where Σ^+ is the Moore–Penrose inverse, and that its expectation is r .

Solution. By [Theorem 2.2.1](#), $\text{Cov}(\mathbf{Z}) = \mathbf{\Lambda}_r^{-1/2}\mathbf{\Gamma}_r^\top\mathbf{\Gamma}_r\mathbf{\Lambda}_r\mathbf{\Gamma}_r^\top\mathbf{\Gamma}_r\mathbf{\Lambda}_r^{-1/2} = \mathbf{I}_r$. By [Theorem 2.2.3\(d\)](#), $\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\Sigma) = \mathcal{C}(\mathbf{\Gamma}_r)$ with probability one, and on that event $\mathbf{Y} -$

$\boldsymbol{\mu} = \boldsymbol{\Gamma}_r \boldsymbol{\Gamma}_r^\top (\mathbf{Y} - \boldsymbol{\mu}) = \boldsymbol{\Gamma}_r \boldsymbol{\Lambda}_r^{1/2} \mathbf{Z}$. The Moore–Penrose inverse is $\boldsymbol{\Sigma}^+ = \boldsymbol{\Gamma}_r \boldsymbol{\Lambda}_r^{-1} \boldsymbol{\Gamma}_r^\top$, so $(\mathbf{Y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^+ (\mathbf{Y} - \boldsymbol{\mu}) = \|\boldsymbol{\Lambda}_r^{-1/2} \boldsymbol{\Gamma}_r^\top (\mathbf{Y} - \boldsymbol{\mu})\|^2 = \|\mathbf{Z}\|^2$, with expectation $\text{tr Cov}(\mathbf{Z}) = r$.

2.2.5.3 C. Going deeper

Exercise C1.

In $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ of full column rank, let $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$ with $\mathbf{V}$ positive definite. Find $\text{Cov}(\hat{\boldsymbol{\beta}})$ for $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ and $\text{Cov}(\tilde{\boldsymbol{\beta}})$ for $\tilde{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$. Show that $\text{Cov}(\hat{\boldsymbol{\beta}}) - \text{Cov}(\tilde{\boldsymbol{\beta}})$ is nonnegative definite. *Hint:* show that $\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}$ and $\tilde{\boldsymbol{\beta}}$ are uncorrelated.

Solution. By [Theorem 2.2.1](#),

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}, \quad \text{Cov}(\tilde{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1}.$$

Write $\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}} = \mathbf{D}\mathbf{Y}$ with $\mathbf{D} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top - (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1}$, and note $\mathbf{D}\mathbf{X} = \mathbf{I} - \mathbf{I} = \mathbf{0}$. Then $\text{Cov}(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}, \tilde{\boldsymbol{\beta}}) = \sigma^2 \mathbf{D} \mathbf{V} \mathbf{V}^{-1} \mathbf{X} (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} = \mathbf{0}$, so by [Proposition 2.2.2\(b\)](#), $\text{Cov}(\hat{\boldsymbol{\beta}}) = \text{Cov}(\tilde{\boldsymbol{\beta}}) + \text{Cov}(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}})$, and the last term is nonnegative definite.

2.3 Expectation of quadratic forms

Sums of squares are quadratic forms in the response. The sample variance is $(n-1)^{-1} \mathbf{Y}^\top (\mathbf{I} - n^{-1} \mathbf{J}) \mathbf{Y}$. The residual sum of squares of a regression is $\mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$, where $\mathbf{M}$ is the projection onto the column space of $\mathbf{X}$ ([Chapter 6](#)). Test statistics are ratios of such forms. Their expectations follow from a short trace argument that needs no distributional assumption.

2.3.1 The basic formula

Theorem 2.3.1 — Mean of a quadratic form

Let $\mathbf{Y}$ be a $p \times 1$ random vector with finite second moments, $\text{E}(\mathbf{Y}) = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$, and let $\mathbf{A}$ be a constant $p \times p$ matrix. Then

$$\text{E}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}.$$

Proof. A 1×1 matrix equals its trace, and $\text{tr}(\mathbf{a} \mathbf{b}^\top) = \mathbf{b}^\top \mathbf{a}$ ([Theorem 1.6.1](#)), so $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \text{tr}(\mathbf{A} \mathbf{Y} \mathbf{Y}^\top)$. By [Proposition 2.1.1](#) and [Proposition 2.1.2\(b\)](#),

$$\text{E}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \text{E}(\mathbf{Y} \mathbf{Y}^\top)) = \text{tr}(\mathbf{A}(\boldsymbol{\Sigma} + \boldsymbol{\mu} \boldsymbol{\mu}^\top)) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}.$$

■

The formula has two parts. The trace term is what the form picks up from random variation. The second term is the value of the form at the mean, which is what it would be without any

randomness. The theorem does not require independence, normality or symmetry of $\mathbf{A}$. Since $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \mathbf{Y}^\top \mathbf{A}_s \mathbf{Y}$ with $\mathbf{A}_s = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$, and $\text{tr}(\mathbf{A} \mathbf{\Sigma}) = \text{tr}(\mathbf{A}_s \mathbf{\Sigma})$ when $\mathbf{\Sigma}$ is symmetric, we may assume $\mathbf{A}$ symmetric whenever that is convenient.

Corollary 2.3.2

Under the conditions of [Theorem 2.3.1](#):

- (a) $\mathbb{E}[(\mathbf{Y} - \mathbf{b})^\top \mathbf{A}(\mathbf{Y} - \mathbf{b})] = \text{tr}(\mathbf{A} \mathbf{\Sigma}) + (\boldsymbol{\mu} - \mathbf{b})^\top \mathbf{A}(\boldsymbol{\mu} - \mathbf{b})$ for any constant $\mathbf{b}$;
- (b) if $\mathbf{\Sigma} = \sigma^2 \mathbf{I}$, then $\mathbb{E}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \sigma^2 \text{tr}(\mathbf{A}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$;
- (c) if $\mathbf{\Sigma}$ is positive definite, the squared Mahalanobis distance [Eq. 2.2.2](#) has $\mathbb{E} \Delta^2(\mathbf{Y}) = p$;
- (d) for random vectors $\mathbf{U}$ ($p \times 1$) and $\mathbf{V}$ ($q \times 1$) with finite second moments and a constant $p \times q$ matrix $\mathbf{A}$, $\mathbb{E}(\mathbf{U}^\top \mathbf{A} \mathbf{V}) = \text{tr}(\mathbf{A} \text{Cov}(\mathbf{V}, \mathbf{U})) + \mathbb{E}(\mathbf{U})^\top \mathbf{A} \mathbb{E}(\mathbf{V})$.

Proof.

- (a) applies the theorem to $\mathbf{Y} - \mathbf{b}$, which has covariance matrix $\mathbf{\Sigma}$. (b) is immediate.
- (c) is (a) with $\mathbf{b} = \boldsymbol{\mu}$ and $\mathbf{A} = \mathbf{\Sigma}^{-1}$, so the trace is $\text{tr}(\mathbf{I}_p) = p$. For (d), $\mathbf{U}^\top \mathbf{A} \mathbf{V} = \text{tr}(\mathbf{A} \mathbf{V} \mathbf{U}^\top)$ and $\mathbb{E}(\mathbf{V} \mathbf{U}^\top) = \text{Cov}(\mathbf{V}, \mathbf{U}) + \mathbb{E}(\mathbf{V}) \mathbb{E}(\mathbf{U})^\top$.

■

Example. (Sample variance with unequal means and variances)

Let $Y_1, \dots, Y_n$ be uncorrelated with means μ_i and variances σ_i^2 , and let $S^2 = (n-1)^{-1} \mathbf{Y}^\top \mathbf{C} \mathbf{Y}$ with $\mathbf{C} = \mathbf{I} - n^{-1} \mathbf{J}$. Here $\mathbf{\Sigma} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$, so $\text{tr}(\mathbf{C} \mathbf{\Sigma}) = (1 - 1/n) \sum_i \sigma_i^2$, and $\boldsymbol{\mu}^\top \mathbf{C} \boldsymbol{\mu} = \sum_i (\mu_i - \bar{\mu})^2$. Therefore

$$\mathbb{E}(S^2) = \overline{\sigma^2} + \frac{1}{n-1} \sum_{i=1}^n (\mu_i - \bar{\mu})^2, \quad \overline{\sigma^2} = \frac{1}{n} \sum_{i=1}^n \sigma_i^2.$$

With a common mean, S^2 is unbiased for the *average* variance, even when the variances differ. Differences in the means inflate it. That is exactly why a residual mean square estimates σ^2 only when the mean structure is right.

Example. (Residual sum of squares under a wrong mean)

Let $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$ and $\mathbb{E}(\mathbf{Y}) = \boldsymbol{\theta}$, where $\boldsymbol{\theta}$ need not lie in $\mathcal{C}(\mathbf{X})$. The residual sum of squares is $\mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$, where $\mathbf{M}$ is symmetric and idempotent of rank $r = \text{rank}(\mathbf{X})$, so $\text{tr}(\mathbf{I} - \mathbf{M}) = n - r$ ([Theorem 1.9.2](#)). By [Corollary 2.3.2\(b\)](#),

$$\mathbb{E}(\text{RSS}) = \sigma^2(n - r) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2. \quad (2.3.1)$$

If the model is right, $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{X})$, the second term vanishes and $\text{RSS}/(n - r)$ is unbiased for σ^2 . If the model is wrong, the estimate is biased upwards by the squared distance from the true mean to the model space, divided by $n - r$. For instance, fit a straight

line at $x = 1, \dots, 6$ when the true mean is the parabola $\theta_i = \frac{1}{4}(x_i - 3.5)^2$ and $\sigma^2 = 1$. Then $\sigma^2(n - r) = 4$, $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2 = 2.333$, and $E(\text{RSS}) = 6.333$. Chapter 19 studies such departures systematically.

The theorem uses only two moments, so it holds equally for skewed and heavy-tailed data. The listing checks it for a correlated vector $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{B}\mathbf{U}$, where $\mathbf{U}$ has independent, centred exponential components, which are strongly skewed. With the matrices in the listing, $\text{tr}(\mathbf{A}\boldsymbol{\Sigma}) = 6.9$ and $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} = 17.0$, so $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = 23.9$. The average over 10^6 draws is 23.940, with Monte Carlo standard error 0.018. The script also verifies the formula exactly, by enumerating a discrete distribution with eight atoms.

```
import itertools
import numpy as np
rng = np.random.default_rng(20260917)

mu = np.array([1.0, -1.0, 2.0])
B = np.array([[1.0, 0.0, 0.0],
              [0.5, 1.0, 0.0],
              [-0.5, 0.8, 0.6]])
Sigma = B @ B.T                                     # Cov(mu + B U) when Cov(U) = I
A = np.array([[2.0, 1.0, 0.0],
              [1.0, 1.0, -1.0],
              [0.0, -1.0, 3.0]])

theory = np.trace(A @ Sigma) + mu @ A @ mu          # Theorem: no normality needed

N = 1_000_000
U = rng.exponential(size=(N, 3)) - 1.0             # skewed, mean 0, variance 1
Y = mu + U @ B.T
Q = np.einsum("ij,jk,ik->i", Y, A, Y)             # Y_i' A Y_i for every draw
print(f"theory {theory:.4f}    simulation {Q.mean():.4f}  {Q.std() / np.sqrt(N):.4f}")
)
```

2.3.2 The variance needs fourth moments

The mean of $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ involves only $E(Y_i Y_j)$. Its second moment involves $E(Y_i Y_j Y_k Y_l)$, so no formula for $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$ can be written in terms of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ alone. Two random vectors with the same mean and covariance matrix can give the same quadratic form very different variances. The next theorem handles independent components, the case that matters for residual sums of squares under the classical error assumptions. Chapter 4 gives the normal case with a general $\boldsymbol{\Sigma}$ (Theorem 4.2.3).

Theorem 2.3.3 — Variance of a quadratic form

Let $Y_1, \dots, Y_n$ be independent with means $\theta_1, \dots, \theta_n$ and common central moments $\sigma^2 = E(Y_i - \theta_i)^2$, $\mu_3 = E(Y_i - \theta_i)^3$ and $\mu_4 = E(Y_i - \theta_i)^4 < \infty$. Let $\mathbf{A}$ be a constant symmetric $n \times n$ matrix, and let $\mathbf{a} = (a_{11}, \dots, a_{nn})^\top$ be its diagonal. Then

$$\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = (\mu_4 - 3\sigma^4) \mathbf{a}^\top \mathbf{a} + 2\sigma^4 \text{tr}(\mathbf{A}^2) + 4\sigma^2 \boldsymbol{\theta}^\top \mathbf{A}^2 \boldsymbol{\theta} + 4\mu_3 \boldsymbol{\theta}^\top \mathbf{A} \mathbf{a}.$$

Proof. Write $\mathbf{Y} = \boldsymbol{\theta} + \boldsymbol{\varepsilon}$, where the ε_i are independent with mean zero. Then

$$\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = q + \ell + \boldsymbol{\theta}^\top \mathbf{A} \boldsymbol{\theta}, \quad q = \boldsymbol{\varepsilon}^\top \mathbf{A} \boldsymbol{\varepsilon}, \quad \ell = 2\mathbf{b}^\top \boldsymbol{\varepsilon}, \quad \mathbf{b} = \mathbf{A} \boldsymbol{\theta},$$

so $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{Var}(q) + \text{Var}(\ell) + 2 \text{Cov}(q, \ell)$. We compute the three terms.

The linear term. $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, so $\text{Var}(\ell) = 4\sigma^2 \mathbf{b}^\top \mathbf{b} = 4\sigma^2 \boldsymbol{\theta}^\top \mathbf{A}^2 \boldsymbol{\theta}$.

The cross term. Since $\text{E}(\ell) = 0$, $\text{Cov}(q, \ell) = \text{E}(q\ell) = 2 \sum_{i,j,k} a_{ij} b_k \text{E}(\varepsilon_i \varepsilon_j \varepsilon_k)$. By independence and zero means, $\text{E}(\varepsilon_i \varepsilon_j \varepsilon_k)$ vanishes unless $i = j = k$, when it equals μ_3 . So $\text{Cov}(q, \ell) = 2\mu_3 \sum_i a_{ii} b_i = 2\mu_3 \boldsymbol{\theta}^\top \mathbf{A} \mathbf{a}$.

The quadratic term. $\text{E}(q^2) = \sum_{i,j,k,l} a_{ij} a_{kl} \text{E}(\varepsilon_i \varepsilon_j \varepsilon_k \varepsilon_l)$. The fourth-order moment is μ_4 if all four indices agree. It is σ^4 if the indices form two distinct pairs, which happens in three patterns: $i = j \neq k = l$, $i = k \neq j = l$ and $i = l \neq j = k$. Otherwise some index occurs exactly once, and the moment is zero. Using the symmetry of $\mathbf{A}$,

$$\text{E}(q^2) = \mu_4 \sum_i a_{ii}^2 + \sigma^4 \left[\sum_{i \neq k} a_{ii} a_{kk} + 2 \sum_{i \neq j} a_{ij}^2 \right] = \mu_4 \mathbf{a}^\top \mathbf{a} + \sigma^4 [(\text{tr } \mathbf{A})^2 - \mathbf{a}^\top \mathbf{a} + 2 \text{tr}(\mathbf{A}^2) - 2\mathbf{a}^\top \mathbf{a}],$$

since $\sum_{i,j} a_{ij}^2 = \text{tr}(\mathbf{A}^2)$. By [Theorem 2.3.1](#), $\text{E}(q) = \sigma^2 \text{tr } \mathbf{A}$, and subtracting its square gives $\text{Var}(q) = (\mu_4 - 3\sigma^4) \mathbf{a}^\top \mathbf{a} + 2\sigma^4 \text{tr}(\mathbf{A}^2)$. Adding the three terms proves the formula. ■

For normal errors, $\mu_3 = 0$ and $\mu_4 = 3\sigma^4$, and the formula collapses to $2\sigma^4 \text{tr}(\mathbf{A}^2) + 4\sigma^2 \boldsymbol{\theta}^\top \mathbf{A}^2 \boldsymbol{\theta}$. Both extra terms describe how non-normality leaks into second-order behaviour. The kurtosis term depends on $\mathbf{A}$ only through its diagonal. The skewness term needs both a skewed error distribution and a mean that the form does not annihilate. For the residual sum of squares of a correct model, $\mathbf{A} \boldsymbol{\theta} = (\mathbf{I} - \mathbf{M}) \boldsymbol{\theta} = \mathbf{0}$, so the skewness term disappears but the kurtosis term remains.

Example. (Skewed errors and a misspecified line)

Continue [Example \(Residual sum of squares under a wrong mean\)](#), now with independent centred exponential errors, for which $\sigma^2 = 1$, $\mu_3 = 2$ and $\mu_4 = 9$. [Theorem 2.3.3](#) gives $\text{Var}(\text{RSS}) = 29.80$. Of this, the kurtosis term contributes 16.73 and the skewness term -4.27 . Normal errors with the same mean and variance would give 17.33, so the exponential errors multiply the variance by 1.72 even though the mean of RSS is unchanged. Over 10^6 simulated data sets, the variance of RSS is 30.04. The script also confirms the formula exactly for a two-point error distribution.

```
se = Q.std() / np.sqrt(N)
vals, probs = np.array([-1.0, 2.0]), np.array([2 / 3, 1 / 3])
exact = 0.0
for idx in itertools.product(range(2), repeat=3):
    u = vals[list(idx)]
    y = mu + B @ u
    exact += np.prod(probs[list(idx)]) * (y @ A @ y)
x = np.arange(1.0, 7.0) # n = 6 design points
X = np.column_stack([np.ones_like(x), x])
M = X @ np.linalg.solve(X.T @ X, X.T)
A2 = np.eye(6) - M # RSS = Y'(I - M)Y for a straight
line
```



```

theta = 0.25 * (x - 3.5) ** 2          # true mean is curved: the line is
    wrong
a = np.diag(A2)

def var_quadform(A, theta, m2, m3, m4):
    """Var(Y'AY) for independent Y_i with means theta_i and common central moments.
    """
    a = np.diag(A)
    return ((m4 - 3 * m2**2) * a @ a + 2 * m2**2 * np.trace(A @ A)
            + 4 * m2 * theta @ A @ A @ theta + 4 * m3 * theta @ A @ a)

# A2 = I - M for a straight line at x = 1..6; theta = true (curved) mean
mean_rss = np.trace(A2) + theta @ A2 @ theta          # sigma^2 = 1
var_expo = var_quadform(A2, theta, 1.0, 2.0, 9.0) # centred exponential errors
var_norm = var_quadform(A2, theta, 1.0, 0.0, 3.0) # what normal errors would give

E = rng.exponential(size=(N, 6)) - 1.0
R = np.einsum("ij,jk,ik->i", theta + E, A2, theta + E)
print(f"mean   {mean_rss:.3f} vs {R.mean():.3f}")
print(f"var    {var_expo:.3f} vs {R.var():.3f}   (normal-theory value {var_norm:.3f})")

```

2.3.3 Exercises

2.3.3.1 A. Check your understanding

Exercise A1.

For random vectors $\mathbf{U}$ and $\mathbf{V}$ of the same size, show that $E\|\mathbf{U} - \mathbf{V}\|^2 = \text{tr Cov}(\mathbf{U}) + \text{tr Cov}(\mathbf{V}) - 2 \text{tr Cov}(\mathbf{U}, \mathbf{V}) + \|E\mathbf{U} - E\mathbf{V}\|^2$.

2.3.3.2 B. Practice

Exercise B1.

Let $E(Y_t) = \mu$ and $\text{Cov}(Y_t, Y_s) = \sigma^2 \phi^{|t-s|}$ for $t, s = 1, \dots, n$, with $|\phi| < 1$. Show that the sample variance satisfies

$$E(S^2) = \sigma^2 \left[1 - \frac{2}{n(n-1)} \sum_{k=1}^{n-1} (n-k) \phi^k \right].$$

Is S^2 biased up or down when $\phi > 0$? What happens as $n \rightarrow \infty$ with ϕ fixed?

Solution. With $\mathbf{C} = \mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top$ and $E(\mathbf{Y}) = \mu\mathbf{1}$, $\mathbf{C}E(\mathbf{Y}) = \mathbf{0}$, so $E(\mathbf{Y}^\top \mathbf{C} \mathbf{Y}) = \text{tr}(\mathbf{C}\Sigma) = \text{tr}(\Sigma) - n^{-1}\mathbf{1}^\top \Sigma \mathbf{1}$. Here $\text{tr}(\Sigma) = n\sigma^2$, and $\mathbf{1}^\top \Sigma \mathbf{1}$ sums all entries. There are n diagonal entries and $2(n-k)$ entries at lag k , so $\mathbf{1}^\top \Sigma \mathbf{1} = \sigma^2 [n + 2 \sum_{k=1}^{n-1} (n-k) \phi^k]$. Dividing $n\sigma^2 - n^{-1}\mathbf{1}^\top \Sigma \mathbf{1}$ by $n-1$ gives the formula. For $\phi > 0$ the sum is positive, so S^2 is biased downwards: positively correlated observations cluster, and the sample looks less variable than the process is. As $n \rightarrow \infty$ the sum is $O(n)$, so the bias is $O(1/n)$ and vanishes.

Exercise B2.

Suppose $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$. Show that $E(\text{RSS}) = \sigma^2 \text{tr}((\mathbf{I} - \mathbf{M})\mathbf{V})$. For the equicorrelated $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$ and a model with $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, show that $E(\text{RSS})/(n - r) = \sigma^2(1 - \rho)$.

Exercise B3.

Let $Y_1, \dots, Y_n$ be independent and identically distributed with variance σ^2 and finite fourth central moment μ_4 . Use [Theorem 2.3.3](#) to show that $\text{Var}(S^2) = (\mu_4 - 3\sigma^4)/n + 2\sigma^4/(n - 1)$. Compare with the normal case.

Solution. Take $\mathbf{A} = \mathbf{C}/(n - 1)$ and $\boldsymbol{\theta} = \mu\mathbf{1}$. Then $\mathbf{A}\boldsymbol{\theta} = \mathbf{0}$, so the last two terms of [Theorem 2.3.3](#) vanish. Every diagonal entry of $\mathbf{A}$ is $(1 - 1/n)/(n - 1) = 1/n$, so $\mathbf{a}^\top \mathbf{a} = 1/n$, and $\text{tr}(\mathbf{A}^2) = \text{tr}(\mathbf{C})/(n - 1)^2 = 1/(n - 1)$. This gives the formula. Normal data have $\mu_4 = 3\sigma^4$ and $\text{Var}(S^2) = 2\sigma^4/(n - 1)$. Heavy tails ($\mu_4 > 3\sigma^4$) add a term of the same order $1/n$, so the normal-theory variance is wrong even in large samples.

Exercise B4.

Under the conditions of [Theorem 2.3.3](#) with $\boldsymbol{\theta} = \mathbf{0}$, show that $\text{Cov}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}, \mathbf{Y}^\top \mathbf{B} \mathbf{Y}) = (\mu_4 - 3\sigma^4)\mathbf{a}^\top \mathbf{b} + 2\sigma^4 \text{tr}(\mathbf{A}\mathbf{B})$, where $\mathbf{a}$ and $\mathbf{b}$ are the diagonals of the symmetric matrices $\mathbf{A}$ and $\mathbf{B}$. *Hint:* apply the theorem to $\mathbf{A} + \mathbf{B}$.

Exercise B5.

Let $Y_1, \dots, Y_n$ be independent $N(0, \sigma^2)$. Among symmetric $\mathbf{A}$ with $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \sigma^2$, show that $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$ is minimized by $\mathbf{A} = n^{-1}\mathbf{I}$. *Hint:* $\text{tr}(\mathbf{A}^2) \geq (\text{tr} \mathbf{A})^2/n$.

2.4 Correlation matrices

Covariances carry units: the covariance of income in dollars with age in years is in dollar-years. To compare the strength of linear association across pairs of variables, each covariance is divided by the two standard deviations.

Definition 2.4.1 — Correlation matrix

Let $\mathbf{Y}$ have covariance matrix $\boldsymbol{\Sigma} = (\sigma_{ij})$ with every $\sigma_{ii} > 0$, and let $\mathbf{D}_\sigma = \text{diag}(\sigma_{11}^{1/2}, \dots, \sigma_{pp}^{1/2})$ be the diagonal matrix of standard deviations. The **correlation matrix** of $\mathbf{Y}$ is

$$\mathbf{R} = \mathbf{D}_\sigma^{-1} \boldsymbol{\Sigma} \mathbf{D}_\sigma^{-1}, \quad \text{with entries} \quad \rho_{ij} = \frac{\sigma_{ij}}{\sqrt{\sigma_{ii}\sigma_{jj}}} = \text{corr}(Y_i, Y_j).$$

Equivalently $\boldsymbol{\Sigma} = \mathbf{D}_\sigma \mathbf{R} \mathbf{D}_\sigma$. For random vectors $\mathbf{U}$ and $\mathbf{V}$ with standard-deviation matrices $\mathbf{D}_U$ and $\mathbf{D}_V$, the cross-correlation matrix is $\mathbf{D}_U^{-1} \text{Cov}(\mathbf{U}, \mathbf{V}) \mathbf{D}_V^{-1}$.

Some books write $\mathbf{P}_\rho$ for the correlation matrix. We use $\mathbf{R}$ in this chapter, and $\hat{\mathbf{R}}$ for its sample counterpart in [Section 2.7](#).

Proposition 2.4.1 — Properties of correlation matrices

Let $\mathbf{R}$ be the correlation matrix of $\mathbf{Y}$.

- (a) $\mathbf{R}$ is the covariance matrix of the standardized vector $\mathbf{Z} = \mathbf{D}_\sigma^{-1}(\mathbf{Y} - \boldsymbol{\mu})$. Hence $\mathbf{R}$ is nonnegative definite with unit diagonal, and $\text{tr}(\mathbf{R}) = p$.
- (b) $|\rho_{ij}| \leq 1$, with $|\rho_{ij}| = 1$ iff $Z_j = \rho_{ij}Z_i$ with probability one.
- (c) $\text{rank}(\mathbf{R}) = \text{rank}(\boldsymbol{\Sigma})$. In particular, $\mathbf{R}$ is positive definite iff $\boldsymbol{\Sigma}$ is.
- (d) If $\mathbf{Y}^* = \mathbf{D}\mathbf{Y} + \mathbf{c}$ with $\mathbf{D} = \text{diag}(d_1, \dots, d_p)$ and every $d_i \neq 0$, then $\rho_{ij}^* = \text{sign}(d_i d_j) \rho_{ij}$. Positive rescaling and shifting leave $\mathbf{R}$ unchanged.
- (e) Conversely, every nonnegative definite matrix with unit diagonal is the correlation matrix of some random vector.

Proof.

- (a) [Theorem 2.2.1](#) gives $\text{Cov}(\mathbf{Z}) = \mathbf{D}_\sigma^{-1} \boldsymbol{\Sigma} \mathbf{D}_\sigma^{-1} = \mathbf{R}$, and [Theorem 2.2.3\(a\)](#) applies.
- (b) The 2×2 matrix of rows and columns i, j of $\mathbf{R}$ is the covariance matrix of (Z_i, Z_j) , so it is nonnegative definite and its determinant $1 - \rho_{ij}^2$ is nonnegative. If $\rho_{ij} = \pm 1$, the vector $(\rho_{ij}, -1)^\top$ is in the null space of that 2×2 matrix, so $\rho_{ij}Z_i - Z_j = 0$ with probability one by [Theorem 2.2.3\(c\)](#). Conversely, $Z_j = cZ_i$ with $\text{Var}(Z_i) = \text{Var}(Z_j) = 1$ forces $c = \pm 1$ and $\rho_{ij} = c$.
- (c) $\mathbf{D}_\sigma$ is nonsingular.
- (d) $\text{Cov}(\mathbf{Y}^*) = \mathbf{D} \boldsymbol{\Sigma} \mathbf{D}$ has entries $d_i d_j \sigma_{ij}$ and standard deviations $|d_i| \sigma_{ii}^{1/2}$.
- (e) Such a matrix is a covariance matrix by [Theorem 2.2.3\(b\)](#), and its variances are one.

■

Part (d) is the reason correlation is the natural scale-free summary. Part (e) is less obvious than it looks. A symmetric matrix with unit diagonal and off-diagonal entries in $[-1, 1]$ need not be a correlation matrix, because the correlations must be jointly consistent.

Proposition 2.4.2 — Three correlations

Let $|\rho_{12}| < 1$ and $|\rho_{13}| < 1$. The matrix

$$\mathbf{R} = \begin{pmatrix} 1 & \rho_{12} & \rho_{13} \\ \rho_{12} & 1 & \rho_{23} \\ \rho_{13} & \rho_{23} & 1 \end{pmatrix}$$

is a correlation matrix iff

$$\rho_{12}\rho_{13} - \sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)} \leq \rho_{23} \leq \rho_{12}\rho_{13} + \sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)}.$$

Proof. By [Proposition 2.4.1\(e\)](#) we must decide when $\mathbf{R}$ is nonnegative definite. Write $\mathbf{r} = (\rho_{12}, \rho_{13})^\top$ and let $\mathbf{R}_2$ be the lower right 2×2 block. The nonsingular matrix

$\mathbf{T} = \begin{pmatrix} 1 & \mathbf{0}^\top \\ -\mathbf{r} & \mathbf{I}_2 \end{pmatrix}$ gives

$$\mathbf{T}\mathbf{R}\mathbf{T}^\top = \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{R}_2 - \mathbf{r}\mathbf{r}^\top \end{pmatrix},$$

and $\mathbf{a}^\top \mathbf{R} \mathbf{a} = \mathbf{b}^\top (\mathbf{T}\mathbf{R}\mathbf{T}^\top) \mathbf{b}$ with $\mathbf{b} = (\mathbf{T}^\top)^{-1} \mathbf{a}$ ranges over all vectors as $\mathbf{a}$ does. So $\mathbf{R}$ is nonnegative definite iff the Schur complement

$$\mathbf{R}_2 - \mathbf{r}\mathbf{r}^\top = \begin{pmatrix} 1 - \rho_{12}^2 & \rho_{23} - \rho_{12}\rho_{13} \\ \rho_{23} - \rho_{12}\rho_{13} & 1 - \rho_{13}^2 \end{pmatrix}$$

is. A symmetric 2×2 matrix with positive diagonal is nonnegative definite iff its determinant is nonnegative, because its eigenvalues have positive sum and their product is the determinant. The determinant is $(1 - \rho_{12}^2)(1 - \rho_{13}^2) - (\rho_{23} - \rho_{12}\rho_{13})^2$, and it is nonnegative exactly on the stated interval. ■

If Y_2 and Y_3 both correlate 0.8 with Y_1 , the interval is $[0.28, 1]$. Two variables that are strongly related to a third cannot be negatively correlated with each other, nor even uncorrelated. The quantity $(\rho_{23} - \rho_{12}\rho_{13}) / \sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)}$, which the proof forces into $[-1, 1]$, is the correlation of Y_2 and Y_3 after the linear effect of Y_1 has been removed from both. [Section 2.5](#) develops this residualization, and [Definition 3.6.1](#) names the result the partial correlation.

Correlation matrices do not transform simply under linear maps: the correlation matrix of $\mathbf{A}\mathbf{Y}$ must be recomputed from $\mathbf{A}\mathbf{\Sigma}\mathbf{A}^\top$. Theory is therefore done with covariance matrices, and correlation matrices are used for description.

2.4.1 Exercises

2.4.1.1 A. Check your understanding

Exercise A1.

Show that $(1 - \rho)\mathbf{I}_p + \rho\mathbf{1}\mathbf{1}^\top$ is a correlation matrix iff $-1/(p - 1) \leq \rho \leq 1$. Explain the lower bound through the variance of $\sum_i Z_i$.

Solution. The matrix sends $\mathbf{1}$ to $(1 + (p - 1)\rho)\mathbf{1}$ and every $\mathbf{u} \perp \mathbf{1}$ to $(1 - \rho)\mathbf{u}$. Its eigenvalues are $1 + (p - 1)\rho$ (once) and $1 - \rho$ ($p - 1$ times). By [Proposition 2.4.1\(e\)](#) it is a correlation matrix iff both are nonnegative. If $\mathbf{Z}$ has this correlation matrix, $\text{Var}(\sum_i Z_i) = \mathbf{1}^\top \mathbf{R} \mathbf{1} = p[1 + (p - 1)\rho]$, which must be nonnegative. Many variables cannot all be strongly negatively correlated with one another, because their sum would then have negative variance.

2.4.1.2 B. Practice

Exercise B1.

Show that $\det(\mathbf{\Sigma}) \leq \prod_i \sigma_{ii}$ for every covariance matrix, with equality for positive definite $\mathbf{\Sigma}$ iff $\mathbf{\Sigma}$ is diagonal. *Hint:* factor out $\mathbf{D}_\sigma$ and use the eigenvalues of $\mathbf{R}$.

Solution. If some $\sigma_{ii} = 0$, the i th row of Σ vanishes (apply [Theorem 2.2.3\(c\)](#) with $\mathbf{a}$ the i th coordinate vector) and both sides are zero. Otherwise $\det \Sigma = \det(\mathbf{D}_\sigma)^2 \det \mathbf{R} = \prod_i \sigma_{ii} \cdot \det \mathbf{R}$. The eigenvalues $\lambda_i \geq 0$ of $\mathbf{R}$ sum to $\text{tr} \mathbf{R} = p$, so by the inequality of arithmetic and geometric means $\det \mathbf{R} = \prod_i \lambda_i \leq (p^{-1} \sum_i \lambda_i)^p = 1$. Equality holds iff every $\lambda_i = 1$, that is, iff $\mathbf{R} = \mathbf{I}$, that is, iff Σ is diagonal.

2.4.1.3 C. Going deeper

Exercise C1.

- (a) Show that a $p \times p$ matrix is a correlation matrix iff it is the Gram matrix $(\mathbf{u}_i^\top \mathbf{u}_j)$ of p unit vectors in $\mathbb{R}^p$. (b) For $p = 3$, write $\rho_{ij} = \cos \vartheta_{ij}$ with $\vartheta_{ij} \in [0, \pi]$. Show that [Proposition 2.4.2](#) is equivalent to the spherical triangle inequalities $|\vartheta_{12} - \vartheta_{13}| \leq \vartheta_{23} \leq \min(\vartheta_{12} + \vartheta_{13}, 2\pi - \vartheta_{12} - \vartheta_{13})$.

2.5 Moments of partitioned random vectors

Regression with random regressors studies a response vector $\mathbf{Y}$ ($p \times 1$) together with a regressor vector $\mathbf{X}$ ($q \times 1$) observed on the same unit. In this section $\mathbf{X}$ is a random vector, not a model matrix. We stack the two into one vector and read off what the blocks of its moments mean.

2.5.1 Blocks of the mean and covariance

Theorem 2.5.1 — Partitioned moments

Let $\mathbf{W} = (\mathbf{Y}^\top, \mathbf{X}^\top)^\top$ have finite second moments, and write

$$\mathbf{E}(\mathbf{W}) = \begin{pmatrix} \boldsymbol{\mu}_Y \\ \boldsymbol{\mu}_X \end{pmatrix}, \quad \text{Cov}(\mathbf{W}) = \begin{pmatrix} \Sigma_{YY} & \Sigma_{YX} \\ \Sigma_{XY} & \Sigma_{XX} \end{pmatrix}.$$

- (a) $\boldsymbol{\mu}_Y = \mathbf{E}(\mathbf{Y})$, $\boldsymbol{\mu}_X = \mathbf{E}(\mathbf{X})$, $\Sigma_{YY} = \text{Cov}(\mathbf{Y})$, $\Sigma_{XX} = \text{Cov}(\mathbf{X})$, $\Sigma_{YX} = \text{Cov}(\mathbf{Y}, \mathbf{X})$ is $p \times q$, and $\Sigma_{XY} = \Sigma_{YX}^\top$.
 (b) For constants $\mathbf{A}$, $\mathbf{B}$, $\mathbf{c}$ of conformable sizes, $\text{Cov}(\mathbf{A}\mathbf{Y}, \mathbf{B}\mathbf{X}) = \mathbf{A}\Sigma_{YX}\mathbf{B}^\top$, and

$$\text{Cov}(\mathbf{A}\mathbf{Y} + \mathbf{B}\mathbf{X} + \mathbf{c}) = \mathbf{A}\Sigma_{YY}\mathbf{A}^\top + \mathbf{A}\Sigma_{YX}\mathbf{B}^\top + \mathbf{B}\Sigma_{XY}\mathbf{A}^\top + \mathbf{B}\Sigma_{XX}\mathbf{B}^\top.$$

- (c) $\mathcal{C}(\Sigma_{XY}) \subseteq \mathcal{C}(\Sigma_{XX})$ and $\mathcal{C}(\Sigma_{YX}) \subseteq \mathcal{C}(\Sigma_{YY})$. Equivalently, for every generalized inverse Σ_{XX}^- ,

$$\Sigma_{XX}\Sigma_{XX}^-\Sigma_{XY} = \Sigma_{XY}, \quad \Sigma_{YX}\Sigma_{XX}^-\Sigma_{XX} = \Sigma_{YX},$$

and similarly with the roles of $\mathbf{Y}$ and $\mathbf{X}$ exchanged.

Proof.

- (a) The outer product $(\mathbf{W} - \mathbf{E}\mathbf{W})(\mathbf{W} - \mathbf{E}\mathbf{W})^\top$ has blocks $(\mathbf{Y} - \boldsymbol{\mu}_Y)(\mathbf{Y} - \boldsymbol{\mu}_Y)^\top$,

- $(\mathbf{Y} - \boldsymbol{\mu}_Y)(\mathbf{X} - \boldsymbol{\mu}_X)^\top$, its transpose, and $(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^\top$. Take expectations blockwise. (b) $\mathbf{A}\mathbf{Y} + \mathbf{B}\mathbf{X} + \mathbf{c} = [\mathbf{A}, \mathbf{B}]\mathbf{W} + \mathbf{c}$, and $\mathbf{A}\mathbf{Y} = [\mathbf{A}, \mathbf{0}]\mathbf{W}$, $\mathbf{B}\mathbf{X} = [\mathbf{0}, \mathbf{B}]\mathbf{W}$. Apply [Theorem 2.2.1](#) and multiply out the blocks.
- (b) Let $\mathbf{a} \in \mathcal{N}(\boldsymbol{\Sigma}_{XX})$. By [Theorem 2.2.3\(c\)](#), $\mathbf{a}^\top \mathbf{X}$ is constant with probability one, so its covariance with $\mathbf{Y}$ vanishes: $\boldsymbol{\Sigma}_{YX}\mathbf{a} = \text{Cov}(\mathbf{Y}, \mathbf{a}^\top \mathbf{X}) = \mathbf{0}$. Thus $\mathcal{N}(\boldsymbol{\Sigma}_{XX}) \subseteq \mathcal{N}(\boldsymbol{\Sigma}_{YX})$. Taking orthogonal complements, and using $\mathcal{N}(\mathbf{B})^\perp = \mathcal{C}(\mathbf{B}^\top)$ and the symmetry of $\boldsymbol{\Sigma}_{XX}$, gives $\mathcal{C}(\boldsymbol{\Sigma}_{XY}) \subseteq \mathcal{C}(\boldsymbol{\Sigma}_{XX})$. The same argument with $\mathbf{Y}$ and $\mathbf{X}$ exchanged gives the second inclusion. A column inclusion $\mathcal{C}(\boldsymbol{\Sigma}_{XY}) \subseteq \mathcal{C}(\boldsymbol{\Sigma}_{XX})$ means $\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{XX}\mathbf{K}^\top$ for some $\mathbf{K}$. Then $\boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XX}\mathbf{K}^\top = \boldsymbol{\Sigma}_{XY}$. Conversely, that identity exhibits every column of $\boldsymbol{\Sigma}_{XY}$ as a combination of columns of $\boldsymbol{\Sigma}_{XX}$ ([Theorem 1.4.1](#)). Transposing gives $\boldsymbol{\Sigma}_{YX} = \mathbf{K}\boldsymbol{\Sigma}_{XX}$ and the second identity. ■

Part (c) looks technical, but it removes every difficulty with singular $\boldsymbol{\Sigma}_{XX}$ in what follows. A direction in which $\mathbf{X}$ does not vary cannot covary with anything.

2.5.2 Linear prediction and residual covariance

The most important use of the blocks is to split $\mathbf{Y}$ into a part explained linearly by $\mathbf{X}$ and a part uncorrelated with $\mathbf{X}$.

Theorem 2.5.2 — Best linear predictor

In the setting of [Theorem 2.5.1](#), fix a generalized inverse $\boldsymbol{\Sigma}_{XX}^-$ and put

$$\mathbf{B}_* = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^-, \quad \mathbf{E} = \mathbf{Y} - \boldsymbol{\mu}_Y - \mathbf{B}_*(\mathbf{X} - \boldsymbol{\mu}_X), \quad \boldsymbol{\Sigma}_{YY \cdot X} = \boldsymbol{\Sigma}_{YY} - \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XY}.$$

- (a) $\mathbf{E}(\mathbf{E}) = \mathbf{0}$ and $\text{Cov}(\mathbf{E}, \mathbf{X}) = \mathbf{0}$.
- (b) $\text{Cov}(\mathbf{E}) = \boldsymbol{\Sigma}_{YY \cdot X}$. This matrix is nonnegative definite and does not depend on the choice of generalized inverse. Neither does $\mathbf{B}_*(\mathbf{X} - \boldsymbol{\mu}_X)$, with probability one.
- (c) For every constant $p \times q$ matrix $\mathbf{B}$ and $p \times 1$ vector $\mathbf{c}$,

$$\mathbf{E}[(\mathbf{Y} - \mathbf{c} - \mathbf{B}\mathbf{X})(\mathbf{Y} - \mathbf{c} - \mathbf{B}\mathbf{X})^\top] = \boldsymbol{\Sigma}_{YY \cdot X} + (\mathbf{B} - \mathbf{B}_*)\boldsymbol{\Sigma}_{XX}(\mathbf{B} - \mathbf{B}_*)^\top + \mathbf{d}\mathbf{d}^\top,$$

where $\mathbf{d} = \boldsymbol{\mu}_Y - \mathbf{c} - \mathbf{B}\boldsymbol{\mu}_X$. So the mean squared error matrix is never smaller, in the nonnegative definite ordering, than $\boldsymbol{\Sigma}_{YY \cdot X}$. Equality holds iff $\mathbf{c} = \boldsymbol{\mu}_Y - \mathbf{B}\boldsymbol{\mu}_X$ and $\mathbf{B}\boldsymbol{\Sigma}_{XX} = \boldsymbol{\Sigma}_{YX}$.

Proof. By [Theorem 2.5.1\(c\)](#), write $\boldsymbol{\Sigma}_{YX} = \mathbf{K}\boldsymbol{\Sigma}_{XX}$. Then $\mathbf{B}_*\boldsymbol{\Sigma}_{XX} = \mathbf{K}\boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XX} = \mathbf{K}\boldsymbol{\Sigma}_{XX} = \boldsymbol{\Sigma}_{YX}$.

- (a) The mean is clear. By [Theorem 2.5.1\(b\)](#), $\text{Cov}(\mathbf{E}, \mathbf{X}) = \boldsymbol{\Sigma}_{YX} - \mathbf{B}_*\boldsymbol{\Sigma}_{XX} = \mathbf{0}$.
- (b) Again by [Theorem 2.5.1\(b\)](#), $\text{Cov}(\mathbf{E}) = \boldsymbol{\Sigma}_{YY} - \mathbf{B}_*\boldsymbol{\Sigma}_{XY} - \boldsymbol{\Sigma}_{YX}\mathbf{B}_*^\top + \mathbf{B}_*\boldsymbol{\Sigma}_{XX}\mathbf{B}_*^\top$. Using $\mathbf{B}_*\boldsymbol{\Sigma}_{XX} = \boldsymbol{\Sigma}_{YX}$, the last term is $\boldsymbol{\Sigma}_{YX}\mathbf{B}_*^\top$, which cancels the third, so $\text{Cov}(\mathbf{E}) = \boldsymbol{\Sigma}_{YY} - \mathbf{B}_*\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{YY \cdot X}$. Next, $\boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XY} = \mathbf{K}\boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XX}\mathbf{K}^\top = \mathbf{K}\boldsymbol{\Sigma}_{XX}\mathbf{K}^\top$, which involves no generalized inverse. A covariance matrix is nonnegative definite. Finally, by [Theorem 2.2.3\(d\)](#), $\mathbf{X} - \boldsymbol{\mu}_X = \boldsymbol{\Sigma}_{XX}\mathbf{v}$ for some random $\mathbf{v}$,

with probability one, and then $\mathbf{B}_*(\mathbf{X} - \boldsymbol{\mu}_X) = \boldsymbol{\Sigma}_{YX}\mathbf{v}$ for every choice of generalized inverse.

- (c) Decompose $\mathbf{Y} - \mathbf{c} - \mathbf{B}\mathbf{X} = \mathbf{E} + (\mathbf{B}_* - \mathbf{B})(\mathbf{X} - \boldsymbol{\mu}_X) + \mathbf{d}$. The three pieces are, respectively, mean zero, mean zero and constant, and the first two are uncorrelated by (a). So the cross products have zero expectation, and the expected outer product is $\text{Cov}(\mathbf{E}) + (\mathbf{B}_* - \mathbf{B})\boldsymbol{\Sigma}_{XX}(\mathbf{B}_* - \mathbf{B})^\top + \mathbf{d}\mathbf{d}^\top$. Both added matrices are nonnegative definite. They vanish iff $\mathbf{d} = \mathbf{0}$ and $(\mathbf{B} - \mathbf{B}_*)\boldsymbol{\Sigma}_{XX}^{1/2} = \mathbf{0}$, and the latter is equivalent to $(\mathbf{B} - \mathbf{B}_*)\boldsymbol{\Sigma}_{XX} = \mathbf{0}$, that is, to $\mathbf{B}\boldsymbol{\Sigma}_{XX} = \boldsymbol{\Sigma}_{YX}$.

■

The affine function $\boldsymbol{\mu}_Y + \mathbf{B}_*(\mathbf{X} - \boldsymbol{\mu}_X)$ is the **best linear predictor** of $\mathbf{Y}$ from $\mathbf{X}$. Because the ordering in (c) is the nonnegative definite ordering, it is simultaneously best for every linear combination $\mathbf{a}^\top \mathbf{Y}$ and for the total squared error. The matrix $\boldsymbol{\Sigma}_{YY.X}$ is the **partial covariance matrix** of $\mathbf{Y}$ given $\mathbf{X}$. When $\boldsymbol{\Sigma}_{XX}$ is nonsingular, $\mathbf{B}_* = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^{-1}$ holds the population regression coefficients, and $\boldsymbol{\Sigma}_{YY.X}$ is a Schur complement. Two consequences follow from Chapter 1. If $\text{Cov}(\mathbf{W})$ is positive definite, the YY block of its inverse is $\boldsymbol{\Sigma}_{YY.X}^{-1}$ (Theorem 1.6.4). And $\det \text{Cov}(\mathbf{W}) = \det(\boldsymbol{\Sigma}_{XX})\det(\boldsymbol{\Sigma}_{YY.X})$ (Theorem 1.6.3), so the generalized variance factorizes into the part of $\mathbf{X}$ and the part of $\mathbf{Y}$ that $\mathbf{X}$ leaves unexplained.

Nothing here says that $E(\mathbf{Y} | \mathbf{X})$ is linear. The best linear predictor is best among affine functions, whatever the joint distribution. For jointly normal vectors, the conditional mean happens to be this affine function, and $\boldsymbol{\Sigma}_{YY.X}$ is the conditional covariance matrix (Theorem 3.5.2). The partial correlation of Definition 3.6.1 is the correlation computed from $\boldsymbol{\Sigma}_{YY.X}$. Chapter 6 shows that least squares estimates $\mathbf{B}_*$ in large samples, whether or not the regression function is linear. It also recovers the predictor as an orthogonal projection in the space of square-integrable random variables. For scalar Y , the coefficient vector used there is $\mathbf{B}_*^\top$.

Proposition 2.5.3 — Multiple correlation

Let Y be scalar with $\sigma_{YY} = \text{Var}(Y) > 0$, let $\boldsymbol{\Sigma}_{XX}$ be positive definite, and write $\boldsymbol{\sigma}_{XY} = \text{Cov}(\mathbf{X}, Y)$. Over all $\mathbf{a} \neq \mathbf{0}$,

$$\max_{\mathbf{a}} \text{corr}(Y, \mathbf{a}^\top \mathbf{X})^2 = \rho_{Y.X}^2 = \frac{\boldsymbol{\sigma}_{XY}^\top \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}}{\sigma_{YY}},$$

attained at $\mathbf{a} = \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}$, the coefficients of the best linear predictor (if $\boldsymbol{\sigma}_{XY} = \mathbf{0}$, every $\mathbf{a}$ attains the maximum 0). Moreover $1 - \rho_{Y.X}^2 = \sigma_{YY.X} / \sigma_{YY}$.

Proof. $\text{corr}(Y, \mathbf{a}^\top \mathbf{X})^2 = (\mathbf{a}^\top \boldsymbol{\sigma}_{XY})^2 / (\sigma_{YY} \mathbf{a}^\top \boldsymbol{\Sigma}_{XX} \mathbf{a})$. Write $\mathbf{a}^\top \boldsymbol{\sigma}_{XY} = (\boldsymbol{\Sigma}_{XX}^{1/2} \mathbf{a})^\top (\boldsymbol{\Sigma}_{XX}^{-1/2} \boldsymbol{\sigma}_{XY})$. The Cauchy–Schwarz inequality gives $(\mathbf{a}^\top \boldsymbol{\sigma}_{XY})^2 \leq (\mathbf{a}^\top \boldsymbol{\Sigma}_{XX} \mathbf{a})(\boldsymbol{\sigma}_{XY}^\top \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY})$, with equality iff $\boldsymbol{\Sigma}_{XX}^{1/2} \mathbf{a}$ is proportional to $\boldsymbol{\Sigma}_{XX}^{-1/2} \boldsymbol{\sigma}_{XY}$. Dividing gives the bound and the maximizer. The last identity is the definition of $\sigma_{YY.X} = \sigma_{YY} - \boldsymbol{\sigma}_{XY}^\top \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}$, divided by σ_{YY} .

■

$\rho_{Y.X}^2$ is the population version of R^2 : the fraction of the variance of Y that the best linear predictor accounts for. Chapter 14 studies its sample version.

2.5.3 Conditioning: total expectation and total covariance

When the joint distribution is available, conditioning gives another decomposition of the covariance matrix. Conditional expectations and covariances are taken entry by entry, as in [Definition 2.1.1](#), and $\text{Cov}(\mathbf{Y} \mid \mathbf{X}) = \mathbb{E}[(\mathbf{Y} - \mathbb{E}(\mathbf{Y} \mid \mathbf{X}))(\mathbf{Y} - \mathbb{E}(\mathbf{Y} \mid \mathbf{X}))^\top \mid \mathbf{X}]$.

Proposition 2.5.4 — Laws of total expectation and covariance

If $\mathbf{Y}$ has finite second moments, then $\mathbb{E}(\mathbf{Y}) = \mathbb{E}[\mathbb{E}(\mathbf{Y} \mid \mathbf{X})]$ and

$$\text{Cov}(\mathbf{Y}) = \mathbb{E}[\text{Cov}(\mathbf{Y} \mid \mathbf{X})] + \text{Cov}(\mathbb{E}(\mathbf{Y} \mid \mathbf{X})).$$

Proof. The first identity is the tower property of conditional expectation, applied to each component. For the second, let $\mathbf{m}(\mathbf{X}) = \mathbb{E}(\mathbf{Y} \mid \mathbf{X})$ and write $\mathbf{Y} - \boldsymbol{\mu} = (\mathbf{Y} - \mathbf{m}(\mathbf{X})) + (\mathbf{m}(\mathbf{X}) - \boldsymbol{\mu})$. The outer product has two cross terms. The conditional expectation of $(\mathbf{Y} - \mathbf{m}(\mathbf{X}))(\mathbf{m}(\mathbf{X}) - \boldsymbol{\mu})^\top$ given $\mathbf{X}$ is $[\mathbb{E}(\mathbf{Y} \mid \mathbf{X}) - \mathbf{m}(\mathbf{X})](\mathbf{m}(\mathbf{X}) - \boldsymbol{\mu})^\top = \mathbf{0}$, so its expectation is zero, and likewise for the other. The expectation of $(\mathbf{Y} - \mathbf{m}(\mathbf{X}))(\mathbf{Y} - \mathbf{m}(\mathbf{X}))^\top$ is $\mathbb{E}[\text{Cov}(\mathbf{Y} \mid \mathbf{X})]$ by the tower property, and that of $(\mathbf{m}(\mathbf{X}) - \boldsymbol{\mu})(\mathbf{m}(\mathbf{X}) - \boldsymbol{\mu})^\top$ is $\text{Cov}(\mathbf{m}(\mathbf{X}))$ because $\mathbb{E} \mathbf{m}(\mathbf{X}) = \boldsymbol{\mu}$. ■

Example. (A shared random effect)

Measurements $Y_1, \dots, Y_k$ on the same subject share a subject effect: $\mathbf{Y} = \alpha \mathbf{1} + \boldsymbol{\varepsilon}$, where α has variance τ^2 , $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_k$, and $\boldsymbol{\varepsilon}$ is independent of α with mean zero. Given α , the mean is $\alpha \mathbf{1}$ and the covariance is $\sigma^2 \mathbf{I}$. By [Proposition 2.5.4](#),

$$\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_k + \tau^2 \mathbf{1}\mathbf{1}^\top,$$

so every pair of measurements has correlation $\tau^2/(\tau^2 + \sigma^2)$. Independent errors combined with a shared random component produce exactly this *equicorrelation* structure. It is the starting point of Chapter 32. Ordinary least squares ignores it, with consequences that Chapter 33 examines.

2.5.4 Exercises

2.5.4.1 A. Check your understanding

Exercise A1.

For scalar Y and X with $\text{Var}(X) > 0$, show that the best linear predictor has slope σ_{YX}/σ_{XX} , and that its mean squared error is $\sigma_{YY}(1 - \rho^2)$.

Exercise A2.

Let X be uniform on $(-1, 1)$ and $Y = X^2 + \varepsilon$, with ε independent of X , mean 0 and variance σ^2 . Find the best linear predictor of Y from X and its mean squared error, and compare with the mean squared error of $E(Y | X)$.

2.5.4.2 B. Practice**Exercise B1.**

Let (Y_1, Y_2, Y_3) have correlation matrix $\mathbf{R}$. Compute the partial covariance matrix of (Y_2, Y_3) given Y_1 from [Theorem 2.5.2](#), and show that the corresponding correlation is

$$\frac{\rho_{23} - \rho_{12}\rho_{13}}{\sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)}}.$$

Exercise B2.

Let $\mathbf{X} = (X_1, X_1)^\top$ with $\text{Var}(X_1) = 1$, and let $\text{Cov}(Y, X_1) = c$. Find two different generalized inverses of Σ_{XX} that give different $\mathbf{B}_*$, and show that both give the same predictor $\mathbf{B}_*(\mathbf{X} - \mu_X)$.

Exercise B3.

A random label K takes values $1, \dots, g$ with probabilities π_k . Given $K = k$, $\mathbf{Y}$ has mean $\mathbf{m}_k$ and covariance Σ_k . Show that $\text{Cov}(\mathbf{Y}) = \sum_k \pi_k \Sigma_k + \sum_k \pi_k (\mathbf{m}_k - \bar{\mathbf{m}})(\mathbf{m}_k - \bar{\mathbf{m}})^\top$, where $\bar{\mathbf{m}} = \sum_k \pi_k \mathbf{m}_k$. Interpret the two terms as within-group and between-group covariance.

2.5.4.3 C. Going deeper**Exercise C1.**

Let $\mathbf{N}$ be the symmetric partitioned matrix

$$\mathbf{N} = \begin{pmatrix} \Sigma_{YY} & \Sigma_{YX} \\ \Sigma_{XY} & \Sigma_{XX} \end{pmatrix}.$$

Show that $\mathbf{N}$ is nonnegative definite iff Σ_{XX} is nonnegative definite, $\mathcal{C}(\Sigma_{XY}) \subseteq \mathcal{C}(\Sigma_{XX})$, and $\Sigma_{YY} - \Sigma_{YX}\Sigma_{XX}^- \Sigma_{XY}$ is nonnegative definite for some generalized inverse Σ_{XX}^- (equivalently, given the column-space condition, for every one). Show by a 2×2 example that the column-space condition cannot be dropped.

Solution. If the matrix is nonnegative definite, it is the covariance matrix of some $\mathbf{W} = (\mathbf{Y}^\top, \mathbf{X}^\top)^\top$ ([Theorem 2.2.3\(b\)](#)). Then Σ_{XX} is a covariance matrix, the column condition is [Theorem 2.5.1\(c\)](#), and the Schur complement is $\text{Cov}(\mathbf{E})$ by [Theorem 2.5.2\(b\)](#). Conversely, the column condition gives $\Sigma_{YX} = \mathbf{K}\Sigma_{XX}$, and then

$\Sigma_{YX}\Sigma_{XX}^{-}\Sigma_{XY} = K\Sigma_{XX}K^{\top}$. With the nonsingular $T = \begin{pmatrix} I & -K \\ 0 & I \end{pmatrix}$,

$$T \begin{pmatrix} \Sigma_{YY} & \Sigma_{YX} \\ \Sigma_{XY} & \Sigma_{XX} \end{pmatrix} T^{\top} = \begin{pmatrix} \Sigma_{YY} - K\Sigma_{XX}K^{\top} & 0 \\ 0 & \Sigma_{XX} \end{pmatrix},$$

which is nonnegative definite. So is the original matrix, since $\mathbf{a}^{\top}N\mathbf{a} = \mathbf{b}^{\top}(TNT^{\top})\mathbf{b}$ with $\mathbf{b} = (T^{\top})^{-1}\mathbf{a}$. For the example, take $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. Here $\Sigma_{XX} = 0$ is nonnegative definite, and with the generalized inverse 0 the Schur complement is $1 \geq 0$. But the determinant is -1 , so the matrix is not nonnegative definite. The column condition fails, because $\mathcal{C}(1) \not\subseteq \mathcal{C}(0)$. Without that condition the Schur complement also depends on the choice: every scalar t is a generalized inverse of 0, and the Schur complement is then $1 - 1 \cdot t \cdot 1 = 1 - t$, which is negative for $t > 1$. Under the column condition, $\Sigma_{YX}\Sigma_{XX}^{-}\Sigma_{XY} = K\Sigma_{XX}K^{\top}$ for every choice, as shown above.

2.6 Moment generating functions and independence

Means and covariance matrices summarize a distribution. They do not determine it, and they cannot certify independence. For that we need a transform of the whole distribution. This section collects the facts that [Chapter 3](#) and [Chapter 4](#) use.

Definition 2.6.1 — Moment generating function

The **moment generating function** of a $p \times 1$ random vector $\mathbf{Y}$ is

$$M_{\mathbf{Y}}(\mathbf{t}) = E \exp(\mathbf{t}^{\top} \mathbf{Y}), \quad \mathbf{t} \in \mathbb{R}^p,$$

where the expectation may be $+\infty$. We say that $M_{\mathbf{Y}}$ *exists* if it is finite for all $\mathbf{t}$ in some neighbourhood $\{\|\mathbf{t}\| < h\}$ of the origin, $h > 0$.

Theorem 2.6.1 — Properties of moment generating functions

Suppose $M_{\mathbf{Y}}$ exists.

- (a) *Uniqueness.* If $M_{\mathbf{V}}$ also exists and $M_{\mathbf{Y}} = M_{\mathbf{V}}$ on a neighbourhood of $\mathbf{0}$, then $\mathbf{Y}$ and $\mathbf{V}$ have the same distribution.
- (b) *Moments.* All moments of $\mathbf{Y}$ are finite. The gradient of $M_{\mathbf{Y}}$ at $\mathbf{0}$ is $E(\mathbf{Y})$, and its Hessian at $\mathbf{0}$ is $E(\mathbf{Y}\mathbf{Y}^{\top})$. The Hessian of $\log M_{\mathbf{Y}}$ at $\mathbf{0}$ is $\text{Cov}(\mathbf{Y})$.
- (c) *Affine maps.* For a constant $k \times p$ matrix $\mathbf{A}$ and $k \times 1$ vector $\mathbf{b}$, $M_{\mathbf{A}\mathbf{Y}+\mathbf{b}}(\mathbf{s}) = \exp(\mathbf{s}^{\top} \mathbf{b}) M_{\mathbf{Y}}(\mathbf{A}^{\top} \mathbf{s})$.
- (d) *Independence.* Partition $\mathbf{Y} = (\mathbf{Y}_1^{\top}, \mathbf{Y}_2^{\top})^{\top}$ and $\mathbf{t} = (\mathbf{t}_1^{\top}, \mathbf{t}_2^{\top})^{\top}$ conformably. Then $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent iff $M_{\mathbf{Y}}(\mathbf{t}) = M_{\mathbf{Y}_1}(\mathbf{t}_1) M_{\mathbf{Y}_2}(\mathbf{t}_2)$ for all $\mathbf{t}$ in a neighbourhood of $\mathbf{0}$.

Proof. For (a), the scalar case is a theorem of analysis: a moment generating function that is finite near the origin extends analytically, which pins down the character-

istic function and hence the distribution (Billingsley 1995). For vectors, fix $\mathbf{a}$. For small $|s|$, $M_{\mathbf{a}^\top \mathbf{Y}}(s) = M_{\mathbf{Y}}(s\mathbf{a}) = M_{\mathbf{V}}(s\mathbf{a}) = M_{\mathbf{a}^\top \mathbf{V}}(s)$, so $\mathbf{a}^\top \mathbf{Y}$ and $\mathbf{a}^\top \mathbf{V}$ have the same distribution by the scalar case, and [Proposition 2.6.2](#) below finishes the argument. For (b), finiteness near $\mathbf{0}$ implies $E \exp(h'|Y_i|) < \infty$ for some $h' > 0$ and each i , so every polynomial in the components is integrable and differentiation under the expectation is justified. Then $\partial M / \partial t_i = E(Y_i e^{t^\top \mathbf{Y}})$ and $\partial^2 M / \partial t_i \partial t_j = E(Y_i Y_j e^{t^\top \mathbf{Y}})$. At $\mathbf{t} = \mathbf{0}$ these are $E(Y_i)$ and $E(Y_i Y_j)$. For $K = \log M$, the chain rule gives $\nabla^2 K = M^{-1} \nabla^2 M - M^{-2} (\nabla M)(\nabla M)^\top$, which at $\mathbf{0}$, where $M = 1$, is $E(\mathbf{Y} \mathbf{Y}^\top) - \boldsymbol{\mu} \boldsymbol{\mu}^\top = \text{Cov}(\mathbf{Y})$. For (c), $E \exp(\mathbf{s}^\top (\mathbf{A} \mathbf{Y} + \mathbf{b})) = e^{\mathbf{s}^\top \mathbf{b}} E \exp((\mathbf{A}^\top \mathbf{s})^\top \mathbf{Y})$. For (d), independence gives $E(e^{t_1^\top \mathbf{Y}_1} e^{t_2^\top \mathbf{Y}_2}) = E(e^{t_1^\top \mathbf{Y}_1}) E(e^{t_2^\top \mathbf{Y}_2})$. Conversely, let $\mathbf{V}_1$ and $\mathbf{V}_2$ be independent with the distributions of $\mathbf{Y}_1$ and $\mathbf{Y}_2$. The vector $(\mathbf{V}_1^\top, \mathbf{V}_2^\top)^\top$ has moment generating function $M_{\mathbf{Y}_1}(\mathbf{t}_1) M_{\mathbf{Y}_2}(\mathbf{t}_2)$, which by hypothesis equals $M_{\mathbf{Y}}$ near $\mathbf{0}$. By (a) it has the same joint distribution as $\mathbf{Y}$, so $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent. ■

The next fact explains why [Chapter 3](#) can define the multivariate normal through linear combinations.

Proposition 2.6.2 — Distributions are determined by linear combinations

If $\mathbf{a}^\top \mathbf{Y}$ and $\mathbf{a}^\top \mathbf{V}$ have the same distribution for every $\mathbf{a} \in \mathbb{R}^p$, then $\mathbf{Y}$ and $\mathbf{V}$ have the same distribution.

Proof. The characteristic function $\phi_{\mathbf{Y}}(\mathbf{t}) = E \exp(i \mathbf{t}^\top \mathbf{Y})$ exists for every random vector and determines its distribution (Billingsley 1995). Since $\phi_{\mathbf{Y}}(\mathbf{t}) = \phi_{\mathbf{t}^\top \mathbf{Y}}(1) = \phi_{\mathbf{t}^\top \mathbf{V}}(1) = \phi_{\mathbf{V}}(\mathbf{t})$ for every $\mathbf{t}$, the two distributions coincide. ■

If $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent, so are $g(\mathbf{Y}_1)$ and $h(\mathbf{Y}_2)$ for measurable g and h . Independence of statistics is usually proved this way. For normal vectors, [Theorem 3.4.1](#) reduces the independence of two blocks to a zero cross-covariance, and [Theorem 4.4.1](#) and [Theorem 4.4.2](#) build on that.

Independence of *several* vectors means that the joint distribution factors into all the marginals, not merely that each pair is independent.

Example. (Pairwise but not mutually independent)

Let S_1 and S_2 be independent, each equal to ± 1 with probability $\frac{1}{2}$, and put $S_3 = S_1 S_2$. Each S_j is ± 1 with probability $\frac{1}{2}$. Each pair is independent: for instance, $\Pr(S_1 = s, S_3 = u) = \Pr(S_1 = s, S_2 = su) = \frac{1}{4}$ for all signs s, u . So $\text{Cov}(\mathbf{S}) = \mathbf{I}_3$. But the three are not mutually independent, since S_3 is a function of the other two. A third-order moment shows the failure directly: $E(S_1 S_2 S_3) = 1$, whereas mutual independence would force it to be 0. No condition on the covariance matrix can detect this, because covariances involve only pairs.

2.6.1 Exercises

2.6.1.1 A. Check your understanding

Exercise A1.

Find the moment generating function of the multinomial count vector of [Example \(Multinomial counts\)](#), and recover its mean and covariance matrix from [Theorem 2.6.1\(b\)](#).

2.6.1.2 B. Practice

Exercise B1.

Let $S_1, \dots, S_k$ be independent random signs. For each nonempty $A \subseteq \{1, \dots, k\}$ put $T_A = \prod_{i \in A} S_i$. Show that the $2^k - 1$ variables T_A are pairwise independent with covariance matrix $\mathbf{I}$, and find a triple of them that is not mutually independent.

2.7 Sample moments and their computation

The population quantities of this chapter have sample analogues, and the operator rules give their expectations at once. Let $\mathbf{y}_1, \dots, \mathbf{y}_n$ be $p \times 1$ observations, and stack them as the rows of the $n \times p$ data matrix $\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_n]^\top$. The script letter keeps the data matrix apart from a single random vector $\mathbf{Y}$. The rows keep the lowercase $\mathbf{y}_i$ of observed data, but in [Proposition 2.7.1](#) they are treated as random. With $\mathbf{C} = \mathbf{I}_n - n^{-1}\mathbf{1}\mathbf{1}^\top$, the sample mean vector and the **sample covariance matrix** are

$$\bar{\mathbf{y}} = \frac{1}{n} \sum_{i=1}^n \mathbf{y}_i = \frac{1}{n} \mathbf{Y}^\top \mathbf{1}, \quad \mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})^\top = \frac{1}{n-1} \mathbf{Y}^\top \mathbf{C} \mathbf{Y}.$$

The second form of $\mathbf{S}$ holds because $\mathbf{C}\mathbf{Y}$ is the data matrix with its column means subtracted, and $\mathbf{C}$ is symmetric and idempotent. The sample correlation matrix is $\hat{\mathbf{R}} = \hat{\mathbf{D}}^{-1} \mathbf{S} \hat{\mathbf{D}}^{-1}$, where $\hat{\mathbf{D}}$ holds the sample standard deviations.

Proposition 2.7.1 — Moments of sample moments

Suppose the rows $\mathbf{y}_i$ are pairwise uncorrelated random vectors with common mean $\boldsymbol{\mu}$ and common covariance matrix $\boldsymbol{\Sigma}$. Then

- (a) $E(\bar{\mathbf{y}}) = \boldsymbol{\mu}$ and $\text{Cov}(\bar{\mathbf{y}}) = \boldsymbol{\Sigma}/n$;
- (b) $E(\mathbf{S}) = \boldsymbol{\Sigma}$;
- (c) $\mathbf{S}$ is nonnegative definite, and $\text{rank}(\mathbf{S}) \leq \min(p, n-1)$ for every realization.

Proof.

- (a) is [Proposition 2.2.2\(c\)](#) with $c_i = 1/n$. For (b), let $\mathbf{u}_j$ denote the j th column of $\mathbf{Y}$, the n observations of variable j . The (j, k) entry of $\mathbf{Y}^\top \mathbf{C} \mathbf{Y}$ is the bilinear form $\mathbf{u}_j^\top \mathbf{C} \mathbf{u}_k$. Because different rows are uncorrelated, $\text{Cov}(\mathbf{u}_k, \mathbf{u}_j) = \sigma_{kj} \mathbf{I}_n$, and

$E(\mathbf{u}_j) = \mu_j \mathbf{1}$. By [Corollary 2.3.2\(d\)](#),

$$E(\mathbf{u}_j^\top \mathbf{C} \mathbf{u}_k) = \sigma_{kj} \operatorname{tr}(\mathbf{C}) + \mu_j \mu_k \mathbf{1}^\top \mathbf{C} \mathbf{1} = (n-1) \sigma_{jk},$$

since $\operatorname{tr}(\mathbf{C}) = n-1$ and $\mathbf{C} \mathbf{1} = \mathbf{0}$. (c) For any $\mathbf{a}$, $\mathbf{a}^\top \mathbf{S} \mathbf{a} = \|\mathbf{C} \mathbf{y} \mathbf{a}\|^2 / (n-1) \geq 0$. The rank of $\mathbf{y}^\top \mathbf{C} \mathbf{y} = (\mathbf{C} \mathbf{y})^\top (\mathbf{C} \mathbf{y})$ equals $\operatorname{rank}(\mathbf{C} \mathbf{y})$, which is at most $\operatorname{rank}(\mathbf{C}) = n-1$ and at most p . ■

The divisor $n-1$ is the trace of $\mathbf{C}$, the dimension of the space the centred data live in, exactly as $n-r$ is for regression residuals in [Example \(Residual sum of squares under a wrong mean\)](#). Part (c) has a practical side. With $n \leq p$, as in genomics or finance, the sample covariance matrix is always singular, even when $\mathbf{\Sigma}$ is positive definite. It has a linear combination with zero sample variance whether or not one exists in the population. Chapter 28 returns to this.

Computation. In practice $\mathbf{C}$ is never formed: one subtracts column means. The eigenvalues of $\mathbf{S}$ are the squared singular values of $\mathbf{C} \mathbf{y} / \sqrt{n-1}$ ([Theorem 1.8.1](#)), which can be computed without forming a cross-product matrix ([Chapter 10](#)).

Example. (Macroeconomic series)

Longley's data give $n = 16$ annual observations (1947–1962) on $p = 7$ US series: total employment, the GNP deflator, GNP, unemployment, armed forces, population and the year. The eigenvalues of $\mathbf{S}$ run from 9.94×10^9 down to 1.07×10^{-2} , a ratio of 9.3×10^{11} . That spread says little about the structure of the data. It mostly reflects units: the standard deviation of GNP is 99395, that of the deflator 10.79.

The correlation matrix removes the units ([Proposition 2.4.1\(d\)](#)). Its eigenvalues sum to $p = 7$. The largest is 5.533, so one direction carries about 79% of the standardized variation, and the smallest is 2.59×10^{-4} . By [Theorem 1.7.7](#), the unit vector $\mathbf{v}$ belonging to the smallest eigenvalue minimizes the sample variance of $\mathbf{v}^\top \mathbf{z}$ over standardized combinations. Its two largest weights are 0.681 on year and -0.674 on GNP, and the combination has standard deviation 0.016, against 1 for each standardized variable. In these sixteen years GNP is almost an exact linear function of time (their correlation is 0.995). This is the sample version of [Theorem 2.2.3\(c\)](#): a nearly zero eigenvalue marks a nearly constant linear combination. For regression it signals collinearity, which Chapter 26 studies. Finally, the first 5 rows alone give a sample covariance matrix of rank 4, as [Proposition 2.7.1\(c\)](#) requires.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.longley.load_pandas().data
names = list(data.columns)

Y = data.to_numpy(dtype=float)           # n x p data matrix, one row per
    year
n, p = Y.shape
C = np.eye(n) - np.ones((n, n)) / n     # centring matrix I - J/n
S = Y.T @ C @ Y / (n - 1)               # sample covariance matrix
d = np.sqrt(np.diag(S))
R = S / np.outer(d, d)                  # D^{-1} S D^{-1}, D = diag(d)
```

```

lam_S = np.linalg.eigvalsh(S)
lam_R, V = np.linalg.eigh(R)                                # ascending eigenvalues
print("eigenvalues of S:", lam_S)
print("eigenvalues of R:", lam_R)
v = V[:, 0]                                                  # direction of least variance
Z = (Y - Y.mean(0)) / d                                       # standardized variables
print("sample variance of Z v:", np.var(Z @ v, ddof=1))

```

2.7.1 Exercises

2.7.1.1 B. Practice

Exercise B1.

Suppose the rows of $\mathbf{Y}$ are observed in time order, with common mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, and that neighbouring rows are correlated: $\text{Cov}(\mathbf{y}_i, \mathbf{y}_{i+1}) = \rho\boldsymbol{\Sigma}$, while $\text{Cov}(\mathbf{y}_i, \mathbf{y}_j) = \mathbf{0}$ for $|i - j| \geq 2$. Show that $E(\mathbf{S}) = (1 - 2\rho/n)\boldsymbol{\Sigma}$ and $\text{Cov}(\bar{\mathbf{y}}) = n^{-2}[n + 2(n - 1)\rho]\boldsymbol{\Sigma}$. Which of the two effects disappears as $n \rightarrow \infty$?

Solution. As in the proof of Proposition 2.7.1, $\text{Cov}(\mathbf{u}_k, \mathbf{u}_j) = \sigma_{kj}\mathbf{T}$, where $\mathbf{T}$ is the $n \times n$ tridiagonal matrix with ones on the diagonal and ρ next to it. Since $\mathbf{C}\mathbf{1} = \mathbf{0}$, Corollary 2.3.2(d) gives $E(\mathbf{u}_j^\top \mathbf{C} \mathbf{u}_k) = \sigma_{kj} \text{tr}(\mathbf{C}\mathbf{T}) = \sigma_{kj} [\text{tr}(\mathbf{T}) - n^{-1}\mathbf{1}^\top \mathbf{T} \mathbf{1}]$. Here $\text{tr}(\mathbf{T}) = n$ and $\mathbf{1}^\top \mathbf{T} \mathbf{1} = n + 2(n - 1)\rho$, so the bracket is $(n - 1) - 2(n - 1)\rho/n$, and dividing by $n - 1$ gives $E(\mathbf{S}) = (1 - 2\rho/n)\boldsymbol{\Sigma}$. For the mean, $\bar{\mathbf{y}} = n^{-1} \sum_i \mathbf{y}_i$ and Proposition 2.2.2 gives n^{-2} times the sum of all $\text{Cov}(\mathbf{y}_i, \mathbf{y}_j)$: n diagonal terms $\boldsymbol{\Sigma}$ and $2(n - 1)$ neighbouring terms $\rho\boldsymbol{\Sigma}$. The bias of $\mathbf{S}$ is $O(1/n)$ and disappears, but $n \text{Cov}(\bar{\mathbf{y}}) \rightarrow (1 + 2\rho)\boldsymbol{\Sigma}$, not $\boldsymbol{\Sigma}$. For $\rho > 0$ the naive standard error $\sqrt{s_{jj}/n}$ stays too small by a fixed factor however large the sample.

2.8 Summary and Notes

2.8.1 Summary

Idea.

1. Expectation of a random matrix is taken entry by entry, and it commutes with constant linear maps, transposes and traces (Proposition 2.1.1).
2. $\text{Cov}(\mathbf{Y}) = E[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^\top] = E(\mathbf{Y}\mathbf{Y}^\top) - \boldsymbol{\mu}\boldsymbol{\mu}^\top$, and $\text{Cov}(\mathbf{U}, \mathbf{V})$ is the rectangular cross-covariance, with $\text{Cov}(\mathbf{V}, \mathbf{U}) = \text{Cov}(\mathbf{U}, \mathbf{V})^\top$ (Definition 2.1.2 and Proposition 2.1.2). Uncorrelated does not imply independent (Example (A point on a circle)).
3. Affine maps: $E(\mathbf{A}\mathbf{Y} + \mathbf{b}) = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$, $\text{Cov}(\mathbf{A}\mathbf{Y} + \mathbf{b}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top$, $\text{Cov}(\mathbf{A}\mathbf{U}, \mathbf{B}\mathbf{V}) = \mathbf{A} \text{Cov}(\mathbf{U}, \mathbf{V}) \mathbf{B}^\top$ (Theorem 2.2.1). In particular $\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$.
4. Covariance matrices are exactly the nonnegative definite matrices. $\boldsymbol{\Sigma} \mathbf{a} = \mathbf{0}$ iff $\mathbf{a}^\top \mathbf{Y}$ is constant with probability one, and $\mathbf{Y}$ lives in the flat $\boldsymbol{\mu} + \mathcal{C}(\boldsymbol{\Sigma})$ (Theorem 2.2.3).
5. Whitening by $\mathbf{W}$ with $\mathbf{W}\boldsymbol{\Sigma}\mathbf{W}^\top = \mathbf{I}$ is unique up to an orthogonal factor. Ev-

- ery choice gives the same Mahalanobis distance $(\mathbf{Y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu})$ (Proposition 2.2.4).
6. $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ under second moments only (Theorem 2.3.1). Hence $E(\text{RSS}) = \sigma^2(n - r) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2$ (Example (Residual sum of squares under a wrong mean)). The variance of a quadratic form needs third and fourth moments (Theorem 2.3.3).
 7. Correlation matrices are the nonnegative definite matrices with unit diagonal. They are invariant to rescaling, and their entries must be jointly consistent (Proposition 2.4.1 and Proposition 2.4.2).
 8. For partitioned vectors, $\mathcal{C}(\boldsymbol{\Sigma}_{XY}) \subseteq \mathcal{C}(\boldsymbol{\Sigma}_{XX})$. The best linear predictor $\boldsymbol{\mu}_Y + \boldsymbol{\Sigma}_{YX} \boldsymbol{\Sigma}_{XX}^{-1}(\mathbf{X} - \boldsymbol{\mu}_X)$ leaves an error uncorrelated with $\mathbf{X}$ with covariance matrix given by the Schur complement $\boldsymbol{\Sigma}_{YY \cdot X}$ (Theorem 2.5.1 and Theorem 2.5.2).
 9. Moment generating functions determine distributions and factor exactly under independence (Theorem 2.6.1). Pairwise independence is weaker than mutual independence (Example (Pairwise but not mutually independent)).
 10. The sample covariance matrix is unbiased under uncorrelated rows and has rank at most $n - 1$. A tiny eigenvalue of the sample correlation matrix flags a nearly constant combination (Proposition 2.7.1 and Example (Macroeconomic series)).

2.8.2 Notes and sources

The operator calculus. Treating expectation and covariance as operators on random vectors and matrices is standard in linear-model texts. Rencher and Schaalje (2008, chapter 3) develops mean vectors, covariance and correlation matrices, partitioned vectors and linear functions in detail and at a gentle pace. Seber and Lee (2003, §§1.4–1.6) give a compact operator treatment, first and second moments of quadratic forms, and moment generating functions with their use in proving independence. This chapter covers the same ground. Rencher and Schaalje already introduce the standardized distance and the transformation by $\boldsymbol{\Sigma}^{-1/2}$. The chapter adds the characterization of singular covariance matrices through their column space (Theorem 2.2.3), the non-uniqueness of whitening and the comparison of symmetric and Cholesky whitening, the best linear predictor with generalized inverses, and the sample covariance matrix. Christensen (2020, chapter 1) states the covariance rules briefly in the notation of linear models. For the multivariate setting, Anderson (2003) remains the standard reference.

Mahalanobis distance and whitening. The distance Eq. 2.2.2 is due to Mahalanobis (1936), who introduced it to compare populations described by several correlated measurements. Kessy et al. (2018) compare whitening transformations. They show that the symmetric choice $\boldsymbol{\Sigma}^{-1/2}$ is the one closest to the original data in mean squared error (Exercise (B3)), and set it against Cholesky and principal-component whitening.

Quadratic forms. The trace argument for Theorem 2.3.1 is folklore. Theorem 2.3.3 is stated without proof by Atiqullah (1962), who used it to study how non-normal errors affect the residual variance estimate and the F -test. The kurtosis term depends on $\mathbf{A}$ only through its diagonal. For the residual sum of squares of a correct model it equals $(\mu_4 - 3\sigma^4) \sum_i (1 - h_{ii})^2$, which takes its simplest form, $(\mu_4 - 3\sigma^4)(n - r)^2/n$, when all leverages h_{ii} are equal. These are the quadratically balanced designs of Atiqullah's title. Seber and Lee (2003, theorem 1.6) give a proof by direct expansion. The proof here organizes the same expansion by splitting the form into quadratic, linear and constant parts. Mathai and Provost (1992) is a book-length treatment of moments and distributions of quadratic forms, including general covariance structures.

Correlation, partial covariance and prediction. The feasible range in [Proposition 2.4.2](#) is a special case of the fact that a symmetric matrix is nonnegative definite iff a Schur complement is, when the leading block is positive definite. Horn and Johnson (2013) give the general Schur-complement and Loewner-order results used in [Theorem 2.5.2](#). Stating the best linear predictor with a generalized inverse, so that nothing needs Σ_{XX} to be nonsingular, is in the spirit of Rao (1973), who uses generalized inverses throughout.

Probability background. Uniqueness of moment generating functions and characteristic functions, and the device of [Proposition 2.6.2](#), are proved in Billingsley (1995). The Longley data come from Longley (1967), who used them to test the numerical accuracy of regression programs. Their near-collinearity, visible in [Example \(Macroeconomic series\)](#), is the reason.

2.8.3 References

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Chapter 3

The Multivariate Normal Distribution

A random vector is normal when every linear combination of its entries is normal. From that definition come the moment generating function, the density, and closure under all linear maps, singular ones included. Two further properties make the normal distribution the working assumption of linear-model theory: zero covariance between jointly normal blocks means independence, and conditioning one block on another gives a linear regression with constant error variance. Partial correlation, the correlation that remains after linear adjustment, connects these facts to least squares.

What you need. [Chapter 1](#) (spectral theorem, square roots, generalized and partitioned inverses), [Chapter 2](#) (mean vectors and covariance matrices). Some examples refer forward to least squares ([Chapter 5](#) and [Chapter 6](#)).

3.0.1 Roadmap

- [Definition through linear combinations](#)
- [Moment generating function and density](#)
- [Linear transformations and marginals](#)
- [Independence and zero covariance](#)
- [Conditional distributions](#)
- [Partial correlation](#)
- [Summary and notes](#)

3.1 Definition through linear combinations

Nearly every exact distributional statement in the first half of this book has the same form. The response is $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with normal errors, and the object of interest is a linear function of $\mathbf{Y}$ (a coefficient estimate, a fitted value, a residual) or a quadratic function of $\mathbf{Y}$ (a sum of squares). A usable notion of normality for random vectors must therefore do two things. It must survive every linear map, including maps that collapse dimensions. And it must say when two linear functions of the same vector are independent.

The residual vector shows why the first requirement has to include singular cases. In a model with p linearly independent columns its covariance matrix has rank $n - p$ (Chapter 6), so the residuals have no density on $\mathbb{R}^n$. Defining multivariate normality through a density would exclude it. This chapter uses a definition that does not mention densities at all: a random vector is normal when every linear combination of its entries is normal. The density, the moment generating function and the representation $\boldsymbol{\mu} + \mathbf{A}\mathbf{Z}$ all follow as theorems.

3.1.1 The univariate normal, including the degenerate case

For $\mu \in \mathbb{R}$ and $\sigma^2 > 0$, a random variable X has the $N(\mu, \sigma^2)$ distribution when it has density $(2\pi\sigma^2)^{-1/2} \exp\{-(x - \mu)^2/(2\sigma^2)\}$. We also allow $\sigma^2 = 0$, and take $N(\mu, 0)$ to be the point mass at μ . Treating constants as normal variables with zero variance costs nothing and removes a great many special cases later.

Lemma 3.1.1

Let $X \sim N(\mu, \sigma^2)$ with $\sigma^2 \geq 0$.

- (a) $E e^{sX} = \exp(\mu s + \frac{1}{2}\sigma^2 s^2)$ for every real s .
- (b) A random variable whose moment generating function is $\exp(\mu s + \frac{1}{2}\sigma^2 s^2)$ on an interval around 0 has the $N(\mu, \sigma^2)$ distribution.
- (c) $aX + b \sim N(a\mu + b, a^2\sigma^2)$, and if $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent, then $X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Proof.

- (a) For $\sigma^2 = 0$ this is $e^{\mu s}$. Otherwise, completing the square, $sx - (x - \mu)^2/(2\sigma^2) = \mu s + \sigma^2 s^2/2 - (x - \mu - \sigma^2 s)^2/(2\sigma^2)$, and the remaining Gaussian integral equals $(2\pi\sigma^2)^{1/2}$.
- (b) is the uniqueness theorem for moment generating functions (Theorem 2.6.1(a)), in its scalar form.
- (c) The moment generating function of $aX + b$ is $e^{bs} E e^{(as)X} = \exp\{(a\mu + b)s + \frac{1}{2}a^2\sigma^2 s^2\}$. For the sum, independence gives $E e^{s(X_1 + X_2)} = E e^{sX_1} E e^{sX_2}$, which is the normal moment generating function with the added parameters. Apply (b).

■

3.1.2 The definition

Definition 3.1.1 — Multivariate normal distribution

A random vector $\mathbf{Y} = (Y_1, \dots, Y_n)^\top$ has a **multivariate normal distribution** if $\mathbf{a}^\top \mathbf{Y}$ has a univariate normal distribution (possibly degenerate) for every $\mathbf{a} \in \mathbb{R}^n$. Such a $\mathbf{Y}$ is also said to be **jointly normal**, and its entries are said to be jointly normal random

variables.

Taking $\mathbf{a}$ to be a coordinate vector shows that each Y_i is normal, so $E Y_i^2 < \infty$. By the Cauchy–Schwarz inequality every product $Y_i Y_j$ is integrable, so the mean vector $\boldsymbol{\mu} = E \mathbf{Y}$ and the covariance matrix $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y})$ of Definition 2.1.2 exist. By Theorem 2.2.1, $E(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\mu}$ and $\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$, and a normal variable is determined by its mean and variance. Hence

$$\mathbf{a}^\top \mathbf{Y} \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}) \quad \text{for every } \mathbf{a} \in \mathbb{R}^n. \quad (3.1.1)$$

We write $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. The notation claims that the joint distribution of $\mathbf{Y}$ depends only on $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. That is true, but it is not obvious: the definition fixes the law of each one-dimensional projection, and it is not yet clear that these laws pin down the joint law. The claim is proved in Theorem 3.2.1. Until then, $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ should be read as “normal, with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$ ”.

3.1.3 Existence

Theorem 3.1.2 — Existence and the representation $\boldsymbol{\mu} + \mathbf{A}\mathbf{Z}$

Let $Z_1, \dots, Z_m$ be independent $N(0, 1)$ variables and $\mathbf{Z} = (Z_1, \dots, Z_m)^\top$.

- (a) $\mathbf{Z} \sim N_m(\mathbf{0}, \mathbf{I}_m)$.
- (b) For any $n \times m$ matrix $\mathbf{A}$ and $\boldsymbol{\mu} \in \mathbb{R}^n$, $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{A}\mathbf{Z}$ is multivariate normal with mean $\boldsymbol{\mu}$ and covariance $\mathbf{A}\mathbf{A}^\top$.
- (c) For every $\boldsymbol{\mu} \in \mathbb{R}^n$ and every nonnegative definite $n \times n$ matrix $\boldsymbol{\Sigma}$ there is a random vector with the $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ distribution, for example $\boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2} \mathbf{Z}$ with $m = n$.

No other matrices occur: the covariance matrix of a normal vector is nonnegative definite.

Proof.

- (a) $\mathbf{a}^\top \mathbf{Z} = \sum_i a_i Z_i$ is a sum of independent normal variables $a_i Z_i \sim N(0, a_i^2)$, which is normal by Lemma 3.1.1(c).
- (b) $\mathbf{a}^\top \mathbf{Y} = \mathbf{a}^\top \boldsymbol{\mu} + (\mathbf{A}^\top \mathbf{a})^\top \mathbf{Z}$ is a constant plus a linear combination of $\mathbf{Z}$, hence normal by (a). Its moments come from Theorem 2.2.1: $E \mathbf{Y} = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{Y}) = \mathbf{A} \mathbf{I} \mathbf{A}^\top = \mathbf{A} \mathbf{A}^\top$.
- (c) By Theorem 1.7.6 a nonnegative definite $\boldsymbol{\Sigma}$ has a symmetric nonnegative definite square root with $\boldsymbol{\Sigma}^{1/2} \boldsymbol{\Sigma}^{1/2} = \boldsymbol{\Sigma}$. Take $\mathbf{A} = \boldsymbol{\Sigma}^{1/2}$ in (b). The last sentence is Theorem 2.2.3.

■

Any factorization $\boldsymbol{\Sigma} = \mathbf{A}\mathbf{A}^\top$ serves in (b). The symmetric square root is convenient in proofs. The Cholesky factor, a lower triangular $\mathbf{L}$ with $\mathbf{L}\mathbf{L}^\top = \boldsymbol{\Sigma}$, is cheaper to compute and is what software uses. A factor with fewer columns than rows is natural when $\boldsymbol{\Sigma}$ is singular. Once Theorem 3.2.1 is proved, all of these give the same distribution. The first listing checks this by simulation. It draws 400,000 vectors from a three-dimensional normal distribution with each of

two factors. Both sample covariance matrices agree with Σ to within 0.016 in every entry, and the quantiles of a linear combination match the exact normal quantiles.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(301)
mu = np.array([1.0, -2.0, 0.5])
Sigma = np.array([[4.0, 1.2, -0.8],
                  [1.2, 2.0, 0.6],
                  [-0.8, 0.6, 1.5]])
N = 400_000

def draw_normal(mu, A, size, rng):
    """Rows are independent copies of mu + A Z with Z ~ N(0, I)."""
    Z = rng.standard_normal((size, A.shape[1]))
    return mu + Z @ A.T

L = np.linalg.cholesky(Sigma)          # lower triangular, L L^T = Sigma
lam, U = np.linalg.eigh(Sigma)
root = U @ np.diag(np.sqrt(lam)) @ U.T # symmetric square root

Y_chol = draw_normal(mu, L, N, rng)
Y_root = draw_normal(mu, root, N, rng)
for Y in (Y_chol, Y_root):
    print(np.round(np.cov(Y, rowvar=False) - Sigma, 3))
```

3.1.4 Singular normal distributions

When Σ is singular, [Definition 3.1.1](#) still applies, and the distribution is called **singular** or degenerate. If $\Sigma \mathbf{v} = \mathbf{0}$, then $\text{Var}(\mathbf{v}^\top \mathbf{Y}) = \mathbf{v}^\top \Sigma \mathbf{v} = 0$, so $\mathbf{v}^\top \mathbf{Y} = \mathbf{v}^\top \boldsymbol{\mu}$ with probability one. Every vector in $\mathcal{N}(\Sigma)$ gives such a linear constraint, and together they say that $\mathbf{Y}$ lies in the affine subspace $\boldsymbol{\mu} + \mathcal{C}(\Sigma)$ with probability one ([Theorem 2.2.3](#)). A subspace of dimension $\text{rank}(\Sigma) < n$ has Lebesgue measure zero, so no density on $\mathbb{R}^n$ is possible.

Example. (Three overlapping sums)

Let $\mathbf{Z} \sim N_3(\mathbf{0}, \mathbf{I})$ and $\mathbf{U} = (Z_1 + Z_2, Z_2 - Z_3, Z_1 + Z_3)^\top$. Then $\mathbf{U} = \mathbf{A}\mathbf{Z}$ with rows $(1, 1, 0)$, $(0, 1, -1)$ and $(1, 0, 1)$, so $\mathbf{U}$ is normal with mean zero and covariance

$$\mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}.$$

The vector $(1, -1, -1)^\top$ lies in $\mathcal{N}(\mathbf{A}\mathbf{A}^\top)$, and the rank is 2. Correspondingly $U_1 - U_2 - U_3 = 0$ for every outcome, not merely with probability one. Each U_i is $N(0, 2)$, each pair (U_i, U_j) has a nonsingular covariance matrix (determinant 3) and a bivariate density, and yet the triple lives on a plane.

In regression the singular case is the rule. With $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$, the fitted values and the residuals are normal vectors of ranks $\text{rank}(\mathbf{X})$ and $n - \text{rank}(\mathbf{X})$ ([Example \(Least squares under normal errors\)](#)). The last part of the listing draws a normal vector in $\mathbb{R}^3$ from a 3×2 factor. Every draw lies on the plane $\boldsymbol{\mu} + \mathcal{C}(\mathbf{A})$ to within 2.7×10^{-15} , which is rounding error.

3.1.5 Why linear combinations suffice

Definition 3.1.1 describes a joint distribution entirely through its one-dimensional projections. That this loses no information is a general fact about random vectors, the Cramér–Wold device proved with characteristic functions in [Proposition 2.6.2](#): if $\mathbf{a}^\top \mathbf{Y}$ and $\mathbf{a}^\top \mathbf{W}$ have the same distribution for every $\mathbf{a} \in \mathbb{R}^n$, then $\mathbf{Y}$ and $\mathbf{W}$ have the same distribution. We do not need it in this generality. For normal vectors the moment generating function is finite everywhere, and the next section runs the same argument with it. The device explains why the definition is natural. It also warns that all directions $\mathbf{a}$ matter. Normality of the n coordinate variables, which are only n directions, is not enough ([Section 3.3](#) and [Exercise \(B3\)](#)).

3.1.6 Exercises

3.1.6.1 A. Check your understanding

Exercise A1.

Let $Y_1, \dots, Y_n$ be independent with $Y_i \sim N(\mu_i, \sigma_i^2)$. Show directly from [Definition 3.1.1](#) that $\mathbf{Y} = (Y_1, \dots, Y_n)^\top$ is multivariate normal, and find its mean and covariance. Show also that a constant vector $\mathbf{c}$ is $N_n(\mathbf{c}, \mathbf{0})$.

Exercise A2.

For the bivariate normal of [Example \(The bivariate normal\)](#), find the Cholesky factor $\mathbf{L}$ of Σ and write Y_1 and Y_2 in terms of two independent standard normals. Do the same with the symmetric square root when $\sigma_1 = \sigma_2 = 1$. Why do the two representations define the same distribution?

3.1.6.2 B. Practice

Exercise B1.

Let $Z \sim N(0, 1)$ and $\mathbf{Y} = (Z, 2Z - 1, 3 - Z)^\top$. Show that $\mathbf{Y}$ is normal, find $\boldsymbol{\mu}$ and Σ , and find the rank of Σ . Describe the set in which $\mathbf{Y}$ lies, and write it as the solution set of two linear equations.

Solution. $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{A}Z$ with $\boldsymbol{\mu} = (0, -1, 3)^\top$ and $\mathbf{A} = (1, 2, -1)^\top$, so $\mathbf{Y}$ is normal by [Theorem 3.1.2](#) with

$$\Sigma = \mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ -1 & -2 & 1 \end{pmatrix}, \quad \text{rank}(\Sigma) = 1.$$

$\mathbf{Y}$ lies on the line $\{\boldsymbol{\mu} + s\mathbf{A} : s \in \mathbb{R}\}$. The null space of Σ is $\mathbf{A}^\perp$, spanned by $(2, -1, 0)^\top$ and $(1, 0, 1)^\top$. The corresponding constraints $\mathbf{v}^\top \mathbf{Y} = \mathbf{v}^\top \boldsymbol{\mu}$ are $2Y_1 - Y_2 = 1$ and $Y_1 + Y_3 = 3$, and both hold identically.

3.2 Moment generating function and density

3.2.1 Moment generating functions of random vectors

The moment generating function of a random vector $\mathbf{Y} \in \mathbb{R}^n$ is $M_{\mathbf{Y}}(\mathbf{t}) = \mathbb{E} \exp(\mathbf{t}^\top \mathbf{Y})$, and [Theorem 2.6.1](#) proves the two properties we need when it is finite on a neighbourhood of $\mathbf{0}$. *Uniqueness*: two random vectors whose moment generating functions are finite and equal near $\mathbf{0}$ have the same distribution. *Factorization*: two blocks $\mathbf{Y}_1, \mathbf{Y}_2$ are independent iff $M_{\mathbf{Y}}(\mathbf{t}) = M_{\mathbf{Y}_1}(\mathbf{t}_1)M_{\mathbf{Y}_2}(\mathbf{t}_2)$ near $\mathbf{0}$. The second extends to k blocks by induction. If $M_{\mathbf{Y}}$ is the product of the k block functions, setting $\mathbf{t}_1 = \mathbf{0}$ shows that $(\mathbf{Y}_2, \dots, \mathbf{Y}_k)$ has the product of the last $k-1$, so $\mathbf{Y}_1$ is independent of $(\mathbf{Y}_2, \dots, \mathbf{Y}_k)$, whose blocks are mutually independent by the induction hypothesis. The converse is immediate.

Theorem 3.2.1 — Moment generating function

A random vector $\mathbf{Y}$ is $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ iff

$$M_{\mathbf{Y}}(\mathbf{t}) = \exp(\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}) \quad \text{for every } \mathbf{t} \in \mathbb{R}^n. \quad (3.2.1)$$

Consequently the distribution of a normal vector is determined by its mean vector and covariance matrix.

Proof. Suppose $\mathbf{Y}$ is normal with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. By [Eq. 3.1.1](#), $\mathbf{t}^\top \mathbf{Y} \sim N(\mathbf{t}^\top \boldsymbol{\mu}, \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})$, and $M_{\mathbf{Y}}(\mathbf{t})$ is the moment generating function of this scalar evaluated at $s = 1$. [Lemma 3.1.1\(a\)](#) gives [Eq. 3.2.1](#).

Conversely, suppose [Eq. 3.2.1](#) holds for a symmetric $\boldsymbol{\Sigma}$. Fix $\mathbf{a}$. The moment generating function of $\mathbf{a}^\top \mathbf{Y}$ at s is $M_{\mathbf{Y}}(s\mathbf{a}) = \exp\{(\mathbf{a}^\top \boldsymbol{\mu})s + \frac{1}{2}(\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})s^2\}$. The coefficient $\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$ is nonnegative, since otherwise this function would tend to zero as $s \rightarrow \pm\infty$, whereas every random variable X has $\mathbb{E} e^{sX} + \mathbb{E} e^{-sX} \geq \mathbb{E} e^{|sX|} \geq 1$. By [Lemma 3.1.1\(b\)](#), $\mathbf{a}^\top \mathbf{Y} \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$. So $\mathbf{Y}$ is normal, and matching means and variances of all linear combinations shows $\mathbb{E} \mathbf{Y} = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$.

Finally, two normal vectors with the same $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ have the same moment generating function [Eq. 3.2.1](#), which is finite everywhere, so they have the same distribution by the uniqueness property. ■

The last sentence is what makes $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ a well-defined distribution. Combined with [Theorem 3.1.2](#), it says that $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ holds iff $\mathbf{Y}$ has the same distribution as $\boldsymbol{\mu} + \mathbf{A}\mathbf{Z}$ for some, and then every, matrix $\mathbf{A}$ with $\mathbf{A}\mathbf{A}^\top = \boldsymbol{\Sigma}$. The number of columns of $\mathbf{A}$, and so the length of $\mathbf{Z}$, does not matter. Some books take this representation as the definition ([Christensen 2020](#); [Seber and Lee 2003](#)). The proof above shows the two definitions agree.

The moment generating function also encodes moments. Differentiating Eq. 3.2.1 at $\mathbf{t} = \mathbf{0}$ gives $\partial M_{\mathbf{Y}}/\partial \mathbf{t} = \boldsymbol{\mu}$ and $\partial^2 M_{\mathbf{Y}}/\partial \mathbf{t} \partial \mathbf{t}^\top = \boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\top = \mathbf{E}(\mathbf{Y}\mathbf{Y}^\top)$. Higher derivatives give the fourth moments needed for variances of quadratic forms in Chapter 4 (Exercise (B1)).

3.2.2 The density

Theorem 3.2.2 — Density

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

- (a) If $\boldsymbol{\Sigma}$ is positive definite, $\mathbf{Y}$ has the density

$$f(\mathbf{y}) = (2\pi)^{-n/2} \det(\boldsymbol{\Sigma})^{-1/2} \exp\left\{-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu})\right\}, \quad \mathbf{y} \in \mathbb{R}^n. \quad (3.2.2)$$

- (b) Conversely, a random vector with density Eq. 3.2.2, for some positive definite $\boldsymbol{\Sigma}$, is $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.
(c) If $\boldsymbol{\Sigma}$ is singular, $\mathbf{Y}$ has no density on $\mathbb{R}^n$.

Proof.

- (a) By Theorem 3.2.1, $\mathbf{Y}$ has the same distribution as $\boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$ with $\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I})$, so it suffices to find the density of the latter. The entries of $\mathbf{Z}$ are independent standard normals, so $\mathbf{Z}$ has density $\prod_i (2\pi)^{-1/2} e^{-z_i^2/2} = (2\pi)^{-n/2} \exp(-\frac{1}{2}\mathbf{z}^\top \mathbf{z})$. Since $\boldsymbol{\Sigma}$ is positive definite, so is $\boldsymbol{\Sigma}^{1/2}$ (its eigenvalues are the positive square roots of those of $\boldsymbol{\Sigma}$), and the map $\mathbf{z} \mapsto \mathbf{y} = \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{z}$ is an affine bijection of $\mathbb{R}^n$ with inverse $\mathbf{z} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{y} - \boldsymbol{\mu})$. The change-of-variables formula multiplies the density of $\mathbf{Z}$ at $\mathbf{z}(\mathbf{y})$ by $|\det \boldsymbol{\Sigma}^{-1/2}| = \det(\boldsymbol{\Sigma})^{-1/2}$. Finally $\mathbf{z}^\top \mathbf{z} = (\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\Sigma}^{-1/2} (\mathbf{y} - \boldsymbol{\mu}) = (\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \boldsymbol{\mu})$.
(b) A density determines a distribution, and by (a) the $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ distribution has this density.
(c) $\mathbf{Y}$ lies with probability one in $\boldsymbol{\mu} + \mathcal{C}(\boldsymbol{\Sigma})$, an affine subspace of dimension less than n . It has Lebesgue measure zero, so it would receive probability zero under any distribution with a density.

■

A singular normal vector does have a density with respect to the natural measure on the affine subspace where it lives (Proposition 3.3.4 and Exercise (B2)), but that is rarely needed. In this book, arguments are made with the moment generating function or with linear transformations, which work in every case, and densities are used only for pictures and likelihoods.

Example. (The bivariate normal)

For $n = 2$, write $\sigma_1^2, \sigma_2^2 > 0$ for the variances and ρ for the correlation. Then

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}, \quad \det \boldsymbol{\Sigma} = \sigma_1^2\sigma_2^2(1 - \rho^2),$$

so Σ is positive definite iff $|\rho| < 1$. With standardized coordinates $u_i = (y_i - \mu_i)/\sigma_i$, the density Eq. 3.2.2 is

$$f(y_1, y_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{u_1^2 - 2\rho u_1 u_2 + u_2^2}{2(1-\rho^2)}\right\}.$$

When $\rho = 0$ it factors into the two univariate normal densities. When $\rho = \pm 1$ there is no density: $\text{Var}(u_2 \mp u_1) = 0$ in standardized units, so all the probability lies on a line.

3.2.3 Contours and the Mahalanobis distance

For positive definite Σ , the density Eq. 3.2.2 depends on $\mathbf{y}$ only through

$$\Delta^2(\mathbf{y}) = (\mathbf{y} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{y} - \boldsymbol{\mu}),$$

the squared Mahalanobis distance from $\mathbf{y}$ to $\boldsymbol{\mu}$ introduced in Section 2.2. The density is a decreasing function of Δ^2 , so its contours are the ellipsoids $\{\mathbf{y} : \Delta^2(\mathbf{y}) = c^2\}$. Their shape comes from the spectral decomposition $\Sigma = \sum_i \lambda_i \mathbf{u}_i \mathbf{u}_i^\top$ (Theorem 1.7.2). In the coordinates $w_i = \mathbf{u}_i^\top(\mathbf{y} - \boldsymbol{\mu})$ we have $\Delta^2 = \sum_i w_i^2 / \lambda_i$, so the ellipsoid has principal axes along the eigenvectors $\mathbf{u}_i$, with half-lengths $c\sqrt{\lambda_i}$. Large-variance directions are long axes. The Mahalanobis distance measures distance from the centre in units of standard deviation along each principal direction. The volume of $\{\mathbf{y} : \Delta^2(\mathbf{y}) \leq c^2\}$ is proportional to $\prod_i \sqrt{\lambda_i} = \det(\Sigma)^{1/2}$, so the generalized variance $\det \Sigma$ (Section 2.1) measures the size of the region the distribution occupies. The factor $\det(\Sigma)^{-1/2}$ in Eq. 3.2.2 spreads the same total probability over that region: a larger generalized variance means a flatter density.

Proposition 3.2.3

If $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$ with Σ positive definite, then $\Delta^2(\mathbf{Y})$ has the chi-squared distribution with n degrees of freedom, $\chi^2(n)$, which is $\chi^2(n, 0)$ in the notation of Definition 4.1.1. So the ellipsoid $\{\mathbf{y} : \Delta^2(\mathbf{y}) \leq q\}$, where q is the $1 - \alpha$ quantile of $\chi^2(n)$, has probability exactly $1 - \alpha$.

Proof. Put $\mathbf{Z} = \Sigma^{-1/2}(\mathbf{Y} - \boldsymbol{\mu})$. For every $\mathbf{a}$, $\mathbf{a}^\top \mathbf{Z} = (\Sigma^{-1/2} \mathbf{a})^\top \mathbf{Y} - \mathbf{a}^\top \Sigma^{-1/2} \boldsymbol{\mu}$ is a linear combination of $\mathbf{Y}$ minus a constant, hence normal by Definition 3.1.1, so $\mathbf{Z}$ is normal. Its mean is zero and its covariance is $\mathbf{I}$ (Proposition 2.2.4(a)). By Theorem 3.2.1 its moment generating function is $\exp(\frac{1}{2} \mathbf{t}^\top \mathbf{t}) = \prod_i \exp(\frac{1}{2} t_i^2)$, which factorizes, so its entries are independent standard normals. $\Delta^2(\mathbf{Y}) = \mathbf{Z}^\top \mathbf{Z} = \sum_i Z_i^2$ is a sum of n independent squared standard normals, which is the definition of $\chi^2(n)$. ■

Chapter 4 generalizes this considerably, to $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ with a nonzero mean and a general $\mathbf{A}$ (Theorem 4.3.2). Because the density is largest inside these ellipsoids, they are the smallest regions with their probability content, which is why they reappear as confidence regions for $\boldsymbol{\beta}$ in Chapter 12.

Computing with the density. Neither Σ^{-1} nor $\det \Sigma$ should be formed explicitly. With the Cholesky factorization $\Sigma = \mathbf{L} \mathbf{L}^\top$, we have $\Delta^2(\mathbf{y}) = \|\mathbf{L}^{-1}(\mathbf{y} - \boldsymbol{\mu})\|^2$, computed by a triangular solve, and $\log \det \Sigma = 2 \sum_i \log l_{ii}$. The listing evaluates the log density this way, checks it

against `scipy`, and counts how many of the simulated vectors fall inside the 95% ellipsoid. The fraction is 0.9500. The average of Δ^2 over the draws is 3.004 and its variance is 5.994, against the $\chi^2(3)$ values 3 and 6.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(301)
mu = np.array([1.0, -2.0, 0.5])
Sigma = np.array([[4.0, 1.2, -0.8],
                  [1.2, 2.0, 0.6],
                  [-0.8, 0.6, 1.5]])

N = 400_000
a = np.array([0.3, -1.0, 2.0])
q = [0.05, 0.25, 0.5, 0.75, 0.95]
exact = stats.norm.ppf(q, loc=a @ mu, scale=np.sqrt(a @ Sigma @ a))
cov_err = max(np.abs(np.cov(Y, rowvar=False) - Sigma).max() for Y in (Y_chol, Y_root))

def draw_normal(mu, A, size, rng):
    """Rows are independent copies of mu + A Z with Z ~ N(0, I)."""
    Z = rng.standard_normal((size, A.shape[1]))
    return mu + Z @ A.T

L = np.linalg.cholesky(Sigma)                # lower triangular, L L^T = Sigma
lam, U = np.linalg.eigh(Sigma)
root = U @ np.diag(np.sqrt(lam)) @ U.T       # symmetric square root

Y_chol = draw_normal(mu, L, N, rng)
Y_root = draw_normal(mu, root, N, rng)
for Y in (Y_chol, Y_root):
    print(np.round(np.cov(Y, rowvar=False) - Sigma, 3))

def normal_logpdf(y, mu, Sigma):
    """log density of N(mu, Sigma) at the rows of y, Sigma positive definite."""
    L = np.linalg.cholesky(Sigma)
    w = np.linalg.solve(L, (y - mu).T)       # w = L^{-1}(y - mu), so w^T w =
    Delta^2
    D2 = np.sum(w**2, axis=0)
    logdet = 2.0 * np.sum(np.log(np.diag(L)))
    return -0.5 * (len(mu) * np.log(2 * np.pi) + logdet + D2), D2

logf, D2 = normal_logpdf(Y_chol, mu, Sigma)
inside = np.mean(D2 <= stats.chi2.ppf(0.95, df=3))
print(f"fraction inside the 95% ellipsoid: {inside:.4f}")
```

3.2.4 Exercises

3.2.4.1 A. Check your understanding

Exercise A1.

Show that $f(y_1, y_2) = k \exp\{-\frac{1}{2}(2y_1^2 + 2y_1y_2 + y_2^2 - 4y_1 - 2y_2)\}$ is a bivariate normal density for a suitable constant k . Find k , μ , Σ and the correlation.

Solution. The quadratic part is $\mathbf{y}^\top \mathbf{P} \mathbf{y}$ with $\mathbf{P} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, which is positive definite with $\det \mathbf{P} = 1$. Matching $(\mathbf{y} - \boldsymbol{\mu})^\top \mathbf{P} (\mathbf{y} - \boldsymbol{\mu}) = \mathbf{y}^\top \mathbf{P} \mathbf{y} - 2\boldsymbol{\mu}^\top \mathbf{P} \mathbf{y} + \boldsymbol{\mu}^\top \mathbf{P} \boldsymbol{\mu}$ with the linear terms requires $\mathbf{P} \boldsymbol{\mu} = (2, 1)^\top$, so $\boldsymbol{\mu} = (1, 0)^\top$ and $\boldsymbol{\mu}^\top \mathbf{P} \boldsymbol{\mu} = 2$. The exponent is $-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})^\top \mathbf{P} (\mathbf{y} - \boldsymbol{\mu}) + 1$. By [Theorem 3.2.2\(b\)](#) this is the $N_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ density with $\boldsymbol{\Sigma} = \mathbf{P}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ provided $ke = (2\pi)^{-1}(\det \boldsymbol{\Sigma})^{-1/2} = (2\pi)^{-1}$. So $k = e^{-1}/(2\pi)$, and the correlation is $-1/\sqrt{2}$.

Exercise A2.

Let $\boldsymbol{\Sigma}$ have unit variances and correlation ρ , with $|\rho| < 1$. Find the principal axes and half-lengths of the ellipse $\Delta^2 = c^2$, show that its area is $\pi c^2 \sqrt{1 - \rho^2}$, and show that it contains probability $1 - e^{-c^2/2}$.

3.2.4.2 B. Practice

Exercise B1.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathbf{X} = \mathbf{Y} - \boldsymbol{\mu}$. Expand the moment generating function of $\mathbf{X}$ in a power series to show that all odd-order moments of $\mathbf{X}$ vanish and

$$E(X_i X_j X_k X_l) = \sigma_{ij} \sigma_{kl} + \sigma_{ik} \sigma_{jl} + \sigma_{il} \sigma_{jk}.$$

Deduce that $\text{Var}(\mathbf{X}^\top \mathbf{A} \mathbf{X}) = 2 \text{tr}(\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma})$ for symmetric $\mathbf{A}$.

Solution. By [Theorem 3.2.1](#), $M_{\mathbf{X}}(\mathbf{t}) = \exp(\frac{1}{2} \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}) = \sum_{m \geq 0} (\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})^m / (2^m m!)$, a series containing only terms of even degree in $\mathbf{t}$. Mixed moments of order d are the d th partial derivatives at $\mathbf{0}$, which come from the degree- d terms, so odd-order moments vanish. The degree-four term is $p(\mathbf{t}) = \frac{1}{8} (\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})^2 = T(\mathbf{t}, \mathbf{t}, \mathbf{t}, \mathbf{t})$, where $T(\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{x}) = \frac{1}{8} (\mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{v})(\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{x})$ is 4-linear. Differentiating $T(\mathbf{t}, \mathbf{t}, \mathbf{t}, \mathbf{t})$ once in each of the directions $\mathbf{e}_i, \mathbf{e}_j, \mathbf{e}_k, \mathbf{e}_l$ gives the sum of T over the 24 ways of placing these four vectors in the four slots. This holds whether or not the indices are distinct, since it is the product rule applied to a multilinear form. A placement contributes $\frac{1}{8} \sigma_{ab} \sigma_{cd}$, where the first two slots receive the indices a, b and the last two receive c, d . Each of the three ways of splitting the four *directions* into two pairs, $\{ij, kl\}$, $\{ik, jl\}$, $\{il, jk\}$, arises from 8 placements (choose which pair goes first, and the order within each pair), which gives the formula. Then, for symmetric $\mathbf{A}$,

$$E(\mathbf{X}^\top \mathbf{A} \mathbf{X})^2 = \sum_{i,j,k,l} a_{ij} a_{kl} E(X_i X_j X_k X_l) = (\text{tr} \mathbf{A} \boldsymbol{\Sigma})^2 + 2 \text{tr}(\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}),$$

and subtracting $(E \mathbf{X}^\top \mathbf{A} \mathbf{X})^2 = (\text{tr} \mathbf{A} \boldsymbol{\Sigma})^2$ gives the variance.

Exercise B2.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\text{rank}(\boldsymbol{\Sigma}) = r < n$, and use the notation of [Proposition 3.3.4](#). Show that $\mathbf{V} = \mathbf{Q}^\top (\mathbf{Y} - \boldsymbol{\mu})$ has a density on $\mathbb{R}^r$, and that this density can be written as $(2\pi)^{-r/2} (\lambda_1 \cdots \lambda_r)^{-1/2} \exp\{-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^+ (\mathbf{y} - \boldsymbol{\mu})\}$ with $\mathbf{y} = \boldsymbol{\mu} + \mathbf{Q} \mathbf{v}$, where $\boldsymbol{\Sigma}^+ = \mathbf{Q} \boldsymbol{\Lambda}^{-1} \mathbf{Q}^\top$ is the Moore–Penrose inverse.

Exercise B3.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma}$ positive definite and let $\boldsymbol{\nu}$ be a fixed vector. Show that $(\mathbf{Y} - \boldsymbol{\nu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\nu})$ has the noncentral chi-squared distribution $\chi^2(n, \gamma)$ with $\gamma = (\boldsymbol{\mu} - \boldsymbol{\nu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \boldsymbol{\nu})$, in the convention of [Definition 4.1.1](#), and find its mean.

3.2.4.3 C. Going deeper**Exercise C1.**

Let $\mathbf{Y}$ have density [Eq. 3.2.2](#), and let $E = \{\mathbf{y} : \Delta^2(\mathbf{y}) \leq q\}$. Show that every Borel set A with $\Pr(\mathbf{Y} \in A) \geq \Pr(\mathbf{Y} \in E)$ has volume at least that of E . *Hint:* compare $\int_{A \setminus E} f$ with $\int_{E \setminus A} f$.

Solution. The density is a decreasing function of Δ^2 , so with $c = (2\pi)^{-n/2} \det(\boldsymbol{\Sigma})^{-1/2} e^{-q/2} > 0$ we have $f \geq c$ on E and $f < c$ off E . We may assume A has finite volume. Subtracting $\Pr(\mathbf{Y} \in A \cap E)$ from both sides of $\Pr(\mathbf{Y} \in A) \geq \Pr(\mathbf{Y} \in E)$ gives $\int_{A \setminus E} f \geq \int_{E \setminus A} f$. Hence

$$c \operatorname{vol}(A \setminus E) \geq \int_{A \setminus E} f \geq \int_{E \setminus A} f \geq c \operatorname{vol}(E \setminus A),$$

so $\operatorname{vol}(A \setminus E) \geq \operatorname{vol}(E \setminus A)$. Adding $\operatorname{vol}(A \cap E)$ to both sides gives $\operatorname{vol}(A) \geq \operatorname{vol}(E)$. The argument uses only that E is a set on which f is at least as large as anywhere outside it, so it applies to the level sets of any density.

3.3 Linear transformations and marginals

3.3.1 Affine maps

Theorem 3.3.1 — Affine transformations

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, let $\mathbf{A}$ be any $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^m$. Then

$$\mathbf{A}\mathbf{Y} + \mathbf{b} \sim N_m(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top).$$

Proof. For $\mathbf{c} \in \mathbb{R}^m$, $\mathbf{c}^\top(\mathbf{A}\mathbf{Y} + \mathbf{b}) = (\mathbf{A}^\top \mathbf{c})^\top \mathbf{Y} + \mathbf{c}^\top \mathbf{b}$ is a linear combination of $\mathbf{Y}$ plus a constant, which is normal by [Definition 3.1.1](#) and [Lemma 3.1.1\(c\)](#). So $\mathbf{A}\mathbf{Y} + \mathbf{b}$ is normal, and its mean and covariance are given by [Theorem 2.2.1](#). ■

The proof is one line because the definition was chosen to make it one line. There is no rank condition on $\mathbf{A}$. If $\mathbf{A}$ has more rows than $\operatorname{rank}(\boldsymbol{\Sigma})$, the result is a singular normal vector, and that is allowed. With a density-based definition, the same theorem needs $\mathbf{A}$ to have full row rank and $\boldsymbol{\Sigma}$ to be positive definite, and the most useful cases fall outside it. The moment generating function gives a second proof: $E e^{\mathbf{t}^\top(\mathbf{A}\mathbf{Y} + \mathbf{b})} = e^{\mathbf{t}^\top \mathbf{b}} M_{\mathbf{Y}}(\mathbf{A}^\top \mathbf{t}) = \exp\{\mathbf{t}^\top(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}) + \frac{1}{2} \mathbf{t}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top \mathbf{t}\}.$

Corollary 3.3.2 — Marginal distributions

Every subvector of a normal vector is normal. If $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathbf{Y}_I$ consists of the entries with indices in a set I , then $\mathbf{Y}_I \sim N(\boldsymbol{\mu}_I, \boldsymbol{\Sigma}_{II})$, where $\boldsymbol{\mu}_I$ and $\boldsymbol{\Sigma}_{II}$ keep the entries, rows and columns indexed by I . In particular $Y_i \sim N(\mu_i, \sigma_{ii})$.

Proof. $\mathbf{Y}_I = \mathbf{A}\mathbf{Y}$, where the rows of $\mathbf{A}$ are the coordinate vectors $\mathbf{e}_i^\top$, $i \in I$. Then $\mathbf{A}\boldsymbol{\mu} = \boldsymbol{\mu}_I$ and $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top = \boldsymbol{\Sigma}_{II}$. ■

Corollary 3.3.3 — Standardization and rotation

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

- (a) If $\boldsymbol{\Sigma}$ is positive definite and $\boldsymbol{\Sigma} = \mathbf{A}\mathbf{A}^\top$ with $\mathbf{A}$ square, then $\mathbf{A}^{-1}(\mathbf{Y} - \boldsymbol{\mu}) \sim N_n(\mathbf{0}, \mathbf{I})$. Its entries are independent standard normal variables.
- (b) If $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}$ and $\mathbf{Q}$ is an $n \times n$ orthogonal matrix, then $\mathbf{Q}^\top\mathbf{Y} \sim N_n(\mathbf{Q}^\top\boldsymbol{\mu}, \sigma^2\mathbf{I})$.

Proof. Both follow from [Theorem 3.3.1](#): $\mathbf{A}^{-1}\mathbf{A}\mathbf{A}^\top(\mathbf{A}^{-1})^\top = \mathbf{I}$ and $\mathbf{Q}^\top(\sigma^2\mathbf{I})\mathbf{Q} = \sigma^2\mathbf{I}$. Independence in (a) was shown in the proof of [Proposition 3.2.3](#). ■

Part (b) is the probabilistic basis of the canonical form of the linear model. Spherical normal errors look the same in every orthonormal coordinate system, so one may rotate to coordinates adapted to $\mathcal{C}(\mathbf{X})$, in which fitted values and residuals occupy separate independent coordinates. [Chapter 4](#) uses this rotation to find the distributions of sums of squares.

3.3.2 The rank of a normal vector

[Theorem 3.1.2](#) writes a normal vector as an affine function of independent standard normals *in distribution*. When $\boldsymbol{\Sigma}$ is singular the representation can be made exact, with as few standard normals as the rank.

Proposition 3.3.4

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\text{rank}(\boldsymbol{\Sigma}) = r \geq 1$, and let $\boldsymbol{\Sigma} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^\top$, where $\mathbf{Q}$ is $n \times r$ with orthonormal columns and $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_r)$ holds the positive eigenvalues. Put $\mathbf{W} = \boldsymbol{\Lambda}^{-1/2}\mathbf{Q}^\top(\mathbf{Y} - \boldsymbol{\mu})$. Then $\mathbf{W} \sim N_r(\mathbf{0}, \mathbf{I}_r)$, and

$$\mathbf{Y} = \boldsymbol{\mu} + \mathbf{Q}\boldsymbol{\Lambda}^{1/2}\mathbf{W} \quad \text{with probability one.}$$

Proof. By [Theorem 3.3.1](#), $\mathbf{W}$ is normal with covariance $\boldsymbol{\Lambda}^{-1/2}\mathbf{Q}^\top\mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^\top\mathbf{Q}\boldsymbol{\Lambda}^{-1/2} = \mathbf{I}_r$. Next, $\mathbf{Q}\boldsymbol{\Lambda}^{1/2}\mathbf{W} = \mathbf{Q}\mathbf{Q}^\top(\mathbf{Y} - \boldsymbol{\mu})$, and $\mathbf{Q}\mathbf{Q}^\top$ is the symmetric idempotent matrix with range $\mathcal{C}(\boldsymbol{\Sigma})$ (the orthogonal projection onto it, in the language of [Chapter 6](#)). Since

$\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\boldsymbol{\Sigma})$ with probability one (Theorem 2.2.3), $\mathbf{Q}\mathbf{Q}^\top(\mathbf{Y} - \boldsymbol{\mu}) = \mathbf{Y} - \boldsymbol{\mu}$ with probability one. ■

So a normal vector of rank r is exactly r independent standard normals placed in $\mathbb{R}^n$ by an injective affine map. Exercise (B4) gives the same construction for any random vector, with uncorrelated entries in place of independent normal ones. Singular normal vectors are lower-dimensional normal vectors in disguise.

Example. (Least squares under normal errors)

Let $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ with $\mathbf{X}$ of full column rank p , and let $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$, the symmetric idempotent matrix of rank p with $\mathbf{M}\mathbf{X} = \mathbf{X}$ (Chapter 5 and Chapter 6). By Theorem 3.3.1:

- (a) $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y} \sim N_p(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1})$, a nonsingular normal vector;
- (b) $\boldsymbol{\lambda}^\top\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\lambda}^\top\boldsymbol{\beta}, \sigma^2\boldsymbol{\lambda}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\boldsymbol{\lambda})$ for every $\boldsymbol{\lambda} \in \mathbb{R}^p$;
- (c) the fitted values $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{M})$, singular of rank p ;
- (d) the residuals $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{Y} \sim N_n(\mathbf{0}, \sigma^2(\mathbf{I} - \mathbf{M}))$, singular of rank $n - p$.

No density for $\hat{\mathbf{Y}}$ or $\hat{\boldsymbol{\varepsilon}}$ was needed, and none exists.

3.3.3 Independent normal vectors

Lemma 3.3.5

If $\mathbf{Y}_1 \sim N_{n_1}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ and $\mathbf{Y}_2 \sim N_{n_2}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$ are independent, then $(\mathbf{Y}_1^\top, \mathbf{Y}_2^\top)^\top$ is jointly normal with mean $(\boldsymbol{\mu}_1^\top, \boldsymbol{\mu}_2^\top)^\top$ and block-diagonal covariance $\text{diag}(\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2)$. Consequently, if also $n_1 = n_2$, then $\mathbf{Y}_1 + \mathbf{Y}_2 \sim N(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2, \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)$.

Proof. $\mathbf{a}_1^\top\mathbf{Y}_1 + \mathbf{a}_2^\top\mathbf{Y}_2$ is a sum of independent normal variables, which is normal by Lemma 3.1.1(c). Independence makes the cross-covariance zero. The sum is $[\mathbf{I}, \mathbf{I}]$ applied to the stacked vector. ■

Independence is essential in this lemma. Two normal vectors defined on the same probability space need not be jointly normal, as we now see.

3.3.4 Normal marginals are not enough

Corollary 3.3.2 cannot be reversed. A vector whose entries are each normal need not be jointly normal. Two examples make the point, one extreme and one realistic.

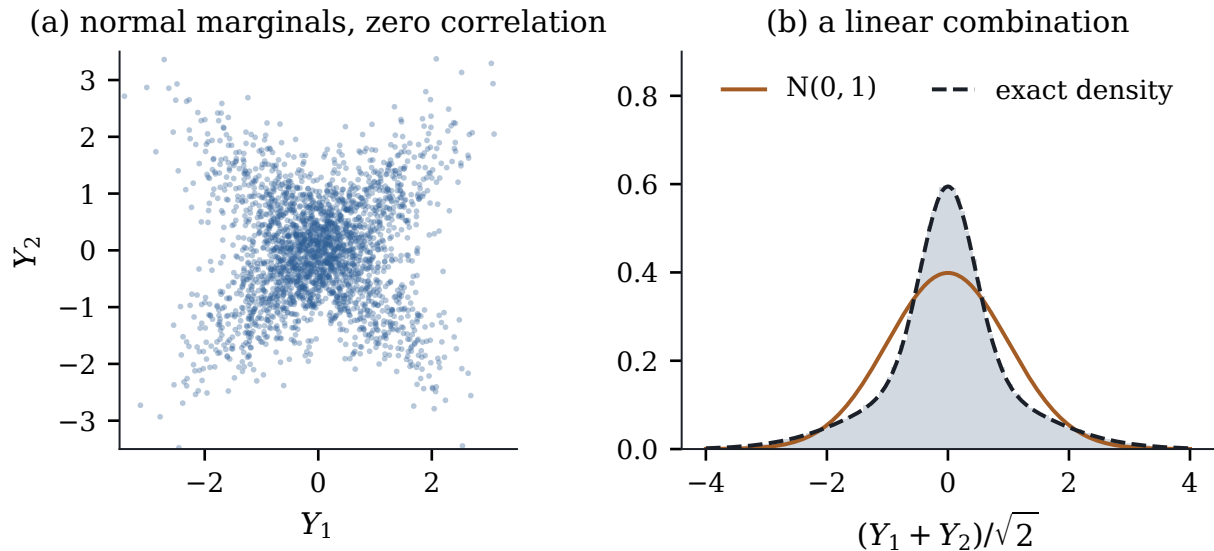


Figure 3.3.1: An equal mixture of bivariate normals with correlations ± 0.8 . (a) Both margins are exactly $N(0, 1)$ and the correlation is zero, but the scatter is an X, not an ellipse. (b) The linear combination $(Y_1 + Y_2)/\sqrt{2}$ has variance one but is sharply peaked and heavy-tailed compared with $N(0, 1)$. By [Definition 3.1.1](#), one non-normal direction is enough to rule out joint normality.

Example. (A random sign)

Let $Z \sim N(0, 1)$ and let S be independent of Z with $\Pr(S = 1) = \Pr(S = -1) = \frac{1}{2}$. Put $W = SZ$. By symmetry of Z , $W \sim N(0, 1)$. Also $\text{Cov}(Z, W) = E(S)E(Z^2) = 0$. But $Z + W = (1 + S)Z$ equals zero with probability $\frac{1}{2}$ and is otherwise $N(0, 4)$. A variable with an atom at zero and a continuous part is not normal, so (Z, W) is not jointly normal. And Z and W are far from independent: $|W| = |Z|$ always.

Example. (A mixture of two correlations)

Fix $0 < \rho < 1$. With probability $\frac{1}{2}$ draw (Y_1, Y_2) from the bivariate normal with unit variances and correlation ρ , and otherwise from the one with correlation $-\rho$. Both components have $N(0, 1)$ margins, so the mixture does too, exactly. The covariance is $\frac{1}{2}\rho + \frac{1}{2}(-\rho) = 0$. The pair is nevertheless dependent: since $E(Y_1^2 Y_2^2) = 1 + 2\rho^2$ for a standard bivariate normal with correlation r , the squares have correlation ρ^2 , not zero. The direction $\mathbf{a} = (1, 1)^\top/\sqrt{2}$ exposes the failure of joint normality. $S = (Y_1 + Y_2)/\sqrt{2}$ is an equal mixture of $N(0, 1 + \rho)$ and $N(0, 1 - \rho)$. It has variance 1 but kurtosis $E S^4 = \frac{3}{2}\{(1 + \rho)^2 + (1 - \rho)^2\} = 3(1 + \rho^2)$, where a normal variable would have 3.

[Figure 3.3.1](#) shows a simulation with $\rho = 0.8$ and 200,000 draws. Kolmogorov–Smirnov tests of normality for the two margins give p -values 0.84 and 0.29. The sample correlation is -0.0033 , the correlation of the squares is 0.640 (theory 0.64), and the kurtosis of S is 4.88 (theory 4.92). The Kolmogorov–Smirnov distance of S from $N(0, 1)$ is 0.067, which with this many draws rejects normality at any conventional level.

```
import numpy as np
from scipy import integrate, stats
rng = np.random.default_rng(302)
```

```

rho, N = 0.8, 200_000

sign = rng.choice([-1.0, 1.0], size=N)          # which component each draw uses
Z1, Z2 = rng.standard_normal((2, N))
Y1 = Z1
Y2 = sign * rho * Z1 + np.sqrt(1 - rho**2) * Z2 # corr(Y1, Y2 | sign) = sign * rho

S = (Y1 + Y2) / np.sqrt(2)                      # a direction that is not normal
print("marginal KS p-values:", stats.kstest(Y1, "norm").pvalue, stats.kstest(Y2, "
    norm").pvalue)
print("correlation:", np.corrcoef(Y1, Y2)[0, 1])
print("corr of squares:", np.corrcoef(Y1**2, Y2**2)[0, 1])
print("kurtosis of S:", stats.kurtosis(S, fisher=False), " theory:", 3 * (1 + rho
    **2))
print("KS p-value of S:", stats.kstest(S, "norm").pvalue)

```

Idea. Normality of a vector is a property of *every direction*, not of the coordinate axes. Normal margins with zero correlation can hide strong dependence. The theorems of the next two sections, which turn zero covariance into independence and give linear conditional means, need *joint* normality. They are not about variables that merely happen to be normal one at a time.

3.3.5 Exercises

3.3.5.1 A. Check your understanding

Exercise A1.

Let $\mathbf{Y} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\mu} = (0, 1, 2)^\top$ and $\boldsymbol{\Sigma} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$. Find the joint distribution of $U = Y_1 + Y_2 + Y_3$, $V = Y_1 - Y_3$ and $W = Y_1 - 2Y_2 + Y_3$. Which of the pairs are independent? Is V independent of (U, W) ?

Exercise A2.

In [Example \(Least squares under normal errors\)](#), show that the i th residual is $N(0, \sigma^2(1 - m_{ii}))$, where m_{ii} is the i th diagonal entry of $\mathbf{M}$, and that it is zero with probability one iff $m_{ii} = 1$.

3.3.5.2 B. Practice

Exercise B1.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}\mathbf{1}, \sigma^2\mathbf{I})$ and let $\mathbf{Q}$ be an orthogonal matrix whose first row is $n^{-1/2}\mathbf{1}^\top$. Show that the entries of $\mathbf{V} = \mathbf{Q}\mathbf{Y}$ are independent, with $V_1 \sim N(\sqrt{n}\boldsymbol{\mu}, \sigma^2)$ and $V_i \sim N(0, \sigma^2)$ for $i \geq 2$. Show that $\sum_i (Y_i - \bar{Y})^2 = \sum_{i \geq 2} V_i^2$, and conclude again that $\bar{Y}$ and S^2 are independent.

Exercise B2.

Let $Z \sim N(0, 1)$ and, for $c > 0$, let $W = Z$ if $|Z| \leq c$ and $W = -Z$ otherwise. Show that $W \sim N(0, 1)$, that there is a c with $\text{Cov}(Z, W) = 0$, and that (Z, W) is not jointly normal for any c .

Solution. Because Z and $-Z$ have the same distribution and the event $\{|Z| > c\}$ is unchanged by the sign change, $\Pr(W \leq w) = \Pr(Z \leq w, |Z| \leq c) + \Pr(-Z \leq w, |Z| > c) = \Pr(Z \leq w)$. Next, $\text{Cov}(Z, W) = E(Z^2; |Z| \leq c) - E(Z^2; |Z| > c) = 2E(Z^2; |Z| \leq c) - 1$, which is continuous and increasing in c , tends to -1 as $c \rightarrow 0$ and to 1 as $c \rightarrow \infty$, and so vanishes at some c . Finally $Z + W = 2Z\mathbb{I}\{|Z| \leq c\}$ equals 0 with probability $\Pr(|Z| > c) \in (0, 1)$ but is not constant. A normal variable is either constant or has a density, so $Z + W$ is not normal.

Exercise B3.

For the pair (Z, W) of [Example \(A random sign\)](#), find every $\mathbf{a} \neq \mathbf{0}$ for which $a_1 Z + a_2 W$ is normal. *Hint:* an equal mixture of $N(0, v_1)$ and $N(0, v_2)$ has fourth moment $\frac{3}{2}(v_1^2 + v_2^2)$. Conclude that (Z, W) is normal along exactly two lines through the origin, the coordinate axes.

3.4 Independence and zero covariance

For any random vectors with finite second moments, independence implies zero cross-covariance, because the expectation of a product of functions of independent variables factors. The converse fails in general, as [Example \(A random sign\)](#) and [Example \(A mixture of two correlations\)](#) show. For jointly normal vectors it holds. This is the single most used property of the normal distribution in linear-model theory. Every independence statement behind a t or F statistic reduces to it.

Theorem 3.4.1 — Independence

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ be partitioned into blocks $\mathbf{Y} = (\mathbf{Y}_1^\top, \dots, \mathbf{Y}_k^\top)^\top$, with $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma} = (\boldsymbol{\Sigma}_{ij})$ partitioned to match.

- (a) The blocks $\mathbf{Y}_1, \dots, \mathbf{Y}_k$ are mutually independent iff $\boldsymbol{\Sigma}_{ij} = \text{Cov}(\mathbf{Y}_i, \mathbf{Y}_j) = \mathbf{0}$ for all $i \neq j$.
- (b) In particular, $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent iff $\text{Cov}(\mathbf{Y}_1, \mathbf{Y}_2) = \mathbf{0}$, and the scalar entries $Y_1, \dots, Y_n$ are mutually independent iff $\boldsymbol{\Sigma}$ is diagonal.

Proof. If the blocks are independent, every cross-covariance is zero. Conversely, suppose $\boldsymbol{\Sigma}_{ij} = \mathbf{0}$ for $i \neq j$, and partition $\mathbf{t}$ to match. Then

$$\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t} = \sum_{i=1}^k \left(\mathbf{t}_i^\top \boldsymbol{\mu}_i + \frac{1}{2} \mathbf{t}_i^\top \boldsymbol{\Sigma}_{ii} \mathbf{t}_i \right),$$

because all the cross terms $\mathbf{t}_i^\top \boldsymbol{\Sigma}_{ij} \mathbf{t}_j$ vanish. Exponentiating, [Theorem 3.2.1](#) gives $M_{\mathbf{Y}}(\mathbf{t}) = \prod_i M_i(\mathbf{t}_i)$, where M_i is the moment generating function of $N(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_{ii})$. By [Corollary 3.3.2](#) that is the distribution of $\mathbf{Y}_i$, so $M_i = M_{\mathbf{Y}_i}$. The factorization property of [Section 3.2](#) gives mutual independence. Part (b) is the cases $k = 2$ and $k = n$. ■

A consequence worth stating explicitly: for jointly normal blocks, *pairwise* independence implies *mutual* independence, because condition (a) involves only pairs. For general random vectors this is false, as the example of three random signs in [Example \(Pairwise but not mutually independent\)](#) shows. Joint normality of the whole vector matters here: joint normality of each pair is not enough ([Exercise \(C1\)](#)).

The theorem is usually applied to two linear functions of the same normal vector.

Corollary 3.4.2

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, and let $\mathbf{A}$ and $\mathbf{B}$ be matrices with n columns. Then $\mathbf{AY}$ and $\mathbf{BY}$ are independent iff $\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^\top = \mathbf{0}$. In that case $g(\mathbf{AY})$ and $h(\mathbf{BY})$ are independent for all measurable functions g and h .

Proof. The stacked vector $((\mathbf{AY})^\top, (\mathbf{BY})^\top)^\top$ is a linear function of $\mathbf{Y}$, so it is normal by [Theorem 3.3.1](#). Its off-diagonal covariance block is $\text{Cov}(\mathbf{AY}, \mathbf{BY}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^\top$ ([Theorem 2.2.1](#)). Apply [Theorem 3.4.1](#). Functions of independent random vectors are independent. ■

The condition $\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^\top = \mathbf{0}$ involves only second moments, but the conclusion is full independence. In particular, any quadratic form in $\mathbf{BY}$, such as $\|\mathbf{BY}\|^2$, is independent of $\mathbf{AY}$. [Chapter 4](#) extends this to quadratic forms that are not written as squared lengths ([Theorem 4.4.1](#)).

Example. (Sample mean and deviations)

Let $\mathbf{Y} \sim N_n(\mu \mathbf{1}, \sigma^2 \mathbf{I})$, so that $Y_1, \dots, Y_n$ is a random sample from $N(\mu, \sigma^2)$. The mean is $\bar{Y} = n^{-1} \mathbf{1}^\top \mathbf{Y}$, and the vector of deviations is $\mathbf{d} = (\mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top) \mathbf{Y}$. As in [Example \(Centring a sample\)](#),

$$\text{Cov}(\bar{Y}, \mathbf{d}) = \sigma^2 n^{-1} \mathbf{1}^\top (\mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top) = \mathbf{0}^\top,$$

By [Corollary 3.4.2](#), $\bar{Y}$ is independent of $\mathbf{d}$, and so of every function of the deviations: the sample variance $S^2 = \|\mathbf{d}\|^2/(n-1)$, the range, the sample skewness. The distribution of $(n-1)S^2/\sigma^2$ is found in [Corollary 4.4.5](#). The computation used $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ only through $\boldsymbol{\Sigma} \mathbf{1} \in \text{span}(\mathbf{1})$, which is the precise condition for the conclusion ([Exercise \(B1\)](#)).

Example. (Coefficients and residuals)

In the setting of [Example \(Least squares under normal errors\)](#),

$$\text{Cov}(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\varepsilon}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\sigma^2 \mathbf{I}) (\mathbf{I} - \mathbf{M}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \{(\mathbf{I} - \mathbf{M}) \mathbf{X}\}^\top = \mathbf{0},$$

since $(\mathbf{I} - \mathbf{M})\mathbf{X} = \mathbf{0}$. Hence $\hat{\boldsymbol{\beta}}$ is independent of the residual vector, and so of the residual sum of squares $\|\hat{\boldsymbol{\varepsilon}}\|^2$ and of the estimate $s^2 = \|\hat{\boldsymbol{\varepsilon}}\|^2/(n-p)$. The same argument shows that the fitted values $\hat{\mathbf{Y}}$ and the residuals are independent. This independence is what allows a t statistic $(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta})/\{s^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}\}^{1/2}$ to have a t distribution (Chapter 11).

Remark (What joint normality buys). The two examples use only a covariance calculation. Without joint normality, zero covariance between $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\varepsilon}}$ still holds under the second-moment assumptions of [Definition 5.3.1](#), but independence does not follow, and neither do exact t and F distributions. Chapter 21 examines what survives approximately.

3.4.1 Exercises

3.4.1.1 A. Check your understanding

Exercise A1.

Let Y_1, Y_2 be independent $N(\mu, \sigma^2)$. Show that $Y_1 + Y_2$ and $Y_1 - Y_2$ are independent. What if the variances differ?

Exercise A2.

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$ and let $\mathbf{u}$ be a unit vector. Show that $\mathbf{u}^\top \mathbf{Y}$ and $\|\mathbf{Y} - (\mathbf{u}^\top \mathbf{Y})\mathbf{u}\|^2$ are independent.

3.4.1.2 B. Practice

Exercise B1.

Let $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma}$ arbitrary. Show that $\bar{Y}$ is independent of the deviations $Y_i - \bar{Y}$ iff $\boldsymbol{\Sigma}\mathbf{1} \in \text{span}(\mathbf{1})$, that is, iff all row sums of $\boldsymbol{\Sigma}$ are equal. Check the condition for the equicorrelation matrix $\sigma^2\{(1-\rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top\}$, and for a two-group matrix: the first m_1 and the last m_2 observations form groups, all variances are 1, the correlation is ρ within a group and τ between groups.

Solution. $\bar{Y} = n^{-1}\mathbf{1}^\top \mathbf{Y}$ and the deviations are $(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{Y}$, so by [Corollary 3.4.2](#) they are independent iff $n^{-1}\mathbf{1}^\top \boldsymbol{\Sigma}(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top) = \mathbf{0}^\top$, that is, $(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\boldsymbol{\Sigma}\mathbf{1} = \mathbf{0}$. The null space of $\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top$ is $\text{span}(\mathbf{1})$, and $\boldsymbol{\Sigma}\mathbf{1}$ is the vector of row sums. For the equicorrelation matrix, $\boldsymbol{\Sigma}\mathbf{1} = \sigma^2(1-\rho+n\rho)\mathbf{1}$, so independence holds for every admissible ρ . In the two-group matrix, a row in the first group sums to $1+(m_1-1)\rho+m_2\tau$ and a row in the second to $1+(m_2-1)\rho+m_1\tau$. The difference is $(m_1-m_2)(\rho-\tau)$, so independence holds iff the groups have equal size or $\rho = \tau$, and the latter is the equicorrelated case.

Exercise B2.

Let $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ with $\text{rank}(\mathbf{X}) = r < p$, let $\mathbf{G}$ be a generalized inverse of $\mathbf{X}^\top \mathbf{X}$

and $\mathbf{M} = \mathbf{XGX}^\top$. Show that the least squares estimate $\mathbf{GX}^\top \mathbf{Y}$ and the residual vector $(\mathbf{I} - \mathbf{M})\mathbf{Y}$ are independent, and that every estimable function $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ with $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ has an estimate independent of the residual sum of squares.

3.4.1.3 C. Going deeper

Exercise C1.

Let φ be the $N(0, 1)$ density, $0 < \varepsilon \leq e^{3/2}$, and

$$f(x, y, z) = \varphi(x)\varphi(y)\varphi(z)\{1 + \varepsilon xyz e^{-(x^2+y^2+z^2)/2}\}.$$

Show that f is a probability density, and that under f every pair of coordinates is $N_2(\mathbf{0}, \mathbf{I})$. So the three variables are pairwise jointly normal and pairwise independent. Show that they are not mutually independent and that (X, Y, Z) is not jointly normal. Which step of [Theorem 3.4.1](#) fails?

Solution. Since $|t|e^{-t^2/2} \leq e^{-1/2}$ for all t , the perturbation satisfies $|\varepsilon xyz e^{-(x^2+y^2+z^2)/2}| \leq \varepsilon e^{-3/2} \leq 1$, so $f \geq 0$. The function $z \mapsto z\varphi(z)e^{-z^2/2}$ is odd and integrable, so integrating f over z removes the perturbation and leaves $\varphi(x)\varphi(y)$. Integrating further shows that f integrates to 1, and by symmetry each pair has density $\varphi \otimes \varphi$, the $N_2(\mathbf{0}, \mathbf{I})$ density. Mutual independence would force $f = \varphi(x)\varphi(y)\varphi(z)$ almost everywhere, but the perturbation is nonzero off the coordinate planes. Directly, $E(XYZ) = \varepsilon \left(\int t^2 \varphi(t) e^{-t^2/2} dt \right)^3 > 0$, whereas independence with mean-zero margins gives 0. If (X, Y, Z) were jointly normal, its covariance matrix would be $\mathbf{I}$ (unit variances from the margins, zero covariances from the pairs), and by [Theorem 3.4.1](#) the coordinates would be independent, which they are not. The theorem's hypothesis is joint normality of the whole vector. Here only the pairs are normal, and the moment generating function of the triple does not factor.

3.5 Conditional distributions

Throughout this section $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ is split into $\mathbf{Y}_1 \in \mathbb{R}^{n_1}$ and $\mathbf{Y}_2 \in \mathbb{R}^{n_2}$, with

$$\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}, \quad \boldsymbol{\Sigma}_{21} = \boldsymbol{\Sigma}_{12}^\top,$$

as in [Theorem 2.5.1](#). The question is what $\mathbf{Y}_1$ looks like once $\mathbf{Y}_2$ has been observed. The answer is the probabilistic core of regression.

3.5.1 Partial covariance with a generalized inverse

The natural answer involves $\boldsymbol{\Sigma}_{22}^{-1}$, but $\boldsymbol{\Sigma}_{22}$ may be singular, for instance when $\mathbf{Y}_2$ contains a sum and its summands. A generalized inverse ([Definition 1.3.1](#)) removes the need for a separate treatment. The facts that make this work were proved for arbitrary random vectors in [Section 2.5](#). We collect them in the notation of this chapter.

Lemma 3.5.1

Let Σ be nonnegative definite and partitioned as above, and let G be any generalized inverse of Σ_{22} . Then:

- (a) $\mathcal{C}(\Sigma_{21}) \subseteq \mathcal{C}(\Sigma_{22})$, so $\Sigma_{21} = \Sigma_{22}H$ for some matrix H ;
- (b) $\Sigma_{12}G\Sigma_{22} = \Sigma_{12}$ and $\Sigma_{22}G\Sigma_{21} = \Sigma_{21}$;
- (c) $\Sigma_{12}G\Sigma_{21} = H^\top \Sigma_{22}H$, which does not depend on the choice of G and is nonnegative definite.

The matrix $\Sigma_{11.2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^-\Sigma_{21}$, the **Schur complement** of Σ_{22} in Σ , is therefore well defined. It is the partial covariance matrix of Y_1 given Y_2 in the sense of [Theorem 2.5.2](#).

Proof. By [Theorem 3.1.2](#), Σ is the covariance matrix of some random vector, so (a) and (b) are [Theorem 2.5.1\(c\)](#). For (c), $\Sigma_{12}G\Sigma_{21} = H^\top \Sigma_{22}G\Sigma_{22}H = H^\top \Sigma_{22}H$. ■

3.5.2 The conditional distribution**Theorem 3.5.2 — Conditional distribution**

Let $Y \sim N_n(\mu, \Sigma)$ be partitioned as above, let G be a generalized inverse of Σ_{22} , and put $B = \Sigma_{12}G$. Then:

- (a) $W = Y_1 - \mu_1 - B(Y_2 - \mu_2)$ is independent of Y_2 and $W \sim N_{n_1}(\mathbf{0}, \Sigma_{11.2})$;
- (b) the conditional distribution of Y_1 given $Y_2 = y_2$ is

$$Y_1 \mid Y_2 = y_2 \sim N_{n_1}\left(\mu_1 + \Sigma_{12}\Sigma_{22}^-(y_2 - \mu_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^-\Sigma_{21}\right), \quad (3.5.1)$$

and for (y_2) in $(\mu_2 + \mathcal{C}(\Sigma_{22}))$, where (Y_2) lies with one, the conditional mean does not depend on the choice of (G) ;

- (c) if Σ is positive definite, then $\Sigma_{22}^- = \Sigma_{22}^{-1}$, $\Sigma_{11.2}$ is positive definite, $\Sigma_{11.2}^{-1}$ is the leading $n_1 \times n_1$ block of Σ^{-1} , and $\det \Sigma = \det \Sigma_{22} \det \Sigma_{11.2}$.

Proof.

- (a) $W = [I, -B](Y - \mu)$, so $(W^\top, Y_2^\top)^\top$ is an affine function of Y and is jointly normal ([Theorem 3.3.1](#)). The vector W is the error of the best linear predictor of Y_1 from Y_2 , so by [Theorem 2.5.2\(a,b\)](#) it has mean zero, $\text{Cov}(W, Y_2) = \mathbf{0}$ and $\text{Cov}(W) = \Sigma_{11.2}$. These are moment calculations, valid for any random vector. What normality adds is that zero covariance becomes independence ([Theorem 3.4.1](#)), and that W , a linear function of Y , is normal.
- (b) Write $Y_1 = \mu_1 + B(Y_2 - \mu_2) + W$. The first two terms are a function of Y_2 , and W is independent of Y_2 . For independent W and Y_2 , the conditional distribution of

$h(\mathbf{W}, \mathbf{Y}_2)$ given $\mathbf{Y}_2 = \mathbf{y}_2$ is the distribution of $h(\mathbf{W}, \mathbf{y}_2)$ (Billingsley 1995). With $h(\mathbf{w}, \mathbf{y}_2) = \boldsymbol{\mu}_1 + \mathbf{B}(\mathbf{y}_2 - \boldsymbol{\mu}_2) + \mathbf{w}$, this is the normal distribution Eq. 3.5.1. If $\mathbf{y}_2 - \boldsymbol{\mu}_2 = \boldsymbol{\Sigma}_{22}\mathbf{k}$, then $\mathbf{B}(\mathbf{y}_2 - \boldsymbol{\mu}_2) = \boldsymbol{\Sigma}_{12}\mathbf{G}\boldsymbol{\Sigma}_{22}\mathbf{k} = \boldsymbol{\Sigma}_{12}\mathbf{k}$, which does not involve $\mathbf{G}$.

- (c) If $\boldsymbol{\Sigma}$ is positive definite, so is its principal submatrix $\boldsymbol{\Sigma}_{22}$, and its only generalized inverse is its inverse. By the block determinant formula (Theorem 1.6.3), which needs only $\boldsymbol{\Sigma}_{22}$ nonsingular, $\det \boldsymbol{\Sigma} = \det \boldsymbol{\Sigma}_{22} \det \boldsymbol{\Sigma}_{11.2}$. The left side and $\det \boldsymbol{\Sigma}_{22}$ are positive, so $\boldsymbol{\Sigma}_{11.2}$ is nonsingular. It is nonnegative definite, being $\text{Cov}(\mathbf{W})$ by (a), hence positive definite. With both $\boldsymbol{\Sigma}_{22}$ and its Schur complement nonsingular, the partitioned-inverse formula (Theorem 1.6.4) gives $\boldsymbol{\Sigma}_{11.2}^{-1}$ as the leading block of $\boldsymbol{\Sigma}^{-1}$.

When $\boldsymbol{\Sigma}$ is positive definite, (b) can also be read off the densities. By (c), $\boldsymbol{\Sigma}_{11.2}$ is positive definite, so $\mathbf{W}$ has a density. The map $(\mathbf{y}_1, \mathbf{y}_2) \mapsto (\mathbf{w}, \mathbf{y}_2)$ has unit Jacobian, so $f_{\mathbf{Y}}(\mathbf{y}_1, \mathbf{y}_2) = f_{\mathbf{W}}(\mathbf{y}_1 - \boldsymbol{\mu}_1 - \mathbf{B}(\mathbf{y}_2 - \boldsymbol{\mu}_2)) f_{\mathbf{Y}_2}(\mathbf{y}_2)$ by independence, and dividing by $f_{\mathbf{Y}_2}(\mathbf{y}_2)$ gives the conditional density.

■

Idea. Every jointly normal vector splits as

$$\mathbf{Y}_1 = \underbrace{\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)}_{\text{function of } \mathbf{Y}_2} + \underbrace{\mathbf{W}}_{\text{independent of } \mathbf{Y}_2}.$$

The first term is a linear regression of $\mathbf{Y}_1$ on $\mathbf{Y}_2$. The second is a normal error whose distribution is the same whatever value $\mathbf{Y}_2$ takes.

3.5.3 What the theorem says

Three features of Eq. 3.5.1 are special to the normal distribution, and each is an assumption of the classical linear model.

- (i) *The regression is linear.* $E(\mathbf{Y}_1 | \mathbf{Y}_2 = \mathbf{y}_2)$ is an affine function of $\mathbf{y}_2$, with coefficient matrix $\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}$.
- (ii) *The errors are homoscedastic.* The conditional covariance $\boldsymbol{\Sigma}_{11.2}$ is the same for every $\mathbf{y}_2$. Observing $\mathbf{Y}_2$ tells us where $\mathbf{Y}_1$ is centred, but not how spread out it is.
- (iii) *Conditioning cannot increase uncertainty.* $\boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{11.2} = \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$ is nonnegative definite (Lemma 3.5.1(c)), so every linear combination $\mathbf{a}^\top \mathbf{Y}_1$ has conditional variance at most its unconditional variance. The reduction is zero iff $\boldsymbol{\Sigma}_{12} = \mathbf{0}$.

None of these holds in general. In the mixture of Example (A mixture of two correlations), given $Y_1 = y_1$ the variable Y_2 is an equal mixture of $N(\rho y_1, 1 - \rho^2)$ and $N(-\rho y_1, 1 - \rho^2)$. Its conditional mean is 0 for every y_1 , but its conditional variance is $1 - \rho^2 + \rho^2 y_1^2$, which grows with $|y_1|$. Linearity and constant variance are not consequences of normal margins or of zero correlation. They are consequences of joint normality.

One response, several predictors. The case used most in this book has a scalar response Y and a vector $\mathbf{X}$ of k predictors, jointly normal, with $\text{Var}(Y) = \sigma_Y^2$, $\boldsymbol{\sigma}_{XY} = \text{Cov}(\mathbf{X}, Y)$ and $\boldsymbol{\Sigma}_{XX} = \text{Cov}(\mathbf{X})$ positive definite. Theorem 3.5.2 says

$$Y = \alpha + \beta^\top \mathbf{X} + e, \quad \beta = \Sigma_{XX}^{-1} \sigma_{XY}, \quad \alpha = \mu_Y - \beta^\top \mu_X, \quad e \sim N(0, \sigma^2) \text{ independent of } \mathbf{X}, \quad (3.5.2)$$

with $\sigma^2 = \sigma_Y^2 - \sigma_{XY}^\top \Sigma_{XX}^{-1} \sigma_{XY} = \sigma_Y^2(1 - \rho_{Y \cdot X}^2)$, where $\rho_{Y \cdot X}^2 = \sigma_{XY}^\top \Sigma_{XX}^{-1} \sigma_{XY} / \sigma_Y^2$ is the population squared multiple correlation ([Proposition 2.5.3](#)). So if observations $(Y_i, \mathbf{X}_i)$ are drawn independently from a joint normal distribution, then conditionally on the predictors, the normal linear model with independent homoscedastic errors holds exactly. Inference that treats the predictors as fixed is valid for random predictors, as Chapter 14 develops.

The bivariate case. With $k = 1$, correlation ρ and standard deviations σ_X, σ_Y ,

$$E(Y | X = x) = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X), \quad \text{Var}(Y | X = x) = \sigma_Y^2(1 - \rho^2).$$

In standard units, $(E(Y | X = x) - \mu_Y) / \sigma_Y = \rho(x - \mu_X) / \sigma_X$. The predicted response is fewer standard deviations from its mean than the predictor is from its mean, whenever $|\rho| < 1$. This is **regression toward the mean**, Galton's observation about the heights of parents and children (Galton 1886), and the origin of the word "regression". It is also a warning. Regressing X on Y is a different line, $E(X | Y = y) = \mu_X + \rho(\sigma_X / \sigma_Y)(y - \mu_Y)$, and when $0 < |\rho| < 1$ neither line is the major axis of the density contours.

Example. (Two regression lines)

Take $\mu_X = 1$, $\mu_Y = 2$, $\sigma_X = 1$, $\sigma_Y = 1.5$ and $\rho = 0.7$. The regression of Y on X has slope 1.05 and residual standard deviation 1.071. Drawn in the same (x, y) plane, the regression of X on Y has slope 2.14, and the major axis of the ellipses lies between them, with slope 1.76. [Figure 3.5.1](#) shows the three lines. The line $E(Y | X = x)$ passes through the points where each contour ellipse has a vertical tangent ([Exercise \(B2\)](#)): on a vertical line $X = x$, the density is highest at the conditional mean. A least squares line fitted to 60 draws has slope 1.12 (standard error 0.16). It estimates the regression of Y on X , not the major axis.

Example. (Holding a third variable fixed)

Let $\mathbf{Y} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with

$$\boldsymbol{\mu} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} 4 & 1 & 2 \\ 1 & 3 & 2 \\ 2 & 2 & 2 \end{pmatrix}.$$

Conditioning Y_1 on Y_2 alone gives slope $\sigma_{12} / \sigma_{22} = \frac{1}{3}$ and conditional variance $\frac{11}{3}$. Conditioning on (Y_2, Y_3) , the listing computes $\mathbf{B} = \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1}$ in exact arithmetic and finds

$$E(Y_1 | Y_2 = y_2, Y_3 = y_3) = 3 - y_2 + 2y_3, \quad \text{Var}(Y_1 | Y_2, Y_3) = 1.$$

The coefficient of y_2 changes sign. Among outcomes with the same Y_3 , larger Y_2 goes with smaller Y_1 . Across all outcomes, larger Y_2 goes with larger Y_1 , because both tend to be large when Y_3 is large. The conditional variance 1 also equals $\det \boldsymbol{\Sigma} / \det \boldsymbol{\Sigma}_{22} = 2/2$, as [Theorem 3.5.2\(c\)](#) predicts.

```
from fractions import Fraction as F
```

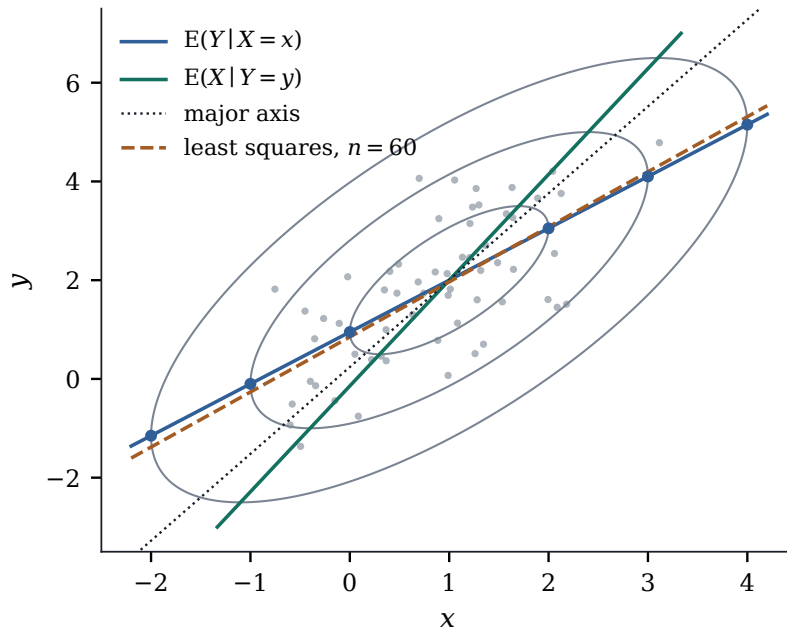


Figure 3.5.1: Contours $\Delta^2 = 1, 4, 9$ of a bivariate normal density with $\rho = 0.7$. The conditional mean line $E(Y | X = x)$ passes through the points of vertical tangency (dots). The line $E(X | Y = y)$ passes through the points of horizontal tangency and is steeper, and the major axis lies between the two. Least squares on a sample of 60 points estimates $E(Y | X = x)$.

```
import numpy as np
rng = np.random.default_rng(305)
mx, my, sx, sy, rho = 1.0, 2.0, 1.0, 1.5, 0.7
Sigma = np.array([[sx**2, rho * sx * sy], [rho * sx * sy, sy**2]])
slope_y_on_x = rho * sy / sx # E(Y | X = x)
slope_x_on_y = rho * sx / sy # E(X | Y = y), as dx/dy
cond_sd = sy * np.sqrt(1 - rho**2)
n = 60
XY = rng.multivariate_normal([mx, my], Sigma, size=n)
b_ls, a_ls = np.polyfit(XY[:, 0], XY[:, 1], 1)
Lin = np.linalg.inv(np.linalg.cholesky(Sigma))
theta = np.linspace(0, 2 * np.pi, 200_001)
circle = np.vstack([np.cos(theta), np.sin(theta)])
L = np.linalg.cholesky(Sigma)
big = rng.multivariate_normal([mx, my], Sigma, size=4_000_000)
x0 = 2.0
s1 = big[np.abs(big[:, 0] - x0) < 0.01, 1]
resid = XY[:, 1] - a_ls - b_ls * XY[:, 0]
se = np.sqrt(resid @ resid / (n - 2) / np.sum((XY[:, 0] - XY[:, 0].mean())**2))
lam, U = np.linalg.eigh(Sigma)
slope_major = U[1, 1] / U[0, 1]
levels = [1, 4, 9]
gx, gy = np.meshgrid(np.linspace(-3, 5, 300), np.linspace(-4, 8, 300))
D = np.stack([gx - mx, gy - my], axis=-1)
D2 = np.einsum("...i,ij,...j->...", D, np.linalg.inv(Sigma), D)
xs = np.linspace(-2.2, 4.2, 2)
ys = np.linspace(-3.0, 7.0, 2)

mu = [F(1), F(0), F(-1)]
S = [[F(4), F(1), F(2)],
      [F(1), F(3), F(2)],
```

```

[F(2), F(2), F(2)]

def inv2(M):
    det = M[0][0] * M[1][1] - M[0][1] * M[1][0]
    return [[M[1][1] / det, -M[0][1] / det], [-M[1][0] / det, M[0][0] / det]]

# Y1 given (Y2, Y3): B = sigma_12 Sigma_22^{-1}, variance sigma_11 - B sigma_21
S22inv = inv2([[S[1][1], S[1][2]], [S[2][1], S[2][2]]])
s12 = [S[0][1], S[0][2]]
B = [s12[0] * S22inv[0][j] + s12[1] * S22inv[1][j] for j in range(2)]
var_1_23 = S[0][0] - (B[0] * s12[0] + B[1] * s12[1])
const = mu[0] - B[0] * mu[1] - B[1] * mu[2]
print("E(Y1 | y2, y3) =", const, "+", B[0], "y2 +", B[1], "y3; variance", var_1_23)

# (Y1, Y2) given Y3, and the partial correlation of Y1 and Y2 given Y3
C = [[S[i][j] - S[i][2] * S[2][j] / S[2][2] for j in range(2)] for i in range(2)]
print("Cov((Y1, Y2) | Y3) =", C)

```

3.5.4 The conditional mean is the best predictor

Section 6.11 defines the best linear predictor $L(\mathbf{Y}_1 | \mathbf{Y}_2)$ as a projection, and shows that the conditional mean is the best predictor among *all* square-integrable functions of $\mathbf{Y}_2$. For a scalar response, the gap between the two is the approximation error of Proposition 6.11.2(c). Under joint normality the gap is zero.

Proposition 3.5.3

Let $\mathbf{Y}$ be jointly normal and partitioned as above, and let $\mathbf{m}(\mathbf{Y}_2) = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$. For every function $\mathbf{g}$ with $E\|\mathbf{g}(\mathbf{Y}_2)\|^2 < \infty$,

$$E\|\mathbf{Y}_1 - \mathbf{g}(\mathbf{Y}_2)\|^2 = \text{tr}(\boldsymbol{\Sigma}_{11.2}) + E\|\mathbf{m}(\mathbf{Y}_2) - \mathbf{g}(\mathbf{Y}_2)\|^2.$$

So $\mathbf{m}(\mathbf{Y}_2)$ minimizes the mean squared prediction error over all predictors, it equals the best linear predictor of Theorem 6.11.1 (entry by entry, when $\boldsymbol{\Sigma}_{22}$ is positive definite), and the minimum is $\text{tr}(\boldsymbol{\Sigma}_{11.2})$.

Proof. $\mathbf{Y}_1 - \mathbf{g}(\mathbf{Y}_2) = \mathbf{W} + (\mathbf{m}(\mathbf{Y}_2) - \mathbf{g}(\mathbf{Y}_2))$ with $\mathbf{W}$ as in Theorem 3.5.2. The cross term $E[\mathbf{W}^\top(\mathbf{m}(\mathbf{Y}_2) - \mathbf{g}(\mathbf{Y}_2))]$ vanishes, because $\mathbf{W}$ has mean zero and is independent of $\mathbf{Y}_2$. And $E\|\mathbf{W}\|^2 = \text{tr Cov}(\mathbf{W}) = \text{tr}(\boldsymbol{\Sigma}_{11.2})$. The minimizer $\mathbf{m}$ is affine, so it is also the best affine predictor, which is the best linear predictor. ■

The proof is short because independence of $\mathbf{W}$ and $\mathbf{Y}_2$ is much stronger than the zero correlation that defines a projection. For non-normal vectors only zero correlation is available, and the best linear predictor is merely the best *linear* predictor.

3.5.5 Exercises

3.5.5.1 A. Check your understanding

Exercise A1.

For $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ of [Example \(Holding a third variable fixed\)](#), find the conditional distribution of (Y_1, Y_2) given $Y_3 = y_3$ and of Y_3 given $(Y_1, Y_2) = (y_1, y_2)$. Verify that the conditional variances are smaller than the unconditional ones.

Exercise A2.

In the setting of [Eq. 3.5.2](#), let $\boldsymbol{\Omega}$ be the inverse of $\text{Cov}((Y, \mathbf{X}^\top)^\top)$. Show that the row of $\boldsymbol{\Omega}$ belonging to Y is $(1, -\boldsymbol{\beta}^\top)/\sigma^2$, where $\sigma^2 = \sigma_Y^2(1 - \rho_{Y \cdot X}^2)$. So the regression coefficients and the error variance can be read off one row of the precision matrix.

Solution. Put Y first and apply [Exercise \(B5\)](#) with the scalar block Y and $\mathbf{Y}_2 = \mathbf{X}$. The conditional variance is $\omega_{YY}^{-1} = \text{Var}(Y | \mathbf{X}) = \sigma^2$, and the coefficient vector of the conditional mean is $-\omega_{YY}^{-1}\boldsymbol{\Omega}_{YX} = \boldsymbol{\beta}^\top$. Hence $\omega_{YX} = 1/\sigma^2$ and $\boldsymbol{\Omega}_{YX} = -\boldsymbol{\beta}^\top/\sigma^2$. By [Eq. 3.5.2](#), $\sigma^2 = \sigma_Y^2(1 - \rho_{Y \cdot X}^2)$.

3.5.5.2 B. Practice

Exercise B1.

Suppose $\mathbf{Y}_2 \sim N(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$ and that, given $\mathbf{Y}_2 = \mathbf{y}_2$, $\mathbf{Y}_1$ is $N(\mathbf{a} + \mathbf{B}\mathbf{y}_2, \mathbf{S})$ with $\mathbf{a}$, $\mathbf{B}$ and $\mathbf{S}$ not depending on $\mathbf{y}_2$. Show that $(\mathbf{Y}_1^\top, \mathbf{Y}_2^\top)^\top$ is jointly normal and find its mean and covariance. Explain why [Example \(A mixture of two correlations\)](#) does not contradict this.

Solution. Condition on $\mathbf{Y}_2$:

$$\mathbb{E} e^{\mathbf{t}_1^\top \mathbf{Y}_1 + \mathbf{t}_2^\top \mathbf{Y}_2} = \mathbb{E} \left[e^{\mathbf{t}_2^\top \mathbf{Y}_2} e^{\mathbf{t}_1^\top (\mathbf{a} + \mathbf{B}\mathbf{Y}_2) + \frac{1}{2} \mathbf{t}_1^\top \mathbf{S} \mathbf{t}_1} \right] = e^{\mathbf{t}_1^\top \mathbf{a} + \frac{1}{2} \mathbf{t}_1^\top \mathbf{S} \mathbf{t}_1} M_{\mathbf{Y}_2}(\mathbf{t}_2 + \mathbf{B}^\top \mathbf{t}_1).$$

Inserting $M_{\mathbf{Y}_2}$ from [Theorem 3.2.1](#), the exponent is linear plus $\frac{1}{2} \mathbf{t}^\top \mathbf{C} \mathbf{t}$ with $\mathbf{C} = \begin{pmatrix} \mathbf{S} + \mathbf{B}\boldsymbol{\Sigma}_{22}\mathbf{B}^\top & \mathbf{B}\boldsymbol{\Sigma}_{22} \\ \boldsymbol{\Sigma}_{22}\mathbf{B}^\top & \boldsymbol{\Sigma}_{22} \end{pmatrix}$, so the pair is normal with mean $((\mathbf{a} + \mathbf{B}\boldsymbol{\mu}_2)^\top, \boldsymbol{\mu}_2^\top)^\top$ and covariance $\mathbf{C}$. In [Example \(A mixture of two correlations\)](#), the conditional distribution of Y_2 given Y_1 is a mixture, not a normal distribution, and its variance depends on y_1 .

Exercise B2.

Show that for a bivariate normal density with $|\rho| < 1$, the points of each contour ellipse at which the tangent is vertical lie on the line $y = \mathbb{E}(Y | X = x)$. Explain the result in terms of the conditional density of Y given $X = x$.

Solution. Write the ellipse as $\{\boldsymbol{\mu} + \mathbf{u} : \mathbf{u}^\top \boldsymbol{\Sigma}^{-1} \mathbf{u} = c^2\}$. At a point with vertical tangent, the normal vector $\boldsymbol{\Sigma}^{-1} \mathbf{u}$ is horizontal, so $\boldsymbol{\Sigma}^{-1} \mathbf{u} = \lambda(1, 0)^\top$ and $\mathbf{u} = \lambda(\sigma_X^2, \rho\sigma_X\sigma_Y)^\top$. The slope $u_2/u_1 = \rho\sigma_Y/\sigma_X$ is that of the conditional mean line, and both such points ($\lambda > 0$

and $\lambda < 0$) lie on it. Along the vertical line $X = x$, the joint density is proportional to the conditional density of Y , which is highest at $E(Y | X = x)$. The contour that just touches the vertical line touches it at the point of highest density on that line.

Exercise B3.

Let $Z_1, \dots, Z_n$ be independent $N(0, 1)$ and $S_k = Z_1 + \dots + Z_k$. Show that $(S_1, \dots, S_n)$ is jointly normal with $\text{Cov}(S_j, S_k) = \min(j, k)$. Find the conditional distribution of S_j given $S_n = s$, and comment on its variance as a function of j .

Solution. $(S_1, \dots, S_n)^\top = \mathbf{L}\mathbf{Z}$ with $\mathbf{L}$ lower triangular and all entries on and below the diagonal equal to one, so the vector is normal with covariance $\mathbf{L}\mathbf{L}^\top$, whose (j, k) entry counts the common indices, $\min(j, k)$. By [Theorem 3.5.2](#) with $\mathbf{Y}_1 = S_j$, $\mathbf{Y}_2 = S_n$,

$$S_j | S_n = s \sim N\left(\frac{j}{n}s, j - \frac{j^2}{n}\right) = N\left(\frac{j}{n}s, \frac{j(n-j)}{n}\right).$$

The conditional mean interpolates linearly between 0 and s . The variance is zero at both ends, where the value is known, and largest in the middle.

Exercise B4.

Let (Y, X_1, X_2) be normal with positive definite covariance, and let $\mathbf{Y}_2 = (X_1, X_2, X_1 + X_2)^\top$. Show that Σ_{22} is singular. Using a generalized inverse, show that the conditional distribution of Y given $\mathbf{Y}_2$ from [Theorem 3.5.2](#) coincides with its conditional distribution given (X_1, X_2) .

Exercise B5.

With $\Omega = \Sigma^{-1}$ partitioned like Σ , show that $E(\mathbf{Y}_1 | \mathbf{Y}_2 = \mathbf{y}_2) = \boldsymbol{\mu}_1 - \Omega_{11}^{-1}\Omega_{12}(\mathbf{y}_2 - \boldsymbol{\mu}_2)$ and $\text{Cov}(\mathbf{Y}_1 | \mathbf{Y}_2) = \Omega_{11}^{-1}$.

Solution. By [Theorem 1.6.4](#), $\Omega_{11} = \Sigma_{11.2}^{-1}$ and $\Omega_{12} = -\Sigma_{11.2}^{-1}\Sigma_{12}\Sigma_{22}^{-1}$. Hence $-\Omega_{11}^{-1}\Omega_{12} = \Sigma_{12}\Sigma_{22}^{-1}$ and $\Omega_{11}^{-1} = \Sigma_{11.2}$. Substitute into [Eq. 3.5.1](#).

3.5.5.3 C. Going deeper

Exercise C1.

Assume Σ is positive definite. Prove [Theorem 3.5.2\(b\)](#) by dividing the joint density by the marginal density of $\mathbf{Y}_2$ directly, using the partitioned inverse and the block determinant formula.

3.6 Partial correlation

Two variables can be strongly correlated only because both depend on a third. [Example \(Holding a third variable fixed\)](#) showed a positive association between Y_1 and Y_2 turning negative once Y_3 is held fixed. The conditional covariance matrix $\Sigma_{11 \cdot 2}$ measures association that remains after conditioning, and its correlation form is the partial correlation.

3.6.1 Definition and interpretations

Definition 3.6.1 — Partial correlation

Let $\mathbf{Y} = (\mathbf{Y}_1^\top, \mathbf{Y}_2^\top)^\top$ be a random vector with covariance matrix Σ , partitioned as in [Section 3.5](#), and write $\Sigma_{11 \cdot 2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} = (\sigma_{ij \cdot 2})$. The **partial covariance** of Y_i and Y_j (two entries of $\mathbf{Y}_1$) given $\mathbf{Y}_2$ is $\sigma_{ij \cdot 2}$, and their **partial correlation** given $\mathbf{Y}_2$ is

$$\rho_{ij \cdot 2} = \frac{\sigma_{ij \cdot 2}}{\sqrt{\sigma_{ii \cdot 2} \sigma_{jj \cdot 2}}},$$

provided $\sigma_{ii \cdot 2}$ and $\sigma_{jj \cdot 2}$ are positive. When the conditioning variables are $Y_k, Y_l, \dots$ we also write $\rho_{ij \cdot kl \dots}$. The matrix of partial correlations is $\mathbf{D}^{-1/2}\Sigma_{11 \cdot 2}\mathbf{D}^{-1/2}$ with $\mathbf{D} = \text{diag}(\Sigma_{11 \cdot 2})$, the correlation matrix ([Definition 2.4.1](#)) built from $\Sigma_{11 \cdot 2}$.

The definition uses only Σ , so partial correlations exist for any random vector with finite second moments. The next result gives them two meanings, one that always applies and one that needs normality.

Proposition 3.6.1

In the setting of [Definition 3.6.1](#):

- (a) $\Sigma_{11 \cdot 2}$ is the covariance matrix of the prediction errors $\mathbf{W} = \mathbf{Y}_1 - \boldsymbol{\mu}_1 - \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$, and $\mathbf{W}$ is uncorrelated with $\mathbf{Y}_2$. If Σ_{22} is positive definite, the entries of $\mathbf{W}$ are the errors of the best linear predictors $L(Y_i | \mathbf{Y}_2)$, so $\rho_{ij \cdot 2}$ is the correlation between $Y_i - L(Y_i | \mathbf{Y}_2)$ and $Y_j - L(Y_j | \mathbf{Y}_2)$.
- (b) If $\mathbf{Y}$ is jointly normal, $\rho_{ij \cdot 2}$ is the correlation of Y_i and Y_j in their conditional distribution given $\mathbf{Y}_2 = \mathbf{y}_2$, the same for every $\mathbf{y}_2$. Moreover $\rho_{ij \cdot 2} = 0$ iff Y_i and Y_j are conditionally independent given $\mathbf{Y}_2$.

Proof.

- (a) This is [Theorem 2.5.2](#), restated: $\text{Cov}(\mathbf{W}, \mathbf{Y}_2) = \mathbf{0}$ and $\text{Cov}(\mathbf{W}) = \Sigma_{11 \cdot 2}$ are the moment calculations in the proof of [Theorem 3.5.2\(a\)](#), which use no normality. When Σ_{22} is positive definite, the i th entry of $\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$ is $L(Y_i | \mathbf{Y}_2)$ by [Theorem 6.11.1](#).
- (b) By [Theorem 3.5.2](#), the conditional distribution of (Y_i, Y_j) given $\mathbf{Y}_2 = \mathbf{y}_2$ is bivariate normal with covariance matrix the corresponding 2×2 block of $\Sigma_{11 \cdot 2}$. Its correlation is $\rho_{ij \cdot 2}$, and by [Theorem 3.4.1](#) applied to this conditional normal distribution, the two are independent iff the off-diagonal entry is zero.

Part (a) is the definition to remember: *a partial correlation is the correlation of two residuals*. Remove from each variable the part that is linearly predictable from the conditioning variables, and correlate what is left. This is the population version of the Frisch–Waugh–Lovell construction ([Theorem 6.6.2](#)).

3.6.2 Computing partial correlations

Two routes avoid computing $\Sigma_{11 \cdot 2}$ in full.

With a single conditioning variable, $\Sigma_{11 \cdot 2}$ is a 2×2 matrix with entries $\sigma_{ij} - \sigma_{i3}\sigma_{j3}/\sigma_{33}$, and in correlation units this is the familiar three-variable formula of [Exercise \(B1\)](#),

$$\rho_{12 \cdot 3} = \frac{\rho_{12} - \rho_{13}\rho_{23}}{\sqrt{(1 - \rho_{13}^2)(1 - \rho_{23}^2)}}, \quad |\rho_{13}| < 1, \quad |\rho_{23}| < 1. \quad (3.6.1)$$

The residual interpretation shows that the same formula can be applied one conditioning variable at a time.

Proposition 3.6.2 — Recursion

Let S be a set of indices and $k \notin S \cup \{1, 2\}$ one more. Suppose $\sigma_{11 \cdot S}$, $\sigma_{22 \cdot S}$ and $\sigma_{kk \cdot S}$ are positive and $|\rho_{1k \cdot S}| < 1$, $|\rho_{2k \cdot S}| < 1$. Then

$$\sigma_{12 \cdot Sk} = \sigma_{12 \cdot S} - \frac{\sigma_{1k \cdot S} \sigma_{2k \cdot S}}{\sigma_{kk \cdot S}}, \quad \rho_{12 \cdot Sk} = \frac{\rho_{12 \cdot S} - \rho_{1k \cdot S} \rho_{2k \cdot S}}{\sqrt{(1 - \rho_{1k \cdot S}^2)(1 - \rho_{2k \cdot S}^2)}}.$$

With S empty this is [Eq. 3.6.1](#).

Proof. No normality is needed. For $i \in \{1, 2, k\}$ let e_i be the error of the best linear predictor of Y_i from $\mathbf{Y}_S$, as in [Proposition 3.6.1\(a\)](#). Then each e_i has mean zero, is uncorrelated with $\mathbf{Y}_S$, and $\text{Cov}(e_i, e_j) = \sigma_{ij \cdot S}$. Put $r_i = e_i - c_i e_k$ with $c_i = \sigma_{ik \cdot S}/\sigma_{kk \cdot S}$, for $i = 1, 2$.

We claim that r_i is the error of the best linear predictor of Y_i from $(\mathbf{Y}_S, Y_k)$. First, $Y_i - r_i$ is an affine function of $(\mathbf{Y}_S, Y_k)$, because e_i and e_k differ from Y_i and Y_k by affine functions of $\mathbf{Y}_S$. Second, r_i has mean zero and is uncorrelated with $\mathbf{Y}_S$ and with e_k , by the choice of c_i , hence with $Y_k = e_k + (\text{affine in } \mathbf{Y}_S)$. Now let r'_i be the error from [Proposition 3.6.1\(a\)](#) for predicting Y_i from $(\mathbf{Y}_S, Y_k)$, which has the same two properties. The difference $d = r_i - r'_i$ is an affine function of $(\mathbf{Y}_S, Y_k)$ with mean zero and is uncorrelated with $(\mathbf{Y}_S, Y_k)$, so $\text{Var}(d) = \text{Cov}(d, d) = 0$. Thus $r_i = r'_i$ with probability one.

Therefore $\sigma_{12 \cdot Sk} = \text{Cov}(r_1, r_2) = \sigma_{12 \cdot S} - c_2 \sigma_{1k \cdot S} - c_1 \sigma_{2k \cdot S} + c_1 c_2 \sigma_{kk \cdot S}$, which simplifies to the first formula, and similarly $\sigma_{ii \cdot Sk} = \sigma_{ii \cdot S}(1 - \rho_{ik \cdot S}^2)$, which is positive by hypothesis. Dividing $\sigma_{12 \cdot Sk}$ by $\sqrt{\sigma_{11 \cdot Sk} \sigma_{22 \cdot Sk}}$ and then numerator and denominator by $\sqrt{\sigma_{11 \cdot S} \sigma_{22 \cdot S}}$ gives the second.

Proposition 3.6.3 — Partial correlations from the precision matrix

Let Σ be positive definite and $\Omega = \Sigma^{-1} = (\omega_{ij})$. The partial correlation of Y_i and Y_j given all the other entries of $\mathbf{Y}$ is

$$\rho_{ij \cdot \text{rest}} = -\frac{\omega_{ij}}{\sqrt{\omega_{ii}\omega_{jj}}},$$

and the conditional variance of Y_i given all the others is $1/\omega_{ii}$.

Proof. Permute $\mathbf{Y}$ so that (Y_i, Y_j) comes first. By [Theorem 3.5.2\(c\)](#) (a statement about matrices, valid whether or not $\mathbf{Y}$ is normal), the leading 2×2 block of Ω is $\Sigma_{11 \cdot 2}^{-1}$. Inverting,

$$\Sigma_{11 \cdot 2} = \begin{pmatrix} \omega_{ii} & \omega_{ij} \\ \omega_{ij} & \omega_{jj} \end{pmatrix}^{-1} = \frac{1}{\omega_{ii}\omega_{jj} - \omega_{ij}^2} \begin{pmatrix} \omega_{jj} & -\omega_{ij} \\ -\omega_{ij} & \omega_{ii} \end{pmatrix},$$

whose correlation is $-\omega_{ij}/\sqrt{\omega_{ii}\omega_{jj}}$. The same argument with a 1×1 leading block gives the conditional variance. ■

[Proposition 3.6.3](#) has a striking consequence for normal vectors: *a zero entry ω_{ij} of the inverse covariance matrix means that Y_i and Y_j are conditionally independent given all the other variables.* Zeros in Σ describe marginal independence, zeros in Σ^{-1} describe conditional independence. The pattern of zeros in Σ^{-1} defines a graph on the variables, which is the starting point of Gaussian graphical models (Dempster 1972; Lauritzen 1996).

Example. (The trivariate example, continued)

For Σ of [Example \(Holding a third variable fixed\)](#), the listing gives the covariance of (Y_1, Y_2) given Y_3 as $\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$. So $\rho_{12} = 0.289$ while $\rho_{12 \cdot 3} = -0.707$. The script confirms that [Eq. 3.6.1](#), [Proposition 3.6.3](#), and the correlation of simulated residuals all give this value.

3.6.3 Sample partial correlations

For data, replace Σ by the sample covariance matrix. Because the empirical distribution of the rows is a probability distribution in its own right ([Section 6.11](#)), all the identities above hold exactly for the sample versions. In particular, for two variables $\mathbf{y}, \mathbf{x}$ and conditioning variables collected with an intercept in $\mathbf{Z}$, the following numbers are *identical*:

- (1) the correlation of the residuals of $\mathbf{y}$ and of $\mathbf{x}$ after least squares regression on $\mathbf{Z}$;
- (2) the recursion [Eq. 3.6.1](#) applied to sample correlations (one conditioning variable) or repeatedly (several);
- (3) $-w_{12}/\sqrt{w_{11}w_{22}}$ computed from the inverse of the sample covariance or correlation matrix of $(\mathbf{y}, \mathbf{x}, \text{conditioning variables})$;
- (4) $\text{sign}(t)\sqrt{t^2/(t^2 + n - p)}$, where t is the t statistic of $\mathbf{x}$ in the regression of $\mathbf{y}$ on $[\mathbf{Z}, \mathbf{x}]$ with p columns ([Exercise \(B3\)](#)).

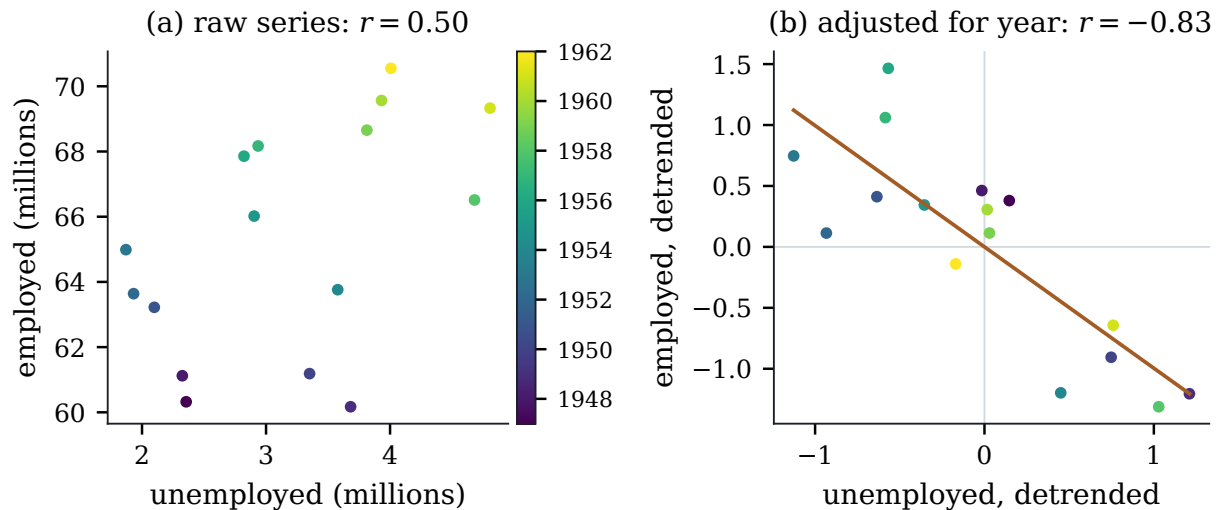


Figure 3.6.1: Employment and unemployment in the United States, 1947–1962, coloured by year. (a) The raw series are positively correlated because both trend upwards. (b) After removing the linear trend in year from each, the residuals are strongly negatively correlated. Their correlation is the partial correlation given year, and the fitted slope is the multiple-regression coefficient.

Example. (Employment and unemployment)

Longley’s macroeconomic series record, among other things, total employment and the number unemployed in the United States for the 16 years 1947–1962. Their correlation is 0.502: years with more people employed had more people unemployed. Both series grew with the population. Their correlations with the calendar year are 0.971 and 0.668. Adjusting for year, the partial correlation is -0.829 . Once the common trend is removed, a year with unemployment above its trend is a year with employment below its trend, as one would expect. Figure 3.6.1 shows both scatterplots. The listing computes the partial correlation in all four ways above. They agree to ten decimal places. The slope in panel (b), -0.996 , is the coefficient of unemployment in the regression of employment on unemployment and year, as Theorem 6.6.2 requires. Its t statistic is -5.34 .

Removing a linear time trend before correlating two series is the problem that Frisch and Waugh (1933) solved, and it is a special case of partial correlation.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.longley.load_pandas().data

emp = data["TOTEMP"].to_numpy() / 1000      # thousands -> millions of persons
unemp = data["UNEMP"].to_numpy() / 1000
year = data["YEAR"].to_numpy()
n = len(emp)
Z = np.column_stack([np.ones(n), year])     # the variables we adjust for

def resid(v, Z):
    coef, *_ = np.linalg.lstsq(Z, v, rcond=None)
    return v - Z @ coef

# 1. correlation of residuals (detrended series)
r_resid = np.corrcoef(resid(emp, Z), resid(unemp, Z))[0, 1]
```

```

# 2. the one-variable recursion from ordinary correlations
R = np.corrcoef(np.vstack([emp, unemp, year]))
r12, r13, r23 = R[0, 1], R[0, 2], R[1, 2]
r_formula = (r12 - r13 * r23) / np.sqrt((1 - r13**2) * (1 - r23**2))

# 3. from the inverse of the correlation (or covariance) matrix
W = np.linalg.inv(R)
r_precision = -W[0, 1] / np.sqrt(W[0, 0] * W[1, 1])

# 4. from the t statistic of unemployment in the regression of emp on (1, unemp,
    year)
fit = sm.OLS(emp, np.column_stack([Z, unemp])).fit()
t = fit.tvalues[2]
r_t = np.sign(t) * np.sqrt(t**2 / (t**2 + n - 3))

print(f"marginal r = {r12:.4f}")
print(f"partial r:  {r_resid:.10f} {r_formula:.10f} {r_precision:.10f} {r_t:.10f}")

```

Remark (Adjusting is not always right). A partial correlation answers a precise question: how are the parts of two variables not linearly explained by $\mathbf{Y}_2$ related? Whether that is the *right* question depends on what $\mathbf{Y}_2$ is. Adjusting for a common cause, as the year stands in for population growth here, removes a spurious association. Adjusting for a common effect can *create* an association between variables that are independent. Chapter 25 treats the difference. Under normality, the sampling distribution of a sample partial correlation given q variables is that of an ordinary sample correlation from $n - q$ observations, a result used for tests in Chapter 14.

3.6.4 Exercises

3.6.4.1 A. Check your understanding

Exercise A1.

Show that $\rho_{12 \cdot 3} = 0$ iff $\rho_{12} = \rho_{13}\rho_{23}$. Find $\rho_{12 \cdot 3}$ when $\rho_{12} = 0$ and $\rho_{13} = \rho_{23} = 0.6$, and interpret.

3.6.4.2 B. Practice

Exercise B1.

Let X , Z and e be independent, with $X, Z \sim N(0, 1)$ and $e \sim N(0, \tau^2)$, and let $C = X + Z + e$. Show that $\rho_{XZ} = 0$ but $\rho_{XZ \cdot C} = -1/(1 + \tau^2)$. What happens as $\tau \rightarrow 0$ and as $\tau \rightarrow \infty$?

Solution. The covariance matrix of (X, Z, C) is $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 + \tau^2 \end{pmatrix}$. Conditioning on C ,

$$\sigma_{XZ \cdot C} = 0 - \frac{1}{2 + \tau^2}, \quad \sigma_{XX \cdot C} = \sigma_{ZZ \cdot C} = 1 - \frac{1}{2 + \tau^2} = \frac{1 + \tau^2}{2 + \tau^2},$$

so $\rho_{XZ \cdot C} = -1/(1 + \tau^2)$. As $\tau \rightarrow 0$ it tends to -1 : if $X + Z$ is known exactly, a larger

X forces a smaller Z . As $\tau \rightarrow \infty$ it tends to 0, since C carries little information about $X + Z$. Two independent causes become dependent once their common effect is held fixed.

Exercise B2.

Let Σ be positive definite. Show that the coefficient of Y_j in the best linear predictor of Y_1 from all the other entries is $-\omega_{1j}/\omega_{11}$. Deduce that $\rho_{1j \cdot \text{rest}}^2$ equals the product of the coefficient of Y_j for predicting Y_1 and the coefficient of Y_1 for predicting Y_j , each from all the other variables.

Solution. Put Y_1 first. By [Exercise \(B5\)](#) with a scalar first block, the coefficient vector of the best linear predictor of Y_1 from the rest is $-\Omega_{12}/\omega_{11}$, whose j th entry is $-\omega_{1j}/\omega_{11}$. Likewise the coefficient of Y_1 in the predictor of Y_j is $-\omega_{j1}/\omega_{jj}$. Their product is $\omega_{1j}^2/(\omega_{11}\omega_{jj})$, which is $\rho_{1j \cdot \text{rest}}^2$ by [Proposition 3.6.3](#).

Exercise B3.

Let D be an $n \times k$ data matrix whose columns are centred, and let $S = D^\top D$. Show, using [Theorem 6.6.2](#) and the partitioned inverse, that $-w_{12}/\sqrt{w_{11}w_{22}}$, computed from $S^{-1} = (w_{ij})$, equals the correlation of the residuals of columns 1 and 2 after regression on the remaining columns.

Exercise B4.

Using the Longley data of [Example \(Employment and unemployment\)](#), compute the partial correlation of employment and unemployment given both year and GNP, in at least two of the four ways. Draw the corresponding residual scatterplot. Does adjusting for GNP as well as year change the conclusion?

3.7 Summary and Notes

3.7.1 Summary

Idea.

1. $\mathbf{Y}$ is multivariate normal when every linear combination $\mathbf{a}^\top \mathbf{Y}$ is normal, constants included ([Definition 3.1.1](#)). The definition covers singular covariance matrices, which regression needs for fitted values and residuals.
2. $\boldsymbol{\mu} + \mathbf{AZ}$, with $\mathbf{Z}$ standard normal, is $N(\boldsymbol{\mu}, \mathbf{AA}^\top)$. Every mean and every nonnegative definite covariance occurs ([Theorem 3.1.2](#)).
3. The moment generating function is $\exp(\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \Sigma \mathbf{t})$, so a normal distribution is determined by $\boldsymbol{\mu}$ and Σ ([Theorem 3.2.1](#)). When Σ is positive definite there is a density whose contours are the ellipsoids of constant Mahalanobis distance, and $\Delta^2(\mathbf{Y}) \sim \chi^2(n)$ ([Theorem 3.2.2](#) and [Proposition 3.2.3](#)).

4. Affine maps of normal vectors are normal, with no rank condition, and so are all subvectors ([Theorem 3.3.1](#) and [Corollary 3.3.2](#)). A normal vector of rank r is an exact affine image of r independent standard normals ([Proposition 3.3.4](#)).
5. Normal margins do not imply joint normality, and zero correlation between normal variables does not imply independence without it ([Example \(A random sign\)](#) and [Example \(A mixture of two correlations\)](#)).
6. For jointly normal blocks, zero cross-covariance is equivalent to independence. $\mathbf{AY}$ and $\mathbf{BY}$ are independent iff $\mathbf{A}\mathbf{\Sigma}\mathbf{B}^\top = \mathbf{0}$ ([Theorem 3.4.1](#) and [Corollary 3.4.2](#)).
7. $\mathbf{Y}_1 \mid \mathbf{Y}_2 = \mathbf{y}_2 \sim \mathcal{N}(\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^-(\mathbf{y}_2 - \boldsymbol{\mu}_2), \boldsymbol{\Sigma}_{11.2})$: linear mean, constant covariance, reduced variance ([Theorem 3.5.2](#)). The conditional mean is the best predictor of all, and coincides with the best linear predictor ([Proposition 3.5.3](#)).
8. The partial correlation is the correlation of the residuals from linear prediction on the conditioning variables. Under normality it is the conditional correlation, and it is zero iff the variables are conditionally independent ([Definition 3.6.1](#) and [Proposition 3.6.1](#)). It can be computed by a recursion, from the inverse covariance matrix, or by FWL residuals ([Proposition 3.6.2](#) and [Proposition 3.6.3](#)).

3.7.2 Notes and sources

Three definitions. Linear-model texts reach the multivariate normal by different routes, and comparing them explains the choices made here. Rencher and Schaalje (2008, chapter 4) start from the positive definite case, obtain the density by transforming independent standard normals with $\boldsymbol{\Sigma}^{1/2}$, and prove the properties of linear functions and independence with moment generating functions, for full-rank transformations. Seber and Lee (2003, chapter 2) also begin with the density, then extend the definition to singular covariance matrices through the representation $\mathbf{AZ} + \boldsymbol{\mu}$ and prove that normality of all linear combinations characterizes the family. Their moment-generating-function proof of the independence theorem is the model for [Section 3.4](#). Christensen (2020, §1.2) takes $\mathbf{AZ} + \boldsymbol{\mu}$ as the definition from the outset, shows with characteristic functions that the law depends only on $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$, and proves the independence theorem by building an explicit independent version. Defining the distribution through linear combinations, as this chapter does, goes back at least to Rao (1973, chapter 8). It has the advantage that the theorem on affine transformations needs no rank condition and a one-line proof.

Cramér–Wold. The device of reducing a vector to its linear combinations is from Cramér and Wold (1936), where it was used to reduce multivariate limit theorems to univariate ones. [Chapter 2](#) proves it, together with the uniqueness and factorization properties of moment generating functions used in [Section 3.2](#). Billingsley (1995) is the reference for these facts and for the conditioning property of independent variables used in the proof of [Theorem 3.5.2](#).

Conditional distributions and partial correlation. The proof of [Theorem 3.5.2](#) through the independent error $\mathbf{W}$ borrows one idea from Rencher and Schaalje (2008) and Seber and Lee (2003): both pass through the transformed vector $\mathbf{Y}_1 - \mathbf{BY}_2$, which is independent of $\mathbf{Y}_2$, although both still obtain the conditional density as a ratio of densities in the positive definite case. The proof here uses only independence and a generalized inverse, so it covers singular $\boldsymbol{\Sigma}_{22}$, an extension that is standard in multivariate analysis (Rao 1973). Anderson (2003) is the classical reference for the multivariate normal in statistics, including the distribution theory of sample partial and multiple correlation coefficients that Chapter 14 uses. Rencher and Schaalje (2008, §4.5) define partial correlation as a conditional correlation, as in [Proposition 3.6.1\(b\)](#).

The residual interpretation (a) does not require normality and is the one that connects to least squares. Regression toward the mean was described by Galton (1886). The link between zeros of the inverse covariance matrix and conditional independence was made the basis of a modelling strategy by Dempster (1972), and Lauritzen (1996) develops the theory of graphical models that grew from it.

Data. [Example \(Employment and unemployment\)](#) uses the macroeconomic series published by Longley (1967) as a test of numerical accuracy, available in `statsmodels`. The same series appear in [Chapter 2](#), as an example of sample moments, and in [Chapter 6](#), to illustrate ill-conditioning.

3.7.3 References

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Chapter 4

Quadratic Forms and Their Distributions

Every sum of squares in a linear model is a quadratic form $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ in a normal vector, and every F and t statistic is a ratio built from such forms. This chapter works out their laws: the noncentral chi-squared, F and t distributions and the monotonicity of their tails; the mean, variance and a canonical representation of an arbitrary quadratic form; exactly when a quadratic form is chi-squared; when two forms, or a form and a linear function, are independent; and Cochran's theorem, which says when a sum of squares splits into independent chi-squared pieces.

What you need. [Chapter 1](#) (spectral theorem, symmetric idempotent matrices, trace), [Chapter 2](#) (means and covariances of random vectors) and [Chapter 3](#) (the multivariate normal distribution and its independence properties). Orthogonal projections appear here only as symmetric idempotent matrices. [Chapter 6](#) develops their geometry.

4.0.1 Roadmap

- [Noncentral chi-squared, t and F distributions](#)
- [Mean and variance of quadratic forms](#)
- [When a quadratic form is chi-squared](#)
- [Independence of quadratic forms and linear forms](#)
- [The Cochran theorem](#)
- [Summary and notes](#)

4.1 Noncentral chi-squared, t and F distributions

A regression analysis reports a handful of numbers whose sampling behaviour decides everything: the residual sum of squares, the reduction in it when a term is added, the ratio of the two. Each

is a function of the response vector of the form

$$\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} Y_i Y_j,$$

a **quadratic form** in $\mathbf{Y}$. The sample variance is one, with $\mathbf{A} = (n-1)^{-1}(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)$. So is the squared length $\|\mathbf{P}\mathbf{Y}\|^2 = \mathbf{Y}^\top \mathbf{P} \mathbf{Y}$ of any symmetric idempotent image of $\mathbf{Y}$, and this is how the sums of squares of [Chapter 6](#) will arise.

Since $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ is a scalar, it equals its own transpose $\mathbf{Y}^\top \mathbf{A}^\top \mathbf{Y}$, and so also $\mathbf{Y}^\top \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top) \mathbf{Y}$. Replacing $\mathbf{A}$ by its symmetric part changes nothing, and *throughout this chapter the matrix of a quadratic form is symmetric*. Several formulas below, the variance formula among them, are false for a non-symmetric $\mathbf{A}$ ([Exercise \(A1\)](#)).

When $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$ and $\mathbf{A}$ is symmetric idempotent, $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ has a chi-squared distribution, which is the null distribution of a test. Under an alternative the mean is not zero, and a slightly larger family appears. We start with that family.

4.1.1 Noncentral chi-squared

Recall that $\chi^2(r)$, the chi-squared distribution with r degrees of freedom, is the law of $Z_1^2 + \cdots + Z_r^2$ for independent standard normal Z_i . It is the gamma distribution with shape $r/2$ and scale 2, with density

$$g_r(x) = \frac{x^{r/2-1} e^{-x/2}}{2^{r/2} \Gamma(r/2)}, \quad x > 0, \quad (4.1.1)$$

mean r , variance $2r$ and moment generating function $(1-2t)^{-r/2}$ for $t < 1/2$. By convention $\chi^2(0)$ is the point mass at zero. Everything about the noncentral version follows from one computation.

Lemma 4.1.1

Let $W \sim N(0, 1)$ and $\nu \in \mathbb{R}$. For $t < 1/2$,

$$\mathbb{E}[e^{t(W+\nu)^2}] = (1-2t)^{-1/2} \exp\left(\frac{\nu^2 t}{1-2t}\right).$$

Proof. With $u = w + \nu$ the expectation is $(2\pi)^{-1/2} \int \exp\{tu^2 - (u-\nu)^2/2\} du$. Completing the square in the exponent,

$$tu^2 - \frac{1}{2}(u-\nu)^2 = -\frac{1-2t}{2} \left(u - \frac{\nu}{1-2t}\right)^2 + \frac{\nu^2}{2} \left(\frac{1}{1-2t} - 1\right).$$

The last term equals $\nu^2 t / (1-2t)$. The remaining Gaussian integral is $(2\pi)^{1/2} (1-2t)^{-1/2}$ when $1-2t > 0$. ■

Definition 4.1.1 — Noncentral chi-squared

Let $\mathbf{Z} \sim N_r(\boldsymbol{\mu}, \mathbf{I}_r)$. The distribution of $\mathbf{Z}^\top \mathbf{Z} = \sum_i Z_i^2$ is the **noncentral chi-squared distribution** with r degrees of freedom and **noncentrality** $\gamma = \boldsymbol{\mu}^\top \boldsymbol{\mu} = \|\boldsymbol{\mu}\|^2$, written $\chi^2(r, \gamma)$.

The definition hides a claim: the law of $\|\mathbf{Z}\|^2$ depends on the mean vector $\boldsymbol{\mu}$ only through its length. [Theorem 4.1.2](#) proves it. Intuitively, a rotation $\mathbf{Q}\mathbf{Z}$ of a spherical normal is again spherical normal, with mean $\mathbf{Q}\boldsymbol{\mu}$, and it has the same length as $\mathbf{Z}$, so every mean vector of a given length gives the same answer.

Remark (Convention). In this book the noncentrality is $\gamma = \boldsymbol{\mu}^\top \boldsymbol{\mu}$, with no factor $\frac{1}{2}$. Then $E \chi^2(r, \gamma) = r + \gamma$, which is easy to remember, and γ agrees with the parameter `nc` of `scipy.stats.ncx2`. Several texts, including Rencher and Schaalje (2008) and Christensen (2020), use $\lambda = \boldsymbol{\mu}^\top \boldsymbol{\mu}/2$ instead. When reading them, replace their λ by $\gamma/2$. The same convention applies to the noncentral F distribution below.

Theorem 4.1.2 — Properties of the noncentral chi-squared

Let $U \sim \chi^2(r, \gamma)$.

- (a) The moment generating function is

$$M_U(t) = (1 - 2t)^{-r/2} \exp\left(\frac{\gamma t}{1 - 2t}\right), \quad t < \frac{1}{2}, \quad (4.1.2)$$

so the law depends on $\boldsymbol{\mu}$ only through γ , and $\chi^2(r, 0) = \chi^2(r)$.

- (b) $E(U) = r + \gamma$ and $\text{Var}(U) = 2r + 4\gamma$.
 (c) If $U_1, \dots, U_k$ are independent with $U_i \sim \chi^2(r_i, \gamma_i)$, then $\sum_i U_i \sim \chi^2(\sum_i r_i, \sum_i \gamma_i)$.
 (d) (*Poisson mixture.*) Let $K \sim \text{Poisson}(\gamma/2)$ and, given $K = k$, let $V \sim \chi^2(r + 2k)$. Then $V \sim \chi^2(r, \gamma)$. Consequently U has density

$$f_{r,\gamma}(x) = \sum_{k=0}^{\infty} e^{-\gamma/2} \frac{(\gamma/2)^k}{k!} g_{r+2k}(x), \quad x > 0. \quad (4.1.3)$$

Proof.

- (a) With $\mathbf{Z} \sim N_r(\boldsymbol{\mu}, \mathbf{I})$, the Z_i are independent, and $Z_i = W_i + \mu_i$ with $W_i \sim N(0, 1)$. By [Lemma 4.1.1](#),

$$E e^{t\mathbf{Z}^\top \mathbf{Z}} = \prod_{i=1}^r (1 - 2t)^{-1/2} \exp\left(\frac{\mu_i^2 t}{1 - 2t}\right) = (1 - 2t)^{-r/2} \exp\left(\frac{\gamma t}{1 - 2t}\right).$$

A moment generating function that is finite on an open interval around zero determines the distribution, so the law depends only on (r, γ) . Setting $\gamma = 0$ gives the central mgf.

- (b) The cumulant generating function is $K(t) = \log M_U(t) = -\frac{r}{2} \log(1 - 2t) + \gamma t/(1 - 2t)$. Differentiating, $K'(t) = r(1 - 2t)^{-1} + \gamma(1 - 2t)^{-2}$ and $K''(t) = 2r(1 - 2t)^{-2} + 4\gamma(1 - 2t)^{-3}$. The mean and variance are $K'(0)$ and $K''(0)$.

- (c) By independence the mgf of the sum is the product of the mgfs [Eq. 4.1.2](#), which has the same form with $\sum r_i$ and $\sum \gamma_i$.
- (d) Conditioning on K ,

$$\mathbb{E} e^{tV} = \sum_{k=0}^{\infty} e^{-\gamma/2} \frac{(\gamma/2)^k}{k!} (1-2t)^{-(r+2k)/2} = (1-2t)^{-r/2} e^{-\gamma/2} \exp\left(\frac{\gamma/2}{1-2t}\right),$$

and $-\frac{\gamma}{2} + \frac{\gamma}{2(1-2t)} = \frac{\gamma t}{1-2t}$. This is [Eq. 4.1.2](#). The density of a mixture is the mixture of the densities. ■

The mixture in (d) is more than a formula. It says that noncentrality acts like a random number of extra degrees of freedom, two at a time, with $\gamma/2$ of them on average. Since each extra pair of degrees of freedom pushes the distribution to the right, a larger γ should give a stochastically larger variable. That is the next result, and it is what makes tests based on these distributions sensible.

Example. (A noncentral chi-squared with four degrees of freedom)

Take $r = 4$ and $\gamma = 6$. The Poisson(3) weights of the central densities g_4, g_6, g_8, g_{10} in [Eq. 4.1.3](#) are 0.050, 0.149, 0.224 and 0.224, and the mean and variance are 10 and 32. The listing checks the mixture against a library implementation of the density. The script also squares normal vectors with a mean of length $\sqrt{6}$ in an arbitrary direction, and their average squared length is 10.002. A test that rejects when a $\chi^2(4)$ statistic exceeds its upper 5% point 9.488 rejects with probability 0.470 when the statistic is in fact $\chi^2(4, 6)$. [Figure 4.1.1](#) shows how the density moves right and spreads out as γ grows.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(20240401)

def ncx2_pdf_mixture(x, r, gamma, terms=200):
    """Density of chi^2(r, gamma) as a Poisson(gamma/2) mixture of central densities
    . """
    k = np.arange(terms)
    weights = stats.poisson.pmf(k, gamma / 2)          # P(K = k)
    dens = stats.chi2.pdf(np.asarray(x)[: , None], r + 2 * k)
    return dens @ weights

r, gamma = 4, 6.0
x = np.linspace(0.05, 40, 400)
print(np.max(np.abs(ncx2_pdf_mixture(x, r, gamma) - stats.ncx2.pdf(x, r, gamma))))
```

Proposition 4.1.3 — Monotonicity in the noncentrality

For fixed $r \geq 1$ and $x > 0$, the tail probability $\gamma \mapsto \Pr\{\chi^2(r, \gamma) > x\}$ is continuous and strictly increasing on $[0, \infty)$, and tends to 1 as $\gamma \rightarrow \infty$.

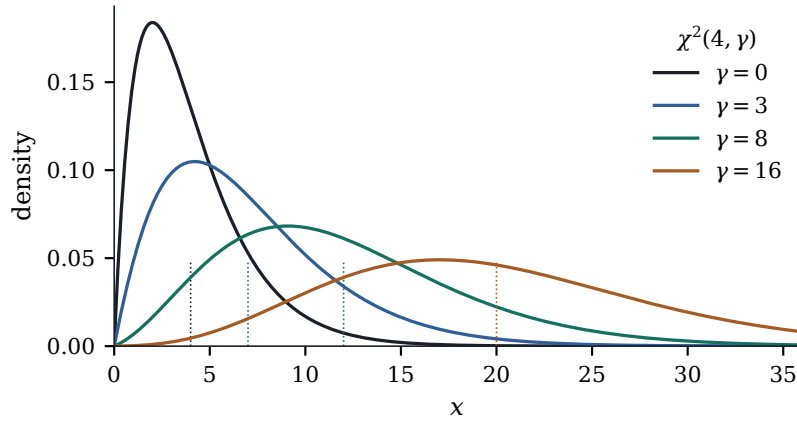


Figure 4.1.1: Densities of $\chi^2(4, \gamma)$. Dotted ticks mark the means $4 + \gamma$. The variance $2r + 4\gamma$ grows with the noncentrality, so the distributions flatten as they move right.

Proof. Let $h(k) = \Pr\{\chi^2(r + 2k) > x\}$ and $p_k(\lambda) = e^{-\lambda}\lambda^k/k!$. By Theorem 4.1.2(d), the tail probability is $\phi(\lambda) = \sum_k p_k(\lambda)h(k)$ with $\lambda = \gamma/2$. First, h is strictly increasing: if $V \sim \chi^2(r + 2k)$ and $E \sim \chi^2(2)$ are independent, then $V + E \sim \chi^2(r + 2k + 2)$ and $h(k + 1) - h(k) = \Pr(V \leq x < V + E) > 0$, since (V, E) has a positive joint density on $(0, \infty)^2$. Next, $p'_k(\lambda) = p_{k-1}(\lambda) - p_k(\lambda)$ (with $p_{-1} = 0$). Because $0 \leq h \leq 1$, the series of derivatives converges uniformly on bounded intervals, so term-by-term differentiation is valid and

$$\phi'(\lambda) = \sum_{k \geq 0} (p_{k-1}(\lambda) - p_k(\lambda))h(k) = \sum_{k \geq 0} p_k(\lambda)(h(k+1) - h(k)) > 0.$$

For the limit, fix $\varepsilon > 0$. Since $\chi^2(r + 2m)$ has mean $r + 2m$ and standard deviation $\sqrt{2r + 4m}$, Chebyshev's inequality gives $h(m) > 1 - \varepsilon$ for some m . Then $\phi(\lambda) \geq h(m) \Pr(K \geq m)$ for $K \sim \text{Poisson}(\lambda)$, and $\Pr(K \geq m) \rightarrow 1$ as $\lambda \rightarrow \infty$. ■

4.1.2 Noncentral F and t

Test statistics divide one quadratic form by another, to remove the unknown scale. The two ratio distributions that result are defined next.

Definition 4.1.2 — Noncentral F

Let $U \sim \chi^2(r, \gamma)$ and $V \sim \chi^2(s)$ be independent, $r, s \geq 1$. The law of

$$F = \frac{U/r}{V/s}$$

is the **noncentral F distribution** $F(r, s, \gamma)$, with numerator and denominator degrees of freedom r and s and noncentrality γ . The central $F(r, s) = F(r, s, 0)$ has upper α point $F_\alpha(r, s)$.

Definition 4.1.3 — Noncentral t

Let $X \sim N(\delta, 1)$ and $V \sim \chi^2(s)$ be independent. The law of $T = X/\sqrt{V/s}$ is the **noncentral t distribution** $t(s, \delta)$, and $t(s) = t(s, 0)$.

Only the numerator is allowed to be noncentral. A noncentral denominator gives a *doubly noncentral* F , which appears when the model used for the error sum of squares is itself wrong (Exercise (B3)). Two relations are immediate. First, $X^2 \sim \chi^2(1, \delta^2)$, so $T^2 \sim F(1, s, \delta^2)$: a two-sided t test is an F test with one numerator degree of freedom. Second, since $E(V^{-1}) = 1/(s-2)$ for $s > 2$, independence gives

$$E(F) = \frac{r + \gamma}{r} \cdot \frac{s}{s - 2}, \quad s > 2.$$

The central F is the null distribution of the tests of Chapter 11, and $F(r, s, \gamma)$ is their distribution under an alternative. The power of such a test is therefore $\Pr\{F(r, s, \gamma) > F_\alpha(r, s)\}$, and the following fact is what guarantees that it behaves as a test should.

Theorem 4.1.4 — Power of the F test increases with γ

For fixed $r, s \geq 1$ and $c > 0$, $\beta(\gamma) = \Pr\{F(r, s, \gamma) > c\}$ is continuous and strictly increasing in $\gamma \geq 0$, and $\beta(\gamma) \rightarrow 1$ as $\gamma \rightarrow \infty$. In particular, the level- α test that rejects when $F > F_\alpha(r, s)$ has power greater than α at every $\gamma > 0$.

Proof. With U, V as in Definition 4.1.2, $F > c$ iff $U > crV/s$. Condition on V , which is independent of U :

$$\beta(\gamma) = E[\psi_\gamma(crV/s)], \quad \psi_\gamma(x) = \Pr\{\chi^2(r, \gamma) > x\}.$$

Since $V > 0$ with probability one, Proposition 4.1.3 gives, for $\gamma < \gamma'$, $\psi_\gamma(crV/s) < \psi_{\gamma'}(crV/s)$ almost surely, so $\beta(\gamma) < \beta(\gamma')$. Continuity and the limit follow from the same proposition by dominated convergence, since $0 \leq \psi_\gamma \leq 1$. ■

Two further monotonicity facts hold for fixed $\gamma > 0$ and fixed level. The power decreases as the numerator degrees of freedom r grow, and increases as the denominator degrees of freedom s grow (Ghosh 1973). The first says that spreading a fixed amount of signal over more dimensions makes it harder to detect. The second says that estimating σ^2 better helps. We use both only qualitatively, and Figure 4.4.1 in Section 4.4 shows the first. Because $T^2 \sim F(1, s, \delta^2)$, Theorem 4.1.4 also shows that the power of the two-sided t test increases with $|\delta|$ (Exercise (A2)).

4.1.3 Exercises**4.1.3.1 A. Check your understanding**

Exercise A1.

Show that $\chi^2(2)$ is the exponential distribution with mean 2. Use [Theorem 4.1.2\(d\)](#) to show that

$$\Pr\{\chi^2(2, \gamma) > x\} = \sum_{k \geq 0} e^{-\gamma/2} \frac{(\gamma/2)^k}{k!} e^{-x/2} \sum_{j=0}^k \frac{(x/2)^j}{j!}.$$

Exercise A2.

Show that $T \sim t(s, \delta)$ implies $T^2 \sim F(1, s, \delta^2)$. Deduce that the power of the two-sided level- α t test depends on δ only through $|\delta|$ and increases with it. Show directly, by conditioning on the denominator, that the power of the one-sided test that rejects for $T > c$ increases with δ .

Solution. $T^2 = X^2/(V/s)$ with $X^2 \sim \chi^2(1, \delta^2)$ independent of V , which is [Definition 4.1.2](#). The two-sided test rejects when $|T| > t_{\alpha/2}(s)$, that is, when $T^2 > F_{\alpha}(1, s)$. Its power is therefore $\Pr\{F(1, s, \delta^2) > F_{\alpha}(1, s)\}$, a strictly increasing function of δ^2 by [Theorem 4.1.4](#). For the one-sided test, $\Pr(T > c) = \mathbb{E} \Pr\{X > c\sqrt{V/s} \mid V\} = \mathbb{E} \Phi(\delta - c\sqrt{V/s})$, and Φ is strictly increasing.

4.1.3.2 B. Practice**Exercise B1.**

Show that the k th cumulant of $\chi^2(r, \gamma)$ is $2^{k-1}(k-1)!(r+k\gamma)$. Deduce that its skewness tends to zero like $3/\sqrt{\gamma}$ as $\gamma \rightarrow \infty$ with r fixed.

Solution. Using $-\frac{1}{2} \log(1-2t) = \sum_{k \geq 1} (2t)^k / (2k)$ and $t/(1-2t) = \sum_{k \geq 1} 2^{k-1} t^k$, the cumulant generating function is

$$K(t) = \sum_{k \geq 1} t^k 2^{k-1} \left(\frac{r}{k} + \gamma \right) = \sum_{k \geq 1} \frac{t^k}{k!} 2^{k-1} (k-1)! (r+k\gamma).$$

The k th cumulant is $k!$ times the coefficient of t^k . In particular $\kappa_2 = 2(r+2\gamma)$ and $\kappa_3 = 8(r+3\gamma)$, so the skewness is $\kappa_3/\kappa_2^{3/2} = 8(r+3\gamma)/\{2(r+2\gamma)\}^{3/2}$. As $\gamma \rightarrow \infty$ this behaves like $24\gamma/(4\gamma)^{3/2} = 3/\sqrt{\gamma}$.

Exercise B2.

Show that $\chi^2(1, \gamma)$ has density $(2\pi x)^{-1/2} e^{-(x+\gamma)/2} \cosh \sqrt{\gamma x}$ for $x > 0$. Verify that expanding $\cosh$ in a power series reproduces [Eq. 4.1.3](#). *Hint:* $\Gamma(k+\frac{1}{2}) = \sqrt{\pi} (2k)!/(4^k k!)$.

Solution. Let $X = (W + \sqrt{\gamma})^2$ with $W \sim N(0, 1)$. For $x > 0$, $X \leq x$ iff $-\sqrt{x} - \sqrt{\gamma} \leq W \leq \sqrt{x} - \sqrt{\gamma}$. Differentiating with ϕ the standard normal density,

$$f(x) = \frac{\phi(\sqrt{x} - \sqrt{\gamma}) + \phi(\sqrt{x} + \sqrt{\gamma})}{2\sqrt{x}} = \frac{e^{-(x+\gamma)/2}}{\sqrt{2\pi x}} \cdot \frac{e^{\sqrt{\gamma x}} + e^{-\sqrt{\gamma x}}}{2}.$$

Expanding, $f(x) = (2\pi x)^{-1/2} e^{-(x+\gamma)/2} \sum_k (\gamma x)^k / (2k)!$. The k th term of Eq. 4.1.3 is

$$\frac{e^{-\gamma/2} (\gamma/2)^k x^{k-1/2} e^{-x/2}}{k! 2^{k+1/2} \Gamma(k + \frac{1}{2})}.$$

By the hint the denominator is $k! 2^{k+1/2} \sqrt{\pi} (2k)! / (4^k k!) = \sqrt{2\pi} 2^{-k} (2k)!$, and the k th term becomes $(2\pi x)^{-1/2} e^{-(x+\gamma)/2} (\gamma x)^k / (2k)!$, as required.

Exercise B3.

Let $U_\gamma \sim \chi^2(r, \gamma)$ with r fixed. Show that $(U_\gamma - r - \gamma) / \sqrt{2r + 4\gamma}$ converges in distribution to $N(0, 1)$ as $\gamma \rightarrow \infty$. *Hint:* U_γ has the law of $(W_1 + \sqrt{\gamma})^2 + W_2^2 + \dots + W_r^2$ with $W_1, \dots, W_r$ independent standard normal.

Exercise B4.

Let $U \sim \chi^2(a)$ and $V \sim \chi^2(b)$ be independent. Show that $B = U/(U + V)$ has the $\text{Beta}(a/2, b/2)$ distribution and is independent of $U + V$, and that $F = (U/a)/(V/b) = (b/a) B/(1 - B)$. Deduce the null distribution of R^2 asked for in Chapter 6.

Solution. The joint density of (U, V) is proportional to $u^{a/2-1} v^{b/2-1} e^{-(u+v)/2}$. Change variables to $w = u/(u + v) \in (0, 1)$ and $s = u + v > 0$, so that $u = ws$, $v = (1 - w)s$ and the Jacobian is s . The density of (W, S) is proportional to

$$w^{a/2-1} (1 - w)^{b/2-1} \cdot s^{(a+b)/2-1} e^{-s/2},$$

a product of a function of w and a function of s . So B and $U + V$ are independent, $B \sim \text{Beta}(a/2, b/2)$ and $U + V \sim \chi^2(a + b)$. Also $U/V = B/(1 - B)$. For R^2 in a model with intercept, $R^2 = \text{SSR}/(\text{SSR} + \text{SSE})$, where $\text{SSR} = \|(\mathbf{M} - \mathbf{P}_1)\mathbf{Y}\|^2$ and $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$ come from orthogonal projections of ranks $r - 1$ and $n - r$. If $\boldsymbol{\mu} = \mu \mathbf{1}$, both noncentralities vanish, and Theorem 4.4.4 with $a = r - 1$, $b = n - r$ gives $R^2 \sim \text{Beta}((r - 1)/2, (n - r)/2)$.

Exercise B5.

Show that $f_{r,\gamma}(x)/g_r(x)$ is strictly increasing in $x > 0$ for every $\gamma > 0$. Conclude that among tests based on one observation $U \sim \chi^2(r, \gamma)$ of $\gamma = 0$ against a fixed $\gamma > 0$, rejecting for large U is most powerful.

4.2 Mean and variance of quadratic forms

4.2.1 The mean needs no normality

Theorem 2.3.1 computed $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$ for any random vector with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$:

$$E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}. \quad (4.2.1)$$

The formula has two parts with distinct roles. The trace term is what the noise contributes. The term $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ is what the signal contributes. For nonnegative definite $\mathbf{A}$ it vanishes exactly when $\mathbf{A} \boldsymbol{\mu} = \mathbf{0}$, which describes the mean vectors a well-designed estimator is meant to ignore.

Example. (Two estimators of σ^2 under a drifting mean)

Let $Y_1, \dots, Y_n$ be uncorrelated with common variance σ^2 , and consider

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2, \quad Q = \frac{1}{2(n-1)} \sum_{i=1}^{n-1} (Y_{i+1} - Y_i)^2,$$

the sample variance and the **mean square successive difference**. In matrix form $S^2 = \mathbf{Y}^\top \mathbf{C} \mathbf{Y} / (n-1)$ with $\mathbf{C} = \mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top$, and $Q = \mathbf{Y}^\top \mathbf{D}^\top \mathbf{D} \mathbf{Y} / \{2(n-1)\}$, where $\mathbf{D}$ is the $(n-1) \times n$ first-difference matrix with rows $(\dots, -1, 1, \dots)$. Both matrices annihilate $\mathbf{1}$, so both estimators ignore a constant mean. Since $\text{tr}(\mathbf{C}) = n-1$ and $\text{tr}(\mathbf{D}^\top \mathbf{D}) = \text{tr}(\mathbf{D} \mathbf{D}^\top) = 2(n-1)$, Eq. 4.2.1 with $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ shows that both are unbiased when the mean is constant.

Now let the mean drift linearly, $\mu_i = \mu_0 + \theta i$. Then $\mathbf{C} \boldsymbol{\mu}$ is the centred trend and $\mathbf{D} \boldsymbol{\mu} = \theta \mathbf{1}_{n-1}$, so

$$E(S^2) = \sigma^2 + \theta^2 \frac{n(n+1)}{12}, \quad E(Q) = \sigma^2 + \frac{\theta^2}{2}.$$

The bias of S^2 grows quadratically with the length of the series, while that of Q does not grow at all. With $n = 20$, $\sigma = 1$ and $\theta = 0.1$, the means are 1.350 and 1.005. Differencing removes smooth trends, and this is why Q was proposed for series whose level may wander (von Neumann 1941). The price of robustness to drift is paid in variance, which we compute below.

4.2.2 A canonical representation

Higher moments and distributions need normality, and one representation answers all such questions at once.

Theorem 4.2.1 — Canonical form of a normal quadratic form

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and let $\mathbf{A}$ be symmetric. Write $\boldsymbol{\Sigma} = \mathbf{L} \mathbf{L}^\top$ with $\mathbf{L}$ of size $n \times k$, and take a spectral decomposition $\mathbf{L}^\top \mathbf{A} \mathbf{L} = \mathbf{Q} \boldsymbol{\Lambda} \mathbf{Q}^\top$ with $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_k)$. Put $\mathbf{b} = \mathbf{Q}^\top \mathbf{L}^\top \mathbf{A} \boldsymbol{\mu}$. Then

$$\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \stackrel{d}{=} \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} + \sum_{j=1}^k (\lambda_j W_j^2 + 2b_j W_j), \quad \mathbf{W} \sim N_k(\mathbf{0}, \mathbf{I}_k). \quad (4.2.2)$$

For $\lambda_j \neq 0$ the j th term equals $\lambda_j (W_j + b_j/\lambda_j)^2 - b_j^2/\lambda_j$. The nonzero λ_j are the nonzero

eigenvalues of $\mathbf{A}\Sigma$.

Proof. Such an $\mathbf{L}$ exists. By [Theorem 1.7.2](#), $\Sigma = \mathbf{U}\mathbf{D}\mathbf{U}^\top$ with $\mathbf{U}$ orthogonal and $\mathbf{D}$ diagonal with entries $d_i \geq 0$, and $\mathbf{L} = \mathbf{U}\mathbf{D}^{1/2}$ works with $k = n$. Let $\mathbf{Z} \sim N_k(\mathbf{0}, \mathbf{I})$. The vector $\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}$ is normal with mean $\boldsymbol{\mu}$ and covariance $\mathbf{L}\mathbf{L}^\top = \Sigma$ ([Theorem 3.3.1](#)). A normal distribution is determined by its mean and covariance ([Theorem 3.2.1](#)), so $\mathbf{Y}$ and $\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}$ have the same law, and so do $\mathbf{Y}^\top \mathbf{A}\mathbf{Y}$ and

$$(\boldsymbol{\mu} + \mathbf{L}\mathbf{Z})^\top \mathbf{A}(\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}) = \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu} + 2\boldsymbol{\mu}^\top \mathbf{A}\mathbf{L}\mathbf{Z} + \mathbf{Z}^\top \mathbf{L}^\top \mathbf{A}\mathbf{L}\mathbf{Z}.$$

Put $\mathbf{W} = \mathbf{Q}^\top \mathbf{Z}$, again $N_k(\mathbf{0}, \mathbf{I})$. Then $\mathbf{Z}^\top \mathbf{L}^\top \mathbf{A}\mathbf{L}\mathbf{Z} = \mathbf{W}^\top \boldsymbol{\Lambda} \mathbf{W} = \sum \lambda_j W_j^2$ and $\boldsymbol{\mu}^\top \mathbf{A}\mathbf{L}\mathbf{Z} = \mathbf{b}^\top \mathbf{W}$, which is [Eq. 4.2.2](#). The completed square is algebra. Finally, $\mathbf{L}^\top \mathbf{A}\mathbf{L} = \mathbf{H}\mathbf{F}$ with $\mathbf{F} = \mathbf{A}\mathbf{L}$ and $\mathbf{H} = \mathbf{L}^\top$, and $\mathbf{F}\mathbf{H} = \mathbf{A}\Sigma$. By [Proposition 1.7.1\(d\)](#) these two products have the same nonzero eigenvalues, with the same multiplicities. ■

So a normal quadratic form is a constant plus a sum of independent pieces. Each piece is either a scaled noncentral $\chi^2(1)$ variable, with scale λ_j , or a normal variable with mean zero, when $\lambda_j = 0$ but $b_j \neq 0$. Because the λ_j may be negative, $\mathbf{Y}^\top \mathbf{A}\mathbf{Y}$ need not be positive. The linear pieces are special to singular covariances. If Σ is positive definite, $\mathbf{L}$ can be taken square and nonsingular, and then $\lambda_j = 0$ gives $\mathbf{A}\mathbf{L}\mathbf{q}_j = \mathbf{0}$ for the j th column $\mathbf{q}_j$ of $\mathbf{Q}$, so $b_j = (\mathbf{A}\mathbf{L}\mathbf{q}_j)^\top \boldsymbol{\mu} = 0$. A linear piece needs a direction in which $\mathbf{A}$ responds to the mean while the noise is absent.

Corollary 4.2.2 — Moment generating function

In the setting of [Theorem 4.2.1](#), for $|t| < 1/(2 \max_j |\lambda_j|)$ (any t if all $\lambda_j = 0$),

$$\mathbb{E} e^{t\mathbf{Y}^\top \mathbf{A}\mathbf{Y}} = e^{t\boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu}} \prod_{j=1}^k (1 - 2\lambda_j t)^{-1/2} \exp\left(\frac{2b_j^2 t^2}{1 - 2\lambda_j t}\right).$$

Proof. The terms of [Eq. 4.2.2](#) are independent. If $\lambda_j = 0$, the term $2b_j W_j$ has mgf $e^{2b_j^2 t^2}$. If $\lambda_j \neq 0$, apply [Lemma 4.1.1](#) to $W_j + \nu_j$ with $\nu_j = b_j/\lambda_j$ at the argument $\lambda_j t$, and multiply by $e^{-tb_j^2/\lambda_j}$. The exponent is $b_j^2 t / \{\lambda_j(1 - 2\lambda_j t)\} - b_j^2 t / \lambda_j = 2b_j^2 t^2 / (1 - 2\lambda_j t)$. ■

4.2.3 Variance and covariances

Theorem 4.2.3 — Moments of normal quadratic forms

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$, let $\mathbf{A}$ and $\mathbf{B}$ be symmetric $n \times n$ matrices, and let $\mathbf{K}$ be an $m \times n$ matrix. Then

- (a) $\mathbb{E}(\mathbf{Y}^\top \mathbf{A}\mathbf{Y}) = \text{tr}(\mathbf{A}\Sigma) + \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu}$;
- (b) $\text{Var}(\mathbf{Y}^\top \mathbf{A}\mathbf{Y}) = 2 \text{tr}\{(\mathbf{A}\Sigma)^2\} + 4\boldsymbol{\mu}^\top \mathbf{A}\Sigma \mathbf{A}\boldsymbol{\mu}$;


```

def qf_moments(A, mu, Sigma):
    """Mean and variance of  $Y^T A Y$  for  $Y \sim N(\mu, \text{Sigma})$ ,  $A$  symmetric."""
    AS = A @ Sigma
    mean = np.trace(AS) + mu @ A @ mu
    var = 2 * np.trace(AS @ AS) + 4 * mu @ AS @ A @ mu
    return mean, var

C = np.eye(n) - np.ones((n, n)) / n          # centring matrix
D = np.diff(np.eye(n), axis=0)               # (n-1) x n first differences
A_s2 = C / (n - 1)
A_q = D.T @ D / (2 * (n - 1))
for name, A in [("S2", A_s2), ("Q", A_q)]:
    print(name, qf_moments(A, mu, Sigma), qf_moments(A, 0 * mu, Sigma))

```

Normality enters (b)–(d) through the third and fourth moments of $\mathbf{Y}$. For independent coordinates with excess kurtosis the variance acquires an extra term that involves only the diagonal of $\mathbf{A}$ (Theorem 2.3.3). Variance formulas for sums of squares derived under normality can be badly wrong for heavy-tailed errors, a point taken up in Chapter 21.

4.2.4 Exercises

4.2.4.1 A. Check your understanding

Exercise A1.

Let $\mathbf{Y} \sim N_2(\mathbf{0}, \mathbf{I})$ and $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Compute $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$ directly, and compare with $2 \text{tr}(\mathbf{A}^2)$ and with $2 \text{tr}(\mathbf{A}_s^2)$, where $\mathbf{A}_s = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$.

4.2.4.2 B. Practice

Exercise B1.

Let $(U_i, W_i), i = 1, \dots, n$, be independent copies of a bivariate normal pair with variances σ_U^2, σ_W^2 and covariance σ_{UW} , and let $s_{UW} = (n-1)^{-1} \sum_i (U_i - \bar{U})(W_i - \bar{W})$. Write $s_{UW} = \mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ for the stacked vector $\mathbf{Y} = (\mathbf{U}^\top, \mathbf{W}^\top)^\top$ and a symmetric $2n \times 2n$ matrix $\mathbf{A}$, and use Theorem 4.2.3 to show that $\text{Var}(s_{UW}) = (\sigma_U^2 \sigma_W^2 + \sigma_{UW}^2) / (n-1)$. Check the answer against $\text{Var}(S^2)$ when $\mathbf{U} = \mathbf{W}$.

Exercise B2.

Let $Y_1, \dots, Y_n$ be independent with mean 0, variance σ^2 and $E Y_i^4 = \mu_4$. By Exercise (B4), for symmetric $\mathbf{A}$ and $\mathbf{B}$,

$$\text{Cov}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}, \mathbf{Y}^\top \mathbf{B} \mathbf{Y}) = (\mu_4 - 3\sigma^4) \sum_i a_{ii} b_{ii} + 2\sigma^4 \text{tr}(\mathbf{A} \mathbf{B}).$$

Show that in a one-way layout with $g \geq 2$ groups, not all of size one, the between-group and within-group sums of squares are correlated unless $\mu_4 = 3\sigma^4$. Which sign does the correlation have for heavy-tailed errors?

Solution. In the one-way layout the between-group matrix is $\mathbf{A} = \mathbf{M} - \mathbf{P}_1$ and the within-group matrix is $\mathbf{B} = \mathbf{I} - \mathbf{M}$. They satisfy $\mathbf{AB} = \mathbf{0}$, so $\text{tr}(\mathbf{AB}) = 0$ and the covariance reduces to $(\mu_4 - 3\sigma^4) \sum_i a_{ii}b_{ii}$. An observation in group ℓ has $a_{ii} = 1/n_\ell - 1/n$ and $b_{ii} = 1 - 1/n_\ell$, so

$$\sum_i a_{ii}b_{ii} = \sum_\ell n_\ell \left(\frac{1}{n_\ell} - \frac{1}{n} \right) \left(1 - \frac{1}{n_\ell} \right) = \sum_\ell \left(1 - \frac{n_\ell}{n} \right) \left(1 - \frac{1}{n_\ell} \right).$$

With $g \geq 2$ every first factor is positive, and the second is positive for any group with two or more observations. So the sum is positive, and the covariance vanishes iff $\mu_4 = 3\sigma^4$. For heavy-tailed errors, $\mu_4 > 3\sigma^4$, the two sums of squares are positively correlated: a large error inflates both.

Exercise B3.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$, and let $\mathbf{M}$ be symmetric idempotent of rank r with $\mathbf{M}\boldsymbol{\mu} \neq \boldsymbol{\mu}$, as when a regression model omits a needed regressor. Show that $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2/\sigma^2 \sim \chi^2(n - r, \|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2/\sigma^2)$, and find the bias of $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2/(n - r)$ as an estimator of σ^2 . In the setting of [Theorem 4.4.6](#) without the assumption $\mathbf{P}\boldsymbol{\mu} = \boldsymbol{\mu}$, which parts of the conclusion survive?

Exercise B4.

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \boldsymbol{\Sigma})$. Use [Corollary 4.2.2](#) and [Proposition 1.7.1\(d\)](#) to show that $E e^{t\mathbf{Y}^\top \mathbf{A} \mathbf{Y}} = \det(\mathbf{I}_n - 2t\mathbf{A}\boldsymbol{\Sigma})^{-1/2}$ for $|t|$ small.

4.2.4.3 C. Going deeper

Exercise C1.

Let $\mathbf{Y} \sim N_n(\mu \mathbf{1}, \sigma^2 \mathbf{I})$ with μ and σ^2 unknown. Among symmetric $\mathbf{A}$ with $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \sigma^2$ for all μ and σ^2 , show that $\mathbf{A} = \mathbf{C}/(n - 1)$, which gives S^2 , uniquely minimizes $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$ at every parameter value. (Compare [Exercise \(B5\)](#), where the mean is known to be zero.)

Solution. Since $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \sigma^2 \text{tr} \mathbf{A} + \mu^2 \mathbf{1}^\top \mathbf{A} \mathbf{1}$, unbiasedness means $\text{tr} \mathbf{A} = 1$ and $\mathbf{1}^\top \mathbf{A} \mathbf{1} = 0$. By [Theorem 4.2.3](#), $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = 2\sigma^4 \text{tr}(\mathbf{A}^2) + 4\mu^2 \sigma^2 \|\mathbf{A} \mathbf{1}\|^2$. The map $\mathbf{A} \mapsto \mathbf{C} \mathbf{A} \mathbf{C}$ is an orthogonal projection for the inner product $\text{tr}(\mathbf{AB})$ on symmetric matrices, so $\text{tr}(\mathbf{A}^2) \geq \text{tr}\{(\mathbf{C} \mathbf{A} \mathbf{C})^2\}$, with equality iff $\mathbf{A} = \mathbf{C} \mathbf{A} \mathbf{C}$. Also $\text{tr}(\mathbf{C} \mathbf{A} \mathbf{C}) = \text{tr} \mathbf{A} - \mathbf{1}^\top \mathbf{A} \mathbf{1}/n = 1$. The symmetric matrix $\mathbf{C} \mathbf{A} \mathbf{C}$ has rank at most $n - 1$. If $\lambda_1, \dots, \lambda_{n-1}$ are its eigenvalues on $\mathbf{1}^\perp$, the Cauchy-Schwarz inequality gives $1 = (\sum \lambda_i)^2 \leq (n - 1) \sum \lambda_i^2$. Hence $\text{tr}(\mathbf{A}^2) \geq 1/(n - 1)$, with equality iff $\mathbf{A} = \mathbf{C} \mathbf{A} \mathbf{C}$ and all $\lambda_i = 1/(n - 1)$, that is, iff $\mathbf{A} = \mathbf{C}/(n - 1)$. That matrix also has $\mathbf{A} \mathbf{1} = \mathbf{0}$, so it minimizes both variance terms at once.

4.3 When a quadratic form is chi-squared

A quadratic form is a weighted sum of independent noncentral $\chi^2(1)$ pieces, and the weights are eigenvalues. It is itself chi-squared when every nonzero weight equals one. This section turns that observation into checkable conditions on $\mathbf{A}$ and Σ , and shows that for the spherical case the condition is necessary as well as sufficient.

4.3.1 Spherical normal vectors

Theorem 4.3.1 — Chi-squared forms in a spherical normal vector

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{I}_n)$, let $\mathbf{A}$ be symmetric, and let $r \geq 1$ be an integer and $\gamma \geq 0$. Then $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \sim \chi^2(r, \gamma)$ if and only if $\mathbf{A}$ is idempotent with $\text{rank}(\mathbf{A}) = r$ and $\gamma = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$.

Proof. Sufficiency. By Theorem 1.9.2, $\mathbf{A} = \mathbf{Q}_1 \mathbf{Q}_1^\top$, where the $n \times r$ matrix $\mathbf{Q}_1$ has orthonormal columns and $r = \text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A})$. Then $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \|\mathbf{Q}_1^\top \mathbf{Y}\|^2$ and $\mathbf{Q}_1^\top \mathbf{Y} \sim N_r(\mathbf{Q}_1^\top \boldsymbol{\mu}, \mathbf{Q}_1^\top \mathbf{Q}_1) = N_r(\mathbf{Q}_1^\top \boldsymbol{\mu}, \mathbf{I}_r)$, so by definition $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \sim \chi^2(r, \|\mathbf{Q}_1^\top \boldsymbol{\mu}\|^2)$, and $\|\mathbf{Q}_1^\top \boldsymbol{\mu}\|^2 = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$.

Necessity. Apply Theorem 4.2.1 with $\mathbf{L} = \mathbf{I}$, so that $\mathbf{A} = \mathbf{Q} \boldsymbol{\Lambda} \mathbf{Q}^\top$. Let $\boldsymbol{\nu} = \mathbf{Q}^\top \boldsymbol{\mu}$. Then $\mathbf{b} = \mathbf{Q}^\top \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\Lambda} \boldsymbol{\nu}$, so $b_j = \lambda_j \nu_j$. The constant left after completing squares is $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} - \sum_{\lambda_j \neq 0} \lambda_j \nu_j^2 = \boldsymbol{\nu}^\top \boldsymbol{\Lambda} \boldsymbol{\nu} - \sum_j \lambda_j \nu_j^2 = 0$. Hence $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ has the law of $\sum_{j \in J} \lambda_j (W_j + \nu_j)^2$, where J indexes the nonzero eigenvalues. J is not empty, because a $\chi^2(r, \gamma)$ variable with $r \geq 1$ is not identically zero.

By Lemma 4.1.1, near $t = 0$ the logarithm of the mgf of this sum is $\sum_{j \in J} \{-\frac{1}{2} \log(1 - 2\lambda_j t) + \lambda_j \nu_j^2 t / (1 - 2\lambda_j t)\}$. Expanding $-\frac{1}{2} \log(1 - x) = \sum_{k \geq 1} x^k / (2k)$ and $x / (1 - x) = \sum_{k \geq 1} x^k$, the coefficient of t^k is $(2^{k-1}/k) \sum_{j \in J} \lambda_j^k (1 + k \nu_j^2)$. The same expansion of Eq. 4.1.2 gives $(2^{k-1}/k)(r + k\gamma)$. Equal distributions have equal mgfs, and two power series that agree on an interval have equal coefficients, so

$$\sum_{j \in J} \lambda_j^k (1 + k \nu_j^2) = r + k\gamma \quad \text{for every } k \geq 1. \quad (4.3.1)$$

Let $\rho = \max_{j \in J} |\lambda_j|$. For even k every term on the left is nonnegative and one of them is at least ρ^k . The right side grows only linearly in k , so $\rho \leq 1$. Split J into $J_+ = \{\lambda_j = 1\}$, $J_- = \{\lambda_j = -1\}$ and $J_0 = \{0 < |\lambda_j| < 1\}$, and write $N_\pm = |J_\pm|$ and $S_\pm = \sum_{j \in J_\pm} \nu_j^2$. The terms over J_0 tend to zero as $k \rightarrow \infty$, so

$$N_+ + (-1)^k N_- + k(S_+ + (-1)^k S_-) - r - k\gamma \rightarrow 0.$$

Dividing by k and letting $k \rightarrow \infty$ through even and through odd values gives $S_+ + S_- = \gamma = S_+ - S_-$. Then the same limit without the factor k gives $N_+ + N_- = r = N_+ - N_-$. So $N_- = S_- = 0$, $N_+ = r$ and $S_+ = \gamma$. The terms over J_+ in Eq. 4.3.1 now account for all of $r + k\gamma$, and with $k = 2$ the remaining sum $\sum_{j \in J_0} \lambda_j^2 (1 + 2\nu_j^2)$ is zero. Every term of that sum is positive, so J_0 is empty. Thus every eigenvalue of $\mathbf{A}$ is 0 or 1, $\mathbf{A}$ is idempotent, $\text{rank}(\mathbf{A}) = N_+ = r$, and $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} = \sum_{j \in J_+} \nu_j^2 = \gamma$.

The necessity half is the one that makes the theorem a tool for *recognizing* distributions. If a quadratic form in a spherical normal vector is chi-squared at all, then it is the squared length of a projection. There is no other way.

4.3.2 General covariance matrices

Theorem 4.3.2 — When a quadratic form is chi-squared

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and let $\mathbf{A}$ be symmetric.

- (a) (*Projections.*) If $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ with $\sigma > 0$ and $\mathbf{P}$ is symmetric and idempotent, then

$$\frac{\|\mathbf{P}\mathbf{Y}\|^2}{\sigma^2} = \frac{\mathbf{Y}^\top \mathbf{P} \mathbf{Y}}{\sigma^2} \sim \chi^2(\text{rank}(\mathbf{P}), \frac{\|\mathbf{P}\boldsymbol{\mu}\|^2}{\sigma^2}).$$

- (b) (*Positive definite $\boldsymbol{\Sigma}$.*) If $\boldsymbol{\Sigma}$ is positive definite, then $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ has a $\chi^2(r, \gamma)$ distribution for some $r \geq 1$ and $\gamma \geq 0$ iff $\mathbf{A}\boldsymbol{\Sigma}$ is idempotent and $\mathbf{A} \neq \mathbf{0}$. In that case $r = \text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}\boldsymbol{\Sigma})$ and $\gamma = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$.

- (c) (*Any $\boldsymbol{\Sigma}$.*) If

$$(i) \ \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} = \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}, \quad (ii) \ \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu}, \quad (iii) \ \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu},$$

then $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \sim \chi^2(\text{tr}(\mathbf{A}\boldsymbol{\Sigma}), \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu})$. The three conditions hold in particular when $\mathbf{A}\boldsymbol{\Sigma} \mathbf{A} = \mathbf{A}$, and when $\mathbf{A}\boldsymbol{\Sigma}$ is idempotent and $\boldsymbol{\mu} \in \mathcal{C}(\boldsymbol{\Sigma})$.

Proof.

- (a) $\sigma^{-1} \mathbf{Y} \sim N_n(\sigma^{-1} \boldsymbol{\mu}, \mathbf{I})$. Apply Theorem 4.3.1 with $\mathbf{A} = \mathbf{P}$, using $\mathbf{P}^\top \mathbf{P} = \mathbf{P}$ to write $\mathbf{Y}^\top \mathbf{P} \mathbf{Y} = \|\mathbf{P}\mathbf{Y}\|^2$ and $\boldsymbol{\mu}^\top \mathbf{P} \boldsymbol{\mu} = \|\mathbf{P}\boldsymbol{\mu}\|^2$.
- (b) Let $\boldsymbol{\Sigma}^{1/2}$ be the positive definite square root (Theorem 1.7.6). Then $\mathbf{Z} = \boldsymbol{\Sigma}^{-1/2} \mathbf{Y} \sim N_n(\boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu}, \mathbf{I})$ and $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \mathbf{Z}^\top \mathbf{B} \mathbf{Z}$ with $\mathbf{B} = \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2}$ symmetric. By Theorem 4.3.1, $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ is noncentral chi-squared iff $\mathbf{B}$ is idempotent and nonzero. Now $\mathbf{B}^2 = \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}^{1/2}$, so $\mathbf{B}^2 = \mathbf{B}$ iff $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} = \mathbf{A}$ (multiply by $\boldsymbol{\Sigma}^{-1/2}$ on both sides) iff $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} = \mathbf{A} \boldsymbol{\Sigma}$ (multiply by $\boldsymbol{\Sigma}$ or $\boldsymbol{\Sigma}^{-1}$ on the right). Also $\mathbf{B} \neq \mathbf{0}$ iff $\mathbf{A} \neq \mathbf{0}$. The degrees of freedom are $\text{rank}(\mathbf{B}) = \text{rank}(\mathbf{A})$, which equals $\text{tr}(\mathbf{B}) = \text{tr}(\mathbf{A}\boldsymbol{\Sigma})$, and the noncentrality is $\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1/2} \mathbf{B} \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu} = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$.
- (c) Take $\mathbf{L}$ of size $n \times k$ with full column rank $k = \text{rank}(\boldsymbol{\Sigma})$ and $\mathbf{L} \mathbf{L}^\top = \boldsymbol{\Sigma}$, for instance $\mathbf{L} = \mathbf{U}_1 \mathbf{D}_1^{1/2}$ from the eigenvectors $\mathbf{U}_1$ for the positive eigenvalues $\mathbf{D}_1$ of $\boldsymbol{\Sigma}$. (If $\boldsymbol{\Sigma} = \mathbf{0}$ the claim is trivial.) As in Theorem 4.2.1, $\mathbf{Y} \stackrel{d}{=} \boldsymbol{\mu} + \mathbf{L} \mathbf{Z}$ with $\mathbf{Z} \sim N_k(\mathbf{0}, \mathbf{I})$. Since $\mathbf{L}$ has full column rank, $\mathbf{L}^\top$ has full row rank, so $\mathbf{L} \mathbf{G} \mathbf{L}^\top = \mathbf{L} \mathbf{H} \mathbf{L}^\top$ implies $\mathbf{G} = \mathbf{H}$, and $\mathbf{L} \mathbf{g} = \mathbf{L} \mathbf{h}$ implies $\mathbf{g} = \mathbf{h}$. Put $\mathbf{B} = \mathbf{L}^\top \mathbf{A} \mathbf{L}$ and $\mathbf{c} = \mathbf{L}^\top \mathbf{A} \boldsymbol{\mu}$. Condition (i) reads $\mathbf{L} \mathbf{B}^2 \mathbf{L}^\top = \mathbf{L} \mathbf{B} \mathbf{L}^\top$, so $\mathbf{B}^2 = \mathbf{B}$. Condition (ii) reads $\mathbf{L} \mathbf{B} \mathbf{c} = \mathbf{L} \mathbf{c}$, so $\mathbf{B} \mathbf{c} = \mathbf{c}$. Condition (iii) reads $\mathbf{c}^\top \mathbf{c} = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$. Hence

$$(\boldsymbol{\mu} + \mathbf{L} \mathbf{Z})^\top \mathbf{A} (\boldsymbol{\mu} + \mathbf{L} \mathbf{Z}) = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} + 2 \mathbf{c}^\top \mathbf{Z} + \mathbf{Z}^\top \mathbf{B} \mathbf{Z} = (\mathbf{Z} + \mathbf{c})^\top \mathbf{B} (\mathbf{Z} + \mathbf{c}),$$

because $(\mathbf{Z} + \mathbf{c})^\top \mathbf{B} (\mathbf{Z} + \mathbf{c}) = \mathbf{Z}^\top \mathbf{B} \mathbf{Z} + 2 \mathbf{c}^\top \mathbf{B} \mathbf{Z} + \mathbf{c}^\top \mathbf{B} \mathbf{c}$ and $\mathbf{B} \mathbf{c} = \mathbf{c}$. Now

$\mathbf{Z} + \mathbf{c} \sim N_k(\mathbf{c}, \mathbf{I})$, and [Theorem 4.3.1](#) gives the law $\chi^2(\text{rank } \mathbf{B}, \mathbf{c}^\top \mathbf{B} \mathbf{c})$, where $\text{rank } \mathbf{B} = \text{tr } \mathbf{B} = \text{tr}(\mathbf{A}\mathbf{\Sigma})$ and $\mathbf{c}^\top \mathbf{B} \mathbf{c} = \mathbf{c}^\top \mathbf{c} = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$. If $\mathbf{B} = \mathbf{0}$, then $\mathbf{c} = \mathbf{B} \mathbf{c} = \mathbf{0}$ and the form is identically 0, which is $\chi^2(0)$. If $\mathbf{A}\mathbf{\Sigma}\mathbf{A} = \mathbf{A}$, all three conditions follow by substitution. If $\mathbf{A}\mathbf{\Sigma}$ is idempotent and $\boldsymbol{\mu} = \mathbf{\Sigma} \mathbf{v}$, then (i) follows from multiplying $\mathbf{A}\mathbf{\Sigma}\mathbf{A}\mathbf{\Sigma} = \mathbf{A}\mathbf{\Sigma}$ by $\mathbf{\Sigma}$ on the left, and (ii) and (iii) from the same identity applied to $\mathbf{v}$.

■

Part (a) is the case used most often in this book. For the linear model $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$ and the projection $\mathbf{M}$ onto $\mathcal{C}(\mathbf{X})$, [Chapter 6](#) shows that the residual sum of squares is $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$. Since $(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$, part (a) gives

$$\text{SSE}/\sigma^2 \sim \chi^2(n - \text{rank}(\mathbf{X})),$$

a central distribution whatever the value of $\boldsymbol{\beta}$. The projections that define test statistics, by contrast, do not annihilate the mean under the alternative, and their noncentrality $\|\mathbf{P}\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2$ measures how far the truth is from the hypothesis in the metric that the test can see.

In part (c) idempotence of $\mathbf{A}\mathbf{\Sigma}$ alone is *not* enough when $\mathbf{\Sigma}$ is singular. A mean with a component outside $\mathcal{C}(\mathbf{\Sigma})$ is a deterministic shift that the form may pick up as a constant or as a linear term ([Exercise \(B2\)](#)). Part (c) also covers the natural generalization of the Mahalanobis distance.

Corollary 4.3.3 — Mahalanobis forms

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{\Sigma})$.

- (a) If $\mathbf{\Sigma}$ is positive definite, $(\mathbf{Y} - \boldsymbol{\mu})^\top \mathbf{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu}) \sim \chi^2(n)$ and $\mathbf{Y}^\top \mathbf{\Sigma}^{-1} \mathbf{Y} \sim \chi^2(n, \boldsymbol{\mu}^\top \mathbf{\Sigma}^{-1} \boldsymbol{\mu})$.
- (b) If $\mathbf{G}$ is a symmetric generalized inverse of $\mathbf{\Sigma}$ and $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{\Sigma})$, then $\mathbf{Y}^\top \mathbf{G} \mathbf{Y} \sim \chi^2(\text{rank}(\mathbf{\Sigma}), \boldsymbol{\mu}^\top \mathbf{G} \boldsymbol{\mu})$.

Proof.

- (a) $\mathbf{A} = \mathbf{\Sigma}^{-1}$ satisfies $\mathbf{A}\mathbf{\Sigma}\mathbf{A} = \mathbf{A}$, and $\mathbf{Y} - \boldsymbol{\mu} \sim N_n(\mathbf{0}, \mathbf{\Sigma})$. The central statement was already obtained by whitening in [Proposition 3.2.3](#).
- (b) $\mathbf{G}\mathbf{\Sigma}$ is idempotent because $\mathbf{\Sigma}\mathbf{G}\mathbf{\Sigma} = \mathbf{\Sigma}$, so part (c) of the theorem applies and the degrees of freedom are $\text{tr}(\mathbf{G}\mathbf{\Sigma}) = \text{tr}(\mathbf{L}^\top \mathbf{G} \mathbf{L})$. With $\mathbf{L}$ as in that proof, $\mathbf{L}\mathbf{L}^\top \mathbf{G} \mathbf{L}\mathbf{L}^\top = \mathbf{L}\mathbf{L}^\top$ gives $\mathbf{L}^\top \mathbf{G} \mathbf{L} = \mathbf{I}_k$, whose trace is $k = \text{rank}(\mathbf{\Sigma})$.

■

A symmetric generalized inverse always exists ([Proposition 1.3.3\(d\)](#)). The value of $\mathbf{Y}^\top \mathbf{G} \mathbf{Y}$ does not depend on which one is chosen, with probability one ([Exercise \(B3\)](#)).

Example. (Correlated observations about a common level)

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{\Sigma})$ with $\mathbf{\Sigma}$ positive definite, and suppose we want to measure how far $\mathbf{Y}$ is from a common level $m\mathbf{1}$. The generalized least squares level is $\hat{m} = \mathbf{1}^\top \mathbf{\Sigma}^{-1} \mathbf{Y} / \mathbf{1}^\top \mathbf{\Sigma}^{-1} \mathbf{1}$,

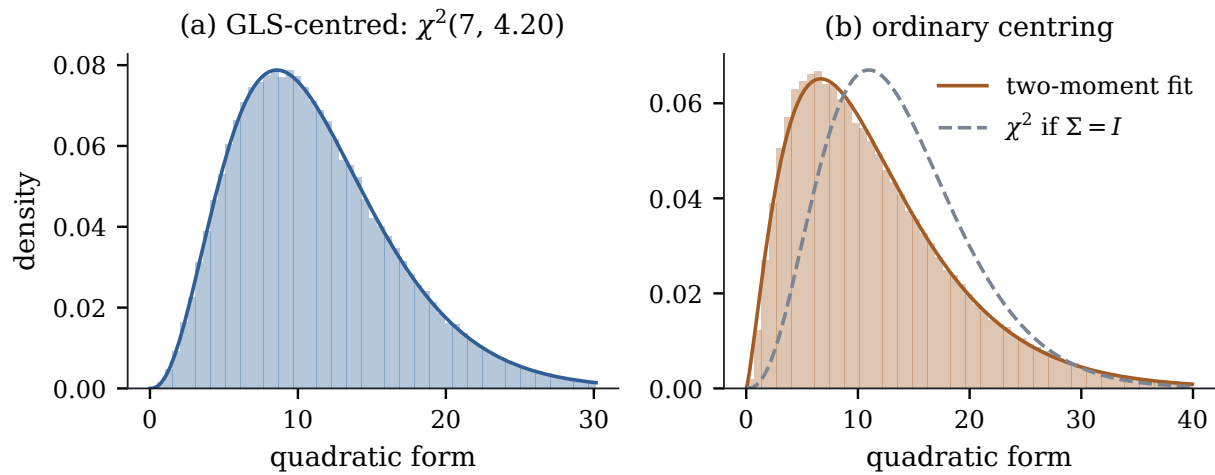


Figure 4.3.1: Two quadratic forms measuring spread about a common level, for $n = 8$ autoregressive observations with a trending mean. (a) With generalized least squares centring, $\mathbf{A}\Sigma$ is idempotent and the form is exactly noncentral chi-squared. (b) With ordinary centring it is a weighted sum of noncentral $\chi^2(1)$ terms. A two-moment scaled chi-squared fits closely. The chi-squared law that would hold for independent observations does not.

and the natural discrepancy is $(\mathbf{Y} - \hat{m}\mathbf{1})^\top \Sigma^{-1}(\mathbf{Y} - \hat{m}\mathbf{1}) = \mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ with

$$\mathbf{A} = \Sigma^{-1} - \frac{\Sigma^{-1} \mathbf{1} \mathbf{1}^\top \Sigma^{-1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}.$$

A direct multiplication gives $\mathbf{A}\Sigma\mathbf{A} = \mathbf{A}$, and $\text{tr}(\mathbf{A}\Sigma) = n - 1$. By Theorem 4.3.2(b) the form is noncentral chi-squared with $n - 1$ degrees of freedom and noncentrality $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$, which is zero exactly when $\boldsymbol{\mu}$ is constant. The listing takes $n = 8$, an autoregressive correlation $\Sigma = (\rho^{|i-j|})$ with $\rho = 0.6$, and a linear trend in the mean, for which $\gamma = 4.200$. Figure 4.3.1(a) compares 200,000 simulated values with the $\chi^2(7, 4.200)$ density.

```
import numpy as np
from scipy import integrate, stats
rng = np.random.default_rng(31415)
n, rho = 8, 0.6
idx = np.arange(n)
Sigma = rho ** np.abs(idx[:, None] - idx[None, :]) # AR(1) correlation matrix
mu = 0.4 * (idx - idx.mean()) # linear trend, mean zero
one = np.ones(n)

W = np.linalg.inv(Sigma)
A = W - np.outer(W @ one, W @ one) / (one @ W @ one) # GLS-centred form
AS = A @ Sigma
print("A Sigma idempotent:", np.allclose(AS @ AS, AS))
df, gamma = np.trace(AS), mu @ A @ mu # chi^2(df, gamma)

L = np.linalg.cholesky(Sigma) # Y = mu + L Z
Z = rng.standard_normal((200_000, n))
Y = mu + Z @ L.T
q_gls = np.einsum("ij,jk,ik->i", Y, A, Y)
print(df, gamma, stats.kstest(q_gls, stats.ncx2(df, gamma).cdf).pvalue)
```

4.3.3 Forms that are not chi-squared

Most quadratic forms fail the conditions of [Theorem 4.3.2](#). The ordinary sum of squares about the mean, $\mathbf{Y}^\top \mathbf{C} \mathbf{Y}$ with $\mathbf{C} = \mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top$, is an example as soon as the observations are correlated. Then $\mathbf{C} \boldsymbol{\Sigma}$ is not idempotent, and by [Theorem 4.2.1](#) the law is that of $\sum_j \lambda_j (W_j + \nu_j)^2$, with weights λ_j equal to the nonzero eigenvalues of $\mathbf{C} \boldsymbol{\Sigma}$. In [Example \(Correlated observations about a common level\)](#) these range from 0.259 to 1.832. Two practical ways to handle such a law are the following.

Exact computation. The characteristic function of the weighted sum is known in closed form (put $t = iu$ in [Corollary 4.2.2](#)), and a distribution function can be recovered from a characteristic function φ by the inversion formula of Gil-Pelaez (1951),

$$\Pr(X > x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \frac{\operatorname{Im}\{e^{-iux} \varphi(u)\}}{u} du.$$

For a weighted sum of k chi-squared terms the integrand decays like $u^{-1-k/2}$, so ordinary quadrature works well. Imhof (1961) gave the real-variable form of this integral that most software uses. The listing implements it directly and checks it against simulation.

Two-moment approximation. Match $a\chi^2(\nu)$ to the mean m and variance v from [Theorem 4.2.3](#). Since $a\chi^2(\nu)$ has mean $a\nu$ and variance $2a^2\nu$, this gives $a = v/(2m)$ and $\nu = 2m^2/v$, with ν usually not an integer (Satterthwaite 1946; Patnaik 1949).

In [Example \(Correlated observations about a common level\)](#), the simulated 95th percentile of $\mathbf{Y}^\top \mathbf{C} \mathbf{Y}$ is 26.51. The exact tail probability beyond it is 0.0490, and the approximation with $a = 2.488$ and $\nu = 4.68$ gives 0.0484. Treating the observations as independent, and so using $\chi^2(7, 6.720)$, gives 0.040. [Figure 4.3.1\(b\)](#) shows the three laws. Here the error from ignoring correlation is modest at the 5% point but visible in the shape. It is not a monotone function of ρ , but strong correlation makes it severe: with $\rho = 0.9$ and the same mean, the naive tail probability at the true 95th percentile is 0.197.

```
def canonical(A, mu, L):
    """Y^T A Y = sum_j lam_j (W_j + nu_j)^2 + c (all lam_j nonzero here), W ~ N(0, I)"""
    lam, Qm = np.linalg.eigh(L.T @ A @ L)
    b = Qm.T @ L.T @ A @ mu
    keep = np.abs(lam) > 1e-10
    assert np.allclose(b[~keep], 0) # no linear terms in this example
    c = mu @ A @ mu - np.sum(b[keep] ** 2 / lam[keep])
    return lam[keep], b[keep] / lam[keep], c

def tail_prob(x, lam, nu, c=0.0):
    """P(sum lam_j (W_j + nu_j)^2 + c > x) by Gil-Pelaez inversion."""
    def integrand(u):
        z = 1 - 2j * lam * u
        phi = np.prod(z ** -0.5 * np.exp(1j * u * lam * nu**2 / z))
        return (np.exp(-1j * u * (x - c)) * phi).imag / u
    edges = np.linspace(0, 400, 401) # integrand decays like u
    # ^(-1-k/2)
    val = sum(integrate.quad(integrand, a, b)[0] for a, b in zip(edges[:-1], edges[1:]))
    return 0.5 + val / np.pi
```

4.3.4 Exercises

4.3.4.1 A. Check your understanding

Exercise A1.

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$ with $n \geq 2$, and $\mathbf{A} = \alpha\mathbf{I} + \beta\mathbf{1}\mathbf{1}^\top$. Find all (α, β) for which $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ has a chi-squared distribution, and identify each form.

Solution. $\mathbf{A}$ has eigenvalue $\alpha + n\beta$ on $\mathbf{1}$ and α on $\mathbf{1}^\perp$. By [Theorem 4.3.1](#), the form is chi-squared iff $\mathbf{A} \neq \mathbf{0}$ and both eigenvalues lie in $\{0, 1\}$. The three solutions are $(1, 0)$, giving $\|\mathbf{Y}\|^2 \sim \chi^2(n)$; $(0, 1/n)$, giving $n\bar{Y}^2 \sim \chi^2(1)$; and $(1, -1/n)$, giving $\sum (Y_i - \bar{Y})^2 \sim \chi^2(n-1)$.

Exercise A2.

Two independent samples of sizes m and k come from $N(\mu_1, \sigma^2)$ and $N(\mu_2, \sigma^2)$. Write the pooled within-sample sum of squares as $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$ for a suitable projection, and show that the two-sample t statistic has the $t(m+k-2, (\mu_1 - \mu_2)/(\sigma\sqrt{1/m + 1/k}))$ distribution.

4.3.4.2 B. Practice

Exercise B1.

Let $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\{(1-\rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top\})$ with $-1/(n-1) < \rho < 1$, and let $\bar{Y}$ and S^2 be the sample mean and variance. Show that $(n-1)S^2/\{\sigma^2(1-\rho)\} \sim \chi^2(n-1)$ and that S^2 is independent of $\bar{Y}$. Deduce that $\sqrt{n}(\bar{Y} - \mu)/S$ has the law of $\{(1 + (n-1)\rho)/(1-\rho)\}^{1/2}$ times a $t(n-1)$ variable, and describe what a positive ρ does to the coverage of the usual 95% t interval for μ .

Exercise B2.

Let $\Sigma = \text{diag}(1, 0)$, $\mu = (0, 1)^\top$, $\mathbf{A} = \mathbf{I}_2$ and $\mathbf{Y} \sim N_2(\mu, \Sigma)$. Show that $\mathbf{A}\Sigma$ is idempotent but $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ is not chi-squared. Which of the conditions of [Theorem 4.3.2\(c\)](#) fail? What changes if $\mu = (m, 0)^\top$? Repeat the first part with $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, where the shift enters through a linear term.

Solution. $\mathbf{A}\Sigma = \Sigma$ is idempotent. But $Y_2 = 1$ with probability one, so $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = Y_1^2 + 1 \geq 1$, while every $\chi^2(r, \gamma)$ with $r \geq 1$ gives positive probability to $(0, 1)$. Conditions (i) and (ii) hold, since $\Sigma^3 = \Sigma$ and $\Sigma^2\mu = \mathbf{0} = \Sigma\mu$. Condition (iii) fails, since $\mu^\top \Sigma\mu = 0 \neq 1 = \mu^\top \mu$. The part of the mean outside $\mathcal{C}(\Sigma)$ adds a constant. If $\mu = (m, 0)^\top \in \mathcal{C}(\Sigma)$, then $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = Y_1^2 \sim \chi^2(1, m^2)$, as [Theorem 4.3.2\(c\)](#) predicts. For $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ and $\mu = (0, 1)^\top$, $\mathbf{A}\Sigma = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$ is again idempotent, and $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = Y_1^2 + 2Y_1Y_2 = Y_1^2 + 2Y_1 = (Y_1 + 1)^2 - 1$, which is negative with positive probability. Conditions (i) and (ii) hold, since $\Sigma\mathbf{A}\Sigma = \Sigma$ and $\Sigma\mathbf{A}\Sigma\mu = \Sigma\mu = (1, 0)^\top$, but (iii) fails because $\mu^\top \mathbf{A}\Sigma\mu = 1 \neq 0 = \mu^\top \mu$. The shift now enters through the linear term $2Y_1$.

Exercise B3.

In [Corollary 4.3.3\(b\)](#), show that $\mathbf{Y}^\top \mathbf{G} \mathbf{Y}$ takes the same value for every generalized inverse $\mathbf{G}$ of Σ , with probability one. *Hint:* [Theorem 2.2.3](#).

Solution. By [Theorem 2.2.3](#), $\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\Sigma)$ with probability one. Since $\boldsymbol{\mu} \in \mathcal{C}(\Sigma)$, also $\mathbf{Y} \in \mathcal{C}(\Sigma)$, so $\mathbf{Y} = \Sigma \mathbf{v}$ for some (random) $\mathbf{v}$. For any generalized inverse $\mathbf{G}$, $\mathbf{Y}^\top \mathbf{G} \mathbf{Y} = \mathbf{v}^\top \Sigma \mathbf{G} \Sigma \mathbf{v} = \mathbf{v}^\top \Sigma \mathbf{v}$, which does not involve $\mathbf{G}$.

Exercise B4.

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \Sigma)$ with Σ positive definite, and let $\mathbf{A} \neq \mathbf{0}$ be nonnegative definite. Show that the two-moment approximation $a\chi^2(\nu)$ to $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ has $\nu = \{\text{tr}(\mathbf{A}\Sigma)\}^2 / \text{tr}\{(\mathbf{A}\Sigma)^2\}$, and that $\nu \leq \text{rank}(\mathbf{A})$, with equality iff all nonzero eigenvalues of $\mathbf{A}\Sigma$ are equal.

4.3.4.3 C. Going deeper

Exercise C1.

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$ with Σ possibly singular, and suppose $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \sim \chi^2(r, \gamma)$ with $r \geq 1$. Show that $\Sigma \mathbf{A} \Sigma \mathbf{A} \Sigma = \Sigma \mathbf{A} \Sigma$. *Hint:* in [Corollary 4.2.2](#), compare the coefficients of t^k for $k \geq 3$ with those of [Eq. 4.1.2](#), and adapt the argument of [Theorem 4.3.1](#).

4.4 Independence of quadratic forms and linear forms

An F or t statistic is a ratio, and its distribution is known only if numerator and denominator are independent. For normal vectors, uncorrelated linear functions are independent ([Theorem 3.4.1](#)). Quadratic forms are not linear, but the following observation brings them within reach of that result.

Idea. A quadratic form is a function of a linear form: for any generalized inverse $\mathbf{A}^-$ of $\mathbf{A}$,

$$\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \mathbf{Y}^\top \mathbf{A} \mathbf{A}^- \mathbf{A} \mathbf{Y} = (\mathbf{A} \mathbf{Y})^\top \mathbf{A}^- (\mathbf{A} \mathbf{Y}).$$

So $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ is independent of anything that is independent of the vector $\mathbf{A} \mathbf{Y}$.

4.4.1 The independence theorems

Theorem 4.4.1 — Linear and quadratic forms

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$, let $\mathbf{B}$ be $m \times n$ and $\mathbf{A}$ symmetric $n \times n$. If $\mathbf{B}\Sigma\mathbf{A} = \mathbf{0}$, then $\mathbf{B}\mathbf{Y}$ and $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ are independent. Conversely, if $\mathbf{B}\mathbf{Y}$ and $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ are independent for every $\boldsymbol{\mu} \in \mathbb{R}^n$, with Σ fixed, then $\mathbf{B}\Sigma\mathbf{A} = \mathbf{0}$.

Proof. The vector $(\mathbf{BY}, \mathbf{AY})$ is a linear function of $\mathbf{Y}$, hence jointly normal (Theorem 3.3.1), and its cross-covariance is $\text{Cov}(\mathbf{BY}, \mathbf{AY}) = \mathbf{B}\Sigma\mathbf{A}^\top = \mathbf{B}\Sigma\mathbf{A}$ (Theorem 2.2.1). If this is zero, $\mathbf{BY}$ and $\mathbf{AY}$ are independent by Theorem 3.4.1, and so are $\mathbf{BY}$ and the function $(\mathbf{AY})^\top \mathbf{A}^-(\mathbf{AY})$ of $\mathbf{AY}$, which is $\mathbf{Y}^\top \mathbf{AY}$ by the key idea above. Conversely, independent variables with finite variances are uncorrelated, so Theorem 4.2.3(c) gives $2\mathbf{B}\Sigma\mathbf{A}\boldsymbol{\mu} = \mathbf{0}$ for every $\boldsymbol{\mu}$, that is, $\mathbf{B}\Sigma\mathbf{A} = \mathbf{0}$. ■

Theorem 4.4.2 — Two quadratic forms

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$ and let $\mathbf{A}, \mathbf{B}$ be symmetric. If $\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$, equivalently $\mathbf{B}\Sigma\mathbf{A} = \mathbf{0}$, then $\mathbf{Y}^\top \mathbf{AY}$ and $\mathbf{Y}^\top \mathbf{BY}$ are independent.

Proof. The two conditions are transposes of each other. As before, $\mathbf{AY}$ and $\mathbf{BY}$ are jointly normal with $\text{Cov}(\mathbf{AY}, \mathbf{BY}) = \mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$, hence independent, and each form is a function of one of them. ■

The condition is sufficient for every $\boldsymbol{\mu}$ and every Σ . When Σ is singular one might hope that the weaker requirement $\Sigma\mathbf{A}\Sigma\mathbf{B}\Sigma = \mathbf{0}$ suffices, since only directions in $\mathcal{C}(\Sigma)$ carry noise. For nonnegative definite $\mathbf{A}$ and $\mathbf{B}$ the two requirements turn out to be equivalent (Exercise (B2)). In the other direction, the condition is necessary under mild assumptions, a result usually credited to Craig (1943). Its general proof is surprisingly delicate, and the history of incorrect published proofs is recounted by Driscoll and Gundberg (1986). The case that matters most for sums of squares has a short proof.

Proposition 4.4.3 — Necessity for nonnegative definite forms

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$ and let $\mathbf{A}, \mathbf{B}$ be nonnegative definite. If $\mathbf{Y}^\top \mathbf{AY}$ and $\mathbf{Y}^\top \mathbf{BY}$ are independent, then $\mathbf{AB} = \mathbf{0}$.

Proof. Independence makes the covariance zero, and by Theorem 4.2.3(d) the covariance is $2\text{tr}(\mathbf{AB})$. With the square roots of Theorem 1.7.6,

$$\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{A}^{1/2}\mathbf{A}^{1/2}\mathbf{B}^{1/2}\mathbf{B}^{1/2}) = \text{tr}\{(\mathbf{A}^{1/2}\mathbf{B}^{1/2})^\top(\mathbf{A}^{1/2}\mathbf{B}^{1/2})\},$$

the sum of squares of the entries of $\mathbf{A}^{1/2}\mathbf{B}^{1/2}$. So $\mathbf{A}^{1/2}\mathbf{B}^{1/2} = \mathbf{0}$, and $\mathbf{AB} = \mathbf{A}^{1/2}(\mathbf{A}^{1/2}\mathbf{B}^{1/2})\mathbf{B}^{1/2} = \mathbf{0}$. ■

For nonnegative definite forms in a standard normal vector, then, zero correlation and independence coincide, just as for jointly normal linear forms. This fails for general symmetric matrices (Exercise (B1)).

4.4.2 Orthogonal projections of a spherical normal vector

The situation in regression is the tidiest one: a spherical normal vector and several projections onto mutually orthogonal subspaces.

Theorem 4.4.4 — Independent projections

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$, and let $\mathbf{P}_1, \dots, \mathbf{P}_k$ be symmetric idempotent matrices with $\mathbf{P}_i \mathbf{P}_j = \mathbf{0}$ for $i \neq j$, and ranks $r_1, \dots, r_k$. Then $\mathbf{P}_1 \mathbf{Y}, \dots, \mathbf{P}_k \mathbf{Y}$ are mutually independent normal vectors, $\mathbf{P}_i \mathbf{Y} \sim N_n(\mathbf{P}_i \boldsymbol{\mu}, \sigma^2 \mathbf{P}_i)$, and the sums of squares

$$\frac{\|\mathbf{P}_i \mathbf{Y}\|^2}{\sigma^2} \sim \chi^2\left(r_i, \frac{\|\mathbf{P}_i \boldsymbol{\mu}\|^2}{\sigma^2}\right), \quad i = 1, \dots, k,$$

are mutually independent.

Proof. Stack the vectors $\mathbf{P}_i \mathbf{Y}$ into one nk -vector. It is a linear function of $\mathbf{Y}$, so it is normal, and the covariance between blocks i and j is $\mathbf{P}_i(\sigma^2 \mathbf{I})\mathbf{P}_j^\top = \sigma^2 \mathbf{P}_i \mathbf{P}_j$. This is $\mathbf{0}$ for $i \neq j$ and $\sigma^2 \mathbf{P}_i$ for $i = j$. By the block form of Theorem 3.4.1 the blocks are mutually independent, and functions of them, the squared lengths, are then mutually independent. Their laws are Theorem 4.3.2(a). ■

In geometric language, which Chapter 6 develops: the coordinates of a spherical normal vector in mutually orthogonal subspaces are independent. It is the reason the rows of an analysis of variance table can be treated separately.

Corollary 4.4.5 — Sample mean and sample variance

Let $Y_1, \dots, Y_n$ ($n \geq 2$) be independent $N(\mu, \sigma^2)$, with sample mean $\bar{Y}$ and sample variance $S^2 = (n-1)^{-1} \sum_i (Y_i - \bar{Y})^2$. Then

- (a) $\bar{Y}$ and S^2 are independent, and $\bar{Y} \sim N(\mu, \sigma^2/n)$;
- (b) $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$;
- (c) for any μ_0 , $T = \sqrt{n}(\bar{Y} - \mu_0)/S \sim t(n-1, \delta)$ with $\delta = \sqrt{n}(\mu - \mu_0)/\sigma$. In particular $T \sim t(n-1)$ when $\mu = \mu_0$.

Proof. Here $\mathbf{Y} \sim N_n(\mu \mathbf{1}, \sigma^2 \mathbf{I})$. Let $\mathbf{P}_1 = n^{-1} \mathbf{1} \mathbf{1}^\top$ and $\mathbf{C} = \mathbf{I} - \mathbf{P}_1$. Both are symmetric and idempotent, $\mathbf{P}_1 \mathbf{C} = \mathbf{0}$, and $\text{rank}(\mathbf{C}) = \text{tr}(\mathbf{C}) = n-1$. Since $\mathbf{P}_1 \mathbf{Y} = \bar{Y} \mathbf{1}$ and $\|\mathbf{C} \mathbf{Y}\|^2 = (n-1)S^2$, Theorem 4.4.4 gives (a), which Example (Sample mean and deviations) obtained directly, and, because $\mathbf{C} \mu \mathbf{1} = \mathbf{0}$, also (b). For (c), $X = \sqrt{n}(\bar{Y} - \mu_0)/\sigma \sim N(\delta, 1)$ and $V = (n-1)S^2/\sigma^2 \sim \chi^2(n-1)$ are independent, and $T = X/\sqrt{V/(n-1)}$. Apply Definition 4.1.3. ■

The independence in (a) characterizes the normal distribution: for $n \geq 2$ independent identically distributed observations with finite variance, $\bar{Y}$ and S^2 are independent only under normality (Lukacs 1942). We will not need this, but it shows that normality is used essentially here. It is not a technicality that more careful arguments could remove.

4.4.3 Preview: the F statistic for nested models

The main use of these results comes in Chapter 11. Its core fits in a few lines.

Theorem 4.4.6 — F statistic for nested subspaces

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$, and let $\mathbf{P}_0$ and $\mathbf{P}$ be symmetric idempotent matrices of ranks $r_0 < r < n$ with $\mathbf{P}\mathbf{P}_0 = \mathbf{P}_0$. Suppose that $\mathbf{P}\boldsymbol{\mu} = \boldsymbol{\mu}$. Then

$$F = \frac{\mathbf{Y}^\top (\mathbf{P} - \mathbf{P}_0) \mathbf{Y} / (r - r_0)}{\mathbf{Y}^\top (\mathbf{I} - \mathbf{P}) \mathbf{Y} / (n - r)} \sim F\left(r - r_0, n - r, \frac{\|(\mathbf{P} - \mathbf{P}_0)\boldsymbol{\mu}\|^2}{\sigma^2}\right),$$

and the noncentrality is zero iff $\mathbf{P}_0\boldsymbol{\mu} = \boldsymbol{\mu}$.

Proof. Transposing $\mathbf{P}\mathbf{P}_0 = \mathbf{P}_0$ gives $\mathbf{P}_0\mathbf{P} = \mathbf{P}_0$. Hence $(\mathbf{P} - \mathbf{P}_0)^2 = \mathbf{P} - 2\mathbf{P}_0 + \mathbf{P}_0 = \mathbf{P} - \mathbf{P}_0$, so $\mathbf{P} - \mathbf{P}_0$ is symmetric idempotent with rank $\text{tr}(\mathbf{P}) - \text{tr}(\mathbf{P}_0) = r - r_0$. Also $(\mathbf{P} - \mathbf{P}_0)(\mathbf{I} - \mathbf{P}) = \mathbf{P} - \mathbf{P} - \mathbf{P}_0 + \mathbf{P}_0 = \mathbf{0}$, and $\mathbf{I} - \mathbf{P}$ is symmetric idempotent of rank $n - r$. By Theorem 4.4.4 the numerator and denominator sums of squares are independent, with σ^{-2} times them distributed as $\chi^2(r - r_0, \|(\mathbf{P} - \mathbf{P}_0)\boldsymbol{\mu}\|^2/\sigma^2)$ and $\chi^2(n - r, \|(\mathbf{I} - \mathbf{P})\boldsymbol{\mu}\|^2/\sigma^2)$. The second noncentrality is 0 since $\mathbf{P}\boldsymbol{\mu} = \boldsymbol{\mu}$. The unknown σ^2 cancels in the ratio, and Definition 4.1.2 applies. Finally, $(\mathbf{P} - \mathbf{P}_0)\boldsymbol{\mu} = \boldsymbol{\mu} - \mathbf{P}_0\boldsymbol{\mu}$. ■

In a linear model, $\mathbf{P} = \mathbf{M}$ projects onto the column space of the full model matrix $\mathbf{X}$ and $\mathbf{P}_0 = \mathbf{M}_0$ onto that of a submodel $\mathbf{X}_0$. The numerator is the drop in residual sum of squares from the submodel to the full model (Chapter 6). The theorem says three things. The test that rejects for large F has an exactly known null distribution. Its power depends on the unknown parameters only through $\gamma = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2$. And, by Theorem 4.1.4, that power increases with γ .

Example. (Detecting curvature)

Let $x_1, \dots, x_n$, $n = 15$, be equally spaced on $[0, 1]$, and suppose $Y_i = 1 + 0.5x_i + \beta_2 x_i^2 + \varepsilon_i$ with independent $N(0, 1)$ errors. To test for curvature, compare the straight line ($\mathbf{X}_0 = [\mathbf{1}, \mathbf{x}]$) with the quadratic ($\mathbf{X} = [\mathbf{1}, \mathbf{x}, \mathbf{x}^2]$). The F statistic has 1 and 12 degrees of freedom, and the 5% critical value is 4.747. For $\beta_2 = 8.0$, the listing finds $\gamma = 6.873$ and power 0.673 from the noncentral F , and the rejection rate in 100,000 simulated data sets is 0.674.

The noncentrality has a clear meaning. It is β_2^2 times the squared length of the part of $\mathbf{x}^2$ orthogonal to $\mathbf{1}$ and $\mathbf{x}$, divided by σ^2 . It grows roughly in proportion to n for a fixed design pattern. Keeping the same design on $[0, 1]$, the smallest n with power at least 0.8 is 21, where $\gamma = 8.97$. Figure 4.4.1 places the example on the power curve and shows the loss of power when the same noncentrality is spread over more numerator degrees

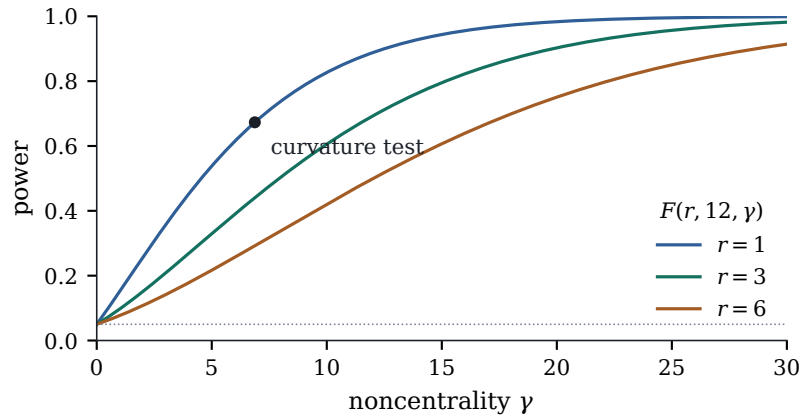


Figure 4.4.1: Power $\Pr\{F(r, 12, \gamma) > F_{0.05}(r, 12)\}$ of the level-0.05 F test as a function of the noncentrality. Every curve starts at 0.05 and increases to 1 (Theorem 4.1.4). For a fixed γ , power falls as the number r of numerator degrees of freedom grows. The dot is the curvature test of Example (Detecting curvature).

of freedom r . At $\gamma = 10$ the power is 0.827, 0.606 and 0.421 for $r = 1, 3, 6$.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(1104)
alpha, s = 0.05, 12
def power(gamma, q, s, alpha=0.05):
    crit = stats.f.ppf(1 - alpha, q, s)
    return stats.ncf.sf(crit, q, s, np.maximum(gamma, 1e-12))
gammas = np.linspace(0, 30, 301)
gamma_ref, rs = 10.0, (1, 3, 6)           # r = numerator degrees of
                                         # freedom
curves = {q: power(gammas, q, s) for q in rs}

n, sigma, beta0, beta1, beta2, reps = 15, 1.0, 1.0, 0.5, 8.0, 100_000
x = np.linspace(0, 1, n)
X0 = np.column_stack([np.ones(n), x])    # straight line
X = np.column_stack([X0, x**2])          # adds curvature

def proj(Z):
    Q, _ = np.linalg.qr(Z)
    return Q @ Q.T

M, M0 = proj(X), proj(X0)
mean = X @ np.array([beta0, beta1, beta2])
gamma = mean @ (M - M0) @ mean / sigma**2  # noncentrality
q, df_resid = 1, n - 3
crit = stats.f.ppf(0.95, q, df_resid)
exact_power = stats.ncf.sf(crit, q, df_resid, gamma)

Y = mean + sigma * rng.standard_normal((reps, n))
num = np.einsum("ij,jk,ik->i", Y, M - M0, Y) / q
den = np.einsum("ij,jk,ik->i", Y, np.eye(n) - M, Y) / df_resid
print(gamma, exact_power, np.mean(num / den > crit))
```

4.4.4 Exercises

4.4.4.1 A. Check your understanding

Exercise A1.

In [Example \(Two estimators of \$\sigma^2\$ under a drifting mean\)](#) let $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\mathbf{I})$ with $n \geq 3$. Use [Theorem 4.2.3\(d\)](#) to show that $\text{Cov}(S^2, Q) = \text{Var}(S^2)$, and deduce that S^2 and Q are dependent with correlation $\{2(n-1)/(3n-4)\}^{1/2}$. Which linear forms $\mathbf{a}^\top \mathbf{Y}$ are independent of Q for every mean vector $\mu \in \mathbb{R}^n$?

Solution. With $\Sigma = \sigma^2\mathbf{I}$, part (d) of [Theorem 4.2.3](#) gives $\text{Cov}(S^2, Q) = 2\sigma^4 \text{tr}(\mathbf{C}\mathbf{D}^\top \mathbf{D})/\{2(n-1)^2\}$. Since $\mathbf{D}\mathbf{1} = \mathbf{0}$, $\mathbf{C}\mathbf{D}^\top \mathbf{D} = \mathbf{D}^\top \mathbf{D}$, whose trace is $2(n-1)$, so $\text{Cov}(S^2, Q) = 2\sigma^4/(n-1) = \text{Var}(S^2)$. The mean terms vanish because $\mathbf{C}\mu = \mathbf{D}\mu = \mathbf{0}$. Using [Var\(Q\)](#) from [Example](#), the correlation is $\{\text{Var}(S^2)/\text{Var}(Q)\}^{1/2} = \{2(n-1)/(3n-4)\}^{1/2}$, which is nonzero, so the forms are dependent. By [Theorem 4.4.1](#), $\mathbf{a}^\top \mathbf{Y}$ is independent of Q for every μ iff $\mathbf{a}^\top \mathbf{D}^\top \mathbf{D} = \mathbf{0}^\top$. Multiplying by $\mathbf{a}$ gives $\|\mathbf{D}\mathbf{a}\|^2 = 0$, so $\mathbf{D}\mathbf{a} = \mathbf{0}$ and $\mathbf{a}$ is a multiple of $\mathbf{1}$; conversely such $\mathbf{a}$ work. The only such linear forms are the multiples of $\bar{Y}$.

4.4.4.2 B. Practice

Exercise B1.

Let $\mathbf{Y} \sim N_2(\mathbf{0}, \mathbf{I})$. Show that $Y_1^2 - Y_2^2$ and $2Y_1Y_2$ are uncorrelated but not independent. Why does [Proposition 4.4.3](#) not apply?

Solution. With $\mathbf{A} = \text{diag}(1, -1)$ and $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\text{tr}(\mathbf{A}\mathbf{B}) = 0$, so the covariance $2\text{tr}(\mathbf{A}\mathbf{B})$ is zero. In polar coordinates $Y_1 = R \cos \Theta$, $Y_2 = R \sin \Theta$, with $R^2 \sim \chi^2(2)$ independent of Θ uniform on $[0, 2\pi)$, the two forms are $R^2 \cos 2\Theta$ and $R^2 \sin 2\Theta$. Now $E(R^4) = 8$ and $E(R^8) = 384$, so $E[(R^2 \cos 2\Theta)^2 (R^2 \sin 2\Theta)^2] = 384/8 = 48$, but the product of the expected squares is $(8/2)^2 = 16$. The forms are dependent. [Proposition 4.4.3](#) needs nonnegative definite matrices, and $\mathbf{A}$ and $\mathbf{B}$ are indefinite.

Exercise B2.

Let $\mathbf{Y} \sim N_n(\mu, \Sigma)$ and let $\mathbf{A}, \mathbf{B}$ be nonnegative definite with $\Sigma\mathbf{A}\Sigma\mathbf{B}\Sigma = \mathbf{0}$. Show that $\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$, so that $\mathbf{Y}^\top \mathbf{A}\mathbf{Y}$ and $\mathbf{Y}^\top \mathbf{B}\mathbf{Y}$ are independent. Show by an example that the implication can fail when $\mathbf{A}$ is not nonnegative definite.

Solution. Write $\mathbf{A} = \mathbf{R}\mathbf{R}^\top$ and $\mathbf{B} = \mathbf{S}\mathbf{S}^\top$, for instance with symmetric square roots, and $\Sigma = \mathbf{L}\mathbf{L}^\top$ with $\mathbf{L}$ of full column rank. Cancelling $\mathbf{L}$ on both sides of $\mathbf{L}(\mathbf{L}^\top \mathbf{A}\mathbf{L})(\mathbf{L}^\top \mathbf{B}\mathbf{L})\mathbf{L}^\top = \mathbf{0}$ gives $\mathbf{L}^\top \mathbf{A}\mathbf{L}\mathbf{L}^\top \mathbf{B}\mathbf{L} = \mathbf{0}$. The trace of this matrix is

$$\text{tr}(\mathbf{L}^\top \mathbf{R}\mathbf{R}^\top \mathbf{L}\mathbf{L}^\top \mathbf{S}\mathbf{S}^\top \mathbf{L}) = \text{tr}\{(\mathbf{R}^\top \Sigma \mathbf{S})(\mathbf{R}^\top \Sigma \mathbf{S})^\top\} = \|\mathbf{R}^\top \Sigma \mathbf{S}\|_F^2,$$

so $\mathbf{R}^\top \Sigma \mathbf{S} = \mathbf{0}$. Hence $\mathbf{A}\Sigma\mathbf{B} = \mathbf{R}(\mathbf{R}^\top \Sigma \mathbf{S})\mathbf{S}^\top = \mathbf{0}$, and [Theorem 4.4.2](#) applies. For nonnegative definite matrices the two conditions are therefore equivalent, since $\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$ trivially implies $\Sigma\mathbf{A}\Sigma\mathbf{B}\Sigma = \mathbf{0}$. Without nonnegative definiteness the argument breaks

at the trace step. Take $\Sigma = \text{diag}(1, 0)$, $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $B = I_2$. Then $\Sigma A \Sigma = \mathbf{0}$, so $\Sigma A \Sigma B \Sigma = \mathbf{0}$, but $A \Sigma B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \neq \mathbf{0}$. With $\mu = (0, m)^\top$ and $m \neq 0$ we have $Y_2 = m$ with probability one, so the forms are $2mY_1$ and $Y_1^2 + m^2$. The second is a nonconstant function of the first, so they are dependent.

Exercise B3.

Let $Y \sim N_n(X\beta, \sigma^2 I)$ with X of full column rank $p < n$, let $\hat{\beta} = (X^\top X)^{-1} X^\top Y$, $M = X(X^\top X)^{-1} X^\top$ and $s^2 = \|(I - M)Y\|^2 / (n - p)$. With v_{jj} the j th diagonal entry of $(X^\top X)^{-1}$, show that $(\hat{\beta}_j - b) / (s\sqrt{v_{jj}}) \sim t(n - p, (\beta_j - b) / (\sigma\sqrt{v_{jj}}))$.

Solution. Here M is symmetric and idempotent with $\text{tr } M = \text{tr}\{(X^\top X)^{-1} X^\top X\} = p$, and $MX = X$. So $(n - p)s^2 / \sigma^2 = \|(I - M)Y\|^2 / \sigma^2 \sim \chi^2(n - p)$ by [Theorem 4.3.2\(a\)](#), because $(I - M)X\beta = \mathbf{0}$. With $B = (X^\top X)^{-1} X^\top$, $B(\sigma^2 I)(I - M) = \sigma^2 (X^\top X)^{-1} (X^\top - X^\top M) = \mathbf{0}$, so $\hat{\beta} = BY$ is independent of s^2 ([Theorem 4.4.1](#)). Since $\hat{\beta} \sim N_p(\beta, \sigma^2 (X^\top X)^{-1})$, the variable $X = (\hat{\beta}_j - b) / (\sigma\sqrt{v_{jj}})$ is $N((\beta_j - b) / (\sigma\sqrt{v_{jj}}), 1)$. The statistic is $X / \sqrt{V / (n - p)}$ with $V = (n - p)s^2 / \sigma^2$, and [Definition 4.1.3](#) applies.

Exercise B4.

Show that $F_\alpha(r, s) \rightarrow \chi_\alpha^2(r) / r$ as $s \rightarrow \infty$, where $\chi_\alpha^2(r)$ is the upper α point of $\chi^2(r)$, and that the power $\Pr\{F(r, s, \gamma) > F_\alpha(r, s)\}$ tends to $\Pr\{\chi^2(r, \gamma) > \chi_\alpha^2(r)\}$, the power of the test that knows σ^2 .

4.5 The Cochran theorem

An analysis of variance starts from an identity

$$Y^\top Y = Y^\top A_1 Y + \cdots + Y^\top A_k Y, \quad A_1 + \cdots + A_k = I,$$

that splits the total sum of squares into pieces attributed to sources of variation. [Theorem 4.4.4](#) says the pieces are independent chi-squared variables when the A_i are mutually orthogonal projections. Checking that directly means checking k idempotence conditions and $\binom{k}{2}$ orthogonality conditions. Cochran's theorem shows that one count of ranks does all of it.

4.5.1 The algebra

The matrix algebra behind this was done in [Chapter 1](#). For symmetric matrices, any two of the three properties “each A_i is idempotent”, “ $A_i A_j = \mathbf{0}$ for $i \neq j$ ” and “the sum is idempotent” imply the third, and symmetric matrices that add to I with ranks adding to n are idempotent and mutually orthogonal ([Theorem 1.9.3](#)). The next theorem adds the rank count for a sum that is idempotent but not necessarily I , which is what nested models need.

Theorem 4.5.1 — Cochran, algebraic form

Let $\mathbf{A}_1, \dots, \mathbf{A}_k$ be symmetric $n \times n$ matrices and $\mathbf{A} = \sum_i \mathbf{A}_i$. Consider

- (a) each $\mathbf{A}_i$ is idempotent;
- (b) $\mathbf{A}_i \mathbf{A}_j = \mathbf{0}$ for all $i \neq j$;
- (c) $\mathbf{A}$ is idempotent;
- (d) $\text{rank}(\mathbf{A}) = \sum_i \text{rank}(\mathbf{A}_i)$.

Then any two of (a), (b), (c) imply all four, and (c) together with (d) implies all four.

Proof. Any two of (a), (b), (c). By Theorem 1.9.3, any two of them imply the third. When all three hold, rank equals trace for symmetric idempotent matrices (Theorem 1.9.2), so $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}) = \sum_i \text{tr}(\mathbf{A}_i) = \sum_i \text{rank}(\mathbf{A}_i)$, which is (d).

(c) and (d) imply (a) and (b). Let $r = \text{rank}(\mathbf{A})$ and $r_i = \text{rank}(\mathbf{A}_i)$. Every $\mathbf{A}\mathbf{v} = \sum_i \mathbf{A}_i \mathbf{v}$ lies in $\mathcal{C}(\mathbf{A}_1) + \dots + \mathcal{C}(\mathbf{A}_k)$, whose dimension is at most $\sum_i r_i = r = \dim \mathcal{C}(\mathbf{A})$. So the two spaces coincide, and in particular $\mathcal{C}(\mathbf{A}_i) \subseteq \mathcal{C}(\mathbf{A})$. By Theorem 1.9.2(a), $\mathbf{A} = \mathbf{Q}\mathbf{Q}^\top$ with $\mathbf{Q}$ of size $n \times r$ and $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r$. Since $\mathbf{A}$ is symmetric and idempotent it fixes every vector of its column space, so $\mathbf{A}\mathbf{A}_i = \mathbf{A}_i$, and transposing, $\mathbf{A}_i \mathbf{A} = \mathbf{A}_i$. Hence $\mathbf{A}_i = \mathbf{A}\mathbf{A}_i\mathbf{A} = \mathbf{Q}\mathbf{B}_i\mathbf{Q}^\top$ with $\mathbf{B}_i = \mathbf{Q}^\top \mathbf{A}_i \mathbf{Q}$. The $r \times r$ matrices $\mathbf{B}_i$ are symmetric, satisfy $\sum_i \mathbf{B}_i = \mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{I}_r$, and have $\text{rank}(\mathbf{B}_i) = r_i$ because $\mathbf{Q}$ has orthonormal columns. By the last statement of Theorem 1.9.3, $\mathbf{B}_i^2 = \mathbf{B}_i$ and $\mathbf{B}_i \mathbf{B}_j = \mathbf{0}$ for $i \neq j$, and multiplying by $\mathbf{Q}$ on the left and $\mathbf{Q}^\top$ on the right transfers both identities to the $\mathbf{A}_i$. ■

4.5.2 The distributional statement**Theorem 4.5.2 — Cochran**

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$, and let $\mathbf{A}_1, \dots, \mathbf{A}_k$ be symmetric with ranks $r_1, \dots, r_k$, whose sum $\mathbf{A} = \sum_i \mathbf{A}_i$ is idempotent of rank r (for example $\mathbf{A} = \mathbf{I}$, $r = n$). The following are equivalent:

- (i) the forms $\mathbf{Y}^\top \mathbf{A}_i \mathbf{Y} / \sigma^2$ are mutually independent, and $\mathbf{Y}^\top \mathbf{A}_i \mathbf{Y} / \sigma^2 \sim \chi^2(r_i, \boldsymbol{\mu}^\top \mathbf{A}_i \boldsymbol{\mu} / \sigma^2)$ for each i ;
- (ii) $r_1 + \dots + r_k = r$;
- (iii) each $\mathbf{A}_i$ is idempotent;
- (iv) $\mathbf{A}_i \mathbf{A}_j = \mathbf{0}$ for all $i \neq j$.

Proof. Condition (c) of Theorem 4.5.1 holds by assumption, so that theorem shows that (ii), (iii) and (iv) are equivalent, and each implies the other two. If (iii) and (iv) hold, Theorem 4.4.4 gives (i). If (i) holds, apply Theorem 4.3.1 to $\sigma^{-1} \mathbf{Y} \sim N_n(\sigma^{-1} \boldsymbol{\mu}, \mathbf{I})$. Each $\mathbf{A}_i$ with $r_i \geq 1$ is idempotent, and an $\mathbf{A}_i$ of rank zero is $\mathbf{0}$. This is (iii).

In words: *if the degrees of freedom add up, the sums of squares are independent chi-squared variables.* Independence in (i) is not needed to reach (iii). The marginal chi-squared laws alone force the whole structure.

Example. (The one-way layout)

Let observations fall into g groups of sizes $n_1, \dots, n_g$, with $Y_{\ell j}$ independent $N(\mu_\ell, \sigma^2)$. Let $\mathbf{M}$ replace each observation by its group mean, and $\mathbf{P}_1 = n^{-1}\mathbf{1}\mathbf{1}^\top$ by the grand mean. Then

$$\mathbf{I} = \mathbf{P}_1 + (\mathbf{M} - \mathbf{P}_1) + (\mathbf{I} - \mathbf{M})$$

splits $\|\mathbf{Y}\|^2$ into $n\bar{Y}^2$, the between-group sum of squares $\sum_\ell n_\ell(\bar{Y}_\ell - \bar{Y})^2$ and the within-group sum of squares $\sum_{\ell,j} (Y_{\ell j} - \bar{Y}_\ell)^2$. The three matrices are symmetric with ranks 1, $g - 1$ and $n - g$, which add to n . Cochran's theorem needs nothing more. The three sums of squares, divided by σ^2 , are independent noncentral chi-squared variables. Their noncentralities are $n\bar{\mu}^2/\sigma^2$, $\sum_\ell n_\ell(\mu_\ell - \bar{\mu})^2/\sigma^2$ with $\bar{\mu} = \sum_\ell n_\ell\mu_\ell/n$, and 0. The last is zero because $\mathbf{M}\boldsymbol{\mu} = \boldsymbol{\mu}$. The ratio of the between and within mean squares is $F(g - 1, n - g, \gamma)$ by [Theorem 4.4.6](#), and it is central iff all group means are equal.

The script uses $g = 3$ groups of sizes 3, 5, 4, means 2, 3, 1.5, and $\sigma = 1$, and draws 200,000 data sets. [Table 4.5.1](#) compares the theoretical means $r_i + \gamma_i$ and variances $2r_i + 4\gamma_i$ with the simulated ones. The largest absolute correlation between the three sums of squares is 0.0013. As a check that goes beyond correlation, the proportion of data sets in which both the between and the within sums of squares exceed their medians is 0.2497, against 0.25 for independent variables.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(5150)

sizes, means, sigma = [3, 5, 4], [2.0, 3.0, 1.5], 1.0
groups = np.repeat(np.arange(3), sizes)
n = len(groups)
Z = (groups[:, None] == np.arange(3)).astype(float) # group indicators
M = Z @ np.diag(1 / np.array(sizes)) @ Z.T # group-mean projection
J = np.ones((n, n)) / n
pieces = [J, M - J, np.eye(n) - M]
ranks = [np.linalg.matrix_rank(P) for P in pieces]
print("ranks", ranks, "sum", sum(ranks), "n", n)

mu = np.array(means)[groups]
gammas = [mu @ P @ mu / sigma**2 for P in pieces]
Y = mu + sigma * rng.standard_normal((200_000, n))
Q = np.stack([np.einsum("ij,jk,ik->i", Y, P, Y) for P in pieces], axis=1) / sigma**2
print(np.corrcoef(Q, rowvar=False).round(3))
```

Table 4.5.1. Cochran decomposition for a one-way layout with $n = 12$: theory against 200,000 simulated data sets.

Sum of squares	r_i	γ_i	mean	simulated	variance	simulated
grand mean	1	60.750	61.750	61.763	245.00	244.95
between groups	2	5.250	7.250	7.230	25.00	24.85
within groups	9	0.000	9.000	8.994	18.00	17.96

Remark (What the theorem does not cover). Two assumptions carry the weight. The first is normality. Without it, [Theorem 4.2.3\(d\)](#) fails, and with independent non-normal errors the between and within sums of squares are in general correlated, with a correlation driven by the fourth cumulant ([Exercise \(B2\)](#)). The second is the spherical covariance $\sigma^2 \mathbf{I}$. With $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$, $\mathbf{V}$ positive definite, apply the theorem to $\mathbf{V}^{-1/2} \mathbf{Y}$, whose forms have matrices $\mathbf{V}^{1/2} \mathbf{A}_i \mathbf{V}^{1/2}$. Their ranks are those of the $\mathbf{A}_i$, so the rank count is unchanged. What changes is the condition on the sum: $\mathbf{V}^{1/2} \mathbf{A} \mathbf{V}^{1/2}$ is idempotent iff $\mathbf{A} \mathbf{V} \mathbf{A} = \mathbf{A}$ (for example $\sum_i \mathbf{A}_i = \mathbf{V}^{-1}$), and conditions (iii) and (iv) become $\mathbf{A}_i \mathbf{V} \mathbf{A}_i = \mathbf{A}_i$ and $\mathbf{A}_i \mathbf{V} \mathbf{A}_j = \mathbf{0}$. These are the generalized least squares decompositions of Chapter 31.

Most analysis of variance tables in this book come from a chain of nested model spaces. The successive differences of their projections are mutually orthogonal projections whose ranks add up, so Cochran's theorem applies without further work ([Exercise \(B2\)](#), [Theorem 9.1.1](#)).

4.5.3 Exercises

4.5.3.1 A. Check your understanding

Exercise A1.

Let $\mathbf{A}_1 = \text{diag}(2, 0)$ and $\mathbf{A}_2 = \text{diag}(-1, 1)$, so that $\mathbf{A}_1 + \mathbf{A}_2 = \mathbf{I}_2$. Which hypotheses of [Theorem 4.5.2](#) fail, and what are the laws of the two forms when $\mathbf{Y} \sim N_2(\mathbf{0}, \mathbf{I})$?

Solution. The sum $\mathbf{I}_2$ is idempotent, but every one of the conditions (ii)–(iv) fails. The ranks are 1 and 2, which add to $3 \neq 2$. Neither matrix is idempotent, since $\mathbf{A}_1^2 = \text{diag}(4, 0)$ and $\mathbf{A}_2^2 = \mathbf{I}_2$. And $\mathbf{A}_1 \mathbf{A}_2 = \text{diag}(-2, 0) \neq \mathbf{0}$. The forms are $2Y_1^2$, which is twice a $\chi^2(1)$ variable, and $Y_2^2 - Y_1^2$, the difference of two independent $\chi^2(1)$ variables, which takes negative values. Neither is chi-squared. They are dependent, because

$$\text{Cov}(2Y_1^2, Y_2^2 - Y_1^2) = 2 \text{Cov}(Y_1^2, Y_2^2) - 2 \text{Var}(Y_1^2) = 0 - 2 \cdot 2 = -4 \neq 0.$$

4.5.3.2 B. Practice

Exercise B1.

A two-way layout has one observation Y_{ij} for each of a rows and b columns. Write $\|\mathbf{Y}\|^2$ as the sum of the grand-mean, row, column and residual sums of squares, identify the four matrices, and use [Theorem 4.5.2](#) to show that under $Y_{ij} \sim N(\mu + \alpha_i + \beta_j, \sigma^2)$

independently the four sums of squares are independent, with degrees of freedom 1, $a - 1$, $b - 1$ and $(a - 1)(b - 1)$.

Exercise B2.

Let $P_1, \dots, P_m$ be symmetric idempotent with $\mathcal{C}(P_1) \subset \mathcal{C}(P_2) \subset \dots \subset \mathcal{C}(P_m)$, and put $P_0 = \mathbf{0}$, $P_{m+1} = \mathbf{I}$. Show that the matrices $P_i - P_{i-1}$, $i = 1, \dots, m+1$, satisfy the conditions of [Theorem 4.5.2](#), so that the sequential sums of squares $\|(P_i - P_{i-1})Y\|^2$ are independent noncentral chi-squared variables when $Y \sim N_n(\mu, \sigma^2 \mathbf{I})$.

4.5.3.3 C. Going deeper

Exercise C1.

Let $Y \sim N_n(\mathbf{0}, \mathbf{I})$, and $Q = Q_1 + Q_2$ with $Q = Y^\top \mathbf{A} Y \sim \chi^2(r)$, $Q_1 = Y^\top \mathbf{A}_1 Y \sim \chi^2(r_1)$ and $Q_2 = Y^\top \mathbf{A}_2 Y$ for a nonnegative definite $\mathbf{A}_2$. Show that Q_1 and Q_2 are independent and $Q_2 \sim \chi^2(r - r_1)$.

Solution. By [Theorem 4.3.1](#), $\mathbf{A}$ and $\mathbf{A}_1$ are idempotent, with ranks r and r_1 . If $\mathbf{A}v = \mathbf{0}$, then $0 = v^\top \mathbf{A}_1 v + v^\top \mathbf{A}_2 v$ with both terms nonnegative, so $\|\mathbf{A}_1 v\|^2 = v^\top \mathbf{A}_1 v = 0$. Thus $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{A}_1)$, hence $\mathcal{C}(\mathbf{A}_1) \subseteq \mathcal{C}(\mathbf{A})$ and $\mathbf{A}\mathbf{A}_1 = \mathbf{A}_1 = \mathbf{A}_1\mathbf{A}$. Then $\mathbf{A}_2^2 = (\mathbf{A} - \mathbf{A}_1)^2 = \mathbf{A} - 2\mathbf{A}_1 + \mathbf{A}_1 = \mathbf{A}_2$, so $\mathbf{A}_2$ is idempotent, and $\mathbf{A}_1\mathbf{A}_2 = \mathbf{A}_1\mathbf{A} - \mathbf{A}_1 = \mathbf{0}$. By [Theorem 4.4.4](#), Q_1 and Q_2 are independent and $Q_2 \sim \chi^2(\text{tr } \mathbf{A}_2) = \chi^2(r - r_1)$.

4.6 Summary and Notes

4.6.1 Summary

Idea.

1. If $Z \sim N_r(\mu, \mathbf{I})$ then $\|Z\|^2 \sim \chi^2(r, \gamma)$ with $\gamma = \|\mu\|^2$, mean $r + \gamma$ and variance $2r + 4\gamma$. The law is a $\text{Poisson}(\gamma/2)$ mixture of $\chi^2(r + 2k)$ laws, and its tails increase strictly with γ ([Theorem 4.1.2](#) and [Proposition 4.1.3](#)).
2. $F(r, s, \gamma)$ and $t(s, \delta)$ put a noncentral numerator over an independent central denominator. The power of an F test increases strictly with γ , and $T^2 \sim F(1, s, \delta^2)$ ([Theorem 4.1.4](#)).
3. $E(Y^\top \mathbf{A} Y) = \text{tr}(\mathbf{A}\Sigma) + \mu^\top \mathbf{A}\mu$ for any distribution. Under normality

$$\text{Var}(Y^\top \mathbf{A} Y) = 2 \text{tr}\{(\mathbf{A}\Sigma)^2\} + 4\mu^\top \mathbf{A}\Sigma \mathbf{A}\mu, \quad \text{Cov}(KY, Y^\top \mathbf{A} Y) = 2K\Sigma \mathbf{A}\mu$$

([Theorem 4.2.3](#)).

4. A normal quadratic form is a constant plus independent terms $\lambda_j(W_j + \nu_j)^2$, and for singular Σ possibly normal terms. The weights λ_j are the nonzero eigenvalues of $\mathbf{A}\Sigma$ ([Theorem 4.2.1](#)).
5. For $Y \sim N_n(\mu, \mathbf{I})$, $Y^\top \mathbf{A} Y$ is chi-squared iff $\mathbf{A}$ is idempotent. Then the degrees of freedom are $\text{rank } \mathbf{A} = \text{tr } \mathbf{A}$ and the noncentrality is $\mu^\top \mathbf{A}\mu$ ([Theorem 4.3.1](#)).

For positive definite Σ the condition is that $A\Sigma$ be idempotent. For projections, $\|PY\|^2/\sigma^2 \sim \chi^2(\text{rank } P, \|P\mu\|^2/\sigma^2)$ (Theorem 4.3.2).

6. BY and $Y^\top AY$ are independent if $B\Sigma A = \mathbf{0}$, and $Y^\top AY$ and $Y^\top BY$ are independent if $A\Sigma B = \mathbf{0}$, because a quadratic form is a function of AY (Theorem 4.4.1 and Theorem 4.4.2).
7. Projections of a spherical normal vector onto mutually orthogonal subspaces are independent. This gives the independence of $\bar{Y}$ and S^2 and the exact $F(r - r_0, n - r, \gamma)$ law of the nested-model F statistic (Theorem 4.4.4, Corollary 4.4.5 and Theorem 4.4.6).
8. Cochran: if symmetric matrices add to an idempotent matrix and their ranks add up, the corresponding sums of squares are independent chi-squared variables (Theorem 4.5.1 and Theorem 4.5.2).

4.6.2 Notes and sources

Source texts. Rencher and Schaalje (2008, chapter 5) cover the material of Section 4.2, Section 4.3, Section 4.4 and Section 4.5 for positive definite covariance matrices. They derive the mgf of a quadratic form by integrating against the normal density and obtain its variance by differentiating the logarithm of the mgf; the converse of the chi-squared theorem is referred to the literature. They use the noncentrality $\mu^\top \mu/2$. Seber and Lee (2003, section 2.4) reduce the general case to a spherical normal vector by an orthogonal diagonalization. They prove necessity of idempotence for central forms by factorizing the moment generating function, and give the eigenvalue criterion for singular covariance matrices. Christensen (2020, section 1.3) states the three conditions of Theorem 4.3.2(c) for arbitrary nonnegative definite Σ , and obtains independence by viewing quadratic forms as functions of linear ones, the approach followed in Section 4.4. The canonical representation (Theorem 4.2.1), the proof of necessity through the coefficients Eq. 4.3.1, the monotonicity proof through the Poisson mixture, and the treatment of the Cochran algebra are this book's own arrangements of standard material.

Noncentral distributions. The noncentral chi-squared distribution first appeared in Fisher (1928). Patnaik (1949) studied its approximation by a scaled central chi-squared variable, and Satterthwaite (1946) used the same two-moment idea for linear combinations of mean squares, which reappears in Chapter 32. Johnson et al. (1995, chapters 29–31) collect the properties of the noncentral chi-squared, F and t families. The monotonicity of power in the noncentrality and in the degrees of freedom is treated systematically by Ghosh (1973).

General quadratic forms. The exact distribution of a quadratic form that is not chi-squared can be computed by numerical inversion of its characteristic function (Gil-Pelaez 1951; Imhof 1961). Mathai and Provost (1992) give a book-length treatment of quadratic forms in normal and non-normal variables. The mean square successive difference of Example (Two estimators of σ^2 under a drifting mean) goes back to von Neumann (1941), who derived the exact distribution of its ratio to S^2 .

Independence and Cochran's theorem. Cochran (1934) proved the decomposition theorem that bears his name, in the context of the analysis of covariance. The necessity of $A\Sigma B = \mathbf{0}$ for independence of two forms is attributed to Craig (1943). Its full proof for arbitrary symmetric matrices and arbitrary means turned out to be subtle, and Driscoll and Gundberg (1986) trace the flawed and correct arguments. Searle (1971) gives general proofs of the independence and Cochran theorems in the style most linear-models texts follow. For nonnegative definite A and B , the condition $\Sigma A \Sigma B \Sigma = \mathbf{0}$ of Christensen (2020, Theorem 1.3.8) is in fact equivalent to

$\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$ (Exercise (B2)), so that theorem adds nothing to Theorem 4.4.2 even for singular Σ , whereas his Theorem 1.3.9, for general symmetric $\mathbf{A}$ and $\mathbf{B}$, does. That the independence of $\bar{Y}$ and S^2 characterizes the normal distribution was shown by Lukacs (1942).

4.6.3 References

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Chapter 5

The Model, Least Squares and the Normal Equations

A regression model is a claim about how the distribution of a response changes with a set of regressors, and the linear model is the simplest useful claim of that kind. It says that the mean of the response is a linear combination of known columns, that the errors around this mean have a common variance and are uncorrelated, and nothing more. This chapter sets the model up and draws out what follows from those assumptions alone. It starts with what the assumptions do and do not say, fits a straight line by calculus, and passes to the matrix form of the model, where polynomials, harmonics, indicator variables and interactions all turn out to be linear models. It then derives the least squares estimator and the normal equations, computes the mean and covariance of the estimator, estimates the error variance without bias, and asks what a fitted coefficient means. The last section surveys the models the rest of the book builds on this one. Throughout, the model matrix has full column rank. [Chapter 6](#) retells the same story geometrically and removes that condition.

What you need. [Chapter 1](#): rank ([Proposition 1.2.3](#)), inverses, nonnegative definite matrices ([Proposition 1.7.5](#)), traces ([Theorem 1.6.1](#)) and the derivatives of linear and quadratic forms ([Theorem 1.11.1](#)). [Chapter 2](#): means and covariances of linear transformations ([Theorem 2.2.1](#)), the mean and variance of a quadratic form ([Theorem 2.3.1](#), [Theorem 2.3.3](#)) and the laws of total expectation and covariance ([Proposition 2.5.4](#)). Normality ([Chapter 3](#)) appears only in remarks.

5.0.1 Roadmap

- [What a regression model claims](#)
- [Simple linear regression as a warm-up](#)
- [The multiple linear regression model](#)
- [Least squares and the normal equations](#)
- [Properties of the estimator under second-moment assumptions](#)
- [Estimating the error variance](#)
- [Interpreting coefficients](#)
- [A panorama of regression models to come](#)
- [Summary and notes](#)

5.1 What a regression model claims

A data set for regression consists of n cases. Each case records a response y_i and a few other measurements, the regressors, on the same unit, such as a patient or a month. The question is how the response depends on the regressors. Two cases with identical regressors almost always have different responses, so a regression model cannot be an equation linking y to the regressors. It is a statement about the *distribution* of the response given the regressors, and usually only about one feature of that distribution, its mean.

This section separates what is automatically true of any response and regressors from what a model actually assumes. The linear model's assumptions come in layers, and each layer buys a different part of the theory.

5.1.1 The regression function

Let Y be a response with finite variance and $\mathbf{X}$ a random vector of regressors, defined on the same probability space. We need only elementary properties of conditional expectation: the tower property $E[E(Y | \mathbf{X})] = E(Y)$, and $E[g(\mathbf{X})Y | \mathbf{X}] = g(\mathbf{X})E(Y | \mathbf{X})$ for functions g of the regressors.

Definition 5.1.1 — Regression function

The **regression function** of Y on $\mathbf{X}$ is the conditional mean

$$m(\mathbf{x}) = E(Y | \mathbf{X} = \mathbf{x}).$$

The **error** is $\varepsilon = Y - m(\mathbf{X})$, so that $Y = m(\mathbf{X}) + \varepsilon$.

The decomposition $Y = m(\mathbf{X}) + \varepsilon$ looks like a model, but it is not one. It holds for every pair $(\mathbf{X}, Y)$ with $EY^2 < \infty$, and the error it defines has two properties that are often listed as “assumptions” even though they are automatic.

Proposition 5.1.1 — The error of the regression function

Let $EY^2 < \infty$, and let m and ε be as in Definition 5.1.1.

- (a) $E(\varepsilon | \mathbf{X}) = 0$, and hence $E(\varepsilon) = 0$.
- (b) $E[\varepsilon g(\mathbf{X})] = 0$ for every function g with $Eg(\mathbf{X})^2 < \infty$. In particular ε is uncorrelated with every such function of the regressors.
- (c) For every such g ,

$$E[(Y - g(\mathbf{X}))^2] = E(\varepsilon^2) + E[(m(\mathbf{X}) - g(\mathbf{X}))^2].$$

So $m(\mathbf{X})$ is the best predictor of Y from $\mathbf{X}$ in mean squared error, and it is the only one up to events of probability zero.

Proof. (a) $E(\varepsilon | \mathbf{X}) = E(Y | \mathbf{X}) - m(\mathbf{X}) = 0$, and the tower property gives $E(\varepsilon) = E[E(\varepsilon | \mathbf{X})] = 0$. (b) The product $\varepsilon g(\mathbf{X})$ is integrable by the Cauchy–Schwarz inequality,

because $E\varepsilon^2 \leq 2EY^2 + 2Em(\mathbf{X})^2$ and $Em(\mathbf{X})^2 \leq EY^2$ by Jensen's inequality for conditional expectations. Then $E[\varepsilon g(\mathbf{X})] = E[g(\mathbf{X}) E(\varepsilon | \mathbf{X})] = 0$. (c) Write $Y - g(\mathbf{X}) = \varepsilon + (m(\mathbf{X}) - g(\mathbf{X}))$ and expand the square. The cross term is $2E[\varepsilon(m(\mathbf{X}) - g(\mathbf{X}))]$, which is zero by (b) applied to the function $m - g$. The second term on the right is zero only if $g(\mathbf{X}) = m(\mathbf{X})$ with probability one. ■

Part (c) is the reason the conditional mean is the natural target. It is what an ideal forecaster would report under squared-error loss. It is also the reason regression is sometimes misread. The function m summarizes the conditional distribution by its centre. Two populations can have the same regression function and very different spreads, shapes and tails around it.

Warning. Part (b) says the error is uncorrelated with the regressors *by construction*. The lack of correlation between errors and regressors therefore cannot be tested from the data or defended as a scientific assumption about the regression function. The real assumption hides elsewhere: in the claim that m has a particular form, or that the quantity of interest is m at all. When someone says “the error is correlated with the regressor”, the error they have in mind is a different object, usually the deviation from a *causal* or *structural* relation rather than from the conditional mean. Chapter 24 and Chapter 25 take this up.

5.1.2 What the linear model adds

A linear regression model makes claims about m and about the errors that $EY^2 < \infty$ does not guarantee. Write $\mathbf{x}_{(i)}$ for the regressors of case i , already transformed into the columns the model uses (a constant 1, the regressors themselves, their squares, indicators of categories, and so on; [Section 5.3](#) gives many examples). The assumptions come in four layers.

- (L1) **Linearity.** $E(Y_i) = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ for an unknown $\boldsymbol{\beta} \in \mathbb{R}^p$. The regression function belongs to a fixed p -dimensional family of functions.
- (L2) **Constant variance.** $\text{Var}(Y_i) = \sigma^2$ for every i , with σ^2 unknown.
- (L3) **Uncorrelated errors.** $\text{Cov}(Y_i, Y_j) = 0$ for $i \neq j$.
- (L4) **Normality.** The errors are jointly normal.

The first three layers are the **second-moment assumptions**. They mention only means, variances and covariances, and they are all that [Section 5.5](#) and [Section 5.6](#) use. Adding (L4) gives the normal linear model, and with it exact sampling distributions for estimates and test statistics, which [Chapter 7](#) and Chapter 11 develop. Each layer is used by a specific part of the theory:

Assumptions	What they deliver	Where
(L1)	the least squares estimator is unbiased	Section 5.5
(L1)–(L3)	its covariance matrix; an unbiased estimate of σ^2 ; Gauss–Markov optimality	Section 5.5 , Section 5.6 , Chapter 7

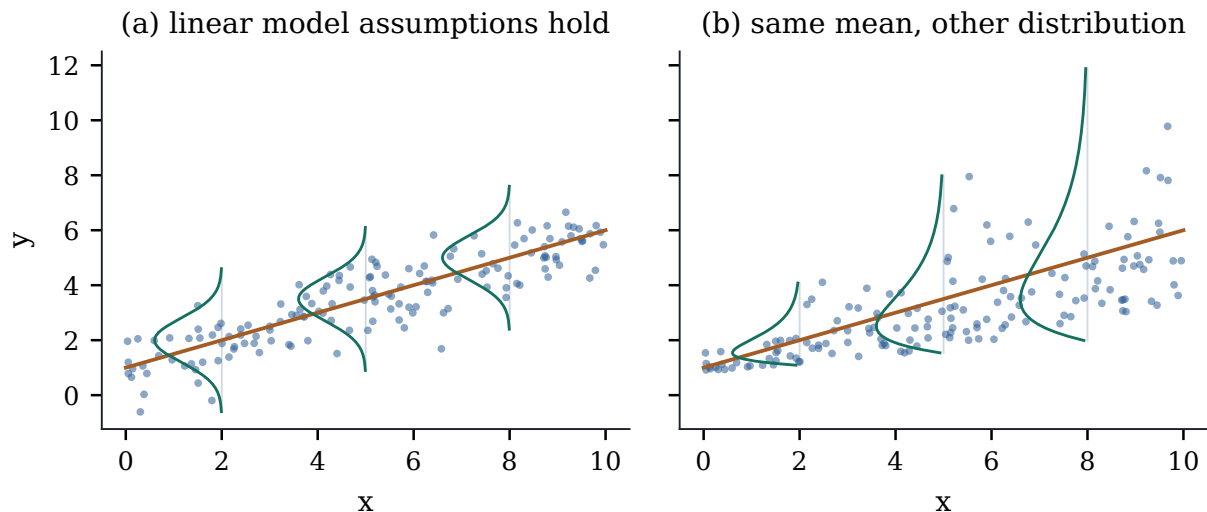


Figure 5.1.1: Two data-generating processes with the same regression function $1 + 0.5x$ (brown line). The curves are conditional densities of Y given x , drawn sideways at $x = 2, 5, 8$. (a) Normal errors with constant variance: all four layers of assumptions hold. (b) Right-skewed errors whose spread grows with x : only linearity holds.

Assumptions	What they deliver	Where
(L1)–(L4)	exact t and F distributions; maximum likelihood; exact intervals	Chapter 7 , Chapter 11, Chapter 12

Least squares itself needs none of them. The estimator of [Section 5.4](#) is defined for any data at all. The assumptions only tell us how good it is.

Example. (Same regression function, different models)

[Figure 5.1.1](#) shows two simulated data sets with the same regression function $m(x) = 1 + 0.5x$. In panel (a) the errors are normal with constant standard deviation 0.8. In panel (b) they are skewed to the right, and their standard deviation grows linearly in x , from 0.15 at $x = 0$ to 2.65 at $x = 10$. The curves show the conditional density of Y at three values of x . Both processes satisfy (L1), so least squares estimates the line without bias in both. Only process (a) satisfies (L2), so only there do the variance formulas of [Section 5.5](#) apply as stated. In process (b) most points lie below the line, because the median of a right-skewed distribution is below its mean. The line is still the conditional mean.

The assumptions concern the errors ε_i , which are never observed. What we see are residuals, the differences between the responses and a fitted function. Whether the assumptions are credible is judged from residuals, from the design of the study, and from knowledge of the subject. Part V of the book is devoted to the first of these and to what goes wrong when the assumptions fail.

5.1.3 Fixed and random regressors

The layers (L1)–(L4) were written with $E(Y_i)$ and $\text{Var}(Y_i)$, as if the regressors were constants. In a designed experiment they are. The experimenter chooses the temperatures, doses or fertilizer levels, and only the responses are random. In an observational study the regressors are measured on units drawn from a population, so the pairs $(x_{(i)}, Y_i)$ are random together.

This book's convention, like that of most linear-model theory, is to treat the regressors as fixed, and to read (L1)–(L4) in the random case as statements *conditional on the regressors*: $E(Y_i | \mathbf{X}) = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, $\text{Var}(Y_i | \mathbf{X}) = \sigma^2$, and so on, where $\mathbf{X}$ collects all the regressors. There are two reasons for this choice.

First, conditional statements imply unconditional ones where it matters. If the estimator is unbiased given every possible $\mathbf{X}$, it is unbiased when $\mathbf{X}$ is random, by the tower property. [Section 5.5](#) makes this precise ([Corollary 5.5.5](#)). Second, conditioning describes the precision of the data actually in hand. A sample whose regressors happen to be spread out estimates a slope better than one whose regressors happen to be bunched together. The conditional variance says so, and the unconditional variance averages the two cases.

Example. (One slope, two sampling schemes)

Take $Y = 1 + 0.5x + \varepsilon$ with standard normal errors and $n = 15$ cases. In the *fixed design*, the regressor values are drawn once from the uniform distribution on $[0, 10]$ and then kept. They have $S_{xx} = \sum_i (x_i - \bar{x})^2 = 103.73$. In the *random design*, a new set of 15 regressor values is drawn for every sample. Over 50,000 simulated samples, the least squares slope (formula in [Section 5.2](#)) averages 0.5007 under the fixed design and 0.5006 under the random one. Both are unbiased. Their variances differ in meaning. Under the fixed design the simulated variance is 0.00960, matching $\sigma^2/S_{xx} = 0.00964$ for this particular design. Under the random design it is 0.00933, matching the *average* $E(\sigma^2/S_{xx}) = 0.00923$ over designs, estimated from the same simulation. Replacing S_{xx} by its expectation 116.67 would give 0.00857, which is too small. By Jensen's inequality $E(1/S_{xx}) \geq 1/E(S_{xx})$, so random bunching of the regressors costs precision on average.

```
import numpy as np

rng = np.random.default_rng(2005)
n, beta0, beta1, sigma = 15, 1.0, 0.5, 1.0
reps = 50_000

def slope(x, y):
    """Least squares slope of y on x (with an intercept)."""
    xc = x - x.mean()
    return xc @ (y - y.mean()) / (xc @ xc)

x_fixed = rng.uniform(0, 10, size=n)          # drawn once, then held fixed
Sxx_fixed = np.sum((x_fixed - x_fixed.mean()) ** 2)
b_fixed = np.empty(reps)
b_random = np.empty(reps)
Sxx_random = np.empty(reps)
for r in range(reps):
    e = sigma * rng.standard_normal(n)
    b_fixed[r] = slope(x_fixed, beta0 + beta1 * x_fixed + e)
    x = rng.uniform(0, 10, size=n)            # a new design for every sample
```



```

Sxx_random[r] = np.sum((x - x.mean()) ** 2)
b_random[r] = slope(x, beta0 + beta1 * x + e)

print(f"fixed design: mean {b_fixed.mean():.4f} var {b_fixed.var():.5f}"
      f" sigma^2/Sxx = {sigma**2 / Sxx_fixed:.5f}")
print(f"random design: mean {b_random.mean():.4f} var {b_random.var():.5f}"
      f" E(sigma^2/Sxx) = {np.mean(sigma**2 / Sxx_random):.5f}")

```

Conditioning on the regressors is not always harmless. It fails when the regressors are measured with error, because then the regression on the *measured* values is not the one of scientific interest (Chapter 24). It fails when a regressor is a lagged value of the response, as in autoregressive time series, because then $\mathbf{X}$ is not determined before the errors are drawn. It also changes the question when the target is a population quantity, such as the best linear predictor of one variable from others in a population where the regression function is not linear (Section 6.11 and Chapter 14).

5.1.4 What regression is used for

The same fitted equation can serve different purposes, and the purpose decides which assumptions matter. For **prediction** of new cases only the accuracy of the fitted mean matters, and a regressor can be useful without any causal role. For **description** the fitted mean summarizes how the response varies across the population sampled. For **estimation of a parameter**, such as an elasticity or a dose-response slope, one coefficient and its uncertainty are the point. For **control**, regressor values are chosen to achieve a desired response, which requires the fitted relation to hold when the regressors are *set* rather than observed: a causal claim.

Warning. The regression function describes how the mean response differs between cases that *are observed* to have different regressor values. It does not by itself describe what would happen if a regressor were *changed*. Units with high values of one regressor usually differ in other ways too. Randomized assignment of the regressor, as in a designed experiment, is the standard way to make the two questions coincide. Without it, the step from association to effect needs assumptions that the data cannot check. Section 5.7 returns to this point, and Chapter 25 treats it in full.

5.1.5 Exercises

5.1.5.1 A. Check your understanding

Exercise A1.

Let X take the values $-1, 0, 1$ with probability $1/3$ each, and let $Y = X^2 + XZ$, where $Z \sim N(0, 1)$ is independent of X . Find the regression function m and the error ε . Show that $E(\varepsilon | X) = 0$ and $\text{Cov}(\varepsilon, X) = 0$, but that ε and X are not independent. Which of the layers (L1)–(L4) fail for a sample from this distribution, with the columns $1, x, x^2$?

Solution. Since $E(XZ | X) = X E(Z) = 0$, $m(x) = x^2$ and $\varepsilon = XZ$. Then $E(\varepsilon | X) = 0$, and $\text{Cov}(\varepsilon, X) = E(X^2Z) = E(X^2)E(Z) = 0$. But $\text{Var}(\varepsilon | X) = X^2$, which is 0 when $X = 0$ and 1 otherwise. The conditional distribution of ε depends on X , so they are not independent. With columns $1, x, x^2$, (L1) holds with $\beta = (0, 0, 1)^\top$ and (L3) holds

for independent cases. (L2) fails, and so does (L4) as stated, because the error of a case with $x = 0$ is identically zero and the joint distribution is not normal with covariance $\sigma^2 \mathbf{I}$.

Exercise A2.

Let (X, Y) be bivariate normal with means μ_X, μ_Y , standard deviations $\sigma_X, \sigma_Y > 0$ and correlation ρ . Use [Theorem 3.5.2](#) to find $m(x)$ and $\text{Var}(Y \mid X = x)$. Conclude that a random sample from this distribution satisfies (L1)–(L4) conditionally on the x_i , with columns 1, x . What are β and σ^2 ?

5.1.5.2 B. Practice

Exercise B1.

Under the conditions of [Proposition 5.1.1](#), show that $\text{Var}(Y) = \text{Var}(m(\mathbf{X})) + \mathbb{E}[\text{Var}(Y \mid \mathbf{X})]$ and that $\mathbb{E}(\varepsilon^2) = \mathbb{E}[\text{Var}(Y \mid \mathbf{X})]$. The ratio $\eta^2 = \text{Var}(m(\mathbf{X})) / \text{Var}(Y)$ is the proportion of the variance of Y that the regression function explains. Compute it for the distribution of [Exercise \(A1\)](#).

Solution. The first identity is the scalar case of [Proposition 2.5.4](#). For the second, $\mathbb{E}(\varepsilon^2 \mid \mathbf{X}) = \mathbb{E}[(Y - m(\mathbf{X}))^2 \mid \mathbf{X}] = \text{Var}(Y \mid \mathbf{X})$; take expectations. In [Exercise \(A1\)](#), $m(X) = X^2$ takes the value 1 with probability 2/3 and 0 with probability 1/3, so $\text{Var}(m(X)) = 2/9$. Also $\mathbb{E}[\text{Var}(Y \mid X)] = \mathbb{E}(X^2) = 2/3$. Hence $\text{Var}(Y) = 8/9$ and $\eta^2 = 1/4$.

Exercise B2.

In [Example \(One slope, two sampling schemes\)](#) the regressors are independent and uniform on $[0, 10]$. Show that $\mathbb{E}(S_{xx}) = (n-1) \cdot 100/12$. Use Jensen's inequality to show that $\mathbb{E}(\sigma^2/S_{xx}) \geq \sigma^2/\mathbb{E}(S_{xx})$, with strict inequality here. Why is the right side of no use for describing the precision of the slope from a given sample?

5.1.5.3 C. Going deeper

Exercise C1.

Let $X \sim N(0, 1)$ and $Y = X^2$. Find the best linear predictor $\beta_0 + \beta_1 X$ of Y ([Theorem 2.5.2](#)) and its error u . Show that $\mathbb{E}(u) = 0$ and $\text{Cov}(u, X) = 0$, but $\mathbb{E}(u \mid X) \neq 0$. Explain why fitting a straight line to a large sample from this distribution estimates a well-defined population quantity even though (L1) fails for the columns 1, x , and why the second-moment theory of this chapter does not describe the sampling variability of the fitted slope.

5.2 Simple linear regression as a warm-up

With a single regressor x and an intercept, the linear model is

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n, \quad (5.2.1)$$

where the x_i are known numbers and the errors satisfy the second-moment assumptions (L1)–(L3) of [Section 5.1](#): $E(\varepsilon_i) = 0$, $\text{Var}(\varepsilon_i) = \sigma^2$, and $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$. The parameter β_1 is the **slope**, the difference in mean response between two values of x one unit apart. The **intercept** β_0 is the mean response at $x = 0$, which may or may not be a value of x that makes sense. Everything in this section is a special case of [Section 5.4](#). We work it out directly because the formulas are short, because they show the structure of the general answer, and because they are the ones used every day.

5.2.1 The least squares line

For a candidate line $b_0 + b_1 x$, the **residual** of case i is $y_i - b_0 - b_1 x_i$, the vertical distance from the data point to the line. The method of least squares chooses the line that minimizes the sum of squared residuals,

$$S(b_0, b_1) = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2.$$

Other criteria are possible, for example the sum of absolute residuals (Chapter 45). Squared error is chosen because it leads to linear equations, because it matches the conditional mean of [Proposition 5.1.1\(c\)](#), and because its statistical properties under (L1)–(L3) can be worked out exactly, which is the business of this chapter.

We use the standard abbreviations

$$\begin{aligned} \bar{x} &= \frac{1}{n} \sum_i x_i, & \bar{y} &= \frac{1}{n} \sum_i y_i, & S_{xx} &= \sum_i (x_i - \bar{x})^2, \\ S_{xy} &= \sum_i (x_i - \bar{x})(y_i - \bar{y}), & S_{yy} &= \sum_i (y_i - \bar{y})^2. \end{aligned} \quad (5.2.2)$$

Two identities are used repeatedly: $\sum_i (x_i - \bar{x}) = 0$, and consequently $S_{xy} = \sum_i (x_i - \bar{x})y_i$ and $S_{xx} = \sum_i (x_i - \bar{x})x_i$. Note that $S_{xx} > 0$ unless all the x_i are equal.

Theorem 5.2.1 — Least squares for a straight line

Suppose the x_i are not all equal. Then $S(b_0, b_1)$ has a unique minimizer $(\hat{\beta}_0, \hat{\beta}_1)$, given by

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}. \quad (5.2.3)$$

It is the unique solution of the two **normal equations**

$$\sum_{i=1}^n (y_i - b_0 - b_1 x_i) = 0, \quad \sum_{i=1}^n x_i (y_i - b_0 - b_1 x_i) = 0. \quad (5.2.4)$$

Moreover, for all b_0, b_1 ,

$$\begin{aligned} S(b_0, b_1) &= S(\hat{\beta}_0, \hat{\beta}_1) + n[(b_0 - \hat{\beta}_0) + (b_1 - \hat{\beta}_1)\bar{x}]^2 \\ &\quad + S_{xx}(b_1 - \hat{\beta}_1)^2. \end{aligned} \quad (5.2.5)$$

Proof. S is a differentiable function of (b_0, b_1) , with

$$\frac{\partial S}{\partial b_0} = -2 \sum_i (y_i - b_0 - b_1 x_i), \quad \frac{\partial S}{\partial b_1} = -2 \sum_i x_i (y_i - b_0 - b_1 x_i).$$

A minimizer must make both derivatives zero, which gives [Eq. 5.2.4](#). The first normal equation divided by n says $\bar{y} - b_0 - b_1 \bar{x} = 0$, so $b_0 = \bar{y} - b_1 \bar{x}$. Substituting into the second, and using $y_i - \bar{y} - b_1(x_i - \bar{x})$ for the residual,

$$\sum_i x_i [(y_i - \bar{y}) - b_1(x_i - \bar{x})] = S_{xy} - b_1 S_{xx} = 0,$$

so $b_1 = S_{xy}/S_{xx}$. This is the only stationary point.

A stationary point need not be a minimum, so we check directly. Write $\hat{\varepsilon}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$, $d_0 = b_0 - \hat{\beta}_0$ and $d_1 = b_1 - \hat{\beta}_1$. Then $y_i - b_0 - b_1 x_i = \hat{\varepsilon}_i - (d_0 + d_1 x_i)$, and

$$S(b_0, b_1) = \sum_i \hat{\varepsilon}_i^2 - 2d_0 \sum_i \hat{\varepsilon}_i - 2d_1 \sum_i x_i \hat{\varepsilon}_i + \sum_i (d_0 + d_1 x_i)^2.$$

The two middle sums vanish by the normal equations. For the last, $d_0 + d_1 x_i = (d_0 + d_1 \bar{x}) + d_1(x_i - \bar{x})$, and the cross term again sums to zero, so $\sum_i (d_0 + d_1 x_i)^2 = n(d_0 + d_1 \bar{x})^2 + d_1^2 S_{xx}$. This proves [Eq. 5.2.5](#). Both added terms are nonnegative, and since $S_{xx} > 0$ they are both zero only when $d_1 = 0$ and then $d_0 = 0$. So $(\hat{\beta}_0, \hat{\beta}_1)$ is the unique global minimizer. ■

The decomposition [Eq. 5.2.5](#) says more than minimality. It shows the level sets of S are ellipses centred at $(\hat{\beta}_0, \hat{\beta}_1)$, whose shape is set by n , $\bar{x}$ and S_{xx} . When $\bar{x}$ is far from zero the ellipses are long and thin, tilted along the direction where $b_0 + b_1 \bar{x}$ stays constant. Many lines then fit almost equally well, provided they pass near the centre of the data. This is a first look at the correlation between the estimated intercept and slope ([Corollary 5.5.4](#)).

The **fitted values** are $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ and the **residuals** are $\hat{\varepsilon}_i = y_i - \hat{y}_i$. The minimum SSE = $\sum_i \hat{\varepsilon}_i^2$ is the **residual sum of squares** (also *error sum of squares*).

Proposition 5.2.2 — Properties of the fitted line

Under the conditions of [Theorem 5.2.1](#):

- (a) $\sum_i \hat{\varepsilon}_i = 0$, $\sum_i x_i \hat{\varepsilon}_i = 0$ and $\sum_i \hat{y}_i \hat{\varepsilon}_i = 0$;

- (b) the line passes through $(\bar{x}, \bar{y})$, and the mean of the fitted values is $\bar{y}$;
 (c) with $SST = S_{yy}$ and $SSR = \sum_i (\hat{y}_i - \bar{y})^2$,

$$SST = SSR + SSE, \quad SSR = \hat{\beta}_1^2 S_{xx} = \frac{S_{xy}^2}{S_{xx}},$$

and so $SSE = S_{yy} - S_{xy}^2/S_{xx}$;

- (d) if also $S_{yy} > 0$, the **coefficient of determination** $R^2 = SSR/SST$ equals r^2 , the square of the sample correlation $r = S_{xy}/\sqrt{S_{xx}S_{yy}}$.

Proof. (a) The first two are the normal equations Eq. 5.2.4. The third follows because $\hat{y}_i$ is a combination of 1 and x_i . (b) The first normal equation is $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$, and averaging $\hat{y}_i = y_i - \hat{\varepsilon}_i$ gives $\bar{y}$. (c) By (b), $\hat{y}_i - \bar{y} = \hat{\beta}_1(x_i - \bar{x})$, so $SSR = \hat{\beta}_1^2 S_{xx} = S_{xy}^2/S_{xx}$. Also $y_i - \bar{y} = (\hat{y}_i - \bar{y}) + \hat{\varepsilon}_i$, and the cross term $\sum_i (\hat{y}_i - \bar{y})\hat{\varepsilon}_i$ vanishes by (a). (d) $R^2 = S_{xy}^2/(S_{xx}S_{yy}) = r^2$. ■

Part (c) is the simplest instance of the analysis of variance: the total variation of the response about its mean splits into a part along the fitted line and a part around it. Chapter 9 develops the general decomposition, and Section 6.7 interprets R^2 as a squared cosine.

5.2.2 Regression toward the mean

Divide S_{xy}/S_{xx} above and below by $n-1$. With the sample standard deviations $s_x = \sqrt{S_{xx}/(n-1)}$ and $s_y = \sqrt{S_{yy}/(n-1)}$, the slope is

$$\hat{\beta}_1 = r \frac{s_y}{s_x}, \quad \text{equivalently} \quad \frac{\hat{y} - \bar{y}}{s_y} = r \frac{x - \bar{x}}{s_x}. \quad (5.2.6)$$

In standard units, the fitted value is the regressor's standardized value multiplied by r . Since $|r| \leq 1$ (the Cauchy–Schwarz inequality, Proposition 1.1.1), a case that is two standard deviations above average in x is predicted to be *less* than two standard deviations above average in y , unless the points lie exactly on a line. This is **regression toward the mean**, and it gave the subject its name: Galton observed it in 1886 for the heights of parents and their adult children.

The effect is purely statistical and involves no mechanism. Regressing x on y instead gives the slope $r s_x/s_y$, which predicts x from y with the same shrinkage toward the mean in the other direction. The two fitted lines differ unless $r^2 = 1$. They answer different questions: the mean of y at given x , and the mean of x at given y . It is a common error to read regression toward the mean as evidence that extreme units become more ordinary over time, or that an intervention given to the most extreme units has worked.

5.2.3 A worked example

Example. (Forecasting December sea temperature)

The Niño 1+2 index is the mean sea surface temperature in a region of the Pacific off

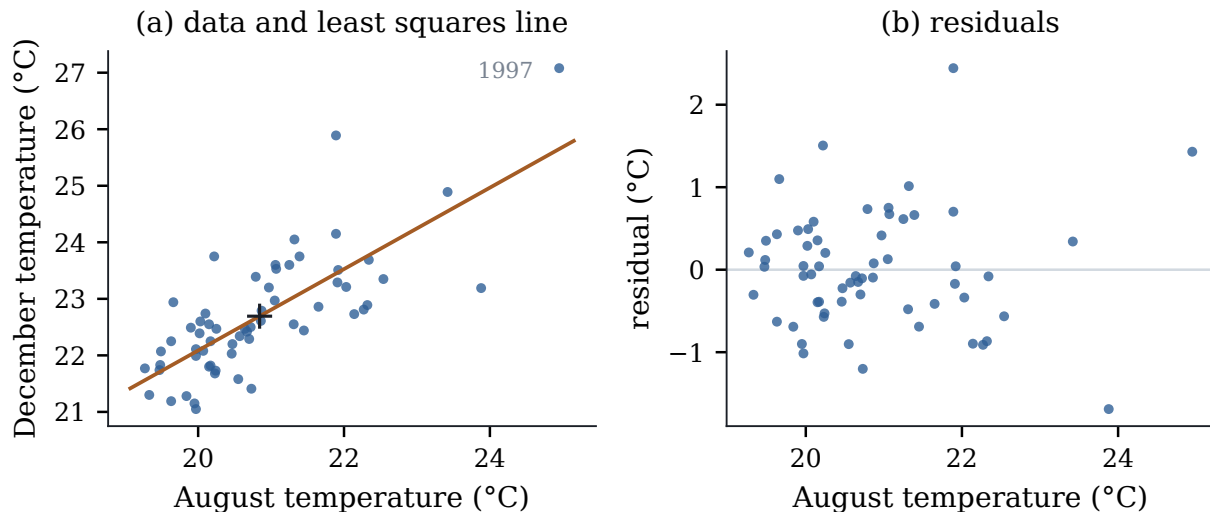


Figure 5.2.1: December against August sea surface temperature in the Niño 1+2 region, 1950–2010. (a) The data and the least squares line, which passes through the point of means (cross). (b) Residuals against the regressor. The El Niño year 1997 is the point at the far right of both panels.

Peru and Ecuador. It is high during El Niño events. We use the monthly values for the 61 years 1950–2010 and ask how well the August value predicts the December value four months later. Here x is the August temperature and y the December temperature, both in degrees Celsius. The summary statistics are

$$\begin{aligned}\bar{x} &= 20.843, & \bar{y} &= 22.693, \\ S_{xx} &= 77.801, & S_{xy} &= 56.008, & S_{yy} &= 70.380,\end{aligned}$$

so by [Theorem 5.2.1](#)

$$\hat{\beta}_1 = \frac{56.008}{77.801} = 0.7199, \quad \hat{\beta}_0 = 22.693 - 0.7199 \times 20.843 = 7.6887.$$

Each extra degree in August goes with 0.7199 of a degree more in the following December, on average. For an August temperature of 22°C the fitted December mean is 23.53°C. The intercept, the fitted December temperature after an August at 0°C, is meaningless: no August in the data is below 19°C.

The residual sum of squares is $SSE = 30.061$, and $R^2 = 0.5729$, the square of the correlation $r = 0.7569$. The slope is less than 1 partly because $r < 1$ and partly because December temperatures vary a little less than August ones: $s_y = 1.083$ against $s_x = 1.139$.

```
import numpy as np
import statsmodels.api as sm
sst = sm.datasets.elnino.load_pandas().data
x = sst["AUG"].to_numpy()           # August temperature
y = sst["DEC"].to_numpy()           # December temperature, same year
n = len(y)

xbar, ybar = x.mean(), y.mean()
Sxx = np.sum((x - xbar) ** 2)
Sxy = np.sum((x - xbar) * (y - ybar))
Syy = np.sum((y - ybar) ** 2)
```

```

b1 = Sxy / Sxx                # slope
b0 = ybar - b1 * xbar         # intercept
fitted = b0 + b1 * x
resid = y - fitted
print(f"n = {n}, slope = {b1:.4f}, intercept = {b0:.4f}")

```

```

print("sum of residuals      ", resid.sum())
print("sum of x * residuals  ", (x * resid).sum())
SSE = resid @ resid
R2 = 1 - SSE / Syy
r = Sxy / np.sqrt(Sxx * Syy)
print(f"R^2 = {R2:.4f}, r^2 = {r**2:.4f}")
print("numpy.polyfit agrees: ", np.polyfit(x, y, 1))

```

The residual plot in [Figure 5.2.1\(b\)](#) is the first diagnostic to draw after any fit. The residuals should show no pattern against x . Here there is no obvious curvature, but the spread is not uniform. The two warmest Augusts, in 1997 and 1983, both fell during strong El Niño events. The 1997 event was still building in December, while the 1982–83 event was over by then, and the two points sit well away from the line on opposite sides. The warmest of all, 1997 (August 24.95°C, December 27.08°C), has a residual of 1.430°C. Because it lies far from $\bar{x}$, it pulls on the slope: without it the slope would be 0.6214. Points with regressor values far from the mean have high *leverage*, a notion made precise in [Section 6.8](#) and Chapter 20.

5.2.4 Regression through the origin

Sometimes theory says the mean response is zero when $x = 0$, and the model $Y_i = \beta_1 x_i + \varepsilon_i$ is fitted without an intercept. Minimizing $\sum_i (y_i - b_1 x_i)^2$ gives one normal equation, $\sum_i x_i (y_i - b_1 x_i) = 0$, with solution

$$\tilde{\beta}_1 = \frac{\sum_i x_i y_i}{\sum_i x_i^2},$$

valid whenever some $x_i \neq 0$. The residuals now satisfy $\sum_i x_i \hat{\varepsilon}_i = 0$ but generally *not* $\sum_i \hat{\varepsilon}_i = 0$, and the decomposition of [Proposition 5.2.2\(c\)](#) about $\bar{y}$ fails. It holds instead about zero, as $\sum_i y_i^2 = \sum_i \hat{y}_i^2 + \sum_i \hat{\varepsilon}_i^2$, which is why software reports an “uncentred” R^2 for such fits. That number is not comparable with the R^2 of a model with an intercept ([Exercise \(A2\)](#)).

Dropping the intercept is a strong assumption. In [Example \(Forecasting December sea temperature\)](#), forcing the line through the origin gives slope 1.0877, a fit much worse than the line with an intercept: the data are nowhere near $x = 0$, and the constraint has nothing to do with how ocean temperatures behave.

5.2.5 Exercises

5.2.5.1 A. Check your understanding

Exercise A1.

For the five points $(x, y) = (0, 1), (1, 1), (2, 3), (3, 4), (4, 6)$, compute $\bar{x}, \bar{y}, S_{xx}, S_{xy}$, the least squares line, the residuals, and R^2 . Check the three identities of [Proposition 5.2.2\(a\)](#).

Solution. $\bar{x} = 2$, $\bar{y} = 3$, $S_{xx} = 10$, $S_{xy} = (-2)(-2) + (-1)(-2) + 0 + 1 \cdot 1 + 2 \cdot 3 = 13$. So $\hat{\beta}_1 = 1.3$ and $\hat{\beta}_0 = 3 - 2.6 = 0.4$. The fitted values are 0.4, 1.7, 3.0, 4.3, 5.6 and the residuals 0.6, -0.7, 0, -0.3, 0.4, which sum to zero. Also $\sum_i x_i \hat{\epsilon}_i = 0 - 0.7 + 0 - 0.9 + 1.6 = 0$. $S_{yy} = 4 + 4 + 0 + 1 + 9 = 18$, $SSE = 0.36 + 0.49 + 0 + 0.09 + 0.16 = 1.1$, and $R^2 = 1 - 1.1/18 = 0.939$, which equals $13^2/(10 \cdot 18)$.

Exercise A2.

For regression through the origin, show that $\sum_i y_i^2 = \sum_i \hat{y}_i^2 + \sum_i \hat{\epsilon}_i^2$, and give a small data set for which $1 - \sum_i \hat{\epsilon}_i^2 / \sum_i y_i^2$ is close to 1 while $1 - \sum_i \hat{\epsilon}_i^2 / S_{yy}$ is negative.

5.2.5.2 B. Practice

Exercise B1.

Show that $\hat{\beta}_1 = \sum_i c_i y_i$ with $c_i = (x_i - \bar{x})/S_{xx}$, and that $\sum_i c_i = 0$, $\sum_i c_i x_i = 1$ and $\sum_i c_i^2 = 1/S_{xx}$. Write $\hat{\beta}_0$ as $\sum_i d_i y_i$ and find $\sum_i d_i$, $\sum_i d_i x_i$ and $\sum_i d_i^2$.

Solution. $S_{xy} = \sum_i (x_i - \bar{x})y_i$, so $\hat{\beta}_1 = \sum_i c_i y_i$. Then $\sum_i c_i = 0$, $\sum_i c_i x_i = \sum_i (x_i - \bar{x})x_i/S_{xx} = 1$, and $\sum_i c_i^2 = S_{xx}/S_{xx}^2 = 1/S_{xx}$. For the intercept, $\hat{\beta}_0 = \bar{y} - \bar{x}\hat{\beta}_1 = \sum_i d_i y_i$ with $d_i = 1/n - \bar{x}c_i$. Hence $\sum_i d_i = 1$, $\sum_i d_i x_i = \bar{x} - \bar{x} = 0$, and $\sum_i d_i^2 = \sum_i (1/n^2 - 2\bar{x}c_i/n + \bar{x}^2 c_i^2) = 1/n + \bar{x}^2/S_{xx}$. These are the coefficients that give the moments in [Corollary 5.5.4](#).

Exercise B2.

Suppose all x_i are distinct. Show that $\hat{\beta}_1$ is a weighted average of the slopes $(y_j - y_i)/(x_j - x_i)$ of the lines through all pairs of points, with weights proportional to $(x_j - x_i)^2$. *Hint:* show that $\sum_{i < j} (x_j - x_i)(y_j - y_i) = nS_{xy}$ and $\sum_{i < j} (x_j - x_i)^2 = nS_{xx}$.

5.2.5.3 C. Going deeper

Exercise C1.

The **least absolute deviations** line minimizes $\sum_i |y_i - b_0 - b_1 x_i|$. Show that some minimizing line passes through at least two of the data points (assume the x_i are not all equal). *Hint:* for fixed b_1 the best b_0 is a median of the $y_i - b_1 x_i$, and the objective is piecewise linear and convex in b_1 . Compare the effect of the 1997 point in [Example \(Forecasting December sea temperature\)](#) on this line and on the least squares line.

5.3 The multiple linear regression model

A response rarely depends on one regressor alone. The money a country's residents hold depends on their income and on interest rates. The number of physician visits depends on health, age and insurance. Even with one measured regressor, a straight line is often too rigid: the curvature

of a trend or the shape of a seasonal cycle needs several columns to describe. Matrix notation handles all of these cases at once.

5.3.1 Matrix form

Suppose case i has regressor values that have been turned into p numbers $x_{i1}, \dots, x_{ip}$, and consider

$$Y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \varepsilon_i, \quad i = 1, \dots, n.$$

Stack the responses, the coefficients and the errors into vectors, and the regressor values into an $n \times p$ matrix:

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}, \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix},$$

$$\mathbf{X} = \begin{pmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & & \vdots \\ x_{n1} & \cdots & x_{np} \end{pmatrix} = \begin{pmatrix} \mathbf{x}_{(1)}^\top \\ \vdots \\ \mathbf{x}_{(n)}^\top \end{pmatrix}.$$

The n equations become the single equation $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$. Row i of $\mathbf{X}$, written $\mathbf{x}_{(i)}^\top$, holds the regressor values of case i , its **design point**. Column j , written $\mathbf{x}_j$, holds the values of the j th regressor across the cases.

Definition 5.3.1 — The linear model

The **linear model** for a random vector $\mathbf{Y} \in \mathbb{R}^n$ is

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n, \quad (5.3.1)$$

where the **model matrix** $\mathbf{X}$ is a known $n \times p$ matrix of constants, and the **coefficient vector** $\boldsymbol{\beta} \in \mathbb{R}^p$ and the **error variance** $\sigma^2 > 0$ are unknown parameters. The conditions on $\boldsymbol{\varepsilon}$ are the **second-moment assumptions**. If in addition $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$, equivalently $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n)$, the model is the **normal linear model**. The model has **full rank** if $\text{rank}(\mathbf{X}) = p$.

The second-moment assumptions are exactly the layers (L1)–(L3) of [Section 5.1](#). By [Theorem 2.2.1](#) they say

$$\mathbf{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}, \quad \text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n. \quad (5.3.2)$$

The first equation is the whole content of (L1). It says the mean vector $\mathbf{E}(\mathbf{Y})$ is a linear combination of the columns of $\mathbf{X}$, with unknown coefficients. The second combines constant variance and uncorrelated errors. When the regressors are random, both equations are read conditionally on $\mathbf{X}$, as explained in [Section 5.1](#). The normal version is a model for the full distribution of $\mathbf{Y}$, in the sense of [Chapter 3](#). Since a normal vector with covariance $\sigma^2 \mathbf{I}$ has independent components ([Theorem 3.4.1](#)), the errors are then independent and not merely uncorrelated.

The definition does not require full rank; most results in this chapter assume it, which forces $p \leq n$, and [Chapter 6](#) and [Chapter 8](#) treat the general case. What must be *known* is $\mathbf{X}$, and

what must be *linear* is the dependence of $E(\mathbf{Y})$ on β . Nor does the model say that the errors are identically distributed, only that they have a common variance.

5.3.2 The intercept

Almost every model in practice includes a column of ones, $\mathbf{x}_1 = \mathbf{1}$, so that

$$E(Y_i) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}, \quad p = k + 1.$$

In this form it is conventional to number the coefficients from 0, and to call k the number of regressors. The coefficient of the column of ones is the **intercept**. Including it lets the fitted surface find its own level, rather than being forced through the origin, and it has two further consequences that [Section 5.4](#) proves: the residuals sum to zero, and the slopes can be computed from centred data. Leaving out the intercept is justified only when theory requires the mean response to vanish at the origin of every regressor, as in the regression through the origin at the end of [Section 5.2](#).

5.3.3 Linear in the parameters, not in the regressors

The word *linear* in “linear model” refers to β . The columns of $\mathbf{X}$ can be any known functions of the measured variables. This turns a remarkably wide class of curved relationships into linear models.

Transformations. If the mean response is thought to depend on $\log z$, the column $\log z_i$ enters $\mathbf{X}$, and $E(Y_i) = \beta_0 + \beta_1 \log z_i$ is linear in (β_0, β_1) . The response can be transformed too, as in the money demand model of [Section 5.4](#), where Y is the logarithm of money holdings. A transformation of the response changes the meaning of the error, and hence the assumptions; Chapter 22 treats this.

Polynomials. A quadratic trend $E(Y_i) = \beta_0 + \beta_1 t_i + \beta_2 t_i^2$ uses the columns $\mathbf{1}$, $\mathbf{t}$ and $\mathbf{t}^2$ (squared entrywise). A polynomial of degree d in one variable needs $d + 1$ columns.

Periodic terms. A seasonal cycle of period one year can be written as $a \cos(2\pi t + \phi)$ with unknown amplitude a and phase ϕ . This is not linear in ϕ . But $a \cos(2\pi t + \phi) = \gamma_1 \cos(2\pi t) + \gamma_2 \sin(2\pi t)$ with $\gamma_1 = a \cos \phi$ and $\gamma_2 = -a \sin \phi$, and this *is* linear in (γ_1, γ_2) . The amplitude is recovered as $a = (\gamma_1^2 + \gamma_2^2)^{1/2}$.

Example. (A curved trend and a seasonal cycle)

The monthly mean concentration of carbon dioxide in the air at Mauna Loa, Hawaii, from March 1958 to December 2001, gives $n = 521$ observations in parts per million. Let t be the time in years, at the middle of each month, and $t_c = t - 1980$. The model

$$\begin{aligned} E(Y) = & \beta_0 + \beta_1 t_c + \beta_2 t_c^2 \\ & + \gamma_1 \cos 2\pi t + \gamma_2 \sin 2\pi t + \gamma_3 \cos 4\pi t + \gamma_4 \sin 4\pi t \end{aligned}$$

has $p = 7$ columns, and the model matrix has full rank. Its least squares fit ([Section 5.4](#)) has

$$\hat{\beta}_0 = 337.6163, \quad \hat{\beta}_1 = 1.3344, \quad \hat{\beta}_2 = 0.0118,$$

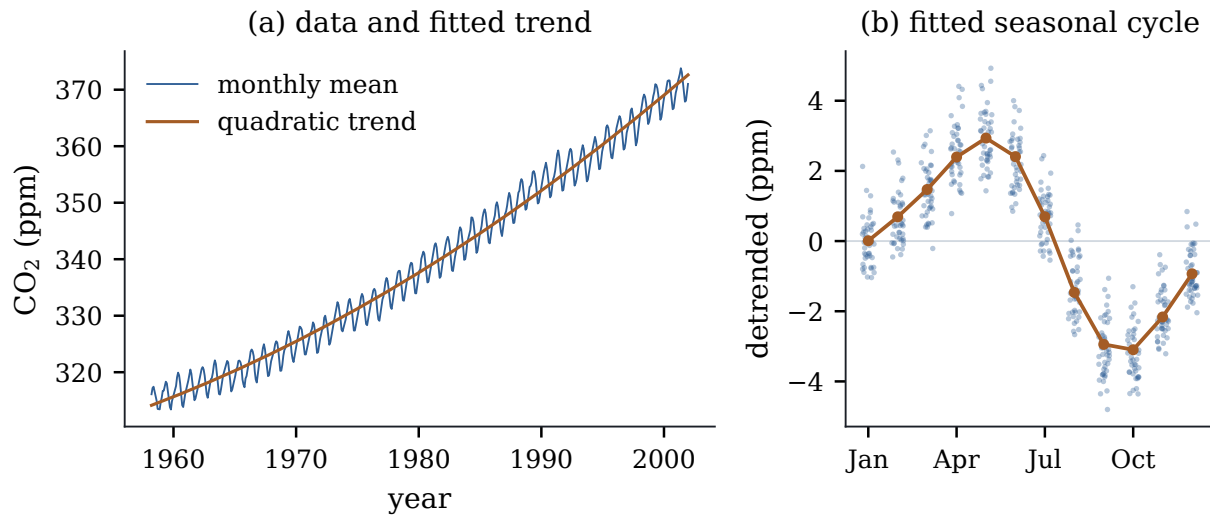


Figure 5.3.1: Monthly carbon dioxide at Mauna Loa, 1958–2001. (a) The data and the fitted quadratic trend. (b) The data with the fitted trend removed, by calendar month, and the fitted seasonal cycle made of two harmonics. The model is curved in time and in the calendar month, but linear in its seven coefficients.

$$\hat{\gamma}_1 = -1.0413, \quad \hat{\gamma}_2 = 2.5641, \quad \hat{\gamma}_3 = 0.6284, \quad \hat{\gamma}_4 = -0.3859.$$

The quadratic term says the rate of increase is itself increasing. The slope of the trend, $\beta_1 + 2\beta_2 t_c$, is estimated as 0.86 ppm per year in 1960 and 1.81 ppm per year in 2000. The annual harmonic has amplitude 2.77 ppm and the half-year harmonic 0.74 ppm. Together they make a seasonal swing of 6.0 ppm from the late-spring peak to the autumn trough, as the vegetation of the northern hemisphere takes up carbon over the summer. The residual standard deviation (estimated as in [Section 5.6](#)) is 0.760 ppm, against 2.743 ppm for a straight line in time alone. [Figure 5.3.1](#) shows the data, the trend and the seasonal cycle.

```
import numpy as np
import statsmodels.api as sm
co2 = sm.datasets.co2.load_pandas().data["co2"]
monthly = co2.resample("MS").mean().dropna()           # weekly values -> monthly means
y = monthly.to_numpy()
t = (monthly.index.year + (monthly.index.month - 0.5) / 12).to_numpy()
tc = t - 1980.0                                         # years since 1980

X = np.column_stack([
    np.ones_like(tc),                                  # intercept
    tc, tc**2,                                         # quadratic trend
    np.cos(2 * np.pi * t), np.sin(2 * np.pi * t),   # annual cycle
    np.cos(4 * np.pi * t), np.sin(4 * np.pi * t),   # half-year harmonic
])
n, p = X.shape
print("n =", n, " p =", p, " rank =", np.linalg.matrix_rank(X))
```

```
beta_hat = np.linalg.solve(X.T @ X, X.T @ y)          # normal equations (fine here)
fitted = X @ beta_hat
resid = y - fitted
print("coefficients:", np.round(beta_hat, 4))
print(f"residual standard deviation {np.sqrt(resid @ resid / (n - p)):.3f} ppm")
```

```
beta_line = np.linalg.solve(X[:, :2].T @ X[:, :2], X[:, :2].T @ y)
print(f"straight line only: slope {beta_line[1]:.3f} ppm per year")
```

The residuals of this model are strongly correlated from month to month: the correlation between consecutive residuals is 0.90, because departures from a smooth trend persist for many months. So (L3) fails and the standard errors of [Section 5.5](#) would be too optimistic. The *fit* is still a sensible description. Chapter 21 treats serially correlated errors.

Models that are not linear. Some relations cannot be rearranged into linear form. Exponential decay toward an unknown asymptote, $E(Y) = \beta_0 + \beta_1 e^{-\beta_2 x}$, is nonlinear in β_2 , and so is a cycle of unknown period. Such models are fitted by iterative nonlinear least squares, which this book does not treat. Others become linear after a transformation: if $Y = \alpha x^\beta \eta$ with a positive multiplicative error η , then $\log Y = \log \alpha + \beta \log x + \log \eta$ is a linear model for $\log Y$, provided the assumptions are made about $\log \eta$ rather than η .

5.3.4 Categorical regressors and indicator variables

A regressor that records a category, such as treatment group, region, or a self-rating on a four-point scale, cannot enter $\mathbf{X}$ as a single column of codes 1, 2, 3, 4. That would force the mean differences between consecutive categories to be equal, which is an arbitrary consequence of the labels. Instead, each category except one gets an **indicator variable** (or *dummy variable*): a column that is 1 for cases in that category and 0 otherwise. The omitted category is the **reference category**.

Example. (Self-rated health)

The RAND Health Insurance Experiment recorded, for 20,190 person-years, the number of outpatient visits to a physician and the person's own rating of their health: excellent, good, fair or poor. With excellent as the reference, the model

$$E(Y_i) = \beta_0 + \beta_1 g_i + \beta_2 f_i + \beta_3 q_i$$

uses three indicator columns $\mathbf{g}, \mathbf{f}, \mathbf{q}$ for good, fair and poor health. For a person in excellent health the mean is β_0 ; in good health, $\beta_0 + \beta_1$; in fair health, $\beta_0 + \beta_2$; in poor health, $\beta_0 + \beta_3$. So $\beta_1, \beta_2, \beta_3$ are the differences in mean visits between each category and the reference. [Proposition 5.4.4](#) shows that the least squares estimates are the corresponding differences of sample means. In these data they are 0.268, 1.058 and 3.160 visits a year, on top of $\hat{\beta}_0 = 2.634$ for excellent health.

Why omit a category? With an indicator for *every* category the columns satisfy $\mathbf{e} + \mathbf{g} + \mathbf{f} + \mathbf{q} = \mathbf{1}$, where $\mathbf{e}$ indicates excellent health. The model matrix $[\mathbf{1}, \mathbf{e}, \mathbf{g}, \mathbf{f}, \mathbf{q}]$ then has five columns and rank four, so it does not have full rank. Such a model is not wrong. It describes the same set of mean vectors, but with one more parameter than it can identify, and the least squares coefficients are no longer unique. This situation is sometimes called the *dummy variable trap*. It is not a trap at all once the right theory is in place: [Section 6.4](#) shows that fitted values are unaffected, and [Chapter 8](#) shows which functions of the coefficients, such as the difference between two categories, are still estimated uniquely.

```
import numpy as np
import statsmodels.api as sm
rand = sm.datasets.randhie.load_pandas().data
```

```

y = rand["mdvis"].to_numpy(dtype=float)
D = rand[["hlthg", "hlthf", "hlthp"]].to_numpy(dtype=float)  # good, fair, poor
X = np.column_stack([np.ones(len(y)), D])                  # excellent =
  reference

excellent = 1.0 - D.sum(axis=1)
X_all = np.column_stack([X, excellent])                    # an indicator for every level, and
  1
print("columns:", X_all.shape[1], " rank:", np.linalg.matrix_rank(X_all))
print("1 - (sum of the four indicators) =", np.abs(X_all[:, 0] - X_all[:, 1:].sum(
  axis=1)).max())

```

Reference coding is one choice among several. **Effect coding** replaces the indicator of category k by a column that is 1 in category k , -1 in the reference category, and 0 elsewhere. Its coefficients measure deviations of category means from their unweighted average rather than from a reference. The fitted values are the same under either coding, because the two model matrices have the same column space ([Proposition 5.7.1](#)). Only the meaning of the coefficients changes.

5.3.5 Interactions

In the model $E(Y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ the slope in x_1 is β_1 whatever the value of x_2 : the effects are **additive**. An **interaction** lets the slope in one variable depend on another. Adding the product column $x_1 x_2$,

$$E(Y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 = (\beta_0 + \beta_2 x_2) + (\beta_1 + \beta_3 x_2) x_1, \quad (5.3.3)$$

gives a slope in x_1 of $\beta_1 + \beta_3 x_2$. The model is still linear in the coefficients. When x_2 is an indicator for a group, [Eq. 5.3.3](#) fits a separate straight line in each group: intercept β_0 and slope β_1 in the reference group, intercept $\beta_0 + \beta_2$ and slope $\beta_1 + \beta_3$ in the other. When both variables are categorical, products of indicators give each combination of categories its own mean, the two-way layout of Chapter 16.

Interactions change what the other coefficients mean. In [Eq. 5.3.3](#), β_1 is the slope in x_1 when $x_2 = 0$, which may be far outside the data. Centring x_2 before forming the product makes β_1 the slope at the average x_2 instead. [Section 5.7](#) returns to this.

5.3.6 When is the model matrix of full rank?

Full rank means the columns of $\mathbf{X}$ are linearly independent: no column is an exact linear combination of the others. By rank-nullity ([Theorem 1.2.2](#)), this holds iff $\mathbf{X}\mathbf{b} = \mathbf{0}$ only for $\mathbf{b} = \mathbf{0}$, that is, iff two different coefficient vectors never give the same mean vector. It fails in three common ways:

1. **Too few cases:** $n < p$. Chapter 28 deals with this setting.
2. **Redundant coding:** an indicator for every category together with an intercept, as above, or the same quantity measured in two units.
3. **Unlucky data:** a design that happens to make columns dependent, such as a quadratic in a variable observed at only two distinct values, or an interaction between two indicators

when some combination of categories never occurs.

Near-failures, where columns are almost dependent, are more common and more troublesome. They make coefficients imprecise (Chapter 26) and computations inaccurate (Chapter 10).

5.3.7 Exercises

5.3.7.1 A. Check your understanding

Exercise A1.

Write out the model matrix for each model, for the data $x = (1, 2, 3, 4)$ and a group label (A, A, B, B) , and give its rank: (a) a quadratic in x ; (b) separate intercepts for the two groups and a common slope; (c) separate lines in the two groups, with reference coding; (d) an intercept, an indicator for A and an indicator for B .

Solution. With g the indicator of group B : (a) rows $(1, x, x^2)$: $(1, 1, 1), (1, 2, 4), (1, 3, 9), (1, 4, 16)$, rank 3. (b) rows $(1, g, x)$: $(1, 0, 1), (1, 0, 2), (1, 1, 3), (1, 1, 4)$, rank 3. (c) rows $(1, g, x, gx)$: $(1, 0, 1, 0), (1, 0, 2, 0), (1, 1, 3, 3), (1, 1, 4, 4)$, rank 4; it is square and nonsingular, so the two lines pass through the data exactly. (d) rows $(1, 1, 0), (1, 1, 0), (1, 0, 1), (1, 0, 1)$, rank 2: the first column is the sum of the other two.

Exercise A2.

Which of these mean functions define linear models in the stated parameters (after a change of parameters if necessary)? (a) $\beta_0 + \beta_1\sqrt{x}$; (b) $\beta_0 e^{\beta_1 x}$; (c) $\beta_0 + \beta_1 x + \beta_2 \max(x - 5, 0)$; (d) $\beta_0 + \beta_1 \sin(x - \beta_2)$; (e) $1/(\beta_0 + \beta_1 x)$.

5.3.7.2 B. Practice

Exercise B1.

For a factor with three levels and an intercept, write the model matrix rows under reference coding (level 3 as reference) and under effect coding. Find the nonsingular 3×3 matrix $\mathbf{K}$ with $\mathbf{X}_{\text{effect}} = \mathbf{X}_{\text{ref}}\mathbf{K}$, and express the effect-coding coefficients in terms of the three level means μ_1, μ_2, μ_3 .

Solution. Reference coding has rows $(1, 1, 0), (1, 0, 1), (1, 0, 0)$ for levels 1, 2, 3. Effect coding has rows $(1, 1, 0), (1, 0, 1), (1, -1, -1)$. Each effect-coded row is the reference-coded row times $\mathbf{K} = \begin{pmatrix} 1 & -1 & -1 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$: for instance $(1, 0, 0)\mathbf{K} = (1, -1, -1)$ and $(1, 1, 0)\mathbf{K} = (1, 1, 0)$. $\det \mathbf{K} = 3 \neq 0$. Under effect coding the level means are $\mu_1 = \alpha_0 + \alpha_1$, $\mu_2 = \alpha_0 + \alpha_2$, $\mu_3 = \alpha_0 - \alpha_1 - \alpha_2$. Adding, $\alpha_0 = (\mu_1 + \mu_2 + \mu_3)/3$, and then $\alpha_k = \mu_k - \alpha_0$ for $k = 1, 2$: deviations from the unweighted average of the means.

Exercise B2.

In Eq. 5.3.3, replace x_2 by $x_2 - c$ in both the main-effect column and the product. Find the new coefficients in terms of the old, and show that the fitted mean function is

unchanged. For which c is the new coefficient of x_1 the slope at the average value of x_2 ?

5.3.7.3 C. Going deeper

Exercise C1.

Let $t_i = i/m$ for $i = 0, \dots, m-1$ (one full period sampled at m equally spaced points, with $m \geq 5$). Show that the columns $\mathbf{1}, \cos 2\pi t, \sin 2\pi t, \cos 4\pi t, \sin 4\pi t$ are mutually orthogonal, and find their squared lengths. Conclude that for a whole number of periods the harmonic coefficients are estimated separately from each other and from the mean. Why is this not exactly the case in [Example \(A curved trend and a seasonal cycle\)](#)?

5.4 Least squares and the normal equations

The straight-line calculation of [Section 5.2](#) extends to any model matrix. The algebra is shorter in matrix form than it was for two coefficients, because the derivatives of linear and quadratic forms were worked out once and for all in [Chapter 1](#). In this section $\mathbf{y}$ is a fixed data vector. Nothing here uses probability. The statistical properties come in [Section 5.5](#).

5.4.1 The least squares estimate

Definition 5.4.1 — Least squares estimate

For an $n \times p$ matrix $\mathbf{X}$ and $\mathbf{y} \in \mathbb{R}^n$, the **least squares criterion** is

$$S(\mathbf{b}) = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \sum_{i=1}^n (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2, \quad \mathbf{b} \in \mathbb{R}^p.$$

A **least squares estimate** of β is any $\hat{\beta}$ that minimizes S . The corresponding **fitted values** are $\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta}$, the **residuals** are $\hat{\epsilon} = \mathbf{y} - \mathbf{X}\hat{\beta}$, and the **residual sum of squares** is $\text{SSE} = S(\hat{\beta}) = \|\hat{\epsilon}\|^2$.

Following the book's notation, we write $\hat{\mathbf{Y}}$ and $\hat{\epsilon}$ in bold for the vectors, and $\hat{y}_i$ and $\hat{\epsilon}_i$ for their entries, whether the data are regarded as fixed numbers or as random.

Theorem 5.4.1 — Least squares for a full-rank model

Let $\mathbf{X}$ be an $n \times p$ matrix of rank p . Then:

- (a) $\mathbf{X}^\top \mathbf{X}$ is positive definite, and in particular nonsingular;
- (b) the **normal equations**

$$\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y} \tag{5.4.1}$$

have exactly one solution,

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}; \quad (5.4.2)$$

(c) for every $\mathbf{b} \in \mathbb{R}^p$,

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{X}\hat{\beta}\|^2 + (\mathbf{b} - \hat{\beta})^\top \mathbf{X}^\top \mathbf{X} (\mathbf{b} - \hat{\beta}), \quad (5.4.3)$$

so $\hat{\beta}$ is the unique least squares estimate.

If instead $\text{rank}(\mathbf{X}) < p$, least squares estimates exist but are never unique.

Proof. (a) $\mathbf{X}^\top \mathbf{X}$ is symmetric, and $\mathbf{b}^\top \mathbf{X}^\top \mathbf{X} \mathbf{b} = \|\mathbf{X}\mathbf{b}\|^2 \geq 0$, with equality iff $\mathbf{X}\mathbf{b} = \mathbf{0}$. Since $\mathbf{X}$ has rank p , rank-nullity (Theorem 1.2.2) gives $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$, so the quadratic form is positive for $\mathbf{b} \neq \mathbf{0}$. A positive definite matrix is nonsingular (Proposition 1.7.5(e)).

(b) This is the computation of Example (The least squares criterion): by Theorem 1.11.1 the gradient of S is $-2\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{b})$ and its Hessian is $2\mathbf{X}^\top \mathbf{X}$. Every minimizer of the differentiable function S is a stationary point, and the stationary points are the solutions of Eq. 5.4.1. By (a) there is exactly one, Eq. 5.4.2.

(c) Put $\mathbf{d} = \mathbf{b} - \hat{\beta}$. Then $\mathbf{y} - \mathbf{X}\mathbf{b} = \hat{\epsilon} - \mathbf{X}\mathbf{d}$, and

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\hat{\epsilon}\|^2 - 2\mathbf{d}^\top \mathbf{X}^\top \hat{\epsilon} + \mathbf{d}^\top \mathbf{X}^\top \mathbf{X} \mathbf{d}.$$

The middle term vanishes because $\mathbf{X}^\top \hat{\epsilon} = \mathbf{X}^\top \mathbf{y} - \mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{0}$ by the normal equations. The last term is positive for $\mathbf{d} \neq \mathbf{0}$ by (a). So $S(\mathbf{b}) > S(\hat{\beta})$ whenever $\mathbf{b} \neq \hat{\beta}$.

For the last statement, suppose $\text{rank}(\mathbf{X}) < p$. The vector $\mathbf{X}^\top \mathbf{y}$ lies in $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$ (Proposition 1.2.3(c)), so by Proposition 1.11.2, applied with $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ and $S(\mathbf{b}) = \mathbf{y}^\top \mathbf{y} + (\mathbf{b}^\top \mathbf{X}^\top \mathbf{X} \mathbf{b} - 2\mathbf{b}^\top \mathbf{X}^\top \mathbf{y})$, the minimizers of S are exactly the solutions of the normal equations, and they exist. But there is a $\mathbf{v} \neq \mathbf{0}$ with $\mathbf{X}\mathbf{v} = \mathbf{0}$, and then $S(\mathbf{b} + t\mathbf{v}) = S(\mathbf{b})$ for every t , so no minimizer is unique.

■

The calculus proof finds the stationary point and then needs a separate argument, Eq. 5.4.3, to show it is the global minimum. The Hessian $2\mathbf{X}^\top \mathbf{X}$ tells the same story: it is positive definite, so S is a strictly convex quadratic bowl with a single lowest point. The name “normal equations” has nothing to do with the normal distribution. It refers to the orthogonality they express: $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\hat{\beta}) = \mathbf{0}$ says the residual vector is *normal*, that is, perpendicular, to every column of $\mathbf{X}$. Chapter 6 makes that geometry the starting point, derives the normal equations again without calculus (Theorem 6.4.2), and treats the rank-deficient case in full.

Remark (The rank-deficient case). When $\text{rank}(\mathbf{X}) < p$, the proof shows that the least squares estimates form an affine set $\hat{\beta} + \mathcal{N}(\mathbf{X})$ of dimension $p - \text{rank}(\mathbf{X})$. Different programs report different members of this set, according to which columns they drop or which constraints they impose. The fitted values $\mathbf{X}\hat{\beta}$ are the same for all of them, since $\mathbf{X}(\hat{\beta} + \mathbf{v}) = \mathbf{X}\hat{\beta}$ for $\mathbf{v} \in \mathcal{N}(\mathbf{X})$. Which linear combinations of the coefficients share this invariance is the question of estimability, answered in Chapter 8 (Definition 8.2.1 and Theorem 8.3.1).

Least squares under known linear restrictions $\mathbf{H}\mathbf{b} = \mathbf{h}$, such as shares that add to one, has a

closed form found by Lagrange multipliers in [Exercise \(B1\)](#); the increase in the residual sum of squares it computes drives the test of the restrictions in Chapter 11.

Example. (The straight line again)

For $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$,

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum_i x_i^2 \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{y} = \begin{pmatrix} n\bar{y} \\ \sum_i x_i y_i \end{pmatrix}.$$

The determinant is $n \sum_i x_i^2 - n^2 \bar{x}^2 = nS_{xx}$, positive iff the x_i are not all equal, which is exactly when $\text{rank}(\mathbf{X}) = 2$. Inverting,

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{nS_{xx}} \begin{pmatrix} \sum_i x_i^2 & -n\bar{x} \\ -n\bar{x} & n \end{pmatrix} = \begin{pmatrix} \frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} & -\frac{\bar{x}}{S_{xx}} \\ -\frac{\bar{x}}{S_{xx}} & \frac{1}{S_{xx}} \end{pmatrix}, \quad (5.4.4)$$

using $\sum_i x_i^2 = S_{xx} + n\bar{x}^2$. The second entry of $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ is $(\sum_i x_i y_i - n\bar{x}\bar{y})/S_{xx} = S_{xy}/S_{xx}$, and the first is $\bar{y} - \bar{x}S_{xy}/S_{xx}$. This recovers [Theorem 5.2.1](#), and [Eq. 5.4.3](#) reduces to [Eq. 5.2.5](#).

5.4.2 Fitted values, residuals and the hat matrix

Substituting [Eq. 5.4.2](#), the fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{H}\mathbf{y}, \quad \mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top, \quad (5.4.5)$$

and the residuals are $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{H})\mathbf{y}$. The $n \times n$ matrix $\mathbf{H}$ is the **hat matrix**, so called because it puts the hat on $\mathbf{y}$. It depends only on the regressors, never on the response.

Proposition 5.4.2 — Algebra of the fit

Let $\mathbf{X}$ have rank p , and let $\mathbf{H}$, $\hat{\mathbf{Y}}$ and $\hat{\boldsymbol{\varepsilon}}$ be as above.

- (a) $\mathbf{H}$ is symmetric and idempotent, $\mathbf{H}\mathbf{X} = \mathbf{X}$ and $(\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{0}$, and $\text{tr } \mathbf{H} = p$, $\text{tr}(\mathbf{I} - \mathbf{H}) = n - p$.
- (b) $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$ and $\hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} = 0$.
- (c) $\|\mathbf{y}\|^2 = \|\hat{\mathbf{Y}}\|^2 + \|\hat{\boldsymbol{\varepsilon}}\|^2$, and $\text{SSE} = \mathbf{y}^\top (\mathbf{I} - \mathbf{H})\mathbf{y} = \mathbf{y}^\top \mathbf{y} - \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$.
- (d) If $\mathbf{1} = \mathbf{X}\mathbf{c}$ for some $\mathbf{c}$ (for example, if $\mathbf{X}$ has an intercept column), then $\sum_i \hat{\varepsilon}_i = 0$, the mean of the fitted values is $\bar{y}$, and

$$\sum_i (y_i - \bar{y})^2 = \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i \hat{\varepsilon}_i^2,$$

that is, $\text{SST} = \text{SSR} + \text{SSE}$.

- (e) If $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ for some $\boldsymbol{\beta}$ and $\boldsymbol{\varepsilon}$, then $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}$ and $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$.

Proof. (a) $\mathbf{H}^\top = \mathbf{X}((\mathbf{X}^\top \mathbf{X})^{-1})^\top \mathbf{X}^\top = \mathbf{H}$, since the inverse of a symmetric matrix is symmetric. $\mathbf{H}\mathbf{X} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} = \mathbf{X}$, and then $\mathbf{H}^2 = \mathbf{H}\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{H}$. By the cyclic property (Theorem 1.6.1), $\text{tr } \mathbf{H} = \text{tr}((\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}) = \text{tr } \mathbf{I}_p = p$. (b) $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{X}^\top (\mathbf{I} - \mathbf{H})\mathbf{y} = ((\mathbf{I} - \mathbf{H})\mathbf{X})^\top \mathbf{y} = \mathbf{0}$, and $\hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = 0$. (c) $\mathbf{y} = \hat{\mathbf{Y}} + \hat{\boldsymbol{\varepsilon}}$ with $\hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} = 0$ gives the first identity. For the second, $(\mathbf{I} - \mathbf{H})$ is symmetric and idempotent, so $\|(\mathbf{I} - \mathbf{H})\mathbf{y}\|^2 = \mathbf{y}^\top (\mathbf{I} - \mathbf{H})\mathbf{y}$; and $\mathbf{y}^\top \mathbf{H}\mathbf{y} = \mathbf{y}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$. (d) $\mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{c}^\top \mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = 0$ by (b). Then $\sum_i \hat{y}_i = \sum_i y_i$. Write $y_i - \bar{y} = (\hat{y}_i - \bar{y}) + \hat{\varepsilon}_i$. The cross term $\sum_i (\hat{y}_i - \bar{y})\hat{\varepsilon}_i = \hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} - \bar{y} \mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = 0$. (e) Substitute $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ into Eq. 5.4.2 and into $(\mathbf{I} - \mathbf{H})\mathbf{y}$, using (a). ■

Part (e) will carry most of the statistical theory. The estimation error $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$ is a fixed linear function of the unobserved errors, and the residual vector does not depend on $\boldsymbol{\beta}$ at all. Part (b) says that the residuals satisfy p independent linear constraints. That is the origin of the divisor $n - p$ in Section 5.6.

For a model with an intercept, part (d) motivates the **coefficient of determination**

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}},$$

the proportion of the variation of the response about its mean that the fitted values reproduce. It generalizes Proposition 5.2.2(d), and lies in $[0, 1]$. It can only increase when a column is added to $\mathbf{X}$ (Exercise (B1)), which is why Chapter 9 introduces an adjusted version (Proposition 9.5.3). Section 6.7 interprets R as the cosine of an angle.

5.4.3 Centring the regressors

For a model with an intercept, write $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$, where $\mathbf{X}_1$ is $n \times k$ with column means $\bar{\mathbf{x}} = \mathbf{X}_1^\top \mathbf{1}/n$. The **centred regressors** are the columns of

$$\mathbf{X}_c = \mathbf{X}_1 - \mathbf{1}\bar{\mathbf{x}}^\top = \left(\mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top\right)\mathbf{X}_1,$$

each of which sums to zero. The next result is the matrix version of the formulas $\hat{\beta}_1 = S_{xy}/S_{xx}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$.

Proposition 5.4.3 — Centred form

Let $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$ have rank $p = k + 1$, and partition $\hat{\boldsymbol{\beta}} = (\hat{\beta}_0, \hat{\boldsymbol{\beta}}_1^\top)^\top$. Then $\mathbf{X}_c$ has rank k , and

$$\hat{\beta}_1 = (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \mathbf{X}_c^\top \mathbf{y} = (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \mathbf{X}_c^\top (\mathbf{y} - \bar{y}\mathbf{1}), \quad \hat{\beta}_0 = \bar{y} - \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_1.$$

Proof. For $\mathbf{b} = (b_0, \mathbf{b}_1^\top)^\top$,

$$\mathbf{X}\mathbf{b} = b_0\mathbf{1} + \mathbf{X}_1\mathbf{b}_1 = (b_0 + \bar{\mathbf{x}}^\top \mathbf{b}_1)\mathbf{1} + \mathbf{X}_c\mathbf{b}_1 = [\mathbf{1}, \mathbf{X}_c] \mathbf{T}\mathbf{b}, \quad \mathbf{T} = \begin{pmatrix} 1 & \bar{\mathbf{x}}^\top \\ \mathbf{0} & \mathbf{I}_k \end{pmatrix}.$$

$\mathbf{T}$ is nonsingular (it is upper triangular with unit diagonal), so $[\mathbf{1}, \mathbf{X}_c] = \mathbf{X}\mathbf{T}^{-1}$ has rank p (Proposition 1.2.3(b)), and hence $\mathbf{X}_c$ has rank k . The criterion satisfies $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 =$

$\|\mathbf{y} - [\mathbf{1}, \mathbf{X}_c]\mathbf{a}\|^2$ with $\mathbf{a} = \mathbf{T}\mathbf{b}$, and $\mathbf{b} \mapsto \mathbf{T}\mathbf{b}$ is a bijection of $\mathbb{R}^p$. So $\hat{\boldsymbol{\beta}}$ minimizes the left side iff $\hat{\mathbf{a}} = \mathbf{T}\hat{\boldsymbol{\beta}}$ minimizes the right side. Because $\mathbf{1}^\top \mathbf{X}_c = \mathbf{0}^\top$, the normal equations for $[\mathbf{1}, \mathbf{X}_c]$ are block diagonal,

$$\begin{pmatrix} n & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{X}_c^\top \mathbf{X}_c \end{pmatrix} \hat{\mathbf{a}} = \begin{pmatrix} n\bar{y} \\ \mathbf{X}_c^\top \mathbf{y} \end{pmatrix},$$

giving $\hat{a}_0 = \bar{y}$ and $\hat{\mathbf{a}}_1 = (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \mathbf{X}_c^\top \mathbf{y}$. Undoing $\mathbf{T}$: $\hat{\beta}_1 = \hat{\mathbf{a}}_1$ and $\hat{\beta}_0 = \hat{a}_0 - \bar{\mathbf{x}}^\top \hat{\mathbf{a}}_1$. Finally $\mathbf{X}_c^\top \mathbf{1} = \mathbf{0}$, so $\mathbf{X}_c^\top \mathbf{y} = \mathbf{X}_c^\top (\mathbf{y} - \bar{y}\mathbf{1})$.

■

In words: to find the slopes, centre everything and fit without an intercept; the intercept then makes the fitted surface pass through the point of means $(\bar{\mathbf{x}}, \bar{y})$. The matrix $\mathbf{X}_c^\top \mathbf{X}_c / (n-1)$ is the sample covariance matrix of the regressors, and $\mathbf{X}_c^\top \mathbf{y} / (n-1)$ their sample covariances with the response. So the slopes are “covariance divided by variance”, as in the straight-line case, with a matrix inverse in place of the division. This is the simplest case of the Frisch–Waugh–Lovell theorem of [Section 6.6](#), with the intercept playing the part of the regressors that are “partialled out”.

5.4.4 Indicator variables

The least squares estimates in a model with only indicator columns are simple averages.

Proposition 5.4.4 — Coefficients of indicator variables

Suppose each case belongs to one of $c \geq 2$ groups, group k has $n_k \geq 1$ cases with mean response $\bar{y}_k$, and $\mathbf{z}_k$ is the indicator column of group k . Let $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_{c-1}]$, so that group c is the reference. Then $\mathbf{X}$ has full rank, and

$$\hat{\beta}_0 = \bar{y}_c, \quad \hat{\beta}_k = \bar{y}_k - \bar{y}_c \quad (k = 1, \dots, c-1).$$

The fitted value of every case is the mean of its group.

Proof. If $\mathbf{X}\mathbf{b} = \mathbf{0}$, the entries for a case in group c give $b_0 = 0$, and those for a case in group $k < c$ give $b_0 + b_k = 0$, so $\mathbf{b} = \mathbf{0}$: the rank is $p = c$. Let $\mathbf{b}$ be the proposed vector. Its fitted value for a case in group $k < c$ is $\bar{y}_c + (\bar{y}_k - \bar{y}_c) = \bar{y}_k$, and for a case in group c it is $\bar{y}_c$. The residuals $y_i - \bar{y}_{k(i)}$ therefore sum to zero within each group, so $\mathbf{z}_k^\top (\mathbf{y} - \mathbf{X}\mathbf{b}) = 0$ for every $k = 1, \dots, c$. Every column of $\mathbf{X}$ is a sum of group indicators, since $\mathbf{1} = \mathbf{z}_1 + \dots + \mathbf{z}_c$, so $\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0}$: the normal equations hold. By [Theorem 5.4.1](#), $\mathbf{b} = \hat{\boldsymbol{\beta}}$.

■

```
import numpy as np
import statsmodels.api as sm
rand = sm.datasets.randhie.load_pandas().data
y = rand["mdvis"].to_numpy(dtype=float)
D = rand[["hlthg", "hlthf", "hlthp"]].to_numpy(dtype=float) # good, fair, poor
X = np.column_stack([np.ones(len(y)), D]) # excellent =
```

```

reference

beta_hat = np.linalg.solve(X.T @ X, X.T @ y)
level = D @ np.array([1, 2, 3])           # 0 = excellent, ...,
      3 = poor
means = np.array([y[level == k].mean() for k in range(4)])
print("coefficients:      ", np.round(beta_hat, 4))
print("group means:      ", np.round(means, 4))
print("means minus reference: ", np.round(means - means[0], 4))

```

For the health data of [Example \(Self-rated health\)](#) the group sizes are 11,019 (excellent), 7,309 (good), 1,560 (fair) and 302 (poor), and the group means are 2.634, 2.902, 3.692 and 5.795 visits. The listing confirms that the coefficients are these means' differences from the reference.

5.4.5 A worked example

Example. (Money demand in Denmark)

Economic theory relates the amount of money people choose to hold to their income, which drives the volume of transactions, and to interest rates, which measure the return given up by holding money rather than bonds. A classic data set records, for $n = 55$ quarters from 1974 to 1987 in Denmark, the logarithm of real money holdings (y), the logarithm of real income, the interest rate on bonds and the interest rate on deposits (rates as fractions per year). The model has $p = 4$ columns: an intercept and the three regressors. Solving the normal equations gives

$$\hat{\beta} = (4.3945, 1.2958, -2.6163, 0.6186)^T,$$

with residual sum of squares $SSE = 0.09253$ and $R^2 = 0.9262$. The listing checks that $\mathbf{X}^T \hat{\epsilon}$ is zero up to rounding error, that the two formulas for SSE in [Proposition 5.4.2\(c\)](#) agree, and that the slopes computed from centred data ([Proposition 5.4.3](#)) are the same. [Section 5.6](#) attaches standard errors to these numbers, and [Section 5.7](#) interprets them.

```

import numpy as np
import statsmodels.api as sm
dk = sm.datasets.danish_data.load_pandas().data
y = dk["lrm"].to_numpy()
X = np.column_stack([np.ones(len(y)), dk["lry"], dk["ibo"], dk["ide"]])
n, p = X.shape
print("n =", n, " p =", p)

```

```

XtX = X.T @ X
Xty = X.T @ y
beta_hat = np.linalg.solve(XtX, Xty)           # solve X^T X b = X^T y
fitted = X @ beta_hat
resid = y - fitted
SSE = resid @ resid

print("beta_hat =", np.round(beta_hat, 4))
print("X^T e     =", X.T @ resid)             # zero up to rounding
print("SSE two ways:", SSE, y @ y - beta_hat @ Xty)

```

```

Z = X[:, 1:] - X[:, 1:].mean(axis=0)          # centred regressors, no intercept column
slopes = np.linalg.solve(Z.T @ Z, Z.T @ (y - y.mean()))
intercept = y.mean() - X[:, 1:].mean(axis=0) @ slopes
print("slopes from centred data:", np.round(slopes, 4))
print("intercept recovered:      ", round(intercept, 4))

```

Warning (Formulas are not algorithms). The formula $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ is for reasoning, not for computing. Forming $\mathbf{X}^\top \mathbf{X}$ squares the condition number of the problem, and inverting it wastes work and accuracy. The listings solve the normal equations with a linear solver, which is acceptable for small, well-conditioned problems like these. Good software computes $\hat{\beta}$ from an orthogonal factorization of $\mathbf{X}$ itself. [Section 6.10](#) shows the loss of accuracy in an example, and [Chapter 10](#) develops the algorithms ([Theorem 10.2.1](#), [Theorem 10.3.2](#)).

5.4.6 Exercises

5.4.6.1 A. Check your understanding

Exercise A1.

Suppose the columns of $\mathbf{X}$ are nonzero and mutually orthogonal. Show that $\hat{\beta}_j = \mathbf{x}_j^\top \mathbf{y} / \mathbf{x}_j^\top \mathbf{x}_j$ for each j , so that each coefficient is the same as in the regression of $\mathbf{y}$ on $\mathbf{x}_j$ alone. Show also that deleting a column does not change the other estimates.

5.4.6.2 B. Practice

Exercise B1.

Let $\mathbf{X} = [\mathbf{X}_1, \mathbf{x}]$ have full rank, and let SSE_1 and SSE be the residual sums of squares from regressing $\mathbf{y}$ on $\mathbf{X}_1$ and on $\mathbf{X}$. Show that $\text{SSE} \leq \text{SSE}_1$, directly from [Definition 5.4.1](#). Deduce that R^2 cannot decrease when a column is added to a model with an intercept.

Solution. Let $\hat{\beta}_1$ be the least squares estimate for $\mathbf{X}_1$. The vector $(\hat{\beta}_1^\top, 0)^\top$ is one candidate in the minimization over $\mathbf{b} \in \mathbb{R}^p$, and it gives the value $\|\mathbf{y} - \mathbf{X}_1 \hat{\beta}_1\|^2 = \text{SSE}_1$. The minimum over all candidates is at most this. With an intercept in $\mathbf{X}_1$, $R^2 = 1 - \text{SSE}/\text{SST}$ and SST does not depend on the model, so R^2 cannot decrease. Equality holds iff the extra coefficient in the larger fit is zero, because by [Eq. 5.4.3](#) the candidate $(\hat{\beta}_1^\top, 0)^\top$ is optimal only if it equals $\hat{\beta}$.

Exercise B2.

Let $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$ with all $w_i > 0$, and $\mathbf{X}$ of full rank. Show that $\sum_i w_i (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2$ has the unique minimizer $(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{y}$, by applying [Theorem 5.4.1](#) to the rescaled data $\mathbf{W}^{1/2} \mathbf{y}$ and $\mathbf{W}^{1/2} \mathbf{X}$. Which cases get most influence when the weights are the reciprocals of the error variances?

Solution. $\sum_i w_i (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2 = \|\mathbf{W}^{1/2} \mathbf{y} - \mathbf{W}^{1/2} \mathbf{X} \mathbf{b}\|^2$. The matrix $\mathbf{W}^{1/2} \mathbf{X}$ has rank p , because $\mathbf{W}^{1/2}$ is nonsingular (Proposition 1.2.3(b)). By Theorem 5.4.1 the unique minimizer is $((\mathbf{W}^{1/2} \mathbf{X})^\top \mathbf{W}^{1/2} \mathbf{X})^{-1} (\mathbf{W}^{1/2} \mathbf{X})^\top \mathbf{W}^{1/2} \mathbf{y} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{y}$. With $w_i = 1/\text{Var}(Y_i)$, the least variable cases count most. This is weighted least squares, the subject of Chapter 21 and, with a general covariance matrix, of Section 6.9.

Exercise B3.

Show that $\hat{\beta} = \mathbf{A} \mathbf{y}$ with $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$, that $\mathbf{A} \mathbf{X} = \mathbf{I}_p$, and that $\mathbf{A} \mathbf{A}^\top = (\mathbf{X}^\top \mathbf{X})^{-1}$. Give an example of a different $p \times n$ matrix $\mathbf{B}$ with $\mathbf{B} \mathbf{X} = \mathbf{I}_p$ when $n > p$. (Chapter 7 compares all such estimators $\mathbf{B} \mathbf{y}$.)

5.4.6.3 C. Going deeper

Exercise C1.

Let $\mathbf{X}$ have full rank and let a new case $(\mathbf{x}_0, y_0)$ be appended. Using Theorem 1.6.5, show that the new estimate is

$$\hat{\beta}_{\text{new}} = \hat{\beta} + \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0 (y_0 - \mathbf{x}_0^\top \hat{\beta})}{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}.$$

Interpret: the estimate moves in proportion to the new case's prediction error. This is the basis of *recursive least squares*; Chapter 10 treats the numerically stable version.

5.5 Properties of the estimator under second-moment assumptions

Least squares was defined for a fixed data vector. Now let the data be random, generated by the linear model of Definition 5.3.1:

$$\mathbf{Y} = \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n,$$

with $\mathbf{X}$ of full rank p . The estimator $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ is then a random vector, and so are $\hat{\mathbf{Y}}$ and $\hat{\boldsymbol{\varepsilon}}$. This section computes their means and covariances. Only the second-moment assumptions are used, so the results hold for skewed, heavy-tailed or discrete errors alike, as long as they have finite variance.

5.5.1 Mean and covariance of the estimator

The key observation is that $\hat{\beta}$ is a *linear* function of $\mathbf{Y}$: $\hat{\beta} = \mathbf{A} \mathbf{Y}$ with the constant matrix $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$. The moments of a linear transformation were worked out in Chapter 2.

Theorem 5.5.1 — Moments of the least squares estimator

Let $\mathbf{X}$ have full rank p and $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$.

- (a) If $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$, then $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$: the least squares estimator is unbiased.
- (b) If moreover $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$, then

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}. \quad (5.5.1)$$

Proof. Let $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$. By Theorem 2.2.1(a), $E(\hat{\boldsymbol{\beta}}) = \mathbf{A} E(\mathbf{Y}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} = \boldsymbol{\beta}$. By Theorem 2.2.1(b),

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = \mathbf{A} (\sigma^2 \mathbf{I}_n) \mathbf{A}^\top = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1},$$

using the symmetry of $(\mathbf{X}^\top \mathbf{X})^{-1}$. ■

The proof is two lines, but the division of labour between the assumptions is worth stating. Unbiasedness needs only (L1): the errors may have unequal variances or be correlated, and $\hat{\boldsymbol{\beta}}$ still centres on $\boldsymbol{\beta}$. The covariance formula needs (L2) and (L3) as well. Neither needs normality. Under normality $\hat{\boldsymbol{\beta}}$ is itself normal, since it is a linear function of a normal vector (Theorem 3.3.1), so $\hat{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$. Chapter 7 builds exact inference on this (Theorem 7.5.1).

The matrix $(\mathbf{X}^\top \mathbf{X})^{-1}$ carries all the information about how the design affects precision. Its diagonal entries give the variances of the individual coefficients, and its off-diagonal entries their covariances. When two columns are strongly positively correlated, their coefficients are typically estimated with a strong *negative* correlation: the data can tell how large the two effects are together, but not how to split the total between them.

Corollary 5.5.2 — Linear functions of the coefficients

Under the assumptions of Theorem 5.5.1(b), let $\mathbf{a} \in \mathbb{R}^p$ and let $\mathbf{L}$ be a constant $q \times p$ matrix. Then

$$E(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) = \mathbf{a}^\top \boldsymbol{\beta}, \quad \text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a},$$

and $\text{Cov}(\mathbf{L}\hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{L} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{L}^\top$. In particular, the estimated mean response at a design point $\mathbf{x}_0$, $\hat{\mu}_0 = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$, is unbiased for $\mu_0 = \mathbf{x}_0^\top \boldsymbol{\beta}$ with $\text{Var}(\hat{\mu}_0) = \sigma^2 \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0$, and $\text{Var}(\hat{\beta}_j) = \sigma^2 [(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}$.

Proof. Apply Theorem 2.2.1 to $\mathbf{L}\hat{\boldsymbol{\beta}}$, with covariance $\mathbf{L} \text{Cov}(\hat{\boldsymbol{\beta}}) \mathbf{L}^\top$, and take $\mathbf{L} = \mathbf{a}^\top$, $\mathbf{L} = \mathbf{x}_0^\top$ and $\mathbf{L} = \mathbf{e}_j^\top$. ■

Linear functions of $\boldsymbol{\beta}$ are what applications usually need: a single coefficient, the difference between two coefficients, the mean response at a new design point, or the change in mean response between two profiles. Chapter 7 shows that for every such function, $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$ has the smallest

variance among all linear unbiased estimators (the Gauss–Markov theorem, [Theorem 7.2.1](#)).

5.5.2 Fitted values and residuals

Proposition 5.5.3 — Moments of fitted values and residuals

Under the linear model with $\mathbf{X}$ of full rank and $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$:

- (a) $E(\hat{\mathbf{Y}}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\hat{\mathbf{Y}}) = \sigma^2 \mathbf{H}$;
- (b) $E(\hat{\boldsymbol{\varepsilon}}) = \mathbf{0}$ and $\text{Cov}(\hat{\boldsymbol{\varepsilon}}) = \sigma^2(\mathbf{I} - \mathbf{H})$; in particular $\text{Var}(\hat{\varepsilon}_i) = \sigma^2(1 - h_{ii})$, where h_{ii} is the i th diagonal entry of $\mathbf{H}$;
- (c) $\text{Cov}(\hat{\mathbf{Y}}, \hat{\boldsymbol{\varepsilon}}) = \mathbf{0}$ and $\text{Cov}(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\varepsilon}}) = \mathbf{0}$;
- (d) $0 \leq h_{ii} \leq 1$ for every i , and $\sum_i h_{ii} = p$.

Proof. $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$ and $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{H})\mathbf{Y}$, with $\mathbf{H}$ symmetric and idempotent and $\mathbf{H}\mathbf{X} = \mathbf{X}$ ([Proposition 5.4.2\(a\)](#)). By [Theorem 2.2.1](#), $E(\hat{\mathbf{Y}}) = \mathbf{H}\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\hat{\mathbf{Y}}) = \sigma^2 \mathbf{H}\mathbf{H}^\top = \sigma^2 \mathbf{H}$; similarly $E(\hat{\boldsymbol{\varepsilon}}) = (\mathbf{I} - \mathbf{H})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$ and $\text{Cov}(\hat{\boldsymbol{\varepsilon}}) = \sigma^2(\mathbf{I} - \mathbf{H})^2 = \sigma^2(\mathbf{I} - \mathbf{H})$. The cross-covariances are $\text{Cov}(\mathbf{H}\mathbf{Y}, (\mathbf{I} - \mathbf{H})\mathbf{Y}) = \sigma^2 \mathbf{H}(\mathbf{I} - \mathbf{H}) = \mathbf{0}$ and $\text{Cov}(\mathbf{A}\mathbf{Y}, (\mathbf{I} - \mathbf{H})\mathbf{Y}) = \sigma^2 \mathbf{A}(\mathbf{I} - \mathbf{H}) = \sigma^2(\mathbf{A} - \mathbf{A}\mathbf{H}) = \mathbf{0}$, since $\mathbf{A}\mathbf{H} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{A}$. For (d), $\sigma^2 h_{ii}$ and $\sigma^2(1 - h_{ii})$ are variances, so both are nonnegative, and $\sum_i h_{ii} = \text{tr } \mathbf{H} = p$. ■

Two consequences deserve emphasis. First, residuals are *not* uncorrelated and *not* of equal variance, even when the errors are both. The residual of a case with h_{ii} near 1 has variance near zero: the fit is pulled so close to such a case that its residual says little about its error. The number h_{ii} is the **leverage** of case i ([Section 6.8](#), Chapter 20). Second, the fitted values and the residuals are uncorrelated. Under normality they are then independent ([Theorem 3.4.1](#)), which is the key to the exact tests of Chapter 11.

5.5.3 The straight line

Specializing to $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$, the inverse [Eq. 5.4.4](#) gives the classical formulas at once.

Corollary 5.5.4 — Moments for a straight line

In the model [Eq. 5.2.1](#) with $S_{xx} > 0$,

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}, \quad \text{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right), \quad \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2 \bar{x}}{S_{xx}},$$

and the estimated mean response at x_0 has

$$\text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_0) = \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right). \quad (5.5.2)$$

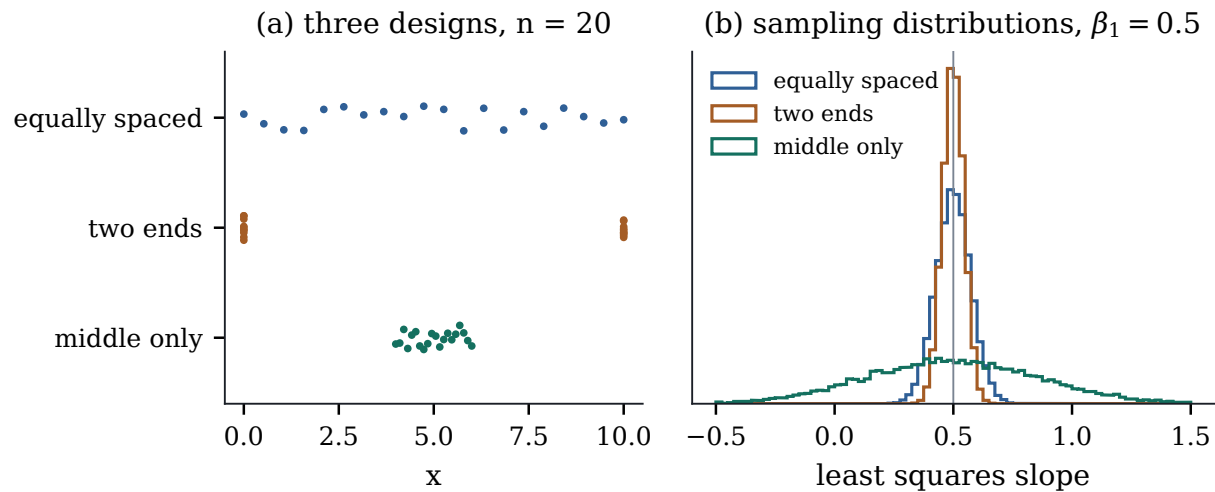


Figure 5.5.1: (a) Three designs with 20 points in $[0, 10]$. (b) Sampling distributions of the least squares slope under each design, with $\beta_1 = 0.5$ and $\sigma = 1$. All three are centred on β_1 ; their spreads are proportional to $S_{xx}^{-1/2}$.

Both estimates are unbiased, and $\bar{Y}$ is uncorrelated with $\hat{\beta}_1$.

Proof. The first three are the entries of $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ in Eq. 5.4.4. For the mean response, Corollary 5.5.2 with $\mathbf{x}_0 = (1, \bar{x})^\top$ gives $\sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} - \frac{2x_0\bar{x}}{S_{xx}} + \frac{x_0^2}{S_{xx}})$, which rearranges to Eq. 5.5.2. Finally $\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1\bar{x}$ (the line passes through the point of means), and $\text{Cov}(\hat{\beta}_0 + \hat{\beta}_1\bar{x}, \hat{\beta}_1) = -\sigma^2\bar{x}/S_{xx} + \bar{x}\sigma^2/S_{xx} = 0$.

■

Three design lessons can be read off. The slope is estimated precisely when the x_i are spread out, since its variance is inversely proportional to S_{xx} . The intercept is estimated precisely only near the data: its variance grows with $\bar{x}^2$, and the estimated intercept and slope are strongly negatively correlated when $\bar{x}$ is large and positive, the tilting of the ellipses noted after Theorem 5.2.1. And the mean response is estimated best at $x_0 = \bar{x}$, with a variance that grows quadratically as x_0 moves away. Extrapolation is doubly risky: the model may fail outside the range of the data, and even if it holds, the estimate is imprecise there.

Example. (Where to put the design points)

Suppose $n = 20$ observations can be taken anywhere in $[0, 10]$ to estimate a slope. Three designs are compared in Figure 5.5.1. Equally spaced points give $S_{xx} = 184.21$ and a slope standard deviation of 0.0737σ . Putting half the points at each end gives the largest possible $S_{xx} = 500.00$ (Exercise (B1)) and 0.0447σ . Bunching the points in the middle, over $[4, 6]$, gives $S_{xx} = 7.37$ and 0.3684σ , more than eight times worse than the two-ends design. The simulated sampling distributions in panel (b), each from 20,000 samples, match these standard deviations. The two-ends design has a price the variance formula does not show: with only two distinct values of x , the data cannot reveal curvature. The equally spaced design gives up some precision for the ability to check the model.

```
import numpy as np
```



```

n = 20
designs = {
    "equally spaced": np.linspace(0, 10, n),
    "two ends": np.repeat([0.0, 10.0], n // 2),
    "middle only": np.linspace(4, 6, n),
}
for name, xd in designs.items():
    Sxx = np.sum((xd - xd.mean()) ** 2)
    print(f"{name:15s} S_xx = {Sxx:7.2f}    sd(slope) = sigma * {1 / np.sqrt(Sxx):.4f}
    ")

```

5.5.4 Only two moments are needed

Example. (Skewed errors)

To see that normality plays no role in [Theorem 5.5.1](#), fit the quadratic $E(Y) = \beta_0 + \beta_1 x + \beta_2 x^2$ at $x = 1, \dots, 12$, with $\beta = (2, 0.5, -0.03)^\top$ and errors that are centred exponential variables scaled to $\sigma = 1.5$. These errors are strongly skewed. Over 100,000 simulated data sets, the averages of the three estimates are 1.9974, 0.5005 and -0.0300 , and the simulated standard deviations are 1.5421, 0.5457 and 0.0409, against 1.5503, 0.5483 and 0.0411 from [Eq. 5.5.1](#). Every entry of the simulated covariance matrix is within 0.011 of the formula, measured as a correlation. The formula also predicts that $\hat{\beta}_1$ and $\hat{\beta}_2$ have correlation -0.973 , because the columns x and x^2 are nearly collinear on $1, \dots, 12$. A slightly larger linear coefficient can be almost exactly compensated by a slightly more negative quadratic one.

```

import numpy as np
rng = np.random.default_rng(5)
x = np.arange(1.0, 13.0)                # twelve design points
X = np.column_stack([np.ones_like(x), x, x**2])
beta = np.array([2.0, 0.5, -0.03])
sigma = 1.5
reps = 100_000

A = np.linalg.solve(X.T @ X, X.T)        # beta_hat = A y
E = sigma * (rng.exponential(size=(reps, len(x))) - 1.0) # skewed, mean 0, sd
sigma
B = (X @ beta + E) @ A.T                # one estimate per row

print("mean of estimates:", np.round(B.mean(axis=0), 4), " true:", beta)
print("Monte Carlo covariance:\n", np.round(np.cov(B.T), 5))
print("sigma^2 (X^T X)^{-1}:\n", np.round(sigma**2 * np.linalg.inv(X.T @ X), 5))

```

5.5.5 Random regressors

When the regressors are random, the assumptions are read conditionally on $\mathbf{X}$ ([Section 5.1](#)). The conditional results then transfer.

Corollary 5.5.5 — Random regressors

Let $(\mathbf{X}, \mathbf{Y})$ be random, with $\text{rank}(\mathbf{X}) = p$ with probability one, and suppose $E(\mathbf{Y} | \mathbf{X}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y} | \mathbf{X}) = \sigma^2 \mathbf{I}_n$. If $E[(\mathbf{X}^\top \mathbf{X})^{-1}]$ is finite, then

$$E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}, \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 E[(\mathbf{X}^\top \mathbf{X})^{-1}].$$

Proof. Conditionally on $\mathbf{X}$, [Theorem 5.5.1](#) gives $E(\hat{\boldsymbol{\beta}} | \mathbf{X}) = \boldsymbol{\beta}$ and $\text{Cov}(\hat{\boldsymbol{\beta}} | \mathbf{X}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$. By the laws of total expectation and covariance ([Proposition 2.5.4](#)), $E(\hat{\boldsymbol{\beta}}) = E(\boldsymbol{\beta}) = \boldsymbol{\beta}$ and $\text{Cov}(\hat{\boldsymbol{\beta}}) = E[\sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}] + \text{Cov}(\boldsymbol{\beta}) = \sigma^2 E[(\mathbf{X}^\top \mathbf{X})^{-1}]$, since $\boldsymbol{\beta}$ is constant. The finiteness assumption guarantees $\hat{\boldsymbol{\beta}}$ has finite second moments, which [Proposition 2.5.4](#) needs. ■

This is what [Example \(One slope, two sampling schemes\)](#) showed by simulation: under the random design, the variance of the slope was $\sigma^2 E(1/S_{xx})$, not σ^2/S_{xx} for any one design. For inference, the conditional covariance $\sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ is the relevant one, because it describes the design actually observed.

5.5.6 Consistency

As the sample grows, a good estimator should approach the truth. For a fixed p and a sequence of model matrices $\mathbf{X}_n$, the mean squared error of $\hat{\boldsymbol{\beta}}$ is

$$E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 = \text{tr Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 \text{tr}(\mathbf{X}_n^\top \mathbf{X}_n)^{-1},$$

by [Theorem 5.5.1](#) and the fact that the trace of a covariance matrix is the sum of the variances.

Proposition 5.5.6 — Consistency in mean square

Under the linear model with $\mathbf{X}_n$ of full rank p for every n , if $\text{tr}(\mathbf{X}_n^\top \mathbf{X}_n)^{-1} \rightarrow 0$, equivalently if the smallest eigenvalue of $\mathbf{X}_n^\top \mathbf{X}_n$ tends to infinity, then $E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 \rightarrow 0$, and hence $\Pr(\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\| > \delta) \rightarrow 0$ for every $\delta > 0$.

Proof. The first claim is the display above. The eigenvalues of $(\mathbf{X}_n^\top \mathbf{X}_n)^{-1}$ are the reciprocals $1/\lambda_1, \dots, 1/\lambda_p$ of those of $\mathbf{X}_n^\top \mathbf{X}_n$, all positive, so the trace is $\sum_j 1/\lambda_j$. It tends to zero iff every $\lambda_j \rightarrow \infty$, iff the smallest does. The last claim is Markov's inequality applied to $\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2$: $\Pr(\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\| > \delta) \leq E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 / \delta^2$. ■

For a straight line the condition is $S_{xx} \rightarrow \infty$ together with $\bar{x}^2/S_{xx} \rightarrow 0$, which holds for any design that keeps spreading, but fails, for instance, for $x_i = 1/i$, where S_{xx} converges to a finite limit. Then even infinitely many observations do not pin down the slope ([Exercise \(C1\)](#)).

5.5.7 What if the assumptions fail?

The proof of [Theorem 5.5.1](#) shows exactly what each assumption contributes, so it also shows what goes wrong without it. Two cases recur throughout the book.

Proposition 5.5.7 — Moments under departures

Let $\mathbf{X}$ have full rank, $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ and $\hat{\boldsymbol{\beta}} = \mathbf{A}\mathbf{Y}$.

- (a) If $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\delta}$ for some $\boldsymbol{\delta} \in \mathbb{R}^n$, then $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\delta}$.
- (b) If $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$ for some covariance matrix $\boldsymbol{\Sigma}$, then $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ and

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Sigma} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}. \quad (5.5.3)$$

Proof. Both are [Theorem 2.2.1](#) applied to $\mathbf{A}\mathbf{Y}$: $E(\mathbf{A}\mathbf{Y}) = \mathbf{A}\mathbf{X}\boldsymbol{\beta} + \mathbf{A}\boldsymbol{\delta}$ and $\text{Cov}(\mathbf{A}\mathbf{Y}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top$. ■

Part (a) says that when the mean is misspecified, least squares estimates not $\boldsymbol{\beta}$ but $\boldsymbol{\beta}$ plus the coefficients from regressing the missing part $\boldsymbol{\delta}$ on $\mathbf{X}$. The most important case is an omitted regressor, which [Section 5.7](#) examines. Part (b) says that with the wrong covariance, the estimator stays unbiased but the formula $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ is wrong. The “sandwich” [Eq. 5.5.3](#) is the correct covariance, and Chapter 21 shows how to estimate it when $\boldsymbol{\Sigma}$ is diagonal but unknown. Chapter 19 studies both departures systematically.

5.5.8 Exercises

5.5.8.1 A. Check your understanding

Exercise A1.

In the model of [Proposition 5.4.4](#) with $c = 2$ groups, show that $\text{Var}(\hat{\beta}_1) = \sigma^2(1/n_1 + 1/n_2)$, both from [Eq. 5.5.1](#) and directly from $\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_2$. For fixed $n = n_1 + n_2$, which split minimizes it?

Exercise A2.

Show that for a straight line $h_{ii} = 1/n + (x_i - \bar{x})^2/S_{xx}$. Which case in [Example \(Forecasting December sea temperature\)](#) has the largest leverage? Check that the leverages sum to 2.

5.5.8.2 B. Practice

Exercise B1.

Suppose n is even and the x_i may be chosen anywhere in $[a, b]$. Show that $S_{xx} \leq n(b-a)^2/4$, with equality iff half the points are at a and half at b . *Hint:* $|x_i - c| \leq (b-a)/2$ with $c = (a+b)/2$, and $S_{xx} = \sum_i (x_i - c)^2 - n(\bar{x} - c)^2$.

Solution. With c the midpoint, $S_{xx} = \sum_i (x_i - c)^2 - n(\bar{x} - c)^2 \leq \sum_i (x_i - c)^2 \leq n(b-a)^2/4$. Equality in the second step needs every $x_i \in \{a, b\}$, and in the first step needs $\bar{x} = c$, that is, equally many points at each end. For $[0, 10]$ and $n = 20$ this gives $S_{xx} = 20 \cdot 25 = 500$, the value in [Example \(Where to put the design points\)](#).

Exercise B2.

In [Example \(Money demand in Denmark\)](#), write the variances of $\hat{\beta}_2 - \hat{\beta}_3$ and $\hat{\beta}_2 + \hat{\beta}_3$, the difference and the sum of the two interest-rate coefficients, in terms of the entries of $\mathbf{G} = (\mathbf{X}^\top \mathbf{X})^{-1}$. Show that if $\hat{\beta}_2$ and $\hat{\beta}_3$ are negatively correlated, the difference is estimated *less* precisely, and the sum *more* precisely, than they would be if the two estimates were uncorrelated. The two rate columns have sample correlation 0.80. Which combination would you expect to be better determined?

Solution. With $\mathbf{a} = (0, 0, 1, \mp 1)^\top$ and indices counting the intercept as the first coefficient, [Corollary 5.5.2](#) gives $\text{Var}(\hat{\beta}_2 \mp \hat{\beta}_3) = \sigma^2(g_{33} + g_{44} \mp 2g_{34})$, and $\text{Cov}(\hat{\beta}_2, \hat{\beta}_3) = \sigma^2 g_{34}$. If $g_{34} < 0$, the variance of the difference exceeds $\sigma^2(g_{33} + g_{44}) = \text{Var}(\hat{\beta}_2) + \text{Var}(\hat{\beta}_3)$, and that of the sum falls below it. Positively correlated columns tend to produce negatively correlated estimates (as in [Example \(Skewed errors\)](#)), because the data can tell how large the combined effect of the two rates is more easily than how to divide it between them. So the sum should be the better determined combination.

5.5.8.3 C. Going deeper
Exercise C1.

Let $x_i = 1/i$ for $i = 1, 2, \dots$ in the straight-line model. Show that S_{xx} converges to a finite limit as $n \rightarrow \infty$, so that $\text{Var}(\hat{\beta}_1)$ does not tend to zero. Then construct a sequence of designs, in which the points may change with n (for example, designs centred farther and farther from 0), with $S_{xx} \rightarrow \infty$ but $\bar{x}^2/S_{xx} \not\rightarrow 0$, and decide whether the slope, the intercept, or both fail to be consistent.

Solution. With $x_i = 1/i$, $S_{xx} = \sum_{i \leq n} 1/i^2 - n\bar{x}^2$. The sum increases to $\pi^2/6$, and $n\bar{x}^2 = H_n^2/n \rightarrow 0$ because the harmonic number $H_n = \sum_{i \leq n} 1/i$ grows like $\log n$. So $S_{xx} \rightarrow \pi^2/6$ and $\text{Var}(\hat{\beta}_1) = \sigma^2/S_{xx}$ stays above $6\sigma^2/\pi^2$.

For even n take $x_i = n + (-1)^i$, $i = 1, \dots, n$: half the points at $n-1$, half at $n+1$. Then $\bar{x} = n$ and $S_{xx} = n$, so $\bar{x}^2/S_{xx} = n \rightarrow \infty$. By [Corollary 5.5.4](#), $\text{Var}(\hat{\beta}_1) = \sigma^2/n \rightarrow 0$, so the slope is consistent (Chebyshev's inequality), while $\text{Var}(\hat{\beta}_0) = \sigma^2(1/n + n) \rightarrow \infty$. The intercept, the mean response at $x = 0$, ever farther from the data, is not consistent: with normal errors, $\hat{\beta}_0 - \beta_0$ is normal with unbounded variance.

5.6 Estimating the error variance

The covariance formula $\text{Cov}(\hat{\beta}) = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ is of little use until σ^2 is estimated. Least squares does not estimate it: the criterion $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ involves only β . But the size of the residuals clearly carries information about the size of the errors. This section finds the right way to turn one into an estimate of the other, and shows why the obvious way is wrong.

5.6.1 The residual sum of squares

Since $\sigma^2 = \text{Var}(\varepsilon_i) = \text{E}(\varepsilon_i^2)$, a natural first idea is to replace each unobserved error by its residual and average: SSE/n . This underestimates σ^2 . The fitted values were chosen to make the residuals as small as possible, so the residuals are systematically smaller than the errors. The next theorem measures exactly how much smaller.

Theorem 5.6.1 — Unbiased estimation of the error variance

In the linear model [Definition 5.3.1](#) with $\text{rank}(\mathbf{X}) = p < n$,

$$\text{E}(\text{SSE}) = \sigma^2(n - p).$$

Hence

$$s^2 = \frac{\text{SSE}}{n - p} = \frac{\|\mathbf{Y} - \mathbf{X}\hat{\beta}\|^2}{n - p} \quad (5.6.1)$$

is an unbiased estimator of σ^2 .

Proof. By [Proposition 5.4.2](#)(e) the residual vector is $\hat{\varepsilon} = (\mathbf{I} - \mathbf{H})\varepsilon$, whatever β is. Since $\mathbf{I} - \mathbf{H}$ is symmetric and idempotent, $\text{SSE} = \hat{\varepsilon}^\top \hat{\varepsilon} = \varepsilon^\top (\mathbf{I} - \mathbf{H})\varepsilon$, a quadratic form in the error vector, which has mean $\mathbf{0}$ and covariance $\sigma^2 \mathbf{I}_n$. By [Corollary 2.3.2](#)(b),

$$\text{E}(\text{SSE}) = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{H}) + \mathbf{0}^\top (\mathbf{I} - \mathbf{H}) \mathbf{0} = \sigma^2(n - p),$$

using $\text{tr}(\mathbf{I} - \mathbf{H}) = n - p$ from [Proposition 5.4.2](#)(a). ■

The computation is the case $\theta = \mathbf{X}\beta$ of [Example \(Residual sum of squares under a wrong mean\)](#) in [Chapter 2](#), specialized to full rank.

The same result follows from [Proposition 5.5.3](#) without quadratic forms. Each residual has mean zero and variance $\sigma^2(1 - h_{ii})$, so $\text{E}(\text{SSE}) = \sum_i \text{E}(\hat{\varepsilon}_i^2) = \sigma^2 \sum_i (1 - h_{ii}) = \sigma^2(n - p)$. This version shows *where* the shortfall comes from: each residual is less variable than its error by the factor $1 - h_{ii}$, and the leverages add up to p .

The divisor $n - p$ is the number of **residual degrees of freedom**. The name reflects the constraints $\mathbf{X}^\top \hat{\varepsilon} = \mathbf{0}$ of [Proposition 5.4.2](#)(b): the n residuals satisfy p independent linear equations, so only $n - p$ of them can vary freely. [Chapter 6](#) makes this precise: the residual vector lies in a subspace of dimension $n - p$, and the degrees of freedom of a sum of squares is

always the dimension of the space it lives in. For the straight line $p = 2$, so $s^2 = \text{SSE}/(n - 2)$. For the sample variance of a single sample, the model matrix is $\mathbf{1}$, $p = 1$, and [Eq. 5.6.1](#) is the familiar divisor $n - 1$.

Corollary 5.6.2 — Estimated covariance and standard errors

Under the conditions of [Theorem 5.6.1](#), $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$ is an unbiased estimator of $\text{Cov}(\hat{\boldsymbol{\beta}})$, and $s^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}$ is an unbiased estimator of $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}})$. The **standard error** of $\hat{\beta}_j$ is

$$\text{se}(\hat{\beta}_j) = s \sqrt{[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}}.$$

Proof. $(\mathbf{X}^\top \mathbf{X})^{-1}$ and $\mathbf{a}$ are constants, so the expectations are $E(s^2)(\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ and similarly for the scalar. ■

The standard error is not unbiased for the standard deviation of $\hat{\beta}_j$. By Jensen's inequality $E(s) \leq \sqrt{E(s^2)} = \sigma$, with strict inequality unless s is constant. The bias is small when $n - p$ is large, and it matters little, since the standard error is used mainly through the ratio $(\hat{\beta}_j - \beta_j)/\text{se}(\hat{\beta}_j)$, whose exact distribution under normality is the t distribution with $n - p$ degrees of freedom ([Chapter 7](#), [Theorem 7.5.1](#)).

Example. (Standard errors for money demand)

For the money demand model of [Example \(Money demand in Denmark\)](#), $n = 55$ and $p = 4$, so

$$s^2 = \frac{0.09253}{51} = 0.001814, \quad s = 0.04259.$$

Since the response is a logarithm, s says that money holdings typically deviate from the fitted relation by about 4 percent. The standard errors of the four coefficients are 0.5811, 0.0940, 0.3282 and 0.6911. The income coefficient 1.2958 is almost fourteen standard errors from zero and the bond rate coefficient -2.6163 about eight, but the deposit rate coefficient 0.6186 is smaller than its standard error. Whether the deposit rate belongs in the model at all is a testing question for Chapter 11.

These standard errors rest on (L2) and (L3). For a quarterly economic series (L3) is doubtful. The correlation between consecutive residuals is 0.62, which suggests that the errors are positively correlated in time, and then the formula $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ usually understates the true variability. Chapter 21 shows how to detect and allow for this.

```
import numpy as np
import statsmodels.api as sm
dk = sm.datasets.danish_data.load_pandas().data
y = dk["lrm"].to_numpy()
X = np.column_stack([np.ones(len(y)), dk["lry"], dk["ibo"], dk["ide"]])
n, p = X.shape
XtX = X.T @ X
Xty = X.T @ y
beta_hat = np.linalg.solve(XtX, Xty)      # solve X^T X b = X^T y
fitted = X @ beta_hat
resid = y - fitted
```

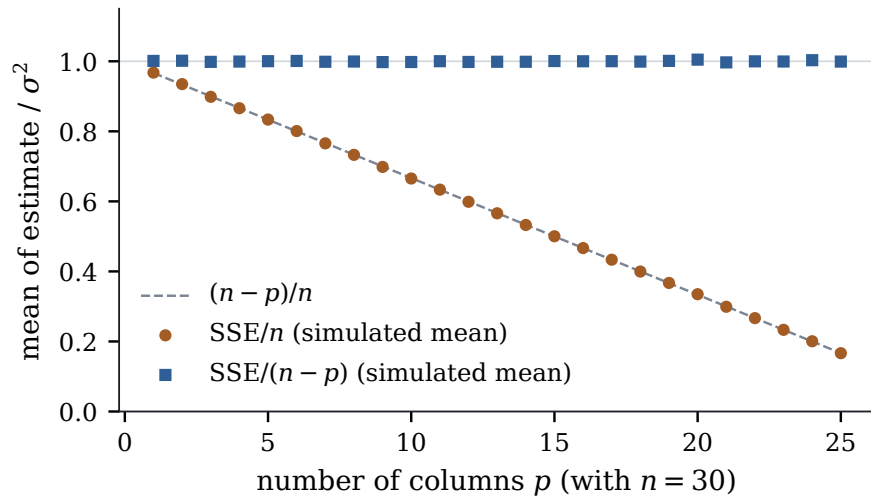


Figure 5.6.1: Average of two estimators of $\sigma^2 = 1$ over 20,000 simulated data sets, with $n = 30$ and p columns. Dividing the residual sum of squares by n underestimates σ^2 by the factor $(n-p)/n$; dividing by $n-p$ is unbiased for every p .

```
SSE = resid @ resid

s2 = SSE / (n - p)                                # unbiased estimate of sigma^2
cov_hat = s2 * np.linalg.inv(XtX)                  # estimated Cov(beta_hat)
se = np.sqrt(np.diag(cov_hat))
print(f"s^2 = {s2:.6f}, s = {np.sqrt(s2):.5f}")
print("standard errors:", np.round(se, 4))
print("SSE / n would give", SSE / n)
```

5.6.2 Dividing by n

The estimator $\hat{\sigma}^2 = SSE/n$ is not unbiased, but it is not arbitrary either: under normality it is the maximum likelihood estimator ([Theorem 7.3.1](#)). Its expectation is

$$E(\hat{\sigma}^2) = \frac{n-p}{n} \sigma^2 = \sigma^2 - \frac{p}{n} \sigma^2.$$

The bias is negligible when p is a small fraction of n , and severe when it is not. For the money demand data the two estimates of σ are 0.04259 and 0.04102, a difference of about 4 percent. But a model with many columns relative to the number of cases, such as a two-way layout with a few replicates per cell or a regression with many indicator variables, can make SSE/n badly misleading.

Example. (How bad is dividing by n ?)

Take $n = 30$ cases and model matrices with an intercept and $p - 1$ further columns of standard normal numbers, for $p = 1, \dots, 25$. For each p , 20,000 error vectors are simulated with $\sigma^2 = 1$. [Figure 5.6.1](#) plots the average of SSE/n and of $SSE/(n-p)$. The first follows the line $(n-p)/n$: at $p = 5$ it averages 0.834, at $p = 15$ it averages 0.500, half the true value, and at $p = 25$ only 0.167. The unbiased estimator averages 1.000, 1.001 and 0.999 at the same values of p .

```

import numpy as np
rng = np.random.default_rng(506)
n, sigma2, reps = 30, 1.0, 20_000
Xfull = np.column_stack([np.ones(n), rng.normal(size=(n, 24))])

ps = np.arange(1, 26)
mean_n, mean_np = [], []
for p in ps:
    X = Xfull[:, :p]
    R = np.eye(n) - X @ np.linalg.solve(X.T @ X, X.T)      # I - H
    E = rng.standard_normal((reps, n))                      # errors, sigma^2 = 1
    SSE = np.einsum("ri,ij,rj->r", E, R, E)                 # e^T (I - H) e
    mean_n.append(SSE.mean() / n)
    mean_np.append(SSE.mean() / (n - p))
print("p      E(SSE/n)  E(SSE/(n-p))")
for p in (1, 5, 15, 25):
    print(f"{p:2d}      {mean_n[p - 1]:.3f}      {mean_np[p - 1]:.3f}")

```

5.6.3 The variance of s^2

Unbiasedness says where s^2 is centred, not how much it varies. Its variance depends on the fourth moment of the errors, so the second-moment assumptions are not enough to find it. With independent errors, [Chapter 2](#) supplies the formula.

Proposition 5.6.3 — Variance of s^2

In the linear model with $\text{rank}(\mathbf{X}) = p < n$, suppose the errors $\varepsilon_1, \dots, \varepsilon_n$ are independent with mean 0, variance σ^2 , common third moment $\mu_3 = E(\varepsilon_i^3)$ and common fourth moment $\mu_4 = E(\varepsilon_i^4) < \infty$. Then

$$\text{Var}(\text{SSE}) = 2\sigma^4(n - p) + (\mu_4 - 3\sigma^4) \sum_{i=1}^n (1 - h_{ii})^2.$$

For normal errors, $\mu_4 = 3\sigma^4$ and $\text{Var}(s^2) = 2\sigma^4/(n - p)$.

Proof. Apply [Theorem 2.3.3](#) to the vector $\boldsymbol{\varepsilon}$, whose means θ_i are all zero, with $\mathbf{A} = \mathbf{I} - \mathbf{H}$. The terms involving $\boldsymbol{\theta}$ vanish (in particular μ_3 enters only multiplied by $\boldsymbol{\theta} = \mathbf{0}$, so its value is irrelevant). The diagonal of $\mathbf{A}$ has entries $1 - h_{ii}$, and $\text{tr}(\mathbf{A}^2) = \text{tr}(\mathbf{A}) = n - p$ because $\mathbf{A}$ is idempotent. For the normal case, the fourth moment of a $N(0, \sigma^2)$ variable is $3\sigma^4$, and $\text{Var}(s^2) = \text{Var}(\text{SSE})/(n - p)^2$. ■

The kurtosis term can be large. Heavy-tailed errors, with μ_4 much larger than $3\sigma^4$, make s^2 much less reliable than the normal-theory formula suggests, even though it remains unbiased. Since $0 \leq 1 - h_{ii} \leq 1$, the sum in the kurtosis term is at most $\sum_i (1 - h_{ii}) = n - p$, which gives the following.

Corollary 5.6.4 — Consistency of s^2

Under the conditions of [Proposition 5.6.3](#),

$$\text{Var}(s^2) \leq \frac{2\sigma^4 + |\mu_4 - 3\sigma^4|}{n - p}.$$

So if $n - p \rightarrow \infty$ along a sequence of models, $s^2 \rightarrow \sigma^2$ in mean square and in probability.

Proof. Divide the formula of [Proposition 5.6.3](#) by $(n - p)^2$ and bound the sum by $n - p$. Since $E(s^2) = \sigma^2$, $E(s^2 - \sigma^2)^2 = \text{Var}(s^2) \rightarrow 0$, and Chebyshev's inequality gives convergence in probability. ■

Example. (Skewed errors inflate the variance of s^2)

With $n = 30$ and the first $p = 6$ columns of the model matrix of [Example \(How bad is dividing by \$n\$?\)](#), the leverages give $\sum_i (1 - h_{ii})^2 = 19.431$. For normal errors with $\sigma^2 = 1$, $\text{Var}(\text{SSE}) = 2(n - p) = 48.00$. For centred exponential errors, which have $\sigma^2 = 1$ and $\mu_4 = 9$, [Proposition 5.6.3](#) gives $48 + 6 \times 19.431 = 164.59$, and a simulation with 200,000 data sets gives 165.19. The standard deviation of s^2 is 0.289 under normal errors and 0.535 under exponential errors, nearly twice as large, although s^2 is unbiased in both cases.

```
p = 6
X = Xfull[:, :p]
H = X @ np.linalg.solve(X.T @ X, X.T)
R = np.eye(n) - H
mu4 = 9.0                                     # fourth moment of a centred Exp(1)
variable
var_normal = 2 * (n - p)
var_exp = (mu4 - 3) * np.sum(np.diag(R) ** 2) + 2 * (n - p)
reps2 = 200_000
E = rng.exponential(size=(reps2, n)) - 1.0
SSE = np.einsum("ri,ij,rj->r", E, R, E)
print(f"Var(SSE): normal errors {var_normal:.2f}, exponential errors {var_exp:.2f},"
      f" simulated {SSE.var():.2f}")
```

Under normality much more is true: SSE/σ^2 has the chi-squared distribution with $n - p$ degrees of freedom and is independent of $\hat{\beta}$ ([Theorem 7.5.1](#), using [Theorem 4.3.2](#) and [Theorem 4.4.1](#)). That is what turns $\hat{\beta}$, s^2 and the standard errors into exact confidence intervals and tests. Among unbiased estimators of σ^2 , s^2 then has the smallest variance ([Theorem 7.4.6](#)).

5.6.4 When the mean is wrong

[Theorem 5.6.1](#) assumed that $E(\mathbf{Y}) = \mathbf{X}\beta$. If instead $E(\mathbf{Y}) = \boldsymbol{\theta}$ is not of this form, [Example \(Residual sum of squares under a wrong mean\)](#) in [Chapter 2](#) shows that

$$E(\text{SSE}) = \sigma^2(n - p) + \|(\mathbf{I} - \mathbf{H})\boldsymbol{\theta}\|^2.$$

The extra term is the squared distance from the true mean vector to the nearest mean vector the model can produce. So an incomplete model *overestimates* σ^2 : variation that the model fails to capture is counted as error. This is one reason to prefer a model that includes all relevant regressors, and it is the basis of the lack-of-fit tests of Chapter 14, which compare s^2 with an estimate of σ^2 that does not depend on the form of the mean.

5.6.5 Exercises

5.6.5.1 A. Check your understanding

Exercise A1.

For regression through the origin, $E(Y_i) = \beta x_i$, what is the unbiased estimator of σ^2 ? Verify $E(\text{SSE}) = \sigma^2(n-1)$ directly, by writing $\text{SSE} = \sum_i \varepsilon_i^2 - (\sum_i x_i \varepsilon_i)^2 / \sum_i x_i^2$.

Exercise A2.

For [Example \(Forecasting December sea temperature\)](#), compute s and the standard error of the slope. Is the slope more than two standard errors away from 1, the value that would mean December simply tracks August one for one?

5.6.5.2 B. Practice

Exercise B1.

Show that $\text{SSE} = \mathbf{y}^\top \mathbf{y} - \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}$, and that for a model with an intercept $\text{SSE} = S_{yy} - \hat{\boldsymbol{\beta}}_1^\top \mathbf{X}_c^\top \mathbf{X}_c \hat{\boldsymbol{\beta}}_1$ in the notation of [Proposition 5.4.3](#). Explain why the second form is numerically preferable when the responses are large and nearly equal, as in [Example \(A curved trend and a seasonal cycle\)](#).

Solution. By [Proposition 5.4.2\(c\)](#), $\text{SSE} = \mathbf{y}^\top \mathbf{y} - \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$, and $\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}$ by the normal equations. For the centred form, the model with columns $[\mathbf{1}, \mathbf{X}_c]$ has the same fitted values (the proof of [Proposition 5.4.3](#)), with coefficients $(\bar{y}, \hat{\boldsymbol{\beta}}_1)$. So $\text{SSE} = \mathbf{y}^\top \mathbf{y} - n\bar{y}^2 - \hat{\boldsymbol{\beta}}_1^\top \mathbf{X}_c^\top \mathbf{X}_c \hat{\boldsymbol{\beta}}_1$, because the cross-product matrix is block diagonal, and $\mathbf{y}^\top \mathbf{y} - n\bar{y}^2 = S_{yy}$. When the responses are about 340 with variation of a few units, $\mathbf{y}^\top \mathbf{y}$ and $\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$ are both near $n \cdot 340^2$ and agree in their leading digits; their difference loses those digits to cancellation. The centred form subtracts numbers of the size of the variation itself.

Exercise B2.

Under normality SSE/σ^2 has mean $n-p$ and variance $2(n-p)$. Among estimators $c \cdot \text{SSE}$, find the c that minimizes the mean squared error $E(c \text{SSE} - \sigma^2)^2$. Compare with $1/(n-p)$ and $1/n$.

Solution. Write $\text{SSE} = \sigma^2 Q$ with $E Q = m = n-p$ and $E Q^2 = 2m + m^2$. Then $E(cQ - 1)^2 \sigma^4 = \sigma^4 (c^2(m^2 + 2m) - 2cm + 1)$, minimized at $c = m/(m^2 + 2m) = 1/(n-p+2)$. The minimum-MSE estimator divides by $n-p+2$: it is biased downward but less variable.

The ordering of divisors is $n - p < n - p + 2$ and, when $p \leq 2$, $n \leq n - p + 2$, so for small p the maximum likelihood divisor n lies between the unbiased and the minimum-MSE choice. Unbiasedness is a convention, not an optimality property; it is kept because it makes s^2 combine cleanly in the t and F statistics.

5.6.5.3 C. Going deeper

Exercise C1.

Show that a quadratic form $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ with $\mathbf{A}$ symmetric is unbiased for σ^2 for every β and σ^2 iff $\mathbf{X}^\top \mathbf{A} \mathbf{X} = \mathbf{0}$ and $\text{tr } \mathbf{A} = 1$. Verify that $\mathbf{A} = (\mathbf{I} - \mathbf{H})/(n - p)$ qualifies, and find another $\mathbf{A}$ that does. (Under normality, [Theorem 7.4.6](#) shows that s^2 has the smallest variance among all unbiased estimators.)

5.7 Interpreting coefficients

A fitted regression is usually reported as a table of coefficients. The short answer, “the change in the mean response per unit change in the regressor, holding the other regressors fixed”, is a useful slogan and a frequent source of error. This section makes the slogan precise within the model, shows how the meaning of a coefficient depends on units, centring, transformations and the other columns of the model, and marks where the slogan stops being true.

5.7.1 What a coefficient measures

Definition 5.7.1 — Interpretation of a coefficient

In a linear model $E(Y | \mathbf{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$, suppose the regressor x_j enters the model only through its own column, so that no other column is a function of x_j . For any two regressor profiles $\mathbf{x}$ and $\mathbf{x}'$ that agree in every regressor except the j th, and have $x'_j = x_j + 1$,

$$E(Y | \mathbf{x}') - E(Y | \mathbf{x}) = \beta_j. \quad (5.7.1)$$

The coefficient β_j is the **partial effect** of x_j *in this model*: the difference in mean response per unit difference in x_j , between profiles that agree on all the other regressors in the model. Its estimate $\hat{\beta}_j$ is the least squares estimate of this difference.

Three qualifications are built into the definition.

1. **It compares profiles, not interventions.** [Eq. 5.7.1](#) compares the mean response of two kinds of case that exist in the population the model describes. Whether changing x_j *in a given case* would change its response by β_j is a causal question, which the model alone cannot answer (Chapter 25).
2. **“The other regressors” means the ones in the model.** Adding or removing a regressor changes the comparison being made, and so changes β_j , unless the regressors are

orthogonal. [Proposition 5.7.2](#) gives the exact relation.

3. **It needs x_j to vary alone.** If x_j also appears in other columns, as in a polynomial or an interaction, there are no two profiles that differ only in the x_j column, and β_j alone does not measure the effect of x_j .

The geometric content of “holding the others fixed” in a data set, where nobody held anything fixed, is the Frisch–Waugh–Lovell theorem of [Section 6.6](#): $\hat{\beta}_j$ is the slope of the response on the part of x_j that is not linearly explained by the other regressors.

5.7.2 Units, rescaling and reparameterization

The coefficient β_j is measured in units of the response per unit of x_j . Changing units rescales it. More generally, any invertible linear recoding of the columns changes the coefficients in a predictable way and leaves the fit untouched.

Proposition 5.7.1 — Reparameterization

Let $\mathbf{X}$ be $n \times p$ of rank p , let $\mathbf{K}$ be a nonsingular $p \times p$ matrix, and put $\mathbf{Z} = \mathbf{X}\mathbf{K}$. Then $\mathbf{Z}$ has rank p , and the least squares estimates from the two model matrices are related by

$$\hat{\beta}_Z = \mathbf{K}^{-1}\hat{\beta}_X.$$

The fitted values, residuals, SSE and s^2 are the same for both, and the estimated covariance matrices satisfy $s^2(\mathbf{Z}^\top \mathbf{Z})^{-1} = \mathbf{K}^{-1}[s^2(\mathbf{X}^\top \mathbf{X})^{-1}](\mathbf{K}^{-1})^\top$.

Proof. $\text{rank}(\mathbf{Z}) = \text{rank}(\mathbf{X}) = p$ by [Proposition 1.2.3\(b\)](#). Then

$$\begin{aligned}\hat{\beta}_Z &= (\mathbf{K}^\top \mathbf{X}^\top \mathbf{X} \mathbf{K})^{-1} \mathbf{K}^\top \mathbf{X}^\top \mathbf{y} \\ &= \mathbf{K}^{-1}(\mathbf{X}^\top \mathbf{X})^{-1}(\mathbf{K}^\top)^{-1} \mathbf{K}^\top \mathbf{X}^\top \mathbf{y} = \mathbf{K}^{-1}\hat{\beta}_X,\end{aligned}$$

and $\mathbf{Z}\hat{\beta}_Z = \mathbf{X}\mathbf{K}\mathbf{K}^{-1}\hat{\beta}_X = \mathbf{X}\hat{\beta}_X$. Equal fitted values give equal residuals, SSE and s^2 . The covariance identity follows from $(\mathbf{Z}^\top \mathbf{Z})^{-1} = \mathbf{K}^{-1}(\mathbf{X}^\top \mathbf{X})^{-1}(\mathbf{K}^{-1})^\top$. ■

With $\mathbf{K} = \text{diag}(1, c_1, \dots, c_k)$, measuring x_j in units $1/c_j$ times as large divides its coefficient and its standard error by c_j , and leaves everything else alone. Centring, effect coding of a factor ([Exercise \(B1\)](#)) and many other changes of coding are also of the form $\mathbf{X}\mathbf{K}$. [Section 6.7](#) shows that what really stays fixed is the column space of $\mathbf{X}$.

The intercept and centring. The intercept β_0 is the mean response at the profile where every regressor is zero. That profile may be impossible (zero income) or far outside the data (an August sea temperature of 0°C in [Example \(Forecasting December sea temperature\)](#)). If the regressors are centred, the intercept becomes the mean response at the average profile $\bar{\mathbf{x}}$, and its estimate is $\bar{y}$ ([Proposition 5.4.3](#)). Centring does not change the slopes. It makes the intercept interpretable, and it makes $\hat{\beta}_0$ uncorrelated with the slopes ([Exercise \(B2\)](#)). With interactions or polynomials, centring also changes what the lower-order coefficients mean, as the next subsections show.

Standardized coefficients. Measuring every variable in units of its own sample standard

deviation gives the **standardized coefficients** $\hat{\beta}_j^* = \hat{\beta}_j s_{x_j}/s_y$: the difference in the response, in standard deviations, per standard deviation of x_j . They make coefficients unit-free and are sometimes used to rank the “importance” of regressors. That use is fragile. A standard deviation is a property of the sample, not of the relation: the same relation observed over a wider range of x_j gives a larger standardized coefficient.

5.7.3 Logarithms

When the response or a regressor is on a logarithmic scale, the coefficients describe proportional rather than absolute differences.

- **Log–log**, $E(\log Y) = \beta_0 + \beta_1 \log x + \dots$: a 1 percent difference in x goes with approximately a β_1 percent difference in Y (exactly, a factor 1.01^{β_1} in the geometric mean). Economists call β_1 an **elasticity**.
- **Log–level**, $E(\log Y) = \beta_0 + \beta_1 x + \dots$: a unit difference in x goes with a factor e^{β_1} in the geometric mean of Y , a difference of $100(e^{\beta_1} - 1)$ percent, approximately $100\beta_1$ percent when β_1 is small.
- **Level–log**, $E(Y) = \beta_0 + \beta_1 \log x + \dots$: a 1 percent difference in x goes with a difference of about $\beta_1/100$ in Y .

The phrase “geometric mean” matters. $E(\log Y)$ is not $\log E(Y)$, so a model for the mean of $\log Y$ is not a model for the mean of Y . Chapter 22 returns to back-transformation.

Example. (Reading the money demand coefficients)

In [Example \(Money demand in Denmark\)](#) the response is log real money and the first regressor is log real income, so $\hat{\beta}_1 = 1.2958$ is an income elasticity. Comparing quarters with the same interest rates, 1 percent more real income goes with about 1.3 percent more real money. The interest rates are measured as fractions, so the bond rate coefficient -2.6163 is per unit of the fraction, that is, per 100 percentage points, an absurdly large change. Rescaling the rate to percentage points ([Proposition 5.7.1](#) with $c = 100$) divides the coefficient by 100: a bond rate one percentage point higher, with income and the deposit rate the same, goes with money holdings lower by the factor $e^{-0.026163}$, a difference of -2.58 percent. The deposit rate has a positive coefficient, which is plausible if deposits are part of the money measured, since a higher deposit rate makes holding them more attractive. But the coefficient is smaller than its standard error.

The standardized coefficients of income, bond rate and deposit rate are 0.619, -0.529 and 0.057. The sample standard deviation of log income is 0.0728, about 7 percent, and that of the bond rate is 0.0308, about 3 percentage points; within those ranges of variation, income and the bond rate move money holdings by similar amounts.

```
import numpy as np
import statsmodels.api as sm
dk = sm.datasets.danish_data.load_pandas().data
y = dk["lrm"].to_numpy()
X = np.column_stack([np.ones(len(y)), dk["lry"], dk["ibo"], dk["ide"]])
XtX = X.T @ X
Xty = X.T @ y
beta_hat = np.linalg.solve(XtX, Xty)          # solve X^T X b = X^T y
```

```

X_pct = X.copy()
X_pct[:, 2:] *= 100                                # rates in percentage points
beta_pct = np.linalg.solve(X_pct.T @ X_pct, X_pct.T @ y)
print("rates as fractions:", np.round(beta_hat, 4))
print("rates in percent:  ", np.round(beta_pct, 5))
print("same fitted values:", np.allclose(X @ beta_hat, X_pct @ beta_pct))

sd = X[:, 1:].std(axis=0, ddof=1)
standardized = beta_hat[1:] * sd / y.std(ddof=1)
print("standardized slopes:", np.round(standardized, 3))

```

5.7.4 Terms that move together

If x enters through both x and x^2 , the mean response $\beta_0 + \beta_1 x + \beta_2 x^2$ changes at the rate

$$\frac{\partial E(Y | x)}{\partial x} = \beta_1 + 2\beta_2 x,$$

which depends on x . The coefficient β_1 alone is this rate at $x = 0$, and β_2 is half the rate at which the slope itself changes. Neither is “the effect of x ”. In [Example \(A curved trend and a seasonal cycle\)](#) the growth rate of carbon dioxide was 0.86 ppm per year in 1960 and 1.81 ppm per year in 2000. The coefficient $\hat{\beta}_1 = 1.3344$ is the growth rate in 1980, the year chosen as the origin of t_c , and would have been different had the origin been chosen differently.

Interactions behave the same way. In [Eq. 5.3.3](#) the slope in x_1 is $\beta_1 + \beta_3 x_2$. Reporting “the effect of x_1 ” requires choosing a value of x_2 , and β_1 is the choice $x_2 = 0$. Centring x_2 at a representative value before forming the product makes β_1 the slope at that value. The fitted surface does not change ([Exercise \(B2\)](#)).

5.7.5 Indicator variables

With reference coding, the coefficient of an indicator is the difference in mean response between its category and the reference category, among profiles that agree on the other regressors. Changing the reference changes every indicator coefficient but not the fit, since the reference coding for one reference category is a reparameterization of the coding for any other.

Example. (Self-rated health, with and without adjustment)

In [Example \(Self-rated health\)](#), people reporting poor health made 3.160 more physician visits a year, on average, than those reporting excellent health. Add two regressors: a chronic-disease score (the variable `disea`) and an indicator of a physical limitation. The coefficient of poor health falls to 1.554, and that of fair health from 1.058 to 0.202. The good-health coefficient changes sign, to -0.081 . The adjusted comparison is between people with the same disease score and limitation status. Much of the difference in visits between health categories goes with differences in measured disease, and the self-rating adds less once disease is taken into account. Neither set of numbers is “the” effect of self-rated health. They answer different questions.

The data record the same people in several years, so the cases are not independent and

the second-moment assumptions do not hold exactly; Chapter 33 treats such clustered data.

```
import numpy as np
import statsmodels.api as sm
rand = sm.datasets.randhie.load_pandas().data
y = rand["mdvis"].to_numpy(dtype=float)
D = rand[["hlthg", "hlthf", "hlthp"]].to_numpy(dtype=float) # good, fair, poor
X = np.column_stack([np.ones(len(y)), D]) # excellent =
    reference

Xa = np.column_stack([X, rand["disea"], rand["physlm"]])
beta_a = np.linalg.solve(Xa.T @ Xa, Xa.T @ y)
print("adjusted for chronic-disease score and physical limitation:")
print(np.round(beta_a, 4))
```

5.7.6 Omitted regressors

How does a coefficient change when other regressors are added or removed? Partition the model matrix as $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$, with $\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{\beta}}_1^\top, \hat{\boldsymbol{\beta}}_2^\top)^\top$ the estimate from the full (“long”) regression, and let $\tilde{\boldsymbol{\beta}}_1 = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{y}$ be the estimate from the “short” regression on $\mathbf{X}_1$ alone.

Proposition 5.7.2 — Omitted regressors

Let $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ have full rank, and let $\boldsymbol{\Pi} = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2$ be the coefficient matrix from regressing each column of $\mathbf{X}_2$ on $\mathbf{X}_1$. Then:

- (a) for every data vector $\mathbf{y}$, $\tilde{\boldsymbol{\beta}}_1 = \hat{\boldsymbol{\beta}}_1 + \boldsymbol{\Pi} \hat{\boldsymbol{\beta}}_2$;
- (b) if $E(\mathbf{Y}) = \mathbf{X}_1 \boldsymbol{\beta}_1 + \mathbf{X}_2 \boldsymbol{\beta}_2$, then $E(\tilde{\boldsymbol{\beta}}_1) = \boldsymbol{\beta}_1 + \boldsymbol{\Pi} \boldsymbol{\beta}_2$.

In particular the short and long coefficients agree, for all $\mathbf{y}$, iff $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$.

Proof. (a) Write $\mathbf{y} = \mathbf{X}_1 \hat{\boldsymbol{\beta}}_1 + \mathbf{X}_2 \hat{\boldsymbol{\beta}}_2 + \hat{\boldsymbol{\varepsilon}}$, where $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$ and so $\mathbf{X}_1^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$. Then

$$\begin{aligned} \tilde{\boldsymbol{\beta}}_1 &= (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{y} \\ &= \hat{\boldsymbol{\beta}}_1 + (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2 \hat{\boldsymbol{\beta}}_2 + (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \hat{\boldsymbol{\varepsilon}} = \hat{\boldsymbol{\beta}}_1 + \boldsymbol{\Pi} \hat{\boldsymbol{\beta}}_2. \end{aligned}$$

(b) Take expectations in (a), using $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ (Theorem 5.5.1), or apply Proposition 5.5.7(a) with $\boldsymbol{\delta} = \mathbf{X}_2 \boldsymbol{\beta}_2$. For the last claim, if $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$ then $\boldsymbol{\Pi} = \mathbf{0}$. Conversely, if $\boldsymbol{\Pi} \neq \mathbf{0}$, take $\mathbf{y} = \mathbf{X}_2 \mathbf{c}$ with $\boldsymbol{\Pi} \mathbf{c} \neq \mathbf{0}$: the long regression gives $\hat{\boldsymbol{\beta}}_2 = \mathbf{c}$, and the two coefficients differ by $\boldsymbol{\Pi} \mathbf{c}$.

■

Part (b) is the **omitted-variable bias** formula. The short regression does not estimate $\boldsymbol{\beta}_1$ but $\boldsymbol{\beta}_1 + \boldsymbol{\Pi} \boldsymbol{\beta}_2$. Each omitted regressor contributes its own coefficient times its association with the included ones. The bias is zero if the omitted regressors have no effect or are uncorrelated with the included ones, in the sense $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$; otherwise it is there, and it can have either sign. Part (a) says the same thing about the numbers actually computed from a data set, with no

model assumptions at all.

Example. (Income without interest rates)

Regressing log money on log income alone, with an intercept, gives an income elasticity of 1.819, against 1.2958 when both interest rates are included. [Proposition 5.7.2\(a\)](#) accounts for the whole difference. Regressing each rate on log income gives slopes -0.2215 (bond rate) and -0.0917 (deposit rate): in these data, quarters with higher income had lower interest rates, and the correlation of log income with the bond rate is -0.52 . The short-regression elasticity therefore absorbs

$$(-0.2215)(-2.6163) + (-0.0917)(0.6186) = 0.5229,$$

mostly the association of higher income with lower bond rates, which in turn go with more money. Which elasticity is the “right” one depends on the question. The long regression compares quarters with equal interest rates. The short one compares quarters as they came, with whatever rates accompanied a given income.

```
X1 = X[:, :2]                                # intercept and log income
X2 = X[:, 2:]                                # the two interest rates
b_short = np.linalg.solve(X1.T @ X1, X1.T @ y)
A = np.linalg.solve(X1.T @ X1, X1.T @ X2)    # regress each rate on (1, lry)
print("short regression slope on income:", round(b_short[1], 4))
print("long slope + A @ (rate coefs):  ", round(beta_hat[1] + A[1] @ beta_hat[2:],
4))
print("auxiliary slopes of the rates on income:", np.round(A[1], 4))
```

Warning (Adjustment is not a cure). [Proposition 5.7.2](#) suggests a tempting strategy: include every variable that might be relevant, so that nothing is omitted. Two problems remain. First, some variables should *not* be held fixed. If a treatment works partly by changing an intermediate variable, adjusting for that variable removes part of the effect one wants to measure. Adjusting for a common consequence of the regressor and the response can create an association where there is none. Second, adding regressors that are nearly collinear with x_j inflates the variance of $\hat{\beta}_j$ (Chapter 26). Which variables to adjust for is a question about how the data were generated, not about the fit. Chapter 25 develops the tools for answering it.

5.7.7 Exercises

5.7.7.1 A. Check your understanding

Exercise A1.

In [Example \(Forecasting December sea temperature\)](#), suppose both temperatures are converted to degrees Fahrenheit, $F = 32 + 1.8C$. What happens to the slope, the intercept, R^2 and s ? Answer using [Proposition 5.7.1](#), noting that the response is also transformed.

5.7.7.2 B. Practice

Exercise B1.

Show that the standardized coefficients are the slopes obtained by regressing the standardized response $(y_i - \bar{y})/s_y$ on the standardized regressors $(x_{ij} - \bar{x}_j)/s_{x_j}$, without an intercept. For a single regressor, show that the standardized slope is the correlation r .

Solution. Let $\mathbf{X}_c$ be the centred regressors and $\mathbf{D} = \text{diag}(s_{x_1}, \dots, s_{x_k})$. The standardized regressors are $\mathbf{X}_c \mathbf{D}^{-1}$ and the standardized response is $(\mathbf{y} - \bar{y}\mathbf{1})/s_y$. By [Proposition 5.4.3](#) the slopes of $\mathbf{y}$ on $\mathbf{X}_c$ are $\hat{\beta}_1$. Dividing the response by s_y divides them by s_y , since least squares is linear in the response; replacing $\mathbf{X}_c$ by $\mathbf{X}_c \mathbf{D}^{-1}$ multiplies them by $\mathbf{D}$ ([Proposition 5.7.1](#) with $\mathbf{K} = \mathbf{D}^{-1}$). The result is $\mathbf{D}\hat{\beta}_1/s_y$, whose entries are $\hat{\beta}_j s_{x_j}/s_y$. For one regressor this is $(S_{xy}/S_{xx})\sqrt{S_{xx}/S_{yy}} = r$, as in [Eq. 5.2.6](#).

Exercise B2.

In the centred parameterization $E(\mathbf{Y}) = \alpha\mathbf{1} + \mathbf{X}_c\beta_1$ of [Proposition 5.4.3](#), show that $\text{Cov}(\hat{\alpha}, \hat{\beta}_1) = \mathbf{0}$, $\text{Var}(\hat{\alpha}) = \sigma^2/n$ and $\text{Cov}(\hat{\beta}_1) = \sigma^2(\mathbf{X}_c^\top \mathbf{X}_c)^{-1}$. Deduce that $\text{Var}(\hat{\beta}_0) = \sigma^2(1/n + \bar{\mathbf{x}}^\top (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \bar{\mathbf{x}})$, which generalizes [Corollary 5.5.4](#).

Solution. The matrix $[\mathbf{1}, \mathbf{X}_c]$ has full rank and $[\mathbf{1}, \mathbf{X}_c]^\top [\mathbf{1}, \mathbf{X}_c] = \text{diag}(n, \mathbf{X}_c^\top \mathbf{X}_c)$, because $\mathbf{1}^\top \mathbf{X}_c = \mathbf{0}^\top$. By [Theorem 5.5.1](#) the covariance of $(\hat{\alpha}, \hat{\beta}_1^\top)^\top$ is $\sigma^2 \text{diag}(1/n, (\mathbf{X}_c^\top \mathbf{X}_c)^{-1})$, which gives all three claims. Since $\hat{\beta}_0 = \hat{\alpha} - \bar{\mathbf{x}}^\top \hat{\beta}_1$, [Corollary 5.5.2](#) gives $\text{Var}(\hat{\beta}_0) = \sigma^2/n + \bar{\mathbf{x}}^\top \sigma^2 (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \bar{\mathbf{x}}$, the cross term being zero. With one regressor, $\mathbf{X}_c^\top \mathbf{X}_c = S_{xx}$.

Exercise B3.

In a model with an intercept and two regressors, let the short regression omit x_2 . Show that the bias of the slope on x_1 is $\beta_2 S_{12}/S_{11}$, where $S_{12} = \sum_i (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2)$ and $S_{11} = \sum_i (x_{i1} - \bar{x}_1)^2$. Tabulate the sign of the bias according to the signs of β_2 and the correlation of the two regressors.

5.7.7.3 C. Going deeper**Exercise C1.**

Construct a data set with $n = 8$ cases, two regressors and an intercept, in which the slope of y on x_1 is positive in the simple regression and negative in the regression on both x_1 and x_2 . Explain the reversal with [Proposition 5.7.2](#), and draw the scatterplot of y against x_1 with the cases labelled by x_2 .

5.8 A panorama of regression models to come

The linear model of this chapter makes three kinds of claim: about the *mean* of the response (a linear combination of known columns), about its *variability* (a common variance, no correlation) and, in the normal version, about its *distribution*. Every model in the rest of the book relaxes

one or more of these claims, or keeps them and asks harder questions about them. This section surveys the route.

5.8.1 Three components of a regression model

It helps to see any regression model as three separate choices.

1. **A distribution for the response given the regressors:** normal, binomial, Poisson, gamma, or something less specified, such as “any distribution with this mean and variance”.
2. **A predictor** that combines the regressors: the linear predictor $\eta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, or a sum of smooth functions, or a sum that includes random terms.
3. **A link** between the predictor and the mean, $g(\mathbb{E}(Y_i)) = \eta_i$, where g is a known monotone function.

The normal linear model chooses the normal distribution with constant variance, the linear predictor, and the identity link $g(\mu) = \mu$. Changing one choice at a time generates the models below.

5.8.2 The ladder

Model	What changes	Where
Linear model, second moments	the starting point: geometry, optimality, estimability, computation	Chapters 5–10
Normal linear model	exact tests, intervals, multiple comparisons, designed experiments	Chapters 11–18
Departures from the assumptions	diagnostics, transformations, resampling, measurement error, causation	Chapters 19–25
Many regressors	collinearity, shrinkage, $p > n$, selection, boosting	Chapters 26–30
General covariance and random effects	$\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$; $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{b} + \boldsymbol{\varepsilon}$	Chapters 31–33
Generalized linear models	non-normal distribution and a link	Chapters 34–41
Nonparametric and additive models	smooth functions in the predictor	Chapters 42–44
Distributional regression	regressors act on more than the mean	Chapter 45

The normal linear model and its designs. With normal errors, the least squares estimator has an exact normal distribution and the residual sum of squares an exact chi-squared distribution ([Theorem 7.5.1](#)). This gives t and F tests and exact confidence regions (Chapters 11 and 12), and procedures for many comparisons at once (Chapter 13). Analysis of variance and designed experiments (Chapters 15–18) are linear models whose columns are indicator variables, often not of full rank; their theory rests on [Chapter 6](#) and [Chapter 8](#).

When the assumptions fail, or there are many regressors. Part V asks what happens when the mean is misspecified, the variance is not constant, the errors are correlated or not normal, the regressors are measured with error, or the question is causal; some remedies, such as robust standard errors based on [Eq. 5.5.3](#) (Chapter 21), stay inside the linear model. Part VI treats large p and nearly dependent columns, where ridge regression, principal components and the lasso trade a little bias for less variance. Regression trees, random forests and neural networks also estimate m , judged by prediction error rather than by the meaning of coefficients; the book does not develop them, but Chapters 29 and 30 compare predictors and treat boosting.

General covariance and mixed models. The general Gauss–Markov model replaces $\sigma^2 \mathbf{I}$ by $\sigma^2 \mathbf{V}$ (Chapter 31), and generalized least squares is least squares in a different inner product ([Section 6.9](#)). Linear mixed models write the response as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{b} + \boldsymbol{\varepsilon},$$

where $\mathbf{b}$ is a vector of random effects shared by related cases, such as repeated measurements on one person or pupils in one school (Chapters 32–33). The RAND data of [Example \(Self-rated health\)](#), with the same people observed in several years, are of this kind.

Generalized linear models. For a binary response the mean is a probability, and a linear predictor can leave $[0, 1]$. For a count the mean must be positive and the variance grows with the mean. Generalized linear models keep the linear predictor, choose a distribution from an exponential family, and connect the two by a link: the logit $\log\{\mu/(1 - \mu)\}$ for binary responses (Chapter 35), the logarithm for counts (Chapter 37). Coefficients act additively on the scale of the link, so in a Poisson model with log link e^{β_j} is the factor by which the mean changes per unit of x_j . Chapters 34–41 develop the theory, overdispersion, correlated responses and missing data.

Nonparametric and additive models. Polynomials, as in [Example \(A curved trend and a seasonal cycle\)](#), force a global shape on a curve. Splines and smoothers let the data choose the shape locally (Chapters 42–43), and additive models replace each term $\beta_j x_j$ by an unknown smooth function $f_j(x_j)$ (Chapter 44). Much of this is still penalized least squares on a model matrix of basis functions.

Beyond the mean. Quantile regression models a chosen quantile of Y , such as the median or the tenth percentile, as a linear function of the regressors, and so describes how the whole conditional distribution shifts. Distributional regression lets the regressors act on the location, the scale and the shape of the response distribution through separate predictors, turning process (b) of [Figure 5.1.1](#) into a model rather than a violation (Chapter 45).

The Bayesian thread. A prior distribution on $(\boldsymbol{\beta}, \sigma^2)$ turns the likelihood into a posterior. For the normal linear model with a conjugate prior the posterior is available in closed form ([Theorem 7.6.2](#)), and it reappears for intervals (Chapter 12) and for generalized linear models (Chapter 39).

5.8.3 A first look beyond the linear model

Example. (Physician visits are counts)

The number of physician visits in the RAND data is a count. It is zero for 31.2 percent of person-years, and its variance is 7.09 times its mean. Group the person-years into ten groups by the chronic-disease score, and compute the mean and variance of visits in each ([Figure 5.8.1](#)). The variance rises with the mean, and in every group it is between

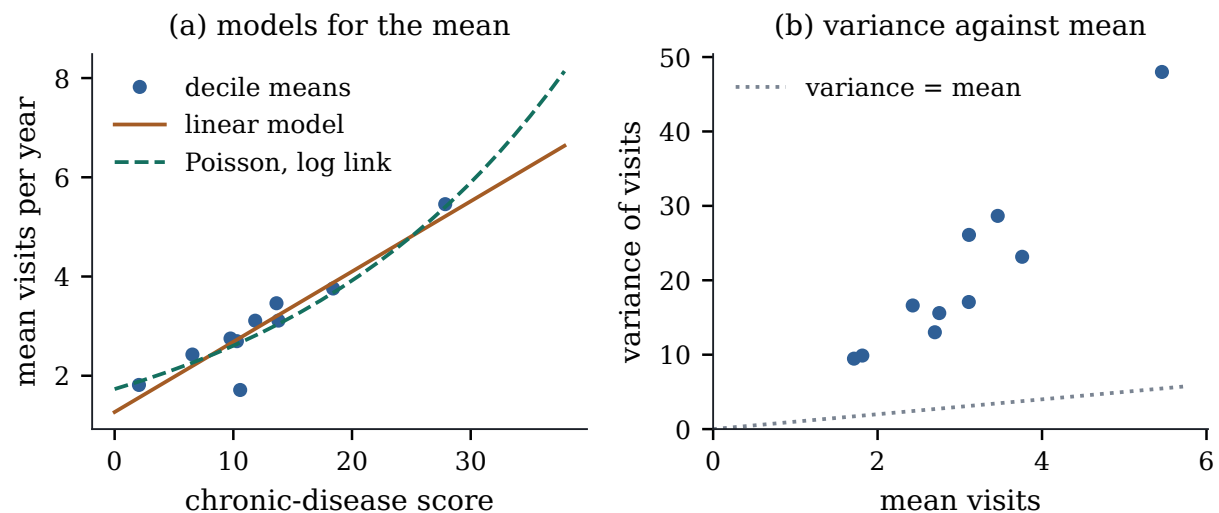


Figure 5.8.1: Annual physician visits in the RAND Health Insurance Experiment, grouped into ten groups by a chronic-disease score. (a) Group means with a linear model and a Poisson regression with log link. (b) Group variances against group means; the dotted line is the Poisson relation variance = mean.

4.8 and 8.8 times the mean. (L2) fails badly.

Two models for the mean are fitted to the disease score x . The linear model gives $1.268 + 0.1416x$ visits. The Poisson regression with log link (Chapter 37) gives $\exp(0.549 + 0.0408x)$: each unit of the index multiplies the mean by $e^{0.0408}$, 4.17 percent more visits. Over the bulk of the data the two curves are close, and both follow the group means. They differ in shape at high values of the index, where the log-linear mean grows ever faster and the linear one at a constant rate, and where the data are too sparse to decide between them. They differ more in what they assume about variability. The linear model assumes a constant variance; the Poisson model assumes variance equal to the mean. The data contradict both. Chapter 38 shows how to keep the log-linear mean and let the variance be a multiple of it.

The Poisson fit is not computed by ordinary least squares, but its estimating equations have the same form as the normal equations: $\mathbf{X}^\top(\mathbf{y} - \hat{\boldsymbol{\mu}}) = \mathbf{0}$, where $\hat{\boldsymbol{\mu}}$ is the vector of fitted means. The residuals are again orthogonal to every column of $\mathbf{X}$. The listing checks this.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
rand = sm.datasets.randhie.load_pandas().data
y = rand["mdvis"].to_numpy(dtype=float)
x = rand["disea"].to_numpy()
X = sm.add_constant(x)

bins = pd.qcut(x, 10, duplicates="drop") # ten groups by disease score
groups = pd.DataFrame({"y": y, "x": x}).groupby(bins, observed=True)
table = groups.agg(x=("x", "mean"), mean=("y", "mean"), var=("y", "var"))
print(table.round(2).to_string(index=False))
print(f"share of zeros {np.mean(y == 0):.3f}; variance/mean overall {y.var() / y.mean():.2f}")

linear = sm.OLS(y, X).fit()
```

```
poisson = sm.GLM(y, X, family=sm.families.Poisson()).fit()
print("linear model:      ", np.round(linear.params, 4))
print("Poisson, log link:", np.round(poisson.params, 4))
mu = poisson.fittedvalues
print("X^T (y - mu_hat) =", X.T @ (y - mu))
```

5.8.4 Exercises

5.8.4.1 A. Check your understanding

Exercise A1.

For each response, name the rung of the ladder you would start from, and say which of the three components (distribution, predictor, link) departs from the normal linear model: (a) whether a loan is repaid; (b) blood pressure measured monthly on the same patients; (c) the number of insurance claims per policy; (d) the 90th percentile of birth weight as a function of the mother's age; (e) crop yield in a randomized block experiment.

5.8.4.2 B. Practice

Exercise B1.

Let Y_i be binary with $\Pr(Y_i = 1) = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ (the *linear probability model*). Show that (L1) holds but (L2) fails, with $\text{Var}(Y_i) = \mu_i(1 - \mu_i)$. Show that least squares is still unbiased, find its covariance matrix from [Proposition 5.5.7\(b\)](#), and explain why fitted values outside $[0, 1]$ are possible.

Solution. $E(Y_i) = \Pr(Y_i = 1) = \mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, which is (L1). $\text{Var}(Y_i) = \mu_i - \mu_i^2$, which varies with i unless all the μ_i are equal or symmetric about $1/2$, so (L2) fails. Unbiasedness needs only (L1) ([Theorem 5.5.1\(a\)](#)). With independent cases, $\boldsymbol{\Sigma} = \text{diag}(\mu_i(1 - \mu_i))$ and [Eq. 5.5.3](#) gives $\text{Cov}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Sigma} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}$. Nothing in least squares constrains $\mathbf{x}^\top \hat{\boldsymbol{\beta}}$ to $[0, 1]$: the estimate is a linear function of $\mathbf{y}$, and at design points far from the centre of the data the fitted line can cross 0 or 1 even when every true probability is inside. The logit link of Chapter 35 removes the problem.

Exercise B2.

For independent $Y_i \sim \text{Poisson}(\mu_i)$ with $\log \mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, write the log-likelihood and show that its gradient is $\mathbf{X}^\top (\mathbf{y} - \boldsymbol{\mu})$. Deduce that when $\mathbf{X}$ contains an intercept the fitted means sum to $\sum_i y_i$, and that for the intercept-only model the fitted mean is $\bar{y}$.

5.8.4.3 C. Going deeper

Exercise C1.

In process (b) of [Example \(Same regression function, different models\)](#) the error standard deviation is $0.15 + 0.25x$. Suppose instead $\log \text{Var}(Y_i) = \gamma_0 + \gamma_1 x_i$ with unknown γ .

Propose a two-step procedure: fit the mean by least squares, then regress $\log \hat{\varepsilon}_i^2$ on x_i . Show that, when the errors are normal and $n \rightarrow \infty$ with p fixed, the slope of the second regression estimates γ_1 consistently, while its intercept estimates $\gamma_0 + E(\log \chi_1^2)$. (Argue first with the errors ε_i in place of the residuals, then explain why the difference does not matter in the limit.) Compute this bias, $-\gamma_E - \log 2 \approx -1.27$, where γ_E is Euler's constant. (Chapter 45 treats models of this kind by likelihood.)

5.9 Summary and Notes

5.9.1 Summary

Idea.

1. The regression function $m(\mathbf{x}) = E(Y \mid \mathbf{X} = \mathbf{x})$ is the best predictor in mean squared error, and its error is automatically uncorrelated with every function of the regressors (Proposition 5.1.1). A regression model is a claim about m and about the distribution around it, made in layers: linearity, constant variance, uncorrelated errors, normality (Section 5.1).
2. The least squares line has slope S_{xy}/S_{xx} and passes through the point of means (Theorem 5.2.1, Proposition 5.2.2). In standard units its slope is the correlation, which is regression toward the mean (Eq. 5.2.6).
3. The linear model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $E(\boldsymbol{\varepsilon}) = \mathbf{0}$, $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, is linear in $\boldsymbol{\beta}$, not in the regressors. Polynomials, harmonics, transformations, indicator variables and interactions are all linear models (Definition 5.3.1).
4. For a full-rank model matrix, least squares has the unique solution $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ of the normal equations $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$, and the criterion exceeds its minimum by $(\mathbf{b} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\mathbf{b} - \hat{\boldsymbol{\beta}})$ (Theorem 5.4.1). The residuals are orthogonal to every column (Proposition 5.4.2). Slopes can be computed from centred data (Proposition 5.4.3), and indicator coefficients are differences of group means (Proposition 5.4.4).
5. Under the second-moment assumptions, $\hat{\boldsymbol{\beta}}$ is unbiased with covariance $\sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$, and so is every linear function $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$ (Theorem 5.5.1, Corollary 5.5.2). Unbiasedness needs only the mean assumption; a wrong mean produces bias and a wrong covariance produces the sandwich formula (Proposition 5.5.7).
6. $s^2 = \text{SSE}/(n - p)$ is unbiased for σ^2 , because the residuals are the errors projected into a space with $n - p$ degrees of freedom (Theorem 5.6.1). Its variance depends on the kurtosis of the errors (Proposition 5.6.3).
7. A coefficient is the difference in mean response per unit of its regressor between profiles that agree on the *other regressors in the model* (Definition 5.7.1). It changes with the units (Proposition 5.7.1) and with the set of other regressors, by exactly the amount given by the omitted-variable formula (Proposition 5.7.2). It is not by itself a causal effect.
8. Everything later in the book changes one of three components of this model: the distribution of the response, the form of the predictor, or the link between predictor and mean (Section 5.8).

5.9.2 Notes and sources

History. Least squares was published by Legendre (1805) as a method for fitting orbits, and claimed by Gauss (1809), who gave it a probabilistic justification through the normal distribution. Half a century earlier Boscovich had fitted a line to measurements of meridian arcs by making the absolute deviations as small as possible, an idea Laplace took up and that returns in this book as quantile regression. The priority dispute and the early history are told by Stigler (1981; 1986). The word *regression* comes from Galton (1886), whose study of the heights of parents and children found the phenomenon now called regression toward the mean. For Galton the slope of a fitted line measured an inherited tendency. Only later did the least squares machinery of the astronomers and the regression of the biometricians become one subject.

The source books. The chapter covers the material of Rencher and Schaalje (2008, chapter 1, chapter 6 and sections 7.1–7.7) and of Sen and Srivastava (1990, chapters 1 and 2), whose presentations of simple and multiple regression by calculus and matrix algebra are the closest to the one here. Sen and Srivastava’s emphasis on the Gauss–Markov conditions as a benchmark, and their discussion of centring and scaling, are reflected in [Section 5.1](#) and [Section 5.7](#). Rencher and Schaalje also treat the geometry, the normal model and R^2 in their chapter 7; in this book those topics are in [Chapter 6](#), [Chapter 7](#) and [Chapter 9](#). The view of a model as a distribution, a linear predictor and a link, and the discussion of indicator variables and model matrices, follow Agresti (2015, chapter 1). The panorama of [Section 5.8](#) is organized like the survey in Fahrmeir, Kneib, Lang and Marx (2021, chapter 2), whose chapter 3 also discusses what the standard assumptions mean in applications and how covariates are coded. Seber and Lee (2003, chapter 3) and Christensen (2020, chapters 1–2) give terser, more general treatments that start from projections.

Fixed and random regressors. Conditioning on the regressors is standard in linear-model theory. Freedman (2009) is a careful discussion of what regression models assume and of the gap between association and causation that [Section 5.1](#) and [Section 5.7](#) point to.

Standard errors under departures. The sandwich formula [Eq. 5.5.3](#), with the unknown covariance matrix estimated from squared residuals, is due to White (1980) in econometrics; it is developed in [Chapter 21](#).

Beyond the linear model. Generalized linear models were introduced by Nelder and Wedderburn (1972) and quantile regression by Koenker and Bassett (1978). Additive models were developed and popularized by Hastie and Tibshirani (1986; 1990), and generalized additive models for location, scale and shape were proposed by Rigby and Stasinopoulos (2005). Each is the subject of a later part of the book.

Data. The Niño 1+2 sea surface temperatures are from the US National Oceanic and Atmospheric Administration. The carbon dioxide record is that of Keeling and Whorf (2004). The Danish money demand data are those analysed by Johansen and Juselius (1990). The RAND Health Insurance Experiment is described by Manning et al. (1987); the extract used here is the one analysed by Cameron and Trivedi (2005). All four data sets are distributed with the Python package statsmodels.

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Chapter 6

Projections: The Geometry of Least Squares

Least squares chooses the point of a subspace that is closest to the data. Everything else in this chapter follows from that one statement: the normal equations, why fitted values are unique when coefficients are not, the orthogonal decompositions behind every sum of squares, the Frisch–Waugh–Lovell theorem, R^2 as a squared cosine, leverage, generalized least squares, and how the projection should actually be computed.

What you need. [Chapter 1](#) (rank, inverses, eigenvalues), [Chapter 5](#) (the linear model and least squares). Random vectors ([Chapter 2](#)) appear only in a few remarks.

6.0.1 Roadmap

- [Least squares as a nearest-point problem](#)
- [The subspaces we need](#)
- [Orthogonal projection](#)
- [Least squares via projection](#)
- [Nested models and sequential projection](#)
- [The Frisch–Waugh–Lovell theorem](#)
- [Reparameterization, centring and \$R^2\$](#)
- [Leverage as geometry](#)
- [Other inner products](#)
- [Computation](#)
- [Projection in the population: best linear prediction](#)
- [Summary and notes](#)

6.1 Least squares as a nearest-point problem

[Chapter 5](#) introduced least squares as a calculus problem: write down the sum of squared errors as a function of the coefficients, differentiate, set the derivative to zero. That route gets the right answer, but it hides the structure of the answer. It treats the coefficient vector β as the basic unknown. It needs $\mathbf{X}^\top \mathbf{X}$ to be invertible before it can say anything. And it gives no hint

of why some quantities, such as fitted values, residuals and sums of squares, behave well in every model, while others, such as individual coefficients, can be fragile or not defined at all.

This chapter starts from a different place. Instead of the p -dimensional space where β lives, it works in the n -dimensional space where the data live.

6.1.1 Two pictures of the same data

A regression data set with n observations and a single regressor is usually drawn as a scatterplot: n points in the plane, one for each observation, with the regressor on one axis and the response on the other. Call this the **variable space** picture. It is the right picture for seeing the shape of a relationship, but it scales badly. With k regressors it needs $k + 1$ dimensions, and it has no natural place for the model's coefficients.

The other picture turns this around. Stack the n responses into a single vector $\mathbf{y} = (y_1, \dots, y_n)^\top$, and regard it as *one* point in $\mathbb{R}^n$. Each column of the model matrix $\mathbf{X}$ is also a vector in $\mathbb{R}^n$: the intercept column is $\mathbf{1}$, and the column for a regressor holds its n observed values. This is the **observation space** picture. Its dimension is the sample size, which is usually far more than three. But it contains only $p + 1$ vectors of interest, and those always span a subspace of dimension at most $p + 1$. Most of the geometry happens in a subspace small enough to draw.

In observation space, the linear model

$$\mathbf{E}(\mathbf{Y}) = \mathbf{X}\beta, \quad \beta \in \mathbb{R}^p,$$

says one thing: the mean vector $\mathbf{E}(\mathbf{Y})$ lies in the set

$$\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\mathbf{b} : \mathbf{b} \in \mathbb{R}^p\},$$

the **column space** of $\mathbf{X}$. That set is a subspace of $\mathbb{R}^n$. The coefficients β are the coordinates of $\mathbf{E}(\mathbf{Y})$ with respect to the columns of $\mathbf{X}$, but the claim itself is only about *where* $\mathbf{E}(\mathbf{Y})$ is. It does not depend on which spanning vectors are used to describe the subspace.

6.1.2 The least squares principle, restated

The observed $\mathbf{y}$ is almost never in $\mathcal{C}(\mathbf{X})$, because noise moves it off the subspace. A natural estimate of $\mathbf{E}(\mathbf{Y})$ is the point of $\mathcal{C}(\mathbf{X})$ that is closest to $\mathbf{y}$. With Euclidean distance, this is exactly least squares:

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \sum_{i=1}^n (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2,$$

where $\mathbf{x}_{(i)}^\top$ is the i th row of $\mathbf{X}$. Minimizing over $\mathbf{b}$ is the same as searching $\mathcal{C}(\mathbf{X})$ for the point nearest to $\mathbf{y}$.

Definition 6.1.1 — Least squares

A vector $\hat{\beta} \in \mathbb{R}^p$ is a **least squares estimate** of β if

$$\|\mathbf{y} - \mathbf{X}\hat{\beta}\| \leq \|\mathbf{y} - \mathbf{X}\mathbf{b}\| \quad \text{for every } \mathbf{b} \in \mathbb{R}^p.$$

The vector $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ is the **fitted value** vector, and $\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \hat{\mathbf{Y}}$ is the **residual** vector.

The definition does not assume that the minimizer is unique, and deliberately so. In [Section 6.4](#) we will see that the *nearest point* $\hat{\mathbf{Y}}$ is always unique. The coefficient vector $\hat{\boldsymbol{\beta}}$ that produces it is unique only when the columns of $\mathbf{X}$ are linearly independent.

6.1.3 A picture with three observations

The smallest case that still shows all the geometry has $n = 3$ and $p = 2$. Then $\mathcal{C}(\mathbf{X})$ is a plane through the origin in $\mathbb{R}^3$, and $\mathbf{y}$ is a point somewhere off the plane.

Example. (Three observations)

Take responses $\mathbf{y} = (1, 4, 2)^\top$ at regressor values $\mathbf{x} = (0, 1, 3)^\top$, with an intercept, so that $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$. The least squares fit (computed by the listing below) is

$$\hat{\mathbf{Y}} = (2.143, 2.286, 2.571)^\top, \quad \hat{\boldsymbol{\varepsilon}} = (-1.143, 1.714, -0.571)^\top,$$

with intercept 2.143 and slope 0.143. Two facts stand out. First, the residual is orthogonal to both columns of $\mathbf{X}$: the entries of $\hat{\boldsymbol{\varepsilon}}$ sum to zero, and $\mathbf{x}^\top \hat{\boldsymbol{\varepsilon}} = 0$. Second, the squared lengths add:

$$\|\mathbf{y}\|^2 = 21.000 = 16.429 + 4.571 = \|\hat{\mathbf{Y}}\|^2 + \|\hat{\boldsymbol{\varepsilon}}\|^2.$$

[Figure 6.1.1](#) shows the configuration. The fitted vector is the foot of the perpendicular dropped from $\mathbf{y}$ to the plane.

```
import numpy as np

y = np.array([1.0, 4.0, 2.0])
X = np.column_stack([np.ones(3), [0.0, 1.0, 3.0]]) # intercept and one regressor

beta_hat, *_ = np.linalg.lstsq(X, y, rcond=None)
y_hat = X @ beta_hat # the point of C(X) nearest to y
e_hat = y - y_hat # the residual vector

print("fitted ", y_hat)
print("residual", e_hat)
print("X^T e ", X.T @ e_hat) # zero: the residual is orthogonal to C(X)
```

The two facts in [Example \(Three observations\)](#) are not coincidences of this data set. They are the whole theory in miniature. The residual is orthogonal to the subspace *because* $\hat{\mathbf{Y}}$ is the nearest point, and the squared lengths add *because* the two pieces are orthogonal. The rest of the chapter makes these two statements precise and follows their consequences.

Idea. Fitted values are determined by the *subspace* $\mathcal{C}(\mathbf{X})$. Coefficients are determined by the *basis* used to describe it. Anything that depends only on $\mathcal{C}(\mathbf{X})$ is unchanged by reparameterization, by rescaling or recoding regressors, and by rank deficiency. This covers fitted values, residuals, sums of squares, R^2 and leverages.

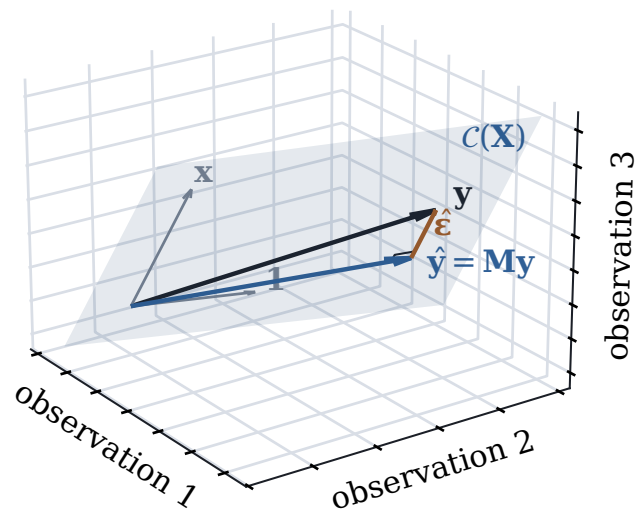


Figure 6.1.1: Least squares with $n = 3$ observations. The model's column space $\mathcal{C}(\mathbf{X})$ is the plane spanned by $\mathbf{1}$ and $\mathbf{x}$. The fitted vector $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$ is the point of the plane nearest to $\mathbf{y}$, and the residual $\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \hat{\mathbf{Y}}$ is perpendicular to the plane.

6.1.4 Plan of the chapter

[Section 6.2](#) collects the facts about subspaces that the geometry needs. [Section 6.3](#) proves that nearest points exist and are unique, and identifies the matrices that compute them: the symmetric idempotent matrices, including $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top$ for every generalized inverse. [Section 6.4](#) applies this to least squares. [Section 6.5](#) studies nested subspaces, which is where sums of squares come from. [Section 6.6](#) proves the Frisch–Waugh–Lovell theorem, which says what a single coefficient in a multiple regression measures. [Section 6.7](#) covers reparameterization and R^2 , and [Section 6.8](#) covers leverage. [Section 6.9](#) replaces the Euclidean inner product with a general one, the step that leads to generalized least squares. [Section 6.10](#) shows how the projection should be computed in floating point.

6.2 The subspaces we need

This section gathers the vector-space facts used in the rest of the chapter. Readers who have worked through [Chapter 1](#) can skim the definitions. The results on orthogonal complements, especially [Lemma 6.2.1](#) and [Theorem 6.2.3](#), are used constantly afterwards and are proved in full.

6.2.1 Subspaces, spans and dimension

A nonempty set $\mathcal{S} \subseteq \mathbb{R}^n$ is a **subspace** if it is closed under addition and scalar multiplication: $\mathbf{u}, \mathbf{v} \in \mathcal{S}$ and $a, b \in \mathbb{R}$ imply $a\mathbf{u} + b\mathbf{v} \in \mathcal{S}$. Every subspace contains $\mathbf{0}$. The **span** of vectors

$\mathbf{v}_1, \dots, \mathbf{v}_k$ is the set of all their linear combinations. It is the smallest subspace containing them. A set of vectors is *linearly independent* if the only linear combination equal to $\mathbf{0}$ is the one with all coefficients zero. A **basis** for $\mathcal{S}$ is a linearly independent set that spans $\mathcal{S}$. All bases of $\mathcal{S}$ have the same number of elements, the **dimension** $\dim \mathcal{S}$.

For an $n \times p$ matrix $\mathbf{A}$ there are two subspaces we use constantly:

$$\begin{aligned}\mathcal{C}(\mathbf{A}) &= \{\mathbf{A}\mathbf{b} : \mathbf{b} \in \mathbb{R}^p\} \subseteq \mathbb{R}^n && \text{(column space),} \\ \mathcal{N}(\mathbf{A}) &= \{\mathbf{b} \in \mathbb{R}^p : \mathbf{A}\mathbf{b} = \mathbf{0}\} \subseteq \mathbb{R}^p && \text{(null space).}\end{aligned}$$

The **rank** of $\mathbf{A}$ is $\text{rank}(\mathbf{A}) = \dim \mathcal{C}(\mathbf{A})$. Two standard facts from [Chapter 1](#): row rank equals column rank, so $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^\top)$; and the rank-nullity theorem

$$\text{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A}) = p. \quad (6.2.1)$$

In regression, $\mathcal{C}(\mathbf{X})$ is the **model space**. $\mathcal{N}(\mathbf{X})$ is the set of directions in which β can move without changing $\mathbf{X}\beta$. It is $\{\mathbf{0}\}$ exactly when $\mathbf{X}$ has full column rank.

Example. (A rank-deficient layout)

Suppose n observations fall into g groups and $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_g]$, where $\mathbf{z}_k$ is the indicator of group k . Since $\mathbf{z}_1 + \dots + \mathbf{z}_g = \mathbf{1}$, the vector $(1, -1, \dots, -1)^\top$ lies in $\mathcal{N}(\mathbf{X})$. The g indicators are linearly independent (their supports are disjoint), so $\text{rank}(\mathbf{X}) = g$ while $p = g + 1$. By [Eq. 6.2.1](#), $\mathcal{N}(\mathbf{X})$ is exactly the line spanned by $(1, -1, \dots, -1)^\top$. The model space $\mathcal{C}(\mathbf{X}) = \mathcal{C}([\mathbf{z}_1, \dots, \mathbf{z}_g])$ is the set of vectors that are constant within groups.

6.2.2 Inner products, orthogonality and complements

Unless stated otherwise $\mathbb{R}^n$ carries the Euclidean inner product $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v}$ and norm $\|\mathbf{u}\| = (\mathbf{u}^\top \mathbf{u})^{1/2}$. Vectors are **orthogonal**, written $\mathbf{u} \perp \mathbf{v}$, if $\mathbf{u}^\top \mathbf{v} = 0$. The Cauchy-Schwarz inequality $|\mathbf{u}^\top \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$, with equality iff one vector is a multiple of the other, lets us define the angle $\theta \in [0, \pi]$ between nonzero vectors by

$$\cos \theta = \frac{\mathbf{u}^\top \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}. \quad (6.2.2)$$

The angle comes back in [Section 6.7](#), where R^2 turns out to be a squared cosine. The Pythagorean identity

$$\mathbf{u} \perp \mathbf{v} \implies \|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \quad (6.2.3)$$

follows by expanding $(\mathbf{u} + \mathbf{v})^\top (\mathbf{u} + \mathbf{v})$. The cross terms are the ones that vanish.

Definition 6.2.1 — Orthogonal complement

The **orthogonal complement** of a subspace $\mathcal{S} \subseteq \mathbb{R}^n$ is

$$\mathcal{S}^\perp = \{\mathbf{v} \in \mathbb{R}^n : \mathbf{v}^\top \mathbf{u} = 0 \text{ for all } \mathbf{u} \in \mathcal{S}\}.$$

It is immediate that $\mathcal{S}^\perp$ is a subspace and that $\mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$: a vector orthogonal to itself has zero length. To test whether $\mathbf{v} \in \mathcal{S}^\perp$ it is enough to check orthogonality to a spanning set of $\mathcal{S}$. In particular, $\mathbf{v} \in \mathcal{C}(\mathbf{A})^\perp$ iff $\mathbf{A}^\top \mathbf{v} = \mathbf{0}$.

Lemma 6.2.1

For any $n \times p$ matrix $\mathbf{A}$,

$$\mathcal{N}(\mathbf{A}^\top) = \mathcal{C}(\mathbf{A})^\perp \quad \text{and} \quad \mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)^\perp.$$

Proof. $\mathbf{v} \in \mathcal{N}(\mathbf{A}^\top)$ means $\mathbf{A}^\top \mathbf{v} = \mathbf{0}$, that is, $\mathbf{v}$ is orthogonal to every column of $\mathbf{A}$, which is the same as $\mathbf{v} \perp \mathcal{C}(\mathbf{A})$. The second identity is the first applied to $\mathbf{A}^\top$. ■

6.2.3 Orthonormal bases and Gram–Schmidt

Vectors $\mathbf{u}_1, \dots, \mathbf{u}_k$ are **orthonormal** if $\mathbf{u}_i^\top \mathbf{u}_j$ is 1 for $i = j$ and 0 otherwise. Orthonormal vectors are linearly independent: taking the inner product of $\sum_j a_j \mathbf{u}_j = \mathbf{0}$ with $\mathbf{u}_i$ gives $a_i = 0$. The coordinates of a vector $\mathbf{v} \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$ are simply the inner products $a_i = \mathbf{u}_i^\top \mathbf{v}$, with no linear system to solve. This is why orthonormal bases are so convenient.

Proposition 6.2.2 — Gram–Schmidt

Let $\mathbf{v}_1, \dots, \mathbf{v}_k$ be linearly independent. Define recursively

$$\mathbf{w}_j = \mathbf{v}_j - \sum_{i < j} (\mathbf{u}_i^\top \mathbf{v}_j) \mathbf{u}_i, \quad \mathbf{u}_j = \mathbf{w}_j / \|\mathbf{w}_j\|, \quad j = 1, \dots, k.$$

Then each $\mathbf{w}_j \neq \mathbf{0}$. The vectors $\mathbf{u}_1, \dots, \mathbf{u}_k$ are orthonormal, and $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_j) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ for every j .

Proof. Induction on j . Suppose the claims hold for $j - 1$. The vector $\mathbf{w}_j$ is $\mathbf{v}_j$ minus an element of $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1}) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$. If $\mathbf{w}_j$ were zero, $\mathbf{v}_j$ would lie in that span, contradicting independence. For $l < j$, $\mathbf{u}_l^\top \mathbf{w}_j = \mathbf{u}_l^\top \mathbf{v}_j - \sum_{i < j} (\mathbf{u}_i^\top \mathbf{v}_j) \mathbf{u}_l^\top \mathbf{u}_i = \mathbf{u}_l^\top \mathbf{v}_j - \mathbf{u}_l^\top \mathbf{v}_j = 0$, so $\mathbf{u}_j$ is orthogonal to its predecessors. Finally, $\mathbf{u}_j \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1}, \mathbf{v}_j) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$ and $\mathbf{v}_j \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_j)$, so the two spans agree. ■

Two consequences are used repeatedly. Every subspace has an orthonormal basis (apply Gram–Schmidt to any basis). And any orthonormal basis of a subspace $\mathcal{S}$ can be extended to an orthonormal basis of $\mathbb{R}^n$: extend it to a basis of $\mathbb{R}^n$ in any way, then apply Gram–Schmidt, which leaves the first vectors unchanged. [Section 6.10](#) shows that Gram–Schmidt, written in matrix form, is the QR factorization.

6.2.4 Direct sums and the orthogonal decomposition

For subspaces $\mathcal{S}, \mathcal{T}$, the sum $\mathcal{S} + \mathcal{T} = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in \mathcal{S}, \mathbf{v} \in \mathcal{T}\}$ is a subspace. When $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$ the sum is *direct*, written $\mathcal{S} \oplus \mathcal{T}$. Every element then decomposes as $\mathbf{u} + \mathbf{v}$ in exactly one way: if $\mathbf{u} + \mathbf{v} = \mathbf{u}' + \mathbf{v}'$ then $\mathbf{u} - \mathbf{u}' = \mathbf{v}' - \mathbf{v}$ lies in both subspaces and so is zero. When in addition $\mathcal{S} \perp \mathcal{T}$, the sum is an *orthogonal* direct sum.

Theorem 6.2.3 — Orthogonal decomposition

For every subspace $\mathcal{S} \subseteq \mathbb{R}^n$,

$$\mathbb{R}^n = \mathcal{S} \oplus \mathcal{S}^\perp, \quad \dim \mathcal{S} + \dim \mathcal{S}^\perp = n, \quad (\mathcal{S}^\perp)^\perp = \mathcal{S}.$$

Hence every $\mathbf{y} \in \mathbb{R}^n$ can be written uniquely as $\mathbf{y} = \mathbf{u} + \mathbf{v}$ with $\mathbf{u} \in \mathcal{S}$ and $\mathbf{v} \perp \mathcal{S}$.

Proof. Let $\mathbf{u}_1, \dots, \mathbf{u}_r$ be an orthonormal basis of $\mathcal{S}$, extended to an orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_n$ of $\mathbb{R}^n$. A vector $\mathbf{y} = \sum_{i=1}^n a_i \mathbf{u}_i$ is orthogonal to $\mathcal{S}$ iff $a_1 = \dots = a_r = 0$, so $\mathcal{S}^\perp = \text{span}(\mathbf{u}_{r+1}, \dots, \mathbf{u}_n)$ has dimension $n - r$. Writing $\mathbf{y} = \sum_{i \leq r} a_i \mathbf{u}_i + \sum_{i > r} a_i \mathbf{u}_i$ exhibits the decomposition, and it is unique because $\mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$. By the same argument with the roles of the two index sets swapped, $(\mathcal{S}^\perp)^\perp = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_r) = \mathcal{S}$. ■

Combining [Theorem 6.2.3](#) with [Lemma 6.2.1](#) gives a matrix fact used in almost every later chapter.

Corollary 6.2.4

For any $n \times p$ matrix $\mathbf{A}$:

- (a) $\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A})$;
- (b) $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)$ and $\mathcal{C}(\mathbf{A} \mathbf{A}^\top) = \mathcal{C}(\mathbf{A})$;
- (c) $\text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{A} \mathbf{A}^\top) = \text{rank}(\mathbf{A})$;
- (d) $\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top)$ and $\mathbb{R}^p = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$.

Proof.

- (a) If $\mathbf{A}\mathbf{b} = \mathbf{0}$ then $\mathbf{A}^\top \mathbf{A}\mathbf{b} = \mathbf{0}$. Conversely, $\mathbf{A}^\top \mathbf{A}\mathbf{b} = \mathbf{0}$ gives $\mathbf{b}^\top \mathbf{A}^\top \mathbf{A}\mathbf{b} = \|\mathbf{A}\mathbf{b}\|^2 = 0$, so $\mathbf{A}\mathbf{b} = \mathbf{0}$.
- (b) By [Lemma 6.2.1](#) applied to the symmetric matrix $\mathbf{A}^\top \mathbf{A}$, and then (a) and [Lemma 6.2.1](#)

again, $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A}^\top \mathbf{A})^\perp = \mathcal{N}(\mathbf{A})^\perp = (\mathcal{C}(\mathbf{A}^\top)^\perp)^\perp = \mathcal{C}(\mathbf{A}^\top)$, using $(\mathcal{S}^\perp)^\perp = \mathcal{S}$ in the last step. The second identity is the first applied to $\mathbf{A}^\top$.

(c) Take dimensions in (b) and use $\text{rank}(\mathbf{A}^\top) = \text{rank}(\mathbf{A})$.

(d) This is [Lemma 6.2.1](#) and [Theorem 6.2.3](#). ■

Part (b) is the reason the normal equations $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$ can always be solved: the right-hand side lies in $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$. We come back to this in [Section 6.4](#).

Lemma 6.2.5

For subspaces $\mathcal{S}_1, \mathcal{S}_2 \subseteq \mathbb{R}^n$,

$$(\mathcal{S}_1 + \mathcal{S}_2)^\perp = \mathcal{S}_1^\perp \cap \mathcal{S}_2^\perp \quad \text{and} \quad (\mathcal{S}_1 \cap \mathcal{S}_2)^\perp = \mathcal{S}_1^\perp + \mathcal{S}_2^\perp.$$

Proof. A vector is orthogonal to $\mathcal{S}_1 + \mathcal{S}_2$ iff it is orthogonal to every $\mathbf{u}_1 + \mathbf{u}_2$, which (taking $\mathbf{u}_2 = \mathbf{0}$ or $\mathbf{u}_1 = \mathbf{0}$) is the case iff it is orthogonal to both subspaces. For the second identity, apply the first to $\mathcal{S}_1^\perp$ and $\mathcal{S}_2^\perp$: $(\mathcal{S}_1^\perp + \mathcal{S}_2^\perp)^\perp = \mathcal{S}_1 \cap \mathcal{S}_2$, and take complements of both sides. ■

[Lemma 6.2.5](#) is what we need to describe reduced models defined by linear constraints ([Theorem 6.5.2](#)).

6.2.5 Exercises

6.2.5.1 A. Check your understanding

Exercise A1.

Show that $\mathcal{S}^\perp$ is a subspace for any set $\mathcal{S}$ (not necessarily a subspace), and that $\mathcal{S} \subseteq \mathcal{T}$ implies $\mathcal{T}^\perp \subseteq \mathcal{S}^\perp$. Show that $\mathcal{S}^\perp = \text{span}(\mathcal{S})^\perp$.

6.2.5.2 B. Practice

Exercise B1.

A one-way layout has three groups of sizes 2, 2, 1 and model matrix $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3]$. Find bases for $\mathcal{C}(\mathbf{X})$, $\mathcal{N}(\mathbf{X})$, $\mathcal{C}(\mathbf{X}^\top)$ and $\mathcal{C}(\mathbf{X})^\perp$, and check that the dimensions agree with [Eq. 6.2.1](#) and [Theorem 6.2.3](#).

Exercise B2.

Show that $\mathcal{C}(\mathbf{AB}) \subseteq \mathcal{C}(\mathbf{A})$, and deduce $\text{rank}(\mathbf{AB}) \leq \min\{\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B})\}$. Show that $\mathcal{C}(\mathbf{AB}) = \mathcal{C}(\mathbf{A})$ whenever $\mathbf{B}$ has full row rank.

6.2.5.3 C. Going deeper

Exercise C1.

For subspaces $\mathcal{S}, \mathcal{T} \subseteq \mathbb{R}^n$, prove $\dim(\mathcal{S} + \mathcal{T}) = \dim \mathcal{S} + \dim \mathcal{T} - \dim(\mathcal{S} \cap \mathcal{T})$. Use it to show that two r -dimensional subspaces of $\mathbb{R}^n$ with $2r > n$ must share a nonzero vector.

6.3 Orthogonal projection

6.3.1 Nearest points

Theorem 6.3.1 — Projection theorem

Let $\mathcal{S}$ be a subspace of $\mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^n$. There is exactly one $\hat{\mathbf{y}} \in \mathcal{S}$ such that

$$\|\mathbf{y} - \hat{\mathbf{y}}\| \leq \|\mathbf{y} - \mathbf{u}\| \quad \text{for all } \mathbf{u} \in \mathcal{S}.$$

It is the unique $\hat{\mathbf{y}} \in \mathcal{S}$ with $\mathbf{y} - \hat{\mathbf{y}} \perp \mathcal{S}$. Moreover, for every $\mathbf{u} \in \mathcal{S}$,

$$\|\mathbf{y} - \mathbf{u}\|^2 = \|\mathbf{y} - \hat{\mathbf{y}}\|^2 + \|\hat{\mathbf{y}} - \mathbf{u}\|^2. \quad (6.3.1)$$

Proof. By Theorem 6.2.3 there is a unique decomposition $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{v}$ with $\hat{\mathbf{y}} \in \mathcal{S}$ and $\mathbf{v} \in \mathcal{S}^\perp$. For $\mathbf{u} \in \mathcal{S}$, $\mathbf{y} - \mathbf{u} = \mathbf{v} + (\hat{\mathbf{y}} - \mathbf{u})$, where $\mathbf{v} \perp \mathcal{S}$ and $\hat{\mathbf{y}} - \mathbf{u} \in \mathcal{S}$. The Pythagorean identity Eq. 6.2.3 then gives Eq. 6.3.1. The right-hand side is at least $\|\mathbf{y} - \hat{\mathbf{y}}\|^2$, with equality iff $\mathbf{u} = \hat{\mathbf{y}}$. So $\hat{\mathbf{y}}$ is the unique nearest point. If some other $\mathbf{w} \in \mathcal{S}$ also had $\mathbf{y} - \mathbf{w} \perp \mathcal{S}$, then $\mathbf{y} = \mathbf{w} + (\mathbf{y} - \mathbf{w})$ would be a second decomposition, contradicting uniqueness. ■

The theorem has two halves, and it helps to keep them apart. The *variational* half says $\hat{\mathbf{y}}$ minimizes a distance. The *orthogonality* half says the error $\mathbf{y} - \hat{\mathbf{y}}$ is perpendicular to the subspace. The first half explains why least squares is a sensible estimation principle. The second is what we compute with, because orthogonality is a set of linear equations.

6.3.2 The projection as a matrix

Write $P_{\mathcal{S}}\mathbf{y}$ for the nearest point $\hat{\mathbf{y}}$. The map $\mathbf{y} \mapsto P_{\mathcal{S}}\mathbf{y}$ is linear: if $\mathbf{y}_k = \hat{\mathbf{y}}_k + \mathbf{v}_k$ are the decompositions of $\mathbf{y}_1$ and $\mathbf{y}_2$, then $a\mathbf{y}_1 + b\mathbf{y}_2 = (a\hat{\mathbf{y}}_1 + b\hat{\mathbf{y}}_2) + (a\mathbf{v}_1 + b\mathbf{v}_2)$ is a decomposition of the same kind, and by uniqueness it is *the* decomposition. A linear map on $\mathbb{R}^n$ is given by a unique $n \times n$ matrix, since two matrices that agree on every vector agree on the standard basis vectors and are therefore equal.

Definition 6.3.1 — Orthogonal projection matrix

The **orthogonal projection** onto a subspace $\mathcal{S}$ is the unique $n \times n$ matrix $P_{\mathcal{S}}$ such that, for every $\mathbf{y}$, $P_{\mathcal{S}}\mathbf{y} \in \mathcal{S}$ and $\mathbf{y} - P_{\mathcal{S}}\mathbf{y} \perp \mathcal{S}$. When $\mathcal{S} = \mathcal{C}(\mathbf{X})$ for the model matrix of a linear model, we write $\mathbf{M} = P_{\mathcal{C}(\mathbf{X})}$.

The most useful formula for $P_{\mathcal{S}}$ uses an orthonormal basis.

Proposition 6.3.2

Let the columns of the $n \times r$ matrix $\mathbf{Q}$ be an orthonormal basis for $\mathcal{S}$, so that $\mathbf{Q}^{\top}\mathbf{Q} = \mathbf{I}_r$. Then

$$P_{\mathcal{S}} = \mathbf{Q}\mathbf{Q}^{\top} = \sum_{i=1}^r \mathbf{u}_i\mathbf{u}_i^{\top},$$

where $\mathbf{u}_1, \dots, \mathbf{u}_r$ are the columns of $\mathbf{Q}$.

Proof. For any $\mathbf{y}$, $\mathbf{Q}\mathbf{Q}^{\top}\mathbf{y} = \mathbf{Q}(\mathbf{Q}^{\top}\mathbf{y}) \in \mathcal{C}(\mathbf{Q}) = \mathcal{S}$, and $\mathbf{Q}^{\top}(\mathbf{y} - \mathbf{Q}\mathbf{Q}^{\top}\mathbf{y}) = \mathbf{Q}^{\top}\mathbf{y} - \mathbf{Q}^{\top}\mathbf{y} = \mathbf{0}$, so the difference is orthogonal to every column of $\mathbf{Q}$ and hence to $\mathcal{S}$. By uniqueness, $\mathbf{Q}\mathbf{Q}^{\top} = P_{\mathcal{S}}$. ■

Example. (Projections onto a line)

For $\mathcal{S} = \text{span}(\mathbf{v})$ with $\mathbf{v} \neq \mathbf{0}$, the single vector $\mathbf{v}/\|\mathbf{v}\|$ is an orthonormal basis, so

$$P_{\mathcal{S}} = \frac{\mathbf{v}\mathbf{v}^{\top}}{\mathbf{v}^{\top}\mathbf{v}}, \quad P_{\mathcal{S}}\mathbf{y} = \frac{\mathbf{v}^{\top}\mathbf{y}}{\mathbf{v}^{\top}\mathbf{v}}\mathbf{v}.$$

With $\mathbf{v} = \mathbf{1}$ this gives $\mathbf{P} = n^{-1}\mathbf{1}\mathbf{1}^{\top}$, the matrix with every entry $1/n$, and $\mathbf{P}\mathbf{y} = \bar{y}\mathbf{1}$. *The sample mean is a projection:* fitting the intercept-only model replaces every observation by $\bar{y}$, and the residuals $y_i - \bar{y}$ form a vector in $\mathbf{1}^{\perp}$, the set of vectors whose entries sum to zero.

6.3.3 Characterizing projection matrices

Which matrices are orthogonal projections? [Proposition 6.3.2](#) shows that every projection has the form $\mathbf{Q}\mathbf{Q}^{\top}$, so it is *symmetric*, and $\mathbf{Q}\mathbf{Q}^{\top}\mathbf{Q}\mathbf{Q}^{\top} = \mathbf{Q}\mathbf{I}\mathbf{Q}^{\top} = \mathbf{Q}\mathbf{Q}^{\top}$, so it is *idempotent*. The converse is the basic recognition theorem.

Theorem 6.3.3

An $n \times n$ matrix $\mathbf{P}$ is the orthogonal projection onto some subspace iff $\mathbf{P}^{\top} = \mathbf{P}$ and $\mathbf{P}^2 = \mathbf{P}$. In that case the subspace is $\mathcal{C}(\mathbf{P})$, and $\mathbf{I} - \mathbf{P}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{P})^{\perp}$.

Proof. Necessity was shown above. For sufficiency, let $\mathbf{P}$ be symmetric and idempotent and let $\mathbf{y} \in \mathbb{R}^n$. Certainly $\mathbf{Py} \in \mathcal{C}(\mathbf{P})$. For any $\mathbf{u} = \mathbf{Pb} \in \mathcal{C}(\mathbf{P})$,

$$\mathbf{u}^\top (\mathbf{y} - \mathbf{Py}) = \mathbf{b}^\top \mathbf{P}^\top (\mathbf{I} - \mathbf{P}) \mathbf{y} = \mathbf{b}^\top (\mathbf{P} - \mathbf{P}^2) \mathbf{y} = 0,$$

so $\mathbf{y} - \mathbf{Py} \perp \mathcal{C}(\mathbf{P})$ and $\mathbf{P} = \mathbf{P}_{\mathcal{C}(\mathbf{P})}$. For the last claim, $\mathbf{I} - \mathbf{P}$ is symmetric and idempotent, since $(\mathbf{I} - \mathbf{P})^2 = \mathbf{I} - 2\mathbf{P} + \mathbf{P}^2 = \mathbf{I} - \mathbf{P}$. It sends $\mathbf{y}$ to $\mathbf{y} - \mathbf{Py} \in \mathcal{C}(\mathbf{P})^\perp$, and $\mathbf{y} - (\mathbf{I} - \mathbf{P})\mathbf{y} = \mathbf{Py} \perp \mathcal{C}(\mathbf{P})^\perp$. ■

Before this theorem, checking that a matrix computes a projection meant thinking about subspaces. Afterwards it is two matrix identities. Also, a projection is determined by its range: if $\mathbf{P}_1, \mathbf{P}_2$ are symmetric idempotent with $\mathcal{C}(\mathbf{P}_1) = \mathcal{C}(\mathbf{P}_2)$, then $\mathbf{P}_1 = \mathbf{P}_2$, because both are the projection onto the same subspace.

Proposition 6.3.4 — Spectrum, rank and trace

Let $\mathbf{P}$ be the orthogonal projection onto an r -dimensional subspace $\mathcal{S}$. Then:

- (a) every eigenvalue of $\mathbf{P}$ is 0 or 1, and 1 has multiplicity r ;
- (b) $\text{rank}(\mathbf{P}) = \text{tr}(\mathbf{P}) = r$;
- (c) $\mathbf{Pv} = \mathbf{v}$ iff $\mathbf{v} \in \mathcal{S}$, and $\mathbf{Pv} = \mathbf{0}$ iff $\mathbf{v} \in \mathcal{S}^\perp$;
- (d) $\mathbf{v}^\top \mathbf{Pv} = \|\mathbf{Pv}\|^2 \leq \|\mathbf{v}\|^2$, so $\mathbf{P}$ and $\mathbf{I} - \mathbf{P}$ are nonnegative definite.

Proof. Write $\mathbf{P} = \mathbf{Q}\mathbf{Q}^\top$ as in Proposition 6.3.2, and extend $\mathbf{Q}$ to an orthogonal matrix $\mathbf{U} = [\mathbf{Q}, \mathbf{Q}_\perp]$. Then $\mathbf{P} = \mathbf{U} \text{diag}(\mathbf{I}_r, \mathbf{0}) \mathbf{U}^\top$ is a spectral decomposition, which gives (a). Rank and trace of a symmetric matrix are the number of nonzero eigenvalues and their sum, which gives (b). Directly, $\text{tr}(\mathbf{Q}\mathbf{Q}^\top) = \text{tr}(\mathbf{Q}^\top \mathbf{Q}) = \text{tr}(\mathbf{I}_r) = r$. Part (c) restates the projection theorem: the nearest point of $\mathcal{S}$ to a vector already in $\mathcal{S}$ is the vector itself, and $\mathbf{v} \perp \mathcal{S}$ iff $\mathbf{0}$ is its nearest point. For (d), $\mathbf{v}^\top \mathbf{Pv} = \mathbf{v}^\top \mathbf{P}^\top \mathbf{Pv}$, and $\|\mathbf{v}\|^2 = \|\mathbf{Pv}\|^2 + \|(\mathbf{I} - \mathbf{P})\mathbf{v}\|^2$ by Pythagoras. ■

The identity $\text{rank} = \text{tr}$ gets a lot of use. The dimension of a subspace, which is the “degrees of freedom” of every sum of squares in this book, can be read off as a sum of diagonal entries without finding a basis.

6.3.4 Generalized inverses

Proposition 6.3.2 needs an orthonormal basis. In statistics the subspace usually comes as the column space of a model matrix $\mathbf{X}$ whose columns are not orthonormal and may not even be independent. To write $\mathbf{M}$ directly in terms of $\mathbf{X}$ we need a substitute for $(\mathbf{X}^\top \mathbf{X})^{-1}$ that exists even when $\mathbf{X}^\top \mathbf{X}$ is singular.

Definition 6.3.2 — Generalized inverse

A **generalized inverse** of an $m \times k$ matrix $\mathbf{A}$ is any $k \times m$ matrix $\mathbf{A}^-$ such that $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$.

If $\mathbf{A}$ is square and nonsingular, multiplying $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$ on both sides by $\mathbf{A}^{-1}$ gives $\mathbf{A}^- = \mathbf{A}^{-1}$, so the generalized inverse is unique and equal to the inverse. When $\mathbf{A}$ is singular there are infinitely many.

Proposition 6.3.5 — Existence

Every matrix has a generalized inverse. If $\mathbf{A}$ has rank r and is permuted so that

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}$$

with $\mathbf{A}_{11}$ an $r \times r$ nonsingular block, then $\mathbf{A}^- = \begin{pmatrix} \mathbf{A}_{11}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$ (permuted back) is a generalized inverse.

Proof. A rank- r matrix has r independent rows and r independent columns, and the submatrix where they cross is nonsingular (Chapter 1). This guarantees the block form. Since the first r columns are a basis for the column space, $\begin{pmatrix} \mathbf{A}_{12} \\ \mathbf{A}_{22} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11} \\ \mathbf{A}_{21} \end{pmatrix} \mathbf{B}$ for some $\mathbf{B}$, so $\mathbf{B} = \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$ and $\mathbf{A}_{22} = \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$. Multiplying out,

$$\mathbf{A}\mathbf{A}^-\mathbf{A} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & \mathbf{0} \end{pmatrix} \mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \end{pmatrix} = \mathbf{A}.$$

■

Among the many generalized inverses, one is singled out by extra symmetry conditions. The **Moore–Penrose inverse** $\mathbf{A}^+$ is the unique matrix satisfying $\mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{A}$, $\mathbf{A}^+\mathbf{A}\mathbf{A}^+ = \mathbf{A}^+$, and both $\mathbf{A}\mathbf{A}^+$ and $\mathbf{A}^+\mathbf{A}$ symmetric (Penrose 1955). Section 6.10 computes it from the singular value decomposition. For projections we will need none of these extra conditions, and that is the point of the next result.

Lemma 6.3.6

Let $\mathbf{X}$ be $n \times p$ and let $\mathbf{G}$ be any generalized inverse of $\mathbf{X}^\top \mathbf{X}$. Then:

- (a) $\mathbf{X}\mathbf{G}\mathbf{X}^\top \mathbf{X} = \mathbf{X}$ and $\mathbf{X}^\top \mathbf{X}\mathbf{G}\mathbf{X}^\top = \mathbf{X}^\top$;
- (b) $\mathbf{X}\mathbf{G}\mathbf{X}^\top$ does not depend on the choice of generalized inverse;
- (c) $\mathbf{X}\mathbf{G}\mathbf{X}^\top$ is symmetric, even when $\mathbf{G}$ is not.

Proof. By Corollary 6.2.4(b), $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$, so there is a $p \times n$ matrix $\mathbf{B}$ with

$\mathbf{X}^\top = \mathbf{X}^\top \mathbf{X} \mathbf{B}$. Transposing, $\mathbf{X} = \mathbf{B}^\top \mathbf{X}^\top \mathbf{X}$. Then

$$\mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{X} = \mathbf{B}^\top (\mathbf{X}^\top \mathbf{X}) \mathbf{G} (\mathbf{X}^\top \mathbf{X}) = \mathbf{B}^\top \mathbf{X}^\top \mathbf{X} = \mathbf{X},$$

which is the first identity in (a). The second follows the same way, using $\mathbf{X}^\top = \mathbf{X}^\top \mathbf{X} \mathbf{B}$ on the right. For (b) and (c),

$$\mathbf{X} \mathbf{G} \mathbf{X}^\top = \mathbf{B}^\top (\mathbf{X}^\top \mathbf{X}) \mathbf{G} (\mathbf{X}^\top \mathbf{X}) \mathbf{B} = \mathbf{B}^\top \mathbf{X}^\top \mathbf{X} \mathbf{B},$$

which involves no $\mathbf{G}$ at all, and $(\mathbf{B}^\top \mathbf{X}^\top \mathbf{X} \mathbf{B})^\top = \mathbf{B}^\top \mathbf{X}^\top \mathbf{X} \mathbf{B}$.

■

Theorem 6.3.7

For any $n \times p$ matrix $\mathbf{X}$ and any generalized inverse $(\mathbf{X}^\top \mathbf{X})^-$,

$$\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top$$

is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$. In particular $\text{rank}(\mathbf{M}) = \text{tr}(\mathbf{M}) = \text{rank}(\mathbf{X})$. If $\mathbf{X}$ has full column rank, $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$.

Proof. By Lemma 6.3.6, $\mathbf{M}$ is symmetric and $\mathbf{M}^2 = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- [\mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top] = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top = \mathbf{M}$. By Theorem 6.3.3, $\mathbf{M}$ is the projection onto $\mathcal{C}(\mathbf{M})$. Clearly $\mathcal{C}(\mathbf{M}) \subseteq \mathcal{C}(\mathbf{X})$, and $\mathbf{M}\mathbf{X} = \mathbf{X}$ (again by the lemma) shows $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{M})$. The rank statement is Proposition 6.3.4(b).

■

It is worth pausing on how strong Theorem 6.3.7 is. The matrix $(\mathbf{X}^\top \mathbf{X})^-$ can be almost anything: symmetric or not, reflexive or not, with entries of any size. The sandwich $\mathbf{X}(\cdot)\mathbf{X}^\top$ removes all of that freedom. We will see in Section 6.4 that this is exactly the difference between coefficients and fitted values, made concrete.

Example. (The one-way layout)

In the setting of Example (A rank-deficient layout), sort the observations by group and let group k have n_k members. The model space is spanned by the orthogonal vectors $\mathbf{z}_1, \dots, \mathbf{z}_g$, so by Proposition 6.3.2 with $\mathbf{u}_k = \mathbf{z}_k / \sqrt{n_k}$,

$$\mathbf{M} = \sum_{k=1}^g \frac{\mathbf{z}_k \mathbf{z}_k^\top}{n_k} = \begin{pmatrix} n_1^{-1} \mathbf{1}_{n_1} \mathbf{1}_{n_1}^\top & & \\ & \ddots & \\ & & n_g^{-1} \mathbf{1}_{n_g} \mathbf{1}_{n_g}^\top \end{pmatrix},$$

a block-diagonal matrix of within-group averaging operators. $\mathbf{M}\mathbf{y}$ replaces each observation by its group mean. The intercept column adds nothing to $\mathcal{C}(\mathbf{X})$, so it does not appear, and $\text{tr}(\mathbf{M}) = \sum_k n_k/n_k = g = \text{rank}(\mathbf{X})$. We got here without a generalized inverse. Theorem 6.3.7 guarantees that any generalized inverse of the singular $(g+1) \times (g+1)$ matrix $\mathbf{X}^\top \mathbf{X}$ produces this same matrix. Section 6.4 checks this numerically.

6.3.5 Sums and products of projections

Complicated subspaces are often built from simpler ones, and their projections are built the same way.

Theorem 6.3.8

Let P_1, P_2 be orthogonal projections. Then $P_1 + P_2$ is an orthogonal projection iff $P_1 P_2 = 0$, that is, iff $\mathcal{C}(P_1) \perp \mathcal{C}(P_2)$. In that case it projects onto $\mathcal{C}(P_1) \oplus \mathcal{C}(P_2)$.

Proof. $P_1 + P_2$ is always symmetric, and $(P_1 + P_2)^2 = P_1 + P_2 + P_1 P_2 + P_2 P_1$. So it is idempotent iff $P_1 P_2 + P_2 P_1 = 0$. If $P_1 P_2 = 0$, then transposing gives $P_2 P_1 = 0$ and the condition holds. Conversely, suppose $P_1 P_2 + P_2 P_1 = 0$. Multiplying on the left by P_1 gives $P_1 P_2 + P_1 P_2 P_1 = 0$, and on the right by P_1 gives $P_1 P_2 P_1 + P_2 P_1 = 0$. Subtracting, $P_1 P_2 = P_2 P_1$, and then $2P_1 P_2 = 0$. The equivalence with orthogonality of ranges holds because $P_1 P_2 = 0$ says exactly that every vector of $\mathcal{C}(P_2)$ is sent to 0 by P_1 , that is (Proposition 6.3.4(c)), lies in $\mathcal{C}(P_1)^\perp$. Finally, $\mathcal{C}(P_1 + P_2) \subseteq \mathcal{C}(P_1) + \mathcal{C}(P_2)$, and for $u = P_1 u + P_2 u$ in that sum we have $(P_1 + P_2)u = u$, so the ranges agree. ■

By induction, if $P_1, \dots, P_k$ are projections with $P_i P_j = 0$ for $i \neq j$, their sum projects onto the orthogonal direct sum of their ranges. When the sum is I , we get an orthogonal decomposition

$$y = P_1 y + \dots + P_k y, \quad \|y\|^2 = \|P_1 y\|^2 + \dots + \|P_k y\|^2.$$

Every analysis of variance table in this book is a table of such a decomposition.

Products behave differently. $P_1 P_2$ need not be symmetric or idempotent, and it is generally not a projection. Exercise (B1) shows that it is one iff P_1 and P_2 commute, in which case it projects onto $\mathcal{C}(P_1) \cap \mathcal{C}(P_2)$. The most important instance is the nested case $\mathcal{C}(P_2) \subseteq \mathcal{C}(P_1)$, treated in Section 6.5.

6.3.6 Exercises

6.3.6.1 A. Check your understanding

Exercise A1.

Let $\mathcal{S} = \text{span}\{(1, 1, 0)^\top, (0, 1, 1)^\top\}$. Compute $P_{\mathcal{S}}$ in two ways: by Gram-Schmidt and Proposition 6.3.2, and by Theorem 6.3.7. Verify symmetry, idempotence and $\text{tr } P_{\mathcal{S}} = 2$. Find $\mathcal{S}^\perp$ and check that $I - P_{\mathcal{S}}$ projects onto it.

Exercise A2.

Show that a symmetric matrix whose eigenvalues all lie in $\{0, 1\}$ is an orthogonal projection. Give a non-symmetric 2×2 matrix with eigenvalues 0 and 1 that is not idempotent, and one that is idempotent.

Exercise A3.

Let $\mathbf{P}$ be an orthogonal projection and a, b real numbers. Show that the eigenvalues of $a\mathbf{P} + b\mathbf{I}$ are b and $a + b$. Show that when $b \neq 0$ and $a + b \neq 0$,

$$(a\mathbf{P} + b\mathbf{I})^{-1} = \frac{1}{b} \left(\mathbf{I} - \frac{a}{a+b} \mathbf{P} \right).$$

Use this to invert the equicorrelation matrix $(1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$.

Exercise A4.

Show that if $\mathbf{G}$ is a generalized inverse of the symmetric matrix $\mathbf{X}^\top \mathbf{X}$, then so is $\mathbf{G}^\top$, and so is $\mathbf{G}\mathbf{X}^\top \mathbf{X}\mathbf{G}^\top$. Which of the four Penrose conditions does the last one satisfy?

6.3.6.2 B. Practice**Exercise B1.**

Let $\mathbf{P}_1, \mathbf{P}_2$ be orthogonal projections. Show that $\mathbf{P}_1\mathbf{P}_2$ is an orthogonal projection iff $\mathbf{P}_1\mathbf{P}_2 = \mathbf{P}_2\mathbf{P}_1$, and that in this case it projects onto $\mathcal{C}(\mathbf{P}_1) \cap \mathcal{C}(\mathbf{P}_2)$. *Hint:* for the range, show $\mathbf{P}_1\mathbf{P}_2\mathbf{u} = \mathbf{u}$ iff $\mathbf{u}$ lies in both subspaces.

Solution. If $\mathbf{P}_1\mathbf{P}_2 = \mathbf{P}_2\mathbf{P}_1$, then $(\mathbf{P}_1\mathbf{P}_2)^\top = \mathbf{P}_2\mathbf{P}_1 = \mathbf{P}_1\mathbf{P}_2$, and $(\mathbf{P}_1\mathbf{P}_2)^2 = \mathbf{P}_1\mathbf{P}_2\mathbf{P}_2\mathbf{P}_1 = \mathbf{P}_1\mathbf{P}_2\mathbf{P}_1 = \mathbf{P}_1\mathbf{P}_1\mathbf{P}_2 = \mathbf{P}_1\mathbf{P}_2$. So the product is symmetric and idempotent. Conversely, if $\mathbf{P}_1\mathbf{P}_2$ is an orthogonal projection, it is symmetric, so $\mathbf{P}_1\mathbf{P}_2 = (\mathbf{P}_1\mathbf{P}_2)^\top = \mathbf{P}_2\mathbf{P}_1$. For the range: if $\mathbf{u}$ lies in both subspaces, then $\mathbf{P}_1\mathbf{P}_2\mathbf{u} = \mathbf{P}_1\mathbf{u} = \mathbf{u}$. If $\mathbf{P}_1\mathbf{P}_2\mathbf{u} = \mathbf{u}$, then $\mathbf{u} \in \mathcal{C}(\mathbf{P}_1)$, and also $\mathbf{u} = \mathbf{P}_2\mathbf{P}_1\mathbf{u} \in \mathcal{C}(\mathbf{P}_2)$. A projection's range is the set of vectors it fixes, so $\mathcal{C}(\mathbf{P}_1\mathbf{P}_2) = \mathcal{C}(\mathbf{P}_1) \cap \mathcal{C}(\mathbf{P}_2)$.

Exercise B2.

Let $\mathbf{G}_0$ be a generalized inverse of $\mathbf{A}$.

- Show that $\mathbf{G}_0 + (\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{U} + \mathbf{V}(\mathbf{I} - \mathbf{A}\mathbf{G}_0)$ is a generalized inverse of $\mathbf{A}$ for all conformable $\mathbf{U}, \mathbf{V}$.
- Show that every generalized inverse $\mathbf{G}$ of $\mathbf{A}$ has this form. *Hint:* take $\mathbf{U} = \mathbf{G} - \mathbf{G}_0$ and $\mathbf{V} = \mathbf{G}_0\mathbf{A}(\mathbf{G} - \mathbf{G}_0)$, and use $\mathbf{A}(\mathbf{G} - \mathbf{G}_0)\mathbf{A} = \mathbf{0}$.
- Conclude that a matrix has a unique generalized inverse iff it is square and non-singular.

Solution. Parts a, b. For (a), multiply out: $\mathbf{A}[\mathbf{G}_0 + (\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{U} + \mathbf{V}(\mathbf{I} - \mathbf{A}\mathbf{G}_0)]\mathbf{A} = \mathbf{A}\mathbf{G}_0\mathbf{A} + (\mathbf{A} - \mathbf{A}\mathbf{G}_0\mathbf{A})\mathbf{U}\mathbf{A} + \mathbf{A}\mathbf{V}(\mathbf{A} - \mathbf{A}\mathbf{G}_0\mathbf{A}) = \mathbf{A}$. For (b), put $\mathbf{\Delta} = \mathbf{G} - \mathbf{G}_0$, so that $\mathbf{A}\mathbf{\Delta}\mathbf{A} = \mathbf{A} - \mathbf{A} = \mathbf{0}$. With $\mathbf{U} = \mathbf{\Delta}$ and $\mathbf{V} = \mathbf{G}_0\mathbf{A}\mathbf{\Delta}$,

$$(\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{\Delta} + \mathbf{G}_0\mathbf{A}\mathbf{\Delta}(\mathbf{I} - \mathbf{A}\mathbf{G}_0) = \mathbf{\Delta} - \mathbf{G}_0\mathbf{A}\mathbf{\Delta} + \mathbf{G}_0\mathbf{A}\mathbf{\Delta} - \mathbf{G}_0(\mathbf{A}\mathbf{\Delta}\mathbf{A})\mathbf{G}_0 = \mathbf{\Delta},$$

so the formula reproduces $\mathbf{G}_0 + \mathbf{\Delta} = \mathbf{G}$.

Exercise B3.

Using the singular value decomposition, show that $(\mathbf{X}^\top \mathbf{X})^+ \mathbf{X}^\top = \mathbf{X}^+$ and $\mathbf{X} \mathbf{X}^+ = \mathbf{M}$. Show also that $\mathbf{X}^+ = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ when $\mathbf{X}$ has full column rank.

6.3.6.3 C. Going deeper**Exercise C1.**

Show that $\mathbf{M}$ is the only $n \times n$ matrix $\mathbf{P}$ satisfying $\mathbf{P}\mathbf{X} = \mathbf{X}$, $\mathbf{P}^\top = \mathbf{P}$ and $\text{rank}(\mathbf{P}) = \text{rank}(\mathbf{X})$. Is the rank condition necessary? Is symmetry?

6.4 Least squares via projection

We now return to the linear model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, with $\mathbf{X}$ an $n \times p$ matrix of rank $r \leq p$, and $\mathbf{M}$ the orthogonal projection onto $\mathcal{C}(\mathbf{X})$. Nothing in this section uses probability. The results hold for every data vector $\mathbf{y}$ and so apply whatever the distribution of the errors.

6.4.1 The characterization of least squares estimates

Theorem 6.4.1 — Least squares is projection

A vector $\hat{\boldsymbol{\beta}}$ is a least squares estimate of $\boldsymbol{\beta}$ iff

$$\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{M}\mathbf{y}.$$

Consequently the fitted values $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$ and residuals $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{y}$ are the same for every least squares estimate.

Proof. Apply the projection theorem (Theorem 6.3.1) with $\mathcal{S} = \mathcal{C}(\mathbf{X})$ and $\mathbf{u} = \mathbf{X}\mathbf{b}$. By Eq. 6.3.1,

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{M}\mathbf{y}\|^2 + \|\mathbf{M}\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 \quad \text{for every } \mathbf{b} \in \mathbb{R}^p.$$

The first term does not involve $\mathbf{b}$, and the second is zero iff $\mathbf{X}\mathbf{b} = \mathbf{M}\mathbf{y}$. Such a $\mathbf{b}$ exists because $\mathbf{M}\mathbf{y} \in \mathcal{C}(\mathbf{X})$. ■

The orthogonality half of the projection theorem turns this into linear equations. The residual $\mathbf{y} - \mathbf{X}\mathbf{b}$ is orthogonal to $\mathcal{C}(\mathbf{X})$ iff it is orthogonal to every column of $\mathbf{X}$, that is, iff $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0}$.

Theorem 6.4.2 — Normal equations

The **normal equations**

$$\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y} \quad (6.4.1)$$

are consistent for every $\mathbf{y}$, and their solutions are exactly the least squares estimates. One solution is $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top \mathbf{y}$, for any generalized inverse, and the set of all solutions is

$$\hat{\boldsymbol{\beta}} + \mathcal{N}(\mathbf{X}) = \{\hat{\boldsymbol{\beta}} + \mathbf{v} : \mathbf{X}\mathbf{v} = \mathbf{0}\}.$$

In particular, the least squares estimate is unique iff $\text{rank}(\mathbf{X}) = p$, and then $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$.

Proof. A vector $\mathbf{b}$ solves Eq. 6.4.1 iff $\mathbf{y} - \mathbf{X}\mathbf{b} \perp \mathcal{C}(\mathbf{X})$. Since $\mathbf{X}\mathbf{b} \in \mathcal{C}(\mathbf{X})$ automatically, this holds iff $\mathbf{y} = \mathbf{X}\mathbf{b} + (\mathbf{y} - \mathbf{X}\mathbf{b})$ is the orthogonal decomposition of $\mathbf{y}$, that is, iff $\mathbf{X}\mathbf{b} = \mathbf{M}\mathbf{y}$. By Theorem 6.4.1 this is the same as $\mathbf{b}$ being a least squares estimate. For the particular solution, $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top \mathbf{y} = \mathbf{M}\mathbf{y}$ by Theorem 6.3.7. Finally, $\mathbf{X}\mathbf{b} = \mathbf{M}\mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}}$ iff $\mathbf{X}(\mathbf{b} - \hat{\boldsymbol{\beta}}) = \mathbf{0}$, and $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$ iff $\text{rank}(\mathbf{X}) = p$ by Eq. 6.2.1. ■

The logic of this proof runs opposite to the usual calculus derivation, and the difference matters. Setting a gradient to zero finds stationary points. One then has to argue separately that they are minima and that one exists. Here existence, minimality and the full solution set all come from the decomposition $\mathbb{R}^n = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\mathbf{X})^\perp$. The consistency of the normal equations, which can look surprising when $\mathbf{X}^\top \mathbf{X}$ is singular, is just the statement $\mathbf{X}^\top \mathbf{y} \in \mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$ from Corollary 6.2.4.

6.4.2 Fitted values and residuals

Collecting the consequences for $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$ and $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{y}$:

Proposition 6.4.3

For any data vector $\mathbf{y}$:

- (a) $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$ and $\hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} = 0$;
- (b) if $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ (for example, the model has an intercept), then $\sum_i \hat{\varepsilon}_i = 0$ and the mean of the fitted values is $\bar{y}$;
- (c) $\|\mathbf{y}\|^2 = \|\hat{\mathbf{Y}}\|^2 + \|\hat{\boldsymbol{\varepsilon}}\|^2$;
- (d) $\|\hat{\boldsymbol{\varepsilon}}\|^2 = \mathbf{y}^\top (\mathbf{I} - \mathbf{M})\mathbf{y}$ and $\|\hat{\mathbf{Y}}\|^2 = \mathbf{y}^\top \mathbf{M}\mathbf{y}$;
- (e) if $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ for some $\boldsymbol{\beta}$, then $\hat{\mathbf{Y}} = \mathbf{X}\boldsymbol{\beta} + \mathbf{M}\boldsymbol{\varepsilon}$ and $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$.

Proof.

- (a) is the orthogonality of $\hat{\boldsymbol{\varepsilon}}$ to $\mathcal{C}(\mathbf{X})$, and $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$. For (b), $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ gives $\mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = 0$, and then $\mathbf{1}^\top \hat{\mathbf{Y}} = \mathbf{1}^\top \mathbf{y}$. (c) is Pythagoras. (d) uses symmetry and idempotence: $\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \mathbf{y}^\top (\mathbf{I} - \mathbf{M})^\top (\mathbf{I} - \mathbf{M})\mathbf{y} = \mathbf{y}^\top (\mathbf{I} - \mathbf{M})\mathbf{y}$. For (e), $\mathbf{M}\mathbf{X} = \mathbf{X}$ and

$$(\mathbf{I} - \mathbf{M})\mathbf{X} = \mathbf{0}.$$

■

Part (e) is simple but important. The residual vector does not depend on β at all. It is the projection of the unobservable error vector onto the $(n - r)$ -dimensional space $\mathcal{C}(\mathbf{X})^\perp$. Everything about σ^2 that the data can reveal must therefore come from $(\mathbf{I} - \mathbf{M})\epsilon$. If $E(\epsilon) = \mathbf{0}$ and $\text{Cov}(\epsilon) = \sigma^2 \mathbf{I}$, the expectation of a quadratic form (Chapter 2) gives

$$E\|\hat{\epsilon}\|^2 = E[\epsilon^\top (\mathbf{I} - \mathbf{M})\epsilon] = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{M}) = \sigma^2(n - r),$$

using Proposition 6.3.4. So $\|\hat{\epsilon}\|^2/(n - r)$ is unbiased for σ^2 . The divisor is the dimension of the space in which the residuals live, not the number of observations and not $n - p$. Chapter 7 develops the sampling theory, and Chapter 9 develops the decomposition.

6.4.3 Coefficients are not unique; fitted values are

When $\text{rank}(\mathbf{X}) < p$, Theorem 6.4.2 gives a whole affine subspace of least squares estimates, one for each generalized inverse and more besides. Statistical software has to pick one. Some programs drop columns that are linearly dependent on earlier ones. Others impose “side conditions” such as a zero coefficient for a reference level or effects that sum to zero. Others report the minimum-norm solution. These choices look very different in the printed coefficient table. The geometry says they are all describing the same fitted vector.

Example. (Four answers, one fit)

Nine observations fall in three groups of sizes 3, 2, 4 with group means 5.700, 7.800 and 4.300. The model matrix $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3]$ has $p = 4$ columns and rank 3, so $\mathbf{X}^\top \mathbf{X}$ is singular. We compute $\hat{\beta} = \mathbf{GX}^\top \mathbf{y}$ for four generalized inverses $\mathbf{G}$ of $\mathbf{X}^\top \mathbf{X}$: the Moore–Penrose inverse; the inverse obtained by deleting the intercept row and column (Proposition 6.3.5); the one obtained by deleting the first group’s row and column; and a deliberately strange, non-symmetric one built from the general formula of Exercise (B2).

Generalized inverse	$\hat{\beta}_0$	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_3$	$\hat{\beta}_1 - \hat{\beta}_2$	$\max \mathbf{M} - \mathbf{M}_{\text{MP}} $
Moore– Penrose	4.450	1.250	3.350	-0.150	-2.100	0
drop intercept	0.000	5.700	7.800	4.300	-2.100	1.1×10^{-16}
drop group 1	5.700	0.000	2.100	-1.400	-2.100	1.1×10^{-16}
random	-9.102	14.802	16.902	13.402	-2.100	1.8×10^{-15}

The coefficient vectors disagree wildly, including in sign. Yet each produces the same projection matrix up to rounding error, and hence the same fitted values (the group means) and residuals. The difference $\hat{\beta}_1 - \hat{\beta}_2$ is the same in every row, and so is $\hat{\beta}_0 + \hat{\beta}_k$ for each k . Individual coefficients are not. The trace of $\mathbf{M}$ is 3.000, the rank of $\mathbf{X}$.

```
import numpy as np

groups = np.array([0, 0, 0, 1, 1, 2, 2, 2, 2]) # three groups, sizes 3, 2, 4
y = np.array([5.1, 6.3, 5.7, 8.2, 7.4, 4.0, 4.9, 3.8, 4.5])
n, g = len(y), 3
X = np.column_stack([np.ones(n), np.eye(g)[groups]]) # [1, indicator columns]
A = X.T @ X
print("n =", n, " p =", X.shape[1], " rank(X) =", np.linalg.matrix_rank(X))
```

```
def g_drop(j):
    """Generalized inverse obtained by deleting row and column j of A."""
    keep = [k for k in range(A.shape[0]) if k != j]
    G = np.zeros_like(A)
    G[np.ix_(keep, keep)] = np.linalg.inv(A[np.ix_(keep, keep)])
    return G

rng = np.random.default_rng(6)
G0 = np.linalg.pinv(A) # Moore-Penrose inverse
U, V = rng.normal(size=A.shape), rng.normal(size=A.shape)
G_random = G0 + (np.eye(4) - G0 @ A) @ U + V @ (np.eye(4) - A @ G0)

ginverses = {
    "Moore-Penrose": G0,
    "drop intercept": g_drop(0),
    "drop group 1": g_drop(1),
    "random": G_random,
}
```

```
M_ref = X @ G0 @ X.T
for name, G in ginverses.items():
    assert np.allclose(A @ G @ A, A) # G really is a g-inverse
    beta = G @ X.T @ y
    M = X @ G @ X.T
    print(f"{name:15s} beta = {np.round(beta, 3)}",
          f" max|M - M_ref| = {np.abs(M - M_ref).max():.1e}",
          f" group1-group2 = {beta[1] - beta[2]:.3f}")
```

Which linear functions of $\hat{\beta}$ come out the same for every least squares estimate? The answer is a first look at estimability, the subject of [Chapter 8](#).

Theorem 6.4.4

For $\lambda \in \mathbb{R}^p$, the value $\lambda^\top \hat{\beta}$ is the same for every least squares estimate $\hat{\beta}$ (and every y) iff $\lambda \in \mathcal{C}(X^\top)$, that is, iff $\lambda^\top = \rho^\top X$ for some $\rho \in \mathbb{R}^n$. In that case $\lambda^\top \hat{\beta} = \rho^\top M y$.

Proof. By [Theorem 6.4.2](#) the least squares estimates form $\hat{\beta} + \mathcal{N}(X)$. So $\lambda^\top \hat{\beta}$ is invariant iff $\lambda^\top v = 0$ for all $v \in \mathcal{N}(X)$, that is, iff $\lambda \in \mathcal{N}(X)^\perp = \mathcal{C}(X^\top)$ ([Corollary 6.2.4](#)). If $\lambda^\top = \rho^\top X$ then $\lambda^\top \hat{\beta} = \rho^\top X \hat{\beta} = \rho^\top M y$. ■

In [Example \(Four answers, one fit\)](#), $\lambda = (0, 1, -1, 0)^\top$ is the difference between any row of X

from group 1, $(1, 1, 0, 0)$, and any row from group 2, $(1, 0, 1, 0)$. So $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ and $\hat{\beta}_1 - \hat{\beta}_2$ is invariant. For $\boldsymbol{\lambda} = (0, 1, 0, 0)^\top$, note that $\boldsymbol{\rho}^\top \mathbf{X}$ has first entry $\sum_i \rho_i$ and remaining entries equal to the sums of ρ_i within each group. Matching $(0, 1, 0, 0)$ would need the group sums to be 1, 0, 0 and their total to be 0, which is impossible. So $\hat{\beta}_1$ is not invariant.

Remark (Side conditions choose coordinates). Imposing $\beta_0 = 0$ or $\beta_1 = 0$ does not add information about the data. It picks one point from the affine set $\hat{\boldsymbol{\beta}} + \mathcal{N}(\mathbf{X})$. A side condition $\mathbf{c}^\top \boldsymbol{\beta} = 0$ picks exactly one point iff $\mathbf{c} \notin \mathcal{C}(\mathbf{X}^\top)$. A condition inside $\mathcal{C}(\mathbf{X}^\top)$ would constrain the fitted values and change the model rather than just its coordinates. [Theorem 8.4.2](#) makes this precise.

6.4.4 Exercises

6.4.4.1 A. Check your understanding

Exercise A1.

For [Example \(Three observations\)](#), write out the 3×3 matrix $\mathbf{M}$ with exact fractional entries, verify $\text{tr } \mathbf{M} = 2$, and recover the fitted values and residuals quoted in the text.

Exercise A2.

Show that the residuals sum to zero whenever $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, even if no column of $\mathbf{X}$ is constant. Give an example of a model without such a column in which this happens, and one in which the residuals do not sum to zero.

6.4.4.2 B. Practice

Exercise B1.

Suppose $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$. Show that $E(\hat{\mathbf{Y}}) = \mathbf{X}\boldsymbol{\beta}$, $\text{Cov}(\hat{\mathbf{Y}}) = \sigma^2 \mathbf{M}$, $\text{Cov}(\hat{\boldsymbol{\varepsilon}}) = \sigma^2(\mathbf{I} - \mathbf{M})$ and $\text{Cov}(\hat{\mathbf{Y}}, \hat{\boldsymbol{\varepsilon}}) = \mathbf{0}$. Which of these statements survive when $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$ for a general positive definite $\mathbf{V}$?

Exercise B2.

Let $\text{rank}(\mathbf{X}) = r < p$ and let $\mathbf{C}$ be a $(p - r) \times p$ matrix. Show that the constrained problem “minimize $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|$ subject to $\mathbf{C}\mathbf{b} = \mathbf{0}$ ” has a unique solution for every $\mathbf{y}$, and that its fitted values are $\mathbf{M}\mathbf{y}$, iff $\text{rank}(\begin{smallmatrix} \mathbf{X} \\ \mathbf{C} \end{smallmatrix}) = p$. Show that the condition implies $\mathcal{C}(\mathbf{C}^\top) \cap \mathcal{C}(\mathbf{X}^\top) = \{\mathbf{0}\}$, and interpret: the constraints must not restrict any estimable function.

6.4.4.3 C. Going deeper

Exercise C1.

Suppose $\mathbf{X}^\top \mathbf{y} \neq \mathbf{0}$. Show that every least squares estimate $\mathbf{b}$ can be written as $\mathbf{G}\mathbf{X}^\top \mathbf{y}$ for some generalized inverse $\mathbf{G}$ of $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$. *Hint:* with $\mathbf{c} = \mathbf{X}^\top \mathbf{y}$, try $\mathbf{G} = \mathbf{G}_0 + (\mathbf{b} -$

$G_0\mathbf{c}\mathbf{c}^\top/(\mathbf{c}^\top\mathbf{c})$. What goes wrong when $\mathbf{X}^\top\mathbf{y} = \mathbf{0}$?

6.5 Nested models and sequential projection

Statistical questions about linear models are almost always comparisons. Does a regressor matter? Are several group means equal? Is a curve needed, or will a line do? Each comparison sets a full model against a *reduced* model whose mean space is a subspace of the full one. This section studies the geometry of such nested pairs. Chapter 9 and Chapter 11 build the analysis of variance and the F test on it.

6.5.1 Two nested subspaces

Let $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$, with ranks $r_0 \leq r$. Write $\mathbf{M}_0$ and $\mathbf{M}$ for the two projections.

Theorem 6.5.1 — Nested projections

If $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$, then:

- (a) $\mathbf{M}\mathbf{M}_0 = \mathbf{M}_0\mathbf{M} = \mathbf{M}_0$;
- (b) $\mathbf{M} - \mathbf{M}_0$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X}_0)^\perp \cap \mathcal{C}(\mathbf{X})$, the orthogonal complement of $\mathcal{C}(\mathbf{X}_0)$ within $\mathcal{C}(\mathbf{X})$;
- (c) $\text{rank}(\mathbf{M} - \mathbf{M}_0) = r - r_0$;
- (d) $(\mathbf{I} - \mathbf{M}_0) - (\mathbf{I} - \mathbf{M}) = \mathbf{M} - \mathbf{M}_0$, and $\mathbf{M}_0(\mathbf{M} - \mathbf{M}_0) = (\mathbf{M} - \mathbf{M}_0)(\mathbf{I} - \mathbf{M}) = \mathbf{0}$.

Proof.

- (a) Every column of $\mathbf{M}_0$ lies in $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ and is left unchanged by $\mathbf{M}$, so $\mathbf{M}\mathbf{M}_0 = \mathbf{M}_0$. Transposing gives $\mathbf{M}_0\mathbf{M} = \mathbf{M}_0$.
- (b) $\mathbf{M} - \mathbf{M}_0$ is symmetric, and by (a) $(\mathbf{M} - \mathbf{M}_0)^2 = \mathbf{M} - \mathbf{M}\mathbf{M}_0 - \mathbf{M}_0\mathbf{M} + \mathbf{M}_0 = \mathbf{M} - \mathbf{M}_0$. By Theorem 6.3.3 it projects onto its column space. If $\mathbf{v} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{v} \perp \mathcal{C}(\mathbf{X}_0)$, then $(\mathbf{M} - \mathbf{M}_0)\mathbf{v} = \mathbf{v} - \mathbf{0} = \mathbf{v}$, so $\mathbf{v} \in \mathcal{C}(\mathbf{M} - \mathbf{M}_0)$. Conversely, a vector $\mathbf{v} = (\mathbf{M} - \mathbf{M}_0)\mathbf{u}$ lies in $\mathcal{C}(\mathbf{X})$ and satisfies $\mathbf{M}_0\mathbf{v} = (\mathbf{M}_0 - \mathbf{M}_0)\mathbf{u} = \mathbf{0}$.
- (c) $\text{tr}(\mathbf{M} - \mathbf{M}_0) = r - r_0$ and Proposition 6.3.4.
- (d) Direct multiplication using (a).

■

The theorem gives a three-way orthogonal decomposition of $\mathbb{R}^n$,

$$\mathbb{R}^n = \underbrace{\mathcal{C}(\mathbf{X}_0)}_{\dim r_0} \oplus \underbrace{\mathcal{C}(\mathbf{X}_0)^\perp \cap \mathcal{C}(\mathbf{X})}_{\dim r - r_0} \oplus \underbrace{\mathcal{C}(\mathbf{X})^\perp}_{\dim n - r},$$

and correspondingly $\mathbf{I} = \mathbf{M}_0 + (\mathbf{M} - \mathbf{M}_0) + (\mathbf{I} - \mathbf{M})$, with the three terms mutually orthogonal projections. Applying the decomposition to $\mathbf{y}$ and using Pythagoras,

$$\|\mathbf{y}\|^2 = \|\mathbf{M}_0\mathbf{y}\|^2 + \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 + \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2. \quad (6.5.1)$$

The middle term has a second reading that is the key to model comparison. Since $(\mathbf{I} - \mathbf{M}_0) = (\mathbf{M} - \mathbf{M}_0) + (\mathbf{I} - \mathbf{M})$ is an orthogonal sum,

$$\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 = \|(\mathbf{I} - \mathbf{M}_0)\mathbf{y}\|^2 - \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \text{SSE}_0 - \text{SSE}. \quad (6.5.2)$$

The *drop in residual sum of squares* from the reduced to the full model is the squared length of a projection onto an $(r - r_0)$ -dimensional space. That is why it has $r - r_0$ degrees of freedom, and why it is independent of SSE under normality: the two lie in orthogonal subspaces (Chapter 4). It also shows that SSE can never increase when the model space grows, with equality iff $\mathbf{M}\mathbf{y} = \mathbf{M}_0\mathbf{y}$.

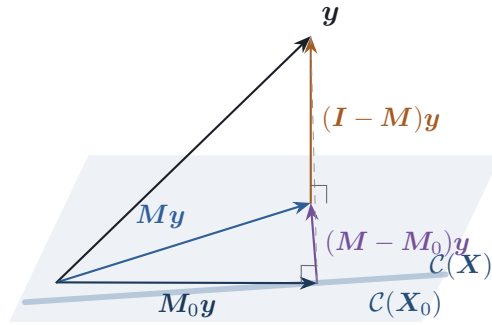


Figure 6.5.1. Nested projections. The fitted vector of the reduced model, $\mathbf{M}_0\mathbf{y}$, is also the projection of $\mathbf{M}\mathbf{y}$ onto $\mathcal{C}(\mathbf{X}_0)$. The three coloured pieces $\mathbf{M}_0\mathbf{y}$, $(\mathbf{M} - \mathbf{M}_0)\mathbf{y}$ and $(\mathbf{I} - \mathbf{M})\mathbf{y}$ are mutually orthogonal. The dashed segment is the reduced model's residual, whose squared length is the sum of the squared lengths of the two segments meeting at $\mathbf{M}\mathbf{y}$.

Part (a) also says $\mathbf{M}_0\mathbf{y} = \mathbf{M}_0(\mathbf{M}\mathbf{y})$. *Fitting the reduced model to the data gives the same result as fitting it to the full model's fitted values.* Projection can be done in stages (Figure 6.5.1).

Example. (Intercept inside simple regression)

Let $\mathbf{X}_0 = \mathbf{1}$ and $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ with $\mathbf{x}$ not constant. Then $\mathcal{C}(\mathbf{X}_0)^\perp \cap \mathcal{C}(\mathbf{X})$ is spanned by the centred regressor $\mathbf{x} - \bar{x}\mathbf{1} = (\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{x}$, which lies in $\mathcal{C}(\mathbf{X})$ and is orthogonal to $\mathbf{1}$. By Example (Projections onto a line),

$$\mathbf{M} - \mathbf{M}_0 = \frac{(\mathbf{x} - \bar{x}\mathbf{1})(\mathbf{x} - \bar{x}\mathbf{1})^\top}{S_{xx}}, \quad S_{xx} = \sum_i (x_i - \bar{x})^2,$$

and so

$$\mathbf{M}\mathbf{y} = \bar{y}\mathbf{1} + \frac{S_{xy}}{S_{xx}}(\mathbf{x} - \bar{x}\mathbf{1}), \quad S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}).$$

Reading off coordinates, the slope is S_{xy}/S_{xx} and the intercept is $\bar{y} - \bar{x} S_{xy}/S_{xx}$. These are the familiar formulas, obtained without solving any equations, and the decomposition Eq. 6.5.1 becomes

$$\sum_i y_i^2 = n\bar{y}^2 + \frac{S_{xy}^2}{S_{xx}} + \text{SSE}.$$

6.5.2 Chains of subspaces

The argument extends to any chain $\mathcal{C}(\mathbf{X}_1) \subseteq \mathcal{C}(\mathbf{X}_2) \subseteq \dots \subseteq \mathcal{C}(\mathbf{X}_k)$ with projections $\mathbf{M}_1, \dots, \mathbf{M}_k$. Setting $\mathbf{M}_0 = \mathbf{0}$ and $\mathbf{M}_{k+1} = \mathbf{I}$, the differences $\mathbf{M}_j - \mathbf{M}_{j-1}$ for $j = 1, \dots, k+1$ are mutually orthogonal projections summing to $\mathbf{I}$. The corresponding lengths

$$\|(\mathbf{M}_j - \mathbf{M}_{j-1})\mathbf{y}\|^2 = \text{SSE}_{j-1} - \text{SSE}_j$$

are the **sequential sums of squares** of [Definition 9.3.1](#). Their values depend on the *order* in which the subspaces are nested. A regressor added early can take credit that it would not get if added last. The Frisch–Waugh–Lovell theorem of [Section 6.6](#) describes what the regressor added last actually contributes.

Example. (Order matters)

For the state data of [Example \(Poverty and murder rates\)](#), enter the three regressors after the intercept in two different orders. The listing computes each squared length $\|(\mathbf{M}_j - \mathbf{M}_{j-1})\mathbf{y}\|^2$ directly from the projections.

Order A		Order B	
poverty	102.49	single-parent	162.84
single-parent	80.04	urban	3.59
urban	0.27	poverty	16.37
residual	101.96	residual	101.96

Entered first, poverty accounts for 102.49 of the corrected total sum of squares 284.76. Entered last, it accounts for only 16.37. The residual sum of squares, and the total explained by all three regressors together (182.80), do not depend on the order, because the final subspace does not. The individual pieces do depend on the order, because the intermediate subspaces differ. Poverty and single parenthood are correlated, so whichever enters first takes the variation they share. The piece for the *last* regressor is the only one with an order-free meaning. By [Theorem 6.6.2](#), it is the squared length of the projection of $\mathbf{y}$ onto the part of that regressor orthogonal to all the others.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
cols = {"poverty": data["poverty"], "single": data["single"], "urban": data["urban"]}

def projection(Z):
    Q, _ = np.linalg.qr(Z)
    return Q @ Q.T
    # fine for n = 50; never do this for large
    n

def sequential_ss(y, columns):
    """Squared lengths ||(M_j - M_{j-1}) y||^2 as columns enter one at a time."""
    n = len(y)
    Z = np.ones((n, 1))
    M_prev = projection(Z)
```

```

pieces = {"mean": y @ M_prev @ y}
for name, x in columns:
    Z = np.column_stack([Z, x])
    M = projection(Z)
    pieces[name] = y @ (M - M_prev) @ y
    M_prev = M
pieces["residual"] = y @ (np.eye(n) - M_prev) @ y
return pieces

order_a = sequential_ss(y, [(k, cols[k]) for k in ["poverty", "single", "urban"]])
order_b = sequential_ss(y, [(k, cols[k]) for k in ["single", "urban", "poverty"]])
for name in ["poverty", "single", "urban", "residual"]:
    print(f"{name:9s}  order A: {order_a[name]:8.2f}  order B: {order_b[name]:8.2f}")
    )

```

6.5.3 Two factors and when the pieces are orthogonal

A sum of projections is a projection exactly when the pieces are orthogonal ([Theorem 6.3.8](#)). In designed experiments, orthogonality comes from balance, and when it fails, it fails in ways that later chapters have to face.

Example. (Additive two-factor model, one observation per cell)

Let factor A have a levels and factor B have b levels, with one observation y_{ij} for each of the $n = ab$ combinations. Let P_A project onto vectors that depend only on the A level, so $P_A \mathbf{y}$ has entries $\bar{y}_{i.}$, and define P_B similarly. Let $P_0 = n^{-1} \mathbf{1} \mathbf{1}^\top$. Both $\mathcal{C}(P_A)$ and $\mathcal{C}(P_B)$ contain $\mathbf{1}$, so $P_A P_0 = P_B P_0 = P_0$ by [Theorem 6.5.1](#).

The crucial computation is $P_A P_B$. Averaging over j with i fixed a vector that depends only on j gives the overall average of that vector. So for every $\mathbf{y}$ the entries of $P_A(P_B \mathbf{y})$ are all equal to $b^{-1} \sum_j \bar{y}_{.j} = \bar{y}_{..}$. Therefore

$$P_A P_B = P_0,$$

and so $(P_A - P_0)(P_B - P_0) = P_0 - P_0 - P_0 + P_0 = \mathbf{0}$. The two *centred* factor spaces are orthogonal. By [Theorem 6.3.8](#), the additive model space has projection $P_0 + (P_A - P_0) + (P_B - P_0) = P_A + P_B - P_0$, and

$$I = P_0 + (P_A - P_0) + (P_B - P_0) + (I - P_A - P_B + P_0)$$

is an orthogonal decomposition with ranks 1, $a-1$, $b-1$ and $(a-1)(b-1)$. The residual projection sends y_{ij} to $y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..}$.

The key step used the fact that every A level appears with every B level equally often. If cells are missing or unequally replicated, $P_A P_B \neq P_0$, the centred factor spaces are no longer orthogonal, and the sum of squares for A depends on whether B was fitted first. Chapter 17 is about that situation.

6.5.4 Reduced models defined by constraints

Reduced models often come as *constraints* on the parameters of the full model, for example $\beta_2 = \beta_3$ or $\beta_4 = 0$, rather than as a smaller model matrix. In mean space, a constraint $\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{0}$ on an estimable function $\mathbf{A}^\top = \mathbf{P}^\top \mathbf{X}$ restricts $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ to the subspace

$$\mathcal{S}_0 = \mathcal{C}(\mathbf{X}) \cap \mathcal{N}(\mathbf{P}^\top) = \{\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X}) : \mathbf{P}^\top \boldsymbol{\mu} = \mathbf{0}\}.$$

To test the constraint we need the piece of $\mathcal{C}(\mathbf{X})$ that $\mathcal{S}_0$ removes.

Theorem 6.5.2

Let $\mathbf{M}$ project onto $\mathcal{C}(\mathbf{X})$ and let $\mathbf{P}$ be an $n \times q$ matrix. If $\mathcal{S}_0 = \mathcal{C}(\mathbf{X}) \cap \mathcal{N}(\mathbf{P}^\top)$, then

$$\mathcal{S}_0^\perp \cap \mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{MP}).$$

Hence the projection onto $\mathcal{S}_0$ is $\mathbf{M} - \mathbf{P}_{\mathcal{C}(\mathbf{MP})}$, and $\dim \mathcal{S}_0 = r - \text{rank}(\mathbf{MP})$.

Proof. By Lemma 6.2.5 and Lemma 6.2.1, $\mathcal{S}_0^\perp = \mathcal{C}(\mathbf{X})^\perp + \mathcal{N}(\mathbf{P}^\top)^\perp = \mathcal{C}(\mathbf{X})^\perp + \mathcal{C}(\mathbf{P})$. If $\mathbf{v} \in \mathcal{S}_0^\perp \cap \mathcal{C}(\mathbf{X})$, write $\mathbf{v} = \mathbf{w} + \mathbf{Pa}$ with $\mathbf{w} \perp \mathcal{C}(\mathbf{X})$. Applying $\mathbf{M}$ gives $\mathbf{v} = \mathbf{Mv} = \mathbf{MPa} \in \mathcal{C}(\mathbf{MP})$. Conversely, $\mathbf{MPa}$ lies in $\mathcal{C}(\mathbf{X})$ and equals $\mathbf{Pa} - (\mathbf{I} - \mathbf{M})\mathbf{Pa} \in \mathcal{C}(\mathbf{P}) + \mathcal{C}(\mathbf{X})^\perp = \mathcal{S}_0^\perp$. The remaining claims follow from Theorem 6.5.1 with the roles $\mathcal{C}(\mathbf{X}_0) = \mathcal{S}_0$. ■

The subspace $\mathcal{C}(\mathbf{MP})$ is the *test space* of the hypothesis. Its dimension $\text{rank}(\mathbf{MP})$ counts how many independent restrictions the constraint really places on the mean. It can be smaller than q if some rows of the constraint are redundant. Chapter 11 uses exactly this space for the numerator of the F statistic.

6.5.5 Exercises

6.5.5.1 A. Check your understanding

Exercise A1.

Construct the projections $\mathbf{P}_A$, $\mathbf{P}_B$ and $\mathbf{P}_0$ of Example (Additive two-factor model, one observation per cell) for $a = 3$, $b = 2$ and verify $\mathbf{P}_A \mathbf{P}_B = \mathbf{P}_0$ numerically. Delete one observation and recompute. Show that the centred factor spaces are no longer orthogonal, and find the largest entry of $(\mathbf{P}_A - \mathbf{P}_0)(\mathbf{P}_B - \mathbf{P}_0)$.

6.5.5.2 B. Practice

Exercise B1.

With $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$, show that $\text{SSE}_0 - \text{SSE} = \|\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_0\|^2$, the squared distance between the two fitted vectors.

Solution. $\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_0 = (\mathbf{M} - \mathbf{M}_0)\mathbf{y}$, and $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 = \text{SSE}_0 - \text{SSE}$ by Eq. 6.5.2.

Exercise B2.

In the model $E(\mathbf{Y}) = \beta_0\mathbf{1} + \beta_1\mathbf{x}_1 + \beta_2\mathbf{x}_2 + \beta_3\mathbf{x}_3$ with full-rank $\mathbf{X}$, the hypothesis $\beta_2 = \beta_3$ defines a reduced model with model matrix $\mathbf{X}_0 = [\mathbf{1}, \mathbf{x}_1, \mathbf{x}_2 + \mathbf{x}_3]$. Show that the test space $\mathcal{C}(\mathbf{X}_0)^\perp \cap \mathcal{C}(\mathbf{X})$ is spanned by $(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_2$, and that $(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_2 = -(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_3$. Verify that this agrees with Theorem 6.5.2 for a suitable $\mathbf{P}$.

Solution. The two model spaces differ in dimension by one, so the test space is a line. The vector $(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_2$ lies in $\mathcal{C}(\mathbf{X})$ and is orthogonal to $\mathcal{C}(\mathbf{X}_0)$. It is nonzero, because $\mathbf{x}_2 \in \mathcal{C}(\mathbf{X}_0)$ would make the columns of $\mathbf{X}$ dependent. Since $\mathbf{x}_2 + \mathbf{x}_3 \in \mathcal{C}(\mathbf{X}_0)$, $(\mathbf{I} - \mathbf{M}_0)(\mathbf{x}_2 + \mathbf{x}_3) = \mathbf{0}$. For Theorem 6.5.2, write the constraint as $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = 0$ with $\boldsymbol{\lambda} = (0, 0, 1, -1)^\top$, and take $\mathbf{P} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\boldsymbol{\lambda}$, so that $\mathbf{P}^\top \mathbf{X} = \boldsymbol{\lambda}^\top$. Then $\mathbf{M}\mathbf{P} = \mathbf{P}$. Each column of $\mathbf{X}_0$ is $\mathbf{X}\mathbf{k}$ with $\boldsymbol{\lambda}^\top \mathbf{k} = 0$, so $\mathbf{P} \perp \mathcal{C}(\mathbf{X}_0)$ and $\mathbf{P}$ spans the same line.

6.5.5.3 C. Going deeper

Exercise C1.

Extend Example (Additive two-factor model, one observation per cell) to $m \geq 2$ observations in every cell and a model with interaction. Find five mutually orthogonal projections summing to $\mathbf{I}$, corresponding to the grand mean, factor A , factor B , the interaction, and within-cell error, and find their ranks. Describe the action of each on $\mathbf{y}$ in terms of cell, row, column and grand means.

6.6 The Frisch–Waugh–Lovell theorem

A coefficient in a multiple regression is often described as the effect of its regressor “holding the other regressors fixed”. In observational data nobody held anything fixed, so the phrase needs a precise meaning. The Frisch–Waugh–Lovell (FWL) theorem supplies one. A coefficient in a multiple regression equals the slope of a simple regression in which the other regressors have been *projected out* of both the response and the regressor of interest.

6.6.1 Statement and proof

Partition the model matrix and coefficients as

$$\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2], \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix},$$

with $\mathbf{X}_1$ of size $n \times p_1$ and $\mathbf{X}_2$ of size $n \times p_2$, and assume $\mathbf{X}$ has full column rank. Let $\mathbf{M}_2$ project onto $\mathcal{C}(\mathbf{X}_2)$, and write

$$\tilde{\mathbf{X}}_1 = (\mathbf{I} - \mathbf{M}_2)\mathbf{X}_1, \quad \tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_2)\mathbf{y}$$

for the parts of $\mathbf{X}_1$ and $\mathbf{y}$ that $\mathbf{X}_2$ cannot explain linearly. Each column of $\tilde{\mathbf{X}}_1$ is the residual vector from regressing the corresponding column of $\mathbf{X}_1$ on $\mathbf{X}_2$.

Lemma 6.6.1 — Orthogonalizing a partitioned model space

Under the assumptions above:

- (a) $\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{X}_2) \oplus \mathcal{C}(\tilde{\mathbf{X}}_1)$, and the two summands are orthogonal;
- (b) $\tilde{\mathbf{X}}_1$ has full column rank;
- (c) $\mathbf{M} = \mathbf{M}_2 + \tilde{\mathbf{M}}_1$, where $\tilde{\mathbf{M}}_1 = \tilde{\mathbf{X}}_1(\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top$.

Proof.

- (a) The columns of $\tilde{\mathbf{X}}_1 = \mathbf{X}_1 - \mathbf{M}_2 \mathbf{X}_1$ lie in $\mathcal{C}(\mathbf{X})$ and are orthogonal to $\mathcal{C}(\mathbf{X}_2)$. Conversely, for any $\mathbf{b} = (\mathbf{b}_1^\top, \mathbf{b}_2^\top)^\top$,

$$\mathbf{X}\mathbf{b} = \mathbf{X}_1\mathbf{b}_1 + \mathbf{X}_2\mathbf{b}_2 = \tilde{\mathbf{X}}_1\mathbf{b}_1 + (\mathbf{M}_2\mathbf{X}_1\mathbf{b}_1 + \mathbf{X}_2\mathbf{b}_2),$$

with the second term in $\mathcal{C}(\mathbf{X}_2)$. So $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\tilde{\mathbf{X}}_1) + \mathcal{C}(\mathbf{X}_2)$.

- (b) If $\tilde{\mathbf{X}}_1\mathbf{a} = \mathbf{0}$, then $\mathbf{X}_1\mathbf{a} = \mathbf{M}_2\mathbf{X}_1\mathbf{a} = \mathbf{X}_2\mathbf{c}$ for some $\mathbf{c}$, so $\mathbf{X}(\mathbf{a}^\top, -\mathbf{c}^\top)^\top = \mathbf{0}$. Full column rank of $\mathbf{X}$ forces $\mathbf{a} = \mathbf{0}$.
- (c) Combine (a) with [Theorem 6.3.8](#) and [Theorem 6.3.7](#).

■

Theorem 6.6.2 — Frisch–Waugh–Lovell

Let $\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{\beta}}_1^\top, \hat{\boldsymbol{\beta}}_2^\top)^\top$ be the least squares estimate in the full model. Then:

- (a) $\hat{\boldsymbol{\beta}}_1 = (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{y}} = (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top \mathbf{y}$. So $\hat{\boldsymbol{\beta}}_1$ is the coefficient from regressing $\tilde{\mathbf{y}}$ (or $\mathbf{y}$ itself) on $\tilde{\mathbf{X}}_1$;
- (b) the residual vector from regressing $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{X}}_1$ equals the residual vector $(\mathbf{I} - \mathbf{M})\mathbf{y}$ of the full model;
- (c) $\hat{\boldsymbol{\beta}}_2 = (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} \mathbf{X}_2^\top (\mathbf{y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1)$.

Proof. Let $\tilde{\mathbf{b}} = (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top \mathbf{y}$. Since $\mathbf{I} - \mathbf{M}_2$ is symmetric and idempotent, $\tilde{\mathbf{X}}_1^\top \mathbf{y} = \mathbf{X}_1^\top (\mathbf{I} - \mathbf{M}_2)\mathbf{y} = \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{y}}$, so the two expressions in (a) agree. By [Lemma 6.6.1\(c\)](#),

$$\mathbf{M}\mathbf{y} = \mathbf{M}_2\mathbf{y} + \tilde{\mathbf{X}}_1\tilde{\mathbf{b}} = \mathbf{M}_2\mathbf{y} + \mathbf{X}_1\tilde{\mathbf{b}} - \mathbf{M}_2\mathbf{X}_1\tilde{\mathbf{b}} = \mathbf{X}_1\tilde{\mathbf{b}} + \mathbf{M}_2(\mathbf{y} - \mathbf{X}_1\tilde{\mathbf{b}}).$$

The last term lies in $\mathcal{C}(\mathbf{X}_2)$, so it equals $\mathbf{X}_2\mathbf{c}$ with $\mathbf{c} = (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} \mathbf{X}_2^\top (\mathbf{y} - \mathbf{X}_1\tilde{\mathbf{b}})$. Thus $\mathbf{M}\mathbf{y} = \mathbf{X}_1\tilde{\mathbf{b}} + \mathbf{X}_2\mathbf{c}$. Because $\mathbf{X}$ has full column rank, the coefficient vector representing $\mathbf{M}\mathbf{y}$ is unique, so $\hat{\boldsymbol{\beta}}_1 = \tilde{\mathbf{b}}$ and $\hat{\boldsymbol{\beta}}_2 = \mathbf{c}$. This proves (a) and (c). For (b), $\mathbf{y} - \mathbf{M}\mathbf{y} = (\mathbf{I} - \mathbf{M}_2)\mathbf{y} - \tilde{\mathbf{M}}_1\mathbf{y} = \tilde{\mathbf{y}} - \tilde{\mathbf{X}}_1\tilde{\mathbf{b}}$, which is the residual of the regression of $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{X}}_1$, because $\tilde{\mathbf{X}}_1\tilde{\mathbf{b}} = \tilde{\mathbf{M}}_1\mathbf{y} = \tilde{\mathbf{M}}_1\tilde{\mathbf{y}}$.

■

The proof uses nothing beyond the orthogonal decomposition of the model space. Part (a) contains a small surprise. Residualizing $\mathbf{y}$ is optional, because the part of $\mathbf{y}$ lying in $\mathcal{C}(\mathbf{X}_2)$ is orthogonal to $\tilde{\mathbf{X}}_1$ and drops out of $\tilde{\mathbf{X}}_1^\top \mathbf{y}$ anyway. Residualizing the regressor of interest is

essential.

Idea. In a multiple regression, $\hat{\beta}_j$ measures how $\mathbf{y}$ moves with *the part of $\mathbf{x}_j$ that is orthogonal to the other regressors*. A regressor that is almost a linear combination of the others has a tiny orthogonal part, so its coefficient rests on very little information. That is collinearity, seen geometrically.

6.6.2 Consequences

Variances. The FWL formula writes $\hat{\beta}_1$ as a linear function of $\mathbf{y}$. If $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, then

$$\text{Cov}(\hat{\beta}_1) = \sigma^2 (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1 (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} = \sigma^2 (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1}.$$

Comparing with $\text{Cov}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ (Theorem 5.5.1) identifies the leading block of the inverse of a partitioned matrix,

$$[(\mathbf{X}^\top \mathbf{X})^{-1}]_{11} = (\mathbf{X}_1^\top (\mathbf{I} - \mathbf{M}_2) \mathbf{X}_1)^{-1} = (\mathbf{X}_1^\top \mathbf{X}_1 - \mathbf{X}_1^\top \mathbf{X}_2 (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} \mathbf{X}_2^\top \mathbf{X}_1)^{-1}, \quad (6.6.1)$$

a Schur-complement identity that Chapter 1 proves by algebra. For a single regressor $\mathbf{x}_j$, with all other columns (including the intercept) in $\mathbf{X}_2$,

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2} = \frac{\sigma^2}{S_{jj}(1 - R_j^2)}, \quad (6.6.2)$$

where $S_{jj} = \sum_i (x_{ij} - \bar{x}_j)^2$ and R_j^2 is the coefficient of determination from regressing $\mathbf{x}_j$ on the other regressors. The second equality is Exercise (B1). The factor $1/(1 - R_j^2)$ is the *variance inflation factor* of Chapter 26.

Centring. Take $\mathbf{X}_2 = \mathbf{1}$. Then $\mathbf{M}_2 = n^{-1} \mathbf{1}\mathbf{1}^\top$ and $\tilde{\mathbf{X}}_1, \tilde{\mathbf{y}}$ are the centred regressors and response. FWL says the slopes in a model with an intercept are the coefficients of a no-intercept regression of centred $\mathbf{y}$ on centred regressors, and part (c) gives the intercept $\bar{y} - \bar{\mathbf{x}}^\top \hat{\beta}_1$. Example (Intercept inside simple regression) was the case $p_1 = 1$.

Detrending and deseasonalizing. The original application was economic time series. Frisch and Waugh (1933) showed that regressing on a time trend together with other variables gives the same slope as first detrending every series. Lovell (1963) proved the analogous statement for seasonal dummy variables, and noted a trap: if only the *response* is seasonally adjusted and the regressors are not, the result is generally wrong. Part (a) of the theorem shows why. What must be residualized is the regressor.

A warning about standard errors. If the residualized regression of $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{X}}_1$ is fitted with ordinary software, the coefficient and residuals are right, but the software divides the residual sum of squares by $n - p_1$ instead of $n - p_1 - p_2$. It does not know that p_2 dimensions were used up in forming $\tilde{\mathbf{y}}$. The reported standard error is too small by the factor $\sqrt{(n - p_1)/(n - p)}$.

Example. (Poverty and murder rates)

For the 50 US states in 2009, regress the murder rate (per 100,000) on the poverty rate, the percentage of single-parent households, and the percentage of the population living in urban areas, with an intercept. The coefficient of poverty is 0.2538. On its own,

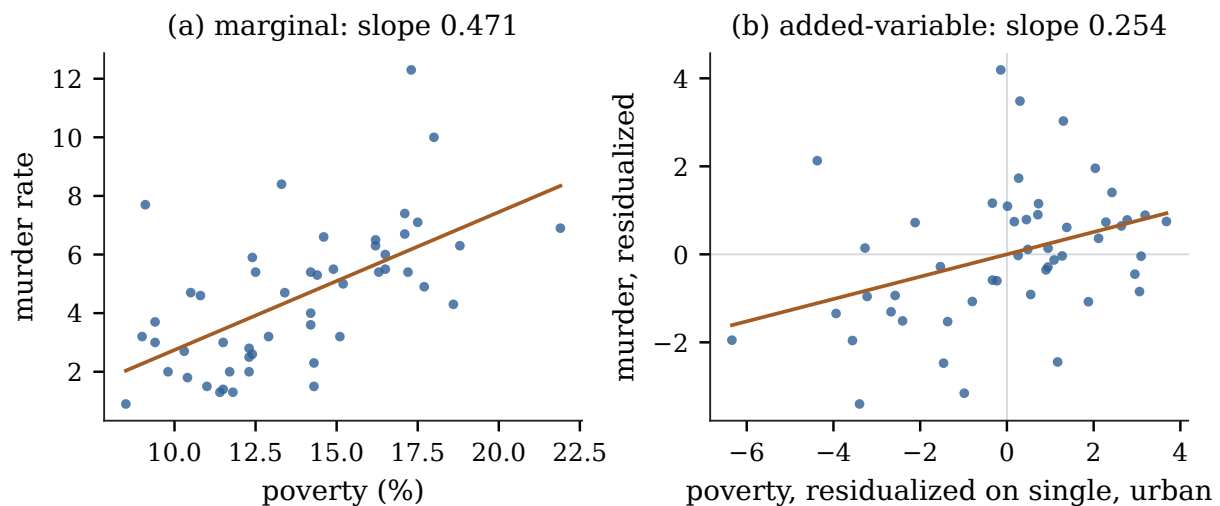


Figure 6.6.1: Murder rate and poverty for 50 US states. (a) The marginal relationship. (b) The added-variable plot: both variables residualized on the single-parent percentage, urban percentage and intercept. The slope in (b) is the multiple-regression coefficient of poverty (Theorem 6.6.2).

a simple regression of murder rate on poverty has slope 0.4706, nearly twice as large. Poverty and single parenthood are correlated, and the simple regression gives poverty credit for part of their shared association.

Figure 6.6.1 shows both regressions. Panel (b) is an **added-variable plot**: the residual of murder rate after regression on the other two regressors and the intercept, against the corresponding residual of poverty. By FWL its least squares slope is exactly the multiple-regression coefficient. The listing confirms the equality to machine precision.

The standard error of the coefficient in the full model is 0.0934. Fitting the residualized regression naively gives 0.0905, smaller by the factor 1.0321, which is $\sqrt{(n-1)/(n-4)}$ as predicted.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statelcrime.load_pandas().data
data = data.drop(index="District of Columbia")           # an extreme point; see the
                                                         exercises of Section 6.6

y = data["murder"].to_numpy()
x1 = data["poverty"].to_numpy()                          # the coefficient we care
                                                         about
X2 = np.column_stack([np.ones(len(y)), data["single"], data["urban"]])
X = np.column_stack([x1, X2])

def resid(v, Z):
    """Residual of v after projecting onto C(Z)."""
    coef, *_ = np.linalg.lstsq(Z, v, rcond=None)
    return v - Z @ coef

beta_full, *_ = np.linalg.lstsq(X, y, rcond=None)        # regress y on everything

y_tilde = resid(y, X2)                                   # (I - M2) y
x_tilde = resid(x1, X2)                                  # (I - M2) x1
```

```

beta_fwl = (x_tilde @ y_tilde) / (x_tilde @ x_tilde) # simple regression, no
            intercept

print(f"full regression:  {beta_full[0]:.6f}")
print(f"FWL regression:   {beta_fwl:.6f}")

```

Remark (What FWL does not say). FWL is an algebraic identity about fitted coefficients. It says what the number $\hat{\beta}_j$ is: an association with the unexplained part of x_j . It does not say that changing x_j in the world, with the other regressors held fixed, would change the response by $\hat{\beta}_j$. That is a causal claim, and it needs assumptions about how the data arose. Chapter 25 takes up this question, including cases where adding a regressor to $\mathbf{X}_2$ makes the coefficient *less* interpretable rather than more.

6.6.3 Fixed effects are a projection

One of the most widely used estimators in applied statistics is the FWL theorem in disguise. Suppose units $k = 1, \dots, K$ are observed repeatedly: firms over years, patients over visits, schools over cohorts. The analyst wants the effect of time-varying regressors $\mathbf{X}_1$ while allowing each unit its own intercept:

$$E(\mathbf{Y}) = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{D}\boldsymbol{\alpha},$$

where $\mathbf{D}$ is the $n \times K$ matrix of unit indicators. Fitting this directly means a regression with K extra columns, and K may be in the thousands.

FWL with $\mathbf{X}_2 = \mathbf{D}$ removes that burden. The projection onto $\mathcal{C}(\mathbf{D})$ replaces each entry by its unit's average (Example (The one-way layout)), so $\mathbf{I} - \mathbf{M}_D$ subtracts unit means. Therefore $\hat{\beta}_1$ is the least squares coefficient from regressing the *within-unit deviations* $y_{kt} - \bar{y}_k$ on $x_{kt} - \bar{x}_k$, with no indicators at all. This is the **within estimator** of panel-data econometrics. Its standard error needs the degrees-of-freedom correction described above, with $p_2 = K$, which is large.

Example. (Grunfeld's investment data)

Grunfeld's classic panel follows 11 US firms for 20 years, giving $n = 220$ observations of gross investment, market value and capital stock. With a separate intercept for each firm, the coefficients of value and capital are 0.1101 and 0.3100. The same numbers come from demeaning every variable within firm and fitting two columns. Ignoring the firms altogether (one common intercept) gives 0.1145 and 0.2275 instead. That comparison uses variation *between* firms, of the sort "large firms invest more", which the within estimator deliberately discards.

```

import numpy as np
import pandas as pd
import statsmodels.api as sm
df = sm.datasets.grunfeld.load_pandas().data
y = df["invest"].to_numpy()
X1 = df[["value", "capital"]].to_numpy()

# (i) least squares with one indicator column per firm ("dummy variables")
D = pd.get_dummies(df["firm"]).to_numpy(dtype=float)
beta_dummies, *_ = np.linalg.lstsq(np.column_stack([X1, D]), y, rcond=None)

```

```

# (ii) FWL: projecting onto  $C(D)$  replaces each value by its firm mean,
#      so  $(I - M_D)$  subtracts firm means -- the "within" transformation
def demean_by(v, groups):
    v = pd.DataFrame(v)
    return (v - v.groupby(groups.to_numpy()).transform("mean")).to_numpy()

X1_within = demean_by(X1, df["firm"])
y_within = demean_by(y, df["firm"]).ravel()
beta_within, *_ = np.linalg.lstsq(X1_within, y_within, rcond=None)

print("with firm indicators:", beta_dummies[:2])
print("within transformation:", beta_within)

```

Whether discarding between-unit variation is wise depends on why units differ. That question is taken up for random-effects alternatives in Chapter 32 and Chapter 33.

Added-variable plots are among the most useful diagnostics for multiple regression. They show the evidence for a single coefficient in two dimensions, together with any points that dominate it. They reappear in Chapter 20.

6.6.4 Exercises

6.6.4.1 A. Check your understanding

Exercise A1.

Refit the model of [Example \(Poverty and murder rates\)](#) with the District of Columbia included. How does the poverty coefficient change? Draw the added-variable plot and locate the District. Relate what you see to its leverage ([Example \(High-leverage states\)](#)).

6.6.4.2 B. Practice

Exercise B1.

Let $\mathbf{x}_j$ be a column of $\mathbf{X}$, let $\mathbf{X}_{(j)}$ be $\mathbf{X}$ with that column removed, and assume $\mathbf{1} \in \mathcal{C}(\mathbf{X}_{(j)})$. Show that $\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2 = S_{jj}(1 - R_j^2)$, where R_j^2 is the R^2 from regressing $\mathbf{x}_j$ on $\mathbf{X}_{(j)}$. Deduce [Eq. 6.6.2](#).

Solution. Since $\mathbf{1} \in \mathcal{C}(\mathbf{X}_{(j)})$, the vector $(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j$ is the residual from regressing $\mathbf{x}_j$ on $\mathbf{X}_{(j)}$. Its squared length is that regression's residual sum of squares, which equals $\text{SST}(1 - R_j^2)$ with $\text{SST} = S_{jj}$ by the definition of R^2 . Substituting into the variance formula from FWL gives [Eq. 6.6.2](#).

Exercise B2.

In [Theorem 6.6.2](#), regressing $\mathbf{y}$ (not $\tilde{\mathbf{y}}$) on $\tilde{\mathbf{X}}_1$ gives the coefficient $\hat{\beta}_1$. Show that its residual vector is $(\mathbf{I} - \mathbf{M})\mathbf{y} + \mathbf{M}_2\mathbf{y}$, so the residuals are generally wrong even though the coefficient is right.

Solution. The fitted vector of the regression of $\mathbf{y}$ on $\tilde{\mathbf{X}}_1$ is $\tilde{\mathbf{M}}_1 \mathbf{y}$. By Lemma 6.6.1, $\mathbf{y} - \tilde{\mathbf{M}}_1 \mathbf{y} = \mathbf{y} - (\mathbf{M} - \mathbf{M}_2) \mathbf{y} = (\mathbf{I} - \mathbf{M}) \mathbf{y} + \mathbf{M}_2 \mathbf{y}$. The extra term $\mathbf{M}_2 \mathbf{y}$ is zero only if $\mathbf{y} \perp \mathcal{C}(\mathbf{X}_2)$.

Exercise B3.

For a single regressor $\mathbf{x}_j$, the *partial correlation* of $\mathbf{y}$ and $\mathbf{x}_j$ given the other regressors is the sample correlation of $\tilde{\mathbf{y}}$ and $\tilde{\mathbf{x}}_j$ (both residualized on $\mathbf{X}_{(j)}$, which contains $\mathbf{1}$). Let $t = \hat{\beta}_j / \text{se}(\hat{\beta}_j)$ with $\text{se}(\hat{\beta}_j)^2 = \text{SSE} / [(n - p) \|\tilde{\mathbf{x}}_j\|^2]$. Show that the squared partial correlation equals $t^2 / (t^2 + n - p)$.

Solution. Write $\tilde{\mathbf{x}} = \tilde{\mathbf{x}}_j$ and $\tilde{\mathbf{y}} = \tilde{\mathbf{y}}$. Both have mean zero because $\mathbf{1} \in \mathcal{C}(\mathbf{X}_{(j)})$, so their correlation is $r = \tilde{\mathbf{x}}^\top \tilde{\mathbf{y}} / (\|\tilde{\mathbf{x}}\| \|\tilde{\mathbf{y}}\|)$. By FWL, $\hat{\beta}_j = \tilde{\mathbf{x}}^\top \tilde{\mathbf{y}} / \|\tilde{\mathbf{x}}\|^2$, and the full-model residual sum of squares equals that of the residualized regression: $\text{SSE} = \|\tilde{\mathbf{y}}\|^2 - (\tilde{\mathbf{x}}^\top \tilde{\mathbf{y}})^2 / \|\tilde{\mathbf{x}}\|^2 = \|\tilde{\mathbf{y}}\|^2 (1 - r^2)$. Hence

$$t^2 = \frac{\hat{\beta}_j^2 (n - p) \|\tilde{\mathbf{x}}\|^2}{\text{SSE}} = \frac{(n - p) r^2}{1 - r^2},$$

and solving gives $r^2 = t^2 / (t^2 + n - p)$.

6.6.4.3 C. Going deeper

Exercise C1.

Drop the full-rank assumption in Theorem 6.6.2. Show that $\mathbf{b}_1$ solves the normal equations of the regression of $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{X}}_1$ iff there exists $\mathbf{b}_2$ such that $(\mathbf{b}_1^\top, \mathbf{b}_2^\top)^\top$ is a least squares estimate in the full model.

6.7 Reparameterization, centring and R^2

6.7.1 Same subspace, same fit

The key idea of Section 6.1 now becomes a theorem.

Theorem 6.7.1 — Invariance under reparameterization

Let $\mathbf{X}$ ($n \times p$) and $\mathbf{W}$ ($n \times q$) satisfy $\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{W})$. Then the two linear models $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $E(\mathbf{Y}) = \mathbf{W}\boldsymbol{\gamma}$ have the same projection $\mathbf{M}$, the same fitted values and residuals for every $\mathbf{y}$, and the same residual sum of squares. There are a $p \times q$ matrix $\mathbf{T}$ with $\mathbf{W} = \mathbf{X}\mathbf{T}$ and a $q \times p$ matrix $\mathbf{S}$ with $\mathbf{X} = \mathbf{W}\mathbf{S}$. If $\hat{\boldsymbol{\gamma}}$ is a least squares estimate for $\mathbf{W}$, then $\mathbf{T}\hat{\boldsymbol{\gamma}}$ is one for $\mathbf{X}$, and $\mathbf{S}\hat{\boldsymbol{\beta}}$ is one for $\mathbf{W}$ whenever $\hat{\boldsymbol{\beta}}$ is one for $\mathbf{X}$. If both matrices have full column rank, then $p = q$, $\mathbf{T} = \mathbf{S}^{-1}$, and $\hat{\boldsymbol{\gamma}} = \mathbf{S}\hat{\boldsymbol{\beta}}$.

Proof. A projection is determined by its range (Theorem 6.3.3), so both models have projection $\mathbf{M}$, and fitted values and residuals follow from Theorem 6.4.1. Each column

of $\mathbf{W}$ lies in $\mathcal{C}(\mathbf{X})$, so $\mathbf{W} = \mathbf{X}\mathbf{T}$ for some $\mathbf{T}$, and likewise $\mathbf{X} = \mathbf{W}\mathbf{S}$. If $\mathbf{W}\hat{\boldsymbol{\gamma}} = \mathbf{M}\mathbf{y}$, then $\mathbf{X}(\mathbf{T}\hat{\boldsymbol{\gamma}}) = \mathbf{M}\mathbf{y}$, so $\mathbf{T}\hat{\boldsymbol{\gamma}}$ is a least squares estimate for $\mathbf{X}$. The other direction is symmetric. With full column rank, $\mathbf{X} = \mathbf{X}\mathbf{T}\mathbf{S}$ gives $\mathbf{X}(\mathbf{I} - \mathbf{T}\mathbf{S}) = \mathbf{0}$ and so $\mathbf{T}\mathbf{S} = \mathbf{I}$, and similarly $\mathbf{S}\mathbf{T} = \mathbf{I}$. Uniqueness of least squares estimates then gives $\hat{\boldsymbol{\gamma}} = \mathbf{S}\hat{\boldsymbol{\beta}}$. ■

Many operations that change the printed coefficient table leave the fit exactly unchanged:

- **Changing units** of a regressor (metres to centimetres, dollars to thousands of dollars) rescales one column: $\mathbf{W} = \mathbf{X}\mathbf{D}$ with $\mathbf{D}$ diagonal. The coefficient is divided by the scale factor, and nothing else changes.
- **Centring** regressors, $\mathbf{x}_j \mapsto \mathbf{x}_j - \bar{x}_j\mathbf{1}$, when the model has an intercept. The slopes are unchanged, and the intercept becomes the fitted value at the mean of the regressors instead of at zero, which is often far outside the data.
- **Recoding a factor**. Reference-level dummies, sum-to-zero effect coding, and cell-means coding without an intercept all span the space of vectors that are constant within levels.
- **Replacing polynomial terms by orthogonal polynomials** (Chapter 42), which span the same space with much better numerical behaviour (Section 6.10).

Operations that *do* change the fit are exactly those that change $\mathcal{C}(\mathbf{X})$. Examples are dropping the intercept when no other combination of columns produces $\mathbf{1}$, replacing a regressor by its logarithm, and adding an interaction.

6.7.2 Centring as a projection

Suppose $\mathbf{1} \in \mathcal{C}(\mathbf{X})$. By Theorem 6.5.1 with $\mathbf{X}_0 = \mathbf{1}$, the fitted vector splits as

$$\mathbf{M}\mathbf{y} = \bar{y}\mathbf{1} + (\mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{y},$$

and the second projection is onto $\mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$, the vectors in the model space with entries summing to zero. By Lemma 6.6.1 with $\mathbf{X}_2 = \mathbf{1}$, that subspace is spanned by the centred columns $(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{X}$. Applying Pythagoras to the orthogonal pieces $\hat{\mathbf{Y}} - \bar{y}\mathbf{1}$ and $\hat{\boldsymbol{\varepsilon}}$ of $\mathbf{y} - \bar{y}\mathbf{1}$ gives the *corrected* decomposition

$$\underbrace{\sum_i (y_i - \bar{y})^2}_{\text{SST}} = \underbrace{\sum_i (\hat{y}_i - \bar{y})^2}_{\text{SSR}} + \underbrace{\sum_i (y_i - \hat{y}_i)^2}_{\text{SSE}}, \quad (6.7.1)$$

with dimensions $n - 1 = (r - 1) + (n - r)$. Chapter 9 builds the analysis of variance on this identity.

6.7.3 R^2 is a squared cosine

The **coefficient of determination** of a model with $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ is

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}.$$

It depends on the model only through $\mathcal{C}(\mathbf{X})$, so by Theorem 6.7.1 it is invariant under every reparameterization. Its geometric meaning is simple.

inside $\mathbf{1}^\perp$: $R^2 = \cos^2 \theta = 0.617$

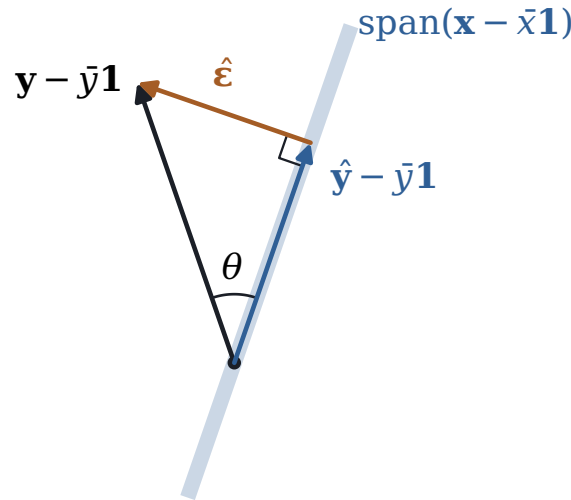


Figure 6.7.1: R^2 as a squared cosine, for $n = 3$ and one regressor. Centred vectors lie in the plane $\mathbf{1}^\perp$, so the whole configuration is two-dimensional. The centred fitted vector is the projection of the centred response onto the line spanned by the centred regressor.

Theorem 6.7.2

Assume $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{y} \notin \text{span}(\mathbf{1})$. Let θ be the angle between the centred response $\mathbf{y} - \bar{y}\mathbf{1}$ and the centred fitted vector $\hat{\mathbf{Y}} - \bar{y}\mathbf{1}$. Then

$$R^2 = \cos^2 \theta = \text{corr}(\mathbf{y}, \hat{\mathbf{Y}})^2,$$

where corr is the sample correlation. For simple regression on one regressor $\mathbf{x}$, $R^2 = \text{corr}(\mathbf{y}, \mathbf{x})^2$.

Proof. Write $\mathbf{u} = \mathbf{y} - \bar{y}\mathbf{1}$ and $\mathbf{v} = \hat{\mathbf{Y}} - \bar{y}\mathbf{1}$. Then $\mathbf{u} = \mathbf{v} + \hat{\boldsymbol{\varepsilon}}$ with $\mathbf{v} \perp \hat{\boldsymbol{\varepsilon}}$, so $\mathbf{u}^\top \mathbf{v} = \|\mathbf{v}\|^2$ and

$$\cos^2 \theta = \frac{(\mathbf{u}^\top \mathbf{v})^2}{\|\mathbf{u}\|^2 \|\mathbf{v}\|^2} = \frac{\|\mathbf{v}\|^2}{\|\mathbf{u}\|^2} = \frac{\text{SSR}}{\text{SST}}.$$

(If $\mathbf{v} = \mathbf{0}$ the angle is undefined, but $R^2 = 0$ and the correlation is taken to be zero.) Since $\hat{\mathbf{Y}}$ has mean $\bar{y}$ (Proposition 6.4.3(b)), $\cos \theta$ is the sample correlation of $\mathbf{y}$ and $\hat{\mathbf{Y}}$. In simple regression $\mathbf{v} = (S_{xy}/S_{xx})(\mathbf{x} - \bar{x}\mathbf{1})$ by Example (Intercept inside simple regression), a scalar multiple of the centred regressor. Rescaling a vector by a nonzero constant does not change a squared cosine, so $\cos^2 \theta$ is the squared correlation of $\mathbf{y}$ and $\mathbf{x}$. When $S_{xy} = 0$ both sides are zero. ■

So R^2 measures how closely the model space lines up with the centred data. Figure 6.7.1 draws the case $n = 3$. For the three-regressor model of Example (Poverty and murder rates), $R^2 = 0.6419$, meaning the centred response makes an angle of 36.8° with the model space. The simple regression on poverty alone has $R^2 = 0.3599$, the square of the correlation 0.5999.

```

import numpy as np
import statsmodels.api as sm

def r_squared_as_cosine(y, X):
    """X must contain the intercept column. Returns (R^2 from sums of squares, cos^2
    of angle)."""
    beta, *_ = np.linalg.lstsq(X, y, rcond=None)
    y_hat = X @ beta
    yc, yhc = y - y.mean(), y_hat - y.mean()          # centre both vectors
    r2 = 1 - np.sum((y - y_hat) ** 2) / np.sum(yc ** 2)
    cos = (yc @ yhc) / (np.linalg.norm(yc) * np.linalg.norm(yhc))
    return r2, cos ** 2

```

Three consequences follow directly from the geometry.

1. R^2 never decreases when the model space grows. If $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ and both contain $\mathbf{1}$, then $\text{SSE}_0 - \text{SSE} = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 \geq 0$ by Eq. 6.5.2. Adding a regressor of pure noise increases R^2 almost surely. This is why R^2 cannot be used for model selection (Chapter 29).
2. $R^2 = 1$ once the model space contains $\mathbf{y}$, which always happens when $\text{rank}(\mathbf{X}) = n$. A model with as many free parameters as observations interpolates, and its R^2 says nothing.
3. Without an intercept, the definition changes. If $\mathbf{1} \notin \mathcal{C}(\mathbf{X})$, the corrected decomposition Eq. 6.7.1 fails, because $\hat{\mathbf{Y}} - \bar{y}\mathbf{1}$ and $\hat{\mathbf{e}}$ need not be orthogonal. Software then usually reports the *uncentred* $R_0^2 = \|\hat{\mathbf{Y}}\|^2 / \|\mathbf{y}\|^2$, the squared cosine of the angle between $\mathbf{y}$ and $\mathcal{C}(\mathbf{X})$ itself. It is typically much larger and cannot be compared with the centred R^2 of a model with an intercept (Exercise (B1)).

6.7.4 Exercises

6.7.4.1 A. Check your understanding

Exercise A1.

A factor with three levels is coded with an intercept and indicators for levels 2 and 3. Write down the matrices $\mathbf{T}$ and $\mathbf{S}$ of Theorem 6.7.1 that convert to the coding in which level 3, rather than level 1, is the omitted baseline, and to sum-to-zero effect coding. Express each new coefficient in terms of the old ones.

Exercise A2.

Show that, for a model with $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, R^2 is unchanged when $\mathbf{y}$ is replaced by $a\mathbf{1} + b\mathbf{y}$ with $b \neq 0$, and when any column of $\mathbf{X}$ is replaced by an affine function of itself. Is it unchanged when a regressor is replaced by its logarithm?

6.7.4.2 B. Practice

Exercise B1.

Show that if $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ then $R_0^2 = \|\hat{\mathbf{Y}}\|^2 / \|\mathbf{y}\|^2 = (n\bar{y}^2 + \text{SSR}) / (n\bar{y}^2 + \text{SST}) \geq R^2$.

Construct a data set and a model without an intercept for which $R_0^2 > 0.95$ but the model's residual sum of squares exceeds that of the intercept-only model.

Solution. With $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, the fitted vector splits orthogonally as $\bar{y}\mathbf{1} + (\hat{\mathbf{Y}} - \bar{y}\mathbf{1})$, so $\|\hat{\mathbf{Y}}\|^2 = n\bar{y}^2 + \text{SSR}$, and likewise $\|\mathbf{y}\|^2 = n\bar{y}^2 + \text{SST}$. For $a \geq 0$ and $0 \leq \text{SSR} \leq \text{SST}$, $(a + \text{SSR})/(a + \text{SST}) \geq \text{SSR}/\text{SST}$. For the example, take responses near 100 with small noise and a regressor unrelated to them, and fit through the origin. The fitted line captures the large mean, so R_0^2 is near 1, but its residuals exceed those around $\bar{y}$.

6.7.4.3 C. Going deeper

Exercise C1.

Suppose $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\mathbf{I})$ and $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ with $\text{rank}(\mathbf{X}) = r$. Using the results of Chapter 4, show that $R^2 \sim \text{Beta}((r-1)/2, (n-r)/2)$, so that $E(R^2) = (r-1)/(n-1)$ even though no regressor is related to the response. Use this to motivate the adjusted $R^2 = 1 - (1 - R^2)(n-1)/(n-r)$.

6.8 Leverage as geometry

The projection $\mathbf{M}$ is an $n \times n$ matrix whose entries have a direct interpretation. Since $\hat{y}_i = \sum_j m_{ij}y_j$, the entry m_{ij} is the weight observation j receives in the fitted value for observation i . So

$$m_{ij} = \frac{\partial \hat{y}_i}{\partial y_j}.$$

The diagonal entries measure how strongly each observation pulls the fitted surface towards itself.

Definition 6.8.1 — Leverage

The **leverage** of observation i is $h_{ii} = m_{ii}$, the i th diagonal entry of $\mathbf{M}$. In this context $\mathbf{M}$ is often called the **hat matrix** and written $\mathbf{H}$, because it “puts the hat on” $\mathbf{y}$.

Leverage is a property of the design, not of the response. It depends on $\mathbf{y}$ not at all, and on $\mathbf{X}$ only through $\mathcal{C}(\mathbf{X})$, so it is unchanged by reparameterization.

Proposition 6.8.1 — Properties of leverage

Let $\text{rank}(\mathbf{X}) = r$, and let $\mathbf{e}_i$ denote the i th standard basis vector. Then:

- (a) $h_{ii} = \|\mathbf{M}\mathbf{e}_i\|^2 = \sum_j m_{ij}^2$, the squared length of the projection of $\mathbf{e}_i$ onto $\mathcal{C}(\mathbf{X})$;
- (b) $0 \leq h_{ii} \leq 1$, and $\sum_i h_{ii} = r$, so the average leverage is r/n ;
- (c) if $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, then $h_{ii} \geq 1/n$;
- (d) $h_{ii} = 1$ iff $\mathbf{e}_i \in \mathcal{C}(\mathbf{X})$. In that case $\hat{y}_i = y_i$ for every $\mathbf{y}$, the i th residual is always zero, and $m_{ij} = 0$ for $j \neq i$;

- (e) if $\mathbf{X} = \mathbf{Q}\mathbf{R}$ with $\mathbf{Q}$ having orthonormal columns spanning $\mathcal{C}(\mathbf{X})$, then h_{ii} is the squared length of the i th row of $\mathbf{Q}$.

Proof.

- (a) $h_{ii} = \mathbf{e}_i^\top \mathbf{M} \mathbf{e}_i = \mathbf{e}_i^\top \mathbf{M}^\top \mathbf{M} \mathbf{e}_i$, and $\mathbf{M} \mathbf{e}_i$ is the i th column of $\mathbf{M}$.
 (b) By Proposition 6.3.4(d), $0 \leq \|\mathbf{M} \mathbf{e}_i\|^2 \leq \|\mathbf{e}_i\|^2 = 1$. The sum is $\text{tr}(\mathbf{M}) = r$.
 (c) With $\mathbf{M}_0 = n^{-1} \mathbf{1} \mathbf{1}^\top$, the difference $\mathbf{M} - \mathbf{M}_0$ is a projection (Theorem 6.5.1), hence nonnegative definite, so $h_{ii} \geq (\mathbf{M}_0)_{ii} = 1/n$.
 (d) Equality holds in $\|\mathbf{M} \mathbf{e}_i\| \leq \|\mathbf{e}_i\|$ iff $(\mathbf{I} - \mathbf{M}) \mathbf{e}_i = \mathbf{0}$, that is, iff $\mathbf{e}_i \in \mathcal{C}(\mathbf{X})$. Then $\mathbf{M} \mathbf{e}_i = \mathbf{e}_i$, so column i of $\mathbf{M}$ is $\mathbf{e}_i$, and by symmetry so is row i . Hence $\hat{y}_i = y_i$.
 (e) $\mathbf{M} = \mathbf{Q}\mathbf{Q}^\top$, so $m_{ii} = \sum_k q_{ik}^2$.

■

Part (d) describes the extreme case. An observation with leverage one has a coordinate direction of its own in the model space. This happens, for example, when it is the only member of a group in a one-way layout, or when a column of $\mathbf{X}$ is an indicator for that single observation. The model then fits the observation exactly, whatever its value, and the data give no check on it. High leverage is a weaker form of the same thing. If h_{ii} is close to one, then $\hat{y}_i \approx y_i$, and a gross error in y_i barely shows up in its residual.

This shows up in the residual variances. If $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, then by Proposition 6.4.3(e)

$$\text{Var}(\hat{\varepsilon}_i) = \sigma^2(1 - h_{ii}),$$

so residuals at high-leverage points are *less* variable than others, even though those are exactly the points where an error would do the most damage. Chapter 20 standardizes residuals to correct for this.

6.8.1 Leverage as distance from the centre

When the model has an intercept, leverage has a second description in variable space: distance from the centroid of the regressors. Write $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$, let $\bar{\mathbf{x}}$ be the vector of column means of $\mathbf{X}_1$, and let $\tilde{\mathbf{X}}_1 = (\mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top) \mathbf{X}_1$ be the centred regressors, assumed to have full column rank.

Proposition 6.8.2

Under these assumptions,

$$h_{ii} = \frac{1}{n} + (\mathbf{x}_{(i)} - \bar{\mathbf{x}})^\top (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} (\mathbf{x}_{(i)} - \bar{\mathbf{x}}),$$

where $\mathbf{x}_{(i)}$ is the i th row of $\mathbf{X}_1$, written as a column vector.

Proof. By Lemma 6.6.1 with $\mathbf{X}_2 = \mathbf{1}$, $\mathbf{M} = n^{-1} \mathbf{1} \mathbf{1}^\top + \tilde{\mathbf{X}}_1 (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top$. The i th row of $\tilde{\mathbf{X}}_1$ is $(\mathbf{x}_{(i)} - \bar{\mathbf{x}})^\top$.

■

The quadratic form is $(n-1)^{-1}$ times the squared Mahalanobis distance of $\mathbf{x}_{(i)}$ from the centroid, measured in the metric of the sample covariance matrix of the regressors. Leverage is therefore high for points that are unusual *jointly*, even when no single coordinate is extreme. A state with moderate poverty and moderate single parenthood, but an unusual *combination* of the two, can have high leverage. For simple regression the formula reduces to $h_{ii} = n^{-1} + (x_i - \bar{x})^2 / S_{xx}$.

Example. (High-leverage states)

Return to the state data, now with all 51 jurisdictions including the District of Columbia, and the model of [Example \(Poverty and murder rates\)](#). The average leverage is $p/n = 0.078$, and the smallest is 0.021, just above the bound $1/n$. The three largest are

Jurisdiction	h_{ii}	multiple of average
District of Columbia	0.506	6.5
Mississippi	0.212	2.7
Alaska	0.177	2.3

The District of Columbia is entirely urban, with by far the highest proportion of single-parent households. It sits at a corner of the regressor cloud that no state comes near, and its leverage is about half the maximum possible. It was omitted from [Example \(Poverty and murder rates\)](#). [Exercise \(A1\)](#) asks what happens to the poverty coefficient when it is put back.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statelcrime.load_pandas().data           # District of Columbia
kept here
X = np.column_stack([np.ones(len(data)), data["poverty"], data["single"], data["urban"]])

Q, _ = np.linalg.qr(X)                                     # orthonormal basis for C(X)
h = np.sum(Q ** 2, axis=1)                                 # h_ii = m_ii = ||row i of Q||^2
n, p = X.shape
print("sum of leverages:", h.sum(), " (= p =", p, ")")
print("range:", h.min(), h.max(), " bounds:", 1 / n, 1)
top = np.argsort(h)[::-1][:3]
for i in top:
    print(f"{data.index[i]:22s} h = {h[i]:.3f}    ({h[i] / (p / n):.1f} x average)")
```

The listing computes leverages from a QR factorization rather than by forming $\mathbf{M}$. For large n the $n \times n$ matrix $\mathbf{M}$ should never be formed. [Section 6.10](#) explains why.

6.8.2 Exercises

6.8.2.1 A. Check your understanding

Exercise A1.

For simple regression, derive $h_{ii} = n^{-1} + (x_i - \bar{x})^2 / S_{xx}$ directly from [Example \(Intercept](#)

inside simple regression), and show that the largest leverage belongs to the observation with x_i farthest from $\bar{x}$.

6.8.2.2 B. Practice

Exercise B1.

Show that $m_{ij}^2 \leq m_{ii}(1 - m_{ii})$ for $i \neq j$, and hence $|m_{ij}| \leq 1/2$. When is the bound attained?

Solution. From Proposition 6.8.1(a), $m_{ii} = \sum_k m_{ik}^2 \geq m_{ii}^2 + m_{ij}^2$. So $m_{ij}^2 \leq m_{ii}(1 - m_{ii}) \leq 1/4$. Equality needs $m_{ii} = 1/2$ and m_{ij} the only other nonzero entry of row i . The intercept-only model with $n = 2$, where every entry of $\mathbf{M}$ is $1/2$, attains it.

Exercise B2.

Show that adding a column to $\mathbf{X}$ cannot decrease any leverage. Show that the leverages in a one-way layout are $1/n_k$ for members of group k , and explain why an observation forming a group on its own has leverage one.

6.9 Other inner products

Everything so far has used the Euclidean inner product $\mathbf{u}^\top \mathbf{v}$. That choice treats the n coordinates of observation space symmetrically: an error of a given size counts the same in every observation, and errors in different observations are unrelated. When observations have different variances or are correlated, Euclidean distance is the wrong way to measure how far $\mathbf{y}$ is from the model space. This section shows that the whole theory carries over to any inner product, and that the result is generalized least squares.

6.9.1 Projection in a general inner product

Let $\mathbf{A}$ be an $n \times n$ symmetric positive definite matrix, and define

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{A}} = \mathbf{u}^\top \mathbf{A} \mathbf{v}, \quad \|\mathbf{u}\|_{\mathbf{A}} = (\mathbf{u}^\top \mathbf{A} \mathbf{u})^{1/2}.$$

This is an inner product: bilinear, symmetric, and positive on nonzero vectors. Say $\mathbf{u}$ and $\mathbf{v}$ are $\mathbf{A}$ -orthogonal if $\mathbf{u}^\top \mathbf{A} \mathbf{v} = 0$. Nothing in the proofs of Proposition 6.2.2, Theorem 6.2.3 or Theorem 6.3.1 used anything about $\mathbf{u}^\top \mathbf{v}$ except these three properties. So those results hold verbatim with $\langle \cdot, \cdot \rangle_{\mathbf{A}}$ in place of the Euclidean inner product. In particular, every $\mathbf{y}$ has a unique $\mathbf{A}$ -nearest point in a subspace $\mathcal{S}$, characterized by $\mathbf{A}$ -orthogonality of the error to $\mathcal{S}$. The map sending $\mathbf{y}$ to it is linear, and we write it $\mathbf{P}_{\mathcal{S}, \mathbf{A}}$.

Theorem 6.9.1

Let $\mathbf{A}$ be positive definite and $\mathbf{X}$ any $n \times p$ matrix. The $\mathbf{A}$ -orthogonal projection onto

$\mathcal{C}(\mathbf{X})$ is

$$\mathbf{P}_{\mathbf{X},\mathbf{A}} = \mathbf{X}(\mathbf{X}^\top \mathbf{A} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{A},$$

for any generalized inverse. It satisfies $\mathbf{P}_{\mathbf{X},\mathbf{A}}^2 = \mathbf{P}_{\mathbf{X},\mathbf{A}}$ and $\mathcal{C}(\mathbf{P}_{\mathbf{X},\mathbf{A}}) = \mathcal{C}(\mathbf{X})$, and the matrix $\mathbf{A}\mathbf{P}_{\mathbf{X},\mathbf{A}}$ is symmetric. The matrix $\mathbf{P}_{\mathbf{X},\mathbf{A}}$ itself is symmetric iff $\mathcal{A}\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$, and in general it is not.

Proof. Let $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$ with $\mathbf{L}$ nonsingular (for instance the Cholesky factor), and put $\mathbf{X}_* = \mathbf{L}^\top \mathbf{X}$ and $\mathbf{y}_* = \mathbf{L}^\top \mathbf{y}$. Then $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|_{\mathbf{A}}^2 = \|\mathbf{y}_* - \mathbf{X}_*\mathbf{b}\|^2$, so $\mathbf{b}$ minimizes the $\mathbf{A}$ -distance iff it is an ordinary least squares estimate for $(\mathbf{y}_*, \mathbf{X}_*)$. By [Theorem 6.3.7](#) and [Theorem 6.4.2](#), one such estimate is $(\mathbf{X}_*^\top \mathbf{X}_*)^{-1} \mathbf{X}_*^\top \mathbf{y}_* = (\mathbf{X}^\top \mathbf{A} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{A} \mathbf{y}$, and $\mathbf{X}_*(\mathbf{X}_*^\top \mathbf{X}_*)^{-1} \mathbf{X}_*^\top$ is invariant to the choice of generalized inverse. Multiplying on the left by $\mathbf{L}^{-\top}$ and on the right by $\mathbf{L}^\top$ shows that $\mathbf{P}_{\mathbf{X},\mathbf{A}} = \mathbf{L}^{-\top} \mathbf{M}_* \mathbf{L}^\top$, where $\mathbf{M}_*$ projects onto $\mathcal{C}(\mathbf{X}_*)$. Idempotence and $\mathcal{C}(\mathbf{P}_{\mathbf{X},\mathbf{A}}) = \mathbf{L}^{-\top} \mathcal{C}(\mathbf{X}_*) = \mathcal{C}(\mathbf{X})$ follow, and $\mathbf{A}\mathbf{P}_{\mathbf{X},\mathbf{A}} = \mathbf{L}\mathbf{M}_*\mathbf{L}^\top$ is symmetric. For the last claim, see [Exercise \(B1\)](#). ■

The proof also gives the general principle: *$\mathbf{A}$ -geometry is Euclidean geometry after the change of variables $\mathbf{y} \mapsto \mathbf{L}^\top \mathbf{y}$.* Any fact about orthogonal projections has an $\mathbf{A}$ -version, obtained by transporting it through $\mathbf{L}$.

Symmetry is the Euclidean signature of an orthogonal projection. The $\mathbf{A}$ -version of [Theorem 6.3.3](#) replaces it with self-adjointness.

Proposition 6.9.2

An idempotent matrix $\mathbf{P}$ is the $\mathbf{A}$ -orthogonal projection onto $\mathcal{C}(\mathbf{P})$ iff $\mathbf{A}\mathbf{P}$ is symmetric, that is, iff $\langle \mathbf{P}\mathbf{u}, \mathbf{v} \rangle_{\mathbf{A}} = \langle \mathbf{u}, \mathbf{P}\mathbf{v} \rangle_{\mathbf{A}}$ for all $\mathbf{u}, \mathbf{v}$.

Proof. For idempotent $\mathbf{P}$, the residual $(\mathbf{I} - \mathbf{P})\mathbf{y}$ is $\mathbf{A}$ -orthogonal to $\mathcal{C}(\mathbf{P})$ for every $\mathbf{y}$ iff $\mathbf{P}^\top \mathbf{A}(\mathbf{I} - \mathbf{P}) = \mathbf{0}$, that is, iff $\mathbf{P}^\top \mathbf{A} = \mathbf{P}^\top \mathbf{A} \mathbf{P}$. If $\mathbf{A}\mathbf{P}$ is symmetric, then $\mathbf{P}^\top \mathbf{A} = \mathbf{A}\mathbf{P}$, and $\mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{A}\mathbf{P}\mathbf{P} = \mathbf{A}\mathbf{P} = \mathbf{P}^\top \mathbf{A}$. Conversely, $\mathbf{P}^\top \mathbf{A} = \mathbf{P}^\top \mathbf{A} \mathbf{P}$ has a symmetric right-hand side, so $\mathbf{P}^\top \mathbf{A}$ and hence its transpose $\mathbf{A}\mathbf{P}$ are symmetric. ■

6.9.2 Generalized least squares

Suppose $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$ with $\mathbf{V}$ known and positive definite. Observations with large variance should count for less, and correlated observations partly repeat each other. The natural distance is the one that makes the errors look uncorrelated: $\|\cdot\|_{\mathbf{V}^{-1}}$. If $\mathbf{V} = \mathbf{L}\mathbf{L}^\top$, then $\mathbf{L}^{-1}\boldsymbol{\varepsilon}$ has covariance $\sigma^2 \mathbf{I}$, and $\|\mathbf{u}\|_{\mathbf{V}^{-1}} = \|\mathbf{L}^{-1}\mathbf{u}\|$.

Definition 6.9.1 — Generalized least squares

A **generalized least squares** (GLS) estimate minimizes $(\mathbf{y} - \mathbf{X}\mathbf{b})^\top \mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\mathbf{b})$. Equivalently, $\mathbf{X}\hat{\boldsymbol{\beta}}_{\text{GLS}} = \mathbf{P}_{\mathbf{X},\mathbf{V}^{-1}}\mathbf{y}$, and $\hat{\boldsymbol{\beta}}_{\text{GLS}}$ solves the generalized normal equations

$$\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{y}.$$

Theorem 6.9.1 with $\mathbf{A} = \mathbf{V}^{-1}$ is the whole of the algebra. Computationally, GLS is ordinary least squares applied to the *whitened* data $(\mathbf{L}^{-1}\mathbf{y}, \mathbf{L}^{-1}\mathbf{X})$. Aitken (1935) showed that GLS gives the best linear unbiased estimator under this covariance, which is the Gauss–Markov theorem transported through $\mathbf{L}$. Chapter 31 proves this, and handles singular $\mathbf{V}$ and the more realistic case where $\mathbf{V}$ must be estimated.

When is the ordinary least squares fit already the GLS fit? Geometrically, the two projections onto $\mathcal{C}(\mathbf{X})$ coincide iff the Euclidean and the $\mathbf{V}^{-1}$ notions of orthogonality agree on this particular subspace.

Theorem 6.9.3 — Kruskal

For positive definite $\mathbf{V}$, $\mathbf{M}\mathbf{y} = \mathbf{P}_{\mathbf{X}, \mathbf{V}^{-1}}\mathbf{y}$ for every $\mathbf{y}$ iff $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$.

Proof. The two projections have the same range. They agree iff they have the same null space, since an idempotent matrix is determined by its range and null space (**Proposition 6.9.4**). The null space of $\mathbf{M}$ is $\mathcal{C}(\mathbf{X})^\perp = \mathcal{N}(\mathbf{X}^\top)$. The null space of $\mathbf{P}_{\mathbf{X}, \mathbf{V}^{-1}}$ is the set of $\mathbf{u}$ with $\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{u} = \mathbf{0}$, that is, $\mathbf{V} \mathcal{N}(\mathbf{X}^\top)$. So the projections agree iff $\mathbf{V} \mathcal{N}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X}^\top)$, and since $\mathbf{V}$ is nonsingular and the spaces have equal dimension, iff $\mathbf{V} \mathcal{N}(\mathbf{X}^\top) \subseteq \mathcal{N}(\mathbf{X}^\top)$. By symmetry of $\mathbf{V}$, this says $\mathbf{X}^\top \mathbf{V} \mathbf{u} = \mathbf{0}$ whenever $\mathbf{X}^\top \mathbf{u} = \mathbf{0}$, that is, $\mathcal{C}(\mathbf{V}\mathbf{X}) \perp \mathcal{C}(\mathbf{X})^\perp$, which is $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$. ■

The condition holds, for example, for equicorrelated errors $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$ whenever $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ (**Exercise A1**). It also underlies the analysis of balanced split-plot designs in Chapter 33. The result is due to Kruskal (1968), and the coordinate-free argument is his.

Example. (Autocorrelated errors)

Take eight equally spaced observations with a linear trend and error covariance $v_{st} = 0.7^{|s-t|}$. The $\mathbf{V}^{-1}$ -projection onto $\mathcal{C}(\mathbf{X})$ is idempotent and self-adjoint for $\langle \cdot, \cdot \rangle_{\mathbf{V}^{-1}}$ but not symmetric: its largest asymmetry $\max |p_{st} - p_{ts}|$ is 0.444. The ordinary and generalized slope estimates are 0.648 and 0.650. The listing checks each claim, including that the GLS fit is ordinary least squares on whitened data.

```
import numpy as np
rng = np.random.default_rng(9)
n = 8
X = np.column_stack([np.ones(n), np.arange(n, dtype=float)])
y = X @ np.array([1.0, 0.5]) + rng.normal(size=n)
t = np.arange(n)
V = 0.7 ** np.abs(t[:, None] - t[None, :])           # AR(1)-type covariance

Vinv = np.linalg.inv(V)
P_V = X @ np.linalg.solve(X.T @ Vinv @ X, X.T @ Vinv) # V^{-1}-orthogonal
               projection onto C(X)

print("idempotent:", np.allclose(P_V @ P_V, P_V))
```

```

print("symmetric: ", np.allclose(P_V, P_V.T))                # False: oblique in
                    the usual geometry
print("self-adjoint for <u,v> = u'V^{-1}v:", np.allclose(Vinv @ P_V, (Vinv @ P_V).T)
    )

# GLS = ordinary least squares after whitening with a square root of V^{-1}
L = np.linalg.cholesky(V)                                    # V = L L'
Xw, yw = np.linalg.solve(L, X), np.linalg.solve(L, y)
beta_whitened, *_ = np.linalg.lstsq(Xw, yw, rcond=None)
beta_gls = np.linalg.solve(X.T @ Vinv @ X, X.T @ Vinv @ y)
print("GLS via P_V equals OLS on whitened data:", np.allclose(beta_gls,
    beta_whitened))

```

6.9.3 Weighted least squares

The simplest non-Euclidean geometry is a diagonal one. If the errors are uncorrelated with variances σ^2/w_i , then $\mathbf{V}^{-1} = \mathbf{W} = \text{diag}(w_1, \dots, w_n)$, and GLS minimizes $\sum_i w_i (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2$. Observations believed to be noisier get less weight in the distance. Whitening multiplies row i of $\mathbf{X}$ and $\mathbf{y}$ by $\sqrt{w_i}$. The fitted values are $\mathbf{P}_{\mathbf{X}, \mathbf{W}} \mathbf{y}$, and the natural leverages are the diagonal entries of $\mathbf{P}_{\mathbf{X}, \mathbf{W}}$. These equal the diagonal of the orthogonal projection onto $\mathcal{C}(\mathbf{W}^{1/2} \mathbf{X})$ and still sum to $\text{rank}(\mathbf{X})$ (Exercise (B3)).

Example. (Engel's food expenditure data)

Engel's 1857 survey of 235 Belgian working-class households records annual income and food expenditure. The spread of expenditure grows with income, as Figure 6.9.1(a) shows. Weighting by $w_i = 1/\text{income}_i^2$ (standard deviation proportional to income) changes the fitted slope from 0.4852 to 0.5740. The change in geometry shows up more clearly in the leverages, panel (b). Under ordinary least squares the richest households have the most leverage, with a maximum of 0.255. In the weighted geometry their large variance shrinks their influence, and no household has leverage above 0.037. Whether this is the *right* geometry depends on whether the variance model holds. Chapter 21 discusses how to check it, and how to get valid standard errors when it fails.

```

import numpy as np
import statsmodels.api as sm
df = sm.datasets.engel.load_pandas().data
x, y = df["income"].to_numpy(), df["foodexp"].to_numpy()
X = np.column_stack([np.ones(len(x)), x])

w = 1.0 / x**2                # working model: sd proportional to
    income
sw = np.sqrt(w)
beta_ols, *_ = np.linalg.lstsq(X, y, rcond=None)
beta_wls, *_ = np.linalg.lstsq(X * sw[:, None], y * sw, rcond=None) # OLS on (W
    ^{1/2} X, W^{1/2} y)

# leverages in the weighted geometry: squared row norms of an orthonormal basis of C
    (W^{1/2} X)
Qw, _ = np.linalg.qr(X * sw[:, None])
h_wls = np.sum(Qw**2, axis=1)
Q, _ = np.linalg.qr(X)

```

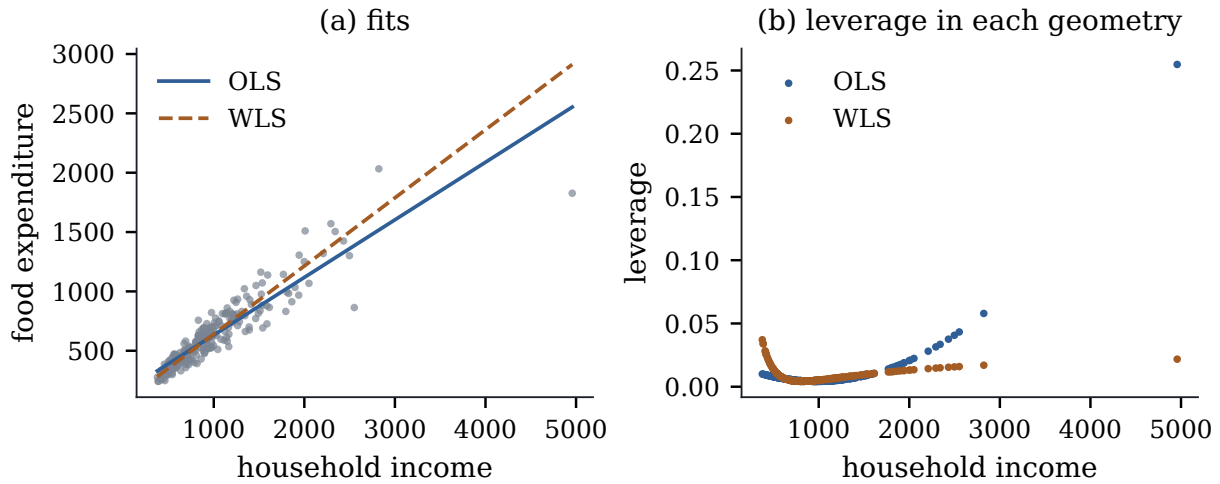


Figure 6.9.1: Engel's data. (a) Ordinary and weighted least squares fits, with weights proportional to $1/\text{income}^2$. (b) Leverages under each geometry. The weighted projection spreads influence away from the high-variance, high-income households.

```
h_ols = np.sum(Q**2, axis=1)
print("OLS:", beta_ols, " WLS:", beta_wls)
print("max leverage OLS %.3f, WLS %.3f" % (h_ols.max(), h_wls.max()))
```

6.9.4 Oblique projections

The matrices $P_{X,A}$ are idempotent without being symmetric. In the Euclidean geometry of $\mathbb{R}^n$ they are **oblique projections**. Idempotent matrices in general have a clean description.

Proposition 6.9.4

Let P be idempotent. Then $\mathbb{R}^n = \mathcal{C}(P) \oplus \mathcal{N}(P)$, and P maps each $y = u + v$ ($u \in \mathcal{C}(P)$, $v \in \mathcal{N}(P)$) to u . Conversely, for any decomposition $\mathbb{R}^n = \mathcal{S} \oplus \mathcal{T}$ there is exactly one idempotent P with $\mathcal{C}(P) = \mathcal{S}$ and $\mathcal{N}(P) = \mathcal{T}$, the *projection onto $\mathcal{S}$ along $\mathcal{T}$* . It is an orthogonal projection iff $\mathcal{T} = \mathcal{S}^\perp$, that is, iff P is symmetric.

Proof. Write $y = Py + (I - P)y$. The first term lies in $\mathcal{C}(P)$, and $P(I - P)y = 0$ puts the second in $\mathcal{N}(P)$. If $u = Pw \in \mathcal{N}(P)$ then $u = P^2w = Pu = 0$, so the sum is direct. For $u \in \mathcal{C}(P)$, $Pu = u$ (write $u = Pw$), so $P(u + v) = u$. Conversely, given $\mathcal{S} \oplus \mathcal{T}$, the map $u + v \mapsto u$ is well defined and linear by uniqueness of the decomposition, and it is idempotent with the required range and null space. It is unique because its values are forced on all of $\mathbb{R}^n$. If $\mathcal{T} = \mathcal{S}^\perp$, the map is the orthogonal projection, which is symmetric. If P is symmetric, then $\mathcal{N}(P) = \mathcal{N}(P^\top) = \mathcal{C}(P)^\perp$ by Lemma 6.2.1. ■

Figure 6.9.2 shows the difference. The nearest-point property belongs to the orthogonal projection alone. An oblique projection can move a vector *farther* away: its operator norm exceeds one unless it is orthogonal (Exercise (B2)). The GLS projection is still a nearest-point map, but in the V^{-1} norm, not the Euclidean one. Which geometry is “right” is not a mathematical

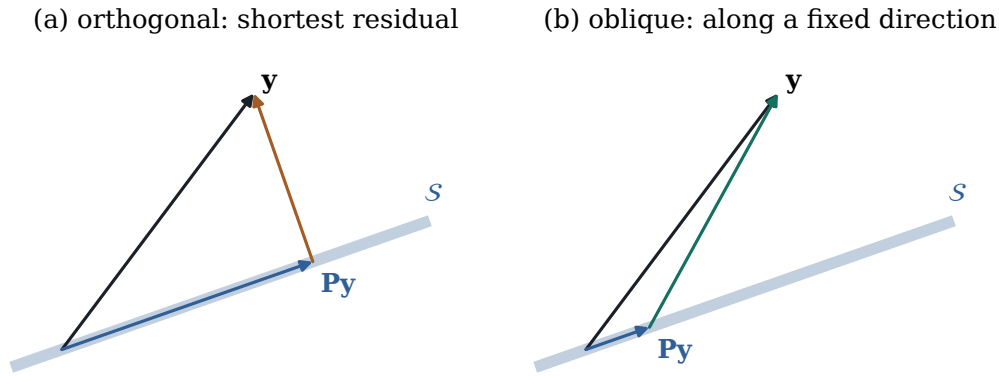


Figure 6.9.2: Projections onto a line $\mathcal{S}$ in $\mathbb{R}^2$. (a) The orthogonal projection: the residual is perpendicular to $\mathcal{S}$, and $\mathbf{Py}$ is the nearest point. (b) An oblique projection moves $\mathbf{y}$ along a fixed direction that is not perpendicular to $\mathcal{S}$. The result is idempotent but is not the nearest point, and the residual is longer.

question. It is decided by the covariance of the errors.

6.9.5 Exercises

6.9.5.1 A. Check your understanding

Exercise A1.

Let $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$ with $-1/(n-1) < \rho < 1$. Show that $\mathbf{V}$ is positive definite and that $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ whenever $\mathbf{1} \in \mathcal{C}(\mathbf{X})$. Conclude that ordinary least squares is generalized least squares for equicorrelated errors in any model with an intercept.

Solution. $\mathbf{V}\mathbf{1} = (1 + (n-1)\rho)\mathbf{1}$ and $\mathbf{V}\mathbf{u} = (1 - \rho)\mathbf{u}$ for $\mathbf{u} \perp \mathbf{1}$. So the eigenvalues are $1 + (n-1)\rho$ and $1 - \rho$, both positive exactly when $-1/(n-1) < \rho < 1$. Then $\mathbf{V}\mathbf{X} = (1 - \rho)\mathbf{X} + \rho\mathbf{1}(\mathbf{1}^\top\mathbf{X})$, whose columns lie in $\mathcal{C}(\mathbf{X})$ when $\mathbf{1}$ does. [Theorem 6.9.3](#) applies.

6.9.5.2 B. Practice

Exercise B1.

Complete the proof of [Theorem 6.9.1](#). Show that $\mathbf{P}_{\mathbf{X},\mathbf{A}}$ is symmetric iff it equals $\mathbf{M}$, and deduce from [Theorem 6.9.3](#) (with $\mathbf{V} = \mathbf{A}^{-1}$) that this happens iff $\mathcal{AC}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$.

Exercise B2.

Let $\mathbf{P} \neq \mathbf{0}$ be idempotent. Show that the operator norm $\|\mathbf{P}\|_2 = \max_{\|\mathbf{u}\|=1} \|\mathbf{P}\mathbf{u}\|$ is at least 1, with equality iff $\mathbf{P}$ is symmetric. *Hint:* if $\mathbf{P}$ is not symmetric, find $\mathbf{u} \in \mathcal{C}(\mathbf{P})$ and $\mathbf{v} \in \mathcal{N}(\mathbf{P})$ with $\mathbf{u}^\top \mathbf{v} \neq 0$, and consider $\mathbf{u} + t\mathbf{v}$ for small t .

Solution. For $\mathbf{u} \in \mathcal{C}(\mathbf{P})$, $\mathbf{P}\mathbf{u} = \mathbf{u}$, so $\|\mathbf{P}\|_2 \geq 1$. If $\mathbf{P}$ is symmetric, $\|\mathbf{P}\mathbf{u}\| \leq \|\mathbf{u}\|$ by Proposition 6.3.4(d). Suppose $\mathbf{P}$ is not symmetric. Then $\mathcal{N}(\mathbf{P}) \neq \mathcal{C}(\mathbf{P})^\perp$ by Proposition 6.9.4. The two spaces have the same dimension, so some $\mathbf{v} \in \mathcal{N}(\mathbf{P})$ is not orthogonal to $\mathcal{C}(\mathbf{P})$. Choose a unit $\mathbf{u} \in \mathcal{C}(\mathbf{P})$ with $\mathbf{u}^\top \mathbf{v} > 0$. For $t > 0$, $\mathbf{P}(\mathbf{u} - t\mathbf{v}) = \mathbf{u}$, while $\|\mathbf{u} - t\mathbf{v}\|^2 = 1 - 2t\mathbf{u}^\top \mathbf{v} + t^2\|\mathbf{v}\|^2 < 1$ for small t . So $\mathbf{P}$ stretches $\mathbf{u} - t\mathbf{v}$, and $\|\mathbf{P}\|_2 > 1$.

Exercise B3.

For weighted least squares, $\mathbf{V} = \text{diag}(1/w_1, \dots, 1/w_n)$ with $w_i > 0$. Show that the diagonal entries of $\mathbf{P}_{\mathbf{X}, \mathbf{V}^{-1}}$ equal those of the orthogonal projection onto $\mathcal{C}(\mathbf{W}^{1/2}\mathbf{X})$, where $\mathbf{W} = \text{diag}(w_i)$, and that they sum to $\text{rank}(\mathbf{X})$. Interpret them as leverages.

6.10 Computation

The formulas $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ and $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ are the right way to *reason* about least squares and the wrong way to *compute* it. There are two problems. $\mathbf{M}$ is $n \times n$, so forming it costs $O(n^2)$ memory for a quantity that is never needed in full. And forming $\mathbf{X}^\top \mathbf{X}$ squares the condition number of the problem, which can throw away half of the available floating-point digits before any equation is solved. This section shows how the geometry itself suggests the fix, which is to build an orthonormal basis for $\mathcal{C}(\mathbf{X})$. Chapter 10 develops the numerical linear algebra in full.

6.10.1 QR: Gram–Schmidt as a factorization

Apply Gram–Schmidt (Proposition 6.2.2) to the columns $\mathbf{x}_1, \dots, \mathbf{x}_p$ of a full-column-rank $\mathbf{X}$. Each $\mathbf{x}_j$ is a combination of $\mathbf{u}_1, \dots, \mathbf{u}_j$ with coefficients $r_{ij} = \mathbf{u}_i^\top \mathbf{x}_j$ for $i < j$ and $r_{jj} = \|\mathbf{w}_j\| > 0$. In matrix form,

$$\mathbf{X} = \mathbf{Q}\mathbf{R},$$

where $\mathbf{Q} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$ is $n \times p$ with $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$, and $\mathbf{R}$ is $p \times p$ upper triangular with positive diagonal. This is the **thin QR factorization**. Since $\mathcal{C}(\mathbf{Q}) = \mathcal{C}(\mathbf{X})$, Proposition 6.3.2 gives $\mathbf{M} = \mathbf{Q}\mathbf{Q}^\top$ at once, and the rest of least squares follows:

$$\hat{\mathbf{Y}} = \mathbf{Q}(\mathbf{Q}^\top \mathbf{y}), \quad \mathbf{R}\hat{\boldsymbol{\beta}} = \mathbf{Q}^\top \mathbf{y}, \quad \text{SSE} = \|\mathbf{y}\|^2 - \|\mathbf{Q}^\top \mathbf{y}\|^2, \quad h_{ii} = \sum_k q_{ik}^2. \quad (6.10.1)$$

The coefficient equation holds because the normal equations become $\mathbf{R}^\top \mathbf{R}\mathbf{b} = \mathbf{R}^\top \mathbf{Q}^\top \mathbf{y}$, and $\mathbf{R}^\top$ is nonsingular. It is solved by back substitution, with no matrix inverse. Nothing of size $n \times n$ is ever formed.

In floating point, Gram–Schmidt as written in Proposition 6.2.2 gradually loses orthogonality among the computed $\mathbf{u}_j$. Production software computes the same factorization with Householder reflections, which are backward stable (Golub and Van Loan 2013; Higham 2002). The geometry does not change: $\mathbf{Q}$ is an orthonormal basis for the model space.

6.10.2 Conditioning

The sensitivity of least squares to rounding is governed by the **condition number**

$$\kappa(\mathbf{X}) = \frac{\sigma_{\max}(\mathbf{X})}{\sigma_{\min}(\mathbf{X})},$$

the ratio of the largest and smallest singular values of $\mathbf{X}$ (Chapter 1). Geometrically, $\kappa(\mathbf{X})$ is large when some combination of columns of unit coefficient norm is nearly zero. The columns are then close to linearly dependent, and the coordinates of a point in $\mathcal{C}(\mathbf{X})$ are poorly determined, even though the point itself is not. The singular values of $\mathbf{X}^\top \mathbf{X}$ are the squares of those of $\mathbf{X}$, so

$$\kappa(\mathbf{X}^\top \mathbf{X}) = \kappa(\mathbf{X})^2.$$

A useful rule of thumb from numerical analysis says that solving a linear system with condition number κ in double precision loses about $\log_{10} \kappa$ of the roughly 16 available significant digits. The normal equations solve a system whose matrix is $\mathbf{X}^\top \mathbf{X}$, so they lose about $2 \log_{10} \kappa(\mathbf{X})$ digits. QR solves the triangular system $\mathbf{R}\mathbf{b} = \mathbf{Q}^\top \mathbf{y}$, and $\kappa(\mathbf{R}) = \kappa(\mathbf{X})$ because $\mathbf{Q}$ has orthonormal columns, so it loses about $\log_{10} \kappa(\mathbf{X})$. The precise perturbation theory has one more subtlety. When the residual is large, the least squares *problem* itself has a sensitivity term proportional to $\kappa(\mathbf{X})^2 \|\hat{\epsilon}\|$, and no algorithm can avoid it (Higham 2002, ch. 20). QR's advantage is largest when the model fits well.

Example. (Polynomial regression)

The columns $1, t, t^2, \dots, t^d$ evaluated at 200 equally spaced points in $[0, 1]$ become nearly dependent as d grows. To isolate the numerical error, the listing sets $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}$ exactly for a random $\boldsymbol{\beta}$, so that the true least squares solution is $\boldsymbol{\beta}$ itself. For $d = 8$ the condition number is 6.8×10^5 . The normal equations return $\boldsymbol{\beta}$ with relative error 1.0×10^{-5} , while QR achieves 1.7×10^{-11} . At $d = 13$, with $\kappa(\mathbf{X}) = 4.1 \times 10^9$, the normal-equation answer has relative error 9.3×10^{-1} and not one correct digit, while QR still gets 4.4×10^{-9} . Figure 6.10.1 plots the whole sequence against the reference lines $\kappa\epsilon$ and $\kappa^2\epsilon$, where $\epsilon \approx 2.2 \times 10^{-16}$ is machine precision.

```
import numpy as np
import statsmodels.api as sm

def fit_normal_equations(X, y):
    return np.linalg.solve(X.T @ X, X.T @ y)

def fit_qr(X, y):
    Q, R = np.linalg.qr(X)                # X = QR, Q has orthonormal columns
    return np.linalg.solve(R, Q.T @ y)     # triangular system R b = Q'y
```

```
rng = np.random.default_rng(10)
n = 200
t = np.linspace(0, 1, n)
rows = []

for degree in range(1, 14):
    X = np.vander(t, degree + 1, increasing=True)    # 1, t, t^2, ..., t^degree
    beta = rng.normal(size=degree + 1)
```

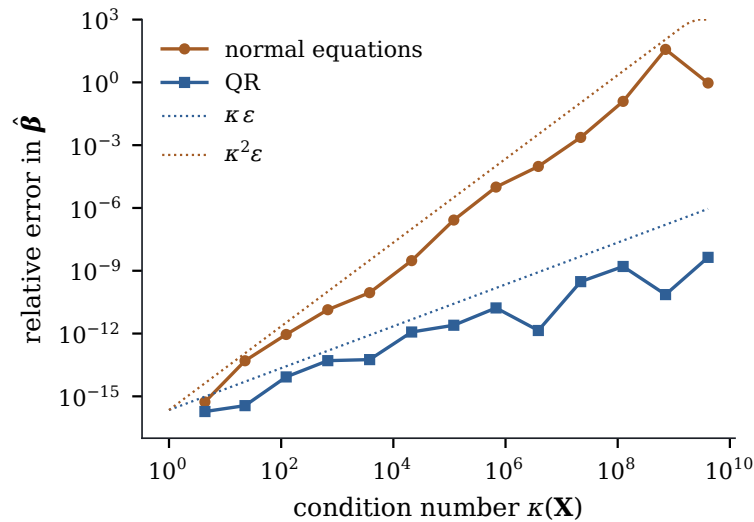


Figure 6.10.1: Accuracy of least squares coefficients for polynomial regressions of degree 1 to 13, against the condition number of the model matrix. The error of the normal equations tracks $\kappa^2\epsilon$ until it reaches order one. The error of QR tracks $\kappa\epsilon$.

```

y = X @ beta                                     # exact fit: the true
answer is beta
kappa = np.linalg.cond(X)
err_ne = np.linalg.norm(fit_normal_equations(X, y) - beta) / np.linalg.norm(beta)
)
err_qr = np.linalg.norm(fit_qr(X, y) - beta) / np.linalg.norm(beta)
rows.append((degree, kappa, err_ne, err_qr))
print(f"degree {degree:2d}   cond(X) = {kappa:9.2e}   normal eq. {err_ne:8.1e}
      QR {err_qr:8.1e}")

```

The polynomial example is extreme by design, but ill-conditioning is common in real data. The macroeconomic series of Longley's classic test problem (in `statsmodels.datasets.longley`) give a model matrix with $\kappa(\mathbf{X}) \approx 4.86 \times 10^9$. Then $\kappa(\mathbf{X}^\top \mathbf{X}) \approx 2.36 \times 10^{19}$, which is larger than $1/\epsilon$. The rule of thumb predicts about 6 correct digits from QR and about 0 from the normal equations. Part of that condition number is only a matter of units: one column is a year, another is population in thousands. Rescaling columns ([Theorem 6.7.1](#)) changes κ without changing the fit ([Exercise \(B1\)](#)). The part that remains after rescaling reflects genuine near-dependence among the regressors, the subject of Chapter 26.

Two practical lessons follow. First, never compute $(\mathbf{X}^\top \mathbf{X})^{-1}$ just to obtain $\hat{\beta}$. Solve with QR, or call a least squares routine that does. Second, reparameterizations that leave $\mathcal{C}(\mathbf{X})$ unchanged, such as centring, scaling and orthogonal polynomials, are statistically neutral but can be numerically decisive.

6.10.3 Rank deficiency and the singular value decomposition

QR with positive diagonal needs full column rank. For rank-deficient or nearly rank-deficient $\mathbf{X}$, the most informative tool is the **singular value decomposition**

$$\mathbf{X} = \mathbf{U} \mathbf{D} \mathbf{V}^\top = \mathbf{U}_r \mathbf{D}_r \mathbf{V}_r^\top,$$

where $\mathbf{U}$ ($n \times n$) and $\mathbf{V}$ ($p \times p$) are orthogonal, $\mathbf{D}$ is $n \times p$ with nonnegative diagonal entries $\sigma_1 \geq \cdots \geq \sigma_r > 0$, and the subscript r keeps only the first r columns. Then:

- $\mathbf{U}_r$ is an orthonormal basis for $\mathcal{C}(\mathbf{X})$, so $\mathbf{M} = \mathbf{U}_r \mathbf{U}_r^\top$;
- the last $p - r$ columns of $\mathbf{V}$ form an orthonormal basis for $\mathcal{N}(\mathbf{X})$;
- the Moore–Penrose inverse is $\mathbf{X}^+ = \mathbf{V}_r \mathbf{D}_r^{-1} \mathbf{U}_r^\top$, and $\mathbf{M} = \mathbf{X} \mathbf{X}^+$.

Proposition 6.10.1 — Minimum-norm least squares

$\hat{\boldsymbol{\beta}}^+ = \mathbf{X}^+ \mathbf{y}$ is a least squares estimate, and it has the smallest Euclidean norm among all least squares estimates.

Proof. $\mathbf{X} \mathbf{X}^+ \mathbf{y} = \mathbf{U}_r \mathbf{D}_r \mathbf{V}_r^\top \mathbf{V}_r \mathbf{D}_r^{-1} \mathbf{U}_r^\top \mathbf{y} = \mathbf{U}_r \mathbf{U}_r^\top \mathbf{y} = \mathbf{M} \mathbf{y}$, so $\hat{\boldsymbol{\beta}}^+$ is a least squares estimate. It lies in $\mathcal{C}(\mathbf{V}_r) = \mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$. Every other least squares estimate is $\hat{\boldsymbol{\beta}}^+ + \mathbf{v}$ with $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ (Theorem 6.4.2), and $\|\hat{\boldsymbol{\beta}}^+ + \mathbf{v}\|^2 = \|\hat{\boldsymbol{\beta}}^+\|^2 + \|\mathbf{v}\|^2$. ■

In Example (Four answers, one fit), the Moore–Penrose row is this minimum-norm solution. In floating point, “ $\sigma_j > 0$ ” becomes “ σ_j larger than a tolerance”, and choosing the tolerance decides the numerical rank. That choice is a statistical decision in disguise. Chapter 10 discusses it, together with pivoted QR (Theorem 10.7.2), which gives most of the SVD’s rank-revealing ability at lower cost.

6.10.4 Exercises

6.10.4.1 A. Check your understanding

Exercise A1.

Prove the four identities in Eq. 6.10.1.

6.10.4.2 B. Practice

Exercise B1.

Compute $\kappa(\mathbf{X})$ for the Longley data (with intercept) as given, after scaling every non-intercept column to unit Euclidean norm, and after centring and scaling. Confirm that the fitted values agree to the precision you expect in all three cases. Which part of the original condition number is “just units”?

Exercise B2.

Implement classical Gram–Schmidt as in Proposition 6.2.2, and the *modified* version that subtracts each projection from the working vector before computing the next inner

product. For the degree-10 polynomial design of [Example \(Polynomial regression\)](#), compare $\|Q^\top Q - I\|$ for the two and for `numpy.linalg.qr`.

6.10.4.3 C. Going deeper

Exercise C1.

Let $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ with $\bar{x} = c$ and $S_{xx} = s^2 n$. Find $\kappa(\mathbf{X})$ as a function of c , s and n , and show that it grows without bound as $c \rightarrow \infty$ with s fixed. Show that centring the regressor, a reparameterization, reduces the condition number to $\max(1, s)/\min(1, s)$.

Solution. Write $\mathbf{x} = c\mathbf{1} + \mathbf{d}$ with $\mathbf{d} \perp \mathbf{1}$ and $\|\mathbf{d}\|^2 = ns^2$. Then $\mathbf{X}^\top \mathbf{X} = n \begin{pmatrix} 1 & c \\ c & c^2 + s^2 \end{pmatrix}$. The matrix in parentheses has trace $T = 1 + c^2 + s^2$ and determinant s^2 , so its eigenvalues are $\lambda_{\pm} = (T \pm \sqrt{T^2 - 4s^2})/2$, with $\lambda_+ \lambda_- = s^2$ and $\lambda_+ \geq T/2$. Hence

$$\kappa(\mathbf{X})^2 = \frac{\lambda_+}{\lambda_-} = \frac{\lambda_+^2}{s^2} \geq \frac{T^2}{4s^2} \rightarrow \infty \quad \text{as } c \rightarrow \infty.$$

After centring, $\mathbf{X} = [\mathbf{1}, \mathbf{d}]$ has orthogonal columns of lengths $\sqrt{n}$ and $s\sqrt{n}$, so its singular values are these lengths and $\kappa = \max(1, s)/\min(1, s)$.

6.11 Projection in the population: best linear prediction

So far the geometry has lived in observation space $\mathbb{R}^n$, and every statement was about one fixed data vector. There is a parallel geometry one level up, in which the vectors are *random variables* and the inner product is an expectation. In that geometry, least squares estimates something even when the linear model is false, namely the population projection of Y onto a span of regressors. The conditional expectation turns out to be a projection too. This section sets up the correspondence. Chapter 14 develops prediction theory from it, and Chapter 19 uses it to study what least squares estimates under misspecification.

6.11.1 Random variables as vectors

Consider random variables with finite second moments on a common probability space. They form a vector space, and

$$\langle U, V \rangle = E(UV), \quad \|U\| = (E U^2)^{1/2}$$

is an inner product, provided we identify variables that are equal with probability one, so that $\|U\| = 0$ forces $U = 0$. This space is usually written L^2 . Two random variables are orthogonal when $E(UV) = 0$. For variables with mean zero, this is the same as being uncorrelated. The Cauchy–Schwarz inequality [Eq. 6.2.2](#) becomes the statement that correlations lie in $[-1, 1]$. The *cosine* of the angle between two centred random variables is their correlation.

For a random k -vector $\mathbf{X}$ with finite second moments, the set $\mathcal{L} = \{a + \mathbf{b}^\top \mathbf{X} : a \in \mathbb{R}, \mathbf{b} \in \mathbb{R}^k\}$ is a subspace of dimension at most $k + 1$. The proofs of [Proposition 6.2.2](#), [Theorem 6.2.3](#) and

Theorem 6.3.1 used only the axioms of an inner product and a finite spanning set, so they apply unchanged. Every Y with $EY^2 < \infty$ has a unique nearest point in $\mathcal{L}$, characterized by orthogonality of the error to $\mathcal{L}$.

Definition 6.11.1 — Best linear predictor

The **best linear predictor** of Y given $\mathbf{X}$, written $L(Y | \mathbf{X})$, is the projection of Y onto $\mathcal{L}$. It is the $a + \mathbf{b}^\top \mathbf{X}$ that minimizes the mean squared error $E(Y - a - \mathbf{b}^\top \mathbf{X})^2$.

Theorem 6.11.1

Let $\boldsymbol{\mu}_X = E\mathbf{X}$, $\mu_Y = EY$, $\boldsymbol{\Sigma}_{XX} = \text{Cov}(\mathbf{X})$ and $\boldsymbol{\sigma}_{XY} = \text{Cov}(\mathbf{X}, Y)$, and suppose $\boldsymbol{\Sigma}_{XX}$ is positive definite. Then

$$L(Y | \mathbf{X}) = \alpha^* + \boldsymbol{\beta}^{*\top} \mathbf{X}, \quad \boldsymbol{\beta}^* = \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}, \quad \alpha^* = \mu_Y - \boldsymbol{\beta}^{*\top} \boldsymbol{\mu}_X.$$

The prediction error $U = Y - L(Y | \mathbf{X})$ has mean zero and is uncorrelated with every component of $\mathbf{X}$. Moreover

$$\text{Var}(Y) = \text{Var}(L(Y | \mathbf{X})) + E U^2, \quad \frac{\text{Var}(L(Y | \mathbf{X}))}{\text{Var}(Y)} = \frac{\boldsymbol{\sigma}_{XY}^\top \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}}{\text{Var}(Y)}. \quad (6.11.1)$$

Proof. Orthogonality of $U = Y - a - \mathbf{b}^\top \mathbf{X}$ to the spanning set $\{1, X_1, \dots, X_k\}$ of $\mathcal{L}$ gives the *population normal equations*

$$E(U) = 0, \quad E(\mathbf{X}U) = \mathbf{0}.$$

The first gives $a = \mu_Y - \mathbf{b}^\top \boldsymbol{\mu}_X$. Substituting into the second and subtracting $\boldsymbol{\mu}_X E(U) = \mathbf{0}$ gives $\text{Cov}(\mathbf{X}, Y) - \text{Cov}(\mathbf{X})\mathbf{b} = \mathbf{0}$, so $\mathbf{b} = \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}$. By the projection theorem the solution is the unique minimizer. Since U has mean zero and is uncorrelated with $\mathbf{X}$, it is uncorrelated with $L(Y | \mathbf{X})$, so the variances add. Finally, $\text{Var}(\boldsymbol{\beta}^{*\top} \mathbf{X}) = \boldsymbol{\beta}^{*\top} \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta}^* = \boldsymbol{\sigma}_{XY}^\top \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}$. ■

The ratio in Eq. 6.11.1 is the population *squared multiple correlation*, the population counterpart of R^2 . By the same cosine argument as Theorem 6.7.2, it is the squared correlation between Y and $L(Y | \mathbf{X})$.

6.11.2 Conditional expectation is also a projection

The linear span $\mathcal{L}$ is one subspace of L^2 . A much larger one is $\mathcal{G} = \{g(\mathbf{X}) : E g(\mathbf{X})^2 < \infty\}$, all square-integrable functions of $\mathbf{X}$. It is infinite-dimensional, but it is closed, and the projection theorem still holds in it. We accept that fact from measure-theoretic probability and only use its conclusion.

Proposition 6.11.2

Let $m(\mathbf{X}) = E(Y | \mathbf{X})$ with $E Y^2 < \infty$. Then:

- (a) $m(\mathbf{X})$ is the projection of Y onto $\mathcal{G}$. It minimizes $E(Y - g(\mathbf{X}))^2$ over g , and $E[(Y - m(\mathbf{X}))g(\mathbf{X})] = 0$ for all $g(\mathbf{X}) \in \mathcal{G}$;
- (b) $L(Y | \mathbf{X}) = L(m(\mathbf{X}) | \mathbf{X})$: the best linear predictor of Y equals the best linear predictor of the regression function;
- (c) $E(Y - L(Y | \mathbf{X}))^2 = E(Y - m(\mathbf{X}))^2 + E(m(\mathbf{X}) - L(Y | \mathbf{X}))^2$;
- (d) if m is affine, $m(\mathbf{X}) = L(Y | \mathbf{X})$.

Proof.

- (a) The orthogonality property follows from the tower rule: $E[(Y - m(\mathbf{X}))g(\mathbf{X})] = E[g(\mathbf{X}) E(Y - m(\mathbf{X}) | \mathbf{X})] = 0$. Orthogonality to the subspace characterizes the nearest point, by the argument of Theorem 6.3.1, which needs only Pythagoras.
- (b) This is Theorem 6.5.1(a) in L^2 . Since $\mathcal{L} \subseteq \mathcal{G}$, projecting onto $\mathcal{L}$ can be done in two stages.
- (c) $Y - L = (Y - m) + (m - L)$, and $m - L \in \mathcal{G}$ is orthogonal to $Y - m$ by (a).
- (d) If $m \in \mathcal{L}$, the second term in (c) is zero.

■

Part (c) is the population version of Figure 6.5.1. The error of the best linear predictor splits into two orthogonal parts. One is irreducible noise, which no function of $\mathbf{X}$ can remove. The other is *approximation error*, the price of insisting on linearity. Chapters in Part IX reduce the second part by enlarging $\mathcal{L}$ towards $\mathcal{G}$.

6.11.3 Least squares estimates the population projection

Here is the connection with the rest of the chapter. Given data $(\mathbf{x}_{(i)}, y_i)$, put probability $1/n$ on each row. For this *empirical distribution*, the L^2 inner product of two variables is $n^{-1} \sum_i u_i v_i = n^{-1} \mathbf{u}^\top \mathbf{v}$, which is the Euclidean inner product of observation space up to the factor $1/n$. So the best linear predictor under the empirical distribution is the least squares fit, and the population normal equations become the normal equations Eq. 6.4.1. Observation-space geometry is L^2 geometry for the empirical distribution.

As the sample grows, the empirical distribution approaches the population distribution, and the projections converge with it.

Proposition 6.11.3 — Consistency for the projection

Let $(\mathbf{X}_i, Y_i)$, $i = 1, 2, \dots$, be independent copies of $(\mathbf{X}, Y)$ with finite second moments and Σ_{XX} positive definite. Let $(\hat{\alpha}_n, \hat{\beta}_n)$ be the least squares coefficients from the first n observations, with an intercept. Then $(\hat{\alpha}_n, \hat{\beta}_n) \rightarrow (\alpha^*, \beta^*)$ with probability one. No

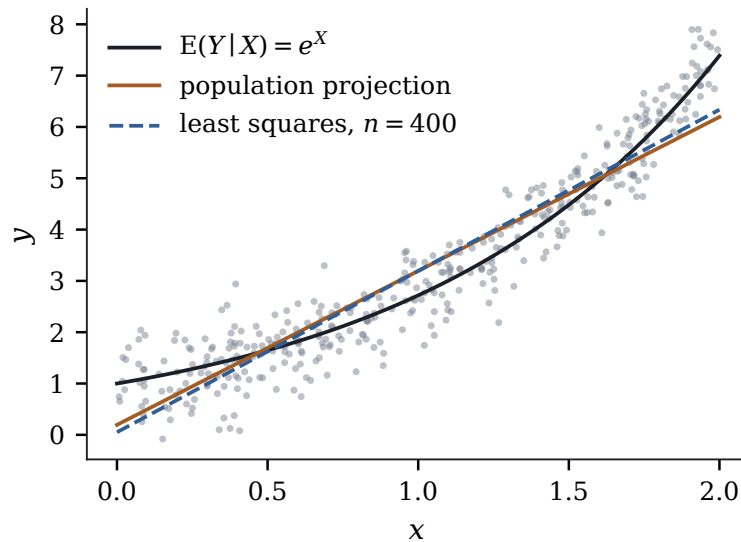


Figure 6.11.1: When the regression function is curved, least squares estimates its best linear approximation in the L^2 sense, the population projection. The approximation error is largest where e^x bends away from the line.

assumption about the form of $E(Y | \mathbf{X})$, the distribution of the errors, or their variance is needed.

Proof. For large n , $\hat{\beta}_n = \mathbf{S}_{XX}^{-1} \mathbf{s}_{XY}$, where $\mathbf{S}_{XX}$ and $\mathbf{s}_{XY}$ are sample covariances with divisor n (Section 6.6, centring). By the strong law of large numbers the sample means, second moments and cross-moments converge almost surely to their population values. So $\mathbf{S}_{XX} \rightarrow \mathbf{\Sigma}_{XX}$ and $\mathbf{s}_{XY} \rightarrow \mathbf{\sigma}_{XY}$. Matrix inversion is continuous at the nonsingular $\mathbf{\Sigma}_{XX}$, so eventually $\mathbf{S}_{XX}$ is invertible and $\hat{\beta}_n \rightarrow \beta^*$. The intercept follows the same way. ■

Example. (A curved mean)

Let $X \sim \text{Uniform}(0, 2)$ and $Y = e^X + \varepsilon$ with independent $\varepsilon \sim N(0, 0.25)$. The regression function is not linear, but Theorem 6.11.1 still identifies a best line. A short calculation gives $\text{Cov}(X, Y) = 1$ and $\text{Var}(X) = 1/3$, so the population projection has slope 3.000 and intercept 0.195. The listing fits least squares lines to samples of increasing size. The estimates settle on these values, not on any feature of e^x . Figure 6.11.1 shows the regression function, the population projection, and a least squares fit.

```
import numpy as np
rng = np.random.default_rng(611)

# X ~ Uniform(0, 2), E(Y | X) = exp(X), Var(Y | X) = 0.25.
# Population projection of Y onto span{1, X}: slope Cov(X, Y)/Var(X), intercept E Y
# - slope * E X.
EX, VarX = 1.0, 4.0 / 12.0
EY = (np.exp(2) - 1) / 2
EXY = (np.exp(2) + 1) / 2
slope_pop = (EXY - EX * EY) / VarX
```

E exp(X)
E X exp(X)

```

intercept_pop = EY - slope_pop * EX

for n in [20, 200, 2000, 20000]:
    x = rng.uniform(0, 2, n)
    y = np.exp(x) + rng.normal(0, 0.5, n)
    b, *_ = np.linalg.lstsq(np.column_stack([np.ones(n), x]), y, rcond=None)
    print(f"n = {n:6d}    intercept {b[0]:.3f}    slope {b[1]:.3f}")
print(f"population    intercept {intercept_pop:.3f}    slope {slope_pop:.3f}")

```

Remark (What changes under misspecification). [Proposition 6.11.3](#) is reassuring about the *target*. A least squares line always estimates a well-defined population quantity. It says nothing about the *uncertainty* of the estimate. When $E(Y | \mathbf{X})$ is not linear, or the variance is not constant, the approximation error $m(\mathbf{X}) - L(Y | \mathbf{X})$ behaves like extra noise that depends on $\mathbf{X}$. The textbook formula $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ for the covariance of $\hat{\beta}$ is then wrong, even in large samples. The “sandwich” covariance estimators of Chapter 21 are designed for exactly this situation.

6.12 Summary and Notes

6.12.1 Summary

Idea.

1. Least squares is the nearest-point problem in observation space. The fitted vector $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$ is the orthogonal projection of $\mathbf{y}$ onto $\mathcal{C}(\mathbf{X})$, and the residual is orthogonal to $\mathcal{C}(\mathbf{X})$ ([Theorem 6.4.1](#)).
2. Orthogonal projections are exactly the symmetric idempotent matrices. Their rank equals their trace ([Theorem 6.3.3](#) and [Proposition 6.3.4](#)).
3. $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top$ for *every* generalized inverse ([Theorem 6.3.7](#)). Fitted values, residuals and every function of them are unique, whatever the rank. Coefficients are unique only when $\text{rank}(\mathbf{X}) = p$, and $\boldsymbol{\lambda}^\top \hat{\beta}$ is invariant iff $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ ([Theorem 6.4.2](#) and [Theorem 6.4.4](#)).
4. Nested model spaces give orthogonal decompositions $\mathbf{I} = \mathbf{M}_0 + (\mathbf{M} - \mathbf{M}_0) + (\mathbf{I} - \mathbf{M})$. The drop in residual sum of squares is $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2$, with $r - r_0$ degrees of freedom ([Theorem 6.5.1](#)).
5. A multiple-regression coefficient is the slope on the part of its regressor orthogonal to the others. Fixed-effects estimators and added-variable plots are applications ([Theorem 6.6.2](#)).
6. Anything that depends only on $\mathcal{C}(\mathbf{X})$ is invariant to reparameterization. R^2 is the squared cosine between centred data and centred fit ([Theorem 6.7.1](#) and [Theorem 6.7.2](#)).
7. Leverage $h_{ii} = m_{ii}$ is the squared length of the projection of a coordinate vector. It lies in $[1/n, 1]$ with an intercept and averages r/n ([Proposition 6.8.1](#)).
8. Replacing the inner product by $\mathbf{u}^\top \mathbf{V}^{-1} \mathbf{v}$ turns least squares into generalized least squares. The projection is then idempotent and $\mathbf{V}^{-1}$ -self-adjoint, but not symmetric ([Theorem 6.9.1](#) and [Theorem 6.9.3](#)).

9. Compute projections with orthonormal bases (QR, SVD), never by inverting $\mathbf{X}^\top \mathbf{X}$, which squares the condition number (Section 6.10).
10. In the population, least squares estimates the L^2 projection of Y onto linear functions of $\mathbf{X}$, whether or not the regression function is linear (Theorem 6.11.1 and Proposition 6.11.3).

6.12.2 Notes and sources

The geometric tradition. Seeing least squares as projection is as old as least squares itself, but it became the organizing principle of linear-model theory only in the twentieth century. Herr (1980) traces the history, from Fisher’s geometric derivations of sampling distributions through the coordinate-free treatments of the 1960s. Book-length geometric developments at different levels include Saville and Wood (1991), written for a first course, and Wickens (1995), for multivariate methods. Christensen (2020) is the most sustained vector-space treatment of linear model theory. Its chapter 2 and appendices A–B cover the material of Section 6.2, Section 6.3 and Section 6.4 for the rank-deficient case throughout, and influenced this chapter’s choice to state results for general rank from the start. Seber and Lee (2003, appendix B) gives a compact account of projections, including the constraint-space result that appears here as Theorem 6.5.2. Agresti (2015, chapter 2) develops projection matrices through worked null-model, one-way and two-way examples, and is a gentle companion to Section 6.5. For the underlying linear algebra, Halmos (1958) remains a model of coordinate-free exposition.

Generalized inverses. The four-condition inverse is due to Penrose (1955). Its statistical uses, and the theory of general $\mathbf{A}^-$ including the characterization in Exercise (B2), are developed by Rao (1973). The invariance of $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top$ (Lemma 6.3.6) is the fact that makes generalized inverses harmless in linear models. Every quantity of statistical interest is sandwiched in this way.

Frisch–Waugh–Lovell. The theorem is due to Frisch and Waugh (1933) for time trends and Lovell (1963) for seasonal dummies and general partitions. It is also known under Lovell’s name alone, and in econometrics as the “partialling-out” result. It underlies added-variable plots, double/debiased machine-learning estimators, and the fixed-effects (“within”) estimator for panel data.

Leverage. Hoaglin and Welsch (1978) introduced the name “hat matrix” into common use and established the basic properties in Proposition 6.8.1. Diagnostic use of leverage is taken up in Chapter 20.

Other inner products. Generalized least squares is due to Aitken (1935). The coordinate-free condition for equality of ordinary and generalized least squares (Theorem 6.9.3) is from Kruskal (1968). Christensen (2020, chapter 10) develops the consequences for the general Gauss–Markov model, which Chapter 31 follows.

Computation. Golub and Van Loan (2013) is the standard reference for QR, the SVD and their perturbation theory. Higham (2002) gives the precise statement of the conditioning of least squares problems, including the $\kappa^2 \|\hat{\epsilon}\|$ term mentioned in Section 6.10. Seber and Lee (2003, chapter 11) surveys algorithms from a statistician’s perspective, and Chapter 10 builds on it.

6.12.3 References

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Chapter 7

Optimality: Gauss–Markov, Maximum Likelihood and Minimum Variance

Least squares was introduced as a way of fitting, and [Chapter 6](#) showed that it is a projection. Neither fact says it is a *good* way of estimating. This chapter asks in what precise senses least squares is the best one can do, and where those senses end. With only means and covariances assumed, least squares has the smallest variance among linear unbiased estimators (Gauss–Markov), and the proof is one more application of the projection theorem. With normal errors three more things hold. Least squares is maximum likelihood. It has the smallest variance among *all* unbiased estimators (Lehmann–Scheffé). And its sampling distribution is known exactly, which is what Part III builds tests and intervals on. The same assumptions also show where optimality stops. Nonlinear estimators can win when the errors are not normal, and biased estimators can win on mean squared error. A Bayesian analysis with a conjugate prior is one principled biased estimator, and it recovers least squares as the prior becomes flat.

What you need. Covariances of linear functions and the mean of a quadratic form ([Chapter 2: Theorem 2.2.1](#) and [Theorem 2.3.1](#)). The multivariate normal and its density ([Chapter 3: Theorem 3.3.1](#) and [Theorem 3.2.2](#)). Chi-squared quadratic forms and their independence ([Chapter 4: Theorem 4.3.2](#) and [Theorem 4.4.1](#)). The linear model and its first two moments ([Chapter 5: Definition 5.3.1](#) and [Theorem 5.5.1](#)). Least squares as projection, including the rank-deficient case ([Chapter 6: Theorem 6.4.1](#), [Theorem 6.4.2](#) and [Theorem 6.4.4](#)).

7.0.1 Roadmap

- [Linear unbiased estimation](#)
- [The Gauss–Markov theorem](#)
- [Maximum likelihood under normality](#)
- [Sufficiency, completeness and minimum variance](#)
- [Sampling distributions of the estimates](#)
- [Bayesian thread I: the conjugate linear model](#)
- [Summary and notes](#)

7.1 Linear unbiased estimation

To call an estimator the best, we have to say what it is compared with and by what criterion. The comparison can't be made against every estimator. The constant estimator $\tilde{\beta}_1 \equiv 0.5$ of a slope has variance zero, and it is perfect when the true slope happens to be 0.5 and useless otherwise. No estimator has smaller variance, yet nobody would call it good. Some restriction on the class of competitors is unavoidable. The classical restriction has two parts. The estimator must be *unbiased*, which rules out estimators that buy precision by ignoring the data. And it must be *linear* in the response, which makes its variance computable from the first two moments of the errors alone. This section describes the class these two conditions carve out. The description is geometric, and the Gauss–Markov theorem of [Section 7.2](#) will follow from it in a few lines.

7.1.1 The setting

Throughout this chapter the response satisfies the linear model of [Chapter 5](#) ([Definition 5.3.1](#)),

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}, \quad (7.1.1)$$

with $\mathbf{X}$ a fixed $n \times p$ matrix of rank $r \leq p$, $\boldsymbol{\beta} \in \mathbb{R}^p$ unknown and $\sigma^2 > 0$ unknown. These are the **second-moment assumptions**. The errors have mean zero, a common variance, and are uncorrelated. Nothing is assumed about their distribution beyond that. From [Section 7.3](#) on we add normality, and we will say so each time. As in [Chapter 6](#), $\mathbf{M}$ denotes the orthogonal projection onto $\mathcal{C}(\mathbf{X})$, $\hat{\boldsymbol{\beta}}$ is any least squares estimate, and $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$ is the residual sum of squares.

The quantities we want to estimate are linear functions $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, for fixed $\boldsymbol{\lambda} \in \mathbb{R}^p$. This covers a single coefficient ($\boldsymbol{\lambda}$ a coordinate vector), a difference of two coefficients, and the mean response $\mathbf{x}_0^\top \boldsymbol{\beta}$ at a new point $\mathbf{x}_0$. Several such functions at once are collected as $\boldsymbol{\Lambda}^\top \boldsymbol{\beta}$, where $\boldsymbol{\Lambda}$ is a $p \times q$ matrix.

7.1.2 Linear and unbiased

Definition 7.1.1 — Linear unbiased and best linear unbiased estimators

Let $\boldsymbol{\lambda} \in \mathbb{R}^p$.

- (a) A **linear estimator** is a statistic of the form $c + \mathbf{a}^\top \mathbf{Y}$, with a constant $c \in \mathbb{R}$ and a constant vector $\mathbf{a} \in \mathbb{R}^n$ (constant means not depending on $\mathbf{Y}$ or on the parameters).
- (b) A linear estimator is a **linear unbiased estimator** (LUE) of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ if $\mathbf{E}(c + \mathbf{a}^\top \mathbf{Y}) = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$ for every $\boldsymbol{\beta} \in \mathbb{R}^p$ and every $\sigma^2 > 0$.
- (c) An LUE $c + \mathbf{a}^\top \mathbf{Y}$ of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is a **best linear unbiased estimator** (BLUE) if $\text{Var}(c + \mathbf{a}^\top \mathbf{Y}) \leq \text{Var}(d + \mathbf{b}^\top \mathbf{Y})$ for every LUE $d + \mathbf{b}^\top \mathbf{Y}$ of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, for every $\boldsymbol{\beta}$ and every $\sigma^2 > 0$.
- (d) For q functions $\boldsymbol{\Lambda}^\top \boldsymbol{\beta}$, a **linear estimator** is $c + \mathbf{A}\mathbf{Y}$ with a constant $q \times n$ matrix $\mathbf{A}$

and $\mathbf{c} \in \mathbb{R}^q$. It is an LUE of $\mathbf{\Lambda}^\top \boldsymbol{\beta}$ if each coordinate is an LUE of the corresponding coordinate of $\mathbf{\Lambda}^\top \boldsymbol{\beta}$, and a BLUE if $\text{Cov}(\mathbf{BY} + \mathbf{d}) - \text{Cov}(\mathbf{AY} + \mathbf{c})$ is nonnegative definite for every LUE $\mathbf{BY} + \mathbf{d}$ (Definition 1.7.1).

The words “for every $\boldsymbol{\beta}$ ” in (b) carry all the weight. An estimator that is unbiased at one parameter value only, like the constant 0.5 when $\beta_1 = 0.5$, does not qualify. The first result (Proposition 7.1.1) turns unbiasedness into linear equations.

Proposition 7.1.1 — Unbiasedness as a linear equation

The linear estimator $c + \mathbf{a}^\top \mathbf{Y}$ is an LUE of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ iff $c = 0$ and $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$. Consequently $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ has an LUE iff $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$.

Proof. By Theorem 2.2.1, $E(c + \mathbf{a}^\top \mathbf{Y}) = c + \mathbf{a}^\top \mathbf{X}\boldsymbol{\beta}$. This equals $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ for all $\boldsymbol{\beta}$ iff the affine function $\boldsymbol{\beta} \mapsto c + (\mathbf{X}^\top \mathbf{a} - \boldsymbol{\lambda})^\top \boldsymbol{\beta}$ vanishes identically. Putting $\boldsymbol{\beta} = \mathbf{0}$ gives $c = 0$. Putting $\boldsymbol{\beta}$ equal to each coordinate vector in turn then gives $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$. The converse is immediate. An LUE therefore exists iff the equation $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$ has a solution, that is, iff $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$. ■

So the constant c is always zero, and from now on we drop it. A linear function $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ with $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ is called **estimable**. For these functions the data carry linear information. For the others no linear combination of the observations is centred on the target. Chapter 8 takes this as the definition of estimability (Definition 8.2.1) and develops its many equivalent forms. For now we need only Proposition 7.1.1 and one comparison. Theorem 6.4.4 of Chapter 6 found the same condition $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ as the answer to a different question: when is $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ the same for every least squares estimate $\hat{\boldsymbol{\beta}}$? The functions that least squares estimates unambiguously are exactly the functions that have linear unbiased estimators at all. When $\mathbf{X}$ has full column rank, $\mathcal{C}(\mathbf{X}^\top) = \mathbb{R}^p$, so every linear function of $\boldsymbol{\beta}$ is estimable.

7.1.3 The geometry of the unbiased class

Fix an estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ and write $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$ for some $\boldsymbol{\rho} \in \mathbb{R}^n$. The coefficient vectors of its LUEs form the set

$$\mathcal{A}_\lambda = \{\mathbf{a} \in \mathbb{R}^n : \mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}\}.$$

This set is the key object of the chapter’s first half.

Proposition 7.1.2 — The set of unbiased coefficient vectors

Let $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$. Then:

- (a) $\mathcal{A}_\lambda = \mathbf{M}\boldsymbol{\rho} + \mathcal{C}(\mathbf{X})^\perp$, an affine subspace of $\mathbb{R}^n$ of dimension $n - r$, parallel to $\mathcal{C}(\mathbf{X})^\perp$;
- (b) every $\mathbf{a} \in \mathcal{A}_\lambda$ has the same projection $\mathbf{M}\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$, and $\mathbf{M}\boldsymbol{\rho}$ is the only element of $\mathcal{A}_\lambda$ that lies in $\mathcal{C}(\mathbf{X})$;
- (c) $\mathbf{M}\boldsymbol{\rho} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \boldsymbol{\lambda}$ for every generalized inverse, and $(\mathbf{M}\boldsymbol{\rho})^\top \mathbf{Y} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ for every

least squares estimate $\hat{\beta}$.

Proof.

- (a) By [Proposition 7.1.1](#), $\mathbf{a} \in \mathcal{A}_\lambda$ means $\mathbf{X}^\top \mathbf{a} = \lambda$. Now $\mathbf{X}^\top \mathbf{a} = \mathbf{X}^\top \boldsymbol{\rho}$ iff $\mathbf{X}^\top (\mathbf{a} - \boldsymbol{\rho}) = \mathbf{0}$, iff $\mathbf{a} - \boldsymbol{\rho} \in \mathcal{N}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})^\perp$ ([Lemma 6.2.1](#)). So $\mathcal{A}_\lambda = \boldsymbol{\rho} + \mathcal{C}(\mathbf{X})^\perp$. Since $\boldsymbol{\rho} - \mathbf{M}\boldsymbol{\rho} = (\mathbf{I} - \mathbf{M})\boldsymbol{\rho} \in \mathcal{C}(\mathbf{X})^\perp$, the same set is $\mathbf{M}\boldsymbol{\rho} + \mathcal{C}(\mathbf{X})^\perp$. Its dimension is $\dim \mathcal{C}(\mathbf{X})^\perp = n - r$.
- (b) If $\mathbf{a} = \mathbf{M}\boldsymbol{\rho} + \mathbf{v}$ with $\mathbf{v} \in \mathcal{C}(\mathbf{X})^\perp$, then $\mathbf{M}\mathbf{a} = \mathbf{M}\boldsymbol{\rho} + \mathbf{M}\mathbf{v} = \mathbf{M}\boldsymbol{\rho}$. If moreover $\mathbf{a} \in \mathcal{C}(\mathbf{X})$, then $\mathbf{a} = \mathbf{M}\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$.
- (c) By [Theorem 6.3.7](#), $\mathbf{M}\boldsymbol{\rho} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\rho} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \lambda$. By [Theorem 6.4.4](#), $\lambda^\top \hat{\beta} = \boldsymbol{\rho}^\top \mathbf{M}\mathbf{Y} = (\mathbf{M}\boldsymbol{\rho})^\top \mathbf{Y}$.

■

Part (c) says that least squares is itself one of the linear unbiased estimators. Its coefficient vector is $\mathbf{a}_* = \mathbf{M}\boldsymbol{\rho}$, the one member of the class that lies in the model space. It does not depend on which $\boldsymbol{\rho}$ was used to represent λ , because every such $\boldsymbol{\rho}$ has the same projection. That $\mathbf{a}_*^\top \mathbf{Y}$ is unbiased can also be seen directly: $E(\mathbf{a}_*^\top \mathbf{Y}) = \boldsymbol{\rho}^\top \mathbf{M}\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\rho}^\top \mathbf{X}\boldsymbol{\beta} = \lambda^\top \boldsymbol{\beta}$.

The variance of any linear estimator follows from [Theorem 2.2.1](#):

$$\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top (\sigma^2 \mathbf{I}) \mathbf{a} = \sigma^2 \|\mathbf{a}\|^2. \quad (7.1.2)$$

Under the second-moment assumptions, then, the variance of a linear estimator is σ^2 times the squared length of its coefficient vector. Finding the BLUE means finding the shortest vector in the affine set $\mathcal{A}_\lambda$.

Idea. The linear unbiased estimators of $\lambda^\top \boldsymbol{\beta}$ correspond to the points of a flat $\mathcal{A}_\lambda = \mathbf{a}_* + \mathcal{C}(\mathbf{X})^\perp$ in $\mathbb{R}^n$. The variance of each is σ^2 times its squared distance from the origin. The question “which LUE is best?” is a nearest-point problem, the same kind of problem that least squares solves in [Chapter 6](#), now posed in the space of coefficient vectors.

A second description of the class is useful too. Every $\mathbf{a} \in \mathcal{A}_\lambda$ is $\mathbf{a}_* + (\mathbf{I} - \mathbf{M})\mathbf{w}$ for some $\mathbf{w}$, so every LUE has the form

$$\mathbf{a}^\top \mathbf{Y} = \lambda^\top \hat{\beta} + \mathbf{w}^\top (\mathbf{I} - \mathbf{M})\mathbf{Y} = \lambda^\top \hat{\beta} + \mathbf{w}^\top \hat{\epsilon}. \quad (7.1.3)$$

An LUE is the least squares estimate plus a linear function of the residuals. The linear functions of the residuals are exactly the linear unbiased estimators of zero ([Exercise \(B1\)](#)). Adding one to an estimator keeps it unbiased and can only add noise. [Section 7.2](#) makes “can only add noise” precise.

7.1.4 Three estimators of a slope

Example. (Three linear unbiased estimators of a slope)

A straight line $E(Y_i) = \beta_0 + \beta_1 x_i$ is fitted to $n = 12$ observations at the regressor values

$x = 1, 2, 2.5, 3, 4, 5, 5.5, 7, 8, 9, 10.5, 12$, for which $S_{xx} = \sum_i (x_i - \bar{x})^2 = 137.229$. Here $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ has full rank and the target is $\boldsymbol{\lambda} = (0, 1)^\top$. Three natural estimators of the slope are linear in $\mathbf{Y}$.

- (i) *Least squares*, $\hat{\beta}_1 = \sum_i (x_i - \bar{x})Y_i / S_{xx}$ (Theorem 5.2.1), with $a_i = (x_i - \bar{x}) / S_{xx}$.
- (ii) *End points*, the slope of the line through the first and last observations, $(Y_{12} - Y_1) / (x_{12} - x_1)$.
- (iii) *Group means*, which splits the data into the six smallest and six largest x values and takes $(\bar{Y}_{\text{high}} - \bar{Y}_{\text{low}}) / (\bar{x}_{\text{high}} - \bar{x}_{\text{low}})$.

Each coefficient vector satisfies $\mathbf{1}^\top \mathbf{a} = 0$ and $\mathbf{x}^\top \mathbf{a} = 1$, which is the condition $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$ of Proposition 7.1.1. So all three are unbiased, whatever the distribution of the errors. By Eq. 7.1.2 their variances are $\sigma^2 \|\mathbf{a}\|^2$:

$$0.00729 \sigma^2, \quad 0.01653 \sigma^2, \quad 0.01008 \sigma^2.$$

Least squares has the smallest, and the other two have efficiencies 0.441 and 0.723 relative to it. The listing also confirms what Proposition 7.1.2(b) predicts: the three coefficient vectors differ, but their projections onto $\mathcal{C}(\mathbf{X})$ all equal the least squares vector.

```
import numpy as np

x = np.array([1.0, 2.0, 2.5, 3.0, 4.0, 5.0, 5.5, 7.0, 8.0, 9.0, 10.5, 12.0])
n = len(x)
X = np.column_stack([np.ones(n), x])
lam = np.array([0.0, 1.0]) # target: the slope beta_1
M = X @ np.linalg.solve(X.T @ X, X.T) # projection onto C(X)

# coefficient vectors a of three linear estimators a^T y of the slope
a_ls = X @ np.linalg.solve(X.T @ X, lam) # least squares: a = X (X^T X)^{-1}
    lambda
a_end = np.zeros(n)
a_end[[0, -1]] = [-1.0, 1.0]
a_end /= x[-1] - x[0] # slope through the two end points
low, high = np.arange(n) < n // 2, np.arange(n) >= n // 2
a_grp = (high / high.sum() - low / low.sum()) / (x[high].mean() - x[low].mean()) #
    group means

for name, a in [("least squares", a_ls), ("end points", a_end), ("group means",
    a_grp)]:
    print(f"{name:14s} X^T a = {X.T @ a}    Var/sigma^2 = {a @ a:.5f}    Ma = a_ls: {
    np.allclose(M @ a, a_ls)}")
```

The end-point estimator throws away ten of the twelve observations, so its poor showing is no surprise. The group-means estimator uses every observation, and it is still clearly worse than least squares. It weights all observations in a half equally, while least squares weights each observation in proportion to its distance from $\bar{x}$. Estimators of this grouping type were proposed by Wald (1940) for regressions in which x itself is measured with error. Chapter 24 returns to that problem.

7.1.5 Why these restrictions?

Linearity and unbiasedness are restrictions of convenience as well as of principle, and it is worth being clear about what each one does.

Unbiasedness removes estimators that are good only near one parameter value. It is not sacred. [Section 7.2](#) shows biased estimators with smaller mean squared error, and [Section 7.6](#) derives a whole family of them from prior information. Unbiasedness can also be too demanding: a nonestimable $\lambda^\top \beta$ has no linear unbiased estimator, and under normality no unbiased estimator of any kind ([Exercise \(C1\)](#)).

Linearity is what makes the second-moment assumptions enough. The variance of a nonlinear estimator, such as a median or a trimmed mean, depends on the whole error distribution. Its variance can't be compared with that of least squares until that distribution is specified. For linear estimators [Eq. 7.1.2](#) needs only σ^2 . The restriction therefore has a cost: optimality among linear estimators says nothing about nonlinear ones. [Section 7.2](#) gives an error distribution under which a simple nonlinear estimator is far better than least squares. [Section 7.4](#) shows that under normality this cannot happen.

7.1.6 Exercises

7.1.6.1 A. Check your understanding

Exercise A1.

In the location model $Y_i = \mu + \varepsilon_i$ ($\mathbf{X} = \mathbf{1}$), show that the LUEs of μ are the weighted averages $\sum_i a_i Y_i$ with $\sum_i a_i = 1$. Identify $\mathcal{A}_\lambda$ and $\mathbf{a}_*$ and find the LUE of smallest variance directly.

Exercise A2.

In the one-way layout $E(Y_{ij}) = \mu + \alpha_i$ with three groups and an intercept ([Example \(Four answers, one fit\)](#)), show that α_1 has no linear unbiased estimator. Write down two different LUEs of $\alpha_1 - \alpha_2$, and verify that their coefficient vectors have the same projection onto $\mathcal{C}(\mathbf{X})$.

7.1.6.2 B. Practice

Exercise B1.

A linear estimator $\mathbf{a}^\top \mathbf{Y}$ is an **unbiased estimator of zero** if $E(\mathbf{a}^\top \mathbf{Y}) = 0$ for every β . Show that this holds iff $\mathbf{a} \in \mathcal{C}(\mathbf{X})^\perp$, iff $\mathbf{a}^\top \mathbf{Y}$ is a linear function of the residual vector $\hat{\varepsilon}$. Show that every unbiased estimator of zero is uncorrelated with $\lambda^\top \hat{\beta}$ for every estimable $\lambda^\top \beta$.

Solution. By [Proposition 7.1.1](#) with $\lambda = \mathbf{0}$, $\mathbf{a}^\top \mathbf{Y}$ is unbiased for zero iff $\mathbf{X}^\top \mathbf{a} = \mathbf{0}$, iff $\mathbf{a} \in \mathcal{C}(\mathbf{X})^\perp$. If so, $\mathbf{a} = (\mathbf{I} - \mathbf{M})\mathbf{a}$ and $\mathbf{a}^\top \mathbf{Y} = \mathbf{a}^\top (\mathbf{I} - \mathbf{M})\mathbf{Y} = \mathbf{a}^\top \hat{\varepsilon}$. Conversely, $\mathbf{w}^\top \hat{\varepsilon} = ((\mathbf{I} - \mathbf{M})\mathbf{w})^\top \mathbf{Y}$ has coefficient vector in $\mathcal{C}(\mathbf{X})^\perp$. For the last claim, write $\lambda^\top \hat{\beta} = \rho^\top \mathbf{M}\mathbf{Y}$. By [Theorem 2.2.1](#), $\text{Cov}(\rho^\top \mathbf{M}\mathbf{Y}, \mathbf{a}^\top \mathbf{Y}) = \sigma^2 \rho^\top \mathbf{M}\mathbf{a} = 0$, because

$$M\mathbf{a} = \mathbf{0}.$$

Exercise B2.

Let $\mathbf{A}$ be $p \times q$ with $\mathcal{C}(\mathbf{A}) \subseteq \mathcal{C}(\mathbf{X}^\top)$. Show that $\mathbf{A}\mathbf{Y}$ is an LUE of $\mathbf{A}^\top\boldsymbol{\beta}$ iff $\mathbf{A}\mathbf{X} = \mathbf{A}^\top$, and that the set of such $\mathbf{A}$ is $\{\mathbf{A}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top + \mathbf{W}(\mathbf{I} - \mathbf{M}) : \mathbf{W} \in \mathbb{R}^{q \times n}\}$.

7.1.6.3 C. Going deeper

Exercise C1.

Assume $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ and $\boldsymbol{\lambda} \notin \mathcal{C}(\mathbf{X}^\top)$. Show that no statistic $\delta(\mathbf{Y})$, linear or not, satisfies $E\delta(\mathbf{Y}) = \boldsymbol{\lambda}^\top\boldsymbol{\beta}$ for all $\boldsymbol{\beta}$. *Hint:* the distribution of $\mathbf{Y}$ depends on $\boldsymbol{\beta}$ only through $\mathbf{X}\boldsymbol{\beta}$.

Solution. Since $\boldsymbol{\lambda} \notin \mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$, there is $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ with $\boldsymbol{\lambda}^\top\mathbf{v} \neq 0$. The parameters $\boldsymbol{\beta}$ and $\boldsymbol{\beta} + \mathbf{v}$ give the same mean $\mathbf{X}\boldsymbol{\beta}$, hence the same distribution of $\mathbf{Y}$, hence the same value of $E\delta(\mathbf{Y})$. An unbiased δ would need $\boldsymbol{\lambda}^\top\boldsymbol{\beta} = \boldsymbol{\lambda}^\top(\boldsymbol{\beta} + \mathbf{v})$, which is false. The argument uses only that the law of $\mathbf{Y}$ is a function of $\mathbf{X}\boldsymbol{\beta}$ and σ^2 . [Chapter 8](#) calls this *nonidentifiability* ([Definition 8.1.1](#)).

Exercise C2.

For the group-means estimator of [Example \(Three linear unbiased estimators of a slope\)](#) with general $x_1 \leq \dots \leq x_n$ ($n = 2m$), find its variance in terms of $\bar{x}_{\text{high}} - \bar{x}_{\text{low}}$. Show that its efficiency relative to least squares is $(\bar{x}_{\text{high}} - \bar{x}_{\text{low}})^2 n / (4S_{xx})$, and that this is 1 iff the x values take only two distinct values, with half of the observations at each.

7.2 The Gauss–Markov theorem

[Section 7.1](#) turned the search for a best linear unbiased estimator into a geometric problem. The unbiased coefficient vectors form the flat $\mathcal{A}_{\boldsymbol{\lambda}} = \mathbf{a}_* + \mathcal{C}(\mathbf{X})^\perp$, and the variance of each is σ^2 times its squared length. The shortest vector in a flat parallel to $\mathcal{C}(\mathbf{X})^\perp$ is the one in $\mathcal{C}(\mathbf{X})$, and that vector is the least squares coefficient vector $\mathbf{a}_*$. The Gauss–Markov theorem says exactly this, together with its matrix form and its consequences.

7.2.1 The theorem

We keep the second-moment assumptions [Eq. 7.1.1](#), with $\mathbf{X}$ of any rank r , and write $(\mathbf{X}^\top\mathbf{X})^-$ for any generalized inverse of $\mathbf{X}^\top\mathbf{X}$.

Theorem 7.2.1 — Gauss–Markov

Assume [Eq. 7.1.1](#). Let $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$, say $\boldsymbol{\lambda} = \mathbf{X}^\top\boldsymbol{\rho}$, and let $\hat{\boldsymbol{\beta}}$ be any least squares

estimate.

- (a) $\lambda^\top \hat{\beta}$ is a linear unbiased estimator of $\lambda^\top \beta$, with

$$\text{Var}(\lambda^\top \hat{\beta}) = \sigma^2 \lambda^\top (X^\top X)^{-1} \lambda.$$

- (b) For every linear unbiased estimator $a^\top Y$ of $\lambda^\top \beta$,

$$\text{Var}(a^\top Y) = \text{Var}(\lambda^\top \hat{\beta}) + \sigma^2 \|(I - M)a\|^2. \quad (7.2.1)$$

- (c) Consequently $\lambda^\top \hat{\beta}$ is a BLUE of $\lambda^\top \beta$, and it is the only one: an LUE $a^\top Y$ is a BLUE iff $a = X(X^\top X)^{-1} \lambda$, that is, iff $a^\top y = \lambda^\top \hat{\beta}$ for every y .

- (d) (*Matrix form.*) Let Λ be $p \times q$ with $\mathcal{C}(\Lambda) \subseteq \mathcal{C}(X^\top)$. For every linear unbiased estimator AY of $\Lambda^\top \beta$,

$$\text{Cov}(AY) = \text{Cov}(\Lambda^\top \hat{\beta}) + \sigma^2 A(I - M)A^\top, \quad (7.2.2)$$

where $\text{Cov}(\Lambda^\top \hat{\beta}) = \sigma^2 \Lambda^\top (X^\top X)^{-1} \Lambda$. So $\text{Cov}(AY) - \text{Cov}(\Lambda^\top \hat{\beta})$ is nonnegative definite, and it is zero iff $AY = \Lambda^\top \hat{\beta}$ for every y .

Proof. Put $a_* = M\rho = X(X^\top X)^{-1} \lambda$. By Proposition 7.1.2(c), $a_*^\top Y = \lambda^\top \hat{\beta}$. Since $a_* \in \mathcal{A}_\lambda$, it is an LUE by Proposition 7.1.1. By Eq. 7.1.2 its variance is

$$\sigma^2 \|M\rho\|^2 = \sigma^2 \rho^\top M\rho = \sigma^2 \rho^\top X(X^\top X)^{-1} X^\top \rho = \sigma^2 \lambda^\top (X^\top X)^{-1} \lambda,$$

using Theorem 6.3.7. This proves (a).

For (b), let $a^\top Y$ be any LUE. Then $a \in \mathcal{A}_\lambda$, so $Ma = a_*$ by Proposition 7.1.2(b). The decomposition $a = Ma + (I - M)a = a_* + (I - M)a$ is orthogonal, so $\|a\|^2 = \|a_*\|^2 + \|(I - M)a\|^2$. Multiplying by σ^2 and using Eq. 7.1.2 twice gives Eq. 7.2.1.

For (c), the second term of Eq. 7.2.1 is nonnegative, so $\lambda^\top \hat{\beta}$ is a BLUE. Because $\sigma^2 > 0$, equality holds iff $(I - M)a = 0$, that is, iff $a \in \mathcal{C}(X)$. By Proposition 7.1.2(b) the only point of $\mathcal{A}_\lambda$ in $\mathcal{C}(X)$ is a_* . Two linear statistics $a^\top y$ and $a_*^\top y$ agree for every y iff $a = a_*$.

For (d), write $\Lambda = X^\top P$ for an $n \times q$ matrix P , and put $A_* = P^\top M$, so that $A_* Y = \Lambda^\top \hat{\beta}$ by Proposition 7.1.2(c) applied column by column. Each row $a_k^\top$ of an LUE matrix A has $a_k \in \mathcal{A}_{\lambda_k}$, where λ_k is the k th column of Λ . So $Ma_k = Mp_k$, and hence $AM = A_*$ (the k th row of AM is $(Ma_k)^\top$, since M is symmetric). Now split the identity:

$$\begin{aligned} AA^\top &= AMA^\top + A(I - M)A^\top \\ &= (AM)(AM)^\top + A(I - M)A^\top = A_* A_*^\top + A(I - M)A^\top, \end{aligned}$$

using $M = MM^\top$. Multiplying by σ^2 and using Theorem 2.2.1 gives $\text{Cov}(AY) = \text{Cov}(A_* Y) + \sigma^2 A(I - M)A^\top$. Also $A_* A_*^\top = P^\top M P = \Lambda^\top (X^\top X)^{-1} \Lambda$, as in (a). The difference $\sigma^2 A(I - M)A^\top = \sigma^2 [A(I - M)][A(I - M)]^\top$ is nonnegative definite. It is zero iff $A(I - M) = 0$, iff $A = AM = A_*$. ■

Three features of the theorem deserve comment.

It is distribution-free. The proof used $E(\varepsilon) = \mathbf{0}$ and $\text{Cov}(\varepsilon) = \sigma^2 \mathbf{I}$, and nothing else. The errors may be skewed, heavy-tailed, discrete, or dependent in any way that leaves them uncorrelated. They need not even be identically distributed, as long as their variances are equal.

It holds in every rank. When $\mathbf{X}$ is rank deficient, $\hat{\beta}$ is not unique and neither is the generalized inverse. But for estimable λ , both $\lambda^\top \hat{\beta}$ and $\lambda^\top (\mathbf{X}^\top \mathbf{X})^- \lambda$ are the same for every choice (Theorem 6.4.4 and Lemma 6.3.6). A nonestimable $\lambda^\top \beta$ has no LUE at all, so the theorem covers every function for which the question makes sense.

The proof is a projection. Equation Eq. 7.2.1 is Pythagoras in $\mathbb{R}^n$, applied to the coefficient vector and not to the data.

Idea. The least squares coefficient vector $\mathbf{a}_*$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$ of the coefficient vector of *any* linear unbiased estimator. Projecting a coefficient vector onto $\mathcal{C}(\mathbf{X})$ keeps the estimator unbiased and removes the part that only adds noise. That part, $(\mathbf{I} - \mathbf{M})\mathbf{a}$, multiplies the residuals, and the residuals carry no information about β .

The decomposition Eq. 7.1.3 gives the same result in the language of estimators. Every LUE is $\lambda^\top \hat{\beta} + \mathbf{w}^\top \hat{\varepsilon}$, and the two terms are uncorrelated (Exercise (B1)), so their variances add. An estimator that uses the residuals only adds uncorrelated noise to least squares.

7.2.2 What the matrix form buys

The matrix form Eq. 7.2.2 is stronger than the scalar statement applied coordinate by coordinate, and it is the form usually quoted for vector estimators.

Corollary 7.2.2 — Consequences of the Gauss–Markov theorem

Assume Eq. 7.1.1, let $\mathcal{C}(\Lambda) \subseteq \mathcal{C}(\mathbf{X}^\top)$, and let $\tilde{\theta} = \mathbf{A}\mathbf{Y}$ be any LUE of $\theta = \Lambda^\top \beta$, with $\hat{\theta} = \Lambda^\top \hat{\beta}$.

- (a) $\text{Var}(\mathbf{b}^\top \tilde{\theta}) \geq \text{Var}(\mathbf{b}^\top \hat{\theta})$ for every $\mathbf{b} \in \mathbb{R}^q$; in particular each coordinate of $\hat{\theta}$ has the smallest variance.
- (b) $E\|\tilde{\theta} - \theta\|^2 \geq E\|\hat{\theta} - \theta\|^2$.
- (c) $\det \text{Cov}(\tilde{\theta}) \geq \det \text{Cov}(\hat{\theta})$.
- (d) If $\text{rank}(\mathbf{X}) = p$, then $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ is the BLUE of β , and $\text{Cov}(\tilde{\beta}) - \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ is nonnegative definite for every LUE $\tilde{\beta}$ of β .
- (e) The fitted vector $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{Y}$ is the BLUE of $\mathbf{X}\beta$, and the fitted value $\hat{y}_i$ is the BLUE of $\mathbf{x}_{(i)}^\top \beta$, with variance $\sigma^2 h_{ii}$, where h_{ii} is the leverage.

Proof. Let $\mathbf{D} = \text{Cov}(\tilde{\theta}) - \text{Cov}(\hat{\theta}) \succeq \mathbf{0}$ (Theorem 7.2.1(d)). (a) $\text{Var}(\mathbf{b}^\top \tilde{\theta}) - \text{Var}(\mathbf{b}^\top \hat{\theta}) = \mathbf{b}^\top \mathbf{D} \mathbf{b} \geq 0$ by Definition 1.7.1. The coordinates are the case $\mathbf{b} = \mathbf{e}_k$. (b) Both estimators are unbiased, so the mean squared errors are the traces of the covariance matrices, and $\text{tr } \mathbf{D} = \sum_k \mathbf{e}_k^\top \mathbf{D} \mathbf{e}_k \geq 0$. (c) Write $\mathbf{C} = \text{Cov}(\hat{\theta})$. If $\mathbf{C}$ is singular, its determinant is zero and there is nothing to prove, because covariance matrices are nonnegative definite (Theorem 2.2.3). Otherwise $\mathbf{C}$ is positive definite, with a symmetric positive definite square root (Theorem 1.7.6). The matrix $\mathbf{C}^{-1/2} \text{Cov}(\tilde{\theta}) \mathbf{C}^{-1/2} = \mathbf{I} + \mathbf{C}^{-1/2} \mathbf{D} \mathbf{C}^{-1/2}$ has all eigenvalues at least 1: the matrix $\mathbf{C}^{-1/2} \mathbf{D} \mathbf{C}^{-1/2}$ is nonnegative definite by Proposi-

tion 1.7.5(c), so its eigenvalues are nonnegative by [Theorem 1.7.4](#), and adding $\mathbf{I}$ adds 1 to each of them. Its determinant $\det \text{Cov}(\tilde{\boldsymbol{\theta}}) / \det \mathbf{C}$ is therefore at least 1. (d) Take $\boldsymbol{\Lambda} = \mathbf{I}_p$, which is allowed because $\mathcal{C}(\mathbf{X}^\top) = \mathbb{R}^p$, and use $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ ([Theorem 5.5.1](#)). (e) Take $\boldsymbol{\Lambda} = \mathbf{X}^\top$. Then $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{M}\mathbf{Y}$ ([Theorem 6.4.1](#)), with covariance $\sigma^2 \mathbf{M}$. Its i th diagonal entry is $\sigma^2 h_{ii}$ ([Definition 6.8.1](#)).

■

Part (b) says least squares minimizes the total mean squared error among unbiased linear estimators. Part (c) has a geometric meaning that Chapter 12 will use. For normal errors, the confidence ellipsoid for $\boldsymbol{\theta}$ has volume proportional to $\sqrt{\det \text{Cov}}$, so least squares gives the smallest ellipsoids.

Example. (The slope estimators under skewed errors)

Return to the three slope estimators of [Example \(Three linear unbiased estimators of a slope\)](#) and simulate 200,000 data sets with $\beta_0 = 2$, $\beta_1 = 0.5$ and errors $\varepsilon_i = E_i - 1$, where the E_i are independent exponential with mean one. These errors have mean zero and variance one, but they are strongly skewed: most are slightly negative and a few are large and positive. The average estimates are 0.5000, 0.4998 and 0.5001, and the simulated variances are 0.00724, 0.01643 and 0.01004, against the exact values 0.00729, 0.01653 and 0.01008. [Figure 7.2.1\(a\)](#) shows the three sampling distributions. All three are centred on the truth, and least squares is the most concentrated: its ± 2 standard deviation bar is the shortest. The end-point estimator, a difference of just two skewed errors, has a sharp peak but long tails, so the height of a density is a poor guide to variance. The errors are far from normal and the theorem does not care.

```
import numpy as np

x = np.array([1.0, 2.0, 2.5, 3.0, 4.0, 5.0, 5.5, 7.0, 8.0, 9.0, 10.5, 12.0])
n = len(x)
X = np.column_stack([np.ones(n), x])
lam = np.array([0.0, 1.0])
a_ls = X @ np.linalg.solve(X.T @ X, lam)           # target: the slope beta_1
            lambda                                     # least squares: a = X (X^T X)^{-1}
a_end = np.zeros(n)
a_end[[0, -1]] = [-1.0, 1.0]
a_end /= x[-1] - x[0]                               # slope through the two end points
low, high = np.arange(n) < n // 2, np.arange(n) >= n // 2
a_grp = (high / high.sum() - low / low.sum()) / (x[high].mean() - x[low].mean()) #
            group means

estimators = {"ls": a_ls, "end": a_end, "grp": a_grp}
rng = np.random.default_rng(7)
beta, sigma, reps = np.array([2.0, 0.5]), 1.0, 200_000
E = sigma * (rng.exponential(size=(reps, n)) - 1.0) # skewed errors, mean 0,
            variance sigma^2
Y = X @ beta + E                                     # one simulated data set per
            row
sims = {name: Y @ a for name, a in estimators.items()}
for name, est in sims.items():
    print(f"{name:4s} mean {est.mean():.4f}    variance {est.var():.4f}")
```

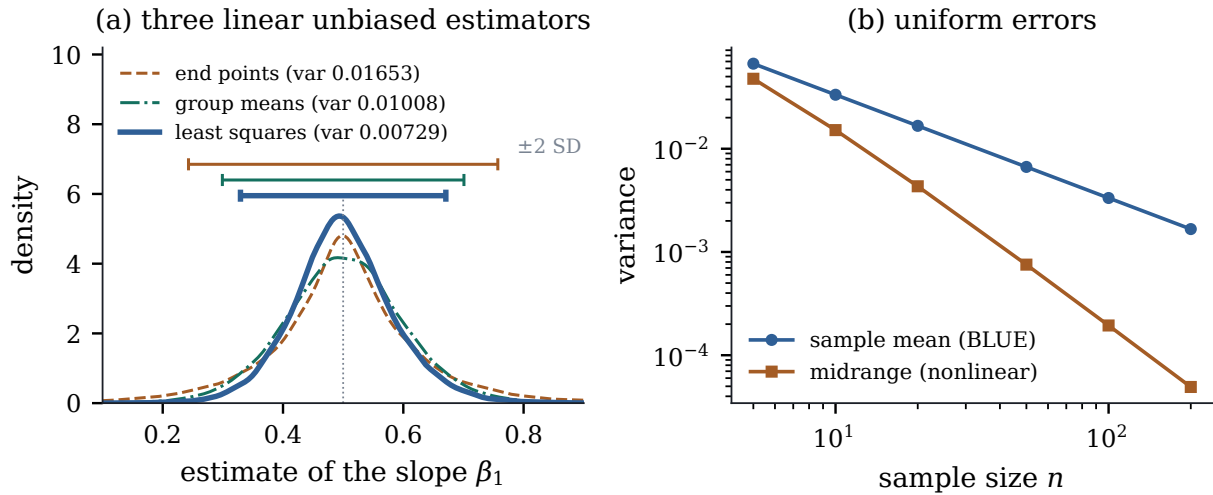


Figure 7.2.1: (a) Sampling distributions of three linear unbiased estimators of a slope, with skewed errors (200,000 simulated data sets), as smoothed densities, with bars marking the true value ± 2 standard deviations. Least squares has the smallest variance, as the Gauss–Markov theorem requires. (b) In the location model with uniform errors, the midrange, which is unbiased but not linear, has far smaller variance than the sample mean, which is the BLUE.

7.2.3 Correlated errors: Aitken’s theorem

If $\text{Cov}(\epsilon) = \sigma^2 \mathbf{V}$ with $\mathbf{V}$ known and positive definite, the theorem applies after a change of coordinates, and the BLUE becomes the generalized least squares estimate of Definition 6.9.1.

Corollary 7.2.3 — Aitken

Assume $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$, with $\mathbf{V}$ known and positive definite. For $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$, the BLUE of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\text{GLS}}$, where $\hat{\boldsymbol{\beta}}_{\text{GLS}}$ is any solution of $\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$. Its variance is $\sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \boldsymbol{\lambda}$.

Proof. Let $\mathbf{V}^{1/2}$ be the symmetric positive definite square root of $\mathbf{V}$ (Theorem 1.7.6). Put $\mathbf{Z} = \mathbf{V}^{-1/2} \mathbf{Y}$ and $\mathbf{X}_* = \mathbf{V}^{-1/2} \mathbf{X}$. By Theorem 2.2.1, $E(\mathbf{Z}) = \mathbf{X}_* \boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Z}) = \sigma^2 \mathbf{I}$, so $\mathbf{Z}$ satisfies Eq. 7.1.1 with model matrix $\mathbf{X}_*$. Since $\mathbf{V}^{-1/2}$ is nonsingular, $\mathcal{C}(\mathbf{X}_*^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{V}^{-1/2}) = \mathcal{C}(\mathbf{X}^\top)$, so the same functions are estimable. The map $\mathbf{a}^\top \mathbf{Y} = (\mathbf{V}^{1/2} \mathbf{a})^\top \mathbf{Z}$ is a bijection between linear estimators based on $\mathbf{Y}$ and those based on $\mathbf{Z}$. It preserves means and variances, since both sides are the same random variable. So a BLUE in one model is a BLUE in the other. By Theorem 7.2.1 applied to $\mathbf{Z}$, the BLUE is $\boldsymbol{\lambda}^\top \mathbf{b}$ for any least squares solution $\mathbf{b}$ of the $\mathbf{Z}$ problem. That is, $\mathbf{b}$ minimizes $\|\mathbf{Z} - \mathbf{X}_* \mathbf{b}\|^2 = (\mathbf{Y} - \mathbf{X} \mathbf{b})^\top \mathbf{V}^{-1} (\mathbf{Y} - \mathbf{X} \mathbf{b})$, which is the generalized least squares criterion. Its normal equations (Theorem 6.4.2) are $\mathbf{X}_*^\top \mathbf{X}_* \mathbf{b} = \mathbf{X}_*^\top \mathbf{Z}$, that is, $\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$. The variance is part (a) of the theorem with $\mathbf{X}_*$ in place of $\mathbf{X}$. ■

Ordinary least squares remains unbiased when $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$, but in general it is no longer best. Theorem 6.9.3 characterized the models in which the two estimators coincide, and Exercise (C3) shows that exactly these models keep ordinary least squares optimal. Chapter 31 develops

the general theory, including singular V .

The whitening in the proof carries the rest of the least squares toolkit with it. The fitted vector $\mathbf{X}\hat{\beta}_{\text{GLS}} = \mathbf{V}^{1/2}\mathbf{M}_*\mathbf{Z}$, where $\mathbf{M}_*$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X}_*)$, is the generalized least squares fit of Definition 6.9.1: it is idempotent but not symmetric, an oblique projection onto $\mathcal{C}(\mathbf{X})$ in the sense of Proposition 6.9.4. The residual sum of squares of the whitened problem is $\|(\mathbf{I} - \mathbf{M}_*)\mathbf{Z}\|^2 = (\mathbf{Y} - \mathbf{X}\hat{\beta}_{\text{GLS}})^\top \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\hat{\beta}_{\text{GLS}})$, and since $\text{rank}(\mathbf{X}_*) = r$, Theorem 2.3.1 and Proposition 6.3.4 give it expectation $\sigma^2(n - r)$. So

$$s_{\text{GLS}}^2 = \frac{(\mathbf{Y} - \mathbf{X}\hat{\beta}_{\text{GLS}})^\top \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\hat{\beta}_{\text{GLS}})}{n - r}$$

is unbiased for σ^2 .

7.2.4 Design: when is a coefficient estimated best?

The theorem fixes $\mathbf{X}$ and optimizes over estimators. The variance it leaves, $\sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$, still depends on $\mathbf{X}$. When the regressor values can be chosen, one can ask which designs make it small. The Frisch–Waugh–Lovell theorem gives a clean answer for single coefficients.

Proposition 7.2.4 — Orthogonal columns are best

Let $\mathbf{X}$ have full column rank, let $\mathbf{x}_j$ be its j th column and $\mathbf{X}_{(j)}$ the other columns, with projection $\mathbf{M}_{(j)}$ onto $\mathcal{C}(\mathbf{X}_{(j)})$. Under Eq. 7.1.1,

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2} \geq \frac{\sigma^2}{\|\mathbf{x}_j\|^2},$$

with equality iff $\mathbf{x}_j$ is orthogonal to every other column. If all columns are mutually orthogonal, then $\hat{\beta}_j = \mathbf{x}_j^\top \mathbf{Y} / \|\mathbf{x}_j\|^2$, and it is unchanged when any other columns are deleted from the model.

Proof. Put $\tilde{\mathbf{x}}_j = (\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j$. By Theorem 6.6.2(a), with $\mathbf{X}_1 = \mathbf{x}_j$ and $\mathbf{X}_2 = \mathbf{X}_{(j)}$, $\hat{\beta}_j = \tilde{\mathbf{x}}_j^\top \mathbf{Y} / \|\tilde{\mathbf{x}}_j\|^2$. So $\text{Var}(\hat{\beta}_j) = \sigma^2 / \|\tilde{\mathbf{x}}_j\|^2$ by Theorem 2.2.1. By Proposition 6.3.4(d) applied to $\mathbf{I} - \mathbf{M}_{(j)}$, $\|\tilde{\mathbf{x}}_j\| \leq \|\mathbf{x}_j\|$, with equality iff $\mathbf{M}_{(j)}\mathbf{x}_j = \mathbf{0}$, that is, iff $\mathbf{x}_j \perp \mathcal{C}(\mathbf{X}_{(j)})$. If all columns are orthogonal, $\tilde{\mathbf{x}}_j = \mathbf{x}_j$, and the formula for $\hat{\beta}_j$ involves no other column, so deleting columns does not change it. ■

So among designs whose columns have given lengths, orthogonal designs minimize the variance of every coefficient at once. This is one reason designed experiments are built from orthogonal contrasts (Part IV), and why collinearity, meaning near-dependence among columns, inflates variances (Chapter 26). Exercise (C1) applies the proposition to a problem of weighing objects on a balance.

7.2.5 What the theorem does not say

The Gauss–Markov theorem is often quoted as “least squares is optimal”. The theorem is narrower than that. It compares least squares only with estimators that are linear and unbiased, and it assumes the model’s mean and covariance are right. Relax any of these and least squares can lose.

Nonlinear estimators. When the errors are not normal, a nonlinear unbiased estimator can beat least squares by a wide margin.

Example. (Uniform errors and the midrange)

In the location model $Y_i = \mu + \varepsilon_i$, let the errors be independent and uniform on $(-1, 1)$, so $\sigma^2 = 1/3$. The BLUE of μ is $\bar{Y}$, with variance $1/(3n)$. The **midrange** $(Y_{(1)} + Y_{(n)})/2$, the average of the smallest and largest observation, is not linear. It is unbiased, because the error distribution is symmetric about zero, and its variance is $2/\{(n+1)(n+2)\}$ ([Exercise \(C2\)](#)). For $n = 20$ the two variances are 0.0167 and 0.00433, a ratio of 3.9. For $n = 100$ they are 0.0033 and 0.00019, a ratio of 17.2. The variance of the mean falls like $1/n$ and that of the midrange like $1/n^2$ ([Figure 7.2.1\(b\)](#)).

```
ns = np.array([5, 10, 20, 50, 100, 200])
var_mean = 1 / (3 * ns)                                # errors uniform on (-1, 1):
    variance 1/3
var_mid = 2 / ((ns + 1) * (ns + 2))                    # exact variance of the midrange
for m, v1, v2 in zip(ns, var_mean, var_mid):
    print(f"n = {m:3d}    Var(mean) = {v1:.5f}    Var(midrange) = {v2:.6f}    ratio = {
        v1 / v2:.1f}")
```

Uniform errors have sharp edges, and the extreme observations locate those edges very precisely. A linear estimator must give every observation a fixed weight whatever the sample looks like, so it cannot exploit this. Heavy-tailed errors produce the opposite phenomenon: there the extremes are unreliable, and estimators that downweight them, such as medians and M-estimators, beat least squares ([Exercise \(C1\)](#); Chapter 20 and Chapter 45 return to such estimators). [Section 7.4](#) shows that under *normal* errors neither phenomenon occurs, and least squares is best among all unbiased estimators.

Biased estimators. Mean squared error is variance plus squared bias. Accepting a little bias can reduce variance by more. For an estimator $\tilde{\beta}$ of β , the natural matrix criterion is

$$E[(\tilde{\beta} - \beta)(\tilde{\beta} - \beta)^\top] = \text{Cov}(\tilde{\beta}) + (E\tilde{\beta} - \beta)(E\tilde{\beta} - \beta)^\top,$$

and the Gauss–Markov theorem ranks estimators by it only within the unbiased class. The ridge estimator $(\mathbf{X}^\top \mathbf{X} + k\mathbf{I})^{-1} \mathbf{X}^\top \mathbf{Y}$ is linear but biased, and some $k > 0$ always gives a smaller total mean squared error than least squares ([Exercise \(B1\)](#)). Stein (1956) and James and Stein (1961) showed more: in the model $\mathbf{Y} \sim N_n(\beta, \sigma^2 \mathbf{I})$ with known σ^2 and $n \geq 3$, a nonlinear shrinkage estimator has smaller total mean squared error than $\mathbf{Y}$ itself *for every* β . Chapter 27 develops shrinkage. [Section 7.6](#) derives a family of biased linear estimators from prior information.

Wrong assumptions. If $\text{Cov}(\varepsilon) \neq \sigma^2 \mathbf{I}$, [Corollary 7.2.3](#) replaces least squares by generalized least squares. If the mean is misspecified, so that $E(\mathbf{Y}) \notin \mathcal{C}(\mathbf{X})$, then $E(\hat{\mathbf{Y}}) = \mathbf{M} E(\mathbf{Y})$ is the projection of the true mean. Least squares then estimates the best approximation within the model, not the truth (compare [Theorem 6.11.1](#)). Part V is about diagnosing and repairing these departures.

7.2.6 Exercises

7.2.6.1 A. Check your understanding

Exercise A1.

Show that $\sum_i \text{Var}(\hat{y}_i) = \sigma^2 r$ under Eq. 7.1.1. Using Corollary 7.2.2(e), deduce that no linear unbiased estimator of $\mathbf{X}\boldsymbol{\beta}$ has total variance less than $\sigma^2 r$. Interpret this as “each estimated dimension of the mean costs one σ^2 ”.

7.2.6.2 B. Practice

Exercise B1.

Let $\mathbf{X}$ have full column rank and, for $k \geq 0$, let $\tilde{\boldsymbol{\beta}}_k = (\mathbf{X}^\top \mathbf{X} + k\mathbf{I})^{-1} \mathbf{X}^\top \mathbf{Y}$. Under Eq. 7.1.1, find the bias and covariance matrix of $\tilde{\boldsymbol{\beta}}_k$. Using a spectral decomposition $\mathbf{X}^\top \mathbf{X} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\top$, show that the total mean squared error is

$$f(k) = \sum_{j=1}^p \frac{\sigma^2 d_j + k^2 \gamma_j^2}{(d_j + k)^2}, \quad \boldsymbol{\gamma} = \mathbf{Q}^\top \boldsymbol{\beta}, \quad \mathbf{D} = \text{diag}(d_1, \dots, d_p),$$

and that $f'(0) < 0$. Conclude that some $k > 0$ beats least squares in total mean squared error, for every $\boldsymbol{\beta}$ and σ^2 , although the best k depends on them.

Solution. Write $\mathbf{S} = \mathbf{X}^\top \mathbf{X}$ and $\mathbf{W}_k = (\mathbf{S} + k\mathbf{I})^{-1}$. Then $E\tilde{\boldsymbol{\beta}}_k = \mathbf{W}_k \mathbf{S} \boldsymbol{\beta} = \boldsymbol{\beta} - k\mathbf{W}_k \boldsymbol{\beta}$, so the bias is $-k\mathbf{W}_k \boldsymbol{\beta}$, and $\text{Cov}(\tilde{\boldsymbol{\beta}}_k) = \sigma^2 \mathbf{W}_k \mathbf{S} \mathbf{W}_k$. With $\mathbf{S} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\top$, $\mathbf{W}_k = \mathbf{Q}(\mathbf{D} + k\mathbf{I})^{-1} \mathbf{Q}^\top$, so $\text{tr Cov} = \sigma^2 \sum_j d_j / (d_j + k)^2$ and $\|k\mathbf{W}_k \boldsymbol{\beta}\|^2 = k^2 \sum_j \gamma_j^2 / (d_j + k)^2$. Adding gives $f(k)$. Differentiating,

$$f'(k) = \sum_j \frac{2d_j(k\gamma_j^2 - \sigma^2)}{(d_j + k)^3}, \quad f'(0) = -2\sigma^2 \sum_j d_j^{-2} < 0.$$

Since f is differentiable at 0 with negative derivative, $f(k) < f(0)$ for all small $k > 0$, and $f(0)$ is the total mean squared error of $\hat{\boldsymbol{\beta}}$. This does not contradict Theorem 7.2.1, because $\tilde{\boldsymbol{\beta}}_k$ is biased for $k > 0$.

Exercise B2.

Suppose the design grows with n and $\mathbf{X}^\top \mathbf{X}$ has full rank for all large n . Show that $\text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) \leq \sigma^2 \|\boldsymbol{\lambda}\|^2 / \lambda_{\min}(\mathbf{X}^\top \mathbf{X})$. Deduce that if the smallest eigenvalue of $\mathbf{X}^\top \mathbf{X}$ tends to infinity, then $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} \rightarrow \boldsymbol{\lambda}^\top \boldsymbol{\beta}$ in mean square, for every $\boldsymbol{\lambda}$.

Exercise B3.

In the location model with $\text{Cov}(\mathbf{Y}) = \sigma^2 \text{diag}(v_1, \dots, v_n)$, compare the variance of the ordinary mean $\bar{Y}$ with that of the BLUE given by Corollary 7.2.3. Show that the ratio is at least one, with equality iff all v_i are equal.

7.2.6.3 C. Going deeper

Exercise C1.

Each of k objects with unknown weights $\beta_1, \dots, \beta_k$ is weighed in n weighings on a two-pan balance. In weighing i every object is put on the left pan or the right pan, and the reading Y_i is the difference of the pan loads plus an error. So $\mathbf{X}$ is $n \times k$ with entries ± 1 , and Eq. 7.1.1 holds.

- (a) Show that $\text{Var}(\hat{\beta}_j) \geq \sigma^2/n$ for every j , with equality for all j iff $\mathbf{X}^\top \mathbf{X} = n\mathbf{I}$.
 (b) Take $k = 3$, $n = 4$, and the two designs whose rows are

$$\mathbf{X}_1 : (1, 1, 1), (1, -1, -1), (-1, 1, -1), (-1, -1, 1); \quad \mathbf{X}_2 : (1, 1, 1), (1, 1, -1), (1, -1, 1), (1, 1, 1).$$

For each, compute $\text{Cov}(\hat{\beta})/\sigma^2$, its trace and its determinant. Which design is better by each criterion, and what do the criteria measure (compare Corollary 7.2.2(b) and (c))?

- (c) Show that if a ± 1 matrix with $\mathbf{X}^\top \mathbf{X} = n\mathbf{I}$ exists with $k = n \geq 3$, then n is a multiple of 4. (Whether every multiple of 4 occurs is the Hadamard conjecture, which is open.)

Solution.

- (a) Every column has $\|\mathbf{x}_j\|^2 = n$, so Proposition 7.2.4 gives $\text{Var}(\hat{\beta}_j) \geq \sigma^2/n$, with equality iff $\mathbf{x}_j$ is orthogonal to the other columns. Equality for all j means all columns are mutually orthogonal, that is, $\mathbf{X}^\top \mathbf{X} = n\mathbf{I}$.
 (b) $\mathbf{X}_1^\top \mathbf{X}_1 = 4\mathbf{I}$, so $\text{Cov}(\hat{\beta})/\sigma^2 = \mathbf{I}/4$, with trace $3/4$ and determinant $1/64$. For $\mathbf{X}_2$, the columns are $(1, 1, 1, 1)$, $(1, 1, -1, 1)$, $(1, -1, 1, 1)$, so

$$\mathbf{X}_2^\top \mathbf{X}_2 = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 4 & 0 \\ 2 & 0 & 4 \end{pmatrix}, \quad (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} = \frac{1}{32} \begin{pmatrix} 16 & -8 & -8 \\ -8 & 12 & 4 \\ -8 & 4 & 12 \end{pmatrix},$$

with trace $40/32 = 5/4$ and determinant $1/32$. $\mathbf{X}_1$ wins on both counts, and on every single variance ($1/4$ against $1/2, 3/8, 3/8$). The trace is the total mean squared error of $\hat{\beta}$ divided by σ^2 , and the determinant governs the volume of confidence ellipsoids.

- (c) Here $\mathbf{X}$ is square and $\mathbf{X}^\top \mathbf{X} = n\mathbf{I}$ gives $\mathbf{X}\mathbf{X}^\top = n\mathbf{I}$, so the rows are mutually orthogonal too. Changing the sign of a column keeps this property, so we may take the first row to be all ones. For the first three rows, let a, b, c, d count the columns in which rows 2 and 3 have signs $(+, +)$, $(+, -)$, $(-, +)$, $(-, -)$. Then $a + b + c + d = n$, orthogonality of row 1 to rows 2 and 3 gives $a + b - c - d = 0$ and $a - b + c - d = 0$, and orthogonality of rows 2 and 3 gives $a - b - c + d = 0$. Adding the four equations gives $4a = n$.

Exercise C2.

Let $U_1, \dots, U_n$ be independent and uniform on $(0, 1)$. The k th smallest has a $\text{Beta}(k, n - k + 1)$ distribution, and $\text{Cov}(U_{(1)}, U_{(n)}) = 1/\{(n+1)^2(n+2)\}$. Derive the variance $2/\{(n+1)(n+2)\}$ of the midrange in Example (Uniform errors and the midrange). Show also that the midrange is unbiased for μ whenever the errors have a distribution

symmetric about zero with finite mean.

Solution. $\text{Var}(U_{(1)}) = \text{Var}(U_{(n)}) = n/\{(n+1)^2(n+2)\}$ from the beta variance. Hence

$$\text{Var}\left(\frac{U_{(1)} + U_{(n)}}{2}\right) = \frac{1}{4} \cdot \frac{2n+2}{(n+1)^2(n+2)} = \frac{1}{2(n+1)(n+2)}.$$

Errors uniform on $(-1, 1)$ are $2U_i - 1$, which multiplies variances by 4 and gives $2/\{(n+1)(n+2)\}$. For unbiasedness, if ε and $-\varepsilon$ have the same distribution, then $\varepsilon_{(1)} + \varepsilon_{(n)}$ and $-(\varepsilon_{(n)} + \varepsilon_{(1)})$ have the same distribution. So the midrange minus μ is symmetric about zero, and it has mean zero because $|\varepsilon_{(1)} + \varepsilon_{(n)}| \leq 2 \max_i |\varepsilon_i|$ is integrable.

Exercise C3.

Let $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$ with $\mathbf{V}$ positive definite. Show that an LUE $\mathbf{a}^\top \mathbf{Y}$ of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is a BLUE iff it is uncorrelated with every linear unbiased estimator of zero. Deduce that $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ (ordinary least squares) is the BLUE for every estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ iff $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$, in agreement with [Theorem 6.9.3](#).

Solution. Every LUE is $\mathbf{a}^\top \mathbf{Y} + \mathbf{u}^\top \mathbf{Y}$ with $\mathbf{u}^\top \mathbf{Y}$ an unbiased estimator of zero, and $\text{Var}(\mathbf{a}^\top \mathbf{Y} + t\mathbf{u}^\top \mathbf{Y}) = \text{Var}(\mathbf{a}^\top \mathbf{Y}) + 2t \text{Cov}(\mathbf{a}^\top \mathbf{Y}, \mathbf{u}^\top \mathbf{Y}) + t^2 \text{Var}(\mathbf{u}^\top \mathbf{Y})$ for every real t . If $\mathbf{a}^\top \mathbf{Y}$ is a BLUE, this quadratic in t is minimized at $t = 0$, which forces the covariance to vanish. Conversely, if every such covariance vanishes, the variance of every competitor is $\text{Var}(\mathbf{a}^\top \mathbf{Y}) + \text{Var}(\mathbf{u}^\top \mathbf{Y}) \geq \text{Var}(\mathbf{a}^\top \mathbf{Y})$. By [Exercise \(B1\)](#), the unbiased estimators of zero are $\mathbf{w}^\top (\mathbf{I} - \mathbf{M})\mathbf{Y}$. So ordinary least squares $\boldsymbol{\rho}^\top \mathbf{M}\mathbf{Y}$ is a BLUE for every $\boldsymbol{\rho}$ iff $\sigma^2 \boldsymbol{\rho}^\top \mathbf{M}\mathbf{V}(\mathbf{I} - \mathbf{M})\mathbf{w} = 0$ for all $\boldsymbol{\rho}, \mathbf{w}$, that is, iff $\mathbf{M}\mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$. Transposing, this says $(\mathbf{I} - \mathbf{M})\mathbf{V}\mathbf{M} = \mathbf{0}$, so every column of $\mathbf{V}\mathbf{M}$ lies in $\mathcal{C}(\mathbf{X})$. Since $\mathcal{C}(\mathbf{V}\mathbf{M}) = \mathcal{C}(\mathbf{V}\mathbf{X})$, this is $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$.

7.3 Maximum likelihood under normality

The Gauss–Markov theorem needs only two moments. That is its strength, and also the reason it can say nothing about σ^2 , nothing about nonlinear estimators, and nothing about the distribution of the estimates. From now on we assume more: the errors are normal. The linear model becomes a fully specified family of distributions,

$$\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}), \quad \boldsymbol{\beta} \in \mathbb{R}^p, \sigma^2 > 0, \quad (7.3.1)$$

the normal linear model of [Definition 5.3.1](#). The first general-purpose method of estimation in such a family is maximum likelihood. This section shows that it gives back least squares for $\boldsymbol{\beta}$, and gives a slightly different estimate of σ^2 .

7.3.1 The likelihood

By [Theorem 3.2.2](#) with $\Sigma = \sigma^2 \mathbf{I}$, which is positive definite with $\det(\sigma^2 \mathbf{I}) = \sigma^{2n}$, the density of $\mathbf{Y}$ at the observed $\mathbf{y}$ is

$$L(\boldsymbol{\beta}, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2}{2\sigma^2}\right\}.$$

As a function of the parameters with $\mathbf{y}$ held fixed, this is the **likelihood**. Its logarithm is

$$\ell(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2}{2\sigma^2}. \quad (7.3.2)$$

The data enter the log-likelihood only through the sum of squares $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$. That is the first sign that normality and least squares belong together. The Gaussian density penalizes a residual by its square, so the most likely parameter values are those with the smallest squared residuals.

One technical point matters. If $\mathbf{y} \in \mathcal{C}(\mathbf{X})$, the residual sum of squares at the least squares fit is zero, and $\ell \rightarrow \infty$ as $\sigma^2 \rightarrow 0$, so no maximum exists. This happens with probability zero when $n > r$. The subspace $\mathcal{C}(\mathbf{X})$ has dimension $r < n$, so it has Lebesgue measure zero, and $\mathbf{Y}$ has a density. We therefore assume $n > r$ throughout, and $\text{SSE} > 0$ for the observed data.

7.3.2 The maximum likelihood estimates

Theorem 7.3.1 — Maximum likelihood in the normal linear model

Assume [Eq. 7.3.1](#) with $n > r$, and let $\mathbf{y}$ be such that $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 > 0$. Then $(\tilde{\boldsymbol{\beta}}, \tilde{\sigma}^2)$ maximizes the likelihood iff $\tilde{\boldsymbol{\beta}}$ is a least squares estimate and

$$\tilde{\sigma}^2 = \hat{\sigma}_{\text{ML}}^2 = \frac{\text{SSE}}{n}.$$

The maximized log-likelihood is $\ell_{\max} = -\frac{n}{2} \{\log(2\pi \text{SSE}/n) + 1\}$. The maximum likelihood estimate of $\mathbf{X}\boldsymbol{\beta}$ is $\mathbf{M}\mathbf{y}$, and for estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ it is $\boldsymbol{\lambda}^\top \tilde{\boldsymbol{\beta}}$. Both are unique.

Proof. Fix $\sigma^2 > 0$. By [Eq. 7.3.2](#), $\ell(\mathbf{b}, \sigma^2)$ is a decreasing function of $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$. So it is maximized over $\mathbf{b}$ exactly at the least squares estimates, and there $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \text{SSE}$ ([Theorem 6.4.1](#)). This gives

$$\begin{aligned} \ell(\mathbf{b}, \sigma^2) &\leq \ell(\tilde{\boldsymbol{\beta}}, \sigma^2) = g(\sigma^2) \\ &:= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{\text{SSE}}{2\sigma^2}, \end{aligned}$$

with equality iff $\mathbf{b}$ is a least squares estimate. Now put $v_0 = \text{SSE}/n$ and $u = v_0/\sigma^2 > 0$. Then $\text{SSE}/(2\sigma^2) = nu/2$ and $\log \sigma^2 = \log v_0 - \log u$, so

$$g(v_0) - g(\sigma^2) = \frac{n}{2} (u - 1 - \log u) \geq 0,$$

because $\log u \leq u - 1$ for all $u > 0$, with equality only at $u = 1$. Hence $\ell(\mathbf{b}, \sigma^2) \leq g(\sigma^2) \leq g(v_0)$, with equality throughout iff $\mathbf{b}$ is a least squares estimate and $\sigma^2 = v_0$. The value

$g(v_0)$ is the stated maximum. The least squares estimates all give the fitted vector $\mathbf{M}\mathbf{y}$ (Theorem 6.4.1), and all give the same value of $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ for $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ (Theorem 6.4.4). ■

The proof finds a global maximum without solving any equations. Calculus gives the same answer and is the usual route in a first course: set the gradient $\partial\ell/\partial\boldsymbol{\beta} = \mathbf{X}^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})/\sigma^2$ to zero, which gives the normal equations, and $\partial\ell/\partial\sigma^2 = -n/(2\sigma^2) + \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2/(2\sigma^4)$ to zero. But the stationarity argument still has to be completed by showing that the stationary point is a maximum (Exercise (B1)). The inequality $\log u \leq u - 1$ handles existence, maximality and uniqueness together.

When $\mathbf{X}$ is rank deficient, the likelihood is constant along the affine set of least squares estimates $\hat{\boldsymbol{\beta}} + \mathcal{N}(\mathbf{X})$, so the MLE of $\boldsymbol{\beta}$ is not unique. This is a general fact about nonidentified parameters. If $\mathbf{X}\boldsymbol{\beta}_1 = \mathbf{X}\boldsymbol{\beta}_2$, the two parameter values give the same distribution of $\mathbf{Y}$ and hence the same likelihood for every data set. Maximum likelihood cannot separate what the model itself does not separate. What it does determine uniquely, $\mathbf{X}\boldsymbol{\beta}$ and the estimable functions, is exactly what least squares determines uniquely.

Example. (Stack loss)

Brownlee's stack loss data record $n = 21$ days of operation of a plant that oxidizes ammonia to nitric acid. The response is the percentage of ammonia lost (times ten). The regressors are air flow, the temperature of the cooling water, and the acid concentration. With an intercept, $p = 4$, and least squares gives

$$\hat{\boldsymbol{\beta}} = (-39.9197, 0.7156, 1.2953, -0.1521)^\top, \quad \text{SSE} = 178.830.$$

By Theorem 7.3.1 these are also the maximum likelihood estimates of $\boldsymbol{\beta}$. The MLE of σ^2 is $178.830/21 = 8.516$, while the unbiased estimate is $s^2 = 178.830/17 = 10.519$. The maximized log-likelihood is -52.288 . A general-purpose numerical optimizer, started far from the answer and knowing nothing about least squares, arrives at the same point.

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, stats

data = sm.datasets.stackloss.load_pandas().data
y = data["STACKLOSS"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["AIRFLOW", "WATERTEMP", "ACIDCONC"]]])
n, p = X.shape

beta_hat, *_ = np.linalg.lstsq(X, y, rcond=None)
sse = np.sum((y - X @ beta_hat) ** 2)
sigma2_mle, s2 = sse / n, sse / (n - p)
loglik_max = -n / 2 * (np.log(2 * np.pi * sigma2_mle) + 1)
print("beta_hat", np.round(beta_hat, 4))
print(f"SSE = {sse:.3f}    MLE of sigma^2 = {sigma2_mle:.3f}    s^2 = {s2:.3f}")
print(f"maximized log-likelihood = {loglik_max:.3f}")
```

```
def negloglik(theta):
    """Minus the normal log-likelihood; theta = (beta, log sigma^2)."""
    b, log_s2 = theta[:p], theta[p]
    r = y - X @ b
```

```

    return n / 2 * (np.log(2 * np.pi) + log_s2) + r @ r / (2 * np.exp(log_s2))

start = np.r_[np.zeros(p), np.log(np.var(y))]
opt = optimize.minimize(negloglik, start, method="BFGS", options={"gtol": 1e-8})
print("numerical MLE of beta  ", np.round(opt.x[:p], 4))
print("numerical MLE sigma^2  ", round(float(np.exp(opt.x[p])), 3))

```

7.3.3 The bias of the variance estimate

Unlike least squares for β , the MLE of σ^2 is biased. Since $E(\text{SSE}) = \sigma^2(n - r)$ (Theorem 5.6.1 in full rank, and Section 6.4 in general),

$$E \hat{\sigma}_{\text{ML}}^2 = \frac{n - r}{n} \sigma^2.$$

The bias is downward, by the fraction r/n . The residual vector lives in the $(n - r)$ -dimensional space $\mathcal{C}(\mathbf{X})^\perp$, and the MLE divides its squared length by n as if it had n dimensions. The r dimensions spent on fitting the mean are the dimensions in which the fit has pulled the data toward itself. With a few regressors and many observations the bias is negligible. It is not negligible when the number of parameters grows with the sample size.

Example. (Many means, one variance)

Suppose m laboratories each make two measurements, $Y_{i1}, Y_{i2} \stackrel{\text{iid}}{\sim} N(\mu_i, \sigma^2)$, with a different μ_i for each laboratory and a common measurement variance. This is a linear model with $n = 2m$ and $r = m$, one indicator column per laboratory. The fitted value for each pair is its mean, and

$$\begin{aligned} \hat{\sigma}_{\text{ML}}^2 &= \frac{1}{2m} \sum_{i=1}^m \sum_{j=1}^2 (Y_{ij} - \bar{Y}_{i\cdot})^2 = \frac{1}{4m} \sum_{i=1}^m (Y_{i1} - Y_{i2})^2, \\ E \hat{\sigma}_{\text{ML}}^2 &= \frac{\sigma^2}{2}. \end{aligned}$$

The bias does not disappear as $m \rightarrow \infty$. By the law of large numbers, $\hat{\sigma}_{\text{ML}}^2 \rightarrow \sigma^2/2$ with probability one, so the MLE is *inconsistent*. The unbiased $s^2 = \text{SSE}/(n - r)$ is exactly twice as large and is consistent. This is the example of Neyman and Scott (1948). It shows that the good large-sample behaviour of maximum likelihood can fail when the number of nuisance parameters grows with the data. Restricted maximum likelihood, which removes the mean before estimating the variance (Exercise (C2)), repairs it, and Chapter 32 builds variance-component estimation on that idea.

Unbiasedness is not the only possible criterion for choosing a divisor. Under normality we can compare estimators of the form SSE/d by mean squared error.

Proposition 7.3.2 — The best multiple of SSE

Assume Eq. 7.3.1 with $k = n - r \geq 1$. Among estimators SSE/d with $d > 0$,

$$\frac{E(\text{SSE}/d - \sigma^2)^2}{\sigma^4} = \frac{2k}{d^2} + \left(\frac{k}{d} - 1\right)^2,$$

which is minimized at $d = k + 2$, with minimum value $2/(k + 2)$. The unbiased choice $d = k$ gives $2/k$, and the maximum likelihood choice $d = n$ gives $(2k + r^2)/n^2$.

Proof. Write $\text{SSE} = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$. By [Theorem 4.2.3](#) with $\mathbf{A} = \mathbf{I} - \mathbf{M}$, $\Sigma = \sigma^2 \mathbf{I}$ and $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$, for which $\mathbf{A}\boldsymbol{\mu} = \mathbf{0}$,

$$\text{E}(\text{SSE}) = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{M}) = \sigma^2 k, \quad \text{Var}(\text{SSE}) = 2\sigma^4 \text{tr}((\mathbf{I} - \mathbf{M})^2) = 2\sigma^4 k.$$

The mean squared error is variance plus squared bias: $2\sigma^4 k/d^2 + (k/d - 1)^2 \sigma^4$. As a function of $t = 1/d$, this is $\sigma^4 \{(2k + k^2)t^2 - 2kt + 1\}$, a convex quadratic minimized at $t = k/(2k + k^2) = 1/(k + 2)$. Substituting $d = k + 2$, k and $n = k + r$ gives the three values. For the last, $2k/n^2 + (k - n)^2/n^2 = (2k + r^2)/n^2$. ■

For the stack loss data, $k = 17$, and the three relative mean squared errors are 0.1176 for s^2 , 0.1134 for the MLE and 0.1053 for $\text{SSE}/(k + 2)$. The unbiased estimator is the worst of the three by this criterion. Comparing $(2k + r^2)/n^2$ with $2/k$ shows that the MLE beats s^2 iff $(r - 4)k < 2r$. So it always does when $r \leq 4$, and it loses once many parameters are fitted relative to the residual degrees of freedom, as in [Example \(Many means, one variance\)](#). In practice s^2 is used anyway, because it is what makes the t and F statistics of [Section 7.5](#) exact. The choice of divisor matters for point estimation of σ^2 and for nothing else. Any fixed multiple of SSE leads to the same tests and intervals once the matching distribution is used.

7.3.4 Profile likelihood

In most problems only some parameters are of interest. The others, called **nuisance parameters**, still appear in the likelihood. A simple and general way to eliminate them is to maximize them out.

Definition 7.3.1 — Profile likelihood

Let the parameter be $\boldsymbol{\theta} = (\boldsymbol{\psi}, \boldsymbol{\eta})$, where $\boldsymbol{\psi}$ is of interest and $\boldsymbol{\eta}$ is a nuisance parameter. The **profile log-likelihood** of $\boldsymbol{\psi}$ is $\ell_p(\boldsymbol{\psi}) = \sup_{\boldsymbol{\eta}} \ell(\boldsymbol{\psi}, \boldsymbol{\eta})$.

The profile log-likelihood is maximized at the MLE of $\boldsymbol{\psi}$, with maximum $\ell_{\max}$. The drop $\ell_{\max} - \ell_p(\boldsymbol{\psi})$ measures how much less plausible the value $\boldsymbol{\psi}$ is than the best value. In the linear model the profiles have closed forms. The proof of [Theorem 7.3.1](#) computed two of them. With σ^2 as nuisance, the profile of $\boldsymbol{\beta}$ is

$$\ell_p(\boldsymbol{\beta}) = -\frac{n}{2} \left\{ \log \left(\frac{2\pi \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2}{n} \right) + 1 \right\},$$

a decreasing function of $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$. Its level sets are the level sets of the residual sum of squares. When $\mathbf{X}$ has full rank they are ellipsoids centred at $\hat{\boldsymbol{\beta}}$, and they are the basis of the confidence ellipsoids of Chapter 12. With $\boldsymbol{\beta}$ as nuisance, the profile of σ^2 is the function g in the proof.

The most useful case is a single estimable function. Here the profile likelihood turns out to be a function of the familiar t statistic.

Proposition 7.3.3 — Profile likelihood of an estimable function

Assume Eq. 7.3.1 with $n > r$, and let $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$, $\boldsymbol{\lambda} \neq \mathbf{0}$. For $\psi \in \mathbb{R}$, put

$$t(\psi) = \frac{\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \psi}{s \sqrt{\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}}}, \quad s^2 = \frac{\text{SSE}}{n - r}.$$

Then the smallest residual sum of squares subject to $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \psi$ is

$$\text{SSE}(\psi) = \text{SSE} + \frac{(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \psi)^2}{\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}}, \quad (7.3.3)$$

and the profile log-likelihood of $\psi = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$, with $(\boldsymbol{\beta}, \sigma^2)$ otherwise free, satisfies

$$2\{\ell_{\max} - \ell_p(\psi)\} = n \log \left(1 + \frac{t(\psi)^2}{n - r} \right). \quad (7.3.4)$$

Proof. Write $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$ and $\mathbf{c} = \mathbf{M}\boldsymbol{\rho}$. Then $\mathbf{c} \neq \mathbf{0}$, since $\mathbf{X}^\top \mathbf{c} = \mathbf{X}^\top \boldsymbol{\rho} = \boldsymbol{\lambda} \neq \mathbf{0}$, and $\|\mathbf{c}\|^2 = \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$ as in the proof of Theorem 7.2.1(a). For any $\mathbf{b}$, Eq. 6.3.1 gives $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \text{SSE} + \|\mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{X}\mathbf{b}\|^2$. Put $\mathbf{u} = \mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b}) \in \mathcal{C}(\mathbf{X})$. The constraint $\boldsymbol{\lambda}^\top \mathbf{b} = \psi$ reads $\boldsymbol{\rho}^\top \mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b}) = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \psi =: \delta$, that is, $\boldsymbol{\rho}^\top \mathbf{u} = \delta$. Since $\mathbf{u} \in \mathcal{C}(\mathbf{X})$, $\boldsymbol{\rho}^\top \mathbf{u} = \boldsymbol{\rho}^\top \mathbf{M}\mathbf{u} = \mathbf{c}^\top \mathbf{u}$. By the Cauchy–Schwarz inequality, $|\delta| = |\mathbf{c}^\top \mathbf{u}| \leq \|\mathbf{c}\| \|\mathbf{u}\|$, so $\|\mathbf{u}\|^2 \geq \delta^2 / \|\mathbf{c}\|^2$. Equality holds for $\mathbf{u} = \delta \mathbf{c} / \|\mathbf{c}\|^2$, which lies in $\mathcal{C}(\mathbf{X})$ and so equals $\mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b})$ for some $\mathbf{b}$. That $\mathbf{b}$ satisfies the constraint. This proves Eq. 7.3.3. Maximizing over σ^2 as in the proof of Theorem 7.3.1, with SSE replaced by SSE(ψ), gives $\ell_p(\psi) = -\frac{n}{2} \{\log(2\pi \text{SSE}(\psi)/n) + 1\}$. Hence

$$\begin{aligned} 2\{\ell_{\max} - \ell_p(\psi)\} &= n \log \frac{\text{SSE}(\psi)}{\text{SSE}} = n \log \left(1 + \frac{\delta^2}{\|\mathbf{c}\|^2 \text{SSE}} \right) \\ &= n \log \left(1 + \frac{t(\psi)^2}{n - r} \right), \end{aligned}$$

since $\delta^2 / (\|\mathbf{c}\|^2 \text{SSE}) = t(\psi)^2 / (n - r)$. ■

Two consequences show up immediately. First, any procedure based on the profile likelihood of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is a procedure based on $|t|$, because Eq. 7.3.4 is increasing in t^2 . The likelihood ratio test of $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \psi_0$ is therefore the t test, a fact Chapter 11 extends to general hypotheses. Second, the large-sample calibration of the likelihood ratio, “reject when $2\{\ell_{\max} - \ell_p\}$ exceeds a $\chi^2(1)$ quantile”, is only approximate. The exact calibration uses the $t(n - r)$ distribution of Section 7.5.

Example. (Profile likelihood for the air-flow coefficient)

For the stack loss model, Figure 7.3.1 plots $\ell_p(\beta_1) - \ell_{\max}$ for the air-flow coefficient β_1 , computed by refitting the model with β_1 fixed on a grid. The listing confirms Eq. 7.3.4 to rounding error. The standard error of $\hat{\beta}_1$ is 0.1349. The set of β_1 with $2\{\ell_{\max} - \ell_p\} \leq 3.841$, the 95% point of $\chi^2(1)$, is the likelihood interval (0.467, 0.965).

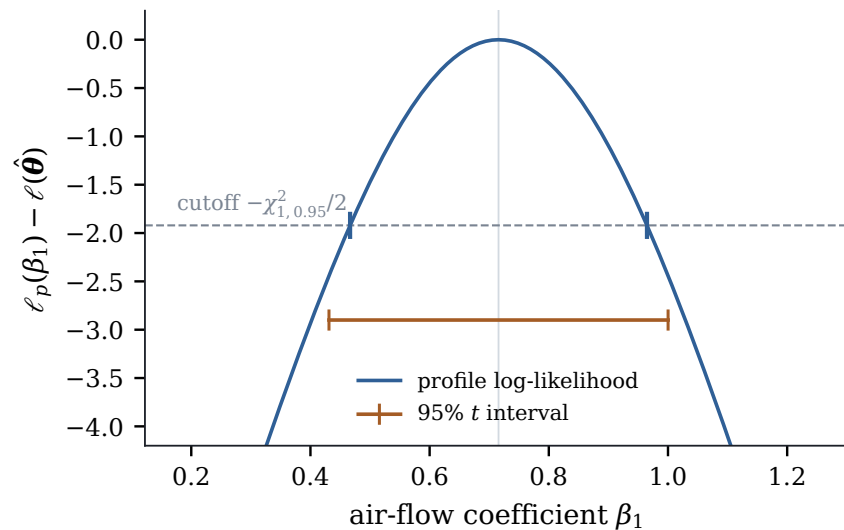


Figure 7.3.1: Profile log-likelihood of the air-flow coefficient in the stack loss regression, relative to its maximum. The ticks on the dashed cutoff line mark the χ^2 -calibrated likelihood interval, and the bar below it marks the exact 95% t interval, which is wider.

By Eq. 7.3.4 it consists of the values with $|t| \leq 1.847$. The exact 95% interval uses the $t(17)$ quantile 2.110 instead, and is (0.431, 1.000). With only 17 residual degrees of freedom, the χ^2 calibration makes the interval about 12% too short.

```
def profile_loglik(b1):
    """Profile log-likelihood of the air-flow coefficient: maximize over everything
    else."""
    others = np.delete(X, 1, axis=1)
    z = y - b1 * X[:, 1]
    coef, *_ = np.linalg.lstsq(others, z, rcond=None)
    sse_b = np.sum((z - others @ coef) ** 2)
    with beta_1 = b1
    return -n / 2 * (np.log(2 * np.pi * sse_b / n) + 1)

se1 = np.sqrt(s2 * np.linalg.inv(X.T @ X)[1, 1])
grid = np.linspace(beta_hat[1] - 4 * se1, beta_hat[1] + 4 * se1, 201)
lp = np.array([profile_loglik(b) for b in grid])
t = (beta_hat[1] - grid) / se1
print("max |2(l_max - l_p) - n log(1 + t^2/(n-p))| =",
      np.abs(2 * (loglik_max - lp) - n * np.log1p(t ** 2 / (n - p))).max())
```

Remark (Other error distributions). The agreement between maximum likelihood and least squares is special to normal errors. If the errors have the Laplace density $(2\tau)^{-1}e^{-|\varepsilon|/\tau}$, the log-likelihood is $-n \log(2\tau) - \sum_i |y_i - \mathbf{x}_i^T \boldsymbol{\beta}|/\tau$, and maximum likelihood minimizes the sum of *absolute* residuals; Exercise (C1) works out the location case. For errors with a t distribution it downweights large residuals. For each error distribution the likelihood picks its own loss function. Least squares is the one that goes with the Gaussian. Chapter 20 uses robust fitting as a diagnostic, and Chapter 45 treats median and quantile regression.

7.3.5 Exercises

7.3.5.1 A. Check your understanding

Exercise A1.

For $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, write the model in the form [Eq. 7.3.1](#) and read off the MLEs from [Theorem 7.3.1](#). Show that $\hat{\sigma}_{\text{ML}}^2 = (n-1)s^2/n$, where s^2 is the sample variance.

Exercise A2.

Let $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ with residual sums of squares $\text{SSE}_0 \geq \text{SSE}$. Show that the ratio of maximized likelihoods is $L_{0,\text{max}}/L_{\text{max}} = (\text{SSE}/\text{SSE}_0)^{n/2}$, and write it as a decreasing function of $(\text{SSE}_0 - \text{SSE})/\text{SSE}$.

Solution. By [Theorem 7.3.1](#), $L_{\text{max}} = (2\pi \text{SSE}/n)^{-n/2} e^{-n/2}$, and similarly for the smaller model. The ratio is $(\text{SSE}_0/\text{SSE})^{-n/2} = \{1 + (\text{SSE}_0 - \text{SSE})/\text{SSE}\}^{-n/2}$, which decreases in $(\text{SSE}_0 - \text{SSE})/\text{SSE}$. That quantity is, up to degrees of freedom, the F statistic of [Theorem 4.4.6](#). So the likelihood ratio test for nested normal linear models is the F test. Chapter 11 develops this.

7.3.5.2 B. Practice

Exercise B1.

Assume $\text{rank}(\mathbf{X}) = p$. Compute the gradient and the Hessian of $\ell(\boldsymbol{\beta}, \sigma^2)$, show that the only stationary point is $(\hat{\boldsymbol{\beta}}, \text{SSE}/n)$, and show that the Hessian there is negative definite. Why does this not by itself prove that the stationary point is a global maximum? What additional argument completes the proof?

Exercise B2.

In [Example \(Many means, one variance\)](#), verify the identity $\sum_j (Y_{ij} - \bar{Y}_{i\cdot})^2 = (Y_{i1} - Y_{i2})^2/2$, compute $E \hat{\sigma}_{\text{ML}}^2$ and $\text{Var} \hat{\sigma}_{\text{ML}}^2$, and prove that $\hat{\sigma}_{\text{ML}}^2 \rightarrow \sigma^2/2$ in probability as $m \rightarrow \infty$.

Solution. $Y_{i1} - \bar{Y}_{i\cdot} = (Y_{i1} - Y_{i2})/2 = -(Y_{i2} - \bar{Y}_{i\cdot})$, so the sum of squares is $2 \cdot (Y_{i1} - Y_{i2})^2/4$. Now $D_i = Y_{i1} - Y_{i2} \sim N(0, 2\sigma^2)$ independently, so $D_i^2/(2\sigma^2) \sim \chi^2(1)$, with mean 1 and variance 2. Hence $\hat{\sigma}_{\text{ML}}^2 = \frac{1}{4m} \sum_i D_i^2$ has mean $2\sigma^2/4 = \sigma^2/2$ and variance $m \cdot 4\sigma^4 \cdot 2/(16m^2) = \sigma^4/(2m)$. The variance tends to zero, so by Chebyshev's inequality $\hat{\sigma}_{\text{ML}}^2 \rightarrow \sigma^2/2$ in probability.

Exercise B3.

Suppose $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{V})$ with $\mathbf{V}$ known and positive definite and $\mathbf{X}$ of full rank. Find the MLEs of $\boldsymbol{\beta}$ and σ^2 .

7.3.5.3 C. Going deeper

Exercise C1.

In the location model $Y_i = \mu + \varepsilon_i$, $i = 1, \dots, n$, let the errors be independent with the Laplace density $(2\tau)^{-1}e^{-|\varepsilon|/\tau}$, so $\text{Var}(\varepsilon_i) = 2\tau^2$.

- Show that every sample median maximizes the likelihood in μ , whatever τ is, and that the MLE of τ is the mean absolute deviation $n^{-1} \sum_i |Y_i - \tilde{\mu}|$ about a median $\tilde{\mu}$.
- The sample median of n observations from a density f that is positive and continuous at its median μ has asymptotic variance $1/\{4nf(\mu)^2\}$. Compute this for the Laplace density and compare it with $\text{Var}(\bar{Y})$. Which is larger, and by what factor?
- $\bar{Y}$ is the BLUE of μ (Corollary 7.2.2). Explain why (b) does not contradict Theorem 7.2.1, and contrast the answer with Example (Uniform errors and the midrange), where the winning nonlinear estimator used the extremes rather than the middle of the sample.

Solution.

- The log-likelihood is $-n \log(2\tau) - \tau^{-1} \sum_i |Y_i - \mu|$. For fixed τ it is maximized by minimizing $g(\mu) = \sum_i |Y_i - \mu|$. The function g is convex and piecewise linear, with slope $\#\{i : Y_i < \mu\} - \#\{i : Y_i > \mu\}$ between data points. The slope is ≤ 0 to the left of every median and ≥ 0 to the right, so the minimizers are exactly the sample medians. Substituting $\tilde{\mu}$ and maximizing $-n \log \tau - g(\tilde{\mu})/\tau$ over $\tau > 0$ gives $\hat{\tau} = g(\tilde{\mu})/n$.
- $f(\mu) = 1/(2\tau)$, so the median has asymptotic variance τ^2/n , while $\text{Var}(\bar{Y}) = 2\tau^2/n$. The mean has twice the variance of the median.
- The median is not a linear function of $\mathbf{Y}$: its weights depend on the ordering of the sample. The Gauss–Markov theorem compares $\bar{Y}$ only with linear unbiased estimators. With Laplace errors, which have heavier tails than the normal, the extremes are unreliable and the median gains by ignoring them; with uniform errors the extremes are the most informative observations and the midrange gains by using only them.

Exercise C2.

Let $\mathbf{Q}_2$ be an $n \times (n-r)$ matrix whose columns are an orthonormal basis of $\mathcal{C}(\mathbf{X})^\perp$, and let $\mathbf{Z} = \mathbf{Q}_2^\top \mathbf{Y}$ (“error contrasts”). Show that $\mathbf{Z} \sim N_{n-r}(\mathbf{0}, \sigma^2 \mathbf{I})$, whatever β is, and that $\|\mathbf{Z}\|^2 = \text{SSE}$. Show that the MLE of σ^2 based on $\mathbf{Z}$ alone is $\text{SSE}/(n-r)$. This is **restricted** (or residual) maximum likelihood.

Solution. $\mathbf{Q}_2^\top \mathbf{X} = \mathbf{0}$, so $\mathbf{Z} = \mathbf{Q}_2^\top \varepsilon \sim N_{n-r}(\mathbf{0}, \sigma^2 \mathbf{Q}_2^\top \mathbf{Q}_2) = N_{n-r}(\mathbf{0}, \sigma^2 \mathbf{I})$ by Theorem 3.3.1. Also $\mathbf{Q}_2 \mathbf{Q}_2^\top = \mathbf{I} - \mathbf{M}$ (Proposition 6.3.2), so $\|\mathbf{Z}\|^2 = \mathbf{Y}^\top \mathbf{Q}_2 \mathbf{Q}_2^\top \mathbf{Y} = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y} = \text{SSE}$. The log-likelihood of $\mathbf{Z}$ is $-\frac{n-r}{2} \log(2\pi\sigma^2) - \|\mathbf{Z}\|^2/(2\sigma^2)$, which is the function g of the proof of Theorem 7.3.1 with n replaced by $n-r$. It is maximized at $\|\mathbf{Z}\|^2/(n-r) = \text{SSE}/(n-r)$. In Example (Many means, one variance) this gives s^2 , which is consistent.

7.4 Sufficiency, completeness and minimum variance

The Gauss–Markov theorem compares least squares only with *linear* unbiased estimators, and [Example \(Uniform errors and the midrange\)](#) showed that the restriction matters: with uniform errors a nonlinear estimator wins easily. This section shows that under normality the restriction can be dropped. In the normal linear model, $\lambda^\top \hat{\beta}$ has the smallest variance among *all* unbiased estimators of an estimable $\lambda^\top \beta$, and s^2 has the smallest variance among all unbiased estimators of σ^2 . The argument is the classical one through sufficiency and completeness. It also produces best unbiased estimators of nonlinear functions, such as σ itself. At the end we compare the variances with the Cramér–Rao lower bound, which least squares attains for β and s^2 does not attain for σ^2 .

Two facts from mathematical statistics are quoted without proof: the factorization criterion for sufficiency and the completeness of full-rank exponential families. Both are proved in standard texts such as Lehmann and Casella (1998) and Lehmann and Romano (2005). Everything built on them is proved here.

7.4.1 Sufficiency

Definition 7.4.1 — Sufficient and complete statistics

Let $\mathbf{Y}$ have a distribution P_θ from a family indexed by $\theta \in \Theta$, and let $\mathbf{T} = \mathbf{T}(\mathbf{Y})$ be a statistic.

- (a) $\mathbf{T}$ is **sufficient** for θ if the conditional law of $\mathbf{Y}$ given $\mathbf{T}$ is the same for every θ .
- (b) $\mathbf{T}$ is **complete** if, for every function h with $E_\theta |h(\mathbf{T})| < \infty$ and $E_\theta h(\mathbf{T}) = 0$ for all $\theta \in \Theta$, we have $\Pr_\theta\{h(\mathbf{T}) = 0\} = 1$ for all θ .

A sufficient statistic carries all the information about θ in the data. Once $\mathbf{T}$ is known, the rest of $\mathbf{Y}$ is noise whose distribution does not involve the parameter. Completeness says that $\mathbf{T}$ carries no redundant information either: no nontrivial function of $\mathbf{T}$ has expectation zero for every parameter value. Equivalently, an unbiased estimator that is a function of $\mathbf{T}$ is unique.

Lemma 7.4.1 — Factorization criterion

Suppose $\mathbf{Y}$ has a density $f_\theta(\mathbf{y})$ on $\mathbb{R}^n$. Then $\mathbf{T}(\mathbf{Y})$ is sufficient iff there are functions g_θ and h , with h not depending on θ , such that $f_\theta(\mathbf{y}) = g_\theta(\mathbf{T}(\mathbf{y})) h(\mathbf{y})$ for all $\mathbf{y}$ and θ .

Lemma 7.4.2 — Completeness in exponential families

Suppose $\mathbf{Y}$ has a density of the form

$$f_\eta(\mathbf{y}) = c(\eta) h(\mathbf{y}) \exp\{\eta^\top \mathbf{T}(\mathbf{y})\}, \quad \eta \in \mathcal{E} \subseteq \mathbb{R}^k,$$

with $\mathbf{T}(\mathbf{y}) \in \mathbb{R}^k$. If $\mathcal{E}$ contains a nonempty open subset of $\mathbb{R}^k$, then $\mathbf{T}(\mathbf{Y})$ is complete

and sufficient.

Sufficiency follows from [Lemma 7.4.1](#) with $g_{\boldsymbol{\eta}}(\mathbf{T}) = c(\boldsymbol{\eta})e^{\boldsymbol{\eta}^\top \mathbf{T}}$. Completeness rests on the uniqueness of Laplace transforms: $E_{\boldsymbol{\eta}} h(\mathbf{T}) = 0$ on an open set of $\boldsymbol{\eta}$ forces a certain two-sided Laplace transform to vanish on an open set, and hence $h(\mathbf{T}) = 0$ almost surely.

For the normal linear model, expand the exponent of the density:

$$\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \mathbf{y}^\top \mathbf{y} - 2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}.$$

So the density of [Eq. 7.3.1](#) is

$$f(\mathbf{y}) = (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}}{2\sigma^2}\right\} \exp\left\{\frac{\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y}}{\sigma^2} - \frac{\mathbf{y}^\top \mathbf{y}}{2\sigma^2}\right\}, \quad (7.4.1)$$

and the data enter only through $\mathbf{X}^\top \mathbf{y}$ and $\mathbf{y}^\top \mathbf{y}$.

Proposition 7.4.3 — A complete sufficient statistic

In the normal linear model [Eq. 7.3.1](#), let $\mathbf{Q}$ be an $n \times r$ matrix whose columns are an orthonormal basis of $\mathcal{C}(\mathbf{X})$. Then $\mathbf{T} = (\mathbf{Q}^\top \mathbf{Y}, \mathbf{Y}^\top \mathbf{Y})$ is complete and sufficient for $(\boldsymbol{\beta}, \sigma^2)$. The statistics $(\mathbf{X}^\top \mathbf{Y}, \mathbf{Y}^\top \mathbf{Y})$ and $(\mathbf{MY}, \text{SSE})$ are one-to-one functions of $\mathbf{T}$, so they are complete and sufficient too.

Proof. Since $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X})$, $\mathbf{X}\boldsymbol{\beta} = \mathbf{Q}\boldsymbol{\gamma}$ with $\boldsymbol{\gamma} = \mathbf{Q}^\top \mathbf{X}\boldsymbol{\beta} \in \mathbb{R}^r$, and $\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y} = \boldsymbol{\gamma}^\top \mathbf{Q}^\top \mathbf{y}$. By [Eq. 7.4.1](#) the density has the form of [Lemma 7.4.2](#) with $k = r + 1$,

$$\mathbf{T}(\mathbf{y}) = (\mathbf{Q}^\top \mathbf{y}, \mathbf{y}^\top \mathbf{y}), \quad \boldsymbol{\eta} = (\boldsymbol{\gamma}/\sigma^2, -1/(2\sigma^2)).$$

As $\boldsymbol{\beta}$ ranges over $\mathbb{R}^p$, $\boldsymbol{\gamma} = \mathbf{Q}^\top \mathbf{X}\boldsymbol{\beta}$ ranges over all of $\mathbb{R}^r$: the $r \times p$ matrix $\mathbf{Q}^\top \mathbf{X}$ has rank r , because $\mathbf{Q}\mathbf{Q}^\top \mathbf{X} = \mathbf{M}\mathbf{X} = \mathbf{X}$ has rank r . Hence $\mathcal{E} = \mathbb{R}^r \times (-\infty, 0)$, which is open, and [Lemma 7.4.2](#) applies. For the equivalent statistics: $\mathbf{MY} = \mathbf{Q}\mathbf{Q}^\top \mathbf{Y}$ ([Proposition 6.3.2](#)) and $\text{SSE} = \mathbf{Y}^\top \mathbf{Y} - \|\mathbf{Q}^\top \mathbf{Y}\|^2$ are functions of $\mathbf{T}$, and conversely $\mathbf{Q}^\top \mathbf{Y} = \mathbf{Q}^\top \mathbf{MY}$ and $\mathbf{Y}^\top \mathbf{Y} = \text{SSE} + \|\mathbf{MY}\|^2$. Also $\mathbf{X}^\top \mathbf{Y} = \mathbf{X}^\top \mathbf{Q}(\mathbf{Q}^\top \mathbf{Y})$, and $\mathbf{Q} = \mathbf{X}\mathbf{K}$ for some $\mathbf{K}$, since the columns of $\mathbf{Q}$ lie in $\mathcal{C}(\mathbf{X})$, so $\mathbf{Q}^\top \mathbf{Y} = \mathbf{K}^\top \mathbf{X}^\top \mathbf{Y}$. A one-to-one function of a complete sufficient statistic is again complete and sufficient, since both definitions refer only to the information (the σ -field) the statistic generates. ■

The reduction is dramatic. An n -dimensional data vector is replaced by $r+1$ numbers: the fitted values (equivalently, the least squares estimates of the estimable functions) and the residual sum of squares. Under normality the residual *vector*, apart from its length, carries no information about the parameters. Its direction is uniformly distributed over the unit sphere of $\mathcal{C}(\mathbf{X})^\perp$ whatever $(\boldsymbol{\beta}, \sigma^2)$ is ([Exercise \(C1\)](#)). This is the precise sense in which the residuals are “just noise”, and it is also why they are useful for checking the model. Their pattern can reveal departures from normality precisely because, under normality, it has a known and parameter-free distribution.

7.4.2 Rao–Blackwell and Lehmann–Scheffé

Theorem 7.4.4 — Rao–Blackwell

Let $\mathbf{T}$ be sufficient for $\boldsymbol{\theta}$, and let $\delta(\mathbf{Y})$ be an unbiased estimator of $g(\boldsymbol{\theta})$ with $\text{Var}_{\boldsymbol{\theta}} \delta < \infty$ for all $\boldsymbol{\theta}$. Then $\delta^* = E(\delta \mid \mathbf{T})$ is a statistic, it is unbiased for $g(\boldsymbol{\theta})$, and

$$\begin{aligned}\text{Var}_{\boldsymbol{\theta}}(\delta) &= \text{Var}_{\boldsymbol{\theta}}(\delta^*) + E_{\boldsymbol{\theta}}[(\delta - \delta^*)^2] \\ &\geq \text{Var}_{\boldsymbol{\theta}}(\delta^*)\end{aligned}$$

for every $\boldsymbol{\theta}$, with equality iff $\delta = \delta^*$ with probability one.

Proof. The law of $\mathbf{Y}$ given $\mathbf{T}$ does not depend on $\boldsymbol{\theta}$, so neither does $E(\delta(\mathbf{Y}) \mid \mathbf{T})$. It is therefore computable from the data, which makes it a statistic. By the law of total expectation (Proposition 2.5.4), $E \delta^* = E \delta = g(\boldsymbol{\theta})$. By the law of total variance in the same result, $\text{Var} \delta = E\{\text{Var}(\delta \mid \mathbf{T})\} + \text{Var}(\delta^*)$, and $E\{\text{Var}(\delta \mid \mathbf{T})\} = E\{E[(\delta - \delta^*)^2 \mid \mathbf{T}]\} = E(\delta - \delta^*)^2$. This last term is zero iff $\delta = \delta^*$ almost surely. ■

Conditioning on a sufficient statistic averages out the part of an estimator that responds to parameter-free noise. It never hurts, and it helps unless the estimator was already a function of $\mathbf{T}$.

Theorem 7.4.5 — Lehmann–Scheffé

Let $\mathbf{T}$ be complete and sufficient for $\boldsymbol{\theta}$. If $\phi(\mathbf{T})$ is unbiased for $g(\boldsymbol{\theta})$ and has finite variance, then $\phi(\mathbf{T})$ is a **uniformly minimum variance unbiased estimator** (UMVUE) of $g(\boldsymbol{\theta})$: $\text{Var}_{\boldsymbol{\theta}} \phi(\mathbf{T}) \leq \text{Var}_{\boldsymbol{\theta}} \delta$ for every unbiased δ and every $\boldsymbol{\theta}$. It is the only unbiased function of $\mathbf{T}$ and, up to sets of probability zero, the only UMVUE.

Proof. If $\phi_1(\mathbf{T})$ and $\phi_2(\mathbf{T})$ are both unbiased, then $E_{\boldsymbol{\theta}}\{\phi_1(\mathbf{T}) - \phi_2(\mathbf{T})\} = 0$ for all $\boldsymbol{\theta}$, so $\phi_1(\mathbf{T}) = \phi_2(\mathbf{T})$ almost surely by completeness. Now let δ be any unbiased estimator with finite variance. By Theorem 7.4.4, $E(\delta \mid \mathbf{T})$ is an unbiased function of $\mathbf{T}$, so it equals $\phi(\mathbf{T})$ almost surely, and $\text{Var} \phi(\mathbf{T}) \leq \text{Var} \delta$, with equality iff $\delta = \phi(\mathbf{T})$ almost surely. (If δ has infinite variance there is nothing to prove.) ■

7.4.3 Best unbiased estimators in the normal linear model

Theorem 7.4.6 — Minimum variance unbiased estimation

Assume the normal linear model Eq. 7.3.1 with $n > r$, and let $s^2 = \text{SSE}/(n - r)$.

- (a) For every $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$, $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ is the UMVUE of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$.
- (b) s^2 is the UMVUE of σ^2 .

- (c) More generally, every estimator of the form $\phi(\mathbf{MY}, \text{SSE})$ with finite variance is the UMVUE of its expectation. In particular, for $k = n - r$,

$$b_k = \sqrt{\frac{2}{k}} \frac{\Gamma(\frac{k+1}{2})}{\Gamma(\frac{k}{2})},$$

s/b_k is the UMVUE of σ , and $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} + z s/b_k$ is the UMVUE of $\mathbf{x}_0^\top \boldsymbol{\beta} + z\sigma$ for every constant z and every $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$.

Proof. By Proposition 7.4.3, $(\mathbf{MY}, \text{SSE})$ is complete and sufficient, so by Theorem 7.4.5 it suffices to show that each estimator is an unbiased function of it with finite variance.

(a) Write $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$. Then $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\rho}^\top \mathbf{MY}$ is a function of $\mathbf{MY}$, it is unbiased by Theorem 7.2.1(a), and its variance is finite. (b) s^2 is a function of SSE, it is unbiased because $E(\text{SSE}) = \sigma^2(n - r)$ (Theorem 2.3.1), and it has finite variance. (c) The first statement is Theorem 7.4.5 again. For σ , note that $\text{SSE}/\sigma^2 = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y} / \sigma^2 \sim \chi^2(k)$ by Theorem 4.3.2(a), since $\mathbf{I} - \mathbf{M}$ is a projection of rank k and $(\mathbf{I} - \mathbf{M}) \mathbf{X} \boldsymbol{\beta} = \mathbf{0}$. If $W \sim \chi^2(k)$, with density $w^{k/2-1} e^{-w/2} / \{2^{k/2} \Gamma(k/2)\}$, then

$$\begin{aligned} E \sqrt{W} &= \frac{1}{2^{k/2} \Gamma(k/2)} \int_0^\infty w^{(k+1)/2-1} e^{-w/2} dw \\ &= \frac{2^{(k+1)/2} \Gamma(\frac{k+1}{2})}{2^{k/2} \Gamma(\frac{k}{2})} = \sqrt{2} \frac{\Gamma(\frac{k+1}{2})}{\Gamma(\frac{k}{2})}. \end{aligned}$$

Hence $E s = \sigma E \sqrt{W/k} = b_k \sigma$, and s/b_k is unbiased for σ , with variance at most $E s^2 / b_k^2 < \infty$. The last estimator is a sum of two unbiased functions of $(\mathbf{MY}, \text{SSE})$. ■

Part (a) is the promised strengthening of Gauss–Markov. Under normality no unbiased estimator of $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, linear or not, beats least squares. The midrange of Example (Uniform errors and the midrange) cannot help when the errors are normal. Part (c) shows the method at work beyond linear estimators. The obvious estimate s of σ is biased, because $E s < \sqrt{E s^2}$ by Jensen’s inequality, and dividing by b_k removes the bias.

Example. (Estimating σ without bias)

The constant $b_k = E(s)/\sigma$ depends only on the residual degrees of freedom. For the stack loss regression, $k = 17$ and $b_{17} = 0.9854$, so the UMVUE of σ is $s/b_{17} = 1.0148 s$. The correction is 1.5% here. It is large for very small k , with $b_2 = 0.8862$, and negligible for large k , with $b_{200} = 0.9988$.

```
import numpy as np
from scipy import special

def mean_s_factor(k):
    """b_k = E(s)/sigma when k s^2 / sigma^2 ~ chi^2(k)."""
    return np.sqrt(2 / k) * np.exp(special.gammaln((k + 1) / 2) - special.gammaln(k / 2))

for k in [2, 5, 10, 17, 50, 200]:
    b = mean_s_factor(k)
```

```
print(f"k = {k:3d}    E(s)/sigma = {b:.4f}    UMVUE of sigma = s / {b:.4f}")
```

7.4.4 Rao–Blackwellization by reflection

The Rao–Blackwell theorem is not only a device for proofs. In the normal linear model we can carry out the conditioning explicitly, and it takes every linear unbiased estimator to least squares.

Example. (Conditioning on the sufficient statistic)

Let $\mathbf{a}^\top \mathbf{Y}$ be any LUE of an estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, such as the end-point slope of [Example \(Three linear unbiased estimators of a slope\)](#). Let $\mathbf{R} = 2\mathbf{M} - \mathbf{I}$, the reflection that fixes $\mathcal{C}(\mathbf{X})$ and reverses $\mathcal{C}(\mathbf{X})^\perp$. It is symmetric and orthogonal ($\mathbf{R}^2 = \mathbf{I}$) and $\mathbf{R}\mathbf{X} = \mathbf{X}$. Consequently:

- $\mathbf{R}\mathbf{Y} \sim N_n(\mathbf{R}\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{R}\mathbf{R}^\top) = N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$ by [Theorem 3.3.1](#), so $\mathbf{R}\mathbf{Y}$ has the same distribution as $\mathbf{Y}$;
- the sufficient statistic is unchanged: $\mathbf{X}^\top \mathbf{R}\mathbf{Y} = \mathbf{X}^\top \mathbf{Y}$ and $\|\mathbf{R}\mathbf{Y}\| = \|\mathbf{Y}\|$.

So the pairs $(\mathbf{a}^\top \mathbf{Y}, \mathbf{T}(\mathbf{Y}))$ and $(\mathbf{a}^\top \mathbf{R}\mathbf{Y}, \mathbf{T}(\mathbf{Y}))$ have the same joint distribution, where $\mathbf{T} = (\mathbf{X}^\top \mathbf{Y}, \mathbf{Y}^\top \mathbf{Y})$. Hence $E(\mathbf{a}^\top \mathbf{Y} \mid \mathbf{T}) = E(\mathbf{a}^\top \mathbf{R}\mathbf{Y} \mid \mathbf{T})$, and averaging,

$$E(\mathbf{a}^\top \mathbf{Y} \mid \mathbf{T}) = E\left(\mathbf{a}^\top \frac{\mathbf{Y} + \mathbf{R}\mathbf{Y}}{2} \mid \mathbf{T}\right) = E(\mathbf{a}^\top \mathbf{M}\mathbf{Y} \mid \mathbf{T}) = \mathbf{a}^\top \mathbf{M}\mathbf{Y} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}},$$

because $\mathbf{a}^\top \mathbf{M}\mathbf{Y}$ is a function of $\mathbf{T}$ and $\mathbf{M}\mathbf{a} = \mathbf{a}_*$ ([Proposition 7.1.2](#)). The Rao–Blackwell improvement of *every* LUE is the least squares estimate. The listing checks the identity $\frac{1}{2}(\mathbf{a}^\top \mathbf{y} + \mathbf{a}^\top \mathbf{R}\mathbf{y}) = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ for simulated data and the end-point slope.

```
import numpy as np

x = np.array([1.0, 2.0, 2.5, 3.0, 4.0, 5.0, 5.5, 7.0, 8.0, 9.0, 10.5, 12.0])
n = len(x)
X = np.column_stack([np.ones(n), x])
M = X @ np.linalg.solve(X.T @ X, X.T)
R = 2 * M - np.eye(n)                                # reflection: keeps C(X), flips C(X)
                                                         )-perp

rng = np.random.default_rng(71)
y = X @ np.array([2.0, 0.5]) + rng.normal(size=n)
y_ref = R @ y
print("X^T y unchanged:", np.allclose(X.T @ y_ref, X.T @ y))
print("y^T y unchanged:", np.isclose(y_ref @ y_ref, y @ y))

a_end = np.zeros(n)
a_end[[0, -1]] = [-1.0, 1.0]
a_end /= x[-1] - x[0]                                # the end-point slope estimator
a_ls = X @ np.linalg.solve(X.T @ X, [0.0, 1.0])
print("average of a^T y over the two points:", (a_end @ y + a_end @ y_ref) / 2)
print("least squares slope:", a_ls @ y)
```

The reflection argument uses the normal distribution only through its invariance under orthogo-

nal maps that fix the mean. It makes concrete what “the residuals are noise” means. Reversing the residual vector changes neither the likelihood nor the sufficient statistic, so an estimator should not depend on the residuals’ direction. An LUE that uses the residuals is replaced by the average of its value at $\mathbf{y}$ and at the reflected data, and that average no longer involves the residuals.

7.4.5 The Cramér–Rao bound

A different route to optimality bounds the variance of every unbiased estimator from below and checks whether the bound is attained. We state the bound for the full-rank normal linear model, where every ingredient can be computed.

Write $\boldsymbol{\theta} = (\boldsymbol{\beta}^\top, v)^\top$ with $v = \sigma^2$. The **score** is the gradient of the log-likelihood Eq. 7.3.2, evaluated at the random $\mathbf{Y}$:

$$\mathbf{S} = \begin{pmatrix} \partial \ell / \partial \boldsymbol{\beta} \\ \partial \ell / \partial v \end{pmatrix} = \begin{pmatrix} \mathbf{X}^\top \boldsymbol{\varepsilon} / v \\ -\frac{n}{2v} + \frac{\boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon}}{2v^2} \end{pmatrix}, \quad \boldsymbol{\varepsilon} = \mathbf{Y} - \mathbf{X}\boldsymbol{\beta}.$$

Proposition 7.4.7 — Score and information

In Eq. 7.3.1 with $\text{rank}(\mathbf{X}) = p$, $\mathbb{E} \mathbf{S} = \mathbf{0}$ and

$$\mathcal{I}(\boldsymbol{\theta}) = \text{Cov}(\mathbf{S}) = \begin{pmatrix} \mathbf{X}^\top \mathbf{X} / \sigma^2 & \mathbf{0} \\ \mathbf{0}^\top & n / (2\sigma^4) \end{pmatrix}.$$

Proof. $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, v\mathbf{I})$. Hence $\mathbb{E}(\mathbf{X}^\top \boldsymbol{\varepsilon} / v) = \mathbf{0}$ and $\text{Cov}(\mathbf{X}^\top \boldsymbol{\varepsilon} / v) = \mathbf{X}^\top \mathbf{X} / v$ by Theorem 2.2.1. Next, $\boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon} / v \sim \chi^2(n)$ by Theorem 4.3.2(a) with $\mathbf{P} = \mathbf{I}$, so it has mean n and variance $2n$. So the last coordinate has mean $-n/(2v) + nv/(2v^2) = 0$ and variance $2n/(4v^2) = n/(2v^2)$. Finally, by Theorem 4.2.3(c) with $\mathbf{K} = \mathbf{X}^\top$, $\mathbf{A} = \mathbf{I}$ and $\boldsymbol{\mu} = \mathbf{0}$, $\text{Cov}(\mathbf{X}^\top \boldsymbol{\varepsilon}, \boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon}) = \mathbf{0}$. ■

Theorem 7.4.8 — Cramér–Rao bound in the normal linear model

Assume Eq. 7.3.1 with $\text{rank}(\mathbf{X}) = p$. Let $\delta(\mathbf{Y})$ have finite variance and satisfy $\mathbb{E}_{\boldsymbol{\theta}} \delta = g(\boldsymbol{\theta})$ for all $\boldsymbol{\theta}$, with g differentiable. Then

$$\begin{aligned} \text{Var}_{\boldsymbol{\theta}}(\delta) &\geq \nabla g(\boldsymbol{\theta})^\top \mathcal{I}(\boldsymbol{\theta})^{-1} \nabla g(\boldsymbol{\theta}) \\ &= \sigma^2 \mathbf{g}_\beta^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{g}_\beta + \frac{2\sigma^4}{n} g_v^2, \end{aligned}$$

where $\mathbf{g}_\beta = \partial g / \partial \boldsymbol{\beta}$ and $g_v = \partial g / \partial v$.

Proof. Write $f_{\boldsymbol{\theta}}$ for the density. Differentiating $g(\boldsymbol{\theta}) = \int \delta(\mathbf{y}) f_{\boldsymbol{\theta}}(\mathbf{y}) d\mathbf{y}$ under the integral sign, which is justified because the normal family is an exponential family (Lehmann

and Casella 1998, chapter 1), gives

$$\begin{aligned}\nabla g(\boldsymbol{\theta}) &= \int \delta(\mathbf{y}) \nabla \log f_{\boldsymbol{\theta}}(\mathbf{y}) f_{\boldsymbol{\theta}}(\mathbf{y}) d\mathbf{y} \\ &= E(\delta \mathbf{S}) = \text{Cov}(\delta, \mathbf{S}),\end{aligned}$$

the last step because $E \mathbf{S} = \mathbf{0}$. For any constant vector $\mathbf{c}$, $0 \leq \text{Var}(\delta - \mathbf{c}^\top \mathbf{S}) = \text{Var} \delta - 2\mathbf{c}^\top \nabla g + \mathbf{c}^\top \mathbf{I} \mathbf{c}$. Choosing $\mathbf{c} = \mathbf{I}^{-1} \nabla g$ gives $\text{Var} \delta \geq \nabla g^\top \mathbf{I}^{-1} \nabla g$. The block-diagonal form of Proposition 7.4.7 gives the right-hand side.

■

For $g(\boldsymbol{\theta}) = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$ the bound is $\sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$, exactly the variance of $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$. Least squares **attains** the bound, which gives a second proof of Theorem 7.4.6(a) in full rank. For $g = \sigma^2$ the bound is $2\sigma^4/n$, while

$$\text{Var}(s^2) = \frac{\sigma^4}{(n-p)^2} \text{Var}\{\chi^2(n-p)\} = \frac{2\sigma^4}{n-p} > \frac{2\sigma^4}{n}.$$

So s^2 does not attain the bound, although by Theorem 7.4.6(b) no unbiased estimator does better. The bound is simply not sharp here. Its ratio to the attained minimum, the **efficiency** of s^2 , is $(n-p)/n$. That is 0.810 for the stack loss regression, and it tends to one as $n \rightarrow \infty$ with p fixed. The same happens for σ : the UMVUE s/b_k has variance $0.0298 \sigma^2$ when $k = 17$, against the bound $\sigma^2/(2n) = 0.0238 \sigma^2$, an efficiency of 0.798. The equality condition in the proof explains why. The bound is attained iff $\text{Var}(\delta - \mathbf{c}^\top \mathbf{S}) = 0$, that is, iff $\delta = g(\boldsymbol{\theta}) + \nabla g^\top \mathbf{I}^{-1} \mathbf{S}$ with probability one. For $g = \sigma^2$ the right-hand side is $\sigma^2 + (2\sigma^4/n)\{-n/(2\sigma^2) + \boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon}/(2\sigma^4)\} = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2/n$. This is not a statistic, because it involves the unknown $\boldsymbol{\beta}$, so no unbiased estimator of σ^2 attains the bound.

Warning. The Cramér–Rao bound is a lower bound, not a target. An estimator that fails to attain it may still be the best unbiased estimator, as s^2 is. When an estimator does attain it, the bound certifies optimality without any appeal to completeness. The Lehmann–Scheffé route is stronger in one respect: it needs no differentiability and covers every estimable function in every rank.

7.4.6 What changes without normality

Every result of this section uses the normal density. Without it the conclusions can fail. Under uniform errors the midrange beats $\bar{Y}$ (Example (Uniform errors and the midrange)). Under Laplace errors the sample median has asymptotic variance $\sigma^2/(2n)$, half that of $\bar{Y}$. Under the second-moment assumptions alone, the best that can be said for s^2 is a result in the spirit of Gauss–Markov. Among quadratic unbiased estimators $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ of σ^2 with $\mathbf{A}$ nonnegative definite, s^2 has the smallest variance when the errors have zero excess kurtosis, or when all leverages are equal. Exercise (C2) outlines this result of Atiqullah (1962).

7.4.7 Exercises

7.4.7.1 A. Check your understanding

Exercise A1.

Assume $\text{rank}(\mathbf{X}) = p$. Use [Lemma 7.4.1](#) and the identity $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \text{SSE} + (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^\top \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ to show directly that $(\hat{\boldsymbol{\beta}}, \text{SSE})$ is sufficient for $(\boldsymbol{\beta}, \sigma^2)$. Is $\hat{\boldsymbol{\beta}}$ alone sufficient?

Exercise A2.

Suppose $\sigma^2 = \sigma_0^2$ is known. Show that $\mathbf{X}^\top \mathbf{Y}$ alone is complete and sufficient for $\boldsymbol{\beta}$, and that $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ is still the UMVUE of every estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$. Why is $\mathbf{X}^\top \mathbf{Y}$ not sufficient when σ^2 is unknown?

7.4.7.2 B. Practice**Exercise B1.**

For estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, find the UMVUE of $(\boldsymbol{\lambda}^\top \boldsymbol{\beta})^2$ in the normal linear model. Can it be negative? Comment.

Solution. $E(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})^2 = (\boldsymbol{\lambda}^\top \boldsymbol{\beta})^2 + \sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$ and $E s^2 = \sigma^2$, so $(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})^2 - s^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$ is unbiased. It is a function of $(\mathbf{M}\mathbf{Y}, \text{SSE})$, so it is the UMVUE by [Theorem 7.4.6\(c\)](#). It is negative whenever $|t| < 1$ for the t statistic of [Proposition 7.3.3](#) at $\psi = 0$, which has positive probability. An estimator of a nonnegative quantity that can be negative is clearly improvable, for example by its positive part. The positive part is biased but has smaller mean squared error. Unbiasedness can be a costly requirement.

Exercise B2.

Compute the observed information $-\nabla^2 \ell$ at $(\boldsymbol{\beta}, v)$, take its expectation, and confirm [Proposition 7.4.7](#) by this second route.

Exercise B3.

For the stack loss model, the 90% quantile of the response at a new operating condition $\mathbf{x}_0$ is $\mathbf{x}_0^\top \boldsymbol{\beta} + 1.2816 \sigma$. Write down its UMVUE, and explain why $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} + 1.2816 s$ is biased. By how much, in units of σ ?

7.4.7.3 C. Going deeper**Exercise C1.**

In the normal linear model with $n - r \geq 2$, let $\mathbf{U} = (\mathbf{I} - \mathbf{M})\mathbf{Y} / \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|$. Show that $\mathbf{U}$ is independent of $(\mathbf{M}\mathbf{Y}, \text{SSE})$, and that its distribution (uniform on the unit sphere of $\mathcal{C}(\mathbf{X})^\perp$) does not depend on $(\boldsymbol{\beta}, \sigma^2)$. *Hint:* write $(\mathbf{I} - \mathbf{M})\mathbf{Y} = \mathbf{Q}_2 \mathbf{Z}$ as in [Exercise \(C2\)](#), and use the rotation invariance of $N(\mathbf{0}, \sigma^2 \mathbf{I})$.

Solution. With $\mathbf{Q}_2$ as in [Exercise \(C2\)](#), $(\mathbf{I} - \mathbf{M})\mathbf{Y} = \mathbf{Q}_2\mathbf{Z}$ with $\mathbf{Z} \sim N_{n-r}(\mathbf{0}, \sigma^2\mathbf{I})$, and $\mathbf{Z}$ is independent of $\mathbf{MY}$ by [Theorem 3.4.1](#), since $\text{Cov}(\mathbf{MY}, \mathbf{Z}) = \sigma^2\mathbf{MQ}_2 = \mathbf{0}$. Then $\text{SSE} = \|\mathbf{Z}\|^2$ and $\mathbf{U} = \mathbf{Q}_2\mathbf{Z}/\|\mathbf{Z}\|$. For any orthogonal $\mathbf{H}$, $\mathbf{HZ}$ has the same law as $\mathbf{Z}$, so the law of $\mathbf{Z}/\|\mathbf{Z}\|$ is invariant under all rotations. The only such law on the sphere is the uniform one. The independence of the direction $\mathbf{Z}/\|\mathbf{Z}\|$ and the length $\|\mathbf{Z}\|$ follows from the polar form of the density of $\mathbf{Z}$, which is a function of $\|\mathbf{z}\|$ only. Hence $\mathbf{U}$ is independent of $(\mathbf{MY}, \text{SSE})$, and its law involves neither $\boldsymbol{\beta}$ nor σ^2 .

Exercise C2.

Assume only that the errors are independent with common variance σ^2 , zero third moment and fourth moment μ_4 . Let $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ be an unbiased estimator of σ^2 with $\mathbf{A}$ symmetric nonnegative definite. (a) Show that $\mathbf{A}\mathbf{X} = \mathbf{0}$ and $\text{tr } \mathbf{A} = 1$. (b) Using [Theorem 2.3.3](#), show that $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = (\mu_4 - 3\sigma^4) \sum_i a_{ii}^2 + 2\sigma^4 \text{tr}(\mathbf{A}^2)$. (c) Show that $\text{tr}(\mathbf{A}^2) \geq 1/(n-r)$, with equality iff $\mathbf{A} = (\mathbf{I} - \mathbf{M})/(n-r)$. (d) Conclude that s^2 has the smallest variance in this class when $\mu_4 = 3\sigma^4$. *Hint for (c):* $\mathbf{A} = (\mathbf{I} - \mathbf{M})\mathbf{A}(\mathbf{I} - \mathbf{M})$, so the eigenvalues of $\mathbf{A}$ live on $\mathcal{C}(\mathbf{X})^\perp$.

7.5 Sampling distributions of the estimates

A point estimate is of little use without a measure of its uncertainty. Under the second-moment assumptions we know the mean and variance of $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ and the mean of s^2 , but not their distributions. Under normality we know everything. The estimates of the estimable functions are jointly normal, the residual sum of squares is a scaled chi-squared variable, and the two are independent. These three facts are what the whole of Part III rests on. Tests, confidence intervals, confidence ellipsoids and prediction intervals are all built from them. The machinery was assembled in [Chapter 3](#) and [Chapter 4](#). Here we apply it.

7.5.1 The main theorem

Theorem 7.5.1 — Sampling distributions in the normal linear model

Assume [Eq. 7.3.1](#) with $r = \text{rank}(\mathbf{X}) < n$. Let $\boldsymbol{\Lambda}$ be $p \times q$ with $\mathcal{C}(\boldsymbol{\Lambda}) \subseteq \mathcal{C}(\mathbf{X}^\top)$, let $(\mathbf{X}^\top \mathbf{X})^-$ be any generalized inverse, and let $s^2 = \text{SSE}/(n-r)$.

- (a) $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} \sim N_q(\boldsymbol{\Lambda}^\top \boldsymbol{\beta}, \sigma^2 \boldsymbol{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^- \boldsymbol{\Lambda})$. In particular $\hat{\mathbf{Y}} = \mathbf{MY} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{M})$, and if $\text{rank}(\mathbf{X}) = p$, $\hat{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$.
- (b) $\text{SSE}/\sigma^2 = (n-r)s^2/\sigma^2 \sim \chi^2(n-r)$.
- (c) $\mathbf{MY}$ and SSE are independent. Hence $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}}$, and every function of the fitted values, is independent of s^2 .
- (d) $E(s^2) = \sigma^2$ and $\text{Var}(s^2) = 2\sigma^4/(n-r)$.

Proof. Write $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{P}$ for an $n \times q$ matrix $\mathbf{P}$. As in the proof of [Theorem 7.2.1\(d\)](#), $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{P}^\top \mathbf{M} \mathbf{Y}$ for every least squares estimate $\hat{\boldsymbol{\beta}}$.

- (a) By [Theorem 3.3.1](#), $\mathbf{P}^\top \mathbf{M} \mathbf{Y}$ is normal with mean $\mathbf{P}^\top \mathbf{M} \mathbf{X} \boldsymbol{\beta} = \mathbf{P}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{\Lambda}^\top \boldsymbol{\beta}$ and covariance $\sigma^2 \mathbf{P}^\top \mathbf{M} \mathbf{M}^\top \mathbf{P} = \sigma^2 \mathbf{P}^\top \mathbf{M} \mathbf{P} = \sigma^2 \mathbf{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{\Lambda}$, by [Theorem 6.3.7](#). The special cases are $\mathbf{\Lambda} = \mathbf{X}^\top$, with $\mathbf{P} = \mathbf{I}$, and $\mathbf{\Lambda} = \mathbf{I}_p$ in full rank.
- (b) $\text{SSE} = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$, and $\mathbf{I} - \mathbf{M}$ is symmetric and idempotent of rank $n - r$ ([Theorem 6.3.3](#) and [Proposition 6.3.4](#)). By [Theorem 4.3.2\(a\)](#), $\text{SSE}/\sigma^2 \sim \chi^2(n - r, \|(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2)$, and the noncentrality is zero because $(\mathbf{I} - \mathbf{M})\mathbf{X} = \mathbf{0}$.
- (c) Apply [Theorem 4.4.1](#) with $\mathbf{B} = \mathbf{M}$, $\mathbf{A} = \mathbf{I} - \mathbf{M}$ and $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$. Then $\mathbf{B} \boldsymbol{\Sigma} \mathbf{A} = \sigma^2 \mathbf{M}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$, so $\mathbf{M} \mathbf{Y}$ and $\mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$ are independent. Functions of independent random vectors are independent.
- (d) A $\chi^2(k)$ variable has mean k and variance $2k$ ([Theorem 4.1.2\(b\)](#) with $\gamma = 0$). So $\text{E SSE} = \sigma^2(n - r)$ and $\text{Var}(\text{SSE}) = 2\sigma^4(n - r)$. Divide by $n - r$ and $(n - r)^2$.

■

The proof is short because the work was done earlier. It is worth seeing where each ingredient comes from geometrically. The fitted vector $\mathbf{M} \mathbf{Y}$ and the residual vector $(\mathbf{I} - \mathbf{M}) \mathbf{Y}$ are projections of $\mathbf{Y}$ onto orthogonal subspaces. For a spherical normal vector, orthogonal projections are independent ([Theorem 4.4.4](#)). Everything about $\boldsymbol{\beta}$ lives in the first projection. The second is pure noise, and its squared length, measured in units of σ^2 , counts $n - r$ independent squared standard normals.

The independence in (c) has no analogue under the second-moment assumptions. If the errors are independent with zero third moment, $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}$ and s^2 are still uncorrelated, but in general they are not independent. The simulation at the end of the section shows the difference.

7.5.2 Standardizing: the t statistic

The distribution in (a) involves the unknown σ . Since $\text{E}(s^2) = \sigma^2$, the matrix $s^2 \mathbf{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{\Lambda}$ is an unbiased estimate of $\text{Cov}(\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}})$, and in full rank $s^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ is unbiased for $\text{Cov}(\hat{\boldsymbol{\beta}})$; this needs only the second-moment assumptions. The square roots of its diagonal entries are the standard errors reported by regression software. Replacing σ by s changes the normal distribution into a t distribution, and independence is what makes the change exact.

Corollary 7.5.2 — The t statistic for an estimable function

Assume [Eq. 7.3.1](#) with $r < n$, let $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ with $\boldsymbol{\lambda} \neq \mathbf{0}$, and put $\text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = s \sqrt{\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}}$. For every constant d ,

$$T_d = \frac{\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - d}{\text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})} \sim t(n - r, \delta), \quad \delta = \frac{\boldsymbol{\lambda}^\top \boldsymbol{\beta} - d}{\sigma \sqrt{\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}}}.$$

In particular $T = (\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta}) / \text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) \sim t(n - r)$, and

$$\Pr\{|\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta}| \leq t_{n-r, \alpha/2} \text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})\} = 1 - \alpha, \quad (7.5.1)$$

where $t_{n-r,\alpha/2}$ is the upper $\alpha/2$ point of $t(n-r)$.

Proof. Put $c^2 = \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$. Then $c > 0$, since $c^2 = \|\mathbf{M}\boldsymbol{\rho}\|^2$ with $\mathbf{X}^\top \mathbf{M}\boldsymbol{\rho} = \boldsymbol{\lambda} \neq \mathbf{0}$. By Theorem 7.5.1(a), $Z = (\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - d)/(\sigma c) \sim N(\delta, 1)$. By (b), $V = (n-r)s^2/\sigma^2 \sim \chi^2(n-r)$, and by (c) Z and V are independent. Now $T_d = Z/\sqrt{V/(n-r)}$, which is $t(n-r, \delta)$ by Definition 4.1.3. With $d = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$, $\delta = 0$, and Eq. 7.5.1 restates $\Pr\{|T| \leq t_{n-r,\alpha/2}\} = 1 - \alpha$. ■

Eq. 7.5.1 is the familiar confidence interval $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} \pm t_{n-r,\alpha/2} \text{se}$, and T_d is the statistic for testing $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = d$. Its noncentrality δ governs the power. Chapter 11 and Chapter 12 develop both. For the air-flow coefficient in the stack loss regression this interval was (0.431, 1.000) (Example (Profile likelihood for the air-flow coefficient)). Two features of the corollary are easy to overlook. The degrees of freedom are $n-r$, the dimension of the residual space. They are not n , and when $\mathbf{X}$ is rank deficient they are not $n-p$ either. And the interval is exact for every sample size, which no large-sample argument delivers.

7.5.3 Several functions at once

For a vector of estimable functions, the natural summary of Theorem 7.5.1(a) is a quadratic form. The covariance matrix $\boldsymbol{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\Lambda}$ may be singular when the columns of $\boldsymbol{\Lambda}$ are linearly dependent, so a generalized inverse is needed.

Corollary 7.5.3 — A chi-squared pivot for several functions

In the setting of Theorem 7.5.1, let $\mathbf{W} = \boldsymbol{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\Lambda}$, let $\mathbf{W}^-$ be any generalized inverse of $\mathbf{W}$, and put $\mathbf{D} = \boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\Lambda}^\top \boldsymbol{\beta}$. Then

$$Q = \frac{\mathbf{D}^\top \mathbf{W}^- \mathbf{D}}{\sigma^2} \sim \chi^2(\text{rank}(\boldsymbol{\Lambda})),$$

Q does not depend on the choice of $\mathbf{W}^-$ (with probability one), and Q is independent of SSE. Consequently

$$\frac{\mathbf{D}^\top \mathbf{W}^- \mathbf{D} / \text{rank}(\boldsymbol{\Lambda})}{s^2} \sim F(\text{rank}(\boldsymbol{\Lambda}), n-r).$$

Proof. By Theorem 7.5.1(a), $\mathbf{D} \sim N_q(\mathbf{0}, \sigma^2 \mathbf{W})$. The quadratic form is unchanged if $\mathbf{W}^-$ is replaced by its symmetric part $\frac{1}{2}(\mathbf{W}^- + \mathbf{W}^{-\top})$, which is again a generalized inverse of the symmetric matrix $\mathbf{W}$ (Proposition 1.3.3(d)). So we may assume $\mathbf{W}^-$ symmetric. Apply Theorem 4.3.2(c) with $\boldsymbol{\Sigma} = \sigma^2 \mathbf{W}$, $\boldsymbol{\mu} = \mathbf{0}$ and $\mathbf{A} = \mathbf{W}^-/\sigma^2$. Conditions (ii) and (iii) hold trivially because $\boldsymbol{\mu} = \mathbf{0}$. Condition (i) is $\boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} = \sigma^2 \mathbf{W} \mathbf{W}^- \mathbf{W} \mathbf{W}^- \mathbf{W} = \sigma^2 \mathbf{W} = \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}$, by two uses of $\mathbf{W} \mathbf{W}^- \mathbf{W} = \mathbf{W}$. So $Q \sim \chi^2(\text{tr}(\mathbf{A} \boldsymbol{\Sigma}))$, where $\text{tr}(\mathbf{A} \boldsymbol{\Sigma}) = \text{tr}(\mathbf{W}^- \mathbf{W}) = \text{rank}(\mathbf{W})$, because $\mathbf{W}^- \mathbf{W}$ is idempotent with the same rank as $\mathbf{W}$ (Proposition 1.3.3(a)), and the trace of an idempotent matrix is its rank (Proposition 1.9.1).

It remains to show $\text{rank}(\mathbf{W}) = \text{rank}(\boldsymbol{\Lambda})$. With $\boldsymbol{\Lambda} = \mathbf{X}^\top \mathbf{P}$, $\mathbf{W} = \mathbf{P}^\top \mathbf{M} \mathbf{P} = (\mathbf{M} \mathbf{P})^\top (\mathbf{M} \mathbf{P})$, so $\text{rank} \mathbf{W} = \text{rank}(\mathbf{M} \mathbf{P})$. Also $\boldsymbol{\Lambda} = \mathbf{X}^\top \mathbf{P} = \mathbf{X}^\top \mathbf{M} \mathbf{P}$. If $\mathbf{X}^\top \mathbf{M} \mathbf{P} \mathbf{u} = \mathbf{0}$, then $\mathbf{M} \mathbf{P} \mathbf{u} \in \mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{X})^\perp = \{\mathbf{0}\}$, so $\mathbf{X}^\top$ is one-to-one on $\mathcal{C}(\mathbf{M} \mathbf{P})$ and $\text{rank} \boldsymbol{\Lambda} = \text{rank}(\mathbf{M} \mathbf{P})$.

For invariance: $D \in \mathcal{C}(W)$ with probability one by [Theorem 2.2.3\(d\)](#). If $D = Wu$, then $D^\top W^- D = u^\top W W^- W u = u^\top W u$, whatever W^- is. Independence follows from [Theorem 7.5.1\(c\)](#), since D is a function of MY . The F ratio is then a ratio of independent chi-squared variables divided by their degrees of freedom ([Definition 4.1.2](#)). ■

The set of $\Lambda^\top \beta$ values for which this F ratio stays below an F quantile is a confidence ellipsoid. Chapter 12 studies these regions, and Chapter 11 turns the same statistic into the test of the general linear hypothesis. Here we note only that the degrees of freedom in the numerator are the number of *linearly independent* functions being estimated, not the number of rows of $\Lambda^\top$.

Example. (A rank-deficient model, simulated)

Consider a one-way layout with three groups of sizes 4, 5, 6, an intercept, one indicator column per group, and a covariate. So $n = 15$ and $p = 5$, but the rank is $r = 4$, because the indicators add up to the intercept. The contrast $\lambda^\top \beta = \alpha_1 - \alpha_2$, with $\lambda = (0, 1, -1, 0, 0)^\top$, is estimable. With $\sigma = 2$, [Theorem 7.5.1](#) gives $\text{Var}(\lambda^\top \hat{\beta}) = \sigma^2 \lambda^\top (X^\top X)^- \lambda = 1.9007$, computed with the Moore–Penrose inverse. In 100,000 simulated data sets the variance of the estimated contrast is 1.9017, and the average of SSE/σ^2 is 11.001, against $n - r = 11$. The sample correlation between the contrast estimate and s^2 is 0.0036. The standardized contrast T falls within ± 2.201 , the 97.5% point of $t(11)$, in a proportion 0.9498 of the samples. [Figure 7.5.1](#) compares the simulated distributions with the theory. Near the centre the $t(11)$ and $N(0, 1)$ densities are hard to tell apart, so panel (b) uses a logarithmic density scale, on which the tails separate: the simulated points follow $t(11)$, and the normal density falls away too fast. The tail mass says the same in numbers. A proportion 0.0758 of the samples have $|T| > 1.96$, against 0.05 under $N(0, 1)$ and 0.0758 under $t(11)$. A nominal 95% interval built with the normal point 1.96 would therefore undercover.

```
import numpy as np
from scipy import stats

groups = np.repeat([0, 1, 2], [4, 5, 6])
z = np.array([2.1, 3.4, 1.8, 4.0, 2.9, 3.3, 5.1, 2.2, 3.8, 1.5, 4.4, 2.7, 3.9, 5.0,
              3.1])
n = len(z)
X = np.column_stack([np.ones(n), np.eye(3)[groups], z]) # p = 5 columns, rank 4
r = np.linalg.matrix_rank(X)
G = np.linalg.pinv(X.T @ X) # one generalized inverse
M = X @ G @ X.T
lam = np.array([0.0, 1.0, -1.0, 0.0, 0.0]) # group 1 minus group 2:
    estimable
beta, sigma = np.array([1.0, 0.5, -0.5, 0.0, 0.8]), 2.0
var_lam = sigma ** 2 * lam @ G @ lam # Var(lambda^T beta_hat)
print(f"n = {n}, p = {X.shape[1]}, rank = {r}, Var(lambda^T beta_hat) = {var_lam:.4f}
      ")
```

```
rng = np.random.default_rng(75)
reps = 100_000
Y = X @ beta + sigma * rng.normal(size=(reps, n)) # one data set per row
est = Y @ (X @ G @ X.T) # lambda^T beta_hat =
    lambda^T G X^T y
```

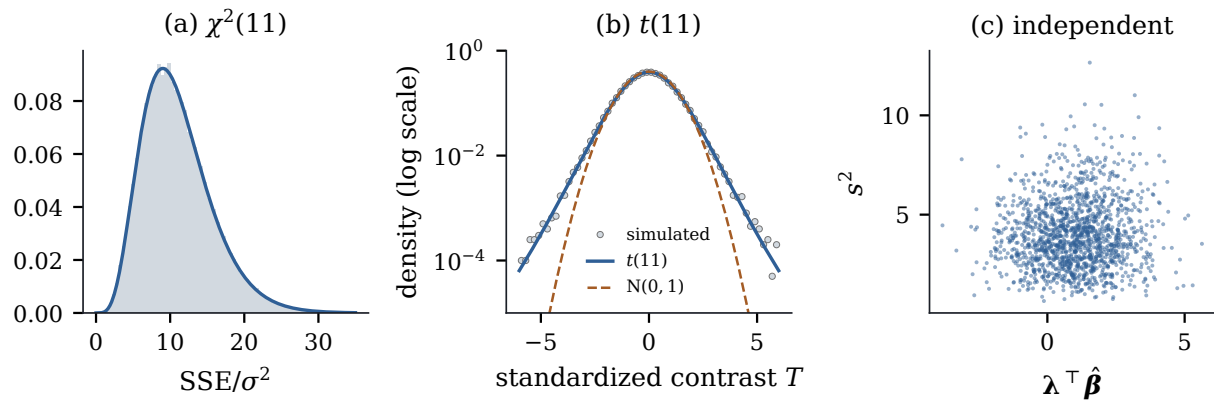


Figure 7.5.1: Sampling distributions in a rank-deficient normal linear model ($n = 15$, $r = 4$; 100,000 simulated data sets). (a) SSE/σ^2 with the $\chi^2(11)$ density. (b) The standardized contrast (binned density, points) with the $t(11)$ and standard normal densities, on a logarithmic scale. (c) The contrast estimate against s^2 : no relationship, as independence requires.

```
sse = np.sum((Y - Y @ M) ** 2, axis=1) # y^T (I - M) y
s2 = sse / (n - r)
t = (est - lam @ beta) / np.sqrt(s2 * (lam @ G @ lam))
print(f"mean {est.mean():.4f}, variance {est.var():.4f}")
print(f"E(SSE)/sigma^2 = {np.mean(sse) / sigma**2:.3f} (n - r = {n - r})")
print(f"corr(estimate, s^2) = {np.corrcoef(est, s2)[0, 1]:.4f}")
print("KS p-value, SSE/sigma^2 vs chi^2(n-r):", round(stats.kstest(sse / sigma**2, "chi2", args=(n - r,)).pvalue, 3))
print("KS p-value, t vs t(n-r):", round(stats.kstest(t, "t", args=(n - r,)).pvalue, 3))
```

7.5.4 Without normality

Which parts of [Theorem 7.5.1](#) survive when the errors are merely uncorrelated with common variance? The first two moments of $\lambda^\top \hat{\beta}$ do, by [Theorem 7.2.1](#), and so does $E(s^2) = \sigma^2$. The normality of $\lambda^\top \hat{\beta}$ is lost, but it often returns approximately. $\lambda^\top \hat{\beta}$ is a weighted sum of independent errors, and a central limit theorem applies when no single observation dominates the weights. Chapter 19 makes this precise. The distribution of SSE is less forgiving, because its variance depends on the fourth moment of the errors.

Proposition 7.5.4 — Variance of SSE without normality

Let the errors be independent with mean zero, variance σ^2 , third moment μ_3 and fourth moment μ_4 , and let h_{ii} be the leverages. Then

$$\text{Var}(\text{SSE}) = 2\sigma^4(n - r) + (\mu_4 - 3\sigma^4) \sum_{i=1}^n (1 - h_{ii})^2.$$

Proof. Apply [Theorem 2.3.3](#) to $\text{SSE} = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$, with $\boldsymbol{\theta} = \mathbf{X}\boldsymbol{\beta}$ and $\mathbf{A} = \mathbf{I} - \mathbf{M}$, whose diagonal entries are $1 - h_{ii}$. The terms $4\sigma^2 \boldsymbol{\theta}^\top \mathbf{A}^2 \boldsymbol{\theta}$ and $4\mu_3 \boldsymbol{\theta}^\top \mathbf{A} \mathbf{a}$ vanish because

$$\mathbf{A}\boldsymbol{\theta} = (\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}, \text{ and } \text{tr}(\mathbf{A}^2) = \text{tr}(\mathbf{A}) = n - r.$$

■

The factor $\mu_4 - 3\sigma^4$ is σ^4 times the excess kurtosis. For normal errors it vanishes and we recover $2\sigma^4(n - r)$. For heavy-tailed errors it is positive, and SSE is more variable than the chi-squared theory says. Intervals for σ^2 built from [Theorem 7.5.1\(b\)](#) are then too short, and this does not improve with sample size: the ratio of the true variance to the nominal one tends to $1 + \frac{1}{2}(\mu_4/\sigma^4 - 3)$ when the leverages are small. The t interval for $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is much less affected, because s^2 enters only through s , and s converges to σ whatever the kurtosis.

Example. (Laplace errors)

Rerun the simulation of [Example \(A rank-deficient model, simulated\)](#) with Laplace (double-exponential) errors of the same variance $\sigma^2 = 4$. For these $\mu_4 = 6\sigma^4$. The contrast still has variance 1.9007. For SSE, $\sum_i (1 - h_{ii})^2 = 8.138$, and [Proposition 7.5.4](#) gives $\text{Var}(\text{SSE}) = 742.6$, more than twice the normal-theory value 352.0. The simulated variance is 741.5. Independence fails too. The contrast and SSE are still uncorrelated, because the errors are symmetric, but the *squared* error of the contrast has correlation 0.211 with SSE. Samples with a few large errors inflate both.

```
E_lap = rng.laplace(scale=sigma / np.sqrt(2), size=(reps, n)) # variance sigma^2,
mu_4 = 6 * sigma^4
Y_lap = X @ beta + E_lap
est_lap = Y_lap @ (X @ G @ lam)
sse_lap = np.sum((Y_lap - Y_lap @ M) ** 2, axis=1)
h = np.diag(np.eye(n) - M)
var_sse_lap = (6 - 3) * sigma**4 * h @ h + 2 * sigma**4 * (n - r)
print(f"Var(SSE): normal theory {2 * sigma**4 * (n - r):.1f}, Laplace formula {
    var_sse_lap:.1f}, "
      f"simulated {sse_lap.var():.1f}")
print(f"corr(squared error of estimate, SSE) under Laplace errors: "
      f"{np.corrcoef((est_lap - lam @ beta) ** 2, sse_lap)[0, 1]:.3f}")
```

In the listing, $\mathbf{h}$ holds the diagonal of $\mathbf{I} - \mathbf{M}$, that is, $1 - h_{ii}$.

7.5.5 Exercises

7.5.5.1 A. Check your understanding

Exercise A1.

Under [Eq. 7.3.1](#), find the joint distribution of $(\hat{\mathbf{Y}}, \hat{\boldsymbol{\varepsilon}})$. Show that $\hat{\mathbf{Y}}$ and $\hat{\boldsymbol{\varepsilon}}$ are independent, and that $\hat{\boldsymbol{\varepsilon}} \sim N_n(\mathbf{0}, \sigma^2(\mathbf{I} - \mathbf{M}))$ whatever $\boldsymbol{\beta}$ is. Is $\hat{\boldsymbol{\varepsilon}}$ independent of SSE?

Exercise A2.

In simple linear regression, write out T of [Corollary 7.5.2](#) for the slope, in terms of S_{xx} and s , and state its distribution. What is r ?

7.5.5.2 B. Practice

Exercise B1.

Let $\lambda_1, \lambda_2 \in \mathcal{C}(\mathbf{X}^\top)$. Show that $\text{Cov}(\lambda_1^\top \hat{\beta}, \lambda_2^\top \hat{\beta}) = \sigma^2 \lambda_1^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda_2$, and that under normality the two estimates are independent iff $\lambda_1^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda_2 = 0$. In a one-way layout with a groups of m observations each and cell-means parameterization $E(Y_{ij}) = \mu_i$, show that the estimates of $\sum_i c_i \mu_i$ and $\sum_i d_i \mu_i$ are independent iff $\sum_i c_i d_i = 0$.

Solution. The covariance is the off-diagonal entry of [Theorem 7.5.1\(a\)](#) with $\Lambda = [\lambda_1, \lambda_2]$. Under normality the pair is bivariate normal, so zero covariance is equivalent to independence ([Theorem 3.4.1](#)). In the cell-means model $\mathbf{X}^\top \mathbf{X} = m\mathbf{I}_a$, so $\lambda_1^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda_2 = \sum_i c_i d_i / m$, which vanishes iff $\sum_i c_i d_i = 0$.

Exercise B2.

A new observation $Y_0 = \mathbf{x}_0^\top \beta + \varepsilon_0$ will be taken, with $\varepsilon_0 \sim N(0, \sigma^2)$ independent of $\mathbf{Y}$ and $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$. Show that

$$\frac{Y_0 - \mathbf{x}_0^\top \hat{\beta}}{s \sqrt{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}} \sim t(n - r),$$

and deduce a prediction interval for Y_0 . Why is the extra 1 in the denominator never negligible?

Solution. Y_0 is independent of $\mathbf{Y}$, so $Y_0 - \mathbf{x}_0^\top \hat{\beta}$ is normal with mean $\mathbf{x}_0^\top \beta - \mathbf{x}_0^\top \beta = 0$ and variance $\sigma^2 + \sigma^2 \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0$. It is a function of $(Y_0, \mathbf{M}\mathbf{Y})$, which is independent of SSE. Dividing by $\sigma \sqrt{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}$ gives a standard normal, independent of $(n - r)s^2/\sigma^2 \sim \chi^2(n - r)$, and the ratio is $t(n - r)$ by [Definition 4.1.3](#). The interval is $\mathbf{x}_0^\top \hat{\beta} \pm t_{n-r, \alpha/2} s \sqrt{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}$. The 1 is the variance of the new error itself, which no amount of data about β can reduce. The second term tends to zero as the design grows. The first does not.

Exercise B3.

Use [Theorem 7.5.1\(b\)](#) to construct an exact $1 - \alpha$ confidence interval for σ^2 . Then, using [Proposition 7.5.4](#) with small leverages, explain why its coverage does not tend to $1 - \alpha$ as $n \rightarrow \infty$ when the errors have nonzero excess kurtosis.

7.5.5.3 C. Going deeper

Exercise C1.

Give a second proof of [Theorem 7.5.1\(b\)](#) and (c) using [Theorem 4.4.4](#) with the two projections $\mathbf{M}$ and $\mathbf{I} - \mathbf{M}$. Extend it: if $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$, show that $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{Y}\|^2$ and SSE are independent, and find the distribution of $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{Y}\|^2/\sigma^2$ when $E(\mathbf{Y}) = \mathbf{X}\beta$ is arbitrary.

Exercise C2.

For the test that rejects $\lambda^\top \beta = d$ when $|T_d| > t_{n-r, \alpha/2}$, show that the power depends on the parameters only through $|\delta|$ of [Corollary 7.5.2](#). For a fixed design, how does the power change when the sample is replicated m times ($\mathbf{X}$ replaced by m stacked copies)?

7.6 Bayesian thread I: the conjugate linear model

The previous sections judged estimators by their behaviour over repeated samples, for each fixed value of the parameters. The Bayesian approach treats (β, σ^2) as uncertain quantities with a probability distribution. It combines a **prior** distribution, expressing what is known before the data, with the likelihood of [Section 7.3](#), and the result is a **posterior** distribution. For the normal linear model there is a family of priors for which the posterior is available in closed form, and it connects directly to the rest of the chapter. The posterior mean is a matrix-weighted combination of the prior mean and the least squares estimate. It is biased in the frequentist sense, which is how it escapes the Gauss–Markov bound. And as the prior becomes flat, the posterior reproduces least squares and the t intervals of [Section 7.5](#) exactly. This is the first of several Bayesian threads in the book. Chapter 12 continues it with credible regions and predictive distributions, Chapter 27 with shrinkage priors, and Chapter 39 with generalized linear models.

7.6.1 The normal-inverse-gamma family

Definition 7.6.1 — Inverse gamma and normal-inverse-gamma distributions (a)

A positive random variable V has the **inverse gamma** distribution $\text{IG}(a, b)$, with $a, b > 0$, if $1/V$ has the gamma distribution with shape a and rate b . Its density is

$$\pi(v) = \frac{b^a}{\Gamma(a)} v^{-a-1} e^{-b/v}, \quad v > 0.$$

- (b) The pair (β, σ^2) has the **normal-inverse-gamma** distribution $\text{NIG}(\mathbf{m}, \mathbf{V}, a, b)$, with $\mathbf{m} \in \mathbb{R}^p$, $\mathbf{V}$ positive definite and $a, b > 0$, if $\sigma^2 \sim \text{IG}(a, b)$ and, given σ^2 , $\beta \sim \text{N}_p(\mathbf{m}, \sigma^2 \mathbf{V})$. Its joint density is

$$\begin{aligned} \pi(\beta, \sigma^2) &= \frac{b^a}{\Gamma(a)(2\pi)^{p/2} \det(\mathbf{V})^{1/2}} (\sigma^2)^{-(a+p/2)-1} \\ &\quad \times \exp\left\{-\frac{2b + (\beta - \mathbf{m})^\top \mathbf{V}^{-1}(\beta - \mathbf{m})}{2\sigma^2}\right\}. \end{aligned} \quad (7.6.1)$$

The prior on β is scaled by σ^2 : prior uncertainty about the coefficients is measured in units of the noise level. This is a modelling choice, and it is what makes the family conjugate. It also makes $\mathbf{V}$ dimensionless relative to the likelihood's $(\mathbf{X}^\top \mathbf{X})^{-1}$, which is the comparison that matters below. The marginal distribution of β is a multivariate t .

Definition 7.6.2 — Multivariate t

Let $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{S})$ and $W \sim \chi^2(\nu)$ be independent, with $\mathbf{S}$ positive definite. The law of $\mathbf{m} + \mathbf{Z}/\sqrt{W/\nu}$ is the **multivariate t distribution** $t_\nu(\mathbf{m}, \mathbf{S})$.

For $p = 1$ and $\mathbf{S} = 1$, this is the $t(\nu)$ distribution of Definition 4.1.3 shifted by $\mathbf{m}$. For every $\mathbf{c} \neq \mathbf{0}$, $\mathbf{c}^\top \mathbf{u}$ with $\mathbf{u} \sim t_\nu(\mathbf{m}, \mathbf{S})$ is $\mathbf{c}^\top \mathbf{m} + \sqrt{\mathbf{c}^\top \mathbf{S} \mathbf{c}} T$ with $T \sim t(\nu)$, because $\mathbf{c}^\top \mathbf{Z}/\sqrt{\mathbf{c}^\top \mathbf{S} \mathbf{c}} \sim N(0, 1)$ is independent of W .

Lemma 7.6.1 — Marginals of the normal-inverse-gamma

If $(\boldsymbol{\beta}, \sigma^2) \sim \text{NIG}(\mathbf{m}, \mathbf{V}, a, b)$, then $2b/\sigma^2 \sim \chi^2(2a)$, $E(\sigma^2) = b/(a - 1)$ for $a > 1$, and

$$\boldsymbol{\beta} \sim t_{2a}(\mathbf{m}, (b/a)\mathbf{V}).$$

In particular, for every $\boldsymbol{\lambda} \neq \mathbf{0}$, $(\boldsymbol{\lambda}^\top \boldsymbol{\beta} - \boldsymbol{\lambda}^\top \mathbf{m})/\sqrt{(b/a)\boldsymbol{\lambda}^\top \mathbf{V} \boldsymbol{\lambda}} \sim t(2a)$.

Proof. Put $W = 2b/\sigma^2$. Changing variables $v = 2b/w$ in the inverse gamma density, with $|dv/dw| = 2b/w^2$, gives the density of W :

$$\frac{b^a}{\Gamma(a)} \left(\frac{2b}{w}\right)^{-a-1} e^{-w/2} \frac{2b}{w^2} = \frac{w^{a-1} e^{-w/2}}{2^a \Gamma(a)}, \quad w > 0,$$

which is the $\chi^2(2a)$ density (Eq. 4.1.1). Then $E\sigma^2 = 2b E(1/W) = 2b/(2a - 2)$, since $E(1/W) = 1/(\nu - 2)$ for $W \sim \chi^2(\nu)$, $\nu > 2$. Next let $\mathbf{Z}_0 = (\boldsymbol{\beta} - \mathbf{m})/\sigma$. Given σ^2 , $\mathbf{Z}_0 \sim N_p(\mathbf{0}, \mathbf{V})$, a law that does not depend on σ^2 . So $\mathbf{Z}_0$ is independent of σ^2 , and hence of W , and

$$\boldsymbol{\beta} = \mathbf{m} + \sigma \mathbf{Z}_0 = \mathbf{m} + \sqrt{\frac{2b}{W}} \mathbf{Z}_0 = \mathbf{m} + \frac{\sqrt{b/a} \mathbf{Z}_0}{\sqrt{W/(2a)}}.$$

Since $\sqrt{b/a} \mathbf{Z}_0 \sim N_p(\mathbf{0}, (b/a)\mathbf{V})$, this is $t_{2a}(\mathbf{m}, (b/a)\mathbf{V})$ by Definition 7.6.2. The last statement is the remark after Definition 7.6.2. ■

7.6.2 The conjugate update**Theorem 7.6.2 — Conjugate posterior**

Let $\mathbf{y}$ follow the normal linear model Eq. 7.3.1, with $\mathbf{X}$ of any rank, and let the prior be $(\boldsymbol{\beta}, \sigma^2) \sim \text{NIG}(\mathbf{m}_0, \mathbf{V}_0, a_0, b_0)$. Then the posterior is $\text{NIG}(\mathbf{m}_n, \mathbf{V}_n, a_n, b_n)$, where

$$\begin{aligned} \mathbf{V}_n &= (\mathbf{V}_0^{-1} + \mathbf{X}^\top \mathbf{X})^{-1}, & \mathbf{m}_n &= \mathbf{V}_n(\mathbf{V}_0^{-1} \mathbf{m}_0 + \mathbf{X}^\top \mathbf{y}), \\ a_n &= a_0 + \frac{n}{2}, \end{aligned} \tag{7.6.2}$$

$$\begin{aligned}
b_n &= b_0 + \frac{1}{2}(\mathbf{y}^\top \mathbf{y} + \mathbf{m}_0^\top \mathbf{V}_0^{-1} \mathbf{m}_0 - \mathbf{m}_n^\top \mathbf{V}_n^{-1} \mathbf{m}_n) \\
&= b_0 + \frac{1}{2}\{\|\mathbf{y} - \mathbf{X}\mathbf{m}_n\|^2 + (\mathbf{m}_n - \mathbf{m}_0)^\top \mathbf{V}_0^{-1}(\mathbf{m}_n - \mathbf{m}_0)\}.
\end{aligned} \tag{7.6.3}$$

Consequently $\beta \mid \mathbf{y} \sim t_{2a_n}(\mathbf{m}_n, (b_n/a_n)\mathbf{V}_n)$, $\sigma^2 \mid \mathbf{y} \sim \text{IG}(a_n, b_n)$, and $\text{E}(\beta \mid \mathbf{y}) = \mathbf{m}_n$.

Proof. Write $\mathbf{P}_0 = \mathbf{V}_0^{-1}$ and $\mathbf{P}_n = \mathbf{P}_0 + \mathbf{X}^\top \mathbf{X}$. The matrix $\mathbf{P}_n$ is positive definite, because $\mathbf{P}_0$ is and $\mathbf{X}^\top \mathbf{X}$ is nonnegative definite. This holds whatever the rank of $\mathbf{X}$. Expanding both sides and using $\mathbf{P}_n \mathbf{m}_n = \mathbf{P}_0 \mathbf{m}_0 + \mathbf{X}^\top \mathbf{y}$ gives the completion of the square

$$\|\mathbf{y} - \mathbf{X}\beta\|^2 + (\beta - \mathbf{m}_0)^\top \mathbf{P}_0(\beta - \mathbf{m}_0) = (\beta - \mathbf{m}_n)^\top \mathbf{P}_n(\beta - \mathbf{m}_n) + C,$$

with $C = \mathbf{y}^\top \mathbf{y} + \mathbf{m}_0^\top \mathbf{P}_0 \mathbf{m}_0 - \mathbf{m}_n^\top \mathbf{P}_n \mathbf{m}_n$. Indeed, the left side is $\beta^\top \mathbf{P}_n \beta - 2\beta^\top (\mathbf{X}^\top \mathbf{y} + \mathbf{P}_0 \mathbf{m}_0) + \mathbf{y}^\top \mathbf{y} + \mathbf{m}_0^\top \mathbf{P}_0 \mathbf{m}_0$, and $(\beta - \mathbf{m}_n)^\top \mathbf{P}_n(\beta - \mathbf{m}_n) = \beta^\top \mathbf{P}_n \beta - 2\beta^\top \mathbf{P}_n \mathbf{m}_n + \mathbf{m}_n^\top \mathbf{P}_n \mathbf{m}_n$. Putting $\beta = \mathbf{m}_n$ in the identity shows that $C = \|\mathbf{y} - \mathbf{X}\mathbf{m}_n\|^2 + (\mathbf{m}_n - \mathbf{m}_0)^\top \mathbf{P}_0(\mathbf{m}_n - \mathbf{m}_0) \geq 0$, which is the second form of [Eq. 7.6.3](#).

By Bayes' theorem the posterior density is proportional, as a function of (β, σ^2) , to prior times likelihood. Using [Eq. 7.6.1](#) and dropping factors free of the parameters,

$$\begin{aligned}
\pi(\beta, \sigma^2 \mid \mathbf{y}) &\propto (\sigma^2)^{-(a_0+p/2)-1} e^{-\{2b_0 + (\beta - \mathbf{m}_0)^\top \mathbf{P}_0(\beta - \mathbf{m}_0)\}/(2\sigma^2)} \\
&\quad \times (\sigma^2)^{-n/2} e^{-\|\mathbf{y} - \mathbf{X}\beta\|^2/(2\sigma^2)}.
\end{aligned}$$

By the completion of the square, the exponent is $-\{2b_0 + C + (\beta - \mathbf{m}_n)^\top \mathbf{P}_n(\beta - \mathbf{m}_n)\}/(2\sigma^2)$, and $2b_0 + C = 2b_n$. The power of σ^2 is $-(a_0 + n/2 + p/2) - 1 = -(a_n + p/2) - 1$. So the posterior is proportional to the $\text{NIG}(\mathbf{m}_n, \mathbf{V}_n, a_n, b_n)$ density [Eq. 7.6.1](#). Two densities that are proportional are equal, since both integrate to one. The marginals follow from [Lemma 7.6.1](#). The posterior mean of β is $\mathbf{m}_n$: by the tower property, $\text{E}(\beta \mid \mathbf{y}) = \text{E}\{\text{E}(\beta \mid \sigma^2, \mathbf{y}) \mid \mathbf{y}\} = \mathbf{m}_n$. ■

The update has a simple structure. **Precisions add:** the posterior precision (per unit σ^2) is the prior precision $\mathbf{V}_0^{-1}$ plus the data precision $\mathbf{X}^\top \mathbf{X}$. **Shape parameters count observations:** each observation adds $1/2$ to a . **Scale parameters accumulate squared discrepancies:** b_n adds half of the residual sum of squares at the posterior mean and half of the prior-weighted distance between the posterior and prior means. Because $\mathbf{P}_n$ is positive definite even when $\mathbf{X}$ is rank deficient, the posterior is proper for every design. The prior supplies information about the directions of $\mathcal{N}(\mathbf{X})$ that the data cannot, and there the posterior equals the prior ([Exercise \(B2\)](#)).

7.6.3 The posterior mean as shrinkage

Proposition 7.6.3 — Posterior mean as a combination

In [Theorem 7.6.2](#), let $\mathbf{X}$ have full column rank and put $\mathbf{W} = \mathbf{V}_n \mathbf{V}_0^{-1}$. Then:

- (a) $\mathbf{m}_n = \mathbf{W}\mathbf{m}_0 + (\mathbf{I} - \mathbf{W})\hat{\beta}$, with $\mathbf{I} - \mathbf{W} = \mathbf{V}_n \mathbf{X}^\top \mathbf{X}$. Equivalently $\mathbf{V}_n^{-1} \mathbf{m}_n =$

$\mathbf{V}_0^{-1}\mathbf{m}_0 + (\mathbf{X}^\top \mathbf{X})\hat{\boldsymbol{\beta}}$: the posterior mean is the precision-weighted average of the prior mean and the least squares estimate.

- (b) $b_n = b_0 + \frac{1}{2}\{\text{SSE} + (\hat{\boldsymbol{\beta}} - \mathbf{m}_0)^\top (\mathbf{V}_0 + (\mathbf{X}^\top \mathbf{X})^{-1})^{-1}(\hat{\boldsymbol{\beta}} - \mathbf{m}_0)\}$.
- (c) (*g-prior.*) If $\mathbf{V}_0 = g(\mathbf{X}^\top \mathbf{X})^{-1}$ with $g > 0$, then $\mathbf{W} = \mathbf{I}/(1 + g)$ and $\mathbf{m}_n = (\mathbf{m}_0 + g\hat{\boldsymbol{\beta}})/(1 + g)$.
- (d) (*Ridge.*) If $\mathbf{m}_0 = \mathbf{0}$ and $\mathbf{V}_0 = k^{-1}\mathbf{I}$ with $k > 0$, then $\mathbf{m}_n = (\mathbf{X}^\top \mathbf{X} + k\mathbf{I})^{-1}\mathbf{X}^\top \mathbf{y}$.

Proof.

- (a) $\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}}$ by the normal equations (Theorem 6.4.2), so $\mathbf{m}_n = \mathbf{V}_n \mathbf{V}_0^{-1} \mathbf{m}_0 + \mathbf{V}_n \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}$, and $\mathbf{V}_n \mathbf{V}_0^{-1} + \mathbf{V}_n \mathbf{X}^\top \mathbf{X} = \mathbf{V}_n \mathbf{V}_n^{-1} = \mathbf{I}$.
- (b) Write $\mathbf{P}_0 = \mathbf{V}_0^{-1}$ and $\mathbf{K} = \mathbf{X}^\top \mathbf{X}$. The completion of the square in the proof of Theorem 7.6.2 shows that $2(b_n - b_0)$ is the minimum over $\boldsymbol{\beta}$ of $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + (\boldsymbol{\beta} - \mathbf{m}_0)^\top \mathbf{P}_0(\boldsymbol{\beta} - \mathbf{m}_0)$, because the remaining term $(\boldsymbol{\beta} - \mathbf{m}_n)^\top \mathbf{P}_n(\boldsymbol{\beta} - \mathbf{m}_n)$ is nonnegative and vanishes at $\mathbf{m}_n$. By Eq. 6.3.1, $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \text{SSE} + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{K}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})$. Put $\mathbf{d} = \hat{\boldsymbol{\beta}} - \mathbf{m}_0$ and $\boldsymbol{\beta} = \mathbf{m}_0 + \mathbf{u}$. The quantity to minimize becomes $\text{SSE} + (\mathbf{u} - \mathbf{d})^\top \mathbf{K}(\mathbf{u} - \mathbf{d}) + \mathbf{u}^\top \mathbf{P}_0 \mathbf{u}$, whose minimum over $\mathbf{u}$, attained at $\mathbf{u} = (\mathbf{K} + \mathbf{P}_0)^{-1} \mathbf{K} \mathbf{d}$, is $\text{SSE} + \mathbf{d}^\top \{\mathbf{K} - \mathbf{K}(\mathbf{K} + \mathbf{P}_0)^{-1} \mathbf{K}\} \mathbf{d}$. By Theorem 1.6.5, $\mathbf{K} - \mathbf{K}(\mathbf{K} + \mathbf{P}_0)^{-1} \mathbf{K} = (\mathbf{K}^{-1} + \mathbf{P}_0^{-1})^{-1} = (\mathbf{V}_0 + (\mathbf{X}^\top \mathbf{X})^{-1})^{-1}$.
- (c) With $\mathbf{V}_0^{-1} = \mathbf{X}^\top \mathbf{X}/g$, $\mathbf{V}_n^{-1} = (1 + 1/g)\mathbf{X}^\top \mathbf{X}$, so $\mathbf{W} = \mathbf{V}_n \mathbf{V}_0^{-1} = \{g/(1 + g)\}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}/g = \mathbf{I}/(1 + g)$. (d) Substitute into Eq. 7.6.2.

■

Part (a) is the Bayesian counterpart of Gauss–Markov. The posterior mean is linear in $\mathbf{y}$, and it is biased unless $\mathbf{m}_0 = \boldsymbol{\beta}$: $\mathbf{E}(\mathbf{m}_n) = \mathbf{W}\mathbf{m}_0 + (\mathbf{I} - \mathbf{W})\boldsymbol{\beta}$. So the Gauss–Markov theorem does not apply to it, and it can have smaller mean squared error than least squares when the prior mean is close to the truth. Part (d) identifies ridge regression, whose mean squared error was studied in Exercise (B1), as a posterior mean. Chapter 27 develops the connection. When the prior is weak relative to the data, $\mathbf{W}$ is close to $\mathbf{0}$ and the posterior mean is close to $\hat{\boldsymbol{\beta}}$. When it is strong, $\mathbf{W}$ is close to $\mathbf{I}$ and the data barely move it.

In general $\mathbf{W}$ is a full matrix, not a multiple of the identity. Shrinkage then happens along the eigenvectors of the problem, not coordinate by coordinate. A single coefficient's posterior mean need not lie between its prior mean and its least squares estimate, as the next example shows. Only the *g*-prior of part (c) shrinks every coordinate by the same factor. Its prior covariance copies the shape of the sampling covariance of $\hat{\boldsymbol{\beta}}$, and that makes it convenient for model comparison. Zellner (1986) introduced it.

7.6.4 The flat-prior limit

What happens when the prior carries no information? Letting $\mathbf{V}_0^{-1} \rightarrow \mathbf{0}$ in Eq. 7.6.2 gives $\mathbf{m}_n \rightarrow \hat{\boldsymbol{\beta}}$ when $\mathbf{X}$ has full rank. The shape parameter needs more care. The NIG density Eq. 7.6.1 carries the factor $(\sigma^2)^{-(a_0 + p/2)-1}$, so letting only $a_0, b_0 \rightarrow 0$ leads to the prior $(\sigma^2)^{-p/2-1}$, with $a_n = n/2$ and t marginals on n degrees of freedom rather than $n - p$. The improper prior $\pi(\boldsymbol{\beta}, \sigma^2) \propto 1/\sigma^2$, uniform in $\boldsymbol{\beta}$ and in $\log \sigma$, is the limit $\mathbf{V}_0^{-1} \rightarrow \mathbf{0}$, $a_0 \rightarrow -p/2$, $b_0 \rightarrow 0$. A negative shape $a_0 = -p/2$ lies outside the proper NIG family, so this prior cannot be reached inside Theorem 7.6.2 and has to be handled directly. The result is clean.

Corollary 7.6.4 — The flat prior reproduces least squares

Let $\text{rank}(\mathbf{X}) = p < n$, let $\text{SSE} > 0$, and take the improper prior $\pi(\boldsymbol{\beta}, \sigma^2) \propto 1/\sigma^2$. Then the posterior is proper and equals $\text{NIG}(\hat{\boldsymbol{\beta}}, (\mathbf{X}^\top \mathbf{X})^{-1}, (n-p)/2, \text{SSE}/2)$. In particular, for every $\boldsymbol{\lambda} \neq \mathbf{0}$,

$$\frac{\boldsymbol{\lambda}^\top \boldsymbol{\beta} - \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}}{s \sqrt{\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}}} \mid \mathbf{y} \sim t(n-p),$$

so the $1 - \alpha$ equal-tailed credible interval for $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is the t interval [Eq. 7.5.1](#).

Proof. By [Eq. 6.3.1](#), $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \text{SSE} + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})$. So prior times likelihood is proportional to

$$\begin{aligned} & (\sigma^2)^{-1-n/2} \exp\left\{-\frac{\text{SSE} + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})}{2\sigma^2}\right\} \\ &= (\sigma^2)^{-(a+p/2)-1} \exp\left\{-\frac{2b + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})}{2\sigma^2}\right\}, \end{aligned}$$

with $a = (n-p)/2 > 0$ and $b = \text{SSE}/2 > 0$. This is proportional to the density [Eq. 7.6.1](#) of $\text{NIG}(\hat{\boldsymbol{\beta}}, (\mathbf{X}^\top \mathbf{X})^{-1}, a, b)$, which is integrable. So the posterior is proper and equal to it. By [Lemma 7.6.1](#) with $2a = n-p$ and $b/a = \text{SSE}/(n-p) = s^2$, the stated ratio has the $t(n-p)$ distribution. ■

The frequentist and Bayesian statements now read alike, but they say different things. The t interval of [Section 7.5](#) has probability $1 - \alpha$ of covering the fixed $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ *before the data are seen*. The credible interval has posterior probability $1 - \alpha$ of containing the random $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ *given the observed data*. That the two coincide is a special property of this model and this prior. It is one reason why $1/\sigma^2$ is regarded as the natural “noninformative” prior for the normal linear model.

7.6.5 A worked example

Example. (A prior for the stack loss plant)

Return to the stack loss regression of [Example \(Stack loss\)](#). Suppose an engineer, before seeing these 21 days, believed from the chemistry of the process that stack loss should rise roughly one for one with air flow and with cooling water temperature, and should not depend on acid concentration. This belief is hypothetical and is used only for illustration. We encode it as

$$\mathbf{m}_0 = (0, 1, 1, 0)^\top, \quad \mathbf{V}_0 = \text{diag}(100^2, 0.1^2, 0.2^2, 0.1^2), \quad a_0 = 3, \quad b_0 = 18.$$

So the prior standard deviations of the slopes are 0.1σ , 0.2σ and 0.1σ , the intercept is left essentially free, and the prior mean of σ^2 is $b_0/(a_0 - 1) = 9$.

The posterior has $a_n = 13.5$, so the marginal posteriors are t with 27 degrees of freedom, and $b_n = 112.523$. The posterior means of the coefficients are

$$\mathbf{m}_n = (-41.1490, 0.7616, 1.1606, -0.1371)^\top,$$

against the least squares estimates $(-39.9197, 0.7156, 1.2953, -0.1521)$. Each slope has moved toward its prior mean. The intercept has moved too, though its prior is vague, because it is strongly correlated with the slopes when the regressors are not centred. The 95% credible interval for the air-flow coefficient, using the $t(27)$ quantile 2.052, is $(0.550, 0.973)$, narrower than the least squares t interval $(0.431, 1.000)$, because the prior adds information. For water temperature and acid concentration the intervals are $(0.594, 1.727)$ and $(-0.392, 0.118)$. The posterior mean of σ^2 is 9.002, almost exactly the prior mean 9, although $s^2 = 10.519$. The two parts of Eq. 7.6.3 explain why. The lack of fit at the posterior mean is $\|\mathbf{y} - \mathbf{X}\mathbf{m}_n\|^2 = 180.667$, and on its own, with b_0 , it would give $(b_0 + 180.667/2)/(a_n - 1) = 8.667$: the shape $a_n = a_0 + n/2$ counts all n observations, while the residuals have spent some of them on β . The prior-data conflict term $(\mathbf{m}_n - \mathbf{m}_0)^\top \mathbf{V}_0^{-1}(\mathbf{m}_n - \mathbf{m}_0) = 8.378$ adds to the scale and makes up the difference, so the data barely move σ^2 from where the prior put it.

Figure 7.6.1(a) shows the three marginal densities for the air-flow coefficient. Panel (b) traces the posterior means of the slopes as the prior precision of the slopes is multiplied by a factor τ , from 10^{-3} (essentially least squares) to 10^3 (essentially the prior mean). The water-temperature coefficient does not move monotonically. It overshoots its prior mean of 1, down to 0.946, before returning. Matrix shrinkage acts on combinations of coefficients, and a single coordinate can pass beyond its prior mean on the way.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.stackloss.load_pandas().data
y = data["STACKLOSS"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["AIRFLOW", "WATERTEMP", "ACIDCONC"]]])
n, p = X.shape
XtX, Xty = X.T @ X, X.T @ y
beta_hat = np.linalg.solve(XtX, Xty)
sse = np.sum((y - X @ beta_hat) ** 2)

def nig_posterior(m0, V0, a0, b0):
    """Conjugate update of the normal-inverse-gamma prior NIG(m0, V0, a0, b0)."""
    P0 = np.linalg.inv(V0) # prior precision (per unit sigma^2)
    Vn = np.linalg.inv(P0 + XtX)
    mn = Vn @ (P0 @ m0 + Xty)
    an = a0 + n / 2
    bn = b0 + (y @ y + m0 @ P0 @ m0 - mn @ (P0 + XtX) @ mn) / 2
    return mn, Vn, an, bn

# prior: vague intercept; slopes centred on 1, 1, 0 with sd 0.1, 0.2, 0.1 in units
# of sigma
m0 = np.array([0.0, 1.0, 1.0, 0.0])
V0 = np.diag([100.0**2, 0.1**2, 0.2**2, 0.1**2])
a0, b0 = 3.0, 18.0 # prior mean of sigma^2 is b0/(a0-1) = 9
mn, Vn, an, bn = nig_posterior(m0, V0, a0, b0)
scale = np.sqrt(np.diag(Vn) * bn / an) # marginal posterior: t with 2
# a_n d.f.
q = stats.t.ppf(0.975, 2 * an)
for j, name in enumerate(["intercept", "air flow", "water temp", "acid conc"]):
    print(f"{name:10s} LS {beta_hat[j]:8.4f} posterior mean {mn[j]:8.4f} ")
```

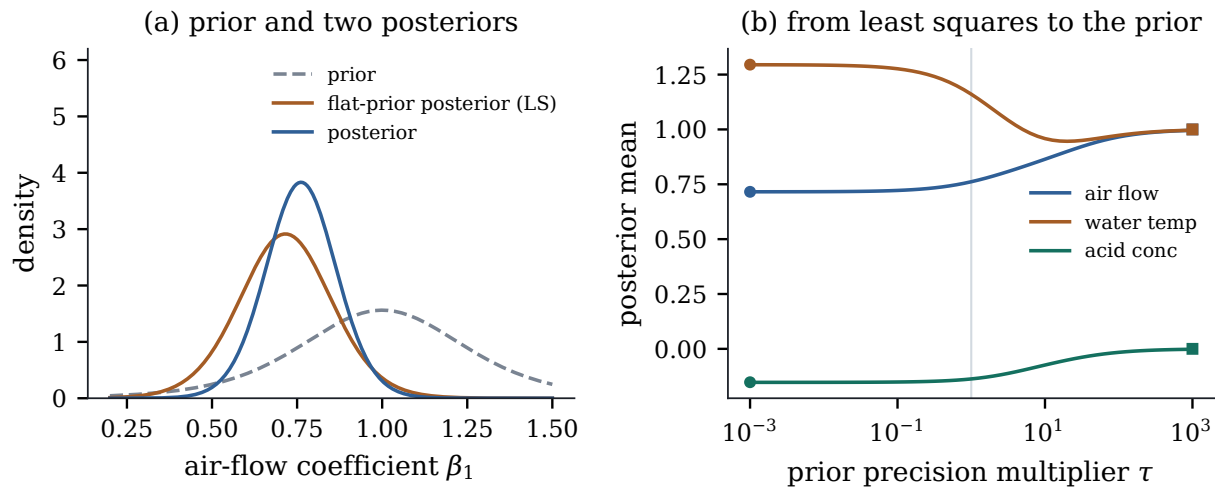


Figure 7.6.1: Conjugate analysis of the stack loss regression. (a) Marginal prior, flat-prior posterior (which is the least squares t distribution) and posterior for the air-flow coefficient. (b) Posterior means of the three slopes as the prior precision of the slopes is scaled by τ . Circles mark least squares, squares mark the prior means, and the grey line marks the prior of the example ($\tau = 1$).

```
f"95% interval ({mn[j] - q * scale[j]:.3f}, {mn[j] + q * scale[j]:.3f})"
print(f"posterior mean of sigma^2 = {bn / (an - 1):.3f}")
```

The script behind this example also checks the theorem numerically. The log posterior computed from Eq. 7.6.2 differs from log prior plus log-likelihood by the same constant at 50 random parameter values. The two forms of b_n agree. A Monte Carlo sample drawn from σ^2 and then $\beta \mid \sigma^2$ reproduces the marginal t quantiles. And a nearly flat prior reproduces least squares and the t interval.

Remark (How the Bayesian answer relates to the optimality results). Seen from the frequentist side, the posterior mean is a linear estimator with bias $\mathbf{W}(\mathbf{m}_0 - \beta)$ and covariance $\sigma^2(\mathbf{I} - \mathbf{W})(\mathbf{X}^\top \mathbf{X})^{-1}(\mathbf{I} - \mathbf{W})^\top$, which is smaller than that of $\hat{\beta}$ in the non-negative definite order. Whether its mean squared error is smaller depends on how far $\mathbf{m}_0$ is from β , in the metric that $\mathbf{W}$ defines. Averaged over the prior, the posterior mean is the estimator with the smallest expected squared error, because a conditional mean minimizes mean squared error. Different criteria single out different estimators. The Gauss–Markov theorem, the Lehmann–Scheffé theorem and the conjugate posterior (Theorem 7.6.2) answer three different questions, and in the flat-prior limit their answers agree.

7.6.6 Exercises

7.6.6.1 A. Check your understanding

Exercise A1.

Show that the $\text{IG}(a, b)$ distribution has mean $b/(a - 1)$ for $a > 1$ and mode $b/(a + 1)$. What are the mean and mode of $\sigma^2 \mid \mathbf{y}$ under the flat prior of Corollary 7.6.4, and how do they compare with s^2 and with the MLE of Theorem 7.3.1?

Exercise A2.

Suppose σ^2 is known and the prior is $\beta \sim N_p(\mathbf{m}_0, \sigma^2 \mathbf{V}_0)$. Show that the posterior is $N_p(\mathbf{m}_n, \sigma^2 \mathbf{V}_n)$ with $\mathbf{m}_n$ and $\mathbf{V}_n$ as in Eq. 7.6.2.

7.6.6.2 B. Practice**Exercise B1.**

Under the g -prior of Proposition 7.6.3(c) with $\mathbf{m}_0 = \mathbf{0}$, show that $b_n = b_0 + \frac{1}{2}\{\text{SSE} + \hat{\beta}^\top \mathbf{X}^\top \mathbf{X} \hat{\beta} / (1 + g)\}$. Show that as $g \rightarrow \infty$ the posterior mean tends to $\hat{\beta}$, and that for finite g the frequentist mean squared error of $\mathbf{m}_n$ is $\{g^2 \sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1} + \|\beta\|^2\} / (1 + g)^2$.

Solution. By Proposition 7.6.3(b), with $\mathbf{V}_0 + (\mathbf{X}^\top \mathbf{X})^{-1} = (1 + g)(\mathbf{X}^\top \mathbf{X})^{-1}$, the quadratic term is $\hat{\beta}^\top \mathbf{X}^\top \mathbf{X} \hat{\beta} / (1 + g)$. As $g \rightarrow \infty$, $\mathbf{m}_n = g\hat{\beta} / (1 + g) \rightarrow \hat{\beta}$. For the mean squared error, $\mathbf{m}_n - \beta = \{g(\hat{\beta} - \beta) - \beta\} / (1 + g)$, and $\hat{\beta} - \beta$ has mean zero and covariance $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$. So the expected squared norm is $\{g^2 \sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1} + \|\beta\|^2\} / (1 + g)^2$. It is smaller than $\sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1}$, the value for $\hat{\beta}$, iff $\|\beta\|^2 < (1 + 2g)\sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1}$.

Exercise B2.

Let $\mathbf{X}$ be rank deficient, $\mathbf{m}_0 = \mathbf{0}$ and $\mathbf{V}_0 = \tau^{-1} \mathbf{I}$. For $\mathbf{v} \in \mathcal{N}(\mathbf{X})$, show that $\mathbf{v}^\top \mathbf{m}_n = 0$ and $\mathbf{v}^\top \mathbf{V}_n \mathbf{v} = \|\mathbf{v}\|^2 / \tau$. So the conditional posterior of $\mathbf{v}^\top \beta$ given σ^2 equals its prior: the data do not update nonidentified directions. What happens to $\lambda^\top \mathbf{m}_n$ for $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ as $\tau \rightarrow 0$?

Solution. $\mathbf{V}_n^{-1} \mathbf{v} = (\tau \mathbf{I} + \mathbf{X}^\top \mathbf{X}) \mathbf{v} = \tau \mathbf{v}$, so $\mathbf{V}_n \mathbf{v} = \mathbf{v} / \tau$. Then $\mathbf{v}^\top \mathbf{m}_n = \mathbf{v}^\top \mathbf{V}_n \mathbf{X}^\top \mathbf{y} = (\mathbf{V}_n \mathbf{v})^\top \mathbf{X}^\top \mathbf{y} = (\mathbf{X} \mathbf{v})^\top \mathbf{y} / \tau = 0$, and $\mathbf{v}^\top \mathbf{V}_n \mathbf{v} = \|\mathbf{v}\|^2 / \tau$. These are the prior mean and variance (per unit σ^2) of $\mathbf{v}^\top \beta$. For the second question, $\mathbf{m}_n = (\mathbf{X}^\top \mathbf{X} + \tau \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$ converges as $\tau \rightarrow 0$ to the minimum-norm least squares estimate $\mathbf{X}^+ \mathbf{y}$ (Proposition 6.10.1). This follows from the spectral decomposition of $\mathbf{X}^\top \mathbf{X}$: eigenvectors with eigenvalue $d > 0$ get the factor $1/(d + \tau) \rightarrow 1/d$, and those with eigenvalue 0 get zero. Hence $\lambda^\top \mathbf{m}_n \rightarrow \lambda^\top \mathbf{X}^+ \mathbf{y} = \lambda^\top \hat{\beta}$, the least squares estimate of the estimable function.

7.6.6.3 C. Going deeper**Exercise C1.**

Under the posterior of Theorem 7.6.2, let $Y_0 = \mathbf{x}_0^\top \beta + \varepsilon_0$ be a future observation, with $\varepsilon_0 \sim N(0, \sigma^2)$ independent of everything given σ^2 . Show that the posterior predictive distribution of Y_0 is $t_{2a_n}(\mathbf{x}_0^\top \mathbf{m}_n, (b_n/a_n)(1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0))$. Compare it with the frequentist prediction interval of Exercise (B2) under the flat prior.

Solution. Given σ^2 and $\mathbf{y}$, $\mathbf{x}_0^\top \beta \sim N(\mathbf{x}_0^\top \mathbf{m}_n, \sigma^2 \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0)$, and ε_0 is independent of it. So $Y_0 \mid \sigma^2, \mathbf{y} \sim N(\mathbf{x}_0^\top \mathbf{m}_n, \sigma^2(1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0))$. Thus $(Y_0, \sigma^2) \mid \mathbf{y}$ is normal-inverse-gamma with $p = 1$, mean $\mathbf{x}_0^\top \mathbf{m}_n$ and $\mathbf{V} = 1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0$, and Lemma 7.6.1 gives the t_{2a_n} law. Under the flat prior, $\mathbf{m}_n = \hat{\beta}$, $\mathbf{V}_n = (\mathbf{X}^\top \mathbf{X})^{-1}$, $2a_n = n - p$ and $b_n/a_n = s^2$. The equal-tailed predictive interval is then exactly the prediction interval of Exercise (B2).

Exercise C2.

Split the data into two batches $(\mathbf{y}_1, \mathbf{X}_1)$ and $(\mathbf{y}_2, \mathbf{X}_2)$. Show that updating the prior with the first batch, and then using that posterior as the prior for the second batch, gives the same posterior as a single update with all the data. Which quantities in [Eq. 7.6.2](#) make this obvious?

7.7 Summary and Notes

7.7.1 Summary

Idea.

1. A linear estimator $\mathbf{a}^\top \mathbf{Y}$ is unbiased for $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ iff $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$. Such estimators exist iff $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$, the same condition under which least squares estimates $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ uniquely. Their coefficient vectors form the flat $\mathbf{M}\boldsymbol{\rho} + \mathcal{C}(\mathbf{X})^\perp$ ([Proposition 7.1.1](#) and [Proposition 7.1.2](#)).
2. **Gauss–Markov.** Under $E(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, in any rank, $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ is the unique BLUE of every estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$. For any LUE, $\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) + \sigma^2 \|(\mathbf{I} - \mathbf{M})\mathbf{a}\|^2$. In matrix form, $\text{Cov}(\mathbf{A}\mathbf{Y}) - \text{Cov}(\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{A}(\mathbf{I} - \mathbf{M})\mathbf{A}^\top \succeq \mathbf{0}$ ([Theorem 7.2.1](#) and [Corollary 7.2.2](#)). With $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$, generalized least squares is the BLUE ([Corollary 7.2.3](#)).
3. The theorem does not cover nonlinear estimators (the midrange beats the mean under uniform errors, [Example \(Uniform errors and the midrange\)](#)) or biased ones (ridge, [Exercise \(B1\)](#)). Orthogonal columns minimize coefficient variances ([Proposition 7.2.4](#)).
4. Under normality, the MLE of $\boldsymbol{\beta}$ is least squares and the MLE of σ^2 is SSE/n , biased by the factor $(n - r)/n$ and inconsistent when r grows with n ([Theorem 7.3.1](#) and [Example \(Many means, one variance\)](#)). The profile likelihood of an estimable function is a decreasing function of t^2 ([Proposition 7.3.3](#)).
5. $(\mathbf{M}\mathbf{Y}, \text{SSE})$ is complete and sufficient. By Lehmann–Scheffé, $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ and s^2 are the UMVUEs, with no linearity restriction ([Proposition 7.4.3](#) and [Theorem 7.4.6](#)). Least squares attains the Cramér–Rao bound for $\boldsymbol{\beta}$, and s^2 has efficiency $(n - p)/n$ ([Theorem 7.4.8](#)).
6. Under normality, $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\Lambda}^\top \boldsymbol{\beta}, \sigma^2 \boldsymbol{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\Lambda})$, $\text{SSE}/\sigma^2 \sim \chi^2(n - r)$, the two are independent, and standardized estimable functions are $t(n - r)$ ([Theorem 7.5.1](#), [Corollary 7.5.2](#) and [Corollary 7.5.3](#)). Without normality the variance of SSE depends on the kurtosis ([Proposition 7.5.4](#)).
7. The normal-inverse-gamma prior is conjugate. The posterior mean is a precision-weighted combination of the prior mean and $\hat{\boldsymbol{\beta}}$, and the flat prior $1/\sigma^2$ reproduces least squares and the t intervals ([Theorem 7.6.2](#), [Proposition 7.6.3](#) and [Corollary 7.6.4](#)).

7.7.2 Notes and sources

Gauss and Markov. Gauss (1823) proved that least squares gives the linear unbiased combination of observations with the smallest variance, without assuming normal errors. It was his second justification of the method, after the normal-theory argument of 1809. Markov's lectures on probability (1900 and later editions) restated the result, and Neyman (1934) attached Markov's name to it. The history is traced by Plackett (1949, 1972) and Stigler (1986). The projection proof of [Section 7.2](#) splits the coefficient vector of a competing estimator by Pythagoras, so that its excess variance is $\sigma^2 \|(\mathbf{I} - \mathbf{M})\mathbf{a}\|^2$. This is the proof of Seber and Lee (2003, section 3.2, Theorem 3.2), stated there for linear functions of the mean vector; we apply it to estimable functions in the rank-deficient case. Christensen (2020, section 2.3) argues instead through the covariance with $\boldsymbol{\rho}^\top \mathbf{M}\mathbf{Y}$. Rencher and Schaalje (2008, section 7.3) give the matrix form for full-rank models, and Agresti (2015, section 2.7) discusses its scope and its extension to generalized least squares. The generalized least squares version ([Corollary 7.2.3](#)) is due to Aitken (1935). The characterization of models in which ordinary least squares remains best ([Exercise \(C3\)](#)) is due to Zyskind (1967) and, in coordinate-free form, to Kruskal (1968).

What Gauss–Markov does not cover. That shrinkage can improve on unbiased estimation in three or more dimensions was discovered by Stein (1956), and James and Stein (1961) gave the explicit estimator. Hoerl and Kennard (1970) introduced ridge regression. The existence of a better ridge estimator ([Exercise \(B1\)](#)) is their theorem. Wald (1940) proposed the group-means slope estimator of [Example \(Three linear unbiased estimators of a slope\)](#) for regressions with errors in the regressor. The optimality of orthogonal weighing designs goes back to Hotelling (1944).

Likelihood. The inconsistency of the maximum likelihood estimate of a variance in the presence of many nuisance means is the example of Neyman and Scott (1948). Restricted maximum likelihood ([Exercise \(C2\)](#)) was developed by Patterson and Thompson (1971) for variance components. Christensen (2020, section 2.4), Seber and Lee (2003, section 3.5) and Rencher and Schaalje (2008, section 7.6) derive the normal-theory MLEs. The maximization through $\log u \leq u - 1$ used in [Theorem 7.3.1](#) follows Seber and Lee (2003, section 3.5).

Minimum variance. The Rao–Blackwell theorem is due to Rao (1945) and Blackwell (1947). Completeness and the uniqueness theorem are due to Lehmann and Scheffé (1950). The two results we quoted, the factorization criterion and the completeness of full-rank exponential families, are proved in Lehmann and Casella (1998) and Lehmann and Romano (2005). The same variance bound was found independently by Rao (1945) and Cramér (1946). Christensen (2020, section 2.5) outlines the minimum variance unbiased argument for the linear model by reparameterizing to full rank. Seber and Lee (2003, section 3.5) compare the Cramér–Rao bound with the variance of s^2 . The optimality of s^2 among nonnegative quadratic unbiased estimators ([Exercise \(C2\)](#)) is due to Atiqullah (1962). That the divisor $k + 2$ minimizes the mean squared error among multiples of SSE ([Proposition 7.3.2](#)) is due to Theil and Schweitzer (1961). The reflection argument of [Example \(Conditioning on the sufficient statistic\)](#) is our own illustration.

Sampling distributions. The distribution theory of [Section 7.5](#) follows Christensen (2020, section 2.6), Seber and Lee (2003, section 3.4) and Rencher and Schaalje (2008, section 7.6.3), with the rank-deficient forms throughout. The effect of kurtosis on inference about variances, and the relative robustness of inference about means, were emphasized by Box (1953).

Bayesian thread. The conjugate normal-inverse-gamma analysis goes back to Raiffa and Schlaifer (1961) and is developed for regression by Zellner (1971) and Box and Tiao (1973). O'Hagan and Forster (2004) give a modern account. Zellner (1986) introduced the g -prior.

Seber and Lee (2003, section 3.12), Christensen (2020, section 2.10) and Rencher and Schaalje (2008, chapter 11) treat Bayesian estimation in the linear model, and Fahrmeir, Kneib, Lang and Marx (2021, section 4.4) present the conjugate analysis as the starting point for richer priors. For an applied account of the Gauss–Markov conditions and what follows from them, see Sen and Srivastava (1990, section 2.5).

Data. The stack loss data are from Brownlee (1965) and are distributed with statsmodels.

7.7.3 References

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Chapter 8

Estimability, Identifiability and Indicator Variables

When the columns of the model matrix are linearly dependent, the data cannot tell some coefficient vectors apart, because different vectors produce the same mean. This chapter asks which questions about β still have answers. The answer is the same from every direction. A linear function $\lambda^\top \beta$ can be estimated without bias exactly when it is determined by the mean vector $\mathbf{X}\beta$, exactly when λ lies in the row space of $\mathbf{X}$, and exactly when every least squares solution gives it the same value. The rest of the chapter uses this criterion. Reparameterizations and side conditions turn out to be two ways of choosing coordinates, and neither can change what is estimable. Factors enter a regression through indicator variables, and each coding of a factor amounts to a choice of which linear functions of the cell means the coefficients report. Contrasts are the estimable functions of a factor, and orthogonal sets of contrasts split the factor's sum of squares into separate pieces.

What you need. [Chapter 1](#) (rank, generalized inverses and the consistency of linear systems, especially [Theorem 1.4.1](#) and [Corollary 1.4.2](#)), [Chapter 2](#) (means and covariances of linear functions, [Theorem 2.2.1](#)), [Chapter 4](#) (for the distribution of contrast sums of squares under normality, [Theorem 4.4.4](#)), [Chapter 5](#) (the linear model, [Definition 5.3.1](#)), and [Chapter 6](#), above all [Section 6.2](#) and [Section 6.4](#), where rank-deficient least squares and the invariance of $\lambda^\top \hat{\beta}$ for $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ ([Theorem 6.4.4](#)) first appeared. A few remarks use the Gauss–Markov theorem of [Chapter 7](#).

8.0.1 Roadmap

- [Identifiability](#)
- [Estimable functions](#)
- [Characterizations of estimability](#)
- [Reparameterization and side conditions](#)
- [Indicator variables and the coding of a factor](#)
- [Several factors, interactions and covariates](#)
- [Contrasts](#)
- [Summary and notes](#)

8.1 Identifiability

Section 6.4 ended with a table of four coefficient vectors that disagreed in every entry, some even in sign, and still produced the same fitted values. Nothing had gone wrong with the computation. The model matrix had linearly dependent columns, and so the least squares problem had a whole affine subspace of solutions. This chapter asks what can be learned about β in that situation. We begin with a question that comes before estimation and does not involve data at all. Suppose we knew the distribution of $\mathbf{Y}$ exactly, as if we had infinitely many replications. Would we then know β ? If not, which features of β would we know?

8.1.1 What the distribution of the data can reveal

Throughout, $\mathbf{Y} = \mathbf{X}\beta + \varepsilon$ is the linear model of Definition 5.3.1: $\mathbf{X}$ is a fixed $n \times p$ matrix of rank r , and $E(\varepsilon) = \mathbf{0}$. We do not assume $r = p$. The coefficient vector enters the model only through the product $\mathbf{X}\beta$. Suppose the distribution of the error vector ε is allowed to depend on other parameters, such as σ^2 , but not on β . Then the distribution of $\mathbf{Y}$ is the distribution of ε shifted by the vector $\mathbf{X}\beta$. Two coefficient vectors with the same product $\mathbf{X}\beta$ therefore give the same distribution of $\mathbf{Y}$, with the other parameters held fixed. Conversely, the distribution of $\mathbf{Y}$ determines its mean $E(\mathbf{Y}) = \mathbf{X}\beta$. So the most we can learn about β from the distribution is the value of $\mathbf{X}\beta$. The mean vector

$$\mu = \mathbf{X}\beta \in \mathcal{C}(\mathbf{X})$$

is the object the data speak about. A value of β is one way of writing μ in coordinates.

This observation reduces the question to linear algebra. The map $\beta \mapsto \mathbf{X}\beta$ from $\mathbb{R}^p$ to $\mathbb{R}^n$ is linear. Two coefficient vectors have the same image iff their difference lies in the null space $\mathcal{N}(\mathbf{X})$. So

$$\{\mathbf{b} \in \mathbb{R}^p : \mathbf{X}\mathbf{b} = \mathbf{X}\beta\} = \beta + \mathcal{N}(\mathbf{X}), \quad (8.1.1)$$

an affine subspace of dimension $p - r$ by rank-nullity (Eq. 6.2.1). All vectors in Eq. 8.1.1 are observationally equivalent: no experiment with this model matrix can distinguish between them.

Definition 8.1.1 — Identifiability

In the linear model $E(\mathbf{Y}) = \mathbf{X}\beta$, $\beta \in \mathbb{R}^p$, the parameter β is **identifiable** if $\mathbf{X}\beta_1 = \mathbf{X}\beta_2$ implies $\beta_1 = \beta_2$. A function g defined on $\mathbb{R}^p$, with values in any set, is **identifiable** if $\mathbf{X}\beta_1 = \mathbf{X}\beta_2$ implies $g(\beta_1) = g(\beta_2)$.

The definition is phrased through the mean, not the whole distribution. By the argument above the two versions agree whenever the error distribution does not involve β , which is the situation throughout this book until the generalized linear models of Part VIII. There the mean is $h(\mathbf{X}\beta)$ for a known one-to-one function h , and the same definition applies unchanged. It says that $g(\beta)$ is a well-defined function of the point μ , whichever coordinates were used to describe that point.

Theorem 8.1.1 — Identifiable means “a function of the mean”

A function g on $\mathbb{R}^p$ is identifiable iff there is a function h on $\mathcal{C}(\mathbf{X})$ with

$$g(\beta) = h(\mathbf{X}\beta) \quad \text{for every } \beta \in \mathbb{R}^p.$$

Equivalently, g is constant on each affine subspace $\beta + \mathcal{N}(\mathbf{X})$.

Proof. If $g = h \circ \mathbf{X}$ and $\mathbf{X}\beta_1 = \mathbf{X}\beta_2$, then $g(\beta_1) = h(\mathbf{X}\beta_1) = h(\mathbf{X}\beta_2) = g(\beta_2)$. Conversely, let g be identifiable. For $\mu \in \mathcal{C}(\mathbf{X})$ choose any β with $\mathbf{X}\beta = \mu$ and put $h(\mu) = g(\beta)$. The value does not depend on the choice, because any two choices have the same image under $\mathbf{X}$. By construction $g(\beta) = h(\mathbf{X}\beta)$. The last sentence restates the definition using [Eq. 8.1.1](#). ■

The theorem makes identifiability a statement about information. A function that can be computed from μ is one that could, in principle, be estimated consistently, provided that μ itself can be estimated. A function that cannot be computed from μ is one about which the model is silent. Its value can be anything along the affine subspace, and more data from the same design cannot help.

8.1.2 When the parameter is identifiable**Proposition 8.1.2**

The following are equivalent: (a) β is identifiable; (b) $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$; (c) $\text{rank}(\mathbf{X}) = p$; (d) $\mathbf{X}^\top \mathbf{X}$ is nonsingular. In that case the function h of [Theorem 8.1.1](#) for $g(\beta) = \beta$ is $h(\mu) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mu$.

Proof. By [Eq. 8.1.1](#), β is identifiable iff every affine subspace $\beta + \mathcal{N}(\mathbf{X})$ is a single point, which is (b). (b) $\Leftrightarrow$ (c) is rank-nullity, and (c) $\Leftrightarrow$ (d) is [Corollary 6.2.4\(c\)](#). If $\mathbf{X}^\top \mathbf{X}$ is nonsingular and $\mu = \mathbf{X}\beta$, then $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mu = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}\beta = \beta$. ■

When $\text{rank}(\mathbf{X}) = r < p$ the parameter is not identifiable, but parts of it may be. The simplest parts are the individual coordinates.

Proposition 8.1.3 — Identifiability of a single coefficient

Write $\mathbf{x}_1, \dots, \mathbf{x}_p$ for the columns of $\mathbf{X}$. For each j the following are equivalent:

- (a) β_j is identifiable;
- (b) every $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ has $v_j = 0$;

- (c) $\mathbf{x}_j$ is not a linear combination of the other columns of $\mathbf{X}$;
- (d) deleting column j from $\mathbf{X}$ lowers the rank.

Proof. (a) $\Leftrightarrow$ (b): by Eq. 8.1.1, the j th coordinates of the vectors observationally equivalent to $\boldsymbol{\beta}$ are $\beta_j + v_j$, $\mathbf{v} \in \mathcal{N}(\mathbf{X})$. They all equal β_j iff $v_j = 0$ for every null vector. (b) $\Leftrightarrow$ (c): a relation $\mathbf{x}_j = \sum_{k \neq j} a_k \mathbf{x}_k$ is the same as a null vector with j th entry -1 , and any null vector with $v_j \neq 0$ can be rescaled to have $v_j = -1$. (c) $\Leftrightarrow$ (d): the column space loses a dimension when $\mathbf{x}_j$ is removed iff $\mathbf{x}_j$ is not in the span of the remaining columns. ■

Part (c) gives a practical test. A coefficient is lost exactly when its regressor is *aliased*, that is, when it can be rebuilt from the others. In a model that is only partly identifiable, some coefficients can still be perfectly well identified. If $\mathbf{X} = [\mathbf{X}_1, \mathbf{x}_j]$ and $\mathbf{x}_j \notin \mathcal{C}(\mathbf{X}_1)$, then β_j is identifiable however badly $\mathbf{X}_1$ is parameterized. A typical case is a numeric covariate added to an overparameterized factor model: the covariate's coefficient is identifiable as long as the covariate is not constant within every level of the factor (Section 8.6).

8.1.3 Where rank deficiency comes from

Linear dependence among the columns of $\mathbf{X}$ arises in three quite different ways, and it helps to distinguish them.

By construction. A factor with g levels entered as an intercept together with all g indicator columns gives $g+1$ columns spanning a g -dimensional space (Example (A rank-deficient layout)). Such *overparameterized* models are written down deliberately, because they treat the levels symmetrically. The dependence is harmless once it is understood, and most of this chapter is about understanding it.

By the logic of the variables. Some sets of variables satisfy an identity that holds for every observation. Examples are shares of a whole that add to one, a total entered together with all its parts, and, the best-known case, age, period and birth cohort.

By the design. Two regressors that are logically unrelated may be exactly collinear in the sample at hand. A factor level may never be observed. The design may have fewer distinct rows than parameters. In each case another data set with the same model could identify the parameter, but this one cannot.

Identifiability is therefore a property of the model matrix, and through it of the design, not of the population being studied. The next example shows the second kind of dependence.

Example. (Age, period and cohort)

A survey is fielded every five years from 2000 to 2020, and its respondents are grouped in eleven five-year age bands from 20 to 70. For each of the 55 combinations we have one average response. A demographer wants to separate three linear trends: in age (people change as they grow older), in period (everyone changes with the times), and in birth cohort (generations differ). With age, period and cohort centred at 45, 2010 and 1965, the model is

$$E(y) = \beta_0 + \beta_A \text{age} + \beta_P \text{period} + \beta_C \text{cohort}.$$

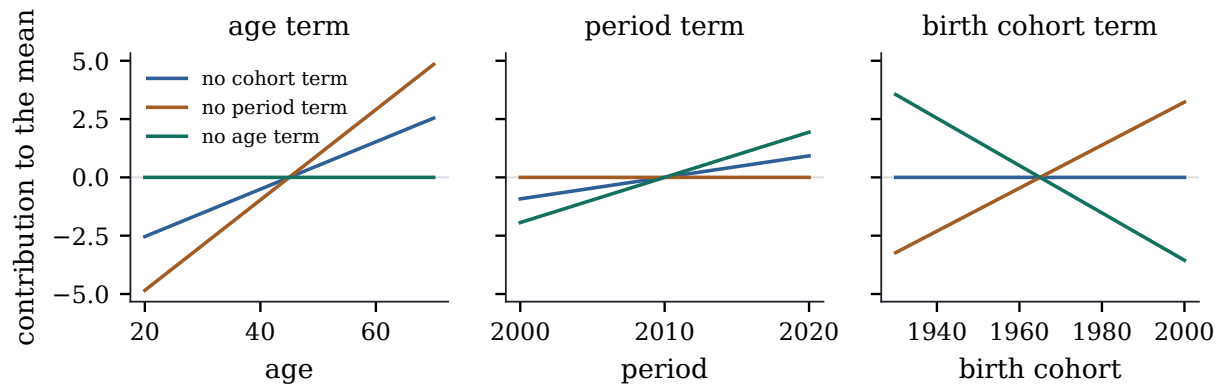


Figure 8.1.1: Three equally good fits of the age–period–cohort model. Each panel shows the contribution of one term to the fitted mean under the three fits of [Example \(Age, period and cohort\)](#). The sum of the three contributions is the same for every respondent, so the data cannot choose between the fits.

But cohort is period minus age, for every respondent. The four columns satisfy age – period + cohort = $\mathbf{0}$, so $\mathbf{v} = (0, 1, -1, 1)^\top$ spans $\mathcal{N}(\mathbf{X})$ and $\text{rank}(\mathbf{X}) = 3$. By [Proposition 8.1.3](#), none of the three slopes is identifiable.

The listing fits the model three times, each time dropping one of the three columns. All three fits have residual sum of squares 157.00, because they have the same column space. They tell three different stories:

Fit	age slope	period slope	cohort slope
no cohort term	0.102	0.092	0.000
no period term	0.194	0.000	0.092
no age term	0.000	0.194	-0.102

One analyst reports a steady ageing effect and no generational change. Another reports a strong ageing effect and a positive cohort trend. A third reports no ageing at all, a strong secular trend, and younger generations scoring lower. The data were in fact simulated with slopes 0.30, -0.10 and 0.20. That vector lies on the parallel line $\boldsymbol{\beta} + t\mathbf{v}$ of equally valid truths, and the data cannot prefer it. Only the identifiable combinations are recovered: the fitted 0.194 and 0.092 estimate the true 0.20 and 0.10.

What the three fits share are the linear functions orthogonal to $\mathbf{v} = (0, 1, -1, 1)^\top$:

$$\beta_A + \beta_P = 0.194, \quad \beta_P + \beta_C = 0.092, \quad \beta_A - \beta_C = 0.102,$$

in every row of the table, and β_0 . Any statement about the separate trends requires information from outside these data, for instance a constraint justified by subject knowledge, and the conclusions then rest on that constraint ([Exercise \(C1\)](#)). [Figure 8.1.1](#) shows how each fit divides the same fitted surface among the three terms.

```
import numpy as np

ages = np.arange(20, 75, 5)           # 11 age bands: 20, 25, ..., 70
periods = np.arange(2000, 2025, 5)   # 5 survey waves: 2000, ..., 2020
A, P = np.meshgrid(ages, periods, indexing="ij")
age, period = A.ravel() - 45.0, P.ravel() - 2010.0 # centred at 45 and 2010
```

```
cohort = period - age                                     # birth year minus 1965, exactly
X = np.column_stack([np.ones(age.size), age, period, cohort])
print("n =", X.shape[0], " p =", X.shape[1], " rank =", np.linalg.matrix_rank(X))
print("X @ (0, 1, -1, 1) =", np.abs(X @ [0, 1, -1, 1]).max())
```

```
rng = np.random.default_rng(8)
truth = np.array([50.0, 0.30, -0.10, 0.20])
y = X @ truth + rng.normal(scale=1.5, size=age.size)

def fit_without(j):
    """Least squares after deleting column j (one of age, period, cohort)."""
    keep = [k for k in range(4) if k != j]
    b = np.zeros(4)
    b[keep] = np.linalg.lstsq(X[:, keep], y, rcond=None)[0]
    return b

stories = {"no cohort term": fit_without(3),
          "no period term": fit_without(2),
          "no age term": fit_without(1)}
for name, b in stories.items():
    sse = np.sum((y - X @ b) ** 2)
    print(f"{name:15s} b = {np.round(b, 3)}   SSE = {sse:.3f}",
          f" age+period = {b[1] + b[2]:.3f}   period+cohort = {b[2] + b[3]:.3f}")
```

8.1.4 Linear and nonlinear identifiable functions

In the example the identifiable linear functions were exactly those whose coefficient vectors are orthogonal to the null space. That is true in general.

Proposition 8.1.4

A linear function $\lambda^\top \beta$ is identifiable iff $\lambda \perp \mathcal{N}(\mathbf{X})$, that is, iff $\lambda \in \mathcal{C}(\mathbf{X}^\top)$. The identifiable linear functions form a subspace of dimension $r = \text{rank}(\mathbf{X})$.

Proof. By [Theorem 8.1.1](#), $\lambda^\top \beta$ is identifiable iff $\lambda^\top (\beta + \mathbf{v}) = \lambda^\top \beta$ for all β and all $\mathbf{v} \in \mathcal{N}(\mathbf{X})$, that is, iff $\lambda^\top \mathbf{v} = 0$ for all $\mathbf{v} \in \mathcal{N}(\mathbf{X})$. By [Corollary 6.2.4\(d\)](#), $\mathcal{N}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{X}^\top)$, which has dimension $\text{rank}(\mathbf{X}^\top) = r$. ■

Identifiable functions need not be linear. If $\lambda_1^\top \beta$ and $\lambda_2^\top \beta$ are identifiable, so are their ratio (where defined), their product, and $\max\{\lambda_1^\top \beta, 0\}$. Every function of μ is identifiable, for example $\|\mu\|^2$ or the largest mean. For instance, in a quadratic regression $E(y) = \beta_0 + \beta_1 x + \beta_2 x^2$ with a full-rank design, the location $-\beta_1/(2\beta_2)$ of the turning point is identifiable. It is not linear, and Chapter 12 has to estimate it with care. The next section shows that among the *linear* functions, identifiability coincides with a much more concrete property, the existence of a linear unbiased estimator.

Idea. The data determine the mean vector $\mu = \mathbf{X}\beta$ and nothing else about β . A function of β is identifiable exactly when it is a function of μ . A *linear* function $\lambda^\top \beta$ is identifiable exactly when λ lies in the row space $\mathcal{C}(\mathbf{X}^\top)$, the orthogonal complement of

the invisible directions $\mathcal{N}(\mathbf{X})$.

8.1.5 Exercises

8.1.5.1 A. Check your understanding

Exercise A1.

In [Example \(Age, period and cohort\)](#), show directly that β_0 , $\beta_A + \beta_P$ and $\beta_P + \beta_C$ are identifiable, that every identifiable linear function is a combination of these three, and that $\beta_A - \beta_C$ is one of them. Is $\beta_A + \beta_C$ identifiable?

Exercise A2.

A household's spending is split into food, housing and other, with shares $s_1 + s_2 + s_3 = 1$. A regression of savings on an intercept and all three shares is proposed. Find $\mathcal{N}(\mathbf{X})$ (assuming the shares vary enough that there is no other relation among the columns), decide which coefficients are identifiable, and describe two standard ways to make the model identifiable.

8.1.5.2 B. Practice

Exercise B1.

Let $\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_g]$ be the indicator matrix of a factor and $\mathbf{x}$ a numeric covariate, and consider $\mathbf{X} = [\mathbf{1}, \mathbf{Z}, \mathbf{x}]$. Show that the slope of $\mathbf{x}$ is identifiable iff $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$, that is, iff $\mathbf{x}$ is not constant within every level. Show that no intercept-type parameter β_0 or α_k is identifiable.

Solution. The columns $\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_g$ satisfy $\mathbf{1} - \sum_k \mathbf{z}_k = \mathbf{0}$, so they are dependent, and $\mathcal{C}([\mathbf{1}, \mathbf{Z}]) = \mathcal{C}(\mathbf{Z})$. By [Proposition 8.1.3\(c\)](#), the slope is identifiable iff $\mathbf{x}$ is not a combination of $\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_g$, that is, iff $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$. A vector lies in $\mathcal{C}(\mathbf{Z})$ iff it is constant within each level. For the other coefficients, $\mathbf{v} = (1, -1, \dots, -1, 0)^\top$ is a null vector with nonzero entries in positions $0, 1, \dots, g$, so by [Proposition 8.1.3\(b\)](#) none of $\beta_0, \alpha_1, \dots, \alpha_g$ is identifiable.

Exercise B2.

Let $\mathbf{P}$ be the orthogonal projection onto $\mathcal{C}(\mathbf{X}^\top)$. Show that g is identifiable iff $g(\boldsymbol{\beta}) = g(\mathbf{P}\boldsymbol{\beta})$ for every $\boldsymbol{\beta}$. Deduce that the identifiable functions are exactly the functions of $\mathbf{P}\boldsymbol{\beta}$, and that $\mathbf{P}\boldsymbol{\beta}$ is the unique coefficient vector in $\mathcal{C}(\mathbf{X}^\top)$ representing the mean $\mathbf{X}\boldsymbol{\beta}$.

Exercise B3.

A design with model matrix $\mathbf{X}$ of rank $r < p$ is augmented by one more observation at a point $\mathbf{x}_0$. Show that the rank increases iff $\mathbf{x}_0 \notin \mathcal{C}(\mathbf{X}^\top)$. Show that, in the age-period-cohort design, no additional observation can make the slopes identifiable, and

explain why in words.

8.1.5.3 C. Going deeper

Exercise C1.

Replace the three linear terms of [Example \(Age, period and cohort\)](#) by three factors: an indicator for each age band, each period and each cohort (with an intercept). Show that the null space of the resulting model matrix has dimension 3 plus exactly one extra dimension. Identify the extra null vector: it assigns the linear sequences $a - \bar{a}$, $-(t - \bar{t})$ and $c - \bar{c}$ to the age, period and cohort effects. Deduce that every second difference of age effects, such as $\alpha_{a+1} - 2\alpha_a + \alpha_{a-1}$, is identifiable, while no linear trend in any single one of the three factors is.

Solution. Each factor's indicators sum to $\mathbf{1}$, which gives the usual three null vectors (intercept minus all indicators of one factor). The identity age – period + cohort = const holds for every observation. Writing the age of an observation as $\sum_a a z_a^A$, and similarly for period and cohort, it becomes a relation $\sum_a a z_a^A - \sum_t t z_t^P + \sum_c c z_c^C - \kappa \mathbf{1} = \mathbf{0}$ among the columns. Subtracting suitable multiples of the three standard relations centres the three sequences, and this gives the stated vector. It is not a combination of the other three, because those are constant on each factor's effects and this one is linear and nonconstant. To see that there are no more, check that the matrix has rank $1 + (A - 1) + (P - 1) + (C - 1) - 1$. This rank check can be done directly or numerically for the design of [Example \(Age, period and cohort\)](#). A function $\sum_a d_a \alpha_a$ is identifiable iff $\mathbf{d}$ is orthogonal to all four null vectors, that is, iff $\sum_a d_a = 0$ and $\sum_a d_a a = 0$. A second difference satisfies both. A linear trend $\sum_a (a - \bar{a}) \alpha_a$ fails the second condition.

8.2 Estimable functions

Identifiability asks whether a quantity is determined by the distribution of the data. Estimation asks for more. We want a statistic whose distribution is centred on the quantity, however the parameter is set. For linear functions of β , the natural candidates are linear functions of $\mathbf{Y}$, and the natural centring requirement is unbiasedness. This leads to the classical notion of an estimable function.

8.2.1 Linear unbiased estimators

Here and below the model is $E(\mathbf{Y}) = \mathbf{X}\beta$ with $\mathbf{X}$ an $n \times p$ matrix of rank r ; no assumption on $\text{Cov}(\mathbf{Y})$ is needed until variances are computed.

Definition 8.2.1 — Estimable function

A linear function $\lambda^\top \beta$, $\lambda \in \mathbb{R}^p$, is **estimable** if there is a vector $\mathbf{a} \in \mathbb{R}^n$ such that

$$E(\mathbf{a}^\top \mathbf{Y}) = \lambda^\top \beta \quad \text{for every } \beta \in \mathbb{R}^p.$$

A vector $\Lambda^\top \beta$ of linear functions, with Λ a $p \times q$ matrix, is estimable if each of its q components is. Equivalently, there is an $n \times q$ matrix A with $E(A^\top Y) = \Lambda^\top \beta$ for all β .

The statistic $a^\top Y$ is a **linear unbiased estimator** of $\lambda^\top \beta$. The requirement “for every β ” is essential. For a single fixed β , almost any statistic can be made unbiased by adding a constant. What the definition demands is one statistic that is centred correctly whatever the true coefficients are. Chapter 7 asked which linear unbiased estimator is best (Definition 7.1.1). Here we ask only whether there is one.

One might allow an *affine* estimator $a_0 + a^\top Y$. This gains nothing, as Chapter 7 already showed.

Lemma 8.2.1

The statistic $a_0 + a^\top Y$ is unbiased for $\lambda^\top \beta$, for every β , iff $a_0 = 0$ and $X^\top a = \lambda$.

Proof. This is Proposition 7.1.1, proved in Chapter 7 by evaluating the expectation $a_0 + a^\top X\beta$ at $\beta = 0$ and at the coordinate vectors. ■

So the linear unbiased estimators of $\lambda^\top \beta$ correspond exactly to the solutions a of the linear system $X^\top a = \lambda$. There is at least one iff the system is consistent, and when it is, the solutions form the affine subspace $a_0 + \mathcal{N}(X^\top)$ (Theorem 1.4.1).

8.2.2 Estimable is the same as identifiable

Theorem 8.2.2

For $\lambda \in \mathbb{R}^p$ the following are equivalent:

- (a) $\lambda^\top \beta$ is estimable;
- (b) $\lambda^\top \beta$ is identifiable;
- (c) $\lambda \in \mathcal{C}(X^\top)$, that is, $\lambda^\top = \rho^\top X$ for some $\rho \in \mathbb{R}^n$.

For a $p \times q$ matrix Λ , $\Lambda^\top \beta$ is estimable iff $\Lambda = X^\top P$ for some $n \times q$ matrix P , that is, iff $\mathcal{C}(\Lambda) \subseteq \mathcal{C}(X^\top)$.

Proof. (a) $\Leftrightarrow$ (c) is Lemma 8.2.1: an unbiased $a^\top Y$ exists iff $X^\top a = \lambda$ has a solution, which is the statement $\lambda \in \mathcal{C}(X^\top)$. (b) $\Leftrightarrow$ (c) is Proposition 8.1.4. The matrix version applies the vector version to each column of Λ and collects the vectors ρ as the columns of P . ■

The theorem connects two different ideas. Identifiability is about what the model could reveal with unlimited data, and it involves no estimator. Estimability is about a concrete property of a concrete statistic. For linear functions they are the same thing, and both reduce to membership

of λ in the row space of $\mathbf{X}$. The row space is spanned by the rows $\mathbf{x}_{(1)}^\top, \dots, \mathbf{x}_{(n)}^\top$, so the theorem says:

A linear function of β is estimable iff it is a linear combination of the expected responses $E(y_i) = \mathbf{x}_{(i)}^\top \beta$.

This reading is the most useful one in practice. To show that $\lambda^\top \beta$ is estimable, write it as a combination of the means of observations that were actually made. To show that it is not, find a direction $\mathbf{v}$ with $\mathbf{X}\mathbf{v} = \mathbf{0}$ and $\lambda^\top \mathbf{v} \neq 0$.

Corollary 8.2.3 — The estimable space

The set of λ for which $\lambda^\top \beta$ is estimable is the subspace $\mathcal{C}(\mathbf{X}^\top)$ of dimension $r = \text{rank}(\mathbf{X})$. In particular:

- (a) every expected response $\mathbf{x}_{(i)}^\top \beta$, and hence the whole mean vector $\mathbf{X}\beta$, is estimable;
- (b) every entry of $\mathbf{X}^\top \mathbf{X}\beta$ is estimable;
- (c) at most r estimable functions can be linearly independent, and a set of r linearly independent ones determines all the others as linear combinations;
- (d) if $r = p$, every linear function of β is estimable.

Proof. The first sentence is [Theorem 8.2.2](#). For (a) take $\rho = \mathbf{e}_i$, the i th coordinate vector. For (b), $\mathbf{X}^\top \mathbf{X}\beta = \mathbf{X}^\top (\mathbf{X}\beta)$ and the rows of $\mathbf{X}^\top \mathbf{X}$ lie in $\mathcal{C}(\mathbf{X}^\top)$. Part (c) is the dimension count, and (d) is the case $\mathcal{C}(\mathbf{X}^\top) = \mathbb{R}^p$. ■

8.2.3 The mean at a new point

A frequent use of a fitted model is to estimate the mean response at a setting $\mathbf{x}_0$ of the regressors, which may or may not have been observed. The quantity $\mathbf{x}_0^\top \beta$ is a linear function of β , so the theorem applies. It is estimable iff $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$. In a full-rank model this is automatic, and the only danger in predicting at a new point is extrapolation, meaning that the model may not hold there. In a rank-deficient model there is a sharper limit. Some settings are *algebraically* out of reach. The data contain no information about the mean there, even if the model is exactly true.

Example. (A fertilizer trial with a fixed ratio)

Ten plots receive nitrogen at 0, 20, 40, 60, 80 kg/ha (two plots each), and phosphorus is always applied at half the nitrogen rate, as in a commercial blend. The model

$$E(y) = \beta_0 + \beta_1 N + \beta_2 P$$

is natural agronomically, but in this design $\mathbf{x}_P = \frac{1}{2}\mathbf{x}_N$, so $\mathbf{v} = (0, 1, -2)^\top$ spans $\mathcal{N}(\mathbf{X})$ and $\text{rank}(\mathbf{X}) = 2$. A linear function is estimable iff its coefficient vector satisfies $\lambda_1 - 2\lambda_2 = 0$. So:

- β_0 , the mean yield without fertilizer, is estimable. Plots with $N = P = 0$ were observed.

- The mean yield at $(N, P) = (40, 20)$ is estimable, since $40 - 2 \cdot 20 = 0$, and so is the mean at any dose of the blend, observed or not, such as $(50, 25)$.
- The mean yield at $(40, 0)$, nitrogen alone, is not estimable, since $40 - 0 \neq 0$.
- Neither β_1 nor β_2 is estimable, but $\beta_1 + \frac{1}{2}\beta_2$, the gain per kilogram of nitrogen *when it comes with its phosphorus*, is.

The listing computes three least squares solutions: the minimum-norm one, the one that omits P , and the one that omits N . They give intercepts 2.9579, 2.9579 and 2.9579, all equal as they must be. The pairs of slopes $(0.0296, 0.0148)$, $(0.0370, 0.0000)$ and $(0.0000, 0.0740)$ all disagree. At the blend dose $(40, 20)$ all three predict 4.439 t/ha. At the nitrogen-only dose $(40, 0)$ they predict 4.143, 4.439 and 2.958. The third value is just $\hat{\beta}_0$, because that solution puts the whole fertilizer effect on phosphorus. Each of these is a legitimate least squares answer, and none of them estimates anything.

The two kinds of prediction differ in kind, not in degree. Nothing in this experiment says what nitrogen does without phosphorus. A larger experiment of the same design would not help. What is needed is at least one plot off the line $P = N/2$ ([Exercise \(B3\)](#)).

```
import numpy as np

N = np.array([0, 0, 20, 20, 40, 40, 60, 60, 80, 80], dtype=float) # kg/ha nitrogen
P = N / 2 # phosphorus,
    always half
X = np.column_stack([np.ones(N.size), N, P])
rng = np.random.default_rng(80)
y = 3.0 + 0.020 * N + 0.030 * P + rng.normal(scale=0.25, size=N.size) # yield, t/
    ha
print("rank(X) =", np.linalg.matrix_rank(X), " null vector (0, 1, -2):", X @ [0, 1,
    -2])
```

```
b_min = np.linalg.pinv(X) @ y # minimum-norm
    solution
b_noP = np.append(np.linalg.lstsq(X[:, :2], y, rcond=None)[0], 0.0) # drop P
b_noN = np.insert(np.linalg.lstsq(X[:, [0, 2]], y, rcond=None)[0], 1, 0.0) # drop N
new_points = {"N=40, P=20 (on the line)": [1, 40, 20],
    "N=40, P=0 (off the line)": [1, 40, 0]}
for name, x0 in new_points.items():
    print(name, [round(float(np.dot(x0, b)), 3) for b in (b_min, b_noP, b_noN)])
```

8.2.4 Many unbiased estimators, one least squares estimator

An estimable function usually has many linear unbiased estimators. In [Example \(A fertilizer trial with a fixed ratio\)](#), the mean at the blend dose $(40, 20)$ is estimated without bias by the average of the two plots that received exactly that dose, $\mathbf{a}_1^\top \mathbf{Y}$ with $\mathbf{a}_1$ putting weight $\frac{1}{2}$ on each. It is also estimated without bias by the value of any least squares fit at that dose. If $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, the first estimator has variance $\sigma^2 \|\mathbf{a}_1\|^2 = 0.500 \sigma^2$, while the least squares estimator has variance $0.100 \sigma^2$, five times smaller, because it uses all ten plots.

The least squares estimator can be written in the same form. If $\mathbf{a}$ is any solution of $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$,

then [Theorem 6.4.4](#) gives

$$\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{a}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{a}^\top \mathbf{M} \mathbf{y} = (\mathbf{M} \mathbf{a})^\top \mathbf{y}, \quad (8.2.1)$$

for every least squares solution $\hat{\boldsymbol{\beta}}$. So the least squares estimator is the linear unbiased estimator whose coefficient vector is the projection $\mathbf{M} \mathbf{a}$ of any one coefficient vector onto $\mathcal{C}(\mathbf{X})$. It is unbiased, because $\mathbf{X}^\top \mathbf{M} \mathbf{a} = \mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$. Every solution of $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$ has the form $\mathbf{M} \mathbf{a} + \mathbf{w}$ with $\mathbf{w} \in \mathcal{N}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})^\perp$, and Pythagoras gives $\|\mathbf{M} \mathbf{a} + \mathbf{w}\|^2 = \|\mathbf{M} \mathbf{a}\|^2 + \|\mathbf{w}\|^2$. This is the Gauss–Markov theorem in its simplest form: among linear unbiased estimators of an estimable function, least squares has the smallest variance when $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$. [Chapter 7](#) states and proves it in general ([Theorem 7.2.1](#)).

8.2.5 What estimability does not depend on

Three remarks keep the concept in its place.

It does not depend on the covariance. Estimability involves only $E(\mathbf{Y}) = \mathbf{X} \boldsymbol{\beta}$, so it is the same whether the errors are uncorrelated, correlated, heteroscedastic or normal, provided $\text{Cov}(\mathbf{Y})$ is free of $\boldsymbol{\beta}$. Generalized least squares in Part VII has the same estimable functions as ordinary least squares.

It is a property of the design, not the response. Whether $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is estimable can be decided before any response is measured, from $\mathbf{X}$ alone. This makes estimability a planning tool. If a comparison matters, the design must contain the rows that make it estimable.

It is a statement about linear functions. A turning point $-\beta_1/(2\beta_2)$ or a quotient of two estimable quantities is not “estimable” in the technical sense, and usually has no unbiased estimator of any kind. Yet when it is identifiable it is a perfectly sensible target, to be estimated by other means. The line that must not be crossed is identifiability, not linearity. In this chapter “nonestimable” is used only for linear functions, and for them it means the same as “not identifiable”.

Warning. Software fitted to a rank-deficient model reports a number for every coefficient and every requested combination, estimable or not. For a nonestimable function the number is an artefact of the particular solution the software chose, as the three predictions at $(40, 0)$ show. It comes with a standard error that is equally arbitrary ([Section 8.3](#)).

8.2.6 Exercises

8.2.6.1 A. Check your understanding

Exercise A1.

In [Example \(A fertilizer trial with a fixed ratio\)](#), decide which of the following are estimable: $\beta_1 - 2\beta_2$; $2\beta_1 + \beta_2$; the mean yield at $(N, P) = (100, 50)$; the mean yield at $(0, 10)$; the difference between the mean yields at $(60, 30)$ and $(20, 10)$. For the estimable ones give a vector $\mathbf{a}$ with $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$.

Exercise A2.

In simple linear regression $E(y_i) = \beta_0 + \beta_1 x_i$, show that β_1 is estimable iff the x_i are not all equal. If all $x_i = c$, which linear functions are estimable, and what is their least squares estimator?

8.2.6.2 B. Practice**Exercise B1.**

Let $\lambda^\top \beta$ be estimable and let $\mathbf{a}^* = \mathbf{M}\mathbf{a}$ as in Eq. 8.2.1. Show that $\mathbf{a}^*$ does not depend on which solution $\mathbf{a}$ of $\mathbf{X}^\top \mathbf{a} = \lambda$ is used, that $\mathbf{a}^* = \mathbf{X}\mathbf{G}\lambda$ for any generalized inverse $\mathbf{G}$ of $\mathbf{X}^\top \mathbf{X}$, and that $\|\mathbf{a}^*\|^2 = \lambda^\top \mathbf{G}\lambda$.

Solution. Two solutions differ by a vector of $\mathcal{N}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})^\perp$, which $\mathbf{M}$ sends to $\mathbf{0}$, so $\mathbf{M}\mathbf{a}$ is the same for both. With $\mathbf{M} = \mathbf{X}\mathbf{G}\mathbf{X}^\top$ (Theorem 6.3.7), $\mathbf{M}\mathbf{a} = \mathbf{X}\mathbf{G}\mathbf{X}^\top \mathbf{a} = \mathbf{X}\mathbf{G}\lambda$. Finally, $\|\mathbf{M}\mathbf{a}\|^2 = \mathbf{a}^\top \mathbf{M}\mathbf{a} = \mathbf{a}^\top \mathbf{X}\mathbf{G}\mathbf{X}^\top \mathbf{a} = \lambda^\top \mathbf{G}\lambda$.

Exercise B2.

Show that $\Lambda^\top \beta$ is estimable iff $\Lambda^\top \mathbf{b}_1 = \Lambda^\top \mathbf{b}_2$ whenever $\mathbf{X}\mathbf{b}_1 = \mathbf{X}\mathbf{b}_2$, and that in that case $\Lambda^\top \hat{\beta} = \mathbf{P}^\top \mathbf{M}\mathbf{y}$ for any $\mathbf{P}$ with $\mathbf{X}^\top \mathbf{P} = \Lambda$. Show also that $\mathbf{P}^\top \mathbf{M}$ does not depend on the choice of $\mathbf{P}$.

Exercise B3.

A study will compare three treatments, A , B and C , in the additive model $E(y) = \mu + \tau_{\text{trt}} + \delta_{\text{site}}$. Two sites are available, and each site can host two of the three treatments. Which allocations make every difference $\tau_s - \tau_t$ estimable? Show that putting $\{A, B\}$ at both sites leaves $\tau_A - \tau_C$ nonestimable, while $\{A, B\}$ at one site and $\{B, C\}$ at the other makes all three differences estimable.

8.2.6.3 C. Going deeper**Exercise C1.**

Suppose the distribution of ε does not depend on β . Show that if some statistic $f(\mathbf{Y})$, linear or not, satisfies $E f(\mathbf{Y}) = g(\beta)$ for every β , then g is identifiable. Conclude that no statistic whatever is unbiased for a nonestimable linear function.

Solution. The distribution of $\mathbf{Y} = \mathbf{X}\beta + \varepsilon$ depends on β only through $\mathbf{X}\beta$. If $\mathbf{X}\beta_1 = \mathbf{X}\beta_2$, then $\mathbf{Y}$ has the same distribution under both, so $g(\beta_1) = E_{\beta_1} f(\mathbf{Y}) = E_{\beta_2} f(\mathbf{Y}) = g(\beta_2)$. A linear function with an unbiased estimator is therefore identifiable, hence estimable by Theorem 8.2.2.

8.3 Characterizations of estimability

[Theorem 8.2.2](#) reduced estimability to a single condition, $\lambda \in \mathcal{C}(\mathbf{X}^\top)$. In practice that condition appears in several forms, depending on what is at hand. We may have the rows of $\mathbf{X}$, the matrix $\mathbf{X}^\top \mathbf{X}$ and one of its generalized inverses, a collection of least squares solutions, or an SVD. This section collects the equivalent forms in one theorem, derives the estimator of an estimable function with its variance, and then turns to the practical question of deciding estimability in floating-point arithmetic, where exact rank is not available.

8.3.1 Eight equivalent conditions

Theorem 8.3.1 — Characterizations of estimability

Let $\mathbf{X}$ be $n \times p$, let $\mathbf{G}$ denote a generalized inverse of $\mathbf{X}^\top \mathbf{X}$, and let $\lambda \in \mathbb{R}^p$. The following are equivalent:

- (a) $\lambda^\top \beta$ is estimable;
- (b) $\lambda \in \mathcal{C}(\mathbf{X}^\top)$: there is $\rho \in \mathbb{R}^n$ with $\lambda^\top = \rho^\top \mathbf{X}$;
- (c) $\lambda \perp \mathcal{N}(\mathbf{X})$: $\lambda^\top \mathbf{v} = 0$ whenever $\mathbf{X}\mathbf{v} = \mathbf{0}$;
- (d) $\lambda \in \mathcal{C}(\mathbf{X}^\top \mathbf{X})$: the system $\mathbf{X}^\top \mathbf{X} \mathbf{r} = \lambda$ has a solution;
- (e) $\lambda^\top \mathbf{G} \mathbf{X}^\top \mathbf{X} = \lambda^\top$ for some generalized inverse $\mathbf{G}$;
- (f) $\lambda^\top \mathbf{G} \mathbf{X}^\top \mathbf{X} = \lambda^\top$ for every generalized inverse $\mathbf{G}$;
- (g) for every $\mathbf{y}$, $\lambda^\top \mathbf{b}$ has the same value for all solutions $\mathbf{b}$ of the normal equations $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$ ([Theorem 6.4.2](#));
- (h) $\text{rank} \begin{pmatrix} \mathbf{X} \\ \lambda^\top \end{pmatrix} = \text{rank}(\mathbf{X})$.

Proof. (a) $\Leftrightarrow$ (b) is [Theorem 8.2.2](#). (b) $\Leftrightarrow$ (c) holds because $\mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$ ([Corollary 6.2.4\(d\)](#)). (b) $\Leftrightarrow$ (d) holds because $\mathcal{C}(\mathbf{X}^\top \mathbf{X}) = \mathcal{C}(\mathbf{X}^\top)$ ([Corollary 6.2.4\(b\)](#)). (d) $\Rightarrow$ (f): if $\lambda = \mathbf{X}^\top \mathbf{X} \mathbf{r}$, then for every generalized inverse $\lambda^\top \mathbf{G} \mathbf{X}^\top \mathbf{X} = \mathbf{r}^\top (\mathbf{X}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{X}) = \mathbf{r}^\top \mathbf{X}^\top \mathbf{X} = \lambda^\top$. (f) $\Rightarrow$ (e) holds because generalized inverses exist ([Proposition 6.3.5](#)). (e) $\Rightarrow$ (d): transposing $\lambda^\top \mathbf{G} \mathbf{X}^\top \mathbf{X} = \lambda^\top$ gives $\mathbf{X}^\top \mathbf{X} (\mathbf{G}^\top \lambda) = \lambda$, so $\mathbf{r} = \mathbf{G}^\top \lambda$ solves the system. (b) $\Leftrightarrow$ (g) is [Theorem 6.4.4](#). (b) $\Leftrightarrow$ (h): the rank of the stacked matrix is the dimension of its row space, $\mathcal{C}([\mathbf{X}^\top, \lambda])$. This contains $\mathcal{C}(\mathbf{X}^\top)$, and it has the same dimension iff it equals $\mathcal{C}(\mathbf{X}^\top)$, that is, iff $\lambda \in \mathcal{C}(\mathbf{X}^\top)$. ■

Conditions (e) and (f) are [Corollary 1.4.2](#) applied to the normal equations. Condition (g) is the one that matters for interpretation: an estimable function is one whose estimate does not depend on the arbitrary choice of solution. Condition (c) is the one to use when proving that something is *not* estimable, since a single null vector $\mathbf{v}$ with $\lambda^\top \mathbf{v} \neq 0$ settles the question. Condition (h) is the most direct computational test, but in floating point it inherits all the difficulties of deciding a rank.

8.3.2 The estimator and its variance

For an estimable function, every least squares solution produces the same estimate, and that estimate has a clean description.

Proposition 8.3.2 — Least squares estimation of estimable functions

Let $\lambda^\top \beta$ be estimable, with $\lambda = X^\top \rho = X^\top X r$, let $\hat{\beta}$ be any least squares solution and G any generalized inverse of $X^\top X$. Then:

- (a) $\lambda^\top \hat{\beta} = \rho^\top M y = r^\top X^\top y = \lambda^\top G X^\top y$;
- (b) $E(\lambda^\top \hat{\beta}) = \lambda^\top \beta$;
- (c) if $\text{Cov}(Y) = \sigma^2 I$, then

$$\text{Var}(\lambda^\top \hat{\beta}) = \sigma^2 \lambda^\top G \lambda = \sigma^2 r^\top \lambda = \sigma^2 \|M\rho\|^2,$$

the same for every G ;

- (d) more generally, if $\Lambda^\top \beta$ is estimable, then $\text{Cov}(\Lambda^\top \hat{\beta}) = \sigma^2 \Lambda^\top G \Lambda$ for every G .

Proof.

- (a) The first equality is [Theorem 6.4.4](#). For the second, the normal equations give $r^\top X^\top y = r^\top X^\top X \hat{\beta} = \lambda^\top \hat{\beta}$. The last is the second with the solution $\hat{\beta} = G X^\top y$ and $r = G^\top \lambda$ (see the proof of [Theorem 8.3.1](#)). (b)–(d) were proved in [Chapter 7](#): unbiasedness and $\text{Var}(\lambda^\top \hat{\beta}) = \sigma^2 \lambda^\top G \lambda$ are [Theorem 7.2.1\(a\)](#), and the matrix form $\text{Cov}(\Lambda^\top \hat{\beta}) = \sigma^2 \Lambda^\top G \Lambda$ is part (d) of the same theorem. Neither depends on G , because $M = X G X^\top$ for every G ([Theorem 6.3.7](#)). The second form in (c) follows from $\lambda^\top G \lambda = r^\top X^\top X G X^\top X r = r^\top X^\top X r = r^\top \lambda$. The third is $\rho^\top M \rho = \|M\rho\|^2$.

■

The formula $\sigma^2 \lambda^\top G \lambda$ is the rank-deficient version of the familiar $\sigma^2 \lambda^\top (X^\top X)^{-1} \lambda$ ([Theorem 5.5.1](#)). The generalized inverse is arbitrary, but the quadratic form in an estimable λ is not. The condition $\lambda \in \mathcal{C}(X^\top)$ under which [Chapter 7](#) proved optimality is exactly estimability, so the Gauss–Markov theorem and the normal theory ([Theorem 7.5.1](#)) apply to every estimable $\lambda^\top \hat{\beta}$ unchanged.

8.3.3 What a generalized inverse reports for a non-estimable function

Software that solves the normal equations with a particular generalized inverse will print $\lambda^\top G X^\top y$ and $\sigma^2 \lambda^\top G \lambda$ for any λ whatever. To see what these numbers mean when λ is not estimable, look at the matrix

$$H_G = G X^\top X. \quad (8.3.1)$$

Proposition 8.3.3 — The matrix $\mathbf{GX}^\top \mathbf{X}$

For any generalized inverse $\mathbf{G}$ of $\mathbf{X}^\top \mathbf{X}$, with $\mathbf{H}_\mathbf{G}$ as in Eq. 8.3.1:

- (a) $\mathbf{H}_\mathbf{G}$ is idempotent, $\mathcal{N}(\mathbf{H}_\mathbf{G}) = \mathcal{N}(\mathbf{X})$, $\text{rank}(\mathbf{H}_\mathbf{G}) = r$, and the rows of $\mathbf{H}_\mathbf{G}$ span $\mathcal{C}(\mathbf{X}^\top)$;
- (b) $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is estimable iff $\boldsymbol{\lambda}^\top \mathbf{H}_\mathbf{G} = \boldsymbol{\lambda}^\top$;
- (c) the solution $\hat{\boldsymbol{\beta}}_\mathbf{G} = \mathbf{GX}^\top \mathbf{Y}$ has $\mathbf{E}(\hat{\boldsymbol{\beta}}_\mathbf{G}) = \mathbf{H}_\mathbf{G} \boldsymbol{\beta}$. So for every $\boldsymbol{\lambda}$, $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_\mathbf{G}$ is unbiased for the estimable function $\boldsymbol{\lambda}^\top \mathbf{H}_\mathbf{G} \boldsymbol{\beta}$, which equals $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ iff $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is estimable;
- (d) for the Moore–Penrose inverse $\mathbf{G} = (\mathbf{X}^\top \mathbf{X})^+$, $\mathbf{H}_\mathbf{G}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X}^\top)$.

Proof.

- (a) $\mathbf{H}_\mathbf{G}^2 = \mathbf{G}(\mathbf{X}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{X}) = \mathbf{GX}^\top \mathbf{X} = \mathbf{H}_\mathbf{G}$. By Proposition 1.3.3(a), $\mathcal{N}(\mathbf{GX}^\top \mathbf{X}) = \mathcal{N}(\mathbf{X}^\top \mathbf{X})$, which is $\mathcal{N}(\mathbf{X})$ by Corollary 6.2.4(a). Rank-nullity gives rank r . The row space of $\mathbf{H}_\mathbf{G}$ is $\mathcal{C}(\mathbf{H}_\mathbf{G}^\top) = \mathcal{N}(\mathbf{H}_\mathbf{G})^\perp = \mathcal{N}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{X}^\top)$.
- (b) is Theorem 8.3.1(e),(f).
- (c) $\mathbf{E}(\mathbf{GX}^\top \mathbf{Y}) = \mathbf{GX}^\top \mathbf{X} \boldsymbol{\beta}$. The coefficient vector $\mathbf{H}_\mathbf{G}^\top \boldsymbol{\lambda}$ lies in the row space of $\mathbf{H}_\mathbf{G}$, which is $\mathcal{C}(\mathbf{X}^\top)$, so the function is estimable. It equals $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ for all $\boldsymbol{\beta}$ iff $\mathbf{H}_\mathbf{G}^\top \boldsymbol{\lambda} = \boldsymbol{\lambda}$, which is (b).
- (d) By the Penrose conditions (Definition 1.3.2), $(\mathbf{X}^\top \mathbf{X})^+ \mathbf{X}^\top \mathbf{X}$ is symmetric and idempotent. It is therefore an orthogonal projection (Theorem 6.3.3), and its null space is $\mathcal{N}(\mathbf{X})$ by (a). So it projects onto $\mathcal{N}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{X}^\top)$.

■

Part (c) answers the question. For a nonestimable $\boldsymbol{\lambda}$, the printed number is a perfectly good unbiased estimate, but of a *different* function, $\boldsymbol{\lambda}^\top \mathbf{H}_\mathbf{G} \boldsymbol{\beta}$, and which function depends on the software's choice of $\mathbf{G}$. The printed “variance” $\sigma^2 \boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}$ is the variance of that statistic when $\mathbf{G}$ is symmetric and reflexive, as the Moore–Penrose inverse is, but in general it need not be the variance of anything (Exercise (C1)). Part (d) gives the cleanest picture. With the Moore–Penrose inverse, $\boldsymbol{\beta} = \mathbf{H} \boldsymbol{\beta} + (\mathbf{I} - \mathbf{H}) \boldsymbol{\beta}$ splits the coefficient vector into its component in the estimable space, which the data determine, and its component in $\mathcal{N}(\mathbf{X})$, which they cannot see. The minimum-norm solution estimates the first component and sets the second to zero.

Example. (Checking estimability in a one-way layout)

Eighteen observations fall into four groups of sizes 4, 6, 3, 5, with $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3, \mathbf{z}_4]$ and $\boldsymbol{\beta} = (\mu, \alpha_1, \dots, \alpha_4)^\top$, so $\text{rank}(\mathbf{X}) = 4$ and $\mathcal{N}(\mathbf{X})$ is spanned by $(1, -1, -1, -1, -1)^\top$. By condition (c), $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is estimable iff $\lambda_0 = \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4$. The listing tests five vectors in two ways: condition (c) through the SVD of $\mathbf{X}$, and condition (e) with three generalized inverses (Moore–Penrose, the one that deletes the intercept row and column, and a non-symmetric one built as in Proposition 1.3.3(e)). The two tests always agree, and condition (e) holds for all three inverses or for none, as (f) requires.

$\lambda^\top \beta$	estimable	$\lambda^\top G \lambda$:		
		Moore–Penrose	drop intercept	non-symmetric
$\mu + \alpha_1$	yes	0.250	0.250	0.250
$\alpha_1 - \alpha_2$	yes	0.417	0.417	0.417
$\alpha_1 + \alpha_2 - 2\alpha_3$	yes	1.750	1.750	1.750
α_1	no	0.188	0.250	0.630
$\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4$	no	0.038	0.950	2.767

For the estimable rows the quadratic form is the same for all three and matches the direct computation: $\text{Var}(\bar{y}_1 - \bar{y}_2) = \sigma^2(\frac{1}{4} + \frac{1}{6})$, for instance. For α_1 the “variances” disagree. With the drop-intercept inverse, the solution is $\hat{\beta}_G = (0, \bar{y}_1, \dots, \bar{y}_4)^\top$, so the reported “ $\hat{\alpha}_1$ ” is $\bar{y}_1$, which estimates $\mu + \alpha_1$. Its reported variance $0.250 \sigma^2$ is the variance of $\bar{y}_1$, which is correct, but $\bar{y}_1$ estimates $\mu + \alpha_1$, not α_1 .

```
import numpy as np

groups = np.repeat([0, 1, 2, 3], [4, 6, 3, 5])          # four groups, sizes 4, 6, 3,
5
X = np.column_stack([np.ones(groups.size), np.eye(4)[groups]]) # [1, z1, z2, z3, z4
]
A = X.T @ X

def is_estimable(lam, X, tol=None):
    """lambda is estimable iff it is orthogonal to N(X); use the SVD of X."""
    U, s, Vt = np.linalg.svd(X)
    if tol is None:
        tol = max(X.shape) * np.finfo(float).eps * s[0]
    r = int(np.sum(s > tol))
    V0 = Vt[r:].T                                     # orthonormal basis of N(X)
    return np.linalg.norm(V0.T @ lam) <= 1e-8 * max(1.0, np.linalg.norm(lam))

G_mp = np.linalg.pinv(A)                             # Moore-Penrose inverse of X^T X
G_ref = np.zeros((5, 5))
G_ref[1:, 1:] = np.linalg.inv(A[1:, 1:])             # delete the intercept row and
column
rng = np.random.default_rng(3)
G_odd = G_mp + (np.eye(5) - G_mp @ A) @ rng.normal(size=(5, 5)) # non-symmetric g-
inverse

candidates = {"mu + alpha_1": [1, 1, 0, 0, 0],
              "alpha_1 - alpha_2": [0, 1, -1, 0, 0],
              "alpha_1 + alpha_2 - 2 alpha_3": [0, 1, 1, -2, 0],
              "alpha_1": [0, 1, 0, 0, 0],
              "alpha_1 + ... + alpha_4": [0, 1, 1, 1, 1]}
for name, lam in candidates.items():
    lam = np.array(lam, dtype=float)
    H_checks = [np.allclose(lam @ G @ A, lam) for G in (G_mp, G_ref, G_odd)]
    variances = [lam @ G @ lam for G in (G_mp, G_ref, G_odd)]
    print(f"{name:30s} SVD test: {is_estimable(lam, X)!s:5s}",
          f" lambda'GX'X = lambda': {H_checks}",
          f" lambda'G lambda: {np.round(variances, 3)}")
```

8.3.4 Deciding estimability in floating point

Every condition in [Theorem 8.3.1](#) is a statement of exact linear algebra. On a computer, the rank of $\mathbf{X}$ must be decided with a tolerance, and the answer to “is λ estimable?” inherits that tolerance. The most reliable test works with $\mathbf{X}$ itself, not with $\mathbf{X}^\top \mathbf{X}$. Compute the singular value decomposition $\mathbf{X} = \mathbf{U} \mathbf{D} \mathbf{V}^\top$. Declare the singular values below a threshold τ to be zero. With the usual choice $\tau = \max(n, p) \epsilon d_1$, where d_1 is the largest singular value and $\epsilon \approx 2.2 \times 10^{-16}$ is the machine precision, the columns $\mathbf{V}_0$ of $\mathbf{V}$ that belong to the discarded singular values form an orthonormal basis of the numerical null space. Then λ is declared estimable iff $\|\mathbf{V}_0^\top \lambda\|$ is negligible relative to $\|\lambda\|$. This is condition (c), and it is what the function `is_estimable` in the listing does. [Chapter 10](#) treats the SVD and its rank decisions in detail ([Theorem 10.4.1](#)).

Two things can go wrong, and both matter.

Exact dependence can be blurred by rounding. If a column is computed, say as a sum of other columns, rounding may leave a singular value of order 10^{-15} rather than zero. The tolerance exists to absorb this.

Near dependence is not dependence. If the columns are nearly but not exactly dependent, every λ is estimable in the exact sense, but some are estimable only with enormous variance. Estimability is then a matter of degree, not a yes-or-no property.

Example. (Estimable in principle, not in practice)

Take twelve equally spaced values t_i on $[-1, 1]$ and the model matrix with columns $\mathbf{1}$, $\mathbf{t}$ and $\mathbf{t} + \delta \mathbf{w}$, where $w_i = \cos(\pi t_i)$ and δ is a small perturbation. At $\delta = 0$ the last two columns coincide. Then $\beta_1 + \beta_2$ is estimable and β_1 is not. For $\delta > 0$ the matrix has full rank, so everything is estimable. But [Figure 8.3.1](#) shows how the variance of $\hat{\beta}_1$ grows as $\delta \rightarrow 0$: in proportion to $1/\delta^2$ (the fitted slope on the log-log scale is -2.00). At $\delta = 10^{-4}$, the smallest singular value of $\mathbf{X}$ is 1.8×10^{-4} , and $\text{Var}(\hat{\beta}_1) = 1.56 \times 10^7 \sigma^2$, while $\text{Var}(\hat{\beta}_1 + \hat{\beta}_2) = 0.212 \sigma^2$ is unaffected. The function that is estimable at $\delta = 0$ stays well estimated, and the one that is not becomes useless long before it becomes nonestimable.

The listing adds a warning about computation. At $\delta = 10^{-8}$, the SVD of $\mathbf{X}$ still sees three nonzero singular values, so β_1 is (barely) estimable. But the matrix $\mathbf{X}^\top \mathbf{X}$ has eigenvalues equal to the *squared* singular values. Its smallest eigenvalue is about 3×10^{-16} , below any sensible tolerance, and `pinv(X.T @ X)` silently treats $\mathbf{X}^\top \mathbf{X}$ as having rank 2. It then reports 0.0529 as the “variance” of $\hat{\beta}_1$, which is the Moore–Penrose quadratic form of a nonestimable function. The true variance, computed from the SVD of $\mathbf{X}$ itself, is more than 10^{16} times larger. Forming $\mathbf{X}^\top \mathbf{X}$ squares the condition number, as [Section 6.10](#) showed, and here that turns a nearly nonestimable function into a numerically nonestimable one.

```
t = np.linspace(-1, 1, 12)
w = np.cos(np.pi * t)                                     # a fixed direction to perturb
                        along
def near_design(delta):
    """Columns 1, t and t + delta*w: exactly rank deficient at delta = 0."""
    return np.column_stack([np.ones_like(t), t, t + delta * w])

for delta in [1.0, 1e-2, 1e-4, 1e-8, 0.0]:
    Xd = near_design(delta)
```

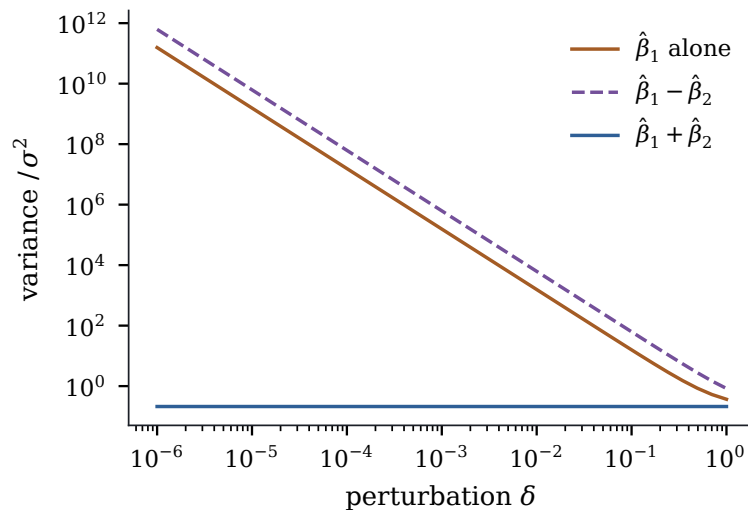


Figure 8.3.1: Variances of three linear functions of $\hat{\beta}$ in the design with columns $\mathbf{1}, \mathbf{t}, \mathbf{t} + \delta \mathbf{w}$. As $\delta \rightarrow 0$, the functions that are not estimable at $\delta = 0$ (β_1 alone and $\beta_1 - \beta_2$) have variances growing like δ^{-2} , while $\beta_1 + \beta_2$, which is estimable at $\delta = 0$, is unaffected.

```
s = np.linalg.svd(Xd, compute_uv=False)
G = np.linalg.pinv(Xd.T @ Xd)
v_sum = np.array([0, 1, 1]) @ G @ [0, 1, 1]      # beta_1 + beta_2
v_one = np.array([0, 1, 0]) @ G @ [0, 1, 0]      # beta_1 alone
print(f"delta={delta:7.0e}  smallest singular value {s[-1]:8.1e}",
      f" Var(b1+b2)/s2 = {v_sum:8.3f}  Var(b1)/s2 = {v_one:9.2e}",
      f" beta_1 estimable? {is_estimable(np.array([0., 1, 0]), Xd)}")
```

The practical rule has two parts. Decide estimability with the SVD of $\mathbf{X}$, never with $\mathbf{X}^\top \mathbf{X}$. Then look at the variance, not only the verdict. A function that passes the test with a variance of $10^7 \sigma^2$ has, for practical purposes, not been estimated. Collinearity, in Chapter 26, is the study of this middle ground.

8.3.5 Exercises

8.3.5.1 A. Check your understanding

Exercise A1.

In [Example \(Checking estimability in a one-way layout\)](#), decide which of the following are estimable, using condition (c): $2\mu + \alpha_1 + \alpha_2$; $\alpha_1 + \alpha_2 - \alpha_3 - \alpha_4$; μ ; $4\mu + \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4$. For each estimable one give its least squares estimator in terms of the group means and its variance.

8.3.5.2 B. Practice

Exercise B1.

In [Example \(Checking estimability in a one-way layout\)](#), compute $\mathbf{H}_G$ for the drop-intercept generalized inverse, and identify the estimable function $\boldsymbol{\lambda}^\top \mathbf{H}_G \boldsymbol{\beta}$ that the soft-

ware is really estimating when it reports “ $\hat{\alpha}_k$ ”. Do the same for the generalized inverse that deletes the row and column of α_4 .

Solution. With $\mathbf{G}$ holding $\text{diag}(1/n_1, \dots, 1/n_4)$ in its lower right block, $\mathbf{G}\mathbf{X}^\top\mathbf{X}$ has first row zero and row $k+1$ equal to $(1, \mathbf{e}_k^\top)$, because the row of $\mathbf{X}^\top\mathbf{X}$ for α_k is $n_k(1, \mathbf{e}_k^\top)$. So $E(\hat{\alpha}_k) = \mu + \alpha_k$: the reported “effect” estimates the group mean. If instead α_4 is deleted, the solution sets $\hat{\alpha}_4 = 0$, $\hat{\mu} = \bar{y}_4$ and $\hat{\alpha}_k = \bar{y}_k - \bar{y}_4$, so $E(\hat{\mu}) = \mu + \alpha_4$ and $E(\hat{\alpha}_k) = \alpha_k - \alpha_4$ for $k \leq 3$.

Exercise B2.

Let $\mathbf{\Lambda}^\top\boldsymbol{\beta}$ be estimable with $\mathbf{\Lambda}$ of full column rank q , and suppose $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$ with $\sigma^2 > 0$. Show that $\text{Cov}(\mathbf{\Lambda}^\top\hat{\boldsymbol{\beta}}) = \sigma^2\mathbf{\Lambda}^\top\mathbf{G}\mathbf{\Lambda}$ is positive definite, whichever generalized inverse is used. *Hint:* write $\mathbf{\Lambda} = \mathbf{X}^\top\mathbf{P}$ and show $\mathbf{M}\mathbf{P}$ has full column rank.

8.3.5.3 C. Going deeper

Exercise C1.

Let $\mathbf{\lambda}^\top\boldsymbol{\beta}$ be nonestimable. Show that as $\mathbf{G}$ ranges over all generalized inverses of $\mathbf{X}^\top\mathbf{X}$, the number $\mathbf{\lambda}^\top\mathbf{G}\boldsymbol{\lambda}$ takes *every* real value, including negative ones. So the “variance” that software attaches to a nonestimable function can be anything at all.

Solution. Let $\mathbf{A} = \mathbf{X}^\top\mathbf{X}$ and $\mathbf{G}_0$ be one generalized inverse. By Proposition 1.3.3(e), $\mathbf{G} = \mathbf{G}_0 + (\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{U}$ is a generalized inverse for every $\mathbf{U}$, and

$$\mathbf{\lambda}^\top\mathbf{G}\boldsymbol{\lambda} = \mathbf{\lambda}^\top\mathbf{G}_0\boldsymbol{\lambda} + \mathbf{u}^\top\mathbf{U}\boldsymbol{\lambda}, \quad \mathbf{u} = (\mathbf{I} - \mathbf{G}_0\mathbf{A})^\top\boldsymbol{\lambda}.$$

Because $\boldsymbol{\lambda}$ is not estimable, $\mathbf{\lambda}^\top\mathbf{G}_0\mathbf{A} \neq \mathbf{\lambda}^\top$ (by (a) $\Leftrightarrow$ (e) of Theorem 8.3.1), so $\mathbf{u} \neq \mathbf{0}$. Also $\boldsymbol{\lambda} \neq \mathbf{0}$, since $\mathbf{0}$ is in $\mathcal{C}(\mathbf{X}^\top)$. Take $\mathbf{U} = t\mathbf{u}\boldsymbol{\lambda}^\top/(\|\mathbf{u}\|^2\|\boldsymbol{\lambda}\|^2)$. Then $\mathbf{\lambda}^\top\mathbf{G}\boldsymbol{\lambda} = \mathbf{\lambda}^\top\mathbf{G}_0\boldsymbol{\lambda} + t$, which takes every real value as t varies.

Exercise C2.

Let $\mathbf{P}$ be the orthogonal projection onto $\mathcal{C}(\mathbf{X}^\top)$. Show that the minimum-norm least squares solution $\hat{\boldsymbol{\beta}}^+$ satisfies $E(\hat{\boldsymbol{\beta}}^+) = \mathbf{P}\boldsymbol{\beta}$, and that for any $\boldsymbol{\lambda}$, $\mathbf{\lambda}^\top\hat{\boldsymbol{\beta}}^+$ is the least squares estimator of the estimable function $(\mathbf{P}\boldsymbol{\lambda})^\top\boldsymbol{\beta}$. Interpret $\mathbf{P}\boldsymbol{\lambda}$ as the estimable function “closest” to $\mathbf{\lambda}^\top\boldsymbol{\beta}$.

8.4 Reparameterization and side conditions

A rank-deficient model can be handled in three ways. One can work with it as it stands, estimating only estimable functions, which is what the last two sections did. One can *reparameterize*, replacing $\mathbf{X}$ by a full-rank matrix with the same column space. Or one can impose *side conditions*, linear restrictions on $\boldsymbol{\beta}$ that select a single coefficient vector from each set $\boldsymbol{\beta} + \mathcal{N}(\mathbf{X})$. This section shows that the second and third are two descriptions of the same act: choosing

coordinates for the mean vector. Neither changes any estimable quantity. It also records what the most common software does by default.

8.4.1 Full-rank reparameterizations

By [Corollary 8.2.3](#), at most r estimable functions are linearly independent. A set of exactly r of them can serve as a new parameter vector.

Theorem 8.4.1 — Reparameterizing to full rank

Let $\text{rank}(\mathbf{X}) = r$. Let $\mathbf{U}$ be an $r \times p$ matrix of rank r whose rows lie in $\mathcal{C}(\mathbf{X}^\top)$, so that $\boldsymbol{\gamma} = \mathbf{U}\boldsymbol{\beta}$ is a vector of r linearly independent estimable functions, and put

$$\mathbf{Z} = \mathbf{X}\mathbf{U}^\top(\mathbf{U}\mathbf{U}^\top)^{-1}. \quad (8.4.1)$$

Then:

- (a) $\mathbf{X} = \mathbf{Z}\mathbf{U}$, $\mathbf{Z}$ has full column rank r , and $\mathcal{C}(\mathbf{Z}) = \mathcal{C}(\mathbf{X})$;
- (b) the models $\text{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$, $\boldsymbol{\beta} \in \mathbb{R}^p$, and $\text{E}(\mathbf{Y}) = \mathbf{Z}\boldsymbol{\gamma}$, $\boldsymbol{\gamma} \in \mathbb{R}^r$, describe the same set of mean vectors, with $\boldsymbol{\gamma} = \mathbf{U}\boldsymbol{\beta}$;
- (c) the least squares estimate in the new model is $\hat{\boldsymbol{\gamma}} = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{y} = \mathbf{U}\hat{\boldsymbol{\beta}}$ for every least squares solution $\hat{\boldsymbol{\beta}}$ of the old one;
- (d) $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is estimable iff $\boldsymbol{\lambda} = \mathbf{U}^\top \mathbf{c}$ for some $\mathbf{c} \in \mathbb{R}^r$, which is then unique, and in that case $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{c}^\top \hat{\boldsymbol{\gamma}}$;
- (e) conversely, if $\mathbf{Z}$ is any $n \times r$ matrix of full column rank with $\mathcal{C}(\mathbf{Z}) = \mathcal{C}(\mathbf{X})$, then $\mathbf{X} = \mathbf{Z}\mathbf{U}$ for a unique $\mathbf{U}$, and $\mathbf{U}$ has rank r and estimable rows.

Proof. Since $\mathcal{C}(\mathbf{U}^\top) \subseteq \mathcal{C}(\mathbf{X}^\top)$ and both have dimension r , $\mathcal{C}(\mathbf{U}^\top) = \mathcal{C}(\mathbf{X}^\top)$. The matrix $\mathbf{Q} = \mathbf{U}^\top(\mathbf{U}\mathbf{U}^\top)^{-1}\mathbf{U}$ is symmetric and idempotent with column space $\mathcal{C}(\mathbf{U}^\top)$, so it is the orthogonal projection onto $\mathcal{C}(\mathbf{X}^\top)$ ([Theorem 6.3.3](#)). Hence $\mathbf{I} - \mathbf{Q}$ projects onto $\mathcal{C}(\mathbf{X}^\top)^\perp = \mathcal{N}(\mathbf{X})$, and $\mathbf{X}(\mathbf{I} - \mathbf{Q}) = \mathbf{0}$, that is, $\mathbf{X} = \mathbf{X}\mathbf{Q} = \mathbf{Z}\mathbf{U}$. (a) Now $r = \text{rank}(\mathbf{X}) = \text{rank}(\mathbf{Z}\mathbf{U}) \leq \text{rank}(\mathbf{Z}) \leq r$, so $\mathbf{Z}$ has full column rank. $\mathcal{C}(\mathbf{Z}) \subseteq \mathcal{C}(\mathbf{X})$ because $\mathbf{Z} = \mathbf{X}(\cdot)$, and $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$ because $\mathbf{X} = \mathbf{Z}\mathbf{U}$. (b) $\mathbf{X}\boldsymbol{\beta} = \mathbf{Z}(\mathbf{U}\boldsymbol{\beta})$, and $\boldsymbol{\beta} \mapsto \mathbf{U}\boldsymbol{\beta}$ maps $\mathbb{R}^p$ onto $\mathbb{R}^r$ because $\mathbf{U}$ has full row rank. So both models describe $\mathcal{C}(\mathbf{X})$. (c) $\mathbf{Z}\hat{\boldsymbol{\gamma}} = \mathbf{M}\mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{Z}(\mathbf{U}\hat{\boldsymbol{\beta}})$ by [Theorem 6.4.1](#), and $\mathbf{Z}$ has full column rank. (d) $\boldsymbol{\lambda}$ is estimable iff $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{U}^\top)$. The vector $\mathbf{c}$ is unique because $\mathbf{U}^\top$ has full column rank, and $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{c}^\top \mathbf{U}\hat{\boldsymbol{\beta}} = \mathbf{c}^\top \hat{\boldsymbol{\gamma}}$. (e) Each column of $\mathbf{X}$ lies in $\mathcal{C}(\mathbf{Z})$, so $\mathbf{X} = \mathbf{Z}\mathbf{U}$ with $\mathbf{U} = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{X}$, unique because $\mathbf{Z}$ has full column rank. Then $\mathbf{U}\boldsymbol{\beta} = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top (\mathbf{X}\boldsymbol{\beta})$ is a linear function of $\mathbf{X}\boldsymbol{\beta}$, so each row of $\mathbf{U}$ is estimable. Finally $r = \text{rank}(\mathbf{X}) \leq \text{rank}(\mathbf{U}) \leq r$. ■

So reparameterization is the same thing as choosing a basis for the estimable space. [Theorem 6.7.1](#) of [Chapter 6](#) already showed that fitted values and residuals do not care which basis is chosen. Part (d) adds that estimable functions do not care either: each is a fixed linear combination of the new parameters, and its estimate is the same combination of their estimates.

Everything in the full-rank theory of [Chapter 5](#) now applies to $\hat{\gamma}$, including [Theorem 5.5.1](#) and the variance estimator of [Theorem 5.6.1](#), with p replaced by r .

For the one-way layout with $\beta = (\mu, \alpha_1, \dots, \alpha_g)^\top$, two choices of U stand out. The rows $(1, e_k^\top)$, $k = 1, \dots, g$, give $\gamma = (\mu + \alpha_1, \dots, \mu + \alpha_g)^\top$, the cell means, and Z is the indicator matrix. The rows $(1, e_1^\top)$ and $(0, e_k^\top - e_1^\top)$, $k = 2, \dots, g$, give the first group's mean and the differences from it. This is the reference coding of [Section 8.5](#).

8.4.2 Side conditions

The other classical device leaves X alone and adds equations. A **side condition** is a set of linear restrictions $Tb = 0$, with T a $q \times p$ matrix, imposed on the solutions of the normal equations. The aim is to single out one solution. Examples in the one-way layout are $\alpha_1 = 0$, $\sum_k \alpha_k = 0$ and $\mu = 0$. The theorem says exactly which restrictions achieve this.

Theorem 8.4.2 — Side conditions

Let X be $n \times p$ of rank r and T be $q \times p$. The following are equivalent:

- (a) for every $y \in \mathbb{R}^n$ there is exactly one b with $X^\top Xb = X^\top y$ and $Tb = 0$;
- (b) for every $\mu \in \mathcal{C}(X)$ there is exactly one b with $Xb = \mu$ and $Tb = 0$;
- (c) $\mathbb{R}^p = \mathcal{N}(X) \oplus \mathcal{N}(T)$;
- (d) $\text{rank} \begin{pmatrix} X \\ T \end{pmatrix} = p$ and $\mathcal{C}(T^\top) \cap \mathcal{C}(X^\top) = \{0\}$;
- (e) $\text{rank}(T) = p - r$, and no nonzero linear combination of the rows of T is estimable.

When these hold, $X^\top X + T^\top T$ is nonsingular, the solution in (a) is

$$\hat{\beta}_T = (X^\top X + T^\top T)^{-1} X^\top y, \quad (8.4.2)$$

and $(X^\top X + T^\top T)^{-1}$ is a generalized inverse of $X^\top X$.

Proof. (a) $\Leftrightarrow$ (b). The solutions of the normal equations for y are the b with $Xb = My$ ([Theorem 6.4.2](#)), and My runs over all of $\mathcal{C}(X)$ as y does.

(b) $\Leftrightarrow$ (c). The set $\{b : Xb = \mu\}$ is a coset $b_0 + \mathcal{N}(X)$, and every $b_0 \in \mathbb{R}^p$ arises (take $\mu = Xb_0$). So (b) says that every coset of $\mathcal{N}(X)$ meets $\mathcal{N}(T)$ in exactly one point. Some point $b_0 + v$ of the coset lies in $\mathcal{N}(T)$ iff $b_0 \in \mathcal{N}(T) + \mathcal{N}(X)$, so existence for every coset means $\mathbb{R}^p = \mathcal{N}(X) + \mathcal{N}(T)$. Two points of the same coset that both lie in $\mathcal{N}(T)$ differ by an element of $\mathcal{N}(X) \cap \mathcal{N}(T)$, and a nonzero element of that intersection added to one such point gives a second. So uniqueness means $\mathcal{N}(X) \cap \mathcal{N}(T) = \{0\}$. Together, these say the sum is direct.

(c) $\Leftrightarrow$ (d). By [Lemma 6.2.1](#) and [Lemma 6.2.5](#),

$$(\mathcal{N}(X) + \mathcal{N}(T))^\perp = \mathcal{C}(X^\top) \cap \mathcal{C}(T^\top), \quad (\mathcal{N}(X) \cap \mathcal{N}(T))^\perp = \mathcal{C}(X^\top) + \mathcal{C}(T^\top).$$

So $\mathcal{N}(X) + \mathcal{N}(T) = \mathbb{R}^p$ iff $\mathcal{C}(X^\top) \cap \mathcal{C}(T^\top) = \{0\}$, and $\mathcal{N}(X) \cap \mathcal{N}(T) = \{0\}$ iff $\mathcal{C}(X^\top) + \mathcal{C}(T^\top) = \mathbb{R}^p$. The last space is the row space of the stacked matrix, so this

holds iff that matrix has rank p .

(d) $\Leftrightarrow$ (e). A vector in $\mathcal{C}(\mathbf{T}^\top)$ is a combination of the rows of $\mathbf{T}$, and it is estimable iff it lies in $\mathcal{C}(\mathbf{X}^\top)$. So the second halves of (d) and (e) say the same thing. When they hold, the sum $\mathcal{C}(\mathbf{X}^\top) + \mathcal{C}(\mathbf{T}^\top)$ is direct, with dimension $r + \text{rank}(\mathbf{T})$. This equals p iff $\text{rank}(\mathbf{T}) = p - r$.

For the formula, suppose (d). If $(\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})\mathbf{v} = \mathbf{0}$, then $0 = \mathbf{v}^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} + \mathbf{v}^\top \mathbf{T}^\top \mathbf{T} \mathbf{v} = \|\mathbf{X} \mathbf{v}\|^2 + \|\mathbf{T} \mathbf{v}\|^2$, so $\mathbf{v}$ lies in $\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}) = \{\mathbf{0}\}$. Hence the matrix is nonsingular. The solution $\mathbf{b}$ of (a) satisfies $(\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})\mathbf{b} = \mathbf{X}^\top \mathbf{y} + \mathbf{T}^\top \mathbf{0}$, which gives Eq. 8.4.2. Finally, write $\mathbf{G}_\mathbf{T} = (\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})^{-1}$. For every $\mathbf{y}$, $\mathbf{G}_\mathbf{T} \mathbf{X}^\top \mathbf{y}$ solves the normal equations, so $\mathbf{X}^\top \mathbf{X} \mathbf{G}_\mathbf{T} \mathbf{X}^\top = \mathbf{X}^\top$. Multiplying on the right by $\mathbf{X}$ gives $\mathbf{X}^\top \mathbf{X} \mathbf{G}_\mathbf{T} \mathbf{X}^\top \mathbf{X} = \mathbf{X}^\top \mathbf{X}$. ■

Condition (e) is the familiar rule of thumb, made exact: a side condition must supply exactly as many restrictions as the rank deficiency, and none of them, singly or in combination, may be estimable. The reason for the second requirement is clear from (d). An estimable restriction constrains the mean vector itself, and so it changes the model instead of merely choosing coordinates.

Remark (Existence and uniqueness separately). The proof shows more than the theorem states. The condition $\mathcal{C}(\mathbf{T}^\top) \cap \mathcal{C}(\mathbf{X}^\top) = \{\mathbf{0}\}$ alone guarantees that the restricted normal equations always have *at least* one solution. The condition $\text{rank}(\frac{\mathbf{X}}{\mathbf{T}}) = p$ alone guarantees *at most* one. A constraint that is too weak, such as a single restriction when $p - r = 2$, leaves many solutions. One with an estimable part leaves none for most $\mathbf{y}$ (Exercise (C2)).

8.4.3 What the constrained coefficients mean

Under a side condition every coefficient becomes identifiable, and it acquires a meaning.

Proposition 8.4.3 — Constrained coefficients are estimable functions

Suppose $\mathbf{T}$ satisfies the conditions of Theorem 8.4.2, and let $\mathbf{P}_\mathbf{T} = (\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})^{-1} \mathbf{X}^\top \mathbf{X}$. For each β , $\mathbf{P}_\mathbf{T} \beta$ is the unique vector $\mathbf{b}$ with $\mathbf{X} \mathbf{b} = \mathbf{X} \beta$ and $\mathbf{T} \mathbf{b} = \mathbf{0}$. The matrix $\mathbf{P}_\mathbf{T}$ is idempotent, with range $\mathcal{N}(\mathbf{T})$ and null space $\mathcal{N}(\mathbf{X})$. Every row of $\mathbf{P}_\mathbf{T}$ lies in $\mathcal{C}(\mathbf{X}^\top)$. So each constrained coefficient $\mathbf{e}_j^\top \mathbf{P}_\mathbf{T} \beta$ is an estimable function of the original parameters, and $\hat{\beta}_\mathbf{T}$ estimates it without bias.

Proof. With $\mathbf{y} = \mathbf{X} \beta$, the vector $\mathbf{P}_\mathbf{T} \beta = \mathbf{G}_\mathbf{T} \mathbf{X}^\top (\mathbf{X} \beta)$ is the unique solution of Theorem 8.4.2(a) for that $\mathbf{y}$. It satisfies $\mathbf{X} \mathbf{b} = \mathbf{M} \mathbf{X} \beta = \mathbf{X} \beta$ and $\mathbf{T} \mathbf{b} = \mathbf{0}$. The matrix $\mathbf{P}_\mathbf{T}$ is $\mathbf{H}_\mathbf{G}$ of Eq. 8.3.1 for the generalized inverse $\mathbf{G}_\mathbf{T}$, so Proposition 8.3.3 gives idempotence, the null space $\mathcal{N}(\mathbf{X})$ and rows in $\mathcal{C}(\mathbf{X}^\top)$. Its range lies in $\mathcal{N}(\mathbf{T})$. The range has dimension r , and so does $\mathcal{N}(\mathbf{T})$, because $\text{rank}(\mathbf{T}) = p - r$. So the two coincide. Unbiasedness is Proposition 8.3.3(c). ■

In words, $\mathbf{P}_T$ is the projection of $\mathbb{R}^p$ onto $\mathcal{N}(T)$ along $\mathcal{N}(X)$, generally an oblique one. It sends the true β to the unique representative of $\beta + \mathcal{N}(X)$ that satisfies the side condition. That representative's coordinates are functions of the mean, so they can be estimated. Their *meaning* is fixed by T . In the one-way layout, the condition $\sum_k \alpha_k = 0$ makes μ the unweighted average of the g cell means and α_k the deviation of cell k from it. The condition $\sum_k n_k \alpha_k = 0$ makes μ the size-weighted average instead. The condition $\alpha_1 = 0$ makes μ the first cell mean. The data do not choose among these. The analyst chooses, by deciding which question the coefficients should answer.

One side condition deserves special mention, because it is the one many programs impose without saying so.

Proposition 8.4.4 — Minimum norm as a side condition

Let the columns of the $p \times (p-r)$ matrix V_0 be a basis of $\mathcal{N}(X)$. Then $T = V_0^\top$ satisfies the conditions of Theorem 8.4.2, and the resulting solution $\hat{\beta}_T$ is the minimum-norm least squares solution X^+y .

Proof. $\mathcal{C}(T^\top) = \mathcal{C}(V_0) = \mathcal{N}(X)$, which has dimension $p-r$ and meets $\mathcal{C}(X^\top) = \mathcal{N}(X)^\perp$ only in $\mathbf{0}$. So (d) holds. The least squares solutions are $\hat{\beta}_T + \mathcal{N}(X)$, and $\hat{\beta}_T \in \mathcal{N}(T) = \mathcal{C}(V_0)^\perp = \mathcal{N}(X)^\perp$. For $v \in \mathcal{N}(X)$, Pythagoras gives $\|\hat{\beta}_T + v\|^2 = \|\hat{\beta}_T\|^2 + \|v\|^2$, so $\hat{\beta}_T$ has the smallest norm. By Proposition 6.10.1 this is X^+y . ■

In the one-way layout the null space is spanned by $(1, -1, \dots, -1)^\top$, so the minimum-norm solution is the one satisfying $\mu = \alpha_1 + \dots + \alpha_g$. No one would choose this restriction on purpose, and its coefficients have no natural interpretation.

Example. (Five side conditions, one fit)

Eighteen simulated observations fall into four groups of sizes 5, 3, 6, 4, with group means 11.111, 15.204, 10.558 and 13.624. In the model $E(y_{kj}) = \mu + \alpha_k$, $p-r=1$, so a side condition is a single nonestimable restriction. The listing solves Eq. 8.4.2 for five of them and computes the minimum-norm solution.

Side condition	$\hat{\mu}$	$\hat{\alpha}_1$	$\hat{\alpha}_2$	$\hat{\alpha}_3$	$\hat{\alpha}_4$
$\alpha_1 = 0$	11.111	0.000	4.093	-0.553	2.513
$\alpha_4 = 0$	13.624	-2.513	1.580	-3.066	0.000
$\sum_k \alpha_k = 0$	12.624	-1.513	2.580	-2.066	1.000
$\sum_k n_k \alpha_k = 0$	12.167	-1.056	3.037	-1.609	1.457
$\mu = 0$	0.000	11.111	15.204	10.558	13.624
minimum norm ($\mu = \sum_k \alpha_k$)	10.100	1.012	5.105	0.459	3.525

Every row solves the normal equations. In every row $\hat{\mu} + \hat{\alpha}_k$ is the k th group mean, and $\hat{\alpha}_1 - \hat{\alpha}_2$ is the same. The value of $\hat{\mu}$ is, in turn, the first group mean, the last group mean, the average of the four group means (12.624), the overall mean $\bar{y} = 12.167$, zero, and an uninterpretable 10.100.

Compare a restriction that *is* estimable, $\alpha_1 = \alpha_2$. It is not a side condition. Imposing it (by solving the Lagrange system for least squares under a linear restriction) forces the first two group means to be equal. The residual sum of squares rises from 14.665 to 46.078. The fit has changed, which is what testing the hypothesis $\alpha_1 = \alpha_2$ in Chapter 11 measures.

```
import numpy as np

sizes = [5, 3, 6, 4]
groups = np.repeat(np.arange(4), sizes)
rng = np.random.default_rng(84)
y = np.array([12.0, 15.0, 11.0, 14.0])[groups] + rng.normal(scale=1.0, size=groups.size)
X = np.column_stack([np.ones(groups.size), np.eye(4)[groups]]) # (mu, alpha_1..
                      alpha_4)
A, c = X.T @ X, X.T @ y
```

```
n_k = np.array(sizes, dtype=float)
side = {"alpha_1 = 0": [0, 1, 0, 0, 0],
        "alpha_4 = 0": [0, 0, 0, 0, 1],
        "sum alpha_k = 0": [0, 1, 1, 1, 1],
        "sum n_k alpha_k = 0": [0, *n_k],
        "mu = 0": [1, 0, 0, 0, 0]}
solutions = {}
for name, t in side.items():
    T = np.array([t], dtype=float)
    G = np.linalg.inv(A + T.T @ T) # (X'X + T'T)^{-1}: a g-inverse of X'X
    b = G @ c
    solutions[name] = b
    print(f"{name:20s} b = {np.round(b, 3)}   T b = {(T @ b)[0]:.1e}")
b_mn = np.linalg.pinv(X) @ y
print(f"{'minimum norm':20s} b = {np.round(b_mn, 3)}",
      f"   mu - sum(alpha) = {b_mn[0] - b_mn[1:].sum():.1e}")
```

```
# "alpha_1 = alpha_2" is a restriction on an estimable function: it changes the
  model
T_bad = np.array([[0, 1, -1, 0, 0]], dtype=float)
K = np.block([[A, T_bad.T], [T_bad, np.zeros((1, 1))]]) # Lagrange (bordered)
                      system
b_bad = np.linalg.lstsq(K, np.append(c, 0.0), rcond=None)[0][:5]
sse_free = np.sum((y - X @ solutions["alpha_1 = 0"]) ** 2)
sse_bad = np.sum((y - X @ b_bad) ** 2)
print(f"SSE without restriction {sse_free:.3f};   with alpha_1 = alpha_2 {sse_bad:.3f}
      ")
```

Reparameterization and side conditions are two languages for the same choice. Deleting the column of α_1 from $\mathbf{X}$ and fitting the full-rank remainder gives exactly the solution with $\alpha_1 = 0$. More generally, deleting a set of columns whose removal leaves a full-rank matrix with the same column space is the side condition that sets those coefficients to zero, and it satisfies Theorem 8.4.2(c) (Exercise (B1)).

8.4.4 What software does

Programs treat a rank-deficient $\mathbf{X}$ differently.

- *Code the factor at full rank before fitting.* Formula interfaces such as R's `lm()` and statsmodels' formula API expand a factor with g levels into an intercept and $g - 1$ columns, by default the reference coding of [Section 8.5](#). The overparameterized matrix never appears. This is a reparameterization in the sense of [Theorem 8.4.1](#).
- *Drop columns found to be dependent.* If the columns supplied are dependent anyway, R's `lm()` computes a QR decomposition in which a column found to be (numerically) a combination of earlier ones is moved to the end and left out. Its coefficient is reported as NA. This is the side condition setting that coefficient to zero, and which column is dropped depends on the order of the columns. SAS's GLM procedure similarly solves the normal equations with a generalized inverse that zeroes the coefficients of dependent columns, and it flags the resulting estimates as not unique.
- *Return the minimum-norm solution.* `numpy.linalg.lstsq` returns $\mathbf{X}^+\mathbf{y}$ silently (it reports the numerical rank only as a second output), and statsmodels' OLS, whose `fit` uses the pseudoinverse by default, returns the same vector with a `SingularMatrixWarning`. This is the side condition of [Proposition 8.4.4](#), and it does not depend on the order of the columns.
- *Pivot greedily.* A QR decomposition with column pivoting ([Chapter 10](#), [Theorem 10.7.2](#)) chooses at each step the remaining column with the largest component orthogonal to those already chosen, and the columns left at the end are dropped. In [Example \(Five side conditions, one fit\)](#) it takes the intercept, then α_3 (the largest group), α_1 and α_4 , and drops α_2 , the smallest group, a choice that depends on the group sizes.

```
import scipy.linalg
import statsmodels.api as sm
b_sm = sm.OLS(y, X).fit().params          # statsmodels default: method="pinv"
"
b_np = np.linalg.lstsq(X, y, rcond=None)[0]  # LAPACK gelsd: minimum-norm
solution
_, R, piv = scipy.linalg.qr(X, mode="economic", pivoting=True)
print("statsmodels:", np.round(b_sm, 3))
print("numpy lstsq:", np.round(b_np, 3))
print("pivoted QR column order:", piv, " |R_kk| =", np.round(np.abs(np.diag(R)), 3))
```

The lesson of this section is that all these outputs are correct, and none is more correct than the others. They are different coordinate systems for one fitted mean vector. Any conclusion that changes when the software changes is a conclusion about a nonestimable function.

Warning. Before interpreting a coefficient from a rank-deficient fit, find out which side condition produced it. A coefficient reported as “the effect of level 3” may be a difference from level 1, a deviation from an average of cell means, a deviation from the overall mean, or, under the minimum-norm convention, a quantity with no interpretation at all.

8.4.5 Exercises

8.4.5.1 A. Check your understanding

Exercise A1.

In the one-way model with four groups and $\boldsymbol{\beta} = (\mu, \alpha_1, \dots, \alpha_4)^\top$, decide which of the following are side conditions in the sense of [Theorem 8.4.2](#): (i) $\alpha_2 = 0$; (ii) $\mu = \alpha_3$; (iii) $\alpha_1 - \alpha_2 = 0$; (iv) $\alpha_1 = 0$ together with $\alpha_2 = 0$; (v) $2\mu + \alpha_1 + \alpha_2 = 0$.

8.4.5.2 B. Practice**Exercise B1.**

Let $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$, where $\mathbf{X}_1$ has full column rank and $\mathcal{C}(\mathbf{X}_1) = \mathcal{C}(\mathbf{X})$. Show that the least squares fit using $\mathbf{X}_1$ alone, padded with zeros for the coefficients of $\mathbf{X}_2$, is the solution of the side condition $\mathbf{T} = [\mathbf{0}, \mathbf{I}]$. Check the conditions of [Theorem 8.4.2](#) directly.

Solution. The padded vector $(\hat{\boldsymbol{\beta}}_1^\top, \mathbf{0}^\top)^\top$ satisfies $\mathbf{X}(\hat{\boldsymbol{\beta}}_1^\top, \mathbf{0}^\top)^\top = \mathbf{X}_1\hat{\boldsymbol{\beta}}_1 = \mathbf{M}\mathbf{y}$, because $\mathcal{C}(\mathbf{X}_1) = \mathcal{C}(\mathbf{X})$. So it solves the normal equations and satisfies $\mathbf{T}\mathbf{b} = \mathbf{0}$. For the conditions: $\text{rank}(\mathbf{T}) = p_2$, and $\text{rank}(\mathbf{X}) = \text{rank}(\mathbf{X}_1) = p_1$, so $\text{rank}(\mathbf{T}) = p - r$. If $\mathbf{T}^\top \mathbf{c} = (\mathbf{0}^\top, \mathbf{c}^\top)^\top$ were in $\mathcal{C}(\mathbf{X}^\top)$, say $\mathbf{X}_1^\top \boldsymbol{\rho} = \mathbf{0}$ and $\mathbf{X}_2^\top \boldsymbol{\rho} = \mathbf{c}$, then $\boldsymbol{\rho} \perp \mathcal{C}(\mathbf{X}_1) = \mathcal{C}(\mathbf{X}) \supseteq \mathcal{C}(\mathbf{X}_2)$, so $\mathbf{c} = \mathbf{0}$. Condition (e) holds.

Exercise B2.

For the additive model $E(y_{ij}) = \mu + \alpha_i + \beta_j$ with $i = 1, 2, j = 1, 2, 3$ and one observation per cell, take $\mathbf{U}$ with rows giving $\mu + \alpha_1 + \beta_1, \alpha_2 - \alpha_1, \beta_2 - \beta_1$ and $\beta_3 - \beta_1$. Compute $\mathbf{Z}$ from [Eq. 8.4.1](#), and check that it is the reference-coded model matrix.

Exercise B3.

Let $\mathbf{T}$ satisfy the conditions of [Theorem 8.4.2](#) and have full row rank $p - r$. Show that the bordered matrix $\begin{pmatrix} \mathbf{X}_T^\top \mathbf{X} & \mathbf{T}^\top \\ \mathbf{X}^\top \mathbf{y} & \mathbf{0} \end{pmatrix}$ is nonsingular, that the solution of the bordered system with right-hand side $(\mathbf{X}^\top \mathbf{y}, \mathbf{0})$ has first block $\hat{\boldsymbol{\beta}}_T$ and second block $\mathbf{0}$, and interpret the zero Lagrange multiplier. Relate this to [Exercise \(B2\)](#).

8.4.5.3 C. Going deeper**Exercise C1.**

Replace $\mathbf{X}$ by $\mathbf{X}\mathbf{D}$ for a nonsingular diagonal $\mathbf{D}$ (a change of units). Show that the minimum-norm solution for $\mathbf{X}\mathbf{D}$, converted back to the original units, is the solution for the side condition $\mathbf{V}_0^\top \mathbf{D}^{-2} \mathbf{b} = \mathbf{0}$, which differs in general from $\mathbf{V}_0^\top \mathbf{b} = \mathbf{0}$. Conclude that the minimum-norm convention depends on the units of measurement, while no estimable function does.

Exercise C2.

Suppose $\text{rank}(\mathbf{T}) = p - r$ but $\mathcal{C}(\mathbf{T}^\top) \cap \mathcal{C}(\mathbf{X}^\top) \neq \{\mathbf{0}\}$. Show that for some $\mathbf{y}$ the restricted

normal equations of [Theorem 8.4.2\(a\)](#) have no solution, and that whenever they have one they have infinitely many.

Solution. By the dimension count, $\mathcal{C}(\mathbf{X}^\top) + \mathcal{C}(\mathbf{T}^\top)$ has dimension less than $r + (p - r) = p$. So $\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}) \neq \{\mathbf{0}\}$, and any solution can be moved along a nonzero vector of that intersection. For existence, the proof of [Theorem 8.4.2](#) shows that a solution exists for every $\mathbf{y}$ iff $\mathcal{N}(\mathbf{X}) + \mathcal{N}(\mathbf{T}) = \mathbb{R}^p$, which fails because its orthogonal complement $\mathcal{C}(\mathbf{X}^\top) \cap \mathcal{C}(\mathbf{T}^\top)$ is nonzero. So some coset $\mathbf{b}_0 + \mathcal{N}(\mathbf{X})$ misses $\mathcal{N}(\mathbf{T})$, and the response $\mathbf{y} = \mathbf{X}\mathbf{b}_0$ has no solution.

8.5 Indicator variables and the coding of a factor

Most regressors in practice are not measurements on a continuous scale. They record a category: treatment arm, region, occupation, party, a yes-or-no answer. Such a variable is a **factor**, and its possible values are its **levels**. A factor enters a linear model through *indicator variables* (also called dummy variables), columns of zeros and ones that mark membership of a level. There are many ways to build these columns, and they give quite different coefficient tables. This section shows that they all describe the same fitted model. Each choice of coding is a choice of which linear functions of the level means the coefficients report. It gives a single formula that says exactly which ones.

8.5.1 One binary regressor

Suppose each observation belongs to one of two groups, and let $d_i = 1$ if observation i is in group 1 and $d_i = 0$ if it is in group 0. The model

$$E(y_i) = \beta_0 + \beta_1 d_i$$

says that the mean is $\mu_0 = \beta_0$ in group 0 and $\mu_1 = \beta_0 + \beta_1$ in group 1. So $\beta_1 = \mu_1 - \mu_0$ is the difference between the two group means. The least squares fit sets the fitted value in each group to that group's mean (the model space is the set of vectors constant within groups, [Example \(The one-way layout\)](#)), so $\hat{\beta}_0 = \bar{y}_0$ and $\hat{\beta}_1 = \bar{y}_1 - \bar{y}_0$. The t statistic for $\beta_1 = 0$ is the pooled two-sample t statistic ([Exercise \(A1\)](#)). The familiar two-sample comparison is a regression on one indicator.

It is tempting to add a second indicator, $1 - d_i$, for group 0. Then $\mathbf{X} = [\mathbf{1}, \mathbf{d}, \mathbf{1} - \mathbf{d}]$ has three columns, but they satisfy $\mathbf{1} - \mathbf{d} - (\mathbf{1} - \mathbf{d}) = \mathbf{0}$, so $\text{rank}(\mathbf{X}) = 2$. This is the **dummy variable trap**: an intercept together with an indicator for every level produces an exactly collinear model matrix. By [Section 8.1](#), none of the three coefficients is then identifiable. The trap is not a mistake in the model, which still describes two means, but in the coordinates.

8.5.2 A factor with g levels

Let a factor have levels $1, \dots, g$, with $n_k \geq 1$ observations at level k . Its **indicator matrix** is the $n \times g$ matrix $\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_g]$ with $z_{ik} = 1$ if observation i is at level k and 0 otherwise.

Each row of $\mathbf{Z}$ contains a single 1, so

$$\mathbf{Z}\mathbf{1}_g = \mathbf{1}_n, \quad \mathbf{Z}^\top \mathbf{Z} = \mathbf{D} = \text{diag}(n_1, \dots, n_g), \quad \mathbf{Z}^\top \mathbf{y} = (y_{1\cdot}, \dots, y_{g\cdot})^\top, \quad (8.5.1)$$

where $y_{k\cdot}$ is the total of the responses at level k . The columns have disjoint supports, so $\mathbf{Z}$ has full column rank g , and $\mathcal{C}(\mathbf{Z})$ is the set of vectors that are constant within levels.

The *cell-means model* $\mathbf{E}(\mathbf{Y}) = \mathbf{Z}\boldsymbol{\mu}$, $\boldsymbol{\mu} = (\mu_1, \dots, \mu_g)^\top$, gives each level its own mean. It has full rank, and by [Eq. 8.5.1](#)

$$\hat{\boldsymbol{\mu}} = \mathbf{D}^{-1} \mathbf{Z}^\top \mathbf{y} = (\bar{y}_1, \dots, \bar{y}_g)^\top,$$

the vector of level means. Any other model containing $\mathcal{C}(\mathbf{Z})$ and nothing more has the same fit. The overparameterized model $[\mathbf{1}, \mathbf{Z}]$ of [Example \(A rank-deficient layout\)](#) is one example. The full-rank codings below are others.

Definition 8.5.1 — Coding of a factor

A **coding** of a factor with g levels is a nonsingular $g \times g$ matrix $\mathbf{K}$. The coded model matrix is $\mathbf{Z}\mathbf{K}$, and the model is $\mathbf{E}(\mathbf{Y}) = \mathbf{Z}\mathbf{K}\boldsymbol{\gamma}$. When the first column of $\mathbf{K}$ is $\mathbf{1}_g$, so that $\mathbf{K} = [\mathbf{1}_g, \mathbf{C}]$, the coded matrix is $[\mathbf{1}_n, \mathbf{Z}\mathbf{C}]$: an intercept and the $g - 1$ columns $\mathbf{Z}\mathbf{C}$. The $g \times (g - 1)$ matrix $\mathbf{C}$ is then called the **coding matrix** of the factor.

Row k of $\mathbf{K}$ lists the values that the coded columns take for an observation at level k . In the software of [Section 8.4](#) the matrix $\mathbf{C}$ is what one specifies (in R, the “contrasts” attribute of a factor), and the program builds the columns $\mathbf{Z}\mathbf{C}$ from it.

Proposition 8.5.1 — What a coding estimates

Let $\mathbf{K}$ be a coding and $\hat{\boldsymbol{\mu}} = \mathbf{D}^{-1} \mathbf{Z}^\top \mathbf{y}$ the vector of level means. Then:

- (a) $\mathcal{C}(\mathbf{Z}\mathbf{K}) = \mathcal{C}(\mathbf{Z})$, so every coding has the same fitted values, residuals and residual sum of squares;
- (b) the coefficients are $\boldsymbol{\gamma} = \mathbf{K}^{-1} \boldsymbol{\mu}$, and their least squares estimates are $\hat{\boldsymbol{\gamma}} = \mathbf{K}^{-1} \hat{\boldsymbol{\mu}}$;
- (c) $[\mathbf{1}_g, \mathbf{C}]$ is nonsingular iff the columns of $\mathbf{C}$ are linearly independent and $\mathbf{1}_g \notin \mathcal{C}(\mathbf{C})$;
- (d) to make the coefficients estimate prescribed linear functions $\mathbf{L}\boldsymbol{\mu}$ of the level means, with $\mathbf{L}$ a nonsingular $g \times g$ matrix, use the coding $\mathbf{K} = \mathbf{L}^{-1}$.

Proof.

- (a) The columns of $\mathbf{Z}\mathbf{K}$ lie in $\mathcal{C}(\mathbf{Z})$, and $\text{rank}(\mathbf{Z}\mathbf{K}) = \text{rank}(\mathbf{Z}) = g$ because $\mathbf{K}$ is nonsingular ([Proposition 1.2.3\(b\)](#)). A subspace of $\mathcal{C}(\mathbf{Z})$ of the same dimension is $\mathcal{C}(\mathbf{Z})$ itself. The fit depends only on the column space ([Theorem 6.7.1](#)).
- (b) The two models describe the same means iff $\mathbf{Z}\mathbf{K}\boldsymbol{\gamma} = \mathbf{Z}\boldsymbol{\mu}$, and since $\mathbf{Z}$ has full column rank this means $\mathbf{K}\boldsymbol{\gamma} = \boldsymbol{\mu}$. For the estimates, $\mathbf{Z}\mathbf{K}\hat{\boldsymbol{\gamma}} = \mathbf{M}\mathbf{y} = \mathbf{Z}\hat{\boldsymbol{\mu}}$, and the same cancellation gives $\mathbf{K}\hat{\boldsymbol{\gamma}} = \hat{\boldsymbol{\mu}}$.
- (c) A square matrix is nonsingular iff its columns are linearly independent. The columns $\mathbf{1}_g, \mathbf{c}_1, \dots, \mathbf{c}_{g-1}$ are independent iff the $\mathbf{c}_j$ are independent and $\mathbf{1}_g$ is not in their span.
- (d) With $\mathbf{K} = \mathbf{L}^{-1}$, (b) gives $\boldsymbol{\gamma} = \mathbf{L}\boldsymbol{\mu}$.

Part (b) is the whole story. Whatever the coding, the coefficients are fixed linear combinations of the level means, and these combinations are the *rows of* $\mathbf{K}^{-1}$, not the columns of $\mathbf{K}$. Each such combination is estimable in the overparameterized model, since it is a function of the means. Part (d) reverses the logic, and in practice it is the useful direction. Decide which questions the coefficients should answer, write them as the rows of $\mathbf{L}$, and invert.

8.5.3 The standard codings

For a factor with g levels, the codings in common use are the following. For each we give the meaning of the intercept γ_0 and of the other coefficients $\gamma_1, \dots, \gamma_{g-1}$, that is, the rows of $\mathbf{K}^{-1}$.

- **Reference (treatment) coding.** $\mathbf{C} = \begin{pmatrix} \mathbf{0}^\top \\ \mathbf{I}_{g-1}^\top \end{pmatrix}$: the columns are the indicators of levels $2, \dots, g$. Then $\gamma_0 = \mu_1$, and $\gamma_j = \mu_{j+1} - \mu_1$ is the difference between level $j+1$ and the *reference level* 1. The level chosen as reference is arbitrary.
- **Sum-to-zero (deviation or effect) coding.** $\mathbf{C} = \begin{pmatrix} \mathbf{I}_{g-1}^\top \\ -\mathbf{1}^\top \end{pmatrix}$. Level $k < g$ has mean $\gamma_0 + \gamma_k$ and level g has mean $\gamma_0 - \sum_{j < g} \gamma_j$. Summing over the g levels gives $\gamma_0 = \bar{\mu} = g^{-1} \sum_k \mu_k$, the unweighted average of the level means. Then $\gamma_k = \mu_k - \bar{\mu}$ is the deviation of level k from that average. The deviation of the last level is $-\sum_{j < g} \gamma_j$.
- **Weighted effect coding.** $\mathbf{C} = \begin{pmatrix} \mathbf{I}_{g-1}^\top \\ -(n_1, \dots, n_{g-1})/n_g \end{pmatrix}$. Now $\sum_k n_k \mu_k = n\gamma_0$, so $\gamma_0 = \sum_k n_k \mu_k / n$, the size-weighted average, estimated by the overall mean $\bar{y}$. The coefficient $\gamma_k = \mu_k - \gamma_0$ is the deviation from it.
- **Successive differences.** With $\mathbf{L}$ having first row $g^{-1}\mathbf{1}^\top$ and rows $\mathbf{e}_{j+1}^\top - \mathbf{e}_j^\top$, part (d) gives a coding in which $\gamma_0 = \bar{\mu}$ and $\gamma_j = \mu_{j+1} - \mu_j$. This is natural for ordered levels.
- **Helmert and orthogonal polynomial codings**, whose columns are contrasts, are treated in [Section 8.7](#).

The first three are the side conditions of [Example \(Five side conditions, one fit\)](#) in another language. In the overparameterized model $\mu + \alpha_k$, reference coding is the side condition $\alpha_1 = 0$, sum-to-zero coding is $\sum_k \alpha_k = 0$, and weighted effect coding is $\sum_k n_k \alpha_k = 0$. In each case the constrained $(\mu, \alpha_2, \dots, \alpha_g)$ or $(\mu, \alpha_1, \dots, \alpha_{g-1})$ equals γ ([Exercise \(C1\)](#)).

Example. (Party identification and ideology)

An extract of the 1996 American National Election Study, distributed with statsmodels as a public-domain data set, contains 944 respondents who placed themselves on a seven-point scale from extremely liberal (1) to extremely conservative (7). It also records their party identification in seven levels from strong Democrat to strong Republican. The level means and sizes are

party identifi- cation	strong Dem	weak Dem	ind. Dem	ind. independenRep	weak Rep	strong Rep
n_k	200	180	108	37	94	150
$\bar{y}_k$	3.335	3.617	3.685	4.054	4.957	5.691

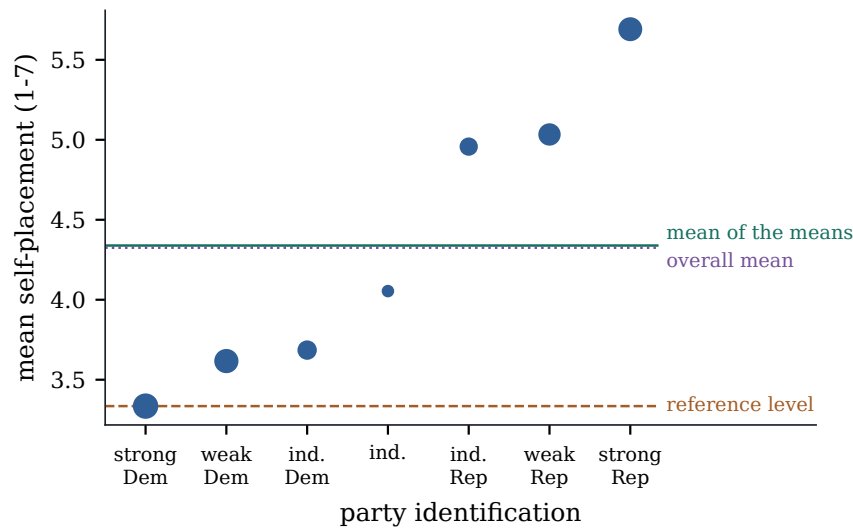


Figure 8.5.1: Mean self-placement by party identification in the 1996 ANES (point area proportional to group size), with the three baselines used by the codings of [Example \(Party identification and ideology\)](#): the reference level, the unweighted mean of the seven means, and the overall mean.

Here are the coefficients of four codings of the same factor ($\gamma_1, \dots, \gamma_6$ refer to the columns of $\mathbf{C}$ in the order defined above; for the two deviation codings they belong to the first six levels):

coding	$\hat{\gamma}_0$	$\hat{\gamma}_1$	$\hat{\gamma}_2$	$\hat{\gamma}_3$	$\hat{\gamma}_4$	$\hat{\gamma}_5$	$\hat{\gamma}_6$
reference (strong Dem)	3.335	0.282	0.350	0.719	1.622	1.698	2.356
sum to zero	4.339	-1.004	-0.722	-0.654	-0.285	0.618	0.694
weighted effect	4.325	-0.990	-0.709	-0.640	-0.271	0.632	0.708
successive differences	4.339	0.282	0.069	0.369	0.903	0.076	0.658

Each row is $\mathbf{K}^{-1}\hat{\boldsymbol{\mu}}$ for its own $\mathbf{K}$, as [Proposition 8.5.1\(b\)](#) says, and all four have residual sum of squares 1178.3. The intercepts are the reference mean 3.335, the unweighted mean of the seven means 4.339, the overall mean $\bar{y} = 4.325$ and again 4.339. [Figure 8.5.1](#) shows the three reference points. The successive differences show that the largest step in ideology, 0.903, falls between pure independents and independents leaning Republican, and the step between weak and strong Republicans is 0.658. The reference coding cannot show this directly. The two deviation codings nearly agree here because the groups are of similar size, apart from the small group of pure independents.

```
import numpy as np
import statsmodels.api as sm

anes = sm.datasets.anes96.load_pandas().data
y = anes["selfLR"].to_numpy()      # 1 = very liberal ... 7 = very conservative
```

```

level = anes["PID"].to_numpy().astype(int)    # 0 = strong Dem. ... 6 = strong Rep.
g = 7
Z = np.eye(g)[level]                         # n x 7 indicator (cell-means) matrix
n_k = Z.sum(axis=0)
cell_means = (Z.T @ y) / n_k
print("group sizes:", n_k.astype(int))
print("cell means: ", np.round(cell_means, 3))

```

```

def coding_matrices(n_k):
    """Each coding is  $X = ZK$  for a nonsingular  $g \times g$   $K$ ; the coefficients are  $K^{-1}\mu$ ."""
    g = len(n_k)
    one = np.ones((g, 1))
    reference = np.vstack([np.zeros(g - 1), np.eye(g - 1)])    # level 0 is baseline
    deviation = np.vstack([np.eye(g - 1), -np.ones(g - 1)])    # sum-to-zero
    weighted = np.vstack([np.eye(g - 1), -n_k[:-1] / n_k[-1]])    # sum  $n_k \times$ 
    effect = 0
    successive = np.array([(j + 1 - g if i <= j else j + 1) / g for j in range(g - 1)])
                                for i in range(g)])    # backward differences
    return {"cell means": np.eye(g),
            "reference": np.hstack([one, reference]),
            "sum to zero": np.hstack([one, deviation]),
            "weighted effect": np.hstack([one, weighted]),
            "successive differences": np.hstack([one, successive])}

for name, K in coding_matrices(n_k).items():
    Xc = Z @ K
    gamma = np.linalg.lstsq(Xc, y, rcond=None)[0]
    print(f"{name:23s}", np.round(gamma, 3))

```

8.5.4 Choosing a coding

Since every coding gives the same fit, the choice is about communication and, to a lesser extent, precision. Four points are worth keeping in mind.

Anything that depends on the fit is invariant. This includes fitted values, residuals, R^2 , $\hat{\sigma}^2$, and the F test that all levels have the same mean. That test compares $\mathcal{C}(Z)$ with $\mathcal{C}(1)$, and the coding does not enter it (Chapter 11).

Coefficient tests answer different questions. The t test for a reference-coded coefficient tests whether level k differs from the reference level. The t test for a sum-coded coefficient tests whether level k differs from the average of all levels, a question that changes if a level is added. Neither question is wrong, but they are different questions. A table of coefficients from an unfamiliar coding should never be read as if it came from the familiar one.

Precision depends on the reference. In reference coding $\text{Var}(\hat{\gamma}_j) = \sigma^2(1/n_{j+1} + 1/n_1)$. A small reference group inflates the variance of every coefficient. In [Example \(Party identification and ideology\)](#), comparing each of the other five levels with the 37 pure independents instead of the 200 strong Democrats would multiply the standard errors by factors between 1.55 and 1.76. A common default is to use the largest group, or a natural control condition, as the reference.

With other regressors, coefficients are adjusted comparisons. When the model also contains other variables, a reference-coded coefficient compares level k with the reference level at equal values of those variables. That is the “holding the others fixed” reading of [Definition 5.7.1](#), with all its caveats. [Section 8.6](#) gives an example in which adjustment changes the answer a great deal.

Idea. A coding is a choice of coordinates for the g -dimensional space of level means. The coefficients are $\mathbf{K}^{-1}\hat{\boldsymbol{\mu}}$. To get the comparisons you want, write them down first as the rows of a nonsingular matrix $\mathbf{L}$, then code with $\mathbf{K} = \mathbf{L}^{-1}$.

For an ordered factor, a useful instance of this principle is the *cumulative* coding, with columns $\mathbf{1}\{\text{level} \geq k\}$ for $k = 2, \dots, g$. Then $\mathbf{K}$ is lower triangular with ones on and below the diagonal. Its inverse takes successive differences, so the intercept is μ_1 and the coefficient of the k th column is $\mu_k - \mu_{k-1}$ ([Exercise \(B2\)](#)). An observation’s mean is built up as a sum of steps, which is often how an ordered factor is best described.

8.5.5 Exercises

8.5.5.1 A. Check your understanding

Exercise A1.

In the model $E(y_i) = \beta_0 + \beta_1 d_i$, with n_0 observations in group 0 (level 1) and n_1 in group 1 (level 2), write down the coding matrix $\mathbf{K}$ with $\boldsymbol{\mu} = \mathbf{K}\boldsymbol{\beta}$. Use [Proposition 8.5.1\(b\)](#), not an inverse of $\mathbf{X}^\top \mathbf{X}$, to find $\hat{\boldsymbol{\beta}}$. Then show that $\text{Var}(\hat{\beta}_1) = \sigma^2(1/n_0 + 1/n_1)$ and that $\hat{\beta}_1 / \text{se}(\hat{\beta}_1)$ is the pooled two-sample t statistic, where $s^2 = \text{SSE}/(n - 2)$ replaces σ^2 .

Solution. The two means are $\mu_0 = \beta_0$ and $\mu_1 = \beta_0 + \beta_1$, so $\mathbf{K} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ and $\mathbf{K}^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$. By [Proposition 8.5.1\(b\)](#), $\hat{\boldsymbol{\beta}} = \mathbf{K}^{-1}\hat{\boldsymbol{\mu}} = (\bar{y}_0, \bar{y}_1 - \bar{y}_0)^\top$. The level means are the estimates of the full-rank cell-means model, with $\text{Cov}(\hat{\boldsymbol{\mu}}) = \sigma^2 \mathbf{D}^{-1} = \sigma^2 \text{diag}(1/n_0, 1/n_1)$ ([Theorem 5.5.1](#)), so $\text{Var}(\hat{\beta}_1) = (-1, 1) \sigma^2 \mathbf{D}^{-1} (-1, 1)^\top = \sigma^2(1/n_0 + 1/n_1)$. The residuals are deviations from the group means ([Proposition 8.5.1\(a\)](#)), so s^2 is the pooled variance, and the ratio is $(\bar{y}_1 - \bar{y}_0) / \{s\sqrt{1/n_0 + 1/n_1}\}$.

Exercise A2.

For a factor with three levels, write down the null space of $[\mathbf{1}, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3]$. Show that adding a second factor’s full set of indicators, $[\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B]$, creates a null space of dimension at least two, and give two independent null vectors.

8.5.5.2 B. Practice

Exercise B1.

Let $\mathbf{X}$ have full column rank, let $\mathbf{e}_i$ be the i th coordinate vector with $\mathbf{e}_i \notin \mathcal{C}(\mathbf{X})$, and fit $[\mathbf{X}, \mathbf{e}_i]$ with coefficients $(\boldsymbol{\beta}, \delta)$. Write $\hat{e}_i$ and h_{ii} for the residual and the leverage ([Definition 6.8.1](#)) of observation i in the fit of $\mathbf{X}$ alone.

- (a) Use [Theorem 6.6.2](#), with $\mathbf{e}_i$ as the regressor of interest and $\mathbf{X}$ as the one partialled

out, to show that $\hat{\delta} = \hat{e}_i/(1 - h_{ii})$, and explain why $h_{ii} < 1$.

- (b) Show that the enlarged fit reproduces y_i exactly, and that its $\hat{\beta}$ is the estimate $\hat{\beta}_{(i)}$ without observation i .
- (c) Deduce the deletion identity $y_i - \mathbf{x}_{(i)}^\top \hat{\beta}_{(i)} = \hat{e}_i/(1 - h_{ii})$: the error in predicting y_i from the other observations is the ordinary residual inflated by $1/(1 - h_{ii})$.

Solution.

- (a) Apply [Theorem 6.6.2\(a\)](#) with $\mathbf{X}_1 = \mathbf{e}_i$ and $\mathbf{X}_2 = \mathbf{X}$, and let $\mathbf{M}$ project onto $\mathcal{C}(\mathbf{X})$. Then $\hat{\delta}$ is the coefficient in the regression of $\mathbf{y}$ on $(\mathbf{I} - \mathbf{M})\mathbf{e}_i$: $\hat{\delta} = \mathbf{e}_i^\top (\mathbf{I} - \mathbf{M})\mathbf{y} / \mathbf{e}_i^\top (\mathbf{I} - \mathbf{M})\mathbf{e}_i = \hat{e}_i/(1 - h_{ii})$. The denominator is $\|(\mathbf{I} - \mathbf{M})\mathbf{e}_i\|^2$, which is positive because $\mathbf{e}_i \notin \mathcal{C}(\mathbf{X})$.
- (b) The residual vector of the enlarged fit is orthogonal to every column, in particular to $\mathbf{e}_i$, so its i th entry is zero. For fixed β , the residual sum of squares $\sum_{j \neq i} (y_j - \mathbf{x}_{(j)}^\top \beta)^2 + (y_i - \mathbf{x}_{(i)}^\top \beta - \delta)^2$ is minimized over δ by making the last term zero. What remains is the deletion criterion, and its minimizer $\hat{\beta}_{(i)}$ is unique: if $\mathbf{X}\mathbf{b} = c\mathbf{e}_i$ with $\mathbf{b} \neq \mathbf{0}$, then $c \neq 0$ by full rank, and $\mathbf{e}_i \in \mathcal{C}(\mathbf{X})$. So $\hat{\beta} = \hat{\beta}_{(i)}$.
- (c) By (b), $\hat{\delta} = y_i - \mathbf{x}_{(i)}^\top \hat{\beta}_{(i)}$; compare with (a). The t statistic for δ is the externally studentized residual of Chapter 20.

Exercise B2.

For an ordered factor with g levels, let the coded columns be the indicators of $\{\text{level} \geq k\}$, $k = 2, \dots, g$, together with an intercept. Show that $\mathbf{K}$ is lower triangular with ones on and below the diagonal, that $\mathbf{K}^{-1}$ is the first-difference matrix, and hence that the coefficients are μ_1 and $\mu_k - \mu_{k-1}$.

Exercise B3.

On each of m days a warehouse runs N_{1i} machines of type 1 and N_{2i} of type 2, and meters its total energy use V_i . Suppose each running machine of type k uses an amount of energy with mean θ_k and variance τ_k^2 , independently of the other machines and the same on every day. Show that $E(V_i) = \theta_1 N_{1i} + \theta_2 N_{2i}$, a linear model *without* intercept, and that (θ_1, θ_2) is identifiable iff the vectors (N_{1i}) and (N_{2i}) are not proportional. Find $\text{Cov}(\mathbf{V})$; why is it not a multiple of $\mathbf{I}$? Why is it essential that θ_k does not vary by day?

8.5.5.3 C. Going deeper

Exercise C1.

Show that fitting the reference-coded model is the same as solving the normal equations of the overparameterized one-way model under the side condition $\alpha_1 = 0$, with $\gamma_0 = \mu$ and $\gamma_j = \alpha_{j+1}$. State and prove the corresponding facts for the sum-to-zero and weighted effect codings.

8.6 Several factors, interactions and covariates

With one factor, rank deficiency is a nuisance, and a coding removes it once and for all. With two or more factors, or a factor together with a numeric regressor, it becomes informative. The rank of the model matrix now depends on which combinations of levels were observed. Whether a comparison is estimable depends on how the design links the levels involved. This section works out the three cases that occur most often: two factors without interaction, two factors with interaction, and a factor with a numeric covariate.

8.6.1 Two factors without interaction

Let factor A have levels $i = 1, \dots, a$ and factor B levels $j = 1, \dots, b$. Call a pair (i, j) a **cell**. A cell is *occupied* if at least one observation has that combination, and we assume every level of each factor occurs in some occupied cell. The *additive* model is

$$E(y) = \mu + \alpha_i + \beta_j \quad \text{for an observation in cell } (i, j), \quad (8.6.1)$$

with model matrix $\mathbf{X} = [\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B]$ built from the two indicator matrices. The number of parameters is $p = 1 + a + b$. Since $\mathbf{Z}_A \mathbf{1}_a = \mathbf{Z}_B \mathbf{1}_b = \mathbf{1}$, the vectors $(1, -\mathbf{1}_a^\top, \mathbf{0}^\top)^\top$ and $(1, \mathbf{0}^\top, -\mathbf{1}_b^\top)^\top$ always lie in $\mathcal{N}(\mathbf{X})$, so $\text{rank}(\mathbf{X}) \leq a + b - 1$. Whether equality holds, and which differences $\alpha_i - \alpha_k$ are estimable, depends on the pattern of occupied cells.

Picture the design as a graph. The vertices are the a levels of A and the b levels of B , and each occupied cell (i, j) is an edge joining level i of A to level j of B . The design is **connected** if this graph is connected, that is, if one can walk from any level to any other through occupied cells.

Theorem 8.6.1 — Connectedness and estimability in the additive model

In the additive model Eq. 8.6.1, let the design graph have c connected components. Then:

- (a) $\text{rank}(\mathbf{X}) = a + b - c$;
- (b) $\alpha_i - \alpha_k$ is estimable iff levels i and k of A lie in the same component, and likewise for $\beta_j - \beta_l$;
- (c) if the design is connected, the estimable functions are exactly the linear functions of the cell means $\mu + \alpha_i + \beta_j$ over *all* cells, occupied or not. In particular every difference between levels of the same factor is estimable.

Proof. A vector $(m, \mathbf{u}^\top, \mathbf{v}^\top)^\top$ lies in $\mathcal{N}(\mathbf{X})$ iff $m + u_i + v_j = 0$ for every occupied cell (i, j) . Fix such a vector. If (i, j) and (k, j) are both occupied, then $u_i = u_k$. If (i, j) and (i, l) are both occupied, then $v_j = v_l$. Following edges, u is constant on the A -levels of each component, say $u_i = t_C$ for all A -levels i in component C , and then $v_j = -m - t_C$ for the B -levels of C . Conversely, any choice of m and of numbers $t_1, \dots, t_c$, one for each component, defines in this way a vector satisfying all the equations. So $\mathcal{N}(\mathbf{X})$

is parameterized linearly and injectively by $(m, t_1, \dots, t_c)$, and it has dimension $c + 1$. Rank-nullity gives $\text{rank}(\mathbf{X}) = 1 + a + b - (c + 1)$, which is (a).

For (b), by [Theorem 8.3.1\(c\)](#), $\alpha_i - \alpha_k$ is estimable iff $u_i - u_k = 0$ for every null vector, that is, iff $t_{C(i)} = t_{C(k)}$ for all choices of the t 's. This holds iff $C(i) = C(k)$. The argument for B is the same.

For (c), if $c = 1$, then (b) makes every $\alpha_i - \alpha_k$ and $\beta_j - \beta_l$ estimable. Pick an occupied cell (i_0, j_0) . Then $\mu + \alpha_i + \beta_j = (\mu + \alpha_{i_0} + \beta_{j_0}) + (\alpha_i - \alpha_{i_0}) + (\beta_j - \beta_{j_0})$ is estimable for every (i, j) . Conversely, every estimable function is a combination of the means of occupied cells ([Theorem 8.2.2](#)), and these are among the cell means.

■

Part (c) contains a point worth stating separately. In a connected additive design, the mean of an *empty* cell is estimable. The data never observed that combination, but additivity lets it be pieced together from cells that were observed: $\mu + \alpha_i + \beta_j$ is the mean of an occupied cell (i, j') , plus the difference $\beta_j - \beta_{j'}$ learned elsewhere. The estimate is only as good as the additivity assumption, which cannot be checked at the empty cell. This is extrapolation of a subtle kind.

Example. (Two islands)

Take $a = 4, b = 5$ and seven occupied cells: $(1, 1), (1, 2), (2, 2), (2, 3)$ and $(3, 4), (4, 4), (4, 5)$ (levels numbered from 1). The design graph has two components: rows 1–2 with columns 1–3, and rows 3–4 with columns 4–5. By [Theorem 8.6.1](#), $\text{rank}(\mathbf{X}) = 4 + 5 - 2 = 7$, one less than $a + b - 1 = 8$. The difference $\alpha_1 - \alpha_2$ is estimable, since rows 1 and 2 are linked through column 2, but $\alpha_1 - \alpha_3$ is not. The two islands could sit at any vertical offset from each other without changing any fitted value. Occupying a single further cell, $(3, 3)$, joins the islands. The rank rises to 8, and $\alpha_1 - \alpha_3$ becomes estimable. The listing checks each statement with the rank test [Theorem 8.3.1\(h\)](#). The script also checks the rank formula on 200 random designs.

```
import numpy as np

def additive_matrix(cells, a, b):
    """Model matrix [1, row indicators, column indicators] for occupied (i, j) cells
    . """
    rows = np.array([i for i, j in cells])
    cols = np.array([j for i, j in cells])
    return np.column_stack([np.ones(len(cells)), np.eye(a)[rows], np.eye(b)[cols]])

def estimable(lam, X):
    """lambda is estimable iff appending it to the rows of X does not raise the rank
    . """
    return np.linalg.matrix_rank(np.vstack([X, lam])) == np.linalg.matrix_rank(X)

a, b = 4, 5
cells = [(0, 0), (0, 1), (1, 1), (1, 2), (2, 3), (3, 3), (3, 4)] # two separate
    blocks
X = additive_matrix(cells, a, b)
row_diff = lambda i, k: np.r_[0, np.eye(a)[i] - np.eye(a)[k], np.zeros(b)]
print("rank:", np.linalg.matrix_rank(X), " (a + b - 1 =", a + b - 1, ")")
print("alpha_1 - alpha_2 estimable:", estimable(row_diff(0, 1), X))
print("alpha_1 - alpha_3 estimable:", estimable(row_diff(0, 2), X))
```

```

X_linked = additive_matrix(cells + [(2, 2)], a, b)                                # one more
      occupied cell
print("after adding cell (3, 3): rank", np.linalg.matrix_rank(X_linked),
      " alpha_1 - alpha_3 estimable:", estimable(row_diff(0, 2), X_linked))

```

A complete design, with every cell occupied, is connected. So is any design in which one level of B appears with every level of A and one level of A with every level of B . Disconnected designs arise in practice more often than one might expect: in observational data with many sparse levels (workers and firms, students and schools, players and teams), and in experiments where blocks were filled carelessly. The estimable functions of a disconnected design are characterized in [Exercise \(C1\)](#).

8.6.2 Two factors with interaction

The additive model assumes that the difference between two levels of A is the same at every level of B . The *interaction model* drops that assumption:

$$E(y) = \mu + \alpha_i + \beta_j + \gamma_{ij} \quad \text{for an observation in cell } (i, j), \quad (8.6.2)$$

with an interaction parameter γ_{ij} for each of the ab cells. Write for the $n \times ab$ indicator matrix of the cells. The column of an empty cell is zero. The model matrix is $\mathbf{X} = [\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B, \gamma]$, with $p = 1 + a + b + ab$ columns, and m denotes the number of occupied cells.

Proposition 8.6.2 — Estimability in the interaction model

In the interaction model [Eq. 8.6.2](#) with m occupied cells, write $\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$ for the mean of an occupied cell.

- (a) $\mathcal{C}(\mathbf{X}) = \mathcal{C}()$ and $\text{rank}(\mathbf{X}) = m$: the model is the cell-means model in disguise.
- (b) The estimable functions are exactly the linear combinations $\sum c_{ij}\mu_{ij}$ over occupied cells.
- (c) An interaction contrast $\sum_{i,j} d_{ij}\gamma_{ij}$ whose coefficient table (d_{ij}) has zero row and column sums is estimable iff $d_{ij} = 0$ for every empty cell. In particular, $\gamma_{ij} - \gamma_{il} - \gamma_{kj} + \gamma_{kl}$ is estimable iff all four cells are occupied.

Proof.

- (a) Every column of $\mathbf{1}$, $\mathbf{Z}_A$ and $\mathbf{Z}_B$ is a sum of columns of γ . For example, the indicator of level i of A is the sum of the indicators of the cells in row i . So $\mathcal{C}(\mathbf{X}) = \mathcal{C}()$. The nonzero columns of γ are the m indicators of occupied cells, which have disjoint supports.
- (b) By [Theorem 8.2.2](#), the estimable functions are the combinations of the expected responses, and these are the μ_{ij} .
- (c) If $d_{ij} = 0$ off the occupied cells, then $\sum_{\text{occ}} d_{ij}\mu_{ij} = \sum d_{ij}\mu + \sum_i \alpha_i \sum_j d_{ij} + \sum_j \beta_j \sum_i d_{ij} + \sum d_{ij}\gamma_{ij} = \sum d_{ij}\gamma_{ij}$, because the row and column sums vanish. So the contrast is estimable. Conversely, if $d_{i_0j_0} \neq 0$ for an empty cell (i_0, j_0) , the column of $\gamma_{i_0j_0}$ in $\mathbf{X}$ is zero, so the coordinate vector $\mathbf{v}$ of that parameter lies in $\mathcal{N}(\mathbf{X})$. The contrast's coefficient vector has inner product $d_{i_0j_0} \neq 0$ with $\mathbf{v}$, so it

is not estimable ([Theorem 8.3.1\(c\)](#)).

■

In the interaction model the fitted value of each occupied cell is its own mean, and an empty cell has no estimable mean at all. The two models treat empty cells in opposite ways. The additive model fills them by assumption, while the interaction model declines to say anything about them. Which is appropriate is a substantive question, and Chapter 17 returns to it.

Example. (Empty cells in a survey)

Cross-classify the 944 ANES respondents of [Example \(Party identification and ideology\)](#) by party identification (seven levels) and education (seven levels, labelled in the statsmodels documentation from “grades 1–8” to “PhD”). Two of the 49 cells are empty: no pure independent and no weak Republican in the sample has only an elementary-school education. Other cells are thin: only 3 pure independents report some high school. The overparameterized interaction model has $p = 64$ columns and rank 47, the number of occupied cells, as [Proposition 8.6.2\(a\)](#) says. The interaction contrast comparing strong Democrats with pure independents across “some high school” and “high-school graduate” involves four occupied cells and is estimable. It rests on only 3 respondents in one cell, so it is imprecise, but estimable. The same comparison across “grades 1–8” and “high-school graduate” involves an empty cell and is not estimable. No amount of computation will produce it, and software that prints a number for it is printing an artefact of its side condition. The additive model, with rank $7 + 7 - 1 = 13$, treats both comparisons as zero by assumption.

```
import statsmodels.api as sm

anes = sm.datasets.anes96.load_pandas().data
pid = anes["PID"].to_numpy().astype(int)           # 7 levels
educ = anes["educ"].to_numpy().astype(int) - 1     # 7 levels, now 0..6
counts = np.zeros((7, 7), dtype=int)
np.add.at(counts, (pid, educ), 1)
print("empty (PID, educ) cells:",
      [(int(i), int(j) + 1) for i, j in zip(*np.nonzero(counts == 0))])

n = len(pid)
cell = pid * 7 + educ
X_int = np.column_stack([np.ones(n), np.eye(7)[pid], np.eye(7)[educ], np.eye(49)[
    cell]])
print("interaction model: p =", X_int.shape[1], " rank =", np.linalg.matrix_rank(
    X_int),
      " occupied cells =", int((counts > 0).sum()))

def interaction_contrast(i, k, j, l):
    """gamma_ij - gamma_il - gamma_kj + gamma_kl, as a coefficient vector on X_int.
    """
    lam = np.zeros(X_int.shape[1])
    for (r, c), s in [((i, j), 1), ((i, l), -1), ((k, j), -1), ((k, l), 1)]:
        lam[15 + r * 7 + c] += s
    return lam

print("PID 0 vs 3, educ 2 vs 3:", estimable(interaction_contrast(0, 3, 1, 2), X_int)
      )
```



```
print("PID 0 vs 3, educ 1 vs 3:", estimable(interaction_contrast(0, 3, 0, 2), X_int)
)
```

8.6.3 A factor and a numeric covariate

Now let a factor with g levels, with indicator matrix $\mathbf{Z}$, share the model with a numeric covariate $\mathbf{x}$. The two basic models are the **parallel lines** model

$$E(y) = \mu_k + \theta x \quad \text{for an observation at level } k, \quad (8.6.3)$$

with model matrix $[\mathbf{Z}, \mathbf{x}]$, and the **separate lines** model

$$E(y) = \mu_k + \theta_k x, \quad (8.6.4)$$

with model matrix $[\mathbf{Z}, \mathbf{Z} \circ \mathbf{x}]$, where $\mathbf{Z} \circ \mathbf{x}$ multiplies each indicator column entrywise by $\mathbf{x}$, so its k th column is $\mathbf{x}$ on level k and zero elsewhere. The same models can be written with an intercept and any coding of the factor. For example, the separate lines model in reference coding has columns $\mathbf{1}$, $\mathbf{Z}\mathbf{C}$, $\mathbf{x}$, $(\mathbf{Z}\mathbf{C}) \circ \mathbf{x}$. The coefficients of the last block are then differences of slopes from the reference level, and the column space is unchanged.

By [Exercise \(B1\)](#), the common slope θ is identifiable iff $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$, that is, iff x varies within at least one level. In the separate lines model, the slope θ_k is identifiable iff x varies within level k , since the k th column of $\mathbf{Z} \circ \mathbf{x}$ is a multiple of $\mathbf{z}_k$ exactly when x is constant on level k . When the slopes are identifiable, the separate lines model is simply g simple regressions fitted at once. Its coefficients are those of the separate fits, because the columns for different levels have disjoint supports.

The parallel lines model is where adjustment happens.

Proposition 8.6.3 — Adjusted differences

In the parallel lines model [Eq. 8.6.3](#) with $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$, let $\bar{y}_k$ and $\bar{x}_k$ be the level means. Then

$$\hat{\theta} = \frac{\sum_k \sum_{i \in k} (x_i - \bar{x}_k)(y_i - \bar{y}_k)}{\sum_k \sum_{i \in k} (x_i - \bar{x}_k)^2}, \quad \hat{\mu}_k - \hat{\mu}_l = (\bar{y}_k - \bar{y}_l) - \hat{\theta}(\bar{x}_k - \bar{x}_l).$$

Proof. The model matrix $[\mathbf{Z}, \mathbf{x}]$ has full column rank because $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$. Apply [Theorem 6.6.2](#) with $\mathbf{X}_1 = \mathbf{x}$ and $\mathbf{X}_2 = \mathbf{Z}$. Projecting onto $\mathcal{C}(\mathbf{Z})$ replaces each entry by its level mean ([Example \(The one-way layout\)](#)), so $(\mathbf{I} - \mathbf{M}_Z)\mathbf{x}$ and $(\mathbf{I} - \mathbf{M}_Z)\mathbf{y}$ are the within-level deviations, and $\hat{\theta}$ is the no-intercept slope of one on the other. This is the first formula. By [Theorem 6.6.2\(c\)](#), $\hat{\boldsymbol{\mu}} = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top (\mathbf{y} - \hat{\theta} \mathbf{x})$, whose k th entry is $\bar{y}_k - \hat{\theta} \bar{x}_k$. Subtracting gives the second formula. ■

The difference $\bar{y}_k - \bar{y}_l$ compares the levels as they happen to be. The adjusted difference compares them at equal values of the covariate, correcting for the fact that the groups differ in x . The

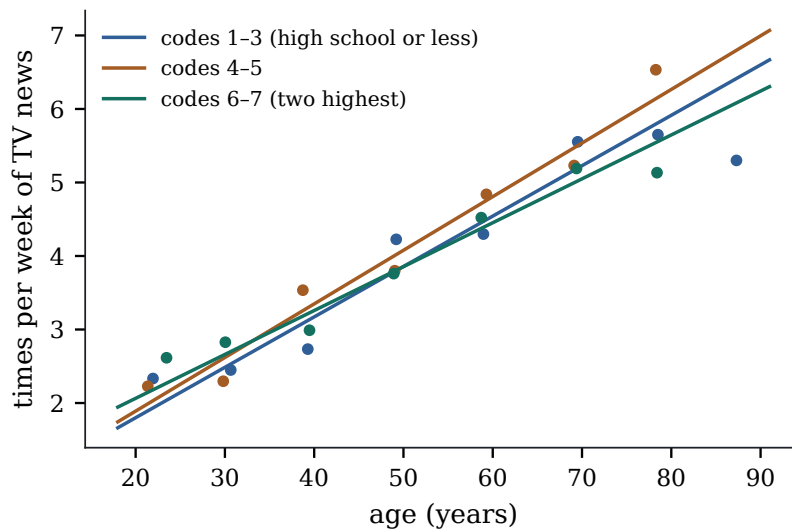


Figure 8.6.1: Times per week of television news against age for three education groups in the 1996 ANES: separate least squares lines (Eq. 8.6.4) and the group means within ten-year age bins (points).

correction is the slope times the difference in covariate means. It is large when the groups are unbalanced on a covariate that matters. The estimated slope uses only variation *within* levels, so that differences between levels in both x and y cannot be mistaken for an effect of x .

Example. (Education and television news, adjusted for age)

In the ANES data, respondents report how many times per week they watch television news. Coding education with seven levels and “high-school graduate” as the reference, respondents with some high school but no diploma watch 1.005 times a week more than high-school graduates. They are also older, with mean ages 59.8 and 48.2, and news watching rises with age. Adding age to the model, the within-level slope is 0.0674 times per week per year of age, and the adjusted difference shrinks to 0.224. More than three quarters of the raw difference is an age difference, and Proposition 8.6.3 accounts for it exactly. Whether the adjusted figure is the “effect of education” is a different question. Here age plausibly influences both schooling (older cohorts had less of it) and viewing habits, which argues for adjusting, but Chapter 25 shows that adjustment is not always appropriate.

Allowing separate age slopes for three education groups (codes 1–3, high school or less, 313 respondents; codes 4–5, 277; and the two highest levels, codes 6–7, 354) gives slopes 0.0686, 0.0729 and 0.0597, close to the common slope 0.0672 of the parallel lines model with the same three groups. The residual sum of squares falls only from 5623.6 to 5616.7 for the two extra parameters. Figure 8.6.1 shows the three fitted lines with binned means. The lines are nearly parallel, and the highest education group starts higher and rises a little more slowly.

```
tv = anes["TVnews"].to_numpy()                                # times per week watching TV
news
age = anes["age"].to_numpy()
E = np.eye(7)[educ][:, [0, 1, 3, 4, 5, 6]]                  # reference: high-school
graduate
unadjusted = np.linalg.lstsq(np.column_stack([np.ones(n), E]), tv, rcond=None)[0]
```

```
adjusted = np.linalg.lstsq(np.column_stack([np.ones(n), E, age]), tv, rcond=None)[0]
print("some high school minus high-school graduate:",
      f"unadjusted {unadjusted[2]:.3f}, adjusted for age {adjusted[2]:.3f}")
print("mean age: some high school", age[educ == 1].mean().round(1),
      " high-school graduate", age[educ == 2].mean().round(1),
      " common age slope", adjusted[-1].round(4))
```

```
grp = np.select([educ <= 2, educ <= 4], [0, 1], 2) # education codes 1-3 / 4-5 /
6-7
D = np.eye(3)[grp]
X_par = np.column_stack([D, age]) # parallel lines: 3 intercepts,
one slope
X_sep = np.column_stack([D, D * age[:, None]]) # separate lines: 3 intercepts,
3 slopes
b_par = np.linalg.lstsq(X_par, tv, rcond=None)[0]
b_sep = np.linalg.lstsq(X_sep, tv, rcond=None)[0]
sse_par = np.sum((tv - X_par @ b_par) ** 2)
sse_sep = np.sum((tv - X_sep @ b_sep) ** 2)
print("separate slopes:", b_sep[3:].round(4),
      f" SSE parallel {sse_par:.1f}, separate {sse_sep:.1f}")
```

Warning. In a separate lines model the difference between two levels is $(\mu_k - \mu_l) + (\theta_k - \theta_l)x$, which depends on x . The coefficient that software labels as the “main effect” of level k is this difference at $x = 0$. For a covariate like age, that is a comparison of newborns, far outside the data. Centre the covariate at a meaningful value before reading such coefficients, or report the difference at several values of x .

8.6.4 Broken lines

Indicators can also switch a slope on. Let c be a known value of x and put $(x-c)_+ = \max(x-c, 0)$, which is $x - c$ times the indicator of $x > c$. The model

$$E(y) = \beta_0 + \beta_1 x + \beta_2 (x - c)_+$$

is a continuous line with a bend at c : slope β_1 to the left of c and $\beta_1 + \beta_2$ to the right. Adding the indicator $\mathbf{1}\{x > c\}$ itself allows a jump at c as well. With the bend point known, these are ordinary linear models, and all of this chapter applies. For example, β_2 is identifiable iff the data contain values of x on both sides of c (strictly below and strictly above) and at least three distinct values of x in all ([Exercise \(B2\)](#)). When c is unknown the model is nonlinear in c . Such piecewise linear terms are the simplest splines, the subject of Part IX.

8.6.5 Exercises

8.6.5.1 A. Check your understanding

Exercise A1.

In the additive model with $a = b = 2$, suppose cells $(1, 1)$, $(2, 1)$ and $(2, 2)$ are occupied and $(1, 2)$ is empty. Find the rank of $\mathbf{X}$ and a basis of $\mathcal{N}(\mathbf{X})$. Show that $\mu + \alpha_1 + \beta_2$, the mean of the empty cell, is estimable by writing it as a combination of the means of

occupied cells.

Exercise A2.

A factor has three levels, and the covariate x is constant on level 2 but varies on levels 1 and 3. Find the rank of the separate lines model matrix $[\mathbf{Z}, \mathbf{Z} \circ \mathbf{x}]$, and say which of $\mu_1, \mu_2, \mu_3, \theta_1, \theta_2, \theta_3$ are identifiable.

8.6.5.2 B. Practice

Exercise B1.

In the separate lines model, show that the mean at level k and covariate value x_0 , $\mu_k + \theta_k x_0$, is estimable iff either x varies within level k , or x_0 equals the common value of x on level k . Interpret this as a statement about extrapolation.

Solution. The mean is $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ with $\boldsymbol{\lambda}$ having 1 in the position of μ_k and x_0 in the position of θ_k . The columns for μ_k and θ_k are $\mathbf{z}_k$ and $\mathbf{z}_k \circ \mathbf{x}$, and they involve only level k . If x varies on level k , both coefficients are identifiable ([Proposition 8.1.3](#)), so $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is estimable. If $x \equiv x^*$ on level k , then $\mathbf{z}_k \circ \mathbf{x} = x^* \mathbf{z}_k$, and the null space contains the vector with $-x^*$ at μ_k and 1 at θ_k . Then $\boldsymbol{\lambda}^\top \mathbf{v} = x_0 - x^*$, which vanishes iff $x_0 = x^*$. A single covariate value within a level supports no statement about other values, however many observations share it.

Exercise B2.

Write the broken-line model $E(y) = \beta_0 + \beta_1 x + \beta_2 (x - c)_+$ in matrix form and prove the identifiability condition stated in the text. Show that the continuous broken line is the same as the model with two separate lines, one on each side of c , constrained to meet at c , and count parameters to explain the difference of one.

Exercise B3.

In the interaction model, suppose all cells are occupied except $(1, 1)$. Show that the $(a - 1)(b - 1)$ contrasts $\gamma_{ij} - \gamma_{i1} - \gamma_{1j} + \gamma_{11}$ are no longer estimable, but that $\gamma_{ij} - \gamma_{i2} - \gamma_{1j} + \gamma_{12}$ is estimable for $i \geq 2, j \geq 3$. How many linearly independent estimable interaction contrasts remain?

8.6.5.3 C. Going deeper

Exercise C1.

In the additive model, let the design graph have components $C_1, \dots, C_c$. Show that $\lambda_0 \mu + \sum_i \lambda_i \alpha_i + \sum_j \kappa_j \beta_j$ is estimable iff $\lambda_0 = \sum_j \kappa_j$ and, for every component C , $\sum_{i \in C} \lambda_i = \sum_{j \in C} \kappa_j$. Deduce [Theorem 8.6.1\(b\)](#).

Solution. From the proof of [Theorem 8.6.1](#), the null vectors are $(m, \mathbf{u}, \mathbf{v})$ with $u_i = t_{C(i)}$ and $v_j = -m - t_{C(j)}$, for arbitrary $m, t_1, \dots, t_c$. The inner product with the coefficient vector is

$$\lambda_0 m + \sum_i \lambda_i t_{C(i)} - \sum_j \kappa_j (m + t_{C(j)}) = m \left(\lambda_0 - \sum_j \kappa_j \right) + \sum_C t_C \left(\sum_{i \in C} \lambda_i - \sum_{j \in C} \kappa_j \right).$$

This vanishes for all m, t iff every bracket is zero, which is the stated condition ([Theorem 8.3.1\(c\)](#)). For $\alpha_i - \alpha_k$, $\lambda_0 = 0$, all $\kappa_j = 0$, and the component sums of λ are zero iff i and k lie in the same component.

8.7 Contrasts

In the one-way model the individual effects α_k are not estimable, but their differences are. The general form of such a difference is a contrast. Contrasts are the natural targets of inference about a factor. They are estimable, they do not depend on the side condition or coding, and they are what a scientific question about a factor usually asks: does treatment beat control, is there a trend across doses, do two groups of levels differ? This section characterizes them, shows how an orthogonal set of contrasts splits the factor's sum of squares, and constructs the Helmert and polynomial families.

8.7.1 Definition and estimability

Throughout, the one-way model is $E(y_{kj}) = \mu + \alpha_k$ for observation j at level k , with $n_k \geq 1$ observations at level $k = 1, \dots, g$, level means $\mu_k = \mu + \alpha_k$ and sample means $\bar{y}_k$.

Definition 8.7.1 — Contrast

In a factor model, a **contrast** in the effects of a factor is a linear function $\sum_k c_k \alpha_k$ whose coefficients satisfy $\sum_k c_k = 0$, not all zero. In the one-way model it equals $\sum_k c_k \mu_k$, and we call either form a contrast in the level means. Two contrasts are **orthogonal** (for the given design) if $\sum_k c_k d_k / n_k = 0$.

The two forms agree because $\sum_k c_k \mu_k = \mu \sum_k c_k + \sum_k c_k \alpha_k$. This is where the condition $\sum_k c_k = 0$ comes from: it makes the unidentified μ drop out. Simple differences $\alpha_k - \alpha_l$, comparisons of one level with the average of others such as $\alpha_1 - \frac{1}{2}(\alpha_2 + \alpha_3)$, and trend coefficients are all contrasts.

Theorem 8.7.1 — Estimable functions in the one-way model

In the one-way model with every $n_k \geq 1$, the function $\lambda_0 \mu + \sum_k \lambda_k \alpha_k$ is estimable iff $\lambda_0 = \sum_k \lambda_k$. In particular, a linear function of the effects alone, $\sum_k c_k \alpha_k$, is estimable iff it is a contrast (or zero). For a contrast, the least squares estimator and its variance under $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ are

$$\hat{\psi} = \sum_k c_k \bar{y}_k, \quad \text{Var}(\hat{\psi}) = \sigma^2 \sum_k \frac{c_k^2}{n_k}. \quad (8.7.1)$$

Proof. The null space of $\mathbf{X} = [\mathbf{1}, \mathbf{Z}]$ is spanned by $\mathbf{v} = (1, -1, \dots, -1)^\top$ (Example (A rank-deficient layout)). By Theorem 8.3.1(c), the function is estimable iff $\lambda_0 - \sum_k \lambda_k = 0$. For an estimable function, $\lambda_0 \mu + \sum_k \lambda_k \alpha_k = \sum_k \lambda_k (\mu + \alpha_k)$. Every least squares solution has $\hat{\mu} + \hat{\alpha}_k = \bar{y}_k$, since the fitted values are the level means (Example (The one-way layout)). So the estimate is $\sum_k \lambda_k \bar{y}_k$. The level means are uncorrelated with variances σ^2/n_k , by Theorem 2.2.1, which gives the variance. ■

The estimator in Eq. 8.7.1 is the same whatever coding or side condition was used to fit the model. Proposition 8.3.2 guarantees this, and it is the reason contrasts, rather than coefficients, should be reported.

8.7.2 Sums of squares and orthogonal contrasts

Each contrast defines a direction in the observation space, and its estimate is the component of $\mathbf{y}$ along that direction. With $\mathbf{D} = \text{diag}(n_1, \dots, n_g)$ and $\mathbf{Z}$ the indicator matrix, put

$$\mathbf{u}_c = \mathbf{ZD}^{-1}\mathbf{c}, \quad (8.7.2)$$

the vector whose entries equal c_k/n_k for the observations at level k .

Proposition 8.7.2 — Contrasts as projections

Let $\mathbf{c}$ and $\mathbf{d}$ be contrasts, and write $\hat{\psi}_c = \mathbf{c}^\top \hat{\boldsymbol{\mu}}$. Then:

- (a) $\hat{\psi}_c = \mathbf{u}_c^\top \mathbf{y}$, and $\mathbf{u}_c$ is a nonzero vector in $\mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$;
- (b) the **sum of squares for the contrast**,

$$\text{SS}(\mathbf{c}) = \frac{\hat{\psi}_c^2}{\sum_k c_k^2/n_k},$$

equals $\|\mathbf{P}\mathbf{y}\|^2$, where $\mathbf{P}$ is the orthogonal projection onto the line spanned by $\mathbf{u}_c$;

- (c) $\text{Cov}(\hat{\psi}_c, \hat{\psi}_d) = \sigma^2 \sum_k c_k d_k / n_k = \sigma^2 \mathbf{u}_c^\top \mathbf{u}_d$ when $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, so orthogonal contrasts have uncorrelated estimates and orthogonal vectors $\mathbf{u}$;
- (d) if $\mathbf{c}_1, \dots, \mathbf{c}_{g-1}$ are pairwise orthogonal contrasts, then

$$\sum_{j=1}^{g-1} \text{SS}(\mathbf{c}_j) = \sum_{k=1}^g n_k (\bar{y}_k - \bar{y})^2,$$

the sum of squares between levels.

Proof.

- (a) $\hat{\boldsymbol{\mu}} = \mathbf{D}^{-1}\mathbf{Z}^\top \mathbf{y}$ (Eq. 8.5.1), so $\mathbf{c}^\top \hat{\boldsymbol{\mu}} = (\mathbf{Z}\mathbf{D}^{-1}\mathbf{c})^\top \mathbf{y}$. Clearly $\mathbf{u}_\mathbf{c} \in \mathcal{C}(\mathbf{Z})$, and $\mathbf{1}^\top \mathbf{u}_\mathbf{c} = \mathbf{1}^\top \mathbf{Z}\mathbf{D}^{-1}\mathbf{c} = (n_1, \dots, n_g)\mathbf{D}^{-1}\mathbf{c} = \sum_k c_k = 0$. It is nonzero because $\mathbf{Z}\mathbf{D}^{-1}$ has full column rank and $\mathbf{c} \neq \mathbf{0}$.
- (b) By Example (Projections onto a line), $\|\mathbf{P}\mathbf{y}\|^2 = (\mathbf{u}_\mathbf{c}^\top \mathbf{y})^2 / \|\mathbf{u}_\mathbf{c}\|^2$, and $\|\mathbf{u}_\mathbf{c}\|^2 = \mathbf{c}^\top \mathbf{D}^{-1}\mathbf{Z}^\top \mathbf{Z}\mathbf{D}^{-1}\mathbf{c} = \mathbf{c}^\top \mathbf{D}^{-1}\mathbf{c} = \sum_k c_k^2 / n_k$.
- (c) Theorem 2.2.1 gives $\text{Cov}(\mathbf{u}_\mathbf{c}^\top \mathbf{Y}, \mathbf{u}_\mathbf{d}^\top \mathbf{Y}) = \sigma^2 \mathbf{u}_\mathbf{c}^\top \mathbf{u}_\mathbf{d}$, and the same computation as in (b) gives $\mathbf{u}_\mathbf{c}^\top \mathbf{u}_\mathbf{d} = \mathbf{c}^\top \mathbf{D}^{-1}\mathbf{d}$.
- (d) The space $\mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$ has dimension $g - 1$ (Theorem 6.5.1(c), with $\mathcal{C}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{Z})$). By (a) and (c) the vectors $\mathbf{u}_{\mathbf{c}_1}, \dots, \mathbf{u}_{\mathbf{c}_{g-1}}$ are $g - 1$ nonzero, mutually orthogonal vectors in it, so they form an orthogonal basis. By Theorem 6.3.8 the sum of the projections onto the lines they span is the projection onto that space, which is $\mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top$ (Theorem 6.5.1(b)). Applying it to $\mathbf{y}$ and using Pythagoras gives $\sum_j \text{SS}(\mathbf{c}_j) = \|(\mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{y}\|^2$. The vector $(\mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{y}$ has entry $\bar{y}_k - \bar{y}$ at every observation of level k .

■

So a complete set of orthogonal contrasts decomposes the between-level sum of squares into $g - 1$ one-dimensional pieces. Under normality the pieces are independent, each is σ^2 times a noncentral χ^2 with one degree of freedom, and each is independent of the residual sum of squares (Theorem 4.4.4). Each contrast can therefore be tested on its own. Chapter 9 places this decomposition in the general theory of orthogonal sums of squares, and Chapter 15 uses it for the one-way analysis of variance. In a *balanced* design, with all n_k equal, orthogonality reduces to the ordinary $\sum_k c_k d_k = 0$. In an unbalanced design, contrasts that look orthogonal on paper need not be orthogonal for the data.

8.7.3 Helmert contrasts

The Helmert matrix of Example (Helmert matrices) has rows proportional to

$$\mathbf{h}_k = (\underbrace{1, \dots, 1}_{k-1}, -(k-1), 0, \dots, 0), \quad k = 2, \dots, g.$$

The contrast $\mathbf{h}_k^\top \boldsymbol{\mu}$ compares level k with the average of the levels before it, since it is $(k - 1)$ times $\frac{1}{k-1} \sum_{l < k} \mu_l - \mu_k$. The $\mathbf{h}_k$ are mutually orthogonal in the ordinary sense, so in a balanced design they are orthogonal contrasts. They suit factors whose levels are added in a meaningful order, such as successive refinements of a treatment, where each new level is compared with everything before it.

8.7.4 Polynomial contrasts

When the levels of a factor are ordered and equally spaced, such as doses 0, 10, 20, 30 or the seven points of a party scale, the natural questions are about the *shape* of the sequence of means. Is there a linear trend? Curvature? Give the levels scores $s_k = k$. Apply Gram–Schmidt (Proposition 6.2.2) to the vectors $\mathbf{1}, \mathbf{s}, \mathbf{s}^2, \dots, \mathbf{s}^{g-1}$ in $\mathbb{R}^g$, where $\mathbf{s}^j$ has entries s_k^j . This gives orthogonal vectors $\mathbf{p}_0, \dots, \mathbf{p}_{g-1}$, and $\mathbf{p}_j$ is a polynomial of degree j in the score. For $j \geq 1$,

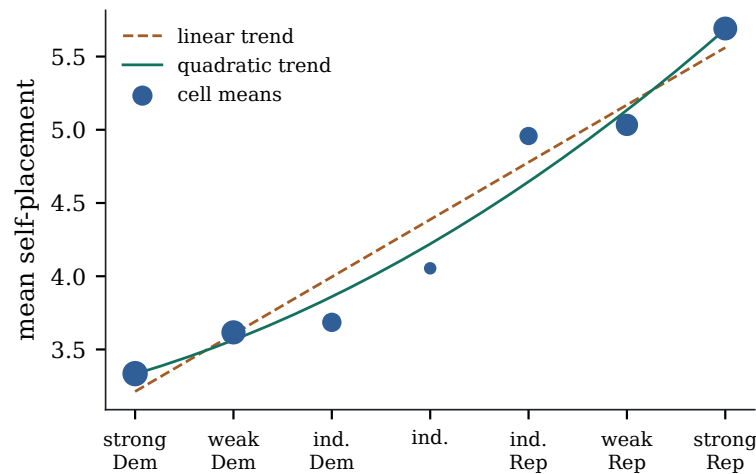


Figure 8.7.1: Mean ideological self-placement by party identification (point area proportional to group size), with the weighted least squares linear and quadratic trends in the score. The linear contrast accounts for almost all of the variation between groups.

$\mathbf{p}_j \perp \mathbf{1}$, so $\mathbf{p}_j$ is a contrast. Rescaled to integers, the first three for $g = 7$ are

linear $(-3, -2, -1, 0, 1, 2, 3)$, quadratic $(5, 0, -3, -4, -3, 0, 5)$, cubic $(-1, 1, 1, 0, -1, -1, 1)$.

By construction $\mathbf{p}_j$ is orthogonal to every polynomial of degree less than j . So if the level means follow a polynomial of degree d in the score, every contrast of degree above d is zero. The linear contrast measures the trend, the quadratic the curvature beyond it, and so on. This is the reason for their use.

With unequal group sizes the classical integer contrasts are no longer orthogonal for the design. The fix follows [Proposition 8.7.2\(c\)](#). Orthogonality is needed in the inner product $\sum_k c_k d_k / n_k$, which suggests setting $\mathbf{c}_j = \mathbf{D}\mathbf{p}_j$, where now the $\mathbf{p}_j$ are orthogonal polynomials in the weighted inner product $\sum_k n_k p_k q_k$. Then $\sum_k c_{jk} = \sum_k n_k p_{jk} = 0$, and $\sum_k c_{jk} c_{lk} / n_k = \sum_k n_k p_{jk} p_{lk} = 0$ for $j \neq l$. In observation space, $\mathbf{u}_{\mathbf{c}_j} = \mathbf{Z}\mathbf{p}_j$ is the polynomial of degree j in each observation's score, orthogonalized over the observations. The weighted linear contrast gives exactly the regression sum of squares of $\mathbf{y}$ on the score ([Exercise \(C1\)](#)).

Example. (Trend in ideology across party identification)

Return to the seven party-identification groups of [Example \(Party identification and ideology\)](#). The between-group sum of squares is 772.87. The classical linear contrast has estimate $\hat{\psi} = \sum_k (k-4)\bar{y}_k = 11.175$, variance $0.1652 \sigma^2$ by [Eq. 8.7.1](#), and sum of squares 755.85, which is 97.8 percent of the between-group sum of squares. The quadratic and cubic contrasts have sums of squares 10.15 and 2.58. Ideology rises almost linearly across the seven-point party scale, with some upward curvature ([Figure 8.7.1](#)).

The group sizes range from 37 to 200, so the six classical contrasts are not orthogonal for this design. Their sums of squares add to 784.02, which exceeds the between-group total. Part of the variation is counted twice. The weighted orthogonal polynomial contrasts decompose the total exactly: 746.57 for the linear trend and 26.30 for the five higher-degree contrasts together.

```
import numpy as np
import statsmodels.api as sm
```



```

anes = sm.datasets.anes96.load_pandas().data
y = anes["selfLR"].to_numpy()
level = anes["PID"].to_numpy().astype(int)          # 0 = strong Dem. ... 6 = strong
Rep.
g = 7
n_k = np.bincount(level).astype(float)
means = np.bincount(level, weights=y) / n_k

```

```

def orthogonal_polynomials(scores, weights):
    """Columns 1..g-1: polynomials in the scores, orthogonal in sum_k w_k u_k v_k.
    """
    V = np.vander(scores, len(scores), increasing=True)      # 1, s, s^2, ...
    W = np.sqrt(weights)[: , None]
    Q, _ = np.linalg.qr(W * V)                               # weighted Gram-Schmidt
    P = Q / W                                                 # back to the original
    scale                                                    #
    P = P / np.abs(P).max(axis=0)                            # cosmetic rescaling
    return P[:, 1:]                                           # drop the constant
    column

def contrast_ss(c, means, n_k):
    """Estimate c'μ-hat and its sum of squares (c'μ-hat)^2 / sum(c_k^2 / n_k)."""
    psi = c @ means
    return psi, psi ** 2 / np.sum(c ** 2 / n_k)

scores = np.arange(g, dtype=float)
C_unw = orthogonal_polynomials(scores, np.ones(g))           # the classical table (
    equal n)
C_wtd = n_k[:, None] * orthogonal_polynomials(scores, n_k)   # c_k = n_k p(k)
ss_between = np.sum(n_k * (means - y.mean()) ** 2)
for name, C in [("unweighted", C_unw), ("weighted", C_wtd)]:
    ss = [contrast_ss(C[:, j], means, n_k)[1] for j in range(g - 1)]
    print(f"{name:10s} SS by degree: {np.round(ss, 2)} total {sum(ss):.2f}",
          f"(between groups {ss_between:.2f})")

```

8.7.5 Coding matrices are not contrast matrices

Section 8.5 coded a factor by an intercept and the columns ZC . When the columns of C are contrasts, as for Helmert or polynomial coding, it is tempting to read the coefficient of column j as the estimate of the contrast $\mathbf{c}_j^\top \boldsymbol{\mu}$. That reading is wrong in general. By Proposition 8.5.1(b) the coefficients are the rows of $\mathbf{K}^{-1}$, with $\mathbf{K} = [\mathbf{1}, \mathbf{C}]$, applied to $\hat{\boldsymbol{\mu}}$. The rows of $\mathbf{K}^{-1}$ and the columns of $\mathbf{K}$ are related simply only when the columns are orthogonal.

Proposition 8.7.3 — Orthogonal codings

If the columns of $\mathbf{K} = [\mathbf{1}_g, \mathbf{c}_1, \dots, \mathbf{c}_{g-1}]$ are mutually orthogonal, then the coefficients of the coded model are

$$\hat{\gamma}_0 = \frac{1}{g} \sum_k \bar{y}_k, \quad \hat{\gamma}_j = \frac{\mathbf{c}_j^\top \hat{\boldsymbol{\mu}}}{\mathbf{c}_j^\top \mathbf{c}_j}, \quad j = 1, \dots, g-1,$$

whatever the group sizes.

Proof. With orthogonal columns, $\mathbf{K}^\top \mathbf{K} = \text{diag}(g, \mathbf{c}_1^\top \mathbf{c}_1, \dots, \mathbf{c}_{g-1}^\top \mathbf{c}_{g-1})$, so $\mathbf{K}^{-1} = (\mathbf{K}^\top \mathbf{K})^{-1} \mathbf{K}^\top$. Apply [Proposition 8.5.1\(b\)](#). ■

The orthogonality required here is the plain one, $\mathbf{c}_j^\top \mathbf{c}_l = 0$, not the design orthogonality of [Definition 8.7.1](#), because $\hat{\boldsymbol{\gamma}} = \mathbf{K}^{-1} \hat{\boldsymbol{\mu}}$ is pure algebra and does not involve the n_k . Helmert coding, in the usual software convention, uses the columns $\mathbf{c}_j = -\mathbf{h}_{j+1}$, which compare level $j+1$ with the levels before it, and $\mathbf{c}_j^\top \mathbf{c}_j = j(j+1)$. In the party data the first three Helmert coefficients are 0.1408, 0.0698 and 0.1271, while the contrasts themselves are 0.2817, 0.4187 and 1.5253, larger by the factors 2, 6 and 12. For a non-orthogonal coding there is no such shortcut. Reference coding has $\mathbf{c}_j = \mathbf{e}_{j+1}$, yet its coefficients are $\mu_{j+1} - \mu_1$, not μ_{j+1} .

```
def helmert(g):
    """Coding matrix whose column k compares level k+1 with the mean of levels 1..k.
    """
    C = np.zeros((g, g - 1))
    for k in range(1, g):
        C[:k, k - 1] = -1.0
        C[k, k - 1] = k
    return C

H = helmert(g)
K = np.column_stack([np.ones(g), H])
gamma = np.linalg.solve(K, means)           # Helmert-coded coefficients
psi = H.T @ means                          # the contrasts c_k' mu-hat
print("coefficients:", np.round(gamma[1:], 4))
print("contrasts:    ", np.round(psi, 4))
print("ratio:        ", np.round(psi / gamma[1:], 1))  # = c_k' c_k = k(k+1)
```

Warning. A coefficient from a contrast-coded regression estimates a contrast only up to a scale factor, and only when the coding columns are orthogonal. To test or report a contrast, compute $\mathbf{c}^\top \hat{\boldsymbol{\mu}}$ and its variance [Eq. 8.7.1](#) directly, or read it from the coded fit using $\mathbf{K}^{-1}$. Do not read it off the coefficient table.

8.7.6 Contrasts in the two-way additive model

In the additive model $\mu + \alpha_i + \beta_j$ of [Section 8.6](#), a **row contrast** is $\sum_i c_i \alpha_i$ with $\sum_i c_i = 0$. It is a combination of the differences $\alpha_i - \alpha_1$, so in a connected design it is estimable ([Theorem 8.6.1](#)). In a balanced design, with $m \geq 1$ observations in every one of the ab cells, the least squares fit is $\bar{y}_{i..} + \bar{y}_{.j.} - \bar{y}_{...}$. This is shown in [Example \(Additive two-factor model, one observation per cell\)](#) for $m = 1$, and the averaging argument there is unchanged for $m > 1$. Hence

$$\widehat{\sum_i c_i \alpha_i} = \sum_i c_i (\bar{y}_{i..} + \bar{y}_{.j.} - \bar{y}_{...}) = \sum_i c_i \bar{y}_{i..}, \quad \text{Var} = \frac{\sigma^2}{bm} \sum_i c_i^2,$$

since the terms not depending on i are multiplied by $\sum_i c_i = 0$. The row contrast is estimated from the row means alone, as if factor B were absent. In an unbalanced design this is false. The row means are then contaminated by the column effects ([Exercise \(B2\)](#)), and the least squares

estimate has to adjust for them. That adjustment is the subject of Chapter 17. Interaction contrasts, and their estimability when cells are empty, were treated in [Proposition 8.6.2](#).

8.7.7 Exercises

8.7.7.1 A. Check your understanding

Exercise A1.

Find the linear, quadratic and cubic orthogonal polynomial contrasts for $g = 4$ equally spaced levels, scaled to integers. Check that they are mutually orthogonal and that the quadratic contrast vanishes when $\mu_k = a + bk$.

Exercise A2.

Show that the Helmert contrasts $\mathbf{h}_2, \dots, \mathbf{h}_g$ are orthogonal for the design iff $n_1 = n_2 = \dots = n_{g-1}$, whatever n_g is. *Hint:* compute $\sum_k h_{jk}h_{lk}/n_k$ for $j < l$.

8.7.7.2 B. Practice

Exercise B1.

Show that for every contrast $\mathbf{c}$, $\text{SS}(\mathbf{c}) \leq \sum_k n_k (\bar{y}_k - \bar{y})^2$, with equality when $c_k = n_k(\bar{y}_k - \bar{y})$. This fact is the basis of Scheffé's method in Chapter 13.

Solution. By [Proposition 8.7.2\(b\)](#), $\text{SS}(\mathbf{c}) = \|\mathbf{P}\mathbf{y}\|^2$ with $\mathbf{P}$ projecting onto a line inside $\mathcal{S} = \mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$. Let $\mathbf{P}_\mathcal{S} = \mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top$. Because the line lies in $\mathcal{S}$, $\mathbf{P}\mathbf{y} = \mathbf{P}\mathbf{P}_\mathcal{S}\mathbf{y}$ ([Theorem 6.5.1\(a\)](#)), and so $\|\mathbf{P}\mathbf{y}\|^2 \leq \|\mathbf{P}_\mathcal{S}\mathbf{y}\|^2 = \sum_k n_k (\bar{y}_k - \bar{y})^2$. Equality holds when the line contains $\mathbf{P}_\mathcal{S}\mathbf{y}$, whose entries are $\bar{y}_k - \bar{y}$ at level k . That vector is $\mathbf{u}_\mathbf{c}$ for $c_k = n_k(\bar{y}_k - \bar{y})$, and this $\mathbf{c}$ is a contrast because $\sum_k n_k(\bar{y}_k - \bar{y}) = 0$.

Exercise B2.

In the additive two-way model with n_{ij} observations in cell (i, j) , show that $E(\bar{y}_{i..}) = \mu + \alpha_i + \sum_j n_{ij}\beta_j/n_{i..}$. Deduce that $\sum_i c_i \bar{y}_{i..}$ is unbiased for every row contrast and every β iff the proportions $n_{ij}/n_{i..}$ do not depend on i , that is, iff the cell counts are *proportional*, $n_{ij} = n_{i..}n_{.j}/n$.

8.7.7.3 C. Going deeper

Exercise C1.

Let $\mathbf{x}$ be the n -vector giving each observation's score s_k . Show that the sum of squares of the weighted linear contrast $\mathbf{c}_1 = \mathbf{D}\mathbf{p}_1$ equals the regression sum of squares $\|(\mathbf{M}_1 - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{y}\|^2$ of the simple regression of $\mathbf{y}$ on $\mathbf{x}$ with intercept, where $\mathbf{M}_1$ projects onto

$\mathcal{C}([1, \mathbf{x}])$. Conclude that the remaining weighted contrasts together measure the lack of fit of the straight line to the level means.

Solution. In the weighted inner product, $\mathbf{p}_1$ is $\mathbf{s}$ minus its weighted mean, $\mathbf{s} - \bar{s}_w \mathbf{1}$ with $\bar{s}_w = \sum_k n_k s_k / n$, up to scale. Then $\mathbf{u}_{c_1} = \mathbf{ZD}^{-1} \mathbf{Dp}_1 = \mathbf{Zp}_1$, whose entries are $s_k - \bar{s}_w$. That is $\mathbf{x} - \bar{x} \mathbf{1}$, the centred score vector. By Proposition 8.7.2(b), $\text{SS}(\mathbf{c}_1)$ is the squared length of the projection of $\mathbf{y}$ onto the line spanned by $\mathbf{x} - \bar{x} \mathbf{1}$. By Example (Intercept inside simple regression) this line is $\mathcal{C}([1, \mathbf{x}]) \cap \mathbf{1}^\perp$, and the projection onto it is $\mathbf{M}_1 - n^{-1} \mathbf{1} \mathbf{1}^\top$. The remaining weighted contrasts span the orthogonal complement of that line in $\mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$, which is $\mathcal{C}(\mathbf{Z}) \cap \mathcal{C}([1, \mathbf{x}])^\perp$. Their total sum of squares is $\|(\mathbf{M} - \mathbf{M}_1) \mathbf{y}\|^2$, the reduction in residual sum of squares from the straight-line model to the model with a separate mean for each level.

8.8 Summary and Notes

8.8.1 Summary

Idea.

1. The data determine the mean vector $\mathbf{X}\boldsymbol{\beta}$ and nothing more. A function of $\boldsymbol{\beta}$ is identifiable iff it is a function of $\mathbf{X}\boldsymbol{\beta}$, and $\boldsymbol{\beta}$ itself is identifiable iff $\text{rank}(\mathbf{X}) = p$ (Definition 8.1.1, Theorem 8.1.1 and Proposition 8.1.2).
2. A single coefficient is identifiable iff its column is not a combination of the others (Proposition 8.1.3). Rank deficiency comes from construction, from identities among variables (age, period and cohort), or from the design (Example (Age, period and cohort)).
3. A linear function $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ has a linear unbiased estimator iff it is identifiable iff $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ (Definition 8.2.1 and Theorem 8.2.2). The estimable functions form a space of dimension $\text{rank}(\mathbf{X})$ (Corollary 8.2.3).
4. Eight equivalent tests of estimability, including $\boldsymbol{\lambda} \perp \mathcal{N}(\mathbf{X})$, $\boldsymbol{\lambda}^\top \mathbf{G} \mathbf{X}^\top \mathbf{X} = \boldsymbol{\lambda}^\top$, and invariance over least squares solutions (Theorem 8.3.1). The estimate is $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ for any solution, with variance $\sigma^2 \boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}$ for any generalized inverse (Proposition 8.3.2).
5. For a nonestimable $\boldsymbol{\lambda}$, a generalized-inverse solution estimates a different, estimable function $\boldsymbol{\lambda}^\top \mathbf{G} \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}$, and the reported variance can be any number (Proposition 8.3.3 and Exercise (C1)). Decide estimability with the SVD of $\mathbf{X}$, and treat near-nonestimable functions as unestimated (Example (Estimable in principle, not in practice)).
6. Full-rank reparameterizations are choices of basis for the estimable space (Theorem 8.4.1). Side conditions $\mathbf{T}\mathbf{b} = \mathbf{0}$ select a unique solution iff $\text{rank}(\mathbf{T}) = p - r$ and no combination of them is estimable (Theorem 8.4.2). The constrained coefficients are then particular estimable functions (Proposition 8.4.3). The minimum-norm solution is itself a side condition (Proposition 8.4.4).
7. A coding of a factor is a nonsingular matrix $\mathbf{K}$, and its coefficients estimate $\mathbf{K}^{-1} \boldsymbol{\mu}$, the rows of $\mathbf{K}^{-1}$ applied to the level means (Definition 8.5.1 and Proposition 8.5.1).
8. In the additive two-way model the rank is $a + b$ minus the number of connected

components of the design, and a difference of levels is estimable iff the levels are connected ([Theorem 8.6.1](#)). In the interaction model, exactly the combinations of occupied cell means are estimable ([Proposition 8.6.2](#)).

9. With a covariate, adjusted differences between levels subtract the slope times the difference in covariate means, the slope being estimated within levels ([Proposition 8.6.3](#)).
10. Contrasts are the estimable functions of a factor's effects ([Definition 8.7.1](#) and [Theorem 8.7.1](#)). Orthogonal contrasts, in the design inner product $\sum_k c_k d_k / n_k$, decompose the between-level sum of squares ([Proposition 8.7.2](#)). Coding coefficients equal scaled contrasts only for orthogonal codings ([Proposition 8.7.3](#)).

8.8.2 Notes and sources

Identifiability and estimability. Christensen (2020, §2.1) develops estimability from identifiability, as this chapter does, and stresses that the estimable functions are exactly the identifiable linear ones. Agresti (2015, §1.4) gives a short treatment aimed at generalized linear models, where the same definitions apply to the linear predictor. Rencher and Schaalje (2008, ch. 12) is a detailed textbook account of the non-full-rank model: estimable functions and their tests, the estimators $\mathbf{r}^\top \mathbf{X}^\top \mathbf{y}$ and $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$, reparameterization and side conditions, with small worked layouts. The notion of an estimable function goes back to R. C. Bose's work on the design of experiments in the 1940s. Scheffé (1959) and Searle (1971) made it standard, and Rao (1973) treats it through generalized inverses. The characterization [Theorem 8.3.1](#) collects conditions that are scattered through these sources. That a generalized-inverse solution estimates the estimable function $\boldsymbol{\lambda}^\top \mathbf{G} \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}$ ([Proposition 8.3.3](#)) follows at once from $E(\mathbf{G} \mathbf{X}^\top \mathbf{Y}) = \mathbf{G} \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}$, but its consequence for software output deserves more emphasis than it usually gets.

Side conditions and software. The exact conditions of [Theorem 8.4.2](#), and the formula $(\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})^{-1} \mathbf{X}^\top \mathbf{y}$, are standard (see Rencher and Schaalje 2008, §12.6, and Searle 1971). The coding matrices, called “contrasts”, that S and R use for factors are described by Chambers and Hastie (1992), and Venables and Ripley (2002) discuss the treatment, sum, Helmert and polynomial codings of R. The minimum-norm solution rests on the Moore–Penrose inverse (Penrose 1955), and column-pivoted QR is described in Golub and Van Loan (2013).

Indicator variables. Sen and Srivastava (1990, ch. 4) is a practical account of indicator variables: the two-sample comparison as a regression, polychotomous factors, hierarchical codings, factors with continuous covariates, broken-line regression and aggregated indicators as responses. Where this chapter meets those themes it takes its own route: the two-sample comparison through the coding proposition ([Exercise \(A1\)](#)), the indicator of a single observation through the Frisch–Waugh–Lovell theorem ([Exercise \(B1\)](#)), and count regressors without intercept in a new setting ([Exercise \(B3\)](#)).

Connectedness, empty cells and cohorts. Connectedness of two-way designs, and its role in estimability, is treated at length by Searle (1971, 1987). Searle (1987) is also the standard reference for unbalanced data and empty cells, which Chapter 17 takes up. The age–period–cohort problem of [Example \(Age, period and cohort\)](#) is a well-known instance of a dependency among variables. Holford (1983) showed which functions of the age, period and cohort effects are estimable, among them the second differences of [Exercise \(C1\)](#).

Data. The survey examples use the extract of the 1996 American National Election Study distributed with statsmodels (Seabold and Perktold 2010) as a public-domain data set. The other examples use simulated data, generated by the scripts in `code/ch08/`.

8.8.3 References

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Chapter 9

Sums of Squares and Orthogonal Decomposition

An analysis of variance table looks like bookkeeping. Behind every row there is a subspace, and behind the whole table an orthogonal decomposition of the space in which the data live: sums of squares are squared lengths of projections, and degrees of freedom are dimensions. Sequential and partial sums of squares, and the Type I, II and III sums of squares printed by software, differ only in which subspaces they project onto, and so in which hypotheses they address. When the subspaces are orthogonal, as in balanced designs and with orthogonal polynomials, the distinctions disappear. The coefficient of determination is a ratio of two of these squared lengths, and we mark what it does not measure. Finally, the expectation of each sum of squares shows what an F ratio compares, preparing the tests of Part III.

What you need. Chapter 6 throughout, especially projections and their ranks ([Theorem 6.3.8](#), [Proposition 6.3.4](#)), nested models ([Theorem 6.5.1](#), [Theorem 6.5.2](#)), the Frisch–Waugh–Lovell theorem ([Theorem 6.6.2](#)) and R^2 as a squared cosine ([Theorem 6.7.2](#)). The linear model and the moments of least squares estimates from [Chapter 5](#) ([Definition 5.3.1](#), [Theorem 5.5.1](#)). The mean of a quadratic form ([Theorem 2.3.1](#)) and the distribution theory of [Chapter 4](#) ([Theorem 4.4.4](#), [Theorem 4.5.1](#)). Estimable functions and contrasts from [Chapter 8](#) appear in [Sections 9.2 to 9.4](#).

9.0.1 Roadmap

- [Decomposing the total sum of squares](#)
- [Extra sums of squares](#)
- [Sequential and partial sums of squares](#)
- [Orthogonal designs](#)
- [The coefficient of determination and its limits](#)
- [Expected mean squares](#)
- [Summary and notes](#)

9.1 Decomposing the total sum of squares

Every sum of squares in a linear model is the squared length of a projection of the data. [Chapter 6](#) showed this for the residual sum of squares, $\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2$, and for the drop in residual sum of squares between nested models, $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2$ ([Eq. 6.5.2](#)). This section turns these observations into a general framework. An analysis of variance is a choice of mutually orthogonal subspaces that together fill the model space. Once that choice is made, the whole table follows from Pythagoras: the sums of squares add up because the pieces are orthogonal, and the degrees of freedom add up because dimensions of orthogonal subspaces add.

Throughout the chapter the model is $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ with $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$ ([Definition 5.3.1](#)). $\mathbf{X}$ is $n \times p$ of rank r , which may be less than p , and $\mathbf{M}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$. The identities of the chapter hold for every data vector $\mathbf{y}$. Probability enters only where a covariance, an expectation or a distribution is stated, and systematically in [Section 9.6](#).

9.1.1 Orthogonal decompositions of the model space

Definition 9.1.1 — Analysis of variance decomposition

An **analysis of variance decomposition** of the model space is a list of subspaces $\mathcal{V}_1, \dots, \mathcal{V}_k$ of $\mathcal{C}(\mathbf{X})$ that are mutually orthogonal and whose sum is $\mathcal{C}(\mathbf{X})$:

$$\mathcal{C}(\mathbf{X}) = \mathcal{V}_1 \oplus \mathcal{V}_2 \oplus \dots \oplus \mathcal{V}_k, \quad \mathcal{V}_i \perp \mathcal{V}_j \ (i \neq j).$$

The subspace $\mathcal{V}_{k+1} = \mathcal{C}(\mathbf{X})^\perp$ is the **error space**. Writing $\mathbf{P}_i$ for the orthogonal projection onto $\mathcal{V}_i$, the **sum of squares** for $\mathcal{V}_i$ is $\|\mathbf{P}_i\mathbf{y}\|^2 = \mathbf{y}^\top \mathbf{P}_i \mathbf{y}$, its **degrees of freedom** are $r_i = \dim \mathcal{V}_i$, and its **mean square** is $\|\mathbf{P}_i\mathbf{y}\|^2/r_i$.

The same space can be decomposed in many ways, and the choice is the statistician's: the constants, a regressor of interest, a block of nuisance regressors, the linear and quadratic parts of a trend. The theorem collects what is true of all decompositions.

Theorem 9.1.1 — The analysis of variance identity

Let $\mathcal{V}_1, \dots, \mathcal{V}_k$ be an analysis of variance decomposition of $\mathcal{C}(\mathbf{X})$, with projections $\mathbf{P}_1, \dots, \mathbf{P}_k$ and dimensions $r_1, \dots, r_k$. Then:

- (a) $\mathbf{P}_i\mathbf{P}_j = \mathbf{0}$ for $i \neq j$, $\mathbf{P}_1 + \dots + \mathbf{P}_k = \mathbf{M}$, and

$$\mathbf{I} = \mathbf{P}_1 + \mathbf{P}_2 + \dots + \mathbf{P}_k + (\mathbf{I} - \mathbf{M}),$$

a sum of $k + 1$ mutually orthogonal projections;

- (b) for every $\mathbf{y} \in \mathbb{R}^n$ the vectors $\mathbf{P}_1\mathbf{y}, \dots, \mathbf{P}_k\mathbf{y}, (\mathbf{I} - \mathbf{M})\mathbf{y}$ are mutually orthogonal and

$$\|\mathbf{y}\|^2 = \|\mathbf{P}_1\mathbf{y}\|^2 + \dots + \|\mathbf{P}_k\mathbf{y}\|^2 + \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2; \quad (9.1.1)$$

- (c) $r_i = \text{rank}(\mathbf{P}_i) = \text{tr}(\mathbf{P}_i)$, $r_1 + \dots + r_k = r$, and the error space has dimension $\text{tr}(\mathbf{I} - \mathbf{M}) = n - r$;

- (d) if the columns of $\mathbf{Q}_i$ ($n \times r_i$) form an orthonormal basis of $\mathcal{V}_i$, then $\|\mathbf{P}_i \mathbf{y}\|^2 = \|\mathbf{Q}_i^\top \mathbf{y}\|^2$ is a sum of r_i squares, and together with an orthonormal basis $\mathbf{Q}_{k+1}$ of the error space the matrix $\mathbf{U} = [\mathbf{Q}_1, \dots, \mathbf{Q}_{k+1}]$ is orthogonal;
- (e) the subspaces $\mathcal{W}_j = \mathcal{V}_1 \oplus \dots \oplus \mathcal{V}_j$ form a chain $\mathcal{W}_1 \subseteq \dots \subseteq \mathcal{W}_k = \mathcal{C}(\mathbf{X})$ with projections $\mathbf{N}_j = \mathbf{P}_1 + \dots + \mathbf{P}_j$, and $\mathbf{P}_j = \mathbf{N}_j - \mathbf{N}_{j-1}$ (with $\mathbf{N}_0 = \mathbf{0}$). Conversely, every chain $\{\mathbf{0}\} = \mathcal{W}_0 \subseteq \mathcal{W}_1 \subseteq \dots \subseteq \mathcal{W}_k = \mathcal{C}(\mathbf{X})$ gives an analysis of variance decomposition with $\mathcal{V}_j = \mathcal{W}_{j-1}^\perp \cap \mathcal{W}_j$.

Proof.

- (a) For any $\mathbf{u}$, $\mathbf{P}_j \mathbf{u} \in \mathcal{V}_j \subseteq \mathcal{V}_i^\perp$, so $\mathbf{P}_i \mathbf{P}_j \mathbf{u} = \mathbf{0}$ by Proposition 6.3.4(c). Applying Theorem 6.3.8 repeatedly, the sum $\mathbf{P}_1 + \dots + \mathbf{P}_k$ is the orthogonal projection onto $\mathcal{V}_1 \oplus \dots \oplus \mathcal{V}_k = \mathcal{C}(\mathbf{X})$, and a projection is determined by its range, so the sum is $\mathbf{M}$. Each $\mathbf{P}_i$ satisfies $\mathbf{M} \mathbf{P}_i = \mathbf{P}_i$, because its columns lie in $\mathcal{C}(\mathbf{X})$, and therefore $(\mathbf{I} - \mathbf{M}) \mathbf{P}_i = \mathbf{0}$. The projection $\mathbf{I} - \mathbf{M}$ onto $\mathcal{C}(\mathbf{X})^\perp$ (Theorem 6.3.3) is thus orthogonal to every $\mathbf{P}_i$.
- (b) Multiply $\mathbf{I} = \sum_i \mathbf{P}_i + (\mathbf{I} - \mathbf{M})$ by $\mathbf{y}$. For two different pieces, $(\mathbf{P}_i \mathbf{y})^\top (\mathbf{P}_j \mathbf{y}) = \mathbf{y}^\top \mathbf{P}_i \mathbf{P}_j \mathbf{y} = 0$, using symmetry and (a). Pythagoras gives the identity.
- (c) Rank equals trace for projections (Proposition 6.3.4(b)), and the trace is additive, so $\sum_i r_i = \text{tr}(\mathbf{M}) = r$ by Theorem 6.3.7, and $\text{tr}(\mathbf{I} - \mathbf{M}) = n - r$.
- (d) By Proposition 6.3.2, $\mathbf{P}_i = \mathbf{Q}_i \mathbf{Q}_i^\top$, so $\mathbf{y}^\top \mathbf{P}_i \mathbf{y} = \|\mathbf{Q}_i^\top \mathbf{y}\|^2$. Columns taken from different $\mathbf{Q}_i$ are orthogonal because the subspaces are, so $\mathbf{U}$ has orthonormal columns. It has $r_1 + \dots + r_k + (n - r) = n$ of them, so it is square and hence orthogonal.
- (e) The $\mathbf{N}_j$ are projections onto $\mathcal{W}_j$ by Theorem 6.3.8, and the chain is nested by construction. Conversely, given a chain, Theorem 6.5.1(b) shows that $\mathbf{N}_j - \mathbf{N}_{j-1}$ is the projection onto $\mathcal{V}_j = \mathcal{W}_{j-1}^\perp \cap \mathcal{W}_j$. For $i < j$ we have $\mathcal{V}_i \subseteq \mathcal{W}_i \subseteq \mathcal{W}_{j-1}$, which is orthogonal to $\mathcal{V}_j$. The projections telescope to $\mathbf{N}_k = \mathbf{M}$, so by Theorem 6.3.8 the orthogonal sum of the $\mathcal{V}_j$ is $\mathcal{C}(\mathbf{M}) = \mathcal{C}(\mathbf{X})$.

■

Part (e) says that decompositions and chains are the same thing. Listing the rows of an analysis of variance table from the top is the same as fitting a sequence of ever larger models and recording what each step adds. Section 9.3 builds on this.

9.1.2 Degrees of freedom are dimensions

“Degrees of freedom” can seem like a rule of thumb (“subtract one for every parameter estimated”). Part (d) of Theorem 9.1.1 gives the phrase an exact meaning. Rotate the data into the orthonormal coordinates $\mathbf{z} = \mathbf{U}^\top \mathbf{y}$. The rotation preserves length, and it splits $\mathbf{z}$ into consecutive blocks $\mathbf{Q}_1^\top \mathbf{y}, \dots, \mathbf{Q}_{k+1}^\top \mathbf{y}$ of sizes $r_1, \dots, r_k, n - r$. The sum of squares for $\mathcal{V}_i$ is the sum of the squares of the r_i coordinates in its block. *A sum of squares has r_i degrees of freedom because it is a sum of r_i squares of orthonormal linear combinations of the data, and it cannot be written with fewer.*

The rotation also explains why degrees of freedom govern the size of a sum of squares. If $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, the coordinates $\mathbf{U}^\top \mathbf{Y}$ are uncorrelated with variance σ^2 (Theorem 2.2.1), so a block of r_i of them has expected sum of squares $r_i \sigma^2$ plus the squared length of its mean

(Section 9.6). The divisor $n - 1$ of the sample variance is the case $\mathbb{R}^n = \text{span}(\mathbf{1}) \oplus \mathbf{1}^\perp$: the n deviations $y_i - \bar{y}$ lie in an $(n - 1)$ -dimensional subspace, and the Helmert matrix of [Example \(Helmert matrices\)](#) writes their sum of squares as $n - 1$ squared orthonormal contrasts.

9.1.3 The regression analysis of variance table

The most common decomposition comes from the chain $\text{span}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{X})$, available whenever $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, which is the case for every model with an intercept. Write $\mathbf{P}_1 = n^{-1}\mathbf{1}\mathbf{1}^\top$ for the projection onto the constant vectors ([Example \(Projections onto a line\)](#)). The chain gives the three-way decomposition of [Eq. 6.5.1](#),

$$\mathbf{I} = \mathbf{P}_1 + (\mathbf{M} - \mathbf{P}_1) + (\mathbf{I} - \mathbf{M}),$$

with ranks 1, $r - 1$ and $n - r$. The squared lengths are

$$\|\mathbf{P}_1 \mathbf{y}\|^2 = n\bar{y}^2, \quad \text{SSR} = \|(\mathbf{M} - \mathbf{P}_1) \mathbf{y}\|^2 = \sum_i (\hat{y}_i - \bar{y})^2, \quad \text{SSE} = \|(\mathbf{I} - \mathbf{M}) \mathbf{y}\|^2 = \sum_i (y_i - \hat{y}_i)^2.$$

They are usually arranged as follows.

Source	df	Sum of squares	Mean square
Mean	1	$n\bar{y}^2$	$n\bar{y}^2$
Regression (after the mean)	$r - 1$	SSR	$\text{SSR}/(r - 1)$
Residual	$n - r$	SSE	$s^2 = \text{SSE}/(n - r)$
Total	n	$\mathbf{y}^\top \mathbf{y}$	

This is the **uncorrected** table. Its total is the raw sum of squares $\sum_i y_i^2$, with n degrees of freedom. The row for the mean is rarely of interest, because it records only that the average response is not zero. Deleting it gives the **corrected** table, whose total is $\text{SST} = \sum_i (y_i - \bar{y})^2 = \|(\mathbf{I} - \mathbf{P}_1) \mathbf{y}\|^2$ on $n - 1$ degrees of freedom. The corrected identity $\text{SST} = \text{SSR} + \text{SSE}$ is [Eq. 6.7.1](#). “Correcting for the mean” is thus a projection. We first project the data onto $\mathbf{1}^\perp$ and then decompose what is left.

The ratio of the two mean squares,

$$F = \frac{\text{SSR}/(r - 1)}{\text{SSE}/(n - r)},$$

compares the variation along the regression directions with the variation in the error space. Under normality and when no regressor matters, it has the $F(r - 1, n - r)$ distribution ([Theorem 4.4.6](#)). [Section 9.6](#) explains why this ratio is the right comparison, and Chapter 11 develops the test.

A few formulas make the table easy to compute from any least squares solution.

Proposition 9.1.2 — Computing the sums of squares

Let $\hat{\boldsymbol{\beta}}$ be any least squares estimate. Then:

- (a) $\|\mathbf{M} \mathbf{y}\|^2 = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$ and $\text{SSE} = \mathbf{y}^\top \mathbf{y} - \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$;
- (b) if $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, then $\text{SSR} = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y} - n\bar{y}^2$;

- (c) if $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$ has full column rank and $\tilde{\mathbf{X}}_1 = (\mathbf{I} - \mathbf{P}_1)\mathbf{X}_1$ is the matrix of centred regressors, then $\text{SSR} = \hat{\boldsymbol{\beta}}_1^\top \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1 \hat{\boldsymbol{\beta}}_1 = \hat{\boldsymbol{\beta}}_1^\top \tilde{\mathbf{X}}_1^\top \mathbf{y}$, where $\hat{\boldsymbol{\beta}}_1$ holds the slopes.

Proof.

- (a) $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{M}\mathbf{y}$ (Theorem 6.4.1), so $\|\mathbf{M}\mathbf{y}\|^2 = \mathbf{y}^\top \mathbf{M}\mathbf{y} = \mathbf{y}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$. The second formula follows from $\text{SSE} = \mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{M}\mathbf{y}$ (Proposition 6.4.3). (b) $\mathbf{M} - \mathbf{P}_1$ is a projection (Theorem 6.5.1), so $\text{SSR} = \mathbf{y}^\top \mathbf{M}\mathbf{y} - \mathbf{y}^\top \mathbf{P}_1 \mathbf{y} = \|\mathbf{M}\mathbf{y}\|^2 - n\bar{y}^2$; now use (a). (c) By Lemma 6.6.1 with $\mathbf{X}_2 = \mathbf{1}$, $\mathbf{M} - \mathbf{P}_1$ is the projection onto $\mathcal{C}(\tilde{\mathbf{X}}_1)$, and by Theorem 6.6.2 the slopes are the coefficients of $(\mathbf{M} - \mathbf{P}_1)\mathbf{y}$ in terms of the columns of $\tilde{\mathbf{X}}_1$. So $(\mathbf{M} - \mathbf{P}_1)\mathbf{y} = \tilde{\mathbf{X}}_1 \hat{\boldsymbol{\beta}}_1$ and $\text{SSR} = \|\tilde{\mathbf{X}}_1 \hat{\boldsymbol{\beta}}_1\|^2$. The last equality uses $\tilde{\mathbf{X}}_1^\top \mathbf{y} = \tilde{\mathbf{X}}_1^\top (\mathbf{M} - \mathbf{P}_1)\mathbf{y} = \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1 \hat{\boldsymbol{\beta}}_1$. ■

In (a), $\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$ is the same for every solution of the normal equations, even though $\hat{\boldsymbol{\beta}}$ is not unique when $r < p$. That is no accident. It equals $\|\mathbf{M}\mathbf{y}\|^2$, which depends only on the model space.

Example. (An analysis of variance table for murder rates)

Return to the regression of the 2009 murder rate of the 50 US states on poverty, single parenthood and urbanization, with an intercept (Example (Poverty and murder rates)). Here $r = 4$ and $\bar{y} = 4.514$. The listing computes every sum of squares from the coordinates of $\mathbf{y}$ in an orthonormal basis of $\mathcal{C}(\mathbf{X})$, as in Theorem 9.1.1(d): the first coordinate gives the mean, the next three the regression.

Source	df	Sum of squares	Mean square
Mean	1	1018.81	1018.81
Regression	3	182.80	60.93
Residual	46	101.96	2.217
Total (uncorrected)	50	1303.57	
Total (corrected)	49	284.76	

The regression mean square is 27.49 times the residual mean square. If none of the three regressors were related to the murder rate and the errors were normal, a ratio this large would occur with probability about 2.5×10^{-10} . The proportion of the corrected total that the regression accounts for is $182.80/284.76 = 0.6419$, the R^2 of Theorem 6.7.2.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")

y = data["murder"].to_numpy()
n = len(y)
X = np.column_stack([np.ones(n), data["poverty"], data["single"], data["urban"]])
r = np.linalg.matrix_rank(X)

Q, _ = np.linalg.qr(X) # orthonormal basis of C(X); first column
```

```

    spans 1
z = Q.T @ y                                # coordinates of My in that basis
ss_mean = z[0] ** 2                        # n * ybar^2
ss_reg = np.sum(z[1:] ** 2)                # ||(M - P1) y||^2, r - 1 squared coordinates
ss_res = y @ y - np.sum(z ** 2)           # ||(I - M) y||^2

rows = [("mean", 1, ss_mean), ("regression", r - 1, ss_reg),
        ("residual", n - r, ss_res), ("total", n, y @ y)]
for name, df, ss in rows:
    print(f"{name:11s} df = {df:3d}    SS = {ss:9.2f}    MS = {ss / df:8.3f}")
F = (ss_reg / (r - 1)) / (ss_res / (n - r))
print(f"corrected total = {ss_reg + ss_res:.2f},    F = {F:.2f}")

```

The QR factorization supplies the orthonormal basis of [Theorem 9.1.1\(d\)](#); its first column is $\pm \mathbf{1}/\sqrt{n}$ because Gram–Schmidt starts with the intercept ([Proposition 6.2.2](#)). No $n \times n$ projection matrix is formed.

9.1.4 Models without an intercept

The corrected table depends on $\mathbf{1} \in \mathcal{C}(\mathbf{X})$. Without it, the chain $\text{span}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{X})$ is not available, and the corrected “identity” fails.

Proposition 9.1.3

For any model matrix $\mathbf{X}$, with $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$ and $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{y}$,

$$\sum_i (y_i - \bar{y})^2 = \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i \hat{\varepsilon}_i^2 - 2\bar{y} \sum_i \hat{\varepsilon}_i.$$

The last term vanishes for every $\mathbf{y}$ iff $\mathbf{1} \in \mathcal{C}(\mathbf{X})$.

Proof. Write $\mathbf{y} - \bar{y}\mathbf{1} = (\hat{\mathbf{Y}} - \bar{y}\mathbf{1}) + \hat{\boldsymbol{\varepsilon}}$ and expand the squared length. The cross term is $2(\hat{\mathbf{Y}} - \bar{y}\mathbf{1})^\top \hat{\boldsymbol{\varepsilon}} = 2\hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} - 2\bar{y}\mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = -2\bar{y}\mathbf{1}^\top \hat{\boldsymbol{\varepsilon}}$, since $\hat{\mathbf{Y}} \perp \hat{\boldsymbol{\varepsilon}}$. If $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, then $\mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = 0$ ([Proposition 6.4.3\(b\)](#)). If $\mathbf{1} \notin \mathcal{C}(\mathbf{X})$, take $\mathbf{y} = \mathbf{1}$. Then $\bar{y} = 1$ and $\mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{1}^\top (\mathbf{I} - \mathbf{M})\mathbf{1} = \|(\mathbf{I} - \mathbf{M})\mathbf{1}\|^2 > 0$, so the last term is not zero. ■

Without an intercept the valid decomposition is the uncorrected one, $\|\mathbf{y}\|^2 = \|\mathbf{M}\mathbf{y}\|^2 + \text{SSE}$, whose regression row has r degrees of freedom. That is appropriate only when the mean response is known to vanish when every regressor is zero.

Example. (Forcing the murder regression through the origin)

Refit the model of [Example \(An analysis of variance table for murder rates\)](#) without the intercept. The residual sum of squares rises from 101.96 to 177.73, so the fit is much worse. Yet the residuals now sum to -8.332 , and the would-be corrected identity is off by $-2\bar{y} \sum_i \hat{\varepsilon}_i = 75.22$: the corrected total 284.76 is not $31.81 + 177.73 = 209.54$. The uncentred ratio $\|\hat{\mathbf{Y}}\|^2/\|\mathbf{y}\|^2$, which some software reports as R^2 for such models, is 0.8637, much larger than the 0.6419 of the better model with an intercept ([Exercise](#)

(B1)).

```

X0 = X[:, 1:]                                # drop the intercept column
b0, *_ = np.linalg.lstsq(X0, y, rcond=None)
e0 = y - X0 @ b0
ybar = y.mean()
sst = np.sum((y - ybar) ** 2)
ssr_star = np.sum((X0 @ b0 - ybar) ** 2)
sse0 = e0 @ e0
print(f"SST = {sst:.2f} but SSR* + SSE = {ssr_star + sse0:.2f}")
print(f"sum of residuals = {e0.sum():.3f}, -2 ybar * sum = {-2 * ybar * e0.sum():.2f}")

```

9.1.5 Coarsening and refining

Coarsening merges rows: replacing $\mathcal{V}_1$ and $\mathcal{V}_2$ by $\mathcal{V}_1 \oplus \mathcal{V}_2$ gives another decomposition, whose sums of squares and degrees of freedom are sums of the old ones. The regression row is a coarsening of the table that gives each regressor its own row (Section 9.3). **Refining** splits a row: any subspace of dimension r_i can be cut into r_i orthogonal lines, each carrying one degree of freedom. A refinement is useful only when its lines answer questions of interest. Section 9.4 chooses them with orthogonal contrasts and orthogonal polynomials.

Idea. An analysis of variance table is a list of mutually orthogonal subspaces that fill $\mathbb{R}^n$. Its sums of squares are the squared lengths of the projections of $\mathbf{y}$, its degrees of freedom are the dimensions, and both columns add up because of Pythagoras.

9.1.6 Exercises

9.1.6.1 A. Check your understanding

Exercise A1.

For simple regression, $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ with $\mathbf{x}$ not constant, show that $\text{SSR} = \hat{\beta}_1^2 S_{xx} = S_{xy}^2 / S_{xx}$ and that the F ratio of the corrected table equals t^2 , where $t = \hat{\beta}_1 \sqrt{S_{xx}} / s$.

Exercise A2.

A regression with an intercept has a corrected table in which the regression row has 3 degrees of freedom and sum of squares 120, the residual mean square is 2.5, and the corrected total sum of squares is 172.5. Find n , the residual degrees of freedom, the F ratio and R^2 .

9.1.6.2 B. Practice

Exercise B1.

Let $\mathbf{U} = [\mathbf{Q}_1, \dots, \mathbf{Q}_{k+1}]$ be the orthogonal matrix of Theorem 9.1.1(d) and $\mathbf{z} = \mathbf{U}^\top \mathbf{Y}$. Suppose $\mathbf{E}(\mathbf{Y}) = \boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$. Show that the coordinates of $\mathbf{z}$

are uncorrelated with variance σ^2 , that the last $n - r$ of them have mean zero, and deduce $E(\text{SSE}) = (n - r)\sigma^2$. Show that if in addition $\mathbf{Y}$ is normal, the coordinates are independent.

Solution. By [Theorem 2.2.1](#), $\text{Cov}(\mathbf{z}) = \mathbf{U}^\top(\sigma^2\mathbf{I})\mathbf{U} = \sigma^2\mathbf{I}$ and $E(\mathbf{z}) = \mathbf{U}^\top\boldsymbol{\mu}$. The last block is $\mathbf{Q}_{k+1}^\top\boldsymbol{\mu}$, and $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ is orthogonal to the error space, so it is zero. Then $\text{SSE} = \|\mathbf{Q}_{k+1}^\top\mathbf{Y}\|^2$ is a sum of $n - r$ squares of variables with mean zero and variance σ^2 , so its expectation is $(n - r)\sigma^2$. Under normality $\mathbf{z}$ is normal ([Theorem 3.3.1](#)) with diagonal covariance, and uncorrelated jointly normal variables are independent ([Theorem 3.4.1](#)).

Exercise B2.

For a model without an intercept, the quantity $1 - \text{SSE}/\text{SST}$ is sometimes reported as R^2 . Show that it is negative iff the model fits worse than the intercept-only model. Construct a data set with $n = 4$ and one regressor for which it is below -1 .

9.1.6.3 C. Going deeper

Exercise C1.

Let $\mathcal{V}$ be a subspace with projection $\mathbf{P}$. Show that

$$\|\mathbf{P}\mathbf{y}\|^2 = \max_{\mathbf{v} \in \mathcal{V}, \mathbf{v} \neq \mathbf{0}} \frac{(\mathbf{v}^\top \mathbf{y})^2}{\mathbf{v}^\top \mathbf{v}},$$

with the maximum attained at $\mathbf{v} = \mathbf{P}\mathbf{y}$ when $\mathbf{P}\mathbf{y} \neq \mathbf{0}$. Interpret: the sum of squares of a row with r_i degrees of freedom is the largest one-degree-of-freedom sum of squares obtainable by refining that row. Chapter 13 uses this fact to build Scheffé's simultaneous intervals.

Solution. For $\mathbf{v} \in \mathcal{V}$, $\mathbf{v}^\top \mathbf{y} = \mathbf{v}^\top \mathbf{P}\mathbf{y}$, so by Cauchy–Schwarz ([Proposition 1.1.1](#)) $(\mathbf{v}^\top \mathbf{y})^2 \leq \mathbf{v}^\top \mathbf{v} \|\mathbf{P}\mathbf{y}\|^2$, with equality iff $\mathbf{v}$ is a multiple of $\mathbf{P}\mathbf{y}$. Taking $\mathbf{v} = \mathbf{P}\mathbf{y} \in \mathcal{V}$ attains the bound. A line $\text{span}(\mathbf{v})$ inside $\mathcal{V}$ carries the one-degree-of-freedom sum of squares $(\mathbf{v}^\top \mathbf{y})^2 / \mathbf{v}^\top \mathbf{v}$ ([Example \(Projections onto a line\)](#)), so no refinement of the row gives a line with a larger sum of squares than the whole row.

9.2 Extra sums of squares

Most questions asked of a regression compare two models. Each question names a reduced model inside the full one, and the natural measure of what the full model adds is how much it lowers the residual sum of squares. This section develops that measure, its computation when regressors are added to a fitted model, and a closed form when the reduced model is defined by linear constraints.

9.2.1 Notation

Definition 9.2.1 — Extra sum of squares

Let $\mathbf{X}_1$ and $\mathbf{X}_2$ be matrices with n rows, let $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$, and let $\mathbf{M}_1$ and $\mathbf{M}$ be the projections onto $\mathcal{C}(\mathbf{X}_1)$ and $\mathcal{C}(\mathbf{X})$. The **extra sum of squares** of $\mathbf{X}_2$ given $\mathbf{X}_1$ is

$$\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) = \mathbf{y}^\top (\mathbf{M} - \mathbf{M}_1) \mathbf{y} = \text{SSE}(\mathbf{X}_1) - \text{SSE}(\mathbf{X}),$$

with $\text{rank}(\mathbf{X}) - \text{rank}(\mathbf{X}_1)$ degrees of freedom. Here $\text{SSE}(\cdot)$ is the residual sum of squares of the model with the given model matrix.

The two expressions agree by [Eq. 6.5.2](#). One also writes $\text{SS}(\beta_2 \mid \beta_1)$, or $\text{SS}(B \mid \mu, A)$ in factor models; some books write R , for “reduction”. Four properties follow from [Theorem 6.5.1](#) and [Theorem 6.7.1](#).

1. *It depends only on the two column spaces.* Rescaling, recoding, centring, replacing $\mathbf{X}_2$ by $\mathbf{X}_2 + \mathbf{X}_1 \mathbf{B}$, and rank deficiency change nothing.
2. *It is nonnegative*, and zero iff $\mathbf{M} \mathbf{y} = \mathbf{M}_1 \mathbf{y}$, that is, iff adding $\mathbf{X}_2$ leaves the fitted values unchanged.
3. *It obeys a chain rule:* $\text{SS}(\mathbf{X}_2, \mathbf{X}_3 \mid \mathbf{X}_1) = \text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) + \text{SS}(\mathbf{X}_3 \mid \mathbf{X}_1, \mathbf{X}_2)$, because the projections telescope.
4. *It is the sum of squares of a residualized regression.* The next lemma makes this precise.

Lemma 9.2.1 — Extra sums of squares by residualizing

For any $\mathbf{X}_1$ and $\mathbf{X}_2$, let $\tilde{\mathbf{X}}_2 = (\mathbf{I} - \mathbf{M}_1) \mathbf{X}_2$. Then $\mathcal{C}([\mathbf{X}_1, \mathbf{X}_2]) = \mathcal{C}(\mathbf{X}_1) \oplus \mathcal{C}(\tilde{\mathbf{X}}_2)$ with the two summands orthogonal, so $\mathbf{M} = \mathbf{M}_1 + \mathbf{P}_{\mathcal{C}(\tilde{\mathbf{X}}_2)}$ and

$$\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) = \|\mathbf{P}_{\mathcal{C}(\tilde{\mathbf{X}}_2)} \mathbf{y}\|^2 = \mathbf{y}^\top \tilde{\mathbf{X}}_2 (\tilde{\mathbf{X}}_2^\top \tilde{\mathbf{X}}_2)^{-1} \tilde{\mathbf{X}}_2^\top \mathbf{y}.$$

Proof. The argument for part (a) of [Lemma 6.6.1](#) uses no rank condition, so it applies here: the columns of $\tilde{\mathbf{X}}_2$ lie in $\mathcal{C}(\mathbf{X})$ and are orthogonal to $\mathcal{C}(\mathbf{X}_1)$, and every $\mathbf{X}_1 \mathbf{b}_1 + \mathbf{X}_2 \mathbf{b}_2$ equals $\tilde{\mathbf{X}}_2 \mathbf{b}_2 + (\mathbf{X}_1 \mathbf{b}_1 + \mathbf{M}_1 \mathbf{X}_2 \mathbf{b}_2)$. By [Theorem 6.3.8](#), $\mathbf{M} = \mathbf{M}_1 + \mathbf{P}_{\mathcal{C}(\tilde{\mathbf{X}}_2)}$. The formula is [Theorem 6.3.7](#) applied to $\tilde{\mathbf{X}}_2$. ■

9.2.2 Adding regressors to a fitted model

Suppose $E(\mathbf{Y}) = \mathbf{X} \beta$ has been fitted and we add the columns of an $n \times t$ matrix $\mathbf{Z}$, giving $E(\mathbf{Y}) = \mathbf{X} \beta + \mathbf{Z} \gamma$ with model matrix $\mathbf{W} = [\mathbf{X}, \mathbf{Z}]$. Only a $t \times t$ matrix has to be inverted.

Theorem 9.2.2 — Adding regressors

Let $\mathbf{W} = [\mathbf{X}, \mathbf{Z}]$ have full column rank $p + t$, let $\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1}$, $\hat{\beta} = \mathbf{C} \mathbf{X}^\top \mathbf{y}$, and $\mathbf{R} = \mathbf{I} - \mathbf{X} \mathbf{C} \mathbf{X}^\top$. Then:

- (a) $\mathbf{Z}^\top \mathbf{RZ}$ is positive definite;
 (b) the least squares coefficients of the enlarged model are

$$\hat{\gamma} = (\mathbf{Z}^\top \mathbf{RZ})^{-1} \mathbf{Z}^\top \mathbf{Ry}, \quad \hat{\beta}_W = \hat{\beta} - \mathbf{L}\hat{\gamma}, \quad \mathbf{L} = \mathbf{CX}^\top \mathbf{Z};$$

- (c) the residual sum of squares falls by

$$\text{SS}(\mathbf{Z} \mid \mathbf{X}) = \mathbf{y}^\top \mathbf{RZ}(\mathbf{Z}^\top \mathbf{RZ})^{-1} \mathbf{Z}^\top \mathbf{Ry} = \hat{\gamma}^\top \mathbf{Z}^\top \mathbf{Ry} = \hat{\gamma}^\top (\mathbf{Z}^\top \mathbf{RZ}) \hat{\gamma};$$

- (d) if $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, then $\text{Cov}(\hat{\gamma}) = \sigma^2 (\mathbf{Z}^\top \mathbf{RZ})^{-1}$, $\text{Cov}(\hat{\beta}_W, \hat{\gamma}) = -\mathbf{L} \text{Cov}(\hat{\gamma})$ and

$$\text{Cov}(\hat{\beta}_W) = \sigma^2 [\mathbf{C} + \mathbf{L}(\mathbf{Z}^\top \mathbf{RZ})^{-1} \mathbf{L}^\top].$$

Proof.

- (a) $\mathbf{R}$ is the symmetric idempotent residual projection of the model $\mathbf{X}$, so $\mathbf{a}^\top \mathbf{Z}^\top \mathbf{RZ} \mathbf{a} = \|\mathbf{RZ} \mathbf{a}\|^2 \geq 0$. If $\mathbf{RZ} \mathbf{a} = \mathbf{0}$, then $\mathbf{Z} \mathbf{a} \in \mathcal{C}(\mathbf{X})$, say $\mathbf{Z} \mathbf{a} = \mathbf{X} \mathbf{b}$, so $\mathbf{W}(-\mathbf{b}^\top, \mathbf{a}^\top)^\top = \mathbf{0}$ and full column rank forces $\mathbf{a} = \mathbf{0}$.
 (b) Apply the Frisch–Waugh–Lovell theorem (Theorem 6.6.2) with $\mathbf{X}_1 = \mathbf{Z}$ and $\mathbf{X}_2 = \mathbf{X}$. Then $\mathbf{M}_2 = \mathbf{I} - \mathbf{R}$ and $\tilde{\mathbf{X}}_1 = \mathbf{RZ}$, so part (a) of that theorem gives $\hat{\gamma} = (\mathbf{Z}^\top \mathbf{RZ})^{-1} \mathbf{Z}^\top \mathbf{Ry}$, which is the stated formula because $\mathbf{R}$ is symmetric and idempotent. Part (c) of the theorem gives $\hat{\beta}_W = \mathbf{CX}^\top (\mathbf{y} - \mathbf{Z}\hat{\gamma}) = \hat{\beta} - \mathbf{L}\hat{\gamma}$.
 (c) By Lemma 9.2.1, the extra sum of squares is the squared length of the projection of $\mathbf{y}$ onto $\mathcal{C}(\mathbf{RZ})$, which is $\mathbf{y}^\top \mathbf{RZ}(\mathbf{Z}^\top \mathbf{RZ})^{-1} \mathbf{Z}^\top \mathbf{Ry}$ by Theorem 6.3.7. Substituting (b) gives the other two forms.
 (d) Both $\hat{\gamma}$ and $\hat{\beta}$ are linear in $\mathbf{y}$, and by Theorem 2.2.1

$$\text{Cov}(\hat{\gamma}) = \sigma^2 (\mathbf{Z}^\top \mathbf{RZ})^{-1} \mathbf{Z}^\top \mathbf{R} \mathbf{RZ} (\mathbf{Z}^\top \mathbf{RZ})^{-1} = \sigma^2 (\mathbf{Z}^\top \mathbf{RZ})^{-1}, \quad \text{Cov}(\hat{\beta}, \hat{\gamma}) = \sigma^2 \mathbf{CX}^\top \mathbf{RZ} (\mathbf{Z}^\top \mathbf{RZ})^{-1}$$

since $\mathbf{X}^\top \mathbf{R} = \mathbf{0}$. With $\text{Cov}(\hat{\beta}) = \sigma^2 \mathbf{C}$ (Theorem 5.5.1), the formula $\hat{\beta}_W = \hat{\beta} - \mathbf{L}\hat{\gamma}$ gives $\text{Cov}(\hat{\beta}_W, \hat{\gamma}) = -\mathbf{L} \text{Cov}(\hat{\gamma})$ and $\text{Cov}(\hat{\beta}_W) = \text{Cov}(\hat{\beta}) + \mathbf{L} \text{Cov}(\hat{\gamma}) \mathbf{L}^\top$. ■

Since $\mathbf{L}(\mathbf{Z}^\top \mathbf{RZ})^{-1} \mathbf{L}^\top$ is nonnegative definite, part (d) shows that *adding regressors never decreases the variances of the coefficients already in the model*, when both models are correct (compare Exercise (B2), and see Exercise (C2)). Chapter 26 measures this price of adjustment as variance inflation. In computation one updates a QR factorization instead of using these formulas (Chapter 10).

Example. (Adding poverty and urbanization)

In the state data, fit the murder rate on single parenthood first: $\hat{y} = -8.2146 + 0.5147 x_{\text{single}}$, with residual sum of squares 121.92. Now add $\mathbf{Z} = [\mathbf{x}_{\text{poverty}}, \mathbf{x}_{\text{urban}}]$. The listing computes $\hat{\gamma} = (0.2538, 0.0046)$ from the old residual projection alone, and corrects the old coefficients to -9.0946 and 0.3980 . These agree with a fresh fit of the enlarged model. The residual sum of squares falls by 19.96, on 2 degrees of freedom, to 101.96. Single parenthood's coefficient drops by almost a quarter once poverty is allowed to take its share, the pattern seen in Example (Order matters).


```

import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
n = len(y)
one = np.ones(n)
cols = {"poverty": data["poverty"].to_numpy(), "single": data["single"].to_numpy(),
        "urban": data["urban"].to_numpy()}

X = np.column_stack([one, cols["single"]])           # the model already
    fitted
Z = np.column_stack([cols["poverty"], cols["urban"]]) # the regressors to add
XtX_inv = np.linalg.inv(X.T @ X)
beta_X = XtX_inv @ X.T @ y
R = np.eye(n) - X @ XtX_inv @ X.T                 # residual projection of
    the old model

gamma = np.linalg.solve(Z.T @ R @ Z, Z.T @ R @ y)   # new coefficients
beta_new = beta_X - XtX_inv @ X.T @ Z @ gamma       # old coefficients,
    corrected
extra_ss = gamma @ Z.T @ R @ y                     # SS(Z | X) = SSE_X -
    SSE_W
print("gamma =", gamma.round(4), " corrected beta =", beta_new.round(4))
print(f"SSE_X = {y @ R @ y:.2f}, extra SS = {extra_ss:.2f}")

```

9.2.3 One column at a time

With $t = 1$, [Theorem 9.2.2](#) links the extra sum of squares of a single regressor to its coefficient and standard error.

Corollary 9.2.3 — The extra sum of squares of one column

Let $\mathbf{X}$ have full column rank p , write c_{jj} for the j th diagonal entry of $(\mathbf{X}^\top \mathbf{X})^{-1}$, and let $\mathbf{X}_{(j)}$ be $\mathbf{X}$ without its j th column. Then

$$\text{SS}(\mathbf{x}_j | \mathbf{X}_{(j)}) = \frac{\hat{\beta}_j^2}{c_{jj}} = t_j^2 s^2, \quad t_j = \frac{\hat{\beta}_j}{s\sqrt{c_{jj}}}, \quad s^2 = \frac{\text{SSE}}{n-p}.$$

Consequently, the ratio of this one-degree-of-freedom sum of squares to the residual mean square is t_j^2 .

Proof. Permute the columns so that $\mathbf{x}_j$ comes last; this carries c_{jj} to the last diagonal entry of the inverse. Write $\mathbf{X} = [\mathbf{X}_{(j)}, \mathbf{x}_j]$ and $\mathbf{R} = \mathbf{I} - \mathbf{X}_{(j)}(\mathbf{X}_{(j)}^\top \mathbf{X}_{(j)})^{-1} \mathbf{X}_{(j)}^\top$ for the residual projection of $\mathbf{X}_{(j)}$. By [Theorem 1.6.4](#) the last diagonal entry of $(\mathbf{X}^\top \mathbf{X})^{-1}$ is the inverse of the Schur complement $\mathbf{x}_j^\top \mathbf{x}_j - \mathbf{x}_j^\top \mathbf{X}_{(j)}(\mathbf{X}_{(j)}^\top \mathbf{X}_{(j)})^{-1} \mathbf{X}_{(j)}^\top \mathbf{x}_j = \mathbf{x}_j^\top \mathbf{R} \mathbf{x}_j$, so $\mathbf{x}_j^\top \mathbf{R} \mathbf{x}_j = 1/c_{jj}$. Now apply [Theorem 9.2.2](#) with $\mathbf{X}_{(j)}$ as the old model and $\mathbf{Z} = \mathbf{x}_j$: then $\hat{\gamma} = \hat{\beta}_j$, and part (c) gives $\text{SS}(\mathbf{x}_j | \mathbf{X}_{(j)}) = \hat{\beta}_j^2 \mathbf{x}_j^\top \mathbf{R} \mathbf{x}_j = \hat{\beta}_j^2 / c_{jj}$. The definition of t_j gives the last form. ■

For poverty in the three-regressor model, $\hat{\beta} = 0.2538$ and $c_{jj} = 0.003935$, so the extra sum of squares is $0.2538^2 / 0.003935 = 16.37$. This is exactly the last entry for poverty in Order B of [Example \(Order matters\)](#). The t statistic is 2.718, and $t^2 s^2 = 7.386 \times 2.217 = 16.37$.

```
Xf = np.column_stack([one, cols["poverty"], cols["single"], cols["urban"]])
fit = sm.OLS(y, Xf).fit()
C = np.linalg.inv(Xf.T @ Xf)
j = 1 # poverty
partial_ss = fit.params[j] ** 2 / C[j, j]
print(f"beta = {fit.params[j]:.4f}, c_jj = {C[j, j]:.6f}, beta^2/c_jj = {partial_ss:.2f}")
print(f"t = {fit.tvalues[j]:.3f}, t^2 * s^2 = {fit.tvalues[j] ** 2 * fit.scale:.2f}")
```

9.2.4 Sums of squares for linear hypotheses

Reduced models often come as constraints: two coefficients are equal, a set of coefficients vanishes, a contrast among group means is zero. [Section 6.5](#) identified the test space of such a constraint ([Theorem 6.5.2](#)). The next theorem gives its sum of squares in terms of the least squares estimate, with no reduced model fitted.

Theorem 9.2.4 — Sum of squares of a linear hypothesis

Let $\mathbf{\Lambda}$ be a $p \times q$ matrix of rank q such that each entry of $\mathbf{\Lambda}^\top \boldsymbol{\beta}$ is estimable, so that $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{T}$ for some $n \times q$ matrix $\mathbf{T} = [\boldsymbol{\rho}_1, \dots, \boldsymbol{\rho}_q]$ ([Definition 8.2.1](#), [Theorem 6.4.4](#)). Let $\mathbf{G}$ be any generalized inverse of $\mathbf{X}^\top \mathbf{X}$, let $\hat{\boldsymbol{\beta}}$ be any least squares estimate, and let $\mathbf{d} \in \mathbb{R}^q$. Then:

- (a) $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}$ and $\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$ do not depend on the choices of $\hat{\boldsymbol{\beta}}$ and $\mathbf{G}$, and $\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$ is positive definite;
- (b) the subspace $\mathcal{S}_0 = \{\mathbf{X} \mathbf{b} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{0}\}$ of $\mathcal{C}(\mathbf{X})$ has orthogonal complement in $\mathcal{C}(\mathbf{X})$ equal to $\mathcal{C}(\mathbf{MT})$, of dimension q , and the projection onto $\mathcal{C}(\mathbf{MT})$ is $\mathbf{MT}(\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} \mathbf{T}^\top \mathbf{M}$;
- (c) the minimum of $\|\mathbf{y} - \mathbf{X} \mathbf{b}\|^2$ over $\{\mathbf{b} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}\}$ exceeds SSE by

$$\text{SS}_H = (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}). \quad (9.2.1)$$

Proof.

- (a) $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{T}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{T}^\top \mathbf{M} \mathbf{y}$ and $\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda} = \mathbf{T}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{T} = \mathbf{T}^\top \mathbf{M} \mathbf{T}$ by [Theorem 6.3.7](#), and neither expression involves the choices. Since $\mathbf{X}^\top \mathbf{M} = \mathbf{X}^\top$, we have $\mathbf{X}^\top (\mathbf{MT}) = \mathbf{\Lambda}$, which has rank q , so the q columns of $\mathbf{MT}$ are linearly independent. Hence $\mathbf{T}^\top \mathbf{M} \mathbf{T} = (\mathbf{MT})^\top (\mathbf{MT})$ is positive definite.
- (b) $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{T}^\top \mathbf{X} \mathbf{b}$, so $\mathcal{S}_0 = \mathcal{C}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}^\top)$. [Theorem 6.5.2](#) identifies its orthogonal complement in $\mathcal{C}(\mathbf{X})$ as $\mathcal{C}(\mathbf{MT})$. The columns of $\mathbf{MT}$ are independent, so [Theorem 6.3.7](#) gives the projection $\mathbf{MT}(\mathbf{T}^\top \mathbf{M} \mathbf{T})^{-1} \mathbf{T}^\top \mathbf{M}$, and $\mathbf{T}^\top \mathbf{M} \mathbf{T} = \mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$ by (a).
- (c) First let $\mathbf{d} = \mathbf{0}$. Minimizing over $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{0}$ is minimizing $\|\mathbf{y} - \boldsymbol{\mu}\|^2$ over $\boldsymbol{\mu} \in \mathcal{S}_0$, so by [Eq. 6.5.2](#) the excess over SSE is the squared length of the projection of $\mathbf{y}$ onto $\mathcal{C}(\mathbf{X}) \ominus \mathcal{S}_0 = \mathcal{C}(\mathbf{MT})$, where $\ominus$ denotes the orthogonal complement within

$\mathcal{C}(\mathbf{X})$. By (b) this is $\mathbf{y}^\top \mathbf{M} \mathbf{T} (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} \mathbf{T}^\top \mathbf{M} \mathbf{y}$, and $\mathbf{T}^\top \mathbf{M} \mathbf{y} = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}$. For general $\mathbf{d}$, the rows of $\mathbf{\Lambda}^\top$ are independent, so there is a $\mathbf{b}_0$ with $\mathbf{\Lambda}^\top \mathbf{b}_0 = \mathbf{d}$. Substituting $\mathbf{b} = \mathbf{b}_0 + \mathbf{b}_*$ turns the problem into minimizing $\|\mathbf{y}_* - \mathbf{X} \mathbf{b}_*\|^2$ subject to $\mathbf{\Lambda}^\top \mathbf{b}_* = \mathbf{0}$, with $\mathbf{y}_* = \mathbf{y} - \mathbf{X} \mathbf{b}_0$. The data $\mathbf{y}_*$ have the same residual sum of squares as $\mathbf{y}$, and $\hat{\boldsymbol{\beta}} - \mathbf{b}_0$ is a least squares estimate for them, since $\mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b}_0) = \mathbf{M} \mathbf{y}_*$. Applying the case $\mathbf{d} = \mathbf{0}$ to $\mathbf{y}_*$ gives Eq. 9.2.1, because $\mathbf{\Lambda}^\top (\hat{\boldsymbol{\beta}} - \mathbf{b}_0) = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$.

■

In Eq. 9.2.1, $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$ measures how far the estimate is from the hypothesis, and $\sigma^2 \mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$ is the covariance of $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}$ (Chapter 7), so SS_H / σ^2 is the squared Mahalanobis length of the discrepancy (Proposition 3.2.3). No reduced model has to be fitted.

In the full-rank case the constrained minimizer itself has a closed form, which recovers the classical restricted least squares estimator.

Theorem 9.2.5 — Restricted least squares

Let $\mathbf{X}$ have full column rank, $\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1}$, and let $\mathbf{\Lambda}$ ($p \times q$) have rank q . For any $\mathbf{d} \in \mathbb{R}^q$, the unique minimizer of $\|\mathbf{y} - \mathbf{X} \mathbf{b}\|^2$ subject to $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}$ is

$$\hat{\boldsymbol{\beta}}_H = \hat{\boldsymbol{\beta}} - \mathbf{C} \mathbf{\Lambda} (\mathbf{\Lambda}^\top \mathbf{C} \mathbf{\Lambda})^{-1} (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}),$$

and $\|\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}_H\|^2 = \text{SSE} + \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)\|^2$, where the last term equals SS_H of Eq. 9.2.1 with $\mathbf{G} = \mathbf{C}$.

Proof. $\mathbf{\Lambda}^\top \mathbf{C} \mathbf{\Lambda}$ is positive definite because $\mathbf{C}$ is and $\mathbf{\Lambda}$ has full column rank, and $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}_H = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}) = \mathbf{d}$. For any $\mathbf{b}$, Eq. 6.3.1 with $\mathbf{u} = \mathbf{X} \mathbf{b}$ gives $\|\mathbf{y} - \mathbf{X} \mathbf{b}\|^2 = \text{SSE} + \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b})\|^2$. Every feasible $\mathbf{b}$ has the form $\hat{\boldsymbol{\beta}}_H + \mathbf{v}$ with $\mathbf{\Lambda}^\top \mathbf{v} = \mathbf{0}$, and then

$$\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b})\|^2 = \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)\|^2 - 2(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} + \|\mathbf{X} \mathbf{v}\|^2.$$

The cross term vanishes: $(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} = (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{\Lambda}^\top \mathbf{C} \mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \mathbf{C} \mathbf{X}^\top \mathbf{X} \mathbf{v} = (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{\Lambda}^\top \mathbf{C} \mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \mathbf{v} = 0$. So the objective is $\text{SSE} + \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)\|^2 + \|\mathbf{X} \mathbf{v}\|^2$, minimized exactly when $\mathbf{X} \mathbf{v} = \mathbf{0}$, that is, $\mathbf{v} = \mathbf{0}$. Finally $\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)\|^2 = (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{\Lambda}^\top \mathbf{C} \mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \mathbf{C} \mathbf{X}^\top \mathbf{X} \mathbf{C} \mathbf{\Lambda} (\mathbf{\Lambda}^\top \mathbf{C} \mathbf{\Lambda})^{-1} (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})$, which simplifies to SS_H .

■

The same estimator comes out of Lagrange multipliers (Proposition 1.11.3 with $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$). The proof above shows instead that the constrained fit is the feasible point nearest to $\hat{\boldsymbol{\beta}}$ in the geometry of $\mathbf{X}^\top \mathbf{X}$.

Example. (Equal effects of poverty and single parenthood)

Both poverty and single parenthood are percentages of the population. Is one percentage point of either associated with the same change in the murder rate? The hypothesis is $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$ with $\boldsymbol{\lambda} = (0, 1, -1, 0)^\top$. Formula Eq. 9.2.1 gives $\text{SS}_H = 1.807$ on one degree of freedom, identical to the drop in residual sum of squares when the two regressors are replaced by their sum. Its ratio to the residual mean square is 0.815, so the data are entirely compatible with equal effects. For a hypothesis with a nonzero right-hand side,

$\beta_{\text{single}} = 0.5$, [Theorem 9.2.5](#) gives the constrained coefficients 0.1803 for poverty and -0.0044 for urbanization, and a sum of squares of 3.357, both confirmed by a direct fit of the reduced model.

```
def hypothesis_ss(X, y, Lam, d=None):
    """(L'b - d)' [L' (X'X)^- L]^-1 (L'b - d) for estimable L'beta = d."""
    G = np.linalg.pinv(X.T @ X)
    b = G @ X.T @ y
    u = Lam.T @ b - (0 if d is None else d)
    return u @ np.linalg.solve(Lam.T @ G @ Lam, u)

def sse(Z):
    b, *_ = np.linalg.lstsq(Z, y, rcond=None)
    e = y - Z @ b
    return e @ e

Lam_equal = np.array([[0.0, 1.0, -1.0, 0.0]]).T          # beta_poverty =
    beta_single
ss_equal = hypothesis_ss(Xf, y, Lam_equal)
X_equal = np.column_stack([one, cols["poverty"] + cols["single"], cols["urban"]])
print(f"H: equal slopes    formula {ss_equal:.3f}    SSE_0 - SSE {sse(X_equal) - sse(
    Xf):.3f}")
```

9.2.5 Constraints that are not estimable

Without estimability, the reduced model $\mathcal{S}_0 = \{\mathbf{X}\mathbf{b} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{0}\}$ is still defined, but it may be larger than the number of constraints suggests.

Proposition 9.2.6 — How much a constraint restricts

For any $p \times q$ matrix $\mathbf{\Lambda}$, the subspace $\mathcal{S}_0 = \{\mathbf{X}\mathbf{b} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{0}\}$ has dimension $r - \dim(\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top))$. So the extra sum of squares for the constraint has $\dim(\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top))$ degrees of freedom. It is identically zero iff no nonzero linear combination of the constrained functions is estimable.

Proof. $\mathcal{S}_0$ is the image of $\mathcal{N}(\mathbf{\Lambda}^\top)$ under $\mathbf{b} \mapsto \mathbf{X}\mathbf{b}$, and the kernel of this map restricted to $\mathcal{N}(\mathbf{\Lambda}^\top)$ is $\mathcal{N}(\mathbf{\Lambda}^\top) \cap \mathcal{N}(\mathbf{X})$, the null space of the stacked matrix $[\mathbf{\Lambda}, \mathbf{X}^\top]^\top$. By rank-nullity ([Theorem 1.2.2](#)),

$$\dim \mathcal{S}_0 = \dim \mathcal{N}(\mathbf{\Lambda}^\top) - \dim(\mathcal{N}(\mathbf{\Lambda}^\top) \cap \mathcal{N}(\mathbf{X})) = (p - \text{rank } \mathbf{\Lambda}) - (p - \text{rank}[\mathbf{\Lambda}, \mathbf{X}^\top]).$$

The dimension formula for a sum of subspaces ([Section 6.2](#)) gives $\text{rank}[\mathbf{\Lambda}, \mathbf{X}^\top] = \dim(\mathcal{C}(\mathbf{\Lambda}) + \mathcal{C}(\mathbf{X}^\top)) = \text{rank } \mathbf{\Lambda} + r - \dim(\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top))$. Substituting, $\dim \mathcal{S}_0 = r - \dim(\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top))$. The degrees of freedom of the extra sum of squares are $r - \dim \mathcal{S}_0$. A combination $\mathbf{\Lambda}\mathbf{a}$ is estimable iff it lies in $\mathcal{C}(\mathbf{X}^\top)$ ([Theorem 6.4.4](#)), which gives the last statement. ■

In the one-way layout $E(Y_{ij}) = \mu + \alpha_i$, the constraint $\alpha_1 = \alpha_2$ has one degree of freedom. The constraint $\alpha_1 = 0$ has none: it does not restrict the mean vector, and only picks one of the least squares solutions ([Theorem 8.4.2](#)). Software asked to “test” a nonestimable hypothesis either

refuses or silently tests its estimable part. Chapter 17 meets this with empty cells.

9.2.6 Exercises

9.2.6.1 A. Check your understanding

Exercise A1.

In simple regression, compute SS_H for $\beta_1 = 0$ from [Eq. 9.2.1](#) and show that it equals $\hat{\beta}_1^2 S_{xx}$, in agreement with [Exercise \(A1\)](#).

9.2.6.2 B. Practice

Exercise B1.

Prove the chain rule $SS(\mathbf{X}_2, \mathbf{X}_3 \mid \mathbf{X}_1) = SS(\mathbf{X}_2 \mid \mathbf{X}_1) + SS(\mathbf{X}_3 \mid \mathbf{X}_1, \mathbf{X}_2)$, and deduce $SS(\mathbf{X}_3 \mid \mathbf{X}_1, \mathbf{X}_2) \leq SS(\mathbf{X}_2, \mathbf{X}_3 \mid \mathbf{X}_1)$. Show by an example from [Example \(Adding poverty and urbanization\)](#) or its listing that $SS(\mathbf{X}_3 \mid \mathbf{X}_1, \mathbf{X}_2) \leq SS(\mathbf{X}_3 \mid \mathbf{X}_1)$ can fail.

Solution. Let $\mathbf{M}_1$, $\mathbf{M}_{12}$ and $\mathbf{M}_{123}$ be the projections onto the nested column spaces. Then $\mathbf{M}_{123} - \mathbf{M}_1 = (\mathbf{M}_{12} - \mathbf{M}_1) + (\mathbf{M}_{123} - \mathbf{M}_{12})$, and the chain rule follows by taking $\mathbf{y}^\top(\cdot)\mathbf{y}$. Both terms are nonnegative, which gives the inequality. For the failure, the listing of [Section 9.3](#) reports $SS(\mathbf{x}_{\text{urban}} \mid \mathbf{1}) = 6.30$ but $SS(\mathbf{x}_{\text{urban}} \mid \mathbf{1}, \mathbf{x}_{\text{poverty}}) = 29.26$. Adjusting for poverty more than quadruples the sum of squares for urbanization.

Exercise B2.

In [Theorem 9.2.5](#) with $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, show that

$$\text{Cov}(\hat{\beta}_H) = \sigma^2 [\mathbf{C} - \mathbf{C}\mathbf{\Lambda}(\mathbf{\Lambda}^\top \mathbf{C}\mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \mathbf{C}],$$

that $\text{Cov}(\hat{\beta}) - \text{Cov}(\hat{\beta}_H)$ is nonnegative definite, and that $E(\hat{\beta}_H) = \beta$ when the constraint is true. What is $E(\hat{\beta}_H)$ when it is false?

Solution. Write $\mathbf{K} = \mathbf{C}\mathbf{\Lambda}(\mathbf{\Lambda}^\top \mathbf{C}\mathbf{\Lambda})^{-1}$, so $\hat{\beta}_H = (\mathbf{I} - \mathbf{K}\mathbf{\Lambda}^\top)\hat{\beta} + \mathbf{K}\mathbf{d}$. By [Theorem 2.2.1](#) and $\text{Cov}(\hat{\beta}) = \sigma^2 \mathbf{C}$, $\text{Cov}(\hat{\beta}_H) = \sigma^2 (\mathbf{I} - \mathbf{K}\mathbf{\Lambda}^\top) \mathbf{C} (\mathbf{I} - \mathbf{\Lambda} \mathbf{K}^\top)$. Expanding and using $\mathbf{\Lambda}^\top \mathbf{C}\mathbf{\Lambda}(\mathbf{\Lambda}^\top \mathbf{C}\mathbf{\Lambda})^{-1} = \mathbf{I}$, the two cross terms and the last term all equal $\mathbf{K}\mathbf{\Lambda}^\top \mathbf{C} = \mathbf{C}\mathbf{\Lambda}(\mathbf{\Lambda}^\top \mathbf{C}\mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \mathbf{C}$, with signs $-, -, +$, which gives the formula. The difference $\sigma^2 \mathbf{C}\mathbf{\Lambda}(\mathbf{\Lambda}^\top \mathbf{C}\mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \mathbf{C}$ is of the form $\mathbf{B}\mathbf{A}\mathbf{B}^\top$ with $\mathbf{A}$ positive definite, hence nonnegative definite. Finally $E(\hat{\beta}_H) = \beta - \mathbf{K}(\mathbf{\Lambda}^\top \beta - \mathbf{d})$, which is β iff $\mathbf{\Lambda}^\top \beta = \mathbf{d}$. A false constraint buys the smaller variance at the price of a bias.

Exercise B3.

In the one-way layout with model matrix $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_g]$ and group sizes $n_1, \dots, n_g$, use the generalized inverse $\mathbf{G} = \text{diag}(0, 1/n_1, \dots, 1/n_g)$ in [Eq. 9.2.1](#) to show that the

sum of squares for $\alpha_1 = \alpha_2$ is $(\bar{y}_1 - \bar{y}_2)^2 / (1/n_1 + 1/n_2)$. Verify with [Proposition 9.2.6](#) that the constraint $\alpha_1 = 0$ has zero degrees of freedom.

Solution. $\mathbf{G}$ is a generalized inverse because $\mathbf{X}^\top \mathbf{X}$ has the block form with first row $(n, n_1, \dots, n_g)$ and lower right block $\text{diag}(n_1, \dots, n_g)$, and a direct multiplication gives $\mathbf{X}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{X} = \mathbf{X}^\top \mathbf{X}$. With $\boldsymbol{\lambda} = (0, 1, -1, 0, \dots, 0)^\top$, $\mathbf{G} \mathbf{X}^\top \mathbf{y} = (0, \bar{y}_1, \dots, \bar{y}_g)^\top$, so $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \bar{y}_1 - \bar{y}_2$ and $\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda} = 1/n_1 + 1/n_2$. For $\alpha_1 = 0$, $\boldsymbol{\lambda} = (0, 1, 0, \dots, 0)^\top$ is not in $\mathcal{C}(\mathbf{X}^\top)$ ([Example \(Four answers, one fit\)](#)), and $\mathcal{C}(\boldsymbol{\lambda})$ is a line, so $\mathcal{C}(\boldsymbol{\lambda}) \cap \mathcal{C}(\mathbf{X}^\top) = \{\mathbf{0}\}$.

9.2.6.3 C. Going deeper

Exercise C1.

Using [Exercise \(C1\)](#) and [Lemma 9.2.1](#), show that

$$\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) = \max_{\mathbf{a}} \frac{(\mathbf{a}^\top \tilde{\mathbf{X}}_2^\top \mathbf{y})^2}{\mathbf{a}^\top \tilde{\mathbf{X}}_2^\top \tilde{\mathbf{X}}_2 \mathbf{a}},$$

the maximum being over $\mathbf{a}$ with $\tilde{\mathbf{X}}_2 \mathbf{a} \neq \mathbf{0}$. Deduce that the extra sum of squares of a block is at least that of any single column of the block, given $\mathbf{X}_1$, and that the sum of squares of an estimable hypothesis $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$ equals $\max_{\mathbf{a}} (\mathbf{a}^\top \boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}})^2 / (\mathbf{a}^\top \boldsymbol{\Lambda}^\top \mathbf{G} \boldsymbol{\Lambda} \mathbf{a})$.

Exercise C2.

[Exercise \(B2\)](#) gives the matrix inequality behind the remark after [Theorem 9.2.2](#). Using part (d) of that theorem instead, show that $\text{Var}(\hat{\beta}_{W,j}) = \text{Var}(\hat{\beta}_j)$ iff the j th row of $\mathbf{L}$ is zero, and that this holds for every j iff $\mathbf{X}^\top \mathbf{Z} = \mathbf{0}$. Show that when $\mathbf{X}^\top \mathbf{Z} = \mathbf{0}$ the estimates $\hat{\boldsymbol{\beta}}$ do not change at all.

Exercise C3.

In the additive two-way model $E(Y_{ijk}) = \mu + \alpha_i + \beta_j$ with all cells occupied, use [Proposition 9.2.6](#) to find the degrees of freedom of the extra sums of squares for the constraints (i) $\sum_i \alpha_i = 0$ and $\sum_j \beta_j = 0$; (ii) $\alpha_1 = \alpha_2$; (iii) $\alpha_1 = 0$ and $\alpha_1 = \alpha_2$.

9.3 Sequential and partial sums of squares

How should the variation explained by a model be shared among its terms? A term is a block of columns representing one idea: a regressor, the coding columns of a factor, or an interaction. One answer enters the terms one at a time and credits each with what it adds to those already present. Another credits each term with what it adds when entered last. This section defines both, explains the three “types” of sums of squares that software prints, and identifies the hypothesis that each of them addresses.

9.3.1 Sequential sums of squares

Write the model matrix as a sequence of terms,

$$\mathbf{X} = [\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_k],$$

where $\mathbf{X}_0$ is usually the intercept column $\mathbf{1}$. Let $\mathbf{M}_j$ be the projection onto $\mathcal{C}(\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_j)$, so that $\mathbf{M}_k = \mathbf{M}$.

Definition 9.3.1 — Sequential sums of squares

The **sequential** (or **Type I**) sum of squares of the term $\mathbf{X}_j$, for the order $\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_k$, is

$$\text{SS}(\mathbf{X}_j \mid \mathbf{X}_0, \dots, \mathbf{X}_{j-1}) = \mathbf{y}^\top (\mathbf{M}_j - \mathbf{M}_{j-1}) \mathbf{y}, \quad j = 1, \dots, k,$$

with $d_j = \text{rank}(\mathbf{M}_j) - \text{rank}(\mathbf{M}_{j-1})$ degrees of freedom.

The sequential sums of squares are the analysis of variance decomposition of [Theorem 9.1.1\(e\)](#) for the chain $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X}_0, \mathbf{X}_1) \subseteq \dots \subseteq \mathcal{C}(\mathbf{X})$, after the first piece $\mathbf{M}_0$ has been set aside. They add up to the regression sum of squares $\text{SS}(\mathbf{X}_1, \dots, \mathbf{X}_k \mid \mathbf{X}_0)$, their degrees of freedom add up to $r - \text{rank}(\mathbf{X}_0)$, and under normality they are independent of each other and of the residual sum of squares ([Theorem 4.4.4](#)). A term linearly dependent on earlier ones receives zero degrees of freedom. The price is that each piece depends on the order: $\mathbf{X}_j$ is credited only with the part of its span orthogonal to *the terms that happen to precede it*. For the murder rates of [Section 6.5](#), poverty received 102.49 first and 16.37 last ([Example \(Order matters\)](#)).

The order dependence can be more surprising than “entering first gives more credit”. The listing computes $\text{SS}(\mathbf{x} \mid \mathbf{1}, S)$ for each of the three regressors and every set S of the other two.

Regressor	given $\mathbf{1}$ only	given $\mathbf{1}$ and one other (named)	given $\mathbf{1}$ and the remaining other (named)	given $\mathbf{1}$ and both others
poverty	102.49	19.69 (single)	125.45 (urban)	16.37
single	162.84	80.04 (poverty)	160.13 (urban)	51.06
urban	6.30	29.26 (poverty)	3.59 (single)	0.27

Adjusting for urbanization *increases* the sum of squares of poverty from 102.49 to 125.45, and adjusting for poverty more than quadruples that of urbanization. Urban states have somewhat less poverty but more murder, so holding one fixed sharpens the association of the other; such a regressor is called a **suppressor** ([Exercise \(B1\)](#)). A sequential sum of squares is not a property of a regressor, and not even a monotone function of the set adjusted for.

```
from itertools import combinations
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
n = len(y)
one = np.ones(n)
cols = {"poverty": data["poverty"].to_numpy(), "single": data["single"].to_numpy(),
        "urban": data["urban"].to_numpy()}
```

```

def sse(Z):
    b, *_ = np.linalg.lstsq(Z, y, rcond=None)
    e = y - Z @ b
    return e @ e

table = {}
for name, x in cols.items():
    others = [k for k in cols if k != name]
    for size in range(3):
        for S in combinations(others, size):
            base = np.column_stack([one] + [cols[k] for k in S])
            table[name, S] = sse(base) - sse(np.column_stack([base, x]))
            print(f"SS({name} | 1{''.join(', ' + k for k in S)}) = {table[name, S]
]:7.2f}")

```

9.3.2 Partial sums of squares: Types II and III

The second answer credits a term with what it adds last. For models whose terms are factors and their interactions, “last” needs care, because a main effect is part of every interaction that contains it. Say that a term $\mathbf{X}_i$ **contains** a term $\mathbf{X}_j$ if the factors of $\mathbf{X}_j$ are a proper subset of the factors of $\mathbf{X}_i$. For example the interaction AB contains the main effects A and B .

Definition 9.3.2 — Partial sums of squares; Types II and III

For a model with terms $\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_k$, write $\mathbf{X}_{(-j)}$ for $\mathbf{X}$ with the columns of $\mathbf{X}_j$ removed.

- (a) The **partial** (or **last-in**) sum of squares of $\mathbf{X}_j$ is $SS(\mathbf{X}_j \mid \mathbf{X}_{(-j)}) = \mathbf{y}^\top (\mathbf{M} - \mathbf{M}_{(-j)}) \mathbf{y}$, where $\mathbf{M}_{(-j)}$ projects onto $\mathcal{C}(\mathbf{X}_{(-j)})$.
- (b) The **Type II** sum of squares of $\mathbf{X}_j$ is its extra sum of squares given all terms that do not contain it: $SS(\mathbf{X}_j \mid \mathbf{X}_i : \mathbf{X}_i \text{ does not contain } \mathbf{X}_j, i \neq j)$.
- (c) The **Type III** sum of squares of $\mathbf{X}_j$ is its partial sum of squares (a) computed when every factor is coded by sum-to-zero contrast columns and every interaction by the products of the columns of its factors.

When no term contains another, as in a regression on numerical regressors without interactions, the three coincide, and they equal the sequential sum of squares of $\mathbf{X}_j$ when it is entered last. They also coincide for a term that is not contained in any other, such as the highest-order interaction. They differ for main effects in models with interactions.

Partial sums of squares are not a decomposition. The subspaces $\mathcal{C}(\mathbf{X}) \ominus \mathcal{C}(\mathbf{X}_{(-j)})$ are generally not orthogonal, so the partial sums of squares can add up to more or less than the regression sum of squares ([Exercise \(C1\)](#)), and they need not be independent of each other, though each is independent of the residual sum of squares under normality.

Why does the definition of Type III name a coding? Because in models with interactions the partial sum of squares of a main effect depends on the coding. Deleting the columns of A while keeping those of AB leaves the reduced space $\mathcal{C}(\mathbf{1}, \mathbf{X}_B, \mathbf{X}_{AB})$, which depends on which columns represent AB , although the full space does not. Different codings give different reduced models and so test different hypotheses.

9.3.3 What each sum of squares tests

A sum of squares $\mathbf{y}^\top \mathbf{P} \mathbf{y}$ with $\mathbf{P}$ a projection of rank d has expectation

$$\mathbb{E}(\mathbf{Y}^\top \mathbf{P} \mathbf{Y}) = \sigma^2 d + \|\mathbf{P} \boldsymbol{\mu}\|^2, \quad \boldsymbol{\mu} = \mathbb{E}(\mathbf{Y}),$$

by [Theorem 2.3.1](#) with $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$. So the hypothesis a sum of squares addresses is $\mathbf{P} \boldsymbol{\mu} = \mathbf{0}$: *the mean has no component in the subspace onto which it projects*. Under normality this is exactly the condition for its F ratio to be central ([Section 9.6](#)). We translate it into statements about parameters.

Proposition 9.3.1 — Hypotheses of sequential and partial sums of squares

Let $\mathbf{X} = [\mathbf{X}_0, \dots, \mathbf{X}_k]$ have full column rank and $\boldsymbol{\mu} = \sum_l \mathbf{X}_l \beta_l$.

- (a) The partial sum of squares of $\mathbf{X}_j$ addresses $\beta_j = \mathbf{0}$: $(\mathbf{M} - \mathbf{M}_{(-j)}) \boldsymbol{\mu} = \mathbf{0}$ iff $\beta_j = \mathbf{0}$.
- (b) The sequential sum of squares of $\mathbf{X}_j$ addresses

$$\beta_j + (\tilde{\mathbf{X}}_j^\top \tilde{\mathbf{X}}_j)^{-1} \tilde{\mathbf{X}}_j^\top \sum_{l>j} \mathbf{X}_l \beta_l = \mathbf{0}, \quad \tilde{\mathbf{X}}_j = (\mathbf{I} - \mathbf{M}_{j-1}) \mathbf{X}_j.$$

This is $\beta_j = \mathbf{0}$ whenever the later terms have zero coefficients, or whenever $\tilde{\mathbf{X}}_j^\top \mathbf{X}_l = \mathbf{0}$ for all $l > j$.

Proof.

- (a) For $l \neq j$, $\mathbf{X}_l \beta_l$ lies in $\mathcal{C}(\mathbf{X}_{(-j)}) \subseteq \mathcal{C}(\mathbf{X})$, so $(\mathbf{M} - \mathbf{M}_{(-j)}) \mathbf{X}_l \beta_l = \mathbf{0}$. Let $\tilde{\mathbf{X}}_j = (\mathbf{I} - \mathbf{M}_{(-j)}) \mathbf{X}_j$. By [Lemma 9.2.1](#), $\mathbf{M} - \mathbf{M}_{(-j)}$ projects onto $\mathcal{C}(\tilde{\mathbf{X}}_j)$, so $(\mathbf{M} - \mathbf{M}_{(-j)}) \mathbf{X}_j \beta_j$ is the projection of $\mathbf{X}_j \beta_j$ onto that space. Since $\mathbf{X}_j \beta_j - \tilde{\mathbf{X}}_j \beta_j = \mathbf{M}_{(-j)} \mathbf{X}_j \beta_j$ is orthogonal to $\mathcal{C}(\tilde{\mathbf{X}}_j)$, the projection is $\tilde{\mathbf{X}}_j \beta_j$. The columns of $\tilde{\mathbf{X}}_j$ are independent ([Lemma 6.6.1\(b\)](#)), so it vanishes iff $\beta_j = \mathbf{0}$.
- (b) Terms with $l < j$ are annihilated by $\mathbf{M}_j - \mathbf{M}_{j-1}$, which projects onto $\mathcal{C}(\tilde{\mathbf{X}}_j)$ by [Lemma 9.2.1](#). As in (a), $\mathbf{X}_j \beta_j$ projects to $\tilde{\mathbf{X}}_j \beta_j$. A later term projects to $\tilde{\mathbf{X}}_j (\tilde{\mathbf{X}}_j^\top \tilde{\mathbf{X}}_j)^{-1} \tilde{\mathbf{X}}_j^\top \mathbf{X}_l \beta_l$ by [Theorem 6.3.7](#). Adding, $(\mathbf{M}_j - \mathbf{M}_{j-1}) \boldsymbol{\mu} = \tilde{\mathbf{X}}_j \mathbf{v}$, where $\mathbf{v}$ is the vector in the display, and this is zero iff $\mathbf{v} = \mathbf{0}$ because $\tilde{\mathbf{X}}_j$ has independent columns. ■

So a sequential sum of squares answers the question “does $\mathbf{X}_j$ matter?” only if the terms fitted after it do not matter, or are orthogonal to it after adjusting for the terms before it. Otherwise the effects of later terms leak into it, through the coefficients $(\tilde{\mathbf{X}}_j^\top \tilde{\mathbf{X}}_j)^{-1} \tilde{\mathbf{X}}_j^\top \mathbf{X}_l$ of the regressions of later terms on $\tilde{\mathbf{X}}_j$.

For factors, the natural parameters are the cell means. Consider a two-way layout with factor A at a levels and B at b levels, $n_{ij} \geq 1$ observations in cell (i, j) , row totals $n_{i\cdot}$, and cell means μ_{ij} . The full model with interaction has model space $\mathcal{C}(\mathbf{X}) =$ all vectors that are constant within

cells, of dimension ab . Define the **weighted** and **unweighted** row means

$$\bar{\mu}_i^w = \frac{1}{n_{i\cdot}} \sum_j n_{ij} \mu_{ij}, \quad \bar{\mu}_{i\cdot} = \frac{1}{b} \sum_j \mu_{ij}.$$

Theorem 9.3.2 — What the sums of squares for a main effect test

In the two-way layout with all $n_{ij} \geq 1$:

- (a) the sequential sum of squares $\text{SS}(A \mid \mu)$ of A entered first addresses $\bar{\mu}_1^w = \cdots = \bar{\mu}_a^w$;
- (b) the Type III sum of squares of A addresses $\bar{\mu}_{1\cdot} = \cdots = \bar{\mu}_{a\cdot}$;
- (c) the partial sum of squares of the columns of A under reference-level coding (indicators of levels $2, \dots, a$ of A and $2, \dots, b$ of B , and their products for AB) addresses $\mu_{11} = \mu_{21} = \cdots = \mu_{a1}$, the equality of the A means at the first level of B ;
- (d) the Type II sum of squares $\text{SS}(A \mid \mu, B)$ addresses a hypothesis that, when the cell means are additive ($\mu_{ij} = \alpha_i + \beta_j$), is equivalent to $\alpha_1 = \cdots = \alpha_a$.

Each sum of squares equals the sum of squares Eq. 9.2.1 of the hypothesis it addresses, in the cell-means model.

Proof. Let $\mathbf{P}_0$, $\mathbf{P}_A$, $\mathbf{P}_B$ and $\mathbf{P}_{A+B}$ be the projections onto the constant vectors, the vectors constant within rows, those constant within columns, and their sum. Each hypothesis below defines a subspace $\mathcal{S}_0$ of $\mathcal{C}(\mathbf{X})$, and each sum of squares is $\|\mathbf{P}\mathbf{y}\|^2$ with $\mathbf{P}$ the projection onto $\mathcal{C}(\mathbf{X}) \ominus \mathcal{S}_0$. Then $\mathbf{P}\boldsymbol{\mu} = \mathbf{0}$ iff $\boldsymbol{\mu} \in \mathcal{S}_0$, and the sum of squares equals $\text{SSE}_0 - \text{SSE}$ for $\mathcal{S}_0$, which is the sum of squares of the hypothesis by Theorem 9.2.4.

- (a) $\mathbf{P}_A$ replaces each entry of a vector by the average of its row, so $\mathbf{P}_A\boldsymbol{\mu}$ has entry $\bar{\mu}_i^w$ in row i , and $\mathbf{P}_0\boldsymbol{\mu}$ has every entry equal to the overall weighted mean $\sum_i n_{i\cdot} \bar{\mu}_i^w / n$. So $(\mathbf{P}_A - \mathbf{P}_0)\boldsymbol{\mu} = \mathbf{0}$ iff all $\bar{\mu}_i^w$ equal their weighted average, that is, iff they are all equal. The relevant $\mathcal{S}_0$ is $\{\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X}) : (\mathbf{P}_A - \mathbf{P}_0)\boldsymbol{\mu} = \mathbf{0}\}$, whose orthogonal complement in $\mathcal{C}(\mathbf{X})$ is $\mathcal{C}(\mathbf{P}_A - \mathbf{P}_0)$, because $\mathcal{C}(\mathbf{P}_A - \mathbf{P}_0) \subseteq \mathcal{C}(\mathbf{X})$.
- (b) Any $a \times b$ array of cell means decomposes uniquely as

$$\mu_{ij} = \bar{\mu}_{..} + (\bar{\mu}_{i\cdot} - \bar{\mu}_{..}) + (\bar{\mu}_{\cdot j} - \bar{\mu}_{..}) + (\mu_{ij} - \bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot j} + \bar{\mu}_{..}),$$

where bars denote unweighted averages. The four arrays lie in the mutually orthogonal subspaces (for the unweighted inner product on arrays) of constants, row effects summing to zero, column effects summing to zero, and arrays with all row and column sums zero, of dimensions $1, a-1, b-1, (a-1)(b-1)$. Sum-to-zero coding spans exactly these: the columns $\mathbf{u}_i - \mathbf{u}_1$ ($i = 2, \dots, a$), where $\mathbf{u}_i$ indicates row i , give all row effects α_i with $\sum_i \alpha_i = 0$; similarly for B ; and the $(a-1)(b-1)$ products are linearly independent arrays with zero row and column sums, so they span that subspace. Deleting the columns of A removes exactly the row-effect component, so the reduced model is $\mathcal{S}_0 = \{\boldsymbol{\mu} : \bar{\mu}_{i\cdot} = \bar{\mu}_{..} \text{ for all } i\}$.

- (c) With reference coding, every cell-mean array is uniquely $\mu_{ij} = m + \alpha_i + \beta_j + \gamma_{ij}$ with $\alpha_1 = \beta_1 = 0$ and $\gamma_{1j} = \gamma_{i1} = 0$; explicitly $m = \mu_{11}$, $\alpha_i = \mu_{i1} - \mu_{11}$, $\beta_j = \mu_{1j} - \mu_{11}$ and $\gamma_{ij} = \mu_{ij} - \mu_{i1} - \mu_{1j} + \mu_{11}$. Deleting the columns of A forces $\alpha_i = 0$, that is, $\mu_{i1} = \mu_{11}$ for all i .
- (d) The Type II sum of squares is $\mathbf{y}^\top (\mathbf{P}_{A+B} - \mathbf{P}_B) \mathbf{y}$. If $\boldsymbol{\mu}$ is additive, then $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{P}_{A+B})$ and $(\mathbf{P}_{A+B} - \mathbf{P}_B)\boldsymbol{\mu} = \boldsymbol{\mu} - \mathbf{P}_B\boldsymbol{\mu}$, which is zero iff $\boldsymbol{\mu}$ is constant within columns,

that is, iff α_i does not depend on i .

■

These are four different questions. Part (a) averages over B with the design's weights, so if B matters and the rows have different mixes of B levels, it can detect “an effect of A ” that is entirely an effect of B ([Exercise B3](#)). Part (b) averages with equal weights. Part (c) compares the rows at one level of B : a simple effect, not a main effect, and one that changes with the reference level. Part (d) is the natural test of A without interaction; with interaction its hypothesis has weights that depend on all the n_{ij} .

Warning (“Type III” with the default coding). Many regression programs code factors by reference levels unless told otherwise. Asking them for last-in sums of squares of main effects in a model with interactions then produces the hypotheses of [Theorem 9.3.2\(c\)](#), which concern a single level of the other factor. They are not Type III sums of squares, although they are sometimes reported under that name. Use sum-to-zero coding, or better, write down the hypothesis and use [Eq. 9.2.1](#).

9.3.4 An unbalanced example

Example. (Party, education and political self-placement)

In the 1996 American National Election Study, 944 respondents placed themselves on a seven-point scale from extremely liberal (1) to extremely conservative (7). Classify them by party identification (Democrat, Independent, Republican, counting those who only lean to a party as Independents) and by education (high school or less, college, graduate study). The cell counts and means are:

	high school	college	graduate
Democrat	146, 3.97	108, 3.43	126, 2.92
Independent	80, 4.39	64, 4.28	95, 4.09
Republican	87, 5.40	105, 5.36	133, 5.40

The design is unbalanced, and education is associated with party: 38% of the Democrats but only 27% of the Republicans have at most a high school education. The pattern of means shows an interaction. Among Democrats self-placement moves left with education by more than a scale point, while among Republicans it does not move at all. The sums of squares for the terms are:

Term	df	Type I, party first	Type I, education first	Type II	Type III	reference coding
party	2	647.46	674.42	674.42	650.41	112.27
education	2	43.26	16.31	43.26	32.03	75.12
interaction	4	35.81	35.81	35.81	35.81	35.81
residual	935	1224.62	1224.62	1224.62	1224.62	1224.62

The two Type I columns each add up to the model sum of squares 726.54, up to rounding. The other columns do not: the Type II entries add up to more. Education receives 16.31 when entered first and 43.26 after party. Since the least educated group contains the most Democrats, who are the most liberal, the marginal comparison mixes the leftward drift of educated Democrats with the partisan composition of each education group, and the two partly cancel. The Type III sum of squares 32.03 tests whether the unweighted averages over the three parties of the education means are equal. The last column, computed with reference coding, answers different questions: 75.12 for education measures the education gradient among *Democrats only* (the first party level), which is the steepest, and 112.27 for party compares the parties among respondents with a high school education only. The interaction row is the same in every column because the interaction is the highest-order term.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
d = sm.datasets.anes96.load_pandas().data
party = pd.cut(d["PID"], [-1, 1.5, 4.5, 6], labels=["Dem", "Ind", "Rep"]).rename("party")
educ = pd.cut(d["educ"], [0, 3.5, 5.5, 7], labels=["HS", "College", "Graduate"]).rename("educ")
y = d["selfLR"].to_numpy()
n = len(y)
print(pd.crosstab(party, educ))
print(d.groupby([party, educ], observed=True)["selfLR"].mean().unstack().round(3))
```

```
def proj(*blocks):
    """Orthogonal projection onto the span of the given column blocks."""
    Z = np.column_stack(blocks)
    U, s, _ = np.linalg.svd(Z, full_matrices=False)
    U = U[:, s > 1e-10 * s[0]] # orthonormal basis of C(Z), any
    rank # rank
    return U @ U.T

one = np.ones((n, 1))
A = pd.get_dummies(party).to_numpy(float) # indicators of party
B = pd.get_dummies(educ).to_numpy(float) # indicators of education
AB = np.column_stack([A[:, [i]] * B for i in range(3)]) # the nine cells
PO, PA, PB = proj(one), proj(one, A), proj(one, B)
```

```

PAB, M = proj(one, A, B), proj(AB)                                # additive model, cell-means
                        model
I = np.eye(n)

def ss(P):
    return y @ P @ y

type1_AB = {"party": ss(PA - P0), "educ": ss(PAB - PA), "inter": ss(M - PAB)}
type1_BA = {"educ": ss(PB - P0), "party": ss(PAB - PB), "inter": ss(M - PAB)}
type2 = {"party": ss(PAB - PB), "educ": ss(PAB - PA), "inter": ss(M - PAB)}
sse = ss(I - M)

def coded(labels, kind):
    """Columns for a factor: sum-to-zero ("sum") or reference-level ("treatment")
    coding."""
    D = pd.get_dummies(labels).to_numpy(float)
    return D[:, 1:] - D[:, [0]] if kind == "sum" else D[:, 1:]

type3 = {}
for kind in ["sum", "treatment"]:
    Ca, Cb = coded(party, kind), coded(educ, kind)
    Cab = np.column_stack([Ca[:, [i]] * Cb for i in range(2)])
    full = proj(one, Ca, Cb, Cab)                                # always the cell-means space
    type3[kind] = {"party": ss(full - proj(one, Cb, Cab)),
                  "educ": ss(full - proj(one, Ca, Cab))}
    assert np.allclose(full, M)
for name, table in [("Type I, party first", type1_AB), ("Type I, education first",
type1_BA),
                    ("Type II", type2), ("Type III, sum coding", type3["sum"]),
                    ("Type III, treatment coding", type3["treatment"])] :
    print(f"{name:30s}", {k: round(float(v), 2) for k, v in table.items()})

```

The script behind these cells also checks every entry against statsmodels, and it confirms [Theorem 9.3.2](#) numerically: the party entries of the first, fourth and fifth columns equal [Eq. 9.2.1](#) for the hypotheses of equal weighted row means, equal unweighted row means, and equal means in the high-school column.

9.3.5 Which one to use

There is no single right sum of squares for a main effect in an unbalanced design with interaction, because there is no single main-effect hypothesis. Decide the hypothesis first, from the four above, and compute its sum of squares with [Eq. 9.2.1](#). If A and B interact, a single “main effect of A ” summarizes a heterogeneous comparison with arbitrary weights; Nelder (1977) argued that such tests are rarely meaningful, and Type II sums of squares respect this marginality by adjusting A for B but not for AB . Without interaction the Type II and Type III hypotheses coincide, and Type II, which does not adjust for the interaction columns, generally gives the more powerful test. In regression without interactions all partial types agree; the distinction matters only for terms contained in other terms.

Chapter 17 treats empty cells, where even Type III hypotheses may be nonestimable ([Exercise \(B4\)](#)).

9.3.6 Exercises

9.3.6.1 A. Check your understanding

Exercise A1.

For a two-way model with terms A , B and AB , write each of the Type I sums of squares (both orders), the Type II sums of squares and the partial sums of squares in the notation $SS(\cdot | \cdot)$. Which of them are equal for every data set, whatever the design?

Solution. Type I, A first: $SS(A | \mu)$, $SS(B | \mu, A)$, $SS(AB | \mu, A, B)$. Type I, B first: $SS(B | \mu)$, $SS(A | \mu, B)$, $SS(AB | \mu, A, B)$. Type II: $SS(A | \mu, B)$, $SS(B | \mu, A)$, $SS(AB | \mu, A, B)$. Partial: $SS(A | \mu, B, AB)$, $SS(B | \mu, A, AB)$, $SS(AB | \mu, A, B)$, where the first two depend on the coding. Always equal: the interaction entries of all four lists; the Type II entry for A and the Type I entry for A with B first; the Type II entry for B and the Type I entry for B with A first.

9.3.6.2 B. Practice

Exercise B1.

Let $\mathbf{X} = [\mathbf{1}, \mathbf{x}_1, \mathbf{x}_2]$, and let r_{y1} , r_{y2} and r_{12} be the sample correlations among $\mathbf{y}$, $\mathbf{x}_1$ and $\mathbf{x}_2$, with $|r_{12}| < 1$. Show that

$$\frac{SS(\mathbf{x}_2 | \mathbf{1})}{SST} = r_{y2}^2, \quad \frac{SS(\mathbf{x}_2 | \mathbf{1}, \mathbf{x}_1)}{SST} = \frac{(r_{y2} - r_{y1}r_{12})^2}{1 - r_{12}^2}.$$

Show that adjusting for $\mathbf{x}_1$ increases the sum of squares of $\mathbf{x}_2$ whenever $r_{y2} = 0$ and $r_{y1}r_{12} \neq 0$, and find the condition for it in general.

Solution. Work in $\mathbf{1}^\perp$ with the centred vectors $\mathbf{u} = \mathbf{y} - \bar{y}\mathbf{1}$, $\mathbf{v}_1$, $\mathbf{v}_2$, scaled to unit length, so that inner products are correlations and $SST = \|\mathbf{y} - \bar{y}\mathbf{1}\|^2$. By Lemma 9.2.1, $SS(\mathbf{x}_2 | \mathbf{1}) = SST(\mathbf{u}^\top \mathbf{v}_2)^2 = SST r_{y2}^2$. Adjusting for $\mathbf{x}_1$ replaces $\mathbf{v}_2$ by $\mathbf{w} = \mathbf{v}_2 - r_{12}\mathbf{v}_1$, with $\|\mathbf{w}\|^2 = 1 - r_{12}^2$ and $\mathbf{u}^\top \mathbf{w} = r_{y2} - r_{y1}r_{12}$, so $SS(\mathbf{x}_2 | \mathbf{1}, \mathbf{x}_1) = SST(\mathbf{u}^\top \mathbf{w})^2 / \|\mathbf{w}\|^2$. If $r_{y2} = 0$ the first ratio is zero and the second is $r_{y1}^2 r_{12}^2 / (1 - r_{12}^2) > 0$. In general the sum of squares increases iff $(r_{y2} - r_{y1}r_{12})^2 > r_{y2}^2(1 - r_{12}^2)$.

Exercise B2.

Show that the hypotheses of Theorem 9.3.2(a) and (b) are the same subspace of cell means iff $n_{ij} = n_{i.}/b$ for all i, j , that is, iff within every row all cells have the same count. Show that under proportional frequencies, $n_{ij} = n_{i.}n_{.j}/n$ (Theorem 9.4.3), the hypothesis (a) compares row means weighted by the column proportions $n_{.j}/n$, the same weights in every row.

Solution. Write $w_{ij} = n_{ij}/n_{i.}$, so $\sum_j w_{ij} = 1$, and let $f_i(\boldsymbol{\mu}) = \sum_j w_{ij}\mu_{ij}$ and $g_i(\boldsymbol{\mu}) = \sum_j \mu_{ij}/b$. Hypothesis (a) is the null space of the linear functionals $\sum_i c_i f_i$ with $\sum_i c_i = 0$, and (b) that of the functionals $\sum_i d_i g_i$ with $\sum_i d_i = 0$. Two subspaces defined as null spaces coincide iff the spaces of functionals coincide. The functional $f_i - f_1$ has coefficient

w_{ij} on cell (i, j) and $-w_{1j}$ on cell $(1, j)$. If it equals some $\sum_i d_i g_i$, whose coefficients are constant across each row, then w_{ij} does not depend on j , so $w_{ij} = 1/b$; likewise $w_{1j} = 1/b$. Hence coincidence forces $n_{ij} = n_{i\cdot}/b$. Conversely, if all $w_{ij} = 1/b$, then $f_i = g_i$. Under proportional frequencies $w_{ij} = n_{ij}/n$, which does not depend on i , so (a) compares $\sum_j (n_{\cdot j}/n) \mu_{ij}$ across rows.

Exercise B3.

Take $a = b = 2$, counts $n_{11} = n_{22} = 9$ and $n_{12} = n_{21} = 1$, and additive cell means $\mu_{ij} = \beta_j$ with $\beta_1 = 0$ and $\beta_2 = 1$, so that A has no effect. Compute the weighted row means $\bar{\mu}_i^w$ and show that the hypothesis addressed by the Type I sum of squares of A entered first is false. What do the Type II and Type III hypotheses say here?

Exercise B4.

Suppose that in a 2×2 layout the cell $(2, 2)$ is empty. Show that $\bar{\mu}_1 - \bar{\mu}_2$ is not estimable. Compute what the last-in sum of squares of A under sum-to-zero coding tests in this case. *Hint:* with a cell missing, the interaction columns restricted to the observed cells are no longer independent of the main-effect columns.

9.3.6.3 C. Going deeper

Exercise C1.

With the notation of [Exercise \(B1\)](#), show that

$$\frac{SS(\mathbf{x}_1 \mid \mathbf{1}, \mathbf{x}_2) + SS(\mathbf{x}_2 \mid \mathbf{1}, \mathbf{x}_1) - SSR}{SST} = \frac{r_{12}[r_{12}(r_{y1}^2 + r_{y2}^2) - 2r_{y1}r_{y2}]}{1 - r_{12}^2}.$$

Deduce that the partial sums of squares add up to the regression sum of squares when $r_{12} = 0$, that they add up to less when $r_{y1} = r_{y2} \neq 0$ and $0 < r_{12} < 1$, and give correlations for which they add up to more.

9.4 Orthogonal designs

[Section 9.3](#) showed how much the sums of squares for a term can depend on the other terms. This section asks when they do not. The answer is geometric: after the mean is removed, the terms must span mutually orthogonal subspaces. Balanced experiments achieve this by design and observational data essentially never do, but orthogonality can be manufactured inside a term by a change of basis, as with orthogonal polynomials and orthogonal contrasts.

9.4.1 When the order does not matter

Let the model matrix be $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1, \dots, \mathbf{X}_k]$ with an intercept and k terms. Write $\mathbf{P}_0 = n^{-1}\mathbf{1}\mathbf{1}^\top$, and let

$$\tilde{\mathbf{X}}_j = (\mathbf{I} - \mathbf{P}_0)\mathbf{X}_j, \quad \mathcal{V}_j = \mathcal{C}(\tilde{\mathbf{X}}_j),$$

be the centred columns of term j and the subspace they span. By [Lemma 9.2.1](#) with $\mathbf{X}_1 = \mathbf{1}$, $\mathcal{C}(\mathbf{1}, \mathbf{X}_j) = \text{span}(\mathbf{1}) \oplus \mathcal{V}_j$, so $\text{SS}(\mathbf{X}_j | \mathbf{1}) = \|\mathbf{P}_{\mathcal{V}_j}\mathbf{y}\|^2$, the sum of squares of the term fitted on its own after the mean. Call the design **orthogonal** (for these terms) if the subspaces $\mathcal{V}_1, \dots, \mathcal{V}_k$ are mutually orthogonal, that is, if $\tilde{\mathbf{X}}_i^\top \tilde{\mathbf{X}}_j = \mathbf{0}$ for $i \neq j$. For numerical regressors this says that every pair of regressors from different terms has sample correlation zero.

Theorem 9.4.1 — Orthogonal designs

For a model $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1, \dots, \mathbf{X}_k]$, the following are equivalent:

- (a) the centred subspaces $\mathcal{V}_1, \dots, \mathcal{V}_k$ are mutually orthogonal;
- (b) $\mathbf{M} = \mathbf{P}_0 + \mathbf{P}_{\mathcal{V}_1} + \dots + \mathbf{P}_{\mathcal{V}_k}$;
- (c) for every j and every $\mathbf{y}$, the marginal and partial sums of squares agree: $\text{SS}(\mathbf{X}_j | \mathbf{1}) = \text{SS}(\mathbf{X}_j | \mathbf{X}_{(-j)})$;
- (d) for every $\mathbf{y}$, the marginal sums of squares add up to the regression sum of squares: $\sum_j \text{SS}(\mathbf{X}_j | \mathbf{1}) = \text{SS}(\mathbf{X}_1, \dots, \mathbf{X}_k | \mathbf{1})$.

When they hold, $\text{SS}(\mathbf{X}_j | \mathbf{1}, \mathbf{X}_S) = \|\mathbf{P}_{\mathcal{V}_j}\mathbf{y}\|^2$ for every set S of other terms. In particular the sequential sum of squares of $\mathbf{X}_j$ is the same in every order, and it equals the partial and the Type II sums of squares of $\mathbf{X}_j$.

Proof. (a) implies (b) and the final statement. Let S be any set of terms. The subspaces $\text{span}(\mathbf{1})$ and $\mathcal{V}_l$, $l \in S$, are mutually orthogonal, because each $\mathcal{V}_l \subseteq \mathbf{1}^\perp$, and they sum to $\mathcal{C}(\mathbf{1}, \mathbf{X}_S)$, because each $\mathcal{C}(\mathbf{1}, \mathbf{X}_l) = \text{span}(\mathbf{1}) \oplus \mathcal{V}_l$. By [Theorem 6.3.8](#) the projection onto $\mathcal{C}(\mathbf{1}, \mathbf{X}_S)$ is $\mathbf{P}_0 + \sum_{l \in S} \mathbf{P}_{\mathcal{V}_l}$. With S all terms this is (b). With S and $S \cup \{j\}$, the difference of the two projections is $\mathbf{P}_{\mathcal{V}_j}$, which gives $\text{SS}(\mathbf{X}_j | \mathbf{1}, \mathbf{X}_S) = \|\mathbf{P}_{\mathcal{V}_j}\mathbf{y}\|^2$. Sequential, partial and Type II sums of squares of $\mathbf{X}_j$ are all of this form.

(b) implies (d): $\text{SS}(\mathbf{X}_1, \dots, \mathbf{X}_k | \mathbf{1}) = \mathbf{y}^\top (\mathbf{M} - \mathbf{P}_0) \mathbf{y} = \sum_j \mathbf{y}^\top \mathbf{P}_{\mathcal{V}_j} \mathbf{y}$.

(d) implies (a). The two sides of (d) are the quadratic forms of the symmetric matrices $\sum_j \mathbf{P}_{\mathcal{V}_j}$ and $\mathbf{M} - \mathbf{P}_0$. Two symmetric matrices with the same quadratic form are equal, because $2\mathbf{u}^\top \mathbf{A} \mathbf{v} = (\mathbf{u} + \mathbf{v})^\top \mathbf{A} (\mathbf{u} + \mathbf{v}) - \mathbf{u}^\top \mathbf{A} \mathbf{u} - \mathbf{v}^\top \mathbf{A} \mathbf{v}$. So $\sum_j \mathbf{P}_{\mathcal{V}_j} = \mathbf{M} - \mathbf{P}_0$, which is idempotent ([Theorem 6.5.1](#)). By the algebraic form of Cochran's theorem ([Theorem 4.5.1](#)), projections whose sum is idempotent satisfy $\mathbf{P}_{\mathcal{V}_i} \mathbf{P}_{\mathcal{V}_j} = \mathbf{0}$ for $i \neq j$, which is (a).

(a) implies (c) is part of the final statement.

(c) implies (a). As before, equality of the quadratic forms gives $\mathbf{P}_{\mathcal{V}_j} = \mathbf{M} - \mathbf{M}_{(-j)}$. The right-hand side projects onto $\mathcal{C}(\mathbf{X}) \ominus \mathcal{C}(\mathbf{X}_{(-j)})$ ([Theorem 6.5.1](#)), which is orthogonal to $\mathcal{C}(\mathbf{X}_{(-j)})$. The space $\mathcal{C}(\mathbf{X}_{(-j)})$ contains $\mathcal{V}_i$ for every $i \neq j$, since $\tilde{\mathbf{X}}_i = \mathbf{X}_i - \mathbf{P}_0 \mathbf{X}_i$ has columns in $\mathcal{C}(\mathbf{1}, \mathbf{X}_i)$. So $\mathcal{V}_j \perp \mathcal{V}_i$. ■

Idea. Orthogonality is a property of the *centred* subspaces. The raw columns of two indicator variables are never orthogonal to the intercept, and a regressor and its square are rarely orthogonal. What matters is whether the parts of the terms that vary around their means point in perpendicular directions.

Orthogonality also stabilizes the coefficients.

Corollary 9.4.2 — Coefficients in orthogonal designs

Suppose $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1, \dots, \mathbf{X}_k]$ has full column rank and satisfies the conditions of [Theorem 9.4.1](#). Then the least squares coefficients of the columns of $\mathbf{X}_j$ are the same in every model that contains $\mathbf{1}$ and $\mathbf{X}_j$, whichever other terms it contains. Only the intercept changes.

Proof. In the model with terms $S \ni j$, the fitted vector is $\mathbf{P}_0 \mathbf{y} + \sum_{l \in S} \mathbf{P}_l \mathbf{y}$ by the proof of the theorem, and $\mathbf{P}_l \mathbf{y} = \tilde{\mathbf{X}}_l \mathbf{b}_l$ with $\mathbf{b}_l = (\tilde{\mathbf{X}}_l^\top \tilde{\mathbf{X}}_l)^{-1} \tilde{\mathbf{X}}_l^\top \mathbf{y}$, which does not depend on S . Since $\tilde{\mathbf{X}}_l \mathbf{b}_l = \mathbf{X}_l \mathbf{b}_l - \mathbf{P}_0 \mathbf{X}_l \mathbf{b}_l$, the fitted vector equals $\sum_{l \in S} \mathbf{X}_l \mathbf{b}_l$ plus a multiple of $\mathbf{1}$. The coefficients are unique by full rank, so the coefficient vector of $\mathbf{X}_j$ is $\mathbf{b}_j$ in every such model. ■

With uncorrelated regressors, dropping one leaves the others' estimates as they were, and each regressor has an order-free sum of squares. This is one reason factorial experiments are laid out on orthogonal arrays (Chapter 18).

9.4.2 Balanced and proportional layouts

For two factors the orthogonality condition reduces to a condition on the cell counts.

Theorem 9.4.3 — Proportional frequencies

In a two-way layout with factor A at a levels, factor B at b levels, cell counts n_{ij} , and all row totals $n_{i\cdot}$ and column totals $n_{\cdot j}$ positive, the centred subspaces of A and B are orthogonal iff

$$n_{ij} = \frac{n_{i\cdot} n_{\cdot j}}{n} \quad \text{for all } i, j. \quad (9.4.1)$$

Proof. A vector in the centred subspace of A takes a value α_i on every observation in row i , with $\sum_i n_{i\cdot} \alpha_i = 0$. A vector in that of B takes the value β_j in column j , with $\sum_j n_{\cdot j} \beta_j = 0$. Their inner product is $\sum_{i,j} n_{ij} \alpha_i \beta_j = \boldsymbol{\alpha}^\top \mathbf{N} \boldsymbol{\beta}$, with $\mathbf{N} = (n_{ij})$. Write $\mathbf{r} = (n_{1\cdot}, \dots, n_{a\cdot})^\top$ and $\mathbf{c} = (n_{\cdot 1}, \dots, n_{\cdot b})^\top$. The subspaces are orthogonal iff $\boldsymbol{\alpha}^\top \mathbf{N} \boldsymbol{\beta} = 0$ whenever $\boldsymbol{\alpha}^\top \mathbf{r} = 0$ and $\mathbf{c}^\top \boldsymbol{\beta} = 0$.

If [Eq. 9.4.1](#) holds, $\mathbf{N} = \mathbf{r} \mathbf{c}^\top / n$ and $\boldsymbol{\alpha}^\top \mathbf{N} \boldsymbol{\beta} = (\boldsymbol{\alpha}^\top \mathbf{r})(\mathbf{c}^\top \boldsymbol{\beta}) / n = 0$. Conversely, let $\mathbf{E} = \mathbf{N} - \mathbf{r} \mathbf{c}^\top / n$. Its row sums are $\mathbf{r} - \mathbf{r}(\mathbf{c}^\top \mathbf{1}) / n = \mathbf{0}$ and its column sums are $\mathbf{c}^\top - (\mathbf{1}^\top \mathbf{r}) \mathbf{c}^\top / n = \mathbf{0}^\top$, since $\mathbf{1}^\top \mathbf{r} = \mathbf{c}^\top \mathbf{1} = n$. So $\boldsymbol{\alpha}^\top \mathbf{E} \boldsymbol{\beta}$ does not change when $\boldsymbol{\alpha}$ or $\boldsymbol{\beta}$ is

shifted by a multiple of $\mathbf{1}$. Given any $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, shift them to $\boldsymbol{\alpha}' = \boldsymbol{\alpha} - (\boldsymbol{\alpha}^\top \mathbf{r}/n)\mathbf{1}$ and $\boldsymbol{\beta}' = \boldsymbol{\beta} - (\mathbf{c}^\top \boldsymbol{\beta}/n)\mathbf{1}$, which satisfy $\boldsymbol{\alpha}'^\top \mathbf{r} = 0$ and $\mathbf{c}^\top \boldsymbol{\beta}' = 0$. Then $\boldsymbol{\alpha}^\top \mathbf{E}\boldsymbol{\beta} = \boldsymbol{\alpha}'^\top \mathbf{E}\boldsymbol{\beta}' = \boldsymbol{\alpha}'^\top \mathbf{N}\boldsymbol{\beta}' - 0 = 0$ by assumption. As $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are arbitrary, $\mathbf{E} = \mathbf{0}$.

■

Equal replication, $n_{ij} = m$, is the common case (Example (Additive two-factor model, one observation per cell), Exercise (C1)). If the interaction subspace is defined as the orthogonal complement of the additive model within the cell-constant vectors, then under Eq. 9.4.1 the subspaces for the mean, A , B and AB form a decomposition whatever the order.

Orthogonality settles Type I against Type II, but not Type III. Under proportional frequencies the Type I and Type II sums of squares of A coincide and address equality of row means weighted by the column proportions $n_{.j}/n$ (Exercise (B2)), while Type III weights the columns equally. They agree with equal replication or without interaction.

Example. (Firms and years)

Grunfeld's investment panel (Example (Grunfeld's investment data)) records 11 firms in each of 20 years, one observation per firm and year. As a two-way layout with factors firm and year, it has $n_{ij} = 1$, so Eq. 9.4.1 holds, and the additive model's sums of squares for the logarithm of investment are order-free:

Source	df	Sum of squares
firm	10	473.482
year	19	26.749
residual	190	14.667
corrected total	219	514.898

Firms differ enormously in the scale of their investment, and years matter much less. Adding the logarithm of market value as a covariate destroys the orthogonality, because value varies with both firm and year. The sum of squares for firm then depends on what else is in the model: it is 90.701 after year and value, and 88.443 after value alone. Most of the variation between firms is variation in size, which value already measures. This is the setting of analysis of covariance (Chapter 18).

```
import numpy as np
import pandas as pd
import statsmodels.api as sm

def proj(*blocks):
    Z = np.column_stack(blocks)
    U, s, _ = np.linalg.svd(Z, full_matrices=False)
    U = U[:, s > 1e-10 * s[0]]
    return U @ U.T

g = sm.datasets.grunfeld.load_pandas().data
y = np.log(g["invest"].to_numpy())
n = len(y)
one = np.ones((n, 1))
F = pd.get_dummies(g["firm"]).to_numpy(float)      # 11 firms
T = pd.get_dummies(g["year"]).to_numpy(float)      # 20 years
```

```

P0, PF, PT, PFT = proj(one), proj(one, F), proj(one, T), proj(one, F, T)

print("centred spaces orthogonal:", np.abs((PF - P0) @ (PT - P0)).max() < 1e-10)
print(f"firm:  first {y @ (PF - P0) @ y:8.3f}    after year {y @ (PFT - PT) @ y:8.3f}"
      ")
print(f"year:  first {y @ (PT - P0) @ y:8.3f}    after firm {y @ (PFT - PF) @ y:8.3f}"
      ")

v = np.log(g["value"].to_numpy())[:, None] # log market value
PFv, PTv, PFTv = proj(one, F, v), proj(one, T, v), proj(one, F, T, v)
print(f"with log value: firm after year, value {y @ (PFTv - PTv) @ y:8.3f}"
      f"    firm after value {y @ (PFv - proj(one, v)) @ y:8.3f}")

```

9.4.3 Orthogonal polynomials

A polynomial regression in a single variable x , with distinct values $x_1, \dots, x_n$, has a built-in chain of models,

$$\mathcal{P}_0 \subseteq \mathcal{P}_1 \subseteq \dots \subseteq \mathcal{P}_d, \quad \mathcal{P}_k = \text{span}(\mathbf{1}, \mathbf{x}, \mathbf{x}^2, \dots, \mathbf{x}^k),$$

where $\mathbf{x}^k$ is the vector with entries x_i^k . The decomposition of interest is that of the chain ([Theorem 9.1.1\(e\)](#)). Its k th subspace, $\mathcal{P}_{k-1}^\perp \cap \mathcal{P}_k$, is a line. Applying Gram–Schmidt ([Proposition 6.2.2](#)) to $\mathbf{1}, \mathbf{x}, \mathbf{x}^2, \dots$ produces unit vectors $\mathbf{q}_0, \mathbf{q}_1, \dots, \mathbf{q}_d$ with $\text{span}(\mathbf{q}_0, \dots, \mathbf{q}_k) = \mathcal{P}_k$ for every k . So $\mathbf{q}_k$ spans that line. The vectors $\mathbf{q}_k$, evaluated at the design points, are the **orthogonal polynomials** of the design. Three consequences follow at once.

1. The sequential sum of squares for degree k , the drop in residual sum of squares from degree $k-1$ to degree k , is $(\mathbf{q}_k^\top \mathbf{y})^2$, with one degree of freedom.
2. The columns $\mathbf{q}_1, \dots, \mathbf{q}_d$ are orthogonal and centred, so as separate terms they form an orthogonal design ([Theorem 9.4.1](#)), and their sums of squares do not depend on the order of entry.
3. The coefficient of $\mathbf{q}_k$ is $\mathbf{q}_k^\top \mathbf{y}$ in every fit that includes it ([Corollary 9.4.2](#)). Raising the degree never changes the lower coefficients.

For equally spaced x the orthogonal polynomials have integer multiples that are widely tabulated. For three points they are $(-1, 0, 1)$ and $(1, -2, 1)$. For four points they are $(-3, -1, 1, 3)$, $(1, -1, -1, 1)$ and $(-1, 3, -3, 1)$. They serve as contrasts for a quantitative factor with equally spaced levels, splitting the treatment sum of squares into linear, quadratic and higher components. Their numerical merits and the recurrence that generates them belong to Chapter 42.

Example. (A trend in the Nile)

The annual flow of the Nile at Aswan was recorded for the 100 years 1871–1970. How much of its variation is a smooth trend? The listing builds the orthonormal polynomials $\mathbf{q}_0, \dots, \mathbf{q}_4$ for the years by a QR factorization of the matrix of powers, and reads off the sequential sums of squares (in units of $(10^8 \text{ m}^3)^2$):

Degree	1	2	3	4
Sequential sum of squares	613893	309415	1894	134142

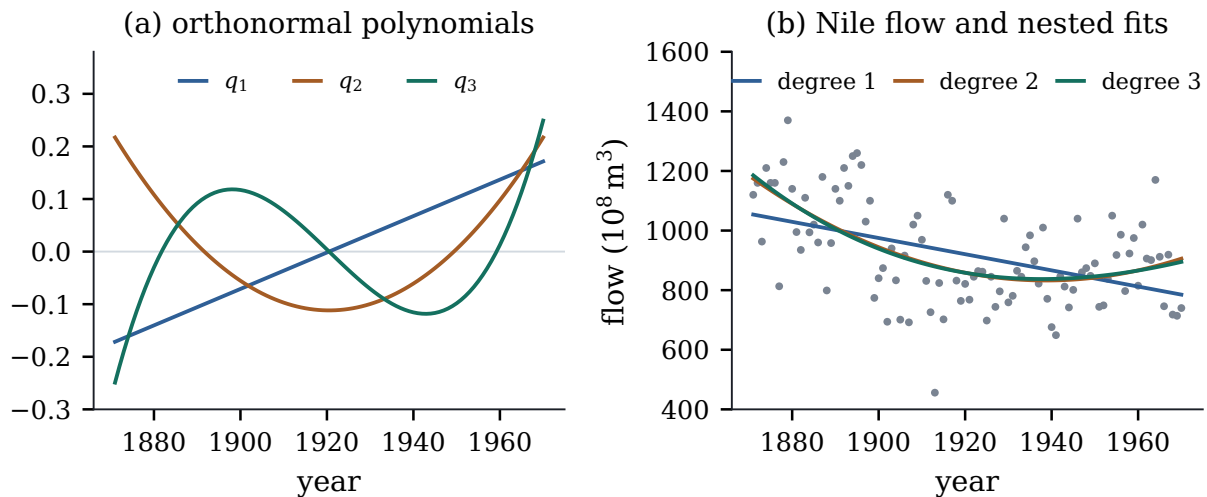


Figure 9.4.1: (a) The orthonormal polynomials q_1, q_2, q_3 for the years 1871–1970. (b) The Nile flow with the nested least squares fits of degrees 1, 2 and 3. The quadratic and cubic fits nearly coincide, because the cubic sum of squares is tiny.

The corrected total is 2835157. The linear term accounts for a fraction 0.217 of it, and the first three terms together for 0.326. Against the residual mean square of the cubic fit, 19895 on 96 degrees of freedom, the quadratic component has $F = 15.55$ and the cubic only 0.10. The quartic sum of squares is large again, a sign that the pattern is not a gentle polynomial at all. The flow dropped abruptly around the end of the nineteenth century (Cobb 1978), and a low-degree polynomial can only approximate a step (Figure 9.4.1).

With the raw powers of the centred and scaled year, the sums of squares depend on the order: the linear column entered after the quadratic and cubic ones receives 74751 instead of 613893. The raw linear coefficient is -134.4 in the linear and quadratic fits, but -117.3 in the cubic and quartic ones. It survives the quadratic term because the years are symmetric about their mean, which makes odd and even centred powers orthogonal (Exercise (C1)), but not the cubic term. The coefficient of q_1 is -783.5 in every fit.

```
import numpy as np
import statsmodels.api as sm
nile = sm.datasets.nile.load_pandas().data
year = nile["year"].to_numpy()
y = nile["volume"].to_numpy()

t = year - year.mean()
V = np.vander(t / t.max(), 5, increasing=True)
Q, R = np.linalg.qr(V)
Q = Q * np.sign(np.diag(R))
# centring only helps rounding
# 1, t, t^2, t^3, t^4 (scaled)
# Gram-Schmidt in matrix form
# make each q_k agree in sign
# coordinates of y on q_0, ...,
# sequential SS: linear, ...,
with t^k
coef = Q.T @ y
q_4
seq_ss = coef[1:] ** 2
quartic
for k, s in enumerate(seq_ss, start=1):
    print(f"degree {k}: sequential SS = {s:10.0f}")
print(f"residual after degree 4: {y @ y - np.sum(coef ** 2):10.0f}")
```

9.4.4 Single-degree-of-freedom decompositions

Orthogonal polynomials split a sum of squares into one-dimensional pieces along directions chosen for their meaning. The same can be done for any hypothesis, by choosing single estimable functions whose estimates are uncorrelated.

Theorem 9.4.4 — Orthogonal single-degree-of-freedom sums of squares

Let $\lambda_1, \dots, \lambda_s$ be nonzero vectors with $\lambda_k^\top \beta$ estimable, so that $\lambda_k = X^\top \rho_k$, and let G be a generalized inverse of $X^\top X$.

- (a) The sum of squares for $\lambda_k^\top \beta = 0$ is

$$SS_k = \frac{(\lambda_k^\top \hat{\beta})^2}{\lambda_k^\top G \lambda_k} = \|P_{v_k} y\|^2, \quad v_k = M \rho_k,$$

the squared length of the projection of y onto the line spanned by v_k .

- (b) $v_h^\top v_k = \lambda_h^\top G \lambda_k$. If $\lambda_h^\top G \lambda_k = 0$ for all $h \neq k$, the lines are mutually orthogonal, and $\sum_k SS_k$ is the sum of squares for the joint hypothesis $\lambda_1^\top \beta = \dots = \lambda_s^\top \beta = 0$, on s degrees of freedom.
- (c) Conversely, every orthonormal basis $u_1, \dots, u_s$ of a subspace of $\mathcal{C}(X)$ arises in this way, with $\lambda_k = X^\top u_k$.

Proof.

- (a) is Theorem 9.2.4 with $q = 1$ and $d = 0$: the test space is spanned by $M \rho_k = v_k$, and the projection of y onto it has squared length $(v_k^\top y)^2 / v_k^\top v_k$, where $v_k^\top y = \rho_k^\top M y = \lambda_k^\top \hat{\beta}$ and $v_k^\top v_k = \lambda_k^\top G \lambda_k$. (b) $v_h^\top v_k = \rho_h^\top M \rho_k = \rho_h^\top X G X^\top \rho_k = \lambda_h^\top G \lambda_k$ by Theorem 6.3.7. Orthogonal nonzero vectors are independent, so $\Lambda = [\lambda_1, \dots, \lambda_s]$ has rank s (because $X^\top V = \Lambda$ with $V = [v_1, \dots, v_s]$, and $X^\top$ is one-to-one on $\mathcal{C}(X)$). The test space of the joint hypothesis is $\mathcal{C}(MT) = \mathcal{C}(V)$, with $T = [\rho_1, \dots, \rho_s]$, by Theorem 9.2.4(b), and the projection onto a span of orthogonal lines is the sum of the projections onto the lines (Theorem 6.3.8). (c) With $\rho_k = u_k$, $v_k = M u_k = u_k$, and $\lambda_h^\top G \lambda_k = u_h^\top u_k$.

■

In the one-way layout with group sizes $n_1, \dots, n_g$, take the generalized inverse of Exercise (B3). A contrast $\sum_i c_i \alpha_i$ with $\sum_i c_i = 0$ (Definition 8.7.1) is estimated by $\sum_i c_i \bar{y}_i$, and $\lambda^\top G \lambda = \sum_i c_i^2 / n_i$. So two contrasts c and c' give orthogonal sums of squares iff

$$\sum_{i=1}^g \frac{c_i c'_i}{n_i} = 0,$$

and $g - 1$ such contrasts split the between-groups sum of squares into single degrees of freedom. With equal group sizes this is ordinary orthogonality, $\sum_i c_i c'_i = 0$; with unequal sizes, contrasts that look orthogonal need not be.

Example. (Splitting the education sum of squares)

Consider the 944 self-placements of [Example \(Party, education and political self-placement\)](#) by education alone. The group sizes are 313, 277 and 354, and the group means are 4.476, 4.357 and 4.167. The between-groups sum of squares is 16.306 on two degrees of freedom. The contrast of college against graduate, $\mathbf{c}_2 = (0, 1, -1)$, has sum of squares 5.653. The contrast of high school against the other two, weighted by their sizes, $\mathbf{c}_1 = (1, -0.439, -0.561)$, is orthogonal to $\mathbf{c}_2$ in the sense above, and has sum of squares 10.652. The two add up to the between-groups sum of squares, up to rounding. The more obvious contrast $(1, -\frac{1}{2}, -\frac{1}{2})$, high school against the plain average of the others, has sum of squares 9.534, and together with $\mathbf{c}_2$ it gives 15.187: with unequal group sizes it is not orthogonal to $\mathbf{c}_2$.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
d = sm.datasets.anes96.load_pandas().data
educ = pd.cut(d["educ"], [0, 3.5, 5.5, 7], labels=["HS", "College", "Graduate"])
y = d["selfLR"].to_numpy()

counts = educ.value_counts(sort=False).to_numpy().astype(float)
means = d.groupby(educ, observed=True)["selfLR"].mean().to_numpy()
grand = y.mean()
ss_treat = np.sum(counts * (means - grand) ** 2)           # SS(educ | 1), 2 df

def contrast_ss(c):
    """(c' ybar)^2 / sum(c_i^2 / n_i): the SS for the hypothesis c' mu = 0."""
    return (c @ means) ** 2 / np.sum(c ** 2 / counts)

def orthogonal(c, h):
    return np.isclose(np.sum(c * h / counts), 0.0)

n2, n3 = counts[1], counts[2]
c1 = np.array([1.0, -n2 / (n2 + n3), -n3 / (n2 + n3)])   # HS against the rest,
    weighted
c2 = np.array([0.0, 1.0, -1.0])                         # College against Graduate
c1_plain = np.array([1.0, -0.5, -0.5])                  # HS against the plain
    average
print("orthogonal:", orthogonal(c1, c2), orthogonal(c1_plain, c2))
print(f"treatment SS {ss_treat:.3f} = {contrast_ss(c1):.3f} + {contrast_ss(c2):.3f}"
      )
print(f"with the plain contrast: {contrast_ss(c1_plain):.3f} + {contrast_ss(c2):.3f}"
      )
```

Single-degree-of-freedom sums of squares reappear in Chapter 13, where many of them are tested at once, and in Chapter 15, where orthogonal contrasts organize the one-way analysis of variance.

9.4.5 Exercises

9.4.5.1 A. Check your understanding

Exercise A1.

With $x = 1, 2, 3, 4$ and $\mathbf{y} = (3, 5, 4, 8)^\top$, use the orthogonal polynomial contrasts for four equally spaced points to compute the linear, quadratic and cubic sums of squares, and check that they add up to the corrected total sum of squares.

Exercise A2.

Which of the 2×2 count tables $\begin{pmatrix} 2 & 4 \\ 3 & 6 \end{pmatrix}$, $\begin{pmatrix} 2 & 4 \\ 4 & 2 \end{pmatrix}$ and $\begin{pmatrix} 1 & 3 \\ 3 & 9 \end{pmatrix}$ give orthogonal centred factor spaces? For the one that does not, compute the inner product of the two centred indicator vectors of the first levels.

Exercise A3.

Show that if the regressors $\mathbf{x}_1, \dots, \mathbf{x}_k$ of a model with an intercept are pairwise uncorrelated, then $R^2 = \sum_j r_{yj}^2$, where r_{yj} is the sample correlation of $\mathbf{y}$ and $\mathbf{x}_j$.

9.4.5.2 B. Practice**Exercise B1.**

In a two-way layout with interaction and $n_{ij} = m \geq 1$ for all cells, show that the Type I sums of squares of A in either order, and its Type II and Type III sums of squares, are all equal to $bm \sum_i (\bar{y}_{i..} - \bar{y}_{...})^2$.

Solution. Equal counts satisfy [Eq. 9.4.1](#), so by [Theorem 9.4.1](#) the Type I sums of squares of A in both orders and its Type II sum of squares equal $\text{SS}(A \mid \mu) = \|(\mathbf{P}_A - \mathbf{P}_0)\mathbf{y}\|^2$. Each row has bm observations, and $\mathbf{P}_A - \mathbf{P}_0$ replaces an observation in row i by $\bar{y}_{i..} - \bar{y}_{...}$, which gives the formula. For Type III, the hypothesis of [Theorem 9.3.2\(b\)](#) is that of (a), because with equal counts weighted and unweighted row means coincide. A sum of squares for a hypothesis depends only on the hypothesis (it is $\text{SSE}_0 - \text{SSE}$), so the Type III sum of squares equals $\text{SS}(A \mid \mu)$ too.

Exercise B2.

Take counts $n_{11} = 1$, $n_{12} = 2$, $n_{21} = 2$, $n_{22} = 4$. Check [Eq. 9.4.1](#). Find cell means for which the Type III hypothesis for A holds but the Type I (and Type II) hypothesis fails.

Exercise B3.

In a one-way layout with g groups of equal size m , show that the Helmert contrasts $\mathbf{c}_k = (1, \dots, 1, -(k-1), 0, \dots, 0)$, with $k-1$ leading ones, $k = 2, \dots, g$, are mutually orthogonal, and write the between-groups sum of squares as a sum of $g-1$ single-degree-of-freedom sums of squares. What goes wrong with unequal group sizes?

9.4.5.3 C. Going deeper

Exercise C1.

Suppose the design points are symmetric about their mean: after sorting, $x_i - \bar{x} = -(x_{n+1-i} - \bar{x})$. Show that every odd power of the centred variable is orthogonal to every even power (including the zeroth), that $\mathbf{q}_k$ is an even or odd vector according to the parity of k , and explain the pattern of raw coefficients in [Example \(A trend in the Nile\)](#).

Solution. Let $\mathbf{t}$ have entries $t_i = x_i - \bar{x}$ and let $\mathbf{J}$ be the reversal permutation, so that $\mathbf{J}\mathbf{t} = -\mathbf{t}$. A vector $\mathbf{u}$ is even if $\mathbf{J}\mathbf{u} = \mathbf{u}$ and odd if $\mathbf{J}\mathbf{u} = -\mathbf{u}$. Powers $\mathbf{t}^k$ have the parity of k , and an even vector $\mathbf{u}$ and an odd vector $\mathbf{v}$ are orthogonal because $\mathbf{u}^\top \mathbf{v} = (\mathbf{J}\mathbf{u})^\top (\mathbf{J}\mathbf{v}) = -\mathbf{u}^\top \mathbf{v}$. Gram–Schmidt on $\mathbf{1}, \mathbf{t}, \mathbf{t}^2, \dots$ subtracts from $\mathbf{t}^k$ only its projections on the earlier $\mathbf{q}_j$ of the same parity, by induction, so $\mathbf{q}_k$ has the parity of k . Hence the column $\mathbf{t}$ is orthogonal to the column $\mathbf{t}^2$ and to $\mathbf{1}$, and adding the quadratic term does not change the linear coefficient ([Corollary 9.4.2](#), applied to the terms $\mathbf{t}$ and $\mathbf{t}^2 - \bar{t}^2 \mathbf{1}$). The cubic column is odd and not orthogonal to $\mathbf{t}$, so it changes the linear coefficient. The quartic column is even. By [Theorem 9.2.2\(b\)](#), adding it changes the cubic fit’s coefficients by $-\mathbf{L}\hat{\gamma}$ with $\mathbf{L} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{t}^4$, where $\mathbf{X} = [\mathbf{1}, \mathbf{t}, \mathbf{t}^2, \mathbf{t}^3]$. Inner products between odd and even columns vanish, so $\mathbf{X}^\top \mathbf{X}$ and its inverse do not link the odd and even columns, and the entries of $\mathbf{X}^\top \mathbf{t}^4$ for the odd columns are zero. Hence the rows of $\mathbf{L}$ for $\mathbf{t}$ and $\mathbf{t}^3$ vanish, and the linear coefficient is unchanged.

9.5 The coefficient of determination and its limits

The coefficient of determination is the most widely reported summary of a regression, and probably the most widely misread. It is a ratio of two rows of the corrected table,

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}},$$

defined when $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{y}$ is not constant. [Section 6.7](#) showed that it is a squared cosine ([Theorem 6.7.2](#)) that cannot decrease as the model space grows. This section adds R^2 as a maximal correlation, its null distribution, the adjusted and partial versions, and what it does not measure.

9.5.1 A maximal correlation

The square root $R = \sqrt{R^2}$ is called the **multiple correlation coefficient**. The name comes from the following property, the sample counterpart of the population result [Proposition 2.5.3](#).

Theorem 9.5.1 — R is the largest correlation with a linear combination

Suppose $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{y}$ is not constant. For every $\mathbf{b}$ such that $\mathbf{X}\mathbf{b}$ is not constant,

$$\text{corr}(\mathbf{y}, \mathbf{X}\mathbf{b})^2 \leq R^2.$$

If $R^2 > 0$, equality holds iff $\mathbf{X}\mathbf{b} = a\hat{\mathbf{Y}} + c\mathbf{1}$ for some $a \neq 0$ and some c , and the correlation $\text{corr}(\mathbf{y}, \mathbf{X}\mathbf{b})$ is maximized, with value R , by the fitted values themselves. If $R^2 = 0$, every such correlation is zero.

Proof. Let $\mathbf{u} = \mathbf{y} - \bar{y}\mathbf{1}$ and $\mathcal{V} = \mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$, whose projection is $\mathbf{M} - \mathbf{P}_1$ (Theorem 6.5.1), and let $\mathbf{v} = \mathbf{X}\mathbf{b} - \overline{\mathbf{X}\mathbf{b}}\mathbf{1}$. Since $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, $\mathbf{v} \in \mathcal{V}$, so $\mathbf{u}^\top \mathbf{v} = \mathbf{u}^\top (\mathbf{M} - \mathbf{P}_1)\mathbf{v} = ((\mathbf{M} - \mathbf{P}_1)\mathbf{u})^\top \mathbf{v}$. The sample correlation is $\mathbf{u}^\top \mathbf{v} / (\|\mathbf{u}\| \|\mathbf{v}\|)$, and by Cauchy–Schwarz (Proposition 1.1.1)

$$\text{corr}(\mathbf{y}, \mathbf{X}\mathbf{b})^2 = \frac{((\mathbf{M} - \mathbf{P}_1)\mathbf{u})^\top \mathbf{v})^2}{\|\mathbf{u}\|^2 \|\mathbf{v}\|^2} \leq \frac{\|(\mathbf{M} - \mathbf{P}_1)\mathbf{u}\|^2}{\|\mathbf{u}\|^2} = \frac{\text{SSR}}{\text{SST}},$$

with equality iff $\mathbf{v}$ and $(\mathbf{M} - \mathbf{P}_1)\mathbf{u} = \hat{\mathbf{Y}} - \bar{y}\mathbf{1}$ are linearly dependent. If $R^2 > 0$ the second vector is nonzero, and since $\mathbf{v} \neq \mathbf{0}$ dependence means $\mathbf{v} = a(\hat{\mathbf{Y}} - \bar{y}\mathbf{1})$ with $a \neq 0$, which is the stated condition; for $a > 0$ the correlation itself is $+R$. If $R^2 = 0$, then $(\mathbf{M} - \mathbf{P}_1)\mathbf{u} = \mathbf{0}$ and $\mathbf{u}^\top \mathbf{v} = 0$. ■

Among all linear combinations of the regressors, the fitted values are the one most correlated with the response. When the regressors are pairwise uncorrelated, R^2 is the sum of the squared simple correlations (Exercise A3). In general it can be smaller or larger than that sum (Exercise C2).

9.5.2 R^2 when nothing is going on

Since R^2 never decreases as regressors are added, a model with many regressors has a sizeable R^2 even if none of them matters. The theory of Chapter 4 says how sizeable.

Theorem 9.5.2 — The distribution of R^2

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$ with $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$, $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ and $1 < r = \text{rank}(\mathbf{X}) < n$. Let $F = [\text{SSR}/(r-1)]/[\text{SSE}/(n-r)]$ and $\gamma = \|(\mathbf{M} - \mathbf{P}_1)\boldsymbol{\mu}\|^2/\sigma^2$. Then:

- (a) $R^2 = \frac{(r-1)F}{(r-1)F + n-r}$, and $F \sim F(r-1, n-r, \gamma)$;
- (b) if $\boldsymbol{\mu} \in \text{span}(\mathbf{1})$, so that no regressor matters, then $R^2 \sim \text{Beta}(\frac{r-1}{2}, \frac{n-r}{2})$ and $E(R^2) = \frac{r-1}{n-1}$.

Proof.

- (a) $R^2/(1-R^2) = \text{SSR}/\text{SSE} = (r-1)F/(n-r)$, and solving for R^2 gives the formula. The projections $\mathbf{M} - \mathbf{P}_1$ and $\mathbf{I} - \mathbf{M}$ are orthogonal with ranks $r-1$ and $n-r$, and $(\mathbf{I} - \mathbf{M})\boldsymbol{\mu} = \mathbf{0}$, so by Theorem 4.4.4 $\text{SSR}/\sigma^2 \sim \chi^2(r-1, \gamma)$ and $\text{SSE}/\sigma^2 \sim \chi^2(n-r)$,

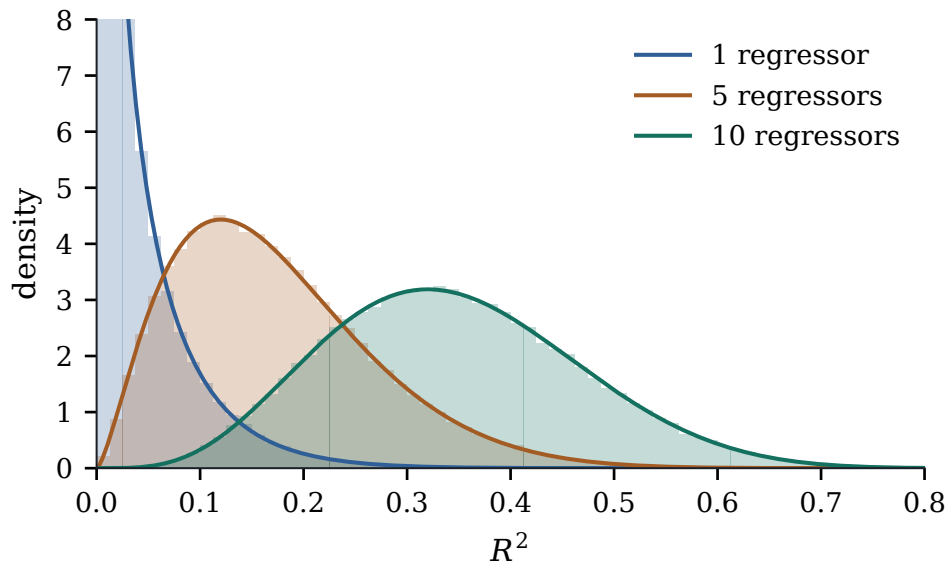


Figure 9.5.1: Null distribution of R^2 for $n = 30$ observations and 1, 5 or 10 regressors unrelated to the response. Histograms: 100,000 simulated data sets each. Curves: the Beta densities of [Theorem 9.5.2](#).

independently. The ratio F is therefore $F(r-1, n-r, \gamma)$ ([Definition 4.1.2](#)).

- (b) Now $\gamma = 0$, and $R^2 = U/(U+V)$ with $U = \text{SSR}/\sigma^2 \sim \chi^2(r-1)$ and $V = \text{SSE}/\sigma^2 \sim \chi^2(n-r)$ independent. By [Exercise \(B4\)](#) this is $\text{Beta}(\frac{r-1}{2}, \frac{n-r}{2})$, whose mean is $\frac{\frac{r-1}{2}}{\frac{r-1}{2} + \frac{n-r}{2}} = (r-1)/(n-1)$.

■

Part (b) answers the exercise [Exercise \(C1\)](#) of [Chapter 6](#). With $k = r - 1$ regressors of pure noise, R^2 is on average $k/(n-1)$. A model with 10 useless regressors fitted to 30 observations has expected R^2 about one third. [Figure 9.5.1](#) compares the Beta densities with 100,000 simulated data sets with $n = 30$ and $k = 1, 5, 10$ regressors drawn at random. The simulated means are 0.0345, 0.1725 and 0.3445, against the exact 0.0345, 0.1724 and 0.3448. The upper 5% points of the three null distributions are 0.130, 0.353 and 0.556, so with ten regressors an R^2 of one half is unremarkable. Part (a) shows that R^2 and F carry the same information: testing R^2 against its null distribution is the overall F test of the regression.

```
import numpy as np
rng = np.random.default_rng(20260919)
n, reps = 30, 100_000
results = {}
for k in (1, 5, 10):
    # number of regressors
    besides 1
    X = np.column_stack([np.ones(n), rng.normal(size=(n, k))])
    r = k + 1
    Q, _ = np.linalg.qr(X)
    Y = 3.0 + rng.normal(size=(n, reps))
    Z = Q.T @ Y
    sst = np.sum(Y ** 2, axis=0) - Z[0] ** 2
    ssr = np.sum(Z[1:] ** 2, axis=0)
    R2 = ssr / sst
    R2adj = 1 - (1 - R2) * (n - 1) / (n - r)
    # no regressor matters
    print(f"k = {k:2d}: mean R^2 = {R2.mean():.4f} (theory {k / (n - 1):.4f}),"
```

```
f" mean adjusted R^2 = {R2adj.mean():+.4f}"
results[k] = R2
```

9.5.3 Adjusted R^2

The bias of R^2 under the null suggests a correction. Write $s^2 = \text{SSE}/(n-r)$ and $s_y^2 = \text{SST}/(n-1)$ for the unbiased estimators of the error variance in the fitted model and in the intercept-only model. Replacing sums of squares by these mean squares defines the **adjusted coefficient of determination**

$$\bar{R}^2 = 1 - \frac{s^2}{s_y^2} = 1 - (1 - R^2) \frac{n-1}{n-r}. \quad (9.5.1)$$

Proposition 9.5.3 — Properties of the adjusted R^2

Let $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, $1 \leq r = \text{rank}(\mathbf{X}) < n$, and let F be the overall F ratio of [Theorem 9.5.2](#).

- (a) $\bar{R}^2 \leq R^2$, with equality iff $r = 1$ or $R^2 = 1$.
- (b) For $r > 1$, $\bar{R}^2 < 0$ iff $F < 1$.
- (c) If a model with rank r_0 is enlarged to one of rank $r = r_0 + q$ containing it, then $\bar{R}^2$ increases iff the F ratio of the added terms, $[\text{SS}(\text{added} \mid \text{old})/q]/s^2$, exceeds 1. For a single added column this means $|t| > 1$.
- (d) Among models for the same response, $\bar{R}^2$ is largest for the model with the smallest residual mean square s^2 .
- (e) Under the normal model with $\boldsymbol{\mu} \in \text{span}(\mathbf{1})$, $E(\bar{R}^2) = 0$.

Proof.

- (a) $1 - \bar{R}^2 = (1 - R^2)(n-1)/(n-r)$, and $(n-1)/(n-r) \geq 1$ with equality iff $r = 1$, so $1 - \bar{R}^2 \geq 1 - R^2$, with equality iff $r = 1$ or $1 - R^2 = 0$.
- (b) $\bar{R}^2 < 0$ iff $s^2 > s_y^2$, that is, $\text{SSE}(n-1) > (\text{SSR} + \text{SSE})(n-r)$, that is, $\text{SSE}(r-1) > \text{SSR}(n-r)$, which for $r > 1$ is $F < 1$.
- (c) s_y^2 does not depend on the model, so $\bar{R}^2$ increases iff s^2 decreases. Write $\text{SSE}_0 = \text{SSE} + D$ with D the extra sum of squares. Then $\text{SSE}/(n-r) < (\text{SSE} + D)/(n-r_0)$ iff $\text{SSE}(n-r_0) < (\text{SSE} + D)(n-r)$, iff $q\text{SSE} < D(n-r)$, iff $(D/q)/s^2 > 1$. For one column the ratio is t^2 ([Corollary 9.2.3](#)).
- (d) is immediate from [Eq. 9.5.1](#).
- (e) By [Theorem 9.5.2\(b\)](#), $E(1 - R^2) = 1 - (r-1)/(n-1) = (n-r)/(n-1)$, so $E(1 - \bar{R}^2) = 1$.

■

Part (e) is the point of the adjustment. In the simulation the averages of $\bar{R}^2$ are 0.0000, 0.0001 and -0.0005 , but $\bar{R}^2$ is negative in a proportion 0.674 of the data sets with one regressor: it is not a proportion of anything. Part (c) shows what selection by $\bar{R}^2$ amounts to: a regressor is

admitted when $|t| > 1$, far below the threshold near two of a 5% test, so the chosen models tend to be too large (Chapter 29).

Example. (Adjusted R^2 for the state data)

For all seven models built from poverty, single parenthood and urbanization (each with an intercept):

Regressors	R^2	$\bar{R}^2$
poverty	0.3599	0.3466
single	0.5719	0.5629
urban	0.0221	0.0017
poverty, single	0.6410	0.6257
poverty, urban	0.4626	0.4398
single, urban	0.5844	0.5668
all three	0.6419	0.6186

R^2 is largest for the full model, as it must be. $\bar{R}^2$ prefers poverty and single parenthood without urbanization. Adding urbanization to that model raises R^2 slightly but lowers $\bar{R}^2$, because its t statistic in the full model is only 0.349, so its F ratio is $0.121 < 1$.

9.5.4 Partial R^2

Applied to what a reduced model leaves unexplained, the idea of R^2 gives a measure for a term.

Definition 9.5.1 — Partial coefficient of determination

Let $\mathcal{C}(\mathbf{X}_1) \subseteq \mathcal{C}(\mathbf{X})$ with $\mathbf{1} \in \mathcal{C}(\mathbf{X}_1)$. The **partial coefficient of determination** of $\mathbf{X}$ over $\mathbf{X}_1$ is

$$R_{\mathbf{X}|\mathbf{X}_1}^2 = \frac{\text{SSE}(\mathbf{X}_1) - \text{SSE}(\mathbf{X})}{\text{SSE}(\mathbf{X}_1)} = \frac{\text{SS}(\mathbf{X} | \mathbf{X}_1)}{\text{SSE}(\mathbf{X}_1)},$$

the proportion of the residual variation of the reduced model that the larger model explains.

Theorem 9.5.4 — Partial R^2 , F and t

With the notation of Definition 9.5.1, let $r_1 = \text{rank}(\mathbf{X}_1)$, $r = \text{rank}(\mathbf{X})$, $q = r - r_1 > 0$, $s^2 = \text{SSE}(\mathbf{X})/(n - r)$, and let $F = [\text{SS}(\mathbf{X} | \mathbf{X}_1)/q]/s^2$. Then:

- (a) $1 - R^2 = (1 - R_1^2)(1 - R_{\mathbf{X}|\mathbf{X}_1}^2)$, where R^2 and R_1^2 belong to the two models;
- (b) $R_{\mathbf{X}|\mathbf{X}_1}^2 = \frac{qF}{qF + n - r}$;
- (c) if $\mathbf{X} = [\mathbf{X}_1, \mathbf{x}_j]$ adds one column, then $R_{\mathbf{X}|\mathbf{X}_1}^2 = \frac{t_j^2}{t_j^2 + n - r}$, and it equals the squared sample correlation between the residuals $(\mathbf{I} - \mathbf{M}_1)\mathbf{y}$ and $(\mathbf{I} - \mathbf{M}_1)\mathbf{x}_j$;

- (d) for a chain $\mathcal{C}(\mathbf{1}) = \mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X}_1) \subseteq \cdots \subseteq \mathcal{C}(\mathbf{X}_k)$, $1 - R_k^2 = \prod_{j=1}^k (1 - R_{\mathbf{X}_j|\mathbf{X}_{j-1}}^2)$.

Proof.

- (a) $1 - R^2 = \text{SSE}(\mathbf{X})/\text{SST} = (\text{SSE}(\mathbf{X}_1)/\text{SST})(\text{SSE}(\mathbf{X})/\text{SSE}(\mathbf{X}_1))$, and the two factors are $1 - R_1^2$ and $1 - R_{\mathbf{X}|\mathbf{X}_1}^2$.
- (b) Write $D = \text{SS}(\mathbf{X} \mid \mathbf{X}_1) = qFs^2$ and $\text{SSE}(\mathbf{X}) = (n - r)s^2$. Then $R_{\mathbf{X}|\mathbf{X}_1}^2 = D/(D + \text{SSE}(\mathbf{X})) = qF/(qF + n - r)$.
- (c) With $q = 1$, $F = t_j^2$ by [Corollary 9.2.3](#), and (b) gives the first formula. For the correlation, let $\tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_1)\mathbf{y}$ and $\tilde{\mathbf{x}} = (\mathbf{I} - \mathbf{M}_1)\mathbf{x}_j$. Then $\text{SSE}(\mathbf{X}_1) = \|\tilde{\mathbf{y}}\|^2$, and by [Lemma 9.2.1](#) $\text{SS}(\mathbf{X} \mid \mathbf{X}_1) = (\tilde{\mathbf{x}}^\top \tilde{\mathbf{y}})^2 / \|\tilde{\mathbf{x}}\|^2$. Their ratio is the squared cosine of the angle between $\tilde{\mathbf{y}}$ and $\tilde{\mathbf{x}}$. Both have mean zero because $\mathbf{1} \in \mathcal{C}(\mathbf{X}_1)$, so the cosine is their sample correlation ([Exercise \(B3\)](#)).
- (d) Apply (a) repeatedly along the chain, with $R_0^2 = 0$ for the intercept-only model.

■

Part (c) identifies the partial R^2 of one regressor with a squared **sample partial correlation**, the sample counterpart of [Definition 3.6.1](#); Chapter 14 studies its sampling distribution. Within one model, ranking regressors by $|t|$ and by partial R^2 is the same thing.

Example. (Partial R^2 in the state data)

In the three-regressor model, with $n - r = 46$:

Regressor	t	partial R^2	R^2 without it
poverty	2.718	0.1383	0.5844
single	4.799	0.3337	0.4626
urban	0.349	0.0026	0.6410

Each partial R^2 equals $t^2/(t^2 + 46)$, and each row satisfies part (a): for poverty, $1 - 0.6419 = (1 - 0.5844)(1 - 0.1383)$. Poverty explains 14% of the variation that single parenthood and urbanization leave unexplained, although on its own it explains 36% of the total ($R^2 = 0.3599$). For the pair poverty and urbanization over single parenthood, the F ratio on 2 and 46 degrees of freedom is 4.502 and the partial R^2 is 0.1637.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")

y = data["murder"].to_numpy()
names = ["poverty", "single", "urban"]
n = len(y)

def fit(cols):
    X = np.column_stack([np.ones(n)] + [data[c].to_numpy() for c in cols])
    return sm.OLS(y, X).fit()

full = fit(names)
```

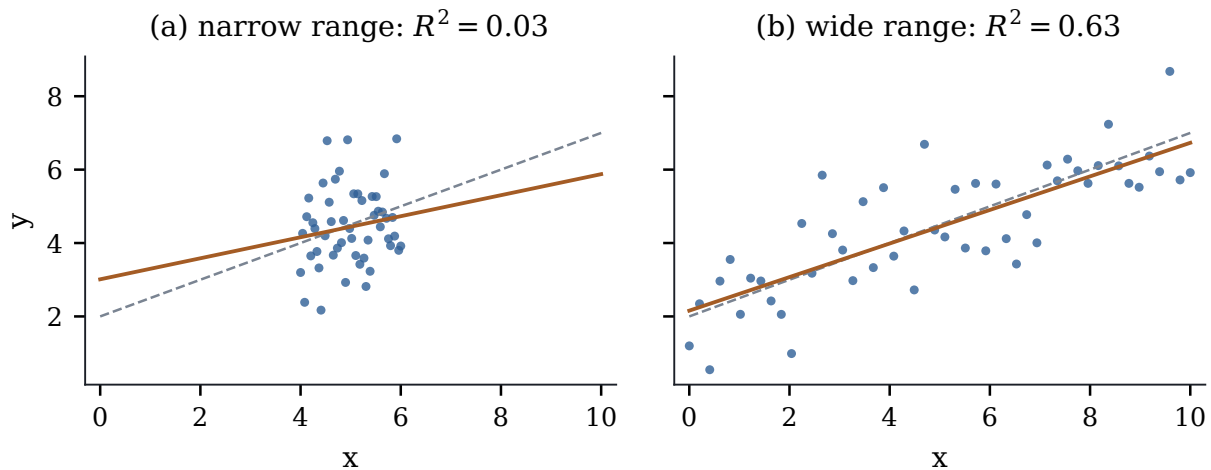


Figure 9.5.2: One line, two designs. The dashed line is the true mean $2 + 0.5x$, the solid line the least squares fit. The errors and the residual mean square are the same in both panels.

```
df_res = full.df_resid
for j, name in enumerate(names, start=1):
    reduced = fit([c for c in names if c != name])
    partial_r2 = (reduced.ssr - full.ssr) / reduced.ssr
    t = full.tvalues[j]
    print(f"{name:8s} partial R^2 = {partial_r2:.4f}, t = {t:6.3f},"
          f" t^2/(t^2 + df) = {t ** 2 / (t ** 2 + df_res):.4f}")
print(f"R^2 = {full.rsquared:.4f}, adjusted R^2 = {full.rsquared_adj:.4f}")
```

9.5.5 What R^2 does not measure

A high R^2 is often read as “a good model” and a low one as “a useless model”; neither reading is safe.

It is not a property of the relationship alone. R^2 depends on the spread of the regressors, which is a property of the design. In simple regression with fixed x values, the expectations of the two sums of squares are (Exercise (B1))

$$E(\text{SSR}) = \sigma^2 + \beta_1^2 S_{xx}, \quad E(\text{SST}) = (n-1)\sigma^2 + \beta_1^2 S_{xx},$$

so R^2 is roughly $\beta_1^2 S_{xx} / (\beta_1^2 S_{xx} + (n-1)\sigma^2)$. In Figure 9.5.2 the line $E(y) = 2 + 0.5x$ and the same 50 errors with $\sigma = 1$ are combined with x values spread evenly over $[4, 6]$ and over $[0, 10]$. The two designs span the same $\mathcal{C}(\mathbf{X})$, so the residuals are identical and the residual mean square is 1.101 in both fits, but R^2 is 0.026 in one case and 0.632 in the other. The approximate values from the formula are 0.081 and 0.689. Restricting the range of a regressor lowers R^2 without changing the relationship.

```
import numpy as np
rng = np.random.default_rng(9)
n, beta0, beta1, sigma = 50, 2.0, 0.5, 1.0
eps = sigma * rng.normal(size=n)
# the same errors in both designs

def fit(x):
    y = beta0 + beta1 * x + eps
```

```

X = np.column_stack([np.ones(n), x])
b, *_ = np.linalg.lstsq(X, y, rcond=None)
sse = np.sum((y - X @ b) ** 2)
r2 = 1 - sse / np.sum((y - y.mean()) ** 2)
return y, b, sse / (n - 2), r2

for label, x in [("narrow", np.linspace(4, 6, n)), ("wide", np.linspace(0, 10, n))]:
    y, b, s2, r2 = fit(x)
    Sxx = np.sum((x - x.mean()) ** 2)
    expected = beta1 ** 2 * Sxx / (beta1 ** 2 * Sxx + (n - 1) * sigma ** 2)
    print(f"{label:6s}: slope {b[1]:.3f}, s^2 {s2:.3f}, R^2 {r2:.3f}, design value {
    expected:.3f}")

```

It is not a test of the model. A line fitted to points on a smooth curve can have a high R^2 while missing the curvature; residual plots and lack-of-fit tests (Chapter 14) reveal it. Conversely, a correct model has a low R^2 whenever the error variance is large, and its coefficients can still be precisely estimated and important.

It is not comparable across responses. Models for y and $\log y$, or for individuals and for group averages, have different denominators SST. Averaging replicates inflates R^2 (Exercise B2).

It is not comparable without an intercept (Proposition 9.1.3, Example (Forcing the murder regression through the origin)).

It is not a measure of prediction or of cause. Computed on the fitting data, it overstates predictive accuracy for new data (Chapter 29), and variation “explained” is linear association, not the effect of changing a regressor (Chapter 25).

9.5.6 Exercises

9.5.6.1 A. Check your understanding

Exercise A1.

A regression with an intercept, $n = 20$ and $r = 6$ has $R^2 = 0.40$. Compute $\bar{R}^2$ and the overall F ratio. Is $\bar{R}^2$ positive? Is it consistent with part (b) of Proposition 9.5.3?

9.5.6.2 B. Practice

Exercise B1.

In simple regression with fixed x values, $E(\mathbf{Y}) = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, use Theorem 2.3.1 to show that $E(\text{SSR}) = \sigma^2 + \beta_1^2 S_{xx}$ and $E(\text{SST}) = (n - 1)\sigma^2 + \beta_1^2 S_{xx}$. Show that the ratio of these expectations tends to 1 as $S_{xx} \rightarrow \infty$ with n fixed, and to $1/(n - 1)$ as $S_{xx} \rightarrow 0$, and interpret the second limit.

Solution. $\text{SSR} = \mathbf{y}^\top (\mathbf{M} - \mathbf{P}_1) \mathbf{y}$, where $\mathbf{M} - \mathbf{P}_1$ projects onto the line spanned by $\mathbf{x} - \bar{x} \mathbf{1}$ (Example (Intercept inside simple regression)) and has trace 1. With $\boldsymbol{\mu} = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}$, $(\mathbf{M} - \mathbf{P}_1) \boldsymbol{\mu} = \beta_1 (\mathbf{x} - \bar{x} \mathbf{1})$, whose squared length is $\beta_1^2 S_{xx}$. Theorem 2.3.1 gives $E(\text{SSR}) = \sigma^2 + \beta_1^2 S_{xx}$. Similarly $\text{SST} = \mathbf{y}^\top (\mathbf{I} - \mathbf{P}_1) \mathbf{y}$, with trace $n - 1$ and $(\mathbf{I} - \mathbf{P}_1) \boldsymbol{\mu} = \beta_1 (\mathbf{x} - \bar{x} \mathbf{1})$.

The ratio $(\sigma^2 + \beta_1^2 S_{xx})/((n-1)\sigma^2 + \beta_1^2 S_{xx})$ tends to 1 as $S_{xx} \rightarrow \infty$ and to $1/(n-1)$ as $S_{xx} \rightarrow 0$. The second limit is the null value of $E(R^2)$ for $r = 2$ (Theorem 9.5.2): with almost no spread in x , the slope contributes almost nothing and the regression sum of squares is almost pure noise.

Exercise B2.

Suppose each of g distinct values $x_1, \dots, x_g$ carries m observations. Show that the least squares line fitted to all gm observations is the same as the line fitted to the g group means, and that its R^2 for the individual data is at most its R^2 for the means, with equality iff every observation equals its group mean.

Solution. The model space for the individual data is spanned by $\mathbf{1}$ and $\mathbf{x}$, both constant within groups, so it lies inside the space of group-constant vectors. By Theorem 6.5.1, projecting $\mathbf{y}$ onto it equals projecting the vector of group means (repeated m times) onto it, which is the fit to the means with each point weighted equally. Write SSR_g and SST_g for the sums over the g means, and W for the within-group sum of squares. For the individual data $\text{SSR} = m \text{SSR}_g$ and $\text{SST} = m \text{SST}_g + W$. Hence $R^2 = m \text{SSR}_g / (m \text{SST}_g + W) \leq \text{SSR}_g / \text{SST}_g$, with equality iff $W = 0$.

Exercise B3.

Using part (d) of Theorem 9.5.4 and the sequential sums of squares of Example (Order matters) (order A), express $1 - R^2$ for the three-regressor model as a product of three factors, and identify each factor as one minus a partial R^2 .

9.5.6.3 C. Going deeper

Exercise C1.

Under the conditions of Theorem 9.5.2, show that for every $c \in (0, 1)$ the probability $\Pr(R^2 > c)$ is a strictly increasing function of the noncentrality γ . *Hint:* part (a) of the theorem and Theorem 4.1.4.

Exercise C2.

For two regressors with sample correlations $r_{y1} = 0.5$, $r_{y2} = 0$ and $r_{12} = 0.6$, compute R^2 and compare it with $r_{y1}^2 + r_{y2}^2$. Use the formulas of Exercise (B1). Explain geometrically why a regressor that is uncorrelated with the response can raise R^2 .

9.6 Expected mean squares

This section computes the expectations of sums of squares when the data are random, using only two moments, and their distributions under normality. Each mean square estimates σ^2 plus a term that vanishes exactly when its hypothesis is true, while the residual mean square estimates σ^2 alone. That is the structure of the F ratio, on which Part III builds.

9.6.1 Expected sums of squares for a decomposition

In this section $E(\mathbf{Y}) = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$. We do not assume at first that $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$, because the effect of a wrong model on the mean squares is part of the story.

Theorem 9.6.1 — Expected mean squares

Let $\mathcal{V}_1, \dots, \mathcal{V}_k$ be an analysis of variance decomposition of $\mathcal{C}(\mathbf{X})$, with projections $\mathbf{P}_i$ and dimensions r_i , and let $E(\mathbf{Y}) = \boldsymbol{\mu}$, $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$.

- (a) $E(\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}) = \sigma^2 r_i + \|\mathbf{P}_i \boldsymbol{\mu}\|^2$, so the mean square of $\mathcal{V}_i$ has expectation

$$E(\text{MS}_i) = \sigma^2 + \frac{\|\mathbf{P}_i \boldsymbol{\mu}\|^2}{r_i}. \quad (9.6.1)$$

- (b) $E(\text{SSE}) = \sigma^2(n - r) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2$, so $E(s^2) = \sigma^2$ iff $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$.
 (c) If $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ with $\mathbf{X} = [\mathbf{X}_0, \dots, \mathbf{X}_k]$ partitioned into terms, the partial sum of squares of $\mathbf{X}_j$ has

$$E \text{ SS}(\mathbf{X}_j \mid \mathbf{X}_{(-j)}) = \sigma^2 d_j + \boldsymbol{\beta}_j^\top \mathbf{X}_j^\top (\mathbf{I} - \mathbf{M}_{(-j)}) \mathbf{X}_j \boldsymbol{\beta}_j,$$

with $d_j = \text{rank}(\mathbf{M} - \mathbf{M}_{(-j)})$, and the sequential sum of squares of $\mathbf{X}_j$ has $E \text{ SS}(\mathbf{X}_j \mid \mathbf{X}_0, \dots, \mathbf{X}_{j-1}) = \sigma^2 d'_j + \|(\mathbf{M}_j - \mathbf{M}_{j-1})\mathbf{X}\boldsymbol{\beta}\|^2$, with $d'_j = \text{rank}(\mathbf{M}_j - \mathbf{M}_{j-1})$.

- (d) If moreover $\mathbf{Y}$ is normal, then $\text{Var}(\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}) = 2\sigma^4 r_i + 4\sigma^2 \|\mathbf{P}_i \boldsymbol{\mu}\|^2$, and the sums of squares of different subspaces, including the error space, are uncorrelated.

Proof.

- (a) By Theorem 2.3.1 with $\mathbf{A} = \mathbf{P}_i$ and $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$, $E(\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}) = \sigma^2 \text{tr}(\mathbf{P}_i) + \boldsymbol{\mu}^\top \mathbf{P}_i \boldsymbol{\mu}$. The trace is r_i (Proposition 6.3.4), and $\boldsymbol{\mu}^\top \mathbf{P}_i \boldsymbol{\mu} = \boldsymbol{\mu}^\top \mathbf{P}_i^\top \mathbf{P}_i \boldsymbol{\mu} = \|\mathbf{P}_i \boldsymbol{\mu}\|^2$. Divide by r_i .
 (b) The same with $\mathbf{A} = \mathbf{I} - \mathbf{M}$, of trace $n - r$. The second term vanishes iff $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$.
 (c) For the partial sum of squares, $(\mathbf{M} - \mathbf{M}_{(-j)})\mathbf{X}_l \boldsymbol{\beta}_l = \mathbf{0}$ for $l \neq j$, because $\mathbf{X}_l \boldsymbol{\beta}_l$ lies in $\mathcal{C}(\mathbf{X}_{(-j)})$ and is fixed by both projections. And $(\mathbf{M} - \mathbf{M}_{(-j)})\mathbf{X}_j \boldsymbol{\beta}_j = \mathbf{X}_j \boldsymbol{\beta}_j - \mathbf{M}_{(-j)}\mathbf{X}_j \boldsymbol{\beta}_j = (\mathbf{I} - \mathbf{M}_{(-j)})\mathbf{X}_j \boldsymbol{\beta}_j$. Its squared length is $\boldsymbol{\beta}_j^\top \mathbf{X}_j^\top (\mathbf{I} - \mathbf{M}_{(-j)}) \mathbf{X}_j \boldsymbol{\beta}_j$, since $\mathbf{I} - \mathbf{M}_{(-j)}$ is symmetric and idempotent. The sequential case is (a) for the chain decomposition (Definition 9.3.1).
 (d) By Theorem 4.2.3(b) with $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$, $\text{Var}(\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}) = 2 \text{tr}\{(\sigma^2 \mathbf{P}_i)^2\} + 4\sigma^2 \boldsymbol{\mu}^\top \mathbf{P}_i \mathbf{P}_i \boldsymbol{\mu} = 2\sigma^4 r_i + 4\sigma^2 \|\mathbf{P}_i \boldsymbol{\mu}\|^2$. By part (d) of the same theorem the covariance of two forms is $2\sigma^4 \text{tr}(\mathbf{P}_i \mathbf{P}_j) + 4\sigma^2 \boldsymbol{\mu}^\top \mathbf{P}_i \mathbf{P}_j \boldsymbol{\mu} = 0$ when $\mathbf{P}_i \mathbf{P}_j = \mathbf{0}$.

■

Parts (a) to (c) use nothing beyond second moments. They hold for skewed or heavy-tailed errors, provided the errors are uncorrelated with a common variance. Part (d) uses normality, and without it both the variances and the zero correlations fail in general (Exercise (C1)).

Part (c) confirms [Proposition 9.3.1](#). For full-rank $\mathbf{X}$ the term $\|(\mathbf{I} - \mathbf{M}_{(-j)})\mathbf{X}_j\boldsymbol{\beta}_j\|^2$ is positive unless $\boldsymbol{\beta}_j = \mathbf{0}$, and it is large only if $\mathbf{X}_j$ has a large part orthogonal to the other terms. A nearly collinear term has a small expected partial sum of squares even when its coefficients are substantial.

Under normality the expectations become full distributions.

Corollary 9.6.2 — Distributions of the mean squares

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$ with $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$, and let $\mathcal{V}_1, \dots, \mathcal{V}_k$ be an analysis of variance decomposition of $\mathcal{C}(\mathbf{X})$. Then the sums of squares $\mathbf{Y}^\top \mathbf{P}_1 \mathbf{Y}, \dots, \mathbf{Y}^\top \mathbf{P}_k \mathbf{Y}$ and SSE are mutually independent,

$$\frac{\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}}{\sigma^2} \sim \chi^2(r_i, \gamma_i), \quad \gamma_i = \frac{\|\mathbf{P}_i \boldsymbol{\mu}\|^2}{\sigma^2}, \quad \frac{\text{SSE}}{\sigma^2} \sim \chi^2(n - r),$$

and $F_i = \text{MS}_i/s^2 \sim F(r_i, n - r, \gamma_i)$, which is central iff $\mathbf{P}_i \boldsymbol{\mu} = \mathbf{0}$. The same holds for a single projection $\mathbf{P}$ onto any subspace of $\mathcal{C}(\mathbf{X})$, such as a partial sum of squares, paired with SSE.

Proof. The projections $\mathbf{P}_1, \dots, \mathbf{P}_k, \mathbf{I} - \mathbf{M}$ are mutually orthogonal ([Theorem 9.1.1\(a\)](#)), so [Theorem 4.4.4](#) gives independence and the noncentral chi-squared laws, with $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2 = 0$ for the error space. The ratio of two independent chi-squared variables, each divided by its degrees of freedom and the numerator noncentral, is the noncentral F ([Definition 4.1.2](#)). A single projection $\mathbf{P}$ onto a subspace of $\mathcal{C}(\mathbf{X})$ satisfies $\mathbf{P}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$, and the same argument applies to the pair. ■

The residual part is the familiar [Theorem 7.5.1](#); the rest applies the same argument to the other pieces.

9.6.2 The logic of the F ratio

By [Eq. 9.6.1](#) every mean square estimates σ^2 plus $\|\mathbf{P}_i \boldsymbol{\mu}\|^2/r_i$, the squared size per dimension of the mean's component in $\mathcal{V}_i$, while under a correct model the residual mean square estimates σ^2 alone. So the ratio

$$F_i = \frac{\text{MS}_i}{s^2}$$

compares two estimates of σ^2 that agree in expectation exactly when $\mathbf{P}_i \boldsymbol{\mu} = \mathbf{0}$. This is the logic of the analysis of variance: *choose a denominator whose expectation equals that of the numerator when the hypothesis holds*. Independence gives the distribution exactly, and the noncentrality $\gamma_i = \|\mathbf{P}_i \boldsymbol{\mu}\|^2/\sigma^2$ governs the power. The probability that F_i exceeds any fixed critical value increases strictly with γ_i ([Theorem 4.1.4](#)). The mean of the ratio follows from independence and $E(1/V) = 1/(\nu - 2)$ for $V \sim \chi^2(\nu)$, $\nu > 2$:

$$E(F_i) = \frac{n - r}{n - r - 2} \cdot \frac{r_i + \gamma_i}{r_i}. \quad (9.6.2)$$

The principle is more general than the choice of s^2 . With random effects the expected mean

squares contain variance components, and the right denominator may be another mean square ([Exercise \(B1\)](#); Chapters 32 and 33). Also, when the model is wrong, [Theorem 9.6.1\(b\)](#) shows that the denominator itself is inflated, by $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2/(n-r)$ (compare [Example \(Residual sum of squares under a wrong mean\)](#)). Omitted terms inflate the denominator of every F ratio, which pushes the ratios towards zero, and they can also shift the numerators, because the omitted part of $\boldsymbol{\mu}$ may have components in the $\mathcal{V}_i$ ([Exercise \(B2\)](#)).

Chapter 11 derives the F test from the likelihood ratio principle and studies its power.

9.6.3 Expected mean squares in an unbalanced layout

In the two-way layout of [Section 9.3](#), [Eq. 9.6.1](#) makes the hypotheses of [Theorem 9.3.2](#) concrete. For the sequential sum of squares of A entered first, $(\mathbf{P}_A - \mathbf{P}_0)\boldsymbol{\mu}$ has entry $\bar{\mu}_i^w - \bar{\mu}^w$ on each of the n_i observations of row i , where $\bar{\mu}^w = \sum_i n_i \bar{\mu}_i^w / n$. So

$$\text{E MS}(A \mid \mu) = \sigma^2 + \frac{1}{a-1} \sum_i n_i (\bar{\mu}_i^w - \bar{\mu}^w)^2.$$

For the Type III sum of squares, the noncentrality is the sum of squares [Eq. 9.2.1](#) evaluated at the true means, divided by σ^2 ([Exercise \(B3\)](#)):

$$\gamma = \frac{(\boldsymbol{\Lambda}^\top \boldsymbol{\mu}_c)^\top (\boldsymbol{\Lambda}^\top \mathbf{N}^{-1} \boldsymbol{\Lambda})^{-1} (\boldsymbol{\Lambda}^\top \boldsymbol{\mu}_c)}{\sigma^2},$$

where $\boldsymbol{\mu}_c$ is the vector of ab cell means, $\mathbf{N} = \text{diag}(n_{ij})$, and the columns of $\boldsymbol{\Lambda}$ express the $a-1$ contrasts of unweighted row means. In the first, the design enters through the weights n_{ij}/n_i , which also define the hypothesis. In the second, the hypothesis does not depend on the design, which enters only through the precision matrix $\mathbf{N}^{-1}$ of the cell means.

Example. (Checking expected mean squares by simulation)

Take the design of [Example \(Party, education and political self-placement\)](#): $n = 944$ respondents in the nine cells of party by education, with the observed counts. Set the true cell means equal to the observed cell means, and $\sigma^2 = 1.21$. The script generates 40,000 data sets with normal errors and 40,000 with centred exponential errors of the same variance, which are strongly skewed. For each it computes the sequential decomposition (party, then education, then interaction), the residual, and the Type III sum of squares for party, and averages the mean squares:

Source	df	γ	E(MS)	normal	skewed	Var(MS)	simulated
mean	1	14594.91	17661.050	17660.423	17661.259	85476.55	86357.17
party	2	535.09	324.942	324.915	324.787	784.89	786.39
education	2	35.75	22.840	22.836	22.891	53.81	53.13
given							
party							
interaction	4	29.60	10.163	10.174	10.179	11.57	11.75
residual	935	0.00	1.210	1.210	1.210	0.0031	0.0031

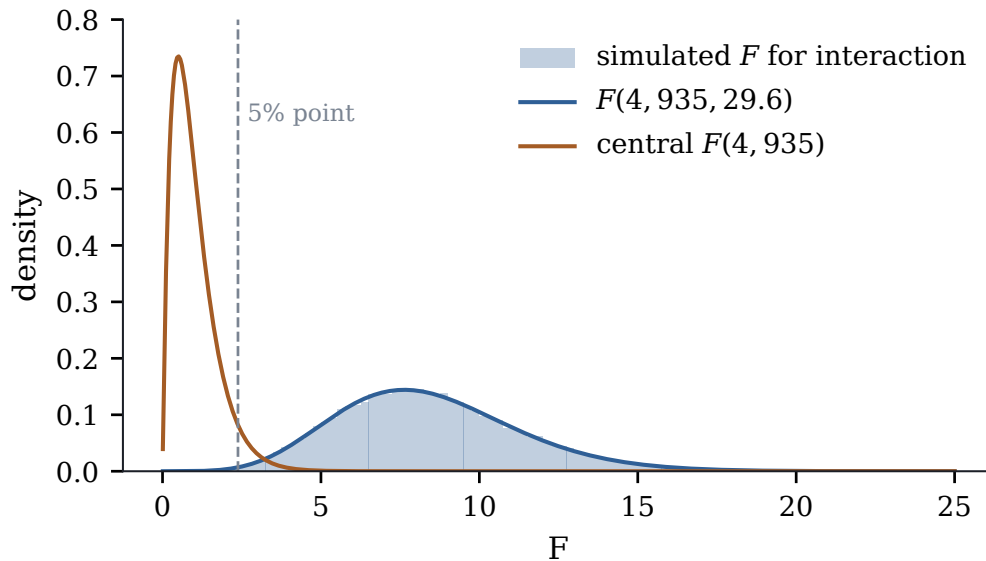


Figure 9.6.1: The F ratio for the interaction in [Example \(Checking expected mean squares by simulation\)](#) over 40,000 simulated data sets with normal errors, with the noncentral $F(4, 935, 29.6)$ density of [Corollary 9.6.2](#) and the central null density $F(4, 935)$ with its upper 5% point.

party,	2	537.53	326.414	326.362	326.259	788.46	789.47
Type III							

The columns $E(\text{MS})$ and $\text{Var}(\text{MS})$ come from [Eq. 9.6.1](#) and [Theorem 9.6.1\(d\)](#), with $\text{Var}(\text{MS}_i) = \sigma^4(2r_i + 4\gamma_i)/r_i^2$; the last column is the simulated variance under normal errors. The averages agree with the expectations under both error distributions, as parts (a) and (b) promise. The variances are another matter. Under skewed errors the variance of the residual mean square is 0.0124, 3.97 times the normal-theory value, because the variance of a quadratic form involves the fourth moment of the errors ([Theorem 2.3.3](#)). And the correlation between the interaction and residual mean squares, 0.008 under normal errors (zero in theory), is 0.020 under skewed errors, against a theoretical 0.025 from the third and fourth moments: the two mean squares are no longer independent.

For the interaction, the noncentrality $\gamma = 29.60$ is large. By [Eq. 9.6.2](#), the F ratio has mean 8.417, and the simulated mean is 8.427. Its 5% critical value is 2.381, and the power is 0.9967 in theory and 0.9967 in the simulation. [Figure 9.6.1](#) shows the simulated distribution with the noncentral F density.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
d = sm.datasets.anes96.load_pandas().data
party = pd.cut(d["PID"], [-1, 1.5, 4.5, 6], labels=["Dem", "Ind", "Rep"])
educ = pd.cut(d["educ"], [0, 3.5, 5.5, 7], labels=["HS", "College", "Graduate"])
cell = party.cat.codes.to_numpy() * 3 + educ.cat.codes.to_numpy()
n = len(cell)

def basis(*blocks):
    Z = np.column_stack(blocks)
    U, s, _ = np.linalg.svd(Z, full_matrices=False)
```

```

    return U[:, s > 1e-10 * s[0]]

one = np.ones((n, 1))
A = np.eye(3)[party.cat.codes.to_numpy()]
B = np.eye(3)[educ.cat.codes.to_numpy()]
Cell = np.eye(9)[cell]
U0, UA, UAB, UC = basis(one), basis(one, A), basis(one, A, B), basis(Cell)

def complement(U_big, U_small):
    """Orthonormal basis of  $C(U\_small)$ -perp within  $C(U\_big)$ ."""
    R = U_big - U_small @ (U_small.T @ U_big)
    return basis(R)

pieces = {"mean": U0, "party": complement(UA, U0),
          "educ | party": complement(UAB, UA), "interaction": complement(UC, UAB)}
Ub = basis(one, B, np.column_stack([(A[:, [i]] - A[:, [0]]) * (B[:, [j]] - B[:,
[0]])
                                     for i in (1, 2) for j in (1, 2)]))
pieces["party (Type III)"] = complement(UC, Ub) # unweighted-means hypothesis

sigma = 1.1
mu = d.groupby(cell)["selfLR"].mean().to_numpy()[cell] # true means: observed cell
means
for name, U in pieces.items():
    r = U.shape[1]
    gamma = np.sum((U.T @ mu) ** 2) / sigma ** 2 # noncentrality  $\|P\mu\|^2 / \sigma^2$ 
    print(f"{name:17s} rank {r}  gamma = {gamma:9.2f}"
          f"  E(MS) = {sigma ** 2 * (1 + gamma / r):9.3f}")

```

```

rng = np.random.default_rng(99)
reps, chunk = 40_000, 4_000
dfe = n - 9
sums = {"normal": [], "skewed": []}
for kind in sums:
    out = []
    for _ in range(reps // chunk):
        if kind == "normal":
            E = sigma * rng.normal(size=(n, chunk))
        else:
            # centred exponential,
            # variance  $\sigma^2$ 
            E = sigma * (rng.exponential(size=(n, chunk)) - 1.0)
        Y = mu[:, None] + E
        ms = {name: np.sum((U.T @ Y) ** 2, axis=0) / U.shape[1] for name, U in
              pieces.items()}
        ms["residual"] = (np.sum(Y ** 2, axis=0) - np.sum((UC.T @ Y) ** 2, axis=0))
        / dfe
        out.append(pd.DataFrame(ms))
    sums[kind] = pd.concat(out, ignore_index=True)
print(pd.DataFrame({k: v.mean() for k, v in sums.items()}).round(3))

```

The simulation never forms an $n \times n$ matrix: each sum of squares comes from the coordinates of $\mathbf{Y}$ in an orthonormal basis of its subspace (Theorem 9.1.1(d)).

9.6.4 Exercises

9.6.4.1 A. Check your understanding

Exercise A1.

In simple regression with $E(\mathbf{Y}) = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}$, show that the regression mean square has expectation $\sigma^2 + \beta_1^2 S_{xx}$, and that the F ratio of the corrected table has noncentrality $\beta_1^2 S_{xx} / \sigma^2$ under normality.

Exercise A2.

In a one-way layout with g groups, group sizes n_i and means μ_i , show that the between-groups mean square has expectation $\sigma^2 + \sum_i n_i (\mu_i - \bar{\mu})^2 / (g - 1)$, where $\bar{\mu} = \sum_i n_i \mu_i / n$. Compare with [Example \(The one-way layout\)](#).

9.6.4.2 B. Practice

Exercise B1.

In a balanced one-way layout with g groups of m observations, let $Y_{ij} = \mu + a_i + \varepsilon_{ij}$, where the a_i and ε_{ij} are uncorrelated random variables with mean zero and variances σ_a^2 and σ^2 . Show that $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z} \mathbf{Z}^\top$, with $\mathbf{Z}$ the matrix of group indicators, and use [Theorem 2.3.1](#) to show that

$$E(\text{MS}_{\text{between}}) = \sigma^2 + m\sigma_a^2, \quad E(\text{MS}_{\text{within}}) = \sigma^2.$$

Which ratio tests $\sigma_a^2 = 0$, and what unbiased estimator of σ_a^2 do the two mean squares suggest?

Solution. $\mathbf{Y} = \mu \mathbf{1} + \mathbf{Z} \mathbf{a} + \boldsymbol{\varepsilon}$, so $E(\mathbf{Y}) = \mu \mathbf{1}$ and $\text{Cov}(\mathbf{Y}) = \mathbf{Z}(\sigma_a^2 \mathbf{I}_g) \mathbf{Z}^\top + \sigma^2 \mathbf{I}$ by [Theorem 2.2.1](#). The between-groups projection is $\mathbf{M} - \mathbf{P}_1$, where $\mathbf{M} = m^{-1} \mathbf{Z} \mathbf{Z}^\top$ averages within groups, and the within-groups projection is $\mathbf{I} - \mathbf{M}$. The mean $\mu \mathbf{1}$ is annihilated by both. By [Theorem 2.3.1](#), $E(\mathbf{Y}^\top (\mathbf{M} - \mathbf{P}_1) \mathbf{Y}) = \text{tr}\{(\mathbf{M} - \mathbf{P}_1)(\sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z} \mathbf{Z}^\top)\}$. Now $\mathbf{Z} \mathbf{Z}^\top = m \mathbf{M}$, and $(\mathbf{M} - \mathbf{P}_1) \mathbf{M} = \mathbf{M} - \mathbf{P}_1$, so the trace is $\sigma^2(g - 1) + \sigma_a^2 m(g - 1)$. For the within-groups projection, $(\mathbf{I} - \mathbf{M}) \mathbf{Z} = \mathbf{0}$, so the trace is $\sigma^2(n - g)$. Dividing by the degrees of freedom gives the two expectations. The ratio $\text{MS}_{\text{between}} / \text{MS}_{\text{within}}$ has expectations that agree iff $\sigma_a^2 = 0$, and $(\text{MS}_{\text{between}} - \text{MS}_{\text{within}}) / m$ is unbiased for σ_a^2 (though it can be negative).

Exercise B2.

Suppose $E(\mathbf{Y}) = \mathbf{X} \boldsymbol{\beta} + \mathbf{z} \delta$ but the model $E(\mathbf{Y}) = \mathbf{X} \boldsymbol{\beta}$ is fitted, with $\mathbf{1} \in \mathcal{C}(\mathbf{X})$. Show that

$$E(\text{SSE}) = \sigma^2(n - r) + \delta^2 \|(\mathbf{I} - \mathbf{M}) \mathbf{z}\|^2, \quad E(\text{SSR}) = \sigma^2(r - 1) + \|(\mathbf{M} - \mathbf{P}_1)(\mathbf{X} \boldsymbol{\beta} + \mathbf{z} \delta)\|^2.$$

Give an example in which $\boldsymbol{\beta}$ has all slopes zero but the regression mean square is inflated.

Exercise B3.

In the setting of [Theorem 9.2.4](#) with $\mathbf{d} = \mathbf{0}$, show that the expected sum of squares of the hypothesis $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$ is $\sigma^2 q + (\mathbf{\Lambda}^\top \boldsymbol{\beta})^\top (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \boldsymbol{\beta}$, so the noncentrality is obtained by evaluating [Eq. 9.2.1](#) at the true parameter.

Solution. By [Theorem 9.2.4](#)(b), the sum of squares is $\mathbf{Y}^\top \mathbf{K} \mathbf{Y}$ with $\mathbf{K} = \mathbf{M} \mathbf{T} (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} \mathbf{T}^\top \mathbf{M}$, a projection of rank q , where $\mathbf{T}$ is the $n \times q$ matrix with $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{T}$. By [Theorem 9.6.1](#)(a) its expectation is $\sigma^2 q + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{K} \mathbf{X} \boldsymbol{\beta}$. Since $\mathbf{T}^\top \mathbf{M} \mathbf{X} \boldsymbol{\beta} = \mathbf{T}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{\Lambda}^\top \boldsymbol{\beta}$, the second term is $(\mathbf{\Lambda}^\top \boldsymbol{\beta})^\top (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \boldsymbol{\beta}$.

9.6.4.3 C. Going deeper**Exercise C1.**

Let the errors be independent with mean zero, variance σ^2 and fourth moment $\mu_4 = \mathbb{E}(\varepsilon_i^4)$, and let $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$. Use [Theorem 2.3.3](#) to show that

$$\text{Var}(\text{SSE}) = (\mu_4 - 3\sigma^4) \sum_i (1 - h_{ii})^2 + 2\sigma^4(n - r),$$

where h_{ii} are the leverages. For the centred exponential errors of [Example \(Checking expected mean squares by simulation\)](#), $\mu_4 = 9\sigma^4$. Show that $\text{Var}(s^2) \approx 8\sigma^4/(n - r)$ when all leverages are small, and compare with the simulated variance of the residual mean square.

Solution. With $\mathbf{A} = \mathbf{I} - \mathbf{M}$ and $\boldsymbol{\theta} = \boldsymbol{\mu}$, the terms of [Theorem 2.3.3](#) involving $\boldsymbol{\theta}$ vanish because $(\mathbf{I} - \mathbf{M})\boldsymbol{\mu} = \mathbf{0}$. The diagonal of $\mathbf{I} - \mathbf{M}$ is $1 - h_{ii}$, and $\text{tr}\{(\mathbf{I} - \mathbf{M})^2\} = n - r$, which gives the formula. If the leverages are small, $\sum_i (1 - h_{ii})^2 \approx n - r$ (it equals $n - 2r + \sum_i h_{ii}^2$), so $\text{Var}(\text{SSE}) \approx (6 + 2)\sigma^4(n - r)$ and $\text{Var}(s^2) \approx 8\sigma^4/(n - r)$, four times the normal value $2\sigma^4/(n - r)$. This is the factor seen in the simulation.

Exercise C2.

For the Type I sum of squares of A entered first in a two-way layout, show that its noncentrality can be positive when $\mu_{ij} = \beta_j$ depends on B only, and find the condition on the counts n_{ij} under which it is zero for all such $\boldsymbol{\mu}$. Relate your answer to [Theorem 9.4.3](#).

9.7 Summary and notes

9.7.1 Summary

Idea.

1. An analysis of variance is a choice of mutually orthogonal subspaces filling the model space. Sums of squares are squared lengths of the projections of $\mathbf{y}$, degrees

- of freedom are dimensions, and both add up by Pythagoras. Decompositions and chains of nested models are the same thing (Definition 9.1.1, Theorem 9.1.1).
2. The regression table comes from the chain $\text{span}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{X})$. The corrected identity $\text{SST} = \text{SSR} + \text{SSE}$ needs $\mathbf{1} \in \mathcal{C}(\mathbf{X})$; without it an extra term $-2\bar{y} \sum_i \hat{\varepsilon}_i$ appears (Proposition 9.1.2, Proposition 9.1.3).
 3. The extra sum of squares $\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) = \mathbf{y}^\top (\mathbf{M} - \mathbf{M}_1) \mathbf{y}$ depends only on column spaces. Adding regressors to a fitted model needs only a $t \times t$ inverse, and never lowers the variances of the old coefficients. For one column it equals $t_j^2 s^2$ (Definition 9.2.1, Theorem 9.2.2, Corollary 9.2.3).
 4. The sum of squares of an estimable hypothesis $\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{d}$ is $(\mathbf{A}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{A}^\top \mathbf{G} \mathbf{A})^{-1} (\mathbf{A}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})$, whatever the rank. A nonestimable constraint restricts only through its estimable part (Theorem 9.2.4, Theorem 9.2.5, Proposition 9.2.6).
 5. Sequential (Type I) sums of squares decompose the regression sum of squares but depend on the order. Partial, Type II and Type III sums of squares do not decompose anything, and in models with interactions they address different hypotheses: weighted row means, additive main effects, unweighted row means, or, with reference coding, means at a single level of the other factor (Definition 9.3.1, Definition 9.3.2, Proposition 9.3.1, Theorem 9.3.2).
 6. Order stops mattering exactly when the centred subspaces of the terms are orthogonal. For two factors this is the proportional-frequency condition. Orthogonal polynomials and orthogonal contrasts manufacture orthogonality inside a single term (Theorem 9.4.1, Theorem 9.4.3, Theorem 9.4.4).
 7. R is the largest correlation of $\mathbf{y}$ with a linear combination of the regressors. Under the null $R^2 \sim \text{Beta}(\frac{r-1}{2}, \frac{n-r}{2})$, with mean $(r-1)/(n-1)$. The adjusted R^2 has mean zero under the null and rises exactly when an added term has $F > 1$. Partial R^2 equals $t^2/(t^2 + n - r)$ (Theorem 9.5.1, Theorem 9.5.2, Proposition 9.5.3, Theorem 9.5.4).
 8. R^2 depends on the design, and it is not a test of the model, a comparison across responses, or a measure of predictive accuracy or of cause (Section 9.5).
 9. Every mean square has expectation $\sigma^2 + \|\mathbf{P}\boldsymbol{\mu}\|^2 / \text{rank}(\mathbf{P})$, using only second moments. Under normality the pieces of a decomposition are independent noncentral chi-squared variables, and an F ratio compares two estimates of σ^2 that agree exactly when its hypothesis $\mathbf{P}\boldsymbol{\mu} = \mathbf{0}$ holds (Theorem 9.6.1, Corollary 9.6.2).

9.7.2 Notes and sources

Analysis of variance. The decomposition of a total sum of squares into components with their own degrees of freedom is Fisher's. An analysis of variance already appears in Fisher and Mackenzie (1923), and Fisher set out the analysis of variance table for general use in *Statistical Methods for Research Workers* (Fisher 1925), after making the notion of degrees of freedom precise in Fisher (1922). The distribution theory that makes the components independent chi-squared variables is Cochran's theorem (Cochran 1934), proved in Chapter 4. The coordinate-free view, in which a table is a list of orthogonal subspaces, follows Christensen (2020). His section 3.6, which splits a test space into orthogonal lines, is the source of Theorem 9.4.4, and his section 7.4 treats proportional numbers. Scheffé (1959) remains a clear account of the classical theory, including expected mean squares for random effects.

Extra sums of squares, adding regressors and restrictions. Seber and Lee (2003, sections

3.7–3.9) give the updating formulas of [Theorem 9.2.2](#), restricted least squares, and their rank-deficient versions. Our proofs go through the Frisch–Waugh–Lovell theorem and Pythagoras instead of Lagrange multipliers. [Proposition 9.2.6](#) makes precise the remark, common to the sources, that only the estimable part of a hypothesis can be tested.

Types of sums of squares. Agresti (2015, sections 2.3–2.4) presents sequential and partial sums of squares through projection matrices, with the decomposition of the regression sum of squares when the regressors are uncorrelated, and notes the Type I, II and III terminology. The problem of main effects in unbalanced layouts with interaction goes back to Yates (1934), whose “weighted squares of means” is closely related to what software now calls Type III. The names of the types come from statistical software, and programs have used them inconsistently, which is why this chapter defines each type by its projection. Nelder (1977) argued that tests of main effects in the presence of interactions violate the marginality of the model terms; his position underlies the preference for Type II sums of squares when interactions are absent. Speed, Hocking and Hackney (1978) identify the cell-mean hypotheses tested by the various sums of squares, and Searle (1987) treats unbalanced data exhaustively in the same cell-mean terms.

The coefficient of determination. The distribution of the sample multiple correlation coefficient, under the null and in general for random normal regressors, is due to Fisher (1928); [Theorem 9.5.2](#) is the simpler fixed-regressor version. Kvralseth (1985) surveys the many different formulas that software has used for R^2 , especially for models without an intercept, and Anscombe (1973) gives four data sets with the same R^2 and very different relationships, a standing warning that R^2 is not a test of the model.

Data. The state crime, election study, Grunfeld and Nile data are public-domain data sets distributed with statsmodels. The Nile series is the one analysed for a change point by Cobb (1978).

What is new here. The organization of the chapter around subspaces, the proofs of [Theorem 9.4.1](#) and [Theorem 9.4.3](#), the reference-coding case of [Theorem 9.3.2](#), and the numerical examples and simulations are this book’s.

9.7.3 References

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Chapter 10

Computation: QR, SVD and Conditioning

Every least squares estimate in this book is written as $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ or as $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-} \mathbf{X}^\top \mathbf{y}$. These are the right formulas to reason with. They are not how a computer should produce the numbers. This chapter explains what a computer should do instead, and how accurate the result is. It separates two questions: how sensitive the least squares problem is to small changes in the data, which depends on $\kappa(\mathbf{X})$ and the size of the residual, and whether an algorithm adds more error than that. The answers give a practical ranking. The Cholesky factorization of $\mathbf{X}^\top \mathbf{X}$ is fast and is enough when $\mathbf{X}$ is well conditioned. Householder or Givens QR is the default. The singular value decomposition is the tool for rank deficiency and near dependence. Pivoted QR decides the rank cheaply, which is how statistical software chooses what to do with aliased columns.

What you need. [Chapter 1](#): the Cholesky factorization ([Theorem 1.7.4](#)), QR ([Theorem 1.5.2](#)), the singular value decomposition ([Theorem 1.8.1](#)), the Eckart–Young inequality ([Proposition 1.8.2](#)) and the condition number ([Definition 1.8.1](#)). [Chapter 5](#): the linear model and the moments of $\hat{\beta}$ ([Theorem 5.5.1](#)). [Chapter 6](#): least squares as projection, and the first look at computation in [Section 6.10](#). A few remarks use the sampling theory of [Chapter 7](#) and the sums of squares of [Chapter 9](#).

10.0.1 Roadmap

- [Why not invert the cross-product matrix](#)
- [The Cholesky factorization](#)
- [Householder and Givens QR](#)
- [The singular value decomposition](#)
- [Conditioning and perturbation bounds](#)
- [Updating and downdating](#)
- [Rank-revealing factorizations](#)
- [Summary and notes](#)

10.1 Why not invert the cross-product matrix

Chapter 5 found the least squares estimate by setting a gradient to zero, which gave $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ when $\mathbf{X}$ has full column rank (Theorem 5.4.1). Read as a recipe, the formula says: form the $p \times p$ matrix $\mathbf{X}^\top \mathbf{X}$, invert it, and multiply. Section 6.10 already showed that the recipe can lose twice as many digits as necessary. This section explains why.

10.1.1 Floating-point numbers

A double-precision number is zero or a number $\pm m \times 2^{e-52}$, where m is an integer with $2^{52} \leq m < 2^{53}$ and e is an integer in a fixed range ($-1022 \leq e \leq 1023$). Between 2^e and 2^{e+1} the doubles are therefore equally spaced, 2^{e-52} apart. Just above 1 the spacing is $\epsilon_M = 2^{-52} \approx 2.22 \times 10^{-16}$, the **machine epsilon**. Rounding a real number x in the range of the doubles to the nearest double gives a number $\text{fl}(x)$ with

$$\text{fl}(x) = x(1 + \delta), \quad |\delta| \leq u = 2^{-53} \approx 1.11 \times 10^{-16}.$$

The **unit roundoff** $u = \epsilon_M/2$ is half the gap, because the nearest double is at most half a gap away. Rounding errors are *relative*: a double represents large and small numbers with the same number of significant digits, about sixteen.

The IEEE 754 standard, which essentially all current hardware follows, requires the basic operations to be correctly rounded. The computed result of $x + y$ is the exact sum, rounded. That is the only property of the hardware that the analysis below uses.

Definition 10.1.1 — Standard model of floating-point arithmetic

For floating-point numbers x, y and $\circ \in \{+, -, \times, /\}$, the computed result satisfies

$$\text{fl}(x \circ y) = (x \circ y)(1 + \delta), \quad |\delta| \leq u,$$

and $\text{fl}(\sqrt{x}) = \sqrt{x}(1 + \delta)$ with $|\delta| \leq u$, provided no result overflows or underflows.

Two consequences are worth seeing at once. The number $1 + u$ lies exactly halfway between the doubles 1 and $1 + \epsilon_M$, and the tie is broken towards 1, so adding u to 1 changes nothing. And most decimal fractions are not doubles, so familiar identities fail at the level of u :

```
import numpy as np

eps = np.finfo(float).eps          # spacing of the doubles just above 1: 2^-52
u = eps / 2                        # unit roundoff: 2^-53
print("eps =", eps, " u =", u)
print("1 + u == 1:", 1.0 + u == 1.0, " 1 + eps == 1:", 1.0 + eps == 1.0)
print("0.1 + 0.2 == 0.3:", 0.1 + 0.2 == 0.3)
```

Trouble comes from accumulation and above all from subtraction. The difference of nearly equal numbers is computed exactly but exposes the errors they already carried: if a and b agree in

ten digits, a rounding error in either becomes a large relative error in $a - b$. This **cancellation** is behind every numerical difficulty in this chapter.

10.1.2 Accumulating rounding errors

Products of many factors $1 + \delta_i$ appear whenever errors accumulate. The following lemma packages them.

Lemma 10.1.1

Let $|\delta_i| \leq u$ and $\rho_i \in \{1, -1\}$ for $i = 1, \dots, k$, and suppose $ku < 1$. Then

$$\prod_{i=1}^k (1 + \delta_i)^{\rho_i} = 1 + \theta_k, \quad |\theta_k| \leq \gamma_k = \frac{ku}{1 - ku}. \quad (10.1.1)$$

Proof. Each factor lies between $1 - u$ and $1/(1 - u)$: for $\rho_i = 1$ because $1 + u \leq 1/(1 - u)$, and for $\rho_i = -1$ because $1/(1 + \delta_i) \geq 1/(1 + u) \geq 1 - u$. So the product lies between $(1 - u)^k$ and $(1 - u)^{-k}$. By Bernoulli's inequality $(1 - u)^k \geq 1 - ku$, which gives the lower end $1 - ku \geq 1 - \gamma_k$, and also the upper end $(1 - u)^{-k} \leq 1/(1 - ku) = 1 + \gamma_k$. ■

The constant γ_k is essentially ku : for $k = 10^6$ it is about 1.1×10^{-10} . This is a worst case; in practice errors of different signs partly cancel and grow more like $\sqrt{k}u$.

The basic step of every least squares algorithm is an inner product.

Proposition 10.1.2 — Rounding error in an inner product

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ have floating-point entries and $nu < 1$. The inner product computed by recursive summation, $s_1 = \text{fl}(x_1 y_1)$ and $s_k = \text{fl}(s_{k-1} + \text{fl}(x_k y_k))$, satisfies

$$s_n = \sum_{i=1}^n x_i y_i (1 + \theta_i), \quad |\theta_i| \leq \gamma_n.$$

Consequently $|s_n - \mathbf{x}^\top \mathbf{y}| \leq \gamma_n \sum_i |x_i y_i|$, and $s_n = (\mathbf{x} + \Delta \mathbf{x})^\top \mathbf{y}$ for a vector $\Delta \mathbf{x}$ with $|\Delta x_i| \leq \gamma_n |x_i|$.

Proof. By the standard model, $\text{fl}(x_k y_k) = x_k y_k (1 + \mu_k)$ and $s_k = (s_{k-1} + x_k y_k (1 + \mu_k))(1 + \sigma_k)$ for $k \geq 2$, with all $|\mu_k|, |\sigma_k| \leq u$. Unrolling the recursion,

$$s_n = \sum_{i=1}^n x_i y_i (1 + \mu_i) \prod_{k=\max(i, 2)}^n (1 + \sigma_k).$$

The i th term carries at most n factors of the form $1 + \delta$, so by Lemma 10.1.1 it equals $x_i y_i (1 + \theta_i)$ with $|\theta_i| \leq \gamma_n$ (since $\gamma_k \leq \gamma_n$ for $k \leq n$). The two consequences follow by taking $\Delta x_i = \theta_i x_i$.

The first conclusion is a *forward* bound, which can be large relative to a result whose terms cancel. The second is a *backward* statement: the computed number is the exact inner product of slightly perturbed data. Over 200 random inner products of length up to 400, checked in rational arithmetic, the largest ratio of error to bound is 0.149.

```
from fractions import Fraction

def gamma(k, u=2.0 ** -53):
    return k * u / (1 - k * u)

def dot_recursive(x, y):
    """Inner product by recursive summation, one rounding per operation."""
    s = 0.0
    for a, b in zip(x, y):
        s = s + a * b
    return s

rng = np.random.default_rng(101)
worst = 0.0
for trial in range(200):
    n = int(rng.integers(2, 400))
    x = rng.normal(size=n) * 10.0 ** rng.integers(-3, 4, size=n)
    y = rng.normal(size=n)
    exact = sum(Fraction(a) * Fraction(b) for a, b in zip(x, y))      # exact
    rational value
    error = abs(Fraction(dot_recursive(x, y)) - exact)
    bound = Fraction(gamma(n)) * sum(abs(Fraction(a) * Fraction(b)) for a, b in zip(
        x, y))
    worst = max(worst, float(error / bound))
print(f"largest error / bound over 200 inner products: {worst:.3f}")
```

10.1.3 Conditioning and stability

A computation turns data d into a result $f(d)$. The computed result can be inaccurate because the *problem* is sensitive, or because the *algorithm* adds more error than the data justify. The first is measured by a **condition number**, the largest factor by which a small relative change in d can be magnified in $f(d)$; the second by the following standard.

Definition 10.1.2 — Backward stability

An algorithm that computes $f(d)$ is **backward stable** if, for every input d , its computed output $\hat{f}(d)$ is the exact result for nearby data: $\hat{f}(d) = f(d + \Delta d)$ with $\|\Delta d\| \leq cu\|d\|$, where c is a modest constant depending only on the dimensions of the problem.

Proposition 10.1.2 says that recursive summation is a backward stable way to compute inner products, with $c = \gamma_n/u \approx n$. Backward stability is the most one can ask, because storing the data as doubles already perturbs them by u . The two notions combine, to first order in u , as

$$\text{relative forward error} \lesssim \text{condition number} \times \text{backward error}. \quad (10.1.2)$$

For least squares the condition number, which involves $\kappa(\mathbf{X})$ and sometimes $\kappa(\mathbf{X})^2$, is the subject of [Section 10.5](#). Sections 10.2 to 10.4 ask whether algorithms are backward stable *for the least squares problem*, with perturbations of $\mathbf{X}$ and $\mathbf{y}$.

10.1.4 What forming the cross-product matrix loses

The normal-equations recipe is not backward stable for least squares, and the damage is done in its very first step. Here is what rounding does to $\mathbf{X}^\top \mathbf{X}$.

Proposition 10.1.3 — Rounding in the cross-product matrix

Let $\mathbf{X}$ be $n \times p$ with full column rank and let $\hat{\mathbf{A}}$ be $\mathbf{X}^\top \mathbf{X}$ computed entry by entry as in [Proposition 10.1.2](#). Then $\hat{\mathbf{A}} = \mathbf{X}^\top \mathbf{X} + \Delta \mathbf{A}$ with $|\Delta \mathbf{A}| \leq \gamma_n |\mathbf{X}|^\top |\mathbf{X}|$ entrywise, and

$$\|\Delta \mathbf{A}\|_2 \leq p \gamma_n \|\mathbf{X}\|_2^2.$$

In particular $\hat{\mathbf{A}}$ is positive definite whenever $p \gamma_n \kappa(\mathbf{X})^2 < 1$.

Proof. The entrywise bound is [Proposition 10.1.2](#) applied to each pair of columns. The spectral norm is at most the Frobenius norm, the Frobenius norm is monotone in the absolute values of the entries and submultiplicative, and $\|\mathbf{X}\|_F^2 = \sum_i \sigma_i^2 \leq p \sigma_1^2$ by [Proposition 1.8.2\(a\)](#). Hence

$$\|\Delta \mathbf{A}\|_2 \leq \gamma_n \|\mathbf{X}^\top |\mathbf{X}|\|_F \leq \gamma_n \|\mathbf{X}\|_F^2 \leq p \gamma_n \sigma_1^2.$$

For a unit vector $\mathbf{v}$, $\mathbf{v}^\top \hat{\mathbf{A}} \mathbf{v} = \|\mathbf{X} \mathbf{v}\|^2 + \mathbf{v}^\top \Delta \mathbf{A} \mathbf{v} \geq \sigma_p^2 - p \gamma_n \sigma_1^2$, which is positive when $p \gamma_n \sigma_1^2 / \sigma_p^2 < 1$. ■

The information about $\mathbf{X}$ that sits in its smallest singular value σ_p appears in $\mathbf{X}^\top \mathbf{X}$ as an eigenvalue σ_p^2 . Rounding perturbs $\mathbf{X}^\top \mathbf{X}$ by an amount of order $u \sigma_1^2$. Once σ_p^2 is below that level, the computed cross-product matrix may not determine the direction $\mathbf{v}_p$ at all. Even rounding the *exact* $\mathbf{X}^\top \mathbf{X}$ to doubles does this much damage. The loss becomes total around $\kappa(\mathbf{X}) \approx u^{-1/2} = 9.5 \times 10^7$, and it is already serious well before that: the normal equations are then a problem with condition number $\kappa(\mathbf{X}^\top \mathbf{X}) = \kappa(\mathbf{X})^2$ ([Eq. 1.8.2](#)), and [Eq. 10.1.2](#) predicts relative errors of order $\kappa(\mathbf{X})^2 u$ in $\hat{\boldsymbol{\beta}}$ whatever method is then used to solve them.

Example. (A cross-product matrix that rounds to a singular one)

Take $\delta = 10^{-9}$ and

$$\mathbf{X} = \begin{pmatrix} 1 & 1 \\ \delta & 0 \\ 0 & \delta \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 1 + \delta^2 & 1 \\ 1 & 1 + \delta^2 \end{pmatrix}.$$

The columns of $\mathbf{X}$ are independent, and $\kappa(\mathbf{X}) = \sqrt{2 + \delta^2} / \delta = 1.41 \times 10^9$. But $\delta^2 = 10^{-18}$ is smaller than u , so $1 + \delta^2$ rounds to 1 and the computed $\mathbf{X}^\top \mathbf{X}$ is the singular matrix of ones. The Cholesky factorization of [Section 10.2](#) fails, and no method applied to the computed normal equations can recover the solution. For $\mathbf{y} = \mathbf{X}(1, 1)^\top$, the exact least

squares estimate is $(1, 1)^\top$ with zero residual. QR, which never forms $\mathbf{X}^\top \mathbf{X}$, returns it with relative error 3.9×10^{-16} .

```
delta = 1e-9
X = np.array([[1.0, 1.0],
              [delta, 0.0],
              [0.0, delta]])
y = X @ np.array([1.0, 1.0])          # exact data: the solution is b = (1, 1),
                                     # residual 0

A = X.T @ X                          # 1 + delta^2 rounds to 1
print("computed X'X =\n", A)
print("cond(X) =", np.linalg.cond(X))
try:
    np.linalg.cholesky(A)
except np.linalg.LinAlgError as err:
    print("Cholesky of the computed X'X fails:", err)

Q, R = np.linalg.qr(X)
b_qr = np.linalg.solve(R, Q.T @ y)
print("QR solution:", b_qr)
```

10.1.5 Why not the inverse either

Even when $\mathbf{X}^\top \mathbf{X}$ is safely positive definite, the inverse costs p^3 flops against $p^3/3$ for a Cholesky factorization plus $2p^2$ for two triangular solves, and it is less accurate. Statistics needs only the diagonal of $\text{Cov}(\hat{\beta}) = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ (Theorem 5.5.1): if $\mathbf{X}^\top \mathbf{X} = \mathbf{R}^\top \mathbf{R}$, its j th entry is σ^2 times the squared length of row j of $\mathbf{R}^{-1}$.

10.1.6 Counting the work

Cost is traditionally counted in floating-point operations (**flops**), each addition, subtraction, multiplication and division counting one. Flop counts ignore memory traffic and parallelism, so they are only a first guide.

Task	Leading flop count	Where
Form $\mathbf{X}^\top \mathbf{X}$ and $\mathbf{X}^\top \mathbf{y}$ (using symmetry)	np^2	this section
Cholesky factorization of $\mathbf{X}^\top \mathbf{X}$	$p^3/3$	Section 10.2
Two triangular solves	$2p^2$	Section 10.2
Householder QR of $\mathbf{X}$, with $\mathbf{Q}^\top \mathbf{y}$	$2np^2 - 2p^3/3$	Section 10.3
Givens QR of $\mathbf{X}$	$3np^2 - p^3$	Section 10.3
Modified Gram–Schmidt	$2np^2$	Section 10.3
Singular values and $\mathbf{V}$, QR first (R-SVD)	$2np^2 + 11p^3$	Section 10.4

When $n \gg p$, as usual in regression, the normal equations cost about np^2 and Householder

QR about $2np^2$: the accurate method is at most twice as expensive. And the one-pass, $O(p^2)$ -memory accumulation of cross-products has a QR counterpart in row-wise Givens QR (Section 10.3).

Idea. Forming $\mathbf{X}^\top \mathbf{X}$ squares the condition number before any equation is solved, and inverting it adds cost and error. Factor $\mathbf{X}$ itself, by an orthogonal transformation, and read everything off the factors.

10.1.7 Exercises

10.1.7.1 A. Check your understanding

Exercise A1.

Show that the doubles in $[1, 2)$ are exactly the numbers $1 + k 2^{-52}$, $k = 0, 1, \dots, 2^{52} - 1$. What is the smallest double greater than 1? What is the largest integer N such that every integer from 0 to N is a double?

Exercise A2.

For $x = 10^{-10}$, the computer evaluates $(1 + x) - 1$. Use the standard model to show that the result is $x + \delta(1 + x)$ for some $|\delta| \leq u$, where the subtraction itself is exact. Estimate the relative error, and check it numerically. Which of the two operations is responsible?

Solution. The addition gives $\text{fl}(1 + x) = (1 + x)(1 + \delta)$. This number is within a factor of two of 1, and the difference of two doubles within a factor of two of each other is itself a double, so the subtraction is exact: $\text{fl}(1 + x) - 1 = x + \delta(1 + x)$. The relative error is $|\delta|(1 + x)/x \approx u/x \approx 10^{-6}$. All of it comes from rounding $1 + x$, which kept only about six significant digits of x . The subtraction then exposed that loss.

10.1.7.2 B. Practice

Exercise B1.

Show that recursive summation of $x_1, \dots, x_n$ returns $\sum_i x_i(1 + \theta_i)$ with $|\theta_i| \leq \gamma_{n-1}$, and that pairwise summation returns such a sum with $|\theta_i| \leq \gamma_{\lceil \log_2 n \rceil}$.

Solution. With $s_1 = x_1$ and $s_k = (s_{k-1} + x_k)(1 + \sigma_k)$, $s_n = \sum_i x_i \prod_{k=\max(i,2)}^n (1 + \sigma_k)$. Term i has at most $n - 1$ factors, so Lemma 10.1.1 gives $|\theta_i| \leq \gamma_{n-1}$. In pairwise summation each x_i passes through one addition per level of the tree, and a balanced tree over n leaves has $\lceil \log_2 n \rceil$ levels. The same unrolling gives at most $\lceil \log_2 n \rceil$ factors per term.

Exercise B2.

Let $\mathbf{A}$ be $m \times n$ and $\mathbf{b} \in \mathbb{R}^n$. Show that the computed product $\text{fl}(\mathbf{A}\mathbf{b})$, each entry an inner product, equals $(\mathbf{A} + \Delta\mathbf{A})\mathbf{b}$ with $|\Delta\mathbf{A}| \leq \gamma_n |\mathbf{A}|$. Deduce $\|\text{fl}(\mathbf{A}\mathbf{b}) - \mathbf{A}\mathbf{b}\| \leq$

$$\gamma_n \sqrt{n} \|\mathbf{A}\|_2 \|\mathbf{b}\|.$$

10.1.7.3 C. Going deeper

Exercise C1.

For $0 < \delta \leq 2^{-27}$ let $\mathbf{X}_\delta$ be the matrix of [Example \(A cross-product matrix that rounds to a singular one\)](#), and let $\mathbf{y} = \mathbf{X}_\delta(3, 1)^\top$ and $\mathbf{y}' = \mathbf{X}_\delta(1, 3)^\top$. Show that the computed cross-product matrices of $[\mathbf{X}_\delta, \mathbf{y}]$ and $[\mathbf{X}_\delta, \mathbf{y}']$ are identical (and do not depend on δ), although the exact least squares estimates are $(3, 1)^\top$ and $(1, 3)^\top$. Conclude that no algorithm that starts from the computed cross-product matrix can solve these problems. Is anything special about 2^{-27} ?

10.2 The Cholesky factorization

If the cross-product matrix is to be used at all, the normal equations should be solved by factoring it as a triangular matrix times its transpose. One factorization of the augmented cross-product matrix then delivers every standard regression quantity.

10.2.1 The algorithm

[Chapter 1](#) proved that a positive definite matrix has a unique factorization $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$ with $\mathbf{L}$ lower triangular and positive on the diagonal ([Theorem 1.7.4\(d\)](#)). Numerical work usually writes it with the upper triangular factor $\mathbf{R} = \mathbf{L}^\top$.

Theorem 10.2.1 — Cholesky factorization

Let $\mathbf{A}$ be a $p \times p$ positive definite matrix.

- (a) There is exactly one upper triangular $\mathbf{R}$ with positive diagonal entries such that $\mathbf{A} = \mathbf{R}^\top \mathbf{R}$.
- (b) Its rows are given, for $j = 1, \dots, p$ in turn, by

$$\begin{aligned} r_{jj} &= \left(a_{jj} - \sum_{k < j} r_{kj}^2 \right)^{1/2}, \\ r_{ji} &= \frac{1}{r_{jj}} \left(a_{ji} - \sum_{k < j} r_{kj} r_{ki} \right), \quad i > j, \end{aligned} \tag{10.2.1}$$

and every radicand in it is positive.

- (c) The computation takes $p^3/3 + O(p^2)$ flops.
- (d) If $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ and $\mathbf{X} = \mathbf{Q}\mathbf{R}_0$ is a QR factorization with $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$ and $\mathbf{R}_0$ upper triangular with positive diagonal, then $\mathbf{R}_0 = \mathbf{R}$.

Proof.

- (a) is [Theorem 1.7.4\(d\)](#) with $\mathbf{R} = \mathbf{L}^\top$.
- (b) Entry (j, i) , $i \geq j$, of $\mathbf{R}^\top \mathbf{R}$ is $\sum_{k \leq j} r_{kj} r_{ki}$, because $r_{kj} = 0$ for $k > j$. Setting it equal to a_{ji} and isolating the $k = j$ term gives the recursion [Eq. 10.2.1](#). So the entries of the unique factor of (a) satisfy [Eq. 10.2.1](#), and [Eq. 10.2.1](#) determines row j from rows $1, \dots, j-1$. By induction on j , the rows computed by [Eq. 10.2.1](#) are those of the factor in (a). In particular the j th radicand is $r_{jj}^2 > 0$.
- (c) Row j needs about $2(j-1)$ flops for each of its $p-j+1$ entries, so the total is $\sum_j 2(j-1)(p-j+1) = p^3/3 + O(p^2)$.
- (d) $\mathbf{X}^\top \mathbf{X} = \mathbf{R}_0^\top \mathbf{Q}^\top \mathbf{Q} \mathbf{R}_0 = \mathbf{R}_0^\top \mathbf{R}_0$, and uniqueness in (a).

■

By part (d), the triangle of a QR factorization of $\mathbf{X}$ and the Cholesky factor of $\mathbf{X}^\top \mathbf{X}$ are the same matrix (see also [Exercise \(B2\)](#)). The two routes of this chapter compute the same object, one from $\mathbf{X}$, the other from $\mathbf{X}^\top \mathbf{X}$ after the small singular values may already have been damaged ([Proposition 10.1.3](#)).

The algorithm itself is backward stable *for the factorization*. The computed $\hat{\mathbf{R}}$ satisfies $\hat{\mathbf{R}}^\top \hat{\mathbf{R}} = \mathbf{A} + \Delta \mathbf{A}$ with $|\Delta \mathbf{A}| \leq \gamma_{p+1} |\hat{\mathbf{R}}|^\top |\hat{\mathbf{R}}|$ entrywise, and it runs to completion whenever $\mathbf{A}$ is not too close to singular (Higham 2002, chapter 10), with no pivoting. The weakness of the normal equations lies entirely in forming $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$, not in factoring it.

```
import numpy as np
import statsmodels.api as sm

def cholesky_upper(A):
    """Upper triangular R with positive diagonal and R'R = A (A positive definite)."""
    A = np.array(A, dtype=float)
    p = A.shape[0]
    R = np.zeros_like(A)
    for j in range(p):
        d = A[j, j] - R[:j, j] @ R[:j, j]
        if d <= 0:
            raise np.linalg.LinAlgError(f"not positive definite at step {j + 1}")
        R[j, j] = np.sqrt(d)
        R[j, j + 1:] = (A[j, j + 1:] - R[:j, j] @ R[:j, j + 1:]) / R[j, j]
    return R

def solve_upper(R, b):
    """Back substitution for R x = b, R upper triangular."""
    x = np.zeros(np.shape(b))
    for i in range(len(b) - 1, -1, -1):
        x[i] = (b[i] - R[i, i + 1:] @ x[i + 1:]) / R[i, i]
    return x

def solve_lower(L, b):
    """Forward substitution for L x = b, L lower triangular."""
    x = np.zeros(np.shape(b))
    for i in range(len(b)):
        x[i] = (b[i] - L[i, :i] @ x[:i]) / L[i, i]
    return x
```

10.2.2 Everything from one factorization

Solving the normal equations $\mathbf{R}^\top \mathbf{R} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$ takes two triangular solves: $\mathbf{R}^\top \mathbf{z} = \mathbf{X}^\top \mathbf{y}$ for $\mathbf{z}$, then $\mathbf{R} \mathbf{b} = \mathbf{z}$. Each costs p^2 flops. It is neater to factor the cross-product matrix of the augmented matrix $\mathbf{Z} = [\mathbf{X}, \mathbf{y}]$. The factor then contains the solution of the first triangular system and the residual sum of squares.

Proposition 10.2.2 — The augmented factorization

Let $\mathbf{X}$ be $n \times p$ of full column rank, suppose $\mathbf{y} \notin \mathcal{C}(\mathbf{X})$, and let

$$\mathbf{Z}^\top \mathbf{Z} = \begin{pmatrix} \mathbf{X}^\top \mathbf{X} & \mathbf{X}^\top \mathbf{y} \\ \mathbf{y}^\top \mathbf{X} & \mathbf{y}^\top \mathbf{y} \end{pmatrix} = \mathbf{T}^\top \mathbf{T}, \quad \mathbf{T} = \begin{pmatrix} \mathbf{R} & \mathbf{z} \\ \mathbf{0}^\top & d \end{pmatrix},$$

be the Cholesky factorization (Theorem 10.2.1) of the cross-product matrix of $\mathbf{Z} = [\mathbf{X}, \mathbf{y}]$. Then:

- (a) $\mathbf{R}$ is the Cholesky factor of $\mathbf{X}^\top \mathbf{X}$, the least squares estimate solves $\mathbf{R} \hat{\boldsymbol{\beta}} = \mathbf{z}$, and $d^2 = \text{SSE} = \|\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}\|^2$;
- (b) for each $k \leq p$, the residual sum of squares of the regression of $\mathbf{y}$ on the first k columns of $\mathbf{X}$ is $d^2 + \sum_{j>k} z_j^2$. The drop in residual sum of squares when column k joins columns $1, \dots, k-1$ is z_k^2 ;
- (c) $\det(\mathbf{X}^\top \mathbf{X}) = \prod_j r_{jj}^2$, $(\mathbf{X}^\top \mathbf{X})^{-1} = \mathbf{R}^{-1} \mathbf{R}^{-\top}$, and the leverage of row $\mathbf{x}_{(i)}^\top$ is $h_{ii} = \|\mathbf{R}^{-\top} \mathbf{x}_{(i)}\|^2$.

Proof. $\mathbf{Z}$ has full column rank because $\mathbf{y} \notin \mathcal{C}(\mathbf{X})$, so $\mathbf{Z}^\top \mathbf{Z}$ is positive definite and $\mathbf{T}$ exists. Multiplying out $\mathbf{T}^\top \mathbf{T}$ and comparing blocks with $\mathbf{Z}^\top \mathbf{Z}$ gives

$$\mathbf{R}^\top \mathbf{R} = \mathbf{X}^\top \mathbf{X}, \quad \mathbf{R}^\top \mathbf{z} = \mathbf{X}^\top \mathbf{y}, \quad \mathbf{z}^\top \mathbf{z} + d^2 = \mathbf{y}^\top \mathbf{y}.$$

The first identity and uniqueness make $\mathbf{R}$ the Cholesky factor of $\mathbf{X}^\top \mathbf{X}$. The second gives $\mathbf{R}^\top \mathbf{R} \hat{\boldsymbol{\beta}} = \mathbf{R}^\top \mathbf{z}$ for the solution of the normal equations, so $\mathbf{R} \hat{\boldsymbol{\beta}} = \mathbf{z}$.

For (b), and for the rest of (a), let $\mathbf{Q} = \mathbf{Z} \mathbf{T}^{-1}$. Then $\mathbf{Q}^\top \mathbf{Q} = \mathbf{T}^{-\top} \mathbf{Z}^\top \mathbf{Z} \mathbf{T}^{-1} = \mathbf{I}$, so the columns $\mathbf{q}_1, \dots, \mathbf{q}_{p+1}$ of $\mathbf{Q}$ are orthonormal, and $\mathbf{Z} = \mathbf{Q} \mathbf{T}$. Because $\mathbf{T}$ is upper triangular with nonzero diagonal, the first k columns of $\mathbf{X}$ span the same subspace as $\mathbf{q}_1, \dots, \mathbf{q}_k$. The last column of $\mathbf{Z} = \mathbf{Q} \mathbf{T}$ reads $\mathbf{y} = \sum_{j \leq p} z_j \mathbf{q}_j + d \mathbf{q}_{p+1}$. By Proposition 6.3.2 the projection of $\mathbf{y}$ onto the span of the first k columns is $\sum_{j \leq k} z_j \mathbf{q}_j$, and the residual sum of squares is $\|\mathbf{y}\|^2 - \sum_{j \leq k} z_j^2 = d^2 + \sum_{j>k} z_j^2$. With $k = p$ this is d^2 , which completes (a). Taking differences gives the rest of (b).

- (c) The determinant of a triangular matrix is the product of its diagonal. The inverse formula follows from $\mathbf{X}^\top \mathbf{X} = \mathbf{R}^\top \mathbf{R}$. Finally $h_{ii} = \mathbf{x}_{(i)}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} = \mathbf{x}_{(i)}^\top \mathbf{R}^{-1} \mathbf{R}^{-\top} \mathbf{x}_{(i)} = \|\mathbf{R}^{-\top} \mathbf{x}_{(i)}\|^2$.

■

Part (b) says that the squares $z_1^2, \dots, z_p^2$ are the *sequential sums of squares* of the regressors in the order of the columns (Definition 9.3.1). A single factorization gives the whole nested

sequence of fits. This is the content of [Theorem 6.5.1](#), now in coordinates. Part (c) costs about np^2 flops for all n leverages, each a triangular solve of p^2 flops.

Example. (Murder rates by the augmented factorization)

For the state data of [Example \(Poverty and murder rates\)](#) ($n = 50$, with an intercept and the percentages in poverty, in single-parent households and urban, so $p = 4$), the augmented factorization ([Proposition 10.2.2](#)) gives the coefficients -9.0946 , 0.2538 , 0.3980 and 0.0046 , and the corner entry $d = 10.0976$, so $\text{SSE} = d^2 = 101.96$. The condition number of $\mathbf{X}$ is only 509, and the solution agrees with a QR solution to a relative difference of 2×10^{-14} . The largest leverage is 0.260, and $\log_{10} \det(\mathbf{X}^\top \mathbf{X}) = 11.14$. The script also checks the standard errors $s \|e_j^\top \mathbf{R}^{-1}\|$ against statsmodels.

```
data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape
```

```
Z = np.column_stack([X, y])          # augmented matrix [X, y]
S = Z.T @ Z                          # (p+1) x (p+1) cross-product matrix
T = cholesky_upper(S)                # T = [[R, z], [0, d]]
R, z, d = T[:p, :p], T[:p, p], T[p, p]

beta = solve_upper(R, z)              # R b = z
sse = d ** 2                         # the last diagonal entry is sqrt(SSE)
K = solve_upper(R, np.eye(p))        # R^{-1}
XtX_inv = K @ K.T                   # (X'X)^{-1} = R^{-1} R^{-T}
h = np.array([np.sum(solve_lower(R.T, x) ** 2) for x in X]) # leverages
print("coefficients:", np.round(beta, 4))
print(f"SSE = {sse:.4f}, d = {d:.4f}")
```

10.2.3 When the normal equations are good enough

[Proposition 10.1.3](#) and [Eq. 10.1.2](#) predict a relative error of order $\kappa^2 u$ in $\hat{\beta}$ from the normal equations, where, as [Section 10.5](#) shows, κ is the condition number of $\mathbf{X}$ after its columns are scaled to equal length. With $\kappa \leq 10^4$ about eight digits survive, more than any coefficient is known statistically. The normal equations are therefore sound for well-conditioned designs, when only cross-products are available, and inside iterations such as the reweighted least squares of Chapter 34. Nearly collinear observational data are where the κ^2 loss bites.

10.2.4 The sweep operator

Before QR became standard, regression programs worked on the augmented cross-product matrix with an operation that adds one variable at a time to the fit. It is still used for fitting many subsets of variables.

Definition 10.2.1 — Sweep

Let $\mathbf{A}$ be symmetric with $a_{kk} \neq 0$. **Sweeping** $\mathbf{A}$ on pivot k produces the symmetric matrix $\mathbf{B}$ with

$$b_{kk} = -\frac{1}{a_{kk}}, \quad b_{ik} = b_{ki} = \frac{a_{ik}}{a_{kk}}, \quad b_{ij} = a_{ij} - \frac{a_{ik}a_{kj}}{a_{kk}} \quad (i, j \neq k).$$

Proposition 10.2.3 — What sweeping computes

Let $\mathbf{S} = \mathbf{Z}^\top \mathbf{Z}$ be the cross-product matrix of $\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_m]$. Let K be a set of column indices such that $\mathbf{Z}_K$ has full column rank, and let L be the remaining indices. Sweeping $\mathbf{S}$ on the pivots in K , one at a time and in any order, meets only positive pivots and produces the matrix $\mathbf{W}$ with blocks

$$\mathbf{W}_{KK} = -(\mathbf{Z}_K^\top \mathbf{Z}_K)^{-1}, \quad \mathbf{W}_{KL} = \mathbf{W}_{LK}^\top = (\mathbf{Z}_K^\top \mathbf{Z}_K)^{-1} \mathbf{Z}_K^\top \mathbf{Z}_L, \quad \mathbf{W}_{LL} = \mathbf{Z}_L^\top (\mathbf{I} - \mathbf{M}_K) \mathbf{Z}_L,$$

where $\mathbf{M}_K$ is the projection onto $\mathcal{C}(\mathbf{Z}_K)$. Column $j \in L$ of $\mathbf{W}_{KL}$ holds the least squares coefficients of $\mathbf{z}_j$ regressed on $\mathbf{Z}_K$, and $\mathbf{W}_{LL}$ holds the residual cross-products. With $\mathbf{Z} = [\mathbf{X}, \mathbf{y}]$ and K the columns of $\mathbf{X}$, the last column of $\mathbf{W}$ contains $\hat{\beta}$ and, in its corner, the residual sum of squares.

Proof. We induct on the size of K ; the case $K = \emptyset$ is $\mathbf{W} = \mathbf{S}$. Add a pivot $m \in L$ with $\mathbf{Z}_{K \cup \{m\}}$ of full column rank, and write $\mathbf{G} = (\mathbf{Z}_K^\top \mathbf{Z}_K)^{-1}$, $\mathbf{B}_j = \mathbf{G} \mathbf{Z}_K^\top \mathbf{z}_j$ and $\mathbf{w} = (\mathbf{I} - \mathbf{M}_K) \mathbf{z}_m$. The pivot is $\mathbf{z}_m^\top (\mathbf{I} - \mathbf{M}_K) \mathbf{z}_m = \|\mathbf{w}\|^2 > 0$. For $i, j \in L \setminus \{m\}$ the new entries are $\mathbf{z}_i^\top (\mathbf{I} - \mathbf{M}_K - \mathbf{w} \mathbf{w}^\top / \|\mathbf{w}\|^2) \mathbf{z}_j = \mathbf{z}_i^\top (\mathbf{I} - \mathbf{M}_{K \cup \{m\}}) \mathbf{z}_j$ by Lemma 6.6.1. The new entries $\mathbf{w}^\top \mathbf{z}_j / \|\mathbf{w}\|^2$ in row m , and $\mathbf{B}_j - \mathbf{B}_m \mathbf{w}^\top \mathbf{z}_j / \|\mathbf{w}\|^2$ in the rows of K , are the coefficients of $\mathbf{z}_j$ on $[\mathbf{Z}_K, \mathbf{z}_m]$ by Theorem 6.6.2. The partitioned inverse (Theorem 1.6.4), with Schur complement $\|\mathbf{w}\|^2$, shows that $(\mathbf{Z}_{K \cup \{m\}}^\top \mathbf{Z}_{K \cup \{m\}})^{-1}$ has blocks $\mathbf{G} + \mathbf{B}_m \mathbf{B}_m^\top / \|\mathbf{w}\|^2$, $-\mathbf{B}_m / \|\mathbf{w}\|^2$ and $1 / \|\mathbf{w}\|^2$, whose negatives are what the sweep produces: $-\mathbf{G} - \mathbf{B}_m \mathbf{B}_m^\top / \|\mathbf{w}\|^2$, $\mathbf{B}_m / \|\mathbf{w}\|^2$ and $-1 / \|\mathbf{w}\|^2$. The result depends only on the set K , so the order of the sweeps does not matter. ■

Sweeping a pivot costs about $(p+1)^2$ flops, so a new variable enters a fitted model for $O(p^2)$ work. The price is the same as for the normal equations: the sweep works with cross-products and inherits their κ^2 sensitivity. For the state data, sweeping all four columns reproduces the coefficients and SSE of Example (Murder rates by the augmented factorization). Sweeping only the intercept and the single-parent column puts the SSE of that two-column model, 121.92, in the corner.

```
def sweep(A, k):
    """Sweep the symmetric matrix A on pivot k."""
    A = np.array(A, dtype=float)
    a = A[k, k]
    B = A - np.outer(A[:, k], A[k, :]) / a
    B[k, :] = A[k, :] / a
    B[:, k] = A[:, k] / a
    B[k, k] = -1 / a
    return B
```

```

W = S.copy()
for k in range(p):
    W = sweep(W, k)
print("coefficients from the sweep:", np.round(W[:p, p], 4))
print(f"SSE from the sweep: {W[p, p]:.4f}")

```

sweep the p columns of X, one at a time

10.2.5 Forming cross-products: the centring problem

With an intercept in the model, the cross-product matrix is usually computed in centred form, $\sum_i (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)$, which by [Theorem 6.7.1](#) changes the parameterization but not the fit. How it is computed matters a great deal. The “textbook” formula $\sum_i x_i^2 - n\bar{x}^2$ subtracts two large, nearly equal numbers whenever the mean is large compared with the spread. The loss is governed by the ratio

$$\kappa_{\mathbf{x}} = \frac{\|\mathbf{x}\|}{\|\mathbf{x} - \bar{x}\mathbf{1}\|} = \left(1 + \frac{n\bar{x}^2}{\sum_i (x_i - \bar{x})^2}\right)^{1/2},$$

which is the condition number of the column, and it is large when the coefficient of variation is small. Three algorithms are in common use:

- *textbook*: accumulate $\sum x_i$ and $\sum x_i^2$ in one pass, then subtract;
- *two-pass*: compute $\bar{x}$ in a first pass, then sum $(x_i - \bar{x})^2$ in a second;
- *updating*: one pass, maintaining the running mean and the running centred sum by the recursions below.

Proposition 10.2.4 — Updating means and centred cross-products

For vectors $\mathbf{x}_1, \mathbf{x}_2, \dots$ in $\mathbb{R}^q$ let $\mathbf{m}_k$ be the mean of the first k and $\mathbf{C}_k = \sum_{i \leq k} (\mathbf{x}_i - \mathbf{m}_k)(\mathbf{x}_i - \mathbf{m}_k)^\top$. With $\mathbf{d}_k = \mathbf{x}_k - \mathbf{m}_{k-1}$,

$$\mathbf{m}_k = \mathbf{m}_{k-1} + \frac{\mathbf{d}_k}{k}, \quad \mathbf{C}_k = \mathbf{C}_{k-1} + \frac{k-1}{k} \mathbf{d}_k \mathbf{d}_k^\top.$$

Proof. The first identity is $k\mathbf{m}_k = (k-1)\mathbf{m}_{k-1} + \mathbf{x}_k$. For the second, write $\mathbf{C}_k = \sum_{i \leq k} \mathbf{x}_i \mathbf{x}_i^\top - k\mathbf{m}_k \mathbf{m}_k^\top$ and put $\mathbf{m} = \mathbf{m}_{k-1}$, $\mathbf{d} = \mathbf{d}_k$. Then

$$\mathbf{C}_k - \mathbf{C}_{k-1} = \mathbf{x}_k \mathbf{x}_k^\top - k\mathbf{m}_k \mathbf{m}_k^\top + (k-1)\mathbf{m} \mathbf{m}^\top.$$

Substituting $\mathbf{x}_k = \mathbf{m} + \mathbf{d}$ and $k\mathbf{m}_k \mathbf{m}_k^\top = k\mathbf{m} \mathbf{m}^\top + \mathbf{m} \mathbf{d}^\top + \mathbf{d} \mathbf{m}^\top + \mathbf{d} \mathbf{d}^\top / k$, all the terms in $\mathbf{m}$ cancel and $\mathbf{d} \mathbf{d}^\top (1 - 1/k)$ remains. ■

The update adds a *centred* rank-one term, so it never subtracts large quantities. It is the cross-product analogue of the row updates of [Section 10.6](#). Chan, Golub and LeVeque (1983) give error bounds of roughly $n\kappa_{\mathbf{x}}^2 u$ for the textbook formula, $n\kappa_{\mathbf{x}} u$ for updating and $nu + n^2 \kappa_{\mathbf{x}}^2 u^2$ for the two-pass algorithm.

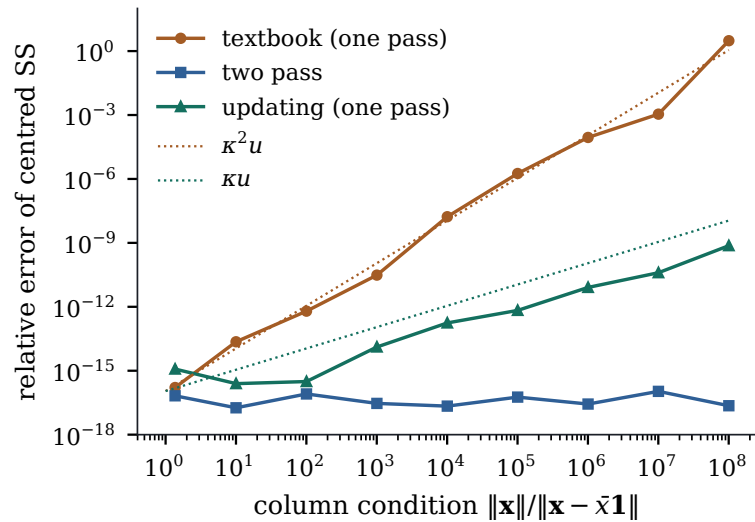


Figure 10.2.1: Relative error of the centred sum of squares of a column of 1000 values with standard deviation one and mean up to 10^8 , against the column condition number κ_x . The one-pass textbook formula loses accuracy like $\kappa^2 u$, the one-pass updating formula like κu , and the two-pass algorithm not at all.

Example. (Centring a column with a large mean)

A sample of 1000 standard normal values is shifted by $c = 1, 10, \dots, 10^8$, and the centred sum of squares is computed by the three algorithms and compared with its exact value (found in rational arithmetic). At $c = 10^6$, where $\kappa_x = 1.0 \times 10^6$, the relative errors are 8.9×10^{-5} for the textbook formula, 8.1×10^{-12} for updating and 2.7×10^{-17} for two passes. At $c = 10^8$ the textbook formula has relative error 3.1: not a single correct digit, and possibly a negative “sum of squares”. Updating gives 7.4×10^{-10} and two passes 2.3×10^{-17} . Figure 10.2.1 shows the three growth rates $\kappa^2 u$, κu and u .

```
def ss_textbook(x):
    return np.sum(x * x) - len(x) * np.mean(x) ** 2      # one pass, then subtract

def ss_two_pass(x):
    m = np.mean(x)                                       # first pass: the mean
    return np.sum((x - m) ** 2)                          # second pass: deviations

def ss_updating(x):
    m, s = 0.0, 0.0
    for k, xk in enumerate(x, start=1):                # one pass, no
        # cancellation
        d = xk - m
        m = m + d / k
        s = s + (k - 1) / k * d * d
    return s
```

Calendar years, population totals and price indices have large means and small spread. A program that forms cross-products should centre them with two passes (or update), solve for the slopes, and recover the intercept as $\bar{y} - \bar{x}^\top \hat{\beta}_{\text{slopes}}$.

10.2.6 Exercises

10.2.6.1 A. Check your understanding

Exercise A1.

Compute the Cholesky factor of $\mathbf{A} = \begin{pmatrix} 4 & 2 & -2 \\ 2 & 10 & 2 \\ -2 & 2 & 6 \end{pmatrix}$ by Eq. 10.2.1, and use it to solve $\mathbf{A}\mathbf{b} = (2, 14, 6)^\top$ and to find $\det \mathbf{A}$.

Solution. Row 1: $r_{11} = 2$, $r_{12} = 1$, $r_{13} = -1$. Row 2: $r_{22} = \sqrt{10-1} = 3$, $r_{23} = (2 - (1)(-1))/3 = 1$. Row 3: $r_{33} = \sqrt{6-1-1} = 2$. So $\mathbf{R} = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ and $\det \mathbf{A} = (2 \cdot 3 \cdot 2)^2 = 144$. Forward substitution in $\mathbf{R}^\top \mathbf{z} = (2, 14, 6)^\top$ gives $z_1 = 1$, $z_2 = (14-1)/3 = 13/3$, $z_3 = (6 + 1 - 13/3)/2 = 4/3$. Back substitution in $\mathbf{R}\mathbf{b} = \mathbf{z}$ gives $b_3 = 2/3$, $b_2 = (13/3 - 2/3)/3 = 11/9$ and $b_1 = (1 - 11/9 + 2/3)/2 = 2/9$. Check: $\mathbf{A}\mathbf{b} = (8/9 + 22/9 - 12/9, 4/9 + 110/9 + 12/9, -4/9 + 22/9 + 36/9) = (2, 14, 6)$.

Exercise A2.

For simple regression, $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$, find the Cholesky factor (Theorem 10.2.1) of $\mathbf{X}^\top \mathbf{X}$ and of the augmented matrix, and show that it reproduces the slope S_{xy}/S_{xx} of Theorem 5.2.1. Where does the centring happen?

Solution. $\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum x_i^2 \end{pmatrix}$ gives $r_{11} = \sqrt{n}$, $r_{12} = \sqrt{n}\bar{x}$ and $r_{22} = (\sum x_i^2 - n\bar{x}^2)^{1/2} = S_{xx}^{1/2}$. The augmented column gives $z_1 = n\bar{y}/\sqrt{n} = \sqrt{n}\bar{y}$ and $z_2 = (\sum x_i y_i - n\bar{x}\bar{y})/S_{xx}^{1/2} = S_{xy}/S_{xx}^{1/2}$. Back substitution gives the slope $b_2 = z_2/r_{22} = S_{xy}/S_{xx}$ and the intercept $b_1 = (z_1 - r_{12}b_2)/r_{11} = \bar{y} - \bar{x}b_2$. The centring happens in the subtraction $a_{22} - r_{12}^2$, which is the textbook formula of Example (Centring a column with a large mean). So the Cholesky factorization of an uncentred cross-product matrix inherits its cancellation.

10.2.6.2 B. Practice

Exercise B1.

Show that the standard error of $\hat{\beta}_j$ is $s \|e_j^\top \mathbf{R}^{-1}\|$, where $s^2 = \text{SSE}/(n-p)$ (Theorem 5.6.1). Show that all p standard errors cost $p^3/3 + O(p^2)$ flops, and all n leverages $np^2 + O(np)$ flops. Why is computing $\mathbf{M}$ itself a bad idea?

Solution. By Theorem 5.5.1 the estimated variance of $\hat{\beta}_j$ is $s^2[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}$, and $[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj} = e_j^\top \mathbf{R}^{-1} \mathbf{R}^{-\top} e_j = \|e_j^\top \mathbf{R}^{-1}\|^2$. The inverse of a triangular matrix is triangular, and column j of $\mathbf{R}^{-1}$ is found by back substitution on its leading $j \times j$ block, at about j^2 flops. The total is $\sum_j j^2 \approx p^3/3$. Each leverage is a triangular solve with $\mathbf{R}^\top$ (p^2 flops) and a squared norm. The projection $\mathbf{M}$ has n^2 entries, which for $n = 10^6$ is too many to store, and only its diagonal is ever needed.

Exercise B2.

Two groups of observations have sizes n_a, n_b , means $\mathbf{m}_a, \mathbf{m}_b$ and centred cross-product matrices $\mathbf{C}_a, \mathbf{C}_b$. Show that the pooled data have centred cross-product matrix

$$\mathbf{C} = \mathbf{C}_a + \mathbf{C}_b + \frac{n_a n_b}{n_a + n_b} (\mathbf{m}_a - \mathbf{m}_b)(\mathbf{m}_a - \mathbf{m}_b)^\top.$$

Show that [Proposition 10.2.4](#) is the special case $n_b = 1$, and explain how the formula allows cross-products to be computed in parallel on separate blocks of data.

10.2.6.3 C. Going deeper**Exercise C1.**

Let $\mathbf{A}$ be nonnegative definite of rank r . Show that the Cholesky recursion with *symmetric pivoting*, which at step j moves the largest remaining diagonal entry of the current Schur complement to position j , produces a permutation $\mathbf{\Pi}$ and $\mathbf{\Pi}^\top \mathbf{A} \mathbf{\Pi} = \mathbf{R}^\top \mathbf{R}$ with $\mathbf{R} = \begin{pmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$ and $\mathbf{R}_{11}$ an $r \times r$ nonsingular upper triangle. (Use [Proposition 1.7.5\(a\)](#).) How is this related to the pivoted QR of [Section 10.7](#)?

10.3 Householder and Givens QR

Orthogonal matrices preserve lengths ([Proposition 1.5.3\(a\)](#)). For any orthogonal $\mathbf{Q}$,

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\| = \|\mathbf{Q}^\top \mathbf{y} - \mathbf{Q}^\top \mathbf{X}\mathbf{b}\| \quad \text{for every } \mathbf{b}.$$

So a least squares problem can be replaced by an equivalent one in which $\mathbf{Q}^\top \mathbf{X}$ is upper triangular, without ever forming $\mathbf{X}^\top \mathbf{X}$. Such a $\mathbf{Q}$ is built from reflections, which zero a whole column below the diagonal at once, or from rotations, which zero one entry at a time.

10.3.1 Householder reflections

A reflection $\mathbf{H} = \mathbf{I} - 2\mathbf{v}\mathbf{v}^\top/(\mathbf{v}^\top \mathbf{v})$ is symmetric and orthogonal, and it maps $\mathbf{v}$ to $-\mathbf{v}$ and fixes every vector orthogonal to $\mathbf{v}$ ([Exercise \(B1\)](#)). The whole numerical content lies in choosing $\mathbf{v}$ so that a given vector is mapped onto a coordinate axis, and in choosing its sign so that no cancellation occurs.

Lemma 10.3.1 — Householder vector

Let $\mathbf{x} \in \mathbb{R}^m$, $\mathbf{x} \neq \mathbf{0}$, let $\alpha = -\text{sign}(x_1)\|\mathbf{x}\|$ (with $\text{sign}(0) = 1$), and let $\mathbf{v} = \mathbf{x} - \alpha\mathbf{e}_1$. Then $\mathbf{v} \neq \mathbf{0}$, and $\mathbf{H} = \mathbf{I} - 2\mathbf{v}\mathbf{v}^\top/(\mathbf{v}^\top \mathbf{v})$ satisfies $\mathbf{H}\mathbf{x} = \alpha\mathbf{e}_1$. The first entry $v_1 = x_1 - \alpha$ is a sum of two numbers of the same sign, so $|v_1| = |x_1| + \|\mathbf{x}\|$. Applying $\mathbf{H}$ to a vector $\mathbf{w}$ as $\mathbf{w} - 2\mathbf{v}(\mathbf{v}^\top \mathbf{w})/(\mathbf{v}^\top \mathbf{v})$ costs about $4m$ flops, and $\mathbf{H}$ is never formed.

Proof. α and x_1 have opposite signs (or $x_1 = 0$), so $v_1 = x_1 - \alpha$ adds two numbers of the same sign and $|v_1| = |x_1| + \|\mathbf{x}\| > 0$. Next, $\mathbf{v}^\top \mathbf{x} = \|\mathbf{x}\|^2 - \alpha x_1$ and $\mathbf{v}^\top \mathbf{v} = \|\mathbf{x}\|^2 - 2\alpha x_1 + \alpha^2 = 2(\|\mathbf{x}\|^2 - \alpha x_1)$, using $\alpha^2 = \|\mathbf{x}\|^2$. Hence $2\mathbf{v}^\top \mathbf{x} / \mathbf{v}^\top \mathbf{v} = 1$ and $\mathbf{H}\mathbf{x} = \mathbf{x} - \mathbf{v} = \alpha \mathbf{e}_1$. ■

The opposite sign choice, $\alpha = +\text{sign}(x_1)\|\mathbf{x}\|$, also gives a reflection onto the axis. But then $v_1 = x_1 - \alpha$ subtracts two nearly equal numbers whenever $\mathbf{x}$ is nearly parallel to $\mathbf{e}_1$. This is the cancellation of [Section 10.1](#), and it can make $\mathbf{v}$ inaccurate in every digit.

Theorem 10.3.2 — Householder QR

Let $\mathbf{X}$ be $n \times p$ with $n \geq p$.

- (a) There are reflections $\mathbf{H}_1, \dots, \mathbf{H}_p$, with $\mathbf{H}_j$ acting only on coordinates $j, \dots, n$ (or equal to $\mathbf{I}$), such that

$$\mathbf{H}_p \cdots \mathbf{H}_1 \mathbf{X} = \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$$

with $\mathbf{R}$ upper triangular of size $p \times p$. Thus $\mathbf{X} = \mathbf{Q} \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$ with $\mathbf{Q} = \mathbf{H}_1 \cdots \mathbf{H}_p$ orthogonal, and $\mathbf{X} = \mathbf{Q}_1 \mathbf{R}$ where $\mathbf{Q}_1$ holds the first p columns of $\mathbf{Q}$. If $\text{rank } \mathbf{X} = p$, then $\mathbf{R}$ is nonsingular, and changing the signs of the rows of $\mathbf{R}$ with negative diagonal entries gives the Cholesky factor of $\mathbf{X}^\top \mathbf{X}$.

- (b) The reduction takes $2np^2 - 2p^3/3$ flops to leading order. Applying $\mathbf{Q}^\top$ (or $\mathbf{Q}$) to a vector takes $4np - 2p^2$, and forming $\mathbf{Q}_1$ explicitly takes another $2np^2 - 2p^3/3$.
- (c) (Backward stability.) In floating-point arithmetic, the computed $\hat{\mathbf{R}}$ is the exact triangular factor of $\mathbf{X} + \Delta \mathbf{X}$, where each column satisfies $\|\Delta \mathbf{x}_j\| \leq c_{np} u \|\mathbf{x}_j\|$. The least squares solution $\hat{\mathbf{b}}$ computed from $\hat{\mathbf{R}}$ and the computed $\mathbf{Q}^\top \mathbf{y}$ is the exact least squares solution for data $(\mathbf{X} + \Delta \mathbf{X}, \mathbf{y} + \Delta \mathbf{y})$ with perturbations of the same kind, $\|\Delta \mathbf{y}\| \leq c_{np} u \|\mathbf{y}\|$. Here c_{np} is a modest multiple of np .

Proof.

- (a) Induction on the column. Suppose $\mathbf{H}_{j-1} \cdots \mathbf{H}_1 \mathbf{X}$ has zeros below the diagonal in its first $j-1$ columns. Let $\mathbf{x}$ be the part of column j in rows $j, \dots, n$. If $\mathbf{x} = \mathbf{0}$, take $\mathbf{H}_j = \mathbf{I}$. Otherwise take the reflection of [Lemma 10.3.1](#) for $\mathbf{x}$, acting on rows $j, \dots, n$. It zeroes column j below the diagonal. It leaves rows $1, \dots, j-1$ alone. It maps the first $j-1$ columns, which are zero in rows $j, \dots, n$, to themselves. After p steps the matrix is $\begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$, and $\mathbf{Q}$ is a product of orthogonal matrices. Orthogonal factors do not change rank, so $\text{rank } \mathbf{R} = \text{rank } \mathbf{X}$. If $\mathbf{D}$ is the diagonal matrix of signs of the diagonal of $\mathbf{R}$, then $\mathbf{X} = (\mathbf{Q}_1 \mathbf{D})(\mathbf{D} \mathbf{R})$ is a QR factorization with positive diagonal, and $\mathbf{D} \mathbf{R}$ is the Cholesky factor by [Theorem 10.2.1\(d\)](#).
- (b) Step j builds $\mathbf{v}$ (about $2(n-j)$ flops) and applies the reflection to the $p-j$ remaining columns of length $n-j+1$, at about $4(n-j)(p-j)$ flops. Summing,

$$\sum_{j=1}^p 4(n-j)(p-j) \approx 4 \int_0^p (n-t)(p-t) dt = 2np^2 - \frac{2}{3}p^3.$$

Applying the p reflections to a vector costs $\sum_j 4(n-j+1) \approx 4np - 2p^2$. Forming

Q_1 means applying the reflections in reverse order to the first p columns of I_n . Reflection j then acts on columns $j, \dots, p$ only, which gives the same integral again.

- (c) This is a theorem of rounding-error analysis, not of linear algebra, and we state it without proof. Its essence is that a reflection applied in floating point is an exact reflection applied to a slightly perturbed matrix, that orthogonal matrices do not amplify perturbations, and that the p steps add their perturbations. See Higham (2002, chapters 19 and 20) and Björck (1996, chapter 2).

■

Part (c) is why QR is the default. Its backward error is small relative to each column separately, which [Section 10.5](#) uses to explain why column scaling does not change the accuracy of QR. Software never forms Q ; it stores the vectors v_j in the zeroed part of X .

```
import numpy as np
import statsmodels.api as sm

def house(x):
    """Unit vector v with (I - 2 v v') x = alpha e_1, alpha = -sign(x_1) ||x||."""
    alpha = -np.copysign(np.linalg.norm(x), x[0])
    v = x.astype(float).copy()
    v[0] -= alpha          # x_1 - alpha adds two numbers of the same
                           sign
    return v / np.linalg.norm(v), alpha

def householder_qr(X):
    """Return the reflection vectors and the p x p triangle R of X = Q [R; 0]."""
    A = np.array(X, dtype=float)
    n, p = A.shape
    vs = []
    for j in range(p):
        v, alpha = house(A[j:, j])
        A[j:, j:] -= 2.0 * np.outer(v, v @ A[j:, j:])    # apply I - 2vv' to the
                                                         trailing block
        vs.append(v)
    return vs, np.triu(A[:p, :])

def apply_Qt(vs, y):
    """Q'y, applying the reflections H_1, ..., H_p in turn (Q itself is never formed)."""
    z = np.array(y, dtype=float)
    for j, v in enumerate(vs):
        z[j:] -= 2.0 * v * (v @ z[j:])
    return z

def apply_Q(vs, z):
    """Q z, applying the reflections in reverse order."""
    w = np.array(z, dtype=float)
    for j in range(len(vs) - 1, -1, -1):
        v = vs[j]
        w[j:] -= 2.0 * v * (v @ w[j:])
    return w
```

10.3.2 Regression quantities from the factorization

Section 6.10 derived the regression quantities from the thin factorization $\mathbf{X} = \mathbf{Q}_1 \mathbf{R}$ (Eq. 6.10.1). The full factorization, with $\mathbf{Q}$ square, adds one thing: the part of $\mathbf{Q}^\top \mathbf{y}$ that the model cannot reach, which is the residual in rotated coordinates.

Proposition 10.3.3 — Least squares from QR

Let $\mathbf{X} = \mathbf{Q} \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$ with $\mathbf{Q}$ orthogonal and $\text{rank } \mathbf{X} = p$, and partition $\mathbf{c} = \mathbf{Q}^\top \mathbf{y}$ as $(\mathbf{c}_1^\top, \mathbf{c}_2^\top)^\top$ with $\mathbf{c}_1 \in \mathbb{R}^p$. Then:

- (a) $\hat{\boldsymbol{\beta}}$ is the solution of the triangular system $\mathbf{R}\mathbf{b} = \mathbf{c}_1$, and $\text{SSE} = \|\mathbf{c}_2\|^2$;
- (b) $\hat{\mathbf{Y}} = \mathbf{Q} \begin{pmatrix} \mathbf{c}_1 \\ \mathbf{0} \end{pmatrix}$ and $\hat{\boldsymbol{\varepsilon}} = \mathbf{Q} \begin{pmatrix} \mathbf{0} \\ \mathbf{c}_2 \end{pmatrix}$;
- (c) $(\mathbf{X}^\top \mathbf{X})^{-1} = \mathbf{R}^{-1} \mathbf{R}^{-\top}$ and $h_{ii} = \|\mathbf{e}_i^\top \mathbf{Q}_1\|^2$;
- (d) c_k^2 is the drop in residual sum of squares when column k is added to columns $1, \dots, k-1$, the sequential sum of squares of Definition 9.3.1.

Proof.

- (a) With $\mathbf{Q}_1$ the first p columns of $\mathbf{Q}$, $\mathbf{X} = \mathbf{Q}_1 \mathbf{R}$ is a thin QR factorization and $\mathbf{c}_1 = \mathbf{Q}_1^\top \mathbf{y}$, so $\mathbf{R}\hat{\boldsymbol{\beta}} = \mathbf{c}_1$ and $\text{SSE} = \|\mathbf{y}\|^2 - \|\mathbf{c}_1\|^2$ are Eq. 6.10.1. What the full $\mathbf{Q}$ adds is $\|\mathbf{c}\| = \|\mathbf{y}\|$, hence $\text{SSE} = \|\mathbf{c}_2\|^2$.
- (b) $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{Q} \begin{pmatrix} \mathbf{R}\hat{\boldsymbol{\beta}} \\ \mathbf{0} \end{pmatrix} = \mathbf{Q} \begin{pmatrix} \mathbf{c}_1 \\ \mathbf{0} \end{pmatrix}$, and $\mathbf{y} = \mathbf{Q}\mathbf{c}$.
- (c) $\mathbf{X}^\top \mathbf{X} = \mathbf{R}^\top \mathbf{Q}_1^\top \mathbf{Q}_1 \mathbf{R} = \mathbf{R}^\top \mathbf{R}$, and the leverage formula is Eq. 6.10.1.
- (d) The first k columns of $\mathbf{X}$ equal the first k columns of $\mathbf{Q}$ times the leading $k \times k$ block of $\mathbf{R}$, which is nonsingular. So they span the same subspace as $\mathbf{q}_1, \dots, \mathbf{q}_k$, and the squared length of the projection of $\mathbf{y}$ onto it is $\sum_{j \leq k} (\mathbf{q}_j^\top \mathbf{y})^2 = \sum_{j \leq k} c_j^2$ (Proposition 6.3.2).

■

Part (d) is the QR form of Proposition 10.2.2(b): by uniqueness of the triangle, $\mathbf{c}_1$ equals the vector $\mathbf{z}$ of that proposition up to signs. The vector $\mathbf{c}_2$ replaces the n correlated residuals by $n - p$ uncorrelated ones with the same sum of squares (Exercise (B1)).

Example. (Sequential sums of squares from one factorization)

For the state data of Example (Murder rates by the augmented factorization), Householder QR (Theorem 10.3.2) gives $\mathbf{R}$ with diagonal $-7.071, 21.511, 21.609$ and 113.413 . The first entry is $-\sqrt{50}$: the sign convention of Lemma 10.3.1 makes it negative. The squares of the first four entries of $\mathbf{Q}^\top \mathbf{y}$ are the sequential sums of squares: 1018.81 for the intercept (which is $n\bar{y}^2$), 102.49 for poverty, 80.04 for single-parent households given poverty, and 0.27 for urbanization given the other two. The remaining 46 entries have squared length $\text{SSE} = 101.96$. The urbanization variable adds almost nothing once the other two are in the model, which is what its small coefficient in Example (Poverty and murder rates) suggested. The script checks every number against nested fits, and checks the leverages against the squared row lengths of $\mathbf{Q}_1$.

```

data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape

vs, R = householder_qr(X)
c = apply_Qt(vs, y)                                # c = Q'y = (c1, c2)
c1, c2 = c[:p], c[p:]
beta = np.linalg.solve(R, c1)                       # triangular system R b = c1
sse = c2 @ c2                                       # SSE = ||c2||^2
resid = apply_Q(vs, np.concatenate([np.zeros(p), c2]))
seq_ss = c1 ** 2                                    # sequential sums of squares, in column
                                                order
print("R diagonal:", np.round(np.diag(R), 4))
print("coefficients:", np.round(beta, 4), f"   SSE = {sse:.4f}")
print("sequential SS (intercept, poverty, single, urban):", np.round(seq_ss, 3))

```

10.3.3 Givens rotations

A reflection zeroes a whole column at once. Sometimes it is better to zero one entry at a time, touching only two rows. This is the case when $\mathbf{X}$ is sparse, when rows arrive one by one, or when an existing factorization must be modified.

- Theorem 10.3.4 — Givens rotations** (a) For $(a, b) \neq (0, 0)$ let $r = (a^2 + b^2)^{1/2}$, $c = a/r$ and $s = b/r$. The rotation $\mathbf{G} = \begin{pmatrix} c & s \\ -s & c \end{pmatrix}$ is orthogonal and $\mathbf{G}(a, b)^\top = (r, 0)^\top$. Embedded in rows i and k of the identity, it changes only rows i and k of a matrix it multiplies, at 6 flops per column.
- (b) Zeroing the subdiagonal of $\mathbf{X}$ column by column, each entry against the row above it, produces $\mathbf{GX} = \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$ with $\mathbf{G}$ orthogonal. It takes $np - p(p+1)/2$ rotations and $3np^2 - p^3$ flops to leading order.
- (c) (Row-wise QR.) Let $\mathbf{R}$ be upper triangular and $\mathbf{x} \in \mathbb{R}^p$. There are p rotations, the k th acting on row k of $\mathbf{R}$ and on the new row, that reduce $\begin{pmatrix} \mathbf{R} \\ \mathbf{x}^\top \end{pmatrix}$ to $\begin{pmatrix} \mathbf{R}_+ \\ \mathbf{0}^\top \end{pmatrix}$ with $\mathbf{R}_+$ upper triangular and $\mathbf{R}_+^\top \mathbf{R}_+ = \mathbf{R}^\top \mathbf{R} + \mathbf{x}\mathbf{x}^\top$. The cost is $3p^2$ flops. If $\mathbf{R}$ and $\mathbf{x}$ are augmented by the response, as in Proposition 10.2.2, the new row ends as $(0, \dots, 0, t)$, and the residual sum of squares increases by t^2 .
- (d) A rotation of rows i and k creates a nonzero in column j only if row i or row k already had one there.

Proof.

- (a) $c^2 + s^2 = 1$ gives $\mathbf{G}^\top \mathbf{G} = \mathbf{I}$. Also $ca + sb = (a^2 + b^2)/r = r$ and $-sa + cb = 0$. Each column of the product needs four multiplications and two additions.
- (b) When entry (i, j) is zeroed against row $i - 1$, working upwards from row n , both rows are already zero in columns $1, \dots, j - 1$, so no earlier zero is destroyed. Column j needs $n - j$ rotations, each acting on $p - j + 1$ columns, so the count is $\sum_j 6(n - j)(p - j + 1) \approx 6 \int_0^p (n - t)(p - t) dt = 3np^2 - p^3$.
- (c) Before rotation k , the new row is zero in columns $1, \dots, k - 1$, and so is row k of $\mathbf{R}$. The rotation chosen by (a) to zero the k th entry of the new row keeps both

rows zero in those columns. After p rotations the new row is zero, and row k of $\mathbf{R}$ has been replaced by a row that is zero before column k . With $\mathbf{G}$ the product of the rotations,

$$\mathbf{R}_+^\top \mathbf{R}_+ = \begin{pmatrix} \mathbf{R} \\ \mathbf{x}^\top \end{pmatrix}^\top \mathbf{G}^\top \mathbf{G} \begin{pmatrix} \mathbf{R} \\ \mathbf{x}^\top \end{pmatrix} = \mathbf{R}^\top \mathbf{R} + \mathbf{x} \mathbf{x}^\top.$$

Rotation k acts on $p - k + 1$ columns, which gives $\sum_k 6(p - k + 1) \approx 3p^2$. In the augmented case, apply p rotations to the augmented $(p + 1) \times (p + 1)$ triangle $\mathbf{T}$ and the row $(\mathbf{x}^\top, y)$. The new row becomes $(\mathbf{0}^\top, t)$, and $\mathbf{T}_+^\top \mathbf{T}_+ = \mathbf{T}^\top \mathbf{T} + (\mathbf{x}^\top, y)^\top (\mathbf{x}^\top, y) - (\mathbf{0}^\top, t)^\top (\mathbf{0}^\top, t)$. The corner entry of $\mathbf{T}_+$ is still d , so the augmented triangle with the new row included is obtained by one more rotation, of d against t . This gives a corner $(d^2 + t^2)^{1/2}$, and the claim follows from [Proposition 10.2.2\(a\)](#).

(d) Each new row is a combination of the two old ones.

■

Part (c) turns QR into a streaming algorithm (Gentleman 1973): feed the rows of $[\mathbf{X}, \mathbf{y}]$ one at a time into $\mathbf{R} = \mathbf{0}$. Memory is $O(p^2)$, one pass suffices, and the result is as accurate as Householder QR, so the one practical advantage of accumulating cross-products is kept without the κ^2 loss. Part (d) keeps $\mathbf{R}$ sparse for sparse designs, such as factors with many levels.

```
def givens(a, b):
    """c, s, r with [[c, s], [-s, c]] @ [a, b] = [r, 0]."""
    if b == 0.0:
        return 1.0, 0.0, a
    r = np.hypot(a, b) # sqrt(a^2 + b^2) without overflow
    return a / r, b / r, r

def add_row(R, z, x, y):
    """Rotate one observation (x, y) into the triangle R and the vector z = (Q'y)[:p].

    Returns the new R, z and the entry that leaves: its square is the increase in SSE.
    """
    R, z, x, y = R.copy(), z.copy(), np.array(x, dtype=float), float(y)
    for k in range(len(x)):
        c, s, r = givens(R[k, k], x[k])
        Rk, xk = R[k, k:].copy(), x[k:].copy()
        R[k, k:], x[k:] = c * Rk + s * xk, -s * Rk + c * xk
        z[k], y = c * z[k] + s * y, -s * z[k] + c * y
    return R, z, y

def rowwise_ls(X, y):
    """Least squares with one pass over the rows and O(p^2) memory."""
    p = X.shape[1]
    R, z, sse = np.zeros((p, p)), np.zeros(p), 0.0
    for xi, yi in zip(X, y):
        R, z, leftover = add_row(R, z, xi, yi)
        sse += leftover ** 2
    return R, z, sse

R_row, z_row, sse_row = rowwise_ls(X, y)
print("coefficients (row-wise Givens):", np.round(np.linalg.solve(R_row, z_row), 4))
```

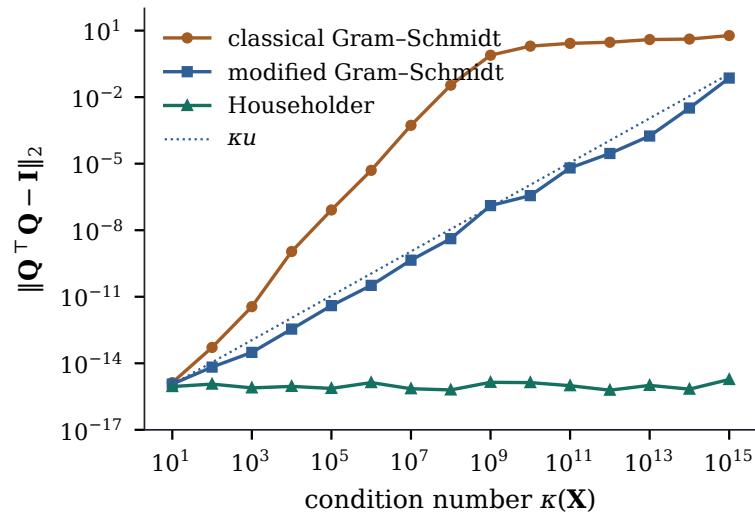


Figure 10.3.1: Departure from orthogonality of the computed Q against the condition number of X , for classical and modified Gram–Schmidt and for Householder QR. The dotted line is κu .

```
print(f"SSE (row-wise Givens): {sse_row:.4f}")
```

The rotation in `givens` uses `hypot`, which computes $(a^2 + b^2)^{1/2}$ without overflow or underflow. Square-root-free “fast” variants (Gentleman 1973) matter less on modern hardware.

10.3.4 Gram–Schmidt revisited

Gram–Schmidt (Theorem 1.5.2) also computes a QR factorization, at $2np^2$ flops, and forms Q_1 explicitly, which gives leverages directly. The classical form computes $r_{kj} = \mathbf{q}_k^\top \mathbf{x}_j$ from the *original* column; the modified form subtracts each new direction from all remaining columns at once. The two agree in exact arithmetic and differ greatly in floating point.

Example. (Loss of orthogonality)

For 100×12 matrices with prescribed condition numbers from 10 to 10^{15} , Figure 10.3.1 plots $\|\hat{Q}^\top \hat{Q} - I\|_2$ for the computed $\hat{Q}$. At $\kappa = 10^8$ classical Gram–Schmidt gives 3.5×10^{-2} , and its columns are no longer an orthonormal basis in any useful sense. Modified Gram–Schmidt gives 4.2×10^{-9} , in line with the bound $O(\kappa u)$ of Björck (1967). The Q accumulated from Householder reflections gives 6.3×10^{-16} , orthogonal to working precision whatever the condition number.

```
def cgs(X):
    """Classical Gram-Schmidt: inner products with the original column."""
    n, p = X.shape
    Q = np.zeros((n, p))
    for j in range(p):
        w = X[:, j] - Q[:, :j] @ (Q[:, :j].T @ X[:, j])
        Q[:, j] = w / np.linalg.norm(w)
    return Q

def mgs(X):
```



```

"""Modified Gram-Schmidt: subtract each direction from the working vector at
once."""
Q = np.array(X, dtype=float)
n, p = Q.shape
for j in range(p):
    Q[:, j] /= np.linalg.norm(Q[:, j])
    Q[:, j + 1:] -= np.outer(Q[:, j], Q[:, j] @ Q[:, j + 1:])
return Q

```

Loss of orthogonality does not make modified Gram–Schmidt useless. It is numerically equivalent to Householder QR applied to $\mathbf{X}$ with p rows of zeros on top (Björck and Paige 1992), so applied to $[\mathbf{X}, \mathbf{y}]$, with $\mathbf{Q}_1^\top \mathbf{y}$ taken from the last column, it gives a backward stable least squares solution. Leverages computed from the rows of $\hat{\mathbf{Q}}_1$ are not reliable when κ is large.

Idea. Orthogonal transformations reduce $\mathbf{X}$ to a triangle while preserving every residual length. Householder reflections do it fastest; Givens rotations do it one entry at a time, which suits streaming, updating and sparse data.

10.3.5 Exercises

10.3.5.1 A. Check your understanding

Exercise A1.

Find the Householder vector and α of Lemma 10.3.1 for $\mathbf{x} = (3, 4)^\top$, and verify $\mathbf{H}\mathbf{x} = \alpha \mathbf{e}_1$. For $\mathbf{x} = (1, 10^{-9})^\top$, compute v_1 with both sign choices in double precision, and explain what goes wrong with the unstable one.

Solution. $\|\mathbf{x}\| = 5$, $\alpha = -5$, $\mathbf{v} = (8, 4)^\top$, $\mathbf{v}^\top \mathbf{v} = 80$ and $\mathbf{v}^\top \mathbf{x} = 40$, so $\mathbf{H}\mathbf{x} = \mathbf{x} - 2 \cdot \frac{40}{80} \mathbf{v} = (3, 4) - (8, 4) = (-5, 0)$. For $\mathbf{x} = (1, 10^{-9})^\top$, $\|\mathbf{x}\| = (1 + 10^{-18})^{1/2}$ rounds to 1. The stable choice gives $v_1 = 1 + 1 = 2$. The unstable one gives $v_1 = 1 - 1 = 0$ in floating point, while the true value is $1 - \|\mathbf{x}\| \approx -5 \times 10^{-19}$. With $\mathbf{v} = (0, 10^{-9})$ the “reflection” swaps the sign of the second coordinate and maps $\mathbf{x}$ to $(1, -10^{-9})$, not onto the axis. An algebraically equivalent formula, $v_1 = -(x_2^2 + \cdots + x_m^2)/(x_1 + \|\mathbf{x}\|)$, rescues that choice.

10.3.5.2 B. Practice

Exercise B1.

In the setting of Proposition 10.3.3, let $\mathbf{Q} = [\mathbf{Q}_1, \mathbf{Q}_2]$. Show that under the linear model with $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, the vector $\mathbf{c}_2 = \mathbf{Q}_2^\top \mathbf{y}$ has mean $\mathbf{0}$ and covariance $\sigma^2 \mathbf{I}_{n-p}$, and that $\hat{\boldsymbol{\varepsilon}} = \mathbf{Q}_2 \mathbf{c}_2$. Under normality, deduce from $\mathbf{c}_2$ alone that $\text{SSE}/\sigma^2 \sim \chi^2(n-p)$ independently of $\hat{\boldsymbol{\beta}}$ (compare Theorem 7.5.1). Is $\mathbf{c}_2$ unique?

Solution. $\mathbf{Q}_2^\top \mathbf{X} = \mathbf{0}$, because $\mathbf{Q}^\top \mathbf{X} = \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$. So $\mathbf{c}_2 = \mathbf{Q}_2^\top \mathbf{X} \boldsymbol{\beta} + \mathbf{Q}_2^\top \boldsymbol{\varepsilon} = \mathbf{Q}_2^\top \boldsymbol{\varepsilon}$, with mean $\mathbf{0}$ and covariance $\sigma^2 \mathbf{Q}_2^\top \mathbf{Q}_2 = \sigma^2 \mathbf{I}$. By Proposition 10.3.3(b), $\hat{\boldsymbol{\varepsilon}} = \mathbf{Q} \begin{pmatrix} \mathbf{0} \\ \mathbf{c}_2 \end{pmatrix} = \mathbf{Q}_2 \mathbf{c}_2$. Under normality, $\mathbf{Q}^\top \boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$ (Theorem 3.3.1), so $\mathbf{c}_1 - \mathbf{R} \boldsymbol{\beta} = \mathbf{Q}_1^\top \boldsymbol{\varepsilon}$ and $\mathbf{c}_2 = \mathbf{Q}_2^\top \boldsymbol{\varepsilon}$ are independent normal vectors with identity covariance times σ^2 . Then $\text{SSE}/\sigma^2 = \|\mathbf{c}_2/\sigma\|^2 \sim \chi^2(n-p)$, and $\hat{\boldsymbol{\beta}} = \mathbf{R}^{-1} \mathbf{c}_1$ is a function of $\mathbf{c}_1$ alone. $\mathbf{c}_2$ is not unique: $\mathbf{Q}_2$

can be replaced by $Q_2 O$ for any orthogonal O , which changes c_2 but not its length or its distribution.

Exercise B2.

Weighted least squares with a diagonal weight matrix W of positive weights is ordinary least squares for $W^{1/2}X$ and $W^{1/2}y$. Show that $\kappa(W^{1/2}X) \leq \kappa(W^{1/2})\kappa(X)$. Give an example with two very different weights in which $\kappa(W^{1/2}X)$ is large although $\kappa(X)$ is small, and one in which it is small although $\kappa(W^{1/2})$ is large.

Solution. $\sigma_{\max}(W^{1/2}X) \leq \sigma_{\max}(W^{1/2})\sigma_{\max}(X)$ and $\|W^{1/2}Xv\| \geq \sigma_{\min}(W^{1/2})\|Xv\| \geq \sigma_{\min}(W^{1/2})\sigma_{\min}(X)$ for unit v , by Proposition 1.8.2. Divide. For the first example take $X = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ (with $\kappa = 1$) and $W = \text{diag}(1, w^2)$ with w large. Then $W^{1/2}X$ has rows $(1, 1)$ and $(w, -w)$, whose singular values are $\sqrt{2}$ and $\sqrt{2}w$, so $\kappa = w$. For the second take $X = \begin{pmatrix} 1 & 0 \\ 0 & 1/w \end{pmatrix}$ with the same W . Then $W^{1/2}X = I$.

10.3.5.3 C. Going deeper

Exercise C1.

Show that, in exact arithmetic, Householder QR (Theorem 10.3.2) applied to $\begin{pmatrix} 0_{p \times p} \\ X \end{pmatrix}$ produces the same R as modified Gram–Schmidt applied to X up to the signs of its rows, and that with the sign convention of Lemma 10.3.1 its reflection vectors are $v_j = \begin{pmatrix} e_j \\ q_j \end{pmatrix} / \sqrt{2}$. This is the observation of Björck and Paige (1992), who take $\alpha = +\|x\|$ and so obtain $\begin{pmatrix} -e_j \\ q_j \end{pmatrix} / \sqrt{2}$.

10.4 The singular value decomposition

QR is the workhorse when X has full column rank. When it does not, or nearly does not, the singular value decomposition tells the whole story: every least squares solution, the one of minimum norm, and, in the smallest singular values, the combinations of the columns that the data cannot resolve. Theorem 1.8.1 established the decomposition, and Proposition 6.10.1 previewed its use for least squares.

10.4.1 Least squares through the SVD

Throughout, X is $n \times p$ of rank $r \geq 1$, with singular value decomposition $X = \sum_{i \leq r} \sigma_i u_i v_i^\top$ (Theorem 1.8.1). We complete $u_1, \dots, u_r$ and $v_1, \dots, v_r$ to orthonormal bases $u_1, \dots, u_n$ of $\mathbb{R}^n$ and $v_1, \dots, v_p$ of $\mathbb{R}^p$, and write $c_i = u_i^\top y$ for the coordinates of the data in the left singular basis.

Theorem 10.4.1 — Least squares via the SVD (a) b is a least squares estimate iff $v_i^\top b = c_i / \sigma_i$ for $i = 1, \dots, r$. The coordinates $v_i^\top b$ for $i > r$ are

arbitrary, and the set of least squares estimates is $\mathbf{X}^+\mathbf{y} + \mathcal{N}(\mathbf{X})$ with $\mathcal{N}(\mathbf{X}) = \text{span}(\mathbf{v}_{r+1}, \dots, \mathbf{v}_p)$.

- (b) $\mathbf{X}^+\mathbf{y} = \sum_{i \leq r} (c_i/\sigma_i) \mathbf{v}_i$ is the unique least squares estimate of smallest Euclidean norm.
- (c) $\hat{\mathbf{Y}} = \sum_{i \leq r} c_i \mathbf{u}_i$ and $\text{SSE} = \sum_{i > r} c_i^2 = \|\mathbf{y}\|^2 - \sum_{i \leq r} c_i^2$.
- (d) If $r = p$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, the estimate $\hat{\beta} = \mathbf{X}^+\mathbf{Y}$ has $\text{Cov}(\hat{\beta}) = \sigma^2 \sum_{i \leq p} \sigma_i^{-2} \mathbf{v}_i \mathbf{v}_i^\top$. In particular $\text{Var}(\mathbf{v}_i^\top \hat{\beta}) = \sigma^2/\sigma_i^2$, and the total variance is $\text{tr Cov}(\hat{\beta}) = \sigma^2 \sum_i \sigma_i^{-2}$.

Proof. Let $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_n]$, an orthogonal matrix. For any $\mathbf{b}$, $\mathbf{X}\mathbf{b} = \sum_{i \leq r} \sigma_i (\mathbf{v}_i^\top \mathbf{b}) \mathbf{u}_i$, so the coordinates of $\mathbf{X}\mathbf{b}$ in the basis $\mathbf{u}_1, \dots, \mathbf{u}_n$ are $\sigma_i \mathbf{v}_i^\top \mathbf{b}$ for $i \leq r$ and zero for $i > r$. By orthogonal invariance,

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{U}^\top \mathbf{y} - \mathbf{U}^\top \mathbf{X}\mathbf{b}\|^2 = \sum_{i \leq r} (c_i - \sigma_i \mathbf{v}_i^\top \mathbf{b})^2 + \sum_{i > r} c_i^2. \quad (10.4.1)$$

The second sum does not involve $\mathbf{b}$, and the first vanishes exactly when $\mathbf{v}_i^\top \mathbf{b} = c_i/\sigma_i$ for $i \leq r$. This proves (a) and (c), since $\mathbf{b} = \sum_i (\mathbf{v}_i^\top \mathbf{b}) \mathbf{v}_i$ and $\mathcal{N}(\mathbf{X})$ is spanned by the $\mathbf{v}_i$ with $i > r$ (Theorem 1.8.1). Part (b) is Proposition 6.10.1. In coordinates, $\|\mathbf{b}\|^2 = \sum_{i \leq p} (\mathbf{v}_i^\top \mathbf{b})^2$, so the minimum-norm estimate is the one whose free coordinates are zero, $\sum_{i \leq r} (c_i/\sigma_i) \mathbf{v}_i$. For (d), $\hat{\beta} = \sum_i \sigma_i^{-1} \mathbf{v}_i \mathbf{u}_i^\top \mathbf{Y}$, and $\text{Cov}(\mathbf{u}_i^\top \mathbf{Y}, \mathbf{u}_j^\top \mathbf{Y}) = \sigma^2 \mathbf{u}_i^\top \mathbf{u}_j$ is σ^2 or 0. By Theorem 2.2.1, $\text{Cov}(\hat{\beta}) = \sigma^2 \sum_i \sigma_i^{-2} \mathbf{v}_i \mathbf{v}_i^\top$. Its trace is $\sigma^2 \sum_i \sigma_i^{-2}$ because $\|\mathbf{v}_i\| = 1$. ■

Part (d) is the statistical face of ill-conditioning. The combination $\mathbf{v}_p^\top \beta$ of the coefficients, along the direction in which the columns of $\mathbf{X}$ are nearly dependent, is estimated with standard deviation σ/σ_p . When σ_p is tiny, the data carry almost no information about that combination, and no algorithm can change this.

```
import numpy as np

def svd_ls(X, y, tol=None):
    """Minimum-norm least squares solution, treating singular values <= tol as zero.
    """
    U, s, Vt = np.linalg.svd(X, full_matrices=False)
    if tol is None:
        tol = max(X.shape) * np.finfo(float).eps * s[0] # the usual default
    r = int(np.sum(s > tol)) # numerical rank
    c = U[:, :r].T @ y # coordinates of the fit
    b = Vt[:r].T @ (c / s[:r])
    sse = y @ y - c @ c
    return b, r, sse, s
```

10.4.2 Computing the SVD

The SVD is not computed from the eigenvalues of $\mathbf{X}^\top \mathbf{X}$, which has already lost the small singular values (Section 10.1). The standard method (Golub and Kahan 1965; Golub and Reinsch

1970) first applies Householder reflections alternately from the left and the right to reduce $\mathbf{X}$ to upper bidiagonal form, at about twice the cost of QR, and then drives the superdiagonal to zero by an implicitly shifted QR-type iteration that never forms a cross-product. When $n \gg p$, it pays to compute $\mathbf{X} = \mathbf{Q}_1 \mathbf{R}$ first and then an SVD $\mathbf{R} = \mathbf{U}_R \mathbf{\Sigma} \mathbf{V}^\top$ of the triangle, since $\mathbf{X} = (\mathbf{Q}_1 \mathbf{U}_R) \mathbf{\Sigma} \mathbf{V}^\top$ (Exercise (B2)). Least squares needs only $\mathbf{U}^\top \mathbf{y}$, never $\mathbf{U}$ itself, which gives the count $2np^2 + 11p^3$ of Section 10.1 (Golub and Van Loan 2013).

The computed singular values are those of a nearby matrix $\mathbf{X} + \mathbf{E}$ with $\|\mathbf{E}\|_2 \leq cu\|\mathbf{X}\|_2$. The next result turns this into accuracy.

Proposition 10.4.2 — Singular values are well conditioned

For any matrices $\mathbf{A}$ and $\mathbf{E}$ of the same size and every k ,

$$|\sigma_k(\mathbf{A} + \mathbf{E}) - \sigma_k(\mathbf{A})| \leq \|\mathbf{E}\|_2.$$

Proof. For any matrix $\mathbf{C}$ and any $\mathbf{B}$ of rank at most $k-1$, $\sigma_k(\mathbf{C}) \leq \|\mathbf{C} - \mathbf{B}\|_2$. When $k-1 < \text{rank } \mathbf{C}$ this is the Eckart–Young inequality (Proposition 1.8.2(c)), and otherwise $\sigma_k(\mathbf{C}) = 0$. Let $\mathbf{A}_{k-1}$ be the truncated SVD of $\mathbf{A}$ with $k-1$ terms, or $\mathbf{A}$ itself if $\text{rank } \mathbf{A} < k$, so that $\|\mathbf{A} - \mathbf{A}_{k-1}\|_2 = \sigma_k(\mathbf{A})$. Then

$$\sigma_k(\mathbf{A} + \mathbf{E}) \leq \|\mathbf{A} + \mathbf{E} - \mathbf{A}_{k-1}\|_2 \leq \sigma_k(\mathbf{A}) + \|\mathbf{E}\|_2.$$

Exchanging the roles of $\mathbf{A}$ and $\mathbf{A} + \mathbf{E}$ gives the other inequality. ■

So a backward stable SVD computes every singular value with *absolute* error of order $u\sigma_1$, and a singular value of that size or below is indistinguishable from zero. An exactly rank-deficient design, such as a one-way layout with an intercept and all its indicators, therefore shows a smallest singular value near $10^{-16}\sigma_1$ rather than zero: 5×10^{-17} times the largest in the script's example.

10.4.3 Numerical rank

In floating point, and with data measured to finite precision, the meaningful question is how many singular values are *clearly* nonzero.

Definition 10.4.1 — Numerical rank

For a tolerance $\tau > 0$, the **numerical rank** of $\mathbf{X}$ is $r_\tau = \#\{i : \sigma_i > \tau\}$. The **truncated SVD** with k terms is $\mathbf{X}_k = \sum_{i \leq k} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$.

By the Eckart–Young inequality, r_τ is the least rank of any matrix within spectral distance τ of $\mathbf{X}$ (Exercise (B1)), so τ should reflect how well $\mathbf{X}$ is known. The default in numpy and LAPACK-based software, $\tau = \max(n, p) \epsilon_M \sigma_1$, accounts for rounding only; data recorded to six significant figures cannot resolve singular values below about $10^{-6}\sigma_1$. The choice matters, because the minimum-norm solution is not a continuous function of $\mathbf{X}$.

Example. (The pseudoinverse jumps)

Let $\mathbf{x}_1 = (1, 2, 3, 4)^\top$, let $\mathbf{d} = (1, -1, -1, 1)^\top$ be orthogonal to it, and let $\mathbf{X}_t = [\mathbf{x}_1, \mathbf{x}_1 + t\mathbf{d}]$, with $\mathbf{y} = (1, 3, 2, 5)^\top$. For every $t \neq 0$ the column space is the same plane $\text{span}(\mathbf{x}_1, \mathbf{d})$, so the fitted values do not depend on t . The coefficients do. At $t = 10^{-6}$ they are -249999 and 250000 , because the coefficients must reproduce a fixed multiple of $\mathbf{d}$ using the small difference $t\mathbf{d}$ of the two columns. At $t = 0$ the rank drops to one. The minimum-norm solution jumps to 0.550 for both columns, and the fitted values jump too, because the $\mathbf{d}$ -direction has disappeared from the model. Treating singular values below a tolerance as zero makes the two answers agree. Whether it should be a statistical question: is the difference between the columns signal, or recording error?

```
x1 = np.array([1.0, 2.0, 3.0, 4.0])
d = np.array([1.0, -1.0, -1.0, 1.0])          # orthogonal to x1
yd = np.array([1.0, 3.0, 2.0, 5.0])
for t in [1e-2, 1e-6, 1e-10, 0.0]:
    Xt = np.column_stack([x1, x1 + t * d])      # second column tends to the first
    b = np.linalg.pinv(Xt, rcond=1e-15) @ yd
    print(f"t = {t:7.0e}    b = {np.round(b, 3)}    fit = {np.round(Xt @ b, 3)}")
```

10.4.4 Truncated SVD as an estimator

The truncated solution $\mathbf{b}_k = \sum_{i \leq k} (c_i / \sigma_i) \mathbf{v}_i$ is more than a numerical safeguard. It is a statistical estimator in its own right, and it trades bias for variance.

Proposition 10.4.3 — Bias and variance of truncated SVD

Suppose $\text{rank } \mathbf{X} = p$, $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, and let $\mathbf{b}_k = \sum_{i \leq k} (\mathbf{u}_i^\top \mathbf{Y} / \sigma_i) \mathbf{v}_i$ for $k \leq p$. Then

$$E(\mathbf{b}_k) = \sum_{i \leq k} (\mathbf{v}_i^\top \boldsymbol{\beta}) \mathbf{v}_i, \quad \text{Cov}(\mathbf{b}_k) = \sigma^2 \sum_{i \leq k} \frac{\mathbf{v}_i \mathbf{v}_i^\top}{\sigma_i^2},$$

$$E\|\mathbf{b}_k - \boldsymbol{\beta}\|^2 = \sum_{i > k} (\mathbf{v}_i^\top \boldsymbol{\beta})^2 + \sigma^2 \sum_{i \leq k} \frac{1}{\sigma_i^2}. \quad (10.4.2)$$

Hence $\mathbf{b}_k$ has smaller mean squared error than the least squares estimate $\mathbf{b}_p$ iff $\sum_{i > k} (\mathbf{v}_i^\top \boldsymbol{\beta})^2 < \sigma^2 \sum_{i > k} \sigma_i^{-2}$.

Proof. $E(\mathbf{u}_i^\top \mathbf{Y}) = \mathbf{u}_i^\top \mathbf{X}\boldsymbol{\beta} = \sigma_i \mathbf{v}_i^\top \boldsymbol{\beta}$, which gives the mean. The covariance is part (d) of [Theorem 10.4.1](#) with the sum stopped at k . For the mean squared error, write $E\|\mathbf{b}_k - \boldsymbol{\beta}\|^2 = \|E\mathbf{b}_k - \boldsymbol{\beta}\|^2 + \text{tr Cov}(\mathbf{b}_k)$. The bias $E\mathbf{b}_k - \boldsymbol{\beta}$ is $-\sum_{i > k} (\mathbf{v}_i^\top \boldsymbol{\beta}) \mathbf{v}_i$, with squared length $\sum_{i > k} (\mathbf{v}_i^\top \boldsymbol{\beta})^2$. Comparing [Eq. 10.4.2](#) for k and p gives the last claim. ■

Each dropped component costs its squared coefficient $(\mathbf{v}_i^\top \boldsymbol{\beta})^2$ in bias and saves σ^2 / σ_i^2 in variance.

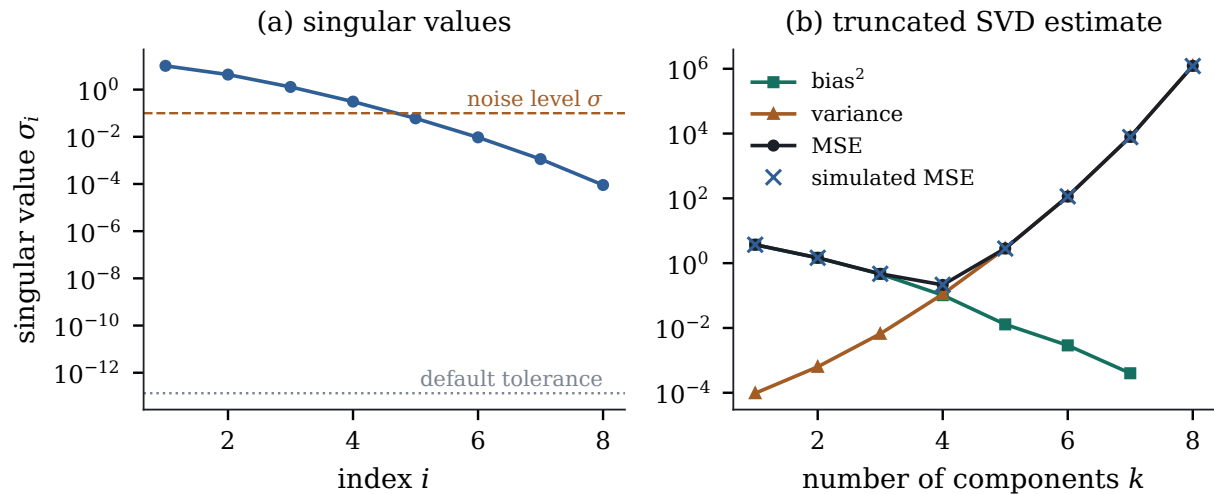


Figure 10.4.1: Truncated SVD for a degree-7 polynomial design. (a) Singular values, with the default numerical-rank tolerance and the noise level σ . (b) Squared bias, variance and mean squared error of the truncated estimate $\mathbf{b}_k$ (Eq. 10.4.2), with simulated values. Least squares is $k = 8$.

It is worth dropping exactly when the signal along $\mathbf{v}_i$ is weaker than the noise in its estimate, that is, when $\sigma_i |\mathbf{v}_i^\top \boldsymbol{\beta}| < \sigma$. The estimator $\mathbf{b}_k$ is principal components regression in its uncentred form. Ridge regression replaces the hard cut-off by weights $\sigma_i^2 / (\sigma_i^2 + \lambda)$. Both are developed in Chapter 27, and choosing k from data is a model-selection problem (Chapter 29).

Example. (Truncating a polynomial design)

The powers $1, t, \dots, t^7$ at 60 equally spaced points in $[0, 1]$ form a design with $\kappa(\mathbf{X}) = 1.1 \times 10^5$ and smallest singular value 9.0×10^{-5} . Take $\sigma = 0.1$ and a coefficient vector whose components along $\mathbf{v}_1, \dots, \mathbf{v}_8$ decay $(2, -1.5, 1, -0.6, 0.3, -0.1, 0.05, -0.02)$, as they do for smooth regression functions. By Eq. 10.4.2, least squares has mean squared error 1231055. The best truncation keeps 4 components and has mean squared error 0.21, made up of squared bias 0.10 and variance 0.11. A simulation with 4000 replicates agrees with the formula at every k (Figure 10.4.1). The default numerical tolerance treats the design as having full rank, which it does to working precision. The statistical “effective rank” is set by the noise level instead.

```
n, p, sigma = 60, 8, 0.1
t = np.linspace(0, 1, n)
X = np.vander(t, p, increasing=True)          # 1, t, ..., t^7: badly conditioned
U, s, Vt = np.linalg.svd(X, full_matrices=False)
a = np.array([2.0, -1.5, 1.0, -0.6, 0.3, -0.1, 0.05, -0.02]) # decaying components
    v_i' beta
beta = Vt.T @ a
print("condition number:", s[0] / s[-1])
print("singular values:", np.array2string(s, precision=2))
```

```
bias2 = np.array([np.sum((Vt[k:] @ beta) ** 2) for k in range(1, p + 1)]) # ||(I -
    V_k V_k') beta||^2
var = np.array([sigma ** 2 * np.sum(1 / s[:k] ** 2) for k in range(1, p + 1)])
mse = bias2 + var
print("k    bias^2    variance    MSE")
for k in range(p):
```

```
print(f"{k + 1} {bias2[k]:10.3e} {var[k]:10.3e} {mse[k]:10.3e}")
```

Idea. The SVD gives every least squares solution, the precision σ/σ_i of each combination $\mathbf{v}_i^\top \boldsymbol{\beta}$, and, relative to a tolerance, how many combinations are determined at all.

10.4.5 Exercises

10.4.5.1 A. Check your understanding

Exercise A1.

Let $\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}$ and $\mathbf{y} = (1, 3, 5)^\top$. Find the SVD of $\mathbf{X}$, the set of least squares estimates, the minimum-norm estimate and the SSE.

Solution. $\mathbf{X} = 2\mathbf{u}_1\mathbf{v}_1^\top$ with $\mathbf{u}_1 = (1, 1, 0)^\top/\sqrt{2}$, $\mathbf{v}_1 = (1, 1)^\top/\sqrt{2}$, $\sigma_1 = 2$ and $r = 1$. Here $c_1 = \mathbf{u}_1^\top \mathbf{y} = 4/\sqrt{2}$, so the least squares estimates are the $\mathbf{b}$ with $\mathbf{v}_1^\top \mathbf{b} = c_1/\sigma_1 = \sqrt{2}$, that is, $b_1 + b_2 = 2$. The minimum-norm estimate is $\sqrt{2}\mathbf{v}_1 = (1, 1)^\top$. The fitted vector is $c_1\mathbf{u}_1 = (2, 2, 0)^\top$, and $\text{SSE} = \|\mathbf{y}\|^2 - c_1^2 = 35 - 8 = 27$, which is $(1-2)^2 + (3-2)^2 + 5^2$.

10.4.5.2 B. Practice

Exercise B1.

Show that if τ is not a singular value of $\mathbf{X}$, then $r_\tau = \min\{\text{rank } \mathbf{B} : \|\mathbf{X} - \mathbf{B}\|_2 \leq \tau\}$, and that the minimum is attained by the truncated SVD $\mathbf{X}_{r_\tau}$.

Solution. $\|\mathbf{X} - \mathbf{X}_k\|_2 = \sigma_{k+1}$. For $k = r_\tau$, $\sigma_{k+1} \leq \tau$ (with $\sigma_{k+1} = 0$ if $k = \text{rank } \mathbf{X}$), so $\mathbf{X}_{r_\tau}$ is within τ and has rank r_τ . If $\text{rank } \mathbf{B} = k < r_\tau$, then by Proposition 1.8.2(c) $\|\mathbf{X} - \mathbf{B}\|_2 \geq \sigma_{k+1} \geq \sigma_{r_\tau} > \tau$. So no matrix of smaller rank is within τ .

Exercise B2.

Let $\mathbf{X} = \mathbf{Q}_1\mathbf{R}$ be a thin QR and $\mathbf{R} = \mathbf{U}_R\boldsymbol{\Sigma}\mathbf{V}^\top$ an SVD of the triangle. Show that $\mathbf{X} = (\mathbf{Q}_1\mathbf{U}_R)\boldsymbol{\Sigma}\mathbf{V}^\top$ is an SVD of $\mathbf{X}$, and that the minimum-norm least squares solution can be computed from $\mathbf{R}$ and $\mathbf{Q}_1^\top \mathbf{y}$ alone. Why is this attractive when n is much larger than p ?

Exercise B3.

Show that $\mathbf{b}_k$ minimizes $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|$ over $\mathbf{b} \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$, and that it is the minimum-norm least squares estimate for the matrix $\mathbf{X}_k$. Interpret $\mathbf{b}_k$ as least squares after deleting the directions in which $\mathbf{X}$ is nearly degenerate.

10.4.5.3 C. Going deeper

Exercise C1.

Show that the ridge estimate $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$ equals $\sum_i \frac{\sigma_i}{\sigma_i^2 + \lambda} c_i \mathbf{v}_i$. Derive its bias and variance in the form of [Proposition 10.4.3](#), and show that for every $\boldsymbol{\beta}$ there is a $\lambda > 0$ (depending on $\boldsymbol{\beta}$ and σ^2) for which its mean squared error is below that of least squares.

10.5 Conditioning and perturbation bounds

[Sections 10.2](#) to [10.4](#) were about algorithms. This section is about the problem itself: if the data $(\mathbf{X}, \mathbf{y})$ change a little, how much can the least squares solution and the residuals change? The answer does not depend on the algorithm. Combined with backward stability through [Eq. 10.1.2](#), it predicts the accuracy of any computed solution.

10.5.1 The perturbation theorem

Throughout this section $\mathbf{X}$ is $n \times p$ with full column rank, $\hat{\boldsymbol{\beta}} = \mathbf{X}^+ \mathbf{y}$ is its least squares solution (assumed nonzero) and $\hat{\boldsymbol{\epsilon}} = \mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}$ its residual. Write $\kappa = \kappa(\mathbf{X}) = \sigma_1 / \sigma_p$. The perturbed data are $\tilde{\mathbf{X}} = \mathbf{X} + \mathbf{E}$ and $\tilde{\mathbf{y}} = \mathbf{y} + \mathbf{f}$, with

$$\|\mathbf{E}\|_2 \leq \epsilon \|\mathbf{X}\|_2, \quad \|\mathbf{f}\| \leq \epsilon \|\mathbf{y}\|. \quad (10.5.1)$$

The solution and residual for the perturbed data are $\tilde{\mathbf{b}}$ and $\tilde{\boldsymbol{\epsilon}}$. A single number measures how badly the model fits relative to the size of the problem:

$$\eta = \frac{\|\hat{\boldsymbol{\epsilon}}\|}{\|\mathbf{X}\|_2 \|\hat{\boldsymbol{\beta}}\|}.$$

Since $\|\mathbf{X} \hat{\boldsymbol{\beta}}\| \leq \|\mathbf{X}\|_2 \|\hat{\boldsymbol{\beta}}\|$, $\eta \leq \tan \theta$, where θ is the angle between $\mathbf{y}$ and $\mathcal{C}(\mathbf{X})$. A model that fits well has small η .

Theorem 10.5.1 — Perturbation of least squares

Under [Eq. 10.5.1](#) with $\kappa\epsilon < 1$, the matrix $\tilde{\mathbf{X}}$ has full column rank, and

$$\frac{\|\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}}\|}{\|\hat{\boldsymbol{\beta}}\|} \leq \frac{\kappa\epsilon}{1 - \kappa\epsilon} (2 + \eta) + \frac{\kappa^2\epsilon}{(1 - \kappa\epsilon)^2} \eta, \quad (10.5.2)$$

$$\frac{\|\tilde{\boldsymbol{\epsilon}} - \hat{\boldsymbol{\epsilon}}\|}{\|\mathbf{y}\|} \leq \epsilon \left(1 + \frac{2\kappa}{1 - \kappa\epsilon} \right). \quad (10.5.3)$$

To first order in ϵ , the coefficient bound is $\epsilon(2\kappa + \kappa(\kappa + 1)\eta)$.

Proof. By [Proposition 10.4.2](#), $\sigma_p(\tilde{\mathbf{X}}) \geq \sigma_p - \|\mathbf{E}\|_2 \geq \sigma_p(1 - \kappa\epsilon) > 0$, so $\tilde{\mathbf{X}}$ has full column rank and $\tilde{\mathbf{X}}^+ \tilde{\mathbf{X}} = \mathbf{I}$. Hence

$$\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}} = \tilde{\mathbf{X}}^+ \tilde{\mathbf{y}} - \tilde{\mathbf{X}}^+ \tilde{\mathbf{X}} \hat{\boldsymbol{\beta}} = \tilde{\mathbf{X}}^+ (\tilde{\mathbf{y}} - \tilde{\mathbf{X}} \hat{\boldsymbol{\beta}}) = \tilde{\mathbf{X}}^+ (\hat{\boldsymbol{\epsilon}} + \mathbf{f} - \mathbf{E} \hat{\boldsymbol{\beta}}).$$

The residual term simplifies because $\mathbf{X}^\top \hat{\boldsymbol{\epsilon}} = \mathbf{0}$: $\tilde{\mathbf{X}}^+ \hat{\boldsymbol{\epsilon}} = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} (\mathbf{X} + \mathbf{E})^\top \hat{\boldsymbol{\epsilon}} = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \mathbf{E}^\top \hat{\boldsymbol{\epsilon}}$. So

$$\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}} = \tilde{\mathbf{X}}^+ (\mathbf{f} - \mathbf{E} \hat{\boldsymbol{\beta}}) + (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \mathbf{E}^\top \hat{\boldsymbol{\epsilon}}. \quad (10.5.4)$$

Now $\|\tilde{\mathbf{X}}^+\|_2 = 1/\sigma_p(\tilde{\mathbf{X}})$ and $\|(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}\|_2 = 1/\sigma_p(\tilde{\mathbf{X}})^2$ ([Theorem 1.8.1](#)). Also $\|\mathbf{y}\| \leq \|\mathbf{X} \hat{\boldsymbol{\beta}}\| + \|\hat{\boldsymbol{\epsilon}}\| \leq \sigma_1 \|\hat{\boldsymbol{\beta}}\| + \|\hat{\boldsymbol{\epsilon}}\|$, so $\|\mathbf{f} - \mathbf{E} \hat{\boldsymbol{\beta}}\| \leq \epsilon(\|\mathbf{y}\| + \sigma_1 \|\hat{\boldsymbol{\beta}}\|) \leq \epsilon \sigma_1 \|\hat{\boldsymbol{\beta}}\| (2 + \eta)$. Therefore

$$\|\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}}\| \leq \frac{\epsilon \sigma_1 \|\hat{\boldsymbol{\beta}}\| (2 + \eta)}{\sigma_p(1 - \kappa\epsilon)} + \frac{\epsilon \sigma_1 \|\hat{\boldsymbol{\epsilon}}\|}{\sigma_p^2(1 - \kappa\epsilon)^2}.$$

Dividing by $\|\hat{\boldsymbol{\beta}}\|$ and using $\|\hat{\boldsymbol{\epsilon}}\| = \eta \sigma_1 \|\hat{\boldsymbol{\beta}}\|$ gives [Eq. 10.5.2](#).

For the residual, $\tilde{\boldsymbol{\epsilon}} = (\mathbf{I} - \tilde{\mathbf{M}}) \tilde{\mathbf{y}}$, where $\tilde{\mathbf{M}}$ projects onto $\mathcal{C}(\tilde{\mathbf{X}})$. Write $\tilde{\mathbf{y}} = \tilde{\mathbf{X}} \hat{\boldsymbol{\beta}} + (\mathbf{f} - \mathbf{E} \hat{\boldsymbol{\beta}}) + \hat{\boldsymbol{\epsilon}}$. The first term is annihilated by $\mathbf{I} - \tilde{\mathbf{M}}$, and $\tilde{\mathbf{M}} \hat{\boldsymbol{\epsilon}} = (\tilde{\mathbf{X}}^+)^{\top} \tilde{\mathbf{X}}^{\top} \hat{\boldsymbol{\epsilon}} = (\tilde{\mathbf{X}}^+)^{\top} \mathbf{E}^{\top} \hat{\boldsymbol{\epsilon}}$ as before. So

$$\tilde{\boldsymbol{\epsilon}} - \hat{\boldsymbol{\epsilon}} = (\mathbf{I} - \tilde{\mathbf{M}})(\mathbf{f} - \mathbf{E} \hat{\boldsymbol{\beta}}) - (\tilde{\mathbf{X}}^+)^{\top} \mathbf{E}^{\top} \hat{\boldsymbol{\epsilon}}.$$

The first term has norm at most $\epsilon(\|\mathbf{y}\| + \sigma_1 \|\hat{\boldsymbol{\beta}}\|) \leq \epsilon(1 + \kappa)\|\mathbf{y}\|$, because $\|\hat{\boldsymbol{\beta}}\| = \|\mathbf{X}^+ \mathbf{y}\| \leq \|\mathbf{y}\|/\sigma_p$. The second has norm at most $\epsilon \sigma_1 \|\hat{\boldsymbol{\epsilon}}\|/(\sigma_p(1 - \kappa\epsilon)) \leq \epsilon \kappa \|\mathbf{y}\|/(1 - \kappa\epsilon)$. Adding and using $1 + \kappa + \kappa/(1 - \kappa\epsilon) \leq 1 + 2\kappa/(1 - \kappa\epsilon)$ gives the bound [Eq. 10.5.3](#). ■

The exact identity [Eq. 10.5.4](#) shows where each term comes from. Perturbing the data along $\mathcal{C}(\mathbf{X})$ moves the solution through $\tilde{\mathbf{X}}^+$, with amplification $1/\sigma_p$. This is the familiar κ of linear systems ([Eq. 1.8.3](#)). The second term is new. A perturbation $\mathbf{E}$ that tilts a column of $\mathbf{X}$ towards the residual direction changes the subspace, and the residual, however large, then “leaks” into the solution through $(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}$, with amplification $1/\sigma_p^2$. The residuals themselves, and hence the fitted values, are only κ -sensitive.

The κ^2 term is not an artefact of the proof.

Example. (The squared condition number is attained)

Let $\delta = 10^{-3}$ and

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 0 & \delta \\ 0 & 0 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 0 \\ \delta \\ 1 \end{pmatrix},$$

so that $\hat{\boldsymbol{\beta}} = (0, 1)^\top$, $\hat{\boldsymbol{\epsilon}} = (0, 0, 1)^\top$, $\kappa = 1000$ and $\eta = 1$. Perturb only the (3, 2) entry of $\mathbf{X}$, by $\epsilon = 10^{-10}$. This tilts the weak second column towards the residual. Its coefficient becomes $(\delta^2 + \epsilon)/(\delta^2 + \epsilon^2)$, and the relative change in $\hat{\boldsymbol{\beta}}$ is 1.00×10^{-4} , equal to $\kappa^2 \epsilon = 1.00 \times 10^{-4}$. A perturbation of relative size 10^{-10} has changed the answer in the fourth digit.

```
import numpy as np
```

```

import statsmodels.api as sm

def ls(X, y):
    return np.linalg.lstsq(X, y, rcond=None)[0]

delta, eps = 1e-3, 1e-10
X = np.array([[1.0, 0.0], [0.0, delta], [0.0, 0.0]])
y = np.array([0.0, delta, 1.0])          # b = (0, 1), residual e = (0, 0, 1)
E = np.array([[0.0, 0.0], [0.0, 0.0], [0.0, eps]]) # ||E|| = eps ||X||
b = ls(X, y)
b_tilde = ls(X + E, y)
print("kappa =", np.linalg.cond(X))
print("relative change in b:", np.linalg.norm(b_tilde - b) / np.linalg.norm(b))
print("kappa^2 * eps      :", np.linalg.cond(X) ** 2 * eps)

```

The script also draws 300 random problems with random perturbations and checks [Eq. 10.5.2](#) on each. The worst ratio of the actual change to the bound is 0.31.

10.5.2 What the theorem says about algorithms

A backward stable algorithm (Householder QR, Givens QR, modified Gram–Schmidt on the augmented matrix, the SVD) returns the exact solution for data perturbed as in [Eq. 10.5.1](#) with $\epsilon \approx cu$, where c grows modestly with the dimensions. Its relative error is therefore about

$$cu(\kappa + \kappa^2\eta).$$

The normal equations incur a relative error of about $cu\kappa^2$ *whatever* η is, because forming $\mathbf{X}^\top \mathbf{X}$ already commits it ([Proposition 10.1.3](#)). Two regimes follow:

- **Small residual** ($\eta \ll 1/\kappa$): QR gains a full factor κ over the normal equations. This is the regime of the experiment in [Section 6.10](#), where $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}$ exactly.
- **Large residual** (η of order one): every method, however stable, loses about $2\log_{10}\kappa$ digits, because the problem is that sensitive. The normal equations are then worse only by a modest factor.

[Figure 10.5.1](#) shows both regimes on test problems with prescribed condition numbers. With zero residual and $\kappa = 10^6$, the normal equations give relative error 1.1×10^{-5} and QR 7.8×10^{-12} . With $\eta = 1$, that is $\|\hat{\epsilon}\| = \|\mathbf{X}\|_2 \|\boldsymbol{\beta}\|$, QR gives 1.9×10^{-7} and the SVD 1.9×10^{-7} . The κ^2 term has caught up with the stable methods.

```

def by_cholesky(X, y):
    L = np.linalg.cholesky(X.T @ X)
    return np.linalg.solve(L.T, np.linalg.solve(L, X.T @ y))

def by_qr(X, y):
    Q, R = np.linalg.qr(X)
    return np.linalg.solve(R, Q.T @ y)

def by_svd(X, y):
    U, s, Vt = np.linalg.svd(X, full_matrices=False)
    return Vt.T @ ((U.T @ y) / s)

```

In regression η is rarely tiny, but QR keeps an advantage: its $\kappa^2\eta$ term is a property of the problem, not of the method.

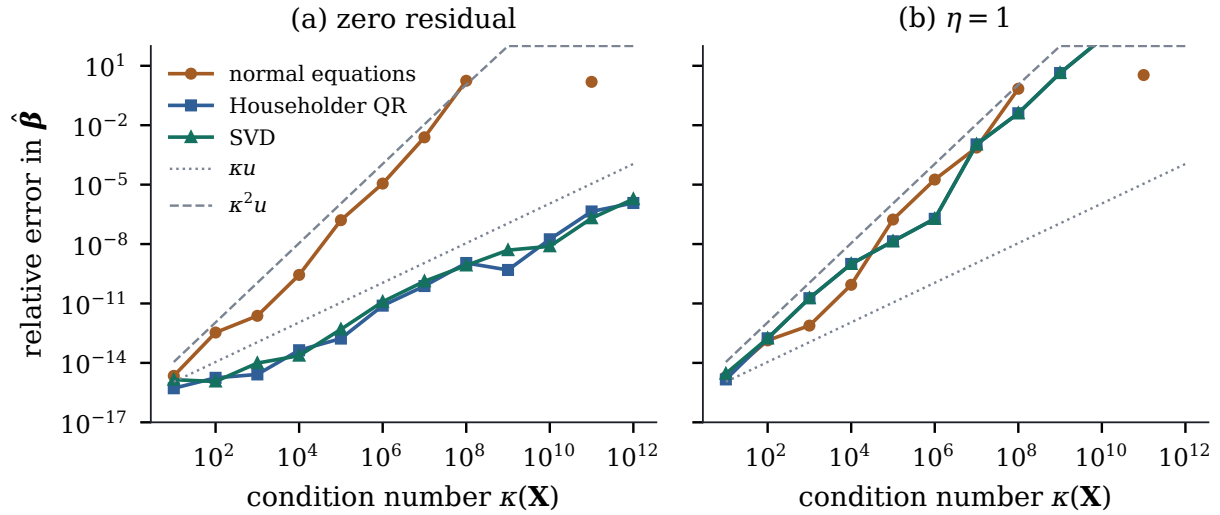


Figure 10.5.1: Relative error of the coefficients computed by the normal equations (Cholesky), Householder QR and the SVD, against $\kappa(\mathbf{X})$. (a) Consistent data, $\hat{\varepsilon} = \mathbf{0}$: QR and SVD track κu , the normal equations track $\kappa^2 u$. (b) A large residual, $\eta = 1$: all three track $\kappa^2 u$, as [Theorem 10.5.1](#) predicts.

10.5.3 Column scaling

The normwise condition number depends on the units of the columns: measuring one regressor in millimetres instead of metres can multiply $\kappa(\mathbf{X})$ by up to 1000 ([Exercise \(B2\)](#)), although the fit is unchanged. Equalizing the column lengths is essentially the best diagonal scaling.

Proposition 10.5.2 — Equilibrating the columns

Let $\mathbf{X}$ have full column rank and let $\mathbf{D}_0 = \text{diag}(1/\|\mathbf{x}_1\|, \dots, 1/\|\mathbf{x}_p\|)$. For every non-singular diagonal $\mathbf{D}$,

$$\kappa(\mathbf{X}\mathbf{D}_0) \leq \sqrt{p} \kappa(\mathbf{X}\mathbf{D}).$$

Proof. The columns of $\mathbf{X}\mathbf{D}_0$ have unit length, so $\sigma_1(\mathbf{X}\mathbf{D}_0) \leq \|\mathbf{X}\mathbf{D}_0\|_F = \sqrt{p}$. For the smallest singular value, let $\mathbf{z}$ be any nonzero vector and put $\mathbf{w} = \mathbf{D}^{-1}\mathbf{D}_0\mathbf{z}$. Then $\|\mathbf{X}\mathbf{D}_0\mathbf{z}\| = \|\mathbf{X}\mathbf{D}\mathbf{w}\| \geq \sigma_p(\mathbf{X}\mathbf{D})\|\mathbf{w}\|$ and $\|\mathbf{w}\| \geq \min_j (1/(|d_j|\|\mathbf{x}_j\|))\|\mathbf{z}\|$. The largest column length of $\mathbf{X}\mathbf{D}$, $\max_j |d_j|\|\mathbf{x}_j\|$, is at most $\sigma_1(\mathbf{X}\mathbf{D})$, because the columns are images of unit vectors. So $\sigma_p(\mathbf{X}\mathbf{D}_0) \geq \sigma_p(\mathbf{X}\mathbf{D})/\sigma_1(\mathbf{X}\mathbf{D}) = 1/\kappa(\mathbf{X}\mathbf{D})$, and $\kappa(\mathbf{X}\mathbf{D}_0) \leq \sqrt{p} \kappa(\mathbf{X}\mathbf{D})$. ■

This is due to van der Sluis (1969). What makes it more than a remark about units is the form of the backward errors. Householder QR has a *columnwise* backward error ([Theorem 10.3.2\(c\)](#)). Rescaling the columns rescales the perturbations with them, so the computed solution is as accurate as if the columns had been equilibrated first. The rounding errors in $\mathbf{X}^\top \mathbf{X}$ are entrywise small relative to $|\mathbf{X}|^\top |\mathbf{X}|$ ([Proposition 10.1.3](#)), which has the same property. Both methods therefore behave according to the condition number of the *scaled* matrix, $\kappa(\mathbf{X}\mathbf{D}_0)$, whatever the units. The rule of thumb should be applied to $\kappa(\mathbf{X}\mathbf{D}_0)$, not to $\kappa(\mathbf{X})$ as recorded.

Example. (Longley’s data, exactly)

Longley (1967) built his macroeconomic data to test regression programs. With an intercept and six regressors, $\kappa(\mathbf{X}) = 4.86 \times 10^9$ (Section 6.10; the data are described in Example (Macroeconomic series)). The data are published as decimals, so the exact least squares solution can be computed in rational arithmetic. The script does so and counts the correct significant digits of the worst coefficient for each method:

Method	Correct digits
Normal equations (Cholesky)	7.2
Householder QR	10.9
SVD	10.8
statsmodels OLS	10.8
Normal equations after scaling the columns to unit length	7.0

The raw κ predicts about 6 digits from QR and none from the normal equations (Section 6.10). Both predictions are far too pessimistic. After scaling the columns to unit length, $\kappa(\mathbf{X}\mathbf{D}_0) = 4.33 \times 10^4$. Then $\kappa u \approx 5 \times 10^{-12}$ predicts about eleven digits from QR, and $\kappa^2 u \approx 2 \times 10^{-7}$ about seven from the normal equations. That is what is observed. Scaling the columns explicitly before forming the normal equations changes nothing, as the argument above predicts. Centring as well reduces the condition number to 1.11×10^2 . That is a reparameterization rather than a scaling (Theorem 6.7.1), and what remains is genuine collinearity among the regressors (Chapter 26).

```
lon = sm.datasets.longley.load_pandas()
XL = sm.add_constant(lon.exog.to_numpy())
yL = lon.endog.to_numpy()
names = ["const"] + list(lon.exog.columns)
print("kappa(X) as recorded:", f"{np.linalg.cond(XL):.3e}")
```

10.5.4 Numerical error and statistical error

The perturbation theorem applies to any perturbation of the data, including the rounding of recorded values, which is far above u . Longley’s GNP deflator is given to one decimal place and the other series to whole units. Perturbing each regressor entry (the year is exact) uniformly within half a unit of its last recorded digit, 1000 times, the coefficients vary with standard deviations between 0.005 and 0.024 of their standard errors. The numerical error of the normal equations, the least accurate method, is at most 4×10^{-8} standard errors. Sampling error dominates, then comes the rounding of the published data, and the arithmetic is last by far. Beaton, Rubin and Barone (1976) made this point about Longley’s data. A problem can be numerically delicate and statistically ordinary, or the reverse.

```
rng = np.random.default_rng(106)
half_unit = np.array([0.0, 0.05, 0.5, 0.5, 0.5, 0.5, 0.0]) # half a unit in the
# last digit
fitL = sm.OLS(yL, XL).fit() # year and constant are
# exact
reps = 1000
shifts = []
```

```

for _ in range(reps):
    Xp = XL + rng.uniform(-1, 1, size=XL.shape) * half_unit
    shifts.append(ls(Xp, yL) - fitL.params)
spread = np.std(shifts, axis=0)
ratio = spread / fitL.bse
for name, r in zip(names, ratio):
    print(f"{name:8s} sd of coefficient under rounding / standard error = {r:.2f}")

```

Idea. Least squares has a κ sensitivity, and a $\kappa^2\eta$ one when the residual is large. Stable algorithms add nothing beyond this; the normal equations add κ^2 regardless. Measure κ after equilibrating the columns.

10.5.5 Exercises

10.5.5.1 A. Check your understanding

Exercise A1.

In [Example \(The squared condition number is attained\)](#), find $\tilde{\mathbf{b}}$ exactly and show that the relative change is $\epsilon(1 - \epsilon)/(\delta^2 + \epsilon^2)$. For which ϵ is it close to $\kappa^2\epsilon$?

10.5.5.2 B. Practice

Exercise B1.

Deduce from [Eq. 10.5.3](#) that the fitted values satisfy $\|\tilde{\mathbf{X}}\tilde{\mathbf{b}} - \mathbf{X}\hat{\boldsymbol{\beta}}\| \leq \epsilon(2 + 2\kappa/(1 - \kappa\epsilon))\|\mathbf{y}\|$. Why are fitted values never κ^2 -sensitive, even though coefficients can be?

Solution. $\tilde{\mathbf{X}}\tilde{\mathbf{b}} = \tilde{\mathbf{y}} - \tilde{\boldsymbol{\epsilon}}$ and $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{y} - \hat{\boldsymbol{\epsilon}}$, so the difference is $\mathbf{f} - (\tilde{\boldsymbol{\epsilon}} - \hat{\boldsymbol{\epsilon}})$, with norm at most $\epsilon\|\mathbf{y}\| + \epsilon(1 + 2\kappa/(1 - \kappa\epsilon))\|\mathbf{y}\|$. The κ^2 term in the coefficients comes from $(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \mathbf{E}^\top \hat{\boldsymbol{\epsilon}}$, a change of coordinates *within* the slightly rotated column space. Multiplying by $\tilde{\mathbf{X}}$ turns the factor $1/\sigma_p^2$ into $1/\sigma_p$. A direction that the columns barely span is a direction in which large coefficient changes produce only small changes in the fit.

Exercise B2.

Show that for consistent data ($\hat{\boldsymbol{\epsilon}} = \mathbf{0}$) the bound reduces to $2\kappa\epsilon/(1 - \kappa\epsilon)$. Construct a perturbation $\mathbf{f}$ of $\mathbf{y}$ alone that changes $\hat{\boldsymbol{\beta}}$ by a relative amount $\kappa\epsilon$, so that the bound is attained up to a factor of two.

10.5.5.3 C. Going deeper

Exercise C1.

Take data with $\hat{\boldsymbol{\beta}} = \mathbf{v}_p$ and $\hat{\boldsymbol{\epsilon}} \neq \mathbf{0}$, and the perturbation $\mathbf{E} = \epsilon\sigma_1 \hat{\boldsymbol{\epsilon}} \mathbf{v}_p^\top / \|\hat{\boldsymbol{\epsilon}}\|$, with $\mathbf{f} = \mathbf{0}$. Use [Eq. 10.5.4](#) to show that, to first order in ϵ , $\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}} = \epsilon\sigma_1 \|\hat{\boldsymbol{\epsilon}}\| \sigma_p^{-2} \mathbf{v}_p$, so that the relative change is $\epsilon\kappa^2\eta$. Compare with the first-order bound $\epsilon(2\kappa + \kappa(\kappa + 1)\eta)$.

10.6 Updating and downdating

A fitted regression often has to be changed slightly: new observations arrive, a suspicious case is set aside, a window slides along a time series, a variable is added or removed. Refitting from scratch costs $O(np^2)$ each time. Modifying the factorization costs $O(p^2)$ for a change of one row, and $O(np)$ for a change of one column. This section develops those modifications and uses them to compute every leave-one-out quantity from a single fit.

10.6.1 Adding and deleting observations

Adding a row is part (c) of [Theorem 10.3.4](#): p rotations fold the new row into the triangle. Removing a row is harder, because orthogonal transformations can only combine rows that are present. The trick is to run the addition backwards.

Theorem 10.6.1 — Updating and downdating a triangular factor

Let $\mathbf{T}$ be an $m \times m$ nonsingular upper triangular matrix with $\mathbf{T}^\top \mathbf{T} = \mathbf{A}$, and let $\mathbf{z} \in \mathbb{R}^m$.

- (a) (*Adding a row.*) There are m rotations, the k th acting on row k of $\mathbf{T}$ and on an extra row, that turn $\begin{pmatrix} \mathbf{T} \\ \mathbf{z}^\top \end{pmatrix}$ into $\begin{pmatrix} \mathbf{T}_+ \\ \mathbf{0}^\top \end{pmatrix}$, with $\mathbf{T}_+$ upper triangular and $\mathbf{T}_+^\top \mathbf{T}_+ = \mathbf{A} + \mathbf{z}\mathbf{z}^\top$.
- (b) (*Deleting a row.*) Let $\mathbf{a}$ solve $\mathbf{T}^\top \mathbf{a} = \mathbf{z}$. The matrix $\mathbf{A} - \mathbf{z}\mathbf{z}^\top$ is positive definite iff $\|\mathbf{a}\| < 1$. In that case put $\alpha = (1 - \|\mathbf{a}\|^2)^{1/2}$. Choose rotations $\mathbf{G}_m, \dots, \mathbf{G}_1$, with $\mathbf{G}_k$ acting on coordinates k and $m+1$, such that $\mathbf{G}_1 \cdots \mathbf{G}_m (\mathbf{a}^\top, \alpha)^\top = (\mathbf{0}^\top, 1)^\top$. Then

$$\mathbf{G}_1 \cdots \mathbf{G}_m \begin{pmatrix} \mathbf{T} \\ \mathbf{0}^\top \end{pmatrix} = \begin{pmatrix} \mathbf{T}_- \\ \mathbf{z}^\top \end{pmatrix}$$

with $\mathbf{T}_-$ upper triangular and $\mathbf{T}_-^\top \mathbf{T}_- = \mathbf{A} - \mathbf{z}\mathbf{z}^\top$.

- (c) Each operation costs $O(m^2)$ flops: $3m^2$ for (a), and $3m^2$ plus a triangular solve of m^2 for (b).

Proof.

- (a) is [Theorem 10.3.4\(c\)](#).
- (b) $\mathbf{A} - \mathbf{z}\mathbf{z}^\top = \mathbf{T}^\top (\mathbf{I} - \mathbf{a}\mathbf{a}^\top) \mathbf{T}$. The matrix $\mathbf{I} - \mathbf{a}\mathbf{a}^\top$ has eigenvalue $1 - \|\mathbf{a}\|^2$ on $\mathbf{a}$ and 1 on $\mathbf{a}^\perp$. So it is positive definite iff $\|\mathbf{a}\| < 1$, and then so is $\mathbf{A} - \mathbf{z}\mathbf{z}^\top$ ([Proposition 1.7.5\(c\)](#)). The rotations exist: $\mathbf{G}_m$ combines a_m with α and leaves 0 in position m and $(a_m^2 + \alpha^2)^{1/2}$ in position $m+1$. Then $\mathbf{G}_{m-1}$ does the same with a_{m-1} , and so on ([Theorem 10.3.4\(a\)](#)). The final vector has norm $(\|\mathbf{a}\|^2 + \alpha^2)^{1/2} = 1$ and is zero except in the last position. That entry is positive, so it is 1. Apply the same rotations, in the same order, to $\begin{pmatrix} \mathbf{T} \\ \mathbf{0}^\top \end{pmatrix}$. Before $\mathbf{G}_k$ is applied, the extra row is zero in columns $1, \dots, k$, because it has only been mixed with rows $k+1, \dots, m$ of $\mathbf{T}$, which are zero there. Row k of $\mathbf{T}$ is zero in columns $1, \dots, k-1$. So $\mathbf{G}_k$ produces a new row k that is zero before column k , and the result is $\begin{pmatrix} \mathbf{T}_- \\ \mathbf{z}^\top \end{pmatrix}$.

with T_- upper triangular. Write $G = G_1 \cdots G_m$, which is orthogonal. Then

$$z = T^\top a = \begin{pmatrix} T \\ 0^\top \end{pmatrix}^\top \begin{pmatrix} a \\ \alpha \end{pmatrix} = \left(G \begin{pmatrix} T \\ 0^\top \end{pmatrix} \right)^\top G \begin{pmatrix} a \\ \alpha \end{pmatrix} = \begin{pmatrix} T_- \\ w^\top \end{pmatrix}^\top \begin{pmatrix} 0 \\ 1 \end{pmatrix} = w,$$

and in the same way $T^\top T = T_-^\top T_- + ww^\top$. Hence $T_-^\top T_- = A - zz^\top$.

(c) The rotations act on rows of length at most m , and there are m of them. ■

If $A = X^\top X$ and $z = x_{(i)}$, then $\|a\|^2 = z^\top A^{-1}z$ is the leverage h_{ii} (Definition 6.8.1): a case can be deleted exactly when the others still determine every coefficient. Downdating the augmented triangle of $[X, y]$ (Proposition 10.2.2) gives the fit without case i , with the square root of the new SSE in the corner. Downdating is less stable than updating: $1 - \|a\|^2$ cancels when $h_{ii} \approx 1$, and the relative error can reach about $u/(1 - h_{ii})$ (Stewart 1979).

```
import numpy as np
import statsmodels.api as sm

def givens(a, b):
    """c, s, r with [[c, s], [-s, c]] @ [a, b] = [r, 0]."""
    if b == 0.0:
        return 1.0, 0.0, a
    r = np.hypot(a, b)
    return a / r, b / r, r

def update(T, z):
    """Triangle of [A; z'] from the triangle T of A: rotate the new row into T."""
    T, z = T.copy(), np.array(z, dtype=float)
    for k in range(len(z)):
        c, s, r = givens(T[k, k], z[k])
        Tk, zk = T[k, k:].copy(), z[k:].copy()
        T[k, k:], z[k:] = c * Tk + s * zk, -s * Tk + c * zk
    return T

def downdate(T, z):
    """Triangle of A with the row z' removed, from the triangle T of A (T'T = A'A)."""
    m = len(z)
    a = np.linalg.solve(T.T, z)          # T'a = z; ||a||^2 is the leverage of z
    alpha2 = 1.0 - a @ a
    if alpha2 <= 0:
        raise np.linalg.LinAlgError("the row cannot be removed (leverage >= 1)")
    alpha = np.sqrt(alpha2)
    T, w = T.copy(), np.zeros(m)        # w: the extra row, initially zero
    for k in range(m - 1, -1, -1):      # rotate (a_k, alpha) into (0, new alpha)
        c, s, r = givens(alpha, a[k])
        alpha = r
        Tk = T[k, k:].copy()
        T[k, k:], w[k:] = c * Tk - s * w[k:], s * Tk + c * w[k:]
    return T                             # on exit w equals z
```

The alternative, **recursive least squares** (Plackett 1950), updates $K = (X^\top X)^{-1}$ by the

Sherman–Morrison formula ([Theorem 1.6.5](#), [Example \(Deleting an observation\)](#)): adding $(\mathbf{x}, y)$ gives

$$\mathbf{K}_+ = \mathbf{K} - \frac{\mathbf{K}\mathbf{x}\mathbf{x}^\top\mathbf{K}}{1 + \mathbf{x}^\top\mathbf{K}\mathbf{x}}, \quad \hat{\boldsymbol{\beta}}_+ = \hat{\boldsymbol{\beta}} + \frac{\mathbf{K}\mathbf{x}(y - \mathbf{x}^\top\hat{\boldsymbol{\beta}})}{1 + \mathbf{x}^\top\mathbf{K}\mathbf{x}}.$$

It also costs $O(p^2)$, but it carries an explicit inverse of the cross-product matrix, with its κ^2 sensitivity, and nothing corrects its accumulated errors ([Example \(Ten-year regressions of consumption\)](#)).

10.6.2 Leave-one-out quantities from one fit

Many diagnostics ask what would happen if observation i were left out. Computing them by n separate fits costs $O(n^2p^2)$, but all of them follow from one fit.

Proposition 10.6.2 — Deletion formulas

Let $\mathbf{X}$ have full column rank, let $h_{ii} < 1$, and let $\hat{\boldsymbol{\beta}}_{(i)}$ and $\text{SSE}_{(i)}$ denote the least squares estimate and residual sum of squares computed without observation i . Then

- (a) $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_{(i)}\hat{\varepsilon}_i/(1 - h_{ii})$;
- (b) $y_i - \mathbf{x}_{(i)}^\top\hat{\boldsymbol{\beta}}_{(i)} = \hat{\varepsilon}_i/(1 - h_{ii})$;
- (c) $\text{SSE}_{(i)} = \text{SSE} - \hat{\varepsilon}_i^2/(1 - h_{ii})$.

Proof.

- (a) is [Exercise \(B1\)](#), which follows from the deletion formula of [Example \(Deleting an observation\)](#).
- (b) Multiply (a) by $\mathbf{x}_{(i)}^\top$: $\mathbf{x}_{(i)}^\top\hat{\boldsymbol{\beta}}_{(i)} = \mathbf{x}_{(i)}^\top\hat{\boldsymbol{\beta}} - h_{ii}\hat{\varepsilon}_i/(1 - h_{ii})$, so $y_i - \mathbf{x}_{(i)}^\top\hat{\boldsymbol{\beta}}_{(i)} = \hat{\varepsilon}_i + h_{ii}\hat{\varepsilon}_i/(1 - h_{ii}) = \hat{\varepsilon}_i/(1 - h_{ii})$.
- (c) Put $g = \hat{\varepsilon}_i/(1 - h_{ii})$ and let $h_{ji} = \mathbf{x}_{(j)}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_{(i)}$ be the entries of $\mathbf{M}$. By (a), $y_j - \mathbf{x}_{(j)}^\top\hat{\boldsymbol{\beta}}_{(i)} = \hat{\varepsilon}_j + h_{ji}g$ for every j , including $j = i$, where it gives g by (b). Summing squares over all j ,

$$\sum_j (\hat{\varepsilon}_j + h_{ji}g)^2 = \text{SSE} + 2g(\mathbf{M}\hat{\boldsymbol{\varepsilon}})_i + g^2(\mathbf{M}^2)_{ii} = \text{SSE} + g^2h_{ii},$$

because $\mathbf{M}\hat{\boldsymbol{\varepsilon}} = \mathbf{0}$ and $\mathbf{M}^2 = \mathbf{M}$. Removing the $j = i$ term, which is g^2 , leaves $\text{SSE}_{(i)} = \text{SSE} - g^2(1 - h_{ii}) = \text{SSE} - \hat{\varepsilon}_i^2/(1 - h_{ii})$. ■

With a QR factorization the leverages cost $O(np)$ ([Proposition 10.3.3\(c\)](#)), and the rest $O(n)$. This gives the PRESS residuals $\hat{\varepsilon}_i/(1 - h_{ii})$, whose sum of squares is leave-one-out cross-validation (Allen 1974; Chapter 29), and the deleted variances $s_{(i)}^2 = \text{SSE}_{(i)}/(n - p - 1)$ behind the externally studentized residuals $t_i = \hat{\varepsilon}_i/(s_{(i)}\sqrt{1 - h_{ii}})$ of Chapter 20.

Example. (Leaving out the District of Columbia)

Refit the state murder-rate regression with all 51 jurisdictions, including the District of Columbia, which [Chapter 6](#) set aside. The residual sum of squares is 142.53 and the PRESS statistic is 230.52. The largest PRESS residual belongs to the District of Columbia. Its ordinary residual is 4.475 and its leverage 0.506, so its prediction error when it is left out is $\hat{\varepsilon}_i/(1 - h_{ii}) = 9.066$. Its externally studentized residual is 4.28. By [Proposition 10.6.2\(c\)](#), deleting it reduces the residual sum of squares to 101.96, the value found for the 50 states in [Example \(Murder rates by the augmented factorization\)](#). The script confirms all 51 deletions three ways: by the formulas, by downdating the augmented triangle, and by refitting.

```
data = sm.datasets.statecrime.load_pandas().data           # all 50 states and DC
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape

Q, R = np.linalg.qr(X)
beta = np.linalg.solve(R, Q.T @ y)
e = y - X @ beta
sse = e @ e
h = np.sum(Q ** 2, axis=1)                                # leverages
press_resid = e / (1 - h)                                  # y_i minus its prediction
                                without case i
sse_loo = sse - e ** 2 / (1 - h)                           # SSE after deleting case i
s_loo = np.sqrt(sse_loo / (n - 1 - p))
t_ext = e / (s_loo * np.sqrt(1 - h))                       # externally studentized
                                residuals
i = int(np.argmax(np.abs(press_resid)))
print(f"largest PRESS residual: {data.index[i]}, e = {e[i]:.3f}, h = {h[i]:.3f}, "
      f" e/(1-h) = {press_resid[i]:.3f}, t = {t_ext[i]:.2f}")
print(f"PRESS = {np.sum(press_resid ** 2):.2f}, SSE = {sse:.2f}")
```

10.6.3 Adding and deleting variables

Changing the columns of $\mathbf{X}$ is a Gram–Schmidt step, or its reverse.

Proposition 10.6.3 — Adding a column

Let $\mathbf{X} = \mathbf{Q}_1 \mathbf{R}$ be a thin QR factorization of a full-column-rank $\mathbf{X}$, and let $\mathbf{z} \notin \mathcal{C}(\mathbf{X})$. Put $\mathbf{r} = \mathbf{Q}_1^\top \mathbf{z}$, $\mathbf{w} = \mathbf{z} - \mathbf{Q}_1 \mathbf{r}$, $\rho = \|\mathbf{w}\|$ and $\mathbf{q} = \mathbf{w}/\rho$. Then

$$[\mathbf{X}, \mathbf{z}] = [\mathbf{Q}_1, \mathbf{q}] \begin{pmatrix} \mathbf{R} & \mathbf{r} \\ \mathbf{0}^\top & \rho \end{pmatrix}$$

is a thin QR factorization. The residual sum of squares falls by $(\mathbf{q}^\top \mathbf{y})^2$ when $\mathbf{z}$ is added, and the coefficient of $\mathbf{z}$ in the enlarged model is $\mathbf{q}^\top \mathbf{y}/\rho = \mathbf{w}^\top \mathbf{y}/\mathbf{w}^\top \mathbf{w}$.

Proof. $\mathbf{w} = (\mathbf{I} - \mathbf{M})\mathbf{z} \neq \mathbf{0}$ because $\mathbf{z} \notin \mathcal{C}(\mathbf{X})$, and $\mathbf{q}$ is a unit vector orthogonal to $\mathcal{C}(\mathbf{Q}_1)$. The product of the factors is $[\mathbf{Q}_1 \mathbf{R}, \mathbf{Q}_1 \mathbf{r} + \rho \mathbf{q}] = [\mathbf{X}, \mathbf{z}]$. By [Proposi-](#)

tion 10.3.3(d), applied to the enlarged factorization, the new column's sequential sum of squares is the square of its coordinate $\mathbf{q}^\top \mathbf{y}$. The last row of the triangular system of [Proposition 10.3.3\(a\)](#) reads $\rho b_z = \mathbf{q}^\top \mathbf{y}$. This is also the Frisch–Waugh–Lovell coefficient $\mathbf{w}^\top \mathbf{y} / \mathbf{w}^\top \mathbf{w}$ of [Theorem 6.6.2](#). ■

If $\mathbf{z}$ is nearly in $\mathcal{C}(\mathbf{X})$, the computed $\mathbf{w}$ loses orthogonality to $\mathbf{Q}_1$; repeating the step once restores it (Björck 1996, chapter 2). Deleting column j of $\mathbf{R}$ leaves an upper Hessenberg block that $p-j$ rotations restore to triangular form, at $O(p^2)$ cost ([Exercise \(B2\)](#)). So stepwise procedures move between models at $O(p^2)$ or $O(np)$ cost per step.

```
def add_column(Q1, R, x):
    """Thin QR of [X, x] from the thin QR of X (one Gram-Schmidt step, repeated once)."""
    r = Q1.T @ x
    w = x - Q1 @ r
    r2 = Q1.T @ w                                     # reorthogonalize: guards against
    cancellation
    w, r = w - Q1 @ r2, r + r2
    rho = np.linalg.norm(w)
    Rn = np.block([[R, r[:, None]], [np.zeros((1, R.shape[1])), np.array([[rho]])]])
    return np.column_stack([Q1, w / rho]), Rn
```

10.6.4 A moving window

Example. (Ten-year regressions of consumption)

Quarterly US real consumption is regressed on an intercept, real disposable income, real investment, population and unemployment (the public-domain macroeconomic series of [Example \(A macroeconomic design\)](#)), in a window of 40 quarters that slides forward one quarter at a time. Each step adds one row and deletes one. Over the 163 steps, whose windows have condition numbers up to 2.9×10^6 , the fit is maintained in two ways. The first applies two Sherman–Morrison updates to $(\mathbf{X}^\top \mathbf{X})^{-1}$. The second updates and then downdates the augmented triangle. After each step both are compared with a fresh QR fit of the window ([Figure 10.6.1](#)). Inverse updating reaches a relative error of 6.2×10^{-7} and ends at 8.4×10^{-8} . Updating the triangle stays below 1.1×10^{-10} and ends at 3.6×10^{-11} . Both are cheap. Only one keeps QR's accuracy.

```
macro = sm.datasets.macrodta.load_pandas().data
yw = macro["realcons"].to_numpy()
Xw = np.column_stack([np.ones(len(yw)), macro[["realdpi", "realinv", "pop", "unemp"]]])
w = 40                                     # ten years of quarterly data
print("kappa of the full design:", f"{np.linalg.cond(Xw):.2e}")
```

```
K = np.linalg.inv(Xw[:w].T @ Xw[:w])      # (a) rank-one updates of the
inverse
g = Xw[:w].T @ yw[:w]
Zw = np.column_stack([Xw, yw])
Tw = np.linalg.qr(Zw[:w], mode="r")      # (b) triangle of [X, y]
```

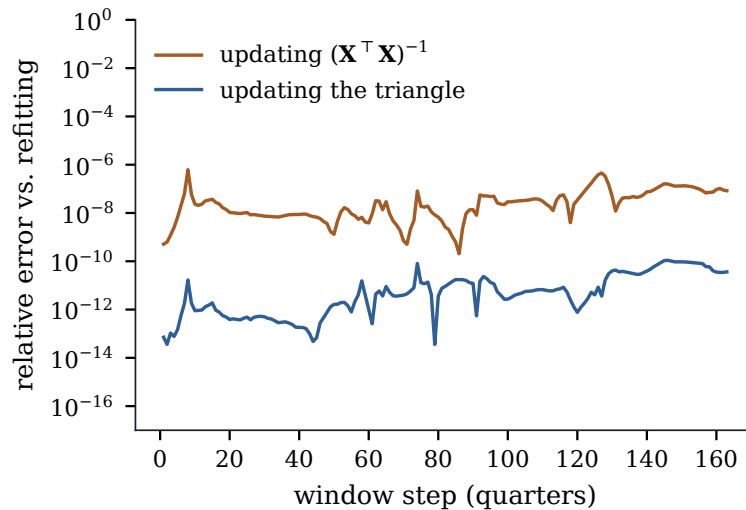


Figure 10.6.1: A moving-window regression maintained by rank-one updates of $(\mathbf{X}^\top \mathbf{X})^{-1}$ and by Givens updating and downdating of the augmented triangle. The vertical axis is the relative difference in the coefficients from a fresh QR fit of each window.

```

p1 = Xw.shape[1]
err_sm, err_giv = [], []
for t in range(w, len(yw)):
    new, old = Xw[t], Xw[t - w]
    Kx = K @ new                                     # add the new quarter (Sherman-
    K = K - np.outer(Kx, Kx) / (1 + new @ Kx)         Morrison)
    Kx = K @ old                                     # delete the oldest quarter
    K = K + np.outer(Kx, Kx) / (1 - old @ Kx)
    g = g + new * yw[t] - old * yw[t - w]
    b_sm = K @ g
    Tw = downdate(update(Tw, Zw[t]), Zw[t - w])
    b_giv = np.linalg.solve(Tw[:p1, :p1], Tw[:p1, p1])
    b_ref = np.linalg.lstsq(Xw[t - w + 1:t + 1], yw[t - w + 1:t + 1], rcond=None)[0]
    # (c) refit
    err_sm.append(np.linalg.norm(b_sm - b_ref) / np.linalg.norm(b_ref))
    err_giv.append(np.linalg.norm(b_giv - b_ref) / np.linalg.norm(b_ref))
print(f"final relative error: inverse updating {err_sm[-1]:.1e}, Givens {err_giv
[-1]:.1e}")

```

Idea. A change of one row or column costs $O(p^2)$ or $O(np)$ on the triangular factor, not $O(np^2)$. All n leave-one-out fits cost one, and the leverage says whether, and how accurately, a case can be removed.

10.6.5 Exercises

10.6.5.1 A. Check your understanding

Exercise A1.

Show that $\text{PRESS} = \sum_i \hat{\epsilon}_i^2 / (1 - h_{ii})^2 \geq \text{SSE}$. Show that if all leverages equal p/n , then $\text{PRESS} = \text{SSE } n^2 / (n - p)^2$. Evaluate this for the state data with $n = 51$ and $p = 4$, and

compare with the actual ratio in [Example \(Leaving out the District of Columbia\)](#).

10.6.5.2 B. Practice

Exercise B1.

Show that the leverage of row i of the augmented matrix $\mathbf{Z} = [\mathbf{X}, \mathbf{y}]$ is $h_{ii} + \hat{\varepsilon}_i^2/\text{SSE}$. Deduce the condition under which the augmented triangle can be downdated, and show that it fails only if $h_{ii} = 1$ or deleting case i leaves a model that fits the remaining data exactly.

Solution. $\mathcal{C}(\mathbf{Z}) = \mathcal{C}(\mathbf{X}) \oplus \text{span}(\hat{\varepsilon})$, with the two summands orthogonal, so by [Theorem 6.3.8](#) the projection onto $\mathcal{C}(\mathbf{Z})$ is $\mathbf{M} + \hat{\varepsilon}\hat{\varepsilon}^\top/\|\hat{\varepsilon}\|^2$. Its i th diagonal entry is $h_{ii} + \hat{\varepsilon}_i^2/\text{SSE}$. Downdating requires this to be below 1. By [Proposition 10.6.2\(c\)](#), $1 - h_{ii} - \hat{\varepsilon}_i^2/\text{SSE} = (1 - h_{ii})\text{SSE}_{(i)}/\text{SSE}$, which vanishes iff $h_{ii} = 1$ or $\text{SSE}_{(i)} = 0$.

Exercise B2.

Let $\mathbf{X} = \mathbf{Q}_1\mathbf{R}$ and let $\mathbf{X}_{-j}$ be $\mathbf{X}$ without column j . Show that $\mathbf{X}_{-j} = \mathbf{Q}_1\mathbf{H}$, where $\mathbf{H}$ is $\mathbf{R}$ without column j , which is upper Hessenberg in its last $p - j$ columns. Show that $p - j$ rotations of adjacent rows reduce $\mathbf{H}$ to $\begin{pmatrix} \mathbf{R}_{-j} \\ \mathbf{0}^\top \end{pmatrix}$, and that $\mathbf{R}_{-j}$ is the triangle of $\mathbf{X}_{-j}$. How much does the residual sum of squares rise?

Solution. Columns of $\mathbf{X}$ are $\mathbf{Q}_1$ times columns of $\mathbf{R}$, so deleting a column of both gives $\mathbf{X}_{-j} = \mathbf{Q}_1\mathbf{H}$. Column $k \geq j$ of $\mathbf{H}$ is column $k + 1$ of $\mathbf{R}$, which has nonzeros down to row $k + 1$: one below the diagonal. Rotate rows $j, j + 1$ to zero entry $(j + 1, j)$. This mixes two rows that are zero before column j , so nothing earlier fills in. Continue with rows $(j + 1, j + 2)$ and so on. After $p - j$ rotations, $\mathbf{GH} = \begin{pmatrix} \mathbf{R}_{-j} \\ \mathbf{0}^\top \end{pmatrix}$ with $\mathbf{R}_{-j}$ upper triangular, and $\mathbf{X}_{-j} = (\mathbf{Q}_1\mathbf{G}^\top) \begin{pmatrix} \mathbf{R}_{-j} \\ \mathbf{0}^\top \end{pmatrix}$, where $\mathbf{Q}_1\mathbf{G}^\top$ has orthonormal columns. Apply the same rotations to $\mathbf{c}_1 = \mathbf{Q}_1^\top \mathbf{y}$. The last entry of the rotated vector is the coordinate of $\mathbf{y}$ along the direction that has dropped out of the model, and its square is the rise in the residual sum of squares. This is the partial sum of squares of column j ([Definition 9.3.2](#)).

10.6.5.3 C. Going deeper

Exercise C1.

Feed the rows of $[\mathbf{X}, \mathbf{y}]$ one at a time into the augmented triangle, as in [Theorem 10.3.4\(c\)](#). For $k > p$, let t_k be the entry left in the new row when case k is added. Show that

$$t_k = \pm \frac{y_k - \mathbf{x}_{(k)}^\top \hat{\boldsymbol{\beta}}_{k-1}}{(1 + \mathbf{x}_{(k)}^\top (\mathbf{X}_{k-1}^\top \mathbf{X}_{k-1})^{-1} \mathbf{x}_{(k)})^{1/2}},$$

where $\hat{\boldsymbol{\beta}}_{k-1}$ and $\mathbf{X}_{k-1}$ use the first $k - 1$ cases. Show that under the normal linear model the t_k are independent $N(0, \sigma^2)$. These are the recursive residuals of Brown, Durbin and Evans (1975), used to test the constancy of a regression over time.

10.7 Rank-revealing factorizations

A model matrix with an intercept and indicators for every level of a factor has dependent, or **aliased**, columns. Software must notice them and decide which to set aside, but in floating point it can only ask whether a column is *nearly* a combination of others, relative to a tolerance. The SVD answers that (Section 10.4) but does not say which columns to drop, and it costs more. The standard tool is QR with column pivoting.

10.7.1 QR with column pivoting

At step k of Householder QR, the trailing columns (rows k to n , columns k to p of the partly reduced matrix) are what remains of the original columns after the parts explained by the first $k - 1$ chosen columns have been removed. **Column pivoting** (Businger and Golub 1965) moves the trailing column of largest norm, the one least explained by the columns already chosen, into position k before reflecting. The result is

$$\mathbf{X}\mathbf{\Pi} = \mathbf{Q} \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{R}_{22} \end{pmatrix},$$

with $\mathbf{\Pi}$ a permutation matrix, $\mathbf{R}_{11}$ the leading $k \times k$ block and $\mathbf{R}_{22}$ the trailing $(p - k) \times (p - k)$ block, for any k . We need one fact about submatrices.

Lemma 10.7.1 — Singular values of a submatrix

If $\mathbf{X}_1$ is formed from some of the columns of $\mathbf{X}$, then $\sigma_i(\mathbf{X}_1) \leq \sigma_i(\mathbf{X})$ for every i up to the number of columns of $\mathbf{X}_1$. The same holds for a submatrix formed from some of the rows.

Proof. Write $\mathbf{X}_1 = \mathbf{X}\mathbf{S}$, where $\mathbf{S}$ holds the corresponding columns of the identity, so $\|\mathbf{S}\|_2 = 1$. Let $\mathbf{X}_{i-1}$ be the truncated SVD with $i - 1$ terms, or $\mathbf{X}$ itself if $\text{rank } \mathbf{X} < i$, so that $\|\mathbf{X} - \mathbf{X}_{i-1}\|_2 = \sigma_i(\mathbf{X})$. The matrix $\mathbf{X}_{i-1}\mathbf{S}$ has rank at most $i - 1$, so by the Eckart–Young inequality, in the form used in the proof of Proposition 10.4.2,

$$\sigma_i(\mathbf{X}_1) \leq \|\mathbf{X}\mathbf{S} - \mathbf{X}_{i-1}\mathbf{S}\|_2 \leq \|\mathbf{X} - \mathbf{X}_{i-1}\|_2 \|\mathbf{S}\|_2 = \sigma_i(\mathbf{X}).$$

For rows, apply this to $\mathbf{X}^\top$, which has the same singular values. ■

Theorem 10.7.2 — Column-pivoted QR

Let $\mathbf{X}$ be $n \times p$ with $n \geq p$, and let $\mathbf{X}\mathbf{\Pi} = \mathbf{Q} \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$ be computed by Householder QR (Theorem 10.3.2) with column pivoting.

- (a) For every $k \leq j$, $r_{kk}^2 \geq \sum_{i=k}^j r_{ij}^2$. In particular $|r_{11}| \geq |r_{22}| \geq \cdots \geq |r_{pp}|$.
- (b) For every $k < p$,

$$\sigma_{k+1}(\mathbf{X}) \leq \|\mathbf{R}_{22}\|_2 \leq \sqrt{p - k} |r_{k+1,k+1}| \quad \text{and} \quad \sigma_{\min}(\mathbf{R}_{11}) \leq \sigma_k(\mathbf{X}).$$

- (c) In exact arithmetic, if $\text{rank } \mathbf{X} = r$, then $r_{kk} \neq 0$ for $k \leq r$ and $\mathbf{R}_{22} = \mathbf{0}$ for $k = r$. The first r columns of $\mathbf{X}\mathbf{\Pi}$ are a basis of $\mathcal{C}(\mathbf{X})$, and $\mathbf{X}\mathbf{\Pi} = \mathbf{Q} \begin{pmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$ with $\mathbf{R}_{11}$ nonsingular.
- (d) The cost is that of the unpivoted factorization, $2np^2 - 2p^3/3$ flops, plus $O(np)$ for keeping track of the column norms.

Proof.

- (a) Reflections preserve the length of each column segment they act on. When column k is chosen, its segment in rows $k, \dots, n$ has the largest norm among the trailing columns, and the reflection maps it to $(r_{kk}, 0, \dots, 0)^\top$, so $|r_{kk}|$ equals that norm. Every column that ends in a position $j > k$ was among the trailing columns at step k , so its segment in rows $k, \dots, n$ had norm at most $|r_{kk}|$. The reflections at steps $k, \dots, p$ act on rows $k, \dots, n$ only, so they preserve this norm, and at the end the segment is $(r_{kj}, \dots, r_{jj}, 0, \dots, 0)^\top$. Hence $\sum_{i=k}^j r_{ij}^2 \leq r_{kk}^2$. With $j = k+1$ this gives $r_{k+1,k+1}^2 \leq r_{kk}^2$.
- (b) $\mathbf{X}\mathbf{\Pi}$ has the same singular values as $\mathbf{X}$, because $\mathbf{\Pi}$ is orthogonal. The matrix $\mathbf{Q} \begin{pmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$ has rank at most k and differs from $\mathbf{X}\mathbf{\Pi}$ by $\mathbf{Q} \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_{22} \end{pmatrix}$, whose norm is $\|\mathbf{R}_{22}\|_2$. The Eckart–Young inequality (Proposition 1.8.2(c)) gives $\sigma_{k+1}(\mathbf{X}) \leq \|\mathbf{R}_{22}\|_2$. Next, $\|\mathbf{R}_{22}\|_2 \leq \|\mathbf{R}_{22}\|_F \leq \sqrt{p-k} \max_j \|\text{column } j \text{ of } \mathbf{R}_{22}\|$, and by (a) with $k+1$ in place of k , each column of $\mathbf{R}_{22}$ has norm at most $|r_{k+1,k+1}|$. Finally, the first k columns of $\mathbf{X}\mathbf{\Pi}$ equal $\mathbf{Q}_1 \begin{pmatrix} \mathbf{R}_{11} \\ \mathbf{0} \end{pmatrix}$, so their singular values are those of $\mathbf{R}_{11}$. Then Lemma 10.7.1 gives $\sigma_k(\mathbf{R}_{11}) \leq \sigma_k(\mathbf{X})$.
- (c) Because $\mathbf{R}_{11}$ is triangular, $\text{rank } \mathbf{R} = k + \text{rank } \mathbf{R}_{22}$ whenever $r_{11}, \dots, r_{kk}$ are nonzero (subtract multiples of the first k columns from the others to clear $\mathbf{R}_{12}$). Also $\text{rank } \mathbf{R} = \text{rank } \mathbf{X} = r$. So while $k < r$, $\mathbf{R}_{22} \neq \mathbf{0}$, some trailing column is nonzero, and the next pivot $r_{k+1,k+1}$ is nonzero. At $k = r$, $\text{rank } \mathbf{R}_{22} = 0$. The first r columns of $\mathbf{X}\mathbf{\Pi}$ are $\mathbf{Q}_1$ times the independent columns $\begin{pmatrix} \mathbf{R}_{11} \\ \mathbf{0} \end{pmatrix}$, so they are independent, and there are $r = \dim \mathcal{C}(\mathbf{X})$ of them.
- (d) The norms of the trailing column segments are updated at each step by subtracting r_{kj}^2 , at $O(p)$ flops per step. When cancellation makes the update inaccurate, the norm is recomputed.

■

Part (b) is the rank-revealing property, and it is one-sided. A small pivot $|r_{k+1,k+1}|$ certifies a small singular value σ_{k+1} : pivoted QR never reports a near dependence that is not there. The converse can fail. A matrix can have a tiny singular value although no pivot is small.

Example. (Kahan's matrix)

Kahan (1966) gave the upper triangular matrix with diagonal $1, s, s^2, \dots, s^{m-1}$ and entries $-cs^{i-1}$ above the diagonal in row i , where $c^2 + s^2 = 1$. Its columns are already in pivoted order (a tiny perturbation of the column scaling breaks ties in favour of that order), so pivoted QR leaves it unchanged. With $m = 100$ and $c = 0.2$, the smallest diagonal entry is 0.133, yet the smallest singular value is 3.7×10^{-9} . No diagonal entry hints at this (Figure 10.7.1(a)). Such matrices are rare in practice. Chan (1987) gave the first rank-revealing QR algorithm; factorizations whose bounds in both directions are polynomial in p exist (Hong and Pan 1992) and can be computed efficiently (Gu

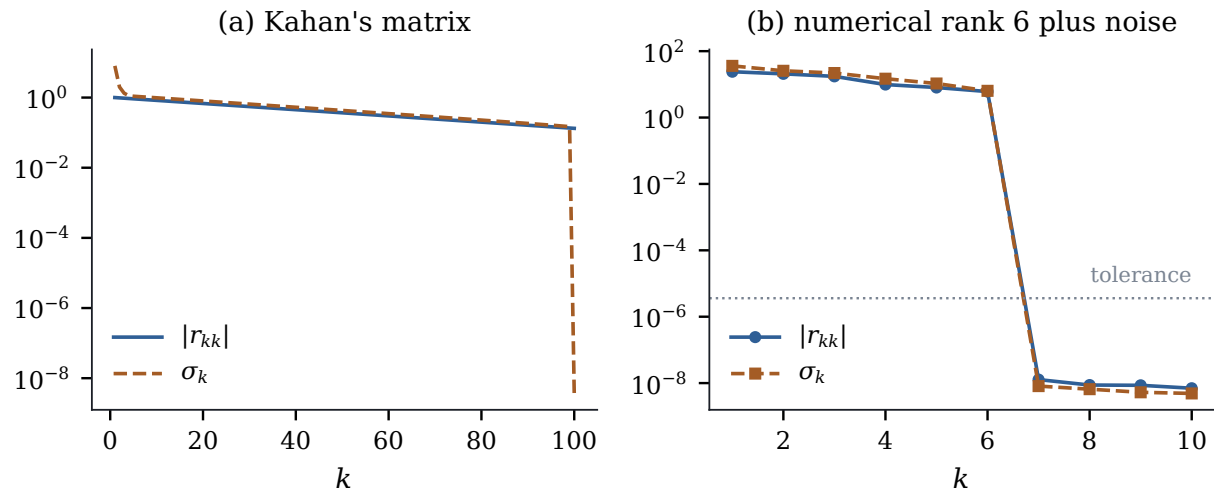


Figure 10.7.1: Diagonal of the pivoted triangle, $|r_{kk}|$, and singular values σ_k . (a) Kahan's matrix: the diagonal decreases slowly while the last singular value collapses. (b) A matrix of numerical rank six: both drop at $k = 7$, well below the tolerance.

and Eisenstat 1996). Panel (b) shows the typical case: a 50×10 matrix of rank six plus noise of relative size 10^{-9} . The pivoted diagonal drops by eight orders of magnitude exactly where the singular values do.

```
import numpy as np
import statsmodels.api as sm

def pivoted_qr(X):
    """Householder QR with column pivoting:  $X[:, \text{perm}] = Q R$ ,  $|r_{11}| \geq |r_{22}| \geq \dots$ """
    A = np.array(X, dtype=float)
    n, p = A.shape
    perm = np.arange(p)
    for k in range(min(n, p)):
        norms = np.sum(A[k:, k:] ** 2, axis=0)  # squared norms of the
        trailing columns
        j = k + int(np.argmax(norms))
        A[:, [k, j]], perm[[k, j]] = A[:, [j, k]], perm[[j, k]]
        x = A[k:, k]
        alpha = -np.copysign(np.linalg.norm(x), x[0])
        if alpha == 0.0:
            break  # the trailing block is
            exactly zero
        v = x.copy()
        v[0] -= alpha
        v /= np.linalg.norm(v)
        A[k:, k:] -= 2.0 * np.outer(v, v @ A[k:, k:])
    return np.triu(A[:p, :]), perm

def numerical_rank(R, tol):
    """Number of diagonal entries of the pivoted triangle above  $\text{tol} * |r_{11}|$ ."""
    d = np.abs(np.diag(R))
    return int(np.sum(d > tol * d[0]))
```

```
m, c = 100, 0.2
sn = np.sqrt(1 - c ** 2)
```

```

K = np.diag(sn ** np.arange(m)) @ (np.eye(m) - c * np.triu(np.ones((m, m)), 1))
K = K @ np.diag(1 - 1e-10 * np.arange(m))          # tiny tilt so that pivoting
                                                    keeps the order
RK, permK = pivoted_qr(K)
sK = np.linalg.svd(K, compute_uv=False)
print("pivot order unchanged:", np.array_equal(permK, np.arange(m)))
print(f"smallest |r_kk| = {np.abs(RK[-1, -1]):.3f}, smallest singular value = {sK
      [-1]:.2e}")

```

10.7.2 Basic solutions and aliased columns

Suppose pivoted QR finds numerical rank $r < p$, so that $\mathbf{X}\Pi \approx \mathbf{Q} \begin{pmatrix} \mathbf{R}_{01} & \mathbf{R}_{02} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$. Write $\mathbf{Q}^\top \mathbf{y} = (\mathbf{c}_1^\top, \mathbf{c}_2^\top)^\top$ with $\mathbf{c}_1 \in \mathbb{R}^r$, and split the permuted coefficients as $(\mathbf{b}_1, \mathbf{b}_2)$. As in [Proposition 10.3.3](#), the least squares estimates are the solutions of

$$\mathbf{R}_{11}\mathbf{b}_1 + \mathbf{R}_{12}\mathbf{b}_2 = \mathbf{c}_1,$$

with $\mathbf{b}_2$ free. Setting $\mathbf{b}_2 = \mathbf{0}$ gives the **basic solution**, which simply drops the last $p - r$ pivoted columns. It has at most r nonzero coefficients. It differs from the minimum-norm solution of [Theorem 10.4.1](#). That solution can also be computed at QR cost, by a second orthogonal reduction from the right that turns $[\mathbf{R}_{11}, \mathbf{R}_{12}]$ into $[\mathbf{T}, \mathbf{0}]$ (a *complete orthogonal decomposition*; Golub and Van Loan 2013). Every least squares solution gives the same fitted values. By [Theorem 6.4.4](#), and in the language of [Chapter 8](#) ([Theorem 8.3.1](#)), every estimable function $\lambda^\top \beta$ has the same estimate under every convention. The coefficients of aliased columns are not estimable.

Software differs in the choice. R's `lm` takes the columns in order, moves to the end any column whose remaining norm falls below a relative tolerance (by default 10^{-7}), and reports its coefficient as NA. statsmodels' OLS uses the pseudoinverse and so reports the minimum-norm solution. Programs built on the sweep operator ([Proposition 10.2.3](#)), such as SAS's GLM procedure, skip numerically zero pivots and set those coefficients to zero, another basic solution.

Example. (Firm effects in investment data)

Grunfeld's investment data ($n = 220$: 11 US firms over 20 years, public domain in statsmodels) are modelled with an intercept, the firm's market value and capital stock, and indicators for all 11 firms: $p = 14$ columns of rank 13. Pivoted QR with a tolerance of 10^{-10} finds rank 13 and leaves out the indicator for General Motors. Three conventions give three different intercepts: -50.67 for the minimum-norm solution (as statsmodels reports), -20.58 for the in-order rule that marks the last indicator as aliased (as R would), and -70.30 for the basic solution from full pivoting. They agree to every printed digit on everything estimable: the coefficient 0.1101 of market value, 0.3100 of capital, the difference -172.20 between the General Motors and US Steel effects, and the residual sum of squares 523719.

```

gr = sm.datasets.grunfeld.load_pandas().data
firms = list(dict.fromkeys(gr["firm"]))
D = np.column_stack([(gr["firm"] == f).to_numpy(float) for f in firms])  # 11
                                indicators
y = gr["invest"].to_numpy()
X = np.column_stack([np.ones(len(y)), gr[["value", "capital"]], D])      # 14
                                columns, rank 13

```



```

n, p = X.shape
R, perm = pivoted_qr(X)
r = numerical_rank(R, 1e-10)
print("p =", p, " numerical rank =", r, " column left out:", perm[r:])

```

```

def basic_solution(X, y, tol=1e-10):
    """Drop the columns that pivoted QR puts last; fit the rest (zeros for dropped
    columns)."""
    R, perm = pivoted_qr(X)
    r = numerical_rank(R, tol)
    b = np.zeros(X.shape[1])
    b[perm[:r]] = np.linalg.lstsq(X[:, perm[:r]], y, rcond=None)[0]
    return b

def in_order_solution(X, y, tol=1e-7):
    """Keep a column only if it is not (nearly) a combination of the columns kept
    before it."""
    kept = []
    for j in range(X.shape[1]):
        Z = X[:, kept + [j]]
        Q, R = np.linalg.qr(Z)
        if abs(R[-1, -1]) > tol * np.linalg.norm(X[:, j]):
            kept.append(j)
    b = np.full(X.shape[1], np.nan) # aliased coefficients reported
    # as missing
    b[kept] = np.linalg.lstsq(X[:, kept], y, rcond=None)[0]
    return b

solutions = {
    "minimum norm (pinv)": np.linalg.pinv(X) @ y,
    "pivoted QR, basic": basic_solution(X, y),
    "in order, aliased = NA": in_order_solution(X, y),
}
for name, b in solutions.items():
    fitted = X @ np.nan_to_num(b)
    print(f"{name:24s} value {b[1]:.5f} capital {b[2]:.5f} const {b[0]:9.3f}"
          f" GM - US Steel {b[3] - b[4]:9.3f} SSE {np.sum((y - fitted) ** 2):.1f}"
          )

```

10.7.3 Choosing a tolerance

Exact aliasing, as in the example, is easy: the offending pivot is at the level of rounding error, far below any reasonable tolerance. The hard case is a column that is nearly, but not exactly, a combination of others. The numerical rank then depends on the tolerance, and so do the coefficients, dramatically.

Example. (A nearly aliased column)

Add to the Grunfeld regressors (intercept, value, capital) a fourth column equal to value plus capital plus noise of relative size 10^{-9} . Its smallest singular value, relative to the largest, is 4.0×10^{-10} . The default tolerance of `numpy.linalg.lstsq`, $\max(n, p) \epsilon_M$ relative to σ_1 , treats the matrix as having full rank. It returns coefficients as large as 8.1×10^5 , which spend one more degree of freedom to reduce the residual sum of squares

from 1768678.4 to 1768252.5, by 0.02 per cent. With the tolerance set to 10^{-8} , the fourth column is treated as aliased, and the largest coefficient is 38.41.

```
rng = np.random.default_rng(108)
z = X[:, 1] + X[:, 2] + 1e-9 * np.linalg.norm(X[:, 1]) * rng.normal(size=n) / np.
    sqrt(n)
Xz = np.column_stack([X[:, :3], z]) # constant, value, capital, value +
    capital + tiny
s = np.linalg.svd(Xz, compute_uv=False)
print("relative singular values:", np.array2string(s / s[0], precision=2))
for tol in [None, 1e-8]:
    b, _, rank, _ = np.linalg.lstsq(Xz, y, rcond=tol)
    print(f"rcond = {tol}: rank {rank}, largest |b_j| = {np.abs(b).max():.3g},"
          f" SSE = {np.sum((y - Xz @ b) ** 2):.1f}")
```

Neither answer is wrong as arithmetic. But a column that differs from a combination of the others only at the level of 10^{-9} of its size carries no information a real measurement could support. Three rules follow. Scale the columns to comparable lengths before deciding the rank (Proposition 10.5.2); R's relative test does this implicitly. Tie the tolerance to the accuracy of the data, not of the machine: regressors recorded to six significant figures cannot resolve near dependences below about 10^{-6} of the column lengths (Proposition 10.4.2, applied to the recording error). And look for a gap, as in Figure 10.7.1(b). Without one the rank decision is fragile, the data are showing collinearity (Chapter 26), and shrinkage is often a better response than a sharp cut-off (Chapter 27).

Idea. Pivoted QR brings the most independent columns to the front. A small trailing pivot proves a near dependence, and setting those coefficients aside changes nothing estimable. The tolerance that decides “small” is a statement about the accuracy of the data.

10.7.4 Exercises

10.7.4.1 A. Check your understanding

Exercise A1.

Apply column-pivoted QR (Theorem 10.7.2) by hand to the matrix with columns $\mathbf{x}_1 = (1, 1, 0, 0)^\top$, $\mathbf{x}_2 = (0, 0, 1, 1)^\top$ and $\mathbf{x}_3 = (1, 1, 1, 1)^\top$ (use Gram–Schmidt on the pivoted columns, which gives the same $|r_{ij}|$). Find the pivot order, $\mathbf{R}$, and the rank. Give the basic solution for $\mathbf{y} = (1, 2, 3, 6)^\top$.

Solution. The norms are $\sqrt{2}, \sqrt{2}, 2$, so $\mathbf{x}_3$ is chosen first, with $|r_{11}| = 2$ and $\mathbf{q}_1 = \mathbf{x}_3/2$. Then $\mathbf{q}_1^\top \mathbf{x}_1 = \mathbf{q}_1^\top \mathbf{x}_2 = 1$, and the remainders are $\mathbf{x}_1 - \mathbf{q}_1 = (\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2})^\top$ and $\mathbf{x}_2 - \mathbf{q}_1$, its negative. Both have norm 1, so the tie goes to $\mathbf{x}_1$, with $|r_{22}| = 1$. The remainder of $\mathbf{x}_2$ is then zero, so $r_{33} = 0$ and the rank is 2. The pivot order is $(3, 1, 2)$, and up to signs $\mathbf{R} = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$. The basic solution regresses $\mathbf{y}$ on $\mathbf{x}_3$ and $\mathbf{x}_1$ with $b_2 = 0$. The fit is the pair of group means, 1.5 for the first two observations and 4.5 for the last two. So $b_3 = 4.5$ and $b_1 = 1.5 - 4.5 = -3$.

10.7.4.2 B. Practice

Exercise B1.

In the setting of this section, show that the basic solution and the minimum-norm solution have the same fitted values. Show that the minimum-norm solution is orthogonal to $\mathcal{N}(\mathbf{X})$, while the basic solution in general is not. For the one-way layout of [Example \(Four answers, one fit\)](#), which generalized inverse in that example gives the basic solution that drops the intercept?

Solution. Both satisfy $\mathbf{R}_{11}\mathbf{b}_1 + \mathbf{R}_{12}\mathbf{b}_2 = \mathbf{c}_1$, so both are least squares estimates ([Proposition 10.3.3](#)), and by [Theorem 6.4.1](#) both have fitted values $\mathbf{M}\mathbf{y}$. The minimum-norm solution lies in $\mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$ ([Theorem 10.4.1\(b\)](#)). The basic solution has zeros in the positions of the dropped columns. A null vector with a nonzero entry in one of those positions is generally not orthogonal to it. For the one-way layout, dropping the intercept column and fitting the three indicators is the “drop intercept” generalized inverse of [Example \(Four answers, one fit\)](#), which gives the group means as coefficients and a zero intercept.

Exercise B2.

Show that $|r_{11}| = \max_j \|\mathbf{x}_j\|$, and deduce $|r_{11}| \leq \sigma_1(\mathbf{X}) \leq \sqrt{p}|r_{11}|$. Combine with [Theorem 10.7.2\(b\)](#) to show that $\kappa(\mathbf{X}) \geq |r_{11}/r_{pp}|$, so that a large ratio of pivots proves ill-conditioning.

10.7.4.3 C. Going deeper

Exercise C1.

For Kahan’s matrix $\mathbf{K}_m(c)$, find a vector $\mathbf{x}$ with $\|\mathbf{x}\| = 1$ and $\|\mathbf{K}_m\mathbf{x}\|$ exponentially small in m when c is fixed. (Solve $\mathbf{K}_m\mathbf{x} = \mathbf{e}_m s^{m-1}$ by back substitution and examine the growth of the entries.) Why does column pivoting not detect this?

10.8 Summary and Notes

10.8.1 Summary

Idea.

1. Floating-point arithmetic commits a relative error of at most $u = 2^{-53}$ per operation ([Definition 10.1.1](#)). Inner products are backward stable ([Proposition 10.1.2](#)), but forming $\mathbf{X}^\top \mathbf{X}$ destroys the information in singular values below about $u^{1/2}\sigma_1$ ([Proposition 10.1.3](#), [Example \(A cross-product matrix that rounds to a singular one\)](#)).
2. The Cholesky factorization of the augmented cross-product matrix gives $\hat{\boldsymbol{\beta}}$, the SSE, the sequential sums of squares, the leverages and the standard errors in

- $np^2 + p^3/3$ flops ([Theorem 10.2.1](#), [Proposition 10.2.2](#)), accurately enough when the column-scaled condition number is modest. Sweeping fits nested models ([Proposition 10.2.3](#)); centring needs two passes or updating ([Proposition 10.2.4](#)).
3. Householder QR reduces $\mathbf{X}$ to a triangle by reflections in $2np^2 - 2p^3/3$ flops and is backward stable with columnwise errors ([Theorem 10.3.2](#), [Proposition 10.3.3](#)). Givens rotations work one row at a time, for streaming and sparse data ([Theorem 10.3.4](#)).
 4. The SVD gives every least squares solution, the minimum-norm one, and the precision σ/σ_i of each combination $\mathbf{v}_i^\top \boldsymbol{\beta}$ ([Theorem 10.4.1](#)). Singular values are perturbed by at most $\|\mathbf{E}\|_2$ ([Proposition 10.4.2](#)), so rank is meaningful only relative to a tolerance ([Definition 10.4.1](#)). Truncated SVD trades bias for variance ([Proposition 10.4.3](#)).
 5. The least squares solution has sensitivity $\kappa + \kappa^2\eta$, where η measures the residual relative to the fit; residuals have sensitivity κ only ([Theorem 10.5.1](#)). The normal equations add κ^2 regardless. Equilibrating the columns is nearly optimal among scalings ([Proposition 10.5.2](#), [Example \(Longley's data, exactly\)](#)).
 6. A triangular factor can be updated or downdated by one row in $O(p^2)$ ([Theorem 10.6.1](#)), and by one column in $O(np)$ ([Proposition 10.6.3](#)). All leave-one-out residuals, coefficients and sums of squares follow from a single fit ([Proposition 10.6.2](#)).
 7. QR with column pivoting orders the columns by how much each adds to the others, and a small trailing pivot certifies a small singular value ([Theorem 10.7.2](#)). Basic and minimum-norm solutions agree on everything estimable ([Example \(Firm effects in investment data\)](#)); the tolerance should reflect the accuracy of the data ([Example \(A nearly aliased column\)](#)).

10.8.2 Notes and sources

Source coverage. The chapter covers Seber and Lee (2003, chapter 11) on solving the normal equations, QR by Gram–Schmidt, Householder and Givens, the SVD, centring and leverages (sections 11.2–11.4, 11.7 and 11.10), with parts of sections 11.6, 11.8 and 11.9 on updating, comparing methods and rank deficiency.

Floating point and stability. The standard model and the γ_n notation follow Higham (2002), whose chapters 3, 10, 19 and 20 are the definitive treatment of inner products, Cholesky, QR and least squares. Goldberg (1991) is a readable introduction to IEEE arithmetic, now specified by IEEE Std 754-2019. The distinction between a problem's condition and an algorithm's backward stability goes back to Wilkinson (1963). Trefethen and Bau (1997) give a short modern account. Läuchli (1961) introduced the matrix of [Example \(A cross-product matrix that rounds to a singular one\)](#) to show the loss in forming normal equations.

Algorithms. Householder (1958) introduced the reflections and Golub (1965) applied them to least squares; Givens (1958) used plane rotations. Gentleman (1973) developed square-root-free rotations and the row-wise algorithm, turned into widely used routines by Miller (1992). Björck (1967) analysed modified Gram–Schmidt, and Björck and Paige (1992) explained its good behaviour for least squares. The SVD algorithm is due to Golub and Kahan (1965) and Golub and Reinsch (1970), column pivoting to Businger and Golub (1965), and the counterexample of [Example \(Kahan's matrix\)](#) to Kahan (1966). Chan (1987) gave the first rank-revealing QR algorithm, Hong and Pan (1992) proved that factorizations with polynomial bounds exist, and

Gu and Eisenstat (1996) showed how to compute them efficiently. Goodnight (1979) describes the sweep operator for statisticians. Welford (1962) gave the updating recursion, and Chan, Golub and LeVeque (1983) compare variance algorithms. Downdating appears in LINPACK (Dongarra et al. 1979) and was analysed by Stewart (1979). The standard monographs are Björck (1996), Lawson and Hanson (1974) and Golub and Van Loan (2013); Thisted (1988) and Kennedy and Gentle (1980) take the statistician's side.

Perturbation theory. Golub and Wilkinson (1966) first observed that the least squares solution can be sensitive to κ^2 when the residual is large. Wedin (1973) gave sharp perturbation bounds for pseudoinverses and least squares. The equilibration result of [Proposition 10.5.2](#) is from van der Sluis (1969). The inequality of [Proposition 10.4.2](#) is the singular-value form of Weyl's (1912) inequality for eigenvalues. Beaton, Rubin and Barone (1976) argued that errors from rounding the data should be the yardstick for computational accuracy, the point made at the end of [Section 10.5](#).

Statistics. Recursive least squares via the Sherman–Morrison formula goes back to Plackett (1950). The PRESS statistic is due to Allen (1974). The recursive residuals of [Exercise \(C1\)](#) are those of Brown, Durbin and Evans (1975). Deletion diagnostics are developed in Chapter 20, collinearity in Chapter 26, and ridge and principal components regression in Chapter 27.

10.8.3 References

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